Core Flight Executive Users Guide

Generated by Doxygen 1.8.13

Contents

1 Core Flight Executive Documentation 2

2 Background 3

3 Applicable Documents 4

4 Dependencies 4

5 Acronyms 5

6 Glossary of Terms 6

7 Version Numbers 7

8 cFE Application Programmer's Interface (API) Reference 9

9 cFE Executive Services Overview 15

9.1 Terminology . . . 16

9.1.1 "Application" and "cFE Application" . . . 16

9.1.2 "Task" . . . 17

9.1.3 "Startup Script" . . . 17

9.2 Software Reset . . . 18

9.3 Reset Types and Subtypes . . . 18

9.4 Exception and Reset (ER) Log . . . 19

9.5 Application and Child Task Management . . . 19

9.6 Starting an Application . . . 19

9.7 Stopping an Application . . . 20

9.8 Restarting an Application . . . 20

9.9 Reloading an Application . . . 20

9.10 Listing Current Applications . . . 21

9.11 Listing Current Tasks . . . 22

9.12 Loading Common Libraries . . . 22
9.13 Basic File System . . . 22
9.14 Performance Data Collection . . . 23
9.14.1 Performance Data Collection Trigger Masks . . . 23
9.14.2 Starting to Collect Performance Data . . . 23
9.14.3 Stopping the Collection of Performance Data . . . 24
9.14.4 Viewing the Collection of Performance Data . . . 24
9.15 Critical Data Store . . . 24
9.16 Memory Pool . . . 25
9.17 System Log . . . 27
9.18 Version Identification . . . 27
9.19 Executive Services Frequently Asked Questions . . . 28

10 cFE Executive Services Commands **28**

11 cFE Executive Services Telemetry **29**

12 cFE Executive Services Configuration Parameters **29**

13 cFE Event Services Overview **33**

13.1 Event Message Format . . . 33
13.2 Local Event Log . . . 34
13.3 Event Message Control . . . 35
13.4 Event Message Filtering . . . 37
13.5 EVS Registry . . . 37
13.6 EVS Counters . . . 38
13.7 Resetting EVS Counters . . . 39
13.8 Effects of a Processor Reset on EVS . . . 39
13.9 Frequently Asked Questions about Event Services . . . 39

14 cFE Event Services Commands **40**

15 cFE Event Services Telemetry **42**

16 cFE Event Services Configuration Parameters **42**

17 cFE Software Bus Overview **43**

17.1 Software Bus Terminology . . . 43

17.1.1 Messages . . . 44

17.1.2 Pipes . . . 44

17.1.3 Subscriptions . . . 45

17.1.4 Memory . . . 45

17.2 Autonomous Actions . . . 46

17.3 Operation of the SB Software . . . 46

17.3.1 Initialization . . . 47

17.3.2 All Resets . . . 47

17.3.3 Message Routing . . . 47

17.3.4 Packet Sequence Values . . . 48

17.3.5 Message Limit Error . . . 48

17.3.6 Pipe Overflow Error . . . 49

17.3.7 SB Event Filtering . . . 49

17.3.8 Diagnostic Data . . . 49

17.3.9 Control of Packet Routing . . . 50

17.3.10 Quality of Service . . . 50

17.3.11 Known Problem . . . 50

17.4 Frequently Asked Questions about Software Bus . . . 50

18 cFE Software Bus Commands **52**

19 cFE Software Bus Telemetry **53**

20 cFE Software Bus Configuration Parameters **53**

21 cFE Table Services Overview **54**

21.1 Managing Tables 55

21.2 cFE Table Types and Table Options 56

21.2.1 Single Buffered Tables 56

21.2.2 Double Buffered Tables 57

21.2.3 Tables with Validation Functions 57

21.2.4 Critical Tables 57

21.2.5 User Defined Address Tables 58

21.2.6 Dump Only Tables 58

21.3 Table Registry 58

21.4 Table Services Telemetry 59

21.5 Effects of Processor Reset on Tables 59

21.6 Frequently Asked Questions about Table Services 60

22 cFE Table Services Commands **61**

23 cFE Table Services Telemetry **62**

24 cFE Table Services Configuration Parameters **62**

25 cFE Time Services Overview **63**

25.1 Time Components 65

25.2 Time Structure 65

25.3 Time Formats 66

25.4 Time Configuration 66

25.4.1 Time Format Selection 67

25.4.2 Enabling Fake Tone Signal 68

25.4.3 Selecting Tone and Data Ordering 68

25.4.4 Specifying Tone and Data Window 69

25.4.5 Specifying Time Server/Client 69

25.4.6 Specifying Time Tone Byte Order . . . 70
25.4.7 Virtual MET . . . 70
25.4.8 Specifying Time Source . . . 70
25.4.9 Specifying Time Signal . . . 71
25.5 Time Format Selection . . . 72
25.6 Enabling Fake Tone Signal . . . 72
25.7 Selecting Tone and Data Ordering . . . 73
25.8 Specifying Tone and Data Window . . . 73
25.9 Specifying Time Server/Client . . . 74
25.10 Specifying Time Tone Byte Order . . . 74
25.11 Virtual MET . . . 74
25.12 Specifying Time Source . . . 75
25.13 Specifying Time Signal . . . 76
25.14 Time Services Paradigm(s) . . . 76
25.15 Flywheeling . . . 77
25.16 Time State . . . 77
25.17 Initialization . . . 78
25.17.1 Power-On Reset . . . 78
25.17.2 Processor Reset . . . 78
25.18 Power-On Reset . . . 79
25.19 Processor Reset . . . 79
25.20 Initialization . . . 79
25.20.1 Power-On Reset . . . 80
25.20.2 Processor Reset . . . 80
25.21 Power-On Reset . . . 80
25.22 Processor Reset . . . 81
25.23 Normal Operation . . . 81
25.23.1 Client . . . 81
25.23.2 Server . . . 82
25.24 Client . . . 83
25.25 Server . . . 84
25.26 Setting Time . . . 85
25.27 Adjusting Time . . . 86
25.28 Setting MET . . . 86
25.29 Frequently Asked Questions . . . 86

26 cFE Time Services Commands **87**

27 cFE Time Services Telemetry **88**

28 cFE Time Services Configuration Parameters **88**

29 cFE Event Message Cross Reference **89**

30 cFE Command Mnemonic Cross Reference **104**

31 cFE Telemetry Mnemonic Cross Reference **108**

32 cFE Mission Configuration Parameters **119**

33 Deprecated List **119**

34 Module Index **120**

34.1 Modules 120

35 Data Structure Index **123**

35.1 Data Structures 123

36 File Index **131**

36.1 File List 131

37 Module Documentation **134**

37.1 cFE Return Code Defines 134

37.1.1 Detailed Description 140

37.1.2 Macro Definition Documentation 140

37.2 cFE Resource ID APIs 174

37.2.1 Detailed Description 174

37.2.2 Function Documentation 174

37.3 cFE Entry/Exit APIs 178

37.3.1 Detailed Description 178

37.3.2 Function Documentation . . . 178
37.4 cFE Application Control APIs . . . 180
37.4.1 Detailed Description . . . 180
37.4.2 Function Documentation . . . 180
37.5 cFE Application Behavior APIs . . . 183
37.5.1 Detailed Description . . . 183
37.5.2 Function Documentation . . . 183
37.6 cFE Information APIs . . . 188
37.6.1 Detailed Description . . . 188
37.6.2 Function Documentation . . . 188
37.7 cFE Child Task APIs . . . 198
37.7.1 Detailed Description . . . 198
37.7.2 Function Documentation . . . 198
37.8 cFE Miscellaneous APIs . . . 204
37.8.1 Detailed Description . . . 204
37.8.2 Function Documentation . . . 204
37.9 cFE Critical Data Store APIs . . . 208
37.9.1 Detailed Description . . . 208
37.9.2 Function Documentation . . . 208
37.10 cFE Memory Manager APIs . . . 213
37.10.1 Detailed Description . . . 213
37.10.2 Function Documentation . . . 213
37.11 cFE Performance Monitor APIs . . . 221
37.11.1 Detailed Description . . . 221
37.11.2 Macro Definition Documentation . . . 221
37.11.3 Function Documentation . . . 222
37.12 cFE Generic Counter APIs . . . 224
37.12.1 Detailed Description . . . 224

37.12.2 Function Documentation 224
37.13cFE Internal Executive Service APIs, internal to CFE core 231
37.13.1 Detailed Description 231
37.13.2 Function Documentation 231
37.14cFE Registration APIs 234
37.14.1 Detailed Description 234
37.14.2 Function Documentation 234
37.15cFE Send Event APIs 237
37.15.1 Detailed Description 237
37.15.2 Function Documentation 237
37.16cFE Reset Event Filter APIs 242
37.16.1 Detailed Description 242
37.16.2 Function Documentation 242
37.17cFE Internal Event Services APIs, internal to CFE core 244
37.17.1 Detailed Description 244
37.17.2 Function Documentation 244
37.18cFE File Header Management APIs 246
37.18.1 Detailed Description 246
37.18.2 Function Documentation 246
37.19cFE File Utility APIs 250
37.19.1 Detailed Description 250
37.19.2 Function Documentation 250
37.20cFE Internal File Service APIs, internal to CFE core 256
37.20.1 Detailed Description 256
37.20.2 Function Documentation 256
37.21cFE Message header APIs 257
37.21.1 Detailed Description 258
37.21.2 Function Documentation 258

37.22cFE Message Id APIs . . . 279
37.22.1 Detailed Description . . . 279
37.22.2 Function Documentation . . . 279
37.23cFE Pipe Management APIs . . . 282
37.23.1 Detailed Description . . . 282
37.23.2 Function Documentation . . . 282
37.24cFE Message Subscription Control APIs . . . 289
37.24.1 Detailed Description . . . 289
37.24.2 Function Documentation . . . 289
37.25cFE Send/Receive Message APIs . . . 294
37.25.1 Detailed Description . . . 294
37.25.2 Function Documentation . . . 294
37.26cFE Zero Copy APIs . . . 299
37.26.1 Detailed Description . . . 299
37.26.2 Function Documentation . . . 299
37.27cFE Setting Message Characteristics APIs . . . 304
37.27.1 Detailed Description . . . 304
37.27.2 Function Documentation . . . 304
37.28cFE Getting Message Characteristics APIs . . . 310
37.28.1 Detailed Description . . . 310
37.28.2 Function Documentation . . . 310
37.29cFE Checksum Control APIs . . . 316
37.29.1 Detailed Description . . . 316
37.29.2 Function Documentation . . . 316
37.30cFE Message ID APIs . . . 319
37.30.1 Detailed Description . . . 319
37.30.2 Function Documentation . . . 319
37.31cFE SB Packet Type Defines . . . 323

37.31.1 Detailed Description . . . 323
37.31.2 Macro Definition Documentation . . . 323
37.32cFE Internal Software Bus APIs, internal to CFE core . . . 324
37.32.1 Detailed Description . . . 324
37.32.2 Function Documentation . . . 324
37.33cFE Registration APIs . . . 326
37.33.1 Detailed Description . . . 326
37.33.2 Function Documentation . . . 326
37.34cFE Manage Table Content APIs . . . 332
37.34.1 Detailed Description . . . 332
37.34.2 Function Documentation . . . 332
37.35cFE Access Table Content APIs . . . 339
37.35.1 Detailed Description . . . 339
37.35.2 Function Documentation . . . 339
37.36cFE Get Table Information APIs . . . 345
37.36.1 Detailed Description . . . 345
37.36.2 Function Documentation . . . 345
37.37cFE Table Type Defines . . . 349
37.37.1 Detailed Description . . . 349
37.37.2 Macro Definition Documentation . . . 349
37.38cFE Internal Table Service APIs, internal to CFE core . . . 352
37.38.1 Detailed Description . . . 352
37.38.2 Function Documentation . . . 352
37.39cFE Get Current Time APIs . . . 354
37.39.1 Detailed Description . . . 354
37.39.2 Function Documentation . . . 354
37.40cFE Get Time Information APIs . . . 358
37.40.1 Detailed Description . . . 358

37.40.2 Function Documentation . . . 358
37.41 cFE Time Arithmetic APIs . . . 361
37.41.1 Detailed Description . . . 361
37.41.2 Function Documentation . . . 361
37.42 cFE Time Conversion APIs . . . 364
37.42.1 Detailed Description . . . 364
37.42.2 Function Documentation . . . 364
37.43 cFE External Time Source APIs . . . 367
37.43.1 Detailed Description . . . 367
37.43.2 Function Documentation . . . 367
37.44 cFE Miscellaneous Time APIs . . . 372
37.44.1 Detailed Description . . . 372
37.44.2 Function Documentation . . . 372
37.45 cFE Internal Time APIs, internal to CFE core . . . 374
37.45.1 Detailed Description . . . 374
37.45.2 Function Documentation . . . 374
37.46 cFE Clock State Flag Defines . . . 376
37.46.1 Detailed Description . . . 376
37.46.2 Macro Definition Documentation . . . 376
37.47 OSAL Semaphore State Defines . . . 379
37.47.1 Detailed Description . . . 379
37.47.2 Macro Definition Documentation . . . 379
37.48 OSAL Binary Semaphore APIs . . . 380
37.48.1 Detailed Description . . . 380
37.48.2 Function Documentation . . . 380
37.49 OSAL BSP low level access APIs . . . 386
37.49.1 Detailed Description . . . 386
37.49.2 Function Documentation . . . 386

37.50 OSAL Real Time Clock APIs . . . 387
37.50.1 Detailed Description . . . 387
37.50.2 Function Documentation . . . 387
37.51 OSAL Core Operation APIs . . . 400
37.51.1 Detailed Description . . . 400
37.51.2 Function Documentation . . . 400
37.52 OSAL Counting Semaphore APIs . . . 405
37.52.1 Detailed Description . . . 405
37.52.2 Function Documentation . . . 405
37.53 OSAL Directory APIs . . . 410
37.53.1 Detailed Description . . . 410
37.53.2 Function Documentation . . . 410
37.54 OSAL Return Code Defines . . . 414
37.54.1 Detailed Description . . . 415
37.54.2 Macro Definition Documentation . . . 415
37.55 OSAL Error Info APIs . . . 424
37.55.1 Detailed Description . . . 424
37.55.2 Function Documentation . . . 424
37.56 OSAL File Access Option Defines . . . 425
37.56.1 Detailed Description . . . 425
37.56.2 Macro Definition Documentation . . . 425
37.57 OSAL Reference Point For Seek Offset Defines . . . 426
37.57.1 Detailed Description . . . 426
37.57.2 Macro Definition Documentation . . . 426
37.58 OSAL Standard File APIs . . . 427
37.58.1 Detailed Description . . . 427
37.58.2 Function Documentation . . . 427
37.59 OSAL File System Level APIs . . . 439

37.59.1 Detailed Description 439
37.59.2 Function Documentation 439
37.60 OSAL Heap APIs 447
37.60.1 Detailed Description 447
37.60.2 Function Documentation 447
37.61 OSAL Object Type Defines 448
37.61.1 Detailed Description 448
37.61.2 Macro Definition Documentation 448
37.62 OSAL Object ID Utility APIs 452
37.62.1 Detailed Description 452
37.62.2 Function Documentation 452
37.63 OSAL Dynamic Loader and Symbol APIs 459
37.63.1 Detailed Description 459
37.63.2 Function Documentation 459
37.64 OSAL Mutex APIs 464
37.64.1 Detailed Description 464
37.64.2 Function Documentation 464
37.65 Network ID APIs 469
37.65.1 Detailed Description 469
37.65.2 Function Documentation 469
37.66 OSAL Printf APIs 471
37.66.1 Detailed Description 471
37.66.2 Function Documentation 471
37.67 OSAL Message Queue APIs 473
37.67.1 Detailed Description 473
37.67.2 Function Documentation 473
37.68 OSAL Select APIs 478
37.68.1 Detailed Description 478

37.68.2 Function Documentation 478

37.69 OSAL Shell APIs 481

37.69.1 Detailed Description 481

37.69.2 Function Documentation 481

37.70 OSAL Socket Address APIs 482

37.70.1 Detailed Description 482

37.70.2 Function Documentation 482

37.71 OSAL Socket Management APIs 486

37.71.1 Detailed Description 486

37.71.2 Function Documentation 486

37.72 OSAL Task APIs 493

37.72.1 Detailed Description 493

37.72.2 Function Documentation 493

37.73 OSAL Time Base APIs 501

37.73.1 Detailed Description 501

37.73.2 Function Documentation 501

37.74 OSAL Timer APIs 506

37.74.1 Detailed Description 506

37.74.2 Function Documentation 506

38 Data Structure Documentation 512

38.1 CCSDS_ExtendedHeader Struct Reference . . . 512

38.1.1 Detailed Description . . . 512

38.1.2 Field Documentation . . . 512

38.2 CCSDS_PrimaryHeader Struct Reference . . . 513

38.2.1 Detailed Description . . . 513

38.2.2 Field Documentation . . . 513

38.3 CFE_ES_AppInfo Struct Reference . . . 514

38.3.1 Detailed Description . . . 515

38.3.2 Field Documentation . . . 515

38.4 CFE_ES_AppNameCmd Struct Reference . . . 520

38.4.1 Detailed Description . . . 521

38.4.2 Field Documentation . . . 521

38.5 CFE_ES_AppNameCmd_Payload Struct Reference . . . 521

38.5.1 Detailed Description . . . 522

38.5.2 Field Documentation . . . 522

38.6 CFE_ES_AppReloadCmd_Payload Struct Reference . . . 522

38.6.1 Detailed Description . . . 523

38.6.2 Field Documentation . . . 523

38.7 CFE_ES_BlockStats Struct Reference . . . 523

38.7.1 Detailed Description . . . 524

38.7.2 Field Documentation . . . 524

38.8 CFE_ES_CDSRegDumpRec Struct Reference . . . 524

38.8.1 Detailed Description . . . 525

38.8.2 Field Documentation . . . 525

38.9 CFE_ES_DeleteCDSCmd Struct Reference . . . 526

38.9.1 Detailed Description . . . 527

38.9.2 Field Documentation . . . 527

38.10 CFE_ES_DeleteCDSCmd_Payload Struct Reference . . . 527
38.10.1 Detailed Description . . . 528
38.10.2 Field Documentation . . . 528
38.11 CFE_ES_DumpCDSRegistryCmd Struct Reference . . . 528
38.11.1 Detailed Description . . . 528
38.11.2 Field Documentation . . . 529
38.12 CFE_ES_DumpCDSRegistryCmd_Payload Struct Reference . . . 529
38.12.1 Detailed Description . . . 529
38.12.2 Field Documentation . . . 530
38.13 CFE_ES_FileNameCmd Struct Reference . . . 530
38.13.1 Detailed Description . . . 530
38.13.2 Field Documentation . . . 530
38.14 CFE_ES_FileNameCmd_Payload Struct Reference . . . 531
38.14.1 Detailed Description . . . 531
38.14.2 Field Documentation . . . 531
38.15 CFE_ES_HousekeepingTlm Struct Reference . . . 532
38.15.1 Detailed Description . . . 532
38.15.2 Field Documentation . . . 532
38.16 CFE_ES_HousekeepingTlm_Payload Struct Reference . . . 532
38.16.1 Detailed Description . . . 534
38.16.2 Field Documentation . . . 534
38.17 CFE_ES_MemPoolStats Struct Reference . . . 544
38.17.1 Detailed Description . . . 545
38.17.2 Field Documentation . . . 545
38.18 CFE_ES_MemStatsTlm Struct Reference . . . 546
38.18.1 Detailed Description . . . 547
38.18.2 Field Documentation . . . 547
38.19 CFE_ES_NoArgsCmd Struct Reference . . . 547

38.19.1 Detailed Description . . . 548
38.19.2 Field Documentation . . . 548
38.20 CFE_ES_OneAppTlm Struct Reference . . . 548
38.20.1 Detailed Description . . . 549
38.20.2 Field Documentation . . . 549
38.21 CFE_ES_OneAppTlm_Payload Struct Reference . . . 549
38.21.1 Detailed Description . . . 549
38.21.2 Field Documentation . . . 550
38.22 CFE_ES_OverWriteSysLogCmd Struct Reference . . . 550
38.22.1 Detailed Description . . . 550
38.22.2 Field Documentation . . . 550
38.23 CFE_ES_OverWriteSysLogCmd_Payload Struct Reference . . . 551
38.23.1 Detailed Description . . . 551
38.23.2 Field Documentation . . . 551
38.24 CFE_ES_PoolAlign Union Reference . . . 552
38.24.1 Detailed Description . . . 552
38.24.2 Field Documentation . . . 552
38.25 CFE_ES_PoolStatsTlm_Payload Struct Reference . . . 553
38.25.1 Detailed Description . . . 553
38.25.2 Field Documentation . . . 553
38.26 CFE_ES_ReloadAppCmd Struct Reference . . . 554
38.26.1 Detailed Description . . . 554
38.26.2 Field Documentation . . . 554
38.27 CFE_ES_RestartCmd Struct Reference . . . 555
38.27.1 Detailed Description . . . 555
38.27.2 Field Documentation . . . 555
38.28 CFE_ES_RestartCmd_Payload Struct Reference . . . 556
38.28.1 Detailed Description . . . 556

38.28.2 Field Documentation 556
38.29 CFE_ES_SendMemPoolStatsCmd Struct Reference 557
38.29.1 Detailed Description 557
38.29.2 Field Documentation 557
38.30 CFE_ES_SendMemPoolStatsCmd_Payload Struct Reference 558
38.30.1 Detailed Description 558
38.30.2 Field Documentation 558
38.31 CFE_ES_SetMaxPRCountCmd Struct Reference 559
38.31.1 Detailed Description 559
38.31.2 Field Documentation 559
38.32 CFE_ES_SetMaxPRCountCmd_Payload Struct Reference 560
38.32.1 Detailed Description 560
38.32.2 Field Documentation 560
38.33 CFE_ES_SetPerfFilterMaskCmd Struct Reference 560
38.33.1 Detailed Description 561
38.33.2 Field Documentation 561
38.34 CFE_ES_SetPerfFilterMaskCmd_Payload Struct Reference 561
38.34.1 Detailed Description 562
38.34.2 Field Documentation 562
38.35 CFE_ES_SetPerfTriggerMaskCmd Struct Reference 562
38.35.1 Detailed Description 563
38.35.2 Field Documentation 563
38.36 CFE_ES_SetPerfTrigMaskCmd_Payload Struct Reference 563
38.36.1 Detailed Description 564
38.36.2 Field Documentation 564
38.37 CFE_ES_StartApp Struct Reference 564
38.37.1 Detailed Description 565
38.37.2 Field Documentation 565

38.38 CFE_ES_StartAppCmd_Payload Struct Reference . . . 565
38.38.1 Detailed Description . . . 566
38.38.2 Field Documentation . . . 566
38.39 CFE_ES_StartPerfCmd_Payload Struct Reference . . . 567
38.39.1 Detailed Description . . . 568
38.39.2 Field Documentation . . . 568
38.40 CFE_ES_StartPerfDataCmd Struct Reference . . . 568
38.40.1 Detailed Description . . . 569
38.40.2 Field Documentation . . . 569
38.41 CFE_ES_StopPerfCmd_Payload Struct Reference . . . 569
38.41.1 Detailed Description . . . 570
38.41.2 Field Documentation . . . 570
38.42 CFE_ES_StopPerfDataCmd Struct Reference . . . 570
38.42.1 Detailed Description . . . 570
38.42.2 Field Documentation . . . 571
38.43 CFE_ES_TaskInfo Struct Reference . . . 571
38.43.1 Detailed Description . . . 572
38.43.2 Field Documentation . . . 572
38.44 CFE_EVS_AppDataCmd_Payload Struct Reference . . . 574
38.44.1 Detailed Description . . . 574
38.44.2 Field Documentation . . . 574
38.45 CFE_EVS_AppNameBitMaskCmd Struct Reference . . . 574
38.45.1 Detailed Description . . . 575
38.45.2 Field Documentation . . . 575
38.46 CFE_EVS_AppNameBitMaskCmd_Payload Struct Reference . . . 575
38.46.1 Detailed Description . . . 576
38.46.2 Field Documentation . . . 576
38.47 CFE_EVS_AppNameCmd Struct Reference . . . 577

38.47.1 Detailed Description 577
38.47.2 Field Documentation 577
38.48 CFE_EVS_AppNameCmd_Payload Struct Reference 578
38.48.1 Detailed Description 578
38.48.2 Field Documentation 578
38.49 CFE_EVS_AppNameEventIDCmd Struct Reference 579
38.49.1 Detailed Description 579
38.49.2 Field Documentation 579
38.50 CFE_EVS_AppNameEventIDCmd_Payload Struct Reference 580
38.50.1 Detailed Description 580
38.50.2 Field Documentation 580
38.51 CFE_EVS_AppNameEventIDMaskCmd Struct Reference 581
38.51.1 Detailed Description 581
38.51.2 Field Documentation 581
38.52 CFE_EVS_AppNameEventIDMaskCmd_Payload Struct Reference 582
38.52.1 Detailed Description 582
38.52.2 Field Documentation 582
38.53 CFE_EVS_AppTlmData Struct Reference 583
38.53.1 Detailed Description 583
38.53.2 Field Documentation 583
38.54 CFE_EVS_BinFilter Struct Reference 584
38.54.1 Detailed Description 585
38.54.2 Field Documentation 585
38.55 CFE_EVS_BitMaskCmd Struct Reference 585
38.55.1 Detailed Description 586
38.55.2 Field Documentation 586
38.56 CFE_EVS_BitMaskCmd_Payload Struct Reference 586
38.56.1 Detailed Description 587

38.56.2 Field Documentation . . . 587
38.57 CFE_EVS_HousekeepingTlm Struct Reference . . . 587
38.57.1 Detailed Description . . . 588
38.57.2 Field Documentation . . . 588
38.58 CFE_EVS_HousekeepingTlm_Payload Struct Reference . . . 588
38.58.1 Detailed Description . . . 589
38.58.2 Field Documentation . . . 589
38.59 CFE_EVS_LogFileCmd_Payload Struct Reference . . . 593
38.59.1 Detailed Description . . . 594
38.59.2 Field Documentation . . . 594
38.60 CFE_EVS_LongEventTlm Struct Reference . . . 594
38.60.1 Detailed Description . . . 594
38.60.2 Field Documentation . . . 595
38.61 CFE_EVS_LongEventTlm_Payload Struct Reference . . . 595
38.61.1 Detailed Description . . . 595
38.61.2 Field Documentation . . . 596
38.62 CFE_EVS_NoArgsCmd Struct Reference . . . 597
38.62.1 Detailed Description . . . 597
38.62.2 Field Documentation . . . 597
38.63 CFE_EVS_PacketID Struct Reference . . . 597
38.63.1 Detailed Description . . . 598
38.63.2 Field Documentation . . . 598
38.64 CFE_EVS_SetEventFormatCode_Payload Struct Reference . . . 599
38.64.1 Detailed Description . . . 599
38.64.2 Field Documentation . . . 600
38.65 CFE_EVS_SetEventFormatModeCmd Struct Reference . . . 600
38.65.1 Detailed Description . . . 600
38.65.2 Field Documentation . . . 601

38.66 CFE_EVS_SetLogMode_Payload Struct Reference . . . 601
38.66.1 Detailed Description . . . 601
38.66.2 Field Documentation . . . 602
38.67 CFE_EVS_SetLogModeCmd Struct Reference . . . 602
38.67.1 Detailed Description . . . 602
38.67.2 Field Documentation . . . 603
38.68 CFE_EVS_ShortEventTlm Struct Reference . . . 603
38.68.1 Detailed Description . . . 603
38.68.2 Field Documentation . . . 603
38.69 CFE_EVS_ShortEventTlm_Payload Struct Reference . . . 604
38.69.1 Detailed Description . . . 604
38.69.2 Field Documentation . . . 604
38.70 CFE_EVS_WriteAppDataFileCmd Struct Reference . . . 605
38.70.1 Detailed Description . . . 605
38.70.2 Field Documentation . . . 605
38.71 CFE_EVS_WriteLogDataFileCmd Struct Reference . . . 606
38.71.1 Detailed Description . . . 606
38.71.2 Field Documentation . . . 606
38.72 CFE_FS_FileWriteMetaData Struct Reference . . . 607
38.72.1 Detailed Description . . . 607
38.72.2 Field Documentation . . . 607
38.73 CFE_FS_Header Struct Reference . . . 609
38.73.1 Detailed Description . . . 609
38.73.2 Field Documentation . . . 609
38.74 CFE_PSP_CommandData_t Struct Reference . . . 611
38.74.1 Detailed Description . . . 612
38.74.2 Field Documentation . . . 612
38.75 CFE_PSP_LinuxReservedAreaFixedLayout_t Struct Reference . . . 614

38.75.1 Detailed Description 614
38.75.2 Field Documentation 614
38.76 CFE_PSP_MemTable_t Struct Reference 615
38.76.1 Detailed Description 615
38.76.2 Field Documentation 615
38.77 CFE_PSP_VersionInfo_t Struct Reference 616
38.77.1 Detailed Description 616
38.77.2 Field Documentation 616
38.78 CFE_SB_AllSubscriptionsTlm Struct Reference 618
38.78.1 Detailed Description 618
38.78.2 Field Documentation 618
38.79 CFE_SB_AllSubscriptionsTlm_Payload Struct Reference 618
38.79.1 Detailed Description 619
38.79.2 Field Documentation 619
38.80 CFE_SB_HousekeepingTlm Struct Reference 620
38.80.1 Detailed Description 620
38.80.2 Field Documentation 620
38.81 CFE_SB_HousekeepingTlm_Payload Struct Reference 621
38.81.1 Detailed Description 622
38.81.2 Field Documentation 622
38.82 CFE_SB_Msg Union Reference 626
38.82.1 Detailed Description 627
38.82.2 Field Documentation 627
38.83 CFE_SB_MsgMapFileEntry Struct Reference 628
38.83.1 Detailed Description 628
38.83.2 Field Documentation 628
38.84 CFE_SB_PipeDepthStats Struct Reference 629
38.84.1 Detailed Description 629

38.84.2 Field Documentation 629
38.85 CFE_SB_PipeInfoEntry Struct Reference 631
38.85.1 Detailed Description 631
38.85.2 Field Documentation 631
38.86 CFE_SB_Qos_t Struct Reference 633
38.86.1 Detailed Description 634
38.86.2 Field Documentation 634
38.87 CFE_SB_RouteCmd Struct Reference 634
38.87.1 Detailed Description 635
38.87.2 Field Documentation 635
38.88 CFE_SB_RouteCmd_Payload Struct Reference 635
38.88.1 Detailed Description 636
38.88.2 Field Documentation 636
38.89 CFE_SB_RoutingFileEntry Struct Reference 637
38.89.1 Detailed Description 637
38.89.2 Field Documentation 637
38.90 CFE_SB_SingleSubscriptionTlm Struct Reference 639
38.90.1 Detailed Description 639
38.90.2 Field Documentation 639
38.91 CFE_SB_SingleSubscriptionTlm_Payload Struct Reference 640
38.91.1 Detailed Description 640
38.91.2 Field Documentation 640
38.92 CFE_SB_StatsTlm Struct Reference 641
38.92.1 Detailed Description 642
38.92.2 Field Documentation 642
38.93 CFE_SB_StatsTlm_Payload Struct Reference 642
38.93.1 Detailed Description 643
38.93.2 Field Documentation 643

38.94 CFE_SB_SubEntries Struct Reference . . . 648
38.94.1 Detailed Description . . . 648
38.94.2 Field Documentation . . . 648
38.95 CFE_SB_WriteFileInfoCmd Struct Reference . . . 649
38.95.1 Detailed Description . . . 649
38.95.2 Field Documentation . . . 649
38.96 CFE_SB_WriteFileInfoCmd_Payload Struct Reference . . . 650
38.96.1 Detailed Description . . . 650
38.96.2 Field Documentation . . . 650
38.97 CFE_SB_ZeroCopyHandle_t Struct Reference . . . 651
38.97.1 Detailed Description . . . 651
38.97.2 Field Documentation . . . 651
38.98 CFE_TBL_AbortLoadCmd Struct Reference . . . 651
38.98.1 Detailed Description . . . 652
38.98.2 Field Documentation . . . 652
38.99 CFE_TBL_AbortLoadCmd_Payload Struct Reference . . . 652
38.99.1 Detailed Description . . . 653
38.99.2 Field Documentation . . . 653
38.100 CFE_TBL_ActivateCmd Struct Reference . . . 653
38.100.1 Detailed Description . . . 653
38.100.2 Field Documentation . . . 654
38.101 CFE_TBL_ActivateCmd_Payload Struct Reference . . . 654
38.101.1 Detailed Description . . . 654
38.101.2 Field Documentation . . . 655
38.102 CFE_TBL_DelCDSCmd_Payload Struct Reference . . . 655
38.102.1 Detailed Description . . . 655
38.102.2 Field Documentation . . . 655
38.103 CFE_TBL_DeleteCDSCmd Struct Reference . . . 656

38.103.1 Detailed Description . . . 656
38.103.2 Field Documentation . . . 656
38.104 CFE_TBL_DumpCmd Struct Reference . . . 657
38.104.1 Detailed Description . . . 657
38.104.2 Field Documentation . . . 657
38.105 CFE_TBL_DumpCmd_Payload Struct Reference . . . 658
38.105.1 Detailed Description . . . 658
38.105.2 Field Documentation . . . 658
38.106 CFE_TBL_DumpRegistryCmd Struct Reference . . . 659
38.106.1 Detailed Description . . . 660
38.106.2 Field Documentation . . . 660
38.107 CFE_TBL_DumpRegistryCmd_Payload Struct Reference . . . 660
38.107.1 Detailed Description . . . 661
38.107.2 Field Documentation . . . 661
38.108 CFE_TBL_File_Hdr Struct Reference . . . 661
38.108.1 Detailed Description . . . 661
38.108.2 Field Documentation . . . 662
38.109 CFE_TBL_FileDef Struct Reference . . . 662
38.109.1 Detailed Description . . . 663
38.109.2 Field Documentation . . . 663
38.110 CFE_TBL_HousekeepingTlm Struct Reference . . . 664
38.110.1 Detailed Description . . . 664
38.110.2 Field Documentation . . . 664
38.111 CFE_TBL_HousekeepingTlm_Payload Struct Reference . . . 665
38.111.1 Detailed Description . . . 666
38.111.2 Field Documentation . . . 666
38.112 CFE_TBL_Info Struct Reference . . . 671
38.112.1 Detailed Description . . . 672

38.112.2 Field Documentation . . . 672
38.113 CFE_TBL_LoadCmd Struct Reference . . . 675
38.113.1 Detailed Description . . . 675
38.113.2 Field Documentation . . . 675
38.114 CFE_TBL_LoadCmd_Payload Struct Reference . . . 676
38.114.1 Detailed Description . . . 676
38.114.2 Field Documentation . . . 676
38.115 CFE_TBL_NoArgsCmd Struct Reference . . . 677
38.115.1 Detailed Description . . . 677
38.115.2 Field Documentation . . . 677
38.116 CFE_TBL_NotifyCmd Struct Reference . . . 677
38.116.1 Detailed Description . . . 678
38.116.2 Field Documentation . . . 678
38.117 CFE_TBL_NotifyCmd_Payload Struct Reference . . . 678
38.117.1 Detailed Description . . . 679
38.117.2 Field Documentation . . . 679
38.118 CFE_TBL_SendRegistryCmd Struct Reference . . . 679
38.118.1 Detailed Description . . . 680
38.118.2 Field Documentation . . . 680
38.119 CFE_TBL_SendRegistryCmd_Payload Struct Reference . . . 680
38.119.1 Detailed Description . . . 681
38.119.2 Field Documentation . . . 681
38.120 CFE_TBL_TableRegistryTlm Struct Reference . . . 681
38.120.1 Detailed Description . . . 681
38.120.2 Field Documentation . . . 682
38.121 CFE_TBL_TblRegPacket_Payload Struct Reference . . . 682
38.121.1 Detailed Description . . . 683
38.121.2 Field Documentation . . . 683

38.122 CFE_TBL_ValidateCmd Struct Reference . . . 688
38.122.1 Detailed Description . . . 688
38.122.2 Field Documentation . . . 688
38.123 CFE_TBL_ValidateCmd_Payload Struct Reference . . . 689
38.123.1 Detailed Description . . . 689
38.123.2 Field Documentation . . . 689
38.124 CFE_TIME_DiagnosticTlm Struct Reference . . . 690
38.124.1 Detailed Description . . . 690
38.124.2 Field Documentation . . . 690
38.125 CFE_TIME_DiagnosticTlm_Payload Struct Reference . . . 690
38.125.1 Detailed Description . . . 692
38.125.2 Field Documentation . . . 693
38.126 CFE_TIME_HousekeepingTlm Struct Reference . . . 703
38.126.1 Detailed Description . . . 703
38.126.2 Field Documentation . . . 704
38.127 CFE_TIME_HousekeepingTlm_Payload Struct Reference . . . 704
38.127.1 Detailed Description . . . 705
38.127.2 Field Documentation . . . 705
38.128 CFE_TIME_LeapsCmd_Payload Struct Reference . . . 708
38.128.1 Detailed Description . . . 709
38.128.2 Field Documentation . . . 709
38.129 CFE_TIME_NoArgsCmd Struct Reference . . . 709
38.129.1 Detailed Description . . . 709
38.129.2 Field Documentation . . . 710
38.130 CFE_TIME_OneHzAdjustmentCmd Struct Reference . . . 710
38.130.1 Detailed Description . . . 710
38.130.2 Field Documentation . . . 710
38.131 CFE_TIME_OneHzAdjustmentCmd_Payload Struct Reference . . . 711

38.131.1 Detailed Description 711
38.131.2 Field Documentation 711
38.132 CFE_TIME_SetLeapSecondsCmd Struct Reference 712
38.132.1 Detailed Description 712
38.132.2 Field Documentation 712
38.133 CFE_TIME_SetSignalCmd Struct Reference 713
38.133.1 Detailed Description 713
38.133.2 Field Documentation 713
38.134 CFE_TIME_SetSourceCmd Struct Reference 714
38.134.1 Detailed Description 714
38.134.2 Field Documentation 714
38.135 CFE_TIME_SetStateCmd Struct Reference 715
38.135.1 Detailed Description 715
38.135.2 Field Documentation 715
38.136 CFE_TIME_SignalCmd_Payload Struct Reference 716
38.136.1 Detailed Description 716
38.136.2 Field Documentation 716
38.137 CFE_TIME_SourceCmd_Payload Struct Reference 717
38.137.1 Detailed Description 717
38.137.2 Field Documentation 717
38.138 CFE_TIME_StateCmd_Payload Struct Reference 717
38.138.1 Detailed Description 718
38.138.2 Field Documentation 718
38.139 CFE_TIME_SysTime Struct Reference 718
38.139.1 Detailed Description 718
38.139.2 Field Documentation 719
38.140 CFE_TIME_TimeCmd Struct Reference 719
38.140.1 Detailed Description 719

38.140.2 Field Documentation . . . 720
38.141 CFE_TIME_TimeCmd_Payload Struct Reference . . . 720
38.141.1 Detailed Description . . . 720
38.141.2 Field Documentation . . . 720
38.142 CFE_TIME_ToneDataCmd Struct Reference . . . 721
38.142.1 Detailed Description . . . 721
38.142.2 Field Documentation . . . 721
38.143 CFE_TIME_ToneDataCmd_Payload Struct Reference . . . 722
38.143.1 Detailed Description . . . 722
38.143.2 Field Documentation . . . 723
38.144 OS_bin_sem_prop_t Struct Reference . . . 724
38.144.1 Detailed Description . . . 724
38.144.2 Field Documentation . . . 724
38.145 OS_count_sem_prop_t Struct Reference . . . 725
38.145.1 Detailed Description . . . 725
38.145.2 Field Documentation . . . 725
38.146 os_dirent_t Struct Reference . . . 726
38.146.1 Detailed Description . . . 726
38.146.2 Field Documentation . . . 726
38.147 OS_FdSet Struct Reference . . . 726
38.147.1 Detailed Description . . . 727
38.147.2 Field Documentation . . . 727
38.148 OS_file_prop_t Struct Reference . . . 727
38.148.1 Detailed Description . . . 727
38.148.2 Field Documentation . . . 728
38.149 os_fsinfo_t Struct Reference . . . 728
38.149.1 Detailed Description . . . 729
38.149.2 Field Documentation . . . 729

38.150 os_fstat_t Struct Reference . . . 730
38.150.1 Detailed Description . . . 730
38.150.2 Field Documentation . . . 730
38.151 OS_heap_prop_t Struct Reference . . . 731
38.151.1 Detailed Description . . . 731
38.151.2 Field Documentation . . . 731
38.152 OS_module_address_t Struct Reference . . . 732
38.152.1 Detailed Description . . . 732
38.152.2 Field Documentation . . . 732
38.153 OS_module_prop_t Struct Reference . . . 734
38.153.1 Detailed Description . . . 734
38.153.2 Field Documentation . . . 734
38.154 OS_mut_sem_prop_t Struct Reference . . . 735
38.154.1 Detailed Description . . . 736
38.154.2 Field Documentation . . . 736
38.155 OS_queue_prop_t Struct Reference . . . 736
38.155.1 Detailed Description . . . 737
38.155.2 Field Documentation . . . 737
38.156 OS_SockAddr_t Struct Reference . . . 737
38.156.1 Detailed Description . . . 738
38.156.2 Field Documentation . . . 738
38.157 OS_SockAddrData_t Union Reference . . . 738
38.157.1 Detailed Description . . . 739
38.157.2 Field Documentation . . . 739
38.158 OS_socket_prop_t Struct Reference . . . 740
38.158.1 Detailed Description . . . 740
38.158.2 Field Documentation . . . 740
38.159 OS_static_symbol_record_t Struct Reference . . . 741

38.159.1 Detailed Description 741
38.159.2 Field Documentation 741
38.160 OS_statvfs_t Struct Reference 742
38.160.1 Detailed Description 742
38.160.2 Field Documentation 742
38.161 OS_task_prop_t Struct Reference 743
38.161.1 Detailed Description 743
38.161.2 Field Documentation 743
38.162 OS_time_t Struct Reference 744
38.162.1 Detailed Description 744
38.162.2 Field Documentation 745
38.163 OS_timebase_prop_t Struct Reference 745
38.163.1 Detailed Description 745
38.163.2 Field Documentation 745
38.164 OS_timer_prop_t Struct Reference 746
38.164.1 Detailed Description 747
38.164.2 Field Documentation 747
38.165 Target_PspConfigData Struct Reference 748
38.165.1 Detailed Description 748
38.165.2 Field Documentation 748

39 File Documentation 750

39.1 build/doc/osconfig-example.h File Reference . . . 750

39.1.1 Macro Definition Documentation . . . 751

39.2 build/native/default_cpu1/inc/cfe_msgids.h File Reference . . . 759

39.3 build/native/default_cpu1/inc/cfe_platform_cfg.h File Reference . . . 759

39.4 build/native/default_cpu1/inc/osconfig.h File Reference . . . 759

39.4.1 Macro Definition Documentation . . . 760

39.5 cpu1_msgids.h File Reference . . . 768

39.5.1 Macro Definition Documentation . . . 770

39.6 cpu1_platform_cfg.h File Reference . . . 776

39.6.1 Macro Definition Documentation . . . 779

39.7 sample_mission_cfg.h File Reference . . . 833

39.7.1 Macro Definition Documentation . . . 835

39.8 sample_perfids.h File Reference . . . 853

39.8.1 Macro Definition Documentation . . . 853

39.9 cfe/docs/src/cfe_api.dox File Reference . . . 856

39.10 cfe/docs/src/cfe_es.dox File Reference . . . 856

39.11 cfe/docs/src/cfe_evs.dox File Reference . . . 856

39.12 cfe/docs/src/cfe_glossary.dox File Reference . . . 856

39.13 cfe/docs/src/cfe_sb.dox File Reference . . . 856

39.14 cfe/docs/src/cfe_tbl.dox File Reference . . . 856

39.15 cfe/docs/src/cfe_time.dox File Reference . . . 856

39.16 cfe/docs/src/cfe_xref.dox File Reference . . . 856

39.17 cfe/docs/src/cfs_versions.dox File Reference . . . 856

39.18 cfe/docs/src/main.dox File Reference . . . 856

39.19 cfe/modules/core_api/fsw/inc/cfe.h File Reference . . . 856

39.20 cfe/modules/core_api/fsw/inc/cfe_endian.h File Reference . . . 857

39.20.1 Macro Definition Documentation . . . 857

39.21 cfe/modules/core_api/fsw/inc/cfe_error.h File Reference . . . 857
39.21.1 Macro Definition Documentation . . . 864
39.21.2 Typedef Documentation . . . 867
39.22 cfe/modules/core_api/fsw/inc/cfe_es.h File Reference . . . 868
39.22.1 Macro Definition Documentation . . . 871
39.23 cfe/modules/core_api/fsw/inc/cfe_es_api_typedefs.h File Reference . . . 871
39.23.1 Macro Definition Documentation . . . 873
39.23.2 Typedef Documentation . . . 877
39.24 cfe/modules/core_api/fsw/inc/cfe_es_core_internal.h File Reference . . . 878
39.25 cfe/modules/core_api/fsw/inc/cfe_es_extern_typedefs.h File Reference . . . 879
39.25.1 Macro Definition Documentation . . . 881
39.25.2 Typedef Documentation . . . 881
39.25.3 Enumeration Type Documentation . . . 887
39.26 cfe/modules/core_api/fsw/inc/cfe_evs.h File Reference . . . 890
39.26.1 Macro Definition Documentation . . . 891
39.27 cfe/modules/core_api/fsw/inc/cfe_evs_api_typedefs.h File Reference . . . 892
39.27.1 Macro Definition Documentation . . . 893
39.27.2 Typedef Documentation . . . 896
39.28 cfe/modules/core_api/fsw/inc/cfe_evs_core_internal.h File Reference . . . 896
39.29 cfe/modules/core_api/fsw/inc/cfe_evs_extern_typedefs.h File Reference . . . 896
39.29.1 Typedef Documentation . . . 897
39.29.2 Enumeration Type Documentation . . . 898
39.30 cfe/modules/core_api/fsw/inc/cfe_fs.h File Reference . . . 900
39.31 cfe/modules/core_api/fsw/inc/cfe_fs_api_typedefs.h File Reference . . . 901
39.31.1 Typedef Documentation . . . 902
39.31.2 Enumeration Type Documentation . . . 902
39.32 cfe/modules/core_api/fsw/inc/cfe_fs_core_internal.h File Reference . . . 903
39.33 cfe/modules/core_api/fsw/inc/cfe_fs_extern_typedefs.h File Reference . . . 903

39.33.1 Macro Definition Documentation 904
39.33.2 Typedef Documentation 905
39.33.3 Enumeration Type Documentation 905
39.34 cfe/modules/core_api/fsw/inc/cfe_msg.h File Reference 907
39.35 cfe/modules/core_api/fsw/inc/cfe_msg_api_typedefs.h File Reference 909
39.35.1 Macro Definition Documentation 910
39.35.2 Typedef Documentation 911
39.35.3 Enumeration Type Documentation 913
39.36 cfe/modules/core_api/fsw/inc/cfe_resourceid.h File Reference 916
39.36.1 Detailed Description 917
39.36.2 Macro Definition Documentation 917
39.36.3 Function Documentation 918
39.37 cfe/modules/core_api/fsw/inc/cfe_resourceid_api_typedefs.h File Reference 923
39.37.1 Detailed Description 924
39.37.2 Macro Definition Documentation 924
39.38 cfe/modules/core_api/fsw/inc/cfe_sb.h File Reference 925
39.38.1 Macro Definition Documentation 927
39.39 cfe/modules/core_api/fsw/inc/cfe_sb_api_typedefs.h File Reference 928
39.39.1 Macro Definition Documentation 930
39.39.2 Typedef Documentation 934
39.40 cfe/modules/core_api/fsw/inc/cfe_sb_core_internal.h File Reference 935
39.41 cfe/modules/core_api/fsw/inc/cfe_sb_extern_typedefs.h File Reference 935
39.41.1 Macro Definition Documentation 936
39.41.2 Typedef Documentation 937
39.41.3 Enumeration Type Documentation 938
39.42 cfe/modules/core_api/fsw/inc/cfe_tbl.h File Reference 939
39.43 cfe/modules/core_api/fsw/inc/cfe_tbl_api_typedefs.h File Reference 940
39.43.1 Macro Definition Documentation 941

39.43.2 Typedef Documentation . . . 942
39.43.3 Enumeration Type Documentation . . . 943
39.44 cfe/modules/core_api/fsw/inc/cfe_tbl_core_internal.h File Reference . . . 943
39.45 cfe/modules/core_api/fsw/inc/cfe_tbl_extern_typedefs.h File Reference . . . 944
39.45.1 Typedef Documentation . . . 944
39.45.2 Enumeration Type Documentation . . . 945
39.46 cfe/modules/core_api/fsw/inc/cfe_tbl_filedef.h File Reference . . . 946
39.46.1 Macro Definition Documentation . . . 946
39.46.2 Typedef Documentation . . . 947
39.47 cfe/modules/core_api/fsw/inc/cfe_time.h File Reference . . . 947
39.47.1 Macro Definition Documentation . . . 949
39.48 cfe/modules/core_api/fsw/inc/cfe_time_api_typedefs.h File Reference . . . 949
39.48.1 Macro Definition Documentation . . . 950
39.48.2 Typedef Documentation . . . 950
39.48.3 Enumeration Type Documentation . . . 950
39.49 cfe/modules/core_api/fsw/inc/cfe_time_core_internal.h File Reference . . . 951
39.50 cfe/modules/core_api/fsw/inc/cfe_time_extern_typedefs.h File Reference . . . 951
39.50.1 Typedef Documentation . . . 953
39.50.2 Enumeration Type Documentation . . . 955
39.51 cfe/modules/core_api/fsw/inc/cfe_version.h File Reference . . . 958
39.51.1 Detailed Description . . . 958
39.51.2 Macro Definition Documentation . . . 958
39.52 cfe/modules/es/fsw/inc/cfe_es_events.h File Reference . . . 961
39.52.1 Macro Definition Documentation . . . 965
39.53 cfe/modules/es/fsw/inc/cfe_es_msg.h File Reference . . . 1008
39.53.1 Macro Definition Documentation . . . 1012
39.53.2 Typedef Documentation . . . 1036
39.54 cfe/modules/evs/fsw/inc/cfe_evs_events.h File Reference . . . 1044

39.54.1 Macro Definition Documentation . . . 1045
39.55 cfe/modules/evs/fsw/inc/cfe_evs_msg.h File Reference . . . 1065
39.55.1 Macro Definition Documentation . . . 1068
39.55.2 Typedef Documentation . . . 1090
39.56 cfe/modules/msg/fsw/inc/ccsds_hdr.h File Reference . . . 1098
39.56.1 Typedef Documentation . . . 1098
39.57 cfe/modules/sb/fsw/inc/cfe_sb_events.h File Reference . . . 1099
39.57.1 Macro Definition Documentation . . . 1101
39.58 cfe/modules/sb/fsw/inc/cfe_sb_msg.h File Reference . . . 1134
39.58.1 Macro Definition Documentation . . . 1136
39.58.2 Typedef Documentation . . . 1147
39.59 cfe/modules/tbl/fsw/inc/cfe_tbl_events.h File Reference . . . 1152
39.59.1 Macro Definition Documentation . . . 1155
39.60 cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h File Reference . . . 1190
39.60.1 Macro Definition Documentation . . . 1192
39.60.2 Typedef Documentation . . . 1202
39.61 cfe/modules/time/fsw/inc/cfe_time_events.h File Reference . . . 1206
39.61.1 Macro Definition Documentation . . . 1208
39.62 cfe/modules/time/fsw/inc/cfe_time_msg.h File Reference . . . 1227
39.62.1 Macro Definition Documentation . . . 1230
39.62.2 Typedef Documentation . . . 1246
39.63 osal/src/os/inc/common_types.h File Reference . . . 1252
39.63.1 Macro Definition Documentation . . . 1253
39.63.2 Typedef Documentation . . . 1255
39.63.3 Function Documentation . . . 1258
39.64 osal/src/os/inc/osapi-binsem.h File Reference . . . 1259
39.65 osal/src/os/inc/osapi-bsp.h File Reference . . . 1260
39.66 osal/src/os/inc/osapi-clock.h File Reference . . . 1260

39.66.1 Enumeration Type Documentation 1262
39.67 osal/src/os/inc/osapi-common.h File Reference 1262
39.67.1 Typedef Documentation 1263
39.67.2 Enumeration Type Documentation 1263
39.68 osal/src/os/inc/osapi-constants.h File Reference 1264
39.68.1 Macro Definition Documentation 1265
39.69 osal/src/os/inc/osapi-countsem.h File Reference 1266
39.70 osal/src/os/inc/osapi-dir.h File Reference 1266
39.70.1 Macro Definition Documentation 1267
39.71 osal/src/os/inc/osapi-error.h File Reference 1267
39.71.1 Macro Definition Documentation 1269
39.71.2 Typedef Documentation 1270
39.72 osal/src/os/inc/osapi-file.h File Reference 1270
39.72.1 Macro Definition Documentation 1272
39.72.2 Enumeration Type Documentation 1273
39.73 osal/src/os/inc/osapi-filesys.h File Reference 1274
39.73.1 Macro Definition Documentation 1275
39.74 osal/src/os/inc/osapi-heap.h File Reference 1275
39.75 osal/src/os/inc/osapi-idmap.h File Reference 1276
39.75.1 Macro Definition Documentation 1277
39.76 osal/src/os/inc/osapi-macros.h File Reference 1278
39.76.1 Macro Definition Documentation 1278
39.77 osal/src/os/inc/osapi-module.h File Reference 1280
39.77.1 Macro Definition Documentation 1280
39.78 osal/src/os/inc/osapi-mutex.h File Reference 1281
39.79 osal/src/os/inc/osapi-network.h File Reference 1282
39.80 osal/src/os/inc/osapi-printf.h File Reference 1282
39.81 osal/src/os/inc/osapi-queue.h File Reference 1283

39.82 osal/src/os/inc/osapi-select.h File Reference . . . 1283
39.82.1 Enumeration Type Documentation . . . 1284
39.83 osal/src/os/inc/osapi-shell.h File Reference . . . 1284
39.84 osal/src/os/inc/osapi-sockets.h File Reference . . . 1285
39.84.1 Macro Definition Documentation . . . 1286
39.84.2 Enumeration Type Documentation . . . 1286
39.85 osal/src/os/inc/osapi-task.h File Reference . . . 1287
39.85.1 Macro Definition Documentation . . . 1288
39.85.2 Typedef Documentation . . . 1289
39.85.3 Function Documentation . . . 1290
39.86 osal/src/os/inc/osapi-timebase.h File Reference . . . 1290
39.86.1 Typedef Documentation . . . 1291
39.87 osal/src/os/inc/osapi-timer.h File Reference . . . 1291
39.87.1 Typedef Documentation . . . 1292
39.88 osal/src/os/inc/osapi-version.h File Reference . . . 1292
39.88.1 Detailed Description . . . 1293
39.88.2 Macro Definition Documentation . . . 1293
39.88.3 Function Documentation . . . 1296
39.89 osal/src/os/inc/osapi.h File Reference . . . 1297
39.90 psp/fsw/inc/cfe_psp.h File Reference . . . 1298
39.90.1 Macro Definition Documentation . . . 1301
39.90.2 Function Documentation . . . 1310
39.91 psp/fsw/inc/cfe_psp_configdata.h File Reference . . . 1325
39.91.1 Detailed Description . . . 1325
39.91.2 Variable Documentation . . . 1325
39.92 psp/fsw/pc-linux/src/cfe_psp_exception.c File Reference . . . 1326
39.92.1 Function Documentation . . . 1326
39.92.2 Variable Documentation . . . 1329

39.93 psp/fsw/pc-linux/src/cfe_psp_memory.c File Reference 1329

39.93.1 Macro Definition Documentation 1330

39.93.2 Function Documentation 1332

39.93.3 Variable Documentation 1338

39.94 psp/fsw/pc-linux/src/cfe_psp_memtab.c File Reference 1340

39.94.1 Variable Documentation 1340

39.95 psp/fsw/pc-linux/src/cfe_psp_ssr.c File Reference 1340

39.95.1 Function Documentation 1341

39.96 psp/fsw/pc-linux/src/cfe_psp_start.c File Reference 1341

39.96.1 Macro Definition Documentation 1342

39.96.2 Function Documentation 1344

39.96.3 Variable Documentation 1346

39.97 psp/fsw/pc-linux/src/cfe_psp_support.c File Reference 1348

39.97.1 Function Documentation 1349

39.97.2 Variable Documentation 1350

39.98 psp/fsw/pc-linux/src/cfe_psp_timer.c File Reference 1351

39.98.1 Macro Definition Documentation 1352

39.98.2 Function Documentation 1352

39.99 psp/fsw/pc-linux/src/cfe_psp_watchdog.c File Reference 1354

39.99.1 Function Documentation 1354

39.99.2 Variable Documentation 1356

Index **1357**

1 Core Flight Executive Documentation

- General Information and Concepts
  - Background
  - Applicable Documents
  - Version Numbers
  - Dependencies
  - Acronyms
  - Glossary of Terms
- Executive Services (ES)
  - cFE Executive Services Overview
  - cFE Executive Services Commands
  - cFE Executive Services Telemetry
  - ES Event Message Reference
  - cFE Executive Services Configuration Parameters
- Events Services (EVS)
  - cFE Event Services Overview
  - cFE Event Services Commands
  - cFE Event Services Telemetry
  - EVS Event Message Reference
  - cFE Event Services Configuration Parameters
- Software Bus Services (SB)
  - cFE Software Bus Overview
  - cFE Software Bus Commands
  - cFE Software Bus Telemetry
  - SB Event Message Reference
  - cFE Software Bus Configuration Parameters
- Table Services (TBL)
  - cFE Table Services Overview
  - cFE Table Services Commands
  - cFE Table Services Telemetry
  - TBL Event Message Reference
  - cFE Table Services Configuration Parameters
- Time Services (TIME)
  - cFE Time Services Overview
  - cFE Time Services Commands
  - cFE Time Services Telemetry
  - TIME Event Message Reference
  - cFE Time Services Configuration Parameters

- cFE Event Message Cross Reference
- cFE Command Mnemonic Cross Reference
- cFE Telemetry Mnemonic Cross Reference
- cFE Application Programmer's Interface (API) Reference

2 Background

The Core Flight Executive (cFE) is an application development and run-time environment. The cFE provides a set of core services including Software Bus (messaging), Time, Event (Alerts), Executive (startup and runtime), and Table services. The cFE defines an application programming interface (API) for each service which serves as the basis for application development.

The cFE Software Bus service provides a publish and subscribe messaging system that allows applications to easily plug and play into the system. Applications subscribe to cFE services at runtime, making system modifications easy. Facilitating rapid prototyping, new applications can be compiled, linked, loaded, and started without requiring the entire system to be rebuilt.

Each service comes complete with a built in application that allows users to interface with each service. To support reuse and project independence, the cFE contains a configurable set of requirements and code. The configurable parameters allow the cFE to be tailored for each environment including desk-top and closed loop simulation environments. This provides the ability to run and test software applications on a developer's desktop and then deploy that same software without changes to the embedded system. In addition the cFE includes the following software development tools:

- Unit Test Framework (UTF) for unit testing applications developed via the cFE
- Software Timing Analyzer that provides visibility into the real-time performance of embedded systems software
- Table Builder
- Command and Telemetry utilities

The cFE is one of the components of the Core Flight System (cFS), a platform and project independent reusable software framework and set of reusable software applications. There are three key aspects to the cFS architecture: a dynamic run-time environment, layered software, and a component based design. The combination of these key aspects along with an implementation targeted to the embedded software domain makes it suitable for reuse on any number of NASA flight projects and/or embedded software systems.

The pivotal design feature, abstracting the software architecture from the hardware and forming the basis of reuse, is component layering. Each layer of the architecture "hides" its implementation and technology details from the other layers by defining and using standard Application Programming Interfaces (APIs). The internals of a layer can be changed without affecting other layers' internals and components.

The layers include an OS Abstraction Layer (OSAL), Platform Support Package (PSP) layer, core Flight Executive (cFE) layer, and an Application layer. The cFE layer runs on top of the PSP and OSAL layers. The cFE comes complete with a build environment, deployment guide, API reference guide, and provides a sample PSP. The OSAL is available open source and once integrated into the cFE build environment, developers will be ready to build and run the system and start developing their mission/project specific applications that easily plug and play into the system.

Core Flight Executive (cFE) Goals

The main long term goal of the cFE is to form the basis for a platform and project independent reusable software framework. The cFE with the OSAL allow the development of portable embedded system software that is independent of a particular Real Time Operating System and hardware platform. A secondary long term goal is to create a standardized, product-line approach for development of embedded aerospace flight software.

Functional and Community Goals

The cFE allows embedded system software to be developed and tested on desktop workstations and ported to the target platform without changing a single line of code, providing a shorter development and debug time. The cFE is an enabler of software collaboration amongst all users promoting the growth of the application and library layers where new applications, libraries, tools, and lessons learned can be contributed and shared.

It is important for application developers to realize the long term and functional goals of the cFE. With a standard set of services providing a standard API, all applications developed with the cFE have an opportunity to become useful on future missions through code reuse. In order to achieve this goal, applications must be written with care to ensure that their code does not have dependencies on specific hardware, software or compilers. The cFE and the underlying generic operating system API (OS API) have been designed to insulate the cFE Application developer from hardware and software dependencies. The developer, however, must make the effort to identify the proper methods through the cFE and OS API to satisfy their software requirements and not be tempted to take a "short-cut" and accomplish their goal with a direct hardware or operating system software interface.

3 Applicable Documents

Document Title	Link
cFE System (L4) Requirements Document	cfe/docs/'cfe requirements.docx'
cFE Functional (L5) Requirements Document	cfe/docs/cFE_FunctionalRequirements.csv
cFE Application Developers Guide	cfe/docs/'cFE Application Developers Guide.md'
cFE User's Guide (includes API)	Autogenerated from code, provided with releases in cFE repository
OS Abstraction Layer (OSAL) API	Autogenerated from code, provided with releases in OSAL repository

4 Dependencies

The Core Flight Executive (cFE) is required to be built with the Operating System Abstraction Layer (OSAL) and Platform Support Package (PSP) componenets of the Core Flight System (cFS). It is always recommended to build with the latest versions of each of the components as backward compatability may not be supported.

Several internal data structures within the cFE use the "char" data type. This data type is typically 1 byte in storage size with a value range -128 to 127 or 0 to 255. The size of the "char" data type and whether or not the type is signed or unsigned can change across platforms. The cFE assumes use of the "char" data type as an **8-bit type.**

5 Acronyms

Acronym	Description
AC	Attitude Control
ACE	Attitude Control Electronics
ACS	Attitude Control System
API	Application Programming Interface
APID	CCSDS Application ID
App	Application
CCSDS	Consultative Committee for Space Data Systems
CDH, C&DH	Command and Data Handling
cFE	core Flight Executive
cFS	core Flight System
CM	Configuration Management
CMD	Command
CPU	Central Processing Unit
EDAC	Error Detection and Correction
EEPROM	Electrically Erasable Programmable Read-Only Memory
ES	Executive Services
EVS	Event Services
FC	Function Code
FDC	Failure Detection and Correction
FSW	Flight Software
HW, H/W	Hardware
ICD	Interface Control Document
MET	Mission Elapsed Time
MID	Message ID
OS	Operating System
OSAL	Operating System Abstraction Layer
PID	Pipeline ID
PKT	Packet
PSP	Platform Support Package
RAM	Random-Access Memory
SB	Software Bus
SDO	Solar Dynamics Observatory
ST5	Space Technology Five
STCF	Spacecraft Time Correlation Factor
SW, S/W	Software
TAI	International Atomic Time
TBD	To Be Determined
TBL	Table Services
TID	Task ID
TIME	Time Services
TLM	Telemetry
UTC	Coordinated Universal Time

6 Glossary of Terms

Term	Definition
Application (or App)	A set of data and functions that is treated as a single entity by the cFE. cFE resources are allocated on a per-Application basis. Applications are made up of a Main Task and zero or more Child Tasks.
Application ID	A processor unique reference to an Application. **NOTE: This is different from a CCSDS Application ID which is referred to as an "APID."**
Application Programmer's Interface (API)	A set of routines, protocols, and tools for building software applications
Platform Support Package (PSP)	A collection of user-provided facilities that interface an OS and the cFE with a specific hardware platform. The PSP is responsible for hardware initialization.
Child Task	A separate thread of execution that is spawned by an Application's Main Task.
Command	A Software Bus Message defined by the receiving Application. Commands can originate from other onboard Applications or from the ground.
Core Flight Executive (cFE)	A runtime environment and a set of services for hosting FSW Applications
Critical Data Store (CDS)	A collection of data that is not modified by the OS or cFE following a Processor Reset.
Cyclic Redundancy Check	A polynomial based method for checking that a data set has remained unchanged from one time period to another.
Developer	Anyone who is coding a cFE Application.
Event Data	Data describing an Event that is supplied to the cFE Event Service. The cFE includes this data in an Event Message.
Event Filter	A numeric value (bit mask) used to determine how frequently to output an application Event Message defined by its Event ID.
Event Format Mode	Defines the Event Message Format downlink option: short or long. The short format is used when there is limited telemetry bandwidth and is binary. The long format is in ASCII and is used for logging to a Local Event Log and to an Event Message Port.
Event ID	A numeric literal used to uniquely name an Application event.
Event Type	A numeric literal used to identify the type of an Application event. An event type may be CFE_EVS_EventType_DEBUG, CFE_EVS_Event↩ Type_INFORMATION, CFE_EVS_EventType_ERROR, or CFE_EVS↩ _EventType_CRITICAL.
Event Message	A data item used to notify the user and/or an external Application of a significant event. Event Messages include a time-stamp of when the message was generated, a processor unique identifier, an Application ID, the Event Type (DEBUG,INFO,ERROR or CRITICAL), and Event Data. An Event Message can either be real-time or playback from a Local Event Log.

7 Version Numbers

Version Number Semantics

The version number is a sequence of four numbers, generally separated by dots when written. These are, in order, the Major number, the Minor number, the Revision number, and the Mission Revision number. Missions may modify the

Mission Revision information as needed to suit their needs.

It is important to note that Major, Minor, and Revision numbers are only updated upon official releases of tagged versions, **NOT** on development builds. We aim to follow the Semantic Versioning v2.0 specification with our versioning.

The MAJOR number shall be incremented on release to indicate when there is a change to an API that may cause existing, correctly-written cFS components to stop working. It may also be incremented for a release that contains changes deemed to be of similar impact, even if there are no actual changes to the API.

The MINOR number shall be incremented on release to indicate the addition of features to the API which do not break the existing code. It may also be incremented for a release that contains changes deemed to be of similar impact, even if there are no actual updates to the API.

The REVISION number shall be incremented on changes that benefit from unique identification such as bug fixes or major documentation updates. The Revision number may also be updated if there are other changes contained within a release that make it desirable for applications to distinguish one release from another. WARNING: The revision number is set to the number 99 in development builds. To distinguish between development builds refer to the BUILD_NUMBER and BUILD_BASELINE detailed in the section "Identifying Development Builds".

The Major, Minor, and Revision numbers are provided in this header file as part of the API definition; this macro must expand to a simple integer value, so that it can be used in simple if directives by the macro preprocessor.

The Mission Version number shall be set to zero in all officially released packages, and is entirely reserved for the use of the mission. The Mission Version is provided as a simple macro defined in the cfe_platform_cfg.h header file.

Version Number Flexibility

The major number may increment when there is no breaking change to the API, if the changes are significant enough to warrant the same level of attention as a breaking API change.

The minor number may increment when there have been no augmentations to the API, if changes are as significant as additions to the public API.

The revision numbers may increment in implementations where no actual implementation-specific code has changed, if there are other changes within the release with similar significance.

How and Where Defined

The Major, Minor, and Revision components of the version are provided as simple macros defined in the cfe_version.h header file as part of the API definition; these macros must expand to simple integer values, so that they can be used in simple if directives by the macro preprocessor.

The Mission Version is provided as a simple macro defined in the cfe_platform_cfg.h header file. As delivered in official releases, these macros must expand to simple integer values, so that they can be used in simple macro preprocessor conditions, but delivered code should not prevent a mission from, for example, deciding that the Mission Version is actually a text string.

Identifying Development Builds

In order to distinguish between development versions, we also provide a BUILD_NUMBER.

The BUILD_NUMBER reflects the number of commits since the BUILD_BASELINE, a baseline git tag, for each particular component. The BUILD_NUMBER integer increases monotonically for a given development cycle. The BUILD_BASELINE identifies the current development cycle and is a git tag with format vX.Y.Z. The Codename used in the version string also refers to the current development cycle. When a new baseline tag and codename are created, the the BUILD_NUMBER resets to zero and begins increasing from a new baseline.

Templates for the version and version string

The following templates are the code to be used in cfe_version.h for either official releases or development builds. The apps and repositories follow the same pattern by replacing the CFE_ prefix with the appropriate name; for example, osal uses OS_, psp uses CFE_PSP_IMPL, and so on.

Template for Official Releases

```
/*<! Official Release Version Number */
#define CFE_SRC_VERSION \
    CFE_STR(CFE_MAJOR_VERSION) "." \
    CFE_STR(CFE_MINOR_VERSION) "." \
    CFE_STR(CFE_REVISION) "."      \
    CFE_STR(CFE_MISSION_REV)

#define CFE_VERSION_STRING \
    "cFE version " CFE_SRC_VERSION
```

Template for Development Builds

```
/*! @brief Development Build Version Number.
 * Baseline git tag + Number of commits since baseline. @n
 * See cfs_versions.dox for format differences between development and release versions.
 */
#define CFE_SRC_VERSION \
    CFE_BUILD_BASELINE CFE_STR(CFE_BUILD_NUMBER)

/*! @brief Development Build Version String.
 * Reports the current development build's baseline, number, and name. Also includes a note about the latest offi
 * See cfs_versions.dox for format differences between development and release versions.
*/
#define CFE_VERSION_STRING                                                    \
    " cFE Development Build "                                                 \
    CFE_SRC_VERSION " (Codename: CONSTELLATION_NAME)"       /* Codename for current development */  \
    ", Last Official Release: cfe vX.Y.Z"   /* For full support please use this version */
```

8 cFE Application Programmer's Interface (API) Reference

Executive Services API

- cFE Entry/Exit APIs
  - CFE_ES_Main - cFE Main Entry Point used by Board Support Package to start cFE

  - CFE_ES_ResetCFE - Reset the cFE Core and all cFE Applications.
- cFE Application Control APIs
  - CFE_ES_RestartApp - Restart a single cFE Application.
  - CFE_ES_ReloadApp - Reload a single cFE Application.
  - CFE_ES_DeleteApp - Delete a cFE Application.
- cFE Application Behavior APIs
  - CFE_ES_RegisterApp - Registers a cFE Application with the Executive Services.
  - CFE_ES_RunLoop - Check for Exit, Restart, or Reload commands.
  - CFE_ES_WaitForStartupSync - Allow an Application to Wait for the "OPERATIONAL" global system state.
  - CFE_ES_WaitForSystemState - Allow an Application to Wait for a minimum global system state.
  - CFE_ES_IncrementTaskCounter - Increments the execution counter for the calling task.
  - CFE_ES_ExitApp - Exit a cFE Application.
- cFE Information APIs
  - CFE_ES_GetResetType - Return the most recent Reset Type.
  - CFE_ES_GetAppID - Get an Application ID for the calling Application.
  - CFE_ES_GetAppIDByName - Get an Application ID associated with a specified Application name.
  - CFE_ES_GetAppName - Get an Application name for a specified Application ID.
  - CFE_ES_GetAppInfo - Get Application Information given a specified App ID.
  - CFE_ES_GetTaskInfo - Get Task Information given a specified Task ID.
- cFE Child Task APIs
  - CFE_ES_RegisterChildTask - Registers a cFE Child task associated with a cFE Application.
  - CFE_ES_CreateChildTask - Creates a new task under an existing Application.
  - CFE_ES_DeleteChildTask - Deletes a task under an existing Application.
  - CFE_ES_ExitChildTask - Exits a child task.
- cFE Critical Data Store APIs
  - CFE_ES_RegisterCDS - Reserve space (or re-obtain previously reserved space) in the Critical Data Store (CDS)
  - CFE_ES_CopyToCDS - Save a block of data in the Critical Data Store (CDS)
  - CFE_ES_RestoreFromCDS - Recover a block of data from the Critical Data Store (CDS)
- cFE Memory Manager APIs
  - CFE_ES_PoolCreate - Initializes a memory pool created by an application while using a semaphore during processing.
  - CFE_ES_PoolCreateEx - Initializes a memory pool created by an application with application specified block sizes.
  - CFE_ES_PoolCreateNoSem - Initializes a memory pool created by an application without using a semaphore during processing.
  - CFE_ES_GetPoolBuf - Gets a buffer from the memory pool created by CFE_ES_PoolCreate or CFE_ES↩_PoolCreateNoSem.
  - CFE_ES_PutPoolBuf - Releases a buffer from the memory pool that was previously allocated via CFE_E↩S_GetPoolBuf.

  - CFE_ES_GetMemPoolStats - Extracts the statistics maintained by the memory pool software.
  - CFE_ES_GetPoolBufInfo - Gets info on a buffer previously allocated via CFE_ES_GetPoolBuf.
- cFE Performance Monitor APIs
  - CFE_ES_PerfLogEntry - Entry marker for use with Software Performance Analysis Tool.
  - CFE_ES_PerfLogExit - Exit marker for use with Software Performance Analysis Tool.
  - CFE_ES_PerfLogAdd - Function called by CFE_ES_PerfLogEntry and CFE_ES_PerfLogExit macros.
- cFE Generic Counter APIs
  - CFE_ES_RegisterGenCounter - Register a generic counter.
  - CFE_ES_DeleteGenCounter - Delete a generic counter.
  - CFE_ES_IncrementGenCounter - Increments the specified generic counter.
  - CFE_ES_SetGenCount - Set the specified generic counter.
  - CFE_ES_GetGenCount - Get the specified generic counter count.
  - CFE_ES_GetGenCounterIDByName - Get the Id associated with a generic counter name.
- cFE Miscellaneous APIs
  - CFE_ES_CalculateCRC - Calculate a CRC on a block of memory.
  - CFE_ES_WriteToSysLog - Write a string to the cFE System Log.
  - CFE_ES_ProcessAsyncEvent - Notification that an asynchronous event was detected by the underlying OS/PSP.

Events Services API

- cFE Registration APIs
  - CFE_EVS_Register - Register an application for receiving event services.
  - CFE_EVS_Unregister - Cleanup internal structures used by the event manager for the calling Application.
- cFE Send Event APIs
  - CFE_EVS_SendEvent - Generate a software event.
  - CFE_EVS_SendEventWithAppID - Generate a software event given the specified Application ID.
  - CFE_EVS_SendTimedEvent - Generate a software event with a specific time tag.
- cFE Reset Event Filter APIs
  - CFE_EVS_ResetFilter - Resets the calling application's event filter for a single event ID.
  - CFE_EVS_ResetAllFilters - Resets all of the calling application's event filters.

File Services API

- cFE File Header Management APIs
  - CFE_FS_ReadHeader - Read the contents of the Standard cFE File Header.
  - CFE_FS_InitHeader - Initializes the contents of the Standard cFE File Header.
  - CFE_FS_WriteHeader - Write the specified Standard cFE File Header to the specified file.
  - CFE_FS_SetTimestamp - Modifies the Time Stamp field in the Standard cFE File Header for the specified file.
- cFE File Utility APIs
  - CFE_FS_ExtractFilenameFromPath - Extracts the filename from a unix style path and filename string.

Software Bus API

- cFE Pipe Management APIs
  - CFE_SB_CreatePipe - Creates a new software bus pipe.
  - CFE_SB_DeletePipe - Delete a software bus pipe.
  - CFE_SB_SetPipeOpts - Set options on a pipe.
  - CFE_SB_GetPipeOpts - Get options on a pipe.
  - CFE_SB_GetPipeName - Get the pipe name for a given id.
  - CFE_SB_GetPipeIdByName - Get pipe id by pipe name.
- cFE Message Subscription Control APIs
  - CFE_SB_Subscribe - Subscribe to a message on the software bus with default parameters.
  - CFE_SB_SubscribeEx - Subscribe to a message on the software bus.
  - CFE_SB_SubscribeLocal - Subscribe to a message while keeping the request local to a cpu.
  - CFE_SB_Unsubscribe - Remove a subscription to a message on the software bus.
  - CFE_SB_UnsubscribeLocal - Remove a subscription to a message on the software bus on the current CPU.
- cFE Send/Receive Message APIs
  - CFE_SB_TransmitMsg - Transmit a message.
  - CFE_SB_ReceiveBuffer - Receive a message from a software bus pipe.
- cFE Zero Copy APIs
  - CFE_SB_ZeroCopyGetPtr - Get a buffer pointer to use for "zero copy" SB sends.
  - CFE_SB_ZeroCopyReleasePtr - Release an unused "zero copy" buffer pointer.
  - CFE_SB_TransmitBuffer - Transmit a buffer.
- cFE Setting Message Characteristics APIs
  - CFE_MSG_Init - Initialize a message.
  - CFE_MSG_SetMsgId - Sets the message id bits in a message.
  - CFE_SB_SetUserDataLength - Sets the length of user data in a software bus message.
  - CFE_MSG_SetSize - Sets the total size of a message.
  - CFE_MSG_SetMsgTime - Sets the time field in a message.
  - CFE_SB_TimeStampMsg - Sets the time field in a software bus message with the current spacecraft time.
  - CFE_MSG_SetFcnCode - Sets the function code field in a message.
  - CFE_MSG_SetSequenceCount - Sets the message sequence count.
  - CFE_SB_MessageStringSet - Copies a string into a software bus message.
- cFE Getting Message Characteristics APIs
  - CFE_SB_GetUserData - Get a pointer to the user data portion of a software bus message.
  - CFE_MSG_GetMsgId - Gets the message id from a message.
  - CFE_SB_GetUserDataLength - Gets the length of user data in a software bus message.
  - CFE_MSG_GetSize - Gets the total size of a message.
  - CFE_MSG_GetMsgTime - Gets the time field from a message.
  - CFE_MSG_GetFcnCode - Gets the function code field from a message.

  - CFE_MSG_GetTypeFromMsgId - Gets message type using message ID.
  - CFE_SB_MessageStringGet - Copies a string out of a software bus message.
- cFE Checksum Control APIs
  - CFE_MSG_GenerateChecksum - Calculates and sets the checksum of a message.
  - CFE_MSG_ValidateChecksum - Validates the checksum of a message.
- cFE Message ID APIs
  - CFE_SB_MsgId_Equal - Identifies whether two CFE_SB_MsgId_t values are equal.
  - CFE_SB_MsgIdToValue - Converts a CFE_SB_MsgId_t to a normal integer.
  - CFE_SB_ValueToMsgId - Converts a normal integer into a CFE_SB_MsgId_t.

Table Services API

- cFE Registration APIs
  - CFE_TBL_Register - Register a table with cFE to obtain Table Management Services.
  - CFE_TBL_Share - Obtain handle of table registered by another application.
  - CFE_TBL_Unregister - Unregister a previously registered table and free associated resources.
- cFE Manage Table Content APIs
  - CFE_TBL_Load - Load a specified table with data from specified source.
  - CFE_TBL_Update - Update contents of a specified table, if an update is pending.
  - CFE_TBL_Validate - Perform steps to validate the contents of a table image.
  - CFE_TBL_Manage - Perform standard operations to maintain a table.
  - CFE_TBL_DumpToBuffer - Copies the contents of a Dump Only Table to a shared buffer.
  - CFE_TBL_Modified - Notify cFE Table Services that table contents have been modified by the Application.
- cFE Access Table Content APIs
  - CFE_TBL_GetAddress - Obtain the current address of the contents of the specified table.
  - CFE_TBL_GetAddresses - Obtain the current addresses of an array of specified tables.
  - CFE_TBL_ReleaseAddress - Release previously obtained pointer to the contents of the specified table.
  - CFE_TBL_ReleaseAddresses - Release the addresses of an array of specified tables.
- cFE Get Table Information APIs
  - CFE_TBL_GetStatus - Obtain current status of pending actions for a table.
  - CFE_TBL_GetInfo - Obtain characteristics/information of/about a specified table.
  - CFE_TBL_NotifyByMessage - Instruct cFE Table Services to notify Application via message when table requires management.

Time Services API

- cFE Get Current Time APIs
  - CFE_TIME_GetTime - Get the current spacecraft time.
  - CFE_TIME_GetTAI - Get the current TAI (MET + SCTF) time.
  - CFE_TIME_GetUTC - Get the current UTC (MET + SCTF - Leap Seconds) time.
  - CFE_TIME_GetMET - Get the current value of the Mission Elapsed Time (MET).
  - CFE_TIME_GetMETseconds - Get the current seconds count of the mission-elapsed time.
  - CFE_TIME_GetMETsubsecs - Get the current sub-seconds count of the mission-elapsed time.
- cFE Get Time Information APIs
  - CFE_TIME_GetSTCF - Get the current value of the spacecraft time correction factor (STCF).
  - CFE_TIME_GetLeapSeconds - Get the current value of the leap seconds counter.
  - CFE_TIME_GetClockState - Get the current state of the spacecraft clock.
  - CFE_TIME_GetClockInfo - Provides information about the spacecraft clock.
- cFE Time Arithmetic APIs
  - CFE_TIME_Add - Adds two time values.
  - CFE_TIME_Subtract - Subtracts two time values.
  - CFE_TIME_Compare - Compares two time values.
- cFE Time Conversion APIs
  - CFE_TIME_MET2SCTime - Convert specified MET into Spacecraft Time.
  - CFE_TIME_Sub2MicroSecs - Converts a sub-seconds count to an equivalent number of microseconds.
  - CFE_TIME_Micro2SubSecs - Converts a number of microseconds to an equivalent sub-seconds count.
- cFE External Time Source APIs
  - CFE_TIME_ExternalTone - Provides the 1 Hz signal from an external source.
  - CFE_TIME_ExternalMET - Provides the Mission Elapsed Time from an external source.
  - CFE_TIME_ExternalGPS - Provide the time from an external source that has data common to GPS receivers.
  - CFE_TIME_ExternalTime - Provide the time from an external source that measures time relative to a known epoch.
  - CFE_TIME_RegisterSynchCallback - Registers a callback function that is called whenever time synchronization occurs.
  - CFE_TIME_UnregisterSynchCallback - Unregisters a callback function that is called whenever time synchronization occurs.
- cFE Miscellaneous Time APIs
  - CFE_TIME_Print - Print a time value as a string.
  - CFE_TIME_Local1HzISR - This function should be called from the system PSP layer once per second.

9 cFE Executive Services Overview

Executive Services (ES) is one of the five core Flight Executive components. ES is the primary interface to the underlying Operating System, providing a high level interface to system control facilities. The ES component is responsible for starting up and restarting the cFE, starting up, shutting down, and restarting cFE Applications, logging errors and performance data, and providing a persistent memory store for cFE Applications.

The interfaces to the ES task include the Ground Interface (commands and telemetry) and the Application Programmer Interfaces (APIs). The ES task interfaces to the OS through the OS Abstraction Layer (OSAL) and platform through the Platform Support Package (PSP).

The functionality provided by the ES task include Software Reset, Application and Child Task Mangement, Basic File System, Performance Data Collection, Critical Data Store, Memory Pool, System Log, Shell Command.

For additional detail on Executive Services, see the following sections:

- Terminology
- Software Reset
  - Reset Types and Subtypes
  - Exception and Reset (ER) Log
- Application and Child Task Management
  - Starting an Application
  - Stopping an Application
  - Restarting an Application
  - Reloading an Application
  - Listing Current Applications
  - Listing Current Tasks
  - Loading Common Libraries
- Basic File System
- Performance Data Collection
- Critical Data Store

- Memory Pool
- System Log
- Version Identification
- Executive Services Frequently Asked Questions

9.1 Terminology

The following sections describe terminology that is very relevant to understanding the Executive Services:

- "Application" and "cFE Application"
- "Task"
- "Startup Script"

Next: "Application" and "cFE Application"
Up To: cFE Executive Services Overview

9.1.1 "Application" and "cFE Application"

Application

The term 'Application' as defined in the Glossary of Terms is *a set of data and functions that is treated as a single entity by the cFE. cFE resources are allocated on a per-Application basis. Applications are made up of a Main Task and zero or more Child Tasks.*

cFE Application

A 'cFE Application' is an application that is external to the cFE and designed to interface to the cFE through the APIs. It is created through an entry in the "Startup Script" (with the 'Object Type' field set to CFE_APP) or by way of the CFE_ES_START_APP_CC ground command.

When referring to one of the five applications internal to the cFE (ES, EVS, SB, TIME or TBL), the term 'Service' or 'Core Application' is typically used.

A listing of cFE applications can be acquired by using the CFE_ES_QUERY_ALL_CC ground command. This listing will include the cFE internal applications as well as cFE applications that are loaded and running.

Next: "Task"
Up To: Terminology

9.1.2 "Task"

A Task is a thread of execution in the operating system, often associated with a cFE Application. Each cFE Application has a Main task providing its CPU context, stack and other OS resources. In addition, each cFE Application can create multiple Child Tasks which are closely associated with the Parent Task and cFE Application.

In a traditional Real Time Operating System such as vxWorks, the cFE Application Main task and child tasks end up being mapped to these OS tasks in the same shared memory space. For example, a Stored Command cFE Application that consists of a cFE Main Task and 10 Relative Time Sequence Child Tasks would have 11 tasks on a vxWorks system. The only association between these tasks exists in the cFE.

In a memory protected process oriented Operating System, the intention is to have a cFE Application implemented as a memory protected process with its own virtual address space. In this Process Model, each cFE Child Task would be a thread in the parent Process, much like a Unix process with multiple threads. In this model, the Stored Command example with a cFE Main Task and 10 Relative Time Sequence Child Tasks would consist of a Unix Process and 10 pthreads, all under the same virtual address space.

Next: "Startup Script"
Prev: "Application" and "cFE Application"
Up To: Terminology

9.1.3 "Startup Script"

The startup script is a text file, written by the user that contains a list of entries (one entry for each application) and is used by the ES application for automating the startup of applications. For a processor reset, ES checks for the CFE_↩ PLATFORM_ES_VOLATILE_STARTUP_FILE first, and if it doesn't exist or for a power on reset ES uses the file passed in to CFE_ES_Main (typically CFE_PLATFORM_ES_NONVOL_STARTUP_FILE but dependent on the PSP).

The fields in a single entry include:

Object Type	`CFE_APP` for an Application, or `CFE_LIB` for a library.
Path/Filename	This is a cFE Virtual filename, not a vxWorks device/pathname
Entry Point	This is the name of the "main" function for App.
CFE Name	The cFE name for the APP or Library
Priority	This is the Priority of the App, not used for a Library
Stack Size	This is the Stack size for the App, not used for a Library
Load Address	This is the Optional Load Address for the App or Library. It is currently not implemented so it should always be 0x0.
Exception Action	This is the Action the cFE should take if the Application has an exception. • 0 = Do a cFE Processor Reset • Non-Zero = Just restart the Application

Immediately after the cFE completes its initialization, the ES Application first looks for the volatile startup script. The location in the file system is defined by the cFE platform configuration parameter named CFE_PLATFORM_ES_V↩ OLATILE_STARTUP_FILE. This configuration parameter contains a path as well as a filename. If the file is found,

ES begins to startup the applications that are listed in the file. If ES does not find the file, it attempts to open the CFE_PLATFORM_ES_NONVOL_STARTUP_FILE.

If ES finds the volatile startup script, the attempt to open the nonvolatile startup script is bypassed.

Any errors encountered in the startup script processing are written to the System Log. The System Log may also contain positive acknowledge messages regarding the startup script processing.

The startup script delivered with the cFE (`cfe_es_startup.scr`) also has some detailed information about the fields and the settings.

Next: Software Reset
Prev: Starting an Application
Up To: Terminology

9.2 Software Reset

The ES Software Reset provides a command to reset the cFE as well as resetting individual applications. Because applications are dependent on the cFE services, it is not possible to reset the cFE without affecting the applications. Therefore, a command to reset the cFE will also reset every application that is running at the time the command is received.

Also include is the Exception and Reset (ER) Log, which has a command for dumping or clearing the log and telemetry to show the number of entries in the log. In addition to the ER log, the user may find information about the most recent reset in the ES task housekeeping telemetry.

The ES Software Reset also provides a command to set the maximum number of processor resets before ES issues a power-on reset. There is a corresponding 'processor resets' counter in ES housekeeping telemetry that may be reset through another ES command.

Next: Reset Types and Subtypes
Prev: Terminology
Up To: cFE Executive Services Overview

9.3 Reset Types and Subtypes

The Reset Type is sent to the ground in the ES housekeeping packet and tells how the current running version of the cFE was invoked. The possible Reset Types expected in the telemetry field are CFE_PSP_RST_TYPE_POWERON and CFE_PSP_RST_TYPE_PROCESSOR. There is a third Reset Type defined in the ES code as CFE_ES_APP_↩RESTART which applies only to restarting an individual application and is covered in more detail in the section titled Application and Child Task.

The Reset Subtype is also sent in the ES housekeeping packet and gives more detail about the type of reset that started the execution of the current running version of the cFE. The possible Reset Subtypes are CFE_PSP_RST_↩SUBTYPE_POWER_CYCLE, CFE_PSP_RST_SUBTYPE_PUSH_BUTTON, CFE_PSP_RST_SUBTYPE_HW_SPE↩CIAL_COMMAND, CFE_PSP_RST_SUBTYPE_HW_WATCHDOG, CFE_PSP_RST_SUBTYPE_RESET_COMMAND, CFE_PSP_RST_SUBTYPE_EXCEPTION, CFE_PSP_RST_SUBTYPE_UNDEFINED_RESET, CFE_PSP_RST_SU↩BTYPE_HWDEBUG_RESET, CFE_PSP_RST_SUBTYPE_BANKSWITCH_RESET.

Next: Exception and Reset (ER) Log
Prev: Software Reset
Up To: cFE Executive Services Overview

9.4 Exception and Reset (ER) Log

The Exception and Reset Log contains detailed information about past resets and exceptions. To view the information the CFE_ES_WRITE_ER_LOG_CC command must be sent. This command will write the log to a binary file. The path and filename may be specified in the command. If the filename command field contains an empty string, the configuration parameter CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE is used to specify the path and filename. Use the ground system to get the file and display the contents. There is also a command to clear the ER log, CFE_ES_CLEAR_ER_LOG_CC.

The size of the ER log is defined by the platform configuration parameter CFE_PLATFORM_ES_ER_LOG_ENTRIES This log is preserved after a processor reset and held in the ES reset area.

A count of the number of entries in the log is present in the ES housekeeping telemetry. This count can be used with the configuration parameter CFE_PLATFORM_ES_ER_LOG_ENTRIES to calculate the fullness of the log.

The information contained in a single log entry is defined by the structure CFE_ES_ERLog_t.

Next: Application and Child Task Management
Prev: Reset Types and Subtypes
Up To: cFE Executive Services Overview

9.5 Application and Child Task Management

The ES Application and Child Task Management provides the user with full control over starting and stopping applications as well as querying information regarding applications, tasks and library routines.

There is no command to start or stop a child task. Child tasks can be controlled (started, stopped or deleted) only by the parent application through an API call.

This provides a way for the user to load a set of library routines, (via the startup script) without starting a corresponding task. See the section related to library routines for more detail.

The ES task maintains a counter for the number of registered applications, number of registered child tasks and the number of registered libraries in the ES housekeeping data.

Next: Starting an Application
Prev: Software Reset
Up To: cFE Executive Services Overview

9.6 Starting an Application

There are two ways to start an application, through the ground command CFE_ES_START_APP_CC or through the startup script. In either case, the object file must be loaded on board before the command is sent or before the startup script is executed. The startup script contains a list of applications and library routines to load and start immediately after the cFE finishes its startup sequence. The parameters in the command, match the elements of an entry in the startup script.

The format of the Start Application command, is defined in the structure CFE_ES_StartAppCmd_t. The members of the structure include, application name, entry point, filename, stack size, load address, exception action and priority.

If the command fails for any reason, an error event will be sent stating the reason for the failure. There may be additional information in the system log that can be viewed by sending the ES command to dump the system log.

After starting an application, the ES task sends an informational event message displaying the application name, filename of the object and the application ID. The new application will then show up in the query list downloaded in response to the CFE_ES_QUERY_ALL_CC command.

Next: Stopping an Application
Up To: Application and Child Task Management

9.7 Stopping an Application

Stopping an application can be done through the ground command CFE_ES_STOP_APP_CC. This command will terminate the application execution and all child tasks created by the application, free the system resources that it allocated and delete the corresponding object file.

The process of stopping an application is done in a controlled manner when the application is properly using the return code from the call to the CFE_ES_RunLoop. When the application properly uses this function, the ES task starts a timer and (via the return code) tells the application to exit at its own convenience. This gives the application time to free its own resources and do any cleanup that may be required before terminating itself by calling CFE_ES_ExitApp. If the timer expires and the application still exists, then ES must 'kill' the application. When the application is killed, ES attempts to cleanup the applications resources as best it could. In this case there is no guarantee that all the system resources are properly released.

The format of the Stop Application command, is defined in the structure CFE_ES_AppNameCmd_t. The only parameter in the command is an application name.

If the command fails for any reason, an error event will be sent stating the reason for the failure. There may be additional information in the system log that can be viewed by sending the ES command to dump the system log.

After stopping an application, the ES task sends a debug message stating the name of the application. After executing the command, the application (or any resources it allocated) should no longer be listed in any cFE tables or files.

Next: Restarting an Application
Prev: Starting an Application
Up To: Application and Child Task Management

9.8 Restarting an Application

The CFE_ES_RESTART_APP_CC command is used to restart an application using the same file name as the last start.

This command checks for file existence, the application is running, and the application is not a core app. If valid, the application restart is requested.

When requested, ES stops the application, unloads the object file, loads the object file using the previous file name, and restarts an application using the parameters defined when the application was previously started, either through the startup script or by way of the CFE_ES_START_APP_CC command.

Next: Reloading an Application
Prev: Stopping an Application
Up To: Application and Child Task Management

9.9 Reloading an Application

The CFE_ES_RELOAD_APP_CC command is used to reload an application using a new file name.

This command performes the same actions as CFE_ES_RESTART_APP_CC only using the new file.

Next: Listing Current Applications
Prev: Restarting an Application
Up To: Application and Child Task Management

9.10 Listing Current Applications

There are two options for receiving information about applications, the CFE_ES_QUERY_ONE_CC command can be used to get details about a single application. This command takes an application name as its only parameter and the application information is sent as a software bus packet that can be telemetered to the ground.

Or the CFE_ES_QUERY_ALL_CC command can be used to get information about all the applications that are currently registered with ES. This command writes the application data to a file and has a one parameter which specifies the path and filename of the output file.

For either command, the following Application information is made available:

- **Application ID** - The Application ID assigned by the cFE to the Application
- **Type Identifier** - Identifies whether the Application is a CORE App or an EXTERNAL App
- **Name** - The Application Name
- **Entry Point** - The symbolic name for the entry point into the Application
- **Filename** - The name of the file the Application was loaded from
- **Stack Size** - The number of bytes allocated for the Application's stack
- **Load Address** - The starting address of memory where the Application was loaded
- **Load Size** - The size, in bytes, of the Application when loaded into memory
- **Start Address** - The physical address that maps to the Entry Point
- **Exception Action** - A flag that identifies whether the the Processor should undergo a Restart or whether just the Application should restart upon an exception condition within the Application
- **Priority** - The assigned priority for the Application
- **Main Task ID** - The Task ID assigned to the main task associated with the Application
- **Main Task Name** - The name of the main task associated with the Application
- **Number of Child Tasks** - The number of child tasks spawned by the main task

For a description of the format in which this data is dumped, see CFE_ES_AppInfo_t.

Next: Listing Current Tasks
Prev: Reloading an Application
Up To: Application and Child Task Management

9.11 Listing Current Tasks

The CFE_ES_QUERY_ALL_TASKS_CC command is used to get a list of child tasks that are currently registered with ES. The following information is provided for each registered task:

- **Task ID** - The Task ID associated with the specified task
- **Task Name** - The name of the Task
- **Application ID** - The ID for the Application the Task is associated with
- **Application Name** - The name of the Application the Task is associated with

Next: Loading Common Libraries
Prev: Listing Current Applications
Up To: Application and Child Task Management

9.12 Loading Common Libraries

Library routines may be loaded only through the startup script. There is an option that allows a library routine initialization function to be executed after the library is loaded. Refer to the cFE Application Developers Guide for more information regarding Library Routines and startup scripts. The startup script delivered with the cFE (`cfe_es_startup.scr`) also has some detailed information about library routines.

Next: Basic File System
Prev: Listing Current Tasks
Up To: Application and Child Task Management

9.13 Basic File System

ES provides minimal functionality to initialize, read, and write cfe File headers.

Next: Performance Data Collection
Prev: Loading Common Libraries
Up To: Application and Child Task Management

9.14 Performance Data Collection

The Performance Data Collection provides precise timing information for each software application similar to how a logic analyzer can trigger and filter data.

API calls are inserted by the development team at key points in the code. The basic operation is to start the data collection, wait some amount of time, then send the command to stop the data collection. When the stop command is received, the ES task writes all the data from the buffer to a file. The file can then be imported to analysis tools for viewing. The size of the buffer is configurable through the CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE platform configuration parameter.

Additional information follows:

- Performance Data Collection Trigger Masks
- Starting to Collect Performance Data
- Stopping the Collection of Performance Data
- Viewing the Collection of Performance Data

Next: Performance Data Collection Trigger Masks
Prev: Basic File System
Up To: cFE Executive Services Overview

9.14.1 Performance Data Collection Trigger Masks

The trigger mask is used to control precisely when to start collecting the data. There is a bit in the trigger mask for every marker used in the code. After a start command is received, the trigger mask is read and dictates when to begin storing data in the buffer.

If the trigger mask is set to all zeros, then the collection will begin immediately after the start command and continue until a stop command is received. In this case the buffer behaves in a 'circular' manner.

Next: Starting to Collect Performance Data
Prev: Performance Data Collection
Up To: Performance Data Collection

9.14.2 Starting to Collect Performance Data

The CFE_ES_START_PERF_DATA_CC command is used to start the data collection process. The ES task sends a debug event when the command is received. It is not possible to start a collection if the buffer-to-file write is in process from an earlier collection. There is an ES telemetry point that can be used to ensure there is not a buffer-to-file write in progress. This ES telemetry point is called 'Perf Data to Write' and begins counting down from 'Data Count' to zero. If this counter is zero, it is ok to send the start command. If any errors are encountered when the start command is received, the details will be displayed in an error event message.

Next: Stopping the Collection of Performance Data
Prev: Performance Data Collection Trigger Masks
Up To: Performance Data Collection

9.14.3 Stopping the Collection of Performance Data

The CFE_ES_STOP_PERF_DATA_CC command is used to stop the data collection process and write the buffer data to a file. The path and filename may be specified in the command. If the filename command field contains an empty string, the configuration parameter CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME is used to specify the path and filename. The number of entries written to the file is determined by the 'data count' variable, which is sent in the ES housekeeping telemetry packet. To ensure cpu hogging does not occur during the write process, ES creates a low priority child task to perform the file write operation. This child task will write a number of entries, then sleep for a short time to give tasks of lower priority a chance to run. The number of entries between delays, and the delay time is displayed in the debug event at the time the stop command is received.

Next: Viewing the Collection of Performance Data
Prev: Starting to Collect Performance Data
Up To: Performance Data Collection

9.14.4 Viewing the Collection of Performance Data

To view the performance data, the file created as a result of the stop command must be transferred to the ground and imported into a viewing tool. See `https://github.com/nasa/perfutils-java` as an example.

Next: Critical Data Store
Prev: Stopping the Collection of Performance Data
Up To: Performance Data Collection

9.15 Critical Data Store

Some missions are required, for health, safety and mission success criteria, to survive Processor Resets. These mission requirements frequently flow down to Attitude Control and/or Command and Data Handling requirements that force an Application developer to design a mechanism for retaining software state information through a Processor Reset. The cFE provides the Critical Data Store to assist the developer in meeting these requirements.

The Critical Data Store is an area of memory that is not cleared during a Processor Reset. In addition, the contents of memory are validated when accessed with a Data Integrity Value that helps to ensure the contents have not been corrupted. Each processor platform, through the design of its Board Support Package, can implement this area of memory in a number of ways to ensure the contents survive a Processor Reset. Applications can allocate a section of this memory for their use in a way similar to the cFE Table Services Overview.

When an Application registers a Critical Data Store (CDS), the Executive Services allocates a section of the Critical Data Store memory for the application's use and assigns the Application specified name to the memory area. The operator can find and learn the characteristics of these Critical Data Stores by using the Dump CDS Registry Command. This command will dump the contents of the CDS Registry maintained by the Executive Services into a file that can be downlinked and examined by the operator.

The CDS Registry dump will identify the following information for each registered CDS:

- **Handle** - the numeric identifier used by an Application to access the contents of the CDS
- **Size** - the number of bytes allocated to the specified CDS

- **Table Flag** - a flag that indicates whether the CDS is associated with a Critical Tables (when non-zero) or not (when equal to zero).
- **Name** - a processor specific name that uniquely identifies the CDS. The name comes in two parts, "`AppName.CDSName`". `AppName` identifies which Application registered the CDS. `CDSName` is the name the Application assigned to the CDS.

The format of the CDS Registry Dump File is a cFE Standard File header (see CFE_FS_Header_t) followed by one or more CDS Registry Dump File Records (see CFE_ES_CDSRegDumpRec_t).

Next: Memory Pool
Prev: Performance Data Collection
Up To: cFE Executive Services Overview

9.16 Memory Pool

Refer to the cFE Application Developers Guide for additional information.

Applications that are designed for generic missions, frequently have to wait until run-time before allocating memory for buffers, data records, etc.

The cFE provides a memory allocation algorithm that may be used by an application to manage its block of memory. The user provides a pointer to its memory block and a list of block sizes and the cFE provides 'get' and 'put' API's to the user for managing its memory pool.

Run-time memory allocation in an embedded system can be risky because of the potential problem of memory fragmentation. Memory fragmentation is also referred to as External Fragmentation and is defined in the wikipedia as:

```
External fragmentation is the phenomenon in which free storage becomes divided
into many small pieces over time. It is a weakness of certain storage
allocation algorithms, occurring when an application allocates and deallocates
("frees") regions of storage of varying sizes, and the allocation algorithm
responds by leaving the allocated and deallocated regions interspersed. The
result is that, although free storage is available, it is effectively unusable
because it is divided into pieces that are too small to satisfy the demands of
the application. The term "external" refers to the fact that the unusable
storage is outside the allocated regions.
```

To help prevent this from happening, the cFE has integrated a memory allocation algorithm that is designed to create blocks at run-time, based on the size of the blocks requested. After a reset, there are no blocks created, the memory pool is said to be unconfigured. As requests for memory blocks are made, the memory pool first tries to use blocks that have been created but are no longer in use. If it cannot find an available block, it will create a new one. The created blocks remain until a reset occurs.

This algorithm is recommended when the size of the requests and the peak rate of requests can be pre-determined. It is highly recommended that adequate margin is designed into the pool size. The memory pool should never get close to being fully configured (i.e. not enough memory to create a new block). If the memory does become fully configured, requests for new size blocks will fail, regardless of whether the created blocks are in-use or not. The margin on the memory pool can be monitored by viewing the 'free bytes' member of the memory pool statistics. The memory pool statistics are dumped only when commanded by way of the ES command CFE_ES_SEND_MEM_POOL_STATS_CC.

A user of the ES memory pool begins by tailoring the memory pool for the particular use, by defining a list of block sizes and allocating a block of memory. These block size definitions simply give the memory pool a set of sizes to choose

from. They do not configure the memory pool in any way and they do not affect the size of the pool. The cFE defines a default set of block sizes in the cfe_platform_cfg.h file.

If the default block sizes are used, the application will create the pool using the simpler CFE_ES_PoolCreate API. This API takes a pointer to the first byte of the memory pool (allocated by the application) and a size parameter. The API returns a handle to be used for the get and put requests.

If the defaults are not sufficient, the user must define the block sizes and use the CFE_ES_PoolCreateEx API.

After receiving a positive response from the PoolCreate API, the memory pool is ready to accept requests, but at this point it is completely unconfigured (meaning there are no blocks created). The first valid request (via CFE_ES_Get↩PoolBuf API) after creating the pool will always cause the memory pool to create a block and return a pointer to the new block. The size of the block depends on the size definitions mentioned earlier. If there is not an exact match between the requested and defined sizes, then the memory pool will create and return the smallest block that meets the following criteria: is a defined size and large enough to hold the request.

If another request for that size comes in before the first block was released through the CFE_ES_PutPoolBuf API, then the memory pool will create a second block of that size and return a pointer to the second block. If both blocks were then released through the CFE_ES_PutPoolBuf API and the memory pool statistics were dumped via the CFE_ES_↩SEND_MEM_POOL_STATS_CC command, the number of blocks created would be two. The number of 'free bytes' in the pool would be the size of the pool minus the sum of the following items:

- the size of the two blocks created (even though they are not 'in-use').
- a buffer descriptor for each of the two blocks created ($2 * 12$ bytes)
- a 168 byte pool descriptor Refer to the cFE Applications Developers Guide for more details.

This allocation algorithm does have its limits. There are certain conditions that can place the memory pool in an undesired state. For instance, if a burst of get requests were received for the same block size, the memory pool may create a large number of blocks of that size. If this is a one-time burst, the memory pool would be configured with this large number of blocks that may no longer be needed. This scenario would use up the 'free bytes' margin in an undesired way. It should be noted that once the blocks are created, they cannot be deleted by any means other than a processor or power-on reset. It is highly recommended that the memory pool statistics be carefully monitored to ensure that the 'free-bytes' margin is sufficient (which is typically dictated by mission requirements).

An operator can obtain information about an Application's Memory Pool by using the Telemeter Memory Pool Statistics Command.

This command will cause Executive Services to extract pertinent statistics from the data used to manage the Memory Pool and telemeter them to the ground in the Memory Pool Statistics Telemetry Packet.

In order to obtain the statistics associated with a memory pool, the operator **MUST** have the correct Memory Handle as reported by the Application who owns the Memory Pool. **It should be noted that an inappropriate Memory Pool Handle can *(and likely will)* cause the system software to crash!** Within the cFE itself, there are three cFE Core Applications that make use of the Executive Services Memory Pool API. These are Software Bus (SB), Event Services (EVS) and Table Services (TBL). Each of these cFE Core Applications report their memory pool handles in telemetry.

The Memory Pool Statistics Telemetry Packet contains the following information:

- **Memory Pool Handle** - the handle, as provided by the operator in the Telemeter Memory Pool Statistics Command. This repeating of the handle in telemetry insures the operator knows which Memory Pool Statistics are being viewed

- **Pool Size** - The total size of the memory pool (in bytes)
- **Number Blocks Requested** - The total number of memory blocks requested for allocation
- **Number of Errors** - The total number of errors encountered when a block was released
- **Number of Free Bytes** - The total number of bytes in the Memory Pool that have never been allocated to a Memory Block
- **Block Statistics** - For each specified size of memory block (of which there are CFE_MISSION_ES_POOL_MAX_BUCKETS), the following statistics are kept
  - **Block Size** - The size, in bytes, of all blocks of this type
  - **Number of Blocks Allocated** - The number of this sized block which are currently allocated and in use
  - **Number of Blocks Free** - The number of this size block which have been in use previously but are no longer being used

Next: System Log
Prev: Critical Data Store
Up To: cFE Executive Services Overview

9.17 System Log

The System Log is an array of bytes that contains back-to-back printf type messages from applications. The cFE internal applications use this log when errors are encountered during initialization before the Event Manager is fully initialized. To view the information the CFE_ES_WRITE_SYSLOG_CC command must be sent. This command will write the log to a binary file. The path and filename may be specified in the command. If the filename command field contains an empty string, the configuration parameter CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE is used to specify the path and filename. Use the ground system to get the file and display the contents. The CFE_ES_CLEAR_SYSLOG_CC is used to clear the System log.

The size of the System log is defined by the platform configuration parameter CFE_PLATFORM_ES_SYSTEM_LOG_SIZE. This log is preserved after a processor reset and held in the ES reset area.

A count of the number of entries in the log is present in the ES housekeeping telemetry.

Next: Version Identification
Prev: Memory Pool
Up To: cFE Executive Services Overview

9.18 Version Identification

Version information is reported at startup, and upon receipt of a No-op command

Next: Executive Services Frequently Asked Questions
Prev: System Log
Up To: cFE Executive Services Overview

9.19 Executive Services Frequently Asked Questions

Prev: Version Identification
Up To: cFE Executive Services Overview

10 cFE Executive Services Commands

The following is a list of commands that are processed by the cFE Executive Services Task.

Global CFE_ES_CLEAR_ER_LOG_CC
Clears the contents of the Exeception and Reset Log

Global CFE_ES_CLEAR_SYSLOG_CC
Clear Executive Services System Log

Global CFE_ES_DELETE_CDS_CC
Delete Critical Data Store

Global CFE_ES_DUMP_CDS_REGISTRY_CC
Dump Critical Data Store Registry to a File

Global CFE_ES_NOOP_CC
Executive Services No-Op

Global CFE_ES_OVER_WRITE_SYSLOG_CC
Set Executive Services System Log Mode to Discard/Overwrite

Global CFE_ES_QUERY_ALL_CC
Writes all Executive Services Information on All Applications to a File

Global CFE_ES_QUERY_ALL_TASKS_CC
Writes a list of All Executive Services Tasks to a File

Global CFE_ES_QUERY_ONE_CC
Request Executive Services Information on a Specified Application

Global CFE_ES_RELOAD_APP_CC
Stops, Unloads, Loads from the command specfied File and Restarts an Application

Global CFE_ES_RESET_COUNTERS_CC
Executive Services Reset Counters

Global CFE_ES_RESET_PR_COUNT_CC
Resets the Processor Reset Counter to Zero

Global CFE_ES_RESTART_APP_CC
Stops, Unloads, Loads using the previous File name, and Restarts an Application

Global CFE_ES_RESTART_CC
Executive Services Processor / Power-On Reset

Global CFE_ES_SEND_MEM_POOL_STATS_CC
Telemeter Memory Pool Statistics

Global CFE_ES_SET_MAX_PR_COUNT_CC
Configure the Maximum Number of Processor Resets before a Power-On Reset

Global CFE_ES_SET_PERF_FILTER_MASK_CC
Set Performance Analyzer's Filter Masks

Global CFE_ES_SET_PERF_TRIGGER_MASK_CC
Set Performance Analyzer's Trigger Masks

Global CFE_ES_START_APP_CC
Load and Start an Application

Global CFE_ES_START_PERF_DATA_CC
Start Performance Analyzer

Global CFE_ES_STOP_APP_CC
Stop and Unload Application

Global CFE_ES_STOP_PERF_DATA_CC
Stop Performance Analyzer

Global CFE_ES_WRITE_ER_LOG_CC
Writes Exeception and Reset Log to a File

Global CFE_ES_WRITE_SYSLOG_CC
Writes contents of Executive Services System Log to a File

11 cFE Executive Services Telemetry

The following are telemetry packets generated by the cFE Executive Services Task.

Global CFE_ES_HousekeepingTlm_Payload_t
Executive Services Housekeeping Packet

Global CFE_ES_HousekeepingTlm_Payload_t
Executive Services Housekeeping Packet

Global CFE_ES_OneAppTlm_Payload_t
Single Application Information Packet

Global CFE_ES_OneAppTlm_Payload_t
Single Application Information Packet

Global CFE_ES_PoolStatsTlm_Payload_t
Memory Pool Statistics Packet

Global CFE_ES_PoolStatsTlm_Payload_t
Memory Pool Statistics Packet

12 cFE Executive Services Configuration Parameters

The following are configuration parameters used to configure the cFE Executive Services either for each platform or for a mission as a whole.

Global CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN
Maximum Length of Full CDS Name in messages

Global CFE_MISSION_ES_CDS_MAX_NAME_LENGTH
Maximum Length of CDS Name

Global CFE_MISSION_ES_DEFAULT_CRC
Mission Default CRC algorithm

Global CFE_MISSION_ES_MAX_APPLICATIONS
Mission Max Apps in a message

Global CFE_MISSION_ES_PERF_MAX_IDS
Define Max Number of Performance IDs for messages

Global CFE_MISSION_ES_POOL_MAX_BUCKETS
Maximum number of block sizes in pool structures

Global CFE_PLATFORM_CORE_MAX_STARTUP_MSEC
CFE core application startup timeout

Global CFE_PLATFORM_ES_APP_KILL_TIMEOUT
Define ES Application Kill Timeout

Global CFE_PLATFORM_ES_APP_SCAN_RATE
Define ES Application Control Scan Rate

Global CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES
Define Maximum Number of Registered CDS Blocks

Global CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01
Define ES Critical Data Store Memory Pool Block Sizes

Global CFE_PLATFORM_ES_CDS_SIZE
Define Critical Data Store Size

Global CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE
Default Application Information Filename

Global CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE
Default Critical Data Store Registry Filename

Global CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE
Default Exception and Reset (ER) Log Filename

Global CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME
Default Performance Data Filename

Global CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE
Define Default System Log Mode following Power On Reset

Global CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE
Define Default System Log Mode following Processor Reset

Global CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
Define Default Stack Size for an Application

Global CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE
Default System Log Filename

Global CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE
Default Application Information Filename

Global CFE_PLATFORM_ES_ER_LOG_ENTRIES

Define Max Number of ER (Exception and Reset) log entries

Global CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE

Maximum size of CPU Context in ES Error Log

Global CFE_PLATFORM_ES_MAX_APPLICATIONS

Define Max Number of Applications

Global CFE_PLATFORM_ES_MAX_GEN_COUNTERS

Define Max Number of Generic Counters

Global CFE_PLATFORM_ES_MAX_LIBRARIES

Define Max Number of Shared libraries

Global CFE_PLATFORM_ES_MAX_MEMORY_POOLS

Maximum number of memory pools

Global CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS

Define Number of Processor Resets Before a Power On Reset

Global CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01

Define Default ES Memory Pool Block Sizes

Global CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN

Define Memory Pool Alignment Size

Global CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING

Default virtual path for persistent storage

Global CFE_PLATFORM_ES_NONVOL_STARTUP_FILE

ES Nonvolatile Startup Filename

Global CFE_PLATFORM_ES_OBJECT_TABLE_SIZE

Define Number of entries in the ES Object table

Global CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY

Define Performance Analyzer Child Task Delay

Global CFE_PLATFORM_ES_PERF_CHILD_PRIORITY

Define Performance Analyzer Child Task Priority

Global CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE

Define Performance Analyzer Child Task Stack Size

Global CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE

Define Max Size of Performance Data Buffer

Global CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS

Define Performance Analyzer Child Task Number of Entries Between Delay

Global CFE_PLATFORM_ES_PERF_FILTMASK_ALL

Define Filter Mask Setting for Enabling All Performance Entries

Global CFE_PLATFORM_ES_PERF_FILTMASK_INIT

Define Default Filter Mask Setting for Performance Data Buffer

Global CFE_PLATFORM_ES_PERF_FILTMASK_NONE

Define Filter Mask Setting for Disabling All Performance Entries

Global CFE_PLATFORM_ES_PERF_TRIGMASK_ALL

Define Filter Trigger Setting for Enabling All Performance Entries

Global CFE_PLATFORM_ES_PERF_TRIGMASK_INIT

Define Default Filter Trigger Setting for Performance Data Buffer

Global CFE_PLATFORM_ES_PERF_TRIGMASK_NONE

Define Default Filter Trigger Setting for Disabling All Performance Entries

Global CFE_PLATFORM_ES_POOL_MAX_BUCKETS

Maximum number of block sizes in pool structures

Global CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING

Default virtual path for volatile storage

Global CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS

ES Ram Disk Number of Sectors

Global CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED

Percentage of Ram Disk Reserved for Decompressing Apps

Global CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE

ES Ram Disk Sector Size

Global CFE_PLATFORM_ES_RESET_AREA_SIZE

Define ES Reset Area Size

Global CFE_PLATFORM_ES_START_TASK_PRIORITY

Define ES Task Priority

Global CFE_PLATFORM_ES_START_TASK_STACK_SIZE

Define ES Task Stack Size

Global CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC

Startup script timeout

Global CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC

Poll timer for startup sync delay

Global CFE_PLATFORM_ES_SYSTEM_LOG_SIZE

Define Size of the cFE System Log.

Global CFE_PLATFORM_ES_USER_RESERVED_SIZE

Define User Reserved Memory Size

Global CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE

ES Volatile Startup Filename

Global CFE_PLATFORM_EVS_START_TASK_PRIORITY

Define EVS Task Priority

Global CFE_PLATFORM_EVS_START_TASK_STACK_SIZE

Define EVS Task Stack Size

Global CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01

Define SB Memory Pool Block Sizes

Global CFE_PLATFORM_SB_START_TASK_PRIORITY

Define SB Task Priority

Global CFE_PLATFORM_SB_START_TASK_STACK_SIZE

Define SB Task Stack Size

Global CFE_PLATFORM_TBL_START_TASK_PRIORITY

Define TBL Task Priority

Global CFE_PLATFORM_TBL_START_TASK_STACK_SIZE

Define TBL Task Stack Size

13 cFE Event Services Overview

Event Services (EVS) provides centralized control for the processing of event messages originating from the EVS task itself, other cFE core applications (ES, SB, TIME, and TBL), and from cFE applications. Event messages are asynchronous messages that are used to inform the operator of a significant event from within the context of a registered application or core service. EVS provides various ways to filter event messages in order to manage event message generation.

Note for messages outside the context of a registered appliction (for example early in app initialization or if registration fails) CFE_ES_WriteToSysLog can be used for reporting.

For more information on cFE Event Services, see the following sections:

- Event Message Format
- Local Event Log
- Event Message Control
- Event Message Filtering
- EVS Registry
- EVS Counters
- Resetting EVS Counters
- Effects of a Processor Reset on EVS
- Frequently Asked Questions about Event Services

13.1 Event Message Format

Event messages are software bus messages that contain the following fields:

- Timestamp
- Event Type
- Spacecraft ID

- Processor ID
- Application Name
- Event ID
- Message

The *Timestamp* corresponds to when the event was generated, in spacecraft time. The *Event Type* is one of the following: DEBUG, INFO, ERROR or CRITICAL. The *Spacecraft ID* and *Processor ID* identify the spacecraft and processor from which the event was generated. Note that the *Spacecraft ID* is defined in the cfe_mission_cfg.h file; The *Processor ID* is defined in the appropriate cfe_platform_cfg.h file. The *Application Name* refers to the Application that issued the event message as specified on application startup (either startup script or app start command). The *Event ID* is an Application unique number that identifies the event. The *Message* is an ASCII text string describing the event. Event messages may have parameters associated with the event message. EVS formats the parameters such that they are part of the ASCII text string that make up the event message.

In order to accommodate missions that have limited telemetry bandwidth, EVS can be configured such that the ASCII text string part of the event message is omitted, thus reducing the size of each event message. This is referred to as *Short Format*; Event messages including the ASCII text string are referred to as *Long Format*. The default setting is specified in the cfe_platform_cfg.h file. EVS also provides commands in order to set the mode (short or long).

Since the design of the cFE's Software Bus is based on run-time registration, no predetermined message routing is defined, hence it is not truly correct to say that events are generated as telemetry. Technically, EVS generates events in the form of software bus messages. Applications such as Telemetry Output and Data Storage can then subscribe to these messages making them telemetry. For the purposes of this document, any references to telemetry assumes that a telemetry application subscribes to the EVS event software bus message and routes it to the ground as telemetry. Note that short format event messages on the Software Bus have different message lengths than long form messages and do not include any part of the long format message string.

The EVS can be configured via ground command to send event messages out one or more message ports. These message ports may include ports such as debug, console, and UART. Messages sent out of the message ports will be in ASCII text format. This is generally used for lab purposes. Note that the event mode (short or long) does affect the event message content sent out these message ports.

Next: Local Event Log
Up To: cFE Event Services Overview

13.2 Local Event Log

In addition to generating a software bus message, EVS logs the event message to a Local Event Log. Note that this is an optional feature that must be enabled via the cfe_platform_cfg.h file. The Local Event Log resides on the same processor as the EVS which is used to store events without relying on an external bus. In multi-processor cFE configurations the Local Event Buffer preserves event messages during non-deterministic processor initialization sequences and during failure scenarios. In order to obtain the contents of the Local Event Log, a command must be sent to write the contents of the buffer to a file which can then be sent to the ground via a file transfer mechanism. Note that event messages stored in the EVS Local Event Log are always long format messages and are not affected by the event mode (short or long).

EVS provides a command in order to clear the Local Event Log .

Local Event Log Mode

EVS can be configured to control the Local Event Log to either discard or overwrite the contents of the log when it becomes full. If the mode is set to overwrite, the log is treated like a circular buffer, overwriting the oldest event message contained in the log first. This control is configured by default in the cfe_platform_cfg.h file but can be modified by a command .

Next: Event Message Control
Prev: Event Message Format
Up To: cFE Event Services Overview

13.3 Event Message Control

In order for an application to be serviced by EVS, it must be registered with EVS. EVS provides various commands in order to control the event messages that are generated as software bus messages.

Event Message Control - By Type

The highest level of event message control that EVS provides is the ability to enable and disable event message types. As mentioned above, there are four event types. They are:

1. DEBUG
2. INFORMATION
3. ERROR
4. CRITICAL

When commands are sent to enable or disable a particular type of event message, ALL event messages of the specified type are affected. Typically, event meesages of type DEBUG are disabled on-orbit. Note that EVS provides the capability to affect multiple types within one command using a bit mask. Note also that the configuration parameter CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG in the cfe_platform_cfg.h file specifies which event message types are enabled/disabled by default.

Event Message Control - By Application

Commands are available to enable and disable the generation of event messages for a particular application. The result is that ALL event messages for the specified Application are affected (i.e. enabled or disabled).

Event Message Control - By Event Type for an Application

EVS also provides the capability to enable / disable an event type for a particular application. Note that EVS provides the capability to affect multiple event types within one command using a bit mask.

Event Message Control - Individual Events

There are two ways to control the generation of individual events depending on whether the application's event message has been registered with EVS or not.

Modifying a registered event message filter

When an application registers with EVS, the application has the option of specifying the events that it wants to register for filtering along with the Event Message Filtering (only the Binary Filtering Scheme exists currently). Note that applications are limited in the number of events that they can register for filtering (see CFE_PLATFORM_EVS_MAX_EVENT_FIL↩TERS in cfe_platform_cfg.h for the mission defined limit). The filtering method uses a mask to determine if the message is forwarded to the software bus, making it available in telemetry (see Event Message Filtering for a description on filtering). Commands are available to modify the filter mask for any registered event.

An on-orbit mission, for example, might be experiencing a problem resulting in an application's event message being repeatedly issued, flooding the downlink. If the event message for the application is registered with EVS, then a command can be issued to set the event message filter to the specified value in order to prevent flooding of the downlink.

Adding/Removing an event message for filtering

Commands are also available to add filtering for those events that are not registered for filtering. Once an event is registered for filtering , the filter can be modified (see above) or removed .

An on-orbit mission, for example, might be experiencing a problem resulting in a event message being repeatedly issued, flooding the downlink. If the event message was not registered with EVS for filtering then the ground can add (i.e. register) the offending application's event for filtering (much like an application registers the event during initialization).

EVS also supports the ability to remove (i.e. unregister) an application's event message. Once it is removed, the event will no longer be filtered. Note that commands issued to disable events by event type, by application or by event type for an application are still valid and could affect this particular event.

Next: Event Message Filtering
Prev: Local Event Log
Up To: cFE Event Services Overview

13.4 Event Message Filtering

EVS uses a hexadecimal bit mask that controls how often a message is filtered. An event's filter mask is bit-wise ANDed with the event's event counter. There is one event counter for each event ID. If the result of the ANDing is zero then the message is sent.

Filter masks can be set so that one out of 1, 2, 4, 8 events are sent. Some examples of masks that use this pattern are: (0x0000, Every one), (0x0001, One of every 2), (0x0003, One of every 4), and (0x0007, One of every 8.

Filter masks can also be set so that only the first n events are sent. For example, the mask 0xFFFF generates one event message and then stops. Note that when the filter counter is reset to zero by command, this will restart the counting and enable n more events to be sent.

Event messages will be filtered until CFE_EVS_MAX_FILTER_COUNT events of the filtered event ID from the application have been received. After this, the filtering will become locked (no more of that event will be received by the ground) until the filter is either reset or deleted by ground command. This is to prevent the counter from rolling over, which would cause some filters to behave improperly. An event message will be sent when this maximum count is reached.

The following shows an example of how filtering works using a filter mask of x'0001', resulting in sending every other event:

	packet x	**packet X+1**	**packet X+2**	**packet X+3**	**packet X+4**	**...**
Event ID counter	x'0000'	x'0001'	x'0002'	x'0003'	x'0004'	
Event Filter mask	x'0001'	x'0001'	x'0001'	x'0001'	x'0001'	
Bitwise AND results	x'0000'	x'0001'	x'0000'	x'0001'	x'0000'	
Send event?	Yes	No	Yes	No	Yes	

In this example, the ground uses a filter mask of x'FFFE' resulting in the first two events being sent and then no more.

	packet x	**packet X+1**	**packet X+2**	**packet X+3**	**packet X+4**	**...**
Event ID counter	x'0000'	x'0001'	x'0002'	x'0003'	x'0004'	
Event Filter mask	x'FFFE'	x'FFFE'	x'FFFE'	x'FFFE'	x'FFFE'	
Bitwise AND results	x'0000'	x'0000'	x'0002'	x'0002'	x'0004'	
Send event?	Yes	Yes	No	No	No	

See cfe_evs.h for predefined macro values which can be used for masks.

Next: EVS Registry
Prev: Event Message Control
Up To: cFE Event Services Overview

13.5 EVS Registry

EVS maintains information on each registered application and all events registered for an application.

The registry contains the following information for each Registered Application:

- Active Flag - If equal to FALSE (0), all events from this Application are Filtered

- Event Count - Total number of events issued by this Application. Note that this value stop incrementing at 65535.

  The following information for each Filtered Event (up to CFE_PLATFORM_EVS_MAX_EVENT_FILTERS)↩ :

- Event ID - Event ID for event whose filter has been defined

- Mask - Binary Filter mask value (see Event Message Filtering for an explanation)

- Count - Current number of times this Event ID has been issued by this Application

Next: EVS Counters
Prev: Event Message Filtering
Up To: cFE Event Services Overview

13.6 EVS Counters

There are 2 types of counters in EVS housekeeping telemetry:

- Total events sent counter

- Number of events sent for each Application

The difference is that the first one is the sum of all of the event messages sent. Both of these represent events that are actually sent (by EVS to the software bus). If an event message is filtered or disabled, neither counter is incremented.

There are other counters available that show how many event messages were generated by an App, however, these are only available for those events that are registered for filtering hence if you have a message that is not registered for filtering and the message type (e.g. DEBUG) is disabled then you won't know if the event was ever issued by an application. These counters are available by sending a command to write the EVS Application Data and transferring the file to the ground.

Next: Resetting EVS Counters
Prev: EVS Registry
Up To: cFE Event Services Overview

13.7 Resetting EVS Counters

As far as reset commands, there are 4 commands available:

1. Reset the total events sent counter

2. Reset the events sent counter for a particular Application - e.g. reset the LC application events counter

3. Reset all of the event counters for a particular registered event for a particular Application - e.g. Reset event counter for Event ID 5 for the LC Application.

4. Reset all of the event counters for ALL registered events for a particular App - e.g. Reset all registered event counters for LC.

Note that there is currently no way to reset ALL of the events sent counters for all of the Apps with one command.

Next: Effects of a Processor Reset on EVS
Prev: EVS Counters
Up To: cFE Event Services Overview

13.8 Effects of a Processor Reset on EVS

On a processor reset, the EVS Registry is cleared such that applications must re-register with EVS in order to use EVS services. All counters are also cleared with the exceptions of those listed below.

On a processor reset, the following EVS data is preserved (if the cFE is configured to include an Local Event Log):

- Local Event Log if the Local Event Log Mode is configured to Discard (1). If the Local Event Log Mode is configured to Overwrite (0), the contents of the log may be overwritten depending on the size and contents of the log prior to the reset.

- Local Event Log Full Flag

- Local Event Log overflow counter

The Local Event Log Mode (overwrite/discard) is set to the configured value specified in the cfe_platform_cfg.h file. The default value is Discard (1). Discard mode will guarantee the contents of the event log are preserved over a processor restart.

This provides the ground with the capability to write the Local Event Log to a file and transfer it to the ground in order to help debug a reset.

Next: Frequently Asked Questions about Event Services
Prev: Resetting EVS Counters
Up To: cFE Event Services Overview

13.9 Frequently Asked Questions about Event Services

(Q) My telemetry stream is being flooded with the same event message. How do I make it stop?	
	The most direct way to stop an event message from flooding your downlink stream is to send a command to EVS to filter the offending event (see Event Message Control or $sc_$cpu_EVS_SetBinFltrMask). In order to stop the event message from being sent, a bit mask of `'0xFFFF'` should be used. If the event is not currently registered for filtering, the event message must be added using the command $sc_$cpu_EVS_AddEvtFltr .
(Q) I filtered an event message and would now like to see it again. What do I do in order to see those events again?	
	If the event message that you are interested is registered with EVS for filtering, then you have 2 options: 1. You can use the $sc_$cpu_EVS_SetBinFltrMask command using a bit mask of `'0x0000'` which will result in getting all of the events for that Event Id **or** 2. You can remove the registration of that event with EVS (see $sc_$cpu_EVS_DelEvtFltr). Note that option (1) is the preferred method.
(Q) What is the purpose of DEBUG event messages?	
	Event message of type "DEBUG" are primarily used during flight software development in order to provide information that is most likely not needed on orbit. Some commands send debug event messages as verification that a command request was received. When writing the EVS local event log to a file, for example, an event message of type DEBUG is issued. On orbit, this event message is probably not needed. Instead, the command counter is used for command verification.
(Q) How do I find out which events are registered for filtering?	
	EVS provides a command ($sc_$cpu_EVS_WriteAppData2File) which generates a file containing all of the applications that have registered with EVS and all of the filters that are registered for each application. Note that EVS merely generates the file. The file must be transferred to the ground in order to view it.
(Q) Why do I see event messages in my console window?	
	By default, the events are configured to transmit out a "port" that shows event messages in the console
(Q) What is the difference between event services and the ES System Log	
	Events are within the context of an App or cFE Service (requires registration with ES). The system log can be written to outside of the Application or cFE Service context, for example during application startup to report errors before registration.

Prev: Effects of a Processor Reset on EVS
Up To: cFE Event Services Overview

14 cFE Event Services Commands

The following is a list of commands that are processed by the cFE Event Services Task.

Global CFE_EVS_ADD_EVENT_FILTER_CC

Add Application Event Filter

Global CFE_EVS_CLEAR_LOG_CC

Clear Event Log

Global CFE_EVS_DELETE_EVENT_FILTER_CC
Delete Application Event Filter

Global CFE_EVS_DISABLE_APP_EVENT_TYPE_CC
Disable Application Event Type

Global CFE_EVS_DISABLE_APP_EVENTS_CC
Disable Event Services for an Application

Global CFE_EVS_DISABLE_EVENT_TYPE_CC
Disable Event Type

Global CFE_EVS_DISABLE_PORTS_CC
Disable Event Services Output Ports

Global CFE_EVS_ENABLE_APP_EVENT_TYPE_CC
Enable Application Event Type

Global CFE_EVS_ENABLE_APP_EVENTS_CC
Enable Event Services for an Application

Global CFE_EVS_ENABLE_EVENT_TYPE_CC
Enable Event Type

Global CFE_EVS_ENABLE_PORTS_CC
Enable Event Services Output Ports

Global CFE_EVS_NOOP_CC
Event Services No-Op

Global CFE_EVS_RESET_ALL_FILTERS_CC
Reset All Event Filters for an Application

Global CFE_EVS_RESET_APP_COUNTER_CC
Reset Application Event Counters

Global CFE_EVS_RESET_COUNTERS_CC
Event Services Reset Counters

Global CFE_EVS_RESET_FILTER_CC
Reset an Event Filter for an Application

Global CFE_EVS_SET_EVENT_FORMAT_MODE_CC
Set Event Format Mode

Global CFE_EVS_SET_FILTER_CC
Set Application Event Filter

Global CFE_EVS_SET_LOG_MODE_CC
Set Logging Mode

Global CFE_EVS_WRITE_APP_DATA_FILE_CC
Write Event Services Application Information to File

Global CFE_EVS_WRITE_LOG_DATA_FILE_CC
Write Event Log to File

15 cFE Event Services Telemetry

The following are telemetry packets generated by the cFE Event Services Task.

Global CFE_EVS_HousekeepingTlm_Payload_t
Event Services Housekeeping Telemetry Packet

Global CFE_EVS_HousekeepingTlm_Payload_t
Event Services Housekeeping Telemetry Packet

Global CFE_EVS_LongEventTlm_Payload_t
Event Message Telemetry Packet (Long format)

Global CFE_EVS_LongEventTlm_Payload_t
Event Message Telemetry Packet (Long format)

Global CFE_EVS_ShortEventTlm_Payload_t
Event Message Telemetry Packet (Short format)

Global CFE_EVS_ShortEventTlm_Payload_t
Event Message Telemetry Packet (Short format)

16 cFE Event Services Configuration Parameters

The following are configuration parameters used to configure the cFE Event Services either for each platform or for a mission as a whole.

Global CFE_MISSION_EVS_MAX_MESSAGE_LENGTH
Maximum Event Message Length

Global CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE
Default EVS Application Data Filename

Global CFE_PLATFORM_EVS_DEFAULT_LOG_FILE
Default Event Log Filename

Global CFE_PLATFORM_EVS_DEFAULT_LOG_MODE
Default EVS Local Event Log Mode

Global CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE
Default EVS Message Format Mode

Global CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG
Default EVS Event Type Filter Mask

Global CFE_PLATFORM_EVS_LOG_MAX
Maximum Number of Events in EVS Local Event Log

Global CFE_PLATFORM_EVS_MAX_EVENT_FILTERS
Define Maximum Number of Event Filters per Application

Global CFE_PLATFORM_EVS_PORT_DEFAULT
Default EVS Output Port State

17 cFE Software Bus Overview

The Software Bus (SB) handles communication between software tasks on a processor. All tasks communicate with each other, with hardware devices, and with the ground by sending command and telemetry messages. The software bus provides an application programming interface (API) to other tasks for sending and receiving messages. This API is independent of the underlying operating system so that tasks can use the same interface regardless of which processor they reside on. Refer to the cFE Application Programmer's Interface (API) Reference for detailed information about the API functions.

The software bus is used internally by the flight software, and normally does not require attention from the ground. However, because of the scalability and the dynamic nature of the software bus, it is strongly recommended that each project carefully review the SB statistics and SB memory pool to be sure adequate margin is met on the configurable items.

The cFE software bus uses a dynamic protocol and builds its routing table at run-time through the SB subscribe API's. Also the cFE software bus pipes are created at run-time through the CFE_SB_CreatePipe API. Because the routing is established, and pipes are created at run-time, it is necessary to have a clear view of the routing details on command. The cFE software bus allows the user to dump the routing table, the pipe table, the message map and the statistics packet. Each of these items are described in detail in the corresponding section of this document.

- Software Bus Terminology
- Autonomous Actions
- Operation of the SB Software
- Frequently Asked Questions about Software Bus

17.1 Software Bus Terminology

In order to fully understand the Software Bus, it is imperative that the basic terms used to describe its features are also understood. Below are the critical terms that help identify what the Software Bus accomplishes for each Application:

- Messages
- Pipes
- Subscriptions
- Memory

Next: Messages
Up To: cFE Software Bus Overview

17.1.1 Messages

The sole purpose of the software bus is to provide applications a way to send messages to each other. The term message and the term packet are used interchangeably throughout this document. A message is a combined set of bytes with a predefined format that is used as the basis of communication on a spacecraft. All commands, telemetry, and other data that are passed between the ground and the spacecraft, and between subsystems of the spacecraft, are considered to be messages. The most common message format is CCSDS (Consultative Committee for Space Data Systems) in `CCSDS Space Packet Protocol`, but can be customized by replacing the message module.

There are two general types of messages - commands (or command packets) and telemetry (or telemetry packets). Command packets are sent to a particular software task from the ground (or another task). Telemetry packets are sent from a particular software task to the ground (or other tasks).

The concept of a message identifier is utilized to provide abstraction from header implementation, often abbreviated as message ID, MsgId, or MID. Header and message identifier values should not be accessed directly to avoid implementation specific dependencies.

Telemetry packets typically contain a timestamp that indicates when the packet was produced. Command packets typically contain a command code that identifies the particular type of command.

The message module provides APIs for 'setting' and 'getting' the fields in the header of the message. The message module was separated from software bus to enable users to customize message headers without requiring clone and own of the entire cfe repository. To customize, remove the built in msg module from the build and replace with custom implementation. See sample target definitions folder for examples.

Following the header is the user defined message data.

Next: Pipes
Up To: Software Bus Terminology

17.1.2 Pipes

The destinations to which messages are sent are called pipes. These are queues that can hold messages until they are read out and processed by a task. Each pipe is created at run-time through the CFE_SB_CreatePipe API. The pipe name and the pipe depth are given as arguments in the API. The pipe identifier (or PipeId) is given back to the caller after the API is executed. Each pipe can be read by only one task, but a task may read more than one pipe. Only the pipe owner is allowed to subscribe to messages on the pipe.

The Pipe IDs are specific to a particular processor (that is, the same ID number may refer to a different pipe on each processor). The pipe information for all pipes that have been created, may be requested at anytime by sending the 'Write Pipe Info' SB command . The software bus also provides a set of figures regarding capacity, current utilization and high water marks relevant to pipes. This information may be requested by sending the command to dump the SB statistics packet .

Next: Subscriptions
Prev: Messages
Up To: Software Bus Terminology

17.1.3 Subscriptions

A subscription is a run-time request for a particular message to be sent to a particular pipe. If the caller of the subscribe API is not the owner of the pipe, the request is rejected and an error event is sent. The application that creates the pipe is considered the owner of the pipe. The pipe specified in the subscription is sometimes referred to as the destination of the message. There are a maximum number of destinations for a particular message. This value is specified by the platform configuration parameter CFE_PLATFORM_SB_MAX_DEST_PER_PKT.

As subscriptions are received, the destinations are added to the head of a linked list. During the sending of a message, the list is traversed beginning at the head of the list. Therefore the message will first be sent to the last subscriber. If an application has timing constraints and needs to receive a message in the shortest possible time, the developer may consider holding off its subscription until other applications have subscribed to the message.

The message limit specifies the maximum number of messages (with the specified Message ID) that are allowed on the specified pipe at any time. This limit is specified by the application at the time of the subscription. If the application uses the CFE_SB_Subscribe API, a message limit default value of four is used. If this default value is not sufficient, the caller would use the CFE_SB_SubscribeEx API that allows the message limit to be specified.

The software bus also provides the user with an option to unsubscribe to a message. The unsubscribe API takes two parameters, Message ID and Pipe ID. Only the owner of a pipe may unsubscribe to messages on that pipe.

Next: Memory
Prev: Pipes
Up To: Software Bus Terminology

17.1.4 Memory

The software bus statically allocates a block of memory for message buffers and subscription blocks. The size of this memory block is defined by the platform configuration parameter CFE_PLATFORM_SB_BUF_MEMORY_BYTES. The memory is managed by the cFE ES memory pool and is used only by the software bus. The ES memory pool allows an application to define the block sizes for the pool at compile time. These sizes are defined by the platform configuration parameters prefixed with CFE_SB_MEM_BLOCK_SIZE (for example, CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01). It is recommended that a project tailor these values for the mission, based on the software bus packet sizes.

At the time a message is sent, two buffers are allocated from the pool. One for a buffer descriptor (CFE_SB_BufferD_t) and one for the size of the packet. Both buffers are returned to the pool when the message has been received by all recipients. More precisely, if there is one recipient for a message, the message buffers will be released on the following call to CFE_SB_ReceiveBuffer for the pipe that received the buffer.

Also when subscriptions are received through the subscribe API's, the software bus allocates a subscription block (CFE_SB_DestinationD_t) from the pool. The subscription blocks are returned to the pool if and when the subscription is nullified through a CFE_SB_Unsubscribe call.

The software bus provides a set of figures regarding memory capacity, current memory utilization and high water marks relevant to the SB memory pool. This information may be requested by sending the command to dump the SB statistics packet. In addition, the current memory utilization value and the 'unmarked memory' value (CFE_PLATFORM_SB_BUF_MEMORY_BYTES minus peak memory in use) are sent in software bus housekeeping telemetry. The unmarked memory value should be monitored regularly to ensure that the value (in bytes) does not continue to decline or approach zero. If this value were to approach zero, there is a possibility that memory requests would fail which may inhibit the sending of a message. The current memory utilization value should also be monitored to ensure the system contains no memory leaks. The value (in bytes) should remain stable under nominal conditions. Refer to the ES users guide for more information regarding the ES Memory Pool.

Next: Autonomous Actions
Prev: Subscriptions
Up To: Software Bus Terminology

17.2 Autonomous Actions

The software bus is primarily a set of library routines that are called by other software tasks to send and receive packets. The software bus does not perform any operations autonomously, except for sending event messages if errors are detected during the transfer of packets.

As do other tasks, the SB task sends out housekeeping telemetry when requested through the 'Send Housekeeping Data' command.

Next: Operation of the SB Software
Prev: Software Bus Terminology
Up To: cFE Software Bus Overview

17.3 Operation of the SB Software

- Initialization
- All Resets
- Message Routing
- Packet Sequence Values
- Message Limit Error
- Pipe Overflow Error
- SB Event Filtering
- Diagnostic Data
- Control of Packet Routing
- Quality of Service
- Known Problem

Next: Initialization
Prev: Autonomous Actions
Up To: cFE Software Bus Overview

17.3.1 Initialization

No action is required by the ground to initialize the software bus. The software bus initializes internal data structures and tables the same way regardless of the type of reset.

Next: All Resets
Up To: Operation of the SB Software

17.3.2 All Resets

The software bus does not preserve any information across a reset of any kind. The software bus initializes internal data structures and tables the same way regardless of the type of reset. The routing is reestablished as the system initializes. It is normal procedure for each task of the system to create the pipe or pipes it needs and do all of its subscriptions during task initialization.

After any reset the following statements are true:

- The routing table is cleared and does not contain any routes.
- All subscriptions are lost and must be regenerated.
- The pipe table contains no data, all pipes must be recreated.
- Any packets in transit at the time of the reset are lost.
- The sequence counters for telemetry packets will begin again with a value of one.

Next: Message Routing
Prev: Initialization
Up To: Operation of the SB Software

17.3.3 Message Routing

In the software bus, all messages are processed in a similar way. The software bus uses the Message ID and the packet length fields (contained in the header) for routing the message to the destination pipe. If either of these two fields do not pass validation, the software bus generates an error event and aborts the delivery process. The software bus performs some validation checks by simply checking message header values against mission or platform configuration parameters. Messages originating from various tasks or instruments are routed to one or more pipes, where they wait until read by a task. The routing configuration for each message is established when applications call one of the SB subscribe APIs. The subscribe APIs take a Message ID and a Pipe ID as parameters. The routing for each packet is stored in SB memory and may be requested at any time by sending the 'Send Routing Info' command. The software bus also provides a set of figures regarding capacity, current utilization and high water marks relevant to the routing. This information may be requested by sending the command to dump the SB statistics packet.

Next: Packet Sequence Values
Prev: All Resets
Up To: Operation of the SB Software

17.3.4 Packet Sequence Values

The sequence count behavior depends on if the message is a command type or telemetry type.

The sequence counter for command messages is not altered by the software bus.

For a telemetry message, the behavior is controlled via API input parameters when sending. When enabled, the software bus will populate the packet sequence counter using an internal counter that gets intialized upon the first subscription to the message (first message will have a packet sequence counter value of 1). From that point on each send request will increment the counter by one, regardless of the number of destinations or if there is an active subscription.

After a rollover condition the sequence counter will be a value of zero for one instance. The sequence counter is incremented after all the checks have passed prior to the actual sending of the message. This includes the parameter checks and the memory allocation check.

When disabled, the original message will not be altered. This method of message delivery is recommended for situations where the sender did not generate the packet, such as a network interface application passing a packet from a remote system to the local software bus.

Next: Message Limit Error
Prev: Message Routing
Up To: Operation of the SB Software

17.3.5 Message Limit Error

Before placing a message on a pipe, the software bus checks the message limit to ensure the maximum number of packets in transit to the destination is not exceeded. If placing the message on the pipe would exceed the message limit, then the action of sending to that pipe is aborted and the 'Message Limit Error' event is sent. This condition will typically occur when an application that receives the packets does not respond quickly enough, or if the sender of the packets produces them too quickly.

This condition occurs often during development and during integration, for example when a remote processor gets reset or a 1553 cable becomes disconnected. Because of the common occurrences, the event may have filtering associated with it. Any filtering for this event would be performed by the cFE Event Services (EVS). Filtering for SB events may be specified in the cFE platform configuration file or may be commanded after the system initializes.

If this error occurs during nominal conditions, it could be an indication that the 'message limit' is not set correctly. The message limit is given at the time of the subscription and given as a parameter in the subscribe API. With the CFE_SB↩_Subscribe API, the SB uses a default message limit value specified by CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT. This constant is currently set to a value of four. If the default value is insufficient, the message limit value can be specified in the CFE_SB_SubscribeEx API.

A related failure is the pipe overflow condition, which can occur if the total number of packets (of all kinds) sent to a particular pipe is too large.

Next: Pipe Overflow Error
Prev: Packet Sequence Values
Up To: Operation of the SB Software

17.3.6 Pipe Overflow Error

Another common error that occurs during the send process is the pipe overflow error. This condition occurs if the total number of packets (of all kinds) sent to a particular pipe is too large. If this error occurs too frequently, it may be an indication that the pipe depth is not set correctly. The pipe depth is given at the time the pipe is created as a parameter in the CFE_SB_CreatePipe API.

Next: SB Event Filtering
Prev: Message Limit Error
Up To: Operation of the SB Software

17.3.7 SB Event Filtering

Most filtering for SB events is performed by the cFE Event Services (EVS). Filtering for SB events may be specified in the cFE platform configuration file or may be commanded after the system initializes. There is no SB event log that limits the number of events based on the capacity of the log, as in the heritage software bus.

There is one case in which events are filtered by the software bus instead of event services. This occurs when the software bus needs to suppress events so that a fatal recursive event condition does not transpire. Because error cases encountered when sending a message generate an event, and events cause a message to be sent a calling sequence could cause a stack overflow if the recursion is not properly terminated. The cFE software bus detects this condition and properly terminates the recursion. This is done by using a set of flags (one flag per event in the Send API) which determine whether an API has relinquished its stack. If the software bus needs to send an event that may cause recursion, the flag is set and the event is sent. If sending the event would cause the same event again, the event call will be bypassed, terminating the recursion. The result is that the user will see only one event instead of the many events that would normally occur without the protection. The heritage software bus did not have this condition because it stored events in the software bus event log and another thread would read them out at a later time.

Next: Diagnostic Data
Prev: Pipe Overflow Error
Up To: Operation of the SB Software

17.3.8 Diagnostic Data

The cFE software bus provides a set of commands to dump SB diagnostic data to help troubleshoot problems or check configuration settings. These commands allow the user to view the routing table, the pipe table or the message map. The message map is a lookup table used during a send operation to give fast access to the routing table index that corresponds to the message being sent.

The software bus also provides a statistics packet that can be used to tune the configuration parameters. This information is sent to the ground in the form of an SB packet when the corresponding command is received. The cFE limits the number of system pipes, unique Message IDs, buffer memory, messages on a pipe and subscriptions per Message ID. These limits are configurable through cFE platform and mission configuration parameters. The statistics packet was designed to let the project verify that these user settings provide the necessary margin to meet requirements.

The SB statistics information shows 'Currently In Use' figures, 'High Water Mark' figures and 'Max Allowed' figures for the following: buffer memory, messages on each pipe (pipe depth stats), System Pipes, Unique Message IDs and total subscriptions.

Depending on the task-scheduling implementation details of the operating system, it is possible to see the peak messages on a pipe occasionally exceed the depth of the pipe. The "Peak Messages In Use" parameter is included in the SB statistics packet under the pipe depth stats.

Next: Control of Packet Routing
Prev: SB Event Filtering
Up To: Operation of the SB Software

17.3.9 Control of Packet Routing

The software bus allows the ground to disable and enable the sending of packets of a specified Message ID to a specified pipe. All destinations that are needed for normal operation are enabled by default. Modifying the routing of packets may be required for the following reasons:

- In flight, one can enable diagnostic packets to see them on the ground.
- During testing, one can disable a destination to simulate an anomaly.

Next: Quality of Service
Prev: Diagnostic Data
Up To: Operation of the SB Software

17.3.10 Quality of Service

The software bus has a parameter in the CFE_SB_SubscribeEx API named Quality, which means Quality of Service (QOS) for off-board routing and is of the type CFE_SB_Qos_t. This structure has two members named priority and reliability. The Quality parameter is currently unused by the software bus. It is a placeholder to be used with the future software bus capability of inter-processor communication. Although currently the software bus does not implement quality of service.

A default quality of services is provided via the CFE_SB_DEFAULT_QOS macro.

Next: Known Problem
Prev: Control of Packet Routing
Up To: Operation of the SB Software

17.3.11 Known Problem

The software bus may perform unexpectedly under an unlikely corner-case scenario. This scenario was revealed in a stress test. The stress test was designed to deplete the Software Bus memory pool by having a high priority application continuously send 1000 byte packets to a lower priority application until the memory pool code returned an error code and sent the following event. "CFE_ES:getPoolBuf err:Request won't fit in remaining memory" At this point the higher priority sending application would stop executing. This would allow the lower priority receiving application to begin receiving the 1000 byte packets. After the receiving app processed all of the packets, the memory was restored to the memory pool as expected. The SB memory-in-use telemetry was zero because there were no software bus packets in transit. At this point any attempt to send a new-sized packet on the software bus was be rejected. The ES memory pool stated that the "... Request won't fit in remaining memory" even though there was currently no memory in use.

The simplest way to prevent this behavior is to ensure that there is margin when sizing the SB memory pool. To check the margin, monitor the "Peak Memory in Use" vs. the configuration parameter CFE_PLATFORM_SB_BUF_MEMO↩RY_BYTES which indicates the amount allocated.

Next: Frequently Asked Questions about Software Bus
Prev: Quality of Service
Up To: Operation of the SB Software

17.4 Frequently Asked Questions about Software Bus

(Q) How is the memory pool handle (sent in SB housekeeping telemetry) intended to be used?

The memory pool handle is used to analyze the SB memory pool statistics. The cFE ES command (CFE_ES_SEND_MEM_POOL_STATS_CC) to dump the memory pool statistics takes the pool handle as a parameter. These statistics tell how the SB memory pool is configured and gives details on margin. An improperly configured SB memory pool may inhibit communication. This may occur if there is not enough margin to create a block of the size needed for a transfer. Refer to the ES memory pool users guide for more details. Memory Pool

(Q) When sending a message, what message header fields are critical for routing the message?

To route the message properly, the software bus uses only the Message ID and packet length fields from the header of the message. If the packet length field is incorrect, then the buffer allocation for the message will also be incorrect. This may appear to the receiver as a truncated message or a message with unknown data added to the end of the message.

(Q) How many copies of the message are performed in a typical message delivery?

There is a single copy of the message performed when sending a message (from the callers memory space) using CFE_SB_TransmitMsg. When transmitting the message, the software bus copies the message from the callers memory space into a buffer in the software bus memory space. There is also the option to request a buffer from SB, write directly to the buffer and send via CFE_SB_TransmitBuffer. This is equivalent to the previous zero copy implementation. The CFE_SB_ReceiveBuffer API gives the user back a pointer to the buffer. When working with the buffers, the additional complexity to be aware of is the buffer is only available to the app from the request to send (on the sending side), or from the receive until the next receive on the same pipe on the receiving side. If the data is required outside that scope, the app needs a local copy.

(Q) When does the software bus free the buffer during a typical message delivery process? Or how long is the message, and the pointer to the buffer in the CFE_SB_ReceiveBuffer valid?

After receiving a buffer by calling CFE_SB_ReceiveBuffer, the buffer received is valid until the next call to CFE_SB_ReceiveBuffer with the same Pipe Id. If the caller needs the message longer than the next call to CFE_SB_ReceiveBuffer, the caller must copy the message to its memory space.

(Q) The first parameter in the CFE_SB_ReceiveBuffer API is a pointer to a pointer which can get confusing. How can I be sure that the correct address is given for this parameter.

Typically a caller declares a ptr of type CFE_SB_Buffer_t (i.e. CFE_SB_Buffer_t *Ptr) then gives the address of that pointer (&Ptr) as this parameter. After a successful call to CFE_SB_ReceiveBuffer, Ptr will point to the first byte of the software bus buffer. This should be used as a read-only pointer. In systems with an MMU, writes to this pointer may cause a memory protection fault.

(Q) Why am I not seeing expected Message Limit error events or Pipe Overflow events?

It is possible the events are being filtered by cFE Event Services. The filtering for this event may be specified in the platform configuration file or it may have been commanded after the system initializes.
There is a corresponding counter for each of these conditions. First verify that the condition is happening by viewing the counter in SB HK telemetry. If the condition is happening, you can view the SB filter information through the EVS App Data Main page by clicking the 'go to' button for SB. The event Id for these events can be learned through a previous event or from the cfe_sb_events.h file.

(Q) Why does the SB provide event filtering through the platform configuration file?

To give the user the ability to filter events before an EVS command can be sent. During system initialization, there are many conditions occurring that can cause a flood of SB events such as No Subscribers, Pipe Overflow and MsgId to Pipe errors. This gives the user a way to limit these events.

(Q) Why does SB have so many debug event messages?

The SB debug messages are positive acknowledgments that an action (like receiving a cmd, creating a pipe or subscribing to a message) has occurred. They are intended to help isolate system problems. For instance, if an expected response to a command is not happening, it may be possible to repeat the scenario with the debug event turned on to verify that the command was successfully received.

(Q) How is the QOS parameter in the CFE_SB_SubscribeEx used by the software bus?

	The QOS parameter is currently unused by the software bus. It is a placeholder to be used with the future software bus capability of inter-processor communication. Setting the QOS as CFE_SB_DEFAULT_QOS will ensure seamless integration when the software bus is expanded to support inter-processor communication.
(Q) Can I confirm my software bus buffer was delivered?	
	There is no built in mechanism for confirming delivery (it could span systems). This could be accomplished by generating a response message from the receiver.

Prev: Operation of the SB Software
Up To: cFE Software Bus Overview

18 cFE Software Bus Commands

The following is a list of commands that are processed by the cFE Software Bus Task.

Global CFE_SB_DISABLE_ROUTE_CC
Disable Software Bus Route

Global CFE_SB_DISABLE_SUB_REPORTING_CC
Disable Subscription Reporting Command

Global CFE_SB_ENABLE_ROUTE_CC
Enable Software Bus Route

Global CFE_SB_ENABLE_SUB_REPORTING_CC
Enable Subscription Reporting Command

Global CFE_SB_NOOP_CC
Software Bus No-Op

Global CFE_SB_RESET_COUNTERS_CC
Software Bus Reset Counters

Global CFE_SB_SEND_PREV_SUBS_CC
Send Previous Subscriptions Command

Global CFE_SB_SEND_SB_STATS_CC
Send Software Bus Statistics

Global CFE_SB_WRITE_MAP_INFO_CC
Write Map Info to a File

Global CFE_SB_WRITE_PIPE_INFO_CC
Write Pipe Info to a File

Global CFE_SB_WRITE_ROUTING_INFO_CC
Write Software Bus Routing Info to a File

19 cFE Software Bus Telemetry

The following are telemetry packets generated by the cFE Software Bus Task.

Global CFE_SB_AllSubscriptionsTlm_Payload_t
SB Previous Subscriptions Packet

Global CFE_SB_AllSubscriptionsTlm_Payload_t
SB Previous Subscriptions Packet

Global CFE_SB_HousekeepingTlm_Payload_t
Software Bus task housekeeping Packet

Global CFE_SB_HousekeepingTlm_Payload_t
Software Bus task housekeeping Packet

Global CFE_SB_SingleSubscriptionTlm_Payload_t
SB Subscription Report Packet

Global CFE_SB_SingleSubscriptionTlm_Payload_t
SB Subscription Report Packet

Global CFE_SB_StatsTlm_Payload_t
SB Statistics Telemetry Packet

Global CFE_SB_StatsTlm_Payload_t
SB Statistics Telemetry Packet

20 cFE Software Bus Configuration Parameters

The following are configuration parameters used to configure the cFE Software Bus either for each platform or for a mission as a whole.

Global CFE_MISSION_SB_MAX_PIPES
Maximum Number of pipes that SB command/telemetry messages may hold

Global CFE_MISSION_SB_MAX_SB_MSG_SIZE
Maximum SB Message Size

Global CFE_PLATFORM_ENDIAN
Platform Endian Indicator

Global CFE_PLATFORM_SB_BUF_MEMORY_BYTES
Size of the SB buffer memory pool

Global CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME
Default Message Map Filename

Global CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT
Default Subscription Message Limit

Global CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME
Default Pipe Information Filename

Global CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME
Default Routing Information Filename

Global CFE_PLATFORM_SB_FILTERED_EVENT1
SB Event Filtering

Global CFE_PLATFORM_SB_HIGHEST_VALID_MSGID
Highest Valid Message Id

Global CFE_PLATFORM_SB_MAX_DEST_PER_PKT
Maximum Number of unique local destinations a single MsgId can have

Global CFE_PLATFORM_SB_MAX_MSG_IDS
Maximum Number of Unique Message IDs SB Routing Table can hold

Global CFE_PLATFORM_SB_MAX_PIPES
Maximum Number of Unique Pipes SB Routing Table can hold

21 cFE Table Services Overview

Applications often organize sets of their parameters into logical units called tables. These are typically constant parameters that can change the behavior of a flight software algorithm and are only intended to be modified by operations personnel. Examples of this would be attitude control gains, sensor scalefactors, telemetry filter settings, etc.

Table Services (TBL) provides a centralized control of flight software tables. Operations personnel would interact with TBL in order to dump the contents of current tables, load new table images, verify the contents of a table image and manage Critical tables.

None of the cFE core applications (EVS, SB, ES, TIME, or TBL) use tables, and it is possible to build cFE without Table Services if not needed or an alternative parameter management mechanism is to be utilized.

For additional detail on Tables and how to manage them, see the following sections:

- Managing Tables
- cFE Table Types and Table Options
- Table Registry
- Table Services Telemetry
- Effects of Processor Reset on Tables
- Frequently Asked Questions about Table Services

21.1 Managing Tables

In order to effectively manage tables, an operator needs to understand how cFE Applications manage tables from their end. There are a number of methods that cFE Applications typically use to manage their tables. Each method is appropriate based upon the nature of the contents of the table.

cFE Applications are required to periodically check to see if their table is to be validated, updated (or in the case of dump-only tables, dumped). Most Applications perform this periodic management at the same time as housekeeping requests are processed. This table management is performed by the cFE Application that "owns" a table (ie - the cFE Application that registered the table with cFE Table Services). It is possible for cFE Applications to "share" a table with other cFE Applications. An Application that shares a table does not typically perform any of the management duties associated with that table.

A table can have one of two different types and a number of different options. These are discussed further in later sections. An operator should understand the chosen type and selected options for a particular table before attempting to modify a table's contents.

To understand the methods of maintaining a table, it is important that the terminology be clear. A table has two images: "Active" and "Inactive". The Active table is the one that a cFE Application is currently accessing when it executes. The Inactive table is a copy of the Active table that an operator (or on-board process such as a stored command processor) can manipulate and change to have a newly desired set of data.

To create an Inactive table image on board, the operator would be required to perform a "Load" to the table. Loads are table images stored in on-board files. The Load can contain either a complete table image or just a part of a table image. If the Load contains just a portion, the Inactive image is first initialized with the contents of the Active image and then the portion identified in the Load file is written on top of the Active image. After the initial Load, an operator can continue to manipulate the Inactive table image with additional partial table load images. This allows the operator to reconfigure the contents of multiple portions of the table before deciding to "Validate" and/or "Activate" it.

Some cFE Applications provide special functions that will examine a table image to determine if the contents are logically sound. This function is referred to as the "Validation Function." When a cFE Application assigns a Validation Function to a table during the table registration process, it is then requiring that a Validation be performed before the table can be Activated. When an operator requests a Validation of a table image, they are sending a request to the owning Application to execute the associated Validation Function on that image. The results of this function are then reported in telemetry. If the Validation is successful, the operator is free to perform a table Activation. If the Validation fails, the operator would be required to make additional changes to the Inactive table image and attempt another Validation before commanding an Activation.

To change an Inactive table image into the Active table image, an operator must Activate a table. When an operator sends the table Activation command, they are notifying the table's owning Application that a new table image is available. It is then up to the Application to determine when is the best time to perform the "Update" of the table. When an Application performs an Update, the contents of the Inactive table image become the Active table image.

Next: cFE Table Types and Table Options
Up To: cFE Table Services Overview

21.2 cFE Table Types and Table Options

A cFE Application Developer has several choices when creating a cFE Application. There are two basic types of tables: single buffered and double buffered. In addition to these two basic types there are a small variety of options possible with each table. These options control special characteristics of the table such as whether it is dump-only, critical or whether it has an application defined location in memory.

Each choice has its advantages and disadvantages. The developer chooses the appropriate type based upon the requirements of the application. Anyone operating a particular cFE Application must understand the nature of the type and options selected for a particular table before they can successfully understand how to perform updates, validations, etc.

For more information on the different types of tables available, see the following sections:

- Table Types
  - Single Buffered Tables
  - Double Buffered Tables
- Table Options
  - Tables with Validation Functions
  - Critical Tables
  - User Defined Address Tables
  - Dump Only Tables

Next: Single Buffered Tables
Prev: Managing Tables
Up To: cFE Table Services Overview

21.2.1 Single Buffered Tables

The default table type for a cFE Application to use is a single buffered table. The principle advantage of a single buffered table is that it can share one of several shared table buffers for uploaded and pending table images. Since many cFE Applications have relatively small tables that are not changed at time critical moments or are not changed very often during a mission, single buffered tables represent the most memory resource efficient method of being managed.

The number of single buffered tables that can have inactive table images being manipulated at one time is specified by a TBL Services configuration parameter (CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS) found in the cfe↩_platform_cfg.h file associated with the processor in question. This parameter identifies the number of shared table buffers that are available.

Since inactive single buffered table images share a common resource, it may not be prudent for an operator to load an image and then delay on the image's activation for an extended period of time.

Single buffered tables are allowed to be critical (see Critical Tables), dump-only (see Dump Only Tables) and/or have a user-defined address (see User Defined Address Tables).

Next: Double Buffered Tables
Up To: cFE Table Types and Table Options

21.2.2 Double Buffered Tables

Under certain conditions, a cFE Application Developer may choose to use a double buffered table type within their application. Double buffered tables retain a dedicated inactive image of the table data. With a dedicated inactive table image available, double buffered tables are then capable of efficiently swapping table contents and/or delaying the activation of a table's contents for an indeterminate amount of time.

Some cFE Applications prefer to delay the Activation of a table until a specified time (e.g. - a Spacecraft Ephemeris). These tables are typically defined as double buffered tables so that the Inactive image can be left sitting untouched for an extended period of time without interfering with shared resources for other tables. Then the Application can perform the Update when the time is right.

Applications which have unusually large tables may decide to conserve memory resources by making them double buffered. This is because the shared buffers used by single buffered tables must be sized to match the largest table. If there is one table that is unusually large, there is little reason to allocate up to CFE_PLATFORM_TBL_MAX_SIM↩ ULTANEOUS_LOADS number of buffers that size. A double buffered table will only allocate ONE extra buffer of that size.

Performance minded Applications that are required to perform processing with tight timing deadlines may choose to use double buffered tables because the Update for a double buffered table is deterministic and quick.

Next: Tables with Validation Functions
Prev: Single Buffered Tables
Up To: cFE Table Types and Table Options

21.2.3 Tables with Validation Functions

Applications that associate Validation Functions with their tables when the tables are registered are effectively requiring that the contents of a table be logically Validated before it is Activated. The cFE will refuse to let a table with an associated Validation Function be Activated until a successful Validation on the Inactive table image has occurred.

Tables that are NOT assigned a Validation Function are assumed to be valid regardless of the contents of the table image. These tables do not require a Validation Command prior to Activation.

Next: Critical Tables
Prev: Double Buffered Tables
Up To: cFE Table Types and Table Options

21.2.4 Critical Tables

Applications that must be able to recover quickly from a Processor Reset may select the "Critical" table option when registering their table. Table Services automatically creates a Critical Data Store for the table and ensures that the contents of the Critical Data Store are updated whenever a Table Activation occurs.

If a Processor Reset happens, when the Application attempts to Register the table again, Table Services automatically locates the associated Critical Data Store and initializes the Table with the saved contents.

Next: User Defined Address Tables
Prev: Tables with Validation Functions
Up To: cFE Table Types and Table Options

21.2.5 User Defined Address Tables

In order to provide a mechanism for Flight Software Maintenance teams to quickly create a table image for dumping contents of memory that isn't normally loaded by the ground, there is an option to create User-Defined Address tables. These tables, when they are first registered, provide a memory address where the Active image of the table is to be maintained. Normally, the address is specified by Table Services from its memory pool.

By specifying the address, the Flight Software Maintenance team can create a Dump-Only table that contains the contents of a data structure that is not normally accessible via telemetry or table dumps. Then, on command, the Flight Software Maintenance team can periodically dump the data structure's contents to an on-board file(s) that can then be transferred to the ground for later analysis.

Next: Dump Only Tables
Prev: Critical Tables
Up To: cFE Table Types and Table Options

21.2.6 Dump Only Tables

On occasion, cFE Applications require a segment of memory in which the Application writes data. The typical cFE Table is not normally modified directly by an Application but only via Load and Activate commands from either the Ground or Stored Command Processor. However, for those situations where an Application wishes to modify the contents of a data structure and the Application is limited in its telemtry bandwidth so that the modified data cannot be telemetered, the Application can create a Dump-Only table.

Dump-Only tables are not allowed to be modified via the Load/Validate/Activate process most other tables are. They are only supposed to be modified by onboard Applications. The Operator can still command a Dump which will be processed by the table's owning Application when it manages its tables. By letting the Application perform the dump, the Operator can feel confident that the table contents are a complete snapshot in time and not corrupted by taking a snapshot while the Application was in the process of modifying its contents.

Next: Table Registry
Prev: User Defined Address Tables
Up To: cFE Table Types and Table Options

21.3 Table Registry

When Applications register tables, Table Services retains pertinent information on the table in the Table Registry. The following information (along with other information that is less important for an operator) is kept for each table:

- The Application ID of the Application that Registered the table
- The full name of the table
- The size, in bytes, of the table
- Pointers to the start addresses of the Table's image buffers, Active and Inactive (if appropriate)
- A pointer to the start address of a Validation Function
- A flag indicating whether a table image has been loaded into an Inactive buffer

- A flag indicating whether the table is Critical and its associated CDS Handle if it is
- A flag indicating whether the table has ever been loaded (initialized)
- A flag indicating whether the table is Dump Only
- A flag indicating whether the table has an Update Pending
- A flag indicating whether the table is double buffered or not
- The System Time when the Table was last Updated
- The filename of the last file loaded into the table
- The File Creation Time for the last file used to load the contents of the table

This information can be obtained by either sending the Dump Registry command which will put all of the information from the Table Registry into an onboard file for later downlink or the operator can send a command to Telemeter the Registry Entry for a single table. This will cause the pertinent registry entry for a single table to be sent via a telemetry packet.

The API function CFE_TBL_Register() returns either CFE_SUCCESS or CFE_TBL_INFO_RECOVERED_TBL to indicate that the table was successfully registered. The difference is whether the table data was recovered from CDS as part of the registration. There are several error return values that describe why the function failed to register the table but nothing related to why the restoration from CDS might have failed. There is, however, a message written to the System Error Log by Table Services that can be dumped by the ground to get this information. Note that failure to restore a table from CDS is not an expected error and requires some sort of data corruption to occur.

Next: Table Services Telemetry
Prev: cFE Table Types and Table Options
Up To: cFE Table Services Overview

21.4 Table Services Telemetry

Table Services produces two different telemetry packets. The first packet, referred to as the Table Services Housekeeping Packet, is routinely produced by Table Services upon receipt of the Housekeeping Request message that is typically sent to all Applications by an on board scheduler. The contents and format of this packet are described in detail at CFE_TBL_HousekeepingTlm_t.

Next: Effects of Processor Reset on Tables
Prev: Table Registry
Up To: cFE Table Services Overview

21.5 Effects of Processor Reset on Tables

When a processor resets, the Table Registry is re-initialized. All Applications must, therefore, re-register and re-initialize their tables. The one exception, however, is if the Application has previously tagged a table as "Critical" during Table Registration, then Table Services will attempt to locate a table image for that table stored in the Critical Data Store. Table Services also attempts to locate the Critical Table Registry which is also maintained in the Critical Data Store.

If Table Services is able to find a valid table image for a Critical table in the Critical Data Store, the contents of the table are automatically loaded into the table and the Application is notified that the table does not require additional initialization.

Next: Frequently Asked Questions about Table Services
Prev: Table Services Telemetry
Up To: cFE Table Services Overview

21.6 Frequently Asked Questions about Table Services

(Q) Is it an error to load a table image that is smaller than the registered size?

Table images that are smaller than the declared size of a table fall into one of two categories.
If the starting offset of the table image (as specified in the Table Image secondary file header) is not equal to zero, then the table image is considered to be a "partial" table load. Partial loads are valid as long as a table has been previously loaded with a non-"partial" table image.
If the starting offset of the table image is zero and the size is less than the declared size of the table, the image is considered "short" but valid. This feature allows application developers to use variable length tables.

(Q) I tried to validate a table and received the following event message that said the event failed:

```
"MyApp validation failed for Inactive 'MyApp.MyTable', Status=0x####"
```

What happened?

The event message indicates the application who owns the table has discovered a problem with the contents of the image. The code number following the 'Status' keyword is defined by the Application. The documentation for the specified Application should be referred to in order to identify the exact nature of the problem.

(Q) What commands do I use to load a table with a new image?

There are a number of steps required to load a table.

1. The operator needs to create a cFE Table Services compatible table image file with the desired data contained in it. This can be accomplished by creating a 'C' source file, compiling it with the appropriate cross compiler for the onboard platform and then running the `elf2cfetbl` utility on the resultant object file.
2. The file needs to be loaded into the onboard processor's filesystem using whichever file transfer protocol is used for that mission.
3. The Load Command is sent next to tell Table Services to load the table image file into the Inactive Table Image Buffer for the table identified in the file.
4. The Validate Command is then sent to validate the contents of the inactive table image. This will ensure the file was not corrupted or improperly defined. The results of the validation are reported in Table Services Housekeeping Telemetry. If a table does not have a validation function associated with it, the operator may wish to compare the computed CRC to verify the table contents match what was intended.
5. Upon successful validation, the operator then sends the Activate Command. The application owning the table should, within a reasonable amount of time, perform a table update and send an event message.

(Q) What causes cFE Table Services to generate the following sys log message:
CFE_TBL:GetAddressInternal-App(%d) attempt to access unowned Tbl Handle=%d

When an application sharing its table(s) with one or more applications is reloaded, the reloaded application's table handle(s) are released. cFE Table Services sees that the table(s) are shared and keeps a 'shadow' version of the table in the Table Services registry. The registry will show the released, shared tables with no name. When the applications sharing the table attempt to access the table via the 'old', released handle, Table Services will return an error code to the applications and generate the sys log message. The applications may then unregister the 'old' handle(s) in order to remove the released, shared table(s) from the Table Services registry and share the newly loaded application table(s).

(Q) When does the Table Services Abort Table Load command need to be issued?

The Abort command should be used whenever a table image has been loaded but the application has not yet activated it and the operator no longer wants the table to be loaded.
The purpose of the Abort command is to free a previously allocated table buffer. It should be noted, however, that multiple table loads to the SAME table without an intervening activation or abort, will simply OVERWRITE the previous table load using the SAME buffer.
Therefore, the most likely scenarios that would lead to a needed abort are as follows:

1. Operator loads a table and realizes immediately that the load is not wanted.
2. Operator loads a table and performs a validation on it. Regardless of whether the table passes or fails the validation, if the operator no longer wants to activate the table, the abort command should be issued.

   It should be noted that a table image that fails activation is retained in the inactive buffer for diagnosis, if necessary. It is NOT released until it is aborted or overwritten and successfully validated and activated.
3. A table image was loaded; the image was successfully validated; the command for activation was sent; but the application fails to perform the activation.

   The Abort command will free the table buffer and clear the activation request.

   This situation can occur when either the application is improperly designed and fails to adequately manage its tables (sometimes seen in the lab during development) or the application is "hung" and not performing as it should.

Prev: Effects of Processor Reset on Tables
Up To: cFE Table Services Overview

22 cFE Table Services Commands

The following is a list of commands that are processed by the cFE Table Services Task.

Global CFE_TBL_ABORT_LOAD_CC
Abort Table Load

Global CFE_TBL_ACTIVATE_CC
Activate Table

Global CFE_TBL_DELETE_CDS_CC
Delete Critical Table from Critical Data Store

Global CFE_TBL_DUMP_CC
Dump Table

Global CFE_TBL_DUMP_REGISTRY_CC
Dump Table Registry

Global CFE_TBL_LOAD_CC
Load Table

Global CFE_TBL_NOOP_CC
Table No-Op

Global CFE_TBL_RESET_COUNTERS_CC
Table Reset Counters

Global CFE_TBL_SEND_REGISTRY_CC

Telemeter One Table Registry Entry

Global CFE_TBL_VALIDATE_CC

Validate Table

23 cFE Table Services Telemetry

The following are telemetry packets generated by the cFE Table Services Task.

Global CFE_TBL_HousekeepingTlm_Payload_t

Table Services Housekeeping Packet

Global CFE_TBL_HousekeepingTlm_Payload_t

Table Services Housekeeping Packet

Global CFE_TBL_TblRegPacket_Payload_t

Table Registry Info Packet

Global CFE_TBL_TblRegPacket_Payload_t

Table Registry Info Packet

24 cFE Table Services Configuration Parameters

The following are configuration parameters used to configure the cFE Table Services either for each platform or for a mission as a whole.

Global CFE_MISSION_TBL_MAX_FULL_NAME_LEN

Maximum Length of Full Table Name in messages

Global CFE_MISSION_TBL_MAX_NAME_LENGTH

Maximum Table Name Length

Global CFE_PLATFORM_TBL_BUF_MEMORY_BYTES

Size of Table Services Table Memory Pool

Global CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE

Default Filename for a Table Registry Dump

Global CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES

Maximum Number of Critical Tables that can be Registered

Global CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE

Maximum Size Allowed for a Double Buffered Table

Global CFE_PLATFORM_TBL_MAX_NUM_HANDLES

Maximum Number of Table Handles

Global CFE_PLATFORM_TBL_MAX_NUM_TABLES

Maximum Number of Tables Allowed to be Registered

Global CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS
Maximum Number of Simultaneous Table Validations

Global CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS
Maximum Number of Simultaneous Loads to Support

Global CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE
Maximum Size Allowed for a Single Buffered Table

Global CFE_PLATFORM_TBL_VALID_PRID_1
Processor ID values used for table load validation

Global CFE_PLATFORM_TBL_VALID_PRID_COUNT
Number of Processor ID's specified for validation

Global CFE_PLATFORM_TBL_VALID_SCID_1
Spacecraft ID values used for table load validation

Global CFE_PLATFORM_TBL_VALID_SCID_COUNT
Number of Spacecraft ID's specified for validation

25 cFE Time Services Overview

The cFE Time Service (TIME) is one of the cFE core services. TIME provides time correlation, distribution and synchronization services. TIME exists in two varieties: a Time Server responsible for maintaining the master time reference for all remote systems, and a Time Client responsible for synchronizing to that master time reference.

Since TIME is a generic implementation aimed to meet the needs of a variety of mission configurations, there are numerous configuration parameters, which dictate the behavior of TIME (see cfe_mission_cfg.h and cfe_platform_cfg.h for the specific mission configuration).

With the exception of those sections specific to Time Clients and Servers, this document assumes the most common physical environment - one instantiation of cFE installed on a single processor. Therefore, TIME represents cFE Time Services configured as a Time Server.

For additional detail on Time Services and how to manage it, see the following sections:

- Time Components
- Time Structure
- Time Formats
- Time Configuration
  - Time Format Selection
  - Enabling Fake Tone Signal

  - Selecting Tone and Data Ordering
  - Specifying Tone and Data Window
  - Specifying Time Server/Client
  - Specifying Time Tone Byte Order
  - Virtual MET
  - Specifying Time Source
  - Specifying Time Signal
- Time Services Paradigm(s)
- Flywheeling
- Time State
- Initialization
  - Power-On Reset
  - Processor Reset
- Initialization
  - Power-On Reset
  - Processor Reset
- Normal Operation
  - Client
  - Server
    * Setting Time
    * Adjusting Time
    * Setting MET
- Frequently Asked Questions

25.1 Time Components

Time knowledge is stored in several pieces, so that the time information can more easily be manipulated and utilized. These components include:

The **Ground Epoch** is an arbitrary date and time that establishes the zero point for spacecraft time calculations. The selection of the epoch is mission specific, although in the past, it was common to select the same epoch as defined for the Operating System used by the computers hosting the ground system software. Recent mission epoch selections have also included using zero seconds after midnight, Jan 1, 2001.

Spacecraft Time is the number of seconds (and fraction of a second) since the ground epoch. Spacecraft time is the sum of **Mission Elapsed Time** (MET) and the **Spacecraft Time Correlation Factor** (STCF). By definition, MET is a measure of time since launch or separation. However, for most missions the MET actually represents the amount of time since powering on the hardware containing the MET timer. The STCF correlates the MET to the ground epoch.

The **Tone** is the signal that MET seconds have incremented. In most hardware configurations, the tone is synonymous with the **1 PPS** signal. The tone signal may be generated by a local hardware timer, or by an external event (G↩ PS receiver, spacewire time tick, 1553 bus signal, etc). TIME may also be configured to simulate the tone for lab environments that do not have the necessary hardware to provide a tone signal. Note that MET sub-seconds will be zero at the instant of the tone.

Time at the Tone is the spacecraft time at the most recent "valid" tone.

Time since the Tone is the amount of time since the tone (usually less than one second). This value is often measured using the local processor clock. Upon detecting the tone signal, TIME stores the contents of the local processor clock to facilitate this measurement.

Thus, **Current Spacecraft Time** is the sum of "time at the tone" and "time since the tone".

Leap Seconds occur to keep clocks correlated to astronomical observations. The modern definition of a second (9,192,631,770 oscillations of a cesium-133 atom) is constant while the earth's rotation has been slow by a small fraction of a second per day. The **International Earth Rotation and Reference System Service** (IERS) maintains the count of leap seconds as a signed whole number that is subject to update twice a year. Although it is possible to have a negative leap second count if the earth rotates too fast, it is highly unlikely. The initial count of leap seconds (10) was established in January of 1972 and the first leap second was added to the initial count in June of 1972. The most recent leap seconds are announced by the International Earth Rotation Service (IERS): https://www.iers.org in IERS Bulletin C (leap second announcements). Search the IERS site for "Bulletin C" to obtain the latest issue/announcement.

Next: Time Structure
Up To: cFE Time Services Overview

25.2 Time Structure

The cFE implementation of the **System Time Structure** is a modified version of the CCSDS Unsegmented Time Code (CUC) which includes 4 bytes of seconds, and 4 bytes of subseconds, where a subsecond is equivalent to 1/(2^32) seconds. The system time structure is used by TIME to store current time, time at the tone, time since the tone, the MET, the STCF and command arguments for time adjustments. Note that typically the 32 bits of seconds and the upper 16 bits of subseconds are used for time stamping Software bus messages, but this is dependent on the underlying defintion.

The system time structure is defined as follows:

```
typedef struct {
      uint32      Seconds;      /* Number of seconds */
      uint32      Subseconds;   /* Number of 2^(-32) subseconds */
} CFE_TIME_SysTime_t;
```

Next: Time Formats
Prev: Time Components
Up To: cFE Time Services Overview

25.3 Time Formats

International Atomic Time (TAI) is one of two time formats supported by cFE TIME. TAI is the number of seconds and sub-seconds elapsed since the ground epoch as measured with the atomic clock previously described. TAI has no reference to leap seconds and is calculated using the following equation:

```
TAI = MET + STCF
```

It should be noted that TAI is only "true" TAI when the selected ground epoch is the same as the TAI epoch (zero seconds after midnight, January 1, 1958). However, nothing precludes configuring cFE TIME to calculate time in the TAI format and setting the STCF to correlate to any other epoch definition.

Coordinated Universal Time (UTC) is the other time format supported by cFE TIME. UTC differs from TAI in the fact that UTC includes a leap seconds adjustment. TIME computes UTC using the following equation:

```
UTC = TAI - Leap Seconds.
```

The preceding UTC equation might seem to imply that TAI includes leap seconds and UTC does not - which is not the case. In fact, the UTC calculation includes a leap seconds adjustment that subtracts leap seconds from the same time components used to create TAI. Alternatively, it might be less confusing to express the UTC equation as follows:

```
UTC = MET + STCF - Leap Seconds
```

Next: Time Configuration
Prev: Time Components
Up To: cFE Time Services Overview

25.4 Time Configuration

All configurations of TIME require a local processor source for a 1Hz interrupt and access to a local clock with a resolution fine enough that it can be used to measure short periods of elapsed time. The local interrupt is used to wake-up TIME at a regular interval for the purpose of verifying that the tone is being received. The local clock is used to measure time since the tone and to provide coarse verification that the tone is occurring at approximately one second intervals. The presumption is that the tone is the most accurate timer in the system and, within reason, is to be trusted. Note that nothing precludes the use of the MET as the local clock, assuming the MET is both local and provides sub-second data. However, the tone must not be used as the source for the local 1Hz interrupt.

Consider the following brief description of three hypothetical hardware configurations. These sample systems may be used as reference examples to help clarify the descriptions of the various TIME configuration selections.

In the first system, there is no MET timer and therefore no tone signal. The MET is a count of the number of "fake" tones generated by TIME software. There is no validation performed regarding the quality of time data. This hardware configuration is a common lab environment using COTS equipment.

In the second system, the MET timer is a hardware register that is directly accessible by TIME. When MET seconds increment, a processor interrupt signals the tone. Upon detecting the tone, TIME can read the MET to establish the time at the tone. To verify that the tone is valid, TIME need only validate that this tone signal occurred approximately one second after the previous tone signal (as measured with the local clock).

In the third system, the MET is located on hardware connected via spacewire. When MET seconds increment, a spacewire time tick triggers a local processor interrupt to signal the tone. Shortly after announcing the tone, the hardware containing the MET also generates a spacewire data packet containing the MET value corresponding to the tone. TIME must wait until both the tone and data packet have been received before validating the tone. The tone must have occurred approximately one second after the previous tone signal and the data packet must have been received within a specified window in time following the tone.

The hardware design choice for how the tone signal is distributed is not material to TIME configuration. The software detecting the tone need only call the cFE API function announcing the arrival of the tone. This function is designed to be called from interrupt handlers.

For detail on each of the individual configuration settings for cFE Time Services, see the following sections:

- Time Format Selection
- Enabling Fake Tone Signal
- Selecting Tone and Data Ordering
- Specifying Tone and Data Window
- Specifying Time Server/Client
- Specifying Time Tone Byte Order
- Virtual MET
- Specifying Time Source
- Specifying Time Signal

Next: Time Services Paradigm(s)
Prev: Time Formats
Up To: cFE Time Services Overview

25.4.1 Time Format Selection

Time format is defined in the mission configuration header file.

This selection defines the default time format as TAI or UTC. The API functions to get time in either specific format are still enabled, but the API function to get time in the default format will follow this selection. Enable one, and **only one**, of the following time format definitions:

```
#define CFE_MISSION_TIME_CFG_DEFAULT_TAI  TRUE
#define CFE_MISSION_TIME_CFG_DEFAULT_UTC  FALSE
```

or

```
#define CFE_MISSION_TIME_CFG_DEFAULT_TAI  FALSE
#define CFE_MISSION_TIME_CFG_DEFAULT_UTC  TRUE
```

The choice of time format is a mission specific decision and is not directly affected by the hardware configuration.

See also

CFE_MISSION_TIME_CFG_DEFAULT_TAI, CFE_MISSION_TIME_CFG_DEFAULT_UTC

Next: Enabling Fake Tone Signal
Up To: Time Configuration

25.4.2 Enabling Fake Tone Signal

The fake tone is defined in the mission configuration header file.

If this selection is set to TRUE, TIME will generate a "fake" tone signal by calling the same API function as would be called upon detection of the "real" tone signal. Enable the fake tone only for hardware configurations that do not provide a tone signal.

```
#define CFE_MISSION_TIME_CFG_FAKE_TONE  TRUE
```

Hypothetical hardware configuration number one (described above) would enable the fake tone signal.

See also

CFE_MISSION_TIME_CFG_FAKE_TONE

Next: Selecting Tone and Data Ordering
Prev: Time Format Selection
Up To: Time Configuration

25.4.3 Selecting Tone and Data Ordering

Tone and data order is defined in the mission configuration header file.

This selection defines which comes first - the tone or the time at the tone data. Does the time data describe the tone that already occurred, or the tone that has not yet occurred? This decision may be driven by the hardware design but can also be arbitrary. Enable one, and only one, of the following:

```
#define CFE_MISSION_TIME_AT_TONE_WAS
#define CFE_MISSION_TIME_AT_TONE_WILL_BE
```

Hypothetical hardware configuration number three (described Time Configuration above) would enable "time at the tone was".

See also

CFE_MISSION_TIME_AT_TONE_WAS, CFE_MISSION_TIME_AT_TONE_WILL_BE

Next: Specifying Tone and Data Window
Prev: Enabling Fake Tone Signal
Up To: Time Configuration

25.4.4 Specifying Tone and Data Window

The tone and data window is defined in the mission configuration header file.

In concert with the definition of tone and data order, this selection defines the valid window in time for the second of the pair to follow the first. Both must be defined, units are micro-seconds.

```
#define CFE_MISSION_TIME_MIN_ELAPSED  0
#define CFE_MISSION_TIME_MAX_ELAPSED  100000
```

Hypothetical hardware configuration number three (described above) might use these values which describe a window that begins immediately after the tone and lasts for one tenth of a second.

See also

CFE_MISSION_TIME_MIN_ELAPSED, CFE_MISSION_TIME_MAX_ELAPSED

Next: Specifying Time Server/Client
Prev: Selecting Tone and Data Ordering
Up To: Time Configuration

25.4.5 Specifying Time Server/Client

Configure TIME as a client only when the target system has multiple processors running separate instantiations of the cFE. One instantiation must be configured as the server and the remainder configured as clients. If the target system has only one processor running the cFE, then TIME must be configured as a server.

Enable one, and only one, of the following definitions in the platform configuration header file:

```
#define CFE_PLATFORM_TIME_CFG_SERVER   TRUE
#define CFE_PLATFORM_TIME_CFG_CLIENT   FALSE
```

or

```
#define CFE_PLATFORM_TIME_CFG_SERVER   FALSE
#define CFE_PLATFORM_TIME_CFG_CLIENT   TRUE
```

See also

CFE_PLATFORM_TIME_CFG_SERVER, CFE_PLATFORM_TIME_CFG_CLIENT

Next: Specifying Time Tone Byte Order
Prev: Specifying Tone and Data Window
Up To: Time Configuration

25.4.6 Specifying Time Tone Byte Order

By default, the CFE time tone message is a payload of integers in platform-endian order (containing the tone's timestamp, the leap seconds, and state information.) In some configurations, it may be better to have the payload produced in big-endian order–particularly in mixed-endian environments.

In order to force the tone message to be in big-endian order, you must define the following:

```
#define CFE_PLATFORM_TIME_CFG_BIGENDIAN
```

Next: Virtual MET
Prev: Specifying Time Server/Client
Up To: Time Configuration

25.4.7 Virtual MET

This configuration option refers to whether the MET is local to this instantiation of TIME. If the MET is not local then TIME must be configured as using a virtual MET.

Therefore, all TIME clients must be configured as using a virtual MET. If the MET was local to any TIME client, then that instantiation of TIME would have to be the server.

TIME servers must be configured as using a virtual MET

Next: Specifying Time Source
Prev: Specifying Time Tone Byte Order
Up To: Time Configuration

25.4.8 Specifying Time Source

TIME configuration provides the ability to specify where the source for time data is originating - either internal or external. In hypothetical system one, the MET is internal. In system two, TIME cannot directly read the MET, therefore time data must be received from an external source.

This selection also enables a command interface to switch between internal and external input. When commanded to use internal time data, TIME will ignore the external data. However, TIME will continue to use the API function as the trigger to generate a "time at the tone" command packet regardless of the internal/external command selection.

Set the following definition to TRUE only for TIME servers using an external time data source.

```
#define CFE_PLATFORM_TIME_CFG_SOURCE  TRUE
```

The remainder of this section pertains only to TIME servers configured to accept external time data.

When configured to accept external time data, TIME requires an additional definition for the type of external data (GPS, MET, spacecraft time, etc.). This selection will enable an API function specific to the selected data type. Regardless of how the time data is received, the receiver need only pass the data to the appropriate API function.

TIME servers using an external time data source must set one, and only one, of the following to TRUE, for example:

```
#define CFE_PLATFORM_TIME_CFG_SRC_MET   TRUE
#define CFE_PLATFORM_TIME_CFG_SRC_GPS   FALSE
#define CFE_PLATFORM_TIME_CFG_SRC_TIME  FALSE
```

configuration definitions for the particular source.

If the cfe_platform_cfg.h file contains "#define CFE_PLATFORM_TIME_CFG_SOURCE TRUE" then time is configured to allow switching between internal and external time sources (see CFE_TIME_SET_SOURCE_CC). If this configuration parameter is set to FALSE then the command to set the source will be rejected.

If this configuration parameter is set to TRUE then ONE and ONLY ONE of the following configuration parameters must also be set TRUE in order to specify the external time source, for example:

```
#define CFE_PLATFORM_TIME_CFG_SRC_MET   TRUE
#define CFE_PLATFORM_TIME_CFG_SRC_GPS   FALSE
#define CFE_PLATFORM_TIME_CFG_SRC_TIME  FALSE
```

Note that Internal MET source depends on available hardware. It may be the local count of tone signals, the contents of a hardware register or an OS specific time function.

Note also that when configured to use an external time source, commands to set the time will be overwritten.

See also

CFE_PLATFORM_TIME_CFG_SRC_MET, CFE_PLATFORM_TIME_CFG_SRC_GPS, CFE_PLATFORM_TI↩ME_CFG_SRC_TIME

Next: Specifying Time Signal
Prev: Virtual MET
Up To: Time Configuration

25.4.9 Specifying Time Signal

Some hardware configurations support a primary and redundant tone signal selection. Setting the following configuration definition to TRUE will result in enabling a TIME command to select the active tone signal.

```
#define CFE_PLATFORM_TIME_CFG_SIGNAL  TRUE
```

Note: this feature requires additional custom software to make the physical signal switch.

See also

CFE_PLATFORM_TIME_CFG_SIGNAL

Next: Time Services Paradigm(s)
Prev: Specifying Time Source
Up To: Time Configuration

25.5 Time Format Selection

Time format is defined in the mission configuration header file.

This selection defines the default time format as TAI or UTC. The API functions to get time in either specific format are still enabled, but the API function to get time in the default format will follow this selection. Enable one, and **only one**, of the following time format definitions:

```
#define CFE_MISSION_TIME_CFG_DEFAULT_TAI  TRUE
#define CFE_MISSION_TIME_CFG_DEFAULT_UTC  FALSE
```

or

```
#define CFE_MISSION_TIME_CFG_DEFAULT_TAI  FALSE
#define CFE_MISSION_TIME_CFG_DEFAULT_UTC  TRUE
```

The choice of time format is a mission specific decision and is not directly affected by the hardware configuration.

See also

CFE_MISSION_TIME_CFG_DEFAULT_TAI, CFE_MISSION_TIME_CFG_DEFAULT_UTC

Next: Enabling Fake Tone Signal
Up To: Time Configuration

25.6 Enabling Fake Tone Signal

The fake tone is defined in the mission configuration header file.

If this selection is set to TRUE, TIME will generate a "fake" tone signal by calling the same API function as would be called upon detection of the "real" tone signal. Enable the fake tone only for hardware configurations that do not provide a tone signal.

```
#define CFE_MISSION_TIME_CFG_FAKE_TONE  TRUE
```

Hypothetical hardware configuration number one (described above) would enable the fake tone signal.

See also

CFE_MISSION_TIME_CFG_FAKE_TONE

Next: Selecting Tone and Data Ordering
Prev: Time Format Selection
Up To: Time Configuration

25.7 Selecting Tone and Data Ordering

Tone and data order is defined in the mission configuration header file.

This selection defines which comes first - the tone or the time at the tone data. Does the time data describe the tone that already occurred, or the tone that has not yet occurred? This decision may be driven by the hardware design but can also be arbitrary. Enable one, and only one, of the following:

```
#define CFE_MISSION_TIME_AT_TONE_WAS
#define CFE_MISSION_TIME_AT_TONE_WILL_BE
```

Hypothetical hardware configuration number three (described Time Configuration above) would enable "time at the tone was".

See also

CFE_MISSION_TIME_AT_TONE_WAS, CFE_MISSION_TIME_AT_TONE_WILL_BE

Next: Specifying Tone and Data Window
Prev: Enabling Fake Tone Signal
Up To: Time Configuration

25.8 Specifying Tone and Data Window

The tone and data window is defined in the mission configuration header file.

In concert with the definition of tone and data order, this selection defines the valid window in time for the second of the pair to follow the first. Both must be defined, units are micro-seconds.

```
#define CFE_MISSION_TIME_MIN_ELAPSED  0
#define CFE_MISSION_TIME_MAX_ELAPSED  100000
```

Hypothetical hardware configuration number three (described above) might use these values which describe a window that begins immediately after the tone and lasts for one tenth of a second.

See also

CFE_MISSION_TIME_MIN_ELAPSED, CFE_MISSION_TIME_MAX_ELAPSED

Next: Specifying Time Server/Client
Prev: Selecting Tone and Data Ordering
Up To: Time Configuration

25.9 Specifying Time Server/Client

Configure TIME as a client only when the target system has multiple processors running separate instantiations of the cFE. One instantiation must be configured as the server and the remainder configured as clients. If the target system has only one processor running the cFE, then TIME must be configured as a server.

Enable one, and only one, of the following definitions in the platform configuration header file:

```
#define CFE_PLATFORM_TIME_CFG_SERVER   TRUE
#define CFE_PLATFORM_TIME_CFG_CLIENT   FALSE
```

or

```
#define CFE_PLATFORM_TIME_CFG_SERVER   FALSE
#define CFE_PLATFORM_TIME_CFG_CLIENT   TRUE
```

See also

CFE_PLATFORM_TIME_CFG_SERVER, CFE_PLATFORM_TIME_CFG_CLIENT

Next: Specifying Time Tone Byte Order
Prev: Specifying Tone and Data Window
Up To: Time Configuration

25.10 Specifying Time Tone Byte Order

By default, the CFE time tone message is a payload of integers in platform-endian order (containing the tone's timestamp, the leap seconds, and state information.) In some configurations, it may be better to have the payload produced in big-endian order–particularly in mixed-endian environments.

In order to force the tone message to be in big-endian order, you must define the following:

```
#define CFE_PLATFORM_TIME_CFG_BIGENDIAN
```

Next: Virtual MET
Prev: Specifying Time Server/Client
Up To: Time Configuration

25.11 Virtual MET

This configuration option refers to whether the MET is local to this instantiation of TIME. If the MET is not local then TIME must be configured as using a virtual MET.

Therefore, all TIME clients must be configured as using a virtual MET. If the MET was local to any TIME client, then that instantiation of TIME would have to be the server.

TIME servers must be configured as using a virtual MET

Next: Specifying Time Source
Prev: Specifying Time Tone Byte Order
Up To: Time Configuration

25.12 Specifying Time Source

TIME configuration provides the ability to specify where the source for time data is originating - either internal or external. In hypothetical system one, the MET is internal. In system two, TIME cannot directly read the MET, therefore time data must be received from an external source.

This selection also enables a command interface to switch between internal and external input. When commanded to use internal time data, TIME will ignore the external data. However, TIME will continue to use the API function as the trigger to generate a "time at the tone" command packet regardless of the internal/external command selection.

Set the following definition to TRUE only for TIME servers using an external time data source.

```
#define CFE_PLATFORM_TIME_CFG_SOURCE  TRUE
```

The remainder of this section pertains only to TIME servers configured to accept external time data.

When configured to accept external time data, TIME requires an additional definition for the type of external data (GPS, MET, spacecraft time, etc.). This selection will enable an API function specific to the selected data type. Regardless of how the time data is received, the receiver need only pass the data to the appropriate API function.

TIME servers using an external time data source must set one, and only one, of the following to TRUE, for example:

```
#define CFE_PLATFORM_TIME_CFG_SRC_MET   TRUE
#define CFE_PLATFORM_TIME_CFG_SRC_GPS   FALSE
#define CFE_PLATFORM_TIME_CFG_SRC_TIME  FALSE
```

configuration definitions for the particular source.

If the cfe_platform_cfg.h file contains "#define CFE_PLATFORM_TIME_CFG_SOURCE TRUE" then time is configured to allow switching between internal and external time sources (see CFE_TIME_SET_SOURCE_CC). If this configuration parameter is set to FALSE then the command to set the source will be rejected.

If this configuration parameter is set to TRUE then ONE and ONLY ONE of the following configuration parameters must also be set TRUE in order to specify the external time source, for example:

```
#define CFE_PLATFORM_TIME_CFG_SRC_MET   TRUE
#define CFE_PLATFORM_TIME_CFG_SRC_GPS   FALSE
#define CFE_PLATFORM_TIME_CFG_SRC_TIME  FALSE
```

Note that Internal MET source depends on available hardware. It may be the local count of tone signals, the contents of a hardware register or an OS specific time function.

Note also that when configured to use an external time source, commands to set the time will be overwritten.

See also

CFE_PLATFORM_TIME_CFG_SRC_MET, CFE_PLATFORM_TIME_CFG_SRC_GPS, CFE_PLATFORM_TI↩ME_CFG_SRC_TIME

Next: Specifying Time Signal
Prev: Virtual MET
Up To: Time Configuration

25.13 Specifying Time Signal

Some hardware configurations support a primary and redundant tone signal selection. Setting the following configuration definition to TRUE will result in enabling a TIME command to select the active tone signal.

```
#define CFE_PLATFORM_TIME_CFG_SIGNAL  TRUE
```

Note: this feature requires additional custom software to make the physical signal switch.

See also

CFE_PLATFORM_TIME_CFG_SIGNAL

Next: Time Services Paradigm(s)
Prev: Specifying Time Source
Up To: Time Configuration

25.14 Time Services Paradigm(s)

In order for the cFE Time Services to work for a particular mission, the methods of obtaining time, distributing time and translating time must follow some standard paradigms used in previous missions. The following describes this expected context:

Mission dependent hardware provides the Tone. When this Tone message is received, TIME latches the local time based on the local clock. Note that in lab environments, a simulated Tone capability exists which uses an SB message. Mission dependent hardware also provides the "time at the tone" message based on the hardware latched time and the reference times stored by TIME Server. The TIME Client then updates its local reference time based on the local hardware latched time at the Tone and the provided Time-at-Tone message packet when certain checks (such as the Validity bit being set) pass.

When used in an environment that includes multiple processors, each running a separate instantiation of cFE software, the presumption is that TIME will be distributed in a client/server relationship. In this model, one processor will have TIME configured as the server and the other processors as clients. The TIME server will maintain the various time components and publish a "time at the tone" message to provide synchronized time to the TIME clients. Environments that have only a single instance of TIME must be configured as a TIME server.

In all configurations, the final step in calculating the time "right now" for any instantiation of TIME is to use a local processor clock to measure the "time since the tone".

The specific MET hardware properties will determine whether the MET value can be modified. However, the cFE design is such that there should never be a need to purposefully change or reset the MET.

Regardless of the physical hardware implementation for the MET (elapsed seconds, elapsed ticks, etc.), cFE TIME will convert the hardware MET value into a System Time Format structure for time calculations and will report the converted value in telemetry. cFE TIME will also maintain and report the STCF in a System Time Format structure.

cFE TIME has no knowledge of the current epoch; it is up to the user to keep time on the spacecraft correlated to an epoch. An exception might appear to be the epoch definition required in the cFE mission configuration definition file. However, this definition is for use only by the API functions that convert spacecraft time and file system time, and the API function that prints spacecraft time as a date and time text string. The cFE "get time" functions are independent of the ground epoch.

The mission configuration parameters, CFE_MISSION_TIME_CFG_DEFAULT_TAI and CFE_MISSION_TIME_CFG↩_DEFAULT_UTC specify the default time format. Applications are encouraged to use the CFE_TIME_GetTime API, which returns time in the format specified by this configuration parameter.

Next: Flywheeling
Prev: Time Components
Up To: cFE Time Services Overview

25.15 Flywheeling

Flywheeling occurs when TIME is not getting a valid tone signal or external "time at the tone" message. While this has minimal impact on internal operations, it can result in the drifting apart of times being stored by different spacecraft systems.

Flywheeling occurs when at least one of the following conditions is true:

- loss of tone signal
- loss of "time at the tone" data packet
- signal and packet not within valid window
- commanded into fly-wheel mode

If the TIME server is in Flywheel mode then the TIME client is also in flywheel mode.

Next: Time State
Prev: Time Services Paradigm(s)
Up To: cFE Time Services Overview

25.16 Time State

Clock state is a combination of factors, most significantly whether the spacecraft time has been accurately set and whether Time Service is operating in FLYWHEEL mode. A ground command is provided to set the state to reflect when the ground has determined the spacecraft time is now correct, or that time is no longer correct. This information will be distributed to Time Clients, and in turn, to any interested sub-systems. If time has not been set then TIME services reports the state of time as invalid, regardless of whether time is flywheeling or not. Also, this command may be used to force a Time Server or Time Client into FLYWHEEL mode. Use of FLYWHEEL mode is mainly for debug purposes although, in extreme circumstances, it may be of value to force Time Service not to rely on normal time updates. Note that when commanded into FLYWHEEL mode, the Time Service will remain so until receipt of another "set state" command setting the state into a mode other than FLYWHEEL. Note also that setting the clock state to VALID or INV↩ ALID on a Time Client that is currently getting time updates from the Time Server will have very limited effect. As soon as the Time Client receives the next time update, the VALID/INVALID selection will be set to that of the Time Server. However, setting a Time Client to FLYWHEEL cannot be overridden by the Time Server since the Time Client will ignore time updates from the Time Server while in FLYWHEEL mode.

Next: Initialization
Prev: Flywheeling
Up To: cFE Time Services Overview

25.17 Initialization

No action is required by the ground to initialize the TIME software; however, time variables in the TIME Server must be set by command to allow correct time to propagate.

For a description of what happens during each type of reset, see below:

- Power-On Reset
- Processor Reset

Next: Power-On Reset
Prev: Time State
Up To: cFE Time Services Overview

25.17.1 Power-On Reset

TIME initializes all counters in housekeeping telemetry, sets the Validity state to Invalid, and initializes the STCF, Leap Seconds, and 1 Hz Adjustment to zero.

Next: Processor Reset
Up To: Initialization

25.17.2 Processor Reset

In the event of a processor reset, the following time values are preserved:

- MET
- STCF
- Leap Seconds
- Clock Signal Selection
- Current Time Client Delay (if applicable)

Note that since it is virtually impossible for TIME services to validate the actual data that is saved across a processor reset, a signature pattern is written to the preserved area. On a processor reset, TIME queries that signature to make sure that it matches what is expected. If the signature does not match, then TIME is initialized as if a cFE power-on reset occurred.

Next: Normal Operation
Prev: Power-On Reset
Up To: Initialization

25.18 Power-On Reset

TIME initializes all counters in housekeeping telemetry, sets the Validity state to Invalid, and initializes the STCF, Leap Seconds, and 1 Hz Adjustment to zero.

Next: Processor Reset
Up To: Initialization

25.19 Processor Reset

In the event of a processor reset, the following time values are preserved:

- MET
- STCF
- Leap Seconds
- Clock Signal Selection
- Current Time Client Delay (if applicable)

Note that since it is virtually impossible for TIME services to validate the actual data that is saved across a processor reset, a signature pattern is written to the preserved area. On a processor reset, TIME queries that signature to make sure that it matches what is expected. If the signature does not match, then TIME is initialized as if a cFE power-on reset occurred.

Next: Normal Operation
Prev: Power-On Reset
Up To: Initialization

25.20 Initialization

No action is required by the ground to initialize the TIME software; however, time variables in the TIME Server must be set by command to allow correct time to propagate.

For a description of what happens during each type of reset, see below:

- Power-On Reset
- Processor Reset

Next: Power-On Reset
Prev: Time State
Up To: cFE Time Services Overview

25.20.1 Power-On Reset

TIME initializes all counters in housekeeping telemetry, sets the Validity state to Invalid, and initializes the STCF, Leap Seconds, and 1 Hz Adjustment to zero.

Next: Processor Reset
Up To: Initialization

25.20.2 Processor Reset

In the event of a processor reset, the following time values are preserved:

- MET
- STCF
- Leap Seconds
- Clock Signal Selection
- Current Time Client Delay (if applicable)

Note that since it is virtually impossible for TIME services to validate the actual data that is saved across a processor reset, a signature pattern is written to the preserved area. On a processor reset, TIME queries that signature to make sure that it matches what is expected. If the signature does not match, then TIME is initialized as if a cFE power-on reset occurred.

Next: Normal Operation
Prev: Power-On Reset
Up To: Initialization

25.21 Power-On Reset

TIME initializes all counters in housekeeping telemetry, sets the Validity state to Invalid, and initializes the STCF, Leap Seconds, and 1 Hz Adjustment to zero.

Next: Processor Reset
Up To: Initialization

25.22 Processor Reset

In the event of a processor reset, the following time values are preserved:

- MET
- STCF
- Leap Seconds
- Clock Signal Selection
- Current Time Client Delay (if applicable)

Note that since it is virtually impossible for TIME services to validate the actual data that is saved across a processor reset, a signature pattern is written to the preserved area. On a processor reset, TIME queries that signature to make sure that it matches what is expected. If the signature does not match, then TIME is initialized as if a cFE power-on reset occurred.

Next: Normal Operation
Prev: Power-On Reset
Up To: Initialization

25.23 Normal Operation

The following sections describe the operator's resposibilities for maintaining time under nominal conditions:

- Client
- Server

Next: Client
Prev: Initialization
Up To: cFE Time Services Overview

25.23.1 Client

Under normal operation, TIME Client systems do not require any attention from the ground, however TIME clients do provide commands to set the persistent latency between the server and client. Latency can be either added or subtracted to the current TIME client time calculation to account for the latency.

Next: Server
Up To: Normal Operation

25.23.2 Server

TIME Servers require maintenance by the operations team to ensure the spacecraft is maintaining a time that can be successfully correlated to other entities. The following sections describe the commands that the operations team can use to help maintain a proper time reference:

- Setting Time
- Adjusting Time
- Setting MET

Next: Setting Time
Prev: Client
Up To: Normal Operation

25.23.2.1 Setting Time

The Time Server provides commands to set time. The new time value represents the desired offset from mission-defined time epoch and takes effect immediately upon execution of this command. Time Service will calculate a new STCF value based on the current MET and the desired new time using one of the following:

If Time Service is configured to compute current time as TAI:

```
STCF = new time - current MET
current time =  current MET + STCF
```

If Time Service is configured to compute current time as UTC:

```
STCF = ((new time) - (current MET)) + Leap Seconds
current time = ((current MET) + STCF) - Leap Seconds
```

See also

CFE_TIME_SET_TIME_CC

Next: Adjusting Time
Up To: Server

25.23.2.2 Adjusting Time

The TIME Server includes commands to set the STCF, Leap Seconds, and Validity state. The STCF should be set implicity using the CFE_TIME_SET_TIME_CC or explicitly using CFE_TIME_SET_STCF_CC. TIME provides the ability to command a one time adjustment (CFE_TIME_ADD_ADJUST_CC and CFE_TIME_SUB_ADJUST_CC) to the current STCF. In addition there is a 1Hz adjustment (CFE_TIME_ADD_1HZ_ADJUSTMENT_CC and CFE_TIME_SUB_1HZ_↩ ADJUSTMENT_CC) that can be made to the STCF to compensate for oscillator drift. Mission specific ground correlation should be used to assist in determining the proper values to use. The Leap Seconds should be set to the current TA↩ I-UTC. Note that the International Earth Rotation and Reference Systems Service Bulletin C, which defines the current difference, reports it as UTC-TAI, and thus that value must be negated. **The Leap Seconds value will always be a positive number.** The Validity state does not have to be set to invalid to change the STCF or Leap Seconds, and should be set to valid at any time that the TIME Server time reference should be synchronized to by the other systems.

See also

CFE_TIME_ADD_ADJUST_CC, CFE_TIME_SUB_ADJUST_CC, CFE_TIME_SET_STCF_CC, CFE_TIME_A↩ DD_1HZ_ADJUSTMENT_CC, CFE_TIME_SUB_1HZ_ADJUSTMENT_CC, CFE_TIME_SET_LEAP_SECOND↩ S_CC

Next: Setting MET
Prev: Setting Time
Up To: Server

25.23.2.3 Setting MET

The TIME Server provides the capability to set the MET. Note that the MET (as implemented for cFE Time Service) is a logical representation and not a physical timer. Thus, setting the MET is not dependent on whether the hardware supports a MET register that can be written to. Note also that Time Service "assumes" that during normal operation, the MET is synchronized to the tone signal. Therefore, unless operating in FLYWHEEL mode, the sub-seconds portion of the MET will be set to zero at the next tone signal interrupt. The new MET takes effect immediately upon execution of this command.

See also

CFE_TIME_SET_MET_CC

Next: Frequently Asked Questions
Prev: Adjusting Time
Up To: Server

25.24 Client

Under normal operation, TIME Client systems do not require any attention from the ground, however TIME clients do provide commands to set the persistent latency between the server and client. Latency can be either added or subtracted to the current TIME client time calculation to account for the latency.

Next: Server
Up To: Normal Operation

25.25 Server

TIME Servers require maintenance by the operations team to ensure the spacecraft is maintaining a time that can be successfully correlated to other entities. The following sections describe the commands that the operations team can use to help maintain a proper time reference:

- Setting Time
- Adjusting Time
- Setting MET

Next: Setting Time
Prev: Client
Up To: Normal Operation

25.25.0.1 Setting Time

The Time Server provides commands to set time. The new time value represents the desired offset from mission-defined time epoch and takes effect immediately upon execution of this command. Time Service will calculate a new STCF value based on the current MET and the desired new time using one of the following:

If Time Service is configured to compute current time as TAI:

```
STCF = new time - current MET
current time =  current MET + STCF
```

If Time Service is configured to compute current time as UTC:

```
STCF = ((new time) - (current MET)) + Leap Seconds
current time = ((current MET) + STCF) - Leap Seconds
```

See also

CFE_TIME_SET_TIME_CC

Next: Adjusting Time
Up To: Server

25.25.0.2 Adjusting Time

The TIME Server includes commands to set the STCF, Leap Seconds, and Validity state. The STCF should be set implicity using the CFE_TIME_SET_TIME_CC or explicitly using CFE_TIME_SET_STCF_CC. TIME provides the ability to command a one time adjustment (CFE_TIME_ADD_ADJUST_CC and CFE_TIME_SUB_ADJUST_CC) to the current STCF. In addition there is a 1Hz adjustment (CFE_TIME_ADD_1HZ_ADJUSTMENT_CC and CFE_TIME_SUB_1HZ_↩ ADJUSTMENT_CC) that can be made to the STCF to compensate for oscillator drift. Mission specific ground correlation should be used to assist in determining the proper values to use. The Leap Seconds should be set to the current TA↩ I-UTC. Note that the International Earth Rotation and Reference Systems Service Bulletin C, which defines the current difference, reports it as UTC-TAI, and thus that value must be negated. **The Leap Seconds value will always be a positive number.** The Validity state does not have to be set to invalid to change the STCF or Leap Seconds, and should be set to valid at any time that the TIME Server time reference should be synchronized to by the other systems.

See also

CFE_TIME_ADD_ADJUST_CC, CFE_TIME_SUB_ADJUST_CC, CFE_TIME_SET_STCF_CC, CFE_TIME_A↩ DD_1HZ_ADJUSTMENT_CC, CFE_TIME_SUB_1HZ_ADJUSTMENT_CC, CFE_TIME_SET_LEAP_SECOND↩ S_CC

Next: Setting MET
Prev: Setting Time
Up To: Server

25.25.0.3 Setting MET

The TIME Server provides the capability to set the MET. Note that the MET (as implemented for cFE Time Service) is a logical representation and not a physical timer. Thus, setting the MET is not dependent on whether the hardware supports a MET register that can be written to. Note also that Time Service "assumes" that during normal operation, the MET is synchronized to the tone signal. Therefore, unless operating in FLYWHEEL mode, the sub-seconds portion of the MET will be set to zero at the next tone signal interrupt. The new MET takes effect immediately upon execution of this command.

See also

CFE_TIME_SET_MET_CC

Next: Frequently Asked Questions
Prev: Adjusting Time
Up To: Server

25.26 Setting Time

The Time Server provides commands to set time. The new time value represents the desired offset from mission-defined time epoch and takes effect immediately upon execution of this command. Time Service will calculate a new STCF value based on the current MET and the desired new time using one of the following:

If Time Service is configured to compute current time as TAI:

```
STCF = new time - current MET
current time =  current MET + STCF
```

If Time Service is configured to compute current time as UTC:

```
STCF = ((new time) - (current MET)) + Leap Seconds
current time = ((current MET) + STCF) - Leap Seconds
```

See also

CFE_TIME_SET_TIME_CC

Next: Adjusting Time
Up To: Server

25.27 Adjusting Time

The TIME Server includes commands to set the STCF, Leap Seconds, and Validity state. The STCF should be set implicity using the CFE_TIME_SET_TIME_CC or explicitly using CFE_TIME_SET_STCF_CC. TIME provides the ability to command a one time adjustment (CFE_TIME_ADD_ADJUST_CC and CFE_TIME_SUB_ADJUST_CC) to the current STCF. In addition there is a 1Hz adjustment (CFE_TIME_ADD_1HZ_ADJUSTMENT_CC and CFE_TIME_SUB_1HZ_↩ADJUSTMENT_CC) that can be made to the STCF to compensate for oscillator drift. Mission specific ground correlation should be used to assist in determining the proper values to use. The Leap Seconds should be set to the current TA↩I-UTC. Note that the International Earth Rotation and Reference Systems Service Bulletin C, which defines the current difference, reports it as UTC-TAI, and thus that value must be negated. **The Leap Seconds value will always be a positive number.** The Validity state does not have to be set to invalid to change the STCF or Leap Seconds, and should be set to valid at any time that the TIME Server time reference should be synchronized to by the other systems.

See also

CFE_TIME_ADD_ADJUST_CC, CFE_TIME_SUB_ADJUST_CC, CFE_TIME_SET_STCF_CC, CFE_TIME_A↩DD_1HZ_ADJUSTMENT_CC, CFE_TIME_SUB_1HZ_ADJUSTMENT_CC, CFE_TIME_SET_LEAP_SECOND↩S_CC

Next: Setting MET
Prev: Setting Time
Up To: Server

25.28 Setting MET

The TIME Server provides the capability to set the MET. Note that the MET (as implemented for cFE Time Service) is a logical representation and not a physical timer. Thus, setting the MET is not dependent on whether the hardware supports a MET register that can be written to. Note also that Time Service "assumes" that during normal operation, the MET is synchronized to the tone signal. Therefore, unless operating in FLYWHEEL mode, the sub-seconds portion of the MET will be set to zero at the next tone signal interrupt. The new MET takes effect immediately upon execution of this command.

See also

CFE_TIME_SET_MET_CC

Next: Frequently Asked Questions
Prev: Adjusting Time
Up To: Server

25.29 Frequently Asked Questions

(Q)	

Prev: Normal Operation
Up To: cFE Time Services Overview

26 cFE Time Services Commands

The following is a list of commands that are processed by the cFE Time Services Task.

Global CFE_TIME_ADD_1HZ_ADJUSTMENT_CC
Add Delta to Spacecraft Time Correlation Factor each 1Hz

Global CFE_TIME_ADD_ADJUST_CC
Add Delta to Spacecraft Time Correlation Factor

Global CFE_TIME_ADD_DELAY_CC
Add Time to Tone Time Delay

Global CFE_TIME_NOOP_CC
Time No-Op

Global CFE_TIME_RESET_COUNTERS_CC
Time Reset Counters

Global CFE_TIME_SEND_DIAGNOSTIC_TLM_CC
Request TIME Diagnostic Telemetry

Global CFE_TIME_SET_LEAP_SECONDS_CC
Set Leap Seconds

Global CFE_TIME_SET_MET_CC
Set Mission Elapsed Time

Global CFE_TIME_SET_SIGNAL_CC
Set Tone Signal Source

Global CFE_TIME_SET_SOURCE_CC
Set Time Source

Global CFE_TIME_SET_STATE_CC
Set Time State

Global CFE_TIME_SET_STCF_CC
Set Spacecraft Time Correlation Factor

Global CFE_TIME_SET_TIME_CC
Set Spacecraft Time

Global CFE_TIME_SUB_1HZ_ADJUSTMENT_CC
Subtract Delta from Spacecraft Time Correlation Factor each 1Hz

Global CFE_TIME_SUB_ADJUST_CC
Subtract Delta from Spacecraft Time Correlation Factor

Global CFE_TIME_SUB_DELAY_CC
Subtract Time from Tone Time Delay

27 cFE Time Services Telemetry

The following are telemetry packets generated by the cFE Time Services Task.

Global CFE_TIME_DiagnosticTlm_Payload_t
Time Services Diagnostics Packet

Global CFE_TIME_DiagnosticTlm_Payload_t
Time Services Diagnostics Packet

Global CFE_TIME_HousekeepingTlm_Payload_t
Time Services Housekeeping Packet

Global CFE_TIME_HousekeepingTlm_Payload_t
Time Services Housekeeping Packet

28 cFE Time Services Configuration Parameters

The following are configuration parameters used to configure the cFE Time Services either for each platform or for a mission as a whole.

Global CFE_MISSION_TIME_AT_TONE_WAS
Default Time and Tone Order

Global CFE_MISSION_TIME_CFG_DEFAULT_TAI
Default Time Format

Global CFE_MISSION_TIME_CFG_FAKE_TONE
Default Time Format

Global CFE_MISSION_TIME_DEF_MET_SECS
Default Time Values

Global CFE_MISSION_TIME_EPOCH_YEAR
Default EPOCH Values

Global CFE_MISSION_TIME_FS_FACTOR
Time File System Factor

Global CFE_MISSION_TIME_MIN_ELAPSED
Min and Max Time Elapsed

Global CFE_PLATFORM_TIME_CFG_LATCH_FLY
Define Periodic Time to Update Local Clock Tone Latch

Global CFE_PLATFORM_TIME_CFG_SERVER
Time Server or Time Client Selection

Global CFE_PLATFORM_TIME_CFG_SIGNAL
Include or Exclude the Primary/Redundant Tone Selection Cmd

Global CFE_PLATFORM_TIME_CFG_SOURCE
Include or Exclude the Internal/External Time Source Selection Cmd

Global CFE_PLATFORM_TIME_CFG_SRC_MET

Choose the External Time Source for Server only

Global CFE_PLATFORM_TIME_CFG_START_FLY

Define Time to Start Flywheel Since Last Tone

Global CFE_PLATFORM_TIME_CFG_TONE_LIMIT

Define Timing Limits From One Tone To The Next

Global CFE_PLATFORM_TIME_CFG_VIRTUAL

Time Tone In Big-Endian Order

Local MET or Virtual MET Selection for Time Servers

Global CFE_PLATFORM_TIME_MAX_DELTA_SECS

Define the Max Delta Limits for Time Servers using an Ext Time Source

Global CFE_PLATFORM_TIME_MAX_LOCAL_SECS

Define the Local Clock Rollover Value in seconds and subseconds

Global CFE_PLATFORM_TIME_START_TASK_PRIORITY

Define TIME Task Priorities

Global CFE_PLATFORM_TIME_START_TASK_STACK_SIZE

Define TIME Task Stack Sizes

29 cFE Event Message Cross Reference

The following cross reference maps the text associated with each cFE Event Message to its Event Message Identifier. A user can search this page for the text of the message they wish to learn more about and then click on the associated Event Message Identifier to obtain more information.

Global CFE_ES_ALL_APPS_EID

```
'App Info file written to %s, Entries=%d, FileSize=%d'
```

Global CFE_ES_BOOT_ERR_EID

```
'Invalid cFE restart type %d'
```

Global CFE_ES_BUILD_INF_EID

```
'Build s by s%s, config s'
```

Global CFE_ES_CC1_ERR_EID

```
'Invalid ground command code:  ID = 0x%X, CC = %d'
```

Global CFE_ES_CDS_DELETE_ERR_EID

```
'Error while deleting '%s' from CDS, See SysLog.(Err=0x%08X)'
```

Global CFE_ES_CDS_DELETE_TBL_ERR_EID

```
'CDS '%s' is a Critical Table CDS. Must be deleted via TBL Command'
```

Global CFE_ES_CDS_DELETED_INFO_EID

```
'Successfully removed '%s' from CDS'
```

Global CFE_ES_CDS_DUMP_ERR_EID

```
'Error writing CDS Registry to '%s', Status=0x%08X'
```

Global CFE_ES_CDS_NAME_ERR_EID

```
'Unable to locate '%s' in CDS Registry'
```

Global CFE_ES_CDS_OWNER_ACTIVE_EID

```
'CDS '%s' not deleted because owning app is active'
```

Global CFE_ES_CDS_REG_DUMP_INF_EID

```
'Successfully dumped CDS Registry to '%s':Size=%d,Entries=%d'
```

Global CFE_ES_CDS_REGISTER_ERR_EID

```
'%s Failed to Register CDS '%s', Status=0x%08X'
```

Global CFE_ES_CREATING_CDS_DUMP_ERR_EID

```
'Error creating CDS dump file '%s', Status=0x%08X'
```

Global CFE_ES_ERLOG1_INF_EID

```
'Cleared mode log data'
```

Global CFE_ES_ERLOG2_EID

```
'%s written:Size=%d'
```

Global CFE_ES_ERLOG2_ERR_EID

```
'Error creating file %s, stat=0x%x'
```

Global CFE_ES_ERLOG_PENDING_ERR_EID

```
'Error log write to file %s already in progress'
```

Global CFE_ES_ERR_SYSLOGMODE_EID

```
'Set OverWriteSysLog Command:  Invalid Mode setting = %d'
```

Global CFE_ES_ERREXIT_APP_ERR_EID

```
'Exit Application %s on Error Failed:  CleanUpApp Error 0x%08X.'
```

Global CFE_ES_ERREXIT_APP_INF_EID

```
'Exit Application %s Completed.'
```

Global CFE_ES_EXIT_APP_ERR_EID

```
'Exit Application %s Failed:  CleanUpApp Error 0x%08X.'
```

Global CFE_ES_EXIT_APP_INF_EID

```
'Exit Application %s Completed.'
```

Global CFE_ES_FILEWRITE_ERR_EID

```
'File write,byte cnt err,file %s,request=%d,actual=%d'
```

Global CFE_ES_INIT_INF_EID

```
'cFE ES Initialized'
```

Global CFE_ES_INITSTATS_INF_EID

```
'cFE Version %d.%d.%d chksm %d, OSAL Version %d.%d'
```

Global CFE_ES_INVALID_POOL_HANDLE_ERR_EID

```
'Cannot telemeter memory pool stats.  Illegal Handle (0x%08X)'
```

Global CFE_ES_LEN_ERR_EID

```
'Invalid cmd length:  ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'
```

Global CFE_ES_MID_ERR_EID

```
'Invalid command pipe message ID: 0x%X'
```

Global CFE_ES_NOOP_INF_EID

```
'No-op command'
```

Global CFE_ES_ONE_APP_EID

```
'Sent %s application data'
```

Global CFE_ES_ONE_APPID_ERR_EID

```
'Failed to send %s application data:  GetAppIDByName Failed, RC = 0x%08X'
```

Global CFE_ES_ONE_ERR_EID

```
'Failed to send %s application data, RC = %08X'
```

Global CFE_ES_OSCREATE_ERR_EID

```
'Failed to write App Info file, OS_OpenCreate returned %d'
```

Global CFE_ES_PCR_ERR1_EID

```
'ES_ProcControlReq:  Invalid State (EXCEPTION) Application %s.'
```

Global CFE_ES_PCR_ERR2_EID

```
'ES_ProcControlReq:  Unknown State ( %d ) Application %s.'
```

Global CFE_ES_PERF_DATAWRITTEN_EID

```
'%s written:Size=%d,EntryCount=%d'
```

Global CFE_ES_PERF_FILTMSKCMD_EID

```
'Set Performance Filter Mask command'
```

Global CFE_ES_PERF_FILTMSKERR_EID

```
'Performance Filter Mask Cmd Error,Index(u)out of range(u)'
```

Global CFE_ES_PERF_LOG_ERR_EID

```
'Error creating file %s, stat=%d'
```

Global CFE_ES_PERF_STARTCMD_EID

```
'Start collecting performance data command, trigger mode = d'
```

Global CFE_ES_PERF_STARTCMD_ERR_EID

```
'Cannot start collecting performance data,perf data write in progress'
```

Global CFE_ES_PERF_STARTCMD_TRIG_ERR_EID

```
'Cannot start collecting performance data, trigger mode (d) out of range (d
to d)'
```

Global CFE_ES_PERF_STOPCMD_EID

```
'Perf Stop Cmd Rcvd,%s will write %d entries.%dmS dly every %d entries'
```

Global CFE_ES_PERF_STOPCMD_ERR2_EID

```
'Stop performance data cmd ignored,perf data write in progress'
```

Global CFE_ES_PERF_TRIGMSKCMD_EID

```
'Set Performance Trigger Mask command'
```

Global CFE_ES_PERF_TRIGMSKERR_EID

```
'Performance Trigger Mask Cmd Error,Index(u)out of range(u)'
```

Global CFE_ES_RELOAD_APP_DBG_EID

```
'Reload Application %s Initiated.'
```

Global CFE_ES_RELOAD_APP_ERR1_EID

```
'Failed to reload Application %s, rc = %08X'
```

Global CFE_ES_RELOAD_APP_ERR2_EID

```
'Reload Application %s, GetAppIDByName failed.  RC = 0x%08X.'
```

Global CFE_ES_RELOAD_APP_ERR3_EID

```
'Reload Application %s Failed:  AppCreate Error 0x%08X.'
```

Global CFE_ES_RELOAD_APP_ERR4_EID

```
'Reload Application %s Failed:  CleanUpApp Error 0x%08X.'
```

Global CFE_ES_RELOAD_APP_INF_EID

```
'Reload Application %s Completed, AppID=lu'
```

Global CFE_ES_RESET_INF_EID

```
'Reset Counters command'
```

Global CFE_ES_RESET_PR_COUNT_EID

```
'Reset Processor Reset Count to Zero'
```

Global CFE_ES_RESTART_APP_DBG_EID

```
'Restart Application %s Initiated.'
```

Global CFE_ES_RESTART_APP_ERR1_EID

```
'Restart Application %s Failed, RC = 0x%08X'
```

Global CFE_ES_RESTART_APP_ERR2_EID

```
'Restart Application %s, GetAppIDByName failed.  RC = 0x%08X.'
```

Global CFE_ES_RESTART_APP_ERR3_EID

```
'Restart Application %s Failed:  AppCreate Error 0x%08X.'
```

Global CFE_ES_RESTART_APP_ERR4_EID

```
'Restart Application %s Failed:  CleanUpApp Error 0x%08X.'
```

Global CFE_ES_RESTART_APP_INF_EID

```
'Restart Application %s Completed, AppID=lu'
```

Global CFE_ES_SET_MAX_PR_COUNT_EID

```
'Maximum Processor Reset Count set to:  %d'
```

Global CFE_ES_START_ERR_EID

```
'Failed to start %s from %s, RC = %08X'
```

Global CFE_ES_START_EXC_ACTION_ERR_EID

```
'CFE_ES_StartAppCmd:  Invalid Exception Action:  %d.'
```

Global CFE_ES_START_INF_EID

```
'Started %s from %s, AppID = %d'
```

Global CFE_ES_START_INVALID_ENTRY_POINT_ERR_EID

```
'CFE_ES_StartAppCmd:  App Entry Point is NULL.'
```

Global CFE_ES_START_INVALID_FILENAME_ERR_EID

```
'CFE_ES_StartAppCmd:  invalid filename:  %s'
```

Global CFE_ES_START_NULL_APP_NAME_ERR_EID

```
'CFE_ES_StartAppCmd:  App Name is NULL.'
```

Global CFE_ES_START_PRIORITY_ERR_EID

```
'CFE_ES_StartAppCmd:  Priority is too large:  %d.'
```

Global CFE_ES_STOP_DBG_EID

```
'Stop Application %s Initiated.'
```

Global CFE_ES_STOP_ERR1_EID

```
'Stop Application %s Failed, RC = 0x%08X'
```

Global CFE_ES_STOP_ERR2_EID

```
'Stop Application %s, GetAppIDByName failed.  RC = 0x%08X.'
```

Global CFE_ES_STOP_ERR3_EID

```
'Stop Application %s Failed:  CleanUpApp Error 0x%08X.'
```

Global CFE_ES_STOP_INF_EID

```
'Stop Application %s Completed.'
```

Global CFE_ES_SYSLOG1_INF_EID

```
'Cleared Executive Services log data'
```

Global CFE_ES_SYSLOG2_EID

```
'%s written:Size=%d,Entries=%d'
```

Global CFE_ES_SYSLOG2_ERR_EID

```
'Error creating file %s, stat=0x%x'
```

Global CFE_ES_SYSLOGMODE_EID

```
'Set OverWriteSysLog Command Received with Mode setting = %d'
```

Global CFE_ES_TASKINFO_EID

```
'Task Info file written to %s, Entries=%d, FileSize=%d'
```

Global CFE_ES_TASKINFO_OSCREATE_ERR_EID

```
'Failed to write Task Info file, OS_OpenCreate returned %d'
```

Global CFE_ES_TASKINFO_WR_ERR_EID

```
'Failed to write Task Info file, Task write RC = 0x%08X, exp %d'
```

Global CFE_ES_TASKINFO_WRHDR_ERR_EID

```
'Failed to write Task Info file, WriteHdr rtnd %08X, exp %d'
```

Global CFE_ES_TASKWR_ERR_EID

```
'Failed to write App Info file, Task write RC = 0x%08X, exp %d'
```

Global CFE_ES_TLM_POOL_STATS_INFO_EID

```
'Successfully telemetered memory pool stats for 0x%08X'
```

Global CFE_ES_VERSION_INF_EID

```
'Version Info:  s, s'
```

Global CFE_ES_WRHDR_ERR_EID

```
'Failed to write App Info file, WriteHdr rtnd %08X, exp %d'
```

Global CFE_ES_WRITE_CFE_HDR_ERR_EID

```
'Error writing cFE File Header to '%s', Status=0x%08X'
```

Global CFE_EVS_ADDFILTER_EID

```
'Add Filter Command Received with AppName = %s, EventID = 0x%08x, Mask =
0x%04x'
```

Global CFE_EVS_DELFILTER_EID

```
'Delete Filter Command Received with AppName = %s, EventID = 0x%08x'
```

Global CFE_EVS_DISAPPENTTYPE_EID

```
'Disable App Event Type Command Received with AppName = %s, EventType Bit
Mask = 0x%02x'
```

Global CFE_EVS_DISAPPEVT_EID

```
'Disable App Events Command Received with AppName = %s'
```

Global CFE_EVS_DISEVTTYPE_EID

'Disable Event Type Command Received with Event Type Bit Mask = 0x%02x'

Global CFE_EVS_DISPORT_EID

'Disable Ports Command Received with Port Bit Mask = 0x%02x'

Global CFE_EVS_ENAAPPEVT_EID

'Enable App Events Command Received with AppName = %s'

Global CFE_EVS_ENAAPPEVTTYPE_EID

'Enable App Event Type Command Received with AppName = %s, EventType Bit Mask = 0x%02x'

Global CFE_EVS_ENAEVTTYPE_EID

'Enable Event Type Command Received with Event Type Bit Mask = 0x%02x'

Global CFE_EVS_ENAPORT_EID

'Enable Ports Command Received with Port Bit Mask = 0x%02x'

Global CFE_EVS_ERR_APPNOREGS_EID

'%s not registered with EVS: CC = %lu'

Global CFE_EVS_ERR_CC_EID

'Invalid command code - ID = 0x%08x, CC = %d'

Global CFE_EVS_ERR_CRDATFILE_EID

'Write App Data Command Error: OS_OpenCreate = 0x%08X, filename = %s'

Global CFE_EVS_ERR_CRLOGFILE_EID

'Write Log File Command Error: OS_OpenCreate = 0x%08X, filename = %s'

Global CFE_EVS_ERR_EVTIDNOREGS_EID

'%s Event ID %d not registered for filtering: CC = %lu'

Global CFE_EVS_ERR_ILLAPPIDRANGE_EID

'Illegal application ID %d retrieved for %s: CC = %lu'

Global CFE_EVS_ERR_ILLEGALFMTMOD_EID

'Set Event Format Mode Command: Invalid Event Format Mode = 0x%02x'

Global CFE_EVS_ERR_INVALID_BITMASK_EID

'Bit Mask = 0x%X out of range: CC = %lu'

Global CFE_EVS_ERR_LOGMODE_EID

'Set Log Mode Command Error: Log Mode = %d

Global CFE_EVS_ERR_MAXREGSFILTER_EID

'Add Filter Command: number of registered filters has reached max = %d'

Global CFE_EVS_ERR_MSGID_EID

'Invalid command packet, Message ID = 0x%08X'

Global CFE_EVS_ERR_NOAPPIDFOUND_EID

'Unable to retrieve application ID for %s: CC = %lu'

Global CFE_EVS_ERR_UNREGISTERED_EVS_APP

'App %s not registered with Event Services. Unable to send event'

Global CFE_EVS_ERR_WRDATFILE_EID

'Write App Data Command Error: OS_write = 0x%08X, filename = %s'

Global CFE_EVS_ERR_WRLOGFILE_EID

```
'Write Log File Command Error:  OS_write = 0x%08X, filename = %s'
```

Global CFE_EVS_EVT_FILTERED_EID

```
'Add Filter Command:AppName = %s, EventID = 0x%08x is already registered for
filtering'
```

Global CFE_EVS_FILTER_MAX_EID

```
'Max filter count reached, AppName = %s, EventID = 0x%08x:  Filter locked
until reset'
```

Global CFE_EVS_LEN_ERR_EID

```
'Invalid cmd length:  ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'
```

Global CFE_EVS_LOGMODE_EID

```
'Set Log Mode Command Error:  Log Mode = %d'
```

Global CFE_EVS_NOOP_EID

```
'No-op command'
```

Global CFE_EVS_RSTALLFILTER_EID

```
'Reset All Filters Command Received with AppName = %s'
```

Global CFE_EVS_RSTCNT_EID

```
'Reset Counters Command Received'
```

Global CFE_EVS_RSTEVTCNT_EID

```
'Reset Event Counter Command Received with AppName = %s'
```

Global CFE_EVS_RSTFILTER_EID

```
'Reset Filter Command Received with AppName = %s, EventID = 0x%08x'
```

Global CFE_EVS_SETEVTFMTMOD_EID

```
'Set Event Format Mode Command Received with Mode = 0x%02x'
```

Global CFE_EVS_SETFILTERMSK_EID

```
'Set Filter Mask Command Received with AppName=%s, EventID=0x%08x, Mask=0x%04x'
```

Global CFE_EVS_STARTUP_EID

```
'cFE EVS Initialized'
```

Global CFE_EVS_WRDAT_EID

```
'Write App Data Command:  %d application data entries written to %s'
```

Global CFE_EVS_WRLOG_EID

```
'Write Log File Command:  %d event log entries written to %s'
```

Global CFE_SB_BAD_CMD_CODE_EID

```
'Invalid Cmd, Unexpected Command Code %d'
```

Global CFE_SB_BAD_MSGID_EID

```
'Invalid Cmd, Unexpected Msg Id:  0x%x'
```

Global CFE_SB_BAD_PIPEID_EID

```
'Rcv Err:PipeId %d does not exist,app %s'
```

Global CFE_SB_CMD0_RCVD_EID

```
'No-op Cmd Rcvd'
```

Global CFE_SB_CMD1_RCVD_EID

```
'Reset Counters Cmd Rcvd'
```

Global CFE_SB_CR_PIPE_BAD_ARG_EID

```
'CreatePipeErr:Bad Input Arg:app=%s,ptr=0x%x,depth=%d,maxdepth=%d'
```

Global CFE_SB_CR_PIPE_ERR_EID

```
'CreatePipeErr:OS_QueueCreate returned %d,app %s'
```

Global CFE_SB_CR_PIPE_NAME_TAKEN_EID

```
'CreatePipeErr:Name Taken:app=%s,ptr=0x%x,depth=%d,maxdepth=%d'
```

Global CFE_SB_CR_PIPE_NO_FREE_EID

```
'CreatePipeErr:No Free:app=%s,ptr=0x%x,depth=%d,maxdepth=%d'
```

Global CFE_SB_DEL_PIPE_ERR1_EID

```
'Pipe Delete Error:Bad Argument,PipedId %d,Requestor %s,Idx %d,Stat %d'
```

Global CFE_SB_DEL_PIPE_ERR2_EID

```
'Pipe Delete Error:Caller(%s) is not the owner of pipe %d'
```

Global CFE_SB_DEST_BLK_ERR_EID

```
'Subscribe Err:Request for Destination Blk failed for Msg 0x%x,Pipe %s'
```

Global CFE_SB_DSBL_RTE1_EID

```
'Disable Route Cmd:Route does not exist,Msg 0x%x,Pipe %d'
```

Global CFE_SB_DSBL_RTE2_EID

```
'Route Disabled,Msg 0x%x,Pipe %d'
```

Global CFE_SB_DSBL_RTE3_EID

```
'Disable Route Cmd:Invalid Param.Msg 0x%x,Pipe %d'
```

Global CFE_SB_DUP_SUBSCRIP_EID

```
'Duplicate Subscription,MsgId 0x%x on %s pipe,app %s'
```

Global CFE_SB_ENBL_RTE1_EID

```
'Enbl Route Cmd:Route does not exist.Msg 0x%x,Pipe %d'
```

Global CFE_SB_ENBL_RTE2_EID

```
'Enabling Route,Msg 0x%x,Pipe %d'
```

Global CFE_SB_ENBL_RTE3_EID

```
'Enbl Route Cmd:Invalid Param.Msg 0x%x,Pipe %d'
```

Global CFE_SB_FILEWRITE_ERR_EID

```
'File write,byte cnt err,file %s,request=%d,actual=%d'
```

Global CFE_SB_FULL_SUB_PKT_EID

```
'Full Sub Pkt %d Sent,Entries=%d,Stat=0x%x
'
```

Global CFE_SB_GET_BUF_ERR_EID

```
'Send Err:Request for Buffer Failed.  MsgId 0x%x,app %s,size %d'
```

Global CFE_SB_GETPIPEIDBYNAME_EID

```
'GetPipeIdByName:  ID retrieved.  Name %s,IdOut 0x%x, app %s'
```

Global CFE_SB_GETPIPEIDBYNAME_NAME_ERR_EID

```
'GetPipeIdByName Err:Name not found,Name %s,IdOut 0xx,App %s'
```

Global CFE_SB_GETPIPEIDBYNAME_NULL_ERR_EID

```
'GetPipeIdByName Err:Bad input argument,Name 0x%x,IdOut 0xx,App %s'
```

Global CFE_SB_GETPIPENAME_EID

```
'GetPipeName:  Name retrieved.  NameOut %s,Id %d, app %s'
```

Global CFE_SB_GETPIPENAME_ID_ERR_EID

```
'GetPipeName:  Id error.  NameOut %s,Id %d, app %s'
```

Global CFE_SB_GETPIPENAME_NULL_PTR_EID

```
'GetPipeName:  Null ptr error.  Id %d, app %s'
```

Global CFE_SB_GETPIPEOPTS_EID

```
'GetPipeOpts:  Options retrieved.  app %s'
```

Global CFE_SB_GETPIPEOPTS_ID_ERR_EID

```
'GetPipeOptsErr:Invalid pipe id (%d).app %s'
```

Global CFE_SB_GETPIPEOPTS_PTR_ERR_EID

```
'GetPipeOptsErr:Invalid opts ptr.app %s'
```

Global CFE_SB_HASHCOLLISION_EID

```
'Msg hash collision:  MsgId = 0x%x, collisions = %u'
```

Global CFE_SB_INIT_EID

```
'cFE SB Initialized'
```

Global CFE_SB_LEN_ERR_EID

```
'Invalid cmd length:  ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'
```

Global CFE_SB_MAX_DESTS_MET_EID

```
'Subscribe Err:Max Dests(%d)In Use For Msg 0x%x,pipe %s,app %s'
```

Global CFE_SB_MAX_MSGS_MET_EID

```
'Subscribe Err:Max Msgs(%d)In Use,MsgId 0x%x,pipe %s,app %s'
```

Global CFE_SB_MAX_PIPES_MET_EID

```
'CreatePipeErr:Max Pipes(%d)In Use.app %s'
```

Global CFE_SB_MSG_TOO_BIG_EID

```
'Send Err:Msg Too Big MsgId=0x%x,app=%s,size=%d,MaxSz=%d'
```

Global CFE_SB_MSGID_LIM_ERR_EID

```
'Send Err:Msg Limit Err MsgId 0x%x,pipe %s,sender %s'
```

Global CFE_SB_PART_SUB_PKT_EID

```
'Partial Sub Pkt %d Sent,Entries=%d,Stat=0x%x'
```

Global CFE_SB_PIPE_ADDED_EID

```
'Pipe Created:name %s,id %d,app %s'
```

Global CFE_SB_PIPE_DELETED_EID

```
'Pipe Deleted:id %d,owner %s'
```

Global CFE_SB_Q_FULL_ERR_EID

```
'Pipe Overflow,MsgId 0x%x,pipe %s,stat 0x%x,app %s'
```

Global CFE_SB_Q_RD_ERR_EID

```
'Pipe Read Err,pipe %s,app %s,stat 0x%x'
```

Global CFE_SB_Q_WR_ERR_EID

```
'Pipe Write Err,MsgId 0x%x,pipe %s,stat 0x%x,app %s'
```

Global CFE_SB_RCV_BAD_ARG_EID

```
'Rcv Err:Bad Input Arg:BufPtr 0x%x,pipe %d,t/o %d,app %s'
```

Global CFE_SB_SEND_BAD_ARG_EID

```
'Send Err:Bad input argument,Arg 0x%x,App %s'
```

Global CFE_SB_SEND_INV_MSGID_EID

```
'Send Err:Invalid msgid in msg,MsgId 0x%x,App %s'
```

Global CFE_SB_SEND_NO_SUBS_EID

```
'No subscribers for MsgId 0x%x,sender %s'
```

Global CFE_SB_SETPIPEOPTS_EID

```
'SetPipeOpts:  Options set (%d).  app %s'
```

Global CFE_SB_SETPIPEOPTS_ID_ERR_EID

```
'SetPipeOptsErr:Invalid pipe id (%d).app %s'
```

Global CFE_SB_SETPIPEOPTS_OWNER_ERR_EID

```
'SetPipeOptsErr:Caller not owner (%d).app %s'
```

Global CFE_SB_SND_RTG_EID

```
'%s written:Size=%d,Entries=%d'
```

Global CFE_SB_SND_RTG_ERR1_EID

```
'Error creating file %s, stat=0x%x'
```

Global CFE_SB_SND_STATS_EID

```
'Software Bus Statistics packet sent'
```

Global CFE_SB_SUB_ARG_ERR_EID

```
'Subscribe Err:Bad Arg,MsgId 0x%x,PipeId %d,app %s,scope %d'
```

Global CFE_SB_SUB_INV_CALLER_EID

```
'Subscribe Err:Caller(%s) is not the owner of pipe %d, Msg=0x%x'
```

Global CFE_SB_SUB_INV_PIPE_EID

```
'Subscribe Err:Invalid Pipe Id,Msg=0x%x,PipeId=%d,App %s'
```

Global CFE_SB_SUBSCRIPTION_RCVD_EID

```
'Subscription Rcvd:MsgId 0x%x on %s(%d),app %s'
```

Global CFE_SB_SUBSCRIPTION_REMOVED_EID

```
'Subscription Removed:Msg 0x%x on pipe %d,app %s'
```

Global CFE_SB_SUBSCRIPTION_RPT_EID

```
'Sending Subscription Report Msg=0x%x,Pipe=%d,Stat=0x%x'
```

Global CFE_SB_UNSUB_ARG_ERR_EID

```
'UnSubscribe Err:Bad Arg,MsgId 0x%x,PipeId %d,app %s,scope %d'
```

Global CFE_SB_UNSUB_INV_CALLER_EID

```
'Unsubscribe Err:Caller(%s) is not the owner of pipe %d,Msg=0x%x'
```

Global CFE_SB_UNSUB_INV_PIPE_EID

```
'Unsubscribe Err:Invalid Pipe Id Msg=0x%x,Pipe=%d,app=%s'
```

Global CFE_SB_UNSUB_NO_SUBS_EID

```
'Unsubscribe Err:No subs for Msg 0x%x on %s,app %s'
```

Global CFE_TBL_ACTIVATE_DUMP_ONLY_ERR_EID

```
'Illegal attempt to activate dump-only table '%s''
```

Global CFE_TBL_ACTIVATE_ERR_EID

```
'Cannot activate table '%s'.  No Inactive image available'
```

Global CFE_TBL_ASSUMED_VALID_INF_EID

'Tbl Services assumes '%s' is valid. No Validation Function has been registered'

Global CFE_TBL_CC1_ERR_EID

'Invalid command code - ID = 0x%X, CC = %d'

Global CFE_TBL_CDS_DELETE_ERR_EID

'Error while deleting '%s' from CDS, See SysLog.(Err=0x%08X)'

Global CFE_TBL_CDS_DELETED_INFO_EID

'Successfully removed '%s' from CDS'

Global CFE_TBL_CDS_NOT_FOUND_ERR_EID

'Unable to locate '%s' in CDS Registry'

Global CFE_TBL_CDS_OWNER_ACTIVE_ERR_EID

'CDS '%s' owning app is still active'

Global CFE_TBL_CREATING_DUMP_FILE_ERR_EID

'Error creating dump file '%s', Status=0x%08X'

Global CFE_TBL_DUMP_PENDING_ERR_EID

'A dump for '%s' is already pending'

Global CFE_TBL_FAIL_HK_SEND_ERR_EID

'Unable to send Hk Packet (Status=0x%08X)'

Global CFE_TBL_FAIL_NOTIFY_SEND_ERR_EID

'Manage Notification Pkt Error(MsgId=0x%08X, CC=0x%04X, Param=0x%08X, Status=0x%08X)'

Global CFE_TBL_FILE_ACCESS_ERR_EID

'Unable to open file '%s' for table load, Status = 0x%08X'

Global CFE_TBL_FILE_INCOMPLETE_ERR_EID

'Incomplete load of '%s' into '%s' working buffer'

Global CFE_TBL_FILE_LOADED_INF_EID

'Successful load of '%s' into '%s' working buffer'

Global CFE_TBL_FILE_STD_HDR_ERR_EID

'Unable to read std header for '%s', Status = 0x%08X'

Global CFE_TBL_FILE_SUBTYPE_ERR_EID

'File subtype for '%s' is wrong. Subtype = 0x%08X'

Global CFE_TBL_FILE_TBL_HDR_ERR_EID

'Unable to read tbl header for '%s', Status = 0x%08X'

Global CFE_TBL_FILE_TOO_BIG_ERR_EID

'File '%s' has more data than Tbl Hdr indicates (%d)'

Global CFE_TBL_FILE_TYPE_ERR_EID

'File '%s' is not a cFE file type, ContentType = 0x%08X'

Global CFE_TBL_HANDLE_ACCESS_ERR_EID

No access to Tbl handle=d

Global CFE_TBL_ILLEGAL_BUFF_PARAM_ERR_EID

'Cmd for Table '%s' had illegal buffer parameter (0x%08X)'

Global CFE_TBL_IN_REGISTRY_ERR_EID

''%s' found in Table Registry. CDS cannot be deleted until table is unregistered'

Global CFE_TBL_INIT_INF_EID

```
'Task Initialized'
```

Global CFE_TBL_INTERNAL_ERROR_ERR_EID

```
'Internal Error (Status=0x%08X)'
```

Global CFE_TBL_LEN_ERR_EID

```
'Invalid cmd pkt - ID = 0x%X, CC = %d, Len = %d'
```

Global CFE_TBL_LOAD_ABORT_ERR_EID

```
'Cannot abort load of '%s'.  No load started.'
```

Global CFE_TBL_LOAD_ABORT_INF_EID

```
'Table Load Aborted for '%s''
```

Global CFE_TBL_LOAD_DUMPONLY_ERR_EID

```
Attempted to load Dump Only Tbl 's'
```

Global CFE_TBL_LOAD_EXCEEDS_SIZE_ERR_EID

```
'Cannot load '%s' (%d) at offset %d in '%s' (%d)'
```

Global CFE_TBL_LOAD_FILENAME_LONG_ERR_EID

```
Filename is too long ('s' (lu) > lu)
```

Global CFE_TBL_LOAD_IN_PROGRESS_ERR_EID

```
Load already in progress for 's'
```

Global CFE_TBL_LOAD_PEND_REQ_INF_EID

```
'Tbl Services notifying App that '%s' has a load pending'
```

Global CFE_TBL_LOAD_SUCCESS_INF_EID

```
'Successfully loaded '%s' from '%s''
```

Global CFE_TBL_LOAD_TBLNAME_MISMATCH_ERR_EID

```
table name mismatch (exp=s, tblfilhdr=s)
```

Global CFE_TBL_LOAD_TYPE_ERR_EID

```
'%s Failed to Load '%s' (Invalid Source Type)" </tt></dd> <dt>\anchor _↩
cfeevents000233 Global \_internalref cfe__tbl__events_8h#a0291d922d547d4c3e7f579c361e6f4
"CFE_TBL_LOADING_A_DUMP_ONLY_ERR_EID" </dt><dd> <tt> 'Attempted to load
DUMP-ONLY table '\%s' from '\%s'' </tt></dd> <dt>\anchor _cfeevents000242
Global \_internalref cfe__tbl__events_8h#a251e3f2575102c2782539826e5acdd9e
"CFE_TBL_LOADING_PENDING_ERR_EID" </dt><dd> <tt> 'Attempted to load table
'\%s' while previous load is still pending' </tt></dd> <dt>\anchor _↩
cfeevents000204 Global \_internalref cfe__tbl__events_8h#a92b85df6bed26d6e37695db70b792c
"CFE_TBL_MID_ERR_EID" </dt><dd> <tt> 'Invalid message ID - ID = 0x\%X'
</tt></dd> <dt>\anchor _cfeevents000220 Global \_internalref cfe__tbl↩
__events_8h#a8d2fa0ab9e0f8cbb42f91ad9ae51ec1f "CFE_TBL_NO_INACTIVE_BUFF↩
ER_ERR_EID" </dt><dd> <tt> 'No Inactive Buffer for Table '\%s' present'
</tt></dd> <dt>\anchor _cfeevents000211 Global \_internalref cfe__tbl_↩
_events_8h#a9bd232cdedd88a404ad836c6d8bbbf28 "CFE_TBL_NO_SUCH_TABLE_ERR_E↩
ID" </dt><dd> <tt> 'Unable to locate '\%s' in Table Registry' </tt></dd>
<dt>\anchor _cfeevents000214 Global \_internalref cfe__tbl__events_8h#aaec327141d014594c
"CFE_TBL_NO_WORK_BUFFERS_ERR_EID" </dt><dd> <tt> 'No working buffers available
for table '\%s'' </tt></dd> <dt>\anchor _cfeevents000192 Global \_internalref
cfe__tbl__events_8h#acc829f747453b863fb05914bbae5bb7e "CFE_TBL_NOOP_INF_EID"
</dt><dd> <tt> 'No-op command' </tt></dd> <dt>\anchor _cfeevents000237 ↩
Global \_internalref cfe__tbl__events_8h#ac6bbd22efd1b2df7128e461e6f33533d
```

```
"CFE_TBL_NOT_CRITICAL_TBL_ERR_EID" </dt><dd> <tt> 'Table '\%s' is in Critical
Table Registry but CDS is not tagged as a table' </tt></dd> <dt>\anchor ↩
_cfeevents000238 Global \_internalref cfe__tbl__events_8h#a8160667895f1793ac33f0aedba370
"CFE_TBL_NOT_IN_CRIT_REG_ERR_EID" </dt><dd> <tt> 'Table '\%s' is not found
in Critical Table Registry' </tt></dd> <dt>\anchor _cfeevents000195 ↩
Global \_internalref cfe__tbl__events_8h#a8a9bff78075d63a98402accbd334885d
"CFE_TBL_OVERWRITE_DUMP_INF_EID" </dt><dd> <tt> 'Successfully overwrote
'\%s' with Table '\%s'' </tt></dd> <dt>\anchor _cfeevents000197 Global \↩
_internalref cfe__tbl__events_8h#aa7bdcbfb91991a8ba1211a4864a3a610 "CFE↩
_TBL_OVERWRITE_REG_DUMP_INF_EID" </dt><dd> <tt> 'Successfully overwrote
'\%s' with Table Registry' </tt></dd> <dt>\anchor _cfeevents000228 Global
\_internalref cfe__tbl__events_8h#a9f3d77cb32582f4ff23e3c92acba4ab7 "CF↩
E_TBL_PARTIAL_LOAD_ERR_EID" </dt><dd> <tt> ''\%s' has partial load for
uninitialized table '\%s'' </tt></dd> <dt>\anchor _cfeevents000255 Global
\_internalref cfe__tbl__events_8h#a557fe388d1cda9389a954e258f112e7d "CFE↩
_TBL_PROCESSOR_ID_ERR_EID" </dt><dd> <tt> 'Unable to verify Processor I↩
D for '\%s', ID = 0x\%08X' </tt></dd> <dt>\anchor _cfeevents000248 Global
\_internalref cfe__tbl__events_8h#a24a3ea7397da377a5ec207f2936e2cb9 "C↩
FE_TBL_REGISTER_ERR_EID" </dt><dd> <tt> '\%s Failed to Register '\%s',
Status=0x\%08X' </tt></dd> <dt>\anchor _cfeevents000193 Global \_internalref
cfe__tbl__events_8h#a285c927ff2a1185ff955fb250382fbe5 "CFE_TBL_RESET_INF_↩
EID" </dt><dd> <tt> 'Reset Counters command' </tt></dd> <dt>\anchor _↩
cfeevents000249 Global \_internalref cfe__tbl__events_8h#a63522653228fb514b11ecfefba22c6
"CFE_TBL_SHARE_ERR_EID" </dt><dd> <tt> '\%s Failed to Share '\%s', Status=0x\%08↩
X' </tt></dd> <dt>\anchor _cfeevents000254 Global \_internalref cfe__↩
tbl__events_8h#a23f09c79f441fcc9876915992bf31d2f "CFE_TBL_SPACECRAFT_ID↩
_ERR_EID" </dt><dd> <tt> 'Unable to verify Spacecraft ID for '\%s', I↩
D = 0x\%08X' </tt></dd> <dt>\anchor _cfeevents000200 Global \_internalref
cfe__tbl__events_8h#ae7b791d49aaf3209aa2b522150956a45 "CFE_TBL_TLM_REG_CMD↩
_INF_EID" </dt><dd> <tt> 'Table Registry entry for '\%s' will be telemetered'
</tt></dd> <dt>\anchor _cfeevents000230 Global \_internalref cfe__tbl__↩
events_8h#a5bde087adb2b94b44c2cc949ea9eb2fe "CFE_TBL_TOO_MANY_DUMPS_ERR_↩
EID" </dt><dd> <tt> 'Too many Dump Only Table Dumps have been requested'
</tt></dd> <dt>\anchor _cfeevents000221 Global \_internalref cfe__tbl__↩
events_8h#a127191cb9457a8002052e92ebf8fe79d "CFE_TBL_TOO_MANY_VALIDATIONS↩
_ERR_EID" </dt><dd> <tt> 'Too many Table Validations have been requested'
</tt></dd> <dt>\anchor _cfeevents000250 Global \_internalref cfe__tbl_↩
_events_8h#a2a595920826545f8f3daf127e6e2b5b9 "CFE_TBL_UNREGISTER_ERR_EID"
</dt><dd> <tt> '\%s Failed to Unregister '\%s', Status=0x\%08X' </tt></dd>
<dt>\anchor _cfeevents000235 Global \_internalref cfe__tbl__events_8h#a28bcda5a746c811fe
"CFE_TBL_UNVALIDATED_ERR_EID" </dt><dd> <tt> 'Cannot activate table '\%s'.
Inactive image not Validated' </tt></dd> <dt>\anchor _cfeevents000252 ↩
Global \_internalref cfe__tbl__events_8h#a5acdcade08086741149d2473c0fdaebb
"CFE_TBL_UPDATE_ERR_EID" </dt><dd> <tt> '\%s Failed to Update '\%s', Status=0x\%08↩
X"
```

Global CFE_TBL_UPDATE_SUCCESS_INF_EID

```
'%s Successfully Updated '%s''
```

Global CFE_TBL_VAL_REQ_MADE_INF_EID

```
'Tbl Services issued validation request for '%s''
```

Global CFE_TBL_VALIDATION_ERR_EID

```
'%s validation failed for Inactive '%s', Status=0x%08X'
```

Global CFE_TBL_VALIDATION_INF_EID

```
'%s validation successful for Inactive '%s''
```

Global CFE_TBL_WRITE_CFE_HDR_ERR_EID

```
'Error writing cFE File Header to '%s', Status=0x%08X'
```

Global CFE_TBL_WRITE_DUMP_INF_EID

```
'Successfully dumped Table '%s' to '%s''
```

Global CFE_TBL_WRITE_REG_DUMP_INF_EID

```
'Successfully dumped Table Registry to '%s':Size=%d,Entries=%d'
```

Global CFE_TBL_WRITE_TBL_HDR_ERR_EID

```
'Error writing Tbl image File Header to '%s', Status=0x%08X'
```

Global CFE_TBL_WRITE_TBL_IMG_ERR_EID

```
'Error writing Tbl image to '%s', Status=0x%08X'
```

Global CFE_TBL_WRITE_TBL_REG_ERR_EID

```
'Error writing Registry to '%s', Status=0x%08X'
```

Global CFE_TBL_ZERO_LENGTH_LOAD_ERR_EID

```
'Table Hdr in '%s' indicates no data in file'
```

Global CFE_TIME_1HZ_CFG_EID

```
'1Hz Adjust commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to
true'
```

Global CFE_TIME_1HZ_EID

```
'STCF 1Hz Adjust - secs = %d, ssecs = 0x%X, dir = %d'
```

Global CFE_TIME_CC_ERR_EID

```
'Invalid command code - ID = 0x%X, CC = %d'
```

Global CFE_TIME_DELAY_CFG_EID

```
'Set Delay commands invalid without CFE_PLATFORM_TIME_CFG_CLIENT set to true'
```

Global CFE_TIME_DELAY_EID

```
'Set Tone Delay - secs = %d, usecs = %d, ssecs = 0x%X, dir = %d'
```

Global CFE_TIME_DELAY_ERR_EID

```
'Invalid Tone Delay - secs = %d, usecs = %d'
```

Global CFE_TIME_DELTA_CFG_EID

```
'STCF Adjust commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to
true'
```

Global CFE_TIME_DELTA_EID

```
'STCF Adjust - secs = %d, usecs = %d, ssecs = 0x%X, dir[1=Positive, 2=Negative]
= %d'
```

Global CFE_TIME_DELTA_ERR_EID

```
'Invalid STCF Adjust - secs = %d, usecs = %d, dir[1=Positive, 2=Negative] =
%d'
```

Global CFE_TIME_DIAG_EID

```
'Request diagnostics command'
```

Global CFE_TIME_FLY_OFF_EID

```
'Stop FLYWHEEL'
```

Global CFE_TIME_FLY_ON_EID

```
'Start FLYWHEEL'
```

Global CFE_TIME_ID_ERR_EID

```
'Invalid message ID - ID = 0x%X'
```

Global CFE_TIME_INIT_EID

```
'cFE TIME Initialized'
```

Global CFE_TIME_LEAPS_CFG_EID

```
'Set Leaps commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'
```

Global CFE_TIME_LEAPS_EID

```
'Set Leap Seconds = %d'
```

Global CFE_TIME_LEN_ERR_EID

```
'Invalid cmd length:  ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'
```

Global CFE_TIME_MET_CFG_EID

```
'Set MET commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'
```

Global CFE_TIME_MET_EID

```
'Set MET - secs = %d, usecs = %d, ssecs = 0x%X'
```

Global CFE_TIME_MET_ERR_EID

```
'Invalid MET - secs = %d, usecs = %d'
```

Global CFE_TIME_NOOP_EID

```
'No-op command'
```

Global CFE_TIME_RESET_EID

```
'Reset Counters command'
```

Global CFE_TIME_SIGNAL_CFG_EID

```
'Set Signal commands invalid without CFE_PLATFORM_TIME_CFG_SIGNAL set to
true'
```

Global CFE_TIME_SIGNAL_EID

```
'Set Tone Source = %s'
```

Global CFE_TIME_SIGNAL_ERR_EID

```
'Invalid Tone Source = 0x%X'
```

Global CFE_TIME_SOURCE_CFG_EID

```
'Set Source commands invalid without CFE_PLATFORM_TIME_CFG_SOURCE set to
true'
```

Global CFE_TIME_SOURCE_EID

```
'Set Time Source = %s'
```

Global CFE_TIME_SOURCE_ERR_EID

```
'Invalid Time Source = 0x%X'
```

Global CFE_TIME_STATE_EID

```
'Set Clock State = %s'
```

Global CFE_TIME_STATE_ERR_EID

```
'Invalid Clock State = 0x%X'
```

Global CFE_TIME_STCF_CFG_EID

```
'Set STCF commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'
```

Global CFE_TIME_STCF_EID

```
'Set STCF - secs = %d, usecs = %d, ssecs = 0x%X'
```

Global CFE_TIME_STCF_ERR_EID

```
'Invalid STCF - secs = %d, usecs = %d'
```

Global CFE_TIME_TIME_CFG_EID

```
'Set Time commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'
```

Global CFE_TIME_TIME_EID

```
'Set Time - secs = %d, usecs = %d, ssecs = 0x%X'
```

Global CFE_TIME_TIME_ERR_EID

```
'Invalid Time - secs = %d, usecs = %d'
```

30 cFE Command Mnemonic Cross Reference

The following cross reference maps the cFE command codes to Command Mnemonics. To learn about the details of a particular command, click on its associated command code.

Global CFE_ES_CLEAR_ER_LOG_CC

$sc_$cpu_ES_ClearERLog

Global CFE_ES_CLEAR_SYSLOG_CC

$sc_$cpu_ES_ClearSysLog

Global CFE_ES_DELETE_CDS_CC

$sc_$cpu_ES_DeleteCDS

Global CFE_ES_DUMP_CDS_REGISTRY_CC

$sc_$cpu_ES_WriteCDS2File

Global CFE_ES_NOOP_CC

$sc_$cpu_ES_NOOP

Global CFE_ES_OVER_WRITE_SYSLOG_CC

$sc_$cpu_ES_OverwriteSysLogMode

Global CFE_ES_QUERY_ALL_CC

$sc_$cpu_ES_WriteAppInfo2File

Global CFE_ES_QUERY_ALL_TASKS_CC

$sc_$cpu_ES_WriteTaskInfo2File

Global CFE_ES_QUERY_ONE_CC

$sc_$cpu_ES_QueryApp

Global CFE_ES_RELOAD_APP_CC

$sc_$cpu_ES_ReloadApp

Global CFE_ES_RESET_COUNTERS_CC

$sc_$cpu_ES_ResetCtrs

Global CFE_ES_RESET_PR_COUNT_CC

$sc_$cpu_ES_ResetPRCnt

Global CFE_ES_RESTART_APP_CC

$sc_$cpu_ES_ResetApp

Global CFE_ES_RESTART_CC

$sc_$cpu_ES_ProcessorReset, $sc_$cpu_ES_PowerOnReset

Global CFE_ES_SEND_MEM_POOL_STATS_CC

$sc_$cpu_ES_PoolStats

Global CFE_ES_SET_MAX_PR_COUNT_CC

$sc_$cpu_ES_SetMaxPRCnt

Global CFE_ES_SET_PERF_FILTER_MASK_CC

$sc_$cpu_ES_LAFilterMask

Global CFE_ES_SET_PERF_TRIGGER_MASK_CC

$sc_$cpu_ES_LATriggerMask

Global CFE_ES_START_APP_CC

$sc_$cpu_ES_StartApp

Global CFE_ES_START_PERF_DATA_CC

$sc_$cpu_ES_StartLAData

Global CFE_ES_STOP_APP_CC

$sc_$cpu_ES_StopApp

Global CFE_ES_STOP_PERF_DATA_CC

$sc_$cpu_ES_StopLAData

Global CFE_ES_WRITE_ER_LOG_CC

$sc_$cpu_ES_WriteERLog2File

Global CFE_ES_WRITE_SYSLOG_CC

$sc_$cpu_ES_WriteSysLog2File

Global CFE_EVS_ADD_EVENT_FILTER_CC

$sc_$cpu_EVS_AddEvtFltr

Global CFE_EVS_CLEAR_LOG_CC

$sc_$cpu_EVS_ClrLog

Global CFE_EVS_DELETE_EVENT_FILTER_CC

$sc_$cpu_EVS_DelEvtFltr

Global CFE_EVS_DISABLE_APP_EVENT_TYPE_CC

$sc_$cpu_EVS_DisAppEvtType, $sc_$cpu_EVS_DisAppEvtTypeMask

Global CFE_EVS_DISABLE_APP_EVENTS_CC

$sc_$cpu_EVS_DisAppEvGen

Global CFE_EVS_DISABLE_EVENT_TYPE_CC

$sc_$cpu_EVS_DisEventType, $sc_$cpu_EVS_DisEventTypeMask

Global CFE_EVS_DISABLE_PORTS_CC

$sc_$cpu_EVS_DisPort, $sc_$cpu_EVS_DisPortMask

Global CFE_EVS_ENABLE_APP_EVENT_TYPE_CC

$sc_$cpu_EVS_EnaAppEvtType, $sc_$cpu_EVS_EnaAppEvtTypeMask

Global CFE_EVS_ENABLE_APP_EVENTS_CC

$sc_$cpu_EVS_EnaAppEvGen

Global CFE_EVS_ENABLE_EVENT_TYPE_CC

$sc_$cpu_EVS_EnaEventType, $sc_$cpu_EVS_EnaEventTypeMask

Global CFE_EVS_ENABLE_PORTS_CC

$sc_$cpu_EVS_EnaPort, $sc_$cpu_EVS_EnaPortMask

Global CFE_EVS_NOOP_CC

$sc_$cpu_EVS_NOOP

Global CFE_EVS_RESET_ALL_FILTERS_CC

$sc_$cpu_EVS_RstAllFltrs

Global CFE_EVS_RESET_APP_COUNTER_CC

$sc_$cpu_EVS_RstAppCtrs

Global CFE_EVS_RESET_COUNTERS_CC

$sc_$cpu_EVS_ResetCtrs

Global CFE_EVS_RESET_FILTER_CC

$sc_$cpu_EVS_RstBinFltrCtr

Global CFE_EVS_SET_EVENT_FORMAT_MODE_CC

$sc_$cpu_EVS_SetEvtFmt

Global CFE_EVS_SET_FILTER_CC

$sc_$cpu_EVS_SetBinFltrMask

Global CFE_EVS_SET_LOG_MODE_CC

$sc_$cpu_EVS_SetLogMode

Global CFE_EVS_WRITE_APP_DATA_FILE_CC

$sc_$cpu_EVS_WriteAppData2File

Global CFE_EVS_WRITE_LOG_DATA_FILE_CC

$sc_$cpu_EVS_WriteLog2File

Global CFE_SB_DISABLE_ROUTE_CC

$sc_$cpu_SB_DisRoute

Global CFE_SB_DISABLE_SUB_REPORTING_CC

$sc_$cpu_SB_DisSubRptg

Global CFE_SB_ENABLE_ROUTE_CC

$sc_$cpu_SB_EnaRoute

Global CFE_SB_ENABLE_SUB_REPORTING_CC

$sc_$cpu_SB_EnaSubRptg

Global CFE_SB_NOOP_CC

$sc_$cpu_SB_NOOP

Global CFE_SB_RESET_COUNTERS_CC

$sc_$cpu_SB_ResetCtrs

Global CFE_SB_SEND_PREV_SUBS_CC

$sc_$cpu_SB_SendPrevSubs

Global CFE_SB_SEND_SB_STATS_CC

$sc_$cpu_SB_DumpStats

Global CFE_SB_WRITE_MAP_INFO_CC

$sc_$cpu_SB_WriteMap2File

Global CFE_SB_WRITE_PIPE_INFO_CC

$sc_$cpu_SB_WritePipe2File

Global CFE_SB_WRITE_ROUTING_INFO_CC

$sc_$cpu_SB_WriteRouting2File

Global CFE_TBL_ABORT_LOAD_CC

$sc_$cpu_TBL_LOADABORT

Global CFE_TBL_ACTIVATE_CC

$sc_$cpu_TBL_ACTIVATE

Global CFE_TBL_DELETE_CDS_CC

$sc_$cpu_TBL_DeleteCDS

Global CFE_TBL_DUMP_CC

$sc_$cpu_TBL_DUMP

Global CFE_TBL_DUMP_REGISTRY_CC

$sc_$cpu_TBL_WriteReg2File

Global CFE_TBL_LOAD_CC

$sc_$cpu_TBL_Load

Global CFE_TBL_NOOP_CC

$sc_$cpu_TBL_NOOP

Global CFE_TBL_RESET_COUNTERS_CC

$sc_$cpu_TBL_ResetCtrs

Global CFE_TBL_SEND_REGISTRY_CC

$sc_$cpu_TBL_TLMReg

Global CFE_TBL_VALIDATE_CC

$sc_$cpu_TBL_VALIDATE

Global CFE_TIME_ADD_1HZ_ADJUSTMENT_CC

$sc_$cpu_TIME_Add1HzSTCF

Global CFE_TIME_ADD_ADJUST_CC

$sc_$cpu_TIME_AddSTCFAdj

Global CFE_TIME_ADD_DELAY_CC

$sc_$cpu_TIME_AddClockLat

Global CFE_TIME_NOOP_CC

$sc_$cpu_TIME_NOOP

Global CFE_TIME_RESET_COUNTERS_CC

$sc_$cpu_TIME_ResetCtrs

Global CFE_TIME_SEND_DIAGNOSTIC_TLM_CC

$sc_$cpu_TIME_RequestDiag

Global CFE_TIME_SET_LEAP_SECONDS_CC

$sc_$cpu_TIME_SetClockLeap

Global CFE_TIME_SET_MET_CC

$sc_$cpu_TIME_SetClockMET

Global CFE_TIME_SET_SIGNAL_CC

$sc_$cpu_TIME_SetSignal

Global CFE_TIME_SET_SOURCE_CC

$sc_$cpu_TIME_SetSource

Global CFE_TIME_SET_STATE_CC

$sc_$cpu_TIME_SetState

Global CFE_TIME_SET_STCF_CC

$sc_$cpu_TIME_SetClockSTCF

Global CFE_TIME_SET_TIME_CC

$sc_$cpu_TIME_SetClock

Global CFE_TIME_SUB_1HZ_ADJUSTMENT_CC

$sc_$cpu_TIME_Sub1HzSTCF

Global CFE_TIME_SUB_ADJUST_CC

$sc_$cpu_TIME_SubSTCFAdj

Global CFE_TIME_SUB_DELAY_CC

$sc_$cpu_TIME_SubClockLat

31 cFE Telemetry Mnemonic Cross Reference

The following cross reference maps the cFE telemetry packet members to their associated ground system telemetry mnemonics.

Global CFE_ES_AppInfo::AddressesAreValid

$sc_$cpu_ES_AddrsValid

Global CFE_ES_AppInfo::BSSAddress

$sc_$cpu_ES_BSSAddress

Global CFE_ES_AppInfo::BSSSize

$sc_$cpu_ES_BSSSize

Global CFE_ES_AppInfo::CodeAddress

$sc_$cpu_ES_CodeAddress

Global CFE_ES_AppInfo::CodeSize

$sc_$cpu_ES_CodeSize

Global CFE_ES_AppInfo::DataAddress

$sc_$cpu_ES_DataAddress

Global CFE_ES_AppInfo::DataSize

$sc_$cpu_ES_DataSize

Global CFE_ES_AppInfo::EntryPoint [CFE_MISSION_MAX_API_LEN]

$sc_$cpu_ES_AppEntryPt[OS_MAX_API_NAME]

Global CFE_ES_AppInfo::ExceptionAction

$sc_$cpu_ES_ExceptnActn

Global CFE_ES_AppInfo::ExecutionCounter

$sc_$cpu_ES_ExecutionCtr

Global CFE_ES_AppInfo::FileName [CFE_MISSION_MAX_PATH_LEN]

$sc_$cpu_ES_AppFilename[OS_MAX_PATH_LEN]

Global CFE_ES_AppInfo::MainTaskId

$sc_$cpu_ES_MainTaskId

Global CFE_ES_AppInfo::MainTaskName [CFE_MISSION_MAX_API_LEN]

$sc_$cpu_ES_MainTaskName[OS_MAX_API_NAME]

Global CFE_ES_AppInfo::Name [CFE_MISSION_MAX_API_LEN]

$sc_$cpu_ES_AppName[OS_MAX_API_NAME]

Global CFE_ES_AppInfo::NumOfChildTasks

$sc_$cpu_ES_ChildTasks

Global CFE_ES_AppInfo::Priority

$sc_$cpu_ES_Priority

Global CFE_ES_AppInfo::ResourceId

$sc_$cpu_ES_AppID

Global CFE_ES_AppInfo::StackSize

$sc_$cpu_ES_StackSize

Global CFE_ES_AppInfo::StartAddress

$sc_$cpu_ES_StartAddr

Global CFE_ES_AppInfo::Type

$sc_$cpu_ES_AppType

Global CFE_ES_HousekeepingTlm_Payload::BootSource

$sc_$cpu_ES_BootSource

Global CFE_ES_HousekeepingTlm_Payload::CFECoreChecksum

$sc_$cpu_ES_CKSUM

Global CFE_ES_HousekeepingTlm_Payload::CFEMajorVersion

$sc_$cpu_ES_CFEMAJORVER

Global CFE_ES_HousekeepingTlm_Payload::CFEMinorVersion

$sc_$cpu_ES_CFEMINORVER

Global CFE_ES_HousekeepingTlm_Payload::CFEMissionRevision

$sc_$cpu_ES_CFEMISSIONREV

Global CFE_ES_HousekeepingTlm_Payload::CFERevision

$sc_$cpu_ES_CFEREVISION

Global CFE_ES_HousekeepingTlm_Payload::CommandCounter

$sc_$cpu_ES_CMDPC

Global CFE_ES_HousekeepingTlm_Payload::CommandErrorCounter

$sc_$cpu_ES_CMDEC

Global CFE_ES_HousekeepingTlm_Payload::ERLogEntries

$sc_$cpu_ES_ERLOGENTRIES

Global CFE_ES_HousekeepingTlm_Payload::ERLogIndex

$sc_$cpu_ES_ERLOGINDEX

Global CFE_ES_HousekeepingTlm_Payload::HeapBlocksFree

$sc_$cpu_ES_HeapBlocksFree

Global CFE_ES_HousekeepingTlm_Payload::HeapBytesFree

$sc_$cpu_ES_HeapBytesFree

Global CFE_ES_HousekeepingTlm_Payload::HeapMaxBlockSize

$sc_$cpu_ES_HeapMaxBlkSize

Global CFE_ES_HousekeepingTlm_Payload::MaxProcessorResets

$sc_$cpu_ES_MaxProcResets

Global CFE_ES_HousekeepingTlm_Payload::OSALMajorVersion

$sc_$cpu_ES_OSMAJORVER

Global CFE_ES_HousekeepingTlm_Payload::OSALMinorVersion

$sc_$cpu_ES_OSMINORVER

Global CFE_ES_HousekeepingTlm_Payload::OSALMissionRevision

$sc_$cpu_ES_OSMISSIONREV

Global CFE_ES_HousekeepingTlm_Payload::OSALRevision

$sc_$cpu_ES_OSREVISION

Global CFE_ES_HousekeepingTlm_Payload::PerfDataCount

$sc_$cpu_ES_PerfDataCnt

Global CFE_ES_HousekeepingTlm_Payload::PerfDataEnd

$sc_$cpu_ES_PerfDataEnd

Global CFE_ES_HousekeepingTlm_Payload::PerfDataStart

$sc_$cpu_ES_PerfDataStart

Global CFE_ES_HousekeepingTlm_Payload::PerfDataToWrite

$sc_$cpu_ES_PerfData2Write

Global CFE_ES_HousekeepingTlm_Payload::PerfFilterMask [CFE_MISSION_ES_PERF_MAX_IDS/32]

$sc_$cpu_ES_PerfFltrMask[MaskCnt]

Global CFE_ES_HousekeepingTlm_Payload::PerfMode

$sc_$cpu_ES_PerfMode

Global CFE_ES_HousekeepingTlm_Payload::PerfState

$sc_$cpu_ES_PerfState

Global CFE_ES_HousekeepingTlm_Payload::PerfTriggerCount

$sc_$cpu_ES_PerfTrigCnt

Global CFE_ES_HousekeepingTlm_Payload::PerfTriggerMask [CFE_MISSION_ES_PERF_MAX_IDS/32]

$sc_$cpu_ES_PerfTrigMask[MaskCnt]

Global CFE_ES_HousekeepingTlm_Payload::ProcessorResets

$sc_$cpu_ES_ProcResetCnt

Global CFE_ES_HousekeepingTlm_Payload::RegisteredCoreApps

$sc_$cpu_ES_RegCoreApps

Global CFE_ES_HousekeepingTlm_Payload::RegisteredExternalApps

$sc_$cpu_ES_RegExtApps

Global CFE_ES_HousekeepingTlm_Payload::RegisteredLibs

$sc_$cpu_ES_RegLibs

Global CFE_ES_HousekeepingTlm_Payload::RegisteredTasks

$sc_$cpu_ES_RegTasks

Global CFE_ES_HousekeepingTlm_Payload::ResetSubtype

$sc_$cpu_ES_ResetSubtype

Global CFE_ES_HousekeepingTlm_Payload::ResetType

$sc_$cpu_ES_ResetType

Global CFE_ES_HousekeepingTlm_Payload::SysLogBytesUsed

$sc_$cpu_ES_SYSLOGBYTEUSED

Global CFE_ES_HousekeepingTlm_Payload::SysLogEntries

$sc_$cpu_ES_SYSLOGENTRIES

Global CFE_ES_HousekeepingTlm_Payload::SysLogMode

$sc_$cpu_ES_SYSLOGMODE

Global CFE_ES_HousekeepingTlm_Payload::SysLogSize

$sc_$cpu_ES_SYSLOGSIZE

Global CFE_ES_MemPoolStats::BlockStats [CFE_MISSION_ES_POOL_MAX_BUCKETS]

$sc_$cpu_ES_BlkStats[BLK_SIZES]

Global CFE_ES_MemPoolStats::CheckErrCtr

$sc_$cpu_ES_BlkErrCTR

Global CFE_ES_MemPoolStats::NumBlocksRequested

$sc_$cpu_ES_BlksREQ

Global CFE_ES_MemPoolStats::NumFreeBytes

$sc_$cpu_ES_FreeBytes

Global CFE_ES_MemPoolStats::PoolSize

$sc_$cpu_ES_PoolSize

Global CFE_ES_PoolStatsTlm_Payload::PoolHandle

$sc_$cpu_ES_PoolHandle

Global CFE_EVS_AppTlmData::AppEnableStatus

$sc_$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].APPENASTAT

Global CFE_EVS_AppTlmData::AppID

$sc_$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].APPID

Global CFE_EVS_AppTlmData::AppMessageSentCounter

$sc_$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].APPMSGSENTC

Global CFE_EVS_AppTlmData::Padding

$sc_$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].SPARE2ALIGN3

Global CFE_EVS_HousekeepingTlm_Payload::AppData [CFE_MISSION_ES_MAX_APPLICATIONS]

$sc_$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS]

Global CFE_EVS_HousekeepingTlm_Payload::CommandCounter

$sc_$cpu_EVS_CMDPC

Global CFE_EVS_HousekeepingTlm_Payload::CommandErrorCounter

$sc_$cpu_EVS_CMDEC

Global CFE_EVS_HousekeepingTlm_Payload::LogEnabled

$sc_$cpu_EVS_LOGENABLED

Global CFE_EVS_HousekeepingTlm_Payload::LogFullFlag

$sc_$cpu_EVS_LOGFULL

Global CFE_EVS_HousekeepingTlm_Payload::LogMode

$sc_$cpu_EVS_LOGMODE

Global CFE_EVS_HousekeepingTlm_Payload::LogOverflowCounter

$sc_$cpu_EVS_LOGOVERFLOWC

Global CFE_EVS_HousekeepingTlm_Payload::MessageFormatMode

$sc_$cpu_EVS_MSGFMTMODE

Global CFE_EVS_HousekeepingTlm_Payload::MessageSendCounter

$sc_$cpu_EVS_MSGSENTC

Global CFE_EVS_HousekeepingTlm_Payload::MessageTruncCounter

$sc_$cpu_EVS_MSGTRUNC

Global CFE_EVS_HousekeepingTlm_Payload::OutputPort

$sc_$cpu_EVS_OUTPUTPORT

Global CFE_EVS_HousekeepingTlm_Payload::Spare1

$sc_$cpu_EVS_HK_SPARE1

Global CFE_EVS_HousekeepingTlm_Payload::Spare2

$sc_$cpu_EVS_HK_SPARE2

Global CFE_EVS_HousekeepingTlm_Payload::Spare3

$sc_$cpu_EVS_HK_SPARE3

Global CFE_EVS_HousekeepingTlm_Payload::UnregisteredAppCounter

$sc_$cpu_EVS_UNREGAPPC

Global CFE_EVS_LongEventTlm_Payload::Message [CFE_MISSION_EVS_MAX_MESSAGE_LENGTH]

$sc_$cpu_EVS_EVENT[CFE_MISSION_EVS_MAX_MESSAGE_LENGTH]

Global CFE_EVS_LongEventTlm_Payload::Spare1

$sc_$cpu_EVS_SPARE1

Global CFE_EVS_LongEventTlm_Payload::Spare2

$sc_$cpu_EVS_SPARE2

Global CFE_EVS_PacketID::AppName [CFE_MISSION_MAX_API_LEN]

$sc_$cpu_EVS_APPNAME[OS_MAX_API_NAME]

Global CFE_EVS_PacketID::EventID

$sc_$cpu_EVS_EVENTID

Global CFE_EVS_PacketID::EventType

$sc_$cpu_EVS_EVENTTYPE

Global CFE_EVS_PacketID::ProcessorID

$sc_$cpu_EVS_PROCESSORID

Global CFE_EVS_PacketID::SpacecraftID

$sc_$cpu_EVS_SCID

Global CFE_SB_HousekeepingTlm_Payload::CommandCounter

$sc_$cpu_SB_CMDPC

Global CFE_SB_HousekeepingTlm_Payload::CommandErrorCounter

$sc_$cpu_SB_CMDEC

Global CFE_SB_HousekeepingTlm_Payload::CreatePipeErrorCounter

$sc_$cpu_SB_NewPipeEC

Global CFE_SB_HousekeepingTlm_Payload::DuplicateSubscriptionsCounter

$sc_$cpu_SB_DupSubCnt

Global CFE_SB_HousekeepingTlm_Payload::GetPipeIdByNameErrorCounter

$sc_$cpu_SB_GetPipeIDByNameEC

Global CFE_SB_HousekeepingTlm_Payload::InternalErrorCounter

$sc_$cpu_SB_InternalEC

Global CFE_SB_HousekeepingTlm_Payload::MemInUse

$sc_$cpu_SB_MemInUse

Global CFE_SB_HousekeepingTlm_Payload::MemPoolHandle

$sc_$cpu_SB_MemPoolHdl

Global CFE_SB_HousekeepingTlm_Payload::MsgLimitErrorCounter

$sc_$cpu_SB_MsgLimEC

Global CFE_SB_HousekeepingTlm_Payload::MsgReceiveErrorCounter

$sc_$cpu_SB_MsgRecEC

Global CFE_SB_HousekeepingTlm_Payload::MsgSendErrorCounter

$sc_$cpu_SB_MsgSndEC

Global CFE_SB_HousekeepingTlm_Payload::NoSubscribersCounter

$sc_$cpu_SB_NoSubEC

Global CFE_SB_HousekeepingTlm_Payload::PipeOptsErrorCounter

$sc_$cpu_SB_PipeOptsEC

Global CFE_SB_HousekeepingTlm_Payload::PipeOverflowErrorCounter

$sc_$cpu_SB_PipeOvrEC

Global CFE_SB_HousekeepingTlm_Payload::Spare2Align [1]

$sc_$cpu_SB_Spare2Align[2]

Global CFE_SB_HousekeepingTlm_Payload::SubscribeErrorCounter

$sc_$cpu_SB_SubscrEC

Global CFE_SB_HousekeepingTlm_Payload::UnmarkedMem

$sc_$cpu_SB_UnMarkedMem

Global CFE_SB_PipeDepthStats::CurrentQueueDepth

$sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDINUSE

Global CFE_SB_PipeDepthStats::MaxQueueDepth

$sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDDEPTH

Global CFE_SB_PipeDepthStats::PeakQueueDepth

$sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDPKINUSE

Global CFE_SB_PipeDepthStats::PipeId

$sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDPIPEID

Global CFE_SB_PipeDepthStats::Spare

$sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDSPARE

Global CFE_SB_StatsTlm_Payload::MaxMemAllowed

$sc_$cpu_SB_Stat.SB_SMMBMALW

Global CFE_SB_StatsTlm_Payload::MaxMsgIdsAllowed

$sc_$cpu_SB_Stat.SB_SMMMIDALW

Global CFE_SB_StatsTlm_Payload::MaxPipeDepthAllowed

$sc_$cpu_SB_Stat.SB_SMMPDALW

Global CFE_SB_StatsTlm_Payload::MaxPipesAllowed

$sc_$cpu_SB_Stat.SB_SMMPALW

Global CFE_SB_StatsTlm_Payload::MaxSubscriptionsAllowed

$sc_$cpu_SB_Stat.SB_SMMSALW

Global CFE_SB_StatsTlm_Payload::MemInUse

$sc_$cpu_SB_Stat.SB_SMBMIU

Global CFE_SB_StatsTlm_Payload::MsgIdsInUse

$sc_$cpu_SB_Stat.SB_SMMIDIU

Global CFE_SB_StatsTlm_Payload::PeakMemInUse

$sc_$cpu_SB_Stat.SB_SMPBMIU

Global CFE_SB_StatsTlm_Payload::PeakMsgIdsInUse

$sc_$cpu_SB_Stat.SB_SMPMIDIU

Global CFE_SB_StatsTlm_Payload::PeakPipesInUse

$sc_$cpu_SB_Stat.SB_SMPPIU

Global CFE_SB_StatsTlm_Payload::PeakSBBuffersInUse

$sc_$cpu_SB_Stat.SB_SMPSBBIU

Global CFE_SB_StatsTlm_Payload::PeakSubscriptionsInUse

$sc_$cpu_SB_Stat.SB_SMPSIU

Global CFE_SB_StatsTlm_Payload::PipeDepthStats [CFE_MISSION_SB_MAX_PIPES]

$sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES]

Global CFE_SB_StatsTlm_Payload::PipesInUse

$sc_$cpu_SB_Stat.SB_SMPIU

Global CFE_SB_StatsTlm_Payload::SBBuffersInUse

$sc_$cpu_SB_Stat.SB_SMSBBIU

Global CFE_SB_StatsTlm_Payload::SubscriptionsInUse

$sc_$cpu_SB_Stat.SB_SMSIU

Global CFE_TBL_HousekeepingTlm_Payload::ActiveBuffer

$sc_$cpu_TBL_LastValBuf

Global CFE_TBL_HousekeepingTlm_Payload::ByteAlignPad1

$sc_$cpu_TBL_ByteAlignPad1

Global CFE_TBL_HousekeepingTlm_Payload::CommandCounter

$sc_$cpu_TBL_CMDPC

Global CFE_TBL_HousekeepingTlm_Payload::CommandErrorCounter

$sc_$cpu_TBL_CMDEC

Global CFE_TBL_HousekeepingTlm_Payload::FailedValCounter

$sc_$cpu_TBL_ValFailedCtr

Global CFE_TBL_HousekeepingTlm_Payload::LastFileDumped [CFE_MISSION_MAX_PATH_LEN]

$sc_$cpu_TBL_LastFileDumped[OS_MAX_PATH_LEN]

Global CFE_TBL_HousekeepingTlm_Payload::LastFileLoaded [CFE_MISSION_MAX_PATH_LEN]

$sc_$cpu_TBL_LastFileLoaded[OS_MAX_PATH_LEN]

Global CFE_TBL_HousekeepingTlm_Payload::LastTableLoaded [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

$sc_$cpu_TBL_LastTableLoaded[CFE_TBL_MAX_FULL_NAME_LEN]

Global CFE_TBL_HousekeepingTlm_Payload::LastUpdatedTable [CFE_MISSION_TBL_MAX_FULL_NAME_L↩EN]

$sc_$cpu_TBL_LastUpdTblName[CFE_TB_MAX_FULL_NAME_LEN]

Global CFE_TBL_HousekeepingTlm_Payload::LastUpdateTime

$sc_$cpu_TBL_LastUpdTime, $sc_$cpu_TBL_SECONDS, $sc_$cpu_TBL_SUBSECONDS

Global CFE_TBL_HousekeepingTlm_Payload::LastValCrc

$sc_$cpu_TBL_LastValCRC

Global CFE_TBL_HousekeepingTlm_Payload::LastValStatus

$sc_$cpu_TBl_LastValS

Global CFE_TBL_HousekeepingTlm_Payload::LastValTableName [CFE_MISSION_TBL_MAX_FULL_NAME_L↩EN]

$sc_$cpu_TBL_LastValTblName[CFE_TB_MAX_FULL_NAME_LEN]

Global CFE_TBL_HousekeepingTlm_Payload::MemPoolHandle

$sc_$cpu_TBL_MemPoolHandle

Global CFE_TBL_HousekeepingTlm_Payload::NumFreeSharedBufs

$sc_$cpu_TBL_NumFreeShrBuf

Global CFE_TBL_HousekeepingTlm_Payload::NumLoadPending

$sc_$cpu_TBL_NumUpdatesPend

Global CFE_TBL_HousekeepingTlm_Payload::NumTables

$sc_$cpu_TBL_NumTables

Global CFE_TBL_HousekeepingTlm_Payload::NumValRequests

$sc_$cpu_TBL_ValReqCtr

Global CFE_TBL_HousekeepingTlm_Payload::SuccessValCounter

$sc_$cpu_TBL_ValSuccessCtr

Global CFE_TBL_HousekeepingTlm_Payload::ValidationCounter

$sc_$cpu_TBL_ValCompltdCtr

Global CFE_TBL_TblRegPacket_Payload::ActiveBufferAddr

$sc_$cpu_TBL_ActBufAdd

Global CFE_TBL_TblRegPacket_Payload::ByteAlign4

$sc_$cpu_TBL_Spare4

Global CFE_TBL_TblRegPacket_Payload::Crc

$sc_$cpu_TBL_CRC

Global CFE_TBL_TblRegPacket_Payload::Critical

$sc_$cpu_TBL_Spare3

Global CFE_TBL_TblRegPacket_Payload::DoubleBuffered

$sc_$cpu_TBL_DblBuffered

Global CFE_TBL_TblRegPacket_Payload::DumpOnly

$sc_$cpu_TBL_DumpOnly

Global CFE_TBL_TblRegPacket_Payload::FileCreateTimeSecs

$sc_$cpu_TBL_FILECSECONDS

Global CFE_TBL_TblRegPacket_Payload::FileCreateTimeSubSecs

$sc_$cpu_TBL_FILECSUBSECONDS

Global CFE_TBL_TblRegPacket_Payload::InactiveBufferAddr

$sc_$cpu_TBL_IActBufAdd

Global CFE_TBL_TblRegPacket_Payload::LastFileLoaded [CFE_MISSION_MAX_PATH_LEN]

$sc_$cpu_TBL_LastFileUpd[OS_MAX_PATH_LEN]

Global CFE_TBL_TblRegPacket_Payload::LoadPending

$sc_$cpu_TBL_UpdatePndng

Global CFE_TBL_TblRegPacket_Payload::Name [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

$sc_$cpu_TBL_Name[CFE_TB_MAX_FULL_NAME_LEN]

Global CFE_TBL_TblRegPacket_Payload::OwnerAppName [CFE_MISSION_MAX_API_LEN]

$sc_$cpu_TBL_OwnerApp[OS_MAX_API_NAME]

Global CFE_TBL_TblRegPacket_Payload::Size

$sc_$cpu_TBL_SIZE

Global CFE_TBL_TblRegPacket_Payload::TableLoadedOnce

$sc_$cpu_TBL_LoadedOnce

Global CFE_TBL_TblRegPacket_Payload::TimeOfLastUpdate

$sc_$cpu_TBL_TimeLastUpd, $sc_$cpu_TBL_TLUSECONDS, $sc_$cpu_TBL_TLUSUBSECONDS

Global CFE_TBL_TblRegPacket_Payload::ValidationFuncPtr

$sc_$cpu_TBL_ValFuncPtr

Global CFE_TIME_DiagnosticTlm_Payload::AtToneDelay

$sc_$cpu_TIME_DLatentS, $sc_$cpu_TIME_DLatentSs

Global CFE_TIME_DiagnosticTlm_Payload::AtToneLatch

$sc_$cpu_TIME_DTValidS, $sc_$cpu_TIME_DTValidSs

Global CFE_TIME_DiagnosticTlm_Payload::AtToneLeapSeconds

$sc_$cpu_TIME_DLeapS

Global CFE_TIME_DiagnosticTlm_Payload::AtToneMET

$sc_$cpu_TIME_DTMETS, $sc_$cpu_TIME_DTMETSs

Global CFE_TIME_DiagnosticTlm_Payload::AtToneSTCF

$sc_$cpu_TIME_DSTCFS, $sc_$cpu_TIME_DSTCFSS

Global CFE_TIME_DiagnosticTlm_Payload::ClockFlyState

$sc_$cpu_TIME_DFlywheel

Global CFE_TIME_DiagnosticTlm_Payload::ClockSetState

$sc_$cpu_TIME_DValid

Global CFE_TIME_DiagnosticTlm_Payload::ClockSignal

$sc_$cpu_TIME_DSignal

Global CFE_TIME_DiagnosticTlm_Payload::ClockSource

$sc_$cpu_TIME_DSource

Global CFE_TIME_DiagnosticTlm_Payload::ClockStateAPI

$sc_$cpu_TIME_DAPIState

Global CFE_TIME_DiagnosticTlm_Payload::ClockStateFlags

$sc_$cpu_TIME_DStateFlags, $sc_$cpu_TIME_DFlagSet, $sc_$cpu_TIME_DFlagFly, $sc_$cpu_TIME_DFlagSrc, $sc_$cpu_TIME_DFlagPri, $sc_$cpu_TIME_DFlagSfly, $sc_$cpu_TIME_DFlagCfly, $sc_$cpu_TIME_DFlagAdjd, $sc_$cpu_TIME_DFlag1Hzd, $sc_$cpu_TIME_DFlagClat, $sc_$cpu_TIME_DFlagSorC, $sc_$cpu_TIME_DFlag↩NIU

Global CFE_TIME_DiagnosticTlm_Payload::CurrentLatch

$sc_$cpu_TIME_DLocalS, $sc_$cpu_TIME_DLocalSs

Global CFE_TIME_DiagnosticTlm_Payload::CurrentMET

$sc_$cpu_TIME_DMETS, $sc_$cpu_TIME_DMETSs

Global CFE_TIME_DiagnosticTlm_Payload::CurrentTAI

$sc_$cpu_TIME_DTAIS, $sc_$cpu_TIME_DTAISS

Global CFE_TIME_DiagnosticTlm_Payload::CurrentUTC

$sc_$cpu_TIME_DUTCS, $sc_$cpu_TIME_DUTCSS

Global CFE_TIME_DiagnosticTlm_Payload::DataStoreStatus

$sc_$cpu_TIME_DataStStat

Global CFE_TIME_DiagnosticTlm_Payload::DelayDirection

$sc_$cpu_TIME_DLatentDir

Global CFE_TIME_DiagnosticTlm_Payload::Forced2Fly

$sc_$cpu_TIME_DCMD2Fly

Global CFE_TIME_DiagnosticTlm_Payload::LocalIntCounter

$sc_$cpu_TIME_D1HzISRCNT

Global CFE_TIME_DiagnosticTlm_Payload::LocalTaskCounter

$sc_$cpu_TIME_D1HzTaskCNT

Global CFE_TIME_DiagnosticTlm_Payload::MaxElapsed

$sc_$cpu_TIME_DMaxWindow

Global CFE_TIME_DiagnosticTlm_Payload::MaxLocalClock

$sc_$cpu_TIME_DWrapS, $sc_$cpu_TIME_DWrapSs

Global CFE_TIME_DiagnosticTlm_Payload::MinElapsed

$sc_$cpu_TIME_DMinWindow

Global CFE_TIME_DiagnosticTlm_Payload::OneHzAdjust

$sc_$cpu_TIME_D1HzAdjS, $sc_$cpu_TIME_D1HzAdjSs

Global CFE_TIME_DiagnosticTlm_Payload::OneHzDirection

$sc_$cpu_TIME_D1HzAdjDir

Global CFE_TIME_DiagnosticTlm_Payload::OneTimeAdjust

$sc_$cpu_TIME_DAdjustS, $sc_$cpu_TIME_DAdjustSs

Global CFE_TIME_DiagnosticTlm_Payload::OneTimeDirection

$sc_$cpu_TIME_DAdjustDir

Global CFE_TIME_DiagnosticTlm_Payload::ServerFlyState

$sc_$cpu_TIME_DSrvFly

Global CFE_TIME_DiagnosticTlm_Payload::TimeSinceTone

$sc_$cpu_TIME_DElapsedS, $sc_$cpu_TIME_DElapsedSs

Global CFE_TIME_DiagnosticTlm_Payload::ToneDataCounter

$sc_$cpu_TIME_DTatTCNT

Global CFE_TIME_DiagnosticTlm_Payload::ToneDataLatch

$sc_$cpu_TIME_DTDS, $sc_$cpu_TIME_DTDSs

Global CFE_TIME_DiagnosticTlm_Payload::ToneIntCounter

$sc_$cpu_TIME_DTsISRCNT

Global CFE_TIME_DiagnosticTlm_Payload::ToneIntErrorCounter

$sc_$cpu_TIME_DTsISRERR

Global CFE_TIME_DiagnosticTlm_Payload::ToneMatchCounter

$sc_$cpu_TIME_DVerifyCNT

Global CFE_TIME_DiagnosticTlm_Payload::ToneMatchErrorCounter

$sc_$cpu_TIME_DVerifyER

Global CFE_TIME_DiagnosticTlm_Payload::ToneOverLimit

$sc_$cpu_TIME_DMaxSs

Global CFE_TIME_DiagnosticTlm_Payload::ToneSignalCounter

$sc_$cpu_TIME_DTSDetCNT

Global CFE_TIME_DiagnosticTlm_Payload::ToneSignalLatch

$sc_$cpu_TIME_DTTS, $sc_$cpu_TIME_DTTSs

Global CFE_TIME_DiagnosticTlm_Payload::ToneTaskCounter

$sc_$cpu_TIME_DTsTaskCNT

Global CFE_TIME_DiagnosticTlm_Payload::ToneUnderLimit

$sc_$cpu_TIME_DMinSs

Global CFE_TIME_DiagnosticTlm_Payload::VersionCounter

$sc_$cpu_TIME_DVersionCNT

Global CFE_TIME_DiagnosticTlm_Payload::VirtualMET

$sc_$cpu_TIME_DLogicalMET

Global CFE_TIME_HousekeepingTlm_Payload::ClockStateAPI

$sc_$cpu_TIME_DAPIState

Global CFE_TIME_HousekeepingTlm_Payload::ClockStateFlags

$sc_$cpu_TIME_StateFlg, $sc_$cpu_TIME_FlagSet, $sc_$cpu_TIME_FlagFly, $sc_$cpu_TIME_FlagSrc, $sc_↩ $cpu_TIME_FlagPri, $sc_$cpu_TIME_FlagSfly, $sc_$cpu_TIME_FlagCfly, $sc_$cpu_TIME_FlagAdjd, $sc_$cpu↩ _TIME_Flag1Hzd, $sc_$cpu_TIME_FlagClat, $sc_$cpu_TIME_FlagSorC, $sc_$cpu_TIME_FlagNIU

Global CFE_TIME_HousekeepingTlm_Payload::CommandCounter

$sc_$cpu_TIME_CMDPC

Global CFE_TIME_HousekeepingTlm_Payload::CommandErrorCounter

$sc_$cpu_TIME_CMDEC

Global CFE_TIME_HousekeepingTlm_Payload::LeapSeconds

$sc_$cpu_TIME_LeapSecs

Global CFE_TIME_HousekeepingTlm_Payload::Seconds1HzAdj

$sc_$cpu_TIME_1HzAdjSecs

Global CFE_TIME_HousekeepingTlm_Payload::SecondsDelay

$sc_$cpu_TIME_1HzAdjSecs

Global CFE_TIME_HousekeepingTlm_Payload::SecondsMET
$sc_$cpu_TIME_METSecs

Global CFE_TIME_HousekeepingTlm_Payload::SecondsSTCF
$sc_$cpu_TIME_STCFSecs

Global CFE_TIME_HousekeepingTlm_Payload::Subsecs1HzAdj
$sc_$cpu_TIME_1HzAdjSSecs

Global CFE_TIME_HousekeepingTlm_Payload::SubsecsDelay
$sc_$cpu_TIME_1HzAdjSSecs

Global CFE_TIME_HousekeepingTlm_Payload::SubsecsMET
$sc_$cpu_TIME_METSubsecs

Global CFE_TIME_HousekeepingTlm_Payload::SubsecsSTCF
$sc_$cpu_TIME_STCFSubsecs

32 cFE Mission Configuration Parameters

Global CFE_MISSION_ES_HK_TLM_MSG
cFE Portable Message Numbers for Telemetry

Global CFE_MISSION_EVS_CMD_MSG
cFE Portable Message Numbers for Commands

Global CFE_MISSION_MAX_API_LEN
cFE Maximum length for API names within data exchange structures

Global CFE_MISSION_MAX_FILE_LEN
cFE Maximum length for filenames within data exchange structures

Global CFE_MISSION_MAX_PATH_LEN
cFE Maximum length for pathnames within data exchange structures

Global CFE_MISSION_TIME_DATA_CMD_MSG
cFE Portable Message Numbers for Global Messages

33 Deprecated List

Global CFE_SB_Default_Qos
use CFE_SB_DEFAULT_QOS

Global CFE_SB_GenerateChecksum (CFE_MSG_Message_t *MsgPtr)
Use CFE_MSG_GenerateChecksum

Global CFE_SB_GetChecksum (CFE_MSG_Message_t *MsgPtr)
No use case

Global CFE_SB_GetCmdCode (CFE_MSG_Message_t *MsgPtr)
Use CFE_MSG_GetFcnCode

Global CFE_SB_GetMsgId (const CFE_MSG_Message_t *MsgPtr)
Use CFE_MSG_GetMsgId

Global CFE_SB_GetMsgTime (CFE_MSG_Message_t *MsgPtr)
Use CFE_MSG_GetMsgTime

Global CFE_SB_GetTotalMsgLength (const CFE_MSG_Message_t *MsgPtr)
Use CFE_MSG_GetSize

Global CFE_SB_InitMsg (void *MsgPtr, CFE_SB_MsgId_t MsgId, size_t Length, bool Clear)
Use CFE_MSG_Init

Global CFE_SB_PassMsg (CFE_MSG_Message_t *MsgPtr)
Use CFE_SB_TransmitMsg

Global CFE_SB_RcvMsg (CFE_SB_Buffer_t **BufPtr, CFE_SB_PipeId_t PipeId, int32 TimeOut)
use CFE_SB_ReceiveBuffer

Global CFE_SB_SendMsg (CFE_MSG_Message_t *MsgPtr)
Use CFE_SB_TransmitMsg

Global CFE_SB_SetCmdCode (CFE_MSG_Message_t *MsgPtr, uint16 CmdCode)
Use CFE_MSG_SetFcnCode

Global CFE_SB_SetMsgId (CFE_MSG_Message_t *MsgPtr, CFE_SB_MsgId_t MsgId)
Use CFE_MSG_SetMsgId

Global CFE_SB_SetMsgTime (CFE_MSG_Message_t *MsgPtr, CFE_TIME_SysTime_t Time)
Use CFE_MSG_SetMsgTime

Global CFE_SB_SetTotalMsgLength (CFE_MSG_Message_t *MsgPtr, size_t TotalLength)
Use CFE_MSG_SetSize

Global CFE_SB_ValidateChecksum (CFE_MSG_Message_t *MsgPtr)
Use CFE_MSG_ValidateChecksum

Global CFE_SB_ZeroCopyPass (CFE_SB_Buffer_t *BufPtr, CFE_SB_ZeroCopyHandle_t BufferHandle)
use CFE_SB_TransmitBuffer

Global CFE_SB_ZeroCopySend (CFE_SB_Buffer_t *BufPtr, CFE_SB_ZeroCopyHandle_t BufferHandle)
use CFE_SB_TransmitBuffer

Global OS_TaskRegister (void)
Explicit registration call no longer needed

34 Module Index

34.1 Modules

Here is a list of all modules:

cFE Return Code Defines 134

cFE Resource ID APIs 174

cFE Entry/Exit APIs 178

cFE Application Control APIs 180

cFE Application Behavior APIs 183

cFE Information APIs 188

cFE Child Task APIs 198

cFE Miscellaneous APIs 204

cFE Critical Data Store APIs 208

cFE Memory Manager APIs 213

cFE Performance Monitor APIs 221

cFE Generic Counter APIs 224

cFE Internal Executive Service APIs, internal to CFE core 231

cFE Registration APIs 234

cFE Send Event APIs 237

cFE Reset Event Filter APIs 242

cFE Internal Event Services APIs, internal to CFE core 244

cFE File Header Management APIs 246

cFE File Utility APIs 250

cFE Internal File Service APIs, internal to CFE core 256

cFE Message header APIs 257

cFE Message Id APIs 279

cFE Pipe Management APIs 282

cFE Message Subscription Control APIs 289

cFE Send/Receive Message APIs 294

cFE Zero Copy APIs 299

cFE Setting Message Characteristics APIs 304

cFE Getting Message Characteristics APIs 310

cFE Checksum Control APIs 316

cFE Message ID APIs 319

cFE SB Packet Type Defines 323

cFE Internal Software Bus APIs, internal to CFE core 324

cFE Registration APIs 326

cFE Manage Table Content APIs 332

cFE Access Table Content APIs 339

cFE Get Table Information APIs 345

cFE Table Type Defines 349

cFE Internal Table Service APIs, internal to CFE core 352

cFE Get Current Time APIs 354

cFE Get Time Information APIs 358

cFE Time Arithmetic APIs 361

cFE Time Conversion APIs 364

cFE External Time Source APIs 367

cFE Miscellaneous Time APIs 372

cFE Internal Time APIs, internal to CFE core 374

cFE Clock State Flag Defines 376

OSAL Semaphore State Defines 379

OSAL Binary Semaphore APIs 380

OSAL BSP low level access APIs 386

OSAL Real Time Clock APIs 387

OSAL Core Operation APIs 400

OSAL Counting Semaphore APIs 405

OSAL Directory APIs 410

OSAL Return Code Defines 414

OSAL Error Info APIs 424

OSAL File Access Option Defines 425

OSAL Reference Point For Seek Offset Defines 426

OSAL Standard File APIs 427

OSAL File System Level APIs 439

OSAL Heap APIs 447

OSAL Object Type Defines 448

OSAL Object ID Utility APIs 452

OSAL Dynamic Loader and Symbol APIs 459

OSAL Mutex APIs 464

Network ID APIs 469

OSAL Printf APIs 471

OSAL Message Queue APIs 473

OSAL Select APIs 478

OSAL Shell APIs 481

OSAL Socket Address APIs 482

OSAL Socket Management APIs 486

OSAL Task APIs 493

OSAL Time Base APIs 501

OSAL Timer APIs 506

35 Data Structure Index

35.1 Data Structures

Here are the data structures with brief descriptions:

CCSDS_ExtendedHeader
CCSDS packet extended header 512

CCSDS_PrimaryHeader
CCSDS packet primary header 513

CFE_ES_AppInfo
Application Information 514

CFE_ES_AppNameCmd
Generic application name command 520

CFE_ES_AppNameCmd_Payload
Generic application name command payload 521

CFE_ES_AppReloadCmd_Payload
Reload Application Command Payload 522

CFE_ES_BlockStats
Block statistics 523

CFE_ES_CDSRegDumpRec
CDS Register Dump Record 524

CFE_ES_DeleteCDSCmd
Delete Critical Data Store Command 526

CFE_ES_DeleteCDSCmd_Payload
Delete Critical Data Store Command Payload 527

CFE_ES_DumpCDSRegistryCmd
Dump CDS Registry Command 528

CFE_ES_DumpCDSRegistryCmd_Payload
Dump CDS Registry Command Payload 529

CFE_ES_FileNameCmd
Generic file name command 530

CFE_ES_FileNameCmd_Payload
Generic file name command payload 531

CFE_ES_HousekeepingTlm 532

CFE_ES_HousekeepingTlm_Payload 532

CFE_ES_MemPoolStats
Memory Pool Statistics 544

CFE_ES_MemStatsTlm 546

CFE_ES_NoArgsCmd
Generic "no arguments" command 547

CFE_ES_OneAppTlm 548

CFE_ES_OneAppTlm_Payload 549

CFE_ES_OverWriteSysLogCmd
Overwrite/Discard System Log Configuration Command Payload 550

CFE_ES_OverWriteSysLogCmd_Payload
Overwrite/Discard System Log Configuration Command Payload 551

CFE_ES_PoolAlign
Pool Alignement 552

CFE_ES_PoolStatsTlm_Payload 553

CFE_ES_ReloadAppCmd
Reload Application Command 554

CFE_ES_RestartCmd
Restart cFE Command 555

CFE_ES_RestartCmd_Payload
Restart cFE Command Payload 556

CFE_ES_SendMemPoolStatsCmd
Send Memory Pool Statistics Command 557

CFE_ES_SendMemPoolStatsCmd_Payload
Send Memory Pool Statistics Command Payload 558

CFE_ES_SetMaxPRCountCmd
Set Maximum Processor Reset Count Command 559

CFE_ES_SetMaxPRCountCmd_Payload
Set Maximum Processor Reset Count Command Payload 560

CFE_ES_SetPerfFilterMaskCmd
Set Performance Analyzer Filter Mask Command 560

CFE_ES_SetPerfFilterMaskCmd_Payload
Set Performance Analyzer Filter Mask Command Payload 561

CFE_ES_SetPerfTriggerMaskCmd
Set Performance Analyzer Trigger Mask Command 562

CFE_ES_SetPerfTrigMaskCmd_Payload
Set Performance Analyzer Trigger Mask Command Payload 563

CFE_ES_StartApp
Start Application Command 564

CFE_ES_StartAppCmd_Payload
Start Application Command Payload 565

CFE_ES_StartPerfCmd_Payload
Start Performance Analyzer Command Payload 567

CFE_ES_StartPerfDataCmd
Start Performance Analyzer Command 568

CFE_ES_StopPerfCmd_Payload
Stop Performance Analyzer Command Payload 569

CFE_ES_StopPerfDataCmd
Stop Performance Analyzer Command 570

CFE_ES_TaskInfo
Task Information 571

CFE_EVS_AppDataCmd_Payload
Write Event Services Application Information to File Command Payload 574

CFE_EVS_AppNameBitMaskCmd
Generic App Name and Bitmask Command 574

CFE_EVS_AppNameBitMaskCmd_Payload
Generic App Name and Bitmask Command Payload 575

CFE_EVS_AppNameCmd
Generic App Name Command 577

CFE_EVS_AppNameCmd_Payload
Generic App Name Command Payload 578

CFE_EVS_AppNameEventIDCmd
Generic App Name and Event ID Command 579

CFE_EVS_AppNameEventIDCmd_Payload
Generic App Name and Event ID Command Payload 580

CFE_EVS_AppNameEventIDMaskCmd
Generic App Name, Event ID, Mask Command 581

CFE_EVS_AppNameEventIDMaskCmd_Payload
Generic App Name, Event ID, Mask Command Payload 582

CFE_EVS_AppTlmData 583

CFE_EVS_BinFilter
Event message filter defintion structure 584

CFE_EVS_BitMaskCmd
Generic Bitmask Command 585

CFE_EVS_BitMaskCmd_Payload
Generic Bitmask Command Payload 586

CFE_EVS_HousekeepingTlm 587

CFE_EVS_HousekeepingTlm_Payload 588

CFE_EVS_LogFileCmd_Payload
Write Event Log to File Command Payload 593

CFE_EVS_LongEventTlm 594

CFE_EVS_LongEventTlm_Payload 595

CFE_EVS_NoArgsCmd
Command with no additional arguments 597

CFE_EVS_PacketID 597

CFE_EVS_SetEventFormatCode_Payload
Set Event Format Mode Command Payload 599

CFE_EVS_SetEventFormatModeCmd
Set Event Format Mode Command 600

CFE_EVS_SetLogMode_Payload
Set Log Mode Command Payload 601

CFE_EVS_SetLogModeCmd
Set Log Mode Command 602

CFE_EVS_ShortEventTlm 603

CFE_EVS_ShortEventTlm_Payload 604

CFE_EVS_WriteAppDataFileCmd
Write Event Services Application Information to File Command 605

CFE_EVS_WriteLogDataFileCmd
Write Event Log to File Command 606

CFE_FS_FileWriteMetaData
External Metadata/State object associated with background file writes 607

CFE_FS_Header
Standard cFE File header structure definition 609

CFE_PSP_CommandData_t 611

CFE_PSP_LinuxReservedAreaFixedLayout_t 614

CFE_PSP_MemTable_t 615

CFE_PSP_VersionInfo_t 616

CFE_SB_AllSubscriptionsTlm 618

CFE_SB_AllSubscriptionsTlm_Payload 618

CFE_SB_HousekeepingTlm 620

CFE_SB_HousekeepingTlm_Payload 621

CFE_SB_Msg
Software Bus generic message 626

CFE_SB_MsgMapFileEntry
SB Map File Entry 628

CFE_SB_PipeDepthStats
SB Pipe Depth Statistics 629

CFE_SB_PipeInfoEntry
SB Pipe Information File Entry 631

CFE_SB_Qos_t
Quality Of Service Type Definition 633

CFE_SB_RouteCmd
Enable/Disable Route Command 634

CFE_SB_RouteCmd_Payload
Enable/Disable Route Command Payload 635

CFE_SB_RoutingFileEntry
SB Routing File Entry 637

CFE_SB_SingleSubscriptionTlm 639

CFE_SB_SingleSubscriptionTlm_Payload 640

CFE_SB_StatsTlm 641

CFE_SB_StatsTlm_Payload 642

CFE_SB_SubEntries
SB Previous Subscriptions Entry 648

CFE_SB_WriteFileInfoCmd
Write File Info Command 649

CFE_SB_WriteFileInfoCmd_Payload
Write File Info Command Payload 650

CFE_SB_ZeroCopyHandle_t
CFE_SB_ZeroCopyHandle_t to primitive type definition 651

CFE_TBL_AbortLoadCmd
Abort Load Command 651

CFE_TBL_AbortLoadCmd_Payload
Abort Load Command Payload 652

CFE_TBL_ActivateCmd
Activate Table Command 653

CFE_TBL_ActivateCmd_Payload
Activate Table Command Payload 654

CFE_TBL_DelCDSCmd_Payload
Delete Critical Table CDS Command Payload 655

CFE_TBL_DeleteCDSCmd
Delete Critical Table CDS Command 656

CFE_TBL_DumpCmd 657

CFE_TBL_DumpCmd_Payload
Dump Table Command Payload 658

CFE_TBL_DumpRegistryCmd
Dump Registry Command 659

CFE_TBL_DumpRegistryCmd_Payload
Dump Registry Command Payload 660

CFE_TBL_File_Hdr
The definition of the header fields that are included in CFE Table Data files 661

CFE_TBL_FileDef 662

CFE_TBL_HousekeepingTlm 664

CFE_TBL_HousekeepingTlm_Payload 665

CFE_TBL_Info
Table Info 671

CFE_TBL_LoadCmd
Load Table Command 675

CFE_TBL_LoadCmd_Payload
Load Table Command Payload 676

CFE_TBL_NoArgsCmd
Generic "no arguments" command 677

CFE_TBL_NotifyCmd 677

CFE_TBL_NotifyCmd_Payload
Table Management Notification Command Payload 678

CFE_TBL_SendRegistryCmd
Send Table Registry Command 679

CFE_TBL_SendRegistryCmd_Payload
Send Table Registry Command Payload 680

CFE_TBL_TableRegistryTlm 681

CFE_TBL_TblRegPacket_Payload 682

CFE_TBL_ValidateCmd
Validate Table Command 688

CFE_TBL_ValidateCmd_Payload
Validate Table Command Payload 689

CFE_TIME_DiagnosticTlm 690

CFE_TIME_DiagnosticTlm_Payload 690

CFE_TIME_HousekeepingTlm 703

CFE_TIME_HousekeepingTlm_Payload 704

CFE_TIME_LeapsCmd_Payload
Set leap seconds command payload 708

CFE_TIME_NoArgsCmd
Generic no argument command 709

CFE_TIME_OneHzAdjustmentCmd
Generic seconds, subseconds adjustment command 710

CFE_TIME_OneHzAdjustmentCmd_Payload
Generic seconds, subseconds command payload 711

CFE_TIME_SetLeapSecondsCmd
Set leap seconds command 712

CFE_TIME_SetSignalCmd
Set tone signal source command 713

CFE_TIME_SetSourceCmd
Set time data source command 714

CFE_TIME_SetStateCmd
Set clock state command 715

CFE_TIME_SignalCmd_Payload
Set tone signal source command payload 716

CFE_TIME_SourceCmd_Payload
Set time data source command payload 717

CFE_TIME_StateCmd_Payload
Set clock state command payload 717

CFE_TIME_SysTime
Data structure used to hold system time values 718

CFE_TIME_TimeCmd
Generic seconds, microseconds argument command 719

CFE_TIME_TimeCmd_Payload
Generic seconds, microseconds command payload 720

CFE_TIME_ToneDataCmd
Time at tone data command 721

CFE_TIME_ToneDataCmd_Payload
Time at tone data command payload 722

OS_bin_sem_prop_t
OSAL binary semaphore properties 724

OS_count_sem_prop_t
OSAL counting semaphore properties 725

os_dirent_t
Directory entry 726

OS_FdSet
An abstract structure capable of holding several OSAL IDs 726

OS_file_prop_t
OSAL file properties 727

os_fsinfo_t
OSAL file system info 728

os_fstat_t
File system status 730

OS_heap_prop_t
OSAL heap properties 731

OS_module_address_t
OSAL module address properties 732

OS_module_prop_t
OSAL module properties 734

OS_mut_sem_prop_t
OSAL mutex properties 735

OS_queue_prop_t
OSAL queue properties 736

OS_SockAddr_t
Encapsulates a generic network address 737

OS_SockAddrData_t
Storage buffer for generic network address 738

OS_socket_prop_t
Encapsulates socket properties 740

OS_static_symbol_record_t
Associates a single symbol name with a memory address 741

OS_statvfs_t 742

OS_task_prop_t
OSAL task properties 743

OS_time_t
OSAL time interval structure 744

OS_timebase_prop_t
Time base properties 745

OS_timer_prop_t
Timer properties 746

Target_PspConfigData 748

36 File Index

36.1 File List

Here is a list of all files with brief descriptions:

build/doc/osconfig-example.h 750

build/native/default_cpu1/inc/cfe_msgids.h 759

build/native/default_cpu1/inc/cfe_platform_cfg.h 759

build/native/default_cpu1/inc/osconfig.h 759

cpu1_msgids.h 768

cpu1_platform_cfg.h 776

sample_mission_cfg.h 833

sample_perfids.h 853

cfe/modules/core_api/fsw/inc/cfe.h 856

cfe/modules/core_api/fsw/inc/cfe_endian.h 857

cfe/modules/core_api/fsw/inc/cfe_error.h 857

cfe/modules/core_api/fsw/inc/cfe_es.h 868

cfe/modules/core_api/fsw/inc/cfe_es_api_typedefs.h 871

cfe/modules/core_api/fsw/inc/cfe_es_core_internal.h 878

cfe/modules/core_api/fsw/inc/cfe_es_extern_typedefs.h 879

cfe/modules/core_api/fsw/inc/cfe_evs.h 890

cfe/modules/core_api/fsw/inc/cfe_evs_api_typedefs.h 892

cfe/modules/core_api/fsw/inc/cfe_evs_core_internal.h 896

cfe/modules/core_api/fsw/inc/cfe_evs_extern_typedefs.h 896

cfe/modules/core_api/fsw/inc/cfe_fs.h 900

cfe/modules/core_api/fsw/inc/cfe_fs_api_typedefs.h 901

cfe/modules/core_api/fsw/inc/cfe_fs_core_internal.h 903

cfe/modules/core_api/fsw/inc/cfe_fs_extern_typedefs.h 903

cfe/modules/core_api/fsw/inc/cfe_msg.h 907

cfe/modules/core_api/fsw/inc/cfe_msg_api_typedefs.h 909

cfe/modules/core_api/fsw/inc/cfe_resourceid.h 916

cfe/modules/core_api/fsw/inc/cfe_resourceid_api_typedefs.h 923

cfe/modules/core_api/fsw/inc/cfe_sb.h 925

cfe/modules/core_api/fsw/inc/cfe_sb_api_typedefs.h 928

cfe/modules/core_api/fsw/inc/cfe_sb_core_internal.h 935

cfe/modules/core_api/fsw/inc/cfe_sb_extern_typedefs.h 935

cfe/modules/core_api/fsw/inc/cfe_tbl.h 939

cfe/modules/core_api/fsw/inc/cfe_tbl_api_typedefs.h 940

cfe/modules/core_api/fsw/inc/cfe_tbl_core_internal.h 943

cfe/modules/core_api/fsw/inc/cfe_tbl_extern_typedefs.h 944

cfe/modules/core_api/fsw/inc/cfe_tbl_filedef.h 946

cfe/modules/core_api/fsw/inc/cfe_time.h 947

cfe/modules/core_api/fsw/inc/cfe_time_api_typedefs.h 949

cfe/modules/core_api/fsw/inc/cfe_time_core_internal.h 951

cfe/modules/core_api/fsw/inc/cfe_time_extern_typedefs.h 951

cfe/modules/core_api/fsw/inc/cfe_version.h
Purpose: Provide version identifiers for the cFE core 958

cfe/modules/es/fsw/inc/cfe_es_events.h 961

cfe/modules/es/fsw/inc/cfe_es_msg.h 1008

cfe/modules/evs/fsw/inc/cfe_evs_events.h 1044

cfe/modules/evs/fsw/inc/cfe_evs_msg.h 1065

cfe/modules/msg/fsw/inc/ccsds_hdr.h 1098

cfe/modules/sb/fsw/inc/cfe_sb_events.h 1099

cfe/modules/sb/fsw/inc/cfe_sb_msg.h 1134

cfe/modules/tbl/fsw/inc/cfe_tbl_events.h 1152

cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h 1190

cfe/modules/time/fsw/inc/cfe_time_events.h 1206

cfe/modules/time/fsw/inc/cfe_time_msg.h 1227

osal/src/os/inc/common_types.h 1252

osal/src/os/inc/osapi-binsem.h 1259

osal/src/os/inc/osapi-bsp.h 1260

osal/src/os/inc/osapi-clock.h 1260

osal/src/os/inc/osapi-common.h 1262

osal/src/os/inc/osapi-constants.h 1264

osal/src/os/inc/osapi-countsem.h 1266

osal/src/os/inc/osapi-dir.h 1266

osal/src/os/inc/osapi-error.h 1267

osal/src/os/inc/osapi-file.h 1270

osal/src/os/inc/osapi-filesys.h 1274

osal/src/os/inc/osapi-heap.h 1275

osal/src/os/inc/osapi-idmap.h 1276

osal/src/os/inc/osapi-macros.h 1278

osal/src/os/inc/osapi-module.h 1280

osal/src/os/inc/osapi-mutex.h 1281

osal/src/os/inc/osapi-network.h 1282

osal/src/os/inc/osapi-printf.h 1282

osal/src/os/inc/osapi-queue.h 1283

osal/src/os/inc/osapi-select.h 1283

osal/src/os/inc/osapi-shell.h 1284

osal/src/os/inc/osapi-sockets.h 1285

osal/src/os/inc/osapi-task.h 1287

osal/src/os/inc/osapi-timebase.h 1290

osal/src/os/inc/osapi-timer.h 1291

osal/src/os/inc/osapi-version.h
Purpose: 1292

osal/src/os/inc/osapi.h 1297

psp/fsw/inc/cfe_psp.h 1298

psp/fsw/inc/cfe_psp_configdata.h 1325

psp/fsw/pc-linux/src/cfe_psp_exception.c 1326

psp/fsw/pc-linux/src/cfe_psp_memory.c 1329

psp/fsw/pc-linux/src/cfe_psp_memtab.c 1340

psp/fsw/pc-linux/src/cfe_psp_ssr.c 1340

psp/fsw/pc-linux/src/cfe_psp_start.c 1341

psp/fsw/pc-linux/src/cfe_psp_support.c 1348

psp/fsw/pc-linux/src/cfe_psp_timer.c 1351

psp/fsw/pc-linux/src/cfe_psp_watchdog.c 1354

37 Module Documentation

37.1 cFE Return Code Defines

Macros

- #define CFE_SUCCESS (0)

  Sucessful execution.
- #define CFE_STATUS_NO_COUNTER_INCREMENT ((CFE_Status_t)0x48000001)

  No Counter Increment.
- #define CFE_STATUS_WRONG_MSG_LENGTH ((CFE_Status_t)0xc8000002)

  Wrong Message Length.

- #define CFE_STATUS_UNKNOWN_MSG_ID ((CFE_Status_t)0xc8000003)

  Unknown Message ID.
- #define CFE_STATUS_BAD_COMMAND_CODE ((CFE_Status_t)0xc8000004)

  Bad Command Code.
- #define CFE_STATUS_EXTERNAL_RESOURCE_FAIL ((CFE_Status_t)0xc8000005)

  External failure.
- #define CFE_STATUS_REQUEST_ALREADY_PENDING ((int32)0xc8000006)

  Request already pending.
- #define CFE_STATUS_NOT_IMPLEMENTED ((CFE_Status_t)0xc800ffff)

  Not Implemented.
- #define CFE_EVS_UNKNOWN_FILTER ((CFE_Status_t)0xc2000001)

  Unknown Filter.
- #define CFE_EVS_APP_NOT_REGISTERED ((CFE_Status_t)0xc2000002)

  Application Not Registered.
- #define CFE_EVS_APP_ILLEGAL_APP_ID ((CFE_Status_t)0xc2000003)

  Illegal Application ID.
- #define CFE_EVS_APP_FILTER_OVERLOAD ((CFE_Status_t)0xc2000004)

  Application Filter Overload.
- #define CFE_EVS_RESET_AREA_POINTER ((CFE_Status_t)0xc2000005)

  Reset Area Pointer Failure.
- #define CFE_EVS_EVT_NOT_REGISTERED ((CFE_Status_t)0xc2000006)

  Event Not Registered.
- #define CFE_EVS_FILE_WRITE_ERROR ((CFE_Status_t)0xc2000007)

  File Write Error.
- #define CFE_EVS_INVALID_PARAMETER ((CFE_Status_t)0xc2000008)

  Invalid Pointer.
- #define CFE_EVS_NOT_IMPLEMENTED ((CFE_Status_t)0xc200ffff)

  Not Implemented.
- #define CFE_ES_ERR_RESOURCEID_NOT_VALID ((CFE_Status_t)0xc4000001)

  Resource ID is not valid.
- #define CFE_ES_ERR_NAME_NOT_FOUND ((CFE_Status_t)0xc4000002)

  Resource Name Error.
- #define CFE_ES_ERR_BUFFER ((CFE_Status_t)0xc4000003)

  Invalid Pointer.
- #define CFE_ES_ERR_APP_CREATE ((CFE_Status_t)0xc4000004)

  Application Create Error.
- #define CFE_ES_ERR_CHILD_TASK_CREATE ((CFE_Status_t)0xc4000005)

  Child Task Create Error.
- #define CFE_ES_ERR_SYS_LOG_FULL ((CFE_Status_t)0xc4000006)

  System Log Full.
- #define CFE_ES_ERR_MEM_BLOCK_SIZE ((CFE_Status_t)0xc4000008)

  Memory Block Size Error.
- #define CFE_ES_ERR_LOAD_LIB ((CFE_Status_t)0xc4000009)

  Load Library Error.
- #define CFE_ES_BAD_ARGUMENT ((CFE_Status_t)0xc400000a)

  Bad Argument.
- #define CFE_ES_ERR_CHILD_TASK_REGISTER ((CFE_Status_t)0xc400000b)

Child Task Register Error.

- #define CFE_ES_ERR_SHELL_CMD ((CFE_Status_t)0xc400000c)

  Shell Command Error.
- #define CFE_ES_CDS_ALREADY_EXISTS ((CFE_Status_t)0x4400000d)

  CDS Already Exists.
- #define CFE_ES_CDS_INSUFFICIENT_MEMORY ((CFE_Status_t)0xc400000e)

  CDS Insufficient Memory.
- #define CFE_ES_CDS_INVALID_NAME ((CFE_Status_t)0xc400000f)

  CDS Invalid Name.
- #define CFE_ES_CDS_INVALID_SIZE ((CFE_Status_t)0xc4000010)

  CDS Invalid Size.
- #define CFE_ES_CDS_INVALID ((CFE_Status_t)0xc4000012)

  CDS Invalid.
- #define CFE_ES_CDS_ACCESS_ERROR ((CFE_Status_t)0xc4000013)

  CDS Access Error.
- #define CFE_ES_FILE_IO_ERR ((CFE_Status_t)0xc4000014)

  File IO Error.
- #define CFE_ES_RST_ACCESS_ERR ((CFE_Status_t)0xc4000015)

  Reset Area Access Error.
- #define CFE_ES_ERR_APP_REGISTER ((CFE_Status_t)0xc4000017)

  Application Register Error.
- #define CFE_ES_ERR_CHILD_TASK_DELETE ((CFE_Status_t)0xc4000018)

  Child Task Delete Error.
- #define CFE_ES_ERR_CHILD_TASK_DELETE_MAIN_TASK ((CFE_Status_t)0xc4000019)

  Child Task Delete Passed Main Task.
- #define CFE_ES_CDS_BLOCK_CRC_ERR ((CFE_Status_t)0xc400001A)

  CDS Block CRC Error.
- #define CFE_ES_MUT_SEM_DELETE_ERR ((CFE_Status_t)0xc400001B)

  Mutex Semaphore Delete Error.
- #define CFE_ES_BIN_SEM_DELETE_ERR ((CFE_Status_t)0xc400001C)

  Binary Semaphore Delete Error.
- #define CFE_ES_COUNT_SEM_DELETE_ERR ((CFE_Status_t)0xc400001D)

  Counte Semaphore Delete Error.
- #define CFE_ES_QUEUE_DELETE_ERR ((CFE_Status_t)0xc400001E)

  Queue Delete Error.
- #define CFE_ES_FILE_CLOSE_ERR ((CFE_Status_t)0xc400001F)

  File Close Error.
- #define CFE_ES_CDS_WRONG_TYPE_ERR ((CFE_Status_t)0xc4000020)

  CDS Wrong Type Error.
- #define CFE_ES_CDS_OWNER_ACTIVE_ERR ((CFE_Status_t)0xc4000022)

  CDS Owner Active Error.
- #define CFE_ES_APP_CLEANUP_ERR ((CFE_Status_t)0xc4000023)

  Application Cleanup Error.
- #define CFE_ES_TIMER_DELETE_ERR ((CFE_Status_t)0xc4000024)

  Timer Delete Error.
- #define CFE_ES_BUFFER_NOT_IN_POOL ((CFE_Status_t)0xc4000025)

  Buffer Not In Pool.

- #define CFE_ES_TASK_DELETE_ERR ((CFE_Status_t)0xc4000026)

  Task Delete Error.
- #define CFE_ES_OPERATION_TIMED_OUT ((CFE_Status_t)0xc4000027)

  Operation Timed Out.
- #define CFE_ES_LIB_ALREADY_LOADED ((CFE_Status_t)0x44000028)

  Library Already Loaded.
- #define CFE_ES_ERR_SYS_LOG_TRUNCATED ((CFE_Status_t)0x44000029)

  System Log Message Truncated.
- #define CFE_ES_NO_RESOURCE_IDS_AVAILABLE ((CFE_Status_t)0xc400002B)

  Resource ID is not available.
- #define CFE_ES_POOL_BLOCK_INVALID ((CFE_Status_t)0xc400002C)

  Invalid pool block.
- #define CFE_ES_POOL_BOUNDS_ERROR ((CFE_Status_t)0xc400002D)

  Invalid pool size or buffer address.
- #define CFE_ES_ERR_DUPLICATE_NAME ((CFE_Status_t)0xc400002E)

  Duplicate Name Error.
- #define CFE_ES_NOT_IMPLEMENTED ((CFE_Status_t)0xc400ffff)

  Not Implemented.
- #define CFE_FS_BAD_ARGUMENT ((CFE_Status_t)0xc6000001)

  Bad Argument.
- #define CFE_FS_INVALID_PATH ((CFE_Status_t)0xc6000002)

  Invalid Path.
- #define CFE_FS_FNAME_TOO_LONG ((CFE_Status_t)0xc6000003)

  Filename Too Long.
- #define CFE_FS_NOT_IMPLEMENTED ((CFE_Status_t)0xc600ffff)

  Not Implemented.
- #define CFE_SB_TIME_OUT ((CFE_Status_t)0xca000001)

  Time Out.
- #define CFE_SB_NO_MESSAGE ((CFE_Status_t)0xca000002)

  No Message.
- #define CFE_SB_BAD_ARGUMENT ((CFE_Status_t)0xca000003)

  Bad Argument.
- #define CFE_SB_MAX_PIPES_MET ((CFE_Status_t)0xca000004)

  Max Pipes Met.
- #define CFE_SB_PIPE_CR_ERR ((CFE_Status_t)0xca000005)

  Pipe Create Error.
- #define CFE_SB_PIPE_RD_ERR ((CFE_Status_t)0xca000006)

  Pipe Read Error.
- #define CFE_SB_MSG_TOO_BIG ((CFE_Status_t)0xca000007)

  Message Too Big.
- #define CFE_SB_BUF_ALOC_ERR ((CFE_Status_t)0xca000008)

  Buffer Allocation Error.
- #define CFE_SB_MAX_MSGS_MET ((CFE_Status_t)0xca000009)

  Max Messages Met.
- #define CFE_SB_MAX_DESTS_MET ((CFE_Status_t)0xca00000a)

  Max Destinations Met.
- #define CFE_SB_NO_SUBSCRIBERS ((CFE_Status_t)0xca00000b)

No Subscribers.

- #define CFE_SB_INTERNAL_ERR ((CFE_Status_t)0xca00000c)

  Internal Error.
- #define CFE_SB_WRONG_MSG_TYPE ((CFE_Status_t)0xca00000d)

  Wrong Message Type.
- #define CFE_SB_BUFFER_INVALID ((CFE_Status_t)0xca00000e)

  Buffer Invalid.
- #define CFE_SB_NO_MSG_RECV ((CFE_Status_t)0xca00000f)

  No Message Recieved.
- #define CFE_SB_NOT_IMPLEMENTED ((CFE_Status_t)0xca00ffff)

  Not Implemented.
- #define CFE_TBL_ERR_INVALID_HANDLE ((CFE_Status_t)0xcc000001)

  Invalid Handle.
- #define CFE_TBL_ERR_INVALID_NAME ((CFE_Status_t)0xcc000002)

  Invalid Name.
- #define CFE_TBL_ERR_INVALID_SIZE ((CFE_Status_t)0xcc000003)

  Invalid Size.
- #define CFE_TBL_INFO_UPDATE_PENDING ((CFE_Status_t)0x4c000004)

  Update Pending.
- #define CFE_TBL_ERR_NEVER_LOADED ((CFE_Status_t)0xcc000005)

  Never Loaded.
- #define CFE_TBL_ERR_REGISTRY_FULL ((CFE_Status_t)0xcc000006)

  Registry Full.
- #define CFE_TBL_WARN_DUPLICATE ((CFE_Status_t)0x4c000007)

  Duplicate Warning.
- #define CFE_TBL_ERR_NO_ACCESS ((CFE_Status_t)0xcc000008)

  No Access.
- #define CFE_TBL_ERR_UNREGISTERED ((CFE_Status_t)0xcc000009)

  Unregistered.
- #define CFE_TBL_ERR_BAD_APP_ID ((CFE_Status_t)0xcc00000A)

  Bad Application ID.
- #define CFE_TBL_ERR_HANDLES_FULL ((CFE_Status_t)0xcc00000B)

  Handles Full.
- #define CFE_TBL_ERR_DUPLICATE_DIFF_SIZE ((CFE_Status_t)0xcc00000C)

  Duplicate Table With Different Size.
- #define CFE_TBL_ERR_DUPLICATE_NOT_OWNED ((CFE_Status_t)0xcc00000D)

  Dupicate Table And Not Owned.
- #define CFE_TBL_INFO_UPDATED ((CFE_Status_t)0x4c00000E)

  Updated.
- #define CFE_TBL_ERR_NO_BUFFER_AVAIL ((CFE_Status_t)0xcc00000F)

  No Buffer Available.
- #define CFE_TBL_ERR_DUMP_ONLY ((CFE_Status_t)0xcc000010)

  Dump Only Error.
- #define CFE_TBL_ERR_ILLEGAL_SRC_TYPE ((CFE_Status_t)0xcc000011)

  Illegal Source Type.
- #define CFE_TBL_ERR_LOAD_IN_PROGRESS ((CFE_Status_t)0xcc000012)

  Load In Progress.

- #define CFE_TBL_ERR_FILE_NOT_FOUND ((CFE_Status_t)0xcc000013)

  File Not Found.
- #define CFE_TBL_ERR_FILE_TOO_LARGE ((CFE_Status_t)0xcc000014)

  File Too Large.
- #define CFE_TBL_WARN_SHORT_FILE ((CFE_Status_t)0x4c000015)

  Short File Warning.
- #define CFE_TBL_ERR_BAD_CONTENT_ID ((CFE_Status_t)0xcc000016)

  Bad Content ID.
- #define CFE_TBL_INFO_NO_UPDATE_PENDING ((CFE_Status_t)0x4c000017)

  No Update Pending.
- #define CFE_TBL_INFO_TABLE_LOCKED ((CFE_Status_t)0x4c000018)

  Table Locked.
- #define CFE_TBL_INFO_VALIDATION_PENDING ((CFE_Status_t)0x4c000019)
- #define CFE_TBL_INFO_NO_VALIDATION_PENDING ((CFE_Status_t)0x4c00001A)
- #define CFE_TBL_ERR_BAD_SUBTYPE_ID ((CFE_Status_t)0xcc00001B)

  Bad Subtype ID.
- #define CFE_TBL_ERR_FILE_SIZE_INCONSISTENT ((CFE_Status_t)0xcc00001C)

  File Size Inconsistent.
- #define CFE_TBL_ERR_NO_STD_HEADER ((CFE_Status_t)0xcc00001D)

  No Standard Header.
- #define CFE_TBL_ERR_NO_TBL_HEADER ((CFE_Status_t)0xcc00001E)

  No Table Header.
- #define CFE_TBL_ERR_FILENAME_TOO_LONG ((CFE_Status_t)0xcc00001F)

  Filename Too Long.
- #define CFE_TBL_ERR_FILE_FOR_WRONG_TABLE ((CFE_Status_t)0xcc000020)

  File For Wrong Table.
- #define CFE_TBL_ERR_LOAD_INCOMPLETE ((CFE_Status_t)0xcc000021)

  Load Incomplete.
- #define CFE_TBL_WARN_PARTIAL_LOAD ((CFE_Status_t)0x4c000022)

  Partial Load Warning.
- #define CFE_TBL_ERR_PARTIAL_LOAD ((CFE_Status_t)0xcc000023)

  Partial Load Error.
- #define CFE_TBL_INFO_DUMP_PENDING ((CFE_Status_t)0x4c000024)

  Dump Pending.
- #define CFE_TBL_ERR_INVALID_OPTIONS ((CFE_Status_t)0xcc000025)

  Invalid Options.
- #define CFE_TBL_WARN_NOT_CRITICAL ((CFE_Status_t)0x4c000026)

  Not Critical Warning.
- #define CFE_TBL_INFO_RECOVERED_TBL ((CFE_Status_t)0x4c000027)

  Recovered Table.
- #define CFE_TBL_ERR_BAD_SPACECRAFT_ID ((CFE_Status_t)0xcc000028)

  Bad Spacecraft ID.
- #define CFE_TBL_ERR_BAD_PROCESSOR_ID ((CFE_Status_t)0xcc000029)

  Bad Processor ID.
- #define CFE_TBL_MESSAGE_ERROR ((CFE_Status_t)0xcc00002a)

  Message Error.
- #define CFE_TBL_ERR_SHORT_FILE ((CFE_Status_t)0xcc00002b)

- #define CFE_TBL_ERR_ACCESS ((CFE_Status_t)0xcc00002c)
- #define CFE_TBL_BAD_ARGUMENT ((CFE_Status_t)0xcc00002d)

  Bad Argument.
- #define CFE_TBL_NOT_IMPLEMENTED ((CFE_Status_t)0xcc00ffff)

  Not Implemented.
- #define CFE_TIME_NOT_IMPLEMENTED ((CFE_Status_t)0xce00ffff)

  Not Implemented.
- #define CFE_TIME_INTERNAL_ONLY ((CFE_Status_t)0xce000001)

  Internal Only.
- #define CFE_TIME_OUT_OF_RANGE ((CFE_Status_t)0xce000002)

  Out Of Range.
- #define CFE_TIME_TOO_MANY_SYNCH_CALLBACKS ((CFE_Status_t)0xce000003)

  Too Many Sync Callbacks.
- #define CFE_TIME_CALLBACK_NOT_REGISTERED ((CFE_Status_t)0xce000004)

  Callback Not Registered.
- #define CFE_TIME_BAD_ARGUMENT ((CFE_Status_t)0xce000005)

  Bad Argument.

37.1.1 Detailed Description

37.1.2 Macro Definition Documentation

37.1.2.1 CFE_ES_APP_CLEANUP_ERR

```
#define CFE_ES_APP_CLEANUP_ERR ((CFE_Status_t)0xc4000023)
```

Application Cleanup Error.

Occurs when an attempt was made to Clean Up an application which involves calling Table, EVS, and SB cleanup functions, then deleting all ES resources, child tasks, and unloading the object module. The approach here is to keep going even though one of these steps had an error. There will be syslog messages detailing each problem.

Definition at line 550 of file cfe_error.h.

37.1.2.2 CFE_ES_BAD_ARGUMENT

```
#define CFE_ES_BAD_ARGUMENT ((CFE_Status_t)0xc400000a)
```

Bad Argument.

Bad parameter passed into an ES API.

Definition at line 353 of file cfe_error.h.

37.1.2.3 CFE_ES_BIN_SEM_DELETE_ERR

```
#define CFE_ES_BIN_SEM_DELETE_ERR ((CFE_Status_t)0xc400001C)
```

Binary Semaphore Delete Error.

Occurs when trying to delete a Binary Semaphore that belongs to a task that ES is cleaning up.

Definition at line 489 of file cfe_error.h.

37.1.2.4 CFE_ES_BUFFER_NOT_IN_POOL

```
#define CFE_ES_BUFFER_NOT_IN_POOL ((CFE_Status_t)0xc4000025)
```

Buffer Not In Pool.

The specified address is not in the memory pool.

Definition at line 567 of file cfe_error.h.

37.1.2.5 CFE_ES_CDS_ACCESS_ERROR

```
#define CFE_ES_CDS_ACCESS_ERROR ((CFE_Status_t)0xc4000013)
```

CDS Access Error.

The CDS was inaccessible

Definition at line 420 of file cfe_error.h.

37.1.2.6 CFE_ES_CDS_ALREADY_EXISTS

```
#define CFE_ES_CDS_ALREADY_EXISTS ((CFE_Status_t)0x4400000d)
```

CDS Already Exists.

The Application is receiving the pointer to a CDS that was already present.

Definition at line 377 of file cfe_error.h.

37.1.2.7 CFE_ES_CDS_BLOCK_CRC_ERR

```
#define CFE_ES_CDS_BLOCK_CRC_ERR ((CFE_Status_t)0xc400001A)
```

CDS Block CRC Error.

Occurs when trying to read a CDS Data block and the CRC of the current data does not match the stored CRC for the data. Either the contents of the CDS Data Block are corrupted or the CDS Control Block is corrupted.

Definition at line 471 of file cfe_error.h.

37.1.2.8 CFE_ES_CDS_INSUFFICIENT_MEMORY

```
#define CFE_ES_CDS_INSUFFICIENT_MEMORY ((CFE_Status_t)0xc400000e)
```

CDS Insufficient Memory.

The Application is requesting a CDS Block that is larger than the remaining CDS memory.

Definition at line 386 of file cfe_error.h.

37.1.2.9 CFE_ES_CDS_INVALID

```
#define CFE_ES_CDS_INVALID ((CFE_Status_t)0xc4000012)
```

CDS Invalid.

The CDS contents are invalid.

Definition at line 412 of file cfe_error.h.

37.1.2.10 CFE_ES_CDS_INVALID_NAME

```
#define CFE_ES_CDS_INVALID_NAME ((CFE_Status_t)0xc400000f)
```

CDS Invalid Name.

The Application is requesting a CDS Block with an invalid ASCII string name. Either the name is too long (> CFE_MI↩SSION_ES_CDS_MAX_NAME_LENGTH) or was an empty string.

Definition at line 395 of file cfe_error.h.

37.1.2.11 CFE_ES_CDS_INVALID_SIZE

```
#define CFE_ES_CDS_INVALID_SIZE ((CFE_Status_t)0xc4000010)
```

CDS Invalid Size.

The Application is requesting a CDS Block or Pool with a size beyond the applicable limits, either too large or too small/zero.

Definition at line 404 of file cfe_error.h.

37.1.2.12 CFE_ES_CDS_OWNER_ACTIVE_ERR

```
#define CFE_ES_CDS_OWNER_ACTIVE_ERR ((CFE_Status_t)0xc4000022)
```

CDS Owner Active Error.

Occurs when an attempt was made to delete a CDS when an application with the same name associated with the CDS is still present. CDSs can ONLY be deleted when Applications that created them are not present in the system.

Definition at line 537 of file cfe_error.h.

37.1.2.13 CFE_ES_CDS_WRONG_TYPE_ERR

```
#define CFE_ES_CDS_WRONG_TYPE_ERR ((CFE_Status_t)0xc4000020)
```

CDS Wrong Type Error.

Occurs when Table Services is trying to delete a Critical Data Store that is not a Critical Table Image or when Executive Services is trying to delete a Critical Table Image.

Definition at line 526 of file cfe_error.h.

37.1.2.14 CFE_ES_COUNT_SEM_DELETE_ERR

```
#define CFE_ES_COUNT_SEM_DELETE_ERR ((CFE_Status_t)0xc400001D)
```

Counte Semaphore Delete Error.

Occurs when trying to delete a Counting Semaphore that belongs to a task that ES is cleaning up.

Definition at line 498 of file cfe_error.h.

37.1.2.15 CFE_ES_ERR_APP_CREATE

```
#define CFE_ES_ERR_APP_CREATE ((CFE_Status_t)0xc4000004)
```

Application Create Error.

There was an error loading or creating the App.

Definition at line 312 of file cfe_error.h.

37.1.2.16 CFE_ES_ERR_APP_REGISTER

```
#define CFE_ES_ERR_APP_REGISTER ((CFE_Status_t)0xc4000017)
```

Application Register Error.

Occurs when the CFE_ES_RegisterApp fails.

Definition at line 444 of file cfe_error.h.

37.1.2.17 CFE_ES_ERR_BUFFER

```
#define CFE_ES_ERR_BUFFER ((CFE_Status_t)0xc4000003)
```

Invalid Pointer.

Invalid pointer argument (NULL)

Definition at line 304 of file cfe_error.h.

37.1.2.18 CFE_ES_ERR_CHILD_TASK_CREATE

```
#define CFE_ES_ERR_CHILD_TASK_CREATE ((CFE_Status_t)0xc4000005)
```

Child Task Create Error.

There was an error creating a child task.

Definition at line 320 of file cfe_error.h.

37.1.2.19 CFE_ES_ERR_CHILD_TASK_DELETE

```
#define CFE_ES_ERR_CHILD_TASK_DELETE ((CFE_Status_t)0xc4000018)
```

Child Task Delete Error.

There was an error deleting a child task.

Definition at line 452 of file cfe_error.h.

37.1.2.20 CFE_ES_ERR_CHILD_TASK_DELETE_MAIN_TASK

```
#define CFE_ES_ERR_CHILD_TASK_DELETE_MAIN_TASK ((CFE_Status_t)0xc4000019)
```

Child Task Delete Passed Main Task.

There was an attempt to delete a cFE App Main Task with the CFE_ES_DeleteChildTask API.

Definition at line 461 of file cfe_error.h.

37.1.2.21 CFE_ES_ERR_CHILD_TASK_REGISTER

```
#define CFE_ES_ERR_CHILD_TASK_REGISTER ((CFE_Status_t)0xc400000b)
```

Child Task Register Error.

Errors occured when trying to register a child task.

Definition at line 361 of file cfe_error.h.

37.1.2.22 CFE_ES_ERR_DUPLICATE_NAME

```
#define CFE_ES_ERR_DUPLICATE_NAME ((CFE_Status_t)0xc400002E)
```

Duplicate Name Error.

Resource creation failed due to the name already existing in the system.

Definition at line 639 of file cfe_error.h.

37.1.2.23 CFE_ES_ERR_LOAD_LIB

```
#define CFE_ES_ERR_LOAD_LIB ((CFE_Status_t)0xc4000009)
```

Load Library Error.

Could not load the shared library.

Definition at line 345 of file cfe_error.h.

37.1.2.24 CFE_ES_ERR_MEM_BLOCK_SIZE

```
#define CFE_ES_ERR_MEM_BLOCK_SIZE ((CFE_Status_t)0xc4000008)
```

Memory Block Size Error.

The block size requested is invalid.

Definition at line 337 of file cfe_error.h.

37.1.2.25 CFE_ES_ERR_NAME_NOT_FOUND

```
#define CFE_ES_ERR_NAME_NOT_FOUND ((CFE_Status_t)0xc4000002)
```

Resource Name Error.

There is no match in the system for the given name.

Definition at line 296 of file cfe_error.h.

37.1.2.26 CFE_ES_ERR_RESOURCEID_NOT_VALID

```
#define CFE_ES_ERR_RESOURCEID_NOT_VALID ((CFE_Status_t)0xc4000001)
```

Resource ID is not valid.

This error indicates that the passed in resource identifier (App ID, Lib ID, Counter ID, etc) did not validate.

Definition at line 288 of file cfe_error.h.

37.1.2.27 CFE_ES_ERR_SHELL_CMD

```
#define CFE_ES_ERR_SHELL_CMD ((CFE_Status_t)0xc400000c)
```

Shell Command Error.

Error occured ehen trying to pass a system call to the OS shell

Definition at line 369 of file cfe_error.h.

37.1.2.28 CFE_ES_ERR_SYS_LOG_FULL

```
#define CFE_ES_ERR_SYS_LOG_FULL ((CFE_Status_t)0xc4000006)
```

System Log Full.

The cFE system Log is full. This error means the message was not logged at all

Definition at line 329 of file cfe_error.h.

37.1.2.29 CFE_ES_ERR_SYS_LOG_TRUNCATED

```
#define CFE_ES_ERR_SYS_LOG_TRUNCATED ((CFE_Status_t)0x44000029)
```

System Log Message Truncated.

This information code means the last syslog message was truncated due to insufficient space in the log buffer.

Definition at line 602 of file cfe_error.h.

37.1.2.30 CFE_ES_FILE_CLOSE_ERR

```
#define CFE_ES_FILE_CLOSE_ERR ((CFE_Status_t)0xc400001F)
```

File Close Error.

Occurs when trying to close a file that belongs to a task that ES is cleaning up.

Definition at line 516 of file cfe_error.h.

37.1.2.31 CFE_ES_FILE_IO_ERR

```
#define CFE_ES_FILE_IO_ERR ((CFE_Status_t)0xc4000014)
```

File IO Error.

Occurs when a file operation fails

Definition at line 428 of file cfe_error.h.

37.1.2.32 CFE_ES_LIB_ALREADY_LOADED

```
#define CFE_ES_LIB_ALREADY_LOADED ((CFE_Status_t)0x44000028)
```

Library Already Loaded.

Occurs if CFE_ES_LoadLibrary detects that the requested library name is already loaded.

Definition at line 593 of file cfe_error.h.

37.1.2.33 CFE_ES_MUT_SEM_DELETE_ERR

```
#define CFE_ES_MUT_SEM_DELETE_ERR ((CFE_Status_t)0xc400001B)
```

Mutex Semaphore Delete Error.

Occurs when trying to delete a Mutex that belongs to a task that ES is cleaning up.

Definition at line 480 of file cfe_error.h.

37.1.2.34 CFE_ES_NO_RESOURCE_IDS_AVAILABLE

```
#define CFE_ES_NO_RESOURCE_IDS_AVAILABLE ((CFE_Status_t)0xc400002B)
```

Resource ID is not available.

This error indicates that the maximum resource identifiers (App ID, Lib ID, Counter ID, etc) has already been reached and a new ID cannot be allocated.

Definition at line 612 of file cfe_error.h.

37.1.2.35 CFE_ES_NOT_IMPLEMENTED

```
#define CFE_ES_NOT_IMPLEMENTED ((CFE_Status_t)0xc400ffff)
```

Not Implemented.

Current version of cFE does not have the function or the feature of the function implemented. This could be due to either an early build of the cFE for this platform or the platform does not support the specified feature.

Definition at line 650 of file cfe_error.h.

37.1.2.36 CFE_ES_OPERATION_TIMED_OUT

```
#define CFE_ES_OPERATION_TIMED_OUT ((CFE_Status_t)0xc4000027)
```

Operation Timed Out.

Occurs if the timeout for a given operation was exceeded

Definition at line 584 of file cfe_error.h.

37.1.2.37 CFE_ES_POOL_BLOCK_INVALID

```
#define CFE_ES_POOL_BLOCK_INVALID ((CFE_Status_t)0xc400002C)
```

Invalid pool block.

Software attempted to "put" a block back into a pool which does not appear to belong to that pool. This may mean the pool has become unusable due to memory corruption.

Definition at line 622 of file cfe_error.h.

37.1.2.38 CFE_ES_POOL_BOUNDS_ERROR

```
#define CFE_ES_POOL_BOUNDS_ERROR ((CFE_Status_t)0xc400002D)
```

Invalid pool size or buffer address.

A specified pool address or size is outside the acceptable bounds for that pool configuration.

Definition at line 631 of file cfe_error.h.

37.1.2.39 CFE_ES_QUEUE_DELETE_ERR

```
#define CFE_ES_QUEUE_DELETE_ERR ((CFE_Status_t)0xc400001E)
```

Queue Delete Error.

Occurs when trying to delete a Queue that belongs to a task that ES is cleaning up.

Definition at line 507 of file cfe_error.h.

37.1.2.40 CFE_ES_RST_ACCESS_ERR

```
#define CFE_ES_RST_ACCESS_ERR ((CFE_Status_t)0xc4000015)
```

Reset Area Access Error.

Occurs when the BSP is not successful in returning the reset area address.

Definition at line 436 of file cfe_error.h.

37.1.2.41 CFE_ES_TASK_DELETE_ERR

```
#define CFE_ES_TASK_DELETE_ERR ((CFE_Status_t)0xc4000026)
```

Task Delete Error.

Occurs when trying to delete a task that ES is cleaning up.

Definition at line 576 of file cfe_error.h.

37.1.2.42 CFE_ES_TIMER_DELETE_ERR

```
#define CFE_ES_TIMER_DELETE_ERR ((CFE_Status_t)0xc4000024)
```

Timer Delete Error.

Occurs when trying to delete a Timer that belongs to a task that ES is cleaning up.

Definition at line 559 of file cfe_error.h.

37.1.2.43 CFE_EVS_APP_FILTER_OVERLOAD

```
#define CFE_EVS_APP_FILTER_OVERLOAD ((CFE_Status_t)0xc2000004)
```

Application Filter Overload.

Number of Application event filters input upon registration is greater than CFE_PLATFORM_EVS_MAX_EVENT_FILTERS

Definition at line 230 of file cfe_error.h.

37.1.2.44 CFE_EVS_APP_ILLEGAL_APP_ID

```
#define CFE_EVS_APP_ILLEGAL_APP_ID ((CFE_Status_t)0xc2000003)
```

Illegal Application ID.

Application ID returned by CFE_ES_GetAppIDByName is greater than CFE_PLATFORM_ES_MAX_APPLICATIONS

Definition at line 221 of file cfe_error.h.

37.1.2.45 CFE_EVS_APP_NOT_REGISTERED

```
#define CFE_EVS_APP_NOT_REGISTERED ((CFE_Status_t)0xc2000002)
```

Application Not Registered.

Calling application never previously called CFE_EVS_Register

Definition at line 212 of file cfe_error.h.

37.1.2.46 CFE_EVS_EVT_NOT_REGISTERED

```
#define CFE_EVS_EVT_NOT_REGISTERED ((CFE_Status_t)0xc2000006)
```

Event Not Registered.

CFE_EVS_ResetFilter EventID argument was not found in any event filter registered by the calling application.

Definition at line 248 of file cfe_error.h.

37.1.2.47 CFE_EVS_FILE_WRITE_ERROR

```
#define CFE_EVS_FILE_WRITE_ERROR ((CFE_Status_t)0xc2000007)
```

File Write Error.

A file write error occurred while processing an EVS command

Definition at line 256 of file cfe_error.h.

37.1.2.48 CFE_EVS_INVALID_PARAMETER

```
#define CFE_EVS_INVALID_PARAMETER ((CFE_Status_t)0xc2000008)
```

Invalid Pointer.

Invalid parameter supplied to EVS command

Definition at line 264 of file cfe_error.h.

37.1.2.49 CFE_EVS_NOT_IMPLEMENTED

```
#define CFE_EVS_NOT_IMPLEMENTED ((CFE_Status_t)0xc200ffff)
```

Not Implemented.

Current version of cFE does not have the function or the feature of the function implemented. This could be due to either an early build of the cFE for this platform or the platform does not support the specified feature.

Definition at line 275 of file cfe_error.h.

37.1.2.50 CFE_EVS_RESET_AREA_POINTER

```
#define CFE_EVS_RESET_AREA_POINTER ((CFE_Status_t)0xc2000005)
```

Reset Area Pointer Failure.

Could not get pointer to the ES Reset area, so we could not get the pointer to the EVS Log.

Definition at line 239 of file cfe_error.h.

37.1.2.51 CFE_EVS_UNKNOWN_FILTER

```
#define CFE_EVS_UNKNOWN_FILTER ((CFE_Status_t)0xc2000001)
```

Unknown Filter.

CFE_EVS_Register FilterScheme parameter was illegal

Definition at line 204 of file cfe_error.h.

37.1.2.52 CFE_FS_BAD_ARGUMENT

```
#define CFE_FS_BAD_ARGUMENT ((CFE_Status_t)0xc6000001)
```

Bad Argument.

A parameter given by a caller to a File Services API did not pass validation checks.

Definition at line 663 of file cfe_error.h.

37.1.2.53 CFE_FS_FNAME_TOO_LONG

```
#define CFE_FS_FNAME_TOO_LONG ((CFE_Status_t)0xc6000003)
```

Filename Too Long.

FS filename string is too long

Definition at line 679 of file cfe_error.h.

37.1.2.54 CFE_FS_INVALID_PATH

```
#define CFE_FS_INVALID_PATH ((CFE_Status_t)0xc6000002)
```

Invalid Path.

FS was unable to extract a filename from a path string

Definition at line 671 of file cfe_error.h.

37.1.2.55 CFE_FS_NOT_IMPLEMENTED

```
#define CFE_FS_NOT_IMPLEMENTED ((CFE_Status_t)0xc600ffff)
```

Not Implemented.

Current version of cFE does not have the function or the feature of the function implemented. This could be due to either an early build of the cFE for this platform or the platform does not support the specified feature.

Definition at line 690 of file cfe_error.h.

37.1.2.56 CFE_SB_BAD_ARGUMENT

```
#define CFE_SB_BAD_ARGUMENT ((CFE_Status_t)0xca000003)
```

Bad Argument.

A parameter given by a caller to a Software Bus API did not pass validation checks.

Definition at line 721 of file cfe_error.h.

37.1.2.57 CFE_SB_BUF_ALOC_ERR

```
#define CFE_SB_BUF_ALOC_ERR ((CFE_Status_t)0xca000008)
```

Buffer Allocation Error.

Returned when the memory in the SB message buffer pool has been depleted. The amount of memory in the pool is dictated by the configuration parameter CFE_PLATFORM_SB_BUF_MEMORY_BYTES specified in the cfe_platform↩_cfg.h file. Also the memory statistics, including current utilization figures and high water marks for the SB Buffer memory pool can be monitored by sending a Software Bus command to send the SB statistics packet.

Definition at line 779 of file cfe_error.h.

37.1.2.58 CFE_SB_BUFFER_INVALID

```
#define CFE_SB_BUFFER_INVALID ((CFE_Status_t)0xca00000e)
```

Buffer Invalid.

This error code will be returned when a request to release or send a zero copy buffer is invalid, such as if the handle or buffer is not correct or the buffer was previously released.

Definition at line 839 of file cfe_error.h.

37.1.2.59 CFE_SB_INTERNAL_ERR

```
#define CFE_SB_INTERNAL_ERR ((CFE_Status_t)0xca00000c)
```

Internal Error.

This error code will be returned by the CFE_SB_Subscribe API if the code detects an internal index is out of range. The most likely cause would be a Single Event Upset.

Definition at line 820 of file cfe_error.h.

37.1.2.60 CFE_SB_MAX_DESTS_MET

```
#define CFE_SB_MAX_DESTS_MET ((CFE_Status_t)0xca00000a)
```

Max Destinations Met.

Will be returned when calling one of the SB subscription API's if the SB routing table cannot accomodate another destination for a particular the given message ID. This occurs when the number of destinations in use meets the platform configuration parameter CFE_PLATFORM_SB_MAX_DEST_PER_PKT.

Definition at line 801 of file cfe_error.h.

37.1.2.61 CFE_SB_MAX_MSGS_MET

```
#define CFE_SB_MAX_MSGS_MET ((CFE_Status_t)0xca000009)
```

Max Messages Met.

Will be returned when calling one of the SB subscription API's if the SB routing table cannot accomodate another unique message ID because the platform configuration parameter CFE_PLATFORM_SB_MAX_MSG_IDS has been met.

Definition at line 789 of file cfe_error.h.

37.1.2.62 CFE_SB_MAX_PIPES_MET

```
#define CFE_SB_MAX_PIPES_MET ((CFE_Status_t)0xca000004)
```

Max Pipes Met.

This error code will be returned from CFE_SB_CreatePipe when the SB cannot accomodate the request to create a pipe because the maximum number of pipes (CFE_PLATFORM_SB_MAX_PIPES) are in use. This configuration parameter is defined in the cfe_platform_cfg.h file.

Definition at line 732 of file cfe_error.h.

37.1.2.63 CFE_SB_MSG_TOO_BIG

```
#define CFE_SB_MSG_TOO_BIG ((CFE_Status_t)0xca000007)
```

Message Too Big.

The size field in the message header indicates the message exceeds the max Software Bus message size. The max size is defined by configuration parameter CFE_MISSION_SB_MAX_SB_MSG_SIZE in cfe_mission_cfg.h

Definition at line 766 of file cfe_error.h.

37.1.2.64 CFE_SB_NO_MESSAGE

```
#define CFE_SB_NO_MESSAGE ((CFE_Status_t)0xca000002)
```

No Message.

When "Polling" a pipe for a message in CFE_SB_ReceiveBuffer, this return value indicates that there was not a message on the pipe.

Definition at line 712 of file cfe_error.h.

37.1.2.65 CFE_SB_NO_MSG_RECV

```
#define CFE_SB_NO_MSG_RECV ((CFE_Status_t)0xca00000f)
```

No Message Recieved.

When trying to determine the last senders ID, this return value indicates that there was not a message recived on the pipe.

Definition at line 848 of file cfe_error.h.

37.1.2.66 CFE_SB_NO_SUBSCRIBERS

```
#define CFE_SB_NO_SUBSCRIBERS ((CFE_Status_t)0xca00000b)
```

No Subscribers.

This error code is returned by the CFE_SB_Unsubscribe API if there has not been an entry in the routing tables for the MsgId/PipeId given as parameters.

Definition at line 810 of file cfe_error.h.

37.1.2.67 CFE_SB_NOT_IMPLEMENTED

```
#define CFE_SB_NOT_IMPLEMENTED ((CFE_Status_t)0xca00ffff)
```

Not Implemented.

Current version of cFE does not have the function or the feature of the function implemented. This could be due to either an early build of the cFE for this platform or the platform does not support the specified feature.

Definition at line 859 of file cfe_error.h.

37.1.2.68 CFE_SB_PIPE_CR_ERR

```
#define CFE_SB_PIPE_CR_ERR ((CFE_Status_t)0xca000005)
```

Pipe Create Error.

The maximum number of queues(OS_MAX_QUEUES) are in use. Or possibly a lower level problem with creating the underlying queue has occurred such as a lack of memory. If the latter is the problem, the status code displayed in the event must be tracked.

Definition at line 743 of file cfe_error.h.

37.1.2.69 CFE_SB_PIPE_RD_ERR

```
#define CFE_SB_PIPE_RD_ERR ((CFE_Status_t)0xca000006)
```

Pipe Read Error.

This return value indicates an error at the Queue read level. This error typically cannot be corrected by the caller. Some possible causes are: queue was not properly initialized or created, the number of bytes read from the queue was not the number of bytes requested in the read. The queue id is invalid. Similar errors regarding the pipe will be caught by higher level code in the Software Bus.

Definition at line 756 of file cfe_error.h.

37.1.2.70 CFE_SB_TIME_OUT

```
#define CFE_SB_TIME_OUT ((CFE_Status_t)0xca000001)
```

Time Out.

In CFE_SB_ReceiveBuffer, this return value indicates that a packet has not been received in the time given in the "timeout" parameter.

Definition at line 703 of file cfe_error.h.

37.1.2.71 CFE_SB_WRONG_MSG_TYPE

```
#define CFE_SB_WRONG_MSG_TYPE ((CFE_Status_t)0xca00000d)
```

Wrong Message Type.

This error code will be returned when a request such as CFE_MSG_SetMsgTime is made on a packet that does not include a field for msg time.

Definition at line 829 of file cfe_error.h.

37.1.2.72 CFE_STATUS_BAD_COMMAND_CODE

```
#define CFE_STATUS_BAD_COMMAND_CODE ((CFE_Status_t)0xc8000004)
```

Bad Command Code.

This error code will be returned when a message identification process determined that the command code is does not correspond to any known value

Definition at line 160 of file cfe_error.h.

37.1.2.73 CFE_STATUS_EXTERNAL_RESOURCE_FAIL

```
#define CFE_STATUS_EXTERNAL_RESOURCE_FAIL ((CFE_Status_t)0xc8000005)
```

External failure.

This error indicates that the operation failed for some reason outside the scope of CFE. The real failure may have been in OSAL, PSP, or another dependent library.

Details of the original failure should be written to syslog and/or a system event before returning this error.

Definition at line 172 of file cfe_error.h.

37.1.2.74 CFE_STATUS_NO_COUNTER_INCREMENT

```
#define CFE_STATUS_NO_COUNTER_INCREMENT ((CFE_Status_t)0x48000001)
```

No Counter Increment.

Informational code indicating that a command was processed successfully but that the command counter should *not* be incremented.

Definition at line 133 of file cfe_error.h.

37.1.2.75 CFE_STATUS_NOT_IMPLEMENTED

```
#define CFE_STATUS_NOT_IMPLEMENTED ((CFE_Status_t)0xc800ffff)
```

Not Implemented.

Current version does not have the function or the feature of the function implemented. This could be due to either an early build for this platform or the platform does not support the specified feature.

Definition at line 192 of file cfe_error.h.

37.1.2.76 CFE_STATUS_REQUEST_ALREADY_PENDING

```
#define CFE_STATUS_REQUEST_ALREADY_PENDING ((int32)0xc8000006)
```

Request already pending.

Commands or requests are already pending or the pending request limit has been reached. No more requests can be made until the current request(s) complete.

Definition at line 181 of file cfe_error.h.

37.1.2.77 CFE_STATUS_UNKNOWN_MSG_ID

```
#define CFE_STATUS_UNKNOWN_MSG_ID ((CFE_Status_t)0xc8000003)
```

Unknown Message ID.

This error code will be returned when a message identification process determined that the message ID does not correspond to a known value

Definition at line 151 of file cfe_error.h.

37.1.2.78 CFE_STATUS_WRONG_MSG_LENGTH

```
#define CFE_STATUS_WRONG_MSG_LENGTH ((CFE_Status_t)0xc8000002)
```

Wrong Message Length.

This error code will be returned when a message validation process determined that the message length is incorrect

Definition at line 142 of file cfe_error.h.

37.1.2.79 CFE_SUCCESS

```
#define CFE_SUCCESS (0)
```

Sucessful execution.

Operation was performed successfully

Definition at line 125 of file cfe_error.h.

37.1.2.80 CFE_TBL_BAD_ARGUMENT

```
#define CFE_TBL_BAD_ARGUMENT ((CFE_Status_t)0xcc00002d)
```

Bad Argument.

A parameter given by a caller to a Table API did not pass validation checks.

Definition at line 1288 of file cfe_error.h.

37.1.2.81 CFE_TBL_ERR_ACCESS

```
#define CFE_TBL_ERR_ACCESS ((CFE_Status_t)0xcc00002c)
```

Error code indicating that the TBL file could not be opened by the OS.

Definition at line 1279 of file cfe_error.h.

37.1.2.82 CFE_TBL_ERR_BAD_APP_ID

```
#define CFE_TBL_ERR_BAD_APP_ID ((CFE_Status_t)0xcc00000A)
```

Bad Application ID.

The calling application does not have a legitimate Application ID. Most likely cause is a failure to register with the cFE via the CFE_ES_RegisterApp function.

Definition at line 957 of file cfe_error.h.

37.1.2.83 CFE_TBL_ERR_BAD_CONTENT_ID

```
#define CFE_TBL_ERR_BAD_CONTENT_ID ((CFE_Status_t)0xcc000016)
```

Bad Content ID.

The calling Application called CFE_TBL_Load with a filename that specified a file whose content ID was not that of a table image.

Definition at line 1071 of file cfe_error.h.

37.1.2.84 CFE_TBL_ERR_BAD_PROCESSOR_ID

```
#define CFE_TBL_ERR_BAD_PROCESSOR_ID ((CFE_Status_t)0xcc000029)
```

Bad Processor ID.

The selected table file failed validation for Processor ID. The platform configuration file has verification of table files enabled for Processor ID and an attempt was made to load a table with an invalid Processor ID in the table file header.

Definition at line 1259 of file cfe_error.h.

37.1.2.85 CFE_TBL_ERR_BAD_SPACECRAFT_ID

```
#define CFE_TBL_ERR_BAD_SPACECRAFT_ID ((CFE_Status_t)0xcc000028)
```

Bad Spacecraft ID.

The selected table file failed validation for Spacecraft ID. The platform configuration file has verification of table files enabled for Spacecraft ID and an attempt was made to load a table with an invalid Spacecraft ID in the table file header.

Definition at line 1248 of file cfe_error.h.

37.1.2.86 CFE_TBL_ERR_BAD_SUBTYPE_ID

```
#define CFE_TBL_ERR_BAD_SUBTYPE_ID ((CFE_Status_t)0xcc00001B)
```

Bad Subtype ID.

The calling Application tried to access a table file whose Subtype identifier indicated it was not a table image file.

Definition at line 1112 of file cfe_error.h.

37.1.2.87 CFE_TBL_ERR_DUMP_ONLY

```
#define CFE_TBL_ERR_DUMP_ONLY ((CFE_Status_t)0xcc000010)
```

Dump Only Error.

The calling Application has attempted to perform a load on a table that was created with "Dump Only" attributes.

Definition at line 1015 of file cfe_error.h.

37.1.2.88 CFE_TBL_ERR_DUPLICATE_DIFF_SIZE

```
#define CFE_TBL_ERR_DUPLICATE_DIFF_SIZE ((CFE_Status_t)0xcc00000C)
```

Duplicate Table With Different Size.

An application attempted to register a table with the same name as a table that is already in the registry. The size of the new table is different from the size already in the registry.

Definition at line 976 of file cfe_error.h.

37.1.2.89 CFE_TBL_ERR_DUPLICATE_NOT_OWNED

```
#define CFE_TBL_ERR_DUPLICATE_NOT_OWNED ((CFE_Status_t)0xcc00000D)
```

Dupicate Table And Not Owned.

An application attempted to register a table with the same name as a table that is already in the registry. The previously registered table is owned by a different application.

Definition at line 986 of file cfe_error.h.

37.1.2.90 CFE_TBL_ERR_FILE_FOR_WRONG_TABLE

```
#define CFE_TBL_ERR_FILE_FOR_WRONG_TABLE ((CFE_Status_t)0xcc000020)
```

File For Wrong Table.

The calling Application tried to load a table using a file whose header indicated that it was for a different table.

Definition at line 1156 of file cfe_error.h.

37.1.2.91 CFE_TBL_ERR_FILE_NOT_FOUND

```
#define CFE_TBL_ERR_FILE_NOT_FOUND ((CFE_Status_t)0xcc000013)
```

File Not Found.

The calling Application called CFE_TBL_Load with a bad filename.

Definition at line 1041 of file cfe_error.h.

37.1.2.92 CFE_TBL_ERR_FILE_SIZE_INCONSISTENT

```
#define CFE_TBL_ERR_FILE_SIZE_INCONSISTENT ((CFE_Status_t)0xcc00001C)
```

File Size Inconsistent.

The calling Application tried to access a table file whose Subtype identifier indicated it was not a table image file.

Definition at line 1121 of file cfe_error.h.

37.1.2.93 CFE_TBL_ERR_FILE_TOO_LARGE

```
#define CFE_TBL_ERR_FILE_TOO_LARGE ((CFE_Status_t)0xcc000014)
```

File Too Large.

The calling Application called CFE_TBL_Load with a filename that specified a file that contained more data than the size of the table OR which contained more data than specified in the table header.

Definition at line 1051 of file cfe_error.h.

37.1.2.94 CFE_TBL_ERR_FILENAME_TOO_LONG

```
#define CFE_TBL_ERR_FILENAME_TOO_LONG ((CFE_Status_t)0xcc00001F)
```

Filename Too Long.

The calling Application tried to load a table using a filename that was too long.

Definition at line 1147 of file cfe_error.h.

37.1.2.95 CFE_TBL_ERR_HANDLES_FULL

```
#define CFE_TBL_ERR_HANDLES_FULL ((CFE_Status_t)0xcc00000B)
```

Handles Full.

An application attempted to create a table and the Table Handle Array already used all CFE_PLATFORM_TBL_MAX↩ _NUM_HANDLES in it.

Definition at line 966 of file cfe_error.h.

37.1.2.96 CFE_TBL_ERR_ILLEGAL_SRC_TYPE

```
#define CFE_TBL_ERR_ILLEGAL_SRC_TYPE ((CFE_Status_t)0xcc000011)
```

Illegal Source Type.

The calling Application called CFE_TBL_Load with an illegal value for the second parameter.

Definition at line 1024 of file cfe_error.h.

37.1.2.97 CFE_TBL_ERR_INVALID_HANDLE

```
#define CFE_TBL_ERR_INVALID_HANDLE ((CFE_Status_t)0xcc000001)
```

Invalid Handle.

The calling Application attempted to pass a Table handle that represented too large an index or identified a Table Access Descriptor that was not used.

Definition at line 873 of file cfe_error.h.

37.1.2.98 CFE_TBL_ERR_INVALID_NAME

```
#define CFE_TBL_ERR_INVALID_NAME ((CFE_Status_t)0xcc000002)
```

Invalid Name.

The calling Application attempted to register a table whose name length exceeded the platform configuration value of CFE_MISSION_TBL_MAX_NAME_LENGTH or was zero characters long.

Definition at line 883 of file cfe_error.h.

37.1.2.99 CFE_TBL_ERR_INVALID_OPTIONS

```
#define CFE_TBL_ERR_INVALID_OPTIONS ((CFE_Status_t)0xcc000025)
```

Invalid Options.

The calling Application has used an illegal combination of table options. A summary of the illegal combinations are as follows:

#CFE_TBL_OPT_USR_DEF_ADDR cannot be combined with any of the following:

1. CFE_TBL_OPT_DBL_BUFFER
2. CFE_TBL_OPT_LOAD_DUMP
3. CFE_TBL_OPT_CRITICAL

#CFE_TBL_OPT_DBL_BUFFER cannot be combined with the following:

1. CFE_TBL_OPT_USR_DEF_ADDR
2. CFE_TBL_OPT_DUMP_ONLY

Definition at line 1213 of file cfe_error.h.

37.1.2.100 CFE_TBL_ERR_INVALID_SIZE

```
#define CFE_TBL_ERR_INVALID_SIZE ((CFE_Status_t)0xcc000003)
```

Invalid Size.

The calling Application attempted to register a table: a) that was a double buffered table with size greater than CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE b) that was a single buffered table with size greater than CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE c) that had a size of zero

Definition at line 894 of file cfe_error.h.

37.1.2.101 CFE_TBL_ERR_LOAD_IN_PROGRESS

```
#define CFE_TBL_ERR_LOAD_IN_PROGRESS ((CFE_Status_t)0xcc000012)
```

Load In Progress.

The calling Application called CFE_TBL_Load when another Application was trying to load the table.

Definition at line 1033 of file cfe_error.h.

37.1.2.102 CFE_TBL_ERR_LOAD_INCOMPLETE

```
#define CFE_TBL_ERR_LOAD_INCOMPLETE ((CFE_Status_t)0xcc000021)
```

Load Incomplete.

The calling Application tried to load a table file whose header claimed the load was larger than what was actually read from the file.

Definition at line 1165 of file cfe_error.h.

37.1.2.103 CFE_TBL_ERR_NEVER_LOADED

```
#define CFE_TBL_ERR_NEVER_LOADED ((CFE_Status_t)0xcc000005)
```

Never Loaded.

Table has not been loaded with data.

Definition at line 910 of file cfe_error.h.

37.1.2.104 CFE_TBL_ERR_NO_ACCESS

```
#define CFE_TBL_ERR_NO_ACCESS ((CFE_Status_t)0xcc000008)
```

No Access.

The calling application either failed when calling CFE_TBL_Register, failed when calling CFE_TBL_Share or forgot to call either one.

Definition at line 938 of file cfe_error.h.

37.1.2.105 CFE_TBL_ERR_NO_BUFFER_AVAIL

```
#define CFE_TBL_ERR_NO_BUFFER_AVAIL ((CFE_Status_t)0xcc00000F)
```

No Buffer Available.

The calling Application has tried to allocate a working buffer but none were available.

Definition at line 1006 of file cfe_error.h.

37.1.2.106 CFE_TBL_ERR_NO_STD_HEADER

```
#define CFE_TBL_ERR_NO_STD_HEADER ((CFE_Status_t)0xcc00001D)
```

No Standard Header.

The calling Application tried to access a table file whose standard cFE File Header was the wrong size, etc.

Definition at line 1129 of file cfe_error.h.

37.1.2.107 CFE_TBL_ERR_NO_TBL_HEADER

```
#define CFE_TBL_ERR_NO_TBL_HEADER ((CFE_Status_t)0xcc00001E)
```

No Table Header.

The calling Application tried to access a table file whose standard cFE Table File Header was the wrong size, etc.

Definition at line 1138 of file cfe_error.h.

37.1.2.108 CFE_TBL_ERR_PARTIAL_LOAD

```
#define CFE_TBL_ERR_PARTIAL_LOAD ((CFE_Status_t)0xcc000023)
```

Partial Load Error.

The calling Application tried to load a table file whose header claimed the load did not start with the first byte and the table image had NEVER been loaded before. Partial loads are not allowed on uninitialized tables. It should be noted that CFE_TBL_WARN_SHORT_FILE also indicates a partial load.

Definition at line 1187 of file cfe_error.h.

37.1.2.109 CFE_TBL_ERR_REGISTRY_FULL

```
#define CFE_TBL_ERR_REGISTRY_FULL ((CFE_Status_t)0xcc000006)
```

Registry Full.

An application attempted to create a table and the Table registry already contained CFE_PLATFORM_TBL_MAX_N↩UM_TABLES in it.

Definition at line 919 of file cfe_error.h.

37.1.2.110 CFE_TBL_ERR_SHORT_FILE

```
#define CFE_TBL_ERR_SHORT_FILE ((CFE_Status_t)0xcc00002b)
```

Error code indicating that the TBL file is shorter than indicated in the file header.

Definition at line 1273 of file cfe_error.h.

37.1.2.111 CFE_TBL_ERR_UNREGISTERED

```
#define CFE_TBL_ERR_UNREGISTERED ((CFE_Status_t)0xcc000009)
```

Unregistered.

The calling application is trying to access a table that has been unregistered.

Definition at line 947 of file cfe_error.h.

37.1.2.112 CFE_TBL_INFO_DUMP_PENDING

```
#define CFE_TBL_INFO_DUMP_PENDING ((CFE_Status_t)0x4c000024)
```

Dump Pending.

The calling Application should call CFE_TBL_Manage for the specified table. The ground has requested a dump of the Dump-Only table and needs to synchronize with the owning application.

Definition at line 1197 of file cfe_error.h.

37.1.2.113 CFE_TBL_INFO_NO_UPDATE_PENDING

```
#define CFE_TBL_INFO_NO_UPDATE_PENDING ((CFE_Status_t)0x4c000017)
```

No Update Pending.

The calling Application has attempted to update a table without a pending load.

Definition at line 1079 of file cfe_error.h.

37.1.2.114 CFE_TBL_INFO_NO_VALIDATION_PENDING

```
#define CFE_TBL_INFO_NO_VALIDATION_PENDING ((CFE_Status_t)0x4c00001A)
```

No Validation Pending

The calling Application tried to validate a table that did not have a validation request pending.

Definition at line 1103 of file cfe_error.h.

37.1.2.115 CFE_TBL_INFO_RECOVERED_TBL

```
#define CFE_TBL_INFO_RECOVERED_TBL ((CFE_Status_t)0x4c000027)
```

Recovered Table.

The calling Application registered a critical table whose previous contents were discovered in the Critical Data Store. The discovered contents were copied back into the newly registered table as the table's initial contents.
NOTE: In this situation, the contents of the table are **NOT** validated using the table's validation function.

Definition at line 1237 of file cfe_error.h.

37.1.2.116 CFE_TBL_INFO_TABLE_LOCKED

```
#define CFE_TBL_INFO_TABLE_LOCKED ((CFE_Status_t)0x4c000018)
```

Table Locked.

The calling Application tried to update a table that is locked by another user.

Definition at line 1087 of file cfe_error.h.

37.1.2.117 CFE_TBL_INFO_UPDATE_PENDING

```
#define CFE_TBL_INFO_UPDATE_PENDING ((CFE_Status_t)0x4c000004)
```

Update Pending.

The calling Application has identified a table that has a load pending.

Definition at line 902 of file cfe_error.h.

37.1.2.118 CFE_TBL_INFO_UPDATED

```
#define CFE_TBL_INFO_UPDATED ((CFE_Status_t)0x4c00000E)
```

Updated.

The calling Application has identified a table that has been updated.
NOTE: This is a nominal return code informing the calling application that the table identified in the call has had its contents updated since the last time the application obtained its address or status.

Definition at line 997 of file cfe_error.h.

37.1.2.119 CFE_TBL_INFO_VALIDATION_PENDING

```
#define CFE_TBL_INFO_VALIDATION_PENDING ((CFE_Status_t)0x4c000019)
```

Validation Pending

The calling Application should call CFE_TBL_Validate for the specified table.

Definition at line 1095 of file cfe_error.h.

37.1.2.120 CFE_TBL_MESSAGE_ERROR

```
#define CFE_TBL_MESSAGE_ERROR ((CFE_Status_t)0xcc00002a)
```

Message Error.

Error code indicating that the TBL command was not processed successfully and that the error counter should be incremented.

Definition at line 1267 of file cfe_error.h.

37.1.2.121 CFE_TBL_NOT_IMPLEMENTED

```
#define CFE_TBL_NOT_IMPLEMENTED ((CFE_Status_t)0xcc00ffff)
```

Not Implemented.

Current version of cFE does not have the function or the feature of the function implemented. This could be due to either an early build of the cFE for this platform or the platform does not support the specified feature.

Definition at line 1299 of file cfe_error.h.

37.1.2.122 CFE_TBL_WARN_DUPLICATE

```
#define CFE_TBL_WARN_DUPLICATE ((CFE_Status_t)0x4c000007)
```

Duplicate Warning.

This is an error that the registration is trying to replace an existing table with the same name. The previous table stays in place and the new table is rejected.

Definition at line 929 of file cfe_error.h.

37.1.2.123 CFE_TBL_WARN_NOT_CRITICAL

```
#define CFE_TBL_WARN_NOT_CRITICAL ((CFE_Status_t)0x4c000026)
```

Not Critical Warning.

The calling Application attempted to register a table as "Critical". Table Services failed to create an appropriate Critical Data Store (See System Log for reason) to save the table contents. The table will be treated as a normal table from now on.

Definition at line 1224 of file cfe_error.h.

37.1.2.124 CFE_TBL_WARN_PARTIAL_LOAD

```
#define CFE_TBL_WARN_PARTIAL_LOAD ((CFE_Status_t)0x4c000022)
```

Partial Load Warning.

The calling Application tried to load a table file whose header claimed the load did not start with the first byteIt should be noted that CFE_TBL_WARN_SHORT_FILE also indicates a partial load.

Definition at line 1175 of file cfe_error.h.

37.1.2.125 CFE_TBL_WARN_SHORT_FILE

```
#define CFE_TBL_WARN_SHORT_FILE ((CFE_Status_t)0x4c000015)
```

Short File Warning.

The calling Application called CFE_TBL_Load with a filename that specified a file that started with the first byte of the table but contained less data than the size of the table. It should be noted that CFE_TBL_WARN_PARTIAL_LOAD also indicates a partial load (one that starts at a non-zero offset).

Definition at line 1062 of file cfe_error.h.

37.1.2.126 CFE_TIME_BAD_ARGUMENT

```
#define CFE_TIME_BAD_ARGUMENT ((CFE_Status_t)0xce000005)
```

Bad Argument.

A parameter given by a caller to a TIME Services API did not pass validation checks.

Definition at line 1371 of file cfe_error.h.

37.1.2.127 CFE_TIME_CALLBACK_NOT_REGISTERED

```
#define CFE_TIME_CALLBACK_NOT_REGISTERED ((CFE_Status_t)0xce000004)
```

Callback Not Registered.

An attempt to unregister a cFE Time Services Synchronization callback has failed because the specified callback function was not located in the Synchronization Callback Registry.

Definition at line 1362 of file cfe_error.h.

37.1.2.128 CFE_TIME_INTERNAL_ONLY

```
#define CFE_TIME_INTERNAL_ONLY ((CFE_Status_t)0xce000001)
```

Internal Only.

One of the TIME Services API functions to set the time with data from an external time source has been called, but TIME Services has been commanded to not accept external time data. However, the command is still a signal for the Time Server to generate a "time at the tone" command packet using internal data.

Definition at line 1326 of file cfe_error.h.

37.1.2.129 CFE_TIME_NOT_IMPLEMENTED

```
#define CFE_TIME_NOT_IMPLEMENTED ((CFE_Status_t)0xce00ffff)
```

Not Implemented.

Current version of cFE does not have the function or the feature of the function implemented. This could be due to either an early build of the cFE for this platform or the platform does not support the specified feature.

Definition at line 1314 of file cfe_error.h.

37.1.2.130 CFE_TIME_OUT_OF_RANGE

```
#define CFE_TIME_OUT_OF_RANGE ((CFE_Status_t)0xce000002)
```

Out Of Range.

One of the TIME Services API functions to set the time with data from an external time source has been called, but TIME Services has determined that the new time data is invalid. However, the command is still a signal for the Time Server to generate a "time at the tone" command packet using internal data.

Note that the test for invalid time update data only occurs if TIME Services has previously been commanded to set the clock state to "valid".

Definition at line 1341 of file cfe_error.h.

37.1.2.131 CFE_TIME_TOO_MANY_SYNCH_CALLBACKS

```
#define CFE_TIME_TOO_MANY_SYNCH_CALLBACKS ((CFE_Status_t)0xce000003)
```

Too Many Sync Callbacks.

An attempt to register too many cFE Time Services Synchronization callbacks has been made. Only one callback function is allowed per application. It is expected that the application itself will distribute the single callback to child threads as needed.

Definition at line 1352 of file cfe_error.h.

37.2 cFE Resource ID APIs

Functions

- CFE_Status_t CFE_ES_AppID_ToIndex (CFE_ES_AppId_t AppID, uint32 *Idx)

  Obtain an index value correlating to an ES Application ID.
- int32 CFE_ES_LibID_ToIndex (CFE_ES_LibId_t LibID, uint32 *Idx)

  Obtain an index value correlating to an ES Library ID.
- CFE_Status_t CFE_ES_TaskID_ToIndex (CFE_ES_TaskId_t TaskID, uint32 *Idx)

  Obtain an index value correlating to an ES Task ID.
- CFE_Status_t CFE_ES_CounterID_ToIndex (CFE_ES_CounterId_t CounterID, uint32 *Idx)

  Obtain an index value correlating to an ES Counter ID.

37.2.1 Detailed Description

37.2.2 Function Documentation

37.2.2.1 CFE_ES_AppID_ToIndex()

```
CFE_Status_t CFE_ES_AppID_ToIndex (
            CFE_ES_AppId_t AppID,
            uint32 * Idx )
```

Obtain an index value correlating to an ES Application ID.

This calculates a zero based integer value that may be used for indexing into a local resource table/array.

Index values are only guaranteed to be unique for resources of the same type. For instance, the indices corresponding to two [valid] application IDs will never overlap, but the index of an application and a library ID may be the same. Furthermore, indices may be reused if a resource is deleted and re-created.

Note

There is no inverse of this function - indices cannot be converted back to the original AppID value. The caller should retain the original ID for future use.

Parameters

in	*AppID*	Application ID to convert
out	*Idx*	Buffer where the calculated index will be stored

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

37.2.2.2 CFE_ES_CounterID_ToIndex()

```
CFE_Status_t CFE_ES_CounterID_ToIndex (
            CFE_ES_CounterId_t CounterID,
            uint32 * Idx )
```

Obtain an index value correlating to an ES Counter ID.

This calculates a zero based integer value that may be used for indexing into a local resource table/array.

Index values are only guaranteed to be unique for resources of the same type. For instance, the indices corresponding to two [valid] Counter IDs will never overlap, but the index of an Counter and a library ID may be the same. Furthermore, indices may be reused if a resource is deleted and re-created.

Note

There is no inverse of this function - indices cannot be converted back to the original CounterID value. The caller should retain the original ID for future use.

Parameters

in	*CounterID*	Counter ID to convert
out	*Idx*	Buffer where the calculated index will be stored

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

37.2.2.3 CFE_ES_LibID_ToIndex()

```
int32 CFE_ES_LibID_ToIndex (
            CFE_ES_LibId_t LibID,
            uint32 * Idx )
```

Obtain an index value correlating to an ES Library ID.

This calculates a zero based integer value that may be used for indexing into a local resource table/array.

Index values are only guaranteed to be unique for resources of the same type. For instance, the indices corresponding to two [valid] Library IDs will never overlap, but the index of an Library and a library ID may be the same. Furthermore, indices may be reused if a resource is deleted and re-created.

Note

There is no inverse of this function - indices cannot be converted back to the original LibID value. The caller should retain the original ID for future use.

Parameters

in	*LibID*	Library ID to convert
out	*Idx*	Buffer where the calculated index will be stored

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

37.2.2.4 CFE_ES_TaskID_ToIndex()

```
CFE_Status_t CFE_ES_TaskID_ToIndex (
            CFE_ES_TaskId_t TaskID,
            uint32 * Idx )
```

Obtain an index value correlating to an ES Task ID.

This calculates a zero based integer value that may be used for indexing into a local resource table/array.

Index values are only guaranteed to be unique for resources of the same type. For instance, the indices corresponding to two [valid] Task IDs will never overlap, but the index of an Task and a library ID may be the same. Furthermore, indices may be reused if a resource is deleted and re-created.

Note

There is no inverse of this function - indices cannot be converted back to the original TaskID value. The caller should retain the original ID for future use.

Parameters

in	*TaskID*	Task ID to convert
out	*Idx*	Buffer where the calculated index will be stored

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

37.3 cFE Entry/Exit APIs

Functions

- void CFE_ES_Main (uint32 StartType, uint32 StartSubtype, uint32 ModeId, const char *StartFilePath)

  cFE Main Entry Point used by Board Support Package to start cFE
- CFE_Status_t CFE_ES_ResetCFE (uint32 ResetType)

  Reset the cFE Core and all cFE Applications.

37.3.1 Detailed Description

37.3.2 Function Documentation

37.3.2.1 CFE_ES_Main()

```
void CFE_ES_Main (
            uint32 StartType,
            uint32 StartSubtype,
            uint32 ModeId,
            const char * StartFilePath )
```

cFE Main Entry Point used by Board Support Package to start cFE

Description

cFE main entry point. This is the entry point into the cFE software. It is called only by the Board Support Package software.

Assumptions, External Events, and Notes:

None

Parameters

in	*StartType*	Identifies whether this was a CFE_PSP_RST_TYPE_POWERON or CFE_PSP_RST_TYPE_PROCESSOR.
in	*StartSubtype*	Specifies, in more detail, what caused the `StartType` identified above. See CFE_PSP_RST_SUBTYPE_POWER_CYCLE for possible examples.
in	*ModeId*	Identifies the source of the Boot as determined by the BSP.
in	*StartFilePath*	Identifies the startup file to use to initialize the cFE apps.

See also

CFE_ES_ResetCFE

37.3.2.2 CFE_ES_ResetCFE()

```
CFE_Status_t CFE_ES_ResetCFE (
            uint32 ResetType )
```

Reset the cFE Core and all cFE Applications.

Description

This API causes an immediate reset of the cFE Kernel and all cFE Applications. The caller can specify whether the reset should clear all memory (CFE_PSP_RST_TYPE_POWERON) or try to retain volatile memory areas (CFE↩_PSP_RST_TYPE_PROCESSOR).

Assumptions, External Events, and Notes:

None

Parameters

in	*ResetType*	Identifies the type of reset desired. Allowable settings are: • CFE_PSP_RST_TYPE_POWERON - Causes all memory to be cleared • CFE_PSP_RST_TYPE_PROCESSOR - Attempts to retain volatile disk, critical data store and user reserved memory.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.
CFE_ES_NOT_IMPLEMENTED	Not Implemented.

See also

CFE_ES_Main

37.4 cFE Application Control APIs

Functions

- CFE_Status_t CFE_ES_RestartApp (CFE_ES_AppId_t AppID)

  Restart a single cFE Application.
- CFE_Status_t CFE_ES_ReloadApp (CFE_ES_AppId_t AppID, const char *AppFileName)

  Reload a single cFE Application.
- CFE_Status_t CFE_ES_DeleteApp (CFE_ES_AppId_t AppID)

  Delete a cFE Application.

37.4.1 Detailed Description

37.4.2 Function Documentation

37.4.2.1 CFE_ES_DeleteApp()

```
CFE_Status_t CFE_ES_DeleteApp (
            CFE_ES_AppId_t AppID )
```

Delete a cFE Application.

Description

This API causes a cFE Application to be stopped deleted.

Assumptions, External Events, and Notes:

None

Parameters

in	*AppID*	Identifies the application to be reset.

Returns

Execution status, see cFE Return Code Defines

See also

CFE_ES_RestartApp, CFE_ES_ReloadApp

37.4.2.2 CFE_ES_ReloadApp()

```
CFE_Status_t CFE_ES_ReloadApp (
            CFE_ES_AppId_t AppID,
            const char * AppFileName )
```

Reload a single cFE Application.

Description

This API causes a cFE Application to be stopped and restarted from the specified file.

Assumptions, External Events, and Notes:

The filename is checked for existance prior to load. A missing file will be reported and the reload operation will be aborted prior to unloading the app.

Goes through the standard CFE_ES_CleanUpApp which unloads, then attempts a load using the specified file name.

In the event that an application cannot be reloaded due to a corrupt file, the application may no longer be reloaded when given a valid load file (it has been deleted and no longer exists). To recover, the application may be started by loading the application via the ES_STARTAPP command (CFE_ES_START_APP_CC).

Parameters

in	*AppID*	Identifies the application to be reset.
in	*AppFileName*	Identifies the new file to start.

Returns

Execution status, see cFE Return Code Defines

See also

CFE_ES_RestartApp, CFE_ES_DeleteApp, CFE_ES_START_APP_CC

37.4.2.3 CFE_ES_RestartApp()

```
CFE_Status_t CFE_ES_RestartApp (
            CFE_ES_AppId_t AppID )
```

Restart a single cFE Application.

Description

This API causes a cFE Application to be unloaded and restarted from the same file name as the last start.

Assumptions, External Events, and Notes:

The filename is checked for existance prior to load. A missing file will be reported and the reload operation will be aborted prior to unloading the app.

Goes through the standard CFE_ES_CleanUpApp which unloads, then attempts a load using the original file name.

In the event that an application cannot be reloaded due to a missing file or any other load issue, the application may no longer be restarted or reloaded when given a valid load file (the app has been deleted and no longer exists). To recover, the application may be started by loading the application via the ES_STARTAPP command (CFE_ES_START_APP↩_CC).

Parameters

in	*AppID*	Identifies the application to be reset.

Returns

Execution status, see cFE Return Code Defines

See also

CFE_ES_ReloadApp, CFE_ES_DeleteApp

37.5 cFE Application Behavior APIs

Functions

- void CFE_ES_ExitApp (uint32 ExitStatus)

  Exit a cFE Application.
- bool CFE_ES_RunLoop (uint32 *ExitStatus)

  Check for Exit, Restart, or Reload commands.
- CFE_Status_t CFE_ES_WaitForSystemState (uint32 MinSystemState, uint32 TimeOutMilliseconds)

  Allow an Application to Wait for a minimum global system state.
- void CFE_ES_WaitForStartupSync (uint32 TimeOutMilliseconds)

  Allow an Application to Wait for the "OPERATIONAL" global system state.
- CFE_Status_t CFE_ES_RegisterApp (void)

  Registers a cFE Application with the Executive Services.
- void CFE_ES_IncrementTaskCounter (void)

  Increments the execution counter for the calling task.

37.5.1 Detailed Description

37.5.2 Function Documentation

37.5.2.1 CFE_ES_ExitApp()

```
void CFE_ES_ExitApp (
            uint32 ExitStatus )
```

Exit a cFE Application.

Description

This API is the "Exit Point" for the cFE application

Assumptions, External Events, and Notes:

None

Parameters

in	*ExitStatus*	Acceptable values are: • CFE_ES_RunStatus_APP_EXIT - Indicates that the Application wants to exit normally. • CFE_ES_RunStatus_APP_ERROR - Indicates that the Application is quitting with an error. • CFE_ES_RunStatus_CORE_APP_INIT_ERROR - Indicates that the Core Application could not Init. • CFE_ES_RunStatus_CORE_APP_RUNTIME_ERROR - Indicates that the Core Application had a runtime failure.

See also

CFE_ES_RunLoop, CFE_ES_RegisterApp

37.5.2.2 CFE_ES_IncrementTaskCounter()

```
void CFE_ES_IncrementTaskCounter (
            void  )
```

Increments the execution counter for the calling task.

Description

This routine increments the execution counter that is stored for the calling task. It can be called from cFE Application main tasks, child tasks, or cFE Core application main tasks. Normally, the call is not necessary from a cFE Application, since the CFE_ES_RunLoop call increments the counter for the Application.

Assumptions, External Events, and Notes:

NOTE: This API is not needed for Appplications that call the CFE_ES_RunLoop call.

See also

CFE_ES_RunLoop

37.5.2.3 CFE_ES_RegisterApp()

```
CFE_Status_t CFE_ES_RegisterApp (
            void  )
```

Registers a cFE Application with the Executive Services.

Description

This API registers the calling Application with the cFE.

Assumptions, External Events, and Notes:

NOTE: This function **MUST** be called before any other cFE API functions are called.

Returns

Execution status, see cFE Return Code Defines

See also

CFE_ES_ExitApp, CFE_ES_RunLoop

37.5.2.4 CFE_ES_RunLoop()

```
bool CFE_ES_RunLoop (
            uint32 * ExitStatus )
```

Check for Exit, Restart, or Reload commands.

Description

This is the API that allows an app to check for exit requests from the system.

Assumptions, External Events, and Notes:

None

Parameters

in	*ExitStatus*	A pointer to a variable containing the Application's desired run status. Acceptable values are: • CFE_ES_RunStatus_APP_RUN - Indicates that the Application should continue to run. • CFE_ES_RunStatus_APP_EXIT - Indicates that the Application wants to exit normally. • CFE_ES_RunStatus_APP_ERROR - Indicates that the Application is quitting with an error.

Returns

Boolean indicating application should continue running

Return values

true	Application should continue running
false	Application should not continue running

See also

CFE_ES_ExitApp, CFE_ES_RegisterApp

37.5.2.5 CFE_ES_WaitForStartupSync()

```
void CFE_ES_WaitForStartupSync (
            uint32 TimeOutMilliseconds )
```

Allow an Application to Wait for the "OPERATIONAL" global system state.

Description

This is the API that allows an app to wait for the rest of the apps to complete their entire initialization before continuing. It is most useful for applications such as Health and Safety or the Scheduler that need to wait until applications exist and are running before sending out packets to them.

This is a specialized wrapper for CFE_ES_WaitForSystemState for compatibility with applications using this API.

Assumptions, External Events, and Notes:

This API should only be called as the last item of an Apps initialization. In addition, this API should only be called by an App that is started from the ES Startup file. It should not be used by an App that is started after the system is running. (Although it will cause no harm)

Parameters

in	*TimeOutMilliseconds*	The timeout value in Milliseconds. This parameter must be at least 1000. Lower values will be rounded up. There is not an option to wait indefinitely to avoid hanging a critical application because a non-critical app did not start.

See also

CFE_ES_RunLoop

37.5.2.6 CFE_ES_WaitForSystemState()

```
CFE_Status_t CFE_ES_WaitForSystemState (
            uint32 MinSystemState,
            uint32 TimeOutMilliseconds )
```

Allow an Application to Wait for a minimum global system state.

Description

This is the API that allows an app to wait for the rest of the apps to complete a given stage of initialization before continuing.

This gives finer grained control than CFE_ES_WaitForStartupSync

Assumptions, External Events, and Notes:

This API assumes that the caller has also been initialized sufficiently to satisfy the global system state it is waiting for, and the apps own state will be updated accordingly.

Parameters

in	*TimeOutMilliseconds*	The timeout value in Milliseconds. This parameter must be at least 1000. Lower values will be rounded up. There is not an option to wait indefinitely to avoid hanging a critical application because a non-critical app did not start.
in	*MinSystemState*	Determine the state of the App

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	State successfully achieved
CFE_ES_OPERATION_TIMED_OUT	Timeout was reached

See also

CFE_ES_RunLoop

37.6 cFE Information APIs

Functions

- int32 CFE_ES_GetResetType (uint32 ∗ResetSubtypePtr)

  Return the most recent Reset Type.
- CFE_Status_t CFE_ES_GetAppID (CFE_ES_AppId_t ∗AppIdPtr)

  Get an Application ID for the calling Application.
- CFE_Status_t CFE_ES_GetTaskID (CFE_ES_TaskId_t ∗TaskIdPtr)

  Get the task ID of the calling context.
- CFE_Status_t CFE_ES_GetAppIDByName (CFE_ES_AppId_t ∗AppIdPtr, const char ∗AppName)

  Get an Application ID associated with a specified Application name.
- CFE_Status_t CFE_ES_GetLibIDByName (CFE_ES_LibId_t ∗LibIdPtr, const char ∗LibName)

  Get a Library ID associated with a specified Library name.
- CFE_Status_t CFE_ES_GetAppName (char ∗AppName, CFE_ES_AppId_t AppId, size_t BufferLength)

  Get an Application name for a specified Application ID.
- CFE_Status_t CFE_ES_GetLibName (char ∗LibName, CFE_ES_LibId_t LibId, size_t BufferLength)

  Get a Library name for a specified Library ID.
- CFE_Status_t CFE_ES_GetAppInfo (CFE_ES_AppInfo_t ∗AppInfo, CFE_ES_AppId_t AppId)

  Get Application Information given a specified App ID.
- CFE_Status_t CFE_ES_GetTaskInfo (CFE_ES_TaskInfo_t ∗TaskInfo, CFE_ES_TaskId_t TaskId)

  Get Task Information given a specified Task ID.
- int32 CFE_ES_GetLibInfo (CFE_ES_AppInfo_t ∗LibInfo, CFE_ES_LibId_t LibId)

  Get Library Information given a specified Resource ID.
- int32 CFE_ES_GetModuleInfo (CFE_ES_AppInfo_t ∗ModuleInfo, CFE_ResourceId_t ResourceId)

  Get Information given a specified Resource ID.

37.6.1 Detailed Description

37.6.2 Function Documentation

37.6.2.1 CFE_ES_GetAppID()

```
CFE_Status_t CFE_ES_GetAppID (
            CFE_ES_AppId_t * AppIdPtr )
```

Get an Application ID for the calling Application.

Description

This routine retrieves the cFE Application ID for the calling Application.

Assumptions, External Events, and Notes:

NOTE: **All** tasks associated with the Application would return the same Application ID.

Parameters

out	*AppIdPtr*	Pointer to variable that is to receive the Application's ID. ∗AppIdPtr will be set to the application ID of the calling Application.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_ERR_BUFFER	Invalid Pointer.

See also

CFE_ES_GetResetType, CFE_ES_GetAppIDByName, CFE_ES_GetAppName, CFE_ES_GetTaskInfo

37.6.2.2 CFE_ES_GetAppIDByName()

```
CFE_Status_t CFE_ES_GetAppIDByName (
            CFE_ES_AppId_t * AppIdPtr,
            const char * AppName )
```

Get an Application ID associated with a specified Application name.

Description

This routine retrieves the cFE Application ID associated with a specified Application name.

Assumptions, External Events, and Notes:

None

Parameters

out	*AppIdPtr*	Pointer to variable that is to receive the Application's ID.
in	*AppName*	Pointer to null terminated character string containing an Application name.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_NAME_NOT_FOUND	Resource Name Error.
CFE_ES_ERR_BUFFER	Invalid Pointer.

See also

CFE_ES_GetAppID, CFE_ES_GetAppName, CFE_ES_GetAppInfo

37.6.2.3 CFE_ES_GetAppInfo()

```
CFE_Status_t CFE_ES_GetAppInfo (
            CFE_ES_AppInfo_t * AppInfo,
            CFE_ES_AppId_t AppId )
```

Get Application Information given a specified App ID.

Description

This routine retrieves the information about an App associated with a specified App ID. The information includes all of the information ES maintains for an application (documented in the CFE_ES_AppInfo_t type)

Assumptions, External Events, and Notes:

None

Parameters

`out`	*AppInfo*	Pointer to a structure that will be filled with resource name and memory addresses information.
`in`	*AppId*	ID of application to obtain information about

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_ERR_BUFFER	Invalid Pointer.

See also

CFE_ES_GetAppID, CFE_ES_GetAppIDByName, CFE_ES_GetAppName

37.6.2.4 CFE_ES_GetAppName()

```
CFE_Status_t CFE_ES_GetAppName (
            char * AppName,
            CFE_ES_AppId_t AppId,
            size_t BufferLength )
```

Get an Application name for a specified Application ID.

Description

This routine retrieves the cFE Application name associated with a specified Application ID.

Assumptions, External Events, and Notes:

In the case of a failure (CFE_ES_ERR_RESOURCEID_NOT_VALID), an empty string is returned.

Parameters

out	*AppName*	Pointer to a character array of at least `BufferLength` in size that will be filled with the appropriate Application name.
in	*AppId*	Application ID of Application whose name is being requested.
in	*BufferLength*	The maximum number of characters, including the null terminator, that can be put into the `AppName` buffer. This routine will truncate the name to this length, if necessary.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

See also

CFE_ES_GetAppID, CFE_ES_GetAppIDByName, CFE_ES_GetAppInfo

37.6.2.5 CFE_ES_GetLibIDByName()

```
CFE_Status_t CFE_ES_GetLibIDByName (
            CFE_ES_LibId_t * LibIdPtr,
            const char * LibName )
```

Get a Library ID associated with a specified Library name.

Description

This routine retrieves the cFE Library ID associated with a specified Library name.

Assumptions, External Events, and Notes:

None

Parameters

out	*LibIdPtr*	Pointer to variable that is to receive the Library's ID.
in	*LibName*	Pointer to null terminated character string containing a Library name.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_NAME_NOT_FOUND	Resource Name Error.
CFE_ES_ERR_BUFFER	Invalid Pointer.

See also

CFE_ES_GetLibName

37.6.2.6 CFE_ES_GetLibInfo()

```
int32 CFE_ES_GetLibInfo (
            CFE_ES_AppInfo_t * LibInfo,
            CFE_ES_LibId_t LibId )
```

Get Library Information given a specified Resource ID.

Description

This routine retrieves the information about a Library associated with a specified ID. The information includes all of the information ES maintains for this resource type (documented in the CFE_ES_AppInfo_t type).

This shares the same output structure as CFE_ES_GetAppInfo, such that informational commands can be executed against either applications or libraries. When applied to a library, the task information in the structure will be omitted, as libraries do not have tasks associated.

Assumptions, External Events, and Notes:

None

Parameters

out	*LibInfo*	Pointer to a structure that will be filled with resource name and memory addresses information.
in	*LibId*	ID of application to obtain information about

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_ERR_BUFFER	Invalid Pointer.

See also

CFE_ES_GetLibIDByName, CFE_ES_GetLibName

37.6.2.7 CFE_ES_GetLibName()

```
CFE_Status_t CFE_ES_GetLibName (
            char * LibName,
            CFE_ES_LibId_t LibId,
            size_t BufferLength )
```

Get a Library name for a specified Library ID.

Description

This routine retrieves the cFE Library name associated with a specified Library ID.

Assumptions, External Events, and Notes:

In the case of a failure (CFE_ES_ERR_RESOURCEID_NOT_VALID), an empty string is returned.

Parameters

out	*LibName*	Pointer to a character array of at least `BufferLength` in size that will be filled with the Library name.
in	*LibId*	Library ID of Library whose name is being requested.
in	*BufferLength*	The maximum number of characters, including the null terminator, that can be put into the `LibName` buffer. This routine will truncate the name to this length, if necessary.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

See also

CFE_ES_GetLibIDByName

37.6.2.8 CFE_ES_GetModuleInfo()

```
int32 CFE_ES_GetModuleInfo (
            CFE_ES_AppInfo_t * ModuleInfo,
            CFE_ResourceId_t ResourceId )
```

Get Information given a specified Resource ID.

Description

This routine retrieves the information about an Application or Library associated with a specified ID.

This is a wrapper API that in turn calls either CFE_ES_GetAppInfo or CFE_ES_GetLibInfo if passed an AppId or LibId, respectively.

This allows commands originally targeted to operate on AppIDs to be easily ported to operate on either Libraries or Applications, where relevant.

Assumptions, External Events, and Notes:

None

Parameters

out	*ModuleInfo*	Pointer to a structure that will be filled with resource name and memory addresses information.
in	*ResourceId*	ID of application or library to obtain information about

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_ERR_BUFFER	Invalid Pointer.

See also

CFE_ES_GetLibInfo, CFE_ES_GetAppInfo

37.6.2.9 CFE_ES_GetResetType()

```
int32 CFE_ES_GetResetType (
            uint32 * ResetSubtypePtr )
```

Return the most recent Reset Type.

Description

Provides the caller with codes that identifies the type of Reset the processor most recently underwent. The caller can also obtain information on what caused the reset by supplying a pointer to a variable that will be filled with the Reset Sub-Type.

Assumptions, External Events, and Notes:

None

Parameters

in,out	*ResetSubtypePtr*	Pointer to `uint32` type variable in which the Reset Sub-Type will be stored. The caller can set this pointer to NULL if the Sub-Type is of no interest. ResetSubtypePtr If the provided pointer was not `NULL`, the Reset Sub-Type is stored at the given address. For a list of possible Sub-Type values, see "Reset Sub-Types" .

Returns

Processor reset type

Return values

CFE_PSP_RST_TYPE_POWERON	
CFE_PSP_RST_TYPE_PROCESSOR	

See also

CFE_ES_GetAppID, CFE_ES_GetAppIDByName, CFE_ES_GetAppName, CFE_ES_GetTaskInfo

37.6.2.10 CFE_ES_GetTaskID()

```
CFE_Status_t CFE_ES_GetTaskID (
            CFE_ES_TaskId_t * TaskIdPtr )
```

Get the task ID of the calling context.

Description

This retrieves the current task context from OSAL

Assumptions, External Events, and Notes:

Applications which desire to call other CFE ES services such as CFE_ES_TaskGetInfo() should use this API rather than getting the ID from OSAL directly via OS_TaskGetId().

Parameters

out	*TaskIdPtr*	Pointer to variable that is to receive the ID. Will be set to the ID of the calling task.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_ERR_BUFFER	Invalid Pointer.

37.6.2.11 CFE_ES_GetTaskInfo()

```
CFE_Status_t CFE_ES_GetTaskInfo (
            CFE_ES_TaskInfo_t * TaskInfo,
            CFE_ES_TaskId_t TaskId )
```

Get Task Information given a specified Task ID.

Description

This routine retrieves the information about a Task associated with a specified Task ID. The information includes Task Name, and Parent/Creator Application ID.

Assumptions, External Events, and Notes:

None

Parameters

out	*TaskInfo*	Pointer to a `CFE_ES_TaskInfo_t` structure that holds the specific task information. *TaskInfo is the filled out `CFE_ES_TaskInfo_t` structure containing the Task Name, Parent App Name, Parent App ID among other fields.
in	*TaskId*	Application ID of Application whose name is being requested.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_ERR_BUFFER	Invalid Pointer.

See also

CFE_ES_GetTaskID, CFE_ES_GetTaskIDByName, CFE_ES_GetTaskName

37.7 cFE Child Task APIs

Functions

- CFE_Status_t CFE_ES_RegisterChildTask (void)

  Registers a cFE Child task associated with a cFE Application.
- CFE_Status_t CFE_ES_CreateChildTask (CFE_ES_TaskId_t *TaskIdPtr, const char *TaskName, CFE_ES_ChildTaskMainFuncPtr_t FunctionPtr, CFE_ES_StackPointer_t StackPtr, size_t StackSize, CFE_ES_TaskPriority_Atom_t Priority, uint32 Flags)

  Creates a new task under an existing Application.
- CFE_Status_t CFE_ES_GetTaskIDByName (CFE_ES_TaskId_t *TaskIdPtr, const char *TaskName)

  Get a Task ID associated with a specified Task name.
- CFE_Status_t CFE_ES_GetTaskName (char *TaskName, CFE_ES_TaskId_t TaskId, size_t BufferLength)

  Get a Task name for a specified Task ID.
- CFE_Status_t CFE_ES_DeleteChildTask (CFE_ES_TaskId_t TaskId)

  Deletes a task under an existing Application.
- void CFE_ES_ExitChildTask (void)

  Exits a child task.

37.7.1 Detailed Description

37.7.2 Function Documentation

37.7.2.1 CFE_ES_CreateChildTask()

```
CFE_Status_t CFE_ES_CreateChildTask (
            CFE_ES_TaskId_t * TaskIdPtr,
            const char * TaskName,
            CFE_ES_ChildTaskMainFuncPtr_t FunctionPtr,
            CFE_ES_StackPointer_t StackPtr,
            size_t StackSize,
            CFE_ES_TaskPriority_Atom_t Priority,
            uint32 Flags )
```

Creates a new task under an existing Application.

Description

This routine creates a new task (a separate execution thread) owned by the calling Application.

Assumptions, External Events, and Notes:

None

Parameters

in,out	*TaskIdPtr*	A pointer to a variable that will be filled in with the new task's ID. ∗TaskIdPtr is the Task ID of the newly created child task.
in	*TaskName*	A pointer to a string containing the desired name of the new task. This can be up to OS_MAX_API_NAME characters, including the trailing null.
in	*FunctionPtr*	A pointer to the function that will be spawned as a new task. This function must have the following signature: uint32 function(void). Input parameters for the new task are not supported.
in	*StackPtr*	A pointer to the location where the child task's stack pointer should start. NOTE: Not all underlying operating systems support this parameter. The CFE_ES_TASK_STACK_ALLOCATE constant may be passed to indicate that the stack should be dynamically allocated.
in	*StackSize*	The number of bytes to allocate for the new task's stack.
in	*Priority*	The priority for the new task. Lower numbers are higher priority, with 0 being the highest priority. Applications cannot create tasks with a higher priority (lower number) than their own priority.
in	*Flags*	Reserved for future expansion.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_CHILD_TASK_CREATE	Child Task Create Error.

See also

CFE_ES_RegisterChildTask, CFE_ES_DeleteChildTask, CFE_ES_ExitChildTask

37.7.2.2 CFE_ES_DeleteChildTask()

```
CFE_Status_t CFE_ES_DeleteChildTask (
            CFE_ES_TaskId_t TaskId )
```

Deletes a task under an existing Application.

Description

This routine deletes a task under an Application specified by the `TaskId` obtained when the child task was created using the CFE_ES_CreateChildTask API.

Assumptions, External Events, and Notes:

None

Parameters

in	*TaskId*	The task ID previously obtained when the Child Task was created with the CFE_ES_CreateChildTask API.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_NOT_IMPLEMENTED	Not Implemented.

See also

CFE_ES_RegisterChildTask, CFE_ES_CreateChildTask, CFE_ES_ExitChildTask

37.7.2.3 CFE_ES_ExitChildTask()

```
void CFE_ES_ExitChildTask (
            void  )
```

Exits a child task.

Description

This routine allows the current executing child task to exit and be deleted by ES.

Assumptions, External Events, and Notes:

This function cannot be called from an Application's Main Task.

Note

This function does not return a value, but if it does return at all, it is assumed that the Task was either unregistered or this function was called from a cFE Application's main task.

See also

CFE_ES_RegisterChildTask, CFE_ES_CreateChildTask, CFE_ES_DeleteChildTask

37.7.2.4 CFE_ES_GetTaskIDByName()

```
CFE_Status_t CFE_ES_GetTaskIDByName (
            CFE_ES_TaskId_t * TaskIdPtr,
            const char * TaskName )
```

Get a Task ID associated with a specified Task name.

Description

This routine retrieves the cFE Task ID associated with a specified Task name.

Assumptions, External Events, and Notes:

None

Parameters

out	*TaskIdPtr*	Pointer to variable that is to receive the Task's ID.
in	*TaskName*	Pointer to null terminated character string containing an Task name.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_NAME_NOT_FOUND	Resource Name Error.
CFE_ES_ERR_BUFFER	Invalid Pointer.

See also

CFE_ES_GetTaskName

37.7.2.5 CFE_ES_GetTaskName()

```
CFE_Status_t CFE_ES_GetTaskName (
            char * TaskName,
            CFE_ES_TaskId_t TaskId,
            size_t BufferLength )
```

Get a Task name for a specified Task ID.

Description

This routine retrieves the cFE Task name associated with a specified Task ID.

Assumptions, External Events, and Notes:

In the case of a failure (CFE_ES_ERR_RESOURCEID_NOT_VALID), an empty string is returned.

Parameters

`out`	*TaskName*	Pointer to a character array of at least `BufferLength` in size that will be filled with the Task name.
`in`	*TaskId*	Task ID of Task whose name is being requested.
`in`	*BufferLength*	The maximum number of characters, including the null terminator, that can be put into the `TaskName` buffer. This routine will truncate the name to this length, if necessary.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

See also

CFE_ES_GetTaskIDByName

37.7.2.6 CFE_ES_RegisterChildTask()

```
CFE_Status_t CFE_ES_RegisterChildTask (
            void  )
```

Registers a cFE Child task associated with a cFE Application.

Description

This routine registers a cFE Child task and associates it with its parent cFE Application.

Assumptions, External Events, and Notes:

NOTE: This API **MUST** be called by the Child Task before any other cFE API calls are made.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_CHILD_TASK_REGISTER	Child Task Register Error.

See also

CFE_ES_CreateChildTask, CFE_ES_DeleteChildTask, CFE_ES_ExitChildTask

37.8 cFE Miscellaneous APIs

Functions

- void CFE_ES_BackgroundWakeup (void)

  Wakes up the CFE background task.
- CFE_Status_t CFE_ES_WriteToSysLog (const char ∗SpecStringPtr,...) OS_PRINTF(1

  Write a string to the cFE System Log.
- CFE_Status_t uint32 CFE_ES_CalculateCRC (const void ∗DataPtr, size_t DataLength, uint32 InputCRC, uint32 TypeCRC)

  Calculate a CRC on a block of memory.
- void CFE_ES_ProcessAsyncEvent (void)

  Notification that an asynchronous event was detected by the underlying OS/PSP.

37.8.1 Detailed Description

37.8.2 Function Documentation

37.8.2.1 CFE_ES_BackgroundWakeup()

```
void CFE_ES_BackgroundWakeup (
            void  )
```

Wakes up the CFE background task.

Description

Normally the ES background task wakes up at a periodic interval. Whenever new background work is added, this can be used to wake the task early, which may reduce the delay between adding the job and the job getting processed.

Assumptions, External Events, and Notes:

Note the amount of work that the background task will perform is pro-rated based on the amount of time elapsed since the last wakeup. Waking the task early will not cause the background task to do more work than it otherwise would - it just reduces the delay before work starts initially.

37.8.2.2 CFE_ES_CalculateCRC()

```
CFE_Status_t uint32 CFE_ES_CalculateCRC (
            const void * DataPtr,
            size_t DataLength,
            uint32 InputCRC,
            uint32 TypeCRC )
```

Calculate a CRC on a block of memory.

Description

This routine calculates a cyclic redundancy check (CRC) on a block of memory. The CRC algorithm used is determined by the last parameter.

Assumptions, External Events, and Notes:

None

Parameters

in	*DataPtr*	Pointer to the base of the memory block.
in	*DataLength*	The number of bytes in the memory block.
in	*InputCRC*	A starting value for use in the CRC calculation. This parameter allows the user to calculate the CRC of non-contiguous blocks as a single value. Nominally, the user should set this value to zero.
in	*TypeCRC*	One of the following CRC algorithm selections: • `CFE_MISSION_ES_CRC_8` - (Not currently implemented) • `CFE_MISSION_ES_CRC_16` - CRC-16/ARC Polynomial: 0x8005 Initialization: 0x0000 Reflect Input/Output: true XorOut: 0x0000 • `CFE_MISSION_ES_CRC_32` - (not currently implemented)

Returns

The result of the CRC calculation on the specified memory block, or error code cFE Return Code Defines

37.8.2.3 CFE_ES_ProcessAsyncEvent()

```
void CFE_ES_ProcessAsyncEvent (
            void  )
```

Notification that an asynchronous event was detected by the underlying OS/PSP.

Description

This hook routine is called from the PSP when an exception or other asynchronous system event occurs

Assumptions, External Events, and Notes:

The PSP must guarantee that this function is only invoked from a context which may use OSAL primitives. In general this means that it shouldn't be *directly* invoked from an ISR/signal context.

37.8.2.4 CFE_ES_WriteToSysLog()

```
CFE_Status_t CFE_ES_WriteToSysLog (
            const char * SpecStringPtr,
             ...  )
```

Write a string to the cFE System Log.

Description

This routine writes a formatted string to the cFE system log. This can be used to record very low-level errors that can't be reported using the Event Services. This function is used in place of printf for flight software. It should be used for significant startup events, critical errors, and conditionally compiled debug software.

Assumptions, External Events, and Notes:

None

Parameters

in	*SpecStringPtr*	The format string for the log message. This is similar to the format string for a printf() call.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_SYS_LOG_FULL	System Log Full.

37.9 cFE Critical Data Store APIs

Functions

- CFE_Status_t CFE_ES_RegisterCDS (CFE_ES_CDSHandle_t ∗HandlePtr, size_t BlockSize, const char ∗Name)

  Reserve space (or re-obtain previously reserved space) in the Critical Data Store (CDS)
- CFE_Status_t CFE_ES_GetCDSBlockIDByName (CFE_ES_CDSHandle_t ∗BlockIdPtr, const char ∗BlockName)

  Get a CDS Block ID associated with a specified CDS Block name.
- CFE_Status_t CFE_ES_GetCDSBlockName (char ∗BlockName, CFE_ES_CDSHandle_t BlockId, size_t Buffer↩ Length)

  Get a Block name for a specified Block ID.
- CFE_Status_t CFE_ES_CopyToCDS (CFE_ES_CDSHandle_t Handle, void ∗DataToCopy)

  Save a block of data in the Critical Data Store (CDS)
- CFE_Status_t CFE_ES_RestoreFromCDS (void ∗RestoreToMemory, CFE_ES_CDSHandle_t Handle)

  Recover a block of data from the Critical Data Store (CDS)

37.9.1 Detailed Description

37.9.2 Function Documentation

37.9.2.1 CFE_ES_CopyToCDS()

```
CFE_Status_t CFE_ES_CopyToCDS (
            CFE_ES_CDSHandle_t Handle,
            void * DataToCopy )
```

Save a block of data in the Critical Data Store (CDS)

Description

This routine copies a specified block of memory into the Critical Data Store that had been previously registered via CFE_ES_RegisterCDS. The block of memory to be copied must be at least as big as the size specified when registering the CDS.

Assumptions, External Events, and Notes:

None

Parameters

in	*Handle*	The handle of the CDS block that was previously obtained from CFE_ES_RegisterCDS.
in	*DataToCopy*	A Pointer to the block of memory to be copied into the CDS.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

See also

CFE_ES_RegisterCDS, CFE_ES_RestoreFromCDS

37.9.2.2 CFE_ES_GetCDSBlockIDByName()

```
CFE_Status_t CFE_ES_GetCDSBlockIDByName (
            CFE_ES_CDSHandle_t * BlockIdPtr,
            const char * BlockName )
```

Get a CDS Block ID associated with a specified CDS Block name.

Description

This routine retrieves the CDS Block ID associated with a specified CDS Block name.

Assumptions, External Events, and Notes:

None

Parameters

out	*BlockIdPtr*	Pointer to variable that is to receive the CDS Block ID.
in	*BlockName*	Pointer to null terminated character string containing a CDS Block name.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_NAME_NOT_FOUND	Resource Name Error.
CFE_ES_ERR_BUFFER	Invalid Pointer.

See also

CFE_ES_GetCDSBlockName

37.9.2.3 CFE_ES_GetCDSBlockName()

```
CFE_Status_t CFE_ES_GetCDSBlockName (
            char * BlockName,
            CFE_ES_CDSHandle_t BlockId,
            size_t BufferLength )
```

Get a Block name for a specified Block ID.

Description

This routine retrieves the cFE Block name associated with a specified Block ID.

Assumptions, External Events, and Notes:

In the case of a failure (CFE_ES_ERR_RESOURCEID_NOT_VALID), an empty string is returned.

Parameters

out	*BlockName*	Pointer to a character array of at least `BufferLength` in size that will be filled with the CDS Block name.
in	*BlockId*	Block ID/Handle of CDS registry entry whose name is being requested.
in	*BufferLength*	The maximum number of characters, including the null terminator, that can be put into the `BlockName` buffer. This routine will truncate the name to this length, if necessary.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

See also

CFE_ES_GetCDSBlockIDByName

37.9.2.4 CFE_ES_RegisterCDS()

```
CFE_Status_t CFE_ES_RegisterCDS (
            CFE_ES_CDSHandle_t * HandlePtr,
            size_t BlockSize,
            const char * Name )
```

Reserve space (or re-obtain previously reserved space) in the Critical Data Store (CDS)

Description

This routine allocates a block of memory in the Critical Data Store and associates it with the calling Application. The memory can survive an Application restart as well as a Processor Reset.

Assumptions, External Events, and Notes:

None

Parameters

in,out	*HandlePtr*	Pointer Application's variable that will contain the CDS Memory Block Handle. HandlePtr is the handle of the CDS block that can be used in CFE_ES_CopyToCDS and CFE_ES_RestoreFromCDS.
in	*BlockSize*	The number of bytes needed in the CDS.
in	*Name*	A pointer to a character string containing an application unique name of CFE_MISSION_ES_CDS_MAX_NAME_LENGTH characters or less.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	The memory block was successfully created in the CDS.
CFE_ES_NOT_IMPLEMENTED	The processor does not support a Critical Data Store.
CFE_ES_CDS_ALREADY_EXISTS	CDS Already Exists.
CFE_ES_CDS_INVALID_SIZE	CDS Invalid Size.
CFE_ES_CDS_INVALID_NAME	CDS Invalid Name.
CFE_ES_CDS_REGISTRY_FULL	

See also

CFE_ES_CopyToCDS, CFE_ES_RestoreFromCDS

37.9.2.5 CFE_ES_RestoreFromCDS()

```
CFE_Status_t CFE_ES_RestoreFromCDS (
            void * RestoreToMemory,
            CFE_ES_CDSHandle_t Handle )
```

Recover a block of data from the Critical Data Store (CDS)

Description

This routine copies data from the Critical Data Store identified with the `Handle` into the area of memory pointed to by the `RestoreToMemory` pointer. The area of memory to be copied into must be at least as big as the size specified when registering the CDS. The recovery will indicate an error if the data integrity check maintained by the CDS indicates the contents of the CDS have changed. However, the contents will still be copied into the specified area of memory.

Assumptions, External Events, and Notes:

None

Parameters

in	*Handle*	The handle of the CDS block that was previously obtained from CFE_ES_RegisterCDS.
in,out	*RestoreToMemory*	A Pointer to the block of memory that is to be restored with the contents of the CDS. ∗RestoreToMemory is the contents of the specified CDS.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_CDS_BLOCK_CRC_ERR	CDS Block CRC Error.

See also

CFE_ES_RegisterCDS, CFE_ES_CopyToCDS

37.10 cFE Memory Manager APIs

Functions

- CFE_Status_t CFE_ES_PoolCreateNoSem (CFE_ES_MemHandle_t ∗PoolID, uint8 ∗MemPtr, size_t Size)

  Initializes a memory pool created by an application without using a semaphore during processing.
- CFE_Status_t CFE_ES_PoolCreate (CFE_ES_MemHandle_t ∗PoolID, uint8 ∗MemPtr, size_t Size)

  Initializes a memory pool created by an application while using a semaphore during processing.
- CFE_Status_t CFE_ES_PoolCreateEx (CFE_ES_MemHandle_t ∗PoolID, uint8 ∗MemPtr, size_t Size, uint16 NumBlockSizes, const size_t ∗BlockSizes, bool UseMutex)

  Initializes a memory pool created by an application with application specified block sizes.
- int32 CFE_ES_PoolDelete (CFE_ES_MemHandle_t PoolID)

  Deletes a memory pool that was previously created.
- int32 CFE_ES_GetPoolBuf (CFE_ES_MemPoolBuf_t ∗BufPtr, CFE_ES_MemHandle_t PoolID, size_t Size)

  Gets a buffer from the memory pool created by CFE_ES_PoolCreate or CFE_ES_PoolCreateNoSem.
- CFE_Status_t CFE_ES_GetPoolBufInfo (CFE_ES_MemHandle_t PoolID, CFE_ES_MemPoolBuf_t BufPtr)

  Gets info on a buffer previously allocated via CFE_ES_GetPoolBuf.
- int32 CFE_ES_PutPoolBuf (CFE_ES_MemHandle_t PoolID, CFE_ES_MemPoolBuf_t BufPtr)

  Releases a buffer from the memory pool that was previously allocated via CFE_ES_GetPoolBuf.
- CFE_Status_t CFE_ES_GetMemPoolStats (CFE_ES_MemPoolStats_t ∗BufPtr, CFE_ES_MemHandle_t Handle)

  Extracts the statistics maintained by the memory pool software.

37.10.1 Detailed Description

37.10.2 Function Documentation

37.10.2.1 CFE_ES_GetMemPoolStats()

```
CFE_Status_t CFE_ES_GetMemPoolStats (
            CFE_ES_MemPoolStats_t * BufPtr,
            CFE_ES_MemHandle_t Handle )
```

Extracts the statistics maintained by the memory pool software.

Description

This routine fills the CFE_ES_MemPoolStats_t data structure with the statistics maintained by the memory pool software. These statistics can then be telemetered by the calling Application.

Assumptions, External Events, and Notes:

None

Parameters

in,out	*BufPtr*	Pointer to CFE_ES_MemPoolStats_t data structure to be filled with memory statistics. ∗BufPtr is the Memory Pool Statistics stored in given data structure.
in	*Handle*	The handle to the memory pool whose statistics are desired.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

See also

CFE_ES_PoolCreate, CFE_ES_PoolCreateNoSem, CFE_ES_PoolCreateEx, CFE_ES_GetPoolBuf, CFE_ES_PutPoolBuf

37.10.2.2 CFE_ES_GetPoolBuf()

```
int32 CFE_ES_GetPoolBuf (
            CFE_ES_MemPoolBuf_t * BufPtr,
            CFE_ES_MemHandle_t PoolID,
            size_t Size )
```

Gets a buffer from the memory pool created by CFE_ES_PoolCreate or CFE_ES_PoolCreateNoSem.

Description

This routine obtains a block of memory from the memory pool supplied by the calling application.

Assumptions, External Events, and Notes:

1. The size allocated from the memory pool is, at a minimum, 12 bytes more than requested.

Parameters

in,out	*BufPtr*	A pointer to the Application's pointer in which will be stored the address of the allocated memory buffer. ∗BufPtr is the address of the requested buffer.
in	*PoolID*	The handle to the memory pool as returned by CFE_ES_PoolCreate or CFE_ES_PoolCreateNoSem.
in	*Size*	The size of the buffer requested. NOTE: The size allocated may be larger.

Returns

Bytes Allocated, or error code cFE Return Code Defines

Return values

CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_ERR_MEM_BLOCK_SIZE	Memory Block Size Error.

See also

CFE_ES_PoolCreate, CFE_ES_PoolCreateNoSem, CFE_ES_PoolCreateEx, CFE_ES_PutPoolBuf, CFE_ES_GetMemPoolStats, CFE_ES_GetPoolBufInfo

37.10.2.3 CFE_ES_GetPoolBufInfo()

```
CFE_Status_t CFE_ES_GetPoolBufInfo (
            CFE_ES_MemHandle_t PoolID,
            CFE_ES_MemPoolBuf_t BufPtr )
```

Gets info on a buffer previously allocated via CFE_ES_GetPoolBuf.

Description

This routine gets info on a buffer in the memory pool.

Assumptions, External Events, and Notes:

None

Parameters

in	*PoolID*	The handle to the memory pool as returned by CFE_ES_PoolCreate or CFE_ES_PoolCreateNoSem.
in	*BufPtr*	A pointer to the memory buffer to provide status for.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BUFFER_NOT_IN_POOL	Buffer Not In Pool.

See also

CFE_ES_PoolCreate, CFE_ES_PoolCreateNoSem, CFE_ES_PoolCreateEx, CFE_ES_GetPoolBuf, CFE_ES_GetMemPoolStats, CFE_ES_PutPoolBuf

37.10.2.4 CFE_ES_PoolCreate()

```
CFE_Status_t CFE_ES_PoolCreate (
            CFE_ES_MemHandle_t * PoolID,
            uint8 * MemPtr,
            size_t Size )
```

Initializes a memory pool created by an application while using a semaphore during processing.

Description

This routine initializes a pool of memory supplied by the calling application. When a memory pool created by this routine is processed, mutex handling will be performed.

Assumptions, External Events, and Notes:

1. The size of the pool must be an integral number of 32-bit words
2. The start address of the pool must be 32-bit aligned
3. 168 bytes are used for internal bookkeeping, therefore, they will not be available for allocation.

Parameters

in,out	*PoolID*	A pointer to the variable the caller wishes to have the memory pool handle kept in. PoolID is the memory pool handle.
in	*MemPtr*	A Pointer to the pool of memory created by the calling application. This address must be on a 32-bit boundary.
in	*Size*	The size of the pool of memory. Note that this must be an integral number of 32 bit words.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

CFE_ES_PoolCreateNoSem, CFE_ES_PoolCreateEx, CFE_ES_GetPoolBuf, CFE_ES_PutPoolBuf, CFE_ES_GetMemPoolStats

37.10.2.5 CFE_ES_PoolCreateEx()

```
CFE_Status_t CFE_ES_PoolCreateEx (
            CFE_ES_MemHandle_t * PoolID,
            uint8 * MemPtr,
            size_t Size,
            uint16 NumBlockSizes,
            const size_t * BlockSizes,
            bool UseMutex )
```

Initializes a memory pool created by an application with application specified block sizes.

Description

This routine initializes a pool of memory supplied by the calling application.

Assumptions, External Events, and Notes:

1. The size of the pool must be an integral number of 32-bit words
2. The start address of the pool must be 32-bit aligned
3. 168 bytes are used for internal bookkeeping, therefore, they will not be available for allocation.

Parameters

in,out	*PoolID*	A pointer to the variable the caller wishes to have the memory pool handle kept in. PoolID is the memory pool handle.
in	*MemPtr*	A Pointer to the pool of memory created by the calling application. This address must be on a 32-bit boundary.
in	*Size*	The size of the pool of memory. Note that this must be an integral number of 32 bit words.
in	*NumBlockSizes*	The number of different block sizes specified in the `BlockSizes` array. If set equal to zero or if greater than 17, then default block sizes are used.
in	*BlockSizes*	Pointer to an array of sizes to be used instead of the default block sizes specified by CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01 through CFE_PLATFORM_ES_MAX_BLOCK_SIZE. If the pointer is equal to NULL, the default block sizes are used.
in	*UseMutex*	Flag indicating whether the new memory pool will be processing with mutex handling or not. Valid parameter values are CFE_ES_USE_MUTEX and CFE_ES_NO_MUTEX

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

CFE_ES_PoolCreate, CFE_ES_PoolCreateNoSem, CFE_ES_GetPoolBuf, CFE_ES_PutPoolBuf, CFE_ES_GetMemPoolStats

37.10.2.6 CFE_ES_PoolCreateNoSem()

```
CFE_Status_t CFE_ES_PoolCreateNoSem (
            CFE_ES_MemHandle_t * PoolID,
            uint8 * MemPtr,
            size_t Size )
```

Initializes a memory pool created by an application without using a semaphore during processing.

Description

This routine initializes a pool of memory supplied by the calling application. When a memory pool created by this routine is processed, no mutex handling is performed.

Assumptions, External Events, and Notes:

1. The size of the pool must be an integral number of 32-bit words
2. The start address of the pool must be 32-bit aligned
3. 168 bytes are used for internal bookkeeping, therefore, they will not be available for allocation.

Parameters

in,out	*PoolID*	A pointer to the variable the caller wishes to have the memory pool handle kept in. PoolID is the memory pool handle.
in	*MemPtr*	A Pointer to the pool of memory created by the calling application. This address must be on a 32-bit boundary.
in	*Size*	The size of the pool of memory. Note that this must be an integral number of 32 bit words.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

CFE_ES_PoolCreate, CFE_ES_PoolCreateEx, CFE_ES_GetPoolBuf, CFE_ES_PutPoolBuf, CFE_ES_Get↩MemPoolStats

37.10.2.7 CFE_ES_PoolDelete()

```
int32 CFE_ES_PoolDelete (
            CFE_ES_MemHandle_t PoolID )
```

Deletes a memory pool that was previously created.

Description

This routine removes the pool ID and frees the global table entry for future re-use.

Assumptions, External Events, and Notes:

All buffers associated with the pool become invalid after this call. The application should ensure that buffers/references to the pool are returned before deleting the pool.

Parameters

in	*PoolID*	The ID of the pool to delete

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

See also

CFE_ES_PoolCreate, CFE_ES_PoolCreateNoSem, CFE_ES_GetPoolBuf, CFE_ES_PutPoolBuf, CFE_ES_↩GetMemPoolStats

37.10.2.8 CFE_ES_PutPoolBuf()

```
int32 CFE_ES_PutPoolBuf (
            CFE_ES_MemHandle_t PoolID,
            CFE_ES_MemPoolBuf_t BufPtr )
```

Releases a buffer from the memory pool that was previously allocated via CFE_ES_GetPoolBuf.

Description

This routine releases a buffer back into the memory pool.

Assumptions, External Events, and Notes:

None

Parameters

in	*PoolID*	The handle to the memory pool as returned by CFE_ES_PoolCreate or CFE_ES_PoolCreateNoSem.
in	*BufPtr*	A pointer to the memory buffer to be released.

Returns

Bytes released, or error code cFE Return Code Defines

Return values

CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

See also

CFE_ES_PoolCreate, CFE_ES_PoolCreateNoSem, CFE_ES_PoolCreateEx, CFE_ES_GetPoolBuf, CFE_ES_GetMemPoolStats, CFE_ES_GetPoolBufInfo

37.11 cFE Performance Monitor APIs

Macros

- #define CFE_ES_PerfLogEntry(id) (CFE_ES_PerfLogAdd(id, 0))

  Entry marker for use with Software Performance Analysis Tool.
- #define CFE_ES_PerfLogExit(id) (CFE_ES_PerfLogAdd(id, 1))

  Exit marker for use with Software Performance Analysis Tool.

Functions

- void CFE_ES_PerfLogAdd (uint32 Marker, uint32 EntryExit)

  Function called by CFE_ES_PerfLogEntry and CFE_ES_PerfLogExit macros.

37.11.1 Detailed Description

37.11.2 Macro Definition Documentation

37.11.2.1 CFE_ES_PerfLogEntry

```
#define CFE_ES_PerfLogEntry(
              id ) (CFE_ES_PerfLogAdd(id, 0))
```

Entry marker for use with Software Performance Analysis Tool.

Description

This macro logs the entry or start event/marker for the specified entry `id`. This macro, in conjunction with the CFE_ES_PerfLogExit, is used by the Software Performance Analysis tool (see section 5.15).

Assumptions, External Events, and Notes:

None

Parameters

in	*id*	Identifier of the specific event or marker.

See also

CFE_ES_PerfLogExit, CFE_ES_PerfLogAdd

Definition at line 1462 of file cfe_es.h.

37.11.2.2 CFE_ES_PerfLogExit

```
#define CFE_ES_PerfLogExit(
            id ) (CFE_ES_PerfLogAdd(id, 1))
```

Exit marker for use with Software Performance Analysis Tool.

Description

This macro logs the exit or end event/marker for the specified entry `id`. This macro, in conjunction with the CFE_ES_PerfLogEntry, is used by the Software Performance Analysis tool (see section 5.15).

Assumptions, External Events, and Notes:

None

Parameters

in	*id*	Identifier of the specific event or marker.

See also

CFE_ES_PerfLogEntry, CFE_ES_PerfLogAdd

Definition at line 1481 of file cfe_es.h.

37.11.3 Function Documentation

37.11.3.1 CFE_ES_PerfLogAdd()

```
void CFE_ES_PerfLogAdd (
            uint32 Marker,
            uint32 EntryExit )
```

Function called by CFE_ES_PerfLogEntry and CFE_ES_PerfLogExit macros.

Description

This function logs the entry and exit marker for the specified `id`. This function is used by the Software Performance Analysis tool (see section 5.15).

Assumptions, External Events, and Notes:

None

Parameters

in	*Marker*	Identifier of the specific event or marker.
in	*EntryExit*	Used to specify Entry(0) or Exit(1)

See also

CFE_ES_PerfLogEntry, CFE_ES_PerfLogExit

37.12 cFE Generic Counter APIs

Functions

- CFE_Status_t CFE_ES_RegisterGenCounter (CFE_ES_CounterId_t *CounterIdPtr, const char *CounterName)

  Register a generic counter.
- CFE_Status_t CFE_ES_DeleteGenCounter (CFE_ES_CounterId_t CounterId)

  Delete a generic counter.
- CFE_Status_t CFE_ES_IncrementGenCounter (CFE_ES_CounterId_t CounterId)

  Increments the specified generic counter.
- CFE_Status_t CFE_ES_SetGenCount (CFE_ES_CounterId_t CounterId, uint32 Count)

  Set the specified generic counter.
- CFE_Status_t CFE_ES_GetGenCount (CFE_ES_CounterId_t CounterId, uint32 *Count)

  Get the specified generic counter count.
- CFE_Status_t CFE_ES_GetGenCounterIDByName (CFE_ES_CounterId_t *CounterIdPtr, const char *CounterName)

  Get the Id associated with a generic counter name.
- CFE_Status_t CFE_ES_GetGenCounterName (char *CounterName, CFE_ES_CounterId_t CounterId, size_t BufferLength)

  Get a Counter name for a specified Counter ID.

37.12.1 Detailed Description

37.12.2 Function Documentation

37.12.2.1 CFE_ES_DeleteGenCounter()

```
CFE_Status_t CFE_ES_DeleteGenCounter (
            CFE_ES_CounterId_t CounterId )
```

Delete a generic counter.

Description

This routine deletes a previously registered generic counter.

Assumptions, External Events, and Notes:

None.

Parameters

in	*CounterId*	The Counter Id of the newly created counter.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

CFE_ES_IncrementGenCounter, CFE_ES_RegisterGenCounter, CFE_ES_SetGenCount, CFE_ES_GetGen↩Count, CFE_ES_GetGenCounterIDByName

37.12.2.2 CFE_ES_GetGenCount()

```
CFE_Status_t CFE_ES_GetGenCount (
            CFE_ES_CounterId_t CounterId,
            uint32 * Count )
```

Get the specified generic counter count.

Description

This routine gets the value of a generic counter.

Assumptions, External Events, and Notes:

None.

Parameters

in	*Counter↩Id*	The Counter to get the value from.
in	∗*Count*	The value of the Counter.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

CFE_ES_RegisterGenCounter, CFE_ES_DeleteGenCounter, CFE_ES_SetGenCount, CFE_ES_Increment↩GenCounter, CFE_ES_GetGenCounterIDByName

37.12.2.3 CFE_ES_GetGenCounterIDByName()

```
CFE_Status_t CFE_ES_GetGenCounterIDByName (
            CFE_ES_CounterId_t * CounterIdPtr,
            const char * CounterName )
```

Get the Id associated with a generic counter name.

Description

This routine gets the Counter Id for a generic counter specified by name.

Assumptions, External Events, and Notes:

None.

Parameters

out	*CounterIdPtr*	Pointer to variable that is to receive the Counter's ID.
in	*CounterName*	Pointer to null terminated character string containing a Counter name.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_NAME_NOT_FOUND	Resource Name Error.
CFE_ES_ERR_BUFFER	Invalid Pointer.

See also

CFE_ES_GetGenCounterName

37.12.2.4 CFE_ES_GetGenCounterName()

```
CFE_Status_t CFE_ES_GetGenCounterName (
            char * CounterName,
```

```
CFE_ES_CounterId_t CounterId,
size_t BufferLength )
```

Get a Counter name for a specified Counter ID.

Description

This routine retrieves the cFE Counter name associated with a specified Counter ID.

Assumptions, External Events, and Notes:

In the case of a failure (CFE_ES_ERR_RESOURCEID_NOT_VALID), an empty string is returned.

Parameters

out	*CounterName*	Pointer to a character array of at least `BufferLength` in size that will be filled with the Counter name.
in	*CounterId*	ID of Counter whose name is being requested.
in	*BufferLength*	The maximum number of characters, including the null terminator, that can be put into the `CounterName` buffer. This routine will truncate the name to this length, if necessary.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

See also

CFE_ES_GetGenCounterIDByName

37.12.2.5 CFE_ES_IncrementGenCounter()

```
CFE_Status_t CFE_ES_IncrementGenCounter (
            CFE_ES_CounterId_t CounterId )
```

Increments the specified generic counter.

Description

This routine increments the specified generic counter.

Assumptions, External Events, and Notes:

None.

Parameters

in	*CounterId*	The Counter to be incremented.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

CFE_ES_RegisterGenCounter, CFE_ES_DeleteGenCounter, CFE_ES_SetGenCount, CFE_ES_GetGenCount, CFE_ES_GetGenCounterIDByName

37.12.2.6 CFE_ES_RegisterGenCounter()

```
CFE_Status_t CFE_ES_RegisterGenCounter (
            CFE_ES_CounterId_t * CounterIdPtr,
            const char * CounterName )
```

Register a generic counter.

Description

This routine registers a generic counter.

Assumptions, External Events, and Notes:

None.

Parameters

in	*CounterName	The Name of the generic counter.
out	*CounterIdPtr	The Counter Id of the newly created counter.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

CFE_ES_IncrementGenCounter, CFE_ES_DeleteGenCounter, CFE_ES_SetGenCount, CFE_ES_GetGen↩Count, CFE_ES_GetGenCounterIDByName

37.12.2.7 CFE_ES_SetGenCount()

```
CFE_Status_t CFE_ES_SetGenCount (
            CFE_ES_CounterId_t CounterId,
            uint32 Count )
```

Set the specified generic counter.

Description

This routine sets the specified generic counter to the specified value.

Assumptions, External Events, and Notes:

None.

Parameters

in	*Counter↩Id*	The Counter to be set.
in	*Count*	The new value of the Counter.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

CFE_ES_RegisterGenCounter, CFE_ES_DeleteGenCounter, CFE_ES_IncrementGenCounter, CFE_ES_Get↩GenCount, CFE_ES_GetGenCounterIDByName

37.13 cFE Internal Executive Service APIs, internal to CFE core

Functions

- void CFE_ES_TaskMain (void)

  Entry Point for cFE Core Application.
- int32 CFE_ES_CDS_EarlyInit (void)

  Initializes the cFE core module API Library.
- int32 CFE_ES_RegisterCDSEx (CFE_ES_CDSHandle_t ∗HandlePtr, size_t UserBlockSize, const char ∗Name, bool CriticalTbl)

  Reserve space (or re-obtain previously reserved space) in the Critical Data Store (CDS)
- int32 CFE_ES_DeleteCDS (const char ∗CDSName, bool CalledByTblServices)

  Deletes the specified CDS from the CDS Registry and frees CDS Memory.

37.13.1 Detailed Description

37.13.2 Function Documentation

37.13.2.1 CFE_ES_CDS_EarlyInit()

```
int32 CFE_ES_CDS_EarlyInit (
            void  )
```

Initializes the cFE core module API Library.

Description

Initializes the cFE core module API Library

Assumptions, External Events, and Notes:

1. This function MUST be called before any module API's are called.

37.13.2.2 CFE_ES_DeleteCDS()

```
int32 CFE_ES_DeleteCDS (
            const char * CDSName,
            bool CalledByTblServices )
```

Deletes the specified CDS from the CDS Registry and frees CDS Memory.

Description

Removes the record of the specified CDS from the CDS Registry and frees the associated CDS memory for future use.

Assumptions, External Events, and Notes:

None

Parameters

in	*CDSName*	- Pointer to character string containing complete CDS Name (of the format "AppName.CDSName").
in	*CalledByTblServices*	- Flag that identifies whether the CDS is supposed to be a Critical Table Image or not.

Returns

CFE_SUCCESS Sucessful execution. Operation was performed successfully
CFE_ES_CDS_WRONG_TYPE_ERR CDS Wrong Type Error. Occurs when Table Services is trying to delete a Critical Data Store that is not a Critical Table Image or when Executive Services is trying to delete a Critical Table Image.
CFE_ES_CDS_OWNER_ACTIVE_ERR CDS Owner Active Error. Occurs when an attempt was made to delete a CDS when an application with the same name associated with the CDS is still present. CDSs can ONLY be deleted when Applications that created them are not present in the system.
CFE_ES_ERR_NAME_NOT_FOUND Resource Name Error. There is no match in the system for the given name.
Any of the return values from CFE_ES_UpdateCDSRegistry
Any of the return values from CFE_ES_GenPoolPutBlock

37.13.2.3 CFE_ES_RegisterCDSEx()

```
int32 CFE_ES_RegisterCDSEx (
            CFE_ES_CDSHandle_t * HandlePtr,
            size_t UserBlockSize,
            const char * Name,
            bool CriticalTbl )
```

Reserve space (or re-obtain previously reserved space) in the Critical Data Store (CDS)

Description

This routine is identical to CFE_ES_RegisterCDS except it identifies the contents of the CDS as a critical table. This is crucial because a critical table CDS must only be deleted by cFE Table Services, not via an ES delete CDS command. Otherwise, Table Services may be out of sync with the contents of the CDS.

Assumptions, External Events, and Notes:

1. This function assumes input parameters are error free and have met size/value restrictions.
2. The calling function is responsible for issuing any event messages associated with errors.

Parameters

in,out	*HandlePtr*	Pointer Application's variable that will contain the CDS Memory Block Handle. HandlePtr is the handle of the CDS block that can be used in CFE_ES_CopyToCDS and CFE_ES_RestoreFromCDS.
in	*UserBlockSize*	The number of bytes needed in the CDS.
in	*Name*	Pointer to character string containing the Application's local name for the CDS.
in	*CriticalTbl*	Indicates whether the CDS is to be used as a Critical Table or not

Returns

See return codes for CFE_ES_RegisterCDS

37.13.2.4 CFE_ES_TaskMain()

```
void CFE_ES_TaskMain (
            void  )
```

Entry Point for cFE Core Application.

Description

This is the entry point to the cFE ES Core Application.

Assumptions, External Events, and Notes:

None

37.14 cFE Registration APIs

Functions

- CFE_Status_t CFE_EVS_Register (const void *Filters, uint16 NumFilteredEvents, uint16 FilterScheme)

  Register an application for receiving event services.
- CFE_Status_t CFE_EVS_Unregister (void)

  Cleanup internal structures used by the event manager for the calling Application.

37.14.1 Detailed Description

37.14.2 Function Documentation

37.14.2.1 CFE_EVS_Register()

```
CFE_Status_t CFE_EVS_Register (
            const void * Filters,
            uint16 NumFilteredEvents,
            uint16 FilterScheme )
```

Register an application for receiving event services.

Description

This routine registers an application with event services and allocates/initializes the internal data structures used to support this application's events. An application may not send events unless it has called this routine. The routine also accepts a filter array structure for applications requiring event filtering. In the current implementation of the EVS, only the binary filtering scheme is supported. See section TBD of the cFE Application Programmer's Guide for a description of the behavior of binary filters. Applications may call CFE_EVS_Register more than once, but each call will wipe out all filters registered by previous calls (filter registration is NOT cumulative).

Assumptions, External Events, and Notes:

Note: Event filters can be added, deleted or modified by ground commands. All filtering schemes include a default setting that results in no filtering (such as CFE_EVS_NO_FILTER for binary filters).

Filter Scheme: Binary
Code: CFE_EVS_EventFilter_BINARY
Filter Structure:

```
typedef struct CFE_EVS_BinFilter {
      uint16  EventID,
      uint16  Mask ;
} CFE_EVS_BinFilter_t;
```

Parameters

in	*Filters*	Pointer to an array of event message filters, or NULL if no filtering is desired. The structure of an event message filter depends on the FilterScheme selected. (see Filter Schemes mentioned above)
in	*NumFilteredEvents*	The number of event message filters included in this call. This must be less than or equal to the maximum number of events allowed per application (CFE_PLATFORM_EVS_MAX_EVENT_FILTERS).
in	*FilterScheme*	The event filtering scheme that this application will use. For the first implementation of the event services, only filter type CFE_EVS_EventFilter_BINARY will be supported.

Returns

Execution status below or from CFE_ES_GetAppID, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_EVS_APP_FILTER_OVERLOAD	Application Filter Overload.
CFE_EVS_UNKNOWN_FILTER	Unknown Filter.
CFE_EVS_APP_ILLEGAL_APP_ID	Illegal Application ID.

See also

CFE_EVS_Unregister

37.14.2.2 CFE_EVS_Unregister()

```
CFE_Status_t CFE_EVS_Unregister (
            void  )
```

Cleanup internal structures used by the event manager for the calling Application.

Description

This routine un-registers the calling application from receiving event services and removes and deletes the calling applications filters and counters from the internal event service filter and counter tables if registered. Applications must call this routine as part of their orderly shutdown process.

Assumptions, External Events, and Notes:

None

Returns

Execution status below or from CFE_ES_GetAppID/CFE_ES_PutPoolBuf, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_EVS_APP_NOT_REGISTERED	Application Not Registered.
CFE_EVS_APP_ILLEGAL_APP_ID	Illegal Application ID.

See also

CFE_EVS_Register

37.15 cFE Send Event APIs

Functions

- CFE_Status_t CFE_EVS_SendEvent (uint16 EventID, uint16 EventType, const char ∗Spec,...) OS_PRINTF(3

  Generate a software event.
- CFE_Status_t CFE_Status_t CFE_EVS_SendEventWithAppID (uint16 EventID, uint16 EventType, CFE_ES_↩ AppId_t AppID, const char ∗Spec,...) OS_PRINTF(4

  Generate a software event given the specified Application ID.
- CFE_Status_t CFE_Status_t CFE_Status_t CFE_EVS_SendTimedEvent (CFE_TIME_SysTime_t Time, uint16 EventID, uint16 EventType, const char ∗Spec,...) OS_PRINTF(4

  Generate a software event with a specific time tag.

37.15.1 Detailed Description

37.15.2 Function Documentation

37.15.2.1 CFE_EVS_SendEvent()

```
CFE_Status_t CFE_EVS_SendEvent (
            uint16 EventID,
            uint16 EventType,
            const char * Spec,
             ...  )
```

Generate a software event.

Description

This routine generates a software event message. If the EventID is not filtered, the event will be sent as a software bus message, optionally logged in the local event log, and optionally sent as an ASCII text string out the enabled output port(s).

Assumptions, External Events, and Notes:

This API only works within the context of a registered application or core service. For messages outside the context of a registered appliction (for example early in app initialization or if registration fails) CFE_ES_WriteToSysLog can be used for reporting.

Parameters

in	*EventID*	A numeric literal used to uniquely identify an application event. The `EventID` is defined and supplied by the application sending the event.

Parameters

in	*EventType*	A numeric literal used to classify an event, one of: • CFE_EVS_EventType_DEBUG • CFE_EVS_EventType_INFORMATION • CFE_EVS_EventType_ERROR • CFE_EVS_EventType_CRITICAL
in	*Spec*	A pointer to a null terminated text string describing the output format for the event. This is the same type of format string used for the ANSI `printf` function. Nominally the post-conversion string is limited to 80 characters, but this limit is configurable through the parameter CFE_MISSION_EVS_MAX_MESSAGE_LENGTH. Characters beyond the limit will be truncated. Do not use floating point conversions (f, e, E, g, and G) in the format string unless your application will be running in a system that supports floating point arithmetic. Do not use non-printable characters (\t, \n, etc.) in the format string; they will mess up the formatting when the events are displayed on the ground system.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_EVS_APP_NOT_REGISTERED	Application Not Registered.
CFE_EVS_APP_ILLEGAL_APP_ID	Illegal Application ID.

See also

CFE_EVS_SendEventWithAppID, CFE_EVS_SendTimedEvent

37.15.2.2 CFE_EVS_SendEventWithAppID()

```
CFE_Status_t CFE_Status_t CFE_EVS_SendEventWithAppID (
            uint16 EventID,
            uint16 EventType,
            CFE_ES_AppId_t AppID,
            const char * Spec,
             ...  )
```

Generate a software event given the specified Application ID.

Description

This routine generates a software event message. If the EventID is not filtered, the event will be sent as a software bus message, optionally logged in the local event log, and optionally sent as an ASCII text string out the enabled output port(s). Note that this function should really only be used from within an API in order to preserve the context of an Application's event. In general, CFE_EVS_SendEvent should be used.

Assumptions, External Events, and Notes:

The Application ID must correspond to a registered application or core service. For messages outside the context of a registered appliction (for example early in app initialization or if registration fails) CFE_ES_WriteToSysLog can be used for reporting.

Parameters

in	*EventID*	A numeric literal used to uniquely identify an application event. The `EventID` is defined and supplied by the application sending the event.
in	*EventType*	A numeric literal used to classify an event, one of: • CFE_EVS_EventType_DEBUG • CFE_EVS_EventType_INFORMATION • CFE_EVS_EventType_ERROR • CFE_EVS_EventType_CRITICAL
in	*AppID*	The Application ID from which the event message should appear.
in	*Spec*	A pointer to a null terminated text string describing the output format for the event. This is the same type of format string used for the ANSI `printf` function. Nominally the post-conversion string is limited to 80 characters, but this limit is configurable through the parameter CFE_MISSION_EVS_MAX_MESSAGE_LENGTH. Characters beyond the limit will be truncated. Do not use floating point conversions (f, e, E, g, and G) in the format string unless your application will be running in a system that supports floating point arithmetic. Do not use non-printable characters (\t, \n, etc.) in the format string; they will mess up the formatting when the events are displayed on the ground system.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_EVS_APP_NOT_REGISTERED	Application Not Registered.
CFE_EVS_APP_ILLEGAL_APP_ID	Illegal Application ID.

See also

CFE_EVS_SendEvent, CFE_EVS_SendTimedEvent

37.15.2.3 CFE_EVS_SendTimedEvent()

```
CFE_Status_t CFE_Status_t CFE_Status_t CFE_EVS_SendTimedEvent (
            CFE_TIME_SysTime_t Time,
            uint16 EventID,
            uint16 EventType,
            const char * Spec,
             ...  )
```

Generate a software event with a specific time tag.

Description

This routine is the same as CFE_EVS_SendEvent except that the caller specifies the event time instead of having the EVS use the current spacecraft time. This routine should be used in situations where an error condition is detected at one time, but the event message is reported at a later time.

Assumptions, External Events, and Notes:

This API only works within the context of a registered application or core service. For messages outside the context of a registered appliction (for example early in app initialization or if registration fails) CFE_ES_WriteToSysLog can be used for reporting.

Parameters

in	*Time*	The time to include in the event. This will usually be a time returned by the function CFE_TIME_GetTime.
in	*EventID*	A numeric literal used to uniquely identify an application event. The `EventID` is defined and supplied by the application sending the event.
in	*EventType*	A numeric literal used to classify an event, one of: • CFE_EVS_EventType_DEBUG • CFE_EVS_EventType_INFORMATION • CFE_EVS_EventType_ERROR • CFE_EVS_EventType_CRITICAL
in	*Spec*	A pointer to a null terminated text string describing the output format for the event. This is the same type of format string used for the ANSI `printf` function. Nominally the post-conversion string is limited to 80 characters, but this limit is configurable through the parameter CFE_MISSION_EVS_MAX_MESSAGE_LENGTH. Characters beyond the limit will be truncated. Do not use floating point conversions (f, e, E, g, and G) in the format string unless your application will be running in a system that supports floating point arithmetic. Do not use non-printable characters (\t, \n, etc.) in the format string; they will mess up the formatting when the events are displayed on the ground system.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_EVS_APP_NOT_REGISTERED	Application Not Registered.
CFE_EVS_APP_ILLEGAL_APP_ID	Illegal Application ID.

See also

CFE_EVS_SendEvent, CFE_EVS_SendEventWithAppID

37.16 cFE Reset Event Filter APIs

Functions

- CFE_Status_t CFE_EVS_ResetFilter (int16 EventID)

  Resets the calling application's event filter for a single event ID.
- CFE_Status_t CFE_EVS_ResetAllFilters (void)

  Resets all of the calling application's event filters.

37.16.1 Detailed Description

37.16.2 Function Documentation

37.16.2.1 CFE_EVS_ResetAllFilters()

```
CFE_Status_t CFE_EVS_ResetAllFilters (
            void  )
```

Resets all of the calling application's event filters.

Description

This routine resets all the calling application's event filter counters to zero, providing a quick and convenient method for resetting event filters.

Assumptions, External Events, and Notes:

None

Returns

Execution status below or from CFE_ES_GetAppID, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_EVS_APP_NOT_REGISTERED	Application Not Registered.
CFE_EVS_APP_ILLEGAL_APP_ID	Illegal Application ID.

See also

CFE_EVS_ResetFilter

37.16.2.2 CFE_EVS_ResetFilter()

```
CFE_Status_t CFE_EVS_ResetFilter (
            int16 EventID )
```

Resets the calling application's event filter for a single event ID.

Description

The effect of resetting an event filter depends on the filter scheme. The CFE_EVS_EventFilter_BINARY scheme resets the filter counter for the specified Event ID.

Assumptions, External Events, and Notes:

None

Parameters

in	*EventID*	A numeric literal used to uniquely identify an application event. The `EventID` is defined and supplied by the application sending the event.

Returns

Execution status below or from CFE_ES_GetAppID, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_EVS_APP_NOT_REGISTERED	Application Not Registered.
CFE_EVS_APP_ILLEGAL_APP_ID	Illegal Application ID.

See also

CFE_EVS_ResetAllFilters

37.17 cFE Internal Event Services APIs, internal to CFE core

Functions

- void CFE_EVS_TaskMain (void)

  Entry Point for cFE Core Application.
- int32 CFE_EVS_EarlyInit (void)

  Initializes the cFE core module API Library.
- int32 CFE_EVS_CleanUpApp (CFE_ES_AppId_t AppId)

  Removes EVS resources associated with specified Application.

37.17.1 Detailed Description

37.17.2 Function Documentation

37.17.2.1 CFE_EVS_CleanUpApp()

```
int32 CFE_EVS_CleanUpApp (
            CFE_ES_AppId_t AppId )
```

Removes EVS resources associated with specified Application.

Description

This function is called by cFE Executive Services to cleanup after an Application has been terminated. It frees resources that have been allocated to the specified Application.

37.17.2.2 CFE_EVS_EarlyInit()

```
int32 CFE_EVS_EarlyInit (
            void )
```

Initializes the cFE core module API Library.

Description

Initializes the cFE core module API Library

Assumptions, External Events, and Notes:

1. This function MUST be called before any module API's are called.

37.17.2.3 CFE_EVS_TaskMain()

```
void CFE_EVS_TaskMain (
            void  )
```

Entry Point for cFE Core Application.

Description

This is the entry point to the cFE EVS Core Application.

Assumptions, External Events, and Notes:

None

37.18 cFE File Header Management APIs

Functions

- CFE_Status_t CFE_FS_ReadHeader (CFE_FS_Header_t *Hdr, osal_id_t FileDes)

  Read the contents of the Standard cFE File Header.
- void CFE_FS_InitHeader (CFE_FS_Header_t *Hdr, const char *Description, uint32 SubType)

  Initializes the contents of the Standard cFE File Header.
- CFE_Status_t CFE_FS_WriteHeader (osal_id_t FileDes, CFE_FS_Header_t *Hdr)

  Write the specified Standard cFE File Header to the specified file.
- CFE_Status_t CFE_FS_SetTimestamp (osal_id_t FileDes, CFE_TIME_SysTime_t NewTimestamp)

  Modifies the Time Stamp field in the Standard cFE File Header for the specified file.

37.18.1 Detailed Description

37.18.2 Function Documentation

37.18.2.1 CFE_FS_InitHeader()

```
void CFE_FS_InitHeader (
            CFE_FS_Header_t * Hdr,
            const char * Description,
            uint32 SubType )
```

Initializes the contents of the Standard cFE File Header.

Description

This API will clear the specified CFE_FS_Header_t variable and initialize the description field with the specified value

Parameters

in	*Hdr*	Pointer to a variable of type CFE_FS_Header_t that will be cleared and initialized
in	**Description*	Initializes Header's Description
in	*SubType*	Initializes Header's SubType

See also

CFE_FS_WriteHeader

37.18.2.2 CFE_FS_ReadHeader()

```
CFE_Status_t CFE_FS_ReadHeader (
            CFE_FS_Header_t * Hdr,
            osal_id_t FileDes )
```

Read the contents of the Standard cFE File Header.

Description

This API will fill the specified CFE_FS_Header_t variable with the contents of the Standard cFE File Header of the file identified by the given File Descriptor.

Assumptions, External Events, and Notes:

1. The File has already been successfully opened using OS_OpenCreate and the caller has a legitimate File Descriptor.

Parameters

in	*FileDes*	File Descriptor obtained from a previous call to OS_OpenCreate that is associated with the file whose header is to be read.
in,out	*Hdr*	Pointer to a variable of type CFE_FS_Header_t that will be filled with the contents of the Standard cFE File Header. ∗Hdr is the contents of the Standard cFE File Header for the specified file.

Returns

Execution status, see cFE Return Code Defines

See also

CFE_FS_WriteHeader

37.18.2.3 CFE_FS_SetTimestamp()

```
CFE_Status_t CFE_FS_SetTimestamp (
            osal_id_t FileDes,
            CFE_TIME_SysTime_t NewTimestamp )
```

Modifies the Time Stamp field in the Standard cFE File Header for the specified file.

Description

This API will modify the timestamp found in the Standard cFE File Header of the specified file. The timestamp will be replaced with the time specified by the caller.

Assumptions, External Events, and Notes:

1. The File has already been successfully opened using OS_OpenCreate and the caller has a legitimate File Descriptor.
2. The `NewTimestamp` field has been filled appropriately by the Application.

Parameters

in	*FileDes*	File Descriptor obtained from a previous call to OS_OpenCreate that is associated with the file whose header is to be read.
in	*NewTimestamp*	A CFE_TIME_SysTime_t data structure containing the desired time to be put into the file's Standard cFE File Header.

Returns

Execution status, see cFE Return Code Defines

37.18.2.4 CFE_FS_WriteHeader()

```
CFE_Status_t CFE_FS_WriteHeader (
            osal_id_t FileDes,
            CFE_FS_Header_t * Hdr )
```

Write the specified Standard cFE File Header to the specified file.

Description

This API will output the specified CFE_FS_Header_t variable, with some fields automatically updated, to the specified file as the Standard cFE File Header. This API will automatically populate the following fields in the specified CFE_FS_Header_t:

1. `ContentType` - Filled with 0x63464531 ('cFE1')
2. `Length` - Filled with the sizeof(CFE_FS_Header_t)
3. `SpacecraftID` - Filled with the Spacecraft ID
4. `ProcessorID` - Filled with the Processor ID
5. `ApplicationID` - Filled with the Application ID
6. `TimeSeconds` - Filled with the Time, in seconds, as obtained by CFE_TIME_GetTime
7. `TimeSubSeconds` - Filled with the Time, subseconds, as obtained by CFE_TIME_GetTime

Assumptions, External Events, and Notes:

1. The File has already been successfully opened using OS_OpenCreate and the caller has a legitimate File Descriptor.
2. The `SubType` field has been filled appropriately by the Application.
3. The `Description` field has been filled appropriately by the Application.

Parameters

in	*FileDes*	File Descriptor obtained from a previous call to OS_OpenCreate that is associated with the file whose header is to be read.
in,out	*Hdr*	Pointer to a variable of type CFE_FS_Header_t that will be filled with the contents of the Standard cFE File Header. ∗Hdr is the contents of the Standard cFE File Header for the specified file.

Returns

Execution status, see cFE Return Code Defines

See also

CFE_FS_ReadHeader

37.19 cFE File Utility APIs

Functions

- const char ∗ CFE_FS_GetDefaultMountPoint (CFE_FS_FileCategory_t FileCategory)

  Get the default virtual mount point for a file category.
- const char ∗ CFE_FS_GetDefaultExtension (CFE_FS_FileCategory_t FileCategory)

  Get the default filename extension for a file category.
- int32 CFE_FS_ParseInputFileNameEx (char ∗OutputBuffer, const char ∗InputBuffer, size_t OutputBufSize, size↩_t InputBufSize, const char ∗DefaultInput, const char ∗DefaultPath, const char ∗DefaultExtension)

  Parse a filename input from an input buffer into a local buffer.
- int32 CFE_FS_ParseInputFileName (char ∗OutputBuffer, const char ∗InputName, size_t OutputBufSize, CFE_↩FS_FileCategory_t FileCategory)

  Parse a filename string from the user into a local buffer.
- CFE_Status_t CFE_FS_ExtractFilenameFromPath (const char ∗OriginalPath, char ∗FileNameOnly)

  Extracts the filename from a unix style path and filename string.
- int32 CFE_FS_BackgroundFileDumpRequest (CFE_FS_FileWriteMetaData_t ∗Meta)

  Register a background file dump request.
- bool CFE_FS_BackgroundFileDumpIsPending (const CFE_FS_FileWriteMetaData_t ∗Meta)

  Query if a background file write request is currently pending.
- bool CFE_FS_RunBackgroundFileDump (uint32 ElapsedTime, void ∗Arg)

  Execute the background file write job(s)

37.19.1 Detailed Description

37.19.2 Function Documentation

37.19.2.1 CFE_FS_BackgroundFileDumpIsPending()

```
bool CFE_FS_BackgroundFileDumpIsPending (
            const CFE_FS_FileWriteMetaData_t * Meta )
```

Query if a background file write request is currently pending.

Description

This returns "true" while the request is on the background work queue This returns "false" once the request is complete and removed from the queue.

Assumptions, External Events, and Notes:

None

Parameters

in,out	*Meta*	The background file write persistent state object

Returns

true if request is already pending, false if not

37.19.2.2 CFE_FS_BackgroundFileDumpRequest()

```
int32 CFE_FS_BackgroundFileDumpRequest (
            CFE_FS_FileWriteMetaData_t * Meta )
```

Register a background file dump request.

Description

Puts the previously-initialized metadata into the pending request queue

Assumptions, External Events, and Notes:

Metadata structure should be stored in a static memory area (not on heap) as it must persist and be accessible by the file writer task throughout the asynchronous job operation.

Parameters

in,out	*Meta*	The background file write persistent state object

Returns

Execution status, see cFE Return Code Defines

37.19.2.3 CFE_FS_ExtractFilenameFromPath()

```
CFE_Status_t CFE_FS_ExtractFilenameFromPath (
            const char * OriginalPath,
            char * FileNameOnly )
```

Extracts the filename from a unix style path and filename string.

Description

This API will take the original unix path/filename combination and extract the base filename. Example: Given the path/filename : "/cf/apps/myapp.o.gz" this function will return the filename: "myapp.o.gz".

Assumptions, External Events, and Notes:

1. The paths and filenames used here are the standard unix style filenames separated by "/" characters.
2. The extracted filename (including terminator) is no longer than OS_MAX_PATH_LEN

Parameters

in	*OriginalPath*	The original path.
out	*FileNameOnly*	The filename that is extracted from the path.

Returns

Execution status, see cFE Return Code Defines

37.19.2.4 CFE_FS_GetDefaultExtension()

```
const char* CFE_FS_GetDefaultExtension (
            CFE_FS_FileCategory_t FileCategory )
```

Get the default filename extension for a file category.

Certain file types may have an extension that varies from system to system. This is primarily an issue for application modules which are ".so" on Linux systems, ".dll" on Windows, ".o" on VxWorks, ".obj" on RTEMS, and so on.

This uses a combination of compile-time configuration and hints from the build environment to get the default/expected extension for a given file category.

Returns

String containing the extension, or NULL if unkown/invalid

37.19.2.5 CFE_FS_GetDefaultMountPoint()

```
const char* CFE_FS_GetDefaultMountPoint (
            CFE_FS_FileCategory_t FileCategory )
```

Get the default virtual mount point for a file category.

Certain classes of files generally reside in a common directory, mainly either the persistent storage (/cf typically) or ram disk (/ram typically).

Ephemeral status files are generally in the ram disk while application modules and scripts are generally in the persistent storage.

This returns the expected directory for a given class of files in the form of a virtual OSAL mount point string.

Returns

String containing the mount point, or NULL if unkown/invalid

37.19.2.6 CFE_FS_ParseInputFileName()

```
int32 CFE_FS_ParseInputFileName (
            char * OutputBuffer,
            const char * InputName,
            size_t OutputBufSize,
            CFE_FS_FileCategory_t FileCategory )
```

Parse a filename string from the user into a local buffer.

Description

Simplified API for CFE_FS_ParseInputFileNameEx() where input is always known to be a non-empty, null terminated string and the fixed-length input buffer not needed. For instance this may be used where the input is a fixed string from cfe_platform_cfg.h or similar.

Assumptions, External Events, and Notes:

The parameters are organized such that this is basically like strncpy() with an extra argument, and existing file name accesses which use a direct copy can easily change to use this instead.

See also

CFE_FS_ParseInputFileNameEx()

Parameters

out	*OutputBuffer*	Buffer to store result.
in	*InputName*	A null terminated input string
in	*OutputBufSize*	Maximum Size of output buffer.
in	*FileCategory*	The generalized category of file (implies default path/extension)

Returns

Execution status, see cFE Return Code Defines

37.19.2.7 CFE_FS_ParseInputFileNameEx()

```
int32 CFE_FS_ParseInputFileNameEx (
            char * OutputBuffer,
            const char * InputBuffer,
            size_t OutputBufSize,
            size_t InputBufSize,
            const char * DefaultInput,
            const char * DefaultPath,
            const char * DefaultExtension )
```

Parse a filename input from an input buffer into a local buffer.

Description

This provides a more user friendly way to specify file names, using default values for the path and extension, which can vary from system to system.

If InputBuffer is null or its length is zero, then DefaultInput is used as if it was the content of the input buffer.

If either the pathname or extension is missing from the input, it will be added from defaults, with the complete fully-qualified filename stored in the output buffer.

Assumptions, External Events, and Notes:

1. The paths and filenames used here are the standard unix style filenames separated by "/" (path) and "." (extension) characters.
2. Input Buffer has a fixed max length. Parsing will not exceed InputBufSize, and does not need to be null terminated. However parsing will stop at the first null char, when the input is shorter than the maximum.

Parameters

out	*OutputBuffer*	Buffer to store result.
in	*InputBuffer*	A input buffer that may contain a file name (e.g. from command).
in	*OutputBufSize*	Maximum Size of output buffer.
in	*InputBufSize*	Maximum Size of input buffer.
in	*DefaultInput*	Default value to use for input if InputBffer is empty
in	*DefaultPath*	Default value to use for pathname if omitted from input
in	*DefaultExtension*	Default value to use for extension if omitted from input

Returns

Execution status, see cFE Return Code Defines

37.19.2.8 CFE_FS_RunBackgroundFileDump()

```
bool CFE_FS_RunBackgroundFileDump (
            uint32 ElapsedTime,
            void * Arg )
```

Execute the background file write job(s)

Description

Runs the state machine associated with background file write requests

Assumptions, External Events, and Notes:

This should only be invoked as a background job from the ES background task, it should not be invoked directly.

Parameters

in	*ElapsedTime*	The amount of time passed since last invocation (ms)
in	*Arg*	Not used/ignored

Returns

true if jobs are pending, false if idle

37.20 cFE Internal File Service APIs, internal to CFE core

Functions

- int32 CFE_FS_EarlyInit (void)

  Initializes the cFE core module API Library.

37.20.1 Detailed Description

37.20.2 Function Documentation

37.20.2.1 CFE_FS_EarlyInit()

```
int32 CFE_FS_EarlyInit (
            void  )
```

Initializes the cFE core module API Library.

Description

Initializes the cFE core module API Library

Assumptions, External Events, and Notes:

1. This function MUST be called before any module API's are called.

37.21 cFE Message header APIs

Functions

- CFE_Status_t CFE_MSG_Init (CFE_MSG_Message_t *MsgPtr, CFE_SB_MsgId_t MsgId, CFE_MSG_Size_t Size)

  Initialize a message.
- CFE_Status_t CFE_MSG_GetSize (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_Size_t *Size)

  Gets the total size of a message.
- CFE_Status_t CFE_MSG_SetSize (CFE_MSG_Message_t *MsgPtr, CFE_MSG_Size_t Size)

  Sets the total size of a message.
- CFE_Status_t CFE_MSG_GetType (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_Type_t *Type)

  Gets the message type.
- CFE_Status_t CFE_MSG_SetType (CFE_MSG_Message_t *MsgPtr, CFE_MSG_Type_t Type)

  Sets the message type.
- CFE_Status_t CFE_MSG_GetHeaderVersion (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_HeaderVersion_t *Version)

  Gets the message header version.
- CFE_Status_t CFE_MSG_SetHeaderVersion (CFE_MSG_Message_t *MsgPtr, CFE_MSG_HeaderVersion_t Version)

  Sets the message header version.
- CFE_Status_t CFE_MSG_GetHasSecondaryHeader (const CFE_MSG_Message_t *MsgPtr, bool *HasSecondary)

  Gets the message secondary header boolean.
- CFE_Status_t CFE_MSG_SetHasSecondaryHeader (CFE_MSG_Message_t *MsgPtr, bool HasSecondary)

  Sets the message secondary header boolean.
- CFE_Status_t CFE_MSG_GetApId (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_ApId_t *ApId)

  Gets the message application ID.
- CFE_Status_t CFE_MSG_SetApId (CFE_MSG_Message_t *MsgPtr, CFE_MSG_ApId_t ApId)

  Sets the message application ID.
- CFE_Status_t CFE_MSG_GetSegmentationFlag (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_SegmentationFlag_t *SegFlag)

  Gets the message segmentation flag.
- CFE_Status_t CFE_MSG_SetSegmentationFlag (CFE_MSG_Message_t *MsgPtr, CFE_MSG_SegmentationFlag_t SegFlag)

  Sets the message segmentation flag.
- CFE_Status_t CFE_MSG_GetSequenceCount (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_SequenceCount_t *SeqCnt)

  Gets the message sequence count.
- CFE_Status_t CFE_MSG_SetSequenceCount (CFE_MSG_Message_t *MsgPtr, CFE_MSG_SequenceCount_t SeqCnt)

  Sets the message sequence count.
- CFE_Status_t CFE_MSG_GetEDSVersion (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_EDSVersion_t *Version)

  Gets the message EDS version.
- CFE_Status_t CFE_MSG_SetEDSVersion (CFE_MSG_Message_t *MsgPtr, CFE_MSG_EDSVersion_t Version)

  Sets the message EDS version.
- CFE_Status_t CFE_MSG_GetEndian (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_Endian_t *Endian)

Gets the message endian.

- CFE_Status_t CFE_MSG_SetEndian (CFE_MSG_Message_t *MsgPtr, CFE_MSG_Endian_t Endian)

  Sets the message endian.
- CFE_Status_t CFE_MSG_GetPlaybackFlag (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_PlaybackFlag↩_t *PlayFlag)

  Gets the message playback flag.
- CFE_Status_t CFE_MSG_SetPlaybackFlag (CFE_MSG_Message_t *MsgPtr, CFE_MSG_PlaybackFlag_↩t PlayFlag)

  Sets the message playback flag.
- CFE_Status_t CFE_MSG_GetSubsystem (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_Subsystem_↩t *Subsystem)

  Gets the message subsystem.
- CFE_Status_t CFE_MSG_SetSubsystem (CFE_MSG_Message_t *MsgPtr, CFE_MSG_Subsystem_t Subsystem)

  Sets the message subsystem.
- CFE_Status_t CFE_MSG_GetSystem (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_System_t *System)

  Gets the message system.
- CFE_Status_t CFE_MSG_SetSystem (CFE_MSG_Message_t *MsgPtr, CFE_MSG_System_t System)

  Sets the message system.
- CFE_Status_t CFE_MSG_GenerateChecksum (CFE_MSG_Message_t *MsgPtr)

  Calculates and sets the checksum of a message.
- CFE_Status_t CFE_MSG_ValidateChecksum (const CFE_MSG_Message_t *MsgPtr, bool *IsValid)

  Validates the checksum of a message.
- CFE_Status_t CFE_MSG_SetFcnCode (CFE_MSG_Message_t *MsgPtr, CFE_MSG_FcnCode_t FcnCode)

  Sets the function code field in a message.
- CFE_Status_t CFE_MSG_GetFcnCode (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_FcnCode_t *Fcn↩Code)

  Gets the function code field from a message.
- CFE_Status_t CFE_MSG_GetMsgTime (const CFE_MSG_Message_t *MsgPtr, CFE_TIME_SysTime_t *Time)

  Gets the time field from a message.
- CFE_Status_t CFE_MSG_SetMsgTime (CFE_MSG_Message_t *MsgPtr, CFE_TIME_SysTime_t Time)

  Sets the time field in a message.

37.21.1 Detailed Description

37.21.2 Function Documentation

37.21.2.1 CFE_MSG_GenerateChecksum()

```
CFE_Status_t CFE_MSG_GenerateChecksum (
            CFE_MSG_Message_t * MsgPtr )
```

Calculates and sets the checksum of a message.

Description

This routine calculates the checksum of a message according to an implementation-defined algorithm. Then, it sets the checksum field in the message with the calculated value. The contents and location of this field will depend on the underlying implementation of messages. It may be a checksum, a CRC, or some other algorithm.

Assumptions, External Events, and Notes:

- If the underlying implementation of messages does not include a checksum field, then this routine will return CFE_MSG_WRONG_MSG_TYPE

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.
CFE_MSG_WRONG_MSG_TYPE	Error - wrong type.

37.21.2.2 CFE_MSG_GetApId()

```
CFE_Status_t CFE_MSG_GetApId (
            const CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_ApId_t * ApId )
```

Gets the message application ID.

Description

This routine gets the message application ID.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*ApId*	Application ID

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.3 CFE_MSG_GetEDSVersion()

```
CFE_Status_t CFE_MSG_GetEDSVersion (
            const CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_EDSVersion_t * Version )
```

Gets the message EDS version.

Description

This routine gets the message EDS version.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*Version*	EDS Version

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.4 CFE_MSG_GetEndian()

```
CFE_Status_t CFE_MSG_GetEndian (
            const CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_Endian_t * Endian )
```

Gets the message endian.

Description

This routine gets the message endian.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*Endian*	Endian

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.5 CFE_MSG_GetFcnCode()

```
CFE_Status_t CFE_MSG_GetFcnCode (
            const CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_FcnCode_t * FcnCode )
```

Gets the function code field from a message.

Description

This routine gets the function code from a message.

Assumptions, External Events, and Notes:

- If the underlying implementation of messages does not include a function code field, then this routine will set FcnCode to zero and return CFE_MSG_WRONG_MSG_TYPE

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*FcnCode*	The function code from the message

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.
CFE_MSG_WRONG_MSG_TYPE	Error - wrong type.

37.21.2.6 CFE_MSG_GetHasSecondaryHeader()

```
CFE_Status_t CFE_MSG_GetHasSecondaryHeader (
            const CFE_MSG_Message_t * MsgPtr,
            bool * HasSecondary )
```

Gets the message secondary header boolean.

Description

This routine gets the message secondary header boolean.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*HasSecondary*	Has secondary header flag

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.7 CFE_MSG_GetHeaderVersion()

```
CFE_Status_t CFE_MSG_GetHeaderVersion (
            const CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_HeaderVersion_t * Version )
```

Gets the message header version.

Description

This routine gets the message header version.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*Version*	Header version

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.8 CFE_MSG_GetMsgTime()

```
CFE_Status_t CFE_MSG_GetMsgTime (
            const CFE_MSG_Message_t * MsgPtr,
            CFE_TIME_SysTime_t * Time )
```

Gets the time field from a message.

Description

This routine gets the time from a message.

Assumptions, External Events, and Notes:

- If the underlying implementation of messages does not include a time field, then this routine will set Time to zero and return CFE_MSG_WRONG_MSG_TYPE
- Note default implementation of command messages do not have a time field.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*Time*	Time from the message

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.

Return values

CFE_MSG_BAD_ARGUMENT	Error - bad argument.
CFE_MSG_WRONG_MSG_TYPE	Error - wrong type.

37.21.2.9 CFE_MSG_GetPlaybackFlag()

```
CFE_Status_t CFE_MSG_GetPlaybackFlag (
            const CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_PlaybackFlag_t * PlayFlag )
```

Gets the message playback flag.

Description

This routine gets the message playback flag.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*PlayFlag*	Playback Flag

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.10 CFE_MSG_GetSegmentationFlag()

```
CFE_Status_t CFE_MSG_GetSegmentationFlag (
            const CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_SegmentationFlag_t * SegFlag )
```

Gets the message segmentation flag.

Description

This routine gets the message segmentation flag

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*SegFlag*	Segmentation flag

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.11 CFE_MSG_GetSequenceCount()

```
CFE_Status_t CFE_MSG_GetSequenceCount (
            const CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_SequenceCount_t * SeqCnt )
```

Gets the message sequence count.

Description

This routine gets the message sequence count.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*SeqCnt*	Sequence count

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.12 CFE_MSG_GetSize()

```
CFE_Status_t CFE_MSG_GetSize (
            const CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_Size_t * Size )
```

Gets the total size of a message.

Description

This routine gets the total size of the message.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*Size*	Total message size

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.13 CFE_MSG_GetSubsystem()

```
CFE_Status_t CFE_MSG_GetSubsystem (
            const CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_Subsystem_t * Subsystem )
```

Gets the message subsystem.

Description

This routine gets the message subsystem

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*Subsystem*	Subsystem

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.14 CFE_MSG_GetSystem()

```
CFE_Status_t CFE_MSG_GetSystem (
            const CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_System_t * System )
```

Gets the message system.

Description

This routine gets the message system id

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*System*	System

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.15 CFE_MSG_GetType()

```
CFE_Status_t CFE_MSG_GetType (
            const CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_Type_t * Type )
```

Gets the message type.

Description

This routine gets the message type.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*Type*	Message type

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.16 CFE_MSG_Init()

```
CFE_Status_t CFE_MSG_Init (
            CFE_MSG_Message_t * MsgPtr,
            CFE_SB_MsgId_t MsgId,
            CFE_MSG_Size_t Size )
```

Initialize a message.

Description

This routine initialize a message. The entire message is set to zero (based on size), defaults are set, then the size and bits from MsgId are set.

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.
in	*MsgId*	MsgId that corresponds to message
in	*Size*	Total size of the mesage (used to set length field)

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.17 CFE_MSG_SetApId()

```
CFE_Status_t CFE_MSG_SetApId (
            CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_ApId_t ApId )
```

Sets the message application ID.

Description

This routine sets the message application ID. Typically set at initialization using the MsgId, but API available to set bits that may not be included in MsgId.

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.
in	*ApId*	Application ID

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.18 CFE_MSG_SetEDSVersion()

```
CFE_Status_t CFE_MSG_SetEDSVersion (
            CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_EDSVersion_t Version )
```

Sets the message EDS version.

Description

This routine sets the message EDS version.

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.
in	*Version*	EDS Version

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.19 CFE_MSG_SetEndian()

```
CFE_Status_t CFE_MSG_SetEndian (
            CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_Endian_t Endian )
```

Sets the message endian.

Description

This routine sets the message endian. Invalid endian selection will set big endian.

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.
in	*Endian*	Endian

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.20 CFE_MSG_SetFcnCode()

```
CFE_Status_t CFE_MSG_SetFcnCode (
            CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_FcnCode_t FcnCode )
```

Sets the function code field in a message.

Description

This routine sets the function code of a message.

Assumptions, External Events, and Notes:

- If the underlying implementation of messages does not include a function code field, then this routine will do nothing to the message contents and will return CFE_MSG_WRONG_MSG_TYPE.

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.
in	*FcnCode*	The function code to include in the message.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.
CFE_MSG_WRONG_MSG_TYPE	Error - wrong type.

37.21.2.21 CFE_MSG_SetHasSecondaryHeader()

```
CFE_Status_t CFE_MSG_SetHasSecondaryHeader (
            CFE_MSG_Message_t * MsgPtr,
            bool HasSecondary )
```

Sets the message secondary header boolean.

Description

This routine sets the message has secondary header boolean. Typically only set within message initialization and not used by APPs.

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.
in	*HasSecondary*	Has secondary header flag

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.22 CFE_MSG_SetHeaderVersion()

```
CFE_Status_t CFE_MSG_SetHeaderVersion (
            CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_HeaderVersion_t Version )
```

Sets the message header version.

Description

This routine sets the message header version. Typically only set within message initialization and not used by APPs.

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.
in	*Version*	Header version

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.23 CFE_MSG_SetMsgTime()

```
CFE_Status_t CFE_MSG_SetMsgTime (
            CFE_MSG_Message_t * MsgPtr,
            CFE_TIME_SysTime_t Time )
```

Sets the time field in a message.

Description

This routine sets the time of a message. Most applications will want to use CFE_SB_TimeStampMsg instead of this function. But, when needed, this API can be used to set multiple messages with identical time stamps.

Assumptions, External Events, and Notes:

- If the underlying implementation of messages does not include a time field, then this routine will do nothing to the message contents and will return CFE_MSG_WRONG_MSG_TYPE.
- Note default implementation of command messages do not have a time field.

Parameters

in,out	*MsgPtr*	A pointer to the message.
in	*Time*	The time to include in the message. This will usually be a time from CFE_TIME_GetTime.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.
CFE_MSG_WRONG_MSG_TYPE	Error - wrong type.

37.21.2.24 CFE_MSG_SetPlaybackFlag()

```
CFE_Status_t CFE_MSG_SetPlaybackFlag (
            CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_PlaybackFlag_t PlayFlag )
```

Sets the message playback flag.

Description

This routine sets the message playback flag.

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.
in	*PlayFlag*	Playback Flag

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.25 CFE_MSG_SetSegmentationFlag()

```
CFE_Status_t CFE_MSG_SetSegmentationFlag (
            CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_SegmentationFlag_t SegFlag )
```

Sets the message segmentation flag.

Description

This routine sets the message segmentation flag.

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.
in	*SegFlag*	Segmentation flag

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.26 CFE_MSG_SetSequenceCount()

```
CFE_Status_t CFE_MSG_SetSequenceCount (
            CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_SequenceCount_t SeqCnt )
```

Sets the message sequence count.

Description

This routine sets the message sequence count.

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.
in	*SeqCnt*	Sequence count

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.27 CFE_MSG_SetSize()

```
CFE_Status_t CFE_MSG_SetSize (
            CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_Size_t Size )
```

Sets the total size of a message.

Description

This routine sets the total size of the message.

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.
in	*Size*	Total message size

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.28 CFE_MSG_SetSubsystem()

```
CFE_Status_t CFE_MSG_SetSubsystem (
            CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_Subsystem_t Subsystem )
```

Sets the message subsystem.

Description

This routine sets the message subsystem. Some bits may be set at initialization using the MsgId, but API available to set bits that may not be included in MsgId.

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.
in	*Subsystem*	Subsystem

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.29 CFE_MSG_SetSystem()

```
CFE_Status_t CFE_MSG_SetSystem (
            CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_System_t System )
```

Sets the message system.

Description

This routine sets the message system id. Some bits may be set at initialization using the MsgId, but API available to set bits that may not be included in MsgId.

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.
in	*System*	System

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.30 CFE_MSG_SetType()

```
CFE_Status_t CFE_MSG_SetType (
            CFE_MSG_Message_t * MsgPtr,
            CFE_MSG_Type_t Type )
```

Sets the message type.

Description

This routine sets the message type.

Parameters

in,out	*MsgPtr*	A pointer to the buffer that contains the message.
in	*Type*	Message type

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.21.2.31 CFE_MSG_ValidateChecksum()

```
CFE_Status_t CFE_MSG_ValidateChecksum (
            const CFE_MSG_Message_t * MsgPtr,
            bool * IsValid )
```

Validates the checksum of a message.

Description

This routine validates the checksum of a message according to an implementation-defined algorithm.

Assumptions, External Events, and Notes:

- If the underlying implementation of messages does not include a checksum field, then this routine will return CFE_MSG_WRONG_MSG_TYPE and set the IsValid parameter false.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message. This must point to the first byte of the message header.
out	*IsValid*	Checksum validation result • true - valid • false - invalid or not supported/implemented

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.
CFE_MSG_WRONG_MSG_TYPE	Error - wrong type.

37.22 cFE Message Id APIs

Functions

- CFE_Status_t CFE_MSG_GetMsgId (const CFE_MSG_Message_t ∗MsgPtr, CFE_SB_MsgId_t ∗MsgId)

  Gets the message id from a message.
- CFE_Status_t CFE_MSG_SetMsgId (CFE_MSG_Message_t ∗MsgPtr, CFE_SB_MsgId_t MsgId)

  Sets the message id bits in a message.
- CFE_Status_t CFE_MSG_GetTypeFromMsgId (CFE_SB_MsgId_t MsgId, CFE_MSG_Type_t ∗Type)

  Gets message type using message ID.

37.22.1 Detailed Description

37.22.2 Function Documentation

37.22.2.1 CFE_MSG_GetMsgId()

```
CFE_Status_t CFE_MSG_GetMsgId (
            const CFE_MSG_Message_t * MsgPtr,
            CFE_SB_MsgId_t * MsgId )
```

Gets the message id from a message.

Description

This routine gets the message id from a message. The message id is a hash of bits in the message header, used by the software bus for routing. Message id needs to be unique for each endpoint in the system.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*MsgId*	Message id

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.22.2.2 CFE_MSG_GetTypeFromMsgId()

```
CFE_Status_t CFE_MSG_GetTypeFromMsgId (
            CFE_SB_MsgId_t MsgId,
            CFE_MSG_Type_t * Type )
```

Gets message type using message ID.

Description

This routine gets the message type using the message ID

Parameters

in	*Msg↩ Id*	Message id
out	*Type*	Message type

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.22.2.3 CFE_MSG_SetMsgId()

```
CFE_Status_t CFE_MSG_SetMsgId (
            CFE_MSG_Message_t * MsgPtr,
            CFE_SB_MsgId_t MsgId )
```

Sets the message id bits in a message.

Description

This routine sets the message id bits in a message. The message id is a hash of bits in the message header, used by the software bus for routing. Message id needs to be unique for each endpoint in the system.

Note

This API only sets the bits in the header that make up the message ID. No other values in the header are modified.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the message.
out	*MsgId*	Message id

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

37.23 cFE Pipe Management APIs

Functions

- CFE_Status_t CFE_SB_CreatePipe (CFE_SB_PipeId_t ∗PipeIdPtr, uint16 Depth, const char ∗PipeName)

  Creates a new software bus pipe.
- CFE_Status_t CFE_SB_DeletePipe (CFE_SB_PipeId_t PipeId)

  Delete a software bus pipe.
- CFE_Status_t CFE_SB_PipeId_ToIndex (CFE_SB_PipeId_t PipeID, uint32 ∗Idx)

  Obtain an index value correlating to an SB Pipe ID.
- CFE_Status_t CFE_SB_SetPipeOpts (CFE_SB_PipeId_t PipeId, uint8 Opts)

  Set options on a pipe.
- CFE_Status_t CFE_SB_GetPipeOpts (CFE_SB_PipeId_t PipeId, uint8 ∗OptPtr)

  Get options on a pipe.
- CFE_Status_t CFE_SB_GetPipeName (char ∗PipeNameBuf, size_t PipeNameSize, CFE_SB_PipeId_t PipeId)

  Get the pipe name for a given id.
- CFE_Status_t CFE_SB_GetPipeIdByName (CFE_SB_PipeId_t ∗PipeIdPtr, const char ∗PipeName)

  Get pipe id by pipe name.

37.23.1 Detailed Description

37.23.2 Function Documentation

37.23.2.1 CFE_SB_CreatePipe()

```
CFE_Status_t CFE_SB_CreatePipe (
            CFE_SB_PipeId_t * PipeIdPtr,
            uint16 Depth,
            const char * PipeName )
```

Creates a new software bus pipe.

Description

This routine creates and initializes an input pipe that the calling application can use to receive software bus messages. By default, no messages are routed to the new pipe. So, the application must use CFE_SB_Subscribe to specify which messages it wants to receive on this pipe.

Assumptions, External Events, and Notes:

None

Parameters

in,out	*PipeIdPtr*	A pointer to a variable of type CFE_SB_PipeId_t, which will be filled in with the pipe ID information by the CFE_SB_CreatePipe routine. ∗PipeIdPtr is the identifier for the created pipe.
in	*Depth*	The maximum number of messages that will be allowed on this pipe at one time.
in	*PipeName*	A string to be used to identify this pipe in error messages and routing information telemetry. The string must be no longer than OS_MAX_API_NAME (including terminator). Longer strings will be truncated.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_MAX_PIPES_MET	Max Pipes Met.
CFE_SB_PIPE_CR_ERR	Pipe Create Error.

See also

CFE_SB_DeletePipe CFE_SB_GetPipeOpts CFE_SB_SetPipeOpts CFE_SB_GetPipeIdByName

37.23.2.2 CFE_SB_DeletePipe()

```
CFE_Status_t CFE_SB_DeletePipe (
            CFE_SB_PipeId_t PipeId )
```

Delete a software bus pipe.

Description

This routine deletes an input pipe and cleans up all data structures associated with the pipe. All subscriptions made for this pipe by calls to CFE_SB_Subscribe will be automatically removed from the SB routing tables. Any messages in the pipe will be discarded.

Applications should not call this routine for all of their SB pipes as part of their orderly shutdown process, as the pipe will be deleted by the support framework at the appropriate time.

Assumptions, External Events, and Notes:

None

Parameters

in	*PipeId*	The pipe ID (obtained previously from CFE_SB_CreatePipe) of the pipe to be deleted.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.

See also

CFE_SB_CreatePipe CFE_SB_GetPipeOpts CFE_SB_SetPipeOpts CFE_SB_GetPipeIdByName

37.23.2.3 CFE_SB_GetPipeIdByName()

```
CFE_Status_t CFE_SB_GetPipeIdByName (
            CFE_SB_PipeId_t * PipeIdPtr,
            const char * PipeName )
```

Get pipe id by pipe name.

Description

This routine finds the pipe id for a pipe name.

Parameters

in	*PipeName*	The name of the pipe.
out	*PipeIdPtr*	The PipeId for that name.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.

See also

CFE_SB_CreatePipe CFE_SB_DeletePipe CFE_SB_SetPipeOpts CFE_SB_PIPEOPTS_IGNOREMINE

37.23.2.4 CFE_SB_GetPipeName()

```
CFE_Status_t CFE_SB_GetPipeName (
            char * PipeNameBuf,
            size_t PipeNameSize,
            CFE_SB_PipeId_t PipeId )
```

Get the pipe name for a given id.

Description

This routine finds the pipe name for a pipe id.

Parameters

out	*PipeNameBuf*	The buffer to receive the pipe name.
in	*PipeNameSize*	The size (in chars) of the PipeName buffer.
in	*PipeId*	The PipeId for that name.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.

See also

CFE_SB_CreatePipe CFE_SB_DeletePipe CFE_SB_SetPipeOpts CFE_SB_GetPipeIdByName

37.23.2.5 CFE_SB_GetPipeOpts()

```
CFE_Status_t CFE_SB_GetPipeOpts (
            CFE_SB_PipeId_t PipeId,
            uint8 * OptPtr )
```

Get options on a pipe.

Description

This routine gets the current options on a pipe.

Parameters

in	*PipeId*	The pipe ID of the pipe to get options from.
out	**OptPtr*	A bit field of options.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.

See also

CFE_SB_CreatePipe CFE_SB_DeletePipe CFE_SB_SetPipeOpts CFE_SB_GetPipeIdByName CFE_SB_PIP↩EOPTS_IGNOREMINE

37.23.2.6 CFE_SB_PipeId_ToIndex()

```
CFE_Status_t CFE_SB_PipeId_ToIndex (
            CFE_SB_PipeId_t PipeID,
            uint32 * Idx )
```

Obtain an index value correlating to an SB Pipe ID.

This calculates a zero based integer value that may be used for indexing into a local resource table/array.

Index values are only guaranteed to be unique for resources of the same type. For instance, the indices corresponding to two [valid] application IDs will never overlap, but the index of a pipe ID and an app ID may be the same. Furthermore, indices may be reused if a resource is deleted and re-created.

Note

There is no inverse of this function - indices cannot be converted back to the original PipeID value. The caller should retain the original ID for future use.

Parameters

in	*PipeID*	Pipe ID to convert
out	*Idx*	Buffer where the calculated index will be stored

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

37.23.2.7 CFE_SB_SetPipeOpts()

```
CFE_Status_t CFE_SB_SetPipeOpts (
            CFE_SB_PipeId_t PipeId,
            uint8 Opts )
```

Set options on a pipe.

Description

This routine sets (or clears) options to alter the pipe's behavior. Options are (re)set every call to this routine.

Parameters

in	*PipeId*	The pipe ID of the pipe to set options on.
in	*Opts*	A bit field of options.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.

See also

CFE_SB_CreatePipe CFE_SB_DeletePipe CFE_SB_GetPipeOpts CFE_SB_GetPipeIdByName CFE_SB_PIPEOPTS_IGNOREMINE

37.24 cFE Message Subscription Control APIs

Functions

- CFE_Status_t CFE_SB_SubscribeEx (CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId, CFE_SB_Qos_t Quality, uint16 MsgLim)

  Subscribe to a message on the software bus.
- CFE_Status_t CFE_SB_Subscribe (CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId)

  Subscribe to a message on the software bus with default parameters.
- CFE_Status_t CFE_SB_SubscribeLocal (CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId, uint16 MsgLim)

  Subscribe to a message while keeping the request local to a cpu.
- CFE_Status_t CFE_SB_Unsubscribe (CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId)

  Remove a subscription to a message on the software bus.
- CFE_Status_t CFE_SB_UnsubscribeLocal (CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId)

  Remove a subscription to a message on the software bus on the current CPU.

37.24.1 Detailed Description

37.24.2 Function Documentation

37.24.2.1 CFE_SB_Subscribe()

```
CFE_Status_t CFE_SB_Subscribe (
            CFE_SB_MsgId_t MsgId,
            CFE_SB_PipeId_t PipeId )
```

Subscribe to a message on the software bus with default parameters.

Description

This routine adds the specified pipe to the destination list for the specified message ID. This is the same as CFE_SB_SubscribeEx with the Quality field set to CFE_SB_DEFAULT_QOS and MsgLim set to CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT (4).

Assumptions, External Events, and Notes:

Note: As subscriptions are received, the destinations are added to the head of a linked list. During the sending of a message, the list is traversed beginning at the head of the list. Therefore the message will first be sent to the last subscriber. If an application has timing constraints and needs to receive a message in the shortest possible time, the developer may consider holding off its subscription until other applications have subscribed to the message.

Parameters

in	*MsgId*	The message ID of the message to be subscribed to.
in	*PipeId*	The pipe ID of the pipe the subscribed message should be sent to.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_MAX_MSGS_MET	Max Messages Met.
CFE_SB_MAX_DESTS_MET	Max Destinations Met.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_BUF_ALOC_ERR	Buffer Allocation Error.

See also

CFE_SB_SubscribeEx, CFE_SB_SubscribeLocal, CFE_SB_Unsubscribe, CFE_SB_UnsubscribeLocal

37.24.2.2 CFE_SB_SubscribeEx()

```
CFE_Status_t CFE_SB_SubscribeEx (
            CFE_SB_MsgId_t MsgId,
            CFE_SB_PipeId_t PipeId,
            CFE_SB_Qos_t Quality,
            uint16 MsgLim )
```

Subscribe to a message on the software bus.

Description

This routine adds the specified pipe to the destination list associated with the specified message ID.

Assumptions, External Events, and Notes:

Note: As subscriptions are received, the destinations are added to the head of a linked list. During the sending of a message, the list is traversed beginning at the head of the list. Therefore the message will first be sent to the last subscriber. If an application has timing constraints and needs to receive a message in the shortest possible time, the developer may consider holding off its subscription until other applications have subscribed to the message.

Parameters

in	*MsgId*	The message ID of the message to be subscribed to.
in	*PipeId*	The pipe ID of the pipe the subscribed message should be sent to.
in	*Quality*	The requested Quality of Service (QoS) required of the messages. Most callers will use CFE_SB_DEFAULT_QOS for this parameter.
in	*MsgLim*	The maximum number of messages with this Message ID to allow in this pipe at the same time.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_MAX_MSGS_MET	Max Messages Met.
CFE_SB_MAX_DESTS_MET	Max Destinations Met.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_BUF_ALOC_ERR	Buffer Allocation Error.

See also

CFE_SB_Subscribe, CFE_SB_SubscribeLocal, CFE_SB_Unsubscribe, CFE_SB_UnsubscribeLocal

37.24.2.3 CFE_SB_SubscribeLocal()

```
CFE_Status_t CFE_SB_SubscribeLocal (
            CFE_SB_MsgId_t MsgId,
            CFE_SB_PipeId_t PipeId,
            uint16 MsgLim )
```

Subscribe to a message while keeping the request local to a cpu.

Description

This routine adds the specified pipe to the destination list for the specified message ID. This is similar to CFE_SB_SubscribeEx with the Quality field set to CFE_SB_DEFAULT_QOS and MsgLim set to CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT, but will not report the subscription. Subscription Reporting is enabled for interprocessor communication by way of the Software Bus Network (SBN) Application.

Assumptions, External Events, and Notes:

- This API is typically only used by Software Bus Network (SBN) Application

Parameters

in	*MsgId*	The message ID of the message to be subscribed to.
in	*PipeId*	The pipe ID of the pipe the subscribed message should be sent to.
in	*MsgLim*	The maximum number of messages with this Message ID to allow in this pipe at the same time.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_MAX_MSGS_MET	Max Messages Met.
CFE_SB_MAX_DESTS_MET	Max Destinations Met.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_BUF_ALOC_ERR	Buffer Allocation Error.

See also

CFE_SB_Subscribe, CFE_SB_SubscribeEx, CFE_SB_Unsubscribe, CFE_SB_UnsubscribeLocal

37.24.2.4 CFE_SB_Unsubscribe()

```
CFE_Status_t CFE_SB_Unsubscribe (
            CFE_SB_MsgId_t MsgId,
            CFE_SB_PipeId_t PipeId )
```

Remove a subscription to a message on the software bus.

Description

This routine removes the specified pipe from the destination list for the specified message ID.

Assumptions, External Events, and Notes:

None

Parameters

in	*MsgId*	The message ID of the message to be unsubscribed.
in	*PipeId*	The pipe ID of the pipe the subscribed message should no longer be sent to.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_NO_SUBSCRIBERS	No Subscribers.
CFE_SB_INTERNAL_ERR	Internal Error.

See also

CFE_SB_Subscribe, CFE_SB_SubscribeEx, CFE_SB_SubscribeLocal, CFE_SB_UnsubscribeLocal

37.24.2.5 CFE_SB_UnsubscribeLocal()

```
CFE_Status_t CFE_SB_UnsubscribeLocal (
            CFE_SB_MsgId_t MsgId,
            CFE_SB_PipeId_t PipeId )
```

Remove a subscription to a message on the software bus on the current CPU.

Description

This routine removes the specified pipe from the destination list for the specified message ID on the current CPU.

Assumptions, External Events, and Notes:

- This API is typically only used by Software Bus Network (SBN) Application

Parameters

in	*Msg↩ Id*	The message ID of the message to be unsubscribed.
in	*Pipe↩ Id*	The pipe ID of the pipe the subscribed message should no longer be sent to.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_NO_SUBSCRIBERS	No Subscribers.
CFE_SB_INTERNAL_ERR	Internal Error.

See also

CFE_SB_Subscribe, CFE_SB_SubscribeEx, CFE_SB_SubscribeLocal, CFE_SB_Unsubscribe

37.25 cFE Send/Receive Message APIs

Functions

- CFE_Status_t CFE_SB_TransmitMsg (CFE_MSG_Message_t ∗MsgPtr, bool IncrementSequenceCount)

  Transmit a message.
- CFE_Status_t CFE_SB_SendMsg (CFE_MSG_Message_t ∗MsgPtr)

  DEPRECATED: Send a software bus message.
- CFE_Status_t CFE_SB_PassMsg (CFE_MSG_Message_t ∗MsgPtr)

  DEPRECATED: Passes a software bus message.
- CFE_Status_t CFE_SB_ReceiveBuffer (CFE_SB_Buffer_t ∗∗BufPtr, CFE_SB_PipeId_t PipeId, int32 TimeOut)

  Receive a message from a software bus pipe.
- CFE_Status_t CFE_SB_RcvMsg (CFE_SB_Buffer_t ∗∗BufPtr, CFE_SB_PipeId_t PipeId, int32 TimeOut)

  DEPRECATED: receive buffer.

37.25.1 Detailed Description

37.25.2 Function Documentation

37.25.2.1 CFE_SB_PassMsg()

```
CFE_Status_t CFE_SB_PassMsg (
            CFE_MSG_Message_t * MsgPtr )
```

DEPRECATED: Passes a software bus message.

Deprecated Use CFE_SB_TransmitMsg

Description

This routine sends the specified message to all subscribers. The software bus will read the message ID from the message header to determine which pipes should receive the message. This routine is intended to pass messages not generated by the sending application.

Assumptions, External Events, and Notes:

- This routine will not normally wait for the receiver tasks to process the message before returning control to the caller's task.
- However, if a higher priority task is pending and subscribed to this message, that task may get to run before CFE_SB_PassMsg returns control to the caller.
- This routine will not modify the sequence counter in the message.

Parameters

in	*MsgPtr*	A pointer to the message to be sent. This must point to the first byte of the message header.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_MSG_TOO_BIG	Message Too Big.
CFE_SB_BUF_ALOC_ERR	Buffer Allocation Error.

37.25.2.2 CFE_SB_RcvMsg()

```
CFE_Status_t CFE_SB_RcvMsg (
            CFE_SB_Buffer_t ** BufPtr,
            CFE_SB_PipeId_t PipeId,
            int32 TimeOut )
```

DEPRECATED: receive buffer.

Deprecated use CFE_SB_ReceiveBuffer

37.25.2.3 CFE_SB_ReceiveBuffer()

```
CFE_Status_t CFE_SB_ReceiveBuffer (
            CFE_SB_Buffer_t ** BufPtr,
            CFE_SB_PipeId_t PipeId,
            int32 TimeOut )
```

Receive a message from a software bus pipe.

Description

This routine retrieves the next message from the specified pipe. If the pipe is empty, this routine will block until either a new message comes in or the timeout value is reached.

Assumptions, External Events, and Notes:

Note - If an error occurs in this API, the ∗BufPtr value may be NULL or random. Therefore, it is recommended that the return code be tested for CFE_SUCCESS before processing the message.

Parameters

in,out	*BufPtr*	A pointer to the software bus buffer to receive to. Typically a caller declares a ptr of type CFE_SB_Buffer_t (i.e. CFE_SB_Buffer_t ∗Ptr) then gives the address of that pointer (&Ptr) as this parmeter. After a successful receipt of a message, ∗BufPtr will point to the first byte of the software bus buffer. This should be used as a read-only pointer (in systems with an MMU, writes to this pointer may cause a memory protection fault). The ∗BufPtr is valid only until the next call to CFE_SB_ReceiveBuffer for the same pipe.
in	*PipeId*	The pipe ID of the pipe containing the message to be obtained.
in	*TimeOut*	The number of milliseconds to wait for a new message if the pipe is empty at the time of the call. This can also be set to CFE_SB_POLL for a non-blocking receive or CFE_SB_PEND_FOREVER to wait forever for a message to arrive.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_TIME_OUT	Time Out.
CFE_SB_PIPE_RD_ERR	Pipe Read Error.
CFE_SB_NO_MESSAGE	No Message.

37.25.2.4 CFE_SB_SendMsg()

```
CFE_Status_t CFE_SB_SendMsg (
            CFE_MSG_Message_t * MsgPtr )
```

DEPRECATED: Send a software bus message.

Deprecated Use CFE_SB_TransmitMsg

Description

This routine sends the specified message to all subscribers. The software bus will read the message ID from the message header to determine which pipes should receive the message.

Assumptions, External Events, and Notes:

- This routine will not normally wait for the receiver tasks to process the message before returning control to the caller's task.
- However, if a higher priority task is pending and subscribed to this message, that task may get to run before returning control to the caller.
- This function tracks and increments the source sequence counter of a telemetry message.

Parameters

in	*MsgPtr*	A pointer to the message to be sent. This must point to the first byte of the message header.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_MSG_TOO_BIG	Message Too Big.
CFE_SB_BUF_ALOC_ERR	Buffer Allocation Error.

37.25.2.5 CFE_SB_TransmitMsg()

```
CFE_Status_t CFE_SB_TransmitMsg (
            CFE_MSG_Message_t * MsgPtr,
            bool IncrementSequenceCount )
```

Transmit a message.

Description

This routine copies the specified message into a software bus buffer which is then transmitted to all subscribers. The software bus will read the message ID from the message header to determine which pipes should receive the message.

Assumptions, External Events, and Notes:

- This routine will not normally wait for the receiver tasks to process the message before returning control to the caller's task.
- However, if a higher priority task is pending and subscribed to this message, that task may get to run before returning control to the caller.

Parameters

in	*MsgPtr*	A pointer to the message to be sent. This must point to the first byte of the message header.
in	*IncrementSequenceCount*	Boolean to increment the internally tracked sequence count and update the message if the buffer contains a telemetry message

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_MSG_TOO_BIG	Message Too Big.
CFE_SB_BUF_ALOC_ERR	Buffer Allocation Error.

37.26 cFE Zero Copy APIs

Functions

- CFE_SB_Buffer_t * CFE_SB_ZeroCopyGetPtr (size_t MsgSize, CFE_SB_ZeroCopyHandle_t *BufferHandle)

  Get a buffer pointer to use for "zero copy" SB sends.
- CFE_Status_t CFE_SB_ZeroCopyReleasePtr (CFE_SB_Buffer_t *Ptr2Release, CFE_SB_ZeroCopyHandle_t ZeroCopyHandle)

  Release an unused "zero copy" buffer pointer.
- CFE_Status_t CFE_SB_TransmitBuffer (CFE_SB_Buffer_t *BufPtr, CFE_SB_ZeroCopyHandle_t ZeroCopyHandle, bool IncrementSequenceCount)

  Transmit a buffer.
- CFE_Status_t CFE_SB_ZeroCopySend (CFE_SB_Buffer_t *BufPtr, CFE_SB_ZeroCopyHandle_t BufferHandle)

  DEPRECATED: Send an SB message in "zero copy" mode.
- CFE_Status_t CFE_SB_ZeroCopyPass (CFE_SB_Buffer_t *BufPtr, CFE_SB_ZeroCopyHandle_t BufferHandle)

  DEPRECATED: Pass an SB message in "zero copy" mode.

37.26.1 Detailed Description

37.26.2 Function Documentation

37.26.2.1 CFE_SB_TransmitBuffer()

```
CFE_Status_t CFE_SB_TransmitBuffer (
            CFE_SB_Buffer_t * BufPtr,
            CFE_SB_ZeroCopyHandle_t ZeroCopyHandle,
            bool IncrementSequenceCount )
```

Transmit a buffer.

Description

This routine sends a message that has been created directly in an internal SB message buffer by an application (after a call to CFE_SB_ZeroCopyGetPtr). This interface is more complicated than the normal CFE_SB_TransmitMsg interface, but it avoids an extra copy of the message from the user's memory buffer to the software bus internal buffer. The "zero copy" interface can be used to improve performance in high-rate, high-volume software bus traffic.

Assumptions, External Events, and Notes:

1. A handle returned by CFE_SB_ZeroCopyGetPtr is "consumed" by a *successful* call to CFE_SB_TransmitBuffer.
2. If this function returns CFE_SUCCESS, this indicates the zero copy handle is now owned by software bus, and is no longer owned by the calling application, and should not be re-used.
3. Howver if this function fails (returns any error status) it does not change the state of the buffer at all, meaning the calling application still owns it. (a failure means the buffer is left in the same state it was before the call).
4. Applications should be written as if CFE_SB_ZeroCopyGetPtr is equivalent to a `malloc()` and a successful call to CFE_SB_TransmitBuffer is equivalent to a `free()`.
5. Applications must not de-reference the message pointer (for reading or writing) after a successful call to CFE_SB_TransmitBuffer.
6. This function will increment and apply the internally tracked sequence counter if set to do so.

Parameters

in	*BufPtr*	A pointer to the buffer to be sent.
in	*ZeroCopyHandle*	The handle supplied by the CFE_SB_ZeroCopyGetPtr call
in	*IncrementSequenceCount*	Boolean to increment the internally tracked sequence count and update the message if the buffer contains a telemetry message

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_MSG_TOO_BIG	Message Too Big.
CFE_SB_BUF_ALOC_ERR	Buffer Allocation Error.

37.26.2.2 CFE_SB_ZeroCopyGetPtr()

```
CFE_SB_Buffer_t* CFE_SB_ZeroCopyGetPtr (
            size_t MsgSize,
            CFE_SB_ZeroCopyHandle_t * BufferHandle )
```

Get a buffer pointer to use for "zero copy" SB sends.

Description

This routine can be used to get a pointer to one of the software bus' internal memory buffers that are used for sending messages. The caller can use this memory buffer to build an SB message, then send it using the CFE_SB_TransmitBuffer function. This interface is more complicated than the normal CFE_SB_TransmitMsg interface, but it avoids an extra copy of the message from the user's memory buffer to the software bus internal buffer. The "zero copy" interface can be used to improve performance in high-rate, high-volume software bus traffic.

Assumptions, External Events, and Notes:

1. The pointer returned by CFE_SB_ZeroCopyGetPtr is only good for one call to CFE_SB_TransmitBuffer.
2. Applications should be written as if CFE_SB_ZeroCopyGetPtr is equivalent to a `malloc()` and CFE_SB_TransmitBuffer is equivalent to a `free()`.
3. Applications must not de-reference the message pointer (for reading or writing) after the call to CFE_SB_TransmitBuffer.

Parameters

in	*MsgSize*	The size of the SB message buffer the caller wants (including the SB message header).
out	*BufferHandle*	A handle that must be supplied when sending or releasing in zero copy mode.

Returns

A pointer to a memory buffer that message data can be written to for use with CFE_SB_TransmitBuffer.

37.26.2.3 CFE_SB_ZeroCopyPass()

```
CFE_Status_t CFE_SB_ZeroCopyPass (
            CFE_SB_Buffer_t * BufPtr,
            CFE_SB_ZeroCopyHandle_t BufferHandle )
```

DEPRECATED: Pass an SB message in "zero copy" mode.

Deprecated use CFE_SB_TransmitBuffer

Description

This routine sends a message that has been created directly in an internal SB message buffer by an application (after a call to CFE_SB_ZeroCopyGetPtr). This interface is more complicated than the normal CFE_SB_Transmit↩Msg interface, but it avoids an extra copy of the message from the user's memory buffer to the software bus internal buffer. The "zero copy" interface can be used to improve performance in high-rate, high-volume software bus traffic. This version is intended to pass messages not generated by the caller (to preserve the source sequence count).

Assumptions, External Events, and Notes:

1. The pointer returned by CFE_SB_ZeroCopyGetPtr is only good for one call to CFE_SB_TransmitBuffer or CFE_SB_ZeroCopyPass.
2. Callers must not use the same SB message buffer for multiple sends.
3. Applications should be written as if CFE_SB_ZeroCopyGetPtr is equivalent to a `malloc()` and CFE_SB↩_ZeroCopyPass is equivalent to a `free()`.
4. Applications must not de-reference the message pointer (for reading or writing) after the call to CFE_SB_↩ZeroCopyPass.
5. This routine will not modify the sequence counter in a telemetry message

Parameters

in	*BufPtr*	A pointer to the SB buffer to be sent.
in	*BufferHandle*	The handle supplied with the CFE_SB_ZeroCopyGetPtr call.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.

Return values

CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_MSG_TOO_BIG	Message Too Big.
CFE_SB_BUF_ALOC_ERR	Buffer Allocation Error.
CFE_SB_BUFFER_INVALID	Buffer Invalid.

37.26.2.4 CFE_SB_ZeroCopyReleasePtr()

```
CFE_Status_t CFE_SB_ZeroCopyReleasePtr (
            CFE_SB_Buffer_t * Ptr2Release,
            CFE_SB_ZeroCopyHandle_t ZeroCopyHandle )
```

Release an unused "zero copy" buffer pointer.

Description

This routine can be used to release a pointer to one of the software bus' internal memory buffers.

Assumptions, External Events, and Notes:

1. This function is not needed for normal "zero copy" transfers. It is needed only for cleanup when an application gets a pointer using CFE_SB_ZeroCopyGetPtr, but (due to some error condition) never uses that pointer for a CFE_SB_TransmitBuffer

Parameters

in	*Ptr2Release*	A pointer to the SB internal buffer. This must be a pointer returned by a call to CFE_SB_ZeroCopyGetPtr, but never used in a call to CFE_SB_TransmitBuffer.
in	*ZeroCopyHandle*	This must be the handle supplied with the pointer when CFE_SB_ZeroCopyGetPtr was called.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_BUFFER_INVALID	Buffer Invalid.

37.26.2.5 CFE_SB_ZeroCopySend()

```
CFE_Status_t CFE_SB_ZeroCopySend (
            CFE_SB_Buffer_t * BufPtr,
            CFE_SB_ZeroCopyHandle_t BufferHandle )
```

DEPRECATED: Send an SB message in "zero copy" mode.

Deprecated use CFE_SB_TransmitBuffer

Description

This routine sends a message that has been created directly in an internal SB message buffer by an application (after a call to CFE_SB_ZeroCopyGetPtr). This interface is more complicated than the normal CFE_SB_TransmitMsg interface, but it avoids an extra copy of the message from the user's memory buffer to the software bus internal buffer. The "zero copy" interface can be used to improve performance in high-rate, high-volume software bus traffic.

Assumptions, External Events, and Notes:

1. The pointer returned by CFE_SB_ZeroCopyGetPtr is only good for one call to CFE_SB_TransmitBuffer.
2. Callers must not use the same SB message buffer for multiple sends.
3. Applications should be written as if CFE_SB_ZeroCopyGetPtr is equivalent to a `malloc()` and CFE_SB_TransmitBuffer is equivalent to a `free()`.
4. Applications must not de-reference the message pointer (for reading or writing) after the call to CFE_SB_TransmitBuffer.
5. This function tracks and increments the source sequence counter of a telemetry message.

Parameters

in	*BufPtr*	A pointer to the SB buffer to be sent.
in	*BufferHandle*	The handle supplied with the CFE_SB_ZeroCopyGetPtr call.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_MSG_TOO_BIG	Message Too Big.
CFE_SB_BUF_ALOC_ERR	Buffer Allocation Error.
CFE_SB_BUFFER_INVALID	Buffer Invalid.

37.27 cFE Setting Message Characteristics APIs

Functions

- void CFE_SB_InitMsg (void ∗MsgPtr, CFE_SB_MsgId_t MsgId, size_t Length, bool Clear)

  DEPRECATED - Initialize a buffer for a software bus message.
- void CFE_SB_SetMsgId (CFE_MSG_Message_t ∗MsgPtr, CFE_SB_MsgId_t MsgId)

  DEPRECATED - Sets the message ID of a software bus message.
- void CFE_SB_SetUserDataLength (CFE_MSG_Message_t ∗MsgPtr, size_t DataLength)

  Sets the length of user data in a software bus message.
- void CFE_SB_SetTotalMsgLength (CFE_MSG_Message_t ∗MsgPtr, size_t TotalLength)

  DEPRECATED: Sets the total length of a software bus message.
- CFE_Status_t CFE_SB_SetMsgTime (CFE_MSG_Message_t ∗MsgPtr, CFE_TIME_SysTime_t Time)

  Sets the time field in a software bus message.
- void CFE_SB_TimeStampMsg (CFE_MSG_Message_t ∗MsgPtr)

  Sets the time field in a software bus message with the current spacecraft time.
- CFE_Status_t CFE_SB_SetCmdCode (CFE_MSG_Message_t ∗MsgPtr, uint16 CmdCode)

  DEPRECATED:Sets the command code field in a software bus message.
- int32 CFE_SB_MessageStringSet (char ∗DestStringPtr, const char ∗SourceStringPtr, size_t DestMaxSize, size↩_t SourceMaxSize)

  Copies a string into a software bus message.

37.27.1 Detailed Description

37.27.2 Function Documentation

37.27.2.1 CFE_SB_InitMsg()

```
void CFE_SB_InitMsg (
            void * MsgPtr,
            CFE_SB_MsgId_t MsgId,
            size_t Length,
            bool Clear )
```

DEPRECATED - Initialize a buffer for a software bus message.

Deprecated Use CFE_MSG_Init

Description

This routine fills in the header information needed to create a valid software bus message.

Assumptions, External Events, and Notes:

None

Parameters

in	*MsgPtr*	A pointer to the buffer that will contain the message. This will point to the first byte of the message header. The `void*` data type allows the calling routine to use any data type when declaring its message buffer.
in	*MsgId*	The message ID to put in the message header.
in	*Length*	The total number of bytes of message data, including the SB message header .
in	*Clear*	A flag indicating whether to clear the rest of the message: • true - fill sequence count and packet data with zeroes. • false - leave sequence count and packet data unchanged.

37.27.2.2 CFE_SB_MessageStringSet()

```
int32 CFE_SB_MessageStringSet (
            char * DestStringPtr,
            const char * SourceStringPtr,
            size_t DestMaxSize,
            size_t SourceMaxSize )
```

Copies a string into a software bus message.

Description

Strings within software bus messages have a defined/fixed maximum length, and may not necessarily be null terminated within the message. This presents a possible issue when using the C library functions to copy strings out of a message.

This performs a very similar function to "strncpy()" except that the sizes of *both* buffers are passed in. Neither buffer is required to be null-terminated, but copying will stop after the first termination character is encountered.

If the destination buffer is not completely filled by the source data (such as if the supplied string was shorter than the allotted length) the destination buffer will be padded with NUL characters up to the size of the buffer, similar to what strncpy() does. This ensures that the entire destination buffer is set.

Note

If the source string buffer is already guaranteed to be null terminated, then there is no difference between the C library "strncpy()" function and this implementation. It is only necessary to use this when termination of the source buffer is not guaranteed.

Parameters

out	*DestStringPtr*	Pointer to destination buffer (component of SB message definition)
in	*SourceStringPtr*	Pointer to source buffer
in	*DestMaxSize*	Size of destination buffer as defined by the message definition
in	*SourceMaxSize*	Size of source buffer

Returns

Number of characters copied or error code, see cFE Return Code Defines

37.27.2.3 CFE_SB_SetCmdCode()

```
CFE_Status_t CFE_SB_SetCmdCode (
            CFE_MSG_Message_t * MsgPtr,
            uint16 CmdCode )
```

DEPRECATED:Sets the command code field in a software bus message.

Deprecated Use CFE_MSG_SetFcnCode

Description

This routine sets the command code of a software bus message (if SB messages are implemented as CCSDS packets, this will be the function code).

Assumptions, External Events, and Notes:

- If the underlying implementation of software bus messages does not include a command code field, then this routine will do nothing to the message contents and will return CFE_SB_WRONG_MSG_TYPE.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message. This must point to the first byte of the message header.
in	*CmdCode*	The command code to include in the message.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_WRONG_MSG_TYPE	Wrong Message Type.

37.27.2.4 CFE_SB_SetMsgId()

```
void CFE_SB_SetMsgId (
            CFE_MSG_Message_t * MsgPtr,
            CFE_SB_MsgId_t MsgId )
```

DEPRECATED - Sets the message ID of a software bus message.

Deprecated Use CFE_MSG_SetMsgId

Description

This routine sets the Message ID in a software bus message header.

Assumptions, External Events, and Notes:

None

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message. This must point to the first byte of the message header.
in	*MsgId*	The message ID to put into the message header.

37.27.2.5 CFE_SB_SetMsgTime()

```
CFE_Status_t CFE_SB_SetMsgTime (
            CFE_MSG_Message_t * MsgPtr,
            CFE_TIME_SysTime_t Time )
```

Sets the time field in a software bus message.

Deprecated Use CFE_MSG_SetMsgTime

Description

This routine sets the time of a software bus message. Most applications will want to use CFE_SB_TimeStampMsg instead of this function. But, when needed, this API can be used to send a group of SB messages with identical time stamps.

Assumptions, External Events, and Notes:

- If the underlying implementation of software bus messages does not include a time field, then this routine will do nothing to the message contents and will return CFE_SB_WRONG_MSG_TYPE.
- Note default implementation of command messages do not have a time field and will trigger the CFE_SB_WRONG_MSG_TYPE error

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message. This must point to the first byte of the message header.
in	*Time*	The time to include in the message. This will usually be a time returned by the function CFE_TIME_GetTime.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_SB_WRONG_MSG_TYPE	Wrong Message Type.

37.27.2.6 CFE_SB_SetTotalMsgLength()

```
void CFE_SB_SetTotalMsgLength (
            CFE_MSG_Message_t * MsgPtr,
            size_t TotalLength )
```

DEPRECATED: Sets the total length of a software bus message.

Deprecated Use CFE_MSG_SetSize

Description

This routine sets the field in the SB message header that determines the total length of the message. SB message header formats can be different for each deployment of the cFE. So, applications should use this function rather than trying to poke a length value directly into their SB message buffers.

Assumptions, External Events, and Notes:

None

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message. This must point to the first byte of the message header.
in	*TotalLength*	The length to set (total size of the message, in bytes, including headers).

37.27.2.7 CFE_SB_SetUserDataLength()

```
void CFE_SB_SetUserDataLength (
            CFE_MSG_Message_t * MsgPtr,
            size_t DataLength )
```

Sets the length of user data in a software bus message.

Description

This routine sets the field in the SB message header that determines the size of the user data in a software bus message. SB message header formats can be different for each deployment of the cFE. So, applications should use this function rather than trying to poke a length value directly into their SB message buffers.

Assumptions, External Events, and Notes:

- You must set a valid message ID in the SB message header before calling this function.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message. This must point to the first byte of the message header.
in	*DataLength*	The length to set (size of the user data, in bytes).

37.27.2.8 CFE_SB_TimeStampMsg()

```
void CFE_SB_TimeStampMsg (
            CFE_MSG_Message_t * MsgPtr )
```

Sets the time field in a software bus message with the current spacecraft time.

Description

This routine sets the time of a software bus message with the current spacecraft time. This will be the same time that is returned by the function CFE_TIME_GetTime.

Assumptions, External Events, and Notes:

- If the underlying implementation of software bus messages does not include a time field, then this routine will do nothing.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message. This must point to the first byte of the message header.

37.28 cFE Getting Message Characteristics APIs

Functions

- void ∗ CFE_SB_GetUserData (CFE_MSG_Message_t ∗MsgPtr)

  Get a pointer to the user data portion of a software bus message.
- CFE_SB_MsgId_t CFE_SB_GetMsgId (const CFE_MSG_Message_t ∗MsgPtr)

  DEPRECATED:Get the message ID of a software bus message.
- size_t CFE_SB_GetUserDataLength (const CFE_MSG_Message_t ∗MsgPtr)

  Gets the length of user data in a software bus message.
- size_t CFE_SB_GetTotalMsgLength (const CFE_MSG_Message_t ∗MsgPtr)

  DEPRECATED: Gets the total length of a software bus message.
- uint16 CFE_SB_GetCmdCode (CFE_MSG_Message_t ∗MsgPtr)

  DEPRECATED: Gets the command code field from a software bus message.
- CFE_TIME_SysTime_t CFE_SB_GetMsgTime (CFE_MSG_Message_t ∗MsgPtr)

  DEPRECATED: Gets the time field from a software bus message.
- int32 CFE_SB_MessageStringGet (char ∗DestStringPtr, const char ∗SourceStringPtr, const char ∗DefaultString, size_t DestMaxSize, size_t SourceMaxSize)

  Copies a string out of a software bus message.

37.28.1 Detailed Description

37.28.2 Function Documentation

37.28.2.1 CFE_SB_GetCmdCode()

```
uint16 CFE_SB_GetCmdCode (
            CFE_MSG_Message_t * MsgPtr )
```

DEPRECATED: Gets the command code field from a software bus message.

Deprecated Use CFE_MSG_GetFcnCode

Description

This routine gets the command code from a software bus message (if SB messages are implemented as CCSDS packets, this will be the function code).

Assumptions, External Events, and Notes:

- If the underlying implementation of software bus messages does not include a command code field, then this routine will return a zero.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message. This must point to the first byte of the message header.

Returns

The command code included in the software bus message header (if present). Otherwise, returns a command code value of zero.

37.28.2.2 CFE_SB_GetMsgId()

```
CFE_SB_MsgId_t CFE_SB_GetMsgId (
            const CFE_MSG_Message_t * MsgPtr )
```

DEPRECATED:Get the message ID of a software bus message.

Deprecated Use CFE_MSG_GetMsgId

Description

This routine returns the message ID from a software bus message.

Assumptions, External Events, and Notes:

None

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message.

Returns

The software bus Message ID from the message header.

37.28.2.3 CFE_SB_GetMsgTime()

```
CFE_TIME_SysTime_t CFE_SB_GetMsgTime (
            CFE_MSG_Message_t * MsgPtr )
```

DEPRECATED: Gets the time field from a software bus message.

Deprecated Use CFE_MSG_GetMsgTime

Description

This routine gets the time from a software bus message.

Assumptions, External Events, and Notes:

- If the underlying implementation of software bus messages does not include a time field, then this routine will return a zero time.
- Note default implementation of command messages do not have a time field.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message. This must point to the first byte of the message header.

Returns

The system time included in the software bus message header (if present), otherwise, returns a time value of zero.

37.28.2.4 CFE_SB_GetTotalMsgLength()

```
size_t CFE_SB_GetTotalMsgLength (
            const CFE_MSG_Message_t * MsgPtr )
```

DEPRECATED: Gets the total length of a software bus message.

Deprecated Use CFE_MSG_GetSize

Description

This routine returns the total size of the software bus message.

Assumptions, External Events, and Notes:

- For the CCSDS implementation of this API, the size is derived from the message header.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message. This must point to the first byte of the message header.

Returns

The total size (in bytes) of the software bus message, including headers.

37.28.2.5 CFE_SB_GetUserData()

```
void* CFE_SB_GetUserData (
            CFE_MSG_Message_t * MsgPtr )
```

Get a pointer to the user data portion of a software bus message.

Description

This routine returns a pointer to the user data portion of a software bus message. SB message header formats can be different for each deployment of the cFE. So, applications should use this function and avoid hard coding offsets into their SB message buffers.

Assumptions, External Events, and Notes:

None

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message.

Returns

A pointer to the first byte of user data within the software bus message.

37.28.2.6 CFE_SB_GetUserDataLength()

```
size_t CFE_SB_GetUserDataLength (
            const CFE_MSG_Message_t * MsgPtr )
```

Gets the length of user data in a software bus message.

Description

This routine returns the size of the user data in a software bus message.

Assumptions, External Events, and Notes:

None

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message. This must point to the first byte of the message header.

Returns

The size (in bytes) of the user data in the software bus message.

37.28.2.7 CFE_SB_MessageStringGet()

```
int32 CFE_SB_MessageStringGet (
            char * DestStringPtr,
            const char * SourceStringPtr,
            const char * DefaultString,
            size_t DestMaxSize,
            size_t SourceMaxSize )
```

Copies a string out of a software bus message.

Description

Strings within software bus messages have a defined/fixed maximum length, and may not necessarily be null terminated within the message. This presents a possible issue when using the C library functions to copy strings out of a message.

This function should replace use of C library functions such as strcpy/strncpy when copying strings out of software bus messages to local storage buffers.

Up to [SourceMaxSize] or [DestMaxSize-1] (whichever is smaller) characters will be coped from the source buffer to the destination buffer, and a NUL termination character will be written to the destination buffer as the last character.

If the DefaultString pointer is non-NULL, it will be used in place of the source string if the source is an empty string. This is typically a string constant that comes from the platform configuration, allowing default values to be assumed for fields that are unspecified.

IMPORTANT - the default string, if specified, must be null terminated. This will be the case if a string literal is passed in (the typical/expected use case).

If the default is NULL, then only the source string will be copied, and the result will be an empty string if the source was empty.

If the destination buffer is too small to store the entire string, it will be truncated, but it will still be null terminated.

Parameters

out	*DestStringPtr*	Pointer to destination buffer
in	*SourceStringPtr*	Pointer to source buffer (component of SB message definition)
in	*DefaultString*	Default string to use if source is empty
in	*DestMaxSize*	Size of destination storage buffer (must be at least 2)
in	*SourceMaxSize*	Size of source buffer as defined by the message definition

Returns

Number of characters copied or error code, see cFE Return Code Defines

37.29 cFE Checksum Control APIs

Functions

- uint16 CFE_SB_GetChecksum (CFE_MSG_Message_t *MsgPtr)

  DEPRECATED:Gets the checksum field from a software bus message.
- void CFE_SB_GenerateChecksum (CFE_MSG_Message_t *MsgPtr)

  DEPRECATED:Calculates and sets the checksum of a software bus message.
- bool CFE_SB_ValidateChecksum (CFE_MSG_Message_t *MsgPtr)

  DEPRECATED:Validates the checksum of a software bus message.

37.29.1 Detailed Description

37.29.2 Function Documentation

37.29.2.1 CFE_SB_GenerateChecksum()

```
void CFE_SB_GenerateChecksum (
            CFE_MSG_Message_t * MsgPtr )
```

DEPRECATED:Calculates and sets the checksum of a software bus message.

Deprecated Use CFE_MSG_GenerateChecksum

Description

This routine calculates the checksum of a software bus message according to an implementation-defined algorithm. Then, it sets the checksum field in the message with the calculated value. The contents and location of this field will depend on the underlying implementation of software bus messages. It may be a checksum, a CRC, or some other algorithm.

Assumptions, External Events, and Notes:

- If the underlying implementation of software bus messages does not include a checksum field, then this routine will do nothing.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message. This must point to the first byte of the message header.

37.29.2.2 CFE_SB_GetChecksum()

```
uint16 CFE_SB_GetChecksum (
            CFE_MSG_Message_t * MsgPtr )
```

DEPRECATED:Gets the checksum field from a software bus message.

Deprecated No use case

Description

This routine gets the checksum (or other message integrity check value) from a software bus message. The contents and location of this field will depend on the underlying implementation of software bus messages. It may be a checksum, a CRC, or some other algorithm. Users should not call this function as part of a message integrity check (call CFE_SB_ValidateChecksum instead).

Assumptions, External Events, and Notes:

- If the underlying implementation of software bus messages does not include a checksum field, then this routine will return a zero.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message. This must point to the first byte of the message header.

Returns

The checksum included in the software bus message header (if present), otherwise, returns a checksum value of zero.

37.29.2.3 CFE_SB_ValidateChecksum()

```
bool CFE_SB_ValidateChecksum (
            CFE_MSG_Message_t * MsgPtr )
```

DEPRECATED:Validates the checksum of a software bus message.

Deprecated Use CFE_MSG_ValidateChecksum

Description

This routine calculates the expected checksum of a software bus message according to an implementation-defined algorithm. Then, it checks the calculated value against the value in the message's checksum. If the checksums do not match, this routine will generate an event message reporting the error.

Assumptions, External Events, and Notes:

- If the underlying implementation of software bus messages does not include a checksum field this routine will always return false.

Parameters

in	*MsgPtr*	A pointer to the buffer that contains the software bus message. This must point to the first byte of the message header.

Returns

Boolean checksum result

Return values

true	The checksum field in the packet is valid.
false	The checksum field in the packet is not valid or the message type is wrong.

37.30 cFE Message ID APIs

Functions

- bool CFE_SB_IsValidMsgId (CFE_SB_MsgId_t MsgId)

  Identifies whether a given CFE_SB_MsgId_t is valid.
- static bool CFE_SB_MsgId_Equal (CFE_SB_MsgId_t MsgId1, CFE_SB_MsgId_t MsgId2)

  Identifies whether two CFE_SB_MsgId_t values are equal.
- static CFE_SB_MsgId_Atom_t CFE_SB_MsgIdToValue (CFE_SB_MsgId_t MsgId)

  Converts a CFE_SB_MsgId_t to a normal integer.
- static CFE_SB_MsgId_t CFE_SB_ValueToMsgId (CFE_SB_MsgId_Atom_t MsgIdValue)

  Converts a normal integer into a CFE_SB_MsgId_t.
- uint32 CFE_SB_GetPktType (CFE_SB_MsgId_t MsgId)

  Identifies packet type given message ID.

37.30.1 Detailed Description

37.30.2 Function Documentation

37.30.2.1 CFE_SB_GetPktType()

```
uint32 CFE_SB_GetPktType (
            CFE_SB_MsgId_t MsgId )
```

Identifies packet type given message ID.

Provides the packet type associated with the given message ID

Returns

Packet type

Return values

CFE_MSG_Type_Cmd	Command packet type
CFE_MSG_Type_Tlm	Telemetry packet type
CFE_MSG_Type_Invalid	Invalid/unknown packet type

Referenced by CFE_SB_ValueToMsgId().

37.30.2.2 CFE_SB_IsValidMsgId()

```
bool CFE_SB_IsValidMsgId (
            CFE_SB_MsgId_t MsgId )
```

Identifies whether a given CFE_SB_MsgId_t is valid.

Description

Implements a basic sanity check on the value provided

Returns

Boolean message ID validity indicator

Return values

true	Message ID is within the valid range
false	Message ID is not within the valid range

37.30.2.3 CFE_SB_MsgId_Equal()

```
static bool CFE_SB_MsgId_Equal (
            CFE_SB_MsgId_t MsgId1,
            CFE_SB_MsgId_t MsgId2 )  [inline], [static]
```

Identifies whether two CFE_SB_MsgId_t values are equal.

Description

In cases where the CFE_SB_MsgId_t type is not a simple integer type, it may not be possible to do a direct equality check. This inline function provides an abstraction for the equality check between two CFE_SB_MsgId_t values.

Applications should transition to using this function to compare MsgId values for equality to remain compatible with future versions of cFE.

Returns

Boolean message ID equality indicator

Return values

true	Message IDs are Equal
false	Message IDs are not Equal

Definition at line 1228 of file cfe_sb.h.

References CFE_SB_MSGID_UNWRAP_VALUE.

37.30.2.4 CFE_SB_MsgIdToValue()

```
static CFE_SB_MsgId_Atom_t CFE_SB_MsgIdToValue (
            CFE_SB_MsgId_t MsgId )  [inline], [static]
```

Converts a CFE_SB_MsgId_t to a normal integer.

Description

In cases where the CFE_SB_MsgId_t type is not a simple integer type, it is not possible to directly display the value in a printf-style statement, use it in a switch() statement, or other similar use cases.

This inline function provides the ability to map a CFE_SB_MsgId_t type back into a simple integer value.

Applications should transition to using this function wherever a CFE_SB_MsgId_t type needs to be used as an integer.

Assumptions and Notes:

This negates the type safety that was gained by using a non- integer type for the CFE_SB_MsgId_t value. This should only be used in specific cases such as UI display (printf, events, etc) where the value is being sent externally. Any internal API calls should be updated to use the CFE_SB_MsgId_t type directly, rather than an integer type.

Returns

Integer representation of the CFE_SB_MsgId_t

Definition at line 1259 of file cfe_sb.h.

References CFE_SB_MSGID_UNWRAP_VALUE.

37.30.2.5 CFE_SB_ValueToMsgId()

```
static CFE_SB_MsgId_t CFE_SB_ValueToMsgId (
            CFE_SB_MsgId_Atom_t MsgIdValue )  [inline], [static]
```

Converts a normal integer into a CFE_SB_MsgId_t.

Description

In cases where the CFE_SB_MsgId_t type is not a simple integer type, it is not possible to directly use an integer value supplied via a define or similar method.

This inline function provides the ability to map an integer value into a corresponding CFE_SB_MsgId_t value.

Applications should transition to using this function wherever an integer needs to be used for a CFE_SB_MsgId_t.

Assumptions and Notes:

This negates the type safety that was gained by using a non- integer type for the CFE_SB_MsgId_t value. This should only be used in specific cases where the value is coming from an external source. Any internal API calls should be updated to return the CFE_SB_MsgId_t type directly, rather than an integer type.

Returns

CFE_SB_MsgId_t representation of the integer

Definition at line 1288 of file cfe_sb.h.

References CFE_SB_GetPktType(), and CFE_SB_MSGID_WRAP_VALUE.

Here is the call graph for this function:

37.31 cFE SB Packet Type Defines

Macros

- #define CFE_SB_PKTTYPE_INVALID CFE_MSG_Type_Invalid

  CFE_SB_GetPktType response if message type can not be determined
- #define CFE_SB_PKTTYPE_CMD CFE_MSG_Type_Cmd

  CFE_SB_GetPktType response for command packets
- #define CFE_SB_PKTTYPE_TLM CFE_MSG_Type_Tlm

  CFE_SB_GetPktType response for telemetry packets

37.31.1 Detailed Description

37.31.2 Macro Definition Documentation

37.31.2.1 CFE_SB_PKTTYPE_CMD

```
#define CFE_SB_PKTTYPE_CMD CFE_MSG_Type_Cmd
```

CFE_SB_GetPktType response for command packets

Definition at line 108 of file cfe_sb_api_typedefs.h.

37.31.2.2 CFE_SB_PKTTYPE_INVALID

```
#define CFE_SB_PKTTYPE_INVALID CFE_MSG_Type_Invalid
```

CFE_SB_GetPktType response if message type can not be determined

Definition at line 106 of file cfe_sb_api_typedefs.h.

37.31.2.3 CFE_SB_PKTTYPE_TLM

```
#define CFE_SB_PKTTYPE_TLM CFE_MSG_Type_Tlm
```

CFE_SB_GetPktType response for telemetry packets

Definition at line 109 of file cfe_sb_api_typedefs.h.

37.32 cFE Internal Software Bus APIs, internal to CFE core

Functions

- void CFE_SB_TaskMain (void)

  Entry Point for cFE Core Application.
- int32 CFE_SB_EarlyInit (void)

  Initializes the cFE core module API Library.
- int32 CFE_SB_CleanUpApp (CFE_ES_AppId_t AppId)

  Removes SB resources associated with specified Application.

37.32.1 Detailed Description

37.32.2 Function Documentation

37.32.2.1 CFE_SB_CleanUpApp()

```
int32 CFE_SB_CleanUpApp (
            CFE_ES_AppId_t AppId )
```

Removes SB resources associated with specified Application.

Description

This function is called by cFE Executive Services to cleanup after an Application has been terminated. It frees resources that have been allocated to the specified Application.

37.32.2.2 CFE_SB_EarlyInit()

```
int32 CFE_SB_EarlyInit (
            void  )
```

Initializes the cFE core module API Library.

Description

Initializes the cFE core module API Library

Assumptions, External Events, and Notes:

1. This function MUST be called before any module API's are called.

37.32.2.3 CFE_SB_TaskMain()

```
void CFE_SB_TaskMain (
            void  )
```

Entry Point for cFE Core Application.

Description

This is the entry point to the cFE SB Core Application.

Assumptions, External Events, and Notes:

None

37.33 cFE Registration APIs

Functions

- CFE_Status_t CFE_TBL_Register (CFE_TBL_Handle_t ∗TblHandlePtr, const char ∗Name, size_t Size, uint16 TblOptionFlags, CFE_TBL_CallbackFuncPtr_t TblValidationFuncPtr)

  Register a table with cFE to obtain Table Management Services.
- CFE_Status_t CFE_TBL_Share (CFE_TBL_Handle_t ∗TblHandlePtr, const char ∗TblName)

  Obtain handle of table registered by another application.
- CFE_Status_t CFE_TBL_Unregister (CFE_TBL_Handle_t TblHandle)

  Unregister a previously registered table and free associated resources.

37.33.1 Detailed Description

37.33.2 Function Documentation

37.33.2.1 CFE_TBL_Register()

```
CFE_Status_t CFE_TBL_Register (
            CFE_TBL_Handle_t * TblHandlePtr,
            const char * Name,
            size_t Size,
            uint16 TblOptionFlags,
            CFE_TBL_CallbackFuncPtr_t TblValidationFuncPtr )
```

Register a table with cFE to obtain Table Management Services.

Description

When an application is created and initialized, it is responsible for creating its table images via the TBL API. The application must inform the Table Service of the table name, table size and selection of optional table features.

Assumptions, External Events, and Notes:

Note: This function call can block. Therefore, interrupt service routines should NOT create their own tables. An application should create any table(s) and provide the handle(s) to the interrupt service routine.

Parameters

in,out	*TblHandlePtr*	a pointer to a CFE_TBL_Handle_t type variable that will be assigned the table's handle. The table handle is required for other API calls when accessing the data contained in the table. ∗TblHandlePtr is the handle used to identify table to cFE when performing Table operations. This value is returned at ddress specified by TblHandlePtr.

Parameters

in	*Name*	The application-specific name. This name will be combined with the name of the application to produce a processor specific name of the form "ApplicationName.TableName". The processor specific name will be used in commands for modifying or viewing the contents of the table.
in	*Size*	The size, in bytes, of the table to be created. This is the size that will be allocated as a shared memory resource between the Table Management Service and the calling application.

Parameters

in	*TblOptionFlags*	Flag bits indicating selected options for table. A bitwise OR of the following option flags: • CFE_TBL_OPT_DEFAULT - The default setting for table options is a combination of CFE_TBL_OPT_SNGL_BUFFER and CFE_TBL_OPT_LOAD_DUMP. See below for a description of these two options. This option is mutually exclusive with the CFE_TBL_OPT_DBL_BUFFER, CFE_TBL_OPT_DUMP_ONLY and CFE_TBL_OPT_USR_DEF_ADDR options. • CFE_TBL_OPT_SNGL_BUFFER - When this option is selected, the table will use a shared session table for performing table modifications and a memory copy from the session table to the "active" table buffer will occur when the table is updated. This is the preferred option since it will minimize memory usage. This option is mutually exclusive with the CFE_TBL_OPT_DBL_BUFFER option • CFE_TBL_OPT_DBL_BUFFER - When this option is selected, two instances of the table are created. One is considered the "active" table and the other the "inactive" table. Whenever table modifications occur, they do not require the use of a common session table. Modifications occur in the "inactive" buffer. Then, when it is time to update the table, the pointer to the "active" table is changed to point to the "inactive" buffer thus making it the new "active" buffer. This feature is most useful for time critical applications (ie - interrupt service routines, etc). This option is mutually exclusive with the CFE_TBL_OPT_SNGL_BUFFER and CFE_TBL_OPT_DEFAULT option. • CFE_TBL_OPT_LOAD_DUMP - When this option is selected, the Table Service is allowed to perform all operations on the specified table. This option is mutually exclusive with the CFE_TBL_OPT_DUMP_ONLY option. • CFE_TBL_OPT_DUMP_ONLY - When this option is selected, the Table Service will not perform table loads to this table. This does not prevent, however, a task from writing to the table via an address obtained with the CFE_TBL_GetAddress API function. This option is mutually exclusive with the CFE_TBL_OPT_LOAD_DUMP and CFE_TBL_OPT_DEFAULT options. If the Application wishes to specify their own block of memory as the Dump Only table, they need to also include the CFE_TBL_OPT_USR_DEF_ADDR option explained below. • CFE_TBL_OPT_NOT_USR_DEF - When this option is selected, Table Services allocates memory for the table and, in the case of a double buffered table, it allocates the same amount of memory again for the second buffer. This option is mutually exclusive with the CFE_TBL_OPT_USR_DEF_ADDR option. • CFE_TBL_OPT_USR_DEF_ADDR- When this option is selected, the Table Service will not allocate memory for the table. Table Services will require the Application to identify the location of the active table buffer via the CFE_TBL_Load function. This option implies the CFE_TBL_OPT_DUMP_ONLY and the CFE_TBL_OPT_SNGL_BUFFER options and is mutually exclusive of the CFE_TBL_OPT_DBL_BUFFER option. • CFE_TBL_OPT_CRITICAL- When this option is selected, the Table
		Service will automatically allocate space in the Critical Data Store (CDS) for the table and insure that the contents in the CDS are the same as the contents of the currently active buffer for the table. This option is mutually exclusive of the CFE_TBL_OPT_USR_DEF_ADDR and CFE_TBL_OPT_DUMP_ONLY options. It should also be noted that the

Parameters

in	*TblValidationFuncPtr*	is a pointer to a function that will be executed in the context of the Table Management Service when the contents of a table need to be validated. If set to NULL, then the Table Management Service will assume any data is valid. If the value is not NULL, it must be a pointer to a function with the following prototype: **int32 CallbackFunc(void ∗TblPtr);** where **TblPtr** will be a pointer to the table data that is to be verified. When the function returns CFE_SUCCESS, the data is considered valid and ready for a commit. When the function returns a negative value, the data is considered invalid and an Event Message will be issued containing the returned value. If the function should return a positive number, the table is considered invalid and the return code is considered invalid. Validation functions **must** return either CFE_SUCCESS or a negative number (whose value is at the developer's discretion). The validation function will be executed in the Application's context so that Event Messages describing the validation failure are possible from within the function.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_INFO_RECOVERED_TBL	Recovered Table.
CFE_TBL_ERR_DUPLICATE_DIFF_SIZE	Duplicate Table With Different Size.
CFE_TBL_ERR_DUPLICATE_NOT_OWNED	Dupicate Table And Not Owned.
CFE_TBL_ERR_REGISTRY_FULL	Registry Full.
CFE_TBL_ERR_HANDLES_FULL	Handles Full.
CFE_TBL_ERR_INVALID_SIZE	Invalid Size.
CFE_TBL_ERR_INVALID_NAME	Invalid Name.
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.

See also

CFE_TBL_Unregister, CFE_TBL_Share

37.33.2.2 CFE_TBL_Share()

```
CFE_Status_t CFE_TBL_Share (
            CFE_TBL_Handle_t * TblHandlePtr,
            const char * TblName )
```

Obtain handle of table registered by another application.

Description

After a table has been created, other applications can gain access to that table via the table handle. In order for two or more applications to share a table, the applications that do not create the table must obtain the handle using this function.

Assumptions, External Events, and Notes:

None

Parameters

in,out	*TblHandlePtr*	A pointer to a CFE_TBL_Handle_t type variable that will be assigned the table's handle. The table handle is required for other API calls when accessing the data contained in the table. ∗TblHandlePtr is the handle used to identify table to cFE when performing Table operations. This value is returned at the address specified by TblHandlePtr.
in	*TblName*	The processor specific name of the table. It is important to note that the processor specific table name is different from the table name specified in the CFE_TBL_Register API call. The processor specific table name includes the name of the application that created the table. The name would be of the form "ApplicationName.TableName". An example of this would be "ACS.TamParams" for a table called "TamParams" that was registered by the application called "ACS".

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_ERR_HANDLES_FULL	Handles Full.
CFE_TBL_ERR_INVALID_NAME	Invalid Name.
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.

See also

CFE_TBL_Unregister, CFE_TBL_Register

37.33.2.3 CFE_TBL_Unregister()

```
CFE_Status_t CFE_TBL_Unregister (
            CFE_TBL_Handle_t TblHandle )
```

Unregister a previously registered table and free associated resources.

Description

When an application is being removed from the system, it should unregister those tables that it created. The application should call this function as a part of its cleanup process. The table will be removed from memory once all table addresses referencing it have been released.

Assumptions, External Events, and Notes:

None

Parameters

in	*TblHandle*	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share, that identifies the Table to be unregistered.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.

See also

CFE_TBL_Share, CFE_TBL_Register

37.34 cFE Manage Table Content APIs

Functions

- CFE_Status_t CFE_TBL_Load (CFE_TBL_Handle_t TblHandle, CFE_TBL_SrcEnum_t SrcType, const void *SrcDataPtr)

  Load a specified table with data from specified source.
- CFE_Status_t CFE_TBL_Update (CFE_TBL_Handle_t TblHandle)

  Update contents of a specified table, if an update is pending.
- CFE_Status_t CFE_TBL_Validate (CFE_TBL_Handle_t TblHandle)

  Perform steps to validate the contents of a table image.
- CFE_Status_t CFE_TBL_Manage (CFE_TBL_Handle_t TblHandle)

  Perform standard operations to maintain a table.
- CFE_Status_t CFE_TBL_DumpToBuffer (CFE_TBL_Handle_t TblHandle)

  Copies the contents of a Dump Only Table to a shared buffer.
- CFE_Status_t CFE_TBL_Modified (CFE_TBL_Handle_t TblHandle)

  Notify cFE Table Services that table contents have been modified by the Application.

37.34.1 Detailed Description

37.34.2 Function Documentation

37.34.2.1 CFE_TBL_DumpToBuffer()

```
CFE_Status_t CFE_TBL_DumpToBuffer (
            CFE_TBL_Handle_t TblHandle )
```

Copies the contents of a Dump Only Table to a shared buffer.

Description

Copies contents of a Dump Only table to a shared buffer so that it can be written to a file by the Table Services routine. This function is called by the Application that owns the table in response to a CFE_TBL_INFO_DUMP_PENDING status obtained via CFE_TBL_GetStatus.

Assumptions, External Events, and Notes:

1. If the table does not have a dump pending status, nothing will occur (no error, no dump)
2. Applications may wish to use this function in lieu of CFE_TBL_Manage for their Dump Only tables

Parameters

in	*TblHandle*	Handle of Table to be dumped.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.

See also

CFE_TBL_Manage

37.34.2.2 CFE_TBL_Load()

```
CFE_Status_t CFE_TBL_Load (
            CFE_TBL_Handle_t TblHandle,
            CFE_TBL_SrcEnum_t SrcType,
            const void * SrcDataPtr )
```

Load a specified table with data from specified source.

Description

Once an application has created a table (CFE_TBL_Register), it must provide the values that initialize the contents of that table. The application accomplishes this with one of two different TBL API calls. This function call initializes the table with values that are held in a data structure.

Assumptions, External Events, and Notes:

This function call can block. Therefore, interrupt service routines should NOT initialize their own tables. An application should initialize any table(s) prior to providing the handle(s) to the interrupt service routine.

Parameters

in	*TblHandle*	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share, that identifies the Table to be loaded.
in	*SrcType*	Flag indicating the nature of the given `SrcDataPtr` below. This value can be any one of the following: • CFE_TBL_SRC_FILE - File source When this option is selected, the `SrcDataPtr` will be interpreted as a pointer to a null terminated character string. The string should specify the full path and filename of the file containing the initial data contents of the table. • CFE_TBL_SRC_ADDRESS - Address source When this option is selected, the `SrcDataPtr` will be interpreted as a pointer to a memory location that is the beginning of the initialization data for loading the table OR, in the case of a "user defined" dump only table, the address of the active table itself. The block of memory is assumed to be of the same size specified in the CFE_TBL_Register function Size parameter.
in	*SrcDataPtr*	Pointer to either a character string specifying a filename or a memory address of a block of binary data to be loaded into a table or, if the table was registered with the CFE_TBL_OPT_USR_DEF_ADDR option, the address of the active table buffer.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_WARN_SHORT_FILE	Short File Warning.
CFE_TBL_WARN_PARTIAL_LOAD	Partial Load Warning.
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.
CFE_TBL_ERR_DUMP_ONLY	Dump Only Error.
CFE_TBL_ERR_ILLEGAL_SRC_TYPE	Illegal Source Type.
CFE_TBL_ERR_LOAD_IN_PROGRESS	Load In Progress.
CFE_TBL_ERR_NO_BUFFER_AVAIL	No Buffer Available.
CFE_TBL_ERR_FILE_NOT_FOUND	File Not Found.
CFE_TBL_ERR_FILE_TOO_LARGE	File Too Large.
CFE_TBL_ERR_BAD_CONTENT_ID	Bad Content ID.
CFE_TBL_ERR_PARTIAL_LOAD	Partial Load Error.

See also

CFE_TBL_Update, CFE_TBL_Validate, CFE_TBL_Manage

37.34.2.3 CFE_TBL_Manage()

```
CFE_Status_t CFE_TBL_Manage (
            CFE_TBL_Handle_t TblHandle )
```

Perform standard operations to maintain a table.

Description

An application is **required** to perform a periodic check for an update or a validation request for all the tables that it creates. Typically, the application that created the table would call this function at the start or conclusion of any routine processing cycle. If a table update or validation request is pending, this function would perform either or both before returning.

Assumptions, External Events, and Notes:

None

Parameters

in	*TblHandle*	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share, that identifies the Table to be managed.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_INFO_UPDATED	Updated.
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.

See also

CFE_TBL_Update, CFE_TBL_Validate, CFE_TBL_Load, CFE_TBL_DumpToBuffer

37.34.2.4 CFE_TBL_Modified()

```
CFE_Status_t CFE_TBL_Modified (
            CFE_TBL_Handle_t TblHandle )
```

Notify cFE Table Services that table contents have been modified by the Application.

Description

This API notifies Table Services that the contents of the specified table has been modified by the Application. This notification is important when a table has been registered as "Critical" because Table Services can then update the contents of the table kept in the Critical Data Store.

Assumptions, External Events, and Notes:

None

Parameters

in	*TblHandle*	Handle of Table that was modified.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.

See also

CFE_TBL_Manage

37.34.2.5 CFE_TBL_Update()

```
CFE_Status_t CFE_TBL_Update (
            CFE_TBL_Handle_t TblHandle )
```

Update contents of a specified table, if an update is pending.

Description

An application is **required** to perform a periodic check for an update for all the tables that it creates. Typically, the application that created the table would call this function at the start or conclusion of any routine processing cycle or at regular intervals. To determine whether an update is pending prior to making this call, the Application can use the CFE_TBL_GetStatus API first. If a table update is pending, it will take place during this function call.

Assumptions, External Events, and Notes:

None

Parameters

in	*TblHandle*	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share, that identifies the Table to be updated.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.

Return values

CFE_TBL_INFO_NO_UPDATE_PENDING	No Update Pending.
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.

See also

CFE_TBL_Load, CFE_TBL_Validate, CFE_TBL_Manage

37.34.2.6 CFE_TBL_Validate()

```
CFE_Status_t CFE_TBL_Validate (
            CFE_TBL_Handle_t TblHandle )
```

Perform steps to validate the contents of a table image.

Description

An application is **required** to perform a periodic check for an update or a validation request for all the tables that it creates. Typically, the application that created the table would call this function at the start or conclusion of any routine processing cycle. To determine whether a validation request is pending prior to making this call, the Application can use the CFE_TBL_GetStatus API first. If a table validation is pending, the Application would call this function to perform the necessary actions.

Assumptions, External Events, and Notes:

None

Parameters

in	*TblHandle*	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share, that identifies the Table to be managed.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_INFO_NO_VALIDATION_PENDING	
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.

See also

CFE_TBL_Update, CFE_TBL_Manage, CFE_TBL_Load

37.35 cFE Access Table Content APIs

Functions

- CFE_Status_t CFE_TBL_GetAddress (void **TblPtr, CFE_TBL_Handle_t TblHandle)

  Obtain the current address of the contents of the specified table.
- CFE_Status_t CFE_TBL_ReleaseAddress (CFE_TBL_Handle_t TblHandle)

  Release previously obtained pointer to the contents of the specified table.
- CFE_Status_t CFE_TBL_GetAddresses (void **TblPtrs[ ], uint16 NumTables, const CFE_TBL_Handle_t Tbl↩Handles[ ])

  Obtain the current addresses of an array of specified tables.
- CFE_Status_t CFE_TBL_ReleaseAddresses (uint16 NumTables, const CFE_TBL_Handle_t TblHandles[ ])

  Release the addresses of an array of specified tables.

37.35.1 Detailed Description

37.35.2 Function Documentation

37.35.2.1 CFE_TBL_GetAddress()

```
CFE_Status_t CFE_TBL_GetAddress (
            void ** TblPtr,
            CFE_TBL_Handle_t TblHandle )
```

Obtain the current address of the contents of the specified table.

Description

When a table has been created and initialized, it is available to any application that can identify it with its unique handle. In order to view the data contained in the table, an application must call this function or CFE_TBL_Get↩Addresses.

Assumptions, External Events, and Notes:

1. This call can be a blocking call when the table is not double buffered and is shared with another application of lower priority that just happens to be in the middle of a table update of the specific table. If this occurs, the application performing the table update will automatically have its priority elevated in order to release the resource as soon as possible.
2. An application must always release the returned table address using the CFE_TBL_ReleaseAddress or CF↩E_TBL_ReleaseAddresses function prior to either a CFE_TBL_Update call or any blocking call (e.g. - pending on software bus message, etc). Table updates cannot occur while table addresses have not been released.
3. CFE_TBL_ERR_NEVER_LOADED will be returned if the table has never been loaded (either from file or from a block of memory), but the function will still return a valid table pointer to a table with all zero content. This pointer mush be released with the CFE_TBL_ReleaseAddress API before the table can be loaded with data.

Parameters

in,out	*TblPtr*	The address of a pointer that will be loaded with the address of the first byte of the table. This pointer can then be typecast by the calling application to the appropriate table data structure. ∗TblPtr is the address of the first byte of data associated with the specified table.
in	*TblHandle*	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share, that identifies the Table whose address is to be returned.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_INFO_UPDATED	Updated.
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.
CFE_TBL_ERR_UNREGISTERED	Unregistered.
CFE_TBL_ERR_NEVER_LOADED	Never Loaded.

See also

CFE_TBL_ReleaseAddress, CFE_TBL_GetAddresses, CFE_TBL_ReleaseAddresses

37.35.2.2 CFE_TBL_GetAddresses()

```
CFE_Status_t CFE_TBL_GetAddresses (
            void ** TblPtrs[],
            uint16 NumTables,
            const CFE_TBL_Handle_t TblHandles[] )
```

Obtain the current addresses of an array of specified tables.

Description

When a table has been created and initialized, it is available to any application that can identify it with its unique handle. In order to view the data contained in the table, an application must call this function or CFE_TBL_Get↩Addresses.

Assumptions, External Events, and Notes:

1. This call can be a blocking call when the table is not double buffered and is shared with another application of lower priority that just happens to be in the middle of a table update of the specific table. If this occurs, the application performing the table update will automatically have its priority elevated in order to release the resource as soon as possible.

2. An application must always release the returned table address using the CFE_TBL_ReleaseAddress or CFE_TBL_ReleaseAddresses function prior to either a CFE_TBL_Update call or any blocking call (e.g. - pending on software bus message, etc). Table updates cannot occur while table addresses have not been released.
3. CFE_TBL_ERR_NEVER_LOADED will be returned if the table has never been loaded (either from file or from a block of memory), but the function will still return a valid table pointer to a table with all zero content. This pointer mush be released with the CFE_TBL_ReleaseAddress API before the table can be loaded with data.

Parameters

in,out	*TblPtrs*	Array of Pointers to variables that calling Application wishes to hold the start addresses of the Tables. ∗TblPtrs is an array of addresses of the first byte of data associated with the specified tables.
in	*NumTables*	Size of TblPtrs and TblHandles arrays.
in	*TblHandles*	Array of Table Handles, previously obtained from CFE_TBL_Register or CFE_TBL_Share, of those tables whose start addresses are to be obtained.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_INFO_UPDATED	Updated.
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.
CFE_TBL_ERR_UNREGISTERED	Unregistered.
CFE_TBL_ERR_NEVER_LOADED	Never Loaded.

See also

CFE_TBL_GetAddress, CFE_TBL_ReleaseAddress, CFE_TBL_ReleaseAddresses

37.35.2.3 CFE_TBL_ReleaseAddress()

```
CFE_Status_t CFE_TBL_ReleaseAddress (
            CFE_TBL_Handle_t TblHandle )
```

Release previously obtained pointer to the contents of the specified table.

Description

Each application is **required** to release a table address obtained through the CFE_TBL_GetAddress function.

Assumptions, External Events, and Notes:

An application must always release the returned table address using the CFE_TBL_ReleaseAddress function prior to either a CFE_TBL_Update call or any blocking call (e.g. - pending on software bus message, etc). Table updates cannot occur while table addresses have not been released.

Parameters

in	*TblHandle*	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share, that identifies the Table whose address is to be released.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_INFO_UPDATED	Updated.
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.
CFE_TBL_ERR_NEVER_LOADED	Never Loaded.

See also

CFE_TBL_GetAddress, CFE_TBL_GetAddresses, CFE_TBL_ReleaseAddresses

37.35.2.4 CFE_TBL_ReleaseAddresses()

```
CFE_Status_t CFE_TBL_ReleaseAddresses (
            uint16 NumTables,
            const CFE_TBL_Handle_t TblHandles[] )
```

Release the addresses of an array of specified tables.

Description

Each application is **required** to release a table address obtained through the CFE_TBL_GetAddress function.

Assumptions, External Events, and Notes:

An application must always release the returned table address using the CFE_TBL_ReleaseAddress function prior to either a CFE_TBL_Update call or any blocking call (e.g. - pending on software bus message, etc). Table updates cannot occur while table addresses have not been released.

Parameters

in	*NumTables*	Size of TblHandles array.
in	*TblHandles*	Array of Table Handles, previously obtained from CFE_TBL_Register or CFE_TBL_Share, of those tables whose start addresses are to be released.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_INFO_UPDATED	Updated.
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.
CFE_TBL_ERR_NEVER_LOADED	Never Loaded.

See also

CFE_TBL_GetAddress, CFE_TBL_ReleaseAddress, CFE_TBL_GetAddresses

37.36 cFE Get Table Information APIs

Functions

- CFE_Status_t CFE_TBL_GetStatus (CFE_TBL_Handle_t TblHandle)

  Obtain current status of pending actions for a table.
- CFE_Status_t CFE_TBL_GetInfo (CFE_TBL_Info_t ∗TblInfoPtr, const char ∗TblName)

  Obtain characteristics/information of/about a specified table.
- CFE_Status_t CFE_TBL_NotifyByMessage (CFE_TBL_Handle_t TblHandle, CFE_SB_MsgId_t MsgId, CFE_↩MSG_FcnCode_t CommandCode, uint32 Parameter)

  Instruct cFE Table Services to notify Application via message when table requires management.

37.36.1 Detailed Description

37.36.2 Function Documentation

37.36.2.1 CFE_TBL_GetInfo()

```
CFE_Status_t CFE_TBL_GetInfo (
            CFE_TBL_Info_t * TblInfoPtr,
            const char * TblName )
```

Obtain characteristics/information of/about a specified table.

Description

This API provides the registry information associated with the specified table. The function fills the given data structure with the data found in the Table Registry.

Assumptions, External Events, and Notes:

None

Parameters

in,out	*TblInfoPtr*	A pointer to a CFE_TBL_Info_t data structure that is to be populated with table characteristics and information. ∗TblInfoPtr is the description of the tables characteristics and registry information stored in the CFE_TBL_Info_t data structure format.
in	*TblName*	The processor specific name of the table. It is important to note that the processor specific table name is different from the table name specified in the CFE_TBL_Register API call. The processor specific table name includes the name of the application that created the table. The name would be of the form "ApplicationName.TableName". An example of this would be "ACS.TamParams" for a table called "TamParams" that was registered by the application called "ACS".

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_ERR_INVALID_NAME	Invalid Name.

See also

CFE_TBL_GetStatus

37.36.2.2 CFE_TBL_GetStatus()

```
CFE_Status_t CFE_TBL_GetStatus (
            CFE_TBL_Handle_t TblHandle )
```

Obtain current status of pending actions for a table.

Description

An application is **required** to perform a periodic check for an update or a validation request for all the tables that it creates. Typically, the application that created the table would call this function at the start or conclusion of any routine processing cycle. If a table update or validation request is pending, the Application should follow up with a call to CFE_TBL_Update or CFE_TBL_Validate respectively.

Assumptions, External Events, and Notes:

None

Parameters

in	*TblHandle*	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share, that identifies the Table to be managed.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_INFO_UPDATE_PENDING	Update Pending.
CFE_TBL_INFO_VALIDATION_PENDING	

Return values

CFE_TBL_INFO_DUMP_PENDING	Dump Pending.
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.

Note

Some status return codes are "success" while being non-zero. This behavior will change in the future.

See also

CFE_TBL_Manage, CFE_TBL_Update, CFE_TBL_Validate, CFE_TBL_GetInfo

37.36.2.3 CFE_TBL_NotifyByMessage()

```
CFE_Status_t CFE_TBL_NotifyByMessage (
            CFE_TBL_Handle_t TblHandle,
            CFE_SB_MsgId_t MsgId,
            CFE_MSG_FcnCode_t CommandCode,
            uint32 Parameter )
```

Instruct cFE Table Services to notify Application via message when table requires management.

Description

This API instructs Table Services to send a message to the calling Application whenever the specified table requires management by the application. This feature allows applications to avoid polling table services via the CFE_TBL_Manage call to determine whether a table requires updates, validation, etc. This API should be called following the CFE_TBL_Register API whenever the owning application requires this feature.

Assumptions, External Events, and Notes:

- Only the application that owns the table is allowed to register a notification message
- Recommend **NOT** using the ground command MID which typically impacts command counters. The typical approach is to use a unique MID for inter-task communications similar to how schedulers typically trigger application housekeeping messages.

Parameters

in	*TblHandle*	Handle of Table with which the message should be associated.
in	*MsgId*	Message ID to be used in notification message sent by Table Services.
in	*CommandCode*	Command Code value to be placed in secondary header of message sent by Table Services.
in	*Parameter*	Application defined value to be passed as a parameter in the message sent by Table Services. Suggested use includes an application's table index that allows the same MsgId and Command Code to be used for all table management notifications.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TBL_ERR_BAD_APP_ID	Bad Application ID.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.

See also

CFE_TBL_Register

37.37 cFE Table Type Defines

Macros

- #define CFE_TBL_OPT_BUFFER_MSK (0x0001)

  Table buffer mask.
- #define CFE_TBL_OPT_SNGL_BUFFER (0x0000)

  Single buffer table.
- #define CFE_TBL_OPT_DBL_BUFFER (0x0001)

  Double buffer table.
- #define CFE_TBL_OPT_LD_DMP_MSK (0x0002)

  Table load/dump mask.
- #define CFE_TBL_OPT_LOAD_DUMP (0x0000)

  Load/Dump table.
- #define CFE_TBL_OPT_DUMP_ONLY (0x0002)

  Dump only table.
- #define CFE_TBL_OPT_USR_DEF_MSK (0x0004)

  Table user defined mask.
- #define CFE_TBL_OPT_NOT_USR_DEF (0x0000)

  Not user defined table.
- #define CFE_TBL_OPT_USR_DEF_ADDR (0x0006)

  User Defined table,.
- #define CFE_TBL_OPT_CRITICAL_MSK (0x0008)

  Table critical mask.
- #define CFE_TBL_OPT_NOT_CRITICAL (0x0000)

  Not critical table.
- #define CFE_TBL_OPT_CRITICAL (0x0008)

  Critical table.
- #define CFE_TBL_OPT_DEFAULT (CFE_TBL_OPT_SNGL_BUFFER | CFE_TBL_OPT_LOAD_DUMP)

  Default table options.

37.37.1 Detailed Description

37.37.2 Macro Definition Documentation

37.37.2.1 CFE_TBL_OPT_BUFFER_MSK

```
#define CFE_TBL_OPT_BUFFER_MSK (0x0001)
```

Table buffer mask.

Definition at line 49 of file cfe_tbl_api_typedefs.h.

37.37.2.2 CFE_TBL_OPT_CRITICAL

```
#define CFE_TBL_OPT_CRITICAL (0x0008)
```

Critical table.

Definition at line 64 of file cfe_tbl_api_typedefs.h.

37.37.2.3 CFE_TBL_OPT_CRITICAL_MSK

```
#define CFE_TBL_OPT_CRITICAL_MSK (0x0008)
```

Table critical mask.

Definition at line 62 of file cfe_tbl_api_typedefs.h.

37.37.2.4 CFE_TBL_OPT_DBL_BUFFER

```
#define CFE_TBL_OPT_DBL_BUFFER (0x0001)
```

Double buffer table.

Definition at line 51 of file cfe_tbl_api_typedefs.h.

37.37.2.5 CFE_TBL_OPT_DEFAULT

```
#define CFE_TBL_OPT_DEFAULT (CFE_TBL_OPT_SNGL_BUFFER | CFE_TBL_OPT_LOAD_DUMP)
```

Default table options.

Definition at line 67 of file cfe_tbl_api_typedefs.h.

37.37.2.6 CFE_TBL_OPT_DUMP_ONLY

```
#define CFE_TBL_OPT_DUMP_ONLY (0x0002)
```

Dump only table.

Definition at line 55 of file cfe_tbl_api_typedefs.h.

37.37.2.7 CFE_TBL_OPT_LD_DMP_MSK

```
#define CFE_TBL_OPT_LD_DMP_MSK (0x0002)
```

Table load/dump mask.

Definition at line 53 of file cfe_tbl_api_typedefs.h.

37.37.2.8 CFE_TBL_OPT_LOAD_DUMP

```
#define CFE_TBL_OPT_LOAD_DUMP (0x0000)
```

Load/Dump table.

Definition at line 54 of file cfe_tbl_api_typedefs.h.

37.37.2.9 CFE_TBL_OPT_NOT_CRITICAL

```
#define CFE_TBL_OPT_NOT_CRITICAL (0x0000)
```

Not critical table.

Definition at line 63 of file cfe_tbl_api_typedefs.h.

37.37.2.10 CFE_TBL_OPT_NOT_USR_DEF

```
#define CFE_TBL_OPT_NOT_USR_DEF (0x0000)
```

Not user defined table.

Definition at line 58 of file cfe_tbl_api_typedefs.h.

37.37.2.11 CFE_TBL_OPT_SNGL_BUFFER

```
#define CFE_TBL_OPT_SNGL_BUFFER (0x0000)
```

Single buffer table.

Definition at line 50 of file cfe_tbl_api_typedefs.h.

37.37.2.12 CFE_TBL_OPT_USR_DEF_ADDR

```
#define CFE_TBL_OPT_USR_DEF_ADDR (0x0006)
```

User Defined table,.

Note

Automatically includes CFE_TBL_OPT_DUMP_ONLY option

Definition at line 59 of file cfe_tbl_api_typedefs.h.

37.37.2.13 CFE_TBL_OPT_USR_DEF_MSK

```
#define CFE_TBL_OPT_USR_DEF_MSK (0x0004)
```

Table user defined mask.

Definition at line 57 of file cfe_tbl_api_typedefs.h.

37.38 cFE Internal Table Service APIs, internal to CFE core

Functions

- void CFE_TBL_TaskMain (void)

  Entry Point for cFE Table Services Core Application.
- int32 CFE_TBL_EarlyInit (void)

  Initializes the Table Services API Library.
- int32 CFE_TBL_CleanUpApp (CFE_ES_AppId_t AppId)

  Removes TBL resources associated with specified Application.

37.38.1 Detailed Description

37.38.2 Function Documentation

37.38.2.1 CFE_TBL_CleanUpApp()

```
int32 CFE_TBL_CleanUpApp (
            CFE_ES_AppId_t AppId )
```

Removes TBL resources associated with specified Application.

Description

This function is called by cFE Executive Services to cleanup after an Application has been terminated. It frees TBL services resources that have been allocated to the specified Application.

Assumptions, External Events, and Notes:

1. This function DOES NOT remove any critical tables associated with the specified application from the Critical Data Store.

37.38.2.2 CFE_TBL_EarlyInit()

```
int32 CFE_TBL_EarlyInit (
            void )
```

Initializes the Table Services API Library.

Description

Initializes the Table Services API Library

Assumptions, External Events, and Notes:

1. This function MUST be called before any TBL API's are called.

37.38.2.3 CFE_TBL_TaskMain()

```
void CFE_TBL_TaskMain (
            void  )
```

Entry Point for cFE Table Services Core Application.

Description

This is the entry point to the cFE Table Services Core Application. This Application provides the ground interface to the cFE Table Services.

Assumptions, External Events, and Notes:

None

37.39 cFE Get Current Time APIs

Functions

- CFE_TIME_SysTime_t CFE_TIME_GetTime (void)

  Get the current spacecraft time.
- CFE_TIME_SysTime_t CFE_TIME_GetTAI (void)

  Get the current TAI (MET + SCTF) time.
- CFE_TIME_SysTime_t CFE_TIME_GetUTC (void)

  Get the current UTC (MET + SCTF - Leap Seconds) time.
- CFE_TIME_SysTime_t CFE_TIME_GetMET (void)

  Get the current value of the Mission Elapsed Time (MET).
- uint32 CFE_TIME_GetMETseconds (void)

  Get the current seconds count of the mission-elapsed time.
- uint32 CFE_TIME_GetMETsubsecs (void)

  Get the current sub-seconds count of the mission-elapsed time.

37.39.1 Detailed Description

37.39.2 Function Documentation

37.39.2.1 CFE_TIME_GetMET()

```
CFE_TIME_SysTime_t CFE_TIME_GetMET (
            void  )
```

Get the current value of the Mission Elapsed Time (MET).

Description

This routine returns the current mission-elapsed time (MET). MET is usually derived from a hardware-based clock that is not adjusted during normal operations. Callers of this routine should not assume that the MET return value has any specific relationship to any ground-based time standard.

Assumptions, External Events, and Notes:

None

Returns

The current MET

See also

CFE_TIME_GetTime, CFE_TIME_GetTAI, CFE_TIME_GetUTC, CFE_TIME_GetMETseconds, CFE_TIME_GetMETsubsecs, CFE_TIME_MET2SCTime

37.39.2.2 CFE_TIME_GetMETseconds()

```
uint32 CFE_TIME_GetMETseconds (
            void  )
```

Get the current seconds count of the mission-elapsed time.

Description

This routine is the same as CFE_TIME_GetMET, except that it returns only the integer seconds portion of the MET time.

Assumptions, External Events, and Notes:

None

Returns

The current MET seconds

See also

CFE_TIME_GetTime, CFE_TIME_GetTAI, CFE_TIME_GetUTC, CFE_TIME_GetMET, CFE_TIME_GetME↩Tsubsecs, CFE_TIME_MET2SCTime

37.39.2.3 CFE_TIME_GetMETsubsecs()

```
uint32 CFE_TIME_GetMETsubsecs (
            void  )
```

Get the current sub-seconds count of the mission-elapsed time.

Description

This routine is the same as CFE_TIME_GetMET, except that it returns only the integer sub-seconds portion of the MET time. Each count is equal to 2^(-32) seconds.

Assumptions, External Events, and Notes:

None

Returns

The current MET sub-seconds

See also

CFE_TIME_GetTime, CFE_TIME_GetTAI, CFE_TIME_GetUTC, CFE_TIME_GetMET, CFE_TIME_GetME↩Tseconds, CFE_TIME_MET2SCTime

37.39.2.4 CFE_TIME_GetTAI()

```
CFE_TIME_SysTime_t CFE_TIME_GetTAI (
            void  )
```

Get the current TAI (MET + SCTF) time.

Description

This routine returns the current TAI time to the caller. TAI is an international time standard that does not include leap seconds. This routine should only be used in situations where TAI is absolutely required. Applications that call CFE_TIME_GetTAI may not be portable to all missions. Maintenance of correct TAI in flight is not guaranteed under all mission operations scenarios. To maintain re-usability across missions, most applications should be using CFE_TIME_GetTime, rather than the specific routines for getting UTC/TAI directly.

Assumptions, External Events, and Notes:

1. The "TAI" time returned is referenced to the mission-defined time epoch, which may or may not be the same as the standard TAI epoch.
2. Even though TAI does not include leap seconds, the time returned by this function can still jump forward or backward without warning when the spacecraft clock is set or adjusted by operators. Applications using this function must be able to handle these time discontinuities gracefully.

Returns

The current spacecraft time in TAI

See also

CFE_TIME_GetTime, CFE_TIME_GetUTC, CFE_TIME_GetMET, CFE_TIME_GetMETseconds, CFE_TIME_GetMETsubsecs

37.39.2.5 CFE_TIME_GetTime()

```
CFE_TIME_SysTime_t CFE_TIME_GetTime (
            void  )
```

Get the current spacecraft time.

Description

This routine returns the current spacecraft time. The time returned is either TAI (no leap seconds) or UTC (including leap seconds). This choice is made in the mission configuration file by defining either CFE_MISSION_TIME_CFG_DEFAULT_TAI or CFE_MISSION_TIME_CFG_DEFAULT_UTC as true at compile time. To maintain re-usability across missions, most applications should be using this function (or CFE_TIME_GetTime) rather than the specific routines for getting UTC/TAI directly.

Assumptions, External Events, and Notes:

None

Returns

The current spacecraft time in default format

See also

CFE_TIME_GetTAI, CFE_TIME_GetUTC, CFE_TIME_GetMET, CFE_TIME_GetMETseconds, CFE_TIME_GetMETsubsecs

37.39.2.6 CFE_TIME_GetUTC()

```
CFE_TIME_SysTime_t CFE_TIME_GetUTC (
            void  )
```

Get the current UTC (MET + SCTF - Leap Seconds) time.

Description

This routine returns the current UTC time to the caller. This routine should only be used in situations where UTC is absolutely required. Applications that call CFE_TIME_GetUTC may not be portable to all missions. Maintenance of correct UTC in flight is not guaranteed under all mission operations scenarios. If UTC is maintained in flight, it will jump backwards occasionally due to leap second adjustments. To maintain re-usability across missions, most applications should be using CFE_TIME_GetTime, rather than the specific routines for getting UTC/TAI directly.

Assumptions, External Events, and Notes:

Note: The "UTC" time returned is referenced to the mission-defined time epoch, which may or may not be the same as the standard UTC epoch.

Returns

The current spacecraft time in UTC

See also

CFE_TIME_GetTime, CFE_TIME_GetTAI, CFE_TIME_GetMET, CFE_TIME_GetMETseconds, CFE_TIME_GetMETsubsecs

37.40 cFE Get Time Information APIs

Functions

- CFE_TIME_SysTime_t CFE_TIME_GetSTCF (void)

  Get the current value of the spacecraft time correction factor (STCF).
- int16 CFE_TIME_GetLeapSeconds (void)

  Get the current value of the leap seconds counter.
- CFE_TIME_ClockState_Enum_t CFE_TIME_GetClockState (void)

  Get the current state of the spacecraft clock.
- uint16 CFE_TIME_GetClockInfo (void)

  Provides information about the spacecraft clock.

37.40.1 Detailed Description

37.40.2 Function Documentation

37.40.2.1 CFE_TIME_GetClockInfo()

```
uint16 CFE_TIME_GetClockInfo (
            void  )
```

Provides information about the spacecraft clock.

Description

This routine returns information on the spacecraft clock in a bit mask.

Assumptions, External Events, and Notes:

None

Returns

Spacecraft clock information, cFE Clock State Flag Defines. To extract the information from the returned value, the flags can be used as in the following:
`if ((ReturnValue & CFE_TIME_FLAG_xxxxxx) == CFE_TIME_FLAG_xxxxxx)` then the following definition of the `CFE_TIME_FLAG_xxxxxx` is true.

See also

CFE_TIME_GetSTCF, CFE_TIME_GetLeapSeconds, CFE_TIME_GetClockState

37.40.2.2 CFE_TIME_GetClockState()

```
CFE_TIME_ClockState_Enum_t CFE_TIME_GetClockState (
            void  )
```

Get the current state of the spacecraft clock.

Description

This routine returns the spacecraft clock state. Applications that are highly dependent on valid time may want to call this routine before taking actions based on the times returned by the various clock routines

Assumptions, External Events, and Notes:

None

Returns

The current spacecraft clock state

See also

CFE_TIME_GetSTCF, CFE_TIME_GetLeapSeconds, CFE_TIME_GetClockInfo

37.40.2.3 CFE_TIME_GetLeapSeconds()

```
int16 CFE_TIME_GetLeapSeconds (
            void  )
```

Get the current value of the leap seconds counter.

Description

This routine returns the current value of the leap seconds counter. This is the delta seconds between international atomic time (TAI) and universal coordinated time (UTC). Applications cannot set or adjust the leap seconds; that can only be done through ground commands. However, science applications may want to include the leap seconds counter in their data products to aid in time correlation during downstream science data processing. Note that some mission operations teams do not maintain the leap seconds count, preferring to adjust the STCF instead. Users of this function should check with their mission ops team to see how they are planning to handle leap seconds.

Assumptions, External Events, and Notes:

None

Returns

The current spacecraft leap seconds.

See also

CFE_TIME_GetSTCF, CFE_TIME_GetClockState, CFE_TIME_GetClockInfo

37.40.2.4 CFE_TIME_GetSTCF()

```
CFE_TIME_SysTime_t CFE_TIME_GetSTCF (
            void  )
```

Get the current value of the spacecraft time correction factor (STCF).

Description

This routine returns the current value of the spacecraft time correction factor. This is the delta time between the MET and the TAI time. Applications cannot set or adjust the STCF; that can only be done through ground commands. However, science applications may want to include the STCF in their data products to aid in time correlation during downstream science data processing.

Assumptions, External Events, and Notes:

Does not include leap seconds

Returns

The current SCTF

See also

CFE_TIME_GetLeapSeconds, CFE_TIME_GetClockState, CFE_TIME_GetClockInfo

37.41 cFE Time Arithmetic APIs

Functions

- CFE_TIME_SysTime_t CFE_TIME_Add (CFE_TIME_SysTime_t Time1, CFE_TIME_SysTime_t Time2)

  Adds two time values.
- CFE_TIME_SysTime_t CFE_TIME_Subtract (CFE_TIME_SysTime_t Time1, CFE_TIME_SysTime_t Time2)

  Subtracts two time values.
- CFE_TIME_Compare_t CFE_TIME_Compare (CFE_TIME_SysTime_t TimeA, CFE_TIME_SysTime_t TimeB)

  Compares two time values.

37.41.1 Detailed Description

37.41.2 Function Documentation

37.41.2.1 CFE_TIME_Add()

```
CFE_TIME_SysTime_t CFE_TIME_Add (
            CFE_TIME_SysTime_t Time1,
            CFE_TIME_SysTime_t Time2 )
```

Adds two time values.

Description

This routine adds the two specified times and returns the result. Normally, at least one of the input times should be a value representing a delta time. Adding two absolute times together will not cause an error, but the result will probably be meaningless.

Assumptions, External Events, and Notes:

None

Parameters

in	*Time1*	The first time to be added.
in	*Time2*	The second time to be added.

Returns

The sum of the two times. If the sum is greater than the maximum value that can be stored in a CFE_TIME_Sys↩Time_t, the result will roll over (this is not considered an error).

See also

CFE_TIME_Subtract, CFE_TIME_Compare

37.41.2.2 CFE_TIME_Compare()

```
CFE_TIME_Compare_t CFE_TIME_Compare (
            CFE_TIME_SysTime_t TimeA,
            CFE_TIME_SysTime_t TimeB )
```

Compares two time values.

Description

This routine compares two time values to see which is "greater". It is important that applications use this function rather than trying to directly compare the component pieces of times. This function will handle roll-over cases seamlessly, which may not be intuitively obvious. The cFE's internal representation of time "rolls over" when the 32 bit seconds count reaches 0xFFFFFFFF. Also, subtracting a delta time from an absolute time close to the epoch could result in "roll under". The strange cases that result from these situations can be handled by defining the comparison function for times as follows: Plot the two times on the circumference of a circle where 0 is at the top and 0x80000000 is at the bottom. If the shortest arc from time A to time B runs clockwise around the circle, then time A is less than time B. If the shortest arc from A to B runs counter-clockwise, then time A is greater than time B.

Assumptions, External Events, and Notes:

None

Parameters

in	*TimeA*	The first time to compare.
in	*TimeB*	The second time to compare.

Returns

The result of comparing the two times.

Return values

CFE_TIME_EQUAL	The two specified times are considered to be equal.
CFE_TIME_A_GT↩ _B	The first specified time is considered to be after the second specified time.
CFE_TIME_A_LT↩ _B	The first specified time is considered to be before the second specified time.

See also

CFE_TIME_Add, CFE_TIME_Subtract

37.41.2.3 CFE_TIME_Subtract()

```
CFE_TIME_SysTime_t CFE_TIME_Subtract (
            CFE_TIME_SysTime_t Time1,
            CFE_TIME_SysTime_t Time2 )
```

Subtracts two time values.

Description

This routine subtracts time2 from time1 and returns the result. The time values can represent either absolute or delta times, but not all combinations make sense.

- AbsTime - AbsTime = DeltaTime
- AbsTime - DeltaTime = AbsTime
- DeltaTime - DeltaTime = DeltaTime
- DeltaTime - AbsTime = garbage

Assumptions, External Events, and Notes:

None

Parameters

in	*Time1*	The base time.
in	*Time2*	The time to be subtracted from the base time.

Returns

The result of subtracting the two times. If the subtraction results in an underflow, the result will roll over (this is not considered an error).

See also

CFE_TIME_Add, CFE_TIME_Compare

37.42 cFE Time Conversion APIs

Functions

- CFE_TIME_SysTime_t CFE_TIME_MET2SCTime (CFE_TIME_SysTime_t METTime)

  Convert specified MET into Spacecraft Time.
- uint32 CFE_TIME_Sub2MicroSecs (uint32 SubSeconds)

  Converts a sub-seconds count to an equivalent number of microseconds.
- uint32 CFE_TIME_Micro2SubSecs (uint32 MicroSeconds)

  Converts a number of microseconds to an equivalent sub-seconds count.

37.42.1 Detailed Description

37.42.2 Function Documentation

37.42.2.1 CFE_TIME_MET2SCTime()

```
CFE_TIME_SysTime_t CFE_TIME_MET2SCTime (
            CFE_TIME_SysTime_t METTime )
```

Convert specified MET into Spacecraft Time.

Description

This function returns Spacecraft Time given MET. Note that Spacecraft Time is returned as either UTC or TAI depeneding on whether the mission configuration parameter CFE_MISSION_TIME_CFG_DEFAULT_UTC or CFE_MISSION_TIME_CFG_DEFAULT_TAI was set to true at compile time.

Assumptions, External Events, and Notes:

None

Parameters

in	*METTime*	The MET to be converted.

Returns

Spacecraft Time (UTC or TAI) corresponding to the specified MET

See also

CFE_TIME_GetMET, CFE_TIME_GetMETseconds, CFE_TIME_GetMETsubsecs, CFE_TIME_Sub2MicroSecs, CFE_TIME_Micro2SubSecs

37.42.2.2 CFE_TIME_Micro2SubSecs()

```
uint32 CFE_TIME_Micro2SubSecs (
            uint32 MicroSeconds )
```

Converts a number of microseconds to an equivalent sub-seconds count.

Description

This routine converts from microseconds (each tick is 1e-06 seconds) to a subseconds count (each tick is 1 / 2^32 seconds).

Assumptions, External Events, and Notes:

None

Parameters

in	*MicroSeconds*	The sub-seconds count to convert.

Returns

The equivalent number of subseconds. If the number of microseconds passed in is greater than one second, (i.e. > 999,999), the return value is equal to `0xffffffff`.

See also

CFE_TIME_MET2SCTime, CFE_TIME_Sub2MicroSecs,

37.42.2.3 CFE_TIME_Sub2MicroSecs()

```
uint32 CFE_TIME_Sub2MicroSecs (
            uint32 SubSeconds )
```

Converts a sub-seconds count to an equivalent number of microseconds.

Description

This routine converts from a sub-seconds count (each tick is 1 / 2^32 seconds) to microseconds (each tick is 1e-06 seconds).

Assumptions, External Events, and Notes:

None

Parameters

in	*SubSeconds*	The sub-seconds count to convert.

Returns

The equivalent number of microseconds.

See also

CFE_TIME_MET2SCTime, CFE_TIME_Micro2SubSecs,

37.43 cFE External Time Source APIs

Functions

- void CFE_TIME_ExternalTone (void)

  Provides the 1 Hz signal from an external source.
- void CFE_TIME_ExternalMET (CFE_TIME_SysTime_t NewMET)

  Provides the Mission Elapsed Time from an external source.
- void CFE_TIME_ExternalGPS (CFE_TIME_SysTime_t NewTime, int16 NewLeaps)

  Provide the time from an external source that has data common to GPS receivers.
- void CFE_TIME_ExternalTime (CFE_TIME_SysTime_t NewTime)

  Provide the time from an external source that measures time relative to a known epoch.
- CFE_Status_t CFE_TIME_RegisterSynchCallback (CFE_TIME_SynchCallbackPtr_t CallbackFuncPtr)

  Registers a callback function that is called whenever time synchronization occurs.
- CFE_Status_t CFE_TIME_UnregisterSynchCallback (CFE_TIME_SynchCallbackPtr_t CallbackFuncPtr)

  Unregisters a callback function that is called whenever time synchronization occurs.

37.43.1 Detailed Description

37.43.2 Function Documentation

37.43.2.1 CFE_TIME_ExternalGPS()

```
void CFE_TIME_ExternalGPS (
            CFE_TIME_SysTime_t NewTime,
            int16 NewLeaps )
```

Provide the time from an external source that has data common to GPS receivers.

Description

This routine provides a method to provide cFE TIME with current time data acquired from an external source. There is a presumption that this function will be called at the appropriate time (relative to the tone) such that this call may be used by cFE TIME as the signal to generate the "time at the tone" data command. The "time at the tone" data command must arrive within the configuration parameter specified window for tone signal and data packet verification.

Internally, cFE TIME will calculate a new STCF as the difference between this new time value and the spacecraft MET value at the tone. This allows cFE TIME to always calculate time as the sum of MET and STCF. The value of STCF will change only as much as the drift factor between spacecraft MET and the external time source.

Assumptions, External Events, and Notes:

- This routine is included in the API only when 3 specific configuration parameters are set to true. The first is CFE_PLATFORM_TIME_CFG_SERVER which defines this instantiation of cFE TIME as a time server (not a client). The second required configuration parameter is CFE_PLATFORM_TIME_CFG_SOURCE which enables time source selection commands to the cFE TIME task, and further enables configuration definitions for the selected type of external time data. The third configuration parameter required for this routine is CFE_PLATFORM_TIME_CFG_SRC_GPS, which indicates that the external time data consists of a time value relative to a known epoch, plus a leap seconds value.

Parameters

in	*NewTime*	The MET value at the next (or previous) 1 Hz tone signal.
in	*NewLeaps*	The Leap Seconds value used to calculate time as UTC.

See also

CFE_TIME_ExternalTone, CFE_TIME_ExternalMET, CFE_TIME_ExternalTime

37.43.2.2 CFE_TIME_ExternalMET()

```
void CFE_TIME_ExternalMET (
            CFE_TIME_SysTime_t NewMET )
```

Provides the Mission Elapsed Time from an external source.

Description

This routine provides a method to provide cFE TIME with MET acquired from an external source. There is a presumption that this function will be called at the appropriate time (relative to the tone) such that this call may be used by cFE TIME as the signal to generate the "time at the tone" data command. The "time at the tone" data command must arrive within the configuration parameter specified window for tone signal and data packet verification.

The MET value at the tone "should" have zero subseconds. Although the interface accepts non-zero values for sub-seconds, it may be harmful to other applications that expect zero subseconds at the moment of the tone. Any decision to use non-zero subseconds should be carefully considered.

Assumptions, External Events, and Notes:

- This routine is included in the API only when 3 specific configuration parameters are set to true. The first is CFE_PLATFORM_TIME_CFG_SERVER which defines this instantiation of cFE TIME as a time server (not a client). The second required configuration parameter is CFE_PLATFORM_TIME_CFG_SOURCE which enables time source selection commands to the cFE TIME task, and further enables configuration definitions for the selected type of external time data. The third configuration parameter required for this routine is CFE_PLATFORM_TIME_CFG_SRC_MET, which indicates that the external time data consists of MET.

Parameters

in	*NewMET*	The MET value at the next (or previous) 1 Hz tone signal.

See also

CFE_TIME_ExternalTone, CFE_TIME_ExternalGPS, CFE_TIME_ExternalTime

37.43.2.3 CFE_TIME_ExternalTime()

```
void CFE_TIME_ExternalTime (
            CFE_TIME_SysTime_t NewTime )
```

Provide the time from an external source that measures time relative to a known epoch.

Description

This routine provides a method to provide cFE TIME with current time data acquired from an external source. There is a presumption that this function will be called at the appropriate time (relative to the tone) such that this call may be used by cFE TIME as the signal to generate the "time at the tone" data command. The "time at the tone" data command must arrive within the configuration specified window for tone signal and data packet verification.

Internally, cFE TIME will calculate a new STCF as the difference between this new time value and the spacecraft MET value at the tone. This allows cFE TIME to always calculate time as the sum of MET and STCF. The value of STCF will change only as much as the drift factor between spacecraft MET and the external time source.

Assumptions, External Events, and Notes:

- This routine is included in the API only when 3 specific configuration parameters are set to true. The first is CFE_PLATFORM_TIME_CFG_SERVER which defines this instanciation of cFE TIME as a time server (not a client). The second required configuration parameter is CFE_PLATFORM_TIME_CFG_SOURCE which enables time source selection commands to the cFE TIME task, and further enables configuration definitions for the selected type of external time data. The third configuration parameter required for this routine is CFE_PLATFORM_TIME_CFG_SRC_TIME, which indicates that the external time data consists of a time value relative to a known epoch.

Parameters

in	*NewTime*	The MET value at the next (or previous) 1 Hz tone signal.

See also

CFE_TIME_ExternalTone, CFE_TIME_ExternalMET, CFE_TIME_ExternalGPS

37.43.2.4 CFE_TIME_ExternalTone()

```
void CFE_TIME_ExternalTone (
            void )
```

Provides the 1 Hz signal from an external source.

Description

This routine provides a method for cFE TIME software to be notified of the occurance of the 1Hz tone signal without knowledge of the specific hardware design. Regardless of the source of the tone, this routine should be called as soon as possible after detection to allow cFE TIME software the opportunity to latch the local clock as close as possible to the instant of the tone.

Assumptions, External Events, and Notes:

- This routine may be called directly from within the context of an interrupt handler.

See also

CFE_TIME_ExternalMET, CFE_TIME_ExternalGPS, CFE_TIME_ExternalTime

37.43.2.5 CFE_TIME_RegisterSynchCallback()

```
CFE_Status_t CFE_TIME_RegisterSynchCallback (
            CFE_TIME_SynchCallbackPtr_t CallbackFuncPtr )
```

Registers a callback function that is called whenever time synchronization occurs.

Description

This routine passes a callback function pointer for an Application that wishes to be notified whenever a legitimate time synchronization signal (typically a 1 Hz) is received.

Assumptions, External Events, and Notes:

Only a single callback per application is supported, and this function should only be called from a single thread within each application (typically the apps main thread). If an application requires triggering multiple child tasks at 1Hz, it should distribute the timing signal internally, rather than registering for multiple callbacks.

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TIME_TOO_MANY_SYNCH_CALLBACKS	Too Many Sync Callbacks.

See also

CFE_TIME_UnregisterSynchCallback

37.43.2.6 CFE_TIME_UnregisterSynchCallback()

```
CFE_Status_t CFE_TIME_UnregisterSynchCallback (
            CFE_TIME_SynchCallbackPtr_t CallbackFuncPtr )
```

Unregisters a callback function that is called whenever time synchronization occurs.

Description

This routine removes the specified callback function pointer from the list of Callback functions that are called whenever a time synchronization (typically the 1Hz signal) is received.

Assumptions, External Events, and Notes:

Only a single callback per application is supported, and this function should only be called from a single thread within each application (typically the apps main thread).

Returns

Execution status, see cFE Return Code Defines

Return values

CFE_SUCCESS	Sucessful execution.
CFE_TIME_CALLBACK_NOT_REGISTERED	Callback Not Registered.

See also

CFE_TIME_RegisterSynchCallback

37.44 cFE Miscellaneous Time APIs

Functions

- void CFE_TIME_Print (char *PrintBuffer, CFE_TIME_SysTime_t TimeToPrint)

  Print a time value as a string.
- void CFE_TIME_Local1HzISR (void)

  This function should be called from the system PSP layer once per second.

37.44.1 Detailed Description

37.44.2 Function Documentation

37.44.2.1 CFE_TIME_Local1HzISR()

```
void CFE_TIME_Local1HzISR (
            void  )
```

This function should be called from the system PSP layer once per second.

Description

Drives the time processing logic from the system PSP layer. This must be called once per second based on a hardware interrupt or OS kernel signal.

Assumptions, External Events, and Notes:

This will update the global data structures accordingly, incrementing each by the 1Hz amount.

37.44.2.2 CFE_TIME_Print()

```
void CFE_TIME_Print (
            char * PrintBuffer,
            CFE_TIME_SysTime_t TimeToPrint )
```

Print a time value as a string.

Description

This routine prints the specified time to the specified string buffer in the following format:

`yyyy-ddd-hh`:mm:ss.xxxxx\0

where:

- `yyyy` = year
- `ddd` = Julian day of the year
- `hh` = hour of the day (0 to 23)
- `mm` = minute (0 to 59)
- `ss` = second (0 to 59)
- `xxxxx` = subsecond formatted as a decimal fraction (1/4 second = 0.25000)
- `\0` = trailing null

Assumptions, External Events, and Notes:

- The value of the time argument is simply added to the configuration definitions for the ground epoch and converted into a fixed length string in the buffer provided by the caller.
- A loss of data during the string conversion will occur if the computed year exceeds 9999. However, a year that large would require an unrealistic definition for the ground epoch since the maximum amount of time represented by a CFE_TIME_SysTime structure is approximately 136 years.

Parameters

`in,out`	*PrintBuffer*	Pointer to a character array of at least CFE_TIME_PRINTED_STRING_SIZE characters in length. ∗PrintBuffer is the time as a character string as described above.
`in`	*TimeToPrint*	The time to print into the character array.

37.45 cFE Internal Time APIs, internal to CFE core

Functions

- void CFE_TIME_TaskMain (void)

  Entry Point for cFE Core Application.
- int32 CFE_TIME_EarlyInit (void)

  Initializes the cFE core module API Library.
- int32 CFE_TIME_CleanUpApp (CFE_ES_AppId_t AppId)

  Removes TIME resources associated with specified Application.

37.45.1 Detailed Description

37.45.2 Function Documentation

37.45.2.1 CFE_TIME_CleanUpApp()

```
int32 CFE_TIME_CleanUpApp (
            CFE_ES_AppId_t AppId )
```

Removes TIME resources associated with specified Application.

Description

This function is called by cFE Executive Services to cleanup after an Application has been terminated. It frees resources that have been allocated to the specified Application.

37.45.2.2 CFE_TIME_EarlyInit()

```
int32 CFE_TIME_EarlyInit (
            void )
```

Initializes the cFE core module API Library.

Description

Initializes the cFE core module API Library

Assumptions, External Events, and Notes:

1. This function MUST be called before any module API's are called.

37.45.2.3 CFE_TIME_TaskMain()

```
void CFE_TIME_TaskMain (
            void  )
```

Entry Point for cFE Core Application.

Description

This is the entry point to the cFE TIME Core Application.

Assumptions, External Events, and Notes:

None

37.46 cFE Clock State Flag Defines

Macros

- #define CFE_TIME_FLAG_CLKSET 0x8000

  The spacecraft time has been set.
- #define CFE_TIME_FLAG_FLYING 0x4000

  This instance of Time Services is flywheeling.
- #define CFE_TIME_FLAG_SRCINT 0x2000

  The clock source is set to "internal".
- #define CFE_TIME_FLAG_SIGPRI 0x1000

  The clock signal is set to "primary".
- #define CFE_TIME_FLAG_SRVFLY 0x0800

  The Time Server is in flywheel mode.
- #define CFE_TIME_FLAG_CMDFLY 0x0400

  This instance of Time Services was commanded into flywheel mode.
- #define CFE_TIME_FLAG_ADDADJ 0x0200

  One time STCF Adjustment is to be done in positive direction.
- #define CFE_TIME_FLAG_ADD1HZ 0x0100

  1 Hz STCF Adjustment is to be done in a positive direction
- #define CFE_TIME_FLAG_ADDTCL 0x0080

  Time Client Latency is applied in a positive direction.
- #define CFE_TIME_FLAG_SERVER 0x0040

  This instance of Time Services is a Time Server.
- #define CFE_TIME_FLAG_GDTONE 0x0020

  The tone received is good compared to the last tone received.
- #define CFE_TIME_FLAG_UNUSED 0x001F

  Reserved flags - should be zero.

37.46.1 Detailed Description

37.46.2 Macro Definition Documentation

37.46.2.1 CFE_TIME_FLAG_ADD1HZ

```
#define CFE_TIME_FLAG_ADD1HZ 0x0100
```

1 Hz STCF Adjustment is to be done in a positive direction

Definition at line 719 of file cfe_time_msg.h.

37.46.2.2 CFE_TIME_FLAG_ADDADJ

```
#define CFE_TIME_FLAG_ADDADJ 0x0200
```

One time STCF Adjustment is to be done in positive direction.

Definition at line 718 of file cfe_time_msg.h.

37.46.2.3 CFE_TIME_FLAG_ADDTCL

```
#define CFE_TIME_FLAG_ADDTCL 0x0080
```

Time Client Latency is applied in a positive direction.

Definition at line 720 of file cfe_time_msg.h.

37.46.2.4 CFE_TIME_FLAG_CLKSET

```
#define CFE_TIME_FLAG_CLKSET 0x8000
```

The spacecraft time has been set.

Definition at line 712 of file cfe_time_msg.h.

37.46.2.5 CFE_TIME_FLAG_CMDFLY

```
#define CFE_TIME_FLAG_CMDFLY 0x0400
```

This instance of Time Services was commanded into flywheel mode.

Definition at line 717 of file cfe_time_msg.h.

37.46.2.6 CFE_TIME_FLAG_FLYING

```
#define CFE_TIME_FLAG_FLYING 0x4000
```

This instance of Time Services is flywheeling.

Definition at line 713 of file cfe_time_msg.h.

37.46.2.7 CFE_TIME_FLAG_GDTONE

```
#define CFE_TIME_FLAG_GDTONE 0x0020
```

The tone received is good compared to the last tone received.

Definition at line 722 of file cfe_time_msg.h.

37.46.2.8 CFE_TIME_FLAG_SERVER

```
#define CFE_TIME_FLAG_SERVER 0x0040
```

This instance of Time Services is a Time Server.

Definition at line 721 of file cfe_time_msg.h.

37.46.2.9 CFE_TIME_FLAG_SIGPRI

```
#define CFE_TIME_FLAG_SIGPRI 0x1000
```

The clock signal is set to "primary".

Definition at line 715 of file cfe_time_msg.h.

37.46.2.10 CFE_TIME_FLAG_SRCINT

```
#define CFE_TIME_FLAG_SRCINT 0x2000
```

The clock source is set to "internal".

Definition at line 714 of file cfe_time_msg.h.

37.46.2.11 CFE_TIME_FLAG_SRVFLY

```
#define CFE_TIME_FLAG_SRVFLY 0x0800
```

The Time Server is in flywheel mode.

Definition at line 716 of file cfe_time_msg.h.

37.46.2.12 CFE_TIME_FLAG_UNUSED

```
#define CFE_TIME_FLAG_UNUSED 0x001F
```

Reserved flags - should be zero.

Definition at line 723 of file cfe_time_msg.h.

37.47 OSAL Semaphore State Defines

Macros

- #define OS_SEM_FULL 1

  Semaphore full state.
- #define OS_SEM_EMPTY 0

  Semaphore empty state.

37.47.1 Detailed Description

37.47.2 Macro Definition Documentation

37.47.2.1 OS_SEM_EMPTY

```
#define OS_SEM_EMPTY 0
```

Semaphore empty state.

Definition at line 35 of file osapi-binsem.h.

37.47.2.2 OS_SEM_FULL

```
#define OS_SEM_FULL 1
```

Semaphore full state.

Definition at line 34 of file osapi-binsem.h.

37.48 OSAL Binary Semaphore APIs

Functions

- int32 OS_BinSemCreate (osal_id_t *sem_id, const char *sem_name, uint32 sem_initial_value, uint32 options)

  Creates a binary semaphore.
- int32 OS_BinSemFlush (osal_id_t sem_id)

  Unblock all tasks pending on the specified semaphore.
- int32 OS_BinSemGive (osal_id_t sem_id)

  Increment the semaphore value.
- int32 OS_BinSemTake (osal_id_t sem_id)

  Decrement the semaphore value.
- int32 OS_BinSemTimedWait (osal_id_t sem_id, uint32 msecs)

  Decrement the semaphore value with a timeout.
- int32 OS_BinSemDelete (osal_id_t sem_id)

  Deletes the specified Binary Semaphore.
- int32 OS_BinSemGetIdByName (osal_id_t *sem_id, const char *sem_name)

  Find an existing semaphore ID by name.
- int32 OS_BinSemGetInfo (osal_id_t sem_id, OS_bin_sem_prop_t *bin_prop)

  Fill a property object buffer with details regarding the resource.

37.48.1 Detailed Description

37.48.2 Function Documentation

37.48.2.1 OS_BinSemCreate()

```
int32 OS_BinSemCreate (
            osal_id_t * sem_id,
            const char * sem_name,
            uint32 sem_initial_value,
            uint32 options )
```

Creates a binary semaphore.

Creates a binary semaphore with initial value specified by sem_initial_value and name specified by sem_name. sem_id will be returned to the caller

Parameters

out	*sem_id*	will be set to the non-zero ID of the newly-created resource
in	*sem_name*	the name of the new resource to create
in	*sem_initial_value*	the initial value of the binary semaphore
in	*options*	Reserved for future use, should be passed as 0.

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if sen name or sem_id are NULL
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NO_FREE_IDS	if all of the semaphore ids are taken
OS_ERR_NAME_TAKEN	if this is already the name of a binary semaphore
OS_SEM_FAILURE	if the OS call failed

37.48.2.2 OS_BinSemDelete()

```
int32 OS_BinSemDelete (
            osal_id_t sem_id )
```

Deletes the specified Binary Semaphore.

This is the function used to delete a binary semaphore in the operating system. This also frees the respective sem_id to be used again when another semaphore is created.

Parameters

in	*sem↩ _id*	The object ID to delete

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid binary semaphore
OS_SEM_FAILURE	the OS call failed

37.48.2.3 OS_BinSemFlush()

```
int32 OS_BinSemFlush (
            osal_id_t sem_id )
```

Unblock all tasks pending on the specified semaphore.

The function unblocks all tasks pending on the specified semaphore. However, this function does not change the state of the semaphore.

Parameters

in	*sem↩ _id*	The object ID to operate on

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a binary semaphore
OS_SEM_FAILURE	if an unspecified failure occurs

37.48.2.4 OS_BinSemGetIdByName()

```
int32 OS_BinSemGetIdByName (
            osal_id_t * sem_id,
            const char * sem_name )
```

Find an existing semaphore ID by name.

This function tries to find a binary sem Id given the name of a bin_sem The id is returned through sem_id

Parameters

out	*sem_id*	will be set to the ID of the existing resource
in	*sem_name*	the name of the existing resource to find

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	is semid or sem_name are NULL pointers
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_NOT_FOUND	if the name was not found in the table

37.48.2.5 OS_BinSemGetInfo()

```
int32 OS_BinSemGetInfo (
            osal_id_t sem_id,
            OS_bin_sem_prop_t * bin_prop )
```

Fill a property object buffer with details regarding the resource.

This function will pass back a pointer to structure that contains all of the relevant info(name and creator) about the specified binary semaphore.

Parameters

in	*sem_id*	The object ID to operate on
out	*bin_prop*	The property object buffer to fill

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid semaphore
OS_INVALID_POINTER	if the bin_prop pointer is null

37.48.2.6 OS_BinSemGive()

```
int32 OS_BinSemGive (
            osal_id_t sem_id )
```

Increment the semaphore value.

The function unlocks the semaphore referenced by sem_id by performing a semaphore unlock operation on that semaphore. If the semaphore value resulting from this operation is positive, then no threads were blocked waiting for the semaphore to become unlocked; the semaphore value is simply incremented for this semaphore.

Parameters

in	*sem↩ _id*	The object ID to operate on

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_SEM_FAILURE	the semaphore was not previously initialized or is not in the array of semaphores defined by the system
OS_ERR_INVALID_ID	if the id passed in is not a binary semaphore

37.48.2.7 OS_BinSemTake()

```
int32 OS_BinSemTake (
            osal_id_t sem_id )
```

Decrement the semaphore value.

The locks the semaphore referenced by sem_id by performing a semaphore lock operation on that semaphore. If the semaphore value is currently zero, then the calling thread shall not return from the call until it either locks the semaphore or the call is interrupted.

Parameters

in	*sem↩ _id*	The object ID to operate on

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	the Id passed in is not a valid binary semaphore
OS_SEM_FAILURE	if the OS call failed

37.48.2.8 OS_BinSemTimedWait()

```
int32 OS_BinSemTimedWait (
            osal_id_t sem_id,
            uint32 msecs )
```

Decrement the semaphore value with a timeout.

The function locks the semaphore referenced by sem_id. However, if the semaphore cannot be locked without waiting for another process or thread to unlock the semaphore, this wait shall be terminated when the specified timeout, msecs, expires.

Parameters

in	*sem↩ _id*	The object ID to operate on
in	*msecs*	The maximum amount of time to block, in milliseconds

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_SEM_TIMEOUT	if semaphore was not relinquished in time
OS_SEM_FAILURE	the semaphore was not previously initialized or is not in the array of semaphores defined by the system
OS_ERR_INVALID_ID	if the ID passed in is not a valid semaphore ID

37.49 OSAL BSP low level access APIs

Functions

- uint32 OS_BSP_GetArgC (void)
- char ∗const ∗ OS_BSP_GetArgV (void)
- void OS_BSP_SetExitCode (int32 code)

37.49.1 Detailed Description

These are for OSAL internal BSP information access to pass any BSP-specific boot/command line/startup arguments through to the application, and return a status code back to the OS after exit.

Not intended for user application use

37.49.2 Function Documentation

37.49.2.1 OS_BSP_GetArgC()

```
uint32 OS_BSP_GetArgC (
            void  )
```

Referenced by OS_Application_Startup().

37.49.2.2 OS_BSP_GetArgV()

```
char* const* OS_BSP_GetArgV (
            void  )
```

Referenced by OS_Application_Startup().

37.49.2.3 OS_BSP_SetExitCode()

```
void OS_BSP_SetExitCode (
            int32 code )
```

37.50 OSAL Real Time Clock APIs

Functions

- int32 OS_GetLocalTime (OS_time_t *time_struct)

  Get the local time.
- int32 OS_SetLocalTime (const OS_time_t *time_struct)

  Set the local time.
- static int64 OS_TimeGetTotalSeconds (OS_time_t tm)

  Get interval from an OS_time_t object normalized to whole number of seconds.
- static int64 OS_TimeGetTotalMilliseconds (OS_time_t tm)

  Get interval from an OS_time_t object normalized to millisecond units.
- static int64 OS_TimeGetTotalMicroseconds (OS_time_t tm)

  Get interval from an OS_time_t object normalized to microsecond units.
- static int64 OS_TimeGetTotalNanoseconds (OS_time_t tm)

  Get interval from an OS_time_t object normalized to nanosecond units.
- static int64 OS_TimeGetFractionalPart (OS_time_t tm)

  Get subseconds portion (fractional part only) from an OS_time_t object.
- static uint32 OS_TimeGetSubsecondsPart (OS_time_t tm)

  Get 32-bit normalized subseconds (fractional part only) from an OS_time_t object.
- static uint32 OS_TimeGetMillisecondsPart (OS_time_t tm)

  Get milliseconds portion (fractional part only) from an OS_time_t object.
- static uint32 OS_TimeGetMicrosecondsPart (OS_time_t tm)

  Get microseconds portion (fractional part only) from an OS_time_t object.
- static uint32 OS_TimeGetNanosecondsPart (OS_time_t tm)

  Get nanoseconds portion (fractional part only) from an OS_time_t object.
- static OS_time_t OS_TimeAssembleFromNanoseconds (int64 seconds, uint32 nanoseconds)

  Assemble/Convert a number of seconds + nanoseconds into an OS_time_t interval.
- static OS_time_t OS_TimeAssembleFromMicroseconds (int64 seconds, uint32 microseconds)

  Assemble/Convert a number of seconds + microseconds into an OS_time_t interval.
- static OS_time_t OS_TimeAssembleFromMilliseconds (int64 seconds, uint32 milliseconds)

  Assemble/Convert a number of seconds + milliseconds into an OS_time_t interval.
- static OS_time_t OS_TimeAssembleFromSubseconds (int64 seconds, uint32 subseconds)

  Assemble/Convert a number of seconds + subseconds into an OS_time_t interval.
- static OS_time_t OS_TimeAdd (OS_time_t time1, OS_time_t time2)

  Computes the sum of two time intervals.
- static OS_time_t OS_TimeSubtract (OS_time_t time1, OS_time_t time2)

  Computes the difference between two time intervals.

37.50.1 Detailed Description

37.50.2 Function Documentation

37.50.2.1 OS_GetLocalTime()

```
int32 OS_GetLocalTime (
            OS_time_t * time_struct )
```

Get the local time.

This function gets the local time from the underlying OS.

Note

Mission time management typically uses the cFE Time Service

Parameters

out	*time_struct*	An OS_time_t that will be set to the current time

Returns

Get local time status, see OSAL Return Code Defines

Referenced by CFE_PSP_Get_Timebase(), and CFE_PSP_GetTime().

37.50.2.2 OS_SetLocalTime()

```
int32 OS_SetLocalTime (
            const OS_time_t * time_struct )
```

Set the local time.

This function sets the local time on the underlying OS.

Note

Mission time management typically uses the cFE Time Services

Parameters

in	*time_struct*	An OS_time_t containing the current time

Returns

Set local time status, see OSAL Return Code Defines

37.50.2.3 OS_TimeAdd()

```
static OS_time_t OS_TimeAdd (
            OS_time_t time1,
            OS_time_t time2 )  [inline], [static]
```

Computes the sum of two time intervals.

Parameters

in	*time1*	The first interval
in	*time2*	The second interval

Returns

The sum of the two intervals (time1 + time2)

Definition at line 384 of file osapi-clock.h.

References OS_time_t::ticks.

37.50.2.4 OS_TimeAssembleFromMicroseconds()

```
static OS_time_t OS_TimeAssembleFromMicroseconds (
            int64 seconds,
            uint32 microseconds )  [inline], [static]
```

Assemble/Convert a number of seconds + microseconds into an OS_time_t interval.

This creates an OS_time_t value using a whole number of seconds and a fractional part in units of microseconds. This is the inverse of OS_TimeGetTotalSeconds() and OS_TimeGetMicrosecondsPart(), and should recreate the original OS_time_t value from these separate values (aside from any potential conversion losses due to limited resolution of the data types/units).

See also

OS_TimeGetTotalSeconds(), OS_TimeGetMicrosecondsPart()

Parameters

in	*seconds*	Whole number of seconds
in	*microseconds*	Number of microseconds (fractional part only)

Returns

The input arguments represented as an OS_time_t interval

Definition at line 319 of file osapi-clock.h.

References OS_TIME_TICKS_PER_SECOND, OS_TIME_TICKS_PER_USEC, and OS_time_t::ticks.

37.50.2.5 OS_TimeAssembleFromMilliseconds()

```
static OS_time_t OS_TimeAssembleFromMilliseconds (
            int64 seconds,
            uint32 milliseconds )  [inline], [static]
```

Assemble/Convert a number of seconds + milliseconds into an OS_time_t interval.

This creates an OS_time_t value using a whole number of seconds and a fractional part in units of milliseconds. This is the inverse of OS_TimeGetTotalSeconds() and OS_TimeGetMillisecondsPart(), and should recreate the original OS_time_t value from these separate values (aside from any potential conversion losses due to limited resolution of the data types/units).

See also

OS_TimeGetTotalSeconds(), OS_TimeGetMillisecondsPart()

Parameters

in	*seconds*	Whole number of seconds
in	*milliseconds*	Number of milliseconds (fractional part only)

Returns

The input arguments represented as an OS_time_t interval

Definition at line 343 of file osapi-clock.h.

References OS_TIME_TICKS_PER_MSEC, OS_TIME_TICKS_PER_SECOND, and OS_time_t::ticks.

37.50.2.6 OS_TimeAssembleFromNanoseconds()

```
static OS_time_t OS_TimeAssembleFromNanoseconds (
            int64 seconds,
            uint32 nanoseconds )  [inline], [static]
```

Assemble/Convert a number of seconds + nanoseconds into an OS_time_t interval.

This creates an OS_time_t value using a whole number of seconds and a fractional part in units of nanoseconds. This is the inverse of OS_TimeGetTotalSeconds() and OS_TimeGetNanosecondsPart(), and should recreate the original OS_time_t value from these separate values (aside from any potential conversion losses due to limited resolution of the data types/units).

See also

OS_TimeGetTotalSeconds(), OS_TimeGetNanosecondsPart()

Parameters

in	*seconds*	Whole number of seconds
in	*nanoseconds*	Number of nanoseconds (fractional part only)

Returns

The input arguments represented as an OS_time_t interval

Definition at line 295 of file osapi-clock.h.

References OS_TIME_TICK_RESOLUTION_NS, OS_TIME_TICKS_PER_SECOND, and OS_time_t::ticks.

37.50.2.7 OS_TimeAssembleFromSubseconds()

```
static OS_time_t OS_TimeAssembleFromSubseconds (
            int64 seconds,
            uint32 subseconds )  [inline], [static]
```

Assemble/Convert a number of seconds + subseconds into an OS_time_t interval.

This creates an OS_time_t value using a whole number of seconds and a fractional part in units of sub-seconds (1/2^32). This is the inverse of OS_TimeGetTotalSeconds() and OS_TimeGetSubsecondsPart(), and should recreate the original OS_time_t value from these separate values (aside from any potential conversion losses due to limited resolution of the data types/units).

See also

OS_TimeGetTotalSeconds(), OS_TimeGetNanosecondsPart()

Parameters

in	*seconds*	Whole number of seconds
in	*subseconds*	Number of subseconds (32 bit fixed point fractional part)

Returns

The input arguments represented as an OS_time_t interval

Definition at line 366 of file osapi-clock.h.

References OS_TIME_TICKS_PER_SECOND, and OS_time_t::ticks.

37.50.2.8 OS_TimeGetFractionalPart()

```
static int64 OS_TimeGetFractionalPart (
            OS_time_t tm )  [inline], [static]
```

Get subseconds portion (fractional part only) from an OS_time_t object.

Extracts the fractional part from a given OS_time_t object. Units returned are in ticks, not normalized to any standard time unit.

Parameters

in	*tm*	Time interval value

Returns

Fractional/subsecond portion of time interval in ticks

Definition at line 187 of file osapi-clock.h.

References OS_TIME_TICKS_PER_SECOND, and OS_time_t::ticks.

Referenced by OS_TimeGetMicrosecondsPart(), OS_TimeGetMillisecondsPart(), OS_TimeGetNanosecondsPart(), and OS_TimeGetSubsecondsPart().

37.50.2.9 OS_TimeGetMicrosecondsPart()

```
static uint32 OS_TimeGetMicrosecondsPart (
            OS_time_t tm )  [inline], [static]
```

Get microseconds portion (fractional part only) from an OS_time_t object.

Extracts the fractional part from a given OS_time_t object normalized to units of microseconds.

This function may be used to adapt applications initially implemented using an older OSAL version where OS_time_t was a structure containing a "seconds" and "microsecs" field.

This function will obtain a value that is compatible with the "microsecs" field of OS_time_t as it was defined in previous versions of OSAL, as well as the "tv_usec" field of POSIX-style "struct timeval" values.

See also

OS_TimeGetTotalSeconds()

Parameters

in	*tm*	Time interval value

Returns

Number of microseconds in time interval

Definition at line 255 of file osapi-clock.h.

References OS_TIME_TICKS_PER_USEC, and OS_TimeGetFractionalPart().

Referenced by CFE_PSP_Get_Timebase().

Here is the call graph for this function:

37.50.2.10 OS_TimeGetMillisecondsPart()

```
static uint32 OS_TimeGetMillisecondsPart (
            OS_time_t tm )  [inline], [static]
```

Get milliseconds portion (fractional part only) from an OS_time_t object.

Extracts the fractional part from a given OS_time_t object normalized to units of milliseconds.

See also

OS_TimeGetTotalSeconds()

Parameters

in	*tm*	Time interval value

Returns

Number of milliseconds in time interval

Definition at line 230 of file osapi-clock.h.

References OS_TIME_TICKS_PER_MSEC, and OS_TimeGetFractionalPart().

Here is the call graph for this function:

37.50.2.11 OS_TimeGetNanosecondsPart()

```
static uint32 OS_TimeGetNanosecondsPart (
            OS_time_t tm ) [inline], [static]
```

Get nanoseconds portion (fractional part only) from an OS_time_t object.

Extracts the only number of nanoseconds from a given OS_time_t object.

This function will obtain a value that is compatible with the "tv_nsec" field of POSIX-style "struct timespec" values.

See also

OS_TimeGetTotalSeconds()

Parameters

in	*tm*	Time interval value

Returns

Number of nanoseconds in time interval

Definition at line 274 of file osapi-clock.h.

References OS_TIME_TICK_RESOLUTION_NS, and OS_TimeGetFractionalPart().

Here is the call graph for this function:

37.50.2.12 OS_TimeGetSubsecondsPart()

```
static uint32 OS_TimeGetSubsecondsPart (
            OS_time_t tm )  [inline], [static]
```

Get 32-bit normalized subseconds (fractional part only) from an OS_time_t object.

Extracts the fractional part from a given OS_time_t object in maximum precision, with units of 2^(-32) sec. This is a base-2 fixed-point fractional value with the point left-justified in the 32-bit value (i.e. left of MSB).

This is (mostly) compatible with the CFE "subseconds" value, where 0x80000000 represents exactly one half second, and 0 represents a full second.

Parameters

in	*tm*	Time interval value

Returns

Fractional/subsecond portion of time interval as 32-bit fixed point value

Definition at line 206 of file osapi-clock.h.

References OS_TIME_TICKS_PER_SECOND, and OS_TimeGetFractionalPart().

Here is the call graph for this function:

37.50.2.13 OS_TimeGetTotalMicroseconds()

```
static int64 OS_TimeGetTotalMicroseconds (
            OS_time_t tm )  [inline], [static]
```

Get interval from an OS_time_t object normalized to microsecond units.

Note this refers to the complete interval, not just the fractional part.

Parameters

in	*tm*	Time interval value

Returns

Whole number of microseconds in time interval

Definition at line 154 of file osapi-clock.h.

References OS_TIME_TICKS_PER_USEC, and OS_time_t::ticks.

37.50.2.14 OS_TimeGetTotalMilliseconds()

```
static int64 OS_TimeGetTotalMilliseconds (
            OS_time_t tm ) [inline], [static]
```

Get interval from an OS_time_t object normalized to millisecond units.

Note this refers to the complete interval, not just the fractional part.

Parameters

in	*tm*	Time interval value

Returns

Whole number of milliseconds in time interval

Definition at line 140 of file osapi-clock.h.

References OS_TIME_TICKS_PER_MSEC, and OS_time_t::ticks.

37.50.2.15 OS_TimeGetTotalNanoseconds()

```
static int64 OS_TimeGetTotalNanoseconds (
            OS_time_t tm ) [inline], [static]
```

Get interval from an OS_time_t object normalized to nanosecond units.

Note this refers to the complete interval, not just the fractional part.

Note

There is no protection against overflow of the 64-bit return value. Applications must use caution to ensure that the interval does not exceed the representable range of a signed 64 bit integer - approximately 140 years.

Parameters

in	*tm*	Time interval value

Returns

Whole number of microseconds in time interval

Definition at line 172 of file osapi-clock.h.

References OS_TIME_TICK_RESOLUTION_NS, and OS_time_t::ticks.

37.50.2.16 OS_TimeGetTotalSeconds()

```
static int64 OS_TimeGetTotalSeconds (
            OS_time_t tm ) [inline], [static]
```

Get interval from an OS_time_t object normalized to whole number of seconds.

Extracts the number of whole seconds from a given OS_time_t object, discarding any fractional component.

This may also replace a direct read of the "seconds" field from the OS_time_t object from previous versions of OSAL, where the structure was defined with separate seconds/microseconds fields.

See also

OS_TimeGetMicrosecondsPart()

Parameters

in	*tm*	Time interval value

Returns

Whole number of seconds in time interval

Definition at line 126 of file osapi-clock.h.

References OS_TIME_TICKS_PER_SECOND, and OS_time_t::ticks.

Referenced by CFE_PSP_Get_Timebase().

37.50.2.17 OS_TimeSubtract()

```
static OS_time_t OS_TimeSubtract (
            OS_time_t time1,
            OS_time_t time2 )  [inline], [static]
```

Computes the difference between two time intervals.

Parameters

in	*time1*	The first interval
in	*time2*	The second interval

Returns

The difference of the two intervals (time1 - time2)

Definition at line 398 of file osapi-clock.h.

References OS_time_t::ticks.

37.51 OSAL Core Operation APIs

Functions

- void OS_Application_Startup (void)

  Application startup.
- void OS_Application_Run (void)

  Application run.
- int32 OS_API_Init (void)

  Initialization of API.
- void OS_IdleLoop (void)

  Background thread implementation - waits forever for events to occur.
- void OS_DeleteAllObjects (void)

  delete all resources created in OSAL.
- void OS_ApplicationShutdown (uint8 flag)

  Initiate orderly shutdown.
- void OS_ApplicationExit (int32 Status)

  Exit/Abort the application.
- int32 OS_RegisterEventHandler (OS_EventHandler_t handler)

  Callback routine registration.

37.51.1 Detailed Description

These are for OSAL core operations for startup/initialization, running, and shutdown. Typically only used in bsps, unit tests, psps, etc.

Not intended for user application use

37.51.2 Function Documentation

37.51.2.1 OS_API_Init()

```
int32 OS_API_Init (
            void  )
```

Initialization of API.

This function returns initializes the internal data structures of the OS Abstraction Layer. It must be called in the application startup code before calling any other OS routines.

Returns

Execution status, see OSAL Return Code Defines. Any error code (negative) means the OSAL can not be initialized. Typical platform specific response is to abort since additional OSAL calls will have undefined behavior.

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	Failed execution.

Referenced by OS_Application_Startup().

37.51.2.2 OS_Application_Run()

```
void OS_Application_Run (
            void  )
```

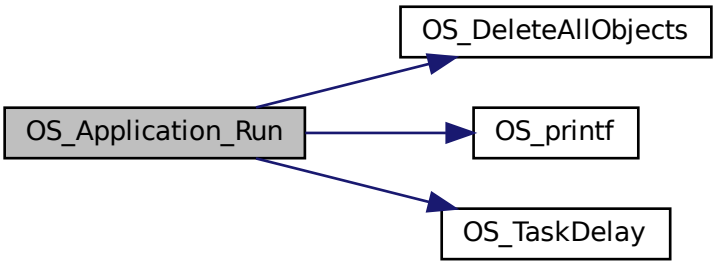
Application run.

Run abstraction such that the same BSP can be used for operations and testing.

Definition at line 467 of file cfe_psp_start.c.

References CFE_PSP_IdleTaskState, OS_DeleteAllObjects(), OS_printf(), and OS_TaskDelay().

Here is the call graph for this function:



37.51.2.3 OS_Application_Startup()

```
void OS_Application_Startup (
            void  )
```

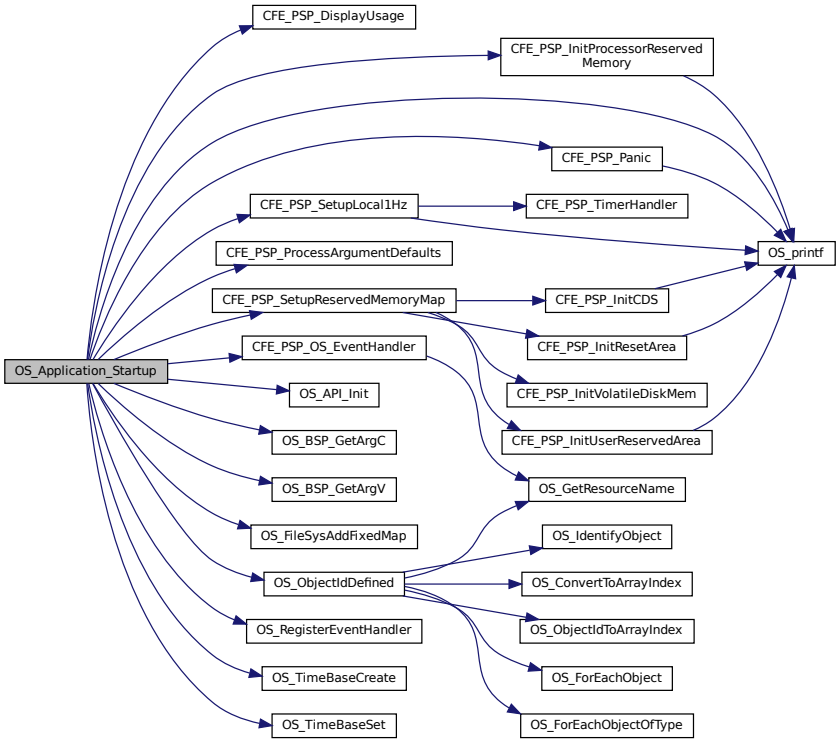
Application startup.

Startup abstraction such that the same BSP can be used for operations and testing.

Definition at line 211 of file cfe_psp_start.c.

References CFE_PSP_CPU_NAME_LENGTH, CFE_PSP_CpuId, CFE_PSP_CpuName, CFE_PSP_DisplayUsage(), CFE_PSP_IdleTaskState, CFE_PSP_InitProcessorReservedMemory(), CFE_PSP_MAIN_FUNCTION, CFE_PS↩P_NONVOL_STARTUP_FILE, CFE_PSP_OS_EventHandler(), CFE_PSP_Panic(), CFE_PSP_ProcessArgument↩Defaults(), CFE_PSP_ReservedMemoryMap, CFE_PSP_RESET_NAME_LENGTH, CFE_PSP_RST_TYPE_POWE↩RON, CFE_PSP_RST_TYPE_PROCESSOR, CFE_PSP_SetupLocal1Hz(), CFE_PSP_SetupReservedMemoryMap(), CFE_PSP_SpacecraftId, CFE_PSP_SUCCESS, CFE_PSP_CommandData_t::CpuId, CFE_PSP_CommandData↩_t::CpuName, CFE_PSP_CommandData_t::GotCpuId, CFE_PSP_CommandData_t::GotCpuName, CFE_PSP_↩CommandData_t::GotResetType, CFE_PSP_CommandData_t::GotSpacecraftId, CFE_PSP_CommandData_t::Got↩SubType, longOpts, optString, OS_API_Init(), OS_BSP_GetArgC(), OS_BSP_GetArgV(), OS_FileSysAddFixedMap(), OS_OBJECT_ID_UNDEFINED, OS_ObjectIdDefined(), OS_printf(), OS_RegisterEventHandler(), OS_SUCCESS, OS_TimeBaseCreate(), OS_TimeBaseSet(), CFE_PSP_CommandData_t::ResetType, CFE_PSP_CommandData_t↩::SpacecraftId, and CFE_PSP_CommandData_t::SubType.

Here is the call graph for this function:



37.51.2.4 OS_ApplicationExit()

```
void OS_ApplicationExit (
            int32 Status )
```

Exit/Abort the application.

Indicates that the OSAL application should exit and return control to the OS This is intended for e.g. scripted unit testing where the test needs to end without user intervention.

This function does not return. Production code typically should not ever call this.

Note

This exits the entire process including tasks that have been created.

37.51.2.5 OS_ApplicationShutdown()

```
void OS_ApplicationShutdown (
            uint8 flag )
```

Initiate orderly shutdown.

Indicates that the OSAL application should perform an orderly shutdown of ALL tasks, clean up all resources, and exit the application.

This allows the task currently blocked in OS_IdleLoop() to wake up, and for that function to return to its caller.

This is preferred over e.g. OS_ApplicationExit() which exits immediately and does not provide for any means to clean up first.

Parameters

in	*flag*	set to true to initiate shutdown, false to cancel

37.51.2.6 OS_DeleteAllObjects()

```
void OS_DeleteAllObjects (
            void )
```

delete all resources created in OSAL.

provides a means to clean up all resources allocated by this instance of OSAL. It would typically be used during an orderly shutdown but may also be helpful for testing purposes.

Referenced by OS_Application_Run().

37.51.2.7 OS_IdleLoop()

```
void OS_IdleLoop (
            void  )
```

Background thread implementation - waits forever for events to occur.

This should be called from the BSP main routine or initial thread after all other board and application initialization has taken place and all other tasks are running.

Typically just waits forever until "OS_shutdown" flag becomes true.

37.51.2.8 OS_RegisterEventHandler()

```
int32 OS_RegisterEventHandler (
            OS_EventHandler_t handler )
```

Callback routine registration.

This hook enables the application code to perform extra platform-specific operations on various system events such as resource creation/deletion.

Note

Some events are invoked while the resource is "locked" and therefore application-defined handlers for these events should not block or attempt to access other OSAL resources.

Parameters

in	*handler*	The application-provided event handler

Returns

Execution status, see OSAL Return Code Defines.

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	Failed execution.

Referenced by OS_Application_Startup().

37.52 OSAL Counting Semaphore APIs

Functions

- int32 OS_CountSemCreate (osal_id_t *sem_id, const char *sem_name, uint32 sem_initial_value, uint32 options)

  Creates a counting semaphore.
- int32 OS_CountSemGive (osal_id_t sem_id)

  Increment the semaphore value.
- int32 OS_CountSemTake (osal_id_t sem_id)

  Decrement the semaphore value.
- int32 OS_CountSemTimedWait (osal_id_t sem_id, uint32 msecs)

  Decrement the semaphore value with timeout.
- int32 OS_CountSemDelete (osal_id_t sem_id)

  Deletes the specified counting Semaphore.
- int32 OS_CountSemGetIdByName (osal_id_t *sem_id, const char *sem_name)

  Find an existing semaphore ID by name.
- int32 OS_CountSemGetInfo (osal_id_t sem_id, OS_count_sem_prop_t *count_prop)

  Fill a property object buffer with details regarding the resource.

37.52.1 Detailed Description

37.52.2 Function Documentation

37.52.2.1 OS_CountSemCreate()

```
int32 OS_CountSemCreate (
            osal_id_t * sem_id,
            const char * sem_name,
            uint32 sem_initial_value,
            uint32 options )
```

Creates a counting semaphore.

Creates a counting semaphore with initial value specified by sem_initial_value and name specified by sem_name. sem_id will be returned to the caller

Parameters

out	*sem_id*	will be set to the non-zero ID of the newly-created resource
in	*sem_name*	the name of the new resource to create
in	*sem_initial_value*	the initial value of the counting semaphore
in	*options*	Reserved for future use, should be passed as 0.

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if sen name or sem_id are NULL
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NO_FREE_IDS	if all of the semaphore ids are taken
OS_ERR_NAME_TAKEN	if this is already the name of a counting semaphore
OS_SEM_FAILURE	if the OS call failed
OS_INVALID_SEM_VALUE	if the semaphore value is too high

37.52.2.2 OS_CountSemDelete()

```
int32 OS_CountSemDelete (
            osal_id_t sem_id )
```

Deletes the specified counting Semaphore.

Parameters

in	*sem↩ _id*	The object ID to delete

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid counting semaphore
OS_SEM_FAILURE	the OS call failed

37.52.2.3 OS_CountSemGetIdByName()

```
int32 OS_CountSemGetIdByName (
            osal_id_t * sem_id,
            const char * sem_name )
```

Find an existing semaphore ID by name.

This function tries to find a counting sem Id given the name of a count_sem The id is returned through sem_id

Parameters

out	*sem_id*	will be set to the ID of the existing resource
in	*sem_name*	the name of the existing resource to find

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	is semid or sem_name are NULL pointers
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_NOT_FOUND	if the name was not found in the table

37.52.2.4 OS_CountSemGetInfo()

```
int32 OS_CountSemGetInfo (
            osal_id_t sem_id,
            OS_count_sem_prop_t * count_prop )
```

Fill a property object buffer with details regarding the resource.

This function will pass back a pointer to structure that contains all of the relevant info(name and creator) about the specified counting semaphore.

Parameters

in	*sem_id*	The object ID to operate on
out	*count_prop*	The property object buffer to fill

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid semaphore
OS_INVALID_POINTER	if the count_prop pointer is null

37.52.2.5 OS_CountSemGive()

```
int32 OS_CountSemGive (
            osal_id_t sem_id )
```

Increment the semaphore value.

The function unlocks the semaphore referenced by sem_id by performing a semaphore unlock operation on that semaphore. If the semaphore value resulting from this operation is positive, then no threads were blocked waiting for the semaphore to become unlocked; the semaphore value is simply incremented for this semaphore.

Parameters

in	*sem↩ _id*	The object ID to operate on

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_SEM_FAILURE	the semaphore was not previously initialized or is not in the array of semaphores defined by the system
OS_ERR_INVALID_ID	if the id passed in is not a counting semaphore

37.52.2.6 OS_CountSemTake()

```
int32 OS_CountSemTake (
            osal_id_t sem_id )
```

Decrement the semaphore value.

The locks the semaphore referenced by sem_id by performing a semaphore lock operation on that semaphore. If the semaphore value is currently zero, then the calling thread shall not return from the call until it either locks the semaphore or the call is interrupted.

Parameters

in	*sem↩ _id*	The object ID to operate on

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	the Id passed in is not a valid counting semaphore
OS_SEM_FAILURE	if the OS call failed

37.52.2.7 OS_CountSemTimedWait()

```
int32 OS_CountSemTimedWait (
            osal_id_t sem_id,
            uint32 msecs )
```

Decrement the semaphore value with timeout.

The function locks the semaphore referenced by sem_id. However, if the semaphore cannot be locked without waiting for another process or thread to unlock the semaphore, this wait shall be terminated when the specified timeout, msecs, expires.

Parameters

in	*sem↩ _id*	The object ID to operate on
in	*msecs*	The maximum amount of time to block, in milliseconds

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_SEM_TIMEOUT	if semaphore was not relinquished in time
OS_SEM_FAILURE	the semaphore was not previously initialized or is not in the array of semaphores defined by the system
OS_ERR_INVALID_ID	if the ID passed in is not a valid semaphore ID

37.53 OSAL Directory APIs

Functions

- int32 OS_DirectoryOpen (osal_id_t *dir_id, const char *path)

  Opens a directory.
- int32 OS_DirectoryClose (osal_id_t dir_id)

  Closes an open directory.
- int32 OS_DirectoryRewind (osal_id_t dir_id)

  Rewinds an open directory.
- int32 OS_DirectoryRead (osal_id_t dir_id, os_dirent_t *dirent)

  Reads the next name in the directory.
- int32 OS_mkdir (const char *path, uint32 access)

  Makes a new directory.
- int32 OS_rmdir (const char *path)

  Removes a directory from the file system.

37.53.1 Detailed Description

37.53.2 Function Documentation

37.53.2.1 OS_DirectoryClose()

```
int32 OS_DirectoryClose (
            osal_id_t dir_id )
```

Closes an open directory.

The directory referred to by dir_id will be closed

Parameters

in	*dir_id*	The handle ID of the directory

Returns

Execution status, see OSAL Return Code Defines

37.53.2.2 OS_DirectoryOpen()

```
int32 OS_DirectoryOpen (
            osal_id_t * dir_id,
            const char * path )
```

Opens a directory.

Prepares for reading the files within a directory

Parameters

out	*dir↩ _id*	The non-zero handle ID of the directory
in	*path*	The directory to open

Returns

Execution status, see OSAL Return Code Defines

37.53.2.3 OS_DirectoryRead()

```
int32 OS_DirectoryRead (
            osal_id_t dir_id,
            os_dirent_t * dirent )
```

Reads the next name in the directory.

Obtains directory entry data for the next file from an open directory

Parameters

in	*dir↩ _id*	The handle ID of the directory
out	*dirent*	Buffer to store directory entry information

Returns

Execution status, see OSAL Return Code Defines

37.53.2.4 OS_DirectoryRewind()

```
int32 OS_DirectoryRewind (
            osal_id_t dir_id )
```

Rewinds an open directory.

Resets a directory read handle back to the first file.

Parameters

in	*dir↩ _id*	The handle ID of the directory

Returns

Execution status, see OSAL Return Code Defines

37.53.2.5 OS_mkdir()

```
int32 OS_mkdir (
            const char * path,
            uint32 access )
```

Makes a new directory.

Makes a directory specified by path.

Parameters

in	*path*	The new directory name
in	*access*	The permissions for the directory (reserved for future use)

Note

Current implementations do not utilize the "access" parameter. Applications should still pass the intended value (OS_READ_WRITE or OS_READ_ONLY) to be compatible with future implementations.

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if path is NULL
OS_FS_ERR_PATH_TOO_LONG	if the path is too long to be stored locally
OS_FS_ERR_PATH_INVALID	if path cannot be parsed
OS_ERROR	if the OS call fails

37.53.2.6 OS_rmdir()

```
int32 OS_rmdir (
            const char * path )
```

Removes a directory from the file system.

Removes a directory from the structure. The directory must be empty prior to this operation.

Parameters

in	*path*	The directory to remove

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if path is NULL
OS_FS_ERR_PATH_INVALID	if path cannot be parsed
OS_FS_ERR_PATH_TOO_LONG	
OS_ERROR	if the directory remove operation failed

37.54 OSAL Return Code Defines

Macros

- #define OS_SUCCESS (0)

  Successful execution.
- #define OS_ERROR (-1)

  Failed execution.
- #define OS_INVALID_POINTER (-2)

  Invalid pointer.
- #define OS_ERROR_ADDRESS_MISALIGNED (-3)

  Address misalignment.
- #define OS_ERROR_TIMEOUT (-4)

  Error timeout.
- #define OS_INVALID_INT_NUM (-5)

  Invalid Interrupt number.
- #define OS_SEM_FAILURE (-6)

  Semaphore failure.
- #define OS_SEM_TIMEOUT (-7)

  Semaphore timeout.
- #define OS_QUEUE_EMPTY (-8)

  Queue empty.
- #define OS_QUEUE_FULL (-9)

  Queue full.
- #define OS_QUEUE_TIMEOUT (-10)

  Queue timeout.
- #define OS_QUEUE_INVALID_SIZE (-11)

  Queue invalid size.
- #define OS_QUEUE_ID_ERROR (-12)

  Queue ID error.
- #define OS_ERR_NAME_TOO_LONG (-13)

  name length including null terminator greater than OS_MAX_API_NAME
- #define OS_ERR_NO_FREE_IDS (-14)

  No free IDs.
- #define OS_ERR_NAME_TAKEN (-15)

  Name taken.
- #define OS_ERR_INVALID_ID (-16)

  Invalid ID.
- #define OS_ERR_NAME_NOT_FOUND (-17)

  Name not found.
- #define OS_ERR_SEM_NOT_FULL (-18)

  Semaphore not full.
- #define OS_ERR_INVALID_PRIORITY (-19)

  Invalid priority.
- #define OS_INVALID_SEM_VALUE (-20)

  Invalid semaphore value.
- #define OS_ERR_FILE (-27)

File error.

- #define OS_ERR_NOT_IMPLEMENTED (-28)

  Not implemented.

- #define OS_TIMER_ERR_INVALID_ARGS (-29)

  Timer invalid arguments.

- #define OS_TIMER_ERR_TIMER_ID (-30)

  Timer ID error.

- #define OS_TIMER_ERR_UNAVAILABLE (-31)

  Timer unavailable.

- #define OS_TIMER_ERR_INTERNAL (-32)

  Timer internal error.

- #define OS_ERR_OBJECT_IN_USE (-33)

  Object in use.

- #define OS_ERR_BAD_ADDRESS (-34)

  Bad address.

- #define OS_ERR_INCORRECT_OBJ_STATE (-35)

  Incorrect object state.

- #define OS_ERR_INCORRECT_OBJ_TYPE (-36)

  Incorrect object type.

- #define OS_ERR_STREAM_DISCONNECTED (-37)

  Stream disconnected.

- #define OS_ERR_OPERATION_NOT_SUPPORTED (-38)

  Requested operation not support on supplied object(s)

- #define OS_ERR_INVALID_SIZE (-40)

  Invalid Size.

- #define OS_FS_ERR_PATH_TOO_LONG (-103)

  FS path too long.

- #define OS_FS_ERR_NAME_TOO_LONG (-104)

  FS name too long.

- #define OS_FS_ERR_DRIVE_NOT_CREATED (-106)

  FS drive not created.

- #define OS_FS_ERR_DEVICE_NOT_FREE (-107)

  FS device not free.

- #define OS_FS_ERR_PATH_INVALID (-108)

  FS path invalid.

37.54.1 Detailed Description

37.54.2 Macro Definition Documentation

37.54.2.1 OS_ERR_BAD_ADDRESS

```
#define OS_ERR_BAD_ADDRESS (-34)
```

Bad address.

Definition at line 81 of file osapi-error.h.

37.54.2.2 OS_ERR_FILE

```
#define OS_ERR_FILE (-27)
```

File error.

Definition at line 74 of file osapi-error.h.

37.54.2.3 OS_ERR_INCORRECT_OBJ_STATE

```
#define OS_ERR_INCORRECT_OBJ_STATE (-35)
```

Incorrect object state.

Definition at line 82 of file osapi-error.h.

37.54.2.4 OS_ERR_INCORRECT_OBJ_TYPE

```
#define OS_ERR_INCORRECT_OBJ_TYPE (-36)
```

Incorrect object type.

Definition at line 83 of file osapi-error.h.

37.54.2.5 OS_ERR_INVALID_ID

```
#define OS_ERR_INVALID_ID (-16)
```

Invalid ID.

Definition at line 69 of file osapi-error.h.

37.54.2.6 OS_ERR_INVALID_PRIORITY

```
#define OS_ERR_INVALID_PRIORITY (-19)
```

Invalid priority.

Definition at line 72 of file osapi-error.h.

37.54.2.7 OS_ERR_INVALID_SIZE

```
#define OS_ERR_INVALID_SIZE (-40)
```

Invalid Size.

Definition at line 86 of file osapi-error.h.

37.54.2.8 OS_ERR_NAME_NOT_FOUND

```
#define OS_ERR_NAME_NOT_FOUND (-17)
```

Name not found.

Definition at line 70 of file osapi-error.h.

37.54.2.9 OS_ERR_NAME_TAKEN

```
#define OS_ERR_NAME_TAKEN (-15)
```

Name taken.

Definition at line 68 of file osapi-error.h.

37.54.2.10 OS_ERR_NAME_TOO_LONG

```
#define OS_ERR_NAME_TOO_LONG (-13)
```

name length including null terminator greater than OS_MAX_API_NAME

Definition at line 66 of file osapi-error.h.

37.54.2.11 OS_ERR_NO_FREE_IDS

```
#define OS_ERR_NO_FREE_IDS (-14)
```

No free IDs.

Definition at line 67 of file osapi-error.h.

37.54.2.12 OS_ERR_NOT_IMPLEMENTED

```
#define OS_ERR_NOT_IMPLEMENTED (-28)
```

Not implemented.

Definition at line 75 of file osapi-error.h.

37.54.2.13 OS_ERR_OBJECT_IN_USE

```
#define OS_ERR_OBJECT_IN_USE (-33)
```

Object in use.

Definition at line 80 of file osapi-error.h.

37.54.2.14 OS_ERR_OPERATION_NOT_SUPPORTED

```
#define OS_ERR_OPERATION_NOT_SUPPORTED (-38)
```

Requested operation not support on supplied object(s)

Definition at line 85 of file osapi-error.h.

37.54.2.15 OS_ERR_SEM_NOT_FULL

```
#define OS_ERR_SEM_NOT_FULL (-18)
```

Semaphore not full.

Definition at line 71 of file osapi-error.h.

37.54.2.16 OS_ERR_STREAM_DISCONNECTED

```
#define OS_ERR_STREAM_DISCONNECTED (-37)
```

Stream disconnected.

Definition at line 84 of file osapi-error.h.

37.54.2.17 OS_ERROR

```
#define OS_ERROR (-1)
```

Failed execution.

Definition at line 54 of file osapi-error.h.

37.54.2.18 OS_ERROR_ADDRESS_MISALIGNED

```
#define OS_ERROR_ADDRESS_MISALIGNED (-3)
```

Address misalignment.

Definition at line 56 of file osapi-error.h.

37.54.2.19 OS_ERROR_TIMEOUT

```
#define OS_ERROR_TIMEOUT (-4)
```

Error timeout.

Definition at line 57 of file osapi-error.h.

37.54.2.20 OS_FS_ERR_DEVICE_NOT_FREE

```
#define OS_FS_ERR_DEVICE_NOT_FREE (-107)
```

FS device not free.

Definition at line 99 of file osapi-error.h.

37.54.2.21 OS_FS_ERR_DRIVE_NOT_CREATED

```
#define OS_FS_ERR_DRIVE_NOT_CREATED (-106)
```

FS drive not created.

Definition at line 98 of file osapi-error.h.

37.54.2.22 OS_FS_ERR_NAME_TOO_LONG

```
#define OS_FS_ERR_NAME_TOO_LONG (-104)
```

FS name too long.

Definition at line 97 of file osapi-error.h.

37.54.2.23 OS_FS_ERR_PATH_INVALID

```
#define OS_FS_ERR_PATH_INVALID (-108)
```

FS path invalid.

Definition at line 100 of file osapi-error.h.

37.54.2.24 OS_FS_ERR_PATH_TOO_LONG

```
#define OS_FS_ERR_PATH_TOO_LONG (-103)
```

FS path too long.

Definition at line 96 of file osapi-error.h.

37.54.2.25 OS_INVALID_INT_NUM

```
#define OS_INVALID_INT_NUM (-5)
```

Invalid Interrupt number.

Definition at line 58 of file osapi-error.h.

37.54.2.26 OS_INVALID_POINTER

```
#define OS_INVALID_POINTER (-2)
```

Invalid pointer.

Definition at line 55 of file osapi-error.h.

37.54.2.27 OS_INVALID_SEM_VALUE

```
#define OS_INVALID_SEM_VALUE (-20)
```

Invalid semaphore value.

Definition at line 73 of file osapi-error.h.

37.54.2.28 OS_QUEUE_EMPTY

```
#define OS_QUEUE_EMPTY (-8)
```

Queue empty.

Definition at line 61 of file osapi-error.h.

37.54.2.29 OS_QUEUE_FULL

```
#define OS_QUEUE_FULL (-9)
```

Queue full.

Definition at line 62 of file osapi-error.h.

37.54.2.30 OS_QUEUE_ID_ERROR

```
#define OS_QUEUE_ID_ERROR (-12)
```

Queue ID error.

Definition at line 65 of file osapi-error.h.

37.54.2.31 OS_QUEUE_INVALID_SIZE

```
#define OS_QUEUE_INVALID_SIZE (-11)
```

Queue invalid size.

Definition at line 64 of file osapi-error.h.

37.54.2.32 OS_QUEUE_TIMEOUT

```
#define OS_QUEUE_TIMEOUT (-10)
```

Queue timeout.

Definition at line 63 of file osapi-error.h.

37.54.2.33 OS_SEM_FAILURE

```
#define OS_SEM_FAILURE (-6)
```

Semaphore failure.

Definition at line 59 of file osapi-error.h.

37.54.2.34 OS_SEM_TIMEOUT

```
#define OS_SEM_TIMEOUT (-7)
```

Semaphore timeout.

Definition at line 60 of file osapi-error.h.

37.54.2.35 OS_SUCCESS

```
#define OS_SUCCESS (0)
```

Successful execution.

Definition at line 53 of file osapi-error.h.

Referenced by CFE_PSP_OS_EventHandler(), and OS_Application_Startup().

37.54.2.36 OS_TIMER_ERR_INTERNAL

```
#define OS_TIMER_ERR_INTERNAL (-32)
```

Timer internal error.

Definition at line 79 of file osapi-error.h.

37.54.2.37 OS_TIMER_ERR_INVALID_ARGS

```
#define OS_TIMER_ERR_INVALID_ARGS (-29)
```

Timer invalid arguments.

Definition at line 76 of file osapi-error.h.

37.54.2.38 OS_TIMER_ERR_TIMER_ID

```
#define OS_TIMER_ERR_TIMER_ID (-30)
```

Timer ID error.

Definition at line 77 of file osapi-error.h.

37.54.2.39 OS_TIMER_ERR_UNAVAILABLE

```
#define OS_TIMER_ERR_UNAVAILABLE (-31)
```

Timer unavailable.

Definition at line 78 of file osapi-error.h.

37.55 OSAL Error Info APIs

Functions

- int32 OS_GetErrorName (int32 error_num, os_err_name_t *err_name)

  Convert an error number to a string.

37.55.1 Detailed Description

37.55.2 Function Documentation

37.55.2.1 OS_GetErrorName()

```
int32 OS_GetErrorName (
            int32 error_num,
            os_err_name_t * err_name )
```

Convert an error number to a string.

Parameters

in	*error_num*	Error number to convert
out	*err_name*	Buffer to store error string

Returns

Execution status, see OSAL Return Code Defines

37.56 OSAL File Access Option Defines

Macros

- #define OS_READ_ONLY 0
- #define OS_WRITE_ONLY 1
- #define OS_READ_WRITE 2

37.56.1 Detailed Description

37.56.2 Macro Definition Documentation

37.56.2.1 OS_READ_ONLY

```
#define OS_READ_ONLY 0
```

Read only file access

Definition at line 35 of file osapi-file.h.

37.56.2.2 OS_READ_WRITE

```
#define OS_READ_WRITE 2
```

Read write file access

Definition at line 37 of file osapi-file.h.

37.56.2.3 OS_WRITE_ONLY

```
#define OS_WRITE_ONLY 1
```

Write only file access

Definition at line 36 of file osapi-file.h.

37.57 OSAL Reference Point For Seek Offset Defines

Macros

- #define OS_SEEK_SET 0
- #define OS_SEEK_CUR 1
- #define OS_SEEK_END 2

37.57.1 Detailed Description

37.57.2 Macro Definition Documentation

37.57.2.1 OS_SEEK_CUR

```
#define OS_SEEK_CUR 1
```

Seek offset current

Definition at line 44 of file osapi-file.h.

37.57.2.2 OS_SEEK_END

```
#define OS_SEEK_END 2
```

Seek offset end

Definition at line 45 of file osapi-file.h.

37.57.2.3 OS_SEEK_SET

```
#define OS_SEEK_SET 0
```

Seek offset set

Definition at line 43 of file osapi-file.h.

37.58 OSAL Standard File APIs

Functions

- int32 OS_OpenCreate (osal_id_t *filedes, const char *path, int32 flags, int32 access)

  Open or create a file.
- int32 OS_close (osal_id_t filedes)

  Closes an open file handle.
- int32 OS_read (osal_id_t filedes, void *buffer, size_t nbytes)

  Read from a file handle.
- int32 OS_write (osal_id_t filedes, const void *buffer, size_t nbytes)

  Write to a file handle.
- int32 OS_TimedRead (osal_id_t filedes, void *buffer, size_t nbytes, int32 timeout)

  File/Stream input read with a timeout.
- int32 OS_TimedWrite (osal_id_t filedes, const void *buffer, size_t nbytes, int32 timeout)

  File/Stream output write with a timeout.
- int32 OS_chmod (const char *path, uint32 access)

  Changes the permissions of a file.
- int32 OS_stat (const char *path, os_fstat_t *filestats)

  Obtain information about a file or directory.
- int32 OS_lseek (osal_id_t filedes, int32 offset, uint32 whence)

  Seeks to the specified position of an open file.
- int32 OS_remove (const char *path)

  Removes a file from the file system.
- int32 OS_rename (const char *old_filename, const char *new_filename)

  Renames a file.
- int32 OS_cp (const char *src, const char *dest)

  Copies a single file from src to dest.
- int32 OS_mv (const char *src, const char *dest)

  Move a single file from src to dest.
- int32 OS_FDGetInfo (osal_id_t filedes, OS_file_prop_t *fd_prop)

  Obtain information about an open file.
- int32 OS_FileOpenCheck (const char *Filename)

  Checks to see if a file is open.
- int32 OS_CloseAllFiles (void)

  Close all open files.
- int32 OS_CloseFileByName (const char *Filename)

  Close a file by filename.

37.58.1 Detailed Description

37.58.2 Function Documentation

37.58.2.1 OS_chmod()

```
int32 OS_chmod (
            const char * path,
            uint32 access )
```

Changes the permissions of a file.

Parameters

in	*path*	File to change
in	*access*	Desired access mode - see OSAL File Access Option Defines

Note

Some file systems do not implement permissions

Returns

Execution status, see OSAL Return Code Defines

37.58.2.2 OS_close()

```
int32 OS_close (
            osal_id_t filedes )
```

Closes an open file handle.

This closes regular file handles and any other file-like resource, such as network streams or pipes.

Parameters

in	*filedes*	The handle ID to operate on

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if file descriptor could not be closed
OS_ERR_INVALID_ID	if the file descriptor passed in is invalid

37.58.2.3 OS_CloseAllFiles()

```
int32 OS_CloseAllFiles (
            void )
```

Close all open files.

Closes All open files that were opened through the OSAL

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if one or more file close returned an error

37.58.2.4 OS_CloseFileByName()

```
int32 OS_CloseFileByName (
            const char * Filename )
```

Close a file by filename.

Allows a file to be closed by name. This will only work if the name passed in is the same name used to open the file.

Parameters

in	*Filename*	The file to close

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_FS_ERR_PATH_INVALID	if the file is not found
OS_ERROR	if the file close returned an error

37.58.2.5 OS_cp()

```
int32 OS_cp (
            const char * src,
            const char * dest )
```

Copies a single file from src to dest.

Note

The behvior of this API on an open file is not defined at the OSAL level due to dependencies on the underlying OS which may or may not allow the related operation based on a varienty of potential configurations. For portability, it is recommended that applications ensure the file is closed prior to removal.

Parameters

in	*src*	The source file to operate on
in	*dest*	The destination file

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the file could not be accessed
OS_INVALID_POINTER	if src or dest are NULL
OS_FS_ERR_PATH_INVALID	if path cannot be parsed
OS_FS_ERR_PATH_TOO_LONG	if the paths given are too long to be stored locally
OS_FS_ERR_NAME_TOO_LONG	if the dest name is too long to be stored locally

37.58.2.6 OS_FDGetInfo()

```
int32 OS_FDGetInfo (
            osal_id_t filedes,
            OS_file_prop_t * fd_prop )
```

Obtain information about an open file.

Copies the information of the given file descriptor into a structure passed in

Parameters

in	*filedes*	The handle ID to operate on
out	*fd_prop*	Storage buffer for file information

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the file descriptor passed in is invalid

37.58.2.7 OS_FileOpenCheck()

```
int32 OS_FileOpenCheck (
            const char * Filename )
```

Checks to see if a file is open.

This function takes a filename and determines if the file is open. The function will return success if the file is open.

Parameters

in	*Filename*	The file to operate on

Returns

OS_SUCCESS if the file is open, or appropriate error code

Return values

OS_ERROR	if the file is not open

37.58.2.8 OS_lseek()

```
int32 OS_lseek (
            osal_id_t filedes,
            int32 offset,
            uint32 whence )
```

Seeks to the specified position of an open file.

Sets the read/write pointer to a specific offset in a specific file.

Parameters

in	*filedes*	The handle ID to operate on
in	*offset*	The file offset to seek to
in	*whence*	The reference point for offset, see OSAL Reference Point For Seek Offset Defines

Returns

Byte offset from the beginning of the file or appropriate error code, see OSAL Return Code Defines

Return values

OS_ERR_INVALID_ID	if the file descriptor passed in is invalid
OS_ERROR	if OS call failed

37.58.2.9 OS_mv()

```
int32 OS_mv (
            const char * src,
            const char * dest )
```

Move a single file from src to dest.

This first attempts to rename the file, which is faster if the source and destination reside on the same file system.

If this fails, it falls back to copying the file and removing the original.

Note

The behvior of this API on an open file is not defined at the OSAL level due to dependencies on the underlying OS which may or may not allow the related operation based on a varienty of potential configurations. For portability, it is recommended that applications ensure the file is closed prior to removal.

Parameters

in	*src*	The source file to operate on
in	*dest*	The destination file

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the file could not be renamed.
OS_INVALID_POINTER	if src or dest are NULL
OS_FS_ERR_PATH_INVALID	if path cannot be parsed
OS_FS_ERR_PATH_TOO_LONG	if the paths given are too long to be stored locally
OS_FS_ERR_NAME_TOO_LONG	if the dest name is too long to be stored locally

37.58.2.10 OS_OpenCreate()

```
int32 OS_OpenCreate (
            osal_id_t * filedes,
            const char * path,
            int32 flags,
            int32 access )
```

Open or create a file.

Implements the same as OS_open/OS_creat but follows the OSAL paradigm of outputting the ID/descriptor separately from the return value, rather than relying on the user to convert it back.

Parameters

out	*filedes*	The handle ID (OS_OBJECT_ID_UNDEFINED on failure)
in	*path*	File name to create or open
in	*flags*	The file permissions - see OS_file_flag_t
in	*access*	Intended access mode - see OSAL File Access Option Defines

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the command was not executed properly

37.58.2.11 OS_read()

```
int32 OS_read (
            osal_id_t filedes,
            void * buffer,
            size_t nbytes )
```

Read from a file handle.

Reads up to nbytes from a file, and puts them into buffer.

Parameters

in	*filedes*	The handle ID to operate on
out	*buffer*	Storage location for file data
in	*nbytes*	Maximum number of bytes to read

Note

All OSAL error codes are negative int32 values. Failure of this call can be checked by testing if the result is less than 0.

Returns

A non-negative byte count or appropriate error code, see OSAL Return Code Defines

Return values

OS_INVALID_POINTER	if buffer is a null pointer
OS_ERROR	if OS call failed
OS_ERR_INVALID_ID	if the file descriptor passed in is invalid

37.58.2.12 OS_remove()

```
int32 OS_remove (
            const char * path )
```

Removes a file from the file system.

Removes a given filename from the drive

Note

The behvior of this API on an open file is not defined at the OSAL level due to dependencies on the underlying OS which may or may not allow the related operation based on a varienty of potential configurations. For portability, it is recommended that applications ensure the file is closed prior to removal.

Parameters

in	*path*	The file to operate on

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if there is no device or the driver returns error
OS_INVALID_POINTER	if path is NULL
OS_FS_ERR_PATH_TOO_LONG	if path is too long to be stored locally
OS_FS_ERR_PATH_INVALID	if path cannot be parsed
OS_FS_ERR_NAME_TOO_LONG	if the name of the file to remove is too long

37.58.2.13 OS_rename()

```
int32 OS_rename (
            const char * old_filename,
            const char * new_filename )
```

Renames a file.

Changes the name of a file, where the source and destination reside on the same file system.

Note

The behvior of this API on an open file is not defined at the OSAL level due to dependencies on the underlying OS which may or may not allow the related operation based on a varienty of potential configurations. For portability, it is recommended that applications ensure the file is closed prior to removal.

Parameters

in	*old_filename*	The original filename
in	*new_filename*	The desired filename

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the file could not be opened or renamed.
OS_INVALID_POINTER	if old or new are NULL
OS_FS_ERR_PATH_INVALID	if path cannot be parsed
OS_FS_ERR_PATH_TOO_LONG	if the paths given are too long to be stored locally
OS_FS_ERR_NAME_TOO_LONG	if the new name is too long to be stored locally

37.58.2.14 OS_stat()

```
int32 OS_stat (
            const char * path,
            os_fstat_t * filestats )
```

Obtain information about a file or directory.

Returns information about a file or directory in a os_fstat_t structure

Parameters

in	*path*	The file to operate on
out	*filestats*	Buffer to store file information

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if path or filestats is NULL
OS_FS_ERR_PATH_TOO_LONG	if the path is too long to be stored locally
OS_FS_ERR_NAME_TOO_LONG	if the name of the file is too long to be stored
OS_FS_ERR_PATH_INVALID	if path cannot be parsed
OS_ERROR	if the OS call failed

37.58.2.15 OS_TimedRead()

```
int32 OS_TimedRead (
            osal_id_t filedes,
            void * buffer,
            size_t nbytes,
            int32 timeout )
```

File/Stream input read with a timeout.

This implements a time-limited read and is primarily intended for use with sockets but may also work with any other stream-like resource that the underlying OS supports.

If data is immediately available on the file/socket, this will return that data along with the actual number of bytes that were immediately available. It will not block.

If no data is immediately available, this will wait up to the given timeout for data to appear. If no data appears within the timeout period, then this returns an error code (not zero).

In all cases this will return successfully as soon as at least 1 byte of actual data is available. It will not attempt to read the entire input buffer.

If an EOF condition occurs prior to timeout, this function returns zero.

Parameters

in	*filedes*	The handle ID to operate on
in	*buffer*	Source location for file data
in	*nbytes*	Maximum number of bytes to read
in	*timeout*	Maximum time to wait, in milliseconds (OS_PEND = forever)

Returns

Byte count on success, zero for timeout, or appropriate error code, see OSAL Return Code Defines

37.58.2.16 OS_TimedWrite()

```
int32 OS_TimedWrite (
            osal_id_t filedes,
            const void * buffer,
            size_t nbytes,
            int32 timeout )
```

File/Stream output write with a timeout.

This implements a time-limited write and is primarily intended for use with sockets but may also work with any other stream-like resource that the underlying OS supports.

If output buffer space is immediately available on the file/socket, this will place data into the buffer and return the actual number of bytes that were queued for output. It will not block.

If no output buffer space is immediately available, this will wait up to the given timeout for space to become available. If no space becomes available within the timeout period, then this returns an error code (not zero).

In all cases this will return successfully as soon as at least 1 byte of actual data is output. It will *not* attempt to write the entire output buffer.

If an EOF condition occurs prior to timeout, this function returns zero.

Parameters

in	*filedes*	The handle ID to operate on
in	*buffer*	Source location for file data
in	*nbytes*	Maximum number of bytes to read
in	*timeout*	Maximum time to wait, in milliseconds (OS_PEND = forever)

Returns

Byte count on success, zero for timeout, or appropriate error code, see OSAL Return Code Defines

37.58.2.17 OS_write()

```
int32 OS_write (
            osal_id_t filedes,
            const void * buffer,
            size_t nbytes )
```

Write to a file handle.

Writes to a file. copies up to a maximum of nbytes of buffer to the file described in filedes

Parameters

in	*filedes*	The handle ID to operate on
in	*buffer*	Source location for file data
in	*nbytes*	Maximum number of bytes to read

Note

All OSAL error codes are negative int32 values. Failure of this call can be checked by testing if the result is less than 0.

Returns

A non-negative byte count or appropriate error code, see OSAL Return Code Defines

Return values

OS_INVALID_POINTER	if buffer is NULL
OS_ERROR	if OS call failed
OS_ERR_INVALID_ID	if the file descriptor passed in is invalid

37.59 OSAL File System Level APIs

Functions

- int32 OS_FileSysAddFixedMap (osal_id_t ∗filesys_id, const char ∗phys_path, const char ∗virt_path)

  Create a fixed mapping between an existing directory and a virtual OSAL mount point.
- int32 OS_mkfs (char ∗address, const char ∗devname, const char ∗volname, size_t blocksize, osal_blockcount_t numblocks)

  Makes a file system on the target.
- int32 OS_mount (const char ∗devname, const char ∗mountpoint)

  Mounts a file system.
- int32 OS_initfs (char ∗address, const char ∗devname, const char ∗volname, size_t blocksize, osal_blockcount_t numblocks)

  Initializes an existing file system.
- int32 OS_rmfs (const char ∗devname)

  Removes a file system.
- int32 OS_unmount (const char ∗mountpoint)

  Unmounts a mounted file system.
- int32 OS_FileSysStatVolume (const char ∗name, OS_statvfs_t ∗statbuf)

  Obtains information about size and free space in a volume.
- int32 OS_chkfs (const char ∗name, bool repair)

  Checks the health of a file system and repairs it if necessary.
- int32 OS_FS_GetPhysDriveName (char ∗PhysDriveName, const char ∗MountPoint)

  Obtains the physical drive name associated with a mount point.
- int32 OS_TranslatePath (const char ∗VirtualPath, char ∗LocalPath)

  Translates a OSAL Virtual file system path to a host Local path.
- int32 OS_GetFsInfo (os_fsinfo_t ∗filesys_info)

  Returns information about the file system.

37.59.1 Detailed Description

37.59.2 Function Documentation

37.59.2.1 OS_chkfs()

```
int32 OS_chkfs (
            const char * name,
            bool repair )
```

Checks the health of a file system and repairs it if necessary.

Checks the drives for inconsistencies and optionally also repairs it

Note

not all operating systems implement this function

Parameters

in	*name*	The device/path to operate on
in	*repair*	Whether to also repair inconsistencies

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	Name is NULL
OS_ERR_NOT_IMPLEMENTED	Not implemented.
OS_ERROR	Failed execution.

37.59.2.2 OS_FileSysAddFixedMap()

```
int32 OS_FileSysAddFixedMap (
            osal_id_t * filesys_id,
            const char * phys_path,
            const char * virt_path )
```

Create a fixed mapping between an existing directory and a virtual OSAL mount point.

This mimics the behavior of a "FS_BASED" entry in the VolumeTable but is registered at runtime. It is intended to be called by the PSP/BSP prior to starting the application.

Note

OSAL virtual mount points are required to be a single, non-empty top-level directory name. Virtual path names always follow the form /<virt_mount_point>/<relative_path>/<file>. Only the relative path may be omitted/empty (i.e. /<virt_mount_point>/<file>) but the virtual mount point must be present and not an empty string. In particular this means it is not possible to directly refer to files in the "root" of the native file system from OSAL. However it is possible to create a virtual map to the root, such as by calling:

```
OS_FileSysAddFixedMap(&fs_id, "/", "/root");
```

Parameters

out	*filesys_id*	A non-zero OSAL ID reflecting the file system
in	*phys_path*	The native system directory (an existing mount point)
in	*virt_path*	The virtual mount point of this filesystem

Returns

Execution status, see OSAL Return Code Defines

Referenced by OS_Application_Startup().

37.59.2.3 OS_FileSysStatVolume()

```
int32 OS_FileSysStatVolume (
            const char * name,
            OS_statvfs_t * statbuf )
```

Obtains information about size and free space in a volume.

Populates the supplied OS_statvfs_t structure, which includes the block size and total/free blocks in a file system volume.

This replaces two older OSAL calls:

OS_fsBlocksFree() is determined by reading the blocks_free output struct member OS_fsBytesFree() is determined by multiplying blocks_free by the block_size member

Parameters

in	*name*	The device/path to operate on
out	*statbuf*	Output structure to populate

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if name or statbuf is NULL
OS_ERROR	if the OS call failed

37.59.2.4 OS_FS_GetPhysDriveName()

```
int32 OS_FS_GetPhysDriveName (
            char * PhysDriveName,
            const char * MountPoint )
```

Obtains the physical drive name associated with a mount point.

Returns the name of the physical volume associated with the drive, when given the OSAL mount point of the drive

Parameters

out	*PhysDriveName*	Buffer to store physical drive name
in	*MountPoint*	OSAL mount point

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if either parameter is NULL
OS_ERROR	if the mountpoint could not be found

37.59.2.5 OS_GetFsInfo()

```
int32 OS_GetFsInfo (
            os_fsinfo_t * filesys_info )
```

Returns information about the file system.

Returns information about the file system in an os_fsinfo_t. This includes the number of open files and file systems

Parameters

out	*filesys_info*	Buffer to store filesystem information

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if filesys_info is NULL

37.59.2.6 OS_initfs()

```
int32 OS_initfs (
            char * address,
```

```
const char * devname,
const char * volname,
size_t blocksize,
osal_blockcount_t numblocks )
```

Initializes an existing file system.

Initializes a file system on the target.

Note

The "volname" parameter of RAM disks should always begin with the string "RAM", e.g. "RAMDISK" or "RA↩ M0","RAM1", etc if multiple devices are created. The underlying implementation uses this to select the correct filesystem type/format, and this may also be used to differentiate between RAM disks and real physical disks.

Parameters

in	*address*	The address at which to start the new disk. If address == 0, then space will be allocated by the OS
in	*devname*	The underlying kernel device to use, if applicable.
in	*volname*	The name of the volume (see note)
in	*blocksize*	The size of a single block on the drive
in	*numblocks*	The number of blocks to allocate for the drive

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if devname or volname are NULL
OS_FS_ERR_PATH_TOO_LONG	if the name is too long
OS_FS_ERR_DEVICE_NOT_FREE	if the volume table is full
OS_FS_ERR_DRIVE_NOT_CREATED	on error

37.59.2.7 OS_mkfs()

```
int32 OS_mkfs (
            char * address,
            const char * devname,
            const char * volname,
            size_t blocksize,
            osal_blockcount_t numblocks )
```

Makes a file system on the target.

Makes a file system on the target. Highly dependent on underlying OS and dependent on OS volume table definition.

Note

The "volname" parameter of RAM disks should always begin with the string "RAM", e.g. "RAMDISK" or "RA↩ M0","RAM1", etc if multiple devices are created. The underlying implementation uses this to select the correct filesystem type/format, and this may also be used to differentiate between RAM disks and real physical disks.

Parameters

in	*address*	The address at which to start the new disk. If address == 0 space will be allocated by the OS.
in	*devname*	The underlying kernel device to use, if applicable.
in	*volname*	The name of the volume (see note)
in	*blocksize*	The size of a single block on the drive
in	*numblocks*	The number of blocks to allocate for the drive

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_INVALID_POINTER	if devname is NULL
OS_FS_ERR_DRIVE_NOT_CREATED	if the OS calls to create the the drive failed
OS_FS_ERR_DEVICE_NOT_FREE	if the volume table is full
OS_SUCCESS	on creating the disk

37.59.2.8 OS_mount()

```
int32 OS_mount (
            const char * devname,
            const char * mountpoint )
```

Mounts a file system.

Mounts a file system / block device at the given mount point.

Parameters

in	*devname*	The name of the drive to mount. devname is the same from OS_mkfs
in	*mountpoint*	The name to call this disk from now on

Returns

Execution status, see OSAL Return Code Defines

37.59.2.9 OS_rmfs()

```
int32 OS_rmfs (
            const char * devname )
```

Removes a file system.

This function will remove or un-map the target file system. Note that this is not the same as un-mounting the file system.

Parameters

in	*devname*	The name of the "generic" drive

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if devname is NULL
OS_ERROR	is the drive specified cannot be located

37.59.2.10 OS_TranslatePath()

```
int32 OS_TranslatePath (
            const char * VirtualPath,
            char * LocalPath )
```

Translates a OSAL Virtual file system path to a host Local path.

Translates a virtual path to an actual system path name

Parameters

in	*VirtualPath*	OSAL virtual path name
out	*LocalPath*	Buffer to store native/translated path name

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if either parameter is NULL

37.59.2.11 OS_unmount()

```
int32 OS_unmount (
            const char * mountpoint )
```

Unmounts a mounted file system.

This function will unmount a drive from the file system and make all open file descriptors useless.

Note

Any open file descriptors referencing this file system should be closed prior to unmounting a drive

Parameters

in	*mountpoint*	The mount point to remove from OS_mount

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if name is NULL
OS_FS_ERR_PATH_TOO_LONG	if the absolute path given is too long
OS_ERROR	if the OS calls failed

37.60 OSAL Heap APIs

Functions

- int32 OS_HeapGetInfo (OS_heap_prop_t *heap_prop)

  Return current info on the heap.

37.60.1 Detailed Description

37.60.2 Function Documentation

37.60.2.1 OS_HeapGetInfo()

```
int32 OS_HeapGetInfo (
            OS_heap_prop_t * heap_prop )
```

Return current info on the heap.

Parameters

out	*heap_prop*	Storage buffer for heap info

Returns

Execution status, see OSAL Return Code Defines

37.61 OSAL Object Type Defines

Macros

- #define OS_OBJECT_TYPE_UNDEFINED 0x00

  Object type undefined.
- #define OS_OBJECT_TYPE_OS_TASK 0x01

  Object task type.
- #define OS_OBJECT_TYPE_OS_QUEUE 0x02

  Object queue type.
- #define OS_OBJECT_TYPE_OS_COUNTSEM 0x03

  Object counting semaphore type.
- #define OS_OBJECT_TYPE_OS_BINSEM 0x04

  Object binary semaphore type.
- #define OS_OBJECT_TYPE_OS_MUTEX 0x05

  Object mutex type.
- #define OS_OBJECT_TYPE_OS_STREAM 0x06

  Object stream type.
- #define OS_OBJECT_TYPE_OS_DIR 0x07

  Object directory type.
- #define OS_OBJECT_TYPE_OS_TIMEBASE 0x08

  Object timebase type.
- #define OS_OBJECT_TYPE_OS_TIMECB 0x09

  Object timer callback type.
- #define OS_OBJECT_TYPE_OS_MODULE 0x0A

  Object module type.
- #define OS_OBJECT_TYPE_OS_FILESYS 0x0B

  Object file system type.
- #define OS_OBJECT_TYPE_OS_CONSOLE 0x0C

  Object console type.
- #define OS_OBJECT_TYPE_USER 0x10

  Object user type.

37.61.1 Detailed Description

37.61.2 Macro Definition Documentation

37.61.2.1 OS_OBJECT_TYPE_OS_BINSEM

```
#define OS_OBJECT_TYPE_OS_BINSEM 0x04
```

Object binary semaphore type.

Definition at line 42 of file osapi-idmap.h.

37.61.2.2 OS_OBJECT_TYPE_OS_CONSOLE

```
#define OS_OBJECT_TYPE_OS_CONSOLE 0x0C
```

Object console type.

Definition at line 50 of file osapi-idmap.h.

37.61.2.3 OS_OBJECT_TYPE_OS_COUNTSEM

```
#define OS_OBJECT_TYPE_OS_COUNTSEM 0x03
```

Object counting semaphore type.

Definition at line 41 of file osapi-idmap.h.

37.61.2.4 OS_OBJECT_TYPE_OS_DIR

```
#define OS_OBJECT_TYPE_OS_DIR 0x07
```

Object directory type.

Definition at line 45 of file osapi-idmap.h.

37.61.2.5 OS_OBJECT_TYPE_OS_FILESYS

```
#define OS_OBJECT_TYPE_OS_FILESYS 0x0B
```

Object file system type.

Definition at line 49 of file osapi-idmap.h.

37.61.2.6 OS_OBJECT_TYPE_OS_MODULE

```
#define OS_OBJECT_TYPE_OS_MODULE 0x0A
```

Object module type.

Definition at line 48 of file osapi-idmap.h.

37.61.2.7 OS_OBJECT_TYPE_OS_MUTEX

```
#define OS_OBJECT_TYPE_OS_MUTEX 0x05
```

Object mutex type.

Definition at line 43 of file osapi-idmap.h.

37.61.2.8 OS_OBJECT_TYPE_OS_QUEUE

```
#define OS_OBJECT_TYPE_OS_QUEUE 0x02
```

Object queue type.

Definition at line 40 of file osapi-idmap.h.

37.61.2.9 OS_OBJECT_TYPE_OS_STREAM

```
#define OS_OBJECT_TYPE_OS_STREAM 0x06
```

Object stream type.

Definition at line 44 of file osapi-idmap.h.

37.61.2.10 OS_OBJECT_TYPE_OS_TASK

```
#define OS_OBJECT_TYPE_OS_TASK 0x01
```

Object task type.

Definition at line 39 of file osapi-idmap.h.

37.61.2.11 OS_OBJECT_TYPE_OS_TIMEBASE

```
#define OS_OBJECT_TYPE_OS_TIMEBASE 0x08
```

Object timebase type.

Definition at line 46 of file osapi-idmap.h.

37.61.2.12 OS_OBJECT_TYPE_OS_TIMECB

```
#define OS_OBJECT_TYPE_OS_TIMECB 0x09
```

Object timer callback type.

Definition at line 47 of file osapi-idmap.h.

37.61.2.13 OS_OBJECT_TYPE_UNDEFINED

```
#define OS_OBJECT_TYPE_UNDEFINED 0x00
```

Object type undefined.

Definition at line 38 of file osapi-idmap.h.

37.61.2.14 OS_OBJECT_TYPE_USER

```
#define OS_OBJECT_TYPE_USER 0x10
```

Object user type.

Definition at line 51 of file osapi-idmap.h.

37.62 OSAL Object ID Utility APIs

Functions

- static unsigned long OS_ObjectIdToInteger (osal_id_t object_id)

  Obtain an integer value corresponding to an object ID.
- static osal_id_t OS_ObjectIdFromInteger (unsigned long value)

  Obtain an osal ID corresponding to an integer value.
- static bool OS_ObjectIdEqual (osal_id_t object_id1, osal_id_t object_id2)

  Check two OSAL object ID values for equality.
- static bool OS_ObjectIdDefined (osal_id_t object_id)

  Check if an object ID is defined.
- int32 OS_GetResourceName (osal_id_t object_id, char ∗buffer, size_t buffer_size)

  Obtain the name of an object given an arbitrary object ID.
- osal_objtype_t OS_IdentifyObject (osal_id_t object_id)

  Obtain the type of an object given an arbitrary object ID.
- int32 OS_ConvertToArrayIndex (osal_id_t object_id, osal_index_t ∗ArrayIndex)

  Converts an abstract ID into a number suitable for use as an array index.
- int32 OS_ObjectIdToArrayIndex (osal_objtype_t idtype, osal_id_t object_id, osal_index_t ∗ArrayIndex)

  Converts an abstract ID into a number suitable for use as an array index.
- void OS_ForEachObject (osal_id_t creator_id, OS_ArgCallback_t callback_ptr, void ∗callback_arg)

  call the supplied callback function for all valid object IDs
- void OS_ForEachObjectOfType (osal_objtype_t objtype, osal_id_t creator_id, OS_ArgCallback_t callback_ptr, void ∗callback_arg)

  call the supplied callback function for valid object IDs of a specific type

37.62.1 Detailed Description

37.62.2 Function Documentation

37.62.2.1 OS_ConvertToArrayIndex()

```
int32 OS_ConvertToArrayIndex (
            osal_id_t object_id,
            osal_index_t * ArrayIndex )
```

Converts an abstract ID into a number suitable for use as an array index.

This will return a unique zero-based integer number in the range of [0,MAX) for any valid object ID. This may be used by application code as an array index for indexing into local tables.

Note

This does NOT verify the validity of the ID, that is left to the caller. This is only the conversion logic.

This routine accepts any object type, and returns a value based on the maximum number of objects for that type. This is equivalent to invoking OS_ObjectIdToArrayIndex() with the idtype set to OS_OBJECT_TYPE_UNDEFINED.

See also

OS_ObjectIdToArrayIndex

Parameters

in	*object_id*	The object ID to operate on
out	**ArrayIndex*	The Index to return

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INCORRECT_OBJ_TYPE	Incorrect object type.

Referenced by OS_ObjectIdDefined().

37.62.2.2 OS_ForEachObject()

```
void OS_ForEachObject (
            osal_id_t creator_id,
            OS_ArgCallback_t callback_ptr,
            void * callback_arg )
```

call the supplied callback function for all valid object IDs

Loops through all defined OSAL objects of all types and calls callback_ptr on each one If creator_id is nonzero then only objects with matching creator id are processed.

Parameters

in	*creator_id*	Filter objects to those created by a specific task This may be passed as OS_OBJECT_CREATOR_ANY to return all objects
in	*callback_ptr*	Function to invoke for each matching object ID
in	*callback_arg*	Opaque Argument to pass to callback function

Referenced by OS_ObjectIdDefined().

37.62.2.3 OS_ForEachObjectOfType()

```
void OS_ForEachObjectOfType (
            osal_objtype_t objtype,
            osal_id_t creator_id,
```

```
            OS_ArgCallback_t callback_ptr,
            void * callback_arg )
```

call the supplied callback function for valid object IDs of a specific type

Loops through all defined OSAL objects of a specific type and calls callback_ptr on each one If creator_id is nonzero then only objects with matching creator id are processed.

Parameters

in	*objtype*	The type of objects to iterate
in	*creator_id*	Filter objects to those created by a specific task This may be passed as OS_OBJECT_CREATOR_ANY to return all objects
in	*callback_ptr*	Function to invoke for each matching object ID
in	*callback_arg*	Opaque Argument to pass to callback function

Referenced by OS_ObjectIdDefined().

37.62.2.4 OS_GetResourceName()

```
int32 OS_GetResourceName (
            osal_id_t object_id,
            char * buffer,
            size_t buffer_size )
```

Obtain the name of an object given an arbitrary object ID.

All OSAL resources generally have a name associated with them. This allows application code to retrieve the name of any valid OSAL object ID.

Parameters

in	*object_id*	The object ID to operate on
out	*buffer*	Buffer in which to store the name
in	*buffer_size*	Size of the output storage buffer

Returns

OS_SUCCESS if successful OS_ERR_INVALID_ID if the passed-in ID is not a valid OSAL ID OS_INVALID↩_POINTER if the passed-in buffer is invalid OS_ERR_NAME_TOO_LONG if the name will not fit in the buffer provided

Referenced by CFE_PSP_OS_EventHandler(), and OS_ObjectIdDefined().

37.62.2.5 OS_IdentifyObject()

```
osal_objtype_t OS_IdentifyObject (
            osal_id_t object_id )
```

Obtain the type of an object given an arbitrary object ID.

Given an arbitrary object ID, get the type of the object

Parameters

in	*object↩ _id*	The object ID to operate on

Returns

The object type portion of the object_id, see OSAL Object Type Defines for expected values

Referenced by OS_ObjectIdDefined().

37.62.2.6 OS_ObjectIdDefined()

```
static bool OS_ObjectIdDefined (
            osal_id_t object_id )  [inline], [static]
```

Check if an object ID is defined.

The OSAL ID values should be treated as abstract values by applications, and not directly manipulated using standard C operators.

This returns false if the ID is NOT a defined resource (i.e. free/empty/invalid).

Note

OS_ObjectIdDefined(OS_OBJECT_ID_UNDEFINED) is always guaranteed to be false.

Parameters

in	*object↩ _id*	The first object ID

Definition at line 139 of file osapi-idmap.h.

References OS_ConvertToArrayIndex(), OS_ForEachObject(), OS_ForEachObjectOfType(), OS_GetResourceName(), OS_IdentifyObject(), and OS_ObjectIdToArrayIndex().

Referenced by OS_Application_Startup().

37.62.2.7 OS_ObjectIdEqual()

```
static bool OS_ObjectIdEqual (
            osal_id_t object_id1,
            osal_id_t object_id2 )  [inline], [static]
```

Check two OSAL object ID values for equality.

The OSAL ID values should be treated as abstract values by applications, and not directly manipulated using standard C operators.

This checks two values for equality, replacing the "==" operator.

Parameters

in	*object_id1*	The first object ID
in	*object_id2*	The second object ID

Returns

true if the object IDs are equal

Definition at line 118 of file osapi-idmap.h.

37.62.2.8 OS_ObjectIdFromInteger()

```
static osal_id_t OS_ObjectIdFromInteger (
            unsigned long value )  [inline], [static]
```

Obtain an osal ID corresponding to an integer value.

Provides the inverse of OS_ObjectIdToInteger(). Reconstitutes the original osal_id_t type from an integer representation.

Parameters

in	*value*	The integer representation of an OSAL ID

Returns

The ID value converted to an osal_id_t

Definition at line 97 of file osapi-idmap.h.

37.62.2.9 OS_ObjectIdToArrayIndex()

```
int32 OS_ObjectIdToArrayIndex (
            osal_objtype_t idtype,
            osal_id_t object_id,
            osal_index_t * ArrayIndex )
```

Converts an abstract ID into a number suitable for use as an array index.

This will return a unique zero-based integer number in the range of [0,MAX) for any valid object ID. This may be used by application code as an array index for indexing into local tables.

This routine operates on a specific object type, and returns a value based on the maximum number of objects for that type.

If the idtype is passed as OS_OBJECT_TYPE_UNDEFINED, then object type verification is skipped and any object ID will be accepted and converted to an index. In this mode, the range of the output depends on the actual passed-in object type.

If the idtype is passed as any other value, the passed-in ID value is first confirmed to be the correct type. This check will guarantee that the output is within an expected range; for instance, if the type is passed as OS_OBJECT_TYPE_OS↩_TASK, then the output index is guaranteed to be between 0 and OS_MAX_TASKS-1 after successful conversion.

Parameters

in	*idtype*	The object type to convert
in	*object_id*	The object ID to operate on
out	**ArrayIndex*	The Index to return

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INCORRECT_OBJ_TYPE	Incorrect object type.

Referenced by OS_ObjectIdDefined().

37.62.2.10 OS_ObjectIdToInteger()

```
static unsigned long OS_ObjectIdToInteger (
            osal_id_t object_id )  [inline], [static]
```

Obtain an integer value corresponding to an object ID.

Obtains an integer representation of an object id, generally for the purpose of printing to the console or system logs.

The returned value is of the type "unsigned long" for direct use with printf-style functions. It is recommended to use the "%lx" conversion specifier as the hexidecimal encoding clearly delineates the internal fields.

Note

This provides the raw integer value and is *not* suitable for use as an array index, as the result is not zero-based. See the OS_ConvertToArrayIndex() to obtain a zero-based index value.

Parameters

in	*object↩ _id*	The object ID

Returns

integer value representation of object ID

Definition at line 79 of file osapi-idmap.h.

37.63 OSAL Dynamic Loader and Symbol APIs

Functions

- int32 OS_SymbolLookup (cpuaddr *symbol_address, const char *symbol_name)

  Find the Address of a Symbol.
- int32 OS_ModuleSymbolLookup (osal_id_t module_id, cpuaddr *symbol_address, const char *symbol_name)

  Find the Address of a Symbol within a module.
- int32 OS_SymbolTableDump (const char *filename, size_t size_limit)

  Dumps the system symbol table to a file.
- int32 OS_ModuleLoad (osal_id_t *module_id, const char *module_name, const char *filename, uint32 flags)

  Loads an object file.
- int32 OS_ModuleUnload (osal_id_t module_id)

  Unloads the module file.
- int32 OS_ModuleInfo (osal_id_t module_id, OS_module_prop_t *module_info)

  Obtain information about a module.

37.63.1 Detailed Description

37.63.2 Function Documentation

37.63.2.1 OS_ModuleInfo()

```
int32 OS_ModuleInfo (
            osal_id_t module_id,
            OS_module_prop_t * module_info )
```

Obtain information about a module.

Returns information about the loadable module

Parameters

in	*module_id*	OSAL ID of the previously the loaded module
out	*module_info*	Buffer to store module information

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the module id invalid
OS_INVALID_POINTER	if the pointer to the ModuleInfo structure is invalid

37.63.2.2 OS_ModuleLoad()

```
int32 OS_ModuleLoad (
            osal_id_t * module_id,
            const char * module_name,
            const char * filename,
            uint32 flags )
```

Loads an object file.

Loads an object file into the running operating system

The "flags" parameter may influence how the loaded module symbols are made available for use in the application. See OS_MODULE_FLAG_LOCAL_SYMBOLS and OS_MODULE_FLAG_GLOBAL_SYMBOLS for descriptions.

Parameters

out	*module_id*	Non-zero OSAL ID corresponding to the loaded module
in	*module_name*	Name of module
in	*filename*	File containing the object code to load
in	*flags*	Options for the loaded module

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the module cannot be loaded
OS_INVALID_POINTER	if one of the parameters is NULL
OS_ERR_NO_FREE_IDS	if the module table is full
OS_ERR_NAME_TAKEN	if the name is in use

37.63.2.3 OS_ModuleSymbolLookup()

```
int32 OS_ModuleSymbolLookup (
            osal_id_t module_id,
            cpuaddr * symbol_address,
            const char * symbol_name )
```

Find the Address of a Symbol within a module.

This is similar to OS_SymbolLookup() but for a specific module ID. This should be used to look up a symbol in a module that has been loaded with the OS_MODULE_FLAG_LOCAL_SYMBOLS flag.

Parameters

in	*module_id*	Module ID that should contain the symbol
out	*symbol_address*	Set to the address of the symbol
in	*symbol_name*	Name of the symbol to look up

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the symbol could not be found
OS_INVALID_POINTER	if one of the pointers passed in are NULL

37.63.2.4 OS_ModuleUnload()

```
int32 OS_ModuleUnload (
            osal_id_t module_id )
```

Unloads the module file.

Unloads the module file from the running operating system

Parameters

in	*module↩ _id*	OSAL ID of the previously the loaded module

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the module is invalid or cannot be unloaded

37.63.2.5 OS_SymbolLookup()

```
int32 OS_SymbolLookup (
```

```
            cpuaddr * symbol_address,
            const char * symbol_name )
```

Find the Address of a Symbol.

This calls to the OS dynamic symbol lookup implementation, and/or checks a static symbol table for a matching symbol name.

The static table is intended to support embedded targets that do not have module loading capability or have it disabled.

Parameters

out	*symbol_address*	Set to the address of the symbol
in	*symbol_name*	Name of the symbol to look up

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the symbol could not be found
OS_INVALID_POINTER	if one of the pointers passed in are NULL

37.63.2.6 OS_SymbolTableDump()

```
int32 OS_SymbolTableDump (
            const char * filename,
            size_t size_limit )
```

Dumps the system symbol table to a file.

Dumps the system symbol table to the specified filename

Parameters

in	*filename*	File to write to
in	*size_limit*	Maximum number of bytes to write

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_NOT_IMPLEMENTED	Not implemented.
OS_ERROR	if the symbol table could not be read or dumped

37.64 OSAL Mutex APIs

Functions

- int32 OS_MutSemCreate (osal_id_t *sem_id, const char *sem_name, uint32 options)

  Creates a mutex semaphore.
- int32 OS_MutSemGive (osal_id_t sem_id)

  Releases the mutex object referenced by sem_id.
- int32 OS_MutSemTake (osal_id_t sem_id)

  Acquire the mutex object referenced by sem_id.
- int32 OS_MutSemDelete (osal_id_t sem_id)

  Deletes the specified Mutex Semaphore.
- int32 OS_MutSemGetIdByName (osal_id_t *sem_id, const char *sem_name)

  Find an existing mutex ID by name.
- int32 OS_MutSemGetInfo (osal_id_t sem_id, OS_mut_sem_prop_t *mut_prop)

  Fill a property object buffer with details regarding the resource.

37.64.1 Detailed Description

37.64.2 Function Documentation

37.64.2.1 OS_MutSemCreate()

```
int32 OS_MutSemCreate (
            osal_id_t * sem_id,
            const char * sem_name,
            uint32 options )
```

Creates a mutex semaphore.

Mutex semaphores are always created in the unlocked (full) state.

Parameters

out	*sem_id*	will be set to the non-zero ID of the newly-created resource
in	*sem_name*	the name of the new resource to create
in	*options*	reserved for future use. Should be passed as 0.

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if sem_id or sem_name are NULL
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NO_FREE_IDS	if there are no more free mutex Ids
OS_ERR_NAME_TAKEN	if there is already a mutex with the same name
OS_SEM_FAILURE	if the OS call failed

37.64.2.2 OS_MutSemDelete()

```
int32 OS_MutSemDelete (
            osal_id_t sem_id )
```

Deletes the specified Mutex Semaphore.

Delete the semaphore. This also frees the respective sem_id such that it can be used again when another is created.

Parameters

in	*sem↩ _id*	The object ID to delete

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid mutex
OS_SEM_FAILURE	if the OS call failed

37.64.2.3 OS_MutSemGetIdByName()

```
int32 OS_MutSemGetIdByName (
            osal_id_t * sem_id,
            const char * sem_name )
```

Find an existing mutex ID by name.

This function tries to find a mutex sem Id given the name of a mut_sem. The id is returned through sem_id

Parameters

out	*sem_id*	will be set to the ID of the existing resource
in	*sem_name*	the name of the existing resource to find

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	is semid or sem_name are NULL pointers
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_NOT_FOUND	if the name was not found in the table

37.64.2.4 OS_MutSemGetInfo()

```
int32 OS_MutSemGetInfo (
            osal_id_t sem_id,
            OS_mut_sem_prop_t * mut_prop )
```

Fill a property object buffer with details regarding the resource.

This function will pass back a pointer to structure that contains all of the relevant info(name and creator) about the specified mutex semaphore.

Parameters

in	*sem_id*	The object ID to operate on
out	*mut_prop*	The property object buffer to fill

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid semaphore
OS_INVALID_POINTER	if the mut_prop pointer is null

37.64.2.5 OS_MutSemGive()

```
int32 OS_MutSemGive (
            osal_id_t sem_id )
```

Releases the mutex object referenced by sem_id.

If there are threads blocked on the mutex object referenced by mutex when this function is called, resulting in the mutex becoming available, the scheduling policy shall determine which thread shall acquire the mutex.

Parameters

in	*sem↩ _id*	The object ID to operate on

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid mutex
OS_SEM_FAILURE	if an unspecified error occurs

37.64.2.6 OS_MutSemTake()

```
int32 OS_MutSemTake (
            osal_id_t sem_id )
```

Acquire the mutex object referenced by sem_id.

If the mutex is already locked, the calling thread shall block until the mutex becomes available. This operation shall return with the mutex object referenced by mutex in the locked state with the calling thread as its owner.

Parameters

in	*sem↩ _id*	The object ID to operate on

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_SEM_FAILURE	if the semaphore was not previously initialized or is not in the array of semaphores defined by the system
OS_ERR_INVALID_ID	the id passed in is not a valid mutex

37.65 Network ID APIs

Functions

- int32 OS_NetworkGetID (void)

  Gets the network ID of the local machine.
- int32 OS_NetworkGetHostName (char ∗host_name, size_t name_len)

  Gets the local machine network host name.

37.65.1 Detailed Description

Provides some basic methods to query a network host name and ID

37.65.2 Function Documentation

37.65.2.1 OS_NetworkGetHostName()

```
int32 OS_NetworkGetHostName (
            char * host_name,
            size_t name_len )
```

Gets the local machine network host name.

If configured in the underlying network stack, this function retrieves the local hostname of the system.

Parameters

out	*host_name*	Buffer to hold name information
in	*name_len*	Maximum length of host name buffer

Returns

Execution status, see OSAL Return Code Defines

37.65.2.2 OS_NetworkGetID()

```
int32 OS_NetworkGetID (
            void  )
```

Gets the network ID of the local machine.

The ID is an implementation-defined value and may not be consistent in meaning across different platform types.

Note

This API may be removed in a future version of OSAL due to inconsistencies between platforms.

Returns

The ID or fixed value of -1 if the host id could not be found. Note it is not possible to differentiate between error codes and valid network IDs here. It is assumed, however, that -1 is never a valid ID.

37.66 OSAL Printf APIs

Functions

- void OS_printf (const char *string,...) OS_PRINTF(1

  Abstraction for the system printf() call.
- void void OS_printf_disable (void)

  This function disables the output from OS_printf.
- void OS_printf_enable (void)

  This function enables the output from OS_printf.

37.66.1 Detailed Description

37.66.2 Function Documentation

37.66.2.1 OS_printf()

```
void OS_printf (
            const char * string,
             ...  )
```

Abstraction for the system printf() call.

This function abstracts out the printf type statements. This is useful for using OS- specific thats that will allow non-polled print statements for the real time systems.

Operates in a manner similar to the printf() call defined by the standard C library and takes all the parameters and formatting options of printf. This abstraction may implement additional buffering, if necessary, to improve the real-time performance of the call.

Strings (including terminator) longer than OS_BUFFER_SIZE will be truncated.

The output of this routine also may be dynamically enabled or disabled by the OS_printf_enable() and OS_printf_disable() calls, respectively.

Parameters

in	*string*	Format string, followed by additional arguments

Referenced by CFE_PSP_AttachExceptions(), CFE_PSP_DeleteCDS(), CFE_PSP_DeleteResetArea(), CFE_PSP_DeleteUserReservedArea(), CFE_PSP_InitCDS(), CFE_PSP_InitProcessorReservedMemory(), CFE_PSP_InitResetArea(), CFE_PSP_InitUserReservedArea(), CFE_PSP_Panic(), CFE_PSP_Restart(), CFE_PSP_SetupLocal1Hz(), OS_Application_Run(), and OS_Application_Startup().

37.66.2.2 OS_printf_disable()

```
void void OS_printf_disable (
            void  )
```

This function disables the output from OS_printf.

37.66.2.3 OS_printf_enable()

```
void OS_printf_enable (
            void  )
```

This function enables the output from OS_printf.

37.67 OSAL Message Queue APIs

Functions

- int32 OS_QueueCreate (osal_id_t *queue_id, const char *queue_name, osal_blockcount_t queue_depth, size_t data_size, uint32 flags)

  Create a message queue.
- int32 OS_QueueDelete (osal_id_t queue_id)

  Deletes the specified message queue.
- int32 OS_QueueGet (osal_id_t queue_id, void *data, size_t size, size_t *size_copied, int32 timeout)

  Receive a message on a message queue.
- int32 OS_QueuePut (osal_id_t queue_id, const void *data, size_t size, uint32 flags)

  Put a message on a message queue.
- int32 OS_QueueGetIdByName (osal_id_t *queue_id, const char *queue_name)

  Find an existing queue ID by name.
- int32 OS_QueueGetInfo (osal_id_t queue_id, OS_queue_prop_t *queue_prop)

  Fill a property object buffer with details regarding the resource.

37.67.1 Detailed Description

37.67.2 Function Documentation

37.67.2.1 OS_QueueCreate()

```
int32 OS_QueueCreate (
            osal_id_t * queue_id,
            const char * queue_name,
            osal_blockcount_t queue_depth,
            size_t data_size,
            uint32 flags )
```

Create a message queue.

This is the function used to create a queue in the operating system. Depending on the underlying operating system, the memory for the queue will be allocated automatically or allocated by the code that sets up the queue. Queue names must be unique; if the name already exists this function fails. Names cannot be NULL.

Parameters

out	*queue_id*	will be set to the non-zero ID of the newly-created resource
in	*queue_name*	the name of the new resource to create
in	*queue_depth*	the maximum depth of the queue
in	*data_size*	the size of each entry in the queue
in	*flags*	options for the queue (reserved for future use, pass as 0)

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if a pointer passed in is NULL
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NO_FREE_IDS	if there are already the max queues created
OS_ERR_NAME_TAKEN	if the name is already being used on another queue
OS_QUEUE_INVALID_SIZE	if the queue depth exceeds the limit
OS_ERROR	if the OS create call fails

37.67.2.2 OS_QueueDelete()

```
int32 OS_QueueDelete (
            osal_id_t queue_id )
```

Deletes the specified message queue.

This is the function used to delete a queue in the operating system. This also frees the respective queue_id to be used again when another queue is created.

Note

If There are messages on the queue, they will be lost and any subsequent calls to QueueGet or QueuePut to this queue will result in errors

Parameters

in	*queue↩ _id*	The object ID to delete

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in does not exist
OS_ERROR	if the OS call to delete the queue fails

37.67.2.3 OS_QueueGet()

```
int32 OS_QueueGet (
            osal_id_t queue_id,
            void * data,
            size_t size,
            size_t * size_copied,
            int32 timeout )
```

Receive a message on a message queue.

If a message is pending, it is returned immediately. Otherwise the calling task will block until a message arrives or the timeout expires.

Parameters

in	*queue_id*	The object ID to operate on
out	*data*	The buffer to store the received message
in	*size*	The size of the data buffer
out	*size_copied*	Set to the actual size of the message
in	*timeout*	The maximum amount of time to block, or OS_PEND to wait forever

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the given ID does not exist
OS_INVALID_POINTER	if a pointer passed in is NULL
OS_QUEUE_EMPTY	if the Queue has no messages on it to be recieved
OS_QUEUE_TIMEOUT	if the timeout was OS_PEND and the time expired
OS_QUEUE_INVALID_SIZE	if the size copied from the queue was not correct

37.67.2.4 OS_QueueGetIdByName()

```
int32 OS_QueueGetIdByName (
            osal_id_t * queue_id,
            const char * queue_name )
```

Find an existing queue ID by name.

This function tries to find a queue Id given the name of the queue. The id of the queue is passed back in queue_id.

Parameters

out	*queue_id*	will be set to the ID of the existing resource
in	*queue_name*	the name of the existing resource to find

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if the name or id pointers are NULL
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_NOT_FOUND	the name was not found in the table

37.67.2.5 OS_QueueGetInfo()

```
int32 OS_QueueGetInfo (
            osal_id_t queue_id,
            OS_queue_prop_t * queue_prop )
```

Fill a property object buffer with details regarding the resource.

This function will pass back a pointer to structure that contains all of the relevant info (name and creator) about the specified queue.

Parameters

in	*queue_id*	The object ID to operate on
out	*queue_prop*	The property object buffer to fill

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if queue_prop is NULL
OS_ERR_INVALID_ID	if the ID given is not a valid queue

37.67.2.6 OS_QueuePut()

```
int32 OS_QueuePut (
            osal_id_t queue_id,
            const void * data,
            size_t size,
            uint32 flags )
```

Put a message on a message queue.

Parameters

in	*queue↩ _id*	The object ID to operate on
in	*data*	The buffer containing the message to put
in	*size*	The size of the data buffer
in	*flags*	Currently reserved/unused, should be passed as 0

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the queue id passed in is not a valid queue
OS_INVALID_POINTER	if the data pointer is NULL
OS_QUEUE_FULL	if the queue cannot accept another message
OS_ERROR	if the OS call returns an error

37.68 OSAL Select APIs

Functions

- int32 OS_SelectMultiple (OS_FdSet ∗ReadSet, OS_FdSet ∗WriteSet, int32 msecs)

  Wait for events across multiple file handles.
- int32 OS_SelectSingle (osal_id_t objid, uint32 ∗StateFlags, int32 msecs)

  Wait for events on a single file handle.
- int32 OS_SelectFdZero (OS_FdSet ∗Set)

  Clear a FdSet structure.
- int32 OS_SelectFdAdd (OS_FdSet ∗Set, osal_id_t objid)

  Add an ID to an FdSet structure.
- int32 OS_SelectFdClear (OS_FdSet ∗Set, osal_id_t objid)

  Clear an ID from an FdSet structure.
- bool OS_SelectFdIsSet (OS_FdSet ∗Set, osal_id_t objid)

  Check if an FdSet structure contains a given ID.

37.68.1 Detailed Description

37.68.2 Function Documentation

37.68.2.1 OS_SelectFdAdd()

```
int32 OS_SelectFdAdd (
            OS_FdSet * Set,
            osal_id_t objid )
```

Add an ID to an FdSet structure.

After this call the set will contain the given OSAL ID

Returns

Execution status, see OSAL Return Code Defines

37.68.2.2 OS_SelectFdClear()

```
int32 OS_SelectFdClear (
            OS_FdSet * Set,
            osal_id_t objid )
```

Clear an ID from an FdSet structure.

After this call the set will no longer contain the given OSAL ID

Returns

Execution status, see OSAL Return Code Defines

37.68.2.3 OS_SelectFdIsSet()

```
bool OS_SelectFdIsSet (
            OS_FdSet * Set,
            osal_id_t objid )
```

Check if an FdSet structure contains a given ID.

Returns

Boolean set status

Return values

true	FdSet structure contains ID
false	FDSet structure does not contain ID

37.68.2.4 OS_SelectFdZero()

```
int32 OS_SelectFdZero (
            OS_FdSet * Set )
```

Clear a FdSet structure.

After this call the set will contain no OSAL IDs

Returns

Execution status, see OSAL Return Code Defines

37.68.2.5 OS_SelectMultiple()

```
int32 OS_SelectMultiple (
            OS_FdSet * ReadSet,
            OS_FdSet * WriteSet,
            int32 msecs )
```

Wait for events across multiple file handles.

Wait for any of the given sets of IDs to be become readable or writable

This function will block until any of the following occurs:

- At least one OSAL ID in the ReadSet is readable
- At least one OSAL ID in the WriteSet is writable
- The timeout has elapsed

The sets are input/output parameters. On entry, these indicate the file handle(s) to wait for. On exit, these are set to the actual file handle(s) that have activity.

If the timeout occurs this returns an error code and all output sets should be empty.

Note

This does not lock or otherwise protect the file handles in the given sets. If a filehandle supplied via one of the FdSet arguments is closed or modified by another while this function is in progress, the results are undefined. Because of this limitation, it is recommended to use OS_SelectSingle() whenever possible.

Returns

Execution status, see OSAL Return Code Defines

37.68.2.6 OS_SelectSingle()

```
int32 OS_SelectSingle (
            osal_id_t objid,
            uint32 * StateFlags,
            int32 msecs )
```

Wait for events on a single file handle.

Wait for a single OSAL filehandle to change state

This function can be used to wait for a single OSAL stream ID to become readable or writable. On entry, the "StateFlags" parameter should be set to the desired state (OS_STREAM_STATE_READABLE and/or OS_STREAM_STATE_WR↩ ITABLE) and upon return the flags will be set to the state actually detected.

As this operates on a single ID, the filehandle is protected during this call, such that another thread accessing the same handle will return an error. However, it is important to note that once the call returns then other threads may then also read/write and affect the state before the current thread can service it.

To mitigate this risk the application may prefer to use the OS_TimedRead/OS_TimedWrite calls.

Returns

Execution status, see OSAL Return Code Defines

37.69 OSAL Shell APIs

Functions

- int32 OS_ShellOutputToFile (const char *Cmd, osal_id_t filedes)

  Executes the command and sends output to a file.

37.69.1 Detailed Description

37.69.2 Function Documentation

37.69.2.1 OS_ShellOutputToFile()

```
int32 OS_ShellOutputToFile (
            const char * Cmd,
            osal_id_t filedes )
```

Executes the command and sends output to a file.

Takes a shell command in and writes the output of that command to the specified file The output file must be opened previously with write access (OS_WRITE_ONLY or OS_READ_WRITE).

Parameters

in	*Cmd*	Command to pass to shell
in	*filedes*	File to send output to.

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the command was not executed properly
OS_ERR_INVALID_ID	if the file descriptor passed in is invalid

37.70 OSAL Socket Address APIs

Functions

- int32 OS_SocketAddrInit (OS_SockAddr_t *Addr, OS_SocketDomain_t Domain)

  Initialize a socket address structure to hold an address of the given family.
- int32 OS_SocketAddrToString (char *buffer, size_t buflen, const OS_SockAddr_t *Addr)

  Get a string representation of a network host address.
- int32 OS_SocketAddrFromString (OS_SockAddr_t *Addr, const char *string)

  Set a network host address from a string representation.
- int32 OS_SocketAddrGetPort (uint16 *PortNum, const OS_SockAddr_t *Addr)

  Get the port number of a network address.
- int32 OS_SocketAddrSetPort (OS_SockAddr_t *Addr, uint16 PortNum)

  Set the port number of a network address.

37.70.1 Detailed Description

These functions provide a means to manipulate network addresses in a manner that is (mostly) agnostic to the actual network address type.

Every network address should be representable as a string (i.e. dotted decimal IP, etc). This can serve as a the "common denominator" to all address types.

37.70.2 Function Documentation

37.70.2.1 OS_SocketAddrFromString()

```
int32 OS_SocketAddrFromString (
            OS_SockAddr_t * Addr,
            const char * string )
```

Set a network host address from a string representation.

The specific format of the output string depends on the address family.

The address structure should have been previously initialized using OS_SocketAddrInit() to set the address family type.

Note

For IPv4, this would typically be the dotted-decimal format (X.X.X.X). It is up to the discretion of the underlying implementation whether to accept hostnames, as this depends on the availability of DNS services. Since many embedded deployments do not have name services, this should not be relied upon.

Parameters

out	*Addr*	The address buffer to initialize
in	*string*	The string to initialize the address from.

Returns

Execution status, see OSAL Return Code Defines

37.70.2.2 OS_SocketAddrGetPort()

```
int32 OS_SocketAddrGetPort (
            uint16 * PortNum,
            const OS_SockAddr_t * Addr )
```

Get the port number of a network address.

For network prototcols that have the concept of a port number (such as TCP/IP and UDP/IP) this function gets the port number from the address structure.

Parameters

out	*PortNum*	Buffer to store the port number
in	*Addr*	The network address buffer

Returns

Execution status, see OSAL Return Code Defines

37.70.2.3 OS_SocketAddrInit()

```
int32 OS_SocketAddrInit (
            OS_SockAddr_t * Addr,
            OS_SocketDomain_t Domain )
```

Initialize a socket address structure to hold an address of the given family.

The address is set to a suitable default value for the family.

Parameters

out	*Addr*	The address buffer to initialize
in	*Domain*	The address family

Returns

Execution status, see OSAL Return Code Defines

37.70.2.4 OS_SocketAddrSetPort()

```
int32 OS_SocketAddrSetPort (
            OS_SockAddr_t * Addr,
            uint16 PortNum )
```

Set the port number of a network address.

For network prototcols that have the concept of a port number (such as TCP/IP and UDP/IP) this function sets the port number from the address structure.

Parameters

in	*PortNum*	The port number to set
out	*Addr*	The network address buffer

Returns

Execution status, see OSAL Return Code Defines

37.70.2.5 OS_SocketAddrToString()

```
int32 OS_SocketAddrToString (
            char * buffer,
            size_t buflen,
            const OS_SockAddr_t * Addr )
```

Get a string representation of a network host address.

The specific format of the output string depends on the address family.

This string should be suitable to pass back into OS_SocketAddrFromString() which should recreate the same network address, and it should also be meaningful to a user of printed or logged as a C string.

Note

For IPv4, this would typically be the dotted-decimal format (X.X.X.X).

Parameters

out	*buffer*	Buffer to hold the output string
in	*buflen*	Maximum length of the output string
in	*Addr*	The network address buffer to convert

Returns

Execution status, see OSAL Return Code Defines

37.71 OSAL Socket Management APIs

Functions

- int32 OS_SocketOpen (osal_id_t *sock_id, OS_SocketDomain_t Domain, OS_SocketType_t Type)

  Opens a socket.
- int32 OS_SocketBind (osal_id_t sock_id, const OS_SockAddr_t *Addr)

  Binds a socket to a given local address.
- int32 OS_SocketConnect (osal_id_t sock_id, const OS_SockAddr_t *Addr, int32 timeout)

  Connects a socket to a given remote address.
- int32 OS_SocketAccept (osal_id_t sock_id, osal_id_t *connsock_id, OS_SockAddr_t *Addr, int32 timeout)

  Waits for and accept the next incoming connection on the given socket.
- int32 OS_SocketRecvFrom (osal_id_t sock_id, void *buffer, size_t buflen, OS_SockAddr_t *RemoteAddr, int32 timeout)

  Reads data from a message-oriented (datagram) socket.
- int32 OS_SocketSendTo (osal_id_t sock_id, const void *buffer, size_t buflen, const OS_SockAddr_t *Remote↩Addr)

  Sends data to a message-oriented (datagram) socket.
- int32 OS_SocketGetIdByName (osal_id_t *sock_id, const char *sock_name)

  Gets an OSAL ID from a given name.
- int32 OS_SocketGetInfo (osal_id_t sock_id, OS_socket_prop_t *sock_prop)

  Gets information about an OSAL Socket ID.

37.71.1 Detailed Description

These functions are loosely related to the BSD Sockets API but made to be more consistent with other OSAL API functions. That is, they operate on OSAL IDs (32-bit opaque number values) and return an OSAL error code.

OSAL Socket IDs are very closely related to File IDs and share the same ID number space. Additionally, the file OS_↩read() / OS_write() / OS_close() calls also work on sockets.

Note that all of functions may return OS_ERR_NOT_IMPLEMENTED if network support is not configured at compile time.

37.71.2 Function Documentation

37.71.2.1 OS_SocketAccept()

```
int32 OS_SocketAccept (
            osal_id_t sock_id,
            osal_id_t * connsock_id,
            OS_SockAddr_t * Addr,
            int32 timeout )
```

Waits for and accept the next incoming connection on the given socket.

This is used for sockets operating in a "server" role. The socket must be a stream type (connection-oriented) and previously bound to a local address using OS_SocketBind(). This will block the caller up to the given timeout or until an incoming connection request occurs, whichever happens first.

The new stream connection is then returned to the caller and the original server socket ID can be reused for the next connection.

Parameters

in	*sock_id*	The server socket ID, previously bound using OS_SocketBind()
out	*connsock↩ _id*	The connection socket, a new ID that can be read/written
in	*Addr*	The remote address of the incoming connection
in	*timeout*	The maximum amount of time to wait, or OS_PEND to wait forever

Returns

Execution status, see OSAL Return Code Defines

37.71.2.2 OS_SocketBind()

```
int32 OS_SocketBind (
            osal_id_t sock_id,
            const OS_SockAddr_t * Addr )
```

Binds a socket to a given local address.

The specified socket will be bound to the local address and port, if available.

If the socket is connectionless, then it only binds to the local address.

If the socket is connection-oriented (stream), then this will also put the socket into a listening state for incoming connections at the local address.

Parameters

in	*sock↩ _id*	The socket ID
in	*Addr*	The local address to bind to

Returns

Execution status, see OSAL Return Code Defines

37.71.2.3 OS_SocketConnect()

```
int32 OS_SocketConnect (
            osal_id_t sock_id,
            const OS_SockAddr_t * Addr,
            int32 timeout )
```

Connects a socket to a given remote address.

The socket will be connected to the remote address and port, if available. This only applies to stream-oriented sockets. Calling this on a datagram socket will return an error (these sockets should use SendTo/RecvFrom).

Parameters

in	*sock↩ _id*	The socket ID
in	*Addr*	The remote address to connect to
in	*timeout*	The maximum amount of time to wait, or OS_PEND to wait forever

Returns

Execution status, see OSAL Return Code Defines

37.71.2.4 OS_SocketGetIdByName()

```
int32 OS_SocketGetIdByName (
            osal_id_t * sock_id,
            const char * sock_name )
```

Gets an OSAL ID from a given name.

Note

OSAL Sockets use generated names according to the address and type.

See also

OS_SocketGetInfo()

Parameters

out	*sock_id*	Buffer to hold result
in	*sock_name*	Name of socket to find

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	is id or name are NULL pointers
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_NOT_FOUND	if the name was not found in the table

37.71.2.5 OS_SocketGetInfo()

```
int32 OS_SocketGetInfo (
            osal_id_t sock_id,
            OS_socket_prop_t * sock_prop )
```

Gets information about an OSAL Socket ID.

OSAL Sockets use generated names according to the address and type. This allows applications to find the name of a given socket.

Parameters

in	*sock_id*	The socket ID
out	*sock_prop*	Buffer to hold socket information

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid semaphore
OS_INVALID_POINTER	if the count_prop pointer is null

37.71.2.6 OS_SocketOpen()

```
int32 OS_SocketOpen (
            osal_id_t * sock_id,
            OS_SocketDomain_t Domain,
            OS_SocketType_t Type )
```

Opens a socket.

A new, unconnected and unbound socket is allocated of the given domain and type.

Parameters

out	*sock↩ _id*	Buffer to hold the non-zero OSAL ID
in	*Domain*	The domain / address family of the socket (INET or INET6, etc)
in	*Type*	The type of the socket (STREAM or DATAGRAM)

Returns

Execution status, see OSAL Return Code Defines

37.71.2.7 OS_SocketRecvFrom()

```
int32 OS_SocketRecvFrom (
            osal_id_t sock_id,
            void * buffer,
            size_t buflen,
            OS_SockAddr_t * RemoteAddr,
            int32 timeout )
```

Reads data from a message-oriented (datagram) socket.

If a message is already available on the socket, this should immediately return that data without blocking. Otherwise, it may block up to the given timeout.

Parameters

in	*sock_id*	The socket ID, previously bound using OS_SocketBind()
out	*buffer*	Pointer to message data receive buffer
in	*buflen*	The maximum length of the message data to receive
out	*RemoteAddr*	Buffer to store the remote network address (may be NULL)
in	*timeout*	The maximum amount of time to wait, or OS_PEND to wait forever

Returns

Count of actual bytes received or error status, see OSAL Return Code Defines

37.71.2.8 OS_SocketSendTo()

```
int32 OS_SocketSendTo (
            osal_id_t sock_id,
            const void * buffer,
            size_t buflen,
            const OS_SockAddr_t * RemoteAddr )
```

Sends data to a message-oriented (datagram) socket.

This sends data in a non-blocking mode. If the socket is not currently able to queue the message, such as if its outbound buffer is full, then this returns an error code.

Parameters

in	*sock_id*	The socket ID, which must be of the datagram type
in	*buffer*	Pointer to message data to send
in	*buflen*	The length of the message data to send
in	*RemoteAddr*	Buffer containing the remote network address to send to

Returns

Count of actual bytes sent or error status, see OSAL Return Code Defines

37.72 OSAL Task APIs

Functions

- int32 OS_TaskCreate (osal_id_t *task_id, const char *task_name, osal_task_entry function_pointer, osal_stackptr_t stack_pointer, size_t stack_size, osal_priority_t priority, uint32 flags)

  Creates a task and starts running it.
- int32 OS_TaskDelete (osal_id_t task_id)

  Deletes the specified Task.
- void OS_TaskExit (void)

  Exits the calling task.
- int32 OS_TaskInstallDeleteHandler (osal_task_entry function_pointer)

  Installs a handler for when the task is deleted.
- int32 OS_TaskDelay (uint32 millisecond)

  Delay a task for specified amount of milliseconds.
- int32 OS_TaskSetPriority (osal_id_t task_id, osal_priority_t new_priority)

  Sets the given task to a new priority.
- int32 OS_TaskRegister (void)

  Obsolete.
- osal_id_t OS_TaskGetId (void)

  Obtain the task id of the calling task.
- int32 OS_TaskGetIdByName (osal_id_t *task_id, const char *task_name)

  Find an existing task ID by name.
- int32 OS_TaskGetInfo (osal_id_t task_id, OS_task_prop_t *task_prop)

  Fill a property object buffer with details regarding the resource.
- int32 OS_TaskFindIdBySystemData (osal_id_t *task_id, const void *sysdata, size_t sysdata_size)

  Reverse-lookup the OSAL task ID from an operating system ID.

37.72.1 Detailed Description

37.72.2 Function Documentation

37.72.2.1 OS_TaskCreate()

```
int32 OS_TaskCreate (
            osal_id_t * task_id,
            const char * task_name,
            osal_task_entry function_pointer,
            osal_stackptr_t stack_pointer,
            size_t stack_size,
            osal_priority_t priority,
            uint32 flags )
```

Creates a task and starts running it.

Creates a task and passes back the id of the task created. Task names must be unique; if the name already exists this function fails. Names cannot be NULL.

Parameters

out	*task_id*	will be set to the non-zero ID of the newly-created resource
in	*task_name*	the name of the new resource to create
in	*function_pointer*	the entry point of the new task
in	*stack_pointer*	pointer to the stack for the task, or NULL to allocate a stack from the system memory heap
in	*stack_size*	the size of the stack, or 0 to use a default stack size.
in	*priority*	initial priority of the new task
in	*flags*	initial options for the new task

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if any of the necessary pointers are NULL
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_INVALID_PRIORITY	if the priority is bad
OS_ERR_NO_FREE_IDS	if there can be no more tasks created
OS_ERR_NAME_TAKEN	if the name specified is already used by a task
OS_ERROR	if an unspecified/other error occurs

37.72.2.2 OS_TaskDelay()

```
int32 OS_TaskDelay (
            uint32 millisecond )
```

Delay a task for specified amount of milliseconds.

Causes the current thread to be suspended from execution for the period of millisecond.

Parameters

in	*millisecond*	Amount of time to delay

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if sleep fails or millisecond = 0

Referenced by CFE_PSP_Restart(), and OS_Application_Run().

37.72.2.3 OS_TaskDelete()

```
int32 OS_TaskDelete (
            osal_id_t task_id )
```

Deletes the specified Task.

The task will be removed from the local tables. and the OS will be configured to stop executing the task at the next opportunity.

Parameters

in	*task↩ _id*	The object ID to operate on

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the ID given to it is invalid
OS_ERROR	if the OS delete call fails

37.72.2.4 OS_TaskExit()

```
void OS_TaskExit (
            void  )
```

Exits the calling task.

The calling thread is terminated. This function does not return.

37.72.2.5 OS_TaskFindIdBySystemData()

```
int32 OS_TaskFindIdBySystemData (
            osal_id_t * task_id,
            const void * sysdata,
            size_t sysdata_size )
```

Reverse-lookup the OSAL task ID from an operating system ID.

This provides a method by which an external entity may find the OSAL task ID corresponding to a system-defined identifier (e.g. TASK_ID, pthread_t, rtems_id, etc).

Normally OSAL does not expose the underlying OS-specific values to the application, but in some circumstances, such as exception handling, the OS may provide this information directly to a BSP handler outside of the normal OSAL API.

Parameters

out	*task_id*	The buffer where the task id output is stored
in	*sysdata*	Pointer to the system-provided identification data
in	*sysdata_size*	Size of the system-provided identification data

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.

37.72.2.6 OS_TaskGetId()

```
osal_id_t OS_TaskGetId (
            void  )
```

Obtain the task id of the calling task.

This function returns the task id of the calling task

Returns

Task ID, or zero if the operation failed (zero is never a valid task ID)

37.72.2.7 OS_TaskGetIdByName()

```
int32 OS_TaskGetIdByName (
            osal_id_t * task_id,
            const char * task_name )
```

Find an existing task ID by name.

This function tries to find a task Id given the name of a task

Parameters

out	*task_id*	will be set to the ID of the existing resource
in	*task_name*	the name of the existing resource to find

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if the pointers passed in are NULL
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_NOT_FOUND	if the name wasn't found in the table

37.72.2.8 OS_TaskGetInfo()

```
int32 OS_TaskGetInfo (
            osal_id_t task_id,
            OS_task_prop_t * task_prop )
```

Fill a property object buffer with details regarding the resource.

This function will pass back a pointer to structure that contains all of the relevant info (creator, stack size, priority, name) about the specified task.

Parameters

in	*task_id*	The object ID to operate on
out	*task_prop*	The property object buffer to fill

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the ID passed to it is invalid
OS_INVALID_POINTER	if the task_prop pointer is NULL

37.72.2.9 OS_TaskInstallDeleteHandler()

```
int32 OS_TaskInstallDeleteHandler (
            osal_task_entry function_pointer )
```

Installs a handler for when the task is deleted.

This function is used to install a callback that is called when the task is deleted. The callback is called when OS_Task↩ Delete is called with the task ID. A task delete handler is useful for cleaning up resources that a task creates, before the task is removed from the system.

Parameters

in	*function_pointer*	function to be called when task exits

Returns

Execution status, see OSAL Return Code Defines

37.72.2.10 OS_TaskRegister()

```
int32 OS_TaskRegister (
            void  )
```

Obsolete.

Deprecated Explicit registration call no longer needed

Obsolete function retained for compatibility purposes. Does Nothing in the current implementation.

Returns

OS_SUCCESS (always), see OSAL Return Code Defines

37.72.2.11 OS_TaskSetPriority()

```
int32 OS_TaskSetPriority (
            osal_id_t task_id,
            osal_priority_t new_priority )
```

Sets the given task to a new priority.

Parameters

in	*task_id*	The object ID to operate on
in	*new_priority*	Set the new priority

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the ID passed to it is invalid
OS_ERR_INVALID_PRIORITY	if the priority is greater than the max allowed
OS_ERROR	if the OS call to change the priority fails

37.73 OSAL Time Base APIs

Functions

- int32 OS_TimeBaseCreate (osal_id_t *timebase_id, const char *timebase_name, OS_TimerSync_t external_↩ sync)

  Create an abstract Time Base resource.
- int32 OS_TimeBaseSet (osal_id_t timebase_id, uint32 start_time, uint32 interval_time)

  Sets the tick period for simulated time base objects.
- int32 OS_TimeBaseDelete (osal_id_t timebase_id)

  Deletes a time base object.
- int32 OS_TimeBaseGetIdByName (osal_id_t *timebase_id, const char *timebase_name)

  Find the ID of an existing time base resource.
- int32 OS_TimeBaseGetInfo (osal_id_t timebase_id, OS_timebase_prop_t *timebase_prop)

  Obtain information about a timebase resource.
- int32 OS_TimeBaseGetFreeRun (osal_id_t timebase_id, uint32 *freerun_val)

  Read the value of the timebase free run counter.

37.73.1 Detailed Description

37.73.2 Function Documentation

37.73.2.1 OS_TimeBaseCreate()

```
int32 OS_TimeBaseCreate (
            osal_id_t * timebase_id,
            const char * timebase_name,
            OS_TimerSync_t external_sync )
```

Create an abstract Time Base resource.

An OSAL time base is an abstraction of a "timer tick" that can, in turn, be used for measurement of elapsed time between events.

Time bases can be simulated by the operating system using the OS kernel-provided timing facilities, or based on a hardware timing source if provided by the BSP.

A time base object has a servicing task associated with it, that runs at elevated priority and will thereby interrupt user-level tasks when timing ticks occur.

If the external_sync function is passed as NULL, the operating system kernel timing resources will be utilized for a simulated timer tick.

If the external_sync function is not NULL, this should point to a BSP-provided function that will block the calling task until the next tick occurs. This can be used for synchronizing with hardware events.

Note

When provisioning a tunable RTOS kernel, such as RTEMS, the kernel should be configured to support at least (OS_MAX_TASKS + OS_MAX_TIMEBASES) threads, to account for the helper threads associated with time base objects.

Parameters

out	*timebase_id*	A non-zero ID corresponding to the timebase resource
in	*timebase_name*	The name of the time base
in	*external_sync*	A synchronization function for BSP hardware-based timer ticks

Returns

Execution status, see OSAL Return Code Defines

Referenced by OS_Application_Startup().

37.73.2.2 OS_TimeBaseDelete()

```
int32 OS_TimeBaseDelete (
            osal_id_t timebase_id )
```

Deletes a time base object.

The helper task and any other resources associated with the time base abstraction will be freed.

Parameters

in	*timebase↩ _id*	The timebase resource to delete

Returns

Execution status, see OSAL Return Code Defines

37.73.2.3 OS_TimeBaseGetFreeRun()

```
int32 OS_TimeBaseGetFreeRun (
            osal_id_t timebase_id,
            uint32 * freerun_val )
```

Read the value of the timebase free run counter.

Poll the timer free-running time counter in a lightweight fashion.

The free run count is a monotonically increasing value reflecting the total time elapsed since the timebase inception. Units are the same as the timebase itself, usually microseconds.

Applications may quickly and efficiently calculate relative time differences by polling this value and subtracting the previous counter value.

The absolute value of this counter is not relevant, because it will "roll over" after 2^32 units of time. For a timebase with microsecond units, this occurs approximately every 4294 seconds, or about 1.2 hours.

Note

To ensure consistency of results, the application should sample the value at a minimum of two times the roll over frequency, and calculate the difference between the consecutive samples.

Parameters

in	*timebase↩ _id*	The timebase to operate on
out	*freerun_val*	Buffer to store the free run counter

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid timebase

37.73.2.4 OS_TimeBaseGetIdByName()

```
int32 OS_TimeBaseGetIdByName (
            osal_id_t * timebase_id,
            const char * timebase_name )
```

Find the ID of an existing time base resource.

Given a time base name, find and output the ID associated with it.

Parameters

out	*timebase_id*	The timebase resource ID
in	*timebase_name*	The name of the timebase resource to find

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if timebase_id or timebase_name are NULL pointers
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_NOT_FOUND	if the name was not found in the table

37.73.2.5 OS_TimeBaseGetInfo()

```
int32 OS_TimeBaseGetInfo (
            osal_id_t timebase_id,
            OS_timebase_prop_t * timebase_prop )
```

Obtain information about a timebase resource.

Fills the buffer referred to by the timebase_prop parameter with relevant information about the time base resource.

This function will pass back a pointer to structure that contains all of the relevant info(name and creator) about the specified timebase.

Parameters

in	*timebase_id*	The timebase resource ID
out	*timebase_prop*	Buffer to store timebase properties

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid timebase
OS_INVALID_POINTER	if the timebase_prop pointer is null

37.73.2.6 OS_TimeBaseSet()

```
int32 OS_TimeBaseSet (
            osal_id_t timebase_id,
            uint32 start_time,
            uint32 interval_time )
```

Sets the tick period for simulated time base objects.

This sets the actual tick period for timing ticks that are simulated by the RTOS kernel (i.e. the "external_sync" parameter on the call to OS_TimeBaseCreate() is NULL).

The RTOS will be configured to wake up the helper thread at the requested interval.

This function has no effect for time bases that are using a BSP-provided external_sync function.

Parameters

in	*timebase_id*	The timebase resource to configure
in	*start_time*	The amount of delay for the first tick, in microseconds.
in	*interval_time*	The amount of delay between ticks, in microseconds.

Returns

Execution status, see OSAL Return Code Defines

Referenced by OS_Application_Startup().

37.74 OSAL Timer APIs

Functions

- int32 OS_TimerCreate (osal_id_t *timer_id, const char *timer_name, uint32 *clock_accuracy, OS_Timer↩ Callback_t callback_ptr)

  Create a timer object.
- int32 OS_TimerAdd (osal_id_t *timer_id, const char *timer_name, osal_id_t timebase_id, OS_ArgCallback_↩ t callback_ptr, void *callback_arg)

  Add a timer object based on an existing TimeBase resource.
- int32 OS_TimerSet (osal_id_t timer_id, uint32 start_time, uint32 interval_time)

  Configures a periodic or one shot timer.
- int32 OS_TimerDelete (osal_id_t timer_id)

  Deletes a timer resource.
- int32 OS_TimerGetIdByName (osal_id_t *timer_id, const char *timer_name)

  Locate an existing timer resource by name.
- int32 OS_TimerGetInfo (osal_id_t timer_id, OS_timer_prop_t *timer_prop)

  Gets information about an existing timer.

37.74.1 Detailed Description

37.74.2 Function Documentation

37.74.2.1 OS_TimerAdd()

```
int32 OS_TimerAdd (
            osal_id_t * timer_id,
            const char * timer_name,
            osal_id_t timebase_id,
            OS_ArgCallback_t callback_ptr,
            void * callback_arg )
```

Add a timer object based on an existing TimeBase resource.

A timer object is a resource that invokes the specified application-provided function upon timer expiration. Timers may be one-shot or periodic in nature.

This function uses an existing time base object to service this timer, which must exist prior to adding the timer. The precision of the timer is the same as that of the underlying time base object. Multiple timer objects can be created referring to a single time base object.

This routine also uses a different callback function prototype from OS_TimerCreate(), allowing a single opaque argument to be passed to the callback routine. The OSAL implementation does not use this parameter, and may be set NULL.

Warning

> Depending on the OS, the callback_ptr function may be similar to an interrupt service routine. Calls that cause the code to block or require an application context (like sending events) are generally not supported.

Parameters

out	*timer_id*	The non-zero resource ID of the timer object
in	*timer_name*	Name of the timer object
in	*timebase↩ _id*	The time base resource to use as a reference
in	*callback_ptr*	Application-provided function to invoke
in	*callback_arg*	Opaque argument to pass to callback function

Returns

Execution status, see OSAL Return Code Defines

37.74.2.2 OS_TimerCreate()

```
int32 OS_TimerCreate (
            osal_id_t * timer_id,
            const char * timer_name,
            uint32 * clock_accuracy,
            OS_TimerCallback_t callback_ptr )
```

Create a timer object.

A timer object is a resource that invokes the specified application-provided function upon timer expiration. Timers may be one-shot or periodic in nature.

This function creates a dedicated (hidden) time base object to service this timer, which is created and deleted with the timer object itself. The internal time base is configured for an OS simulated timer tick at the same interval as the timer.

Note

clock_accuracy comes from the underlying OS tick value. The nearest integer microsecond value is returned, so may not be exact.

Warning

Depending on the OS, the callback_ptr function may be similar to an interrupt service routine. Calls that cause the code to block or require an application context (like sending events) are generally not supported.

Parameters

out	*timer_id*	The non-zero resource ID of the timer object
in	*timer_name*	Name of the timer object
out	*clock_accuracy*	Expected precision of the timer, in microseconds. This is the underlying tick value rounded to the nearest microsecond integer.
in	*callback_ptr*	The function pointer of the timer callback or ISR that will be called by the timer. The user's function is declared as follows: `void timer_callback(uint32 timer_id)` Where the timer_id is passed in to the function by the OSAL

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if any parameters are NULL
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_TAKEN	if the name is already in use by another timer.
OS_ERR_NO_FREE_IDS	if all of the timers are already allocated.
OS_TIMER_ERR_INVALID_ARGS	if the callback pointer is zero.
OS_TIMER_ERR_UNAVAILABLE	if the timer cannot be created.

37.74.2.3 OS_TimerDelete()

```
int32 OS_TimerDelete (
            osal_id_t timer_id )
```

Deletes a timer resource.

The application callback associated with the timer will be stopped, and the resources freed for future use.

Parameters

in	*timer↩ _id*	The timer ID to operate on

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the timer_id is invalid.
OS_TIMER_ERR_INTERNAL	if there was a problem deleting the timer in the host OS.

37.74.2.4 OS_TimerGetIdByName()

```
int32 OS_TimerGetIdByName (
            osal_id_t * timer_id,
            const char * timer_name )
```

Locate an existing timer resource by name.

Outputs the ID associated with the given timer, if it exists.

Parameters

out	*timer_id*	The timer ID corresponding to the name
in	*timer_name*	The timer name to find

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if timer_id or timer_name are NULL pointers
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_NOT_FOUND	if the name was not found in the table

37.74.2.5 OS_TimerGetInfo()

```
int32 OS_TimerGetInfo (
            osal_id_t timer_id,
            OS_timer_prop_t * timer_prop )
```

Gets information about an existing timer.

This function takes timer_id, and looks it up in the OS table. It puts all of the information known about that timer into a structure pointer to by timer_prop.

Parameters

in	*timer_id*	The timer ID to operate on
out	*timer_prop*	Buffer containing timer properties • creator: the OS task ID of the task that created this timer • name: the string name of the timer • start_time: the start time in microseconds, if any • interval_time: the interval time in microseconds, if any • accuracy: the accuracy of the timer in microseconds

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid timer
OS_INVALID_POINTER	if the timer_prop pointer is null

37.74.2.6 OS_TimerSet()

```
int32 OS_TimerSet (
            osal_id_t timer_id,
            uint32 start_time,
            uint32 interval_time )
```

Configures a periodic or one shot timer.

This function programs the timer with a start time and an optional interval time. The start time is the time in microseconds when the user callback function will be called. If the interval time is non-zero, the timer will be reprogrammed with that interval in microseconds to call the user callback function periodically. If the start time and interval time are zero, the function will return an error.

For a "one-shot" timer, the start_time configures the expiration time, and the interval_time should be passed as zero to indicate the timer is not to be automatically reset.

Note

The resolution of the times specified is limited to the clock accuracy returned in the OS_TimerCreate call. If the times specified in the start_msec or interval_msec parameters are less than the accuracy, they will be rounded up to the accuracy of the timer.

Parameters

in	*timer_id*	The timer ID to operate on
in	*start_time*	Time in microseconds to the first expiration
in	*interval_time*	Time in microseconds between subsequent intervals, value of zero will only call the user callback function once after the start_msec time.

Returns

Execution status, see OSAL Return Code Defines

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the timer_id is not valid.
OS_TIMER_ERR_INTERNAL	if there was an error programming the OS timer.
OS_ERROR	if both start time and interval time are zero.

38 Data Structure Documentation

38.1 CCSDS_ExtendedHeader Struct Reference

CCSDS packet extended header.

```
#include <ccsds_hdr.h>
```

Data Fields

- uint8 Subsystem [2]
  subsystem qualifier
- uint8 SystemId [2]
  system qualifier

38.1.1 Detailed Description

CCSDS packet extended header.

Definition at line 75 of file ccsds_hdr.h.

38.1.2 Field Documentation

38.1.2.1 Subsystem

```
uint8 CCSDS_ExtendedHeader::Subsystem[2]
```

subsystem qualifier

Definition at line 78 of file ccsds_hdr.h.

38.1.2.2 SystemId

```
uint8 CCSDS_ExtendedHeader::SystemId[2]
```

system qualifier

Definition at line 85 of file ccsds_hdr.h.

The documentation for this struct was generated from the following file:

- cfe/modules/msg/fsw/inc/ccsds_hdr.h

38.2 CCSDS_PrimaryHeader Struct Reference

CCSDS packet primary header.

```
#include <ccsds_hdr.h>
```

Data Fields

- uint8 StreamId [2]

  packet identifier word (stream ID)
- uint8 Sequence [2]

  packet sequence word
- uint8 Length [2]

  packet length word

38.2.1 Detailed Description

CCSDS packet primary header.

Definition at line 51 of file ccsds_hdr.h.

38.2.2 Field Documentation

38.2.2.1 Length

```
uint8 CCSDS_PrimaryHeader::Length[2]
```

packet length word

Definition at line 66 of file ccsds_hdr.h.

38.2.2.2 Sequence

```
uint8 CCSDS_PrimaryHeader::Sequence[2]
```

packet sequence word

Definition at line 61 of file ccsds_hdr.h.

38.2.2.3 StreamId

```
uint8 CCSDS_PrimaryHeader::StreamId[2]
```

packet identifier word (stream ID)

Definition at line 54 of file ccsds_hdr.h.

The documentation for this struct was generated from the following file:

- cfe/modules/msg/fsw/inc/ccsds_hdr.h

38.3 CFE_ES_AppInfo Struct Reference

Application Information.

```
#include <cfe_es_extern_typedefs.h>
```

Data Fields

- CFE_ResourceId_t ResourceId

  Application or Library ID for this resource.
- uint32 Type

  The type of App: CORE or EXTERNAL.
- char Name [CFE_MISSION_MAX_API_LEN]

  The Registered Name of the Application.
- char EntryPoint [CFE_MISSION_MAX_API_LEN]

  The Entry Point label for the Application.
- char FileName [CFE_MISSION_MAX_PATH_LEN]

  The Filename of the file containing the Application.
- CFE_ES_MemOffset_t StackSize

  The Stack Size of the Application.
- uint32 AddressesAreValid

  Indicates that the Code, Data, and BSS addresses/sizes are valid.
- CFE_ES_MemAddress_t CodeAddress

  The Address of the Application Code Segment.
- CFE_ES_MemOffset_t CodeSize

  The Code Size of the Application.
- CFE_ES_MemAddress_t DataAddress

  The Address of the Application Data Segment.
- CFE_ES_MemOffset_t DataSize

  The Data Size of the Application.
- CFE_ES_MemAddress_t BSSAddress

  The Address of the Application BSS Segment.
- CFE_ES_MemOffset_t BSSSize

  The BSS Size of the Application.

- CFE_ES_MemAddress_t StartAddress

  The Start Address of the Application.
- CFE_ES_ExceptionAction_Enum_t ExceptionAction

  What should occur if Application has an exception (Restart Application OR Restart Processor)
- CFE_ES_TaskPriority_Atom_t Priority

  The Priority of the Application.
- CFE_ES_TaskId_t MainTaskId

  The Application's Main Task ID.
- uint32 ExecutionCounter

  The Application's Main Task Execution Counter.
- char MainTaskName [CFE_MISSION_MAX_API_LEN]

  The Application's Main Task ID.
- uint32 NumOfChildTasks

  Number of Child tasks for an App.

38.3.1 Detailed Description

Application Information.

Structure that is used to provide information about an app. It is primarily used for the QueryOne and QueryAll Commands.

While this structure is primarily intended for Application info, it can also represent Library information where only a subset of the information applies.

Definition at line 445 of file cfe_es_extern_typedefs.h.

38.3.2 Field Documentation

38.3.2.1 AddressesAreValid

```
uint32 CFE_ES_AppInfo::AddressesAreValid
```

Indicates that the Code, Data, and BSS addresses/sizes are valid.

Telemetry Mnemonic(s) $sc_$cpu_ES_AddrsValid

Definition at line 461 of file cfe_es_extern_typedefs.h.

38.3.2.2 BSSAddress

`CFE_ES_MemAddress_t CFE_ES_AppInfo::BSSAddress`

The Address of the Application BSS Segment.

Telemetry Mnemonic(s) $sc_$cpu_ES_BSSAddress

Definition at line 471 of file cfe_es_extern_typedefs.h.

38.3.2.3 BSSSize

`CFE_ES_MemOffset_t CFE_ES_AppInfo::BSSSize`

The BSS Size of the Application.

Telemetry Mnemonic(s) $sc_$cpu_ES_BSSSize

Definition at line 473 of file cfe_es_extern_typedefs.h.

38.3.2.4 CodeAddress

`CFE_ES_MemAddress_t CFE_ES_AppInfo::CodeAddress`

The Address of the Application Code Segment.

Telemetry Mnemonic(s) $sc_$cpu_ES_CodeAddress

Definition at line 463 of file cfe_es_extern_typedefs.h.

38.3.2.5 CodeSize

`CFE_ES_MemOffset_t CFE_ES_AppInfo::CodeSize`

The Code Size of the Application.

Telemetry Mnemonic(s) $sc_$cpu_ES_CodeSize

Definition at line 465 of file cfe_es_extern_typedefs.h.

38.3.2.6 DataAddress

`CFE_ES_MemAddress_t CFE_ES_AppInfo::DataAddress`

The Address of the Application Data Segment.

Telemetry Mnemonic(s) $sc_$cpu_ES_DataAddress

Definition at line 467 of file cfe_es_extern_typedefs.h.

38.3.2.7 DataSize

`CFE_ES_MemOffset_t CFE_ES_AppInfo::DataSize`

The Data Size of the Application.

Telemetry Mnemonic(s) $sc_$cpu_ES_DataSize

Definition at line 469 of file cfe_es_extern_typedefs.h.

38.3.2.8 EntryPoint

`char CFE_ES_AppInfo::EntryPoint[CFE_MISSION_MAX_API_LEN]`

The Entry Point label for the Application.

Telemetry Mnemonic(s) $sc_$cpu_ES_AppEntryPt[OS_MAX_API_NAME]

Definition at line 454 of file cfe_es_extern_typedefs.h.

38.3.2.9 ExceptionAction

`CFE_ES_ExceptionAction_Enum_t CFE_ES_AppInfo::ExceptionAction`

What should occur if Application has an exception (Restart Application OR Restart Processor)

Telemetry Mnemonic(s) $sc_$cpu_ES_ExceptnActn

Definition at line 477 of file cfe_es_extern_typedefs.h.

38.3.2.10 ExecutionCounter

```
uint32 CFE_ES_AppInfo::ExecutionCounter
```

The Application's Main Task Execution Counter.

Telemetry Mnemonic(s) $sc_$cpu_ES_ExecutionCtr

Definition at line 484 of file cfe_es_extern_typedefs.h.

38.3.2.11 FileName

```
char CFE_ES_AppInfo::FileName[CFE_MISSION_MAX_PATH_LEN]
```

The Filename of the file containing the Application.

Telemetry Mnemonic(s) $sc_$cpu_ES_AppFilename[OS_MAX_PATH_LEN]

Definition at line 456 of file cfe_es_extern_typedefs.h.

38.3.2.12 MainTaskId

```
CFE_ES_TaskId_t CFE_ES_AppInfo::MainTaskId
```

The Application's Main Task ID.

Telemetry Mnemonic(s) $sc_$cpu_ES_MainTaskId

Definition at line 482 of file cfe_es_extern_typedefs.h.

38.3.2.13 MainTaskName

```
char CFE_ES_AppInfo::MainTaskName[CFE_MISSION_MAX_API_LEN]
```

The Application's Main Task ID.

Telemetry Mnemonic(s) $sc_$cpu_ES_MainTaskName[OS_MAX_API_NAME]

Definition at line 486 of file cfe_es_extern_typedefs.h.

38.3.2.14 Name

```
char CFE_ES_AppInfo::Name[CFE_MISSION_MAX_API_LEN]
```

The Registered Name of the Application.

Telemetry Mnemonic(s) $sc_$cpu_ES_AppName[OS_MAX_API_NAME]

Definition at line 452 of file cfe_es_extern_typedefs.h.

38.3.2.15 NumOfChildTasks

```
uint32 CFE_ES_AppInfo::NumOfChildTasks
```

Number of Child tasks for an App.

Telemetry Mnemonic(s) $sc_$cpu_ES_ChildTasks

Definition at line 488 of file cfe_es_extern_typedefs.h.

38.3.2.16 Priority

```
CFE_ES_TaskPriority_Atom_t CFE_ES_AppInfo::Priority
```

The Priority of the Application.

Telemetry Mnemonic(s) $sc_$cpu_ES_Priority

Definition at line 480 of file cfe_es_extern_typedefs.h.

38.3.2.17 ResourceId

```
CFE_ResourceId_t CFE_ES_AppInfo::ResourceId
```

Application or Library ID for this resource.

Telemetry Mnemonic(s) $sc_$cpu_ES_AppID

Definition at line 447 of file cfe_es_extern_typedefs.h.

38.3.2.18 StackSize

`CFE_ES_MemOffset_t CFE_ES_AppInfo::StackSize`

The Stack Size of the Application.

Telemetry Mnemonic(s) $sc_$cpu_ES_StackSize

Definition at line 459 of file cfe_es_extern_typedefs.h.

38.3.2.19 StartAddress

`CFE_ES_MemAddress_t CFE_ES_AppInfo::StartAddress`

The Start Address of the Application.

Telemetry Mnemonic(s) $sc_$cpu_ES_StartAddr

Definition at line 475 of file cfe_es_extern_typedefs.h.

38.3.2.20 Type

`uint32 CFE_ES_AppInfo::Type`

The type of App: CORE or EXTERNAL.

Telemetry Mnemonic(s) $sc_$cpu_ES_AppType

Definition at line 449 of file cfe_es_extern_typedefs.h.

The documentation for this struct was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_es_extern_typedefs.h

38.4 CFE_ES_AppNameCmd Struct Reference

Generic application name command.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_ES_AppNameCmd_Payload_t Payload

  Command payload.

38.4.1 Detailed Description

Generic application name command.

Definition at line 1213 of file cfe_es_msg.h.

38.4.2 Field Documentation

38.4.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_ES_AppNameCmd::CmdHeader
```

Command header.

Definition at line 1215 of file cfe_es_msg.h.

38.4.2.2 Payload

```
CFE_ES_AppNameCmd_Payload_t CFE_ES_AppNameCmd::Payload
```

Command payload.

Definition at line 1216 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.5 CFE_ES_AppNameCmd_Payload Struct Reference

Generic application name command payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- char Application [CFE_MISSION_MAX_API_LEN]

  ASCII text string containing Application Name.

38.5.1 Detailed Description

Generic application name command payload.

For command details, see CFE_ES_STOP_APP_CC, CFE_ES_RESTART_APP_CC, CFE_ES_QUERY_ONE_CC

Definition at line 1205 of file cfe_es_msg.h.

38.5.2 Field Documentation

38.5.2.1 Application

```
char CFE_ES_AppNameCmd_Payload::Application[CFE_MISSION_MAX_API_LEN]
```

ASCII text string containing Application Name.

Definition at line 1207 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.6 CFE_ES_AppReloadCmd_Payload Struct Reference

Reload Application Command Payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- char Application [CFE_MISSION_MAX_API_LEN]

  ASCII text string containing Application Name.
- char AppFileName [CFE_MISSION_MAX_PATH_LEN]

  Full path and filename of Application's executable image.

38.6.1 Detailed Description

Reload Application Command Payload.

For command details, see CFE_ES_RELOAD_APP_CC

Definition at line 1234 of file cfe_es_msg.h.

38.6.2 Field Documentation

38.6.2.1 AppFileName

```
char CFE_ES_AppReloadCmd_Payload::AppFileName[CFE_MISSION_MAX_PATH_LEN]
```

Full path and filename of Application's executable image.

Definition at line 1237 of file cfe_es_msg.h.

38.6.2.2 Application

```
char CFE_ES_AppReloadCmd_Payload::Application[CFE_MISSION_MAX_API_LEN]
```

ASCII text string containing Application Name.

Definition at line 1236 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.7 CFE_ES_BlockStats Struct Reference

Block statistics.

```
#include <cfe_es_extern_typedefs.h>
```

Data Fields

- CFE_ES_MemOffset_t BlockSize

  Number of bytes in each of these blocks.
- uint32 NumCreated

  Number of Memory Blocks of this size created.
- uint32 NumFree

  Number of Memory Blocks of this size that are free.

38.7.1 Detailed Description

Block statistics.

Sub-Structure that is used to provide information about a specific block size/bucket within a memory pool.

Definition at line 543 of file cfe_es_extern_typedefs.h.

38.7.2 Field Documentation

38.7.2.1 BlockSize

```
CFE_ES_MemOffset_t CFE_ES_BlockStats::BlockSize
```

Number of bytes in each of these blocks.

Definition at line 545 of file cfe_es_extern_typedefs.h.

38.7.2.2 NumCreated

```
uint32 CFE_ES_BlockStats::NumCreated
```

Number of Memory Blocks of this size created.

Definition at line 546 of file cfe_es_extern_typedefs.h.

38.7.2.3 NumFree

```
uint32 CFE_ES_BlockStats::NumFree
```

Number of Memory Blocks of this size that are free.

Definition at line 547 of file cfe_es_extern_typedefs.h.

The documentation for this struct was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_es_extern_typedefs.h

38.8 CFE_ES_CDSRegDumpRec Struct Reference

CDS Register Dump Record.

```
#include <cfe_es_extern_typedefs.h>
```

Data Fields

- CFE_ES_CDSHandle_t Handle

  Handle of CDS.
- CFE_ES_MemOffset_t Size

  Size, in bytes, of the CDS memory block.
- bool Table

  Flag that indicates whether CDS contains a Critical Table.
- char Name [CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN]

  Processor Unique Name of CDS.
- uint8 ByteAlignSpare [3]

  Spare bytes to ensure structure size is multiple of 4 bytes.

38.8.1 Detailed Description

CDS Register Dump Record.

Structure that is used to provide information about a critical data store. It is primarily used for the Dump CDS registry (CFE_ES_DUMP_CDS_REGISTRY_CC) command.

Note

There is not currently a telemetry message directly containing this data structure, but it does define the format of the data file generated by the Dump CDS registry command. Therefore it should be considered part of the overall telemetry interface.

Definition at line 528 of file cfe_es_extern_typedefs.h.

38.8.2 Field Documentation

38.8.2.1 ByteAlignSpare

```
uint8 CFE_ES_CDSRegDumpRec::ByteAlignSpare[3]
```

Spare bytes to ensure structure size is multiple of 4 bytes.

Definition at line 534 of file cfe_es_extern_typedefs.h.

38.8.2.2 Handle

```
CFE_ES_CDSHandle_t CFE_ES_CDSRegDumpRec::Handle
```

Handle of CDS.

Definition at line 530 of file cfe_es_extern_typedefs.h.

38.8.2.3 Name

```
char CFE_ES_CDSRegDumpRec::Name[CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN]
```

Processor Unique Name of CDS.

Definition at line 533 of file cfe_es_extern_typedefs.h.

38.8.2.4 Size

```
CFE_ES_MemOffset_t CFE_ES_CDSRegDumpRec::Size
```

Size, in bytes, of the CDS memory block.

Definition at line 531 of file cfe_es_extern_typedefs.h.

38.8.2.5 Table

```
bool CFE_ES_CDSRegDumpRec::Table
```

Flag that indicates whether CDS contains a Critical Table.

Definition at line 532 of file cfe_es_extern_typedefs.h.

The documentation for this struct was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_es_extern_typedefs.h

38.9 CFE_ES_DeleteCDSCmd Struct Reference

Delete Critical Data Store Command.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_ES_DeleteCDSCmd_Payload_t Payload

  Command payload.

38.9.1 Detailed Description

Delete Critical Data Store Command.

Definition at line 1287 of file cfe_es_msg.h.

38.9.2 Field Documentation

38.9.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_ES_DeleteCDSCmd::CmdHeader
```

Command header.

Definition at line 1289 of file cfe_es_msg.h.

38.9.2.2 Payload

```
CFE_ES_DeleteCDSCmd_Payload_t CFE_ES_DeleteCDSCmd::Payload
```

Command payload.

Definition at line 1290 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.10 CFE_ES_DeleteCDSCmd_Payload Struct Reference

Delete Critical Data Store Command Payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- char CdsName [CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN]

  ASCII text string containing name of CDS to delete.

38.10.1 Detailed Description

Delete Critical Data Store Command Payload.

For command details, see CFE_ES_DELETE_CDS_CC

Definition at line 1277 of file cfe_es_msg.h.

38.10.2 Field Documentation

38.10.2.1 CdsName

```
char CFE_ES_DeleteCDSCmd_Payload::CdsName[CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN]
```

ASCII text string containing name of CDS to delete.

Definition at line 1280 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.11 CFE_ES_DumpCDSRegistryCmd Struct Reference

Dump CDS Registry Command.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_ES_DumpCDSRegistryCmd_Payload_t Payload
  Command payload.

38.11.1 Detailed Description

Dump CDS Registry Command.

Definition at line 1415 of file cfe_es_msg.h.

38.11.2 Field Documentation

38.11.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_ES_DumpCDSRegistryCmd::CmdHeader`

Command header.

Definition at line 1417 of file cfe_es_msg.h.

38.11.2.2 Payload

`CFE_ES_DumpCDSRegistryCmd_Payload_t CFE_ES_DumpCDSRegistryCmd::Payload`

Command payload.

Definition at line 1418 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.12 CFE_ES_DumpCDSRegistryCmd_Payload Struct Reference

Dump CDS Registry Command Payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- char DumpFilename [CFE_MISSION_MAX_PATH_LEN]
  ASCII text string of full path and filename of file CDS Registry is to be written.

38.12.1 Detailed Description

Dump CDS Registry Command Payload.

For command details, see CFE_ES_DUMP_CDS_REGISTRY_CC

Definition at line 1406 of file cfe_es_msg.h.

38.12.2 Field Documentation

38.12.2.1 DumpFilename

```
char CFE_ES_DumpCDSRegistryCmd_Payload::DumpFilename[CFE_MISSION_MAX_PATH_LEN]
```

ASCII text string of full path and filename of file CDS Registry is to be written.

Definition at line 1408 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.13 CFE_ES_FileNameCmd Struct Reference

Generic file name command.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_ES_FileNameCmd_Payload_t Payload
  Command payload.

38.13.1 Detailed Description

Generic file name command.

Definition at line 1130 of file cfe_es_msg.h.

38.13.2 Field Documentation

38.13.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_ES_FileNameCmd::CmdHeader
```

Command header.

Definition at line 1132 of file cfe_es_msg.h.

38.13.2.2 Payload

`CFE_ES_FileNameCmd_Payload_t CFE_ES_FileNameCmd::Payload`

Command payload.

Definition at line 1133 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.14 CFE_ES_FileNameCmd_Payload Struct Reference

Generic file name command payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- char FileName [CFE_MISSION_MAX_PATH_LEN]
  ASCII text string containing full path and filename of file in which Application data is to be dumped.

38.14.1 Detailed Description

Generic file name command payload.

This format is shared by several executive services commands. For command details, see CFE_ES_QUERY_ALL_CC, CFE_ES_QUERY_ALL_TASKS_CC, CFE_ES_WRITE_SYSLOG_CC, and CFE_ES_WRITE_ER_LOG_CC

Definition at line 1121 of file cfe_es_msg.h.

38.14.2 Field Documentation

38.14.2.1 FileName

`char CFE_ES_FileNameCmd_Payload::FileName[CFE_MISSION_MAX_PATH_LEN]`

ASCII text string containing full path and filename of file in which Application data is to be dumped.

Definition at line 1123 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.15 CFE_ES_HousekeepingTlm Struct Reference

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_TelemetryHeader_t TlmHeader

  Telemetry header.
- CFE_ES_HousekeepingTlm_Payload_t Payload

  Telemetry payload.

38.15.1 Detailed Description

Definition at line 1554 of file cfe_es_msg.h.

38.15.2 Field Documentation

38.15.2.1 Payload

```
CFE_ES_HousekeepingTlm_Payload_t CFE_ES_HousekeepingTlm::Payload
```

Telemetry payload.

Definition at line 1557 of file cfe_es_msg.h.

38.15.2.2 TlmHeader

```
CFE_MSG_TelemetryHeader_t CFE_ES_HousekeepingTlm::TlmHeader
```

Telemetry header.

Definition at line 1556 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.16 CFE_ES_HousekeepingTlm_Payload Struct Reference

```
#include <cfe_es_msg.h>
```

Data Fields

- uint8 CommandCounter

  The ES Application Command Counter.
- uint8 CommandErrorCounter

  The ES Application Command Error Counter.
- uint16 CFECoreChecksum

  Checksum of cFE Core Code.
- uint8 CFEMajorVersion

  Major Version Number of cFE.
- uint8 CFEMinorVersion

  Minor Version Number of cFE.
- uint8 CFERevision

  Sub-Minor Version Number of cFE.
- uint8 CFEMissionRevision

  Mission Version Number of cFE.
- uint8 OSALMajorVersion

  OS Abstraction Layer Major Version Number.
- uint8 OSALMinorVersion

  OS Abstraction Layer Minor Version Number.
- uint8 OSALRevision

  OS Abstraction Layer Revision Number.
- uint8 OSALMissionRevision

  OS Abstraction Layer MissionRevision Number.
- CFE_ES_MemOffset_t SysLogBytesUsed

  Total number of bytes used in system log.
- CFE_ES_MemOffset_t SysLogSize

  Total size of the system log.
- uint32 SysLogEntries

  Number of entries in the system log.
- uint32 SysLogMode

  Write/Overwrite Mode.
- uint32 ERLogIndex

  Current index of the ER Log (wraps around)
- uint32 ERLogEntries

  Number of entries made in the ER Log since the power on.
- uint32 RegisteredCoreApps

  Number of Applications registered with ES.
- uint32 RegisteredExternalApps

  Number of Applications registered with ES.
- uint32 RegisteredTasks

  Number of Tasks (main AND child tasks) registered with ES.
- uint32 RegisteredLibs

  Number of Libraries registered with ES.
- uint32 ResetType

  Reset type (PROCESSOR or POWERON)
- uint32 ResetSubtype

Reset Sub Type.

- uint32 ProcessorResets

  Number of processor resets since last power on.
- uint32 MaxProcessorResets

  Max processor resets before a power on is done.
- uint32 BootSource

  Boot source (as provided from BSP)
- uint32 PerfState

  Current state of Performance Analyzer.
- uint32 PerfMode

  Current mode of Performance Analyzer.
- uint32 PerfTriggerCount

  Number of Times Perfomance Analyzer has Triggered.
- uint32 PerfFilterMask [CFE_MISSION_ES_PERF_MAX_IDS/32]

  Current Setting of Performance Analyzer Filter Masks.
- uint32 PerfTriggerMask [CFE_MISSION_ES_PERF_MAX_IDS/32]

  Current Setting of Performance Analyzer Trigger Masks.
- uint32 PerfDataStart

  Identifies First Stored Entry in Performance Analyzer Log.
- uint32 PerfDataEnd

  Identifies Last Stored Entry in Performance Analyzer Log.
- uint32 PerfDataCount

  Number of Entries Put Into the Performance Analyzer Log.
- uint32 PerfDataToWrite

  Number of Performance Analyzer Log Entries Left to be Written to Log Dump File.
- CFE_ES_MemOffset_t HeapBytesFree

  Number of free bytes remaining in the OS heap.
- CFE_ES_MemOffset_t HeapBlocksFree

  Number of free blocks remaining in the OS heap.
- CFE_ES_MemOffset_t HeapMaxBlockSize

  Number of bytes in the largest free block.

38.16.1 Detailed Description

Name Executive Services Housekeeping Packet

Definition at line 1466 of file cfe_es_msg.h.

38.16.2 Field Documentation

38.16.2.1 BootSource

`uint32 CFE_ES_HousekeepingTlm_Payload::BootSource`

Boot source (as provided from BSP)

Telemetry Mnemonic(s) $sc_$cpu_ES_BootSource

Definition at line 1523 of file cfe_es_msg.h.

38.16.2.2 CFECoreChecksum

`uint16 CFE_ES_HousekeepingTlm_Payload::CFECoreChecksum`

Checksum of cFE Core Code.

Telemetry Mnemonic(s) $sc_$cpu_ES_CKSUM

Definition at line 1473 of file cfe_es_msg.h.

38.16.2.3 CFEMajorVersion

`uint8 CFE_ES_HousekeepingTlm_Payload::CFEMajorVersion`

Major Version Number of cFE.

Telemetry Mnemonic(s) $sc_$cpu_ES_CFEMAJORVER

Definition at line 1475 of file cfe_es_msg.h.

38.16.2.4 CFEMinorVersion

`uint8 CFE_ES_HousekeepingTlm_Payload::CFEMinorVersion`

Minor Version Number of cFE.

Telemetry Mnemonic(s) $sc_$cpu_ES_CFEMINORVER

Definition at line 1477 of file cfe_es_msg.h.

38.16.2.5 CFEMissionRevision

```
uint8 CFE_ES_HousekeepingTlm_Payload::CFEMissionRevision
```

Mission Version Number of cFE.

Telemetry Mnemonic(s) $sc_$cpu_ES_CFEMISSIONREV

Definition at line 1481 of file cfe_es_msg.h.

38.16.2.6 CFERevision

```
uint8 CFE_ES_HousekeepingTlm_Payload::CFERevision
```

Sub-Minor Version Number of cFE.

Telemetry Mnemonic(s) $sc_$cpu_ES_CFEREVISION

Definition at line 1479 of file cfe_es_msg.h.

38.16.2.7 CommandCounter

```
uint8 CFE_ES_HousekeepingTlm_Payload::CommandCounter
```

The ES Application Command Counter.

Telemetry Mnemonic(s) $sc_$cpu_ES_CMDPC

Definition at line 1468 of file cfe_es_msg.h.

38.16.2.8 CommandErrorCounter

```
uint8 CFE_ES_HousekeepingTlm_Payload::CommandErrorCounter
```

The ES Application Command Error Counter.

Telemetry Mnemonic(s) $sc_$cpu_ES_CMDEC

Definition at line 1470 of file cfe_es_msg.h.

38.16.2.9 ERLogEntries

`uint32 CFE_ES_HousekeepingTlm_Payload::ERLogEntries`

Number of entries made in the ER Log since the power on.

Telemetry Mnemonic(s) $sc_$cpu_ES_ERLOGENTRIES

Definition at line 1503 of file cfe_es_msg.h.

38.16.2.10 ERLogIndex

`uint32 CFE_ES_HousekeepingTlm_Payload::ERLogIndex`

Current index of the ER Log (wraps around)

Telemetry Mnemonic(s) $sc_$cpu_ES_ERLOGINDEX

Definition at line 1501 of file cfe_es_msg.h.

38.16.2.11 HeapBlocksFree

`CFE_ES_MemOffset_t CFE_ES_HousekeepingTlm_Payload::HeapBlocksFree`

Number of free blocks remaining in the OS heap.

Telemetry Mnemonic(s) $sc_$cpu_ES_HeapBlocksFree

Definition at line 1548 of file cfe_es_msg.h.

38.16.2.12 HeapBytesFree

`CFE_ES_MemOffset_t CFE_ES_HousekeepingTlm_Payload::HeapBytesFree`

Number of free bytes remaining in the OS heap.

Telemetry Mnemonic(s) $sc_$cpu_ES_HeapBytesFree

Definition at line 1546 of file cfe_es_msg.h.

38.16.2.13 HeapMaxBlockSize

`CFE_ES_MemOffset_t CFE_ES_HousekeepingTlm_Payload::HeapMaxBlockSize`

Number of bytes in the largest free block.

Telemetry Mnemonic(s) $sc_$cpu_ES_HeapMaxBlkSize

Definition at line 1550 of file cfe_es_msg.h.

38.16.2.14 MaxProcessorResets

`uint32 CFE_ES_HousekeepingTlm_Payload::MaxProcessorResets`

Max processor resets before a power on is done.

Telemetry Mnemonic(s) $sc_$cpu_ES_MaxProcResets

Definition at line 1521 of file cfe_es_msg.h.

38.16.2.15 OSALMajorVersion

`uint8 CFE_ES_HousekeepingTlm_Payload::OSALMajorVersion`

OS Abstraction Layer Major Version Number.

Telemetry Mnemonic(s) $sc_$cpu_ES_OSMAJORVER

Definition at line 1483 of file cfe_es_msg.h.

38.16.2.16 OSALMinorVersion

`uint8 CFE_ES_HousekeepingTlm_Payload::OSALMinorVersion`

OS Abstraction Layer Minor Version Number.

Telemetry Mnemonic(s) $sc_$cpu_ES_OSMINORVER

Definition at line 1485 of file cfe_es_msg.h.

38.16.2.17 OSALMissionRevision

```
uint8 CFE_ES_HousekeepingTlm_Payload::OSALMissionRevision
```

OS Abstraction Layer MissionRevision Number.

Telemetry Mnemonic(s) $sc_$cpu_ES_OSMISSIONREV

Definition at line 1489 of file cfe_es_msg.h.

38.16.2.18 OSALRevision

```
uint8 CFE_ES_HousekeepingTlm_Payload::OSALRevision
```

OS Abstraction Layer Revision Number.

Telemetry Mnemonic(s) $sc_$cpu_ES_OSREVISION

Definition at line 1487 of file cfe_es_msg.h.

38.16.2.19 PerfDataCount

```
uint32 CFE_ES_HousekeepingTlm_Payload::PerfDataCount
```

Number of Entries Put Into the Performance Analyzer Log.

Telemetry Mnemonic(s) $sc_$cpu_ES_PerfDataCnt

Definition at line 1541 of file cfe_es_msg.h.

38.16.2.20 PerfDataEnd

```
uint32 CFE_ES_HousekeepingTlm_Payload::PerfDataEnd
```

Identifies Last Stored Entry in Performance Analyzer Log.

Telemetry Mnemonic(s) $sc_$cpu_ES_PerfDataEnd

Definition at line 1539 of file cfe_es_msg.h.

38.16.2.21 PerfDataStart

`uint32 CFE_ES_HousekeepingTlm_Payload::PerfDataStart`

Identifies First Stored Entry in Performance Analyzer Log.

Telemetry Mnemonic(s) $sc_$cpu_ES_PerfDataStart

Definition at line 1537 of file cfe_es_msg.h.

38.16.2.22 PerfDataToWrite

`uint32 CFE_ES_HousekeepingTlm_Payload::PerfDataToWrite`

Number of Performance Analyzer Log Entries Left to be Written to Log Dump File.

Telemetry Mnemonic(s) $sc_$cpu_ES_PerfData2Write

Definition at line 1544 of file cfe_es_msg.h.

38.16.2.23 PerfFilterMask

`uint32 CFE_ES_HousekeepingTlm_Payload::PerfFilterMask[CFE_MISSION_ES_PERF_MAX_IDS/32]`

Current Setting of Performance Analyzer Filter Masks.

Telemetry Mnemonic(s) $sc_$cpu_ES_PerfFltrMask[MaskCnt]

Definition at line 1532 of file cfe_es_msg.h.

38.16.2.24 PerfMode

`uint32 CFE_ES_HousekeepingTlm_Payload::PerfMode`

Current mode of Performance Analyzer.

Telemetry Mnemonic(s) $sc_$cpu_ES_PerfMode

Definition at line 1528 of file cfe_es_msg.h.

38.16.2.25 PerfState

```
uint32 CFE_ES_HousekeepingTlm_Payload::PerfState
```

Current state of Performance Analyzer.

Telemetry Mnemonic(s) $sc_$cpu_ES_PerfState

Definition at line 1526 of file cfe_es_msg.h.

38.16.2.26 PerfTriggerCount

```
uint32 CFE_ES_HousekeepingTlm_Payload::PerfTriggerCount
```

Number of Times Perfomance Analyzer has Triggered.

Telemetry Mnemonic(s) $sc_$cpu_ES_PerfTrigCnt

Definition at line 1530 of file cfe_es_msg.h.

38.16.2.27 PerfTriggerMask

```
uint32 CFE_ES_HousekeepingTlm_Payload::PerfTriggerMask[CFE_MISSION_ES_PERF_MAX_IDS/32]
```

Current Setting of Performance Analyzer Trigger Masks.

Telemetry Mnemonic(s) $sc_$cpu_ES_PerfTrigMask[MaskCnt]

Definition at line 1535 of file cfe_es_msg.h.

38.16.2.28 ProcessorResets

```
uint32 CFE_ES_HousekeepingTlm_Payload::ProcessorResets
```

Number of processor resets since last power on.

Telemetry Mnemonic(s) $sc_$cpu_ES_ProcResetCnt

Definition at line 1519 of file cfe_es_msg.h.

38.16.2.29 RegisteredCoreApps

`uint32 CFE_ES_HousekeepingTlm_Payload::RegisteredCoreApps`

Number of Applications registered with ES.

Telemetry Mnemonic(s) $sc_$cpu_ES_RegCoreApps

Definition at line 1506 of file cfe_es_msg.h.

38.16.2.30 RegisteredExternalApps

`uint32 CFE_ES_HousekeepingTlm_Payload::RegisteredExternalApps`

Number of Applications registered with ES.

Telemetry Mnemonic(s) $sc_$cpu_ES_RegExtApps

Definition at line 1508 of file cfe_es_msg.h.

38.16.2.31 RegisteredLibs

`uint32 CFE_ES_HousekeepingTlm_Payload::RegisteredLibs`

Number of Libraries registered with ES.

Telemetry Mnemonic(s) $sc_$cpu_ES_RegLibs

Definition at line 1512 of file cfe_es_msg.h.

38.16.2.32 RegisteredTasks

`uint32 CFE_ES_HousekeepingTlm_Payload::RegisteredTasks`

Number of Tasks (main AND child tasks) registered with ES.

Telemetry Mnemonic(s) $sc_$cpu_ES_RegTasks

Definition at line 1510 of file cfe_es_msg.h.

38.16.2.33 ResetSubtype

```
uint32 CFE_ES_HousekeepingTlm_Payload::ResetSubtype
```

Reset Sub Type.

Telemetry Mnemonic(s) $sc_$cpu_ES_ResetSubtype

Definition at line 1517 of file cfe_es_msg.h.

38.16.2.34 ResetType

```
uint32 CFE_ES_HousekeepingTlm_Payload::ResetType
```

Reset type (PROCESSOR or POWERON)

Telemetry Mnemonic(s) $sc_$cpu_ES_ResetType

Definition at line 1515 of file cfe_es_msg.h.

38.16.2.35 SysLogBytesUsed

```
CFE_ES_MemOffset_t CFE_ES_HousekeepingTlm_Payload::SysLogBytesUsed
```

Total number of bytes used in system log.

Telemetry Mnemonic(s) $sc_$cpu_ES_SYSLOGBYTEUSED

Definition at line 1492 of file cfe_es_msg.h.

38.16.2.36 SysLogEntries

```
uint32 CFE_ES_HousekeepingTlm_Payload::SysLogEntries
```

Number of entries in the system log.

Telemetry Mnemonic(s) $sc_$cpu_ES_SYSLOGENTRIES

Definition at line 1496 of file cfe_es_msg.h.

38.16.2.37 SysLogMode

```
uint32 CFE_ES_HousekeepingTlm_Payload::SysLogMode
```

Write/Overwrite Mode.

Telemetry Mnemonic(s) $sc_$cpu_ES_SYSLOGMODE

Definition at line 1498 of file cfe_es_msg.h.

38.16.2.38 SysLogSize

```
CFE_ES_MemOffset_t CFE_ES_HousekeepingTlm_Payload::SysLogSize
```

Total size of the system log.

Telemetry Mnemonic(s) $sc_$cpu_ES_SYSLOGSIZE

Definition at line 1494 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.17 CFE_ES_MemPoolStats Struct Reference

Memory Pool Statistics.

```
#include <cfe_es_extern_typedefs.h>
```

Data Fields

- CFE_ES_MemOffset_t PoolSize

  Size of Memory Pool (in bytes)
- uint32 NumBlocksRequested

  Number of times a memory block has been allocated.
- uint32 CheckErrCtr

  Number of errors detected when freeing a memory block.
- CFE_ES_MemOffset_t NumFreeBytes

  Number of bytes never allocated to a block.
- CFE_ES_BlockStats_t BlockStats [CFE_MISSION_ES_POOL_MAX_BUCKETS]

  Contains stats on each block size.

38.17.1 Detailed Description

Memory Pool Statistics.

Structure that is used to provide information about a memory pool. Used by the Memory Pool Stats telemetry message.

See also

CFE_ES_SEND_MEM_POOL_STATS_CC

Definition at line 558 of file cfe_es_extern_typedefs.h.

38.17.2 Field Documentation

38.17.2.1 BlockStats

```
CFE_ES_BlockStats_t CFE_ES_MemPoolStats::BlockStats[CFE_MISSION_ES_POOL_MAX_BUCKETS]
```

Contains stats on each block size.

Telemetry Mnemonic(s) $sc_$cpu_ES_BlkStats[BLK_SIZES]

Definition at line 568 of file cfe_es_extern_typedefs.h.

38.17.2.2 CheckErrCtr

```
uint32 CFE_ES_MemPoolStats::CheckErrCtr
```

Number of errors detected when freeing a memory block.

Telemetry Mnemonic(s) $sc_$cpu_ES_BlkErrCTR

Definition at line 564 of file cfe_es_extern_typedefs.h.

38.17.2.3 NumBlocksRequested

`uint32 CFE_ES_MemPoolStats::NumBlocksRequested`

Number of times a memory block has been allocated.

Telemetry Mnemonic(s) $sc_$cpu_ES_BlksREQ

Definition at line 562 of file cfe_es_extern_typedefs.h.

38.17.2.4 NumFreeBytes

`CFE_ES_MemOffset_t CFE_ES_MemPoolStats::NumFreeBytes`

Number of bytes never allocated to a block.

Telemetry Mnemonic(s) $sc_$cpu_ES_FreeBytes

Definition at line 566 of file cfe_es_extern_typedefs.h.

38.17.2.5 PoolSize

`CFE_ES_MemOffset_t CFE_ES_MemPoolStats::PoolSize`

Size of Memory Pool (in bytes)

Telemetry Mnemonic(s) $sc_$cpu_ES_PoolSize

Definition at line 560 of file cfe_es_extern_typedefs.h.

The documentation for this struct was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_es_extern_typedefs.h

38.18 CFE_ES_MemStatsTlm Struct Reference

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_TelemetryHeader_t TlmHeader

  Telemetry header.
- CFE_ES_PoolStatsTlm_Payload_t Payload

  Telemetry payload.

38.18.1 Detailed Description

Definition at line 1455 of file cfe_es_msg.h.

38.18.2 Field Documentation

38.18.2.1 Payload

```
CFE_ES_PoolStatsTlm_Payload_t CFE_ES_MemStatsTlm::Payload
```

Telemetry payload.

Definition at line 1458 of file cfe_es_msg.h.

38.18.2.2 TlmHeader

```
CFE_MSG_TelemetryHeader_t CFE_ES_MemStatsTlm::TlmHeader
```

Telemetry header.

Definition at line 1457 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.19 CFE_ES_NoArgsCmd Struct Reference

Generic "no arguments" command.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.

38.19.1 Detailed Description

Generic "no arguments" command.

This command structure is used for commands that do not have any parameters. This includes:

1. The Housekeeping Request Message
2. The No-Op Command (For details, see CFE_ES_NOOP_CC)
3. The Reset Counters Command (For details, see CFE_ES_RESET_COUNTERS_CC)

Definition at line 1074 of file cfe_es_msg.h.

38.19.2 Field Documentation

38.19.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_ES_NoArgsCmd::CmdHeader
```

Command header.

Definition at line 1076 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.20 CFE_ES_OneAppTlm Struct Reference

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_TelemetryHeader_t TlmHeader
  Telemetry header.
- CFE_ES_OneAppTlm_Payload_t Payload
  Telemetry payload.

38.20.1 Detailed Description

Definition at line 1439 of file cfe_es_msg.h.

38.20.2 Field Documentation

38.20.2.1 Payload

`CFE_ES_OneAppTlm_Payload_t CFE_ES_OneAppTlm::Payload`

Telemetry payload.

Definition at line 1442 of file cfe_es_msg.h.

38.20.2.2 TlmHeader

`CFE_MSG_TelemetryHeader_t CFE_ES_OneAppTlm::TlmHeader`

Telemetry header.

Definition at line 1441 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.21 CFE_ES_OneAppTlm_Payload Struct Reference

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_ES_AppInfo_t AppInfo
  For more information, see CFE_ES_AppInfo_t.

38.21.1 Detailed Description

Name Single Application Information Packet

Definition at line 1433 of file cfe_es_msg.h.

38.21.2 Field Documentation

38.21.2.1 AppInfo

`CFE_ES_AppInfo_t CFE_ES_OneAppTlm_Payload::AppInfo`

For more information, see CFE_ES_AppInfo_t.

Definition at line 1435 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.22 CFE_ES_OverWriteSysLogCmd Struct Reference

Overwrite/Discard System Log Configuration Command Payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_ES_OverWriteSysLogCmd_Payload_t Payload
  Command payload.

38.22.1 Detailed Description

Overwrite/Discard System Log Configuration Command Payload.

Definition at line 1161 of file cfe_es_msg.h.

38.22.2 Field Documentation

38.22.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_ES_OverWriteSysLogCmd::CmdHeader`

Command header.

Definition at line 1163 of file cfe_es_msg.h.

38.22.2.2 Payload

`CFE_ES_OverWriteSysLogCmd_Payload_t CFE_ES_OverWriteSysLogCmd::Payload`

Command payload.

Definition at line 1164 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.23 CFE_ES_OverWriteSysLogCmd_Payload Struct Reference

Overwrite/Discard System Log Configuration Command Payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- uint32 Mode

  CFE_ES_LogMode_DISCARD=Throw away most recent messages, CFE_ES_LogMode_OVERWRITE=Overwrite oldest with most recent

38.23.1 Detailed Description

Overwrite/Discard System Log Configuration Command Payload.

For command details, see CFE_ES_OVER_WRITE_SYSLOG_CC

Definition at line 1151 of file cfe_es_msg.h.

38.23.2 Field Documentation

38.23.2.1 Mode

`uint32 CFE_ES_OverWriteSysLogCmd_Payload::Mode`

CFE_ES_LogMode_DISCARD=Throw away most recent messages, CFE_ES_LogMode_OVERWRITE=Overwrite oldest with most recent

Definition at line 1153 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.24 CFE_ES_PoolAlign Union Reference

Pool Alignement.

```
#include <cfe_es_api_typedefs.h>
```

Data Fields

- void * Ptr

  Aligned pointer.
- long long int LongInt

  Aligned Long Integer.
- long double LongDouble

  Aligned Long Double.

38.24.1 Detailed Description

Pool Alignement.

Union that can be used for minimum memory alignment of ES memory pools on the target. It contains the longest native data types such that the alignment of this structure should reflect the largest possible alignment requirements for any data on this processor.

Definition at line 95 of file cfe_es_api_typedefs.h.

38.24.2 Field Documentation

38.24.2.1 LongDouble

```
long double CFE_ES_PoolAlign::LongDouble
```

Aligned Long Double.

Definition at line 100 of file cfe_es_api_typedefs.h.

38.24.2.2 LongInt

```
long long int CFE_ES_PoolAlign::LongInt
```

Aligned Long Integer.

Definition at line 99 of file cfe_es_api_typedefs.h.

38.24.2.3 Ptr

```
void* CFE_ES_PoolAlign::Ptr
```

Aligned pointer.

Definition at line 97 of file cfe_es_api_typedefs.h.

The documentation for this union was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_es_api_typedefs.h

38.25 CFE_ES_PoolStatsTlm_Payload Struct Reference

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_ES_MemHandle_t PoolHandle

  Handle of memory pool whose stats are being telemetered.
- CFE_ES_MemPoolStats_t PoolStats

  For more info, see CFE_ES_MemPoolStats_t.

38.25.1 Detailed Description

Name Memory Pool Statistics Packet

Definition at line 1448 of file cfe_es_msg.h.

38.25.2 Field Documentation

38.25.2.1 PoolHandle

```
CFE_ES_MemHandle_t CFE_ES_PoolStatsTlm_Payload::PoolHandle
```

Handle of memory pool whose stats are being telemetered.

Telemetry Mnemonic(s) $sc_$cpu_ES_PoolHandle

Definition at line 1450 of file cfe_es_msg.h.

38.25.2.2 PoolStats

```
CFE_ES_MemPoolStats_t CFE_ES_PoolStatsTlm_Payload::PoolStats
```

For more info, see CFE_ES_MemPoolStats_t.

Definition at line 1452 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.26 CFE_ES_ReloadAppCmd Struct Reference

Reload Application Command.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_ES_AppReloadCmd_Payload_t Payload
  Command payload.

38.26.1 Detailed Description

Reload Application Command.

Definition at line 1244 of file cfe_es_msg.h.

38.26.2 Field Documentation

38.26.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_ES_ReloadAppCmd::CmdHeader
```

Command header.

Definition at line 1246 of file cfe_es_msg.h.

38.26.2.2 Payload

`CFE_ES_AppReloadCmd_Payload_t CFE_ES_ReloadAppCmd::Payload`

Command payload.

Definition at line 1247 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.27 CFE_ES_RestartCmd Struct Reference

Restart cFE Command.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_ES_RestartCmd_Payload_t Payload
  Command payload.

38.27.1 Detailed Description

Restart cFE Command.

Definition at line 1107 of file cfe_es_msg.h.

38.27.2 Field Documentation

38.27.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_ES_RestartCmd::CmdHeader`

Command header.

Definition at line 1109 of file cfe_es_msg.h.

38.27.2.2 Payload

`CFE_ES_RestartCmd_Payload_t CFE_ES_RestartCmd::Payload`

Command payload.

Definition at line 1110 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.28 CFE_ES_RestartCmd_Payload Struct Reference

Restart cFE Command Payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- uint16 RestartType
  CFE_PSP_RST_TYPE_PROCESSOR=Processor Reset or CFE_PSP_RST_TYPE_POWERON=Power-On Reset

38.28.1 Detailed Description

Restart cFE Command Payload.

For command details, see CFE_ES_RESTART_CC

Definition at line 1098 of file cfe_es_msg.h.

38.28.2 Field Documentation

38.28.2.1 RestartType

`uint16 CFE_ES_RestartCmd_Payload::RestartType`

CFE_PSP_RST_TYPE_PROCESSOR=Processor Reset or CFE_PSP_RST_TYPE_POWERON=Power-On Reset

Definition at line 1100 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.29 CFE_ES_SendMemPoolStatsCmd Struct Reference

Send Memory Pool Statistics Command.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_ES_SendMemPoolStatsCmd_Payload_t Payload
  Command payload.

38.29.1 Detailed Description

Send Memory Pool Statistics Command.

Definition at line 1394 of file cfe_es_msg.h.

38.29.2 Field Documentation

38.29.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_ES_SendMemPoolStatsCmd::CmdHeader
```

Command header.

Definition at line 1396 of file cfe_es_msg.h.

38.29.2.2 Payload

```
CFE_ES_SendMemPoolStatsCmd_Payload_t CFE_ES_SendMemPoolStatsCmd::Payload
```

Command payload.

Definition at line 1397 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.30 CFE_ES_SendMemPoolStatsCmd_Payload Struct Reference

Send Memory Pool Statistics Command Payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- char Application [CFE_MISSION_MAX_API_LEN]
  - *RESERVED - should be all zeroes*
- CFE_ES_MemHandle_t PoolHandle

  Handle of Pool whose statistics are to be telemetered.

38.30.1 Detailed Description

Send Memory Pool Statistics Command Payload.

For command details, see CFE_ES_SEND_MEM_POOL_STATS_CC

Definition at line 1384 of file cfe_es_msg.h.

38.30.2 Field Documentation

38.30.2.1 Application

```
char CFE_ES_SendMemPoolStatsCmd_Payload::Application[CFE_MISSION_MAX_API_LEN]
```

- RESERVED - should be all zeroes

Definition at line 1386 of file cfe_es_msg.h.

38.30.2.2 PoolHandle

```
CFE_ES_MemHandle_t CFE_ES_SendMemPoolStatsCmd_Payload::PoolHandle
```

Handle of Pool whose statistics are to be telemetered.

Definition at line 1387 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.31 CFE_ES_SetMaxPRCountCmd Struct Reference

Set Maximum Processor Reset Count Command.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_ES_SetMaxPRCountCmd_Payload_t Payload
  Command payload.

38.31.1 Detailed Description

Set Maximum Processor Reset Count Command.

Definition at line 1265 of file cfe_es_msg.h.

38.31.2 Field Documentation

38.31.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_ES_SetMaxPRCountCmd::CmdHeader
```

Command header.

Definition at line 1267 of file cfe_es_msg.h.

38.31.2.2 Payload

```
CFE_ES_SetMaxPRCountCmd_Payload_t CFE_ES_SetMaxPRCountCmd::Payload
```

Command payload.

Definition at line 1268 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.32 CFE_ES_SetMaxPRCountCmd_Payload Struct Reference

Set Maximum Processor Reset Count Command Payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- uint16 MaxPRCount

  New maximum number of Processor Resets before an automatic Power-On Reset is performed.

38.32.1 Detailed Description

Set Maximum Processor Reset Count Command Payload.

For command details, see CFE_ES_SET_MAX_PR_COUNT_CC

Definition at line 1256 of file cfe_es_msg.h.

38.32.2 Field Documentation

38.32.2.1 MaxPRCount

```
uint16 CFE_ES_SetMaxPRCountCmd_Payload::MaxPRCount
```

New maximum number of Processor Resets before an automatic Power-On Reset is performed.

Definition at line 1258 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.33 CFE_ES_SetPerfFilterMaskCmd Struct Reference

Set Performance Analyzer Filter Mask Command.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_ES_SetPerfFilterMaskCmd_Payload_t Payload

  Command payload.

38.33.1 Detailed Description

Set Performance Analyzer Filter Mask Command.

Definition at line 1350 of file cfe_es_msg.h.

38.33.2 Field Documentation

38.33.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_ES_SetPerfFilterMaskCmd::CmdHeader
```

Command header.

Definition at line 1352 of file cfe_es_msg.h.

38.33.2.2 Payload

```
CFE_ES_SetPerfFilterMaskCmd_Payload_t CFE_ES_SetPerfFilterMaskCmd::Payload
```

Command payload.

Definition at line 1353 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.34 CFE_ES_SetPerfFilterMaskCmd_Payload Struct Reference

Set Performance Analyzer Filter Mask Command Payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- uint32 FilterMaskNum

  Index into array of Filter Masks.
- uint32 FilterMask

  New Mask for specified entry in array of Filter Masks.

38.34.1 Detailed Description

Set Performance Analyzer Filter Mask Command Payload.

For command details, see CFE_ES_SET_PERF_FILTER_MASK_CC

Definition at line 1340 of file cfe_es_msg.h.

38.34.2 Field Documentation

38.34.2.1 FilterMask

```
uint32 CFE_ES_SetPerfFilterMaskCmd_Payload::FilterMask
```

New Mask for specified entry in array of Filter Masks.

Definition at line 1343 of file cfe_es_msg.h.

38.34.2.2 FilterMaskNum

```
uint32 CFE_ES_SetPerfFilterMaskCmd_Payload::FilterMaskNum
```

Index into array of Filter Masks.

Definition at line 1342 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.35 CFE_ES_SetPerfTriggerMaskCmd Struct Reference

Set Performance Analyzer Trigger Mask Command.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_ES_SetPerfTrigMaskCmd_Payload_t Payload

  Command payload.

38.35.1 Detailed Description

Set Performance Analyzer Trigger Mask Command.

Definition at line 1372 of file cfe_es_msg.h.

38.35.2 Field Documentation

38.35.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_ES_SetPerfTriggerMaskCmd::CmdHeader`

Command header.

Definition at line 1374 of file cfe_es_msg.h.

38.35.2.2 Payload

`CFE_ES_SetPerfTrigMaskCmd_Payload_t CFE_ES_SetPerfTriggerMaskCmd::Payload`

Command payload.

Definition at line 1375 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.36 CFE_ES_SetPerfTrigMaskCmd_Payload Struct Reference

Set Performance Analyzer Trigger Mask Command Payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- uint32 TriggerMaskNum

  Index into array of Trigger Masks.
- uint32 TriggerMask

  New Mask for specified entry in array of Trigger Masks.

38.36.1 Detailed Description

Set Performance Analyzer Trigger Mask Command Payload.

For command details, see CFE_ES_SET_PERF_TRIGGER_MASK_CC

Definition at line 1362 of file cfe_es_msg.h.

38.36.2 Field Documentation

38.36.2.1 TriggerMask

```
uint32 CFE_ES_SetPerfTrigMaskCmd_Payload::TriggerMask
```

New Mask for specified entry in array of Trigger Masks.

Definition at line 1365 of file cfe_es_msg.h.

38.36.2.2 TriggerMaskNum

```
uint32 CFE_ES_SetPerfTrigMaskCmd_Payload::TriggerMaskNum
```

Index into array of Trigger Masks.

Definition at line 1364 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.37 CFE_ES_StartApp Struct Reference

Start Application Command.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_ES_StartAppCmd_Payload_t Payload

  Command payload.

38.37.1 Detailed Description

Start Application Command.

Definition at line 1193 of file cfe_es_msg.h.

38.37.2 Field Documentation

38.37.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_ES_StartApp::CmdHeader`

Command header.

Definition at line 1195 of file cfe_es_msg.h.

38.37.2.2 Payload

`CFE_ES_StartAppCmd_Payload_t CFE_ES_StartApp::Payload`

Command payload.

Definition at line 1196 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.38 CFE_ES_StartAppCmd_Payload Struct Reference

Start Application Command Payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- char Application [CFE_MISSION_MAX_API_LEN]

  Name of Application to be started.
- char AppEntryPoint [CFE_MISSION_MAX_API_LEN]

  Symbolic name of Application's entry point.
- char AppFileName [CFE_MISSION_MAX_PATH_LEN]

  Full path and filename of Application's executable image.
- CFE_ES_MemOffset_t StackSize

  Desired stack size for the new application.
- CFE_ES_ExceptionAction_Enum_t ExceptionAction

  CFE_ES_ExceptionAction_RESTART_APP=On exception, restart Application, CFE_ES_ExceptionAction_PROC_RE↩START=On exception, perform a Processor Reset
- CFE_ES_TaskPriority_Atom_t Priority

  The new Applications runtime priority.

38.38.1 Detailed Description

Start Application Command Payload.

For command details, see CFE_ES_START_APP_CC

Definition at line 1173 of file cfe_es_msg.h.

38.38.2 Field Documentation

38.38.2.1 AppEntryPoint

```
char CFE_ES_StartAppCmd_Payload::AppEntryPoint[CFE_MISSION_MAX_API_LEN]
```

Symbolic name of Application's entry point.

Definition at line 1176 of file cfe_es_msg.h.

38.38.2.2 AppFileName

```
char CFE_ES_StartAppCmd_Payload::AppFileName[CFE_MISSION_MAX_PATH_LEN]
```

Full path and filename of Application's executable image.

Definition at line 1177 of file cfe_es_msg.h.

38.38.2.3 Application

```
char CFE_ES_StartAppCmd_Payload::Application[CFE_MISSION_MAX_API_LEN]
```

Name of Application to be started.

Definition at line 1175 of file cfe_es_msg.h.

38.38.2.4 ExceptionAction

```
CFE_ES_ExceptionAction_Enum_t CFE_ES_StartAppCmd_Payload::ExceptionAction
```

CFE_ES_ExceptionAction_RESTART_APP=On exception, restart Application, CFE_ES_ExceptionAction_PROC_R↩ESTART=On exception, perform a Processor Reset

Definition at line 1182 of file cfe_es_msg.h.

38.38.2.5 Priority

```
CFE_ES_TaskPriority_Atom_t CFE_ES_StartAppCmd_Payload::Priority
```

The new Applications runtime priority.

Definition at line 1186 of file cfe_es_msg.h.

38.38.2.6 StackSize

```
CFE_ES_MemOffset_t CFE_ES_StartAppCmd_Payload::StackSize
```

Desired stack size for the new application.

Definition at line 1180 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.39 CFE_ES_StartPerfCmd_Payload Struct Reference

Start Performance Analyzer Command Payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- uint32 TriggerMode

  Desired trigger position (Start, Center, End)

38.39.1 Detailed Description

Start Performance Analyzer Command Payload.

For command details, see CFE_ES_START_PERF_DATA_CC

Definition at line 1299 of file cfe_es_msg.h.

38.39.2 Field Documentation

38.39.2.1 TriggerMode

```
uint32 CFE_ES_StartPerfCmd_Payload::TriggerMode
```

Desired trigger position (Start, Center, End)

Definition at line 1301 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.40 CFE_ES_StartPerfDataCmd Struct Reference

Start Performance Analyzer Command.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_ES_StartPerfCmd_Payload_t Payload

  Command payload.

38.40.1 Detailed Description

Start Performance Analyzer Command.

Definition at line 1307 of file cfe_es_msg.h.

38.40.2 Field Documentation

38.40.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_ES_StartPerfDataCmd::CmdHeader`

Command header.

Definition at line 1309 of file cfe_es_msg.h.

38.40.2.2 Payload

`CFE_ES_StartPerfCmd_Payload_t CFE_ES_StartPerfDataCmd::Payload`

Command payload.

Definition at line 1310 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.41 CFE_ES_StopPerfCmd_Payload Struct Reference

Stop Performance Analyzer Command Payload.

```
#include <cfe_es_msg.h>
```

Data Fields

- char DataFileName [CFE_MISSION_MAX_PATH_LEN]

  ASCII text string of full path and filename of file Performance Analyzer data is to be written.

38.41.1 Detailed Description

Stop Performance Analyzer Command Payload.

For command details, see CFE_ES_STOP_PERF_DATA_CC

Definition at line 1319 of file cfe_es_msg.h.

38.41.2 Field Documentation

38.41.2.1 DataFileName

```
char CFE_ES_StopPerfCmd_Payload::DataFileName[CFE_MISSION_MAX_PATH_LEN]
```

ASCII text string of full path and filename of file Performance Analyzer data is to be written.

Definition at line 1321 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.42 CFE_ES_StopPerfDataCmd Struct Reference

Stop Performance Analyzer Command.

```
#include <cfe_es_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_ES_StopPerfCmd_Payload_t Payload
  Command payload.

38.42.1 Detailed Description

Stop Performance Analyzer Command.

Definition at line 1328 of file cfe_es_msg.h.

38.42.2 Field Documentation

38.42.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_ES_StopPerfDataCmd::CmdHeader`

Command header.

Definition at line 1330 of file cfe_es_msg.h.

38.42.2.2 Payload

`CFE_ES_StopPerfCmd_Payload_t CFE_ES_StopPerfDataCmd::Payload`

Command payload.

Definition at line 1331 of file cfe_es_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/es/fsw/inc/cfe_es_msg.h

38.43 CFE_ES_TaskInfo Struct Reference

Task Information.

```
#include <cfe_es_extern_typedefs.h>
```

Data Fields

- CFE_ES_TaskId_t TaskId

  Task Id.
- uint32 ExecutionCounter

  Task Execution Counter.
- char TaskName [CFE_MISSION_MAX_API_LEN]

  Task Name.
- CFE_ES_AppId_t AppId

  Parent Application ID.
- char AppName [CFE_MISSION_MAX_API_LEN]

  Parent Application Name.
- CFE_ES_MemOffset_t StackSize
- CFE_ES_TaskPriority_Atom_t Priority
- uint8 Spare [2]

38.43.1 Detailed Description

Task Information.

Structure that is used to provide information about a task. It is primarily used for the Query All Tasks (CFE_ES_QUE↩RY_ALL_TASKS_CC) command.

Note

There is not currently a telemetry message directly containing this data structure, but it does define the format of the data file generated by the Query All Tasks command. Therefore it should be considered part of the overall telemetry interface.

Definition at line 504 of file cfe_es_extern_typedefs.h.

38.43.2 Field Documentation

38.43.2.1 AppId

```
CFE_ES_AppId_t CFE_ES_TaskInfo::AppId
```

Parent Application ID.

Definition at line 509 of file cfe_es_extern_typedefs.h.

38.43.2.2 AppName

```
char CFE_ES_TaskInfo::AppName[CFE_MISSION_MAX_API_LEN]
```

Parent Application Name.

Definition at line 510 of file cfe_es_extern_typedefs.h.

38.43.2.3 ExecutionCounter

```
uint32 CFE_ES_TaskInfo::ExecutionCounter
```

Task Execution Counter.

Definition at line 507 of file cfe_es_extern_typedefs.h.

38.43.2.4 Priority

```
CFE_ES_TaskPriority_Atom_t CFE_ES_TaskInfo::Priority
```

Priority of task

Definition at line 512 of file cfe_es_extern_typedefs.h.

38.43.2.5 Spare

```
uint8 CFE_ES_TaskInfo::Spare[2]
```

Spare bytes for alignment

Definition at line 513 of file cfe_es_extern_typedefs.h.

38.43.2.6 StackSize

```
CFE_ES_MemOffset_t CFE_ES_TaskInfo::StackSize
```

Size of task stack

Definition at line 511 of file cfe_es_extern_typedefs.h.

38.43.2.7 TaskId

```
CFE_ES_TaskId_t CFE_ES_TaskInfo::TaskId
```

Task Id.

Definition at line 506 of file cfe_es_extern_typedefs.h.

38.43.2.8 TaskName

```
char CFE_ES_TaskInfo::TaskName[CFE_MISSION_MAX_API_LEN]
```

Task Name.

Definition at line 508 of file cfe_es_extern_typedefs.h.

The documentation for this struct was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_es_extern_typedefs.h

38.44 CFE_EVS_AppDataCmd_Payload Struct Reference

Write Event Services Application Information to File Command Payload.

```
#include <cfe_evs_msg.h>
```

Data Fields

- char AppDataFilename [CFE_MISSION_MAX_PATH_LEN]

  Filename where applicaton data is to be written.

38.44.1 Detailed Description

Write Event Services Application Information to File Command Payload.

For command details, see CFE_EVS_WRITE_APP_DATA_FILE_CC

Definition at line 961 of file cfe_evs_msg.h.

38.44.2 Field Documentation

38.44.2.1 AppDataFilename

```
char CFE_EVS_AppDataCmd_Payload::AppDataFilename[CFE_MISSION_MAX_PATH_LEN]
```

Filename where applicaton data is to be written.

Definition at line 963 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.45 CFE_EVS_AppNameBitMaskCmd Struct Reference

Generic App Name and Bitmask Command.

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_EVS_AppNameBitMaskCmd_Payload_t Payload

  Command payload.

38.45.1 Detailed Description

Generic App Name and Bitmask Command.

Definition at line 1125 of file cfe_evs_msg.h.

38.45.2 Field Documentation

38.45.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_EVS_AppNameBitMaskCmd::CmdHeader`

Command header.

Definition at line 1127 of file cfe_evs_msg.h.

38.45.2.2 Payload

`CFE_EVS_AppNameBitMaskCmd_Payload_t CFE_EVS_AppNameBitMaskCmd::Payload`

Command payload.

Definition at line 1128 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.46 CFE_EVS_AppNameBitMaskCmd_Payload Struct Reference

Generic App Name and Bitmask Command Payload.

```
#include <cfe_evs_msg.h>
```

Data Fields

- char AppName [CFE_MISSION_MAX_API_LEN]

  Application name to use in the command.
- uint8 BitMask

  BitMask to use in the command.
- uint8 Spare

  Pad to even byte.

38.46.1 Detailed Description

Generic App Name and Bitmask Command Payload.

For command details, see CFE_EVS_ENABLE_APP_EVENT_TYPE_CC and/or CFE_EVS_DISABLE_APP_EVENT↩ _TYPE_CC

Definition at line 1115 of file cfe_evs_msg.h.

38.46.2 Field Documentation

38.46.2.1 AppName

```
char CFE_EVS_AppNameBitMaskCmd_Payload::AppName[CFE_MISSION_MAX_API_LEN]
```

Application name to use in the command.

Definition at line 1117 of file cfe_evs_msg.h.

38.46.2.2 BitMask

```
uint8 CFE_EVS_AppNameBitMaskCmd_Payload::BitMask
```

BitMask to use in the command.

Definition at line 1118 of file cfe_evs_msg.h.

38.46.2.3 Spare

```
uint8 CFE_EVS_AppNameBitMaskCmd_Payload::Spare
```

Pad to even byte.

Definition at line 1119 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.47 CFE_EVS_AppNameCmd Struct Reference

Generic App Name Command.

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_EVS_AppNameCmd_Payload_t Payload
  Command payload.

38.47.1 Detailed Description

Generic App Name Command.

Definition at line 1064 of file cfe_evs_msg.h.

38.47.2 Field Documentation

38.47.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_EVS_AppNameCmd::CmdHeader
```

Command header.

Definition at line 1066 of file cfe_evs_msg.h.

38.47.2.2 Payload

```
CFE_EVS_AppNameCmd_Payload_t CFE_EVS_AppNameCmd::Payload
```

Command payload.

Definition at line 1067 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.48 CFE_EVS_AppNameCmd_Payload Struct Reference

Generic App Name Command Payload.

```
#include <cfe_evs_msg.h>
```

Data Fields

- char AppName [CFE_MISSION_MAX_API_LEN]
  Application name to use in the command.

38.48.1 Detailed Description

Generic App Name Command Payload.

For command details, see CFE_EVS_ENABLE_APP_EVENTS_CC, CFE_EVS_DISABLE_APP_EVENTS_CC, CFE↩_EVS_RESET_APP_COUNTER_CC and/or CFE_EVS_RESET_ALL_FILTERS_CC

Definition at line 1056 of file cfe_evs_msg.h.

38.48.2 Field Documentation

38.48.2.1 AppName

```
char CFE_EVS_AppNameCmd_Payload::AppName[CFE_MISSION_MAX_API_LEN]
```

Application name to use in the command.

Definition at line 1058 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.49 CFE_EVS_AppNameEventIDCmd Struct Reference

Generic App Name and Event ID Command.

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_EVS_AppNameEventIDCmd_Payload_t Payload
  Command payload.

38.49.1 Detailed Description

Generic App Name and Event ID Command.

Definition at line 1095 of file cfe_evs_msg.h.

38.49.2 Field Documentation

38.49.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_EVS_AppNameEventIDCmd::CmdHeader
```

Command header.

Definition at line 1097 of file cfe_evs_msg.h.

38.49.2.2 Payload

```
CFE_EVS_AppNameEventIDCmd_Payload_t CFE_EVS_AppNameEventIDCmd::Payload
```

Command payload.

Definition at line 1098 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.50 CFE_EVS_AppNameEventIDCmd_Payload Struct Reference

Generic App Name and Event ID Command Payload.

```
#include <cfe_evs_msg.h>
```

Data Fields

- char AppName [CFE_MISSION_MAX_API_LEN]

  Application name to use in the command.
- uint16 EventID

  Event ID to use in the command.

38.50.1 Detailed Description

Generic App Name and Event ID Command Payload.

For command details, see CFE_EVS_RESET_FILTER_CC and CFE_EVS_DELETE_EVENT_FILTER_CC

Definition at line 1086 of file cfe_evs_msg.h.

38.50.2 Field Documentation

38.50.2.1 AppName

```
char CFE_EVS_AppNameEventIDCmd_Payload::AppName[CFE_MISSION_MAX_API_LEN]
```

Application name to use in the command.

Definition at line 1088 of file cfe_evs_msg.h.

38.50.2.2 EventID

```
uint16 CFE_EVS_AppNameEventIDCmd_Payload::EventID
```

Event ID to use in the command.

Definition at line 1089 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.51 CFE_EVS_AppNameEventIDMaskCmd Struct Reference

Generic App Name, Event ID, Mask Command.

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_EVS_AppNameEventIDMaskCmd_Payload_t Payload
  Command payload.

38.51.1 Detailed Description

Generic App Name, Event ID, Mask Command.

Definition at line 1156 of file cfe_evs_msg.h.

38.51.2 Field Documentation

38.51.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_EVS_AppNameEventIDMaskCmd::CmdHeader
```

Command header.

Definition at line 1158 of file cfe_evs_msg.h.

38.51.2.2 Payload

```
CFE_EVS_AppNameEventIDMaskCmd_Payload_t CFE_EVS_AppNameEventIDMaskCmd::Payload
```

Command payload.

Definition at line 1159 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.52 CFE_EVS_AppNameEventIDMaskCmd_Payload Struct Reference

Generic App Name, Event ID, Mask Command Payload.

```
#include <cfe_evs_msg.h>
```

Data Fields

- char AppName [CFE_MISSION_MAX_API_LEN]

  Application name to use in the command.
- uint16 EventID

  Event ID to use in the command.
- uint16 Mask

  Mask to use in the command.

38.52.1 Detailed Description

Generic App Name, Event ID, Mask Command Payload.

For command details, see CFE_EVS_SET_FILTER_CC, CFE_EVS_ADD_EVENT_FILTER_CC and/or CFE_EVS_↩ DELETE_EVENT_FILTER_CC

Definition at line 1146 of file cfe_evs_msg.h.

38.52.2 Field Documentation

38.52.2.1 AppName

```
char CFE_EVS_AppNameEventIDMaskCmd_Payload::AppName[CFE_MISSION_MAX_API_LEN]
```

Application name to use in the command.

Definition at line 1148 of file cfe_evs_msg.h.

38.52.2.2 EventID

```
uint16 CFE_EVS_AppNameEventIDMaskCmd_Payload::EventID
```

Event ID to use in the command.

Definition at line 1149 of file cfe_evs_msg.h.

38.52.2.3 Mask

```
uint16 CFE_EVS_AppNameEventIDMaskCmd_Payload::Mask
```

Mask to use in the command.

Definition at line 1150 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.53 CFE_EVS_AppTlmData Struct Reference

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_ES_AppId_t AppID
  Numerical application identifier.
- uint16 AppMessageSentCounter
  Application message sent counter.
- uint8 AppEnableStatus
  Application event service enable status.
- uint8 Padding
  Padding for 32 bit boundary.

38.53.1 Detailed Description

Definition at line 1174 of file cfe_evs_msg.h.

38.53.2 Field Documentation

38.53.2.1 AppEnableStatus

```
uint8 CFE_EVS_AppTlmData::AppEnableStatus
```

Application event service enable status.

Telemetry Mnemonic(s) $sc_$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].APPENASTAT

Definition at line 1180 of file cfe_evs_msg.h.

38.53.2.2 AppID

```
CFE_ES_AppId_t CFE_EVS_AppTlmData::AppID
```

Numerical application identifier.

Telemetry Mnemonic(s) $sc_$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].APPID

Definition at line 1176 of file cfe_evs_msg.h.

38.53.2.3 AppMessageSentCounter

```
uint16 CFE_EVS_AppTlmData::AppMessageSentCounter
```

Application message sent counter.

Telemetry Mnemonic(s) $sc_$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].APPMSGSENTC

Definition at line 1178 of file cfe_evs_msg.h.

38.53.2.4 Padding

```
uint8 CFE_EVS_AppTlmData::Padding
```

Padding for 32 bit boundary.

Telemetry Mnemonic(s) $sc_$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].SPARE2ALIGN3

Definition at line 1182 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.54 CFE_EVS_BinFilter Struct Reference

Event message filter defintion structure.

```
#include <cfe_evs_api_typedefs.h>
```

Data Fields

- uint16 EventID

  Numerical event identifier.
- uint16 Mask

  Binary filter mask value.

38.54.1 Detailed Description

Event message filter defintion structure.

Definition at line 60 of file cfe_evs_api_typedefs.h.

38.54.2 Field Documentation

38.54.2.1 EventID

```
uint16 CFE_EVS_BinFilter::EventID
```

Numerical event identifier.

Definition at line 62 of file cfe_evs_api_typedefs.h.

38.54.2.2 Mask

```
uint16 CFE_EVS_BinFilter::Mask
```

Binary filter mask value.

Definition at line 63 of file cfe_evs_api_typedefs.h.

The documentation for this struct was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_evs_api_typedefs.h

38.55 CFE_EVS_BitMaskCmd Struct Reference

Generic Bitmask Command.

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_EVS_BitMaskCmd_Payload_t Payload

  Command payload.

38.55.1 Detailed Description

Generic Bitmask Command.

Definition at line 1033 of file cfe_evs_msg.h.

38.55.2 Field Documentation

38.55.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_EVS_BitMaskCmd::CmdHeader`

Command header.

Definition at line 1035 of file cfe_evs_msg.h.

38.55.2.2 Payload

`CFE_EVS_BitMaskCmd_Payload_t CFE_EVS_BitMaskCmd::Payload`

Command payload.

Definition at line 1036 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.56 CFE_EVS_BitMaskCmd_Payload Struct Reference

Generic Bitmask Command Payload.

```
#include <cfe_evs_msg.h>
```

Data Fields

- uint8 BitMask

  BitMask to use in the command.
- uint8 Spare

  Pad to even byte.

38.56.1 Detailed Description

Generic Bitmask Command Payload.

For command details, see CFE_EVS_ENABLE_EVENT_TYPE_CC, CFE_EVS_DISABLE_EVENT_TYPE_CC, CFE_EVS_ENABLE_PORTS_CC and/or CFE_EVS_DISABLE_PORTS_CC

Definition at line 1024 of file cfe_evs_msg.h.

38.56.2 Field Documentation

38.56.2.1 BitMask

```
uint8 CFE_EVS_BitMaskCmd_Payload::BitMask
```

BitMask to use in the command.

Definition at line 1026 of file cfe_evs_msg.h.

38.56.2.2 Spare

```
uint8 CFE_EVS_BitMaskCmd_Payload::Spare
```

Pad to even byte.

Definition at line 1027 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.57 CFE_EVS_HousekeepingTlm Struct Reference

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_MSG_TelemetryHeader_t TlmHeader

  Telemetry header.
- CFE_EVS_HousekeepingTlm_Payload_t Payload

  Telemetry payload.

38.57.1 Detailed Description

Definition at line 1229 of file cfe_evs_msg.h.

38.57.2 Field Documentation

38.57.2.1 Payload

`CFE_EVS_HousekeepingTlm_Payload_t CFE_EVS_HousekeepingTlm::Payload`

Telemetry payload.

Definition at line 1232 of file cfe_evs_msg.h.

38.57.2.2 TlmHeader

`CFE_MSG_TelemetryHeader_t CFE_EVS_HousekeepingTlm::TlmHeader`

Telemetry header.

Definition at line 1231 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.58 CFE_EVS_HousekeepingTlm_Payload Struct Reference

```
#include <cfe_evs_msg.h>
```

Data Fields

- uint8 CommandCounter

  EVS Command Counter.
- uint8 CommandErrorCounter

  EVS Command Error Counter.
- uint8 MessageFormatMode

  Event message format mode (short/long)
- uint8 MessageTruncCounter

  Event message truncation counter.
- uint8 UnregisteredAppCounter

  Unregistered application message send counter.
- uint8 OutputPort

  Output port mask.
- uint8 LogFullFlag

  Local event log full flag.
- uint8 LogMode

  Local event logging mode (overwrite/discard)
- uint16 MessageSendCounter

  Event message send counter.
- uint16 LogOverflowCounter

  Local event log overflow counter.
- uint8 LogEnabled

  Current event log enable/disable state.
- uint8 Spare1

  Padding for 32 bit boundary.
- uint8 Spare2

  Padding for 32 bit boundary.
- uint8 Spare3

  Padding for 32 bit boundary.
- CFE_EVS_AppTlmData_t AppData [CFE_MISSION_ES_MAX_APPLICATIONS]

  Array of registered application table data.

38.58.1 Detailed Description

Name Event Services Housekeeping Telemetry Packet

Definition at line 1190 of file cfe_evs_msg.h.

38.58.2 Field Documentation

38.58.2.1 AppData

```
CFE_EVS_AppTlmData_t CFE_EVS_HousekeepingTlm_Payload::AppData[CFE_MISSION_ES_MAX_APPLICATIONS]
```

Array of registered application table data.

Telemetry Mnemonic(s) $sc_$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS]

Definition at line 1224 of file cfe_evs_msg.h.

38.58.2.2 CommandCounter

```
uint8 CFE_EVS_HousekeepingTlm_Payload::CommandCounter
```

EVS Command Counter.

Telemetry Mnemonic(s) $sc_$cpu_EVS_CMDPC

Definition at line 1192 of file cfe_evs_msg.h.

38.58.2.3 CommandErrorCounter

```
uint8 CFE_EVS_HousekeepingTlm_Payload::CommandErrorCounter
```

EVS Command Error Counter.

Telemetry Mnemonic(s) $sc_$cpu_EVS_CMDEC

Definition at line 1194 of file cfe_evs_msg.h.

38.58.2.4 LogEnabled

```
uint8 CFE_EVS_HousekeepingTlm_Payload::LogEnabled
```

Current event log enable/disable state.

Telemetry Mnemonic(s) $sc_$cpu_EVS_LOGENABLED

Definition at line 1215 of file cfe_evs_msg.h.

38.58.2.5 LogFullFlag

```
uint8 CFE_EVS_HousekeepingTlm_Payload::LogFullFlag
```

Local event log full flag.

Telemetry Mnemonic(s) $sc_$cpu_EVS_LOGFULL

Definition at line 1205 of file cfe_evs_msg.h.

38.58.2.6 LogMode

```
uint8 CFE_EVS_HousekeepingTlm_Payload::LogMode
```

Local event logging mode (overwrite/discard)

Telemetry Mnemonic(s) $sc_$cpu_EVS_LOGMODE

Definition at line 1207 of file cfe_evs_msg.h.

38.58.2.7 LogOverflowCounter

```
uint16 CFE_EVS_HousekeepingTlm_Payload::LogOverflowCounter
```

Local event log overflow counter.

Telemetry Mnemonic(s) $sc_$cpu_EVS_LOGOVERFLOWC

Definition at line 1212 of file cfe_evs_msg.h.

38.58.2.8 MessageFormatMode

```
uint8 CFE_EVS_HousekeepingTlm_Payload::MessageFormatMode
```

Event message format mode (short/long)

Telemetry Mnemonic(s) $sc_$cpu_EVS_MSGFMTMODE

Definition at line 1196 of file cfe_evs_msg.h.

38.58.2.9 MessageSendCounter

`uint16 CFE_EVS_HousekeepingTlm_Payload::MessageSendCounter`

Event message send counter.

Telemetry Mnemonic(s) $sc_$cpu_EVS_MSGSENTC

Definition at line 1210 of file cfe_evs_msg.h.

38.58.2.10 MessageTruncCounter

`uint8 CFE_EVS_HousekeepingTlm_Payload::MessageTruncCounter`

Event message truncation counter.

Telemetry Mnemonic(s) $sc_$cpu_EVS_MSGTRUNC

Definition at line 1198 of file cfe_evs_msg.h.

38.58.2.11 OutputPort

`uint8 CFE_EVS_HousekeepingTlm_Payload::OutputPort`

Output port mask.

Telemetry Mnemonic(s) $sc_$cpu_EVS_OUTPUTPORT

Definition at line 1203 of file cfe_evs_msg.h.

38.58.2.12 Spare1

`uint8 CFE_EVS_HousekeepingTlm_Payload::Spare1`

Padding for 32 bit boundary.

Telemetry Mnemonic(s) $sc_$cpu_EVS_HK_SPARE1

Definition at line 1217 of file cfe_evs_msg.h.

38.58.2.13 Spare2

```
uint8 CFE_EVS_HousekeepingTlm_Payload::Spare2
```

Padding for 32 bit boundary.

Telemetry Mnemonic(s) $sc_$cpu_EVS_HK_SPARE2

Definition at line 1219 of file cfe_evs_msg.h.

38.58.2.14 Spare3

```
uint8 CFE_EVS_HousekeepingTlm_Payload::Spare3
```

Padding for 32 bit boundary.

Telemetry Mnemonic(s) $sc_$cpu_EVS_HK_SPARE3

Definition at line 1221 of file cfe_evs_msg.h.

38.58.2.15 UnregisteredAppCounter

```
uint8 CFE_EVS_HousekeepingTlm_Payload::UnregisteredAppCounter
```

Unregistered application message send counter.

Telemetry Mnemonic(s) $sc_$cpu_EVS_UNREGAPPC

Definition at line 1201 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.59 CFE_EVS_LogFileCmd_Payload Struct Reference

Write Event Log to File Command Payload.

```
#include <cfe_evs_msg.h>
```

Data Fields

- char LogFilename [CFE_MISSION_MAX_PATH_LEN]

  Filename where log data is to be written.

38.59.1 Detailed Description

Write Event Log to File Command Payload.

For command details, see CFE_EVS_WRITE_LOG_DATA_FILE_CC

Definition at line 941 of file cfe_evs_msg.h.

38.59.2 Field Documentation

38.59.2.1 LogFilename

```
char CFE_EVS_LogFileCmd_Payload::LogFilename[CFE_MISSION_MAX_PATH_LEN]
```

Filename where log data is to be written.

Definition at line 943 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.60 CFE_EVS_LongEventTlm Struct Reference

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_MSG_TelemetryHeader_t TlmHeader

  Telemetry header.
- CFE_EVS_LongEventTlm_Payload_t Payload

  Telemetry payload.

38.60.1 Detailed Description

Definition at line 1275 of file cfe_evs_msg.h.

38.60.2 Field Documentation

38.60.2.1 Payload

`CFE_EVS_LongEventTlm_Payload_t CFE_EVS_LongEventTlm::Payload`

Telemetry payload.

Definition at line 1278 of file cfe_evs_msg.h.

38.60.2.2 TlmHeader

`CFE_MSG_TelemetryHeader_t CFE_EVS_LongEventTlm::TlmHeader`

Telemetry header.

Definition at line 1277 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.61 CFE_EVS_LongEventTlm_Payload Struct Reference

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_EVS_PacketID_t PacketID

  Event packet information.
- char Message [CFE_MISSION_EVS_MAX_MESSAGE_LENGTH]

  Event message string.
- uint8 Spare1

  Structure padding.
- uint8 Spare2

  Structure padding.

38.61.1 Detailed Description

Name Event Message Telemetry Packet (Long format)

Definition at line 1255 of file cfe_evs_msg.h.

38.61.2 Field Documentation

38.61.2.1 Message

`char CFE_EVS_LongEventTlm_Payload::Message[CFE_MISSION_EVS_MAX_MESSAGE_LENGTH]`

Event message string.

Telemetry Mnemonic(s) $sc_$cpu_EVS_EVENT[CFE_MISSION_EVS_MAX_MESSAGE_LENGTH]

Definition at line 1258 of file cfe_evs_msg.h.

38.61.2.2 PacketID

`CFE_EVS_PacketID_t CFE_EVS_LongEventTlm_Payload::PacketID`

Event packet information.

Definition at line 1257 of file cfe_evs_msg.h.

38.61.2.3 Spare1

`uint8 CFE_EVS_LongEventTlm_Payload::Spare1`

Structure padding.

Telemetry Mnemonic(s) $sc_$cpu_EVS_SPARE1

Definition at line 1260 of file cfe_evs_msg.h.

38.61.2.4 Spare2

`uint8 CFE_EVS_LongEventTlm_Payload::Spare2`

Structure padding.

Telemetry Mnemonic(s) $sc_$cpu_EVS_SPARE2

Definition at line 1262 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.62 CFE_EVS_NoArgsCmd Struct Reference

Command with no additional arguments.

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.

38.62.1 Detailed Description

Command with no additional arguments.

Definition at line 921 of file cfe_evs_msg.h.

38.62.2 Field Documentation

38.62.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_EVS_NoArgsCmd::CmdHeader
```

Command header.

Definition at line 923 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.63 CFE_EVS_PacketID Struct Reference

```
#include <cfe_evs_msg.h>
```

Data Fields

- char AppName [CFE_MISSION_MAX_API_LEN]
  Application name.
- uint16 EventID
  Numerical event identifier.
- uint16 EventType
  Numerical event type identifier.
- uint32 SpacecraftID
  Spacecraft identifier.
- uint32 ProcessorID
  Numerical processor identifier.

38.63.1 Detailed Description

Telemetry packet structures

Definition at line 1237 of file cfe_evs_msg.h.

38.63.2 Field Documentation

38.63.2.1 AppName

```
char CFE_EVS_PacketID::AppName[CFE_MISSION_MAX_API_LEN]
```

Application name.

Telemetry Mnemonic(s) $sc_$cpu_EVS_APPNAME[OS_MAX_API_NAME]

Definition at line 1239 of file cfe_evs_msg.h.

38.63.2.2 EventID

```
uint16 CFE_EVS_PacketID::EventID
```

Numerical event identifier.

Telemetry Mnemonic(s) $sc_$cpu_EVS_EVENTID

Definition at line 1241 of file cfe_evs_msg.h.

38.63.2.3 EventType

```
uint16 CFE_EVS_PacketID::EventType
```

Numerical event type identifier.

Telemetry Mnemonic(s) $sc_$cpu_EVS_EVENTTYPE

Definition at line 1243 of file cfe_evs_msg.h.

38.63.2.4 ProcessorID

```
uint32 CFE_EVS_PacketID::ProcessorID
```

Numerical processor identifier.

Telemetry Mnemonic(s) $sc_$cpu_EVS_PROCESSORID

Definition at line 1247 of file cfe_evs_msg.h.

38.63.2.5 SpacecraftID

```
uint32 CFE_EVS_PacketID::SpacecraftID
```

Spacecraft identifier.

Telemetry Mnemonic(s) $sc_$cpu_EVS_SCID

Definition at line 1245 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.64 CFE_EVS_SetEventFormatCode_Payload Struct Reference

Set Event Format Mode Command Payload.

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_EVS_MsgFormat_Enum_t MsgFormat
  Mode to use in the command.
- uint8 Spare
  Pad to even byte.

38.64.1 Detailed Description

Set Event Format Mode Command Payload.

For command details, see CFE_EVS_SET_EVENT_FORMAT_MODE_CC

Definition at line 1002 of file cfe_evs_msg.h.

38.64.2 Field Documentation

38.64.2.1 MsgFormat

```
CFE_EVS_MsgFormat_Enum_t CFE_EVS_SetEventFormatCode_Payload::MsgFormat
```

Mode to use in the command.

Definition at line 1004 of file cfe_evs_msg.h.

38.64.2.2 Spare

```
uint8 CFE_EVS_SetEventFormatCode_Payload::Spare
```

Pad to even byte.

Definition at line 1005 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.65 CFE_EVS_SetEventFormatModeCmd Struct Reference

Set Event Format Mode Command.

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_EVS_SetEventFormatMode_Payload_t Payload

  Command payload.

38.65.1 Detailed Description

Set Event Format Mode Command.

Definition at line 1011 of file cfe_evs_msg.h.

38.65.2 Field Documentation

38.65.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_EVS_SetEventFormatModeCmd::CmdHeader`

Command header.

Definition at line 1013 of file cfe_evs_msg.h.

38.65.2.2 Payload

`CFE_EVS_SetEventFormatMode_Payload_t CFE_EVS_SetEventFormatModeCmd::Payload`

Command payload.

Definition at line 1014 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.66 CFE_EVS_SetLogMode_Payload Struct Reference

Set Log Mode Command Payload.

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_EVS_LogMode_Enum_t LogMode

  Mode to use in the command.
- uint8 Spare

  Pad to even byte.

38.66.1 Detailed Description

Set Log Mode Command Payload.

For command details, see CFE_EVS_SET_LOG_MODE_CC

Definition at line 981 of file cfe_evs_msg.h.

38.66.2 Field Documentation

38.66.2.1 LogMode

`CFE_EVS_LogMode_Enum_t CFE_EVS_SetLogMode_Payload::LogMode`

Mode to use in the command.

Definition at line 983 of file cfe_evs_msg.h.

38.66.2.2 Spare

`uint8 CFE_EVS_SetLogMode_Payload::Spare`

Pad to even byte.

Definition at line 984 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.67 CFE_EVS_SetLogModeCmd Struct Reference

Set Log Mode Command.

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_EVS_SetLogMode_Payload_t Payload

  Command payload.

38.67.1 Detailed Description

Set Log Mode Command.

Definition at line 990 of file cfe_evs_msg.h.

38.67.2 Field Documentation

38.67.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_EVS_SetLogModeCmd::CmdHeader`

Command header.

Definition at line 992 of file cfe_evs_msg.h.

38.67.2.2 Payload

`CFE_EVS_SetLogMode_Payload_t CFE_EVS_SetLogModeCmd::Payload`

Command payload.

Definition at line 993 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.68 CFE_EVS_ShortEventTlm Struct Reference

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_MSG_TelemetryHeader_t TlmHeader
  Telemetry header.
- CFE_EVS_ShortEventTlm_Payload_t Payload
  Telemetry payload.

38.68.1 Detailed Description

Definition at line 1282 of file cfe_evs_msg.h.

38.68.2 Field Documentation

38.68.2.1 Payload

`CFE_EVS_ShortEventTlm_Payload_t CFE_EVS_ShortEventTlm::Payload`

Telemetry payload.

Definition at line 1285 of file cfe_evs_msg.h.

38.68.2.2 TlmHeader

`CFE_MSG_TelemetryHeader_t CFE_EVS_ShortEventTlm::TlmHeader`

Telemetry header.

Definition at line 1284 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.69 CFE_EVS_ShortEventTlm_Payload Struct Reference

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_EVS_PacketID_t PacketID
  Event packet information.

38.69.1 Detailed Description

Name Event Message Telemetry Packet (Short format)

Definition at line 1269 of file cfe_evs_msg.h.

38.69.2 Field Documentation

38.69.2.1 PacketID

```
CFE_EVS_PacketID_t CFE_EVS_ShortEventTlm_Payload::PacketID
```

Event packet information.

Definition at line 1271 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.70 CFE_EVS_WriteAppDataFileCmd Struct Reference

Write Event Services Application Information to File Command.

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_EVS_AppDataCmd_Payload_t Payload
  Command payload.

38.70.1 Detailed Description

Write Event Services Application Information to File Command.

Definition at line 969 of file cfe_evs_msg.h.

38.70.2 Field Documentation

38.70.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_EVS_WriteAppDataFileCmd::CmdHeader
```

Command header.

Definition at line 971 of file cfe_evs_msg.h.

38.70.2.2 Payload

`CFE_EVS_AppDataCmd_Payload_t CFE_EVS_WriteAppDataFileCmd::Payload`

Command payload.

Definition at line 972 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.71 CFE_EVS_WriteLogDataFileCmd Struct Reference

Write Event Log to File Command.

```
#include <cfe_evs_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_EVS_LogFileCmd_Payload_t Payload
  Command payload.

38.71.1 Detailed Description

Write Event Log to File Command.

Definition at line 949 of file cfe_evs_msg.h.

38.71.2 Field Documentation

38.71.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_EVS_WriteLogDataFileCmd::CmdHeader`

Command header.

Definition at line 951 of file cfe_evs_msg.h.

38.71.2.2 Payload

```
CFE_EVS_LogFileCmd_Payload_t CFE_EVS_WriteLogDataFileCmd::Payload
```

Command payload.

Definition at line 952 of file cfe_evs_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/evs/fsw/inc/cfe_evs_msg.h

38.72 CFE_FS_FileWriteMetaData Struct Reference

External Metadata/State object associated with background file writes.

```
#include <cfe_fs_api_typedefs.h>
```

Data Fields

- volatile bool IsPending
- char FileName [OS_MAX_PATH_LEN]
- uint32 FileSubType
- char Description [CFE_FS_HDR_DESC_MAX_LEN]
- CFE_FS_FileWriteGetData_t GetData
- CFE_FS_FileWriteOnEvent_t OnEvent

38.72.1 Detailed Description

External Metadata/State object associated with background file writes.

Applications intending to schedule background file write jobs should instantiate this object in static/global data memory. This keeps track of the state of the file write request(s).

Definition at line 109 of file cfe_fs_api_typedefs.h.

38.72.2 Field Documentation

38.72.2.1 Description

```
char CFE_FS_FileWriteMetaData::Description[CFE_FS_HDR_DESC_MAX_LEN]
```

Description of file (for FS header)

Definition at line 117 of file cfe_fs_api_typedefs.h.

38.72.2.2 FileName

```
char CFE_FS_FileWriteMetaData::FileName[OS_MAX_PATH_LEN]
```

Name of file to write

Definition at line 113 of file cfe_fs_api_typedefs.h.

38.72.2.3 FileSubType

```
uint32 CFE_FS_FileWriteMetaData::FileSubType
```

Type of file to write (for FS header)

Definition at line 116 of file cfe_fs_api_typedefs.h.

38.72.2.4 GetData

```
CFE_FS_FileWriteGetData_t CFE_FS_FileWriteMetaData::GetData
```

Application callback to get a data record

Definition at line 119 of file cfe_fs_api_typedefs.h.

38.72.2.5 IsPending

```
volatile bool CFE_FS_FileWriteMetaData::IsPending
```

Whether request is pending (volatile as it may be checked outside lock)

Definition at line 111 of file cfe_fs_api_typedefs.h.

38.72.2.6 OnEvent

```
CFE_FS_FileWriteOnEvent_t CFE_FS_FileWriteMetaData::OnEvent
```

Application callback for abstract event processing

Definition at line 120 of file cfe_fs_api_typedefs.h.

The documentation for this struct was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_fs_api_typedefs.h

38.73 CFE_FS_Header Struct Reference

Standard cFE File header structure definition.

```
#include <cfe_fs_extern_typedefs.h>
```

Data Fields

- uint32 ContentType

  Identifies the content type (='cFE1'=0x63464531)
- uint32 SubType

  Type of `ContentType`, if necessary.
- uint32 Length

  Length of primary header.
- uint32 SpacecraftID

  Spacecraft that generated the file.
- uint32 ProcessorID

  Processor that generated the file.
- uint32 ApplicationID

  Application that generated the file.
- uint32 TimeSeconds

  File creation timestamp (seconds)
- uint32 TimeSubSeconds

  File creation timestamp (sub-seconds)
- char Description [CFE_FS_HDR_DESC_MAX_LEN]

  File description.

38.73.1 Detailed Description

Standard cFE File header structure definition.

Definition at line 219 of file cfe_fs_extern_typedefs.h.

38.73.2 Field Documentation

38.73.2.1 ApplicationID

```
uint32 CFE_FS_Header::ApplicationID
```

Application that generated the file.

Definition at line 228 of file cfe_fs_extern_typedefs.h.

38.73.2.2 ContentType

```
uint32 CFE_FS_Header::ContentType
```

Identifies the content type (='cFE1'=0x63464531)

Definition at line 221 of file cfe_fs_extern_typedefs.h.

38.73.2.3 Description

```
char CFE_FS_Header::Description[CFE_FS_HDR_DESC_MAX_LEN]
```

File description.

Definition at line 233 of file cfe_fs_extern_typedefs.h.

38.73.2.4 Length

```
uint32 CFE_FS_Header::Length
```

Length of primary header.

Definition at line 225 of file cfe_fs_extern_typedefs.h.

38.73.2.5 ProcessorID

```
uint32 CFE_FS_Header::ProcessorID
```

Processor that generated the file.

Definition at line 227 of file cfe_fs_extern_typedefs.h.

38.73.2.6 SpacecraftID

```
uint32 CFE_FS_Header::SpacecraftID
```

Spacecraft that generated the file.

Definition at line 226 of file cfe_fs_extern_typedefs.h.

38.73.2.7 SubType

```
uint32 CFE_FS_Header::SubType
```

Type of `ContentType`, if necessary.

Standard SubType definitions can be found here

Definition at line 222 of file cfe_fs_extern_typedefs.h.

38.73.2.8 TimeSeconds

```
uint32 CFE_FS_Header::TimeSeconds
```

File creation timestamp (seconds)

Definition at line 230 of file cfe_fs_extern_typedefs.h.

38.73.2.9 TimeSubSeconds

```
uint32 CFE_FS_Header::TimeSubSeconds
```

File creation timestamp (sub-seconds)

Definition at line 231 of file cfe_fs_extern_typedefs.h.

The documentation for this struct was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_fs_extern_typedefs.h

38.74 CFE_PSP_CommandData_t Struct Reference

Data Fields

- char ResetType [CFE_PSP_RESET_NAME_LENGTH]
- uint32 GotResetType
- uint32 SubType
- uint32 GotSubType
- char CpuName [CFE_PSP_CPU_NAME_LENGTH]
- uint32 GotCpuName
- uint32 CpuId
- uint32 GotCpuId
- uint32 SpacecraftId
- uint32 GotSpacecraftId

38.74.1 Detailed Description

Definition at line 94 of file cfe_psp_start.c.

38.74.2 Field Documentation

38.74.2.1 CpuId

```
uint32 CFE_PSP_CommandData_t::CpuId
```

Definition at line 105 of file cfe_psp_start.c.

Referenced by CFE_PSP_ProcessArgumentDefaults(), and OS_Application_Startup().

38.74.2.2 CpuName

```
char CFE_PSP_CommandData_t::CpuName[CFE_PSP_CPU_NAME_LENGTH]
```

Definition at line 102 of file cfe_psp_start.c.

Referenced by CFE_PSP_ProcessArgumentDefaults(), and OS_Application_Startup().

38.74.2.3 GotCpuId

```
uint32 CFE_PSP_CommandData_t::GotCpuId
```

Definition at line 106 of file cfe_psp_start.c.

Referenced by CFE_PSP_ProcessArgumentDefaults(), and OS_Application_Startup().

38.74.2.4 GotCpuName

```
uint32 CFE_PSP_CommandData_t::GotCpuName
```

Definition at line 103 of file cfe_psp_start.c.

Referenced by CFE_PSP_ProcessArgumentDefaults(), and OS_Application_Startup().

38.74.2.5 GotResetType

```
uint32 CFE_PSP_CommandData_t::GotResetType
```

Definition at line 97 of file cfe_psp_start.c.

Referenced by OS_Application_Startup().

38.74.2.6 GotSpacecraftId

```
uint32 CFE_PSP_CommandData_t::GotSpacecraftId
```

Definition at line 109 of file cfe_psp_start.c.

Referenced by CFE_PSP_ProcessArgumentDefaults(), and OS_Application_Startup().

38.74.2.7 GotSubType

```
uint32 CFE_PSP_CommandData_t::GotSubType
```

Definition at line 100 of file cfe_psp_start.c.

Referenced by CFE_PSP_ProcessArgumentDefaults(), and OS_Application_Startup().

38.74.2.8 ResetType

```
char CFE_PSP_CommandData_t::ResetType[CFE_PSP_RESET_NAME_LENGTH]
```

Definition at line 96 of file cfe_psp_start.c.

Referenced by OS_Application_Startup().

38.74.2.9 SpacecraftId

```
uint32 CFE_PSP_CommandData_t::SpacecraftId
```

Definition at line 108 of file cfe_psp_start.c.

Referenced by CFE_PSP_ProcessArgumentDefaults(), and OS_Application_Startup().

38.74.2.10 SubType

```
uint32 CFE_PSP_CommandData_t::SubType
```

Definition at line 99 of file cfe_psp_start.c.

Referenced by CFE_PSP_ProcessArgumentDefaults(), and OS_Application_Startup().

The documentation for this struct was generated from the following file:

- psp/fsw/pc-linux/src/cfe_psp_start.c

38.75 CFE_PSP_LinuxReservedAreaFixedLayout_t Struct Reference

Data Fields

- CFE_PSP_ReservedMemoryBootRecord_t BootRecord
- CFE_PSP_ExceptionStorage_t ExceptionStorage

38.75.1 Detailed Description

Definition at line 81 of file cfe_psp_memory.c.

38.75.2 Field Documentation

38.75.2.1 BootRecord

```
CFE_PSP_ReservedMemoryBootRecord_t CFE_PSP_LinuxReservedAreaFixedLayout_t::BootRecord
```

Definition at line 83 of file cfe_psp_memory.c.

Referenced by CFE_PSP_InitResetArea().

38.75.2.2 ExceptionStorage

```
CFE_PSP_ExceptionStorage_t CFE_PSP_LinuxReservedAreaFixedLayout_t::ExceptionStorage
```

Definition at line 84 of file cfe_psp_memory.c.

Referenced by CFE_PSP_InitResetArea().

The documentation for this struct was generated from the following file:

- psp/fsw/pc-linux/src/cfe_psp_memory.c

38.76 CFE_PSP_MemTable_t Struct Reference

```
#include <cfe_psp.h>
```

Data Fields

- uint32 MemoryType
- uint32 WordSize
- cpuaddr StartAddr
- uint32 Size
- uint32 Attributes

38.76.1 Detailed Description

Definition at line 163 of file cfe_psp.h.

38.76.2 Field Documentation

38.76.2.1 Attributes

```
uint32 CFE_PSP_MemTable_t::Attributes
```

Definition at line 169 of file cfe_psp.h.

38.76.2.2 MemoryType

```
uint32 CFE_PSP_MemTable_t::MemoryType
```

Definition at line 165 of file cfe_psp.h.

38.76.2.3 Size

```
uint32 CFE_PSP_MemTable_t::Size
```

Definition at line 168 of file cfe_psp.h.

38.76.2.4 StartAddr

```
cpuaddr CFE_PSP_MemTable_t::StartAddr
```

Definition at line 167 of file cfe_psp.h.

38.76.2.5 WordSize

```
uint32 CFE_PSP_MemTable_t::WordSize
```

Definition at line 166 of file cfe_psp.h.

The documentation for this struct was generated from the following file:

- psp/fsw/inc/cfe_psp.h

38.77 CFE_PSP_VersionInfo_t Struct Reference

```
#include <cfe_psp_configdata.h>
```

Data Fields

- uint8 MajorVersion
- uint8 MinorVersion
- uint8 Revision
- uint8 MissionRev
- const char * VersionString
- const char * VersionCodeName
- uint32 BuildNumber

38.77.1 Detailed Description

Definition at line 40 of file cfe_psp_configdata.h.

38.77.2 Field Documentation

38.77.2.1 BuildNumber

```
uint32 CFE_PSP_VersionInfo_t::BuildNumber
```

Definition at line 48 of file cfe_psp_configdata.h.

38.77.2.2 MajorVersion

```
uint8 CFE_PSP_VersionInfo_t::MajorVersion
```

Definition at line 42 of file cfe_psp_configdata.h.

38.77.2.3 MinorVersion

```
uint8 CFE_PSP_VersionInfo_t::MinorVersion
```

Definition at line 43 of file cfe_psp_configdata.h.

38.77.2.4 MissionRev

```
uint8 CFE_PSP_VersionInfo_t::MissionRev
```

Definition at line 45 of file cfe_psp_configdata.h.

38.77.2.5 Revision

```
uint8 CFE_PSP_VersionInfo_t::Revision
```

Definition at line 44 of file cfe_psp_configdata.h.

38.77.2.6 VersionCodeName

```
const char* CFE_PSP_VersionInfo_t::VersionCodeName
```

Cross-module compatiblity indicator

Definition at line 47 of file cfe_psp_configdata.h.

38.77.2.7 VersionString

```
const char* CFE_PSP_VersionInfo_t::VersionString
```

The simple semantic version identifier

Definition at line 46 of file cfe_psp_configdata.h.

The documentation for this struct was generated from the following file:

- psp/fsw/inc/cfe_psp_configdata.h

38.78 CFE_SB_AllSubscriptionsTlm Struct Reference

```
#include <cfe_sb_msg.h>
```

Data Fields

- CFE_MSG_TelemetryHeader_t Hdr

  Telemetry header.
- CFE_SB_AllSubscriptionsTlm_Payload_t Payload

  Telemetry payload.

38.78.1 Detailed Description

Definition at line 781 of file cfe_sb_msg.h.

38.78.2 Field Documentation

38.78.2.1 Hdr

```
CFE_MSG_TelemetryHeader_t CFE_SB_AllSubscriptionsTlm::Hdr
```

Telemetry header.

Definition at line 783 of file cfe_sb_msg.h.

38.78.2.2 Payload

```
CFE_SB_AllSubscriptionsTlm_Payload_t CFE_SB_AllSubscriptionsTlm::Payload
```

Telemetry payload.

Definition at line 784 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.79 CFE_SB_AllSubscriptionsTlm_Payload Struct Reference

```
#include <cfe_sb_msg.h>
```

Data Fields

- uint32 PktSegment

  Pkt number(starts at 1) in the series.
- uint32 TotalSegments

  Total number of pkts needed to complete the request.
- uint32 Entries

  Number of entries in the pkt.
- CFE_SB_SubEntries_t Entry [CFE_SB_SUB_ENTRIES_PER_PKT]

  Array of CFE_SB_SubEntries_t entries.

38.79.1 Detailed Description

Name SB Previous Subscriptions Packet

This structure defines the pkt(s) sent by SB that contains a list of all current subscriptions. This pkt is generated on cmd and intended to be used primarily by the Software Bus Networking Application (SBN). Typically, when the cmd is received there are more subscriptions than can fit in one pkt. The complete list of subscriptions is sent via a series of segmented pkts.

Definition at line 772 of file cfe_sb_msg.h.

38.79.2 Field Documentation

38.79.2.1 Entries

```
uint32 CFE_SB_AllSubscriptionsTlm_Payload::Entries
```

Number of entries in the pkt.

Definition at line 777 of file cfe_sb_msg.h.

38.79.2.2 Entry

```
CFE_SB_SubEntries_t CFE_SB_AllSubscriptionsTlm_Payload::Entry[CFE_SB_SUB_ENTRIES_PER_PKT]
```

Array of CFE_SB_SubEntries_t entries.

Definition at line 778 of file cfe_sb_msg.h.

38.79.2.3 PktSegment

```
uint32 CFE_SB_AllSubscriptionsTlm_Payload::PktSegment
```

Pkt number(starts at 1) in the series.

Definition at line 775 of file cfe_sb_msg.h.

38.79.2.4 TotalSegments

```
uint32 CFE_SB_AllSubscriptionsTlm_Payload::TotalSegments
```

Total number of pkts needed to complete the request.

Definition at line 776 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.80 CFE_SB_HousekeepingTlm Struct Reference

```
#include <cfe_sb_msg.h>
```

Data Fields

- CFE_MSG_TelemetryHeader_t Hdr

  Telemetry header.
- CFE_SB_HousekeepingTlm_Payload_t Payload

  Telemetry payload.

38.80.1 Detailed Description

Definition at line 582 of file cfe_sb_msg.h.

38.80.2 Field Documentation

38.80.2.1 Hdr

```
CFE_MSG_TelemetryHeader_t CFE_SB_HousekeepingTlm::Hdr
```

Telemetry header.

Definition at line 584 of file cfe_sb_msg.h.

38.80.2.2 Payload

```
CFE_SB_HousekeepingTlm_Payload_t CFE_SB_HousekeepingTlm::Payload
```

Telemetry payload.

Definition at line 585 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.81 CFE_SB_HousekeepingTlm_Payload Struct Reference

```
#include <cfe_sb_msg.h>
```

Data Fields

- uint8 CommandCounter

  Count of valid commands received.
- uint8 CommandErrorCounter

  Count of invalid commands received.
- uint8 NoSubscribersCounter

  Count pkts sent with no subscribers.
- uint8 MsgSendErrorCounter

  Count of message send errors.
- uint8 MsgReceiveErrorCounter

  Count of message receive errors.
- uint8 InternalErrorCounter

  Count of queue read or write errors.
- uint8 CreatePipeErrorCounter

  Count of errors in create pipe API.
- uint8 SubscribeErrorCounter

  Count of errors in subscribe API.
- uint8 PipeOptsErrorCounter

  Count of errors in set/get pipe options API.
- uint8 DuplicateSubscriptionsCounter

Count of duplicate subscriptions.

- uint8 GetPipeIdByNameErrorCounter

  Count of errors in get pipe id by name API.
- uint8 Spare2Align [1]

  Spare bytes to ensure alignment.
- uint16 PipeOverflowErrorCounter

  Count of pipe overflow errors.
- uint16 MsgLimitErrorCounter

  Count of msg id to pipe errors.
- CFE_ES_MemHandle_t MemPoolHandle

  Handle to SB's Memory Pool.
- uint32 MemInUse

  Memory in use.
- uint32 UnmarkedMem

  cfg param CFE_PLATFORM_SB_BUF_MEMORY_BYTES minus Peak Memory in use

38.81.1 Detailed Description

Name Software Bus task housekeeping Packet

Definition at line 538 of file cfe_sb_msg.h.

38.81.2 Field Documentation

38.81.2.1 CommandCounter

```
uint8 CFE_SB_HousekeepingTlm_Payload::CommandCounter
```

Count of valid commands received.

Telemetry Mnemonic(s) $sc_$cpu_SB_CMDPC

Definition at line 541 of file cfe_sb_msg.h.

38.81.2.2 CommandErrorCounter

```
uint8 CFE_SB_HousekeepingTlm_Payload::CommandErrorCounter
```

Count of invalid commands received.

Telemetry Mnemonic(s) $sc_$cpu_SB_CMDEC

Definition at line 543 of file cfe_sb_msg.h.

38.81.2.3 CreatePipeErrorCounter

```
uint8 CFE_SB_HousekeepingTlm_Payload::CreatePipeErrorCounter
```

Count of errors in create pipe API.

Telemetry Mnemonic(s) $sc_$cpu_SB_NewPipeEC

Definition at line 554 of file cfe_sb_msg.h.

38.81.2.4 DuplicateSubscriptionsCounter

```
uint8 CFE_SB_HousekeepingTlm_Payload::DuplicateSubscriptionsCounter
```

Count of duplicate subscriptions.

Telemetry Mnemonic(s) $sc_$cpu_SB_DupSubCnt

Definition at line 560 of file cfe_sb_msg.h.

38.81.2.5 GetPipeIdByNameErrorCounter

```
uint8 CFE_SB_HousekeepingTlm_Payload::GetPipeIdByNameErrorCounter
```

Count of errors in get pipe id by name API.

Telemetry Mnemonic(s) $sc_$cpu_SB_GetPipeIDByNameEC

Definition at line 562 of file cfe_sb_msg.h.

38.81.2.6 InternalErrorCounter

```
uint8 CFE_SB_HousekeepingTlm_Payload::InternalErrorCounter
```

Count of queue read or write errors.

Telemetry Mnemonic(s) $sc_$cpu_SB_InternalEC

Definition at line 552 of file cfe_sb_msg.h.

38.81.2.7 MemInUse

`uint32 CFE_SB_HousekeepingTlm_Payload::MemInUse`

Memory in use.

Telemetry Mnemonic(s) $sc_$cpu_SB_MemInUse

Definition at line 575 of file cfe_sb_msg.h.

38.81.2.8 MemPoolHandle

`CFE_ES_MemHandle_t CFE_SB_HousekeepingTlm_Payload::MemPoolHandle`

Handle to SB's Memory Pool.

Telemetry Mnemonic(s) $sc_$cpu_SB_MemPoolHdl

Definition at line 572 of file cfe_sb_msg.h.

38.81.2.9 MsgLimitErrorCounter

`uint16 CFE_SB_HousekeepingTlm_Payload::MsgLimitErrorCounter`

Count of msg id to pipe errors.

Telemetry Mnemonic(s) $sc_$cpu_SB_MsgLimEC

Definition at line 569 of file cfe_sb_msg.h.

38.81.2.10 MsgReceiveErrorCounter

`uint8 CFE_SB_HousekeepingTlm_Payload::MsgReceiveErrorCounter`

Count of message receive errors.

Telemetry Mnemonic(s) $sc_$cpu_SB_MsgRecEC

Definition at line 550 of file cfe_sb_msg.h.

38.81.2.11 MsgSendErrorCounter

```
uint8 CFE_SB_HousekeepingTlm_Payload::MsgSendErrorCounter
```

Count of message send errors.

Telemetry Mnemonic(s) $sc_$cpu_SB_MsgSndEC

Definition at line 547 of file cfe_sb_msg.h.

38.81.2.12 NoSubscribersCounter

```
uint8 CFE_SB_HousekeepingTlm_Payload::NoSubscribersCounter
```

Count pkts sent with no subscribers.

Telemetry Mnemonic(s) $sc_$cpu_SB_NoSubEC

Definition at line 545 of file cfe_sb_msg.h.

38.81.2.13 PipeOptsErrorCounter

```
uint8 CFE_SB_HousekeepingTlm_Payload::PipeOptsErrorCounter
```

Count of errors in set/get pipe options API.

Telemetry Mnemonic(s) $sc_$cpu_SB_PipeOptsEC

Definition at line 558 of file cfe_sb_msg.h.

38.81.2.14 PipeOverflowErrorCounter

```
uint16 CFE_SB_HousekeepingTlm_Payload::PipeOverflowErrorCounter
```

Count of pipe overflow errors.

Telemetry Mnemonic(s) $sc_$cpu_SB_PipeOvrEC

Definition at line 567 of file cfe_sb_msg.h.

38.81.2.15 Spare2Align

```
uint8 CFE_SB_HousekeepingTlm_Payload::Spare2Align[1]
```

Spare bytes to ensure alignment.

Telemetry Mnemonic(s) $sc_$cpu_SB_Spare2Align[2]

Definition at line 564 of file cfe_sb_msg.h.

38.81.2.16 SubscribeErrorCounter

```
uint8 CFE_SB_HousekeepingTlm_Payload::SubscribeErrorCounter
```

Count of errors in subscribe API.

Telemetry Mnemonic(s) $sc_$cpu_SB_SubscrEC

Definition at line 556 of file cfe_sb_msg.h.

38.81.2.17 UnmarkedMem

```
uint32 CFE_SB_HousekeepingTlm_Payload::UnmarkedMem
```

cfg param CFE_PLATFORM_SB_BUF_MEMORY_BYTES minus Peak Memory in use

Telemetry Mnemonic(s) $sc_$cpu_SB_UnMarkedMem

Definition at line 578 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.82 CFE_SB_Msg Union Reference

Software Bus generic message.

```
#include <cfe_sb_api_typedefs.h>
```

Data Fields

- CFE_MSG_Message_t Msg

  Base message type without enforced alignment.
- long long int LongInt

  Align to support Long Integer.
- long double LongDouble

  Align to support Long Double.

38.82.1 Detailed Description

Software Bus generic message.

Definition at line 138 of file cfe_sb_api_typedefs.h.

38.82.2 Field Documentation

38.82.2.1 LongDouble

```
long double CFE_SB_Msg::LongDouble
```

Align to support Long Double.

Definition at line 142 of file cfe_sb_api_typedefs.h.

38.82.2.2 LongInt

```
long long int CFE_SB_Msg::LongInt
```

Align to support Long Integer.

Definition at line 141 of file cfe_sb_api_typedefs.h.

38.82.2.3 Msg

```
CFE_MSG_Message_t CFE_SB_Msg::Msg
```

Base message type without enforced alignment.

Definition at line 140 of file cfe_sb_api_typedefs.h.

The documentation for this union was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_sb_api_typedefs.h

38.83 CFE_SB_MsgMapFileEntry Struct Reference

SB Map File Entry.

```
#include <cfe_sb_msg.h>
```

Data Fields

- CFE_SB_MsgId_t MsgId

  Message Id which has been subscribed to.
- CFE_SB_RouteId_Atom_t Index

  Routing raw index value (0 based, not Route ID)

38.83.1 Detailed Description

SB Map File Entry.

Structure of one element of the map information in response to CFE_SB_WRITE_MAP_INFO_CC

Definition at line 714 of file cfe_sb_msg.h.

38.83.2 Field Documentation

38.83.2.1 Index

```
CFE_SB_RouteId_Atom_t CFE_SB_MsgMapFileEntry::Index
```

Routing raw index value (0 based, not Route ID)

Definition at line 717 of file cfe_sb_msg.h.

38.83.2.2 MsgId

```
CFE_SB_MsgId_t CFE_SB_MsgMapFileEntry::MsgId
```

Message Id which has been subscribed to.

Definition at line 716 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.84 CFE_SB_PipeDepthStats Struct Reference

SB Pipe Depth Statistics.

```
#include <cfe_sb_msg.h>
```

Data Fields

- CFE_SB_PipeId_t PipeId

  Pipe Id associated with the stats below.
- uint16 MaxQueueDepth

  Number of messages the pipe can hold.
- uint16 CurrentQueueDepth

  Number of messages currently on the pipe.
- uint16 PeakQueueDepth

  Peak number of messages that have been on the pipe.
- uint16 Spare

  Spare word to ensure alignment.

38.84.1 Detailed Description

SB Pipe Depth Statistics.

Used in SB Statistics Telemetry Packet CFE_SB_StatsTlm_t

Definition at line 593 of file cfe_sb_msg.h.

38.84.2 Field Documentation

38.84.2.1 CurrentQueueDepth

```
uint16 CFE_SB_PipeDepthStats::CurrentQueueDepth
```

Number of messages currently on the pipe.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDINUSE

Definition at line 600 of file cfe_sb_msg.h.

38.84.2.2 MaxQueueDepth

`uint16 CFE_SB_PipeDepthStats::MaxQueueDepth`

Number of messages the pipe can hold.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDDEPTH

Definition at line 598 of file cfe_sb_msg.h.

38.84.2.3 PeakQueueDepth

`uint16 CFE_SB_PipeDepthStats::PeakQueueDepth`

Peak number of messages that have been on the pipe.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDPKINUSE

Definition at line 602 of file cfe_sb_msg.h.

38.84.2.4 PipeId

`CFE_SB_PipeId_t CFE_SB_PipeDepthStats::PipeId`

Pipe Id associated with the stats below.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDPIPEID

Definition at line 596 of file cfe_sb_msg.h.

38.84.2.5 Spare

`uint16 CFE_SB_PipeDepthStats::Spare`

Spare word to ensure alignment.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDSPARE

Definition at line 604 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.85 CFE_SB_PipeInfoEntry Struct Reference

SB Pipe Information File Entry.

```
#include <cfe_sb_msg.h>
```

Data Fields

- CFE_SB_PipeId_t PipeId
- CFE_ES_AppId_t AppId
- char PipeName [CFE_MISSION_MAX_API_LEN]
- char AppName [CFE_MISSION_MAX_API_LEN]
- uint16 MaxQueueDepth
- uint16 CurrentQueueDepth
- uint16 PeakQueueDepth
- uint16 SendErrors
- uint8 Opts
- uint8 Spare [3]

38.85.1 Detailed Description

SB Pipe Information File Entry.

This statistics structure is output as part of the CFE SB "Send Pipe Info" command (CFE_SB_SEND_PIPE_INFO_CC).

Previous versions of CFE simply wrote the internal CFE_SB_PipeD_t object to the file, but this also contains information such as pointers which are not relevant outside the running CFE process.

By defining the pipe info structure separately, it also provides some independence, such that the internal CFE_SB_↩ PipeD_t definition can evolve without changing the binary format of the information file.

Definition at line 624 of file cfe_sb_msg.h.

38.85.2 Field Documentation

38.85.2.1 AppId

```
CFE_ES_AppId_t CFE_SB_PipeInfoEntry::AppId
```

The runtime ID of the application that owns the pipe

Definition at line 627 of file cfe_sb_msg.h.

38.85.2.2 AppName

```
char CFE_SB_PipeInfoEntry::AppName[CFE_MISSION_MAX_API_LEN]
```

The Name of the application that owns the pipe

Definition at line 629 of file cfe_sb_msg.h.

38.85.2.3 CurrentQueueDepth

```
uint16 CFE_SB_PipeInfoEntry::CurrentQueueDepth
```

The current depth of the pipe

Definition at line 631 of file cfe_sb_msg.h.

38.85.2.4 MaxQueueDepth

```
uint16 CFE_SB_PipeInfoEntry::MaxQueueDepth
```

The allocated depth of the pipe (max capacity)

Definition at line 630 of file cfe_sb_msg.h.

38.85.2.5 Opts

```
uint8 CFE_SB_PipeInfoEntry::Opts
```

Pipe options set (bitmask)

Definition at line 634 of file cfe_sb_msg.h.

38.85.2.6 PeakQueueDepth

```
uint16 CFE_SB_PipeInfoEntry::PeakQueueDepth
```

The peak depth of the pipe (high watermark)

Definition at line 632 of file cfe_sb_msg.h.

38.85.2.7 PipeId

```
CFE_SB_PipeId_t CFE_SB_PipeInfoEntry::PipeId
```

The runtime ID of the pipe

Definition at line 626 of file cfe_sb_msg.h.

38.85.2.8 PipeName

```
char CFE_SB_PipeInfoEntry::PipeName[CFE_MISSION_MAX_API_LEN]
```

The Name of the pipe

Definition at line 628 of file cfe_sb_msg.h.

38.85.2.9 SendErrors

```
uint16 CFE_SB_PipeInfoEntry::SendErrors
```

Number of errors when writing to this pipe

Definition at line 633 of file cfe_sb_msg.h.

38.85.2.10 Spare

```
uint8 CFE_SB_PipeInfoEntry::Spare[3]
```

Padding to make this structure a multiple of 4 bytes

Definition at line 635 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.86 CFE_SB_Qos_t Struct Reference

Quality Of Service Type Definition.

```
#include <cfe_sb_extern_typedefs.h>
```

Data Fields

- uint8 Priority

  Specify high(1) or low(0) message priority for off-board routing, currently unused.
- uint8 Reliability

  Specify high(1) or low(0) message transfer reliability for off-board routing, currently unused.

38.86.1 Detailed Description

Quality Of Service Type Definition.

Currently an unused parameter in CFE_SB_SubscribeEx Intended to be used for interprocessor communication only

Definition at line 128 of file cfe_sb_extern_typedefs.h.

38.86.2 Field Documentation

38.86.2.1 Priority

```
uint8 CFE_SB_Qos_t::Priority
```

Specify high(1) or low(0) message priority for off-board routing, currently unused.

Definition at line 130 of file cfe_sb_extern_typedefs.h.

38.86.2.2 Reliability

```
uint8 CFE_SB_Qos_t::Reliability
```

Specify high(1) or low(0) message transfer reliability for off-board routing, currently unused.

Definition at line 131 of file cfe_sb_extern_typedefs.h.

The documentation for this struct was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_sb_extern_typedefs.h

38.87 CFE_SB_RouteCmd Struct Reference

Enable/Disable Route Command.

```
#include <cfe_sb_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t Hdr

  Command header.
- CFE_SB_RouteCmd_Payload_t Payload

  Command payload.

38.87.1 Detailed Description

Enable/Disable Route Command.

Definition at line 519 of file cfe_sb_msg.h.

38.87.2 Field Documentation

38.87.2.1 Hdr

`CFE_MSG_CommandHeader_t CFE_SB_RouteCmd::Hdr`

Command header.

Definition at line 521 of file cfe_sb_msg.h.

38.87.2.2 Payload

`CFE_SB_RouteCmd_Payload_t CFE_SB_RouteCmd::Payload`

Command payload.

Definition at line 522 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.88 CFE_SB_RouteCmd_Payload Struct Reference

Enable/Disable Route Command Payload.

```
#include <cfe_sb_msg.h>
```

Data Fields

- CFE_SB_MsgId_t MsgId

  Message ID of route to be enabled or disabled CFE_SB_MsgId_t.
- CFE_SB_PipeId_t Pipe

  Pipe ID of route to be enabled or disabled CFE_SB_PipeId_t.
- uint8 Spare

  Spare byte to make command even number of bytes.

38.88.1 Detailed Description

Enable/Disable Route Command Payload.

This structure contains a definition used by two SB commands, 'Enable Route' CFE_SB_ENABLE_ROUTE_CC and 'Disable Route' CFE_SB_DISABLE_ROUTE_CC. A route is the destination pipe for a particular message and is therefore defined as a MsgId and PipeId combination.

Definition at line 508 of file cfe_sb_msg.h.

38.88.2 Field Documentation

38.88.2.1 MsgId

```
CFE_SB_MsgId_t CFE_SB_RouteCmd_Payload::MsgId
```

Message ID of route to be enabled or disabled CFE_SB_MsgId_t.

Definition at line 511 of file cfe_sb_msg.h.

38.88.2.2 Pipe

```
CFE_SB_PipeId_t CFE_SB_RouteCmd_Payload::Pipe
```

Pipe ID of route to be enabled or disabled CFE_SB_PipeId_t.

Definition at line 512 of file cfe_sb_msg.h.

38.88.2.3 Spare

```
uint8 CFE_SB_RouteCmd_Payload::Spare
```

Spare byte to make command even number of bytes.

Definition at line 513 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.89 CFE_SB_RoutingFileEntry Struct Reference

SB Routing File Entry.

```
#include <cfe_sb_msg.h>
```

Data Fields

- CFE_SB_MsgId_t MsgId

  Message Id portion of the route.
- CFE_SB_PipeId_t PipeId

  Pipe Id portion of the route.
- uint8 State

  Route Enabled or Disabled.
- uint16 MsgCnt

  Number of msgs with this MsgId sent to this PipeId.
- char AppName [CFE_MISSION_MAX_API_LEN]

  Pipe Depth Statistics.
- char PipeName [CFE_MISSION_MAX_API_LEN]

  Pipe Depth Statistics.

38.89.1 Detailed Description

SB Routing File Entry.

Structure of one element of the routing information in response to CFE_SB_WRITE_ROUTING_INFO_CC

Definition at line 699 of file cfe_sb_msg.h.

38.89.2 Field Documentation

38.89.2.1 AppName

```
char CFE_SB_RoutingFileEntry::AppName[CFE_MISSION_MAX_API_LEN]
```

Pipe Depth Statistics.

Definition at line 705 of file cfe_sb_msg.h.

38.89.2.2 MsgCnt

```
uint16 CFE_SB_RoutingFileEntry::MsgCnt
```

Number of msgs with this MsgId sent to this PipeId.

Definition at line 704 of file cfe_sb_msg.h.

38.89.2.3 MsgId

```
CFE_SB_MsgId_t CFE_SB_RoutingFileEntry::MsgId
```

Message Id portion of the route.

Definition at line 701 of file cfe_sb_msg.h.

38.89.2.4 PipeId

```
CFE_SB_PipeId_t CFE_SB_RoutingFileEntry::PipeId
```

Pipe Id portion of the route.

Definition at line 702 of file cfe_sb_msg.h.

38.89.2.5 PipeName

```
char CFE_SB_RoutingFileEntry::PipeName[CFE_MISSION_MAX_API_LEN]
```

Pipe Depth Statistics.

Definition at line 706 of file cfe_sb_msg.h.

38.89.2.6 State

```
uint8 CFE_SB_RoutingFileEntry::State
```

Route Enabled or Disabled.

Definition at line 703 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.90 CFE_SB_SingleSubscriptionTlm Struct Reference

```
#include <cfe_sb_msg.h>
```

Data Fields

- CFE_MSG_TelemetryHeader_t Hdr

  Telemetry header.
- CFE_SB_SingleSubscriptionTlm_Payload_t Payload

  Telemetry payload.

38.90.1 Detailed Description

Definition at line 740 of file cfe_sb_msg.h.

38.90.2 Field Documentation

38.90.2.1 Hdr

```
CFE_MSG_TelemetryHeader_t CFE_SB_SingleSubscriptionTlm::Hdr
```

Telemetry header.

Definition at line 742 of file cfe_sb_msg.h.

38.90.2.2 Payload

`CFE_SB_SingleSubscriptionTlm_Payload_t CFE_SB_SingleSubscriptionTlm::Payload`

Telemetry payload.

Definition at line 743 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.91 CFE_SB_SingleSubscriptionTlm_Payload Struct Reference

```
#include <cfe_sb_msg.h>
```

Data Fields

- uint8 SubType

  Subscription or Unsubscription.
- CFE_SB_MsgId_t MsgId

  MsgId subscribed or unsubscribe to.
- CFE_SB_Qos_t Qos

  Quality of Service, used only for interprocessor communication.
- CFE_SB_PipeId_t Pipe

  Destination pipe id to send above msg id.

38.91.1 Detailed Description

Name SB Subscription Report Packet

This structure defines the pkt sent by SB when a subscription or a request to unsubscribe is received while subscription reporting is enabled. By default subscription reporting is disabled. This feature is intended to be used primarily by Software Bus Networking Application (SBN)

See also

CFE_SB_ENABLE_SUB_REPORTING_CC, CFE_SB_DISABLE_SUB_REPORTING_CC

Definition at line 730 of file cfe_sb_msg.h.

38.91.2 Field Documentation

38.91.2.1 MsgId

```
CFE_SB_MsgId_t CFE_SB_SingleSubscriptionTlm_Payload::MsgId
```

MsgId subscribed or unsubscribe to.

Definition at line 734 of file cfe_sb_msg.h.

38.91.2.2 Pipe

```
CFE_SB_PipeId_t CFE_SB_SingleSubscriptionTlm_Payload::Pipe
```

Destination pipe id to send above msg id.

Definition at line 736 of file cfe_sb_msg.h.

38.91.2.3 Qos

```
CFE_SB_Qos_t CFE_SB_SingleSubscriptionTlm_Payload::Qos
```

Quality of Service, used only for interprocessor communication.

Definition at line 735 of file cfe_sb_msg.h.

38.91.2.4 SubType

```
uint8 CFE_SB_SingleSubscriptionTlm_Payload::SubType
```

Subscription or Unsubscription.

Definition at line 733 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.92 CFE_SB_StatsTlm Struct Reference

```
#include <cfe_sb_msg.h>
```

Data Fields

- CFE_MSG_TelemetryHeader_t Hdr

  Telemetry header.
- CFE_SB_StatsTlm_Payload_t Payload

  Telemetry payload.

38.92.1 Detailed Description

Definition at line 688 of file cfe_sb_msg.h.

38.92.2 Field Documentation

38.92.2.1 Hdr

```
CFE_MSG_TelemetryHeader_t CFE_SB_StatsTlm::Hdr
```

Telemetry header.

Definition at line 690 of file cfe_sb_msg.h.

38.92.2.2 Payload

```
CFE_SB_StatsTlm_Payload_t CFE_SB_StatsTlm::Payload
```

Telemetry payload.

Definition at line 691 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.93 CFE_SB_StatsTlm_Payload Struct Reference

```
#include <cfe_sb_msg.h>
```

Data Fields

- uint32 MsgIdsInUse

  Current number of MsgIds with a destination.
- uint32 PeakMsgIdsInUse

  Peak number of MsgIds with a destination.
- uint32 MaxMsgIdsAllowed

  cFE Cfg Param CFE_PLATFORM_SB_MAX_MSG_IDS
- uint32 PipesInUse

  Number of pipes currently in use.
- uint32 PeakPipesInUse

  Peak number of pipes since last reboot.
- uint32 MaxPipesAllowed

  cFE Cfg Param CFE_PLATFORM_SB_MAX_PIPES
- uint32 MemInUse

  Memory bytes currently in use for SB msg transfers.
- uint32 PeakMemInUse

  Peak memory bytes in use for SB msg transfers.
- uint32 MaxMemAllowed

  cFE Cfg Param CFE_PLATFORM_SB_BUF_MEMORY_BYTES
- uint32 SubscriptionsInUse

  Number of current subscriptions.
- uint32 PeakSubscriptionsInUse

  Peak number of subscriptions.
- uint32 MaxSubscriptionsAllowed

  product of CFE_PLATFORM_SB_MAX_MSG_IDS and CFE_PLATFORM_SB_MAX_DEST_PER_PKT
- uint32 SBBuffersInUse

  Number of SB message buffers currently in use.
- uint32 PeakSBBuffersInUse

  Max number of SB message buffers in use.
- uint32 MaxPipeDepthAllowed

  Maximum allowed pipe depth.
- CFE_SB_PipeDepthStats_t PipeDepthStats [CFE_MISSION_SB_MAX_PIPES]

  Pipe Depth Statistics CFE_SB_PipeDepthStats_t.

38.93.1 Detailed Description

Name SB Statistics Telemetry Packet

SB Statistics packet sent in response to CFE_SB_SEND_SB_STATS_CC

Definition at line 644 of file cfe_sb_msg.h.

38.93.2 Field Documentation

38.93.2.1 MaxMemAllowed

```
uint32 CFE_SB_StatsTlm_Payload::MaxMemAllowed
```

cFE Cfg Param CFE_PLATFORM_SB_BUF_MEMORY_BYTES

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMMBMALW

Definition at line 665 of file cfe_sb_msg.h.

38.93.2.2 MaxMsgIdsAllowed

```
uint32 CFE_SB_StatsTlm_Payload::MaxMsgIdsAllowed
```

cFE Cfg Param CFE_PLATFORM_SB_MAX_MSG_IDS

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMMMIDALW

Definition at line 651 of file cfe_sb_msg.h.

38.93.2.3 MaxPipeDepthAllowed

```
uint32 CFE_SB_StatsTlm_Payload::MaxPipeDepthAllowed
```

Maximum allowed pipe depth.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMMPDALW

Definition at line 681 of file cfe_sb_msg.h.

38.93.2.4 MaxPipesAllowed

```
uint32 CFE_SB_StatsTlm_Payload::MaxPipesAllowed
```

cFE Cfg Param CFE_PLATFORM_SB_MAX_PIPES

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMMPALW

Definition at line 658 of file cfe_sb_msg.h.

38.93.2.5 MaxSubscriptionsAllowed

```
uint32 CFE_SB_StatsTlm_Payload::MaxSubscriptionsAllowed
```

product of CFE_PLATFORM_SB_MAX_MSG_IDS and CFE_PLATFORM_SB_MAX_DEST_PER_PKT

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMMSALW

Definition at line 672 of file cfe_sb_msg.h.

38.93.2.6 MemInUse

```
uint32 CFE_SB_StatsTlm_Payload::MemInUse
```

Memory bytes currently in use for SB msg transfers.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMBMIU

Definition at line 661 of file cfe_sb_msg.h.

38.93.2.7 MsgIdsInUse

```
uint32 CFE_SB_StatsTlm_Payload::MsgIdsInUse
```

Current number of MsgIds with a destination.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMMIDIU

Definition at line 647 of file cfe_sb_msg.h.

38.93.2.8 PeakMemInUse

```
uint32 CFE_SB_StatsTlm_Payload::PeakMemInUse
```

Peak memory bytes in use for SB msg transfers.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMPBMIU

Definition at line 663 of file cfe_sb_msg.h.

38.93.2.9 PeakMsgIdsInUse

```
uint32 CFE_SB_StatsTlm_Payload::PeakMsgIdsInUse
```

Peak number of MsgIds with a destination.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMPMIDIU

Definition at line 649 of file cfe_sb_msg.h.

38.93.2.10 PeakPipesInUse

```
uint32 CFE_SB_StatsTlm_Payload::PeakPipesInUse
```

Peak number of pipes since last reboot.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMPPIU

Definition at line 656 of file cfe_sb_msg.h.

38.93.2.11 PeakSBBuffersInUse

```
uint32 CFE_SB_StatsTlm_Payload::PeakSBBuffersInUse
```

Max number of SB message buffers in use.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMPSBBIU

Definition at line 678 of file cfe_sb_msg.h.

38.93.2.12 PeakSubscriptionsInUse

```
uint32 CFE_SB_StatsTlm_Payload::PeakSubscriptionsInUse
```

Peak number of subscriptions.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMPSIU

Definition at line 670 of file cfe_sb_msg.h.

38.93.2.13 PipeDepthStats

```
CFE_SB_PipeDepthStats_t CFE_SB_StatsTlm_Payload::PipeDepthStats[CFE_MISSION_SB_MAX_PIPES]
```

Pipe Depth Statistics CFE_SB_PipeDepthStats_t.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES]

Definition at line 684 of file cfe_sb_msg.h.

38.93.2.14 PipesInUse

```
uint32 CFE_SB_StatsTlm_Payload::PipesInUse
```

Number of pipes currently in use.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMPIU

Definition at line 654 of file cfe_sb_msg.h.

38.93.2.15 SBBuffersInUse

```
uint32 CFE_SB_StatsTlm_Payload::SBBuffersInUse
```

Number of SB message buffers currently in use.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMSBBIU

Definition at line 676 of file cfe_sb_msg.h.

38.93.2.16 SubscriptionsInUse

```
uint32 CFE_SB_StatsTlm_Payload::SubscriptionsInUse
```

Number of current subscriptions.

Telemetry Mnemonic(s) $sc_$cpu_SB_Stat.SB_SMSIU

Definition at line 668 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.94 CFE_SB_SubEntries Struct Reference

SB Previous Subscriptions Entry.

```
#include <cfe_sb_msg.h>
```

Data Fields

- CFE_SB_MsgId_t MsgId

  MsgId portion of the subscription.
- CFE_SB_Qos_t Qos

  Qos portion of the subscription.
- CFE_SB_PipeId_t Pipe

  PipeId portion of the subscription.

38.94.1 Detailed Description

SB Previous Subscriptions Entry.

This structure defines an entry used in the CFE_SB_PrevSubsPkt_t Intended to be used primarily by Software Bus Networking Application (SBN)

Used in structure definition CFE_SB_AllSubscriptionsTlm_t

Definition at line 754 of file cfe_sb_msg.h.

38.94.2 Field Documentation

38.94.2.1 MsgId

```
CFE_SB_MsgId_t CFE_SB_SubEntries::MsgId
```

MsgId portion of the subscription.

Definition at line 757 of file cfe_sb_msg.h.

38.94.2.2 Pipe

```
CFE_SB_PipeId_t CFE_SB_SubEntries::Pipe
```

PipeId portion of the subscription.

Definition at line 759 of file cfe_sb_msg.h.

38.94.2.3 Qos

`CFE_SB_Qos_t CFE_SB_SubEntries::Qos`

Qos portion of the subscription.

Definition at line 758 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.95 CFE_SB_WriteFileInfoCmd Struct Reference

Write File Info Command.

```
#include <cfe_sb_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t Hdr
  Command header.
- CFE_SB_WriteFileInfoCmd_Payload_t Payload
  Command payload.

38.95.1 Detailed Description

Write File Info Command.

Definition at line 487 of file cfe_sb_msg.h.

38.95.2 Field Documentation

38.95.2.1 Hdr

`CFE_MSG_CommandHeader_t CFE_SB_WriteFileInfoCmd::Hdr`

Command header.

Definition at line 489 of file cfe_sb_msg.h.

38.95.2.2 Payload

`CFE_SB_WriteFileInfoCmd_Payload_t CFE_SB_WriteFileInfoCmd::Payload`

Command payload.

Definition at line 490 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.96 CFE_SB_WriteFileInfoCmd_Payload Struct Reference

Write File Info Command Payload.

```
#include <cfe_sb_msg.h>
```

Data Fields

- char Filename [CFE_MISSION_MAX_PATH_LEN]
  Path and Filename of data to be loaded.

38.96.1 Detailed Description

Write File Info Command Payload.

This structure contains a generic definition used by SB commands that write to a file

Definition at line 479 of file cfe_sb_msg.h.

38.96.2 Field Documentation

38.96.2.1 Filename

`char CFE_SB_WriteFileInfoCmd_Payload::Filename[CFE_MISSION_MAX_PATH_LEN]`

Path and Filename of data to be loaded.

Definition at line 481 of file cfe_sb_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/sb/fsw/inc/cfe_sb_msg.h

38.97 CFE_SB_ZeroCopyHandle_t Struct Reference

CFE_SB_ZeroCopyHandle_t to primitive type definition.

```
#include <cfe_sb_api_typedefs.h>
```

Data Fields

- struct CFE_SB_BufferD * BufDscPtr

38.97.1 Detailed Description

CFE_SB_ZeroCopyHandle_t to primitive type definition.

Software Zero Copy handle used in many SB APIs

Definition at line 171 of file cfe_sb_api_typedefs.h.

38.97.2 Field Documentation

38.97.2.1 BufDscPtr

```
struct CFE_SB_BufferD* CFE_SB_ZeroCopyHandle_t::BufDscPtr
```

Definition at line 173 of file cfe_sb_api_typedefs.h.

The documentation for this struct was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_sb_api_typedefs.h

38.98 CFE_TBL_AbortLoadCmd Struct Reference

Abort Load Command.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_TBL_AbortLoadCmd_Payload_t Payload
  Command paylod.

38.98.1 Detailed Description

Abort Load Command.

Definition at line 690 of file cfe_tbl_msg.h.

38.98.2 Field Documentation

38.98.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_TBL_AbortLoadCmd::CmdHeader`

Command header.

Definition at line 692 of file cfe_tbl_msg.h.

38.98.2.2 Payload

`CFE_TBL_AbortLoadCmd_Payload_t CFE_TBL_AbortLoadCmd::Payload`

Command paylod.

Definition at line 693 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.99 CFE_TBL_AbortLoadCmd_Payload Struct Reference

Abort Load Command Payload.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- char TableName [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

  Full Name of Table whose load is to be aborted.

38.99.1 Detailed Description

Abort Load Command Payload.

For command details, see CFE_TBL_ABORT_LOAD_CC

Definition at line 680 of file cfe_tbl_msg.h.

38.99.2 Field Documentation

38.99.2.1 TableName

```
char CFE_TBL_AbortLoadCmd_Payload::TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
```

Full Name of Table whose load is to be aborted.

ASCII string containing full table name identifier of a table whose load is to be aborted

Definition at line 682 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.100 CFE_TBL_ActivateCmd Struct Reference

Activate Table Command.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_TBL_ActivateCmd_Payload_t Payload
  Command paylod.

38.100.1 Detailed Description

Activate Table Command.

Definition at line 601 of file cfe_tbl_msg.h.

38.100.2 Field Documentation

38.100.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_TBL_ActivateCmd::CmdHeader`

Command header.

Definition at line 603 of file cfe_tbl_msg.h.

38.100.2.2 Payload

`CFE_TBL_ActivateCmd_Payload_t CFE_TBL_ActivateCmd::Payload`

Command paylod.

Definition at line 604 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.101 CFE_TBL_ActivateCmd_Payload Struct Reference

Activate Table Command Payload.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- char TableName [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
  Full Name of Table to be activated.

38.101.1 Detailed Description

Activate Table Command Payload.

For command details, see CFE_TBL_ACTIVATE_CC

Definition at line 591 of file cfe_tbl_msg.h.

38.101.2 Field Documentation

38.101.2.1 TableName

```
char CFE_TBL_ActivateCmd_Payload::TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
```

Full Name of Table to be activated.

ASCII string containing full table name identifier of table to be activated

Definition at line 593 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.102 CFE_TBL_DelCDSCmd_Payload Struct Reference

Delete Critical Table CDS Command Payload.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- char TableName [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
  Full Name of Table whose CDS is to be deleted.

38.102.1 Detailed Description

Delete Critical Table CDS Command Payload.

For command details, see CFE_TBL_DELETE_CDS_CC

Definition at line 657 of file cfe_tbl_msg.h.

38.102.2 Field Documentation

38.102.2.1 TableName

```
char CFE_TBL_DelCDSCmd_Payload::TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
```

Full Name of Table whose CDS is to be deleted.

ASCII string containing full table name identifier of a critical table whose CDS is to be deleted

Definition at line 659 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.103 CFE_TBL_DeleteCDSCmd Struct Reference

Delete Critical Table CDS Command.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_TBL_DelCDSCmd_Payload_t Payload
  Command payload.

38.103.1 Detailed Description

Delete Critical Table CDS Command.

Definition at line 669 of file cfe_tbl_msg.h.

38.103.2 Field Documentation

38.103.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_TBL_DeleteCDSCmd::CmdHeader
```

Command header.

Definition at line 671 of file cfe_tbl_msg.h.

38.103.2.2 Payload

`CFE_TBL_DelCDSCmd_Payload_t CFE_TBL_DeleteCDSCmd::Payload`

Command payload.

Definition at line 672 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.104 CFE_TBL_DumpCmd Struct Reference

```
#include <cfe_tbl_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_TBL_DumpCmd_Payload_t Payload

  Command payload.

38.104.1 Detailed Description

/brief Dump Table Command

Definition at line 553 of file cfe_tbl_msg.h.

38.104.2 Field Documentation

38.104.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_TBL_DumpCmd::CmdHeader`

Command header.

Definition at line 555 of file cfe_tbl_msg.h.

38.104.2.2 Payload

```
CFE_TBL_DumpCmd_Payload_t CFE_TBL_DumpCmd::Payload
```

Command payload.

Definition at line 556 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.105 CFE_TBL_DumpCmd_Payload Struct Reference

Dump Table Command Payload.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- uint16 ActiveTableFlag

  CFE_TBL_BufferSelect_INACTIVE=Inactive Table, CFE_TBL_BufferSelect_ACTIVE=Active Table
- char TableName [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

  Full name of table to be dumped.
- char DumpFilename [CFE_MISSION_MAX_PATH_LEN]

  Full Filename where data is to be written.

38.105.1 Detailed Description

Dump Table Command Payload.

For command details, see CFE_TBL_DUMP_CC

Definition at line 534 of file cfe_tbl_msg.h.

38.105.2 Field Documentation

38.105.2.1 ActiveTableFlag

```
uint16 CFE_TBL_DumpCmd_Payload::ActiveTableFlag
```

CFE_TBL_BufferSelect_INACTIVE=Inactive Table, CFE_TBL_BufferSelect_ACTIVE=Active Table

Selects either the "Inactive" (CFE_TBL_BufferSelect_INACTIVE) buffer or the "Active" (CFE_TBL_BufferSelect_ACTIVE) buffer to be dumped

Definition at line 536 of file cfe_tbl_msg.h.

38.105.2.2 DumpFilename

```
char CFE_TBL_DumpCmd_Payload::DumpFilename[CFE_MISSION_MAX_PATH_LEN]
```

Full Filename where data is to be written.

ASCII string containing full path of filename where data is to be dumped

Definition at line 545 of file cfe_tbl_msg.h.

38.105.2.3 TableName

```
char CFE_TBL_DumpCmd_Payload::TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
```

Full name of table to be dumped.

ASCII string containing full table name identifier of table to be dumped

Definition at line 542 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.106 CFE_TBL_DumpRegistryCmd Struct Reference

Dump Registry Command.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_TBL_DumpRegistryCmd_Payload_t Payload
  Command payload.

38.106.1 Detailed Description

Dump Registry Command.

Definition at line 623 of file cfe_tbl_msg.h.

38.106.2 Field Documentation

38.106.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_TBL_DumpRegistryCmd::CmdHeader
```

Command header.

Definition at line 625 of file cfe_tbl_msg.h.

38.106.2.2 Payload

```
CFE_TBL_DumpRegistryCmd_Payload_t CFE_TBL_DumpRegistryCmd::Payload
```

Command payload.

Definition at line 626 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.107 CFE_TBL_DumpRegistryCmd_Payload Struct Reference

Dump Registry Command Payload.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- char DumpFilename [CFE_MISSION_MAX_PATH_LEN]

  Full Filename where dumped data is to be written.

38.107.1 Detailed Description

Dump Registry Command Payload.

For command details, see CFE_TBL_DUMP_REGISTRY_CC

Definition at line 612 of file cfe_tbl_msg.h.

38.107.2 Field Documentation

38.107.2.1 DumpFilename

```
char CFE_TBL_DumpRegistryCmd_Payload::DumpFilename[CFE_MISSION_MAX_PATH_LEN]
```

Full Filename where dumped data is to be written.

ASCII string containing full path of filename where registry is to be dumped

Definition at line 614 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.108 CFE_TBL_File_Hdr Struct Reference

The definition of the header fields that are included in CFE Table Data files.

```
#include <cfe_tbl_extern_typedefs.h>
```

Data Fields

- uint32 Reserved
- CFE_ES_MemOffset_t Offset
- CFE_ES_MemOffset_t NumBytes
- char TableName [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

38.108.1 Detailed Description

The definition of the header fields that are included in CFE Table Data files.

This header follows the CFE_FS header and precedes the the actual table data.

Definition at line 68 of file cfe_tbl_extern_typedefs.h.

38.108.2 Field Documentation

38.108.2.1 NumBytes

```
CFE_ES_MemOffset_t CFE_TBL_File_Hdr::NumBytes
```

Number of bytes to load into table

Definition at line 72 of file cfe_tbl_extern_typedefs.h.

38.108.2.2 Offset

```
CFE_ES_MemOffset_t CFE_TBL_File_Hdr::Offset
```

Byte Offset at which load should commence

Definition at line 71 of file cfe_tbl_extern_typedefs.h.

38.108.2.3 Reserved

```
uint32 CFE_TBL_File_Hdr::Reserved
```

Future Use: NumTblSegments in File?

Definition at line 70 of file cfe_tbl_extern_typedefs.h.

38.108.2.4 TableName

```
char CFE_TBL_File_Hdr::TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
```

Fully qualified name of table to load

Definition at line 73 of file cfe_tbl_extern_typedefs.h.

The documentation for this struct was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_tbl_extern_typedefs.h

38.109 CFE_TBL_FileDef Struct Reference

```
#include <cfe_tbl_filedef.h>
```

Data Fields

- char ObjectName [64]

  Name of instantiated variable that contains desired table image.
- char TableName [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

  Name of Table as defined onboard.
- char Description [CFE_FS_HDR_DESC_MAX_LEN]

  Description of table image that is included in cFE File Header.
- char TgtFilename [CFE_MISSION_MAX_FILE_LEN]

  Default filename to be used for output of elf2cfetbl utility.
- uint32 ObjectSize

  Size, in bytes, of instantiated object.

38.109.1 Detailed Description

Definition at line 61 of file cfe_tbl_filedef.h.

38.109.2 Field Documentation

38.109.2.1 Description

```
char CFE_TBL_FileDef::Description[CFE_FS_HDR_DESC_MAX_LEN]
```

Description of table image that is included in cFE File Header.

Definition at line 65 of file cfe_tbl_filedef.h.

38.109.2.2 ObjectName

```
char CFE_TBL_FileDef::ObjectName[64]
```

Name of instantiated variable that contains desired table image.

Definition at line 63 of file cfe_tbl_filedef.h.

38.109.2.3 ObjectSize

```
uint32 CFE_TBL_FileDef::ObjectSize
```

Size, in bytes, of instantiated object.

Definition at line 69 of file cfe_tbl_filedef.h.

38.109.2.4 TableName

```
char CFE_TBL_FileDef::TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
```

Name of Table as defined onboard.

Definition at line 64 of file cfe_tbl_filedef.h.

38.109.2.5 TgtFilename

```
char CFE_TBL_FileDef::TgtFilename[CFE_MISSION_MAX_FILE_LEN]
```

Default filename to be used for output of elf2cfetbl utility.

Definition at line 67 of file cfe_tbl_filedef.h.

The documentation for this struct was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_tbl_filedef.h

38.110 CFE_TBL_HousekeepingTlm Struct Reference

```
#include <cfe_tbl_msg.h>
```

Data Fields

- CFE_MSG_TelemetryHeader_t TlmHeader

  Telemetry header.
- CFE_TBL_HousekeepingTlm_Payload_t Payload

  Telemetry payload.

38.110.1 Detailed Description

Definition at line 789 of file cfe_tbl_msg.h.

38.110.2 Field Documentation

38.110.2.1 Payload

`CFE_TBL_HousekeepingTlm_Payload_t CFE_TBL_HousekeepingTlm::Payload`

Telemetry payload.

Definition at line 792 of file cfe_tbl_msg.h.

38.110.2.2 TlmHeader

`CFE_MSG_TelemetryHeader_t CFE_TBL_HousekeepingTlm::TlmHeader`

Telemetry header.

Definition at line 791 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.111 CFE_TBL_HousekeepingTlm_Payload Struct Reference

```
#include <cfe_tbl_msg.h>
```

Data Fields

- uint8 CommandCounter

  Count of valid commands received.
- uint8 CommandErrorCounter

  Count of invalid commands received.
- uint16 NumTables

  Number of Tables Registered.
- uint16 NumLoadPending

  Number of Tables pending on Applications for their update.
- uint16 ValidationCounter

  Number of completed table validations.
- uint32 LastValCrc

  Data Integrity Value computed for last table validated.
- int32 LastValStatus

  Returned status from validation function for last table validated.
- bool ActiveBuffer

  Indicator of whether table buffer validated was 0=Inactive, 1=Active.
- char LastValTableName [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

  Name of last table validated.
- uint8 SuccessValCounter

Total number of successful table validations.

- uint8 FailedValCounter

  Total number of unsuccessful table validations.

- uint8 NumValRequests

  Number of times Table Services has requested validations from Apps.

- uint8 NumFreeSharedBufs

  Number of free Shared Working Buffers.

- uint8 ByteAlignPad1

  Spare byte to ensure longword alignment.

- CFE_ES_MemHandle_t MemPoolHandle

  Handle to TBL's memory pool.

- CFE_TIME_SysTime_t LastUpdateTime

  Time of last table update.

- char LastUpdatedTable [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

  Name of the last table updated.

- char LastFileLoaded [CFE_MISSION_MAX_PATH_LEN]

  Path and Name of last table image file loaded.

- char LastFileDumped [CFE_MISSION_MAX_PATH_LEN]

  Path and Name of last file dumped to.

- char LastTableLoaded [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

  Name of the last table loaded.

38.111.1 Detailed Description

Name Table Services Housekeeping Packet

Definition at line 730 of file cfe_tbl_msg.h.

38.111.2 Field Documentation

38.111.2.1 ActiveBuffer

```
bool CFE_TBL_HousekeepingTlm_Payload::ActiveBuffer
```

Indicator of whether table buffer validated was 0=Inactive, 1=Active.

Telemetry Mnemonic(s) $sc_$cpu_TBL_LastValBuf

Definition at line 757 of file cfe_tbl_msg.h.

38.111.2.2 ByteAlignPad1

```
uint8 CFE_TBL_HousekeepingTlm_Payload::ByteAlignPad1
```

Spare byte to ensure longword alignment.

Telemetry Mnemonic(s) $sc_$cpu_TBL_ByteAlignPad1

Definition at line 773 of file cfe_tbl_msg.h.

38.111.2.3 CommandCounter

```
uint8 CFE_TBL_HousekeepingTlm_Payload::CommandCounter
```

Count of valid commands received.

Telemetry Mnemonic(s) $sc_$cpu_TBL_CMDPC

Definition at line 735 of file cfe_tbl_msg.h.

38.111.2.4 CommandErrorCounter

```
uint8 CFE_TBL_HousekeepingTlm_Payload::CommandErrorCounter
```

Count of invalid commands received.

Telemetry Mnemonic(s) $sc_$cpu_TBL_CMDEC

Definition at line 737 of file cfe_tbl_msg.h.

38.111.2.5 FailedValCounter

```
uint8 CFE_TBL_HousekeepingTlm_Payload::FailedValCounter
```

Total number of unsuccessful table validations.

Telemetry Mnemonic(s) $sc_$cpu_TBL_ValFailedCtr

Definition at line 763 of file cfe_tbl_msg.h.

38.111.2.6 LastFileDumped

```
char CFE_TBL_HousekeepingTlm_Payload::LastFileDumped[CFE_MISSION_MAX_PATH_LEN]
```

Path and Name of last file dumped to.

Telemetry Mnemonic(s) $sc_$cpu_TBL_LastFileDumped[OS_MAX_PATH_LEN]

Definition at line 783 of file cfe_tbl_msg.h.

38.111.2.7 LastFileLoaded

```
char CFE_TBL_HousekeepingTlm_Payload::LastFileLoaded[CFE_MISSION_MAX_PATH_LEN]
```

Path and Name of last table image file loaded.

Telemetry Mnemonic(s) $sc_$cpu_TBL_LastFileLoaded[OS_MAX_PATH_LEN]

Definition at line 781 of file cfe_tbl_msg.h.

38.111.2.8 LastTableLoaded

```
char CFE_TBL_HousekeepingTlm_Payload::LastTableLoaded[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
```

Name of the last table loaded.

Telemetry Mnemonic(s) $sc_$cpu_TBL_LastTableLoaded[CFE_TBL_MAX_FULL_NAME_LEN]

Definition at line 785 of file cfe_tbl_msg.h.

38.111.2.9 LastUpdatedTable

```
char CFE_TBL_HousekeepingTlm_Payload::LastUpdatedTable[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
```

Name of the last table updated.

Telemetry Mnemonic(s) $sc_$cpu_TBL_LastUpdTblName[CFE_TB_MAX_FULL_NAME_LEN]

Definition at line 779 of file cfe_tbl_msg.h.

38.111.2.10 LastUpdateTime

```
CFE_TIME_SysTime_t CFE_TBL_HousekeepingTlm_Payload::LastUpdateTime
```

Time of last table update.

Telemetry Mnemonic(s) $sc_$cpu_TBL_LastUpdTime, $sc_$cpu_TBL_SECONDS, $sc_$cpu_TBL_SUBSECONDS

Definition at line 777 of file cfe_tbl_msg.h.

38.111.2.11 LastValCrc

```
uint32 CFE_TBL_HousekeepingTlm_Payload::LastValCrc
```

Data Integrity Value computed for last table validated.

Telemetry Mnemonic(s) $sc_$cpu_TBL_LastValCRC

Definition at line 753 of file cfe_tbl_msg.h.

38.111.2.12 LastValStatus

```
int32 CFE_TBL_HousekeepingTlm_Payload::LastValStatus
```

Returned status from validation function for last table validated.

Telemetry Mnemonic(s) $sc_$cpu_TBl_LastValS

Definition at line 755 of file cfe_tbl_msg.h.

38.111.2.13 LastValTableName

```
char CFE_TBL_HousekeepingTlm_Payload::LastValTableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
```

Name of last table validated.

Telemetry Mnemonic(s) $sc_$cpu_TBL_LastValTblName[CFE_TB_MAX_FULL_NAME_LEN]

Definition at line 759 of file cfe_tbl_msg.h.

38.111.2.14 MemPoolHandle

`CFE_ES_MemHandle_t CFE_TBL_HousekeepingTlm_Payload::MemPoolHandle`

Handle to TBL's memory pool.

Telemetry Mnemonic(s) $sc_$cpu_TBL_MemPoolHandle

Definition at line 775 of file cfe_tbl_msg.h.

38.111.2.15 NumFreeSharedBufs

`uint8 CFE_TBL_HousekeepingTlm_Payload::NumFreeSharedBufs`

Number of free Shared Working Buffers.

Telemetry Mnemonic(s) $sc_$cpu_TBL_NumFreeShrBuf

Definition at line 771 of file cfe_tbl_msg.h.

38.111.2.16 NumLoadPending

`uint16 CFE_TBL_HousekeepingTlm_Payload::NumLoadPending`

Number of Tables pending on Applications for their update.

Telemetry Mnemonic(s) $sc_$cpu_TBL_NumUpdatesPend

Definition at line 745 of file cfe_tbl_msg.h.

38.111.2.17 NumTables

`uint16 CFE_TBL_HousekeepingTlm_Payload::NumTables`

Number of Tables Registered.

Telemetry Mnemonic(s) $sc_$cpu_TBL_NumTables

Definition at line 743 of file cfe_tbl_msg.h.

38.111.2.18 NumValRequests

```
uint8 CFE_TBL_HousekeepingTlm_Payload::NumValRequests
```

Number of times Table Services has requested validations from Apps.

Telemetry Mnemonic(s) $sc_$cpu_TBL_ValReqCtr

Definition at line 765 of file cfe_tbl_msg.h.

38.111.2.19 SuccessValCounter

```
uint8 CFE_TBL_HousekeepingTlm_Payload::SuccessValCounter
```

Total number of successful table validations.

Telemetry Mnemonic(s) $sc_$cpu_TBL_ValSuccessCtr

Definition at line 761 of file cfe_tbl_msg.h.

38.111.2.20 ValidationCounter

```
uint16 CFE_TBL_HousekeepingTlm_Payload::ValidationCounter
```

Number of completed table validations.

Telemetry Mnemonic(s) $sc_$cpu_TBL_ValCompltdCtr

Definition at line 751 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.112 CFE_TBL_Info Struct Reference

Table Info.

```
#include <cfe_tbl_api_typedefs.h>
```

Data Fields

- size_t Size

  Size, in bytes, of Table.
- uint32 NumUsers

  Number of Apps with access to the table.
- uint32 FileCreateTimeSecs

  File creation time from last file loaded into table.
- uint32 FileCreateTimeSubSecs

  File creation time from last file loaded into table.
- uint32 Crc

  Most recently calculated CRC by TBL services on table contents.
- CFE_TIME_SysTime_t TimeOfLastUpdate

  Time when Table was last updated.
- bool TableLoadedOnce

  Flag indicating whether table has been loaded once or not.
- bool DumpOnly

  Flag indicating Table is NOT to be loaded.
- bool DoubleBuffered

  Flag indicating Table has a dedicated inactive buffer.
- bool UserDefAddr

  Flag indicating Table address was defined by Owner Application.
- bool Critical

  Flag indicating Table contents are maintained in a CDS.
- char LastFileLoaded [CFE_MISSION_MAX_PATH_LEN]

  Filename of last file loaded into table.

38.112.1 Detailed Description

Table Info.

Definition at line 110 of file cfe_tbl_api_typedefs.h.

38.112.2 Field Documentation

38.112.2.1 Crc

```
uint32 CFE_TBL_Info::Crc
```

Most recently calculated CRC by TBL services on table contents.

Definition at line 116 of file cfe_tbl_api_typedefs.h.

38.112.2.2 Critical

```
bool CFE_TBL_Info::Critical
```

Flag indicating Table contents are maintained in a CDS.

Definition at line 122 of file cfe_tbl_api_typedefs.h.

38.112.2.3 DoubleBuffered

```
bool CFE_TBL_Info::DoubleBuffered
```

Flag indicating Table has a dedicated inactive buffer.

Definition at line 120 of file cfe_tbl_api_typedefs.h.

38.112.2.4 DumpOnly

```
bool CFE_TBL_Info::DumpOnly
```

Flag indicating Table is NOT to be loaded.

Definition at line 119 of file cfe_tbl_api_typedefs.h.

38.112.2.5 FileCreateTimeSecs

```
uint32 CFE_TBL_Info::FileCreateTimeSecs
```

File creation time from last file loaded into table.

Definition at line 114 of file cfe_tbl_api_typedefs.h.

38.112.2.6 FileCreateTimeSubSecs

```
uint32 CFE_TBL_Info::FileCreateTimeSubSecs
```

File creation time from last file loaded into table.

Definition at line 115 of file cfe_tbl_api_typedefs.h.

38.112.2.7 LastFileLoaded

```
char CFE_TBL_Info::LastFileLoaded[CFE_MISSION_MAX_PATH_LEN]
```

Filename of last file loaded into table.

Definition at line 123 of file cfe_tbl_api_typedefs.h.

38.112.2.8 NumUsers

```
uint32 CFE_TBL_Info::NumUsers
```

Number of Apps with access to the table.

Definition at line 113 of file cfe_tbl_api_typedefs.h.

38.112.2.9 Size

```
size_t CFE_TBL_Info::Size
```

Size, in bytes, of Table.

Definition at line 112 of file cfe_tbl_api_typedefs.h.

38.112.2.10 TableLoadedOnce

```
bool CFE_TBL_Info::TableLoadedOnce
```

Flag indicating whether table has been loaded once or not.

Definition at line 118 of file cfe_tbl_api_typedefs.h.

38.112.2.11 TimeOfLastUpdate

```
CFE_TIME_SysTime_t CFE_TBL_Info::TimeOfLastUpdate
```

Time when Table was last updated.

Definition at line 117 of file cfe_tbl_api_typedefs.h.

38.112.2.12 UserDefAddr

```
bool CFE_TBL_Info::UserDefAddr
```

Flag indicating Table address was defined by Owner Application.

Definition at line 121 of file cfe_tbl_api_typedefs.h.

The documentation for this struct was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_tbl_api_typedefs.h

38.113 CFE_TBL_LoadCmd Struct Reference

Load Table Command.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_TBL_LoadCmd_Payload_t Payload
  Command payload.

38.113.1 Detailed Description

Load Table Command.

Definition at line 523 of file cfe_tbl_msg.h.

38.113.2 Field Documentation

38.113.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_TBL_LoadCmd::CmdHeader
```

Command header.

Definition at line 525 of file cfe_tbl_msg.h.

38.113.2.2 Payload

```
CFE_TBL_LoadCmd_Payload_t CFE_TBL_LoadCmd::Payload
```

Command payload.

Definition at line 526 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.114 CFE_TBL_LoadCmd_Payload Struct Reference

Load Table Command Payload.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- char LoadFilename [CFE_MISSION_MAX_PATH_LEN]
  Filename (and path) of data to be loaded.

38.114.1 Detailed Description

Load Table Command Payload.

For command details, see CFE_TBL_LOAD_CC

Definition at line 515 of file cfe_tbl_msg.h.

38.114.2 Field Documentation

38.114.2.1 LoadFilename

```
char CFE_TBL_LoadCmd_Payload::LoadFilename[CFE_MISSION_MAX_PATH_LEN]
```

Filename (and path) of data to be loaded.

Definition at line 517 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.115 CFE_TBL_NoArgsCmd Struct Reference

Generic "no arguments" command.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.

38.115.1 Detailed Description

Generic "no arguments" command.

This command structure is used for commands that do not have any parameters. This includes:

1. The Housekeeping Request Message
2. The No-Op Command (For details, see CFE_TBL_NOOP_CC)
3. The Reset Counters Command (For details, see CFE_TBL_RESET_COUNTERS_CC)

Definition at line 496 of file cfe_tbl_msg.h.

38.115.2 Field Documentation

38.115.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_TBL_NoArgsCmd::CmdHeader
```

Command header.

Definition at line 498 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.116 CFE_TBL_NotifyCmd Struct Reference

```
#include <cfe_tbl_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_TBL_NotifyCmd_Payload_t Payload

  Command payload.

38.116.1 Detailed Description

/brief Table Management Notification Command

Definition at line 717 of file cfe_tbl_msg.h.

38.116.2 Field Documentation

38.116.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_TBL_NotifyCmd::CmdHeader`

Command header.

Definition at line 719 of file cfe_tbl_msg.h.

38.116.2.2 Payload

`CFE_TBL_NotifyCmd_Payload_t CFE_TBL_NotifyCmd::Payload`

Command payload.

Definition at line 720 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.117 CFE_TBL_NotifyCmd_Payload Struct Reference

Table Management Notification Command Payload.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- uint32 Parameter

  Application specified command parameter.

38.117.1 Detailed Description

Table Management Notification Command Payload.

Description

Whenever an application that owns a table calls the CFE_TBL_NotifyByMessage API following the table registration, Table services will generate the following command message with the application specified message ID, command code and parameter whenever the table requires management (e.g. - loads and validations).

Definition at line 709 of file cfe_tbl_msg.h.

38.117.2 Field Documentation

38.117.2.1 Parameter

```
uint32 CFE_TBL_NotifyCmd_Payload::Parameter
```

Application specified command parameter.

Definition at line 711 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.118 CFE_TBL_SendRegistryCmd Struct Reference

Send Table Registry Command.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_TBL_SendRegistryCmd_Payload_t Payload

  Command payload.

38.118.1 Detailed Description

Send Table Registry Command.

Definition at line 646 of file cfe_tbl_msg.h.

38.118.2 Field Documentation

38.118.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_TBL_SendRegistryCmd::CmdHeader`

Command header.

Definition at line 648 of file cfe_tbl_msg.h.

38.118.2.2 Payload

`CFE_TBL_SendRegistryCmd_Payload_t CFE_TBL_SendRegistryCmd::Payload`

Command payload.

Definition at line 649 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.119 CFE_TBL_SendRegistryCmd_Payload Struct Reference

Send Table Registry Command Payload.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- char TableName [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

  Full Name of Table whose registry entry is to be telemetered.

38.119.1 Detailed Description

Send Table Registry Command Payload.

For command details, see CFE_TBL_SEND_REGISTRY_CC

Definition at line 634 of file cfe_tbl_msg.h.

38.119.2 Field Documentation

38.119.2.1 TableName

```
char CFE_TBL_SendRegistryCmd_Payload::TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
```

Full Name of Table whose registry entry is to be telemetered.

ASCII string containing full table name identifier of table whose registry entry is to be telemetered via CFE_TBL_Table↩ RegistryTlm_t

Definition at line 636 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.120 CFE_TBL_TableRegistryTlm Struct Reference

```
#include <cfe_tbl_msg.h>
```

Data Fields

- CFE_MSG_TelemetryHeader_t TlmHeader

  Telemetry header.
- CFE_TBL_TblRegPacket_Payload_t Payload

  Telemetry payload.

38.120.1 Detailed Description

Definition at line 836 of file cfe_tbl_msg.h.

38.120.2 Field Documentation

38.120.2.1 Payload

`CFE_TBL_TblRegPacket_Payload_t CFE_TBL_TableRegistryTlm::Payload`

Telemetry payload.

Definition at line 839 of file cfe_tbl_msg.h.

38.120.2.2 TlmHeader

`CFE_MSG_TelemetryHeader_t CFE_TBL_TableRegistryTlm::TlmHeader`

Telemetry header.

Definition at line 838 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.121 CFE_TBL_TblRegPacket_Payload Struct Reference

```
#include <cfe_tbl_msg.h>
```

Data Fields

- CFE_ES_MemOffset_t Size

  Size, in bytes, of Table.
- uint32 Crc

  Most recently calculated CRC of Table.
- CFE_ES_MemAddress_t ActiveBufferAddr

  Address of Active Buffer.
- CFE_ES_MemAddress_t InactiveBufferAddr

  Address of Inactive Buffer.
- CFE_ES_MemAddress_t ValidationFuncPtr

  Ptr to Owner App's function that validates tbl contents.
- CFE_TIME_SysTime_t TimeOfLastUpdate

  Time when Table was last updated.
- uint32 FileCreateTimeSecs

  File creation time from last file loaded into table.
- uint32 FileCreateTimeSubSecs

File creation time from last file loaded into table.

- bool TableLoadedOnce

  Flag indicating whether table has been loaded once or not.
- bool LoadPending

  Flag indicating an inactive buffer is ready to be copied.
- bool DumpOnly

  Flag indicating Table is NOT to be loaded.
- bool DoubleBuffered

  Flag indicating Table has a dedicated inactive buffer.
- char Name [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

  Processor specific table name.
- char LastFileLoaded [CFE_MISSION_MAX_PATH_LEN]

  Filename of last file loaded into table.
- char OwnerAppName [CFE_MISSION_MAX_API_LEN]

  Name of owning application.
- bool Critical

  Indicates whether table is Critical or not.
- uint8 ByteAlign4

  Spare byte to maintain byte alignment.

38.121.1 Detailed Description

Name Table Registry Info Packet

Definition at line 798 of file cfe_tbl_msg.h.

38.121.2 Field Documentation

38.121.2.1 ActiveBufferAddr

```
CFE_ES_MemAddress_t CFE_TBL_TblRegPacket_Payload::ActiveBufferAddr
```

Address of Active Buffer.

Telemetry Mnemonic(s) $sc_$cpu_TBL_ActBufAdd

Definition at line 804 of file cfe_tbl_msg.h.

38.121.2.2 ByteAlign4

```
uint8 CFE_TBL_TblRegPacket_Payload::ByteAlign4
```

Spare byte to maintain byte alignment.

Telemetry Mnemonic(s) $sc_$cpu_TBL_Spare4

Definition at line 832 of file cfe_tbl_msg.h.

38.121.2.3 Crc

```
uint32 CFE_TBL_TblRegPacket_Payload::Crc
```

Most recently calculated CRC of Table.

Telemetry Mnemonic(s) $sc_$cpu_TBL_CRC

Definition at line 802 of file cfe_tbl_msg.h.

38.121.2.4 Critical

```
bool CFE_TBL_TblRegPacket_Payload::Critical
```

Indicates whether table is Critical or not.

Telemetry Mnemonic(s) $sc_$cpu_TBL_Spare3

Definition at line 830 of file cfe_tbl_msg.h.

38.121.2.5 DoubleBuffered

```
bool CFE_TBL_TblRegPacket_Payload::DoubleBuffered
```

Flag indicating Table has a dedicated inactive buffer.

Telemetry Mnemonic(s) $sc_$cpu_TBL_DblBuffered

Definition at line 822 of file cfe_tbl_msg.h.

38.121.2.6 DumpOnly

```
bool CFE_TBL_TblRegPacket_Payload::DumpOnly
```

Flag indicating Table is NOT to be loaded.

Telemetry Mnemonic(s) $sc_$cpu_TBL_DumpOnly

Definition at line 820 of file cfe_tbl_msg.h.

38.121.2.7 FileCreateTimeSecs

```
uint32 CFE_TBL_TblRegPacket_Payload::FileCreateTimeSecs
```

File creation time from last file loaded into table.

Telemetry Mnemonic(s) $sc_$cpu_TBL_FILECSECONDS

Definition at line 812 of file cfe_tbl_msg.h.

38.121.2.8 FileCreateTimeSubSecs

```
uint32 CFE_TBL_TblRegPacket_Payload::FileCreateTimeSubSecs
```

File creation time from last file loaded into table.

Telemetry Mnemonic(s) $sc_$cpu_TBL_FILECSUBSECONDS

Definition at line 814 of file cfe_tbl_msg.h.

38.121.2.9 InactiveBufferAddr

```
CFE_ES_MemAddress_t CFE_TBL_TblRegPacket_Payload::InactiveBufferAddr
```

Address of Inactive Buffer.

Telemetry Mnemonic(s) $sc_$cpu_TBL_IActBufAdd

Definition at line 806 of file cfe_tbl_msg.h.

38.121.2.10 LastFileLoaded

```
char CFE_TBL_TblRegPacket_Payload::LastFileLoaded[CFE_MISSION_MAX_PATH_LEN]
```

Filename of last file loaded into table.

Telemetry Mnemonic(s) $sc_$cpu_TBL_LastFileUpd[OS_MAX_PATH_LEN]

Definition at line 826 of file cfe_tbl_msg.h.

38.121.2.11 LoadPending

```
bool CFE_TBL_TblRegPacket_Payload::LoadPending
```

Flag indicating an inactive buffer is ready to be copied.

Telemetry Mnemonic(s) $sc_$cpu_TBL_UpdatePndng

Definition at line 818 of file cfe_tbl_msg.h.

38.121.2.12 Name

```
char CFE_TBL_TblRegPacket_Payload::Name[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
```

Processor specific table name.

Telemetry Mnemonic(s) $sc_$cpu_TBL_Name[CFE_TB_MAX_FULL_NAME_LEN]

Definition at line 824 of file cfe_tbl_msg.h.

38.121.2.13 OwnerAppName

```
char CFE_TBL_TblRegPacket_Payload::OwnerAppName[CFE_MISSION_MAX_API_LEN]
```

Name of owning application.

Telemetry Mnemonic(s) $sc_$cpu_TBL_OwnerApp[OS_MAX_API_NAME]

Definition at line 828 of file cfe_tbl_msg.h.

38.121.2.14 Size

`CFE_ES_MemOffset_t CFE_TBL_TblRegPacket_Payload::Size`

Size, in bytes, of Table.

Telemetry Mnemonic(s) $sc_$cpu_TBL_SIZE

Definition at line 800 of file cfe_tbl_msg.h.

38.121.2.15 TableLoadedOnce

`bool CFE_TBL_TblRegPacket_Payload::TableLoadedOnce`

Flag indicating whether table has been loaded once or not.

Telemetry Mnemonic(s) $sc_$cpu_TBL_LoadedOnce

Definition at line 816 of file cfe_tbl_msg.h.

38.121.2.16 TimeOfLastUpdate

`CFE_TIME_SysTime_t CFE_TBL_TblRegPacket_Payload::TimeOfLastUpdate`

Time when Table was last updated.

Telemetry Mnemonic(s) $sc_$cpu_TBL_TimeLastUpd, $sc_$cpu_TBL_TLUSECONDS, $sc_$cpu_TBL_TLUSUB↩ SECONDS

Definition at line 810 of file cfe_tbl_msg.h.

38.121.2.17 ValidationFuncPtr

`CFE_ES_MemAddress_t CFE_TBL_TblRegPacket_Payload::ValidationFuncPtr`

Ptr to Owner App's function that validates tbl contents.

Telemetry Mnemonic(s) $sc_$cpu_TBL_ValFuncPtr

Definition at line 808 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.122 CFE_TBL_ValidateCmd Struct Reference

Validate Table Command.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_TBL_ValidateCmd_Payload_t Payload
  Command payload.

38.122.1 Detailed Description

Validate Table Command.

Definition at line 580 of file cfe_tbl_msg.h.

38.122.2 Field Documentation

38.122.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_TBL_ValidateCmd::CmdHeader
```

Command header.

Definition at line 582 of file cfe_tbl_msg.h.

38.122.2.2 Payload

```
CFE_TBL_ValidateCmd_Payload_t CFE_TBL_ValidateCmd::Payload
```

Command payload.

Definition at line 583 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.123 CFE_TBL_ValidateCmd_Payload Struct Reference

Validate Table Command Payload.

```
#include <cfe_tbl_msg.h>
```

Data Fields

- uint16 ActiveTableFlag

  CFE_TBL_BufferSelect_INACTIVE=Inactive Table, CFE_TBL_BufferSelect_ACTIVE=Active Table
- char TableName [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

  Full Name of Table to be validated.

38.123.1 Detailed Description

Validate Table Command Payload.

For command details, see CFE_TBL_VALIDATE_CC

Definition at line 564 of file cfe_tbl_msg.h.

38.123.2 Field Documentation

38.123.2.1 ActiveTableFlag

```
uint16 CFE_TBL_ValidateCmd_Payload::ActiveTableFlag
```

CFE_TBL_BufferSelect_INACTIVE=Inactive Table, CFE_TBL_BufferSelect_ACTIVE=Active Table

Selects either the "Inactive" (CFE_TBL_BufferSelect_INACTIVE) buffer or the "Active" (CFE_TBL_BufferSelect_ACTIVE) buffer to be validated

Definition at line 566 of file cfe_tbl_msg.h.

38.123.2.2 TableName

```
char CFE_TBL_ValidateCmd_Payload::TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
```

Full Name of Table to be validated.

ASCII string containing full table name identifier of table to be validated

Definition at line 572 of file cfe_tbl_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h

38.124 CFE_TIME_DiagnosticTlm Struct Reference

```
#include <cfe_time_msg.h>
```

Data Fields

- CFE_MSG_TelemetryHeader_t TlmHeader
  Telemetry header.
- CFE_TIME_DiagnosticTlm_Payload_t Payload
  Telemetry payload.

38.124.1 Detailed Description

Definition at line 1123 of file cfe_time_msg.h.

38.124.2 Field Documentation

38.124.2.1 Payload

```
CFE_TIME_DiagnosticTlm_Payload_t CFE_TIME_DiagnosticTlm::Payload
```

Telemetry payload.

Definition at line 1126 of file cfe_time_msg.h.

38.124.2.2 TlmHeader

```
CFE_MSG_TelemetryHeader_t CFE_TIME_DiagnosticTlm::TlmHeader
```

Telemetry header.

Definition at line 1125 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.125 CFE_TIME_DiagnosticTlm_Payload Struct Reference

```
#include <cfe_time_msg.h>
```

Data Fields

- CFE_TIME_SysTime_t AtToneMET

  MET at time of tone.
- CFE_TIME_SysTime_t AtToneSTCF

  STCF at time of tone.
- CFE_TIME_SysTime_t AtToneDelay

  Adjustment for slow tone detection.
- CFE_TIME_SysTime_t AtToneLatch

  Local clock latched at time of tone.
- int16 AtToneLeapSeconds

  Leap Seconds at time of tone.
- int16 ClockStateAPI

  Clock state as per API.
- CFE_TIME_SysTime_t TimeSinceTone

  Time elapsed since the tone.
- CFE_TIME_SysTime_t CurrentLatch

  Local clock latched just "now".
- CFE_TIME_SysTime_t CurrentMET

  MET at this instant.
- CFE_TIME_SysTime_t CurrentTAI

  TAI at this instant.
- CFE_TIME_SysTime_t CurrentUTC

  UTC at this instant.
- int16 ClockSetState

  Time has been "set".
- int16 ClockFlyState

  Current fly-wheel state.
- int16 ClockSource

  Internal vs external, etc.
- int16 ClockSignal

  Primary vs redundant, etc.
- int16 ServerFlyState

  Used by clients only.
- int16 Forced2Fly

  Commanded into fly-wheel.
- uint16 ClockStateFlags

  Clock State Flags.
- int16 OneTimeDirection

  One time STCF adjustment direction (Add = 1, Sub = 2)
- int16 OneHzDirection

  1Hz STCF adjustment direction
- int16 DelayDirection

  Client latency adjustment direction.
- CFE_TIME_SysTime_t OneTimeAdjust

  Previous one-time STCF adjustment.
- CFE_TIME_SysTime_t OneHzAdjust

Current 1Hz STCF adjustment.

- CFE_TIME_SysTime_t ToneSignalLatch

  Local Clock latched at most recent tone signal.
- CFE_TIME_SysTime_t ToneDataLatch

  Local Clock latched at arrival of tone data.
- uint32 ToneMatchCounter

  Tone signal / data verification count.
- uint32 ToneMatchErrorCounter

  Tone signal / data verification error count.
- uint32 ToneSignalCounter

  Tone signal detected SB message count.
- uint32 ToneDataCounter

  Time at the tone data SB message count.
- uint32 ToneIntCounter

  Tone signal ISR execution count.
- uint32 ToneIntErrorCounter

  Tone signal ISR error count.
- uint32 ToneTaskCounter

  Tone task execution count.
- uint32 VersionCounter

  Count of mods to time at tone reference data (version)
- uint32 LocalIntCounter

  Local 1Hz ISR execution count.
- uint32 LocalTaskCounter

  Local 1Hz task execution count.
- uint32 VirtualMET

  Software MET.
- uint32 MinElapsed

  Min tone signal / data pkt arrival window (Sub-seconds)
- uint32 MaxElapsed

  Max tone signal / data pkt arrival window (Sub-seconds)
- CFE_TIME_SysTime_t MaxLocalClock

  Max local clock value before rollover.
- uint32 ToneOverLimit

  Max between tone signal interrupts.
- uint32 ToneUnderLimit

  Min between tone signal interrupts.
- uint32 DataStoreStatus

  Data Store status (preserved across processor reset)

38.125.1 Detailed Description

Name Time Services Diagnostics Packet

Definition at line 977 of file cfe_time_msg.h.

38.125.2 Field Documentation

38.125.2.1 AtToneDelay

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::AtToneDelay`

Adjustment for slow tone detection.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DLatentS, $sc_$cpu_TIME_DLatentSs

Definition at line 986 of file cfe_time_msg.h.

38.125.2.2 AtToneLatch

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::AtToneLatch`

Local clock latched at time of tone.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DTValidS, $sc_$cpu_TIME_DTValidSs

Definition at line 988 of file cfe_time_msg.h.

38.125.2.3 AtToneLeapSeconds

`int16 CFE_TIME_DiagnosticTlm_Payload::AtToneLeapSeconds`

Leap Seconds at time of tone.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DLeapS

Definition at line 991 of file cfe_time_msg.h.

38.125.2.4 AtToneMET

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::AtToneMET`

MET at time of tone.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DTMETS, $sc_$cpu_TIME_DTMETSs

Definition at line 982 of file cfe_time_msg.h.

38.125.2.5 AtToneSTCF

```
CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::AtToneSTCF
```

STCF at time of tone.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DSTCFS, $sc_$cpu_TIME_DSTCFSS

Definition at line 984 of file cfe_time_msg.h.

38.125.2.6 ClockFlyState

```
int16 CFE_TIME_DiagnosticTlm_Payload::ClockFlyState
```

Current fly-wheel state.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DFlywheel

Definition at line 1015 of file cfe_time_msg.h.

38.125.2.7 ClockSetState

```
int16 CFE_TIME_DiagnosticTlm_Payload::ClockSetState
```

Time has been "set".

Telemetry Mnemonic(s) $sc_$cpu_TIME_DValid

Definition at line 1013 of file cfe_time_msg.h.

38.125.2.8 ClockSignal

```
int16 CFE_TIME_DiagnosticTlm_Payload::ClockSignal
```

Primary vs redundant, etc.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DSignal

Definition at line 1019 of file cfe_time_msg.h.

38.125.2.9 ClockSource

```
int16 CFE_TIME_DiagnosticTlm_Payload::ClockSource
```

Internal vs external, etc.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DSource

Definition at line 1017 of file cfe_time_msg.h.

38.125.2.10 ClockStateAPI

```
int16 CFE_TIME_DiagnosticTlm_Payload::ClockStateAPI
```

Clock state as per API.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DAPIState

Definition at line 993 of file cfe_time_msg.h.

38.125.2.11 ClockStateFlags

```
uint16 CFE_TIME_DiagnosticTlm_Payload::ClockStateFlags
```

Clock State Flags.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DStateFlags, $sc_$cpu_TIME_DFlagSet, $sc_$cpu_TIME_DFlagFly, $sc_$cpu_TIME_DFlagSrc, $sc_$cpu_TIME_DFlagPri, $sc_$cpu_TIME_DFlagSfly, $sc_↩ $cpu_TIME_DFlagCfly, $sc_$cpu_TIME_DFlagAdjd, $sc_$cpu_TIME_DFlag1Hzd, $sc_↩ $cpu_TIME_DFlagClat, $sc_$cpu_TIME_DFlagSorC, $sc_$cpu_TIME_DFlagNIU

Definition at line 1029 of file cfe_time_msg.h.

38.125.2.12 CurrentLatch

```
CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::CurrentLatch
```

Local clock latched just "now".

Telemetry Mnemonic(s) $sc_$cpu_TIME_DLocalS, $sc_$cpu_TIME_DLocalSs

Definition at line 1001 of file cfe_time_msg.h.

38.125.2.13 CurrentMET

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::CurrentMET`

MET at this instant.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DMETS, $sc_$cpu_TIME_DMETSs

Definition at line 1003 of file cfe_time_msg.h.

38.125.2.14 CurrentTAI

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::CurrentTAI`

TAI at this instant.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DTAIS, $sc_$cpu_TIME_DTAISS

Definition at line 1005 of file cfe_time_msg.h.

38.125.2.15 CurrentUTC

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::CurrentUTC`

UTC at this instant.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DUTCS, $sc_$cpu_TIME_DUTCSS

Definition at line 1007 of file cfe_time_msg.h.

38.125.2.16 DataStoreStatus

`uint32 CFE_TIME_DiagnosticTlm_Payload::DataStoreStatus`

Data Store status (preserved across processor reset)

Telemetry Mnemonic(s) $sc_$cpu_TIME_DataStStat

Definition at line 1119 of file cfe_time_msg.h.

38.125.2.17 DelayDirection

```
int16 CFE_TIME_DiagnosticTlm_Payload::DelayDirection
```

Client latency adjustment direction.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DLatentDir

Definition at line 1039 of file cfe_time_msg.h.

38.125.2.18 Forced2Fly

```
int16 CFE_TIME_DiagnosticTlm_Payload::Forced2Fly
```

Commanded into fly-wheel.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DCMD2Fly

Definition at line 1023 of file cfe_time_msg.h.

38.125.2.19 LocalIntCounter

```
uint32 CFE_TIME_DiagnosticTlm_Payload::LocalIntCounter
```

Local 1Hz ISR execution count.

Telemetry Mnemonic(s) $sc_$cpu_TIME_D1HzISRCNT

Definition at line 1077 of file cfe_time_msg.h.

38.125.2.20 LocalTaskCounter

```
uint32 CFE_TIME_DiagnosticTlm_Payload::LocalTaskCounter
```

Local 1Hz task execution count.

Telemetry Mnemonic(s) $sc_$cpu_TIME_D1HzTaskCNT

Definition at line 1079 of file cfe_time_msg.h.

38.125.2.21 MaxElapsed

`uint32 CFE_TIME_DiagnosticTlm_Payload::MaxElapsed`

Max tone signal / data pkt arrival window (Sub-seconds)

Telemetry Mnemonic(s) $sc_$cpu_TIME_DMaxWindow

Definition at line 1099 of file cfe_time_msg.h.

38.125.2.22 MaxLocalClock

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::MaxLocalClock`

Max local clock value before rollover.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DWrapS, $sc_$cpu_TIME_DWrapSs

Definition at line 1105 of file cfe_time_msg.h.

38.125.2.23 MinElapsed

`uint32 CFE_TIME_DiagnosticTlm_Payload::MinElapsed`

Min tone signal / data pkt arrival window (Sub-seconds)

Telemetry Mnemonic(s) $sc_$cpu_TIME_DMinWindow

Definition at line 1097 of file cfe_time_msg.h.

38.125.2.24 OneHzAdjust

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::OneHzAdjust`

Current 1Hz STCF adjustment.

Telemetry Mnemonic(s) $sc_$cpu_TIME_D1HzAdjS, $sc_$cpu_TIME_D1HzAdjSs

Definition at line 1047 of file cfe_time_msg.h.

38.125.2.25 OneHzDirection

```
int16 CFE_TIME_DiagnosticTlm_Payload::OneHzDirection
```

1Hz STCF adjustment direction

Telemetry Mnemonic(s) $sc_$cpu_TIME_D1HzAdjDir

Definition at line 1037 of file cfe_time_msg.h.

38.125.2.26 OneTimeAdjust

```
CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::OneTimeAdjust
```

Previous one-time STCF adjustment.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DAdjustS, $sc_$cpu_TIME_DAdjustSs

Definition at line 1045 of file cfe_time_msg.h.

38.125.2.27 OneTimeDirection

```
int16 CFE_TIME_DiagnosticTlm_Payload::OneTimeDirection
```

One time STCF adjustment direction (Add = 1, Sub = 2)

Telemetry Mnemonic(s) $sc_$cpu_TIME_DAdjustDir

Definition at line 1035 of file cfe_time_msg.h.

38.125.2.28 ServerFlyState

```
int16 CFE_TIME_DiagnosticTlm_Payload::ServerFlyState
```

Used by clients only.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DSrvFly

Definition at line 1021 of file cfe_time_msg.h.

38.125.2.29 TimeSinceTone

```
CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::TimeSinceTone
```

Time elapsed since the tone.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DElapsedS, $sc_$cpu_TIME_DElapsedSs

Definition at line 999 of file cfe_time_msg.h.

38.125.2.30 ToneDataCounter

```
uint32 CFE_TIME_DiagnosticTlm_Payload::ToneDataCounter
```

Time at the tone data SB message count.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DTatTCNT

Definition at line 1067 of file cfe_time_msg.h.

38.125.2.31 ToneDataLatch

```
CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::ToneDataLatch
```

Local Clock latched at arrival of tone data.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DTDS, $sc_$cpu_TIME_DTDSs

Definition at line 1055 of file cfe_time_msg.h.

38.125.2.32 ToneIntCounter

```
uint32 CFE_TIME_DiagnosticTlm_Payload::ToneIntCounter
```

Tone signal ISR execution count.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DTsISRCNT

Definition at line 1069 of file cfe_time_msg.h.

38.125.2.33 ToneIntErrorCounter

```
uint32 CFE_TIME_DiagnosticTlm_Payload::ToneIntErrorCounter
```

Tone signal ISR error count.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DTsISRERR

Definition at line 1071 of file cfe_time_msg.h.

38.125.2.34 ToneMatchCounter

```
uint32 CFE_TIME_DiagnosticTlm_Payload::ToneMatchCounter
```

Tone signal / data verification count.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DVerifyCNT

Definition at line 1061 of file cfe_time_msg.h.

38.125.2.35 ToneMatchErrorCounter

```
uint32 CFE_TIME_DiagnosticTlm_Payload::ToneMatchErrorCounter
```

Tone signal / data verification error count.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DVerifyER

Definition at line 1063 of file cfe_time_msg.h.

38.125.2.36 ToneOverLimit

```
uint32 CFE_TIME_DiagnosticTlm_Payload::ToneOverLimit
```

Max between tone signal interrupts.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DMaxSs

Definition at line 1111 of file cfe_time_msg.h.

38.125.2.37 ToneSignalCounter

```
uint32 CFE_TIME_DiagnosticTlm_Payload::ToneSignalCounter
```

Tone signal detected SB message count.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DTSDetCNT

Definition at line 1065 of file cfe_time_msg.h.

38.125.2.38 ToneSignalLatch

```
CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::ToneSignalLatch
```

Local Clock latched at most recent tone signal.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DTTS, $sc_$cpu_TIME_DTTSs

Definition at line 1053 of file cfe_time_msg.h.

38.125.2.39 ToneTaskCounter

```
uint32 CFE_TIME_DiagnosticTlm_Payload::ToneTaskCounter
```

Tone task execution count.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DTsTaskCNT

Definition at line 1073 of file cfe_time_msg.h.

38.125.2.40 ToneUnderLimit

```
uint32 CFE_TIME_DiagnosticTlm_Payload::ToneUnderLimit
```

Min between tone signal interrupts.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DMinSs

Definition at line 1113 of file cfe_time_msg.h.

38.125.2.41 VersionCounter

```
uint32 CFE_TIME_DiagnosticTlm_Payload::VersionCounter
```

Count of mods to time at tone reference data (version)

Telemetry Mnemonic(s) $sc_$cpu_TIME_DVersionCNT

Definition at line 1075 of file cfe_time_msg.h.

38.125.2.42 VirtualMET

```
uint32 CFE_TIME_DiagnosticTlm_Payload::VirtualMET
```

Software MET.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DLogicalMET

Definition at line 1085 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.126 CFE_TIME_HousekeepingTlm Struct Reference

```
#include <cfe_time_msg.h>
```

Data Fields

- CFE_MSG_TelemetryHeader_t TlmHeader

  Telemetry header.
- CFE_TIME_HousekeepingTlm_Payload_t Payload

  Telemetry payload.

38.126.1 Detailed Description

Definition at line 966 of file cfe_time_msg.h.

38.126.2 Field Documentation

38.126.2.1 Payload

`CFE_TIME_HousekeepingTlm_Payload_t CFE_TIME_HousekeepingTlm::Payload`

Telemetry payload.

Definition at line 969 of file cfe_time_msg.h.

38.126.2.2 TlmHeader

`CFE_MSG_TelemetryHeader_t CFE_TIME_HousekeepingTlm::TlmHeader`

Telemetry header.

Definition at line 968 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.127 CFE_TIME_HousekeepingTlm_Payload Struct Reference

```
#include <cfe_time_msg.h>
```

Data Fields

- uint8 CommandCounter

  Time Command Execution Counter.
- uint8 CommandErrorCounter

  Time Command Error Counter.
- uint16 ClockStateFlags

  State Flags.
- int16 ClockStateAPI

  API State.
- int16 LeapSeconds

  Current Leaps Seconds.
- uint32 SecondsMET

  Current MET (seconds)
- uint32 SubsecsMET

  Current MET (sub-seconds)
- uint32 SecondsSTCF

Current STCF (seconds)

- uint32 SubsecsSTCF

  Current STCF (sub-seconds)
- uint32 Seconds1HzAdj

  Current 1 Hz SCTF adjustment (seconds)
- uint32 Subsecs1HzAdj

  Current 1 Hz SCTF adjustment (sub-seconds)
- uint32 SecondsDelay

  Current 1 Hz SCTF Delay (seconds)
- uint32 SubsecsDelay

  Current 1 Hz SCTF Delay (sub-seconds)

38.127.1 Detailed Description

Name Time Services Housekeeping Packet

Definition at line 907 of file cfe_time_msg.h.

38.127.2 Field Documentation

38.127.2.1 ClockStateAPI

```
int16 CFE_TIME_HousekeepingTlm_Payload::ClockStateAPI
```

API State.

Telemetry Mnemonic(s) $sc_$cpu_TIME_DAPIState

Definition at line 922 of file cfe_time_msg.h.

38.127.2.2 ClockStateFlags

```
uint16 CFE_TIME_HousekeepingTlm_Payload::ClockStateFlags
```

State Flags.

Telemetry Mnemonic(s) $sc_$cpu_TIME_StateFlg, $sc_$cpu_TIME_FlagSet, $sc_$cpu_TIME_FlagFly, $sc_$cpu_TIME_FlagSrc, $sc_$cpu_TIME_FlagPri, $sc_$cpu_TIME_FlagSfly, $sc_$cpu_TIME_FlagCfly, $sc_$cpu_TIME_FlagAdjd, $sc_$cpu_TIME_Flag1Hzd, $sc_$cpu_TIME_FlagClat, $sc_$cpu_TIME_FlagSorC, $sc_$cpu_TIME_FlagNIU

Definition at line 920 of file cfe_time_msg.h.

38.127.2.3 CommandCounter

```
uint8 CFE_TIME_HousekeepingTlm_Payload::CommandCounter
```

Time Command Execution Counter.

Telemetry Mnemonic(s) $sc_$cpu_TIME_CMDPC

Definition at line 912 of file cfe_time_msg.h.

38.127.2.4 CommandErrorCounter

```
uint8 CFE_TIME_HousekeepingTlm_Payload::CommandErrorCounter
```

Time Command Error Counter.

Telemetry Mnemonic(s) $sc_$cpu_TIME_CMDEC

Definition at line 914 of file cfe_time_msg.h.

38.127.2.5 LeapSeconds

```
int16 CFE_TIME_HousekeepingTlm_Payload::LeapSeconds
```

Current Leaps Seconds.

Telemetry Mnemonic(s) $sc_$cpu_TIME_LeapSecs

Definition at line 928 of file cfe_time_msg.h.

38.127.2.6 Seconds1HzAdj

```
uint32 CFE_TIME_HousekeepingTlm_Payload::Seconds1HzAdj
```

Current 1 Hz SCTF adjustment (seconds)

Telemetry Mnemonic(s) $sc_$cpu_TIME_1HzAdjSecs

Definition at line 948 of file cfe_time_msg.h.

38.127.2.7 SecondsDelay

```
uint32 CFE_TIME_HousekeepingTlm_Payload::SecondsDelay
```

Current 1 Hz SCTF Delay (seconds)

Telemetry Mnemonic(s) $sc_$cpu_TIME_1HzAdjSecs

Definition at line 958 of file cfe_time_msg.h.

38.127.2.8 SecondsMET

```
uint32 CFE_TIME_HousekeepingTlm_Payload::SecondsMET
```

Current MET (seconds)

Telemetry Mnemonic(s) $sc_$cpu_TIME_METSecs

Definition at line 934 of file cfe_time_msg.h.

38.127.2.9 SecondsSTCF

```
uint32 CFE_TIME_HousekeepingTlm_Payload::SecondsSTCF
```

Current STCF (seconds)

Telemetry Mnemonic(s) $sc_$cpu_TIME_STCFSecs

Definition at line 939 of file cfe_time_msg.h.

38.127.2.10 Subsecs1HzAdj

```
uint32 CFE_TIME_HousekeepingTlm_Payload::Subsecs1HzAdj
```

Current 1 Hz SCTF adjustment (sub-seconds)

Telemetry Mnemonic(s) $sc_$cpu_TIME_1HzAdjSSecs

Definition at line 950 of file cfe_time_msg.h.

38.127.2.11 SubsecsDelay

```
uint32 CFE_TIME_HousekeepingTlm_Payload::SubsecsDelay
```

Current 1 Hz SCTF Delay (sub-seconds)

Telemetry Mnemonic(s) $sc_$cpu_TIME_1HzAdjSSecs

Definition at line 960 of file cfe_time_msg.h.

38.127.2.12 SubsecsMET

```
uint32 CFE_TIME_HousekeepingTlm_Payload::SubsecsMET
```

Current MET (sub-seconds)

Telemetry Mnemonic(s) $sc_$cpu_TIME_METSubsecs

Definition at line 936 of file cfe_time_msg.h.

38.127.2.13 SubsecsSTCF

```
uint32 CFE_TIME_HousekeepingTlm_Payload::SubsecsSTCF
```

Current STCF (sub-seconds)

Telemetry Mnemonic(s) $sc_$cpu_TIME_STCFSubsecs

Definition at line 941 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.128 CFE_TIME_LeapsCmd_Payload Struct Reference

Set leap seconds command payload.

```
#include <cfe_time_msg.h>
```

Data Fields

- int16 LeapSeconds

38.128.1 Detailed Description

Set leap seconds command payload.

Definition at line 752 of file cfe_time_msg.h.

38.128.2 Field Documentation

38.128.2.1 LeapSeconds

```
int16 CFE_TIME_LeapsCmd_Payload::LeapSeconds
```

Definition at line 754 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.129 CFE_TIME_NoArgsCmd Struct Reference

Generic no argument command.

```
#include <cfe_time_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.

38.129.1 Detailed Description

Generic no argument command.

Definition at line 731 of file cfe_time_msg.h.

38.129.2 Field Documentation

38.129.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_TIME_NoArgsCmd::CmdHeader
```

Command header.

Definition at line 733 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.130 CFE_TIME_OneHzAdjustmentCmd Struct Reference

Generic seconds, subseconds adjustment command.

```
#include <cfe_time_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_TIME_OneHzAdjustmentCmd_Payload_t Payload
  Command payload.

38.130.1 Detailed Description

Generic seconds, subseconds adjustment command.

Definition at line 868 of file cfe_time_msg.h.

38.130.2 Field Documentation

38.130.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_TIME_OneHzAdjustmentCmd::CmdHeader
```

Command header.

Definition at line 870 of file cfe_time_msg.h.

38.130.2.2 Payload

```
CFE_TIME_OneHzAdjustmentCmd_Payload_t CFE_TIME_OneHzAdjustmentCmd::Payload
```

Command payload.

Definition at line 871 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.131 CFE_TIME_OneHzAdjustmentCmd_Payload Struct Reference

Generic seconds, subseconds command payload.

```
#include <cfe_time_msg.h>
```

Data Fields

- uint32 Seconds
- uint32 Subseconds

38.131.1 Detailed Description

Generic seconds, subseconds command payload.

Definition at line 858 of file cfe_time_msg.h.

38.131.2 Field Documentation

38.131.2.1 Seconds

```
uint32 CFE_TIME_OneHzAdjustmentCmd_Payload::Seconds
```

Definition at line 860 of file cfe_time_msg.h.

38.131.2.2 Subseconds

```
uint32 CFE_TIME_OneHzAdjustmentCmd_Payload::Subseconds
```

Definition at line 861 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.132 CFE_TIME_SetLeapSecondsCmd Struct Reference

Set leap seconds command.

```
#include <cfe_time_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_TIME_LeapsCmd_Payload_t Payload

  Command payload.

38.132.1 Detailed Description

Set leap seconds command.

Definition at line 760 of file cfe_time_msg.h.

38.132.2 Field Documentation

38.132.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_TIME_SetLeapSecondsCmd::CmdHeader
```

Command header.

Definition at line 762 of file cfe_time_msg.h.

38.132.2.2 Payload

`CFE_TIME_LeapsCmd_Payload_t CFE_TIME_SetLeapSecondsCmd::Payload`

Command payload.

Definition at line 763 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.133 CFE_TIME_SetSignalCmd Struct Reference

Set tone signal source command.

```
#include <cfe_time_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_TIME_SignalCmd_Payload_t Payload
  Command payload.

38.133.1 Detailed Description

Set tone signal source command.

Definition at line 818 of file cfe_time_msg.h.

38.133.2 Field Documentation

38.133.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_TIME_SetSignalCmd::CmdHeader`

Command header.

Definition at line 820 of file cfe_time_msg.h.

38.133.2.2 Payload

`CFE_TIME_SignalCmd_Payload_t CFE_TIME_SetSignalCmd::Payload`

Command payload.

Definition at line 821 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.134 CFE_TIME_SetSourceCmd Struct Reference

Set time data source command.

```
#include <cfe_time_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_TIME_SourceCmd_Payload_t Payload
  Command payload.

38.134.1 Detailed Description

Set time data source command.

Definition at line 799 of file cfe_time_msg.h.

38.134.2 Field Documentation

38.134.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_TIME_SetSourceCmd::CmdHeader`

Command header.

Definition at line 801 of file cfe_time_msg.h.

38.134.2.2 Payload

`CFE_TIME_SourceCmd_Payload_t CFE_TIME_SetSourceCmd::Payload`

Command payload.

Definition at line 802 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.135 CFE_TIME_SetStateCmd Struct Reference

Set clock state command.

```
#include <cfe_time_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_TIME_StateCmd_Payload_t Payload
  Command payload.

38.135.1 Detailed Description

Set clock state command.

Definition at line 780 of file cfe_time_msg.h.

38.135.2 Field Documentation

38.135.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_TIME_SetStateCmd::CmdHeader`

Command header.

Definition at line 782 of file cfe_time_msg.h.

38.135.2.2 Payload

```
CFE_TIME_StateCmd_Payload_t CFE_TIME_SetStateCmd::Payload
```

Command payload.

Definition at line 783 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.136 CFE_TIME_SignalCmd_Payload Struct Reference

Set tone signal source command payload.

```
#include <cfe_time_msg.h>
```

Data Fields

- int16 ToneSource

  CFE_TIME_ToneSignalSelect_PRIMARY=Primary Source, CFE_TIME_ToneSignalSelect_REDUNDANT=Redundant Source

38.136.1 Detailed Description

Set tone signal source command payload.

Definition at line 808 of file cfe_time_msg.h.

38.136.2 Field Documentation

38.136.2.1 ToneSource

```
int16 CFE_TIME_SignalCmd_Payload::ToneSource
```

CFE_TIME_ToneSignalSelect_PRIMARY=Primary Source, CFE_TIME_ToneSignalSelect_REDUNDANT=Redundant Source

Selects either the "Primary" or "Redundant" tone signal source

Definition at line 810 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.137 CFE_TIME_SourceCmd_Payload Struct Reference

Set time data source command payload.

```
#include <cfe_time_msg.h>
```

Data Fields

- int16 TimeSource

  CFE_TIME_SourceSelect_INTERNAL=Internal Source, CFE_TIME_SourceSelect_EXTERNAL=External Source

38.137.1 Detailed Description

Set time data source command payload.

Definition at line 789 of file cfe_time_msg.h.

38.137.2 Field Documentation

38.137.2.1 TimeSource

```
int16 CFE_TIME_SourceCmd_Payload::TimeSource
```

CFE_TIME_SourceSelect_INTERNAL=Internal Source, CFE_TIME_SourceSelect_EXTERNAL=External Source

Selects either the "Internal" and "External" clock source

Definition at line 791 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.138 CFE_TIME_StateCmd_Payload Struct Reference

Set clock state command payload.

```
#include <cfe_time_msg.h>
```

Data Fields

- int16 ClockState

  CFE_TIME_ClockState_INVALID=Spacecraft time has not been accurately set, CFE_TIME_ClockState_VAL↩ ID=Spacecraft clock has been accurately set, CFE_TIME_ClockState_FLYWHEEL=Force into FLYWHEEL mode

38.138.1 Detailed Description

Set clock state command payload.

Definition at line 769 of file cfe_time_msg.h.

38.138.2 Field Documentation

38.138.2.1 ClockState

`int16 CFE_TIME_StateCmd_Payload::ClockState`

CFE_TIME_ClockState_INVALID=Spacecraft time has not been accurately set, CFE_TIME_ClockState_VALID=Spacecraft clock has been accurately set, CFE_TIME_ClockState_FLYWHEEL=Force into FLYWHEEL mode

Selects the current clock state

Definition at line 771 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.139 CFE_TIME_SysTime Struct Reference

Data structure used to hold system time values.

```
#include <cfe_time_extern_typedefs.h>
```

Data Fields

- uint32 Seconds

  Number of seconds since epoch.
- uint32 Subseconds

  Number of subseconds since epoch (LSB = 2^(-32) seconds)

38.139.1 Detailed Description

Data structure used to hold system time values.

Description

The CFE_TIME_SysTime_t data structure is used to hold time values. Time is referred to as the elapsed time (in seconds and subseconds) since a specified epoch time. The subseconds field contains the number of 2^(-32) second intervals that have elapsed since the epoch.

Definition at line 47 of file cfe_time_extern_typedefs.h.

38.139.2 Field Documentation

38.139.2.1 Seconds

```
uint32 CFE_TIME_SysTime::Seconds
```

Number of seconds since epoch.

Definition at line 49 of file cfe_time_extern_typedefs.h.

38.139.2.2 Subseconds

```
uint32 CFE_TIME_SysTime::Subseconds
```

Number of subseconds since epoch (LSB = 2^(-32) seconds)

Definition at line 50 of file cfe_time_extern_typedefs.h.

The documentation for this struct was generated from the following file:

- cfe/modules/core_api/fsw/inc/cfe_time_extern_typedefs.h

38.140 CFE_TIME_TimeCmd Struct Reference

Generic seconds, microseconds argument command.

```
#include <cfe_time_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader
  Command header.
- CFE_TIME_TimeCmd_Payload_t Payload
  Command payload.

38.140.1 Detailed Description

Generic seconds, microseconds argument command.

Definition at line 836 of file cfe_time_msg.h.

38.140.2 Field Documentation

38.140.2.1 CmdHeader

`CFE_MSG_CommandHeader_t CFE_TIME_TimeCmd::CmdHeader`

Command header.

Definition at line 838 of file cfe_time_msg.h.

38.140.2.2 Payload

`CFE_TIME_TimeCmd_Payload_t CFE_TIME_TimeCmd::Payload`

Command payload.

Definition at line 839 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.141 CFE_TIME_TimeCmd_Payload Struct Reference

Generic seconds, microseconds command payload.

```
#include <cfe_time_msg.h>
```

Data Fields

- uint32 Seconds
- uint32 MicroSeconds

38.141.1 Detailed Description

Generic seconds, microseconds command payload.

Definition at line 827 of file cfe_time_msg.h.

38.141.2 Field Documentation

38.141.2.1 MicroSeconds

```
uint32 CFE_TIME_TimeCmd_Payload::MicroSeconds
```

Definition at line 830 of file cfe_time_msg.h.

38.141.2.2 Seconds

```
uint32 CFE_TIME_TimeCmd_Payload::Seconds
```

Definition at line 829 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.142 CFE_TIME_ToneDataCmd Struct Reference

Time at tone data command.

```
#include <cfe_time_msg.h>
```

Data Fields

- CFE_MSG_CommandHeader_t CmdHeader

  Command header.
- CFE_TIME_ToneDataCmd_Payload_t Payload

  Command payload.

38.142.1 Detailed Description

Time at tone data command.

Definition at line 896 of file cfe_time_msg.h.

38.142.2 Field Documentation

38.142.2.1 CmdHeader

```
CFE_MSG_CommandHeader_t CFE_TIME_ToneDataCmd::CmdHeader
```

Command header.

Definition at line 898 of file cfe_time_msg.h.

38.142.2.2 Payload

```
CFE_TIME_ToneDataCmd_Payload_t CFE_TIME_ToneDataCmd::Payload
```

Command payload.

Definition at line 899 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.143 CFE_TIME_ToneDataCmd_Payload Struct Reference

Time at tone data command payload.

```
#include <cfe_time_msg.h>
```

Data Fields

- CFE_TIME_SysTime_t AtToneMET

  MET at time of tone.
- CFE_TIME_SysTime_t AtToneSTCF

  STCF at time of tone.
- int16 AtToneLeapSeconds

  Leap Seconds at time of tone.
- int16 AtToneState

  Clock state at time of tone.

38.143.1 Detailed Description

Time at tone data command payload.

Definition at line 885 of file cfe_time_msg.h.

38.143.2 Field Documentation

38.143.2.1 AtToneLeapSeconds

```
int16 CFE_TIME_ToneDataCmd_Payload::AtToneLeapSeconds
```

Leap Seconds at time of tone.

Definition at line 889 of file cfe_time_msg.h.

38.143.2.2 AtToneMET

```
CFE_TIME_SysTime_t CFE_TIME_ToneDataCmd_Payload::AtToneMET
```

MET at time of tone.

Definition at line 887 of file cfe_time_msg.h.

38.143.2.3 AtToneState

```
int16 CFE_TIME_ToneDataCmd_Payload::AtToneState
```

Clock state at time of tone.

Definition at line 890 of file cfe_time_msg.h.

38.143.2.4 AtToneSTCF

```
CFE_TIME_SysTime_t CFE_TIME_ToneDataCmd_Payload::AtToneSTCF
```

STCF at time of tone.

Definition at line 888 of file cfe_time_msg.h.

The documentation for this struct was generated from the following file:

- cfe/modules/time/fsw/inc/cfe_time_msg.h

38.144 OS_bin_sem_prop_t Struct Reference

OSAL binary semaphore properties.

```
#include <osapi-binsem.h>
```

Data Fields

- char name [OS_MAX_API_NAME]
- osal_id_t creator
- int32 value

38.144.1 Detailed Description

OSAL binary semaphore properties.

Definition at line 39 of file osapi-binsem.h.

38.144.2 Field Documentation

38.144.2.1 creator

```
osal_id_t OS_bin_sem_prop_t::creator
```

Definition at line 42 of file osapi-binsem.h.

38.144.2.2 name

```
char OS_bin_sem_prop_t::name[OS_MAX_API_NAME]
```

Definition at line 41 of file osapi-binsem.h.

38.144.2.3 value

```
int32 OS_bin_sem_prop_t::value
```

Definition at line 43 of file osapi-binsem.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-binsem.h

38.145 OS_count_sem_prop_t Struct Reference

OSAL counting semaphore properties.

```
#include <osapi-countsem.h>
```

Data Fields

- char name [OS_MAX_API_NAME]
- osal_id_t creator
- int32 value

38.145.1 Detailed Description

OSAL counting semaphore properties.

Definition at line 32 of file osapi-countsem.h.

38.145.2 Field Documentation

38.145.2.1 creator

```
osal_id_t OS_count_sem_prop_t::creator
```

Definition at line 35 of file osapi-countsem.h.

38.145.2.2 name

```
char OS_count_sem_prop_t::name[OS_MAX_API_NAME]
```

Definition at line 34 of file osapi-countsem.h.

38.145.2.3 value

```
int32 OS_count_sem_prop_t::value
```

Definition at line 36 of file osapi-countsem.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-countsem.h

38.146 os_dirent_t Struct Reference

Directory entry.

```
#include <osapi-dir.h>
```

Data Fields

- char FileName [OS_MAX_FILE_NAME]

38.146.1 Detailed Description

Directory entry.

Definition at line 32 of file osapi-dir.h.

38.146.2 Field Documentation

38.146.2.1 FileName

```
char os_dirent_t::FileName[OS_MAX_FILE_NAME]
```

Definition at line 34 of file osapi-dir.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-dir.h

38.147 OS_FdSet Struct Reference

An abstract structure capable of holding several OSAL IDs.

```
#include <osapi-select.h>
```

Data Fields

- uint8 object_ids [(OS_MAX_NUM_OPEN_FILES+7)/8]

38.147.1 Detailed Description

An abstract structure capable of holding several OSAL IDs.

This is part of the select API and is manipulated using the related API calls. It should not be modified directly by applications.

See also

OS_SelectFdZero(), OS_SelectFdAdd(), OS_SelectFdClear(), OS_SelectFdIsSet()

Definition at line 39 of file osapi-select.h.

38.147.2 Field Documentation

38.147.2.1 object_ids

```
uint8 OS_FdSet::object_ids[(OS_MAX_NUM_OPEN_FILES+7)/8]
```

Definition at line 41 of file osapi-select.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-select.h

38.148 OS_file_prop_t Struct Reference

OSAL file properties.

```
#include <osapi-file.h>
```

Data Fields

- char Path [OS_MAX_PATH_LEN]
- osal_id_t User
- uint8 IsValid

38.148.1 Detailed Description

OSAL file properties.

Definition at line 49 of file osapi-file.h.

38.148.2 Field Documentation

38.148.2.1 IsValid

```
uint8 OS_file_prop_t::IsValid
```

Definition at line 53 of file osapi-file.h.

38.148.2.2 Path

```
char OS_file_prop_t::Path[OS_MAX_PATH_LEN]
```

Definition at line 51 of file osapi-file.h.

38.148.2.3 User

```
osal_id_t OS_file_prop_t::User
```

Definition at line 52 of file osapi-file.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-file.h

38.149 os_fsinfo_t Struct Reference

OSAL file system info.

```
#include <osapi-filesys.h>
```

Data Fields

- uint32 MaxFds

  Total number of file descriptors.
- uint32 FreeFds

  Total number that are free.
- uint32 MaxVolumes

  Maximum number of volumes.
- uint32 FreeVolumes

  Total number of volumes free.

38.149.1 Detailed Description

OSAL file system info.

Definition at line 35 of file osapi-filesys.h.

38.149.2 Field Documentation

38.149.2.1 FreeFds

```
uint32 os_fsinfo_t::FreeFds
```

Total number that are free.

Definition at line 38 of file osapi-filesys.h.

38.149.2.2 FreeVolumes

```
uint32 os_fsinfo_t::FreeVolumes
```

Total number of volumes free.

Definition at line 40 of file osapi-filesys.h.

38.149.2.3 MaxFds

```
uint32 os_fsinfo_t::MaxFds
```

Total number of file descriptors.

Definition at line 37 of file osapi-filesys.h.

38.149.2.4 MaxVolumes

```
uint32 os_fsinfo_t::MaxVolumes
```

Maximum number of volumes.

Definition at line 39 of file osapi-filesys.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-filesys.h

38.150 os_fstat_t Struct Reference

File system status.

```
#include <osapi-file.h>
```

Data Fields

- uint32 FileModeBits
- OS_time_t FileTime
- size_t FileSize

38.150.1 Detailed Description

File system status.

Note

This used to be directly typedef'ed to the "struct stat" from the C library

Some C libraries (glibc in particular) actually define member names to reference into sub-structures, so attempting to reuse a name like "st_mtime" might not work.

Definition at line 64 of file osapi-file.h.

38.150.2 Field Documentation

38.150.2.1 FileModeBits

```
uint32 os_fstat_t::FileModeBits
```

Definition at line 66 of file osapi-file.h.

38.150.2.2 FileSize

```
size_t os_fstat_t::FileSize
```

Definition at line 68 of file osapi-file.h.

38.150.2.3 FileTime

```
OS_time_t os_fstat_t::FileTime
```

Definition at line 67 of file osapi-file.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-file.h

38.151 OS_heap_prop_t Struct Reference

OSAL heap properties.

```
#include <osapi-heap.h>
```

Data Fields

- size_t free_bytes
- osal_blockcount_t free_blocks
- size_t largest_free_block

38.151.1 Detailed Description

OSAL heap properties.

See also

OS_HeapGetInfo()

Definition at line 36 of file osapi-heap.h.

38.151.2 Field Documentation

38.151.2.1 free_blocks

```
osal_blockcount_t OS_heap_prop_t::free_blocks
```

Definition at line 39 of file osapi-heap.h.

38.151.2.2 free_bytes

```
size_t OS_heap_prop_t::free_bytes
```

Definition at line 38 of file osapi-heap.h.

38.151.2.3 largest_free_block

```
size_t OS_heap_prop_t::largest_free_block
```

Definition at line 40 of file osapi-heap.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-heap.h

38.152 OS_module_address_t Struct Reference

OSAL module address properties.

```
#include <osapi-module.h>
```

Data Fields

- uint32 valid
- uint32 flags
- cpuaddr code_address
- cpuaddr code_size
- cpuaddr data_address
- cpuaddr data_size
- cpuaddr bss_address
- cpuaddr bss_size

38.152.1 Detailed Description

OSAL module address properties.

Definition at line 78 of file osapi-module.h.

38.152.2 Field Documentation

38.152.2.1 bss_address

```
cpuaddr OS_module_address_t::bss_address
```

Definition at line 86 of file osapi-module.h.

38.152.2.2 bss_size

```
cpuaddr OS_module_address_t::bss_size
```

Definition at line 87 of file osapi-module.h.

38.152.2.3 code_address

```
cpuaddr OS_module_address_t::code_address
```

Definition at line 82 of file osapi-module.h.

38.152.2.4 code_size

```
cpuaddr OS_module_address_t::code_size
```

Definition at line 83 of file osapi-module.h.

38.152.2.5 data_address

```
cpuaddr OS_module_address_t::data_address
```

Definition at line 84 of file osapi-module.h.

38.152.2.6 data_size

```
cpuaddr OS_module_address_t::data_size
```

Definition at line 85 of file osapi-module.h.

38.152.2.7 flags

```
uint32 OS_module_address_t::flags
```

Definition at line 81 of file osapi-module.h.

38.152.2.8 valid

```
uint32 OS_module_address_t::valid
```

Definition at line 80 of file osapi-module.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-module.h

38.153 OS_module_prop_t Struct Reference

OSAL module properties.

```
#include <osapi-module.h>
```

Data Fields

- cpuaddr entry_point
- cpuaddr host_module_id
- char filename [OS_MAX_PATH_LEN]
- char name [OS_MAX_API_NAME]
- OS_module_address_t addr

38.153.1 Detailed Description

OSAL module properties.

Definition at line 91 of file osapi-module.h.

38.153.2 Field Documentation

38.153.2.1 addr

```
OS_module_address_t OS_module_prop_t::addr
```

Definition at line 97 of file osapi-module.h.

38.153.2.2 entry_point

```
cpuaddr OS_module_prop_t::entry_point
```

Definition at line 93 of file osapi-module.h.

38.153.2.3 filename

```
char OS_module_prop_t::filename[OS_MAX_PATH_LEN]
```

Definition at line 95 of file osapi-module.h.

38.153.2.4 host_module_id

```
cpuaddr OS_module_prop_t::host_module_id
```

Definition at line 94 of file osapi-module.h.

38.153.2.5 name

```
char OS_module_prop_t::name[OS_MAX_API_NAME]
```

Definition at line 96 of file osapi-module.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-module.h

38.154 OS_mut_sem_prop_t Struct Reference

OSAL mutex properties.

```
#include <osapi-mutex.h>
```

Data Fields

- char name [OS_MAX_API_NAME]
- osal_id_t creator

38.154.1 Detailed Description

OSAL mutex properties.

Definition at line 32 of file osapi-mutex.h.

38.154.2 Field Documentation

38.154.2.1 creator

```
osal_id_t OS_mut_sem_prop_t::creator
```

Definition at line 35 of file osapi-mutex.h.

38.154.2.2 name

```
char OS_mut_sem_prop_t::name[OS_MAX_API_NAME]
```

Definition at line 34 of file osapi-mutex.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-mutex.h

38.155 OS_queue_prop_t Struct Reference

OSAL queue properties.

```
#include <osapi-queue.h>
```

Data Fields

- char name [OS_MAX_API_NAME]
- osal_id_t creator

38.155.1 Detailed Description

OSAL queue properties.

Definition at line 32 of file osapi-queue.h.

38.155.2 Field Documentation

38.155.2.1 creator

```
osal_id_t OS_queue_prop_t::creator
```

Definition at line 35 of file osapi-queue.h.

38.155.2.2 name

```
char OS_queue_prop_t::name[OS_MAX_API_NAME]
```

Definition at line 34 of file osapi-queue.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-queue.h

38.156 OS_SockAddr_t Struct Reference

Encapsulates a generic network address.

```
#include <osapi-sockets.h>
```

Data Fields

- size_t ActualLength

  Length of the actual address data.
- OS_SockAddrData_t AddrData

  Abstract Address data.

38.156.1 Detailed Description

Encapsulates a generic network address.

This is just an abstract buffer type that holds a network address. It is allocated for the worst-case size defined by OS_SOCKADDR_MAX_LEN, and the real size is stored within.

Definition at line 99 of file osapi-sockets.h.

38.156.2 Field Documentation

38.156.2.1 ActualLength

```
size_t OS_SockAddr_t::ActualLength
```

Length of the actual address data.

Definition at line 101 of file osapi-sockets.h.

38.156.2.2 AddrData

```
OS_SockAddrData_t OS_SockAddr_t::AddrData
```

Abstract Address data.

Definition at line 102 of file osapi-sockets.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-sockets.h

38.157 OS_SockAddrData_t Union Reference

Storage buffer for generic network address.

```
#include <osapi-sockets.h>
```

Data Fields

- uint8 Buffer [OS_SOCKADDR_MAX_LEN]

  Ensures length of at least OS_SOCKADDR_MAX_LEN.
- uint32 AlignU32

  Ensures uint32 alignment.
- void * AlignPtr

  Ensures pointer alignment.

38.157.1 Detailed Description

Storage buffer for generic network address.

This is a union type that helps to ensure a minimum alignment value for the data storage, such that it can be cast to the system-specific type without increasing alignment requirements.

Definition at line 85 of file osapi-sockets.h.

38.157.2 Field Documentation

38.157.2.1 AlignPtr

```
void* OS_SockAddrData_t::AlignPtr
```

Ensures pointer alignment.

Definition at line 89 of file osapi-sockets.h.

38.157.2.2 AlignU32

```
uint32 OS_SockAddrData_t::AlignU32
```

Ensures uint32 alignment.

Definition at line 88 of file osapi-sockets.h.

38.157.2.3 Buffer

```
uint8 OS_SockAddrData_t::Buffer[OS_SOCKADDR_MAX_LEN]
```

Ensures length of at least OS_SOCKADDR_MAX_LEN.

Definition at line 87 of file osapi-sockets.h.

The documentation for this union was generated from the following file:

- osal/src/os/inc/osapi-sockets.h

38.158 OS_socket_prop_t Struct Reference

Encapsulates socket properties.

```
#include <osapi-sockets.h>
```

Data Fields

- char name [OS_MAX_API_NAME]

  Name of the socket.
- osal_id_t creator

  OSAL TaskID which opened the socket.

38.158.1 Detailed Description

Encapsulates socket properties.

This is for consistency with other OSAL resource types. Currently no extra properties are exposed here but this could change in a future revision of OSAL as needed.

Definition at line 112 of file osapi-sockets.h.

38.158.2 Field Documentation

38.158.2.1 creator

```
osal_id_t OS_socket_prop_t::creator
```

OSAL TaskID which opened the socket.

Definition at line 115 of file osapi-sockets.h.

38.158.2.2 name

```
char OS_socket_prop_t::name[OS_MAX_API_NAME]
```

Name of the socket.

Definition at line 114 of file osapi-sockets.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-sockets.h

38.159 OS_static_symbol_record_t Struct Reference

Associates a single symbol name with a memory address.

```
#include <osapi-module.h>
```

Data Fields

- const char * Name
- void(* Address)(void)
- const char * Module

38.159.1 Detailed Description

Associates a single symbol name with a memory address.

If the OS_STATIC_SYMBOL_TABLE feature is enabled, then an array of these structures should be provided by the application. When the application needs to find a symbol address, the static table will be checked in addition to (or instead of) the OS/library-provided lookup function.

This static symbol allows systems that do not implement dynamic module loading to maintain the same semantics as dynamically loaded modules.

Definition at line 113 of file osapi-module.h.

38.159.2 Field Documentation

38.159.2.1 Address

```
void(* OS_static_symbol_record_t::Address) (void)
```

Definition at line 116 of file osapi-module.h.

38.159.2.2 Module

```
const char* OS_static_symbol_record_t::Module
```

Definition at line 117 of file osapi-module.h.

38.159.2.3 Name

```
const char* OS_static_symbol_record_t::Name
```

Definition at line 115 of file osapi-module.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-module.h

38.160 OS_statvfs_t Struct Reference

```
#include <osapi-filesys.h>
```

Data Fields

- size_t block_size
- osal_blockcount_t total_blocks
- osal_blockcount_t blocks_free

38.160.1 Detailed Description

Definition at line 49 of file osapi-filesys.h.

38.160.2 Field Documentation

38.160.2.1 block_size

```
size_t OS_statvfs_t::block_size
```

Block size of underlying FS

Definition at line 51 of file osapi-filesys.h.

38.160.2.2 blocks_free

```
osal_blockcount_t OS_statvfs_t::blocks_free
```

Available blocks in underlying FS

Definition at line 53 of file osapi-filesys.h.

38.160.2.3 total_blocks

```
osal_blockcount_t OS_statvfs_t::total_blocks
```

Total blocks in underlying FS

Definition at line 52 of file osapi-filesys.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-filesys.h

38.161 OS_task_prop_t Struct Reference

OSAL task properties.

```
#include <osapi-task.h>
```

Data Fields

- char name [OS_MAX_API_NAME]
- osal_id_t creator
- size_t stack_size
- osal_priority_t priority

38.161.1 Detailed Description

OSAL task properties.

Definition at line 57 of file osapi-task.h.

38.161.2 Field Documentation

38.161.2.1 creator

```
osal_id_t OS_task_prop_t::creator
```

Definition at line 60 of file osapi-task.h.

38.161.2.2 name

```
char OS_task_prop_t::name[OS_MAX_API_NAME]
```

Definition at line 59 of file osapi-task.h.

38.161.2.3 priority

```
osal_priority_t OS_task_prop_t::priority
```

Definition at line 62 of file osapi-task.h.

38.161.2.4 stack_size

```
size_t OS_task_prop_t::stack_size
```

Definition at line 61 of file osapi-task.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-task.h

38.162 OS_time_t Struct Reference

OSAL time interval structure.

```
#include <osapi-clock.h>
```

Data Fields

- int64 ticks

38.162.1 Detailed Description

OSAL time interval structure.

This is used to represent a basic time interval.

When used with OS_GetLocalTime/OS_SetLocalTime, this represents the interval from the OS's epoch point, typically 01 Jan 1970 00:00:00 UTC on systems that have a persistent real time clock (RTC), or the system boot time if there is no RTC available.

Applications should not directly access fields within this structure, as the definition may change in future versions of OSAL. Instead, applications should use the accessor/conversion methods defined below.

Definition at line 45 of file osapi-clock.h.

38.162.2 Field Documentation

38.162.2.1 ticks

```
int64 OS_time_t::ticks
```

Ticks elapsed since reference point

Definition at line 47 of file osapi-clock.h.

Referenced by OS_TimeAdd(), OS_TimeAssembleFromMicroseconds(), OS_TimeAssembleFromMilliseconds(), OS_TimeAssembleFromNanoseconds(), OS_TimeAssembleFromSubseconds(), OS_TimeGetFractionalPart(), OS_TimeGetTotalMicroseconds(), OS_TimeGetTotalMilliseconds(), OS_TimeGetTotalNanoseconds(), OS_TimeGetTotalSeconds(), and OS_TimeSubtract().

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-clock.h

38.163 OS_timebase_prop_t Struct Reference

Time base properties.

```
#include <osapi-timebase.h>
```

Data Fields

- char name [OS_MAX_API_NAME]
- osal_id_t creator
- uint32 nominal_interval_time
- uint32 freerun_time
- uint32 accuracy

38.163.1 Detailed Description

Time base properties.

Definition at line 37 of file osapi-timebase.h.

38.163.2 Field Documentation

38.163.2.1 accuracy

```
uint32 OS_timebase_prop_t::accuracy
```

Definition at line 43 of file osapi-timebase.h.

38.163.2.2 creator

```
osal_id_t OS_timebase_prop_t::creator
```

Definition at line 40 of file osapi-timebase.h.

38.163.2.3 freerun_time

```
uint32 OS_timebase_prop_t::freerun_time
```

Definition at line 42 of file osapi-timebase.h.

38.163.2.4 name

```
char OS_timebase_prop_t::name[OS_MAX_API_NAME]
```

Definition at line 39 of file osapi-timebase.h.

38.163.2.5 nominal_interval_time

```
uint32 OS_timebase_prop_t::nominal_interval_time
```

Definition at line 41 of file osapi-timebase.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-timebase.h

38.164 OS_timer_prop_t Struct Reference

Timer properties.

```
#include <osapi-timer.h>
```

Data Fields

- char name [OS_MAX_API_NAME]
- osal_id_t creator
- uint32 start_time
- uint32 interval_time
- uint32 accuracy

38.164.1 Detailed Description

Timer properties.

Definition at line 37 of file osapi-timer.h.

38.164.2 Field Documentation

38.164.2.1 accuracy

```
uint32 OS_timer_prop_t::accuracy
```

Definition at line 43 of file osapi-timer.h.

38.164.2.2 creator

```
osal_id_t OS_timer_prop_t::creator
```

Definition at line 40 of file osapi-timer.h.

38.164.2.3 interval_time

```
uint32 OS_timer_prop_t::interval_time
```

Definition at line 42 of file osapi-timer.h.

38.164.2.4 name

```
char OS_timer_prop_t::name[OS_MAX_API_NAME]
```

Definition at line 39 of file osapi-timer.h.

38.164.2.5 start_time

```
uint32 OS_timer_prop_t::start_time
```

Definition at line 41 of file osapi-timer.h.

The documentation for this struct was generated from the following file:

- osal/src/os/inc/osapi-timer.h

38.165 Target_PspConfigData Struct Reference

```
#include <cfe_psp_configdata.h>
```

Data Fields

- uint32 PSP_WatchdogMin
- uint32 PSP_WatchdogMax
- uint32 PSP_MemTableSize
- uint32 PSP_ExceptionLogSize
- CFE_PSP_MemTable_t * PSP_MemoryTable
- uint32 HW_NumEepromBanks
- CFE_PSP_VersionInfo_t PSP_VersionInfo

38.165.1 Detailed Description

PSP/Hardware configuration parameters This structure should be instantiated by the PSP according such that other modules do not need to directly include the PSP configuration at compile time.

Definition at line 57 of file cfe_psp_configdata.h.

38.165.2 Field Documentation

38.165.2.1 HW_NumEepromBanks

```
uint32 Target_PspConfigData::HW_NumEepromBanks
```

Number of EEPROM banks on this platform

Definition at line 68 of file cfe_psp_configdata.h.

38.165.2.2 PSP_ExceptionLogSize

```
uint32 Target_PspConfigData::PSP_ExceptionLogSize
```

Size of PSP exception log

Definition at line 62 of file cfe_psp_configdata.h.

38.165.2.3 PSP_MemoryTable

```
CFE_PSP_MemTable_t* Target_PspConfigData::PSP_MemoryTable
```

Pointer to PSP memory table (forward reference)

Definition at line 63 of file cfe_psp_configdata.h.

38.165.2.4 PSP_MemTableSize

```
uint32 Target_PspConfigData::PSP_MemTableSize
```

Size of PSP memory table

Definition at line 61 of file cfe_psp_configdata.h.

38.165.2.5 PSP_VersionInfo

```
CFE_PSP_VersionInfo_t Target_PspConfigData::PSP_VersionInfo
```

Definition at line 70 of file cfe_psp_configdata.h.

38.165.2.6 PSP_WatchdogMax

```
uint32 Target_PspConfigData::PSP_WatchdogMax
```

PSP Maximum watchdog in milliseconds

Definition at line 60 of file cfe_psp_configdata.h.

38.165.2.7 PSP_WatchdogMin

```
uint32 Target_PspConfigData::PSP_WatchdogMin
```

PSP Minimum watchdog in milliseconds

Definition at line 59 of file cfe_psp_configdata.h.

The documentation for this struct was generated from the following file:

- psp/fsw/inc/cfe_psp_configdata.h

39 File Documentation

39.1 build/doc/osconfig-example.h File Reference

Macros

- #define OS_MAX_TASKS

  Configuration file Operating System Abstraction Layer.
- #define OS_MAX_QUEUES

  The maximum number of queues to support.
- #define OS_MAX_COUNT_SEMAPHORES

  The maximum number of counting semaphores to support.
- #define OS_MAX_BIN_SEMAPHORES

  The maximum number of binary semaphores to support.
- #define OS_MAX_MUTEXES

  The maximum number of mutexes to support.
- #define OS_MAX_MODULES

  The maximum number of modules to support.
- #define OS_MAX_TIMEBASES

  The maximum number of timebases to support.
- #define OS_MAX_TIMERS

  The maximum number of timer callbacks to support.
- #define OS_MAX_NUM_OPEN_FILES

  The maximum number of concurrently open files to support.
- #define OS_MAX_NUM_OPEN_DIRS

  The maximum number of concurrently open directories to support.
- #define OS_MAX_FILE_SYSTEMS

  The maximum number of file systems to support.
- #define OS_MAX_SYM_LEN

  The maximum length of symbols.
- #define OS_MAX_FILE_NAME

  The maximum length of OSAL file names.
- #define OS_MAX_PATH_LEN

  The maximum length of OSAL path names.

- #define OS_MAX_API_NAME

  The maximum length of OSAL resource names.
- #define OS_SOCKADDR_MAX_LEN

  The maximum size of the socket address structure.
- #define OS_BUFFER_SIZE

  The maximum size of output produced by a single OS_printf()
- #define OS_BUFFER_MSG_DEPTH

  The maximum number of OS_printf() output strings to buffer.
- #define OS_UTILITYTASK_PRIORITY

  Priority level of the background utility task.
- #define OS_UTILITYTASK_STACK_SIZE

  The stack size of the background utility task.
- #define OS_MAX_CMD_LEN

  The maximum size of a shell command.
- #define OS_QUEUE_MAX_DEPTH

  The maximum depth of OSAL queues.
- #define OS_SHELL_CMD_INPUT_FILE_NAME ""

  The name of the temporary file used to store shell commands.
- #define OS_PRINTF_CONSOLE_NAME ""

  The name of the primary console device.
- #define OS_MAX_CONSOLES 1

  The maximum number of console devices to support.
- #define OS_MODULE_FILE_EXTENSION ".so"

  The system-specific file extension used on loadable module files.
- #define OS_FS_DEV_NAME_LEN 32
- #define OS_FS_PHYS_NAME_LEN 64
- #define OS_FS_VOL_NAME_LEN 32

39.1.1 Macro Definition Documentation

39.1.1.1 OS_BUFFER_MSG_DEPTH

```
#define OS_BUFFER_MSG_DEPTH
```

The maximum number of OS_printf() output strings to buffer.

Based on the OSAL_CONFIG_PRINTF_BUFFER_DEPTH configuration option

Definition at line 199 of file osconfig-example.h.

39.1.1.2 OS_BUFFER_SIZE

```
#define OS_BUFFER_SIZE
```

The maximum size of output produced by a single OS_printf()

Based on the OSAL_CONFIG_PRINTF_BUFFER_SIZE configuration option

Definition at line 192 of file osconfig-example.h.

39.1.1.3 OS_FS_DEV_NAME_LEN

```
#define OS_FS_DEV_NAME_LEN 32
```

Device name length

Definition at line 284 of file osconfig-example.h.

39.1.1.4 OS_FS_PHYS_NAME_LEN

```
#define OS_FS_PHYS_NAME_LEN 64
```

Physical drive name length

Definition at line 285 of file osconfig-example.h.

39.1.1.5 OS_FS_VOL_NAME_LEN

```
#define OS_FS_VOL_NAME_LEN 32
```

Volume name length

Definition at line 286 of file osconfig-example.h.

39.1.1.6 OS_MAX_API_NAME

```
#define OS_MAX_API_NAME
```

The maximum length of OSAL resource names.

Based on the OSAL_CONFIG_MAX_API_NAME configuration option

Note

This value must include a terminating NUL character

Definition at line 175 of file osconfig-example.h.

Referenced by CFE_PSP_OS_EventHandler().

39.1.1.7 OS_MAX_BIN_SEMAPHORES

```
#define OS_MAX_BIN_SEMAPHORES
```

The maximum number of binary semaphores to support.

Based on the OSAL_CONFIG_MAX_BIN_SEMAPHORES configuration option

Definition at line 84 of file osconfig-example.h.

39.1.1.8 OS_MAX_CMD_LEN

```
#define OS_MAX_CMD_LEN
```

The maximum size of a shell command.

This limit is only applicable if shell support is enabled.

Based on the OSAL_CONFIG_MAX_CMD_LEN configuration option

Note

This value must include a terminating NUL character

Definition at line 230 of file osconfig-example.h.

39.1.1.9 OS_MAX_CONSOLES

```
#define OS_MAX_CONSOLES 1
```

The maximum number of console devices to support.

Fixed value based on current OSAL implementation, not user configurable.

Definition at line 272 of file osconfig-example.h.

39.1.1.10 OS_MAX_COUNT_SEMAPHORES

```
#define OS_MAX_COUNT_SEMAPHORES
```

The maximum number of counting semaphores to support.

Based on the OSAL_CONFIG_MAX_COUNT_SEMAPHORES configuration option

Definition at line 77 of file osconfig-example.h.

39.1.1.11 OS_MAX_FILE_NAME

```
#define OS_MAX_FILE_NAME
```

The maximum length of OSAL file names.

This limit applies specifically to the file name portion, not the directory portion, of a path name.

Based on the OSAL_CONFIG_MAX_FILE_NAME configuration option

Note

This value must include a terminating NUL character

Definition at line 154 of file osconfig-example.h.

39.1.1.12 OS_MAX_FILE_SYSTEMS

```
#define OS_MAX_FILE_SYSTEMS
```

The maximum number of file systems to support.

Based on the OSAL_CONFIG_MAX_FILE_SYSTEMS configuration option

Definition at line 133 of file osconfig-example.h.

39.1.1.13 OS_MAX_MODULES

```
#define OS_MAX_MODULES
```

The maximum number of modules to support.

Based on the OSAL_CONFIG_MAX_MODULES configuration option

Definition at line 98 of file osconfig-example.h.

39.1.1.14 OS_MAX_MUTEXES

```
#define OS_MAX_MUTEXES
```

The maximum number of mutexes to support.

Based on the OSAL_CONFIG_MAX_MUTEXES configuration option

Definition at line 91 of file osconfig-example.h.

39.1.1.15 OS_MAX_NUM_OPEN_DIRS

```
#define OS_MAX_NUM_OPEN_DIRS
```

The maximum number of concurrently open directories to support.

Based on the OSAL_CONFIG_MAX_NUM_OPEN_DIRS configuration option

Definition at line 126 of file osconfig-example.h.

39.1.1.16 OS_MAX_NUM_OPEN_FILES

```
#define OS_MAX_NUM_OPEN_FILES
```

The maximum number of concurrently open files to support.

Based on the OSAL_CONFIG_MAX_NUM_OPEN_FILES configuration option

Definition at line 119 of file osconfig-example.h.

39.1.1.17 OS_MAX_PATH_LEN

```
#define OS_MAX_PATH_LEN
```

The maximum length of OSAL path names.

This limit applies to the overall length of a path name, including the file name and directory portions.

Based on the OSAL_CONFIG_MAX_PATH_LEN configuration option

Note

This value must include a terminating NUL character

Definition at line 166 of file osconfig-example.h.

39.1.1.18 OS_MAX_QUEUES

```
#define OS_MAX_QUEUES
```

The maximum number of queues to support.

Based on the OSAL_CONFIG_MAX_QUEUES configuration option

Definition at line 70 of file osconfig-example.h.

39.1.1.19 OS_MAX_SYM_LEN

```
#define OS_MAX_SYM_LEN
```

The maximum length of symbols.

Based on the OSAL_CONFIG_MAX_SYM_LEN configuration option

Note

This value must include a terminating NUL character

Definition at line 142 of file osconfig-example.h.

39.1.1.20 OS_MAX_TASKS

```
#define OS_MAX_TASKS
```

Configuration file Operating System Abstraction Layer.

The specific definitions in this file may only be modified by setting the respective OSAL configuration options in the CMake build.

Any direct modifications to the generated copy will be overwritten each time CMake executes.

Note

This file was automatically generated by CMake from /home/runner/work/cFS/cFS/cfe/default_config.cmake The maximum number of to support

Based on the OSAL_CONFIG_MAX_TASKS configuration option

Definition at line 63 of file osconfig-example.h.

39.1.1.21 OS_MAX_TIMEBASES

```
#define OS_MAX_TIMEBASES
```

The maximum number of timebases to support.

Based on the OSAL_CONFIG_MAX_TIMEBASES configuration option

Definition at line 105 of file osconfig-example.h.

39.1.1.22 OS_MAX_TIMERS

```
#define OS_MAX_TIMERS
```

The maximum number of timer callbacks to support.

Based on the OSAL_CONFIG_MAX_TIMERS configuration option

Definition at line 112 of file osconfig-example.h.

39.1.1.23 OS_MODULE_FILE_EXTENSION

```
#define OS_MODULE_FILE_EXTENSION ".so"
```

The system-specific file extension used on loadable module files.

Fixed value based on system selection, not user configurable.

Definition at line 279 of file osconfig-example.h.

39.1.1.24 OS_PRINTF_CONSOLE_NAME

```
#define OS_PRINTF_CONSOLE_NAME ""
```

The name of the primary console device.

This is the device to which OS_printf() output is written. The output may be configured to tag each line with this prefix for identification.

Based on the OSAL_CONFIG_PRINTF_CONSOLE_NAME configuration option

Definition at line 257 of file osconfig-example.h.

39.1.1.25 OS_QUEUE_MAX_DEPTH

```
#define OS_QUEUE_MAX_DEPTH
```

The maximum depth of OSAL queues.

Based on the OSAL_CONFIG_QUEUE_MAX_DEPTH configuration option

Definition at line 237 of file osconfig-example.h.

39.1.1.26 OS_SHELL_CMD_INPUT_FILE_NAME

```
#define OS_SHELL_CMD_INPUT_FILE_NAME ""
```

The name of the temporary file used to store shell commands.

This configuration is only applicable if shell support is enabled, and only necessary/relevant on some OS implementations.

Based on the OSAL_CONFIG_SHELL_CMD_INPUT_FILE_NAME configuration option

Definition at line 247 of file osconfig-example.h.

39.1.1.27 OS_SOCKADDR_MAX_LEN

```
#define OS_SOCKADDR_MAX_LEN
```

The maximum size of the socket address structure.

This is part of the Socket API, and should be set large enough to hold the largest address type in use on the target system.

Based on the OSAL_CONFIG_SOCKADDR_MAX_LEN configuration option

Definition at line 185 of file osconfig-example.h.

39.1.1.28 OS_UTILITYTASK_PRIORITY

```
#define OS_UTILITYTASK_PRIORITY
```

Priority level of the background utility task.

This task is responsible for writing buffered output of OS_printf to the actual console device, and any other future maintenance task.

Based on the OSAL_CONFIG_UTILITYTASK_PRIORITY configuration option

Definition at line 209 of file osconfig-example.h.

39.1.1.29 OS_UTILITYTASK_STACK_SIZE

```
#define OS_UTILITYTASK_STACK_SIZE
```

The stack size of the background utility task.

This task is responsible for writing buffered output of OS_printf to the actual console device, and any other future maintenance task.

Based on the OSAL_CONFIG_UTILITYTASK_STACK_SIZE configuration option

Definition at line 219 of file osconfig-example.h.

39.2 build/native/default_cpu1/inc/cfe_msgids.h File Reference

```
#include "/home/runner/work/cFS/cFS/sample_defs/cpu1_msgids.h"
```

39.3 build/native/default_cpu1/inc/cfe_platform_cfg.h File Reference

```
#include "/home/runner/work/cFS/cFS/sample_defs/cpu1_platform_cfg.h"
```

39.4 build/native/default_cpu1/inc/osconfig.h File Reference

Macros

- #define OSAL_CONFIG_INCLUDE_DYNAMIC_LOADER

  Configuration file Operating System Abstraction Layer.
- #define OSAL_CONFIG_INCLUDE_NETWORK
- #define OSAL_CONFIG_INCLUDE_STATIC_LOADER
- #define OSAL_CONFIG_DEBUG_PRINTF
- #define OSAL_CONFIG_DEBUG_PERMISSIVE_MODE
- #define OS_MAX_TASKS 64

  The maximum number of to support.
- #define OS_MAX_QUEUES 64

  The maximum number of queues to support.
- #define OS_MAX_COUNT_SEMAPHORES 20

  The maximum number of counting semaphores to support.
- #define OS_MAX_BIN_SEMAPHORES 20

  The maximum number of binary semaphores to support.
- #define OS_MAX_MUTEXES 20

  The maximum number of mutexes to support.
- #define OS_MAX_MODULES 20

  The maximum number of modules to support.
- #define OS_MAX_TIMEBASES 5

  The maximum number of timebases to support.
- #define OS_MAX_TIMERS 10

  The maximum number of timer callbacks to support.
- #define OS_MAX_NUM_OPEN_FILES 50

  The maximum number of concurrently open files to support.
- #define OS_MAX_NUM_OPEN_DIRS 4

  The maximum number of concurrently open directories to support.
- #define OS_MAX_FILE_SYSTEMS 14

  The maximum number of file systems to support.
- #define OS_MAX_SYM_LEN 64

  The maximum length of symbols.
- #define OS_MAX_FILE_NAME 20

The maximum length of OSAL file names.

- #define OS_MAX_PATH_LEN 64

  The maximum length of OSAL path names.

- #define OS_MAX_API_NAME 20

  The maximum length of OSAL resource names.

- #define OS_SOCKADDR_MAX_LEN 28

  The maximum size of the socket address structure.

- #define OS_BUFFER_SIZE 172

  The maximum size of output produced by a single OS_printf()

- #define OS_BUFFER_MSG_DEPTH 100

  The maximum number of OS_printf() output strings to buffer.

- #define OS_UTILITYTASK_PRIORITY 245

  Priority level of the background utility task.

- #define OS_UTILITYTASK_STACK_SIZE 2048

  The stack size of the background utility task.

- #define OS_MAX_CMD_LEN 1000

  The maximum size of a shell command.

- #define OS_QUEUE_MAX_DEPTH 50

  The maximum depth of OSAL queues.

- #define OS_SHELL_CMD_INPUT_FILE_NAME ""

  The name of the temporary file used to store shell commands.

- #define OS_PRINTF_CONSOLE_NAME ""

  The name of the primary console device.

- #define OS_MAX_CONSOLES 1

  The maximum number of console devices to support.

- #define OS_MODULE_FILE_EXTENSION ".so"

  The system-specific file extension used on loadable module files.

- #define OS_FS_DEV_NAME_LEN 32
- #define OS_FS_PHYS_NAME_LEN 64
- #define OS_FS_VOL_NAME_LEN 32

39.4.1 Macro Definition Documentation

39.4.1.1 OS_BUFFER_MSG_DEPTH

```
#define OS_BUFFER_MSG_DEPTH 100
```

The maximum number of OS_printf() output strings to buffer.

Based on the OSAL_CONFIG_PRINTF_BUFFER_DEPTH configuration option

Definition at line 199 of file osconfig.h.

39.4.1.2 OS_BUFFER_SIZE

```
#define OS_BUFFER_SIZE 172
```

The maximum size of output produced by a single OS_printf()

Based on the OSAL_CONFIG_PRINTF_BUFFER_SIZE configuration option

Definition at line 192 of file osconfig.h.

39.4.1.3 OS_FS_DEV_NAME_LEN

```
#define OS_FS_DEV_NAME_LEN 32
```

Device name length

Definition at line 284 of file osconfig.h.

39.4.1.4 OS_FS_PHYS_NAME_LEN

```
#define OS_FS_PHYS_NAME_LEN 64
```

Physical drive name length

Definition at line 285 of file osconfig.h.

39.4.1.5 OS_FS_VOL_NAME_LEN

```
#define OS_FS_VOL_NAME_LEN 32
```

Volume name length

Definition at line 286 of file osconfig.h.

39.4.1.6 OS_MAX_API_NAME

```
#define OS_MAX_API_NAME 20
```

The maximum length of OSAL resource names.

Based on the OSAL_CONFIG_MAX_API_NAME configuration option

Note

This value must include a terminating NUL character

Definition at line 175 of file osconfig.h.

39.4.1.7 OS_MAX_BIN_SEMAPHORES

```
#define OS_MAX_BIN_SEMAPHORES 20
```

The maximum number of binary semaphores to support.

Based on the OSAL_CONFIG_MAX_BIN_SEMAPHORES configuration option

Definition at line 84 of file osconfig.h.

39.4.1.8 OS_MAX_CMD_LEN

```
#define OS_MAX_CMD_LEN 1000
```

The maximum size of a shell command.

This limit is only applicable if shell support is enabled.

Based on the OSAL_CONFIG_MAX_CMD_LEN configuration option

Note

This value must include a terminating NUL character

Definition at line 230 of file osconfig.h.

39.4.1.9 OS_MAX_CONSOLES

```
#define OS_MAX_CONSOLES 1
```

The maximum number of console devices to support.

Fixed value based on current OSAL implementation, not user configurable.

Definition at line 272 of file osconfig.h.

39.4.1.10 OS_MAX_COUNT_SEMAPHORES

```
#define OS_MAX_COUNT_SEMAPHORES 20
```

The maximum number of counting semaphores to support.

Based on the OSAL_CONFIG_MAX_COUNT_SEMAPHORES configuration option

Definition at line 77 of file osconfig.h.

39.4.1.11 OS_MAX_FILE_NAME

```
#define OS_MAX_FILE_NAME 20
```

The maximum length of OSAL file names.

This limit applies specifically to the file name portion, not the directory portion, of a path name.

Based on the OSAL_CONFIG_MAX_FILE_NAME configuration option

Note

This value must include a terminating NUL character

Definition at line 154 of file osconfig.h.

39.4.1.12 OS_MAX_FILE_SYSTEMS

```
#define OS_MAX_FILE_SYSTEMS 14
```

The maximum number of file systems to support.

Based on the OSAL_CONFIG_MAX_FILE_SYSTEMS configuration option

Definition at line 133 of file osconfig.h.

39.4.1.13 OS_MAX_MODULES

```
#define OS_MAX_MODULES 20
```

The maximum number of modules to support.

Based on the OSAL_CONFIG_MAX_MODULES configuration option

Definition at line 98 of file osconfig.h.

39.4.1.14 OS_MAX_MUTEXES

```
#define OS_MAX_MUTEXES 20
```

The maximum number of mutexes to support.

Based on the OSAL_CONFIG_MAX_MUTEXES configuration option

Definition at line 91 of file osconfig.h.

39.4.1.15 OS_MAX_NUM_OPEN_DIRS

```
#define OS_MAX_NUM_OPEN_DIRS 4
```

The maximum number of concurrently open directories to support.

Based on the OSAL_CONFIG_MAX_NUM_OPEN_DIRS configuration option

Definition at line 126 of file osconfig.h.

39.4.1.16 OS_MAX_NUM_OPEN_FILES

```
#define OS_MAX_NUM_OPEN_FILES 50
```

The maximum number of concurrently open files to support.

Based on the OSAL_CONFIG_MAX_NUM_OPEN_FILES configuration option

Definition at line 119 of file osconfig.h.

39.4.1.17 OS_MAX_PATH_LEN

```
#define OS_MAX_PATH_LEN 64
```

The maximum length of OSAL path names.

This limit applies to the overall length of a path name, including the file name and directory portions.

Based on the OSAL_CONFIG_MAX_PATH_LEN configuration option

Note

This value must include a terminating NUL character

Definition at line 166 of file osconfig.h.

39.4.1.18 OS_MAX_QUEUES

```
#define OS_MAX_QUEUES 64
```

The maximum number of queues to support.

Based on the OSAL_CONFIG_MAX_QUEUES configuration option

Definition at line 70 of file osconfig.h.

39.4.1.19 OS_MAX_SYM_LEN

```
#define OS_MAX_SYM_LEN 64
```

The maximum length of symbols.

Based on the OSAL_CONFIG_MAX_SYM_LEN configuration option

Note

This value must include a terminating NUL character

Definition at line 142 of file osconfig.h.

39.4.1.20 OS_MAX_TASKS

```
#define OS_MAX_TASKS 64
```

The maximum number of to support.

Based on the OSAL_CONFIG_MAX_TASKS configuration option

Definition at line 63 of file osconfig.h.

39.4.1.21 OS_MAX_TIMEBASES

```
#define OS_MAX_TIMEBASES 5
```

The maximum number of timebases to support.

Based on the OSAL_CONFIG_MAX_TIMEBASES configuration option

Definition at line 105 of file osconfig.h.

39.4.1.22 OS_MAX_TIMERS

```
#define OS_MAX_TIMERS 10
```

The maximum number of timer callbacks to support.

Based on the OSAL_CONFIG_MAX_TIMERS configuration option

Definition at line 112 of file osconfig.h.

39.4.1.23 OS_MODULE_FILE_EXTENSION

```
#define OS_MODULE_FILE_EXTENSION ".so"
```

The system-specific file extension used on loadable module files.

Fixed value based on system selection, not user configurable.

Definition at line 279 of file osconfig.h.

39.4.1.24 OS_PRINTF_CONSOLE_NAME

```
#define OS_PRINTF_CONSOLE_NAME ""
```

The name of the primary console device.

This is the device to which OS_printf() output is written. The output may be configured to tag each line with this prefix for identification.

Based on the OSAL_CONFIG_PRINTF_CONSOLE_NAME configuration option

Definition at line 257 of file osconfig.h.

39.4.1.25 OS_QUEUE_MAX_DEPTH

```
#define OS_QUEUE_MAX_DEPTH 50
```

The maximum depth of OSAL queues.

Based on the OSAL_CONFIG_QUEUE_MAX_DEPTH configuration option

Definition at line 237 of file osconfig.h.

39.4.1.26 OS_SHELL_CMD_INPUT_FILE_NAME

```
#define OS_SHELL_CMD_INPUT_FILE_NAME ""
```

The name of the temporary file used to store shell commands.

This configuration is only applicable if shell support is enabled, and only necessary/relevant on some OS implementations.

Based on the OSAL_CONFIG_SHELL_CMD_INPUT_FILE_NAME configuration option

Definition at line 247 of file osconfig.h.

39.4.1.27 OS_SOCKADDR_MAX_LEN

```
#define OS_SOCKADDR_MAX_LEN 28
```

The maximum size of the socket address structure.

This is part of the Socket API, and should be set large enough to hold the largest address type in use on the target system.

Based on the OSAL_CONFIG_SOCKADDR_MAX_LEN configuration option

Definition at line 185 of file osconfig.h.

39.4.1.28 OS_UTILITYTASK_PRIORITY

```
#define OS_UTILITYTASK_PRIORITY 245
```

Priority level of the background utility task.

This task is responsible for writing buffered output of OS_printf to the actual console device, and any other future maintenance task.

Based on the OSAL_CONFIG_UTILITYTASK_PRIORITY configuration option

Definition at line 209 of file osconfig.h.

39.4.1.29 OS_UTILITYTASK_STACK_SIZE

```
#define OS_UTILITYTASK_STACK_SIZE 2048
```

The stack size of the background utility task.

This task is responsible for writing buffered output of OS_printf to the actual console device, and any other future maintenance task.

Based on the OSAL_CONFIG_UTILITYTASK_STACK_SIZE configuration option

Definition at line 219 of file osconfig.h.

39.4.1.30 OSAL_CONFIG_DEBUG_PERMISSIVE_MODE

```
#define OSAL_CONFIG_DEBUG_PERMISSIVE_MODE
```

Definition at line 46 of file osconfig.h.

39.4.1.31 OSAL_CONFIG_DEBUG_PRINTF

```
#define OSAL_CONFIG_DEBUG_PRINTF
```

Definition at line 45 of file osconfig.h.

39.4.1.32 OSAL_CONFIG_INCLUDE_DYNAMIC_LOADER

```
#define OSAL_CONFIG_INCLUDE_DYNAMIC_LOADER
```

Configuration file Operating System Abstraction Layer.

The specific definitions in this file may only be modified by setting the respective OSAL configuration options in the CMake build.

Any direct modifications to the generated copy will be overwritten each time CMake executes.

Note

This file was automatically generated by CMake from /home/runner/work/cFS/cFS/osal/default_config.cmake

Definition at line 41 of file osconfig.h.

39.4.1.33 OSAL_CONFIG_INCLUDE_NETWORK

```
#define OSAL_CONFIG_INCLUDE_NETWORK
```

Definition at line 42 of file osconfig.h.

39.4.1.34 OSAL_CONFIG_INCLUDE_STATIC_LOADER

```
#define OSAL_CONFIG_INCLUDE_STATIC_LOADER
```

Definition at line 43 of file osconfig.h.

39.5 cpu1_msgids.h File Reference

```
#include "cfe_mission_cfg.h"
```

Macros

- #define CFE_PLATFORM_CMD_MID_BASE 0x1800

  Platform command message ID base offset.
- #define CFE_PLATFORM_TLM_MID_BASE 0x0800

  Platform telemetry message ID base offset.
- #define CFE_PLATFORM_CMD_MID_BASE_GLOB 0x1860

  "Global" command message ID base offset
- #define CFE_EVS_CMD_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_EVS_CMD_MSG /∗ 0x1801 ∗/
- #define CFE_SB_CMD_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_SB_CMD_MSG /∗ 0x1803 ∗/
- #define CFE_TBL_CMD_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_TBL_CMD_MSG /∗ 0x1804 ∗/
- #define CFE_TIME_CMD_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_TIME_CMD_MSG /∗ 0x1805 ∗/
- #define CFE_ES_CMD_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_ES_CMD_MSG /∗ 0x1806 ∗/
- #define CFE_ES_SEND_HK_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_ES_SEND_HK_MSG /∗ 0x1808 ∗/
- #define CFE_EVS_SEND_HK_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_EVS_SEND_HK_↩MSG /∗ 0x1809 ∗/
- #define CFE_SB_SEND_HK_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_SB_SEND_HK_MSG /∗ 0x180B ∗/
- #define CFE_TBL_SEND_HK_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_TBL_SEND_HK_↩MSG /∗ 0x180C ∗/
- #define CFE_TIME_SEND_HK_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_TIME_SEND_H↩K_MSG /∗ 0x180D ∗/
- #define CFE_SB_SUB_RPT_CTRL_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_SB_SUB_R↩PT_CTRL_MSG /∗ 0x180E ∗/
- #define CFE_TIME_TONE_CMD_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_TIME_TONE_↩CMD_MSG /∗ 0x1810 ∗/
- #define CFE_TIME_1HZ_CMD_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_TIME_1HZ_CM↩D_MSG /∗ 0x1811 ∗/
- #define CFE_TIME_DATA_CMD_MID CFE_PLATFORM_CMD_MID_BASE_GLOB + CFE_MISSION_TIME_↩DATA_CMD_MSG /∗ 0x1860 ∗/
- #define CFE_TIME_SEND_CMD_MID CFE_PLATFORM_CMD_MID_BASE_GLOB + CFE_MISSION_TIME_↩SEND_CMD_MSG /∗ 0x1862 ∗/
- #define CFE_ES_HK_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_ES_HK_TLM_MSG /∗ 0x0800 ∗/
- #define CFE_EVS_HK_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_EVS_HK_TLM_MSG /∗ 0x0801 ∗/
- #define CFE_SB_HK_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_SB_HK_TLM_MSG /∗ 0x0803 ∗/
- #define CFE_TBL_HK_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_TBL_HK_TLM_MSG /∗ 0x0804 ∗/
- #define CFE_TIME_HK_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_TIME_HK_TLM_M↩SG /∗ 0x0805 ∗/
- #define CFE_TIME_DIAG_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_TIME_DIAG_TL↩M_MSG /∗ 0x0806 ∗/
- #define CFE_EVS_LONG_EVENT_MSG_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_EVS_L↩ONG_EVENT_MSG_MSG /∗ 0x0808 ∗/

- #define CFE_EVS_SHORT_EVENT_MSG_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_EVS_↩ SHORT_EVENT_MSG_MSG /∗ 0x0809 ∗/
- #define CFE_SB_STATS_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_SB_STATS_TLM↩ _MSG /∗ 0x080A ∗/
- #define CFE_ES_APP_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_ES_APP_TLM_MSG /∗ 0x080B ∗/
- #define CFE_TBL_REG_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_TBL_REG_TLM_↩ MSG /∗ 0x080C ∗/
- #define CFE_SB_ALLSUBS_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_SB_ALLSUBS↩ _TLM_MSG /∗ 0x080D ∗/
- #define CFE_SB_ONESUB_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_SB_ONESUB_↩ TLM_MSG /∗ 0x080E ∗/
- #define CFE_ES_MEMSTATS_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_ES_MEMS↩ TATS_TLM_MSG /∗ 0x0810 ∗/

39.5.1 Macro Definition Documentation

39.5.1.1 CFE_ES_APP_TLM_MID

```
#define CFE_ES_APP_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_ES_APP_TLM_MSG /* 0x080B */
```

Definition at line 126 of file cpu1_msgids.h.

39.5.1.2 CFE_ES_CMD_MID

```
#define CFE_ES_CMD_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_ES_CMD_MSG /* 0x1806 */
```

Definition at line 93 of file cpu1_msgids.h.

39.5.1.3 CFE_ES_HK_TLM_MID

```
#define CFE_ES_HK_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_ES_HK_TLM_MSG /* 0x0800 */
```

Definition at line 116 of file cpu1_msgids.h.

39.5.1.4 CFE_ES_MEMSTATS_TLM_MID

```
#define CFE_ES_MEMSTATS_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_ES_MEMSTATS_TLM_MSG /*
0x0810 */
```

Definition at line 130 of file cpu1_msgids.h.

39.5.1.5 CFE_ES_SEND_HK_MID

```
#define CFE_ES_SEND_HK_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_ES_SEND_HK_MSG /* 0x1808 */
```

Definition at line 95 of file cpu1_msgids.h.

39.5.1.6 CFE_EVS_CMD_MID

```
#define CFE_EVS_CMD_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_EVS_CMD_MSG /* 0x1801 */
```

Definition at line 88 of file cpu1_msgids.h.

39.5.1.7 CFE_EVS_HK_TLM_MID

```
#define CFE_EVS_HK_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_EVS_HK_TLM_MSG /* 0x0801 */
```

Definition at line 117 of file cpu1_msgids.h.

39.5.1.8 CFE_EVS_LONG_EVENT_MSG_MID

```
#define CFE_EVS_LONG_EVENT_MSG_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_EVS_LONG_EVENT_MSG_MSG
/* 0x0808 */
```

Definition at line 123 of file cpu1_msgids.h.

39.5.1.9 CFE_EVS_SEND_HK_MID

```
#define CFE_EVS_SEND_HK_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_EVS_SEND_HK_MSG /* 0x1809 */
```

Definition at line 96 of file cpu1_msgids.h.

39.5.1.10 CFE_EVS_SHORT_EVENT_MSG_MID

```
#define CFE_EVS_SHORT_EVENT_MSG_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_EVS_SHORT_EVENT_MSG_↩
MSG /* 0x0809 */
```

Definition at line 124 of file cpu1_msgids.h.

39.5.1.11 CFE_PLATFORM_CMD_MID_BASE

```
#define CFE_PLATFORM_CMD_MID_BASE 0x1800
```

Platform command message ID base offset.

Example mechanism for setting default command bits and deconflicting MIDs across multiple platforms in a mission. For any sufficiently complex mission this method is typically replaced by a centralized message ID management scheme.

0x1800 - Nominal value for default message ID implementation (V1). This sets the command field and the secondary header present field. Typical V1 command MID range is 0x1800-1FFF. Additional cpus can deconflict message IDs by incrementing this value to provide sub-allocations (0x1900 for example). 0x0080 - Command bit for MISSION_MSGID_V2 message ID implementation (V2). Although this can be used for the value below due to the relatively small set of MIDs in the framework it will not scale so an alternative method of deconfliction is recommended.

Definition at line 60 of file cpu1_msgids.h.

39.5.1.12 CFE_PLATFORM_CMD_MID_BASE_GLOB

```
#define CFE_PLATFORM_CMD_MID_BASE_GLOB 0x1860
```

"Global" command message ID base offset

0x1860 - Nominal value for message ID V1 0x00E0 - Potential value for MISSION_MSGID_V2, note command bit is 0x0080. Works in limited cases only, alternative method of deconfliction is recommended. See CFE_PLATFORM_CMD_MID_BASE for more information

Definition at line 83 of file cpu1_msgids.h.

39.5.1.13 CFE_PLATFORM_TLM_MID_BASE

```
#define CFE_PLATFORM_TLM_MID_BASE 0x0800
```

Platform telemetry message ID base offset.

0x0800 - Nominal for message ID V1 0x0000 - Potential value for MISSION_MSGID_V2, but limited to a range of 0x0000-0x007F since the command bit is 0x0080. Alternative method of deconfliction is recommended.

See CFE_PLATFORM_CMD_MID_BASE for more information

Definition at line 72 of file cpu1_msgids.h.

39.5.1.14 CFE_SB_ALLSUBS_TLM_MID

```
#define CFE_SB_ALLSUBS_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_SB_ALLSUBS_TLM_MSG /* 0x080↩
D */
```

Definition at line 128 of file cpu1_msgids.h.

39.5.1.15 CFE_SB_CMD_MID

```
#define CFE_SB_CMD_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_SB_CMD_MSG /* 0x1803 */
```

Definition at line 90 of file cpu1_msgids.h.

39.5.1.16 CFE_SB_HK_TLM_MID

```
#define CFE_SB_HK_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_SB_HK_TLM_MSG /* 0x0803 */
```

Definition at line 119 of file cpu1_msgids.h.

39.5.1.17 CFE_SB_ONESUB_TLM_MID

```
#define CFE_SB_ONESUB_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_SB_ONESUB_TLM_MSG /* 0x080E
*/
```

Definition at line 129 of file cpu1_msgids.h.

39.5.1.18 CFE_SB_SEND_HK_MID

```
#define CFE_SB_SEND_HK_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_SB_SEND_HK_MSG /* 0x180B */
```

Definition at line 98 of file cpu1_msgids.h.

39.5.1.19 CFE_SB_STATS_TLM_MID

```
#define CFE_SB_STATS_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_SB_STATS_TLM_MSG /* 0x080A */
```

Definition at line 125 of file cpu1_msgids.h.

39.5.1.20 CFE_SB_SUB_RPT_CTRL_MID

```
#define CFE_SB_SUB_RPT_CTRL_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_SB_SUB_RPT_CTRL_MSG /*
0x180E */
```

Definition at line 102 of file cpu1_msgids.h.

39.5.1.21 CFE_TBL_CMD_MID

```
#define CFE_TBL_CMD_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_TBL_CMD_MSG /* 0x1804 */
```

Definition at line 91 of file cpu1_msgids.h.

39.5.1.22 CFE_TBL_HK_TLM_MID

```
#define CFE_TBL_HK_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_TBL_HK_TLM_MSG /* 0x0804 */
```

Definition at line 120 of file cpu1_msgids.h.

39.5.1.23 CFE_TBL_REG_TLM_MID

```
#define CFE_TBL_REG_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_TBL_REG_TLM_MSG /* 0x080C */
```

Definition at line 127 of file cpu1_msgids.h.

39.5.1.24 CFE_TBL_SEND_HK_MID

```
#define CFE_TBL_SEND_HK_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_TBL_SEND_HK_MSG /* 0x180C */
```

Definition at line 99 of file cpu1_msgids.h.

39.5.1.25 CFE_TIME_1HZ_CMD_MID

```
#define CFE_TIME_1HZ_CMD_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_TIME_1HZ_CMD_MSG /* 0x1811 */
```

Definition at line 105 of file cpu1_msgids.h.

39.5.1.26 CFE_TIME_CMD_MID

```
#define CFE_TIME_CMD_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_TIME_CMD_MSG /* 0x1805 */
```

Definition at line 92 of file cpu1_msgids.h.

39.5.1.27 CFE_TIME_DATA_CMD_MID

```
#define CFE_TIME_DATA_CMD_MID CFE_PLATFORM_CMD_MID_BASE_GLOB + CFE_MISSION_TIME_DATA_CMD_MSG /*
0x1860 */
```

Definition at line 110 of file cpu1_msgids.h.

39.5.1.28 CFE_TIME_DIAG_TLM_MID

```
#define CFE_TIME_DIAG_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_TIME_DIAG_TLM_MSG /* 0x0806
*/
```

Definition at line 122 of file cpu1_msgids.h.

39.5.1.29 CFE_TIME_HK_TLM_MID

```
#define CFE_TIME_HK_TLM_MID CFE_PLATFORM_TLM_MID_BASE + CFE_MISSION_TIME_HK_TLM_MSG /* 0x0805 */
```

Definition at line 121 of file cpu1_msgids.h.

39.5.1.30 CFE_TIME_SEND_CMD_MID

```
#define CFE_TIME_SEND_CMD_MID CFE_PLATFORM_CMD_MID_BASE_GLOB + CFE_MISSION_TIME_SEND_CMD_MSG /*
0x1862 */
```

Definition at line 111 of file cpu1_msgids.h.

39.5.1.31 CFE_TIME_SEND_HK_MID

```
#define CFE_TIME_SEND_HK_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_TIME_SEND_HK_MSG /* 0x180D */
```

Definition at line 100 of file cpu1_msgids.h.

39.5.1.32 CFE_TIME_TONE_CMD_MID

```
#define CFE_TIME_TONE_CMD_MID CFE_PLATFORM_CMD_MID_BASE + CFE_MISSION_TIME_TONE_CMD_MSG /* 0x1810
*/
```

Definition at line 104 of file cpu1_msgids.h.

39.6 cpu1_platform_cfg.h File Reference

Macros

- #define CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING "/cf"
- #define CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING "/ram"
- #define CFE_PLATFORM_SB_MAX_MSG_IDS 256
- #define CFE_PLATFORM_SB_MAX_PIPES 64
- #define CFE_PLATFORM_SB_MAX_DEST_PER_PKT 16
- #define CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT 4
- #define CFE_PLATFORM_SB_BUF_MEMORY_BYTES 524288
- #define CFE_PLATFORM_SB_HIGHEST_VALID_MSGID 0x1FFF
- #define CFE_PLATFORM_ENDIAN CCSDS_LITTLE_ENDIAN
- #define CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME "/ram/cfe_sb_route.dat"
- #define CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME "/ram/cfe_sb_pipe.dat"
- #define CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME "/ram/cfe_sb_msgmap.dat"
- #define CFE_PLATFORM_SB_FILTERED_EVENT1 CFE_SB_SEND_NO_SUBS_EID
- #define CFE_PLATFORM_SB_FILTER_MASK1 CFE_EVS_FIRST_4_STOP
- #define CFE_PLATFORM_SB_FILTERED_EVENT2 CFE_SB_DUP_SUBSCRIP_EID
- #define CFE_PLATFORM_SB_FILTER_MASK2 CFE_EVS_FIRST_4_STOP
- #define CFE_PLATFORM_SB_FILTERED_EVENT3 CFE_SB_MSGID_LIM_ERR_EID
- #define CFE_PLATFORM_SB_FILTER_MASK3 CFE_EVS_FIRST_16_STOP
- #define CFE_PLATFORM_SB_FILTERED_EVENT4 CFE_SB_Q_FULL_ERR_EID
- #define CFE_PLATFORM_SB_FILTER_MASK4 CFE_EVS_FIRST_16_STOP
- #define CFE_PLATFORM_SB_FILTERED_EVENT5 0
- #define CFE_PLATFORM_SB_FILTER_MASK5 CFE_EVS_NO_FILTER
- #define CFE_PLATFORM_SB_FILTERED_EVENT6 0
- #define CFE_PLATFORM_SB_FILTER_MASK6 CFE_EVS_NO_FILTER
- #define CFE_PLATFORM_SB_FILTERED_EVENT7 0
- #define CFE_PLATFORM_SB_FILTER_MASK7 CFE_EVS_NO_FILTER
- #define CFE_PLATFORM_SB_FILTERED_EVENT8 0
- #define CFE_PLATFORM_SB_FILTER_MASK8 CFE_EVS_NO_FILTER
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01 8
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02 16
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03 20
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04 36
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05 64
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06 96
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07 128
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08 160
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09 256
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10 512

- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11 1024
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12 2048
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13 4096
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14 8192
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15 16384
- #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16 32768
- #define CFE_PLATFORM_SB_MAX_BLOCK_SIZE (CFE_MISSION_SB_MAX_SB_MSG_SIZE + 40)
- #define CFE_PLATFORM_TIME_CFG_SERVER true
- #define CFE_PLATFORM_TIME_CFG_CLIENT false
- #define CFE_PLATFORM_TIME_CFG_VIRTUAL true
- #define CFE_PLATFORM_TIME_CFG_SIGNAL false
- #define CFE_PLATFORM_TIME_CFG_SOURCE false
- #define CFE_PLATFORM_TIME_CFG_SRC_MET false
- #define CFE_PLATFORM_TIME_CFG_SRC_GPS false
- #define CFE_PLATFORM_TIME_CFG_SRC_TIME false
- #define CFE_PLATFORM_TIME_MAX_DELTA_SECS 0
- #define CFE_PLATFORM_TIME_MAX_DELTA_SUBS 500000
- #define CFE_PLATFORM_TIME_MAX_LOCAL_SECS 27
- #define CFE_PLATFORM_TIME_MAX_LOCAL_SUBS 0
- #define CFE_PLATFORM_TIME_CFG_TONE_LIMIT 20000
- #define CFE_PLATFORM_TIME_CFG_START_FLY 2
- #define CFE_PLATFORM_TIME_CFG_LATCH_FLY 8
- #define CFE_PLATFORM_ES_MAX_APPLICATIONS 32
- #define CFE_PLATFORM_ES_MAX_LIBRARIES 10
- #define CFE_PLATFORM_ES_ER_LOG_ENTRIES 20
- #define CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE 256
- #define CFE_PLATFORM_ES_SYSTEM_LOG_SIZE 3072
- #define CFE_PLATFORM_ES_OBJECT_TABLE_SIZE 30
- #define CFE_PLATFORM_ES_MAX_GEN_COUNTERS 8
- #define CFE_PLATFORM_ES_APP_SCAN_RATE 1000
- #define CFE_PLATFORM_ES_APP_KILL_TIMEOUT 5
- #define CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE 512
- #define CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS 4096
- #define CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED 30
- #define CFE_PLATFORM_ES_CDS_SIZE (128 ∗ 1024)
- #define CFE_PLATFORM_ES_USER_RESERVED_SIZE (1024 ∗ 1024)
- #define CFE_PLATFORM_ES_RESET_AREA_SIZE (170 ∗ 1024)
- #define CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN 4
- #define CFE_PLATFORM_ES_NONVOL_STARTUP_FILE "/cf/cfe_es_startup.scr"
- #define CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE "/ram/cfe_es_startup.scr"
- #define CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE "/ram/cfe_es_app_info.log"
- #define CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE "/ram/cfe_es_taskinfo.log"
- #define CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE "/ram/cfe_es_syslog.log"
- #define CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE "/ram/cfe_erlog.log"
- #define CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME "/ram/cfe_es_perf.dat"
- #define CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE "/ram/cfe_cds_reg.log"
- #define CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE 0
- #define CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE 1
- #define CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE 10000
- #define CFE_PLATFORM_ES_PERF_FILTMASK_NONE 0
- #define CFE_PLATFORM_ES_PERF_FILTMASK_ALL ∼CFE_PLATFORM_ES_PERF_FILTMASK_NONE

- #define CFE_PLATFORM_ES_PERF_FILTMASK_INIT CFE_PLATFORM_ES_PERF_FILTMASK_ALL
- #define CFE_PLATFORM_ES_PERF_TRIGMASK_NONE 0
- #define CFE_PLATFORM_ES_PERF_TRIGMASK_ALL ~CFE_PLATFORM_ES_PERF_TRIGMASK_NONE
- #define CFE_PLATFORM_ES_PERF_TRIGMASK_INIT CFE_PLATFORM_ES_PERF_TRIGMASK_NONE
- #define CFE_PLATFORM_ES_PERF_CHILD_PRIORITY 200
- #define CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE 4096
- #define CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY 20
- #define CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS 50
- #define CFE_PLATFORM_ES_DEFAULT_STACK_SIZE 8192
- #define CFE_PLATFORM_EVS_START_TASK_PRIORITY 61
- #define CFE_PLATFORM_EVS_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
- #define CFE_PLATFORM_SB_START_TASK_PRIORITY 64
- #define CFE_PLATFORM_SB_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
- #define CFE_PLATFORM_ES_START_TASK_PRIORITY 68
- #define CFE_PLATFORM_ES_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
- #define CFE_PLATFORM_TIME_START_TASK_PRIORITY 60
- #define CFE_PLATFORM_TIME_TONE_TASK_PRIORITY 25
- #define CFE_PLATFORM_TIME_1HZ_TASK_PRIORITY 25
- #define CFE_PLATFORM_TIME_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_S↩ IZE
- #define CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE 4096
- #define CFE_PLATFORM_TIME_1HZ_TASK_STACK_SIZE 8192
- #define CFE_PLATFORM_TBL_START_TASK_PRIORITY 70
- #define CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
- #define CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES 512
- #define CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS 2
- #define CFE_PLATFORM_ES_POOL_MAX_BUCKETS 17
- #define CFE_PLATFORM_ES_MAX_MEMORY_POOLS 10
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01 8
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02 16
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03 32
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04 48
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05 64
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06 96
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07 128
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08 160
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09 256
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10 512
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11 1024
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12 2048
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13 4096
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14 8192
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15 16384
- #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16 32768
- #define CFE_PLATFORM_ES_MAX_BLOCK_SIZE 80000
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01 8
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02 16
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03 32
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04 48
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05 64
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06 96

- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07 128
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08 160
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09 256
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10 512
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11 1024
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12 2048
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13 4096
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14 8192
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15 16384
- #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16 32768
- #define CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE 80000
- #define CFE_PLATFORM_EVS_MAX_EVENT_FILTERS 8
- #define CFE_PLATFORM_EVS_DEFAULT_LOG_FILE "/ram/cfe_evs.log"
- #define CFE_PLATFORM_EVS_LOG_MAX 20
- #define CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE "/ram/cfe_evs_app.dat"
- #define CFE_PLATFORM_EVS_PORT_DEFAULT 0x0001
- #define CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG 0xE
- #define CFE_PLATFORM_EVS_DEFAULT_LOG_MODE 1
- #define CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE CFE_EVS_MsgFormat_LONG
- #define CFE_PLATFORM_TBL_BUF_MEMORY_BYTES 524288
- #define CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE 16384
- #define CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE 16384
- #define CFE_PLATFORM_TBL_MAX_NUM_TABLES 128
- #define CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES 32
- #define CFE_PLATFORM_TBL_MAX_NUM_HANDLES 256
- #define CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS 4
- #define CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS 10
- #define CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE "/ram/cfe_tbl_reg.log"
- #define CFE_PLATFORM_TBL_VALID_SCID_COUNT 0
- #define CFE_PLATFORM_TBL_U32FROM4CHARS(_C1, _C2, _C3, _C4) ((uint32)(_C1) << 24 | (uint32)(_C2) << 16 | (uint32)(_C3) << 8 | (uint32)(_C4))
- #define CFE_PLATFORM_TBL_VALID_SCID_1 (0x42)
- #define CFE_PLATFORM_TBL_VALID_SCID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
- #define CFE_PLATFORM_TBL_VALID_PRID_COUNT 0
- #define CFE_PLATFORM_TBL_VALID_PRID_1 (1)
- #define CFE_PLATFORM_TBL_VALID_PRID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
- #define CFE_PLATFORM_TBL_VALID_PRID_3 0
- #define CFE_PLATFORM_TBL_VALID_PRID_4 0
- #define CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC 50
- #define CFE_PLATFORM_CORE_MAX_STARTUP_MSEC 30000
- #define CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC 1000

39.6.1 Macro Definition Documentation

39.6.1.1 CFE_PLATFORM_CORE_MAX_STARTUP_MSEC

```
#define CFE_PLATFORM_CORE_MAX_STARTUP_MSEC 30000
```

Purpose CFE core application startup timeout

Description:

The upper limit for the amount of time that the cFE core applications (ES, SB, EVS, TIME, TBL) are each alloted to reach their respective "ready" states.

The CFE "main" thread starts individual tasks for each of the core applications (except FS). Each of these must perform some initialization work before the next core application can be started, so the main thread waits to ensure that the application has reached the "ready" state before starting the next application.

If any core application fails to start, then it indicates a major problem with the system and startup is aborted.

Units are in milliseconds

Limits:

Must be defined as an integer value that is greater than or equal to zero.

Definition at line 1691 of file cpu1_platform_cfg.h.

39.6.1.2 CFE_PLATFORM_ENDIAN

```
#define CFE_PLATFORM_ENDIAN CCSDS_LITTLE_ENDIAN
```

Purpose Platform Endian Indicator

Description:

The value of this constant indicates the endianess of the target system

Limits

This parameter has a lower limit of 0 and an upper limit of 1.

Definition at line 192 of file cpu1_platform_cfg.h.

39.6.1.3 CFE_PLATFORM_ES_APP_KILL_TIMEOUT

```
#define CFE_PLATFORM_ES_APP_KILL_TIMEOUT 5
```

Purpose Define ES Application Kill Timeout

Description:

ES Application Kill Timeout. This parameter controls the number of "scan periods" that ES will wait for an application to Exit after getting the signal Delete, Reload or Restart. The sequence works as follows:

1. ES will set the control request for an App to Delete/Restart/Reload and set this kill timer to the value in this parameter.
2. If the App is reponding and Calls it's RunLoop function, it will drop out of it's main loop and call CFE_ES_↩ ExitApp. Once it calls Exit App, then ES can delete, restart, or reload the app the next time it scans the app table.
3. If the App is not responding, the ES App will decrement this Kill Timeout value each time it runs. If the timeout value reaches zero, ES will kill the app.

The Kill timeout value depends on the CFE_PLATFORM_ES_APP_SCAN_RATE. If the Scan Rate is 1000, or 1 second, and this CFE_PLATFORM_ES_APP_KILL_TIMEOUT is set to 5, then it will take 5 seconds to kill a non-responding App. If the Scan Rate is 250, or 1/4 second, and the CFE_PLATFORM_ES_APP_KILL_TIMEOUT is set to 2, then it will take 1/2 second to time out.

Limits

There is a lower limit of 1 and an upper limit of 100 on this configuration paramater. Units are number of CFE_P↩ LATFORM_ES_APP_SCAN_RATE cycles.

Definition at line 622 of file cpu1_platform_cfg.h.

39.6.1.4 CFE_PLATFORM_ES_APP_SCAN_RATE

```
#define CFE_PLATFORM_ES_APP_SCAN_RATE 1000
```

Purpose Define ES Application Control Scan Rate

Description:

ES Application Control Scan Rate. This parameter controls the speed that ES scans the Application Table looking for App Delete/Restart/Reload requests. All Applications are deleted, restarted, or reloaded by the ES Application. ES will periodically scan for control requests to process. The scan rate is controlled by this parameter, which is given in milliseconds. A value of 1000 means that ES will scan the Application Table once per second. Be careful not to set the value of this too low, because ES will use more CPU cycles scanning the table.

Limits

There is a lower limit of 100 and an upper limit of 20000 on this configuration paramater. millisecond units.

Definition at line 593 of file cpu1_platform_cfg.h.

39.6.1.5 CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE

```
#define CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE 80000
```

Definition at line 1327 of file cpu1_platform_cfg.h.

39.6.1.6 CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES

```
#define CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES 512
```

Purpose Define Maximum Number of Registered CDS Blocks

Description:

Maximum number of registered CDS Blocks

Limits

There is a lower limit of 8. There are no restrictions on the upper limit however, the maximum number of CDS entries is system dependent and should be verified.

Definition at line 1217 of file cpu1_platform_cfg.h.

39.6.1.7 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01 8
```

Purpose Define ES Critical Data Store Memory Pool Block Sizes

Description:

Intermediate ES Critical Data Store Memory Pool Block Sizes

Limits

These sizes MUST be increasing and MUST be an integral multiple of 4.

Definition at line 1311 of file cpu1_platform_cfg.h.

39.6.1.8 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02 16
```

Definition at line 1312 of file cpu1_platform_cfg.h.

39.6.1.9 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03 32
```

Definition at line 1313 of file cpu1_platform_cfg.h.

39.6.1.10 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04 48
```

Definition at line 1314 of file cpu1_platform_cfg.h.

39.6.1.11 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05 64
```

Definition at line 1315 of file cpu1_platform_cfg.h.

39.6.1.12 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06 96
```

Definition at line 1316 of file cpu1_platform_cfg.h.

39.6.1.13 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07 128
```

Definition at line 1317 of file cpu1_platform_cfg.h.

39.6.1.14 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08 160
```

Definition at line 1318 of file cpu1_platform_cfg.h.

39.6.1.15 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09 256
```

Definition at line 1319 of file cpu1_platform_cfg.h.

39.6.1.16 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10 512
```

Definition at line 1320 of file cpu1_platform_cfg.h.

39.6.1.17 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11 1024
```

Definition at line 1321 of file cpu1_platform_cfg.h.

39.6.1.18 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12 2048
```

Definition at line 1322 of file cpu1_platform_cfg.h.

39.6.1.19 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13 4096
```

Definition at line 1323 of file cpu1_platform_cfg.h.

39.6.1.20 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14 8192
```

Definition at line 1324 of file cpu1_platform_cfg.h.

39.6.1.21 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15 16384
```

Definition at line 1325 of file cpu1_platform_cfg.h.

39.6.1.22 CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16 32768
```

Definition at line 1326 of file cpu1_platform_cfg.h.

39.6.1.23 CFE_PLATFORM_ES_CDS_SIZE

```
#define CFE_PLATFORM_ES_CDS_SIZE (128 * 1024)
```

Purpose Define Critical Data Store Size

Description:

Defines the Critical Data Store (CDS) area size in bytes size. The CDS is one of four memory areas that are preserved during a processor reset. NOTE: Changing this value changes memory allocation, and may require changes to platform specific values (in CFE_PSP) such as USER_RESERVED_MEM in VxWorks depending on the memory areas being used for preserved data and on OS specific behavior.

Limits

There is a lower limit of 8192 and an upper limit of UINT_MAX (4 Gigabytes) on this configuration paramater.

Definition at line 699 of file cpu1_platform_cfg.h.

39.6.1.24 CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE "/ram/cfe_es_app_info.log"
```

Purpose Default Application Information Filename

Description:

The value of this constant defines the filename used to store information pertaining to all of the Applications that are registered with Executive Services. This filename is used only when no filename is specified in the the command to query all system apps.

Limits

The length of each string, including the NULL terminator cannot exceed the OS_MAX_PATH_LEN value.

Definition at line 804 of file cpu1_platform_cfg.h.

39.6.1.25 CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE "/ram/cfe_cds_reg.log"
```

Purpose Default Critical Data Store Registry Filename

Description:

The value of this constant defines the filename used to store the Critical Data Store Registry. This filename is used only when no filename is specified in the command to stop performance data collecting.

Limits

The length of each string, including the NULL terminator cannot exceed the OS_MAX_PATH_LEN value.

Definition at line 878 of file cpu1_platform_cfg.h.

39.6.1.26 CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE "/ram/cfe_erlog.log"
```

Purpose Default Exception and Reset (ER) Log Filename

Description:

The value of this constant defines the filename used to store the Exception and Reset (ER) Log. This filename is used only when no filename is specified in the command to dump the ER log. No file specified in the cmd means the first character in the cmd filename is a NULL terminator (zero).

Limits

The length of each string, including the NULL terminator cannot exceed the OS_MAX_PATH_LEN value.

Definition at line 850 of file cpu1_platform_cfg.h.

39.6.1.27 CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME

```
#define CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME "/ram/cfe_es_perf.dat"
```

Purpose Default Performance Data Filename

Description:

The value of this constant defines the filename used to store the Performance Data. This filename is used only when no filename is specified in the command to stop performance data collecting.

Limits

The length of each string, including the NULL terminator cannot exceed the OS_MAX_PATH_LEN value.

Definition at line 864 of file cpu1_platform_cfg.h.

39.6.1.28 CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE

```
#define CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE 0
```

Purpose Define Default System Log Mode following Power On Reset

Description:

Defines the default mode for the operation of the ES System log following a power on reset. The log may operate in either Overwrite mode = 0, where once the log becomes full the oldest message in the log will be overwritten, or Discard mode = 1, where once the log becomes full the contents of the log are preserved and the new event is discarded. This constant may hold a value of either 0 or 1 depending on the desired default. Overwrite Mode = 0, Discard Mode = 1.

Limits

There is a lower limit of 0 and an upper limit of 1 on this configuration paramater.

Definition at line 896 of file cpu1_platform_cfg.h.

39.6.1.29 CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE

```
#define CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE 1
```

Purpose Define Default System Log Mode following Processor Reset

Description:

Defines the default mode for the operation of the ES System log following a processor reset. The log may operate in either Overwrite mode = 0, where once the log becomes full the oldest message in the log will be overwritten, or Discard mode = 1, where once the log becomes full the contents of the log are preserved and the new event is discarded. This constant may hold a value of either 0 or 1 depending on the desired default. Overwrite Mode = 0, Discard Mode = 1.

Limits

There is a lower limit of 0 and an upper limit of 1 on this configuration paramater.

Definition at line 914 of file cpu1_platform_cfg.h.

39.6.1.30 CFE_PLATFORM_ES_DEFAULT_STACK_SIZE

```
#define CFE_PLATFORM_ES_DEFAULT_STACK_SIZE 8192
```

Purpose Define Default Stack Size for an Application

Description:

This parameter defines a default stack size. This parameter is used by the cFE Core Applications.

Limits

There is a lower limit of 2048. There are no restrictions on the upper limit however, the maximum stack size size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 1064 of file cpu1_platform_cfg.h.

39.6.1.31 CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE "/ram/cfe_es_syslog.log"
```

Purpose Default System Log Filename

Description:

The value of this constant defines the filename used to store important information (as ASCII text strings) that might not be able to be sent in an Event Message. This filename is used only when no filename is specified in the command to dump the system log. No file specified in the cmd means the first character in the cmd filename is a NULL terminator (zero).

Limits

The length of each string, including the NULL terminator cannot exceed the OS_MAX_PATH_LEN value.

Definition at line 835 of file cpu1_platform_cfg.h.

39.6.1.32 CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE "/ram/cfe_es_taskinfo.log"
```

Purpose Default Application Information Filename

Description:

The value of this constant defines the filename used to store information pertaining to all of the Applications that are registered with Executive Services. This filename is used only when no filename is specified in the the command to query all system tasks.

Limits

The length of each string, including the NULL terminator cannot exceed the OS_MAX_PATH_LEN value.

Definition at line 819 of file cpu1_platform_cfg.h.

39.6.1.33 CFE_PLATFORM_ES_ER_LOG_ENTRIES

```
#define CFE_PLATFORM_ES_ER_LOG_ENTRIES 20
```

Purpose Define Max Number of ER (Exception and Reset) log entries

Description:

Defines the maximum number of ER (Exception and Reset) log entries

Limits

There is a lower limit of 1. There are no restrictions on the upper limit however, the maximum number of log entries is system dependent and should be verified.

Definition at line 520 of file cpu1_platform_cfg.h.

39.6.1.34 CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE

```
#define CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE 256
```

Purpose Maximum size of CPU Context in ES Error Log

Description:

This should be large enough to accommodate the CPU context information supplied by the PSP on the given platform.

Limits:

Must be greater than zero and a multiple of sizeof(uint32). Limited only by the available memory and the number of entries in the error log. Any context information beyond this size will be truncated.

Definition at line 534 of file cpu1_platform_cfg.h.

39.6.1.35 CFE_PLATFORM_ES_MAX_APPLICATIONS

```
#define CFE_PLATFORM_ES_MAX_APPLICATIONS 32
```

Purpose Define Max Number of Applications

Description:

Defines the maximum number of applications that can be loaded into the system. This number does not include child tasks.

Limits

There is a lower limit of 6. The lower limit corresponds to the cFE internal applications. There are no restrictions on the upper limit however, the maximum number of applications is system dependent and should be verified. AppIDs that are checked against this configuration are defined by a 32 bit data word.

Definition at line 493 of file cpu1_platform_cfg.h.

39.6.1.36 CFE_PLATFORM_ES_MAX_BLOCK_SIZE

```
#define CFE_PLATFORM_ES_MAX_BLOCK_SIZE 80000
```

Definition at line 1300 of file cpu1_platform_cfg.h.

39.6.1.37 CFE_PLATFORM_ES_MAX_GEN_COUNTERS

```
#define CFE_PLATFORM_ES_MAX_GEN_COUNTERS 8
```

Purpose Define Max Number of Generic Counters

Description:

Defines the maximum number of Generic Counters that can be registered.

Limits

This parameter has a lower limit of 1 and an upper limit of 65535.

Definition at line 574 of file cpu1_platform_cfg.h.

39.6.1.38 CFE_PLATFORM_ES_MAX_LIBRARIES

```
#define CFE_PLATFORM_ES_MAX_LIBRARIES 10
```

Purpose Define Max Number of Shared libraries

Description:

Defines the maximum number of cFE Shared libraries that can be loaded into the system.

Limits

There is a lower limit of 1. There are no restrictions on the upper limit however, the maximum number of libraries is system dependent and should be verified.

Definition at line 507 of file cpu1_platform_cfg.h.

39.6.1.39 CFE_PLATFORM_ES_MAX_MEMORY_POOLS

```
#define CFE_PLATFORM_ES_MAX_MEMORY_POOLS 10
```

Purpose Maximum number of memory pools

Description:

The upper limit for the number of memory pools than can concurrently exist within the system.

The CFE_SB and CFE_TBL core subsystems each define a memory pool.

Individual applications may also create memory pools, so this value should be set sufficiently high enough to support the applications being used on this platform.

Limits:

Must be at least 2 to support CFE core - SB and TBL pools. No specific upper limit.

Definition at line 1265 of file cpu1_platform_cfg.h.

39.6.1.40 CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS

```
#define CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS 2
```

Purpose Define Number of Processor Resets Before a Power On Reset

Description:

Number of Processor Resets before a Power On Reset is called. If set to 2, then 2 processor resets will occur, and the 3rd processor reset will be a power on reset instead.

Limits

There is a lower limit of 0. There are no restrictions on the upper limit however, the maximum number of processor resets may be system dependent and should be verified.

Definition at line 1232 of file cpu1_platform_cfg.h.

39.6.1.41 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01 8
```

Purpose Define Default ES Memory Pool Block Sizes

Description:

Default Intermediate ES Memory Pool Block Sizes. If an application is using the CFE_ES Memory Pool APIs (CFE_ES_PoolCreate, CFE_ES_PoolCreateNoSem, CFE_ES_GetPoolBuf and CFE_ES_PutPoolBuf) but finds these sizes inappropriate for their use, they may wish to use the CFE_ES_PoolCreateEx API to specify their own intermediate block sizes

Limits

These sizes MUST be increasing and MUST be an integral multiple of 4. Also, CFE_PLATFORM_ES_MAX_BLOCK_SIZE must be larger than CFE_MISSION_SB_MAX_SB_MSG_SIZE and both CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE and CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE. Note that if Table Services have been removed from the CFE, the table size limits are still enforced although the table size definitions may be reduced.

Definition at line 1284 of file cpu1_platform_cfg.h.

39.6.1.42 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02 16
```

Definition at line 1285 of file cpu1_platform_cfg.h.

39.6.1.43 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03 32
```

Definition at line 1286 of file cpu1_platform_cfg.h.

39.6.1.44 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04 48
```

Definition at line 1287 of file cpu1_platform_cfg.h.

39.6.1.45 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05 64
```

Definition at line 1288 of file cpu1_platform_cfg.h.

39.6.1.46 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06 96
```

Definition at line 1289 of file cpu1_platform_cfg.h.

39.6.1.47 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07 128
```

Definition at line 1290 of file cpu1_platform_cfg.h.

39.6.1.48 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08 160
```

Definition at line 1291 of file cpu1_platform_cfg.h.

39.6.1.49 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09 256
```

Definition at line 1292 of file cpu1_platform_cfg.h.

39.6.1.50 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10 512
```

Definition at line 1293 of file cpu1_platform_cfg.h.

39.6.1.51 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11 1024
```

Definition at line 1294 of file cpu1_platform_cfg.h.

39.6.1.52 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12 2048
```

Definition at line 1295 of file cpu1_platform_cfg.h.

39.6.1.53 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13 4096
```

Definition at line 1296 of file cpu1_platform_cfg.h.

39.6.1.54 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14 8192
```

Definition at line 1297 of file cpu1_platform_cfg.h.

39.6.1.55 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15 16384
```

Definition at line 1298 of file cpu1_platform_cfg.h.

39.6.1.56 CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16 32768
```

Definition at line 1299 of file cpu1_platform_cfg.h.

39.6.1.57 CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN

```
#define CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN 4
```

Purpose Define Memory Pool Alignment Size

Description:

Ensures that buffers obtained from a memory pool are aligned to a certain minimum block size. Note the allocator will always align to the minimum required by the CPU architecture. This may be set greater than the CPU requirement as desired for optimal performance.

For some architectures/applications it may be beneficial to set this to the cache line size of the target CPU, or to use special SIMD instructions that require a more stringent memory alignment.

Limits

This must always be a power of 2, as it is used as a binary address mask.

Definition at line 761 of file cpu1_platform_cfg.h.

39.6.1.58 CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING

```
#define CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING "/cf"
```

Purpose Default virtual path for persistent storage

Description:

This configures the default location in the virtual file system for persistent/non-volatile storage. Files such as the startup script, app/library dynamic modules, and configuration tables are expected to be stored in this directory.

Definition at line 49 of file cpu1_platform_cfg.h.

39.6.1.59 CFE_PLATFORM_ES_NONVOL_STARTUP_FILE

```
#define CFE_PLATFORM_ES_NONVOL_STARTUP_FILE "/cf/cfe_es_startup.scr"
```

Purpose ES Nonvolatile Startup Filename

Description:

The value of this constant defines the path and name of the file that contains a list of modules that will be loaded and started by the cFE after the cFE finishes its startup sequence.

Limits

The length of each string, including the NULL terminator cannot exceed the OS_MAX_PATH_LEN value.

Definition at line 775 of file cpu1_platform_cfg.h.

39.6.1.60 CFE_PLATFORM_ES_OBJECT_TABLE_SIZE

```
#define CFE_PLATFORM_ES_OBJECT_TABLE_SIZE 30
```

Purpose Define Number of entries in the ES Object table

Description:

Defines the number of entries in the ES Object table. This table controls the core cFE startup.

Limits

There is a lower limit of 15. There are no restrictions on the upper limit however, the maximum object table size is system dependent and should be verified.

Definition at line 563 of file cpu1_platform_cfg.h.

39.6.1.61 CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY

```
#define CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY 20
```

Purpose Define Performance Analyzer Child Task Delay

Description:

This parameter defines the delay time (in milliseconds) between performance data file writes performed by the Executive Services Performace Analyzer Child Task.

Limits

It is recommended this parameter be greater than or equal to 20ms. This parameter is limited by the maximum value allowed by the data type. In this case, the data type is an unsigned 32-bit integer, so the valid range is 0 to 0xFFFFFFFF.

Definition at line 1038 of file cpu1_platform_cfg.h.

39.6.1.62 CFE_PLATFORM_ES_PERF_CHILD_PRIORITY

```
#define CFE_PLATFORM_ES_PERF_CHILD_PRIORITY 200
```

Purpose Define Performance Analyzer Child Task Priority

Description:

This parameter defines the priority of the child task spawed by the Executive Services to write performance data to a file. Lower numbers are higher priority, with 1 being the highest priority in the case of a child task.

Limits

Valid range for a child task is 1 to 255 however, the priority cannot be higher (lower number) than the ES parent application priority.

Definition at line 1009 of file cpu1_platform_cfg.h.

39.6.1.63 CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE

```
#define CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE 4096
```

Purpose Define Performance Analyzer Child Task Stack Size

Description:

This parameter defines the stack size of the child task spawed by the Executive Services to write performance data to a file.

Limits

It is recommended this parameter be greater than or equal to 4KB. This parameter is limited by the maximum value allowed by the data type. In this case, the data type is an unsigned 32-bit integer, so the valid range is 0 to 0xFFFFFFFF.

Definition at line 1023 of file cpu1_platform_cfg.h.

39.6.1.64 CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE

```
#define CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE 10000
```

Purpose Define Max Size of Performance Data Buffer

Description:

Defines the maximum size of the performance data buffer. Units are number of performance data entries. An entry is defined by a 32 bit data word followed by a 64 bit time stamp.

Limits

There is a lower limit of 1025. There are no restrictions on the upper limit however, the maximum buffer size size is system dependent and should be verified. The units are number of entries. An entry is defined by a 32 bit data word followed by a 64 bit time stamp.

Definition at line 930 of file cpu1_platform_cfg.h.

39.6.1.65 CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS

```
#define CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS 50
```

Purpose Define Performance Analyzer Child Task Number of Entries Between Delay

Description:

This parameter defines the number of performace analyzer entries the Performace Analyzer Child Task will write to the file between delays.

Definition at line 1048 of file cpu1_platform_cfg.h.

39.6.1.66 CFE_PLATFORM_ES_PERF_FILTMASK_ALL

```
#define CFE_PLATFORM_ES_PERF_FILTMASK_ALL ~CFE_PLATFORM_ES_PERF_FILTMASK_NONE
```

Purpose Define Filter Mask Setting for Enabling All Performance Entries

Description:

Defines the filter mask for enabling all performance entries. The value is a bit mask. For each bit, 0 means the corresponding entry is disabled and 1 means it is enabled.

Definition at line 950 of file cpu1_platform_cfg.h.

39.6.1.67 CFE_PLATFORM_ES_PERF_FILTMASK_INIT

```
#define CFE_PLATFORM_ES_PERF_FILTMASK_INIT CFE_PLATFORM_ES_PERF_FILTMASK_ALL
```

Purpose Define Default Filter Mask Setting for Performance Data Buffer

Description:

Defines the default filter mask for the performance data buffer. The value is a bit mask. For each bit, 0 means the corresponding entry is disabled and 1 means it is enabled.

Definition at line 961 of file cpu1_platform_cfg.h.

39.6.1.68 CFE_PLATFORM_ES_PERF_FILTMASK_NONE

```
#define CFE_PLATFORM_ES_PERF_FILTMASK_NONE 0
```

Purpose Define Filter Mask Setting for Disabling All Performance Entries

Description:

Defines the filter mask for disabling all performance entries. The value is a bit mask. For each bit, 0 means the corresponding entry is disabled and 1 means it is enabled.

Definition at line 940 of file cpu1_platform_cfg.h.

39.6.1.69 CFE_PLATFORM_ES_PERF_TRIGMASK_ALL

```
#define CFE_PLATFORM_ES_PERF_TRIGMASK_ALL ~CFE_PLATFORM_ES_PERF_TRIGMASK_NONE
```

Purpose Define Filter Trigger Setting for Enabling All Performance Entries

Description:

Defines the trigger mask for enabling all performance data entries. The value is a bit mask. For each bit, 0 means the trigger for the corresponding entry is disabled and 1 means it is enabled.

Definition at line 983 of file cpu1_platform_cfg.h.

39.6.1.70 CFE_PLATFORM_ES_PERF_TRIGMASK_INIT

```
#define CFE_PLATFORM_ES_PERF_TRIGMASK_INIT CFE_PLATFORM_ES_PERF_TRIGMASK_NONE
```

Purpose Define Default Filter Trigger Setting for Performance Data Buffer

Description:

Defines the default trigger mask for the performance data buffer. The value is a 32-bit mask. For each bit, 0 means the trigger for the corresponding entry is disabled and 1 means it is enabled.

Definition at line 994 of file cpu1_platform_cfg.h.

39.6.1.71 CFE_PLATFORM_ES_PERF_TRIGMASK_NONE

```
#define CFE_PLATFORM_ES_PERF_TRIGMASK_NONE 0
```

Purpose Define Default Filter Trigger Setting for Disabling All Performance Entries

Description:

Defines the default trigger mask for disabling all performance data entries. The value is a bit mask. For each bit, 0 means the trigger for the corresponding entry is disabled and 1 means it is enabled.

Definition at line 972 of file cpu1_platform_cfg.h.

39.6.1.72 CFE_PLATFORM_ES_POOL_MAX_BUCKETS

```
#define CFE_PLATFORM_ES_POOL_MAX_BUCKETS 17
```

Purpose Maximum number of block sizes in pool structures

Description:

The upper limit for the number of block sizes supported in the generic pool implementation, which in turn implements the memory pools and CDS.

Limits:

Must be at least one. No specific upper limit, but the number is anticipated to be reasonably small (i.e. tens, not hundreds). Large values have not been tested.

The ES and CDS block size lists must correlate with this value

Definition at line 1247 of file cpu1_platform_cfg.h.

39.6.1.73 CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING

```
#define CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING "/ram"
```

Purpose Default virtual path for volatile storage

Description:

The CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING parameter is used to set the cFE mount path for the CFE RAM disk. This is a parameter for missions that do not want to use the default value of "/ram", or for missions that need to have a different value for different CPUs or Spacecraft. Note that the vxWorks OSAL cannot currently handle names that have more than one path separator in it. The names "/ram", "/ramdisk", "/disk123" will all work, but "/disks/ram" will not. Multiple separators can be used with the posix or RTEMS ports.

Definition at line 65 of file cpu1_platform_cfg.h.

39.6.1.74 CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS

```
#define CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS 4096
```

Purpose ES Ram Disk Number of Sectors

Description:

Defines the ram disk number of sectors. The ram disk is one of four memory areas that are preserved on a processor reset. NOTE: Changing this value changes memory allocation, and may require changes to platform specific values (in CFE_PSP) such as USER_RESERVED_MEM in VxWorks depending on the memory areas being used for preserved data and on OS specific behavior.

Limits

There is a lower limit of 128. There are no restrictions on the upper limit however, the maximum number of RAM sectors is system dependent and should be verified.

Definition at line 658 of file cpu1_platform_cfg.h.

39.6.1.75 CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED

```
#define CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED 30
```

Purpose Percentage of Ram Disk Reserved for Decompressing Apps

Description:

The CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED parameter is used to make sure that the Volatile (RAM) Disk has a defined amount of free space during a processor reset. The cFE uses the Volatile disk to decompress cFE applications during system startup. If this Volatile disk happens to get filled with logs and misc files, then a processor reset may not work, because there will be no room to decompress cFE apps. To solve that problem, this parameter sets the "Low Water Mark" for disk space on a Processor reset. It should be set to allow the largest cFE Application to be decompressed. During a Processor reset, if there is not sufficient space left on the disk, it will be re-formatted in order to clear up some space.

This feature can be turned OFF by setting the parameter to 0.

Limits

There is a lower limit of 0 and an upper limit of 75 on this configuration paramater.Units are percentage. A setting of zero will turn this feature off.

Definition at line 682 of file cpu1_platform_cfg.h.

39.6.1.76 CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE

```
#define CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE 512
```

Purpose ES Ram Disk Sector Size

Description:

Defines the ram disk sector size. The ram disk is 1 of 4 memory areas that are preserved on a processor reset. NOTE: Changing this value changes memory allocation, and may require changes to platform specific values (in C↩ FE_PSP) such as USER_RESERVED_MEM in VxWorks depending on the memory areas being used for preserved data and on OS specific behavior.

Limits

There is a lower limit of 128. There are no restrictions on the upper limit however, the maximum RAM disk sector size is system dependent and should be verified.

Definition at line 640 of file cpu1_platform_cfg.h.

39.6.1.77 CFE_PLATFORM_ES_RESET_AREA_SIZE

```
#define CFE_PLATFORM_ES_RESET_AREA_SIZE (170 * 1024)
```

Purpose Define ES Reset Area Size

Description:

The ES Reset Area Size. This is the size in bytes of the cFE Reset variable and log area. This is a block of memory used by the cFE to store the system log ER Log and critical reset variables. This is 4 of 4 of the memory areas that are preserved during a processor reset. Note: This area must be sized large enough to hold all of the data structures. It should be automatically sized based on the CFE_ES_ResetData_t type, but circular dependancies in the headers prevent it from being defined this way. NOTE: Changing this value changes memory allocation, and may require changes to platform specific values (in CFE_PSP) such as USER_RESERVED_MEM in VxWorks depending on the memory areas being used for preserved data and on OS specific behavior.

Limits

There is a lower limit of 153600 (150KBytes) and an upper limit of UINT_MAX (4 Gigabytes) on this configuration paramater.

Definition at line 742 of file cpu1_platform_cfg.h.

39.6.1.78 CFE_PLATFORM_ES_START_TASK_PRIORITY

```
#define CFE_PLATFORM_ES_START_TASK_PRIORITY 68
```

Purpose Define ES Task Priority

Description:

Defines the cFE_ES Task priority.

Limits

Not Applicable

Definition at line 1127 of file cpu1_platform_cfg.h.

39.6.1.79 CFE_PLATFORM_ES_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_ES_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
```

Purpose Define ES Task Stack Size

Description:

Defines the cFE_ES Task Stack Size

Limits

There is a lower limit of 2048 on this configuration paramater. There are no restrictions on the upper limit however, the maximum stack size size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 1142 of file cpu1_platform_cfg.h.

39.6.1.80 CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC

```
#define CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC 1000
```

Purpose Startup script timeout

Description:

The upper limit for the total amount of time that all apps listed in the CFE ES startup script may take to all become ready.

Unlike the "core" app timeout, this is a soft limit; if the alloted time is exceeded, it probably indicates an issue with one of the apps, but does not cause CFE ES to take any additional action other than logging the event to the syslog.

Units are in milliseconds

Limits:

Must be defined as an integer value that is greater than or equal to zero.

Definition at line 1709 of file cpu1_platform_cfg.h.

39.6.1.81 CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC

```
#define CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC 50
```

Purpose Poll timer for startup sync delay

Description:

During startup, some tasks may need to synchronize their own initialization with the initialization of other applications in the system.

CFE ES implements an API to accomplish this, that performs a task delay (sleep) while polling the overall system state until other tasks are ready.

This value controls the amount of time that the CFE_ES_ApplicationSyncDelay will sleep between each check of the system state. This should be large enough to allow other tasks to run, but not so large as to noticeably delay the startup completion.

Units are in milliseconds

Limits:

Must be defined as an integer value that is greater than or equal to zero.

Definition at line 1667 of file cpu1_platform_cfg.h.

39.6.1.82 CFE_PLATFORM_ES_SYSTEM_LOG_SIZE

```
#define CFE_PLATFORM_ES_SYSTEM_LOG_SIZE 3072
```

Purpose Define Size of the cFE System Log.

Description:

Defines the size in bytes of the cFE system log. The system log holds variable length strings that are terminated by a linefeed and null character.

Limits

There is a lower limit of 512. There are no restrictions on the upper limit however, the maximum system log size is system dependent and should be verified.

Definition at line 549 of file cpu1_platform_cfg.h.

39.6.1.83 CFE_PLATFORM_ES_USER_RESERVED_SIZE

```
#define CFE_PLATFORM_ES_USER_RESERVED_SIZE (1024 * 1024)
```

Purpose Define User Reserved Memory Size

Description:

User Reserved Memory Size. This is the size in bytes of the cFE User reserved Memory area. This is a block of memory that is available for cFE application use. The address is obtained by calling CFE_PSP_GetUserReservedArea. The User Reserved Memory is one of four memory areas that are preserved during a processor reset. NOTE: Changing this value changes memory allocation, and may require changes to platform specific values (in CFE_PSP) such as USER_RESERVED_MEM in VxWorks depending on the memory areas being used for preserved data and on OS specific behavior.

Limits

There is a lower limit of 1024 and an upper limit of UINT_MAX (4 Gigabytes) on this configuration paramater.

Definition at line 719 of file cpu1_platform_cfg.h.

39.6.1.84 CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE

```
#define CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE "/ram/cfe_es_startup.scr"
```

Purpose ES Volatile Startup Filename

Description:

The value of this constant defines the path and name of the file that contains a list of modules that will be loaded and started by the cFE after the cFE finishes its startup sequence.

Limits

The length of each string, including the NULL terminator cannot exceed the OS_MAX_PATH_LEN value.

Definition at line 789 of file cpu1_platform_cfg.h.

39.6.1.85 CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE

```
#define CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE "/ram/cfe_evs_app.dat"
```

Purpose Default EVS Application Data Filename

Description:

The value of this constant defines the filename used to store the EVS Application Data(event counts/filtering information). This filename is used only when no filename is specified in the command to dump the event log.

Limits

The length of each string, including the NULL terminator cannot exceed the OS_MAX_PATH_LEN value.

Definition at line 1381 of file cpu1_platform_cfg.h.

39.6.1.86 CFE_PLATFORM_EVS_DEFAULT_LOG_FILE

```
#define CFE_PLATFORM_EVS_DEFAULT_LOG_FILE "/ram/cfe_evs.log"
```

Purpose Default Event Log Filename

Description:

The value of this constant defines the filename used to store the Event Services local event log. This filename is used only when no filename is specified in the command to dump the event log.

Limits

The length of each string, including the NULL terminator cannot exceed the OS_MAX_PATH_LEN value.

Definition at line 1354 of file cpu1_platform_cfg.h.

39.6.1.87 CFE_PLATFORM_EVS_DEFAULT_LOG_MODE

```
#define CFE_PLATFORM_EVS_DEFAULT_LOG_MODE 1
```

Purpose Default EVS Local Event Log Mode

Description:

Defines a state of overwrite(0) or discard(1) for the operation of the EVS local event log. The log may operate in either Overwrite mode = 0, where once the log becomes full the oldest event in the log will be overwritten, or Discard mode = 1, where once the log becomes full the contents of the log are preserved and the new event is discarded. Overwrite Mode = 0, Discard Mode = 1.

Limits

The valid settings are 0 or 1

Definition at line 1428 of file cpu1_platform_cfg.h.

39.6.1.88 CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE

```
#define CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE CFE_EVS_MsgFormat_LONG
```

Purpose Default EVS Message Format Mode

Description:

Defines the default message format (long or short) for event messages being sent to the ground. Choose between CFE_EVS_MsgFormat_LONG or CFE_EVS_MsgFormat_SHORT.

Limits

The valid settings are CFE_EVS_MsgFormat_LONG or CFE_EVS_MsgFormat_SHORT

Definition at line 1441 of file cpu1_platform_cfg.h.

39.6.1.89 CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG

```
#define CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG 0xE
```

Purpose Default EVS Event Type Filter Mask

Description:

Defines a state of on or off for all four event types. The term event 'type' refers to the criticality level and may be Debug, Informational, Error or Critical. Each event type has a bit position. (bit 0 = Debug, bit 1 = Info, bit 2 = Error, bit 3 = Critical). This is a global setting, meaning it applies to all applications. To filter an event type, set its bit to zero. For example, 0xE means Debug = OFF, Info = ON, Error = ON, Critical = ON

Limits

The valid settings are 0x0 to 0xF.

Definition at line 1412 of file cpu1_platform_cfg.h.

39.6.1.90 CFE_PLATFORM_EVS_LOG_MAX

```
#define CFE_PLATFORM_EVS_LOG_MAX 20
```

Purpose Maximum Number of Events in EVS Local Event Log

Description:

Dictates the EVS local event log capacity. Units are the number of events.

Limits

There are no restrictions on the lower and upper limits however, the maximum log size is system dependent and should be verified.

Definition at line 1366 of file cpu1_platform_cfg.h.

39.6.1.91 CFE_PLATFORM_EVS_MAX_EVENT_FILTERS

```
#define CFE_PLATFORM_EVS_MAX_EVENT_FILTERS 8
```

Purpose Define Maximum Number of Event Filters per Application

Description:

Maximum number of events that may be filtered per application.

Limits

There are no restrictions on the lower and upper limits however, the maximum number of event filters is system dependent and should be verified.

Definition at line 1340 of file cpu1_platform_cfg.h.

39.6.1.92 CFE_PLATFORM_EVS_PORT_DEFAULT

```
#define CFE_PLATFORM_EVS_PORT_DEFAULT 0x0001
```

Purpose Default EVS Output Port State

Description:

Defines the default port state (enabled or deisabled) for the four output ports defined within the Event Service. Port 1 is usually the uart output terminal. To enable a port, set the proper bit to a 1. Bit 0 is port 1, bit 1 is port2 etc.

Limits

The valid settings are 0x0 to 0xF.

Definition at line 1395 of file cpu1_platform_cfg.h.

39.6.1.93 CFE_PLATFORM_EVS_START_TASK_PRIORITY

```
#define CFE_PLATFORM_EVS_START_TASK_PRIORITY 61
```

Purpose Define EVS Task Priority

Description:

Defines the cFE_EVS Task priority.

Limits

Not Applicable

Definition at line 1075 of file cpu1_platform_cfg.h.

39.6.1.94 CFE_PLATFORM_EVS_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_EVS_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
```

Purpose Define EVS Task Stack Size

Description:

Defines the cFE_EVS Task Stack Size

Limits

There is a lower limit of 2048 on this configuration paramater. There are no restrictions on the upper limit however, the maximum stack size size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 1090 of file cpu1_platform_cfg.h.

39.6.1.95 CFE_PLATFORM_SB_BUF_MEMORY_BYTES

```
#define CFE_PLATFORM_SB_BUF_MEMORY_BYTES 524288
```

Purpose Size of the SB buffer memory pool

Description:

Dictates the size of the SB memory pool. For each message the SB sends, the SB dynamically allocates from this memory pool, the memory needed to process the message. The memory needed to process each message is msg size + msg descriptor(CFE_SB_BufferD_t). This memory pool is also used to allocate destination descriptors (CFE_SB_DestinationD_t) during the subscription process. To see the run-time, high-water mark and the current utilization figures regarding this parameter, send an SB command to 'Send Statistics Pkt'. Some memory statistics have been added to the SB housekeeping packet. NOTE: It is important to monitor these statistics to ensure the desired memory margin is met.

Limits

This parameter has a lower limit of 512 and an upper limit of UINT_MAX (4 Gigabytes).

Definition at line 155 of file cpu1_platform_cfg.h.

39.6.1.96 CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME

```
#define CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME "/ram/cfe_sb_msgmap.dat"
```

Purpose Default Message Map Filename

Description:

The value of this constant defines the filename used to store the software bus message map information. This filename is used only when no filename is specified in the command. The message map is a lookup table (array of 16bit words) that has an element for each possible MsgId value and holds the routing table index for that MsgId. The Msg Map provides fast access to the destinations of a message.

Limits

The length of each string, including the NULL terminator cannot exceed the OS_MAX_PATH_LEN value.

Definition at line 237 of file cpu1_platform_cfg.h.

39.6.1.97 CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT

```
#define CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT 4
```

Purpose Default Subscription Message Limit

Description:

Dictates the default Message Limit when using the CFE_SB_Subscribe API. This will limit the number of messages with a specific message ID that can be received through a subscription. This only changes the default; other message limits can be set on a per subscription basis using CFE_SB_SubscribeEx .

Limits

This parameter has a lower limit of 4 and an upper limit of 65535.

Definition at line 133 of file cpu1_platform_cfg.h.

39.6.1.98 CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME

```
#define CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME "/ram/cfe_sb_pipe.dat"
```

Purpose Default Pipe Information Filename

Description:

The value of this constant defines the filename used to store the software bus pipe information. This filename is used only when no filename is specified in the command.

Limits

The length of each string, including the NULL terminator cannot exceed the OS_MAX_PATH_LEN value.

Definition at line 220 of file cpu1_platform_cfg.h.

39.6.1.99 CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME

```
#define CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME "/ram/cfe_sb_route.dat"
```

Purpose Default Routing Information Filename

Description:

The value of this constant defines the filename used to store the software bus routing information. This filename is used only when no filename is specified in the command.

Limits

The length of each string, including the NULL terminator cannot exceed the OS_MAX_PATH_LEN value.

Definition at line 206 of file cpu1_platform_cfg.h.

39.6.1.100 CFE_PLATFORM_SB_FILTER_MASK1

```
#define CFE_PLATFORM_SB_FILTER_MASK1 CFE_EVS_FIRST_4_STOP
```

Definition at line 255 of file cpu1_platform_cfg.h.

39.6.1.101 CFE_PLATFORM_SB_FILTER_MASK2

```
#define CFE_PLATFORM_SB_FILTER_MASK2 CFE_EVS_FIRST_4_STOP
```

Definition at line 258 of file cpu1_platform_cfg.h.

39.6.1.102 CFE_PLATFORM_SB_FILTER_MASK3

```
#define CFE_PLATFORM_SB_FILTER_MASK3 CFE_EVS_FIRST_16_STOP
```

Definition at line 261 of file cpu1_platform_cfg.h.

39.6.1.103 CFE_PLATFORM_SB_FILTER_MASK4

```
#define CFE_PLATFORM_SB_FILTER_MASK4 CFE_EVS_FIRST_16_STOP
```

Definition at line 264 of file cpu1_platform_cfg.h.

39.6.1.104 CFE_PLATFORM_SB_FILTER_MASK5

```
#define CFE_PLATFORM_SB_FILTER_MASK5 CFE_EVS_NO_FILTER
```

Definition at line 267 of file cpu1_platform_cfg.h.

39.6.1.105 CFE_PLATFORM_SB_FILTER_MASK6

```
#define CFE_PLATFORM_SB_FILTER_MASK6 CFE_EVS_NO_FILTER
```

Definition at line 270 of file cpu1_platform_cfg.h.

39.6.1.106 CFE_PLATFORM_SB_FILTER_MASK7

```
#define CFE_PLATFORM_SB_FILTER_MASK7 CFE_EVS_NO_FILTER
```

Definition at line 273 of file cpu1_platform_cfg.h.

39.6.1.107 CFE_PLATFORM_SB_FILTER_MASK8

```
#define CFE_PLATFORM_SB_FILTER_MASK8 CFE_EVS_NO_FILTER
```

Definition at line 276 of file cpu1_platform_cfg.h.

39.6.1.108 CFE_PLATFORM_SB_FILTERED_EVENT1

```
#define CFE_PLATFORM_SB_FILTERED_EVENT1 CFE_SB_SEND_NO_SUBS_EID
```

Purpose SB Event Filtering

Description:

This group of configuration paramters dictates what SB events will be filtered through EVS. The filtering will begin after the SB task initializes and stay in effect until a cmd to EVS changes it. This allows the operator to set limits on the number of event messages that are sent during system initialization. NOTE: Set all unused event values and mask values to zero

Limits

This filtering applies only to SB events. These parameters have a lower limit of 0 and an upper limit of 65535.

Definition at line 254 of file cpu1_platform_cfg.h.

39.6.1.109 CFE_PLATFORM_SB_FILTERED_EVENT2

```
#define CFE_PLATFORM_SB_FILTERED_EVENT2 CFE_SB_DUP_SUBSCRIP_EID
```

Definition at line 257 of file cpu1_platform_cfg.h.

39.6.1.110 CFE_PLATFORM_SB_FILTERED_EVENT3

```
#define CFE_PLATFORM_SB_FILTERED_EVENT3 CFE_SB_MSGID_LIM_ERR_EID
```

Definition at line 260 of file cpu1_platform_cfg.h.

39.6.1.111 CFE_PLATFORM_SB_FILTERED_EVENT4

```
#define CFE_PLATFORM_SB_FILTERED_EVENT4 CFE_SB_Q_FULL_ERR_EID
```

Definition at line 263 of file cpu1_platform_cfg.h.

39.6.1.112 CFE_PLATFORM_SB_FILTERED_EVENT5

```
#define CFE_PLATFORM_SB_FILTERED_EVENT5 0
```

Definition at line 266 of file cpu1_platform_cfg.h.

39.6.1.113 CFE_PLATFORM_SB_FILTERED_EVENT6

```
#define CFE_PLATFORM_SB_FILTERED_EVENT6 0
```

Definition at line 269 of file cpu1_platform_cfg.h.

39.6.1.114 CFE_PLATFORM_SB_FILTERED_EVENT7

```
#define CFE_PLATFORM_SB_FILTERED_EVENT7 0
```

Definition at line 272 of file cpu1_platform_cfg.h.

39.6.1.115 CFE_PLATFORM_SB_FILTERED_EVENT8

```
#define CFE_PLATFORM_SB_FILTERED_EVENT8 0
```

Definition at line 275 of file cpu1_platform_cfg.h.

39.6.1.116 CFE_PLATFORM_SB_HIGHEST_VALID_MSGID

```
#define CFE_PLATFORM_SB_HIGHEST_VALID_MSGID 0x1FFF
```

Purpose Highest Valid Message Id

Description:

The value of this constant dictates the range of valid message ID's, from 0 to CFE_PLATFORM_SB_HIGHEST_↩ VALID_MSGID (inclusive).

Altough this can be defined differently across platforms, each platform can only publish/subscribe to message ids within their allowable range. Typically this value is set the same across all mission platforms to avoid this complexity.

Limits

CFE_SB_INVALID_MSG is set to the maxumum representable number of type CFE_SB_MsgId_t. CFE_PLATF↩ ORM_SB_HIGHEST_VALID_MSGID lower limit is 1, up to CFE_SB_INVALID_MSG_ID - 1.

When using the direct message map implementation for software bus routing, this value is used to size the map where a value of 0x1FFF results in a 16 KBytes map and 0xFFFF is 128 KBytes.

When using the hash implementation for software bus routing, a multiple of the CFE_PLATFORM_SB_MAX_MSG_IDS is used to size the message map. In that case the range selected here does not impact message map memory use, so it's resonable to use up to the full range supported by the message ID implementation.

Definition at line 181 of file cpu1_platform_cfg.h.

39.6.1.117 CFE_PLATFORM_SB_MAX_BLOCK_SIZE

```
#define CFE_PLATFORM_SB_MAX_BLOCK_SIZE (CFE_MISSION_SB_MAX_SB_MSG_SIZE + 40)
```

Definition at line 305 of file cpu1_platform_cfg.h.

39.6.1.118 CFE_PLATFORM_SB_MAX_DEST_PER_PKT

```
#define CFE_PLATFORM_SB_MAX_DEST_PER_PKT 16
```

Purpose Maximum Number of unique local destinations a single MsgId can have

Description:

Dictates the maximum number of unique local destinations a single MsgId can have.

Limits

This parameter has a lower limit of 1. There are no restrictions on the upper limit however, the maximum number of destinations per packet is system dependent and should be verified. Destination number values that are checked against this configuration are defined by a 16 bit data word.

Definition at line 118 of file cpu1_platform_cfg.h.

39.6.1.119 CFE_PLATFORM_SB_MAX_MSG_IDS

```
#define CFE_PLATFORM_SB_MAX_MSG_IDS 256
```

Purpose Maximum Number of Unique Message IDs SB Routing Table can hold

Description:

Dictates the maximum number of unique MsgIds the SB routing table will hold. This constant has a direct affect on the size of SB's tables and arrays. Keeping this count as low as possible will save memory. To see the run-time, high-water mark and the current utilization figures regarding this parameter, send an SB command to 'Send Statistics Pkt'.

Limits

This must be a power of two if software bus message routing hash implementation is being used. Lower than 64 will cause unit test failures, and telemetry reporting is impacted below 32. There is no hard upper limit, but impacts memory footprint. For software bus message routing search implementation the number of msg ids subscribed to impacts performance.

Definition at line 85 of file cpu1_platform_cfg.h.

39.6.1.120 CFE_PLATFORM_SB_MAX_PIPES

```
#define CFE_PLATFORM_SB_MAX_PIPES 64
```

Purpose Maximum Number of Unique Pipes SB Routing Table can hold

Description:

Dictates the maximum number of unique Pipes the SB routing table will hold. This constant has a direct affect on the size of SB's tables and arrays. Keeping this count as low as possible will save memory. To see the run-time, high-water mark and the current utilization figures regarding this parameter, send an SB command to 'Send Statistics Pkt'.

Limits

This parameter has a lower limit of 1. This parameter must also be less than or equal to OS_MAX_QUEUES.

Definition at line 102 of file cpu1_platform_cfg.h.

39.6.1.121 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01 8
```

Purpose Define SB Memory Pool Block Sizes

Description:

Software Bus Memory Pool Block Sizes

Limits

These sizes MUST be increasing and MUST be an integral multiple of 4. The number of block sizes defined cannot exceed CFE_PLATFORM_ES_POOL_MAX_BUCKETS

Definition at line 289 of file cpu1_platform_cfg.h.

39.6.1.122 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02 16
```

Definition at line 290 of file cpu1_platform_cfg.h.

39.6.1.123 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03 20
```

Definition at line 291 of file cpu1_platform_cfg.h.

39.6.1.124 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04 36
```

Definition at line 292 of file cpu1_platform_cfg.h.

39.6.1.125 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05 64
```

Definition at line 293 of file cpu1_platform_cfg.h.

39.6.1.126 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06 96
```

Definition at line 294 of file cpu1_platform_cfg.h.

39.6.1.127 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07 128
```

Definition at line 295 of file cpu1_platform_cfg.h.

39.6.1.128 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08 160
```

Definition at line 296 of file cpu1_platform_cfg.h.

39.6.1.129 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09 256
```

Definition at line 297 of file cpu1_platform_cfg.h.

39.6.1.130 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10 512
```

Definition at line 298 of file cpu1_platform_cfg.h.

39.6.1.131 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11 1024
```

Definition at line 299 of file cpu1_platform_cfg.h.

39.6.1.132 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12 2048
```

Definition at line 300 of file cpu1_platform_cfg.h.

39.6.1.133 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13 4096
```

Definition at line 301 of file cpu1_platform_cfg.h.

39.6.1.134 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14 8192
```

Definition at line 302 of file cpu1_platform_cfg.h.

39.6.1.135 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15 16384
```

Definition at line 303 of file cpu1_platform_cfg.h.

39.6.1.136 CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16 32768
```

Definition at line 304 of file cpu1_platform_cfg.h.

39.6.1.137 CFE_PLATFORM_SB_START_TASK_PRIORITY

```
#define CFE_PLATFORM_SB_START_TASK_PRIORITY 64
```

Purpose Define SB Task Priority

Description:

Defines the cFE_SB Task priority.

Limits

Not Applicable

Definition at line 1101 of file cpu1_platform_cfg.h.

39.6.1.138 CFE_PLATFORM_SB_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_SB_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
```

Purpose Define SB Task Stack Size

Description:

Defines the cFE_SB Task Stack Size

Limits

There is a lower limit of 2048 on this configuration paramater. There are no restrictions on the upper limit however, the maximum stack size size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 1116 of file cpu1_platform_cfg.h.

39.6.1.139 CFE_PLATFORM_TBL_BUF_MEMORY_BYTES

```
#define CFE_PLATFORM_TBL_BUF_MEMORY_BYTES 524288
```

Purpose Size of Table Services Table Memory Pool

Description:

Defines the TOTAL size of the memory pool that cFE Table Services allocates from the system. The size must be large enough to provide memory for each registered table, the inactive buffers for double buffered tables and for the shared inactive buffers for single buffered tables.

Limits

The cFE does not place a limit on the size of this parameter.

Definition at line 1457 of file cpu1_platform_cfg.h.

39.6.1.140 CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE

```
#define CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE "/ram/cfe_tbl_reg.log"
```

Purpose Default Filename for a Table Registry Dump

Description:

Defines the file name used to store the table registry when no filename is specified in the dump registry command.

Limits

The length of each string, including the NULL terminator cannot exceed the OS_MAX_PATH_LEN value.

Definition at line 1571 of file cpu1_platform_cfg.h.

39.6.1.141 CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES

```
#define CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES 32
```

Purpose Maximum Number of Critical Tables that can be Registered

Description:

Defines the maximum number of critical tables supported by this processor's Table Services.

Limits

This number must be less than 32767. It should be recognized that this parameter determines the size of the Critical Table Registry which is maintained in the Critical Data Store. An excessively high number will waste Critical Data Store memory. Therefore, this number must not exceed the value defined in CFE_ES_CDS_MAX_CRITICAL_T↩ ABLES.

Definition at line 1512 of file cpu1_platform_cfg.h.

39.6.1.142 CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE

```
#define CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE 16384
```

Purpose Maximum Size Allowed for a Double Buffered Table

Description:

Defines the maximum allowed size (in bytes) of a double buffered table.

Limits

The cFE does not place a limit on the size of this parameter but it must be less than half of CFE_PLATFORM_TBL_BUF_MEMORY_BYTES.

Definition at line 1469 of file cpu1_platform_cfg.h.

39.6.1.143 CFE_PLATFORM_TBL_MAX_NUM_HANDLES

```
#define CFE_PLATFORM_TBL_MAX_NUM_HANDLES 256
```

Purpose Maximum Number of Table Handles

Description:

Defines the maximum number of Table Handles.

Limits

This number must be less than 32767. This number must be at least as big as the number of tables (CFE_PLATFORM_TBL_MAX_NUM_TABLES) and should be set higher if tables are shared between applications.

Definition at line 1525 of file cpu1_platform_cfg.h.

39.6.1.144 CFE_PLATFORM_TBL_MAX_NUM_TABLES

```
#define CFE_PLATFORM_TBL_MAX_NUM_TABLES 128
```

Purpose Maximum Number of Tables Allowed to be Registered

Description:

Defines the maximum number of tables supported by this processor's Table Services.

Limits

This number must be less than 32767. It should be recognized that this parameter determines the size of the Table Registry. An excessively high number will waste memory.

Definition at line 1498 of file cpu1_platform_cfg.h.

39.6.1.145 CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS

```
#define CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS 10
```

Purpose Maximum Number of Simultaneous Table Validations

Description:

Defines the maximum number of pending validations that the Table Services can handle at any one time. When a table has a validation function, a validation request is made of the application to perform that validation. This number determines how many of those requests can be outstanding at any one time.

Limits

This number must be less than 32767. An excessively high number will degrade system performance and waste memory. A number less than 20 is suggested but not required.

Definition at line 1558 of file cpu1_platform_cfg.h.

39.6.1.146 CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS

```
#define CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS 4
```

Purpose Maximum Number of Simultaneous Loads to Support

Description:

Defines the maximum number of single buffered tables that can be loaded simultaneously. This number is used to determine the number of shared buffers to allocate.

Limits

This number must be less than 32767. An excessively high number will degrade system performance and waste memory. A number less than 5 is suggested but not required.

Definition at line 1540 of file cpu1_platform_cfg.h.

39.6.1.147 CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE

```
#define CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE 16384
```

Purpose Maximum Size Allowed for a Single Buffered Table

Description:

Defines the maximum allowed size (in bytes) of a single buffered table. **NOTE:** This size determines the size of all shared table buffers. Therefore, this size will be multiplied by CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_L↩OADS below when allocating memory for shared tables.

Limits

The cFE does not place a limit on the size of this parameter but it must be small enough to allow for CFE_PLATF↩ORM_TBL_MAX_SIMULTANEOUS_LOADS number of tables to fit into CFE_PLATFORM_TBL_BUF_MEMOR↩Y_BYTES.

Definition at line 1485 of file cpu1_platform_cfg.h.

39.6.1.148 CFE_PLATFORM_TBL_START_TASK_PRIORITY

```
#define CFE_PLATFORM_TBL_START_TASK_PRIORITY 70
```

Purpose Define TBL Task Priority

Description:

Defines the cFE_TBL Task priority.

Limits

Not Applicable

Definition at line 1189 of file cpu1_platform_cfg.h.

39.6.1.149 CFE_PLATFORM_TBL_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
```

Purpose Define TBL Task Stack Size

Description:

Defines the cFE_TBL Task Stack Size

Limits

There is a lower limit of 2048 on this configuration paramater. There are no restrictions on the upper limit however, the maximum stack size size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 1204 of file cpu1_platform_cfg.h.

39.6.1.150 CFE_PLATFORM_TBL_U32FROM4CHARS

```
#define CFE_PLATFORM_TBL_U32FROM4CHARS(
              _C1,
              _C2,
              _C3,
              _C4 ) ((uint32)(_C1) << 24 | (uint32)(_C2) << 16 | (uint32)(_C3) << 8 | (uint32)(↩
_C4))
```

Definition at line 1593 of file cpu1_platform_cfg.h.

39.6.1.151 CFE_PLATFORM_TBL_VALID_PRID_1

```
#define CFE_PLATFORM_TBL_VALID_PRID_1 (1)
```

Purpose Processor ID values used for table load validation

Description:

Defines the processor ID values used for validating the processor ID field in the table file header. To be valid, the spacecraft ID specified in the table file header must match one of the values defined here.

Limits

This value can be any 32 bit unsigned integer.

Definition at line 1642 of file cpu1_platform_cfg.h.

39.6.1.152 CFE_PLATFORM_TBL_VALID_PRID_2

```
#define CFE_PLATFORM_TBL_VALID_PRID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
```

Definition at line 1643 of file cpu1_platform_cfg.h.

39.6.1.153 CFE_PLATFORM_TBL_VALID_PRID_3

```
#define CFE_PLATFORM_TBL_VALID_PRID_3 0
```

Definition at line 1644 of file cpu1_platform_cfg.h.

39.6.1.154 CFE_PLATFORM_TBL_VALID_PRID_4

```
#define CFE_PLATFORM_TBL_VALID_PRID_4 0
```

Definition at line 1645 of file cpu1_platform_cfg.h.

39.6.1.155 CFE_PLATFORM_TBL_VALID_PRID_COUNT

```
#define CFE_PLATFORM_TBL_VALID_PRID_COUNT 0
```

Purpose Number of Processor ID's specified for validation

Description:

Defines the number of specified processor ID values that are verified during table loads. If the number is zero then no validation of the processor ID field in the table file header is performed when tables are loaded. Non-zero values indicate how many values from the list of processor ID's defined below are compared to the processor ID field in the table file header. The ELF2CFETBL tool may be used to create table files with specified processor ID values.

Limits

This number must be greater than or equal to zero and less than or equal to 4.

Definition at line 1628 of file cpu1_platform_cfg.h.

39.6.1.156 CFE_PLATFORM_TBL_VALID_SCID_1

```
#define CFE_PLATFORM_TBL_VALID_SCID_1 (0x42)
```

Purpose Spacecraft ID values used for table load validation

Description:

Defines the spacecraft ID values used for validating the spacecraft ID field in the table file header. To be valid, the spacecraft ID specified in the table file header must match one of the values defined here.

Limits

This value can be any 32 bit unsigned integer.

Definition at line 1608 of file cpu1_platform_cfg.h.

39.6.1.157 CFE_PLATFORM_TBL_VALID_SCID_2

```
#define CFE_PLATFORM_TBL_VALID_SCID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
```

Definition at line 1609 of file cpu1_platform_cfg.h.

39.6.1.158 CFE_PLATFORM_TBL_VALID_SCID_COUNT

```
#define CFE_PLATFORM_TBL_VALID_SCID_COUNT 0
```

Purpose Number of Spacecraft ID's specified for validation

Description:

Defines the number of specified spacecraft ID values that are verified during table loads. If the number is zero then no validation of the spacecraft ID field in the table file header is performed when tables are loaded. Non-zero values indicate how many values from the list of spacecraft ID's defined below are compared to the spacecraft ID field in the table file header. The ELF2CFETBL tool may be used to create table files with specified spacecraft ID values.

Limits

This number must be greater than or equal to zero and less than or equal to 2.

Definition at line 1590 of file cpu1_platform_cfg.h.

39.6.1.159 CFE_PLATFORM_TIME_1HZ_TASK_PRIORITY

```
#define CFE_PLATFORM_TIME_1HZ_TASK_PRIORITY 25
```

Definition at line 1159 of file cpu1_platform_cfg.h.

39.6.1.160 CFE_PLATFORM_TIME_1HZ_TASK_STACK_SIZE

```
#define CFE_PLATFORM_TIME_1HZ_TASK_STACK_SIZE 8192
```

Definition at line 1178 of file cpu1_platform_cfg.h.

39.6.1.161 CFE_PLATFORM_TIME_CFG_CLIENT

```
#define CFE_PLATFORM_TIME_CFG_CLIENT false
```

Definition at line 320 of file cpu1_platform_cfg.h.

39.6.1.162 CFE_PLATFORM_TIME_CFG_LATCH_FLY

```
#define CFE_PLATFORM_TIME_CFG_LATCH_FLY 8
```

Purpose Define Periodic Time to Update Local Clock Tone Latch

Description:

Define Periodic Time to Update Local Clock Tone Latch. Applies only when in flywheel mode. This define dicates the period at which the simulated 'last tone' time is updated. Units are seconds.

Limits

Not Applicable

Definition at line 477 of file cpu1_platform_cfg.h.

39.6.1.163 CFE_PLATFORM_TIME_CFG_SERVER

```
#define CFE_PLATFORM_TIME_CFG_SERVER true
```

Purpose Time Server or Time Client Selection

Description:

This configuration parameter selects whether the Time task functions as a time "server" or "client". A time server generates the "time at the tone" packet which is received by time clients.

Limits

Enable one, and only one by defining either CFE_PLATFORM_TIME_CFG_SERVER or CFE_PLATFORM_TIM↩E_CFG_CLIENT AS true. The other must be defined as false.

Definition at line 319 of file cpu1_platform_cfg.h.

39.6.1.164 CFE_PLATFORM_TIME_CFG_SIGNAL

```
#define CFE_PLATFORM_TIME_CFG_SIGNAL false
```

Purpose Include or Exclude the Primary/Redundant Tone Selection Cmd

Description:

Depending on the specific hardware system configuration, it may be possible to switch between a primary and redundant tone signal. If supported by hardware, this definitions will enable command interfaces to select the active tone signal. Both Time Clients and Time Servers support this feature. Note: Set the CFE_PLATFORM_TIME_C↩FG_SIGNAL define to true to enable tone signal commands.

Limits

Not Applicable

Definition at line 367 of file cpu1_platform_cfg.h.

39.6.1.165 CFE_PLATFORM_TIME_CFG_SOURCE

```
#define CFE_PLATFORM_TIME_CFG_SOURCE false
```

Purpose Include or Exclude the Internal/External Time Source Selection Cmd

Description:

By default, Time Servers maintain time using an internal MET which may be a h/w register or software counter, depending on available hardware. The following definition enables command interfaces to switch between an internal MET, or external time data received from one of several supported external time sources. Only a Time Server may be configured to use external time data. Note: Set the CFE_PLATFORM_TIME_CFG_SOURCE define to true to include the Time Source Selection Command (command allows selection between the internal or external time source). Then choose the external source with the CFE_TIME_CFG_SRC_??? define.

Limits

Only applies if CFE_PLATFORM_TIME_CFG_SERVER is set to true.

Definition at line 387 of file cpu1_platform_cfg.h.

39.6.1.166 CFE_PLATFORM_TIME_CFG_SRC_GPS

```
#define CFE_PLATFORM_TIME_CFG_SRC_GPS false
```

Definition at line 404 of file cpu1_platform_cfg.h.

39.6.1.167 CFE_PLATFORM_TIME_CFG_SRC_MET

```
#define CFE_PLATFORM_TIME_CFG_SRC_MET false
```

Purpose Choose the External Time Source for Server only

Description:

If CFE_PLATFORM_TIME_CFG_SOURCE is set to true, then one of the following external time source types must also be set to true. Do not set any of the external time source types to true unless CFE_PLATFORM_TIME_CF↩G_SOURCE is set to true.

Limits

1. If CFE_PLATFORM_TIME_CFG_SOURCE is set to true then one and only one of the following three external time sources can and must be set true: CFE_PLATFORM_TIME_CFG_SRC_MET, CFE_PLATFORM_TI↩ME_CFG_SRC_GPS, CFE_PLATFORM_TIME_CFG_SRC_TIME
2. Only applies if CFE_PLATFORM_TIME_CFG_SERVER is set to true.

Definition at line 403 of file cpu1_platform_cfg.h.

39.6.1.168 CFE_PLATFORM_TIME_CFG_SRC_TIME

```
#define CFE_PLATFORM_TIME_CFG_SRC_TIME false
```

Definition at line 405 of file cpu1_platform_cfg.h.

39.6.1.169 CFE_PLATFORM_TIME_CFG_START_FLY

```
#define CFE_PLATFORM_TIME_CFG_START_FLY 2
```

Purpose Define Time to Start Flywheel Since Last Tone

Description:

Define time to enter flywheel mode (in seconds since last tone data update) Units are microseconds as measured with the local clock.

Limits

Not Applicable

Definition at line 464 of file cpu1_platform_cfg.h.

39.6.1.170 CFE_PLATFORM_TIME_CFG_TONE_LIMIT

```
#define CFE_PLATFORM_TIME_CFG_TONE_LIMIT 20000
```

Purpose Define Timing Limits From One Tone To The Next

Description:

Defines limits to the timing of the 1Hz tone signal. A tone signal is valid only if it arrives within one second (plus or minus the tone limit) from the previous tone signal.Units are microseconds as measured with the local clock.

Limits

Not Applicable

Definition at line 452 of file cpu1_platform_cfg.h.

39.6.1.171 CFE_PLATFORM_TIME_CFG_VIRTUAL

```
#define CFE_PLATFORM_TIME_CFG_VIRTUAL true
```

Purpose Time Tone In Big-Endian Order

Description:

If this configuration parameter is defined, the CFE time server will publish time tones with payloads in big-endian order, and time clients will expect the tones to be in big-endian order. This is useful for mixed-endian environments. This will become obsolete once EDS is available and the CFE time tone message is defined.

Purpose Local MET or Virtual MET Selection for Time Servers

Description:

Depending on the specific hardware system configuration, it may be possible for Time Servers to read the "local" MET from a h/w register rather than having to track the MET as the count of tone signal interrupts (virtual MET)

Time Clients must be defined as using a virtual MET. Also, a Time Server cannot be defined as having both a h/w MET and an external time source (they both cannot synchronize to the same tone).

Note: "disable" this define (set to false) only for Time Servers with local hardware that supports a h/w MET that is synchronized to the tone signal !!!

Limits

Only applies if CFE_PLATFORM_TIME_CFG_SERVER is set to true.

Definition at line 352 of file cpu1_platform_cfg.h.

39.6.1.172 CFE_PLATFORM_TIME_MAX_DELTA_SECS

```
#define CFE_PLATFORM_TIME_MAX_DELTA_SECS 0
```

Purpose Define the Max Delta Limits for Time Servers using an Ext Time Source

Description:

If CFE_PLATFORM_TIME_CFG_SOURCE is set to true and one of the external time sources is also set to true, then the delta time limits for range checking is used.

When a new time value is received from an external source, the value is compared against the "expected" time value. If the delta exceeds the following defined amount, then the new time data will be ignored. This range checking is only performed after the clock state has been commanded to "valid". Until then, external time data is accepted unconditionally.

Limits

Applies only if both CFE_PLATFORM_TIME_CFG_SERVER and CFE_PLATFORM_TIME_CFG_SOURCE are set to true.

Definition at line 424 of file cpu1_platform_cfg.h.

39.6.1.173 CFE_PLATFORM_TIME_MAX_DELTA_SUBS

```
#define CFE_PLATFORM_TIME_MAX_DELTA_SUBS 500000
```

Definition at line 425 of file cpu1_platform_cfg.h.

39.6.1.174 CFE_PLATFORM_TIME_MAX_LOCAL_SECS

```
#define CFE_PLATFORM_TIME_MAX_LOCAL_SECS 27
```

Purpose Define the Local Clock Rollover Value in seconds and subseconds

Description:

Specifies the capability of the local clock. Indicates the time at which the local clock rolls over.

Limits

Not Applicable

Definition at line 437 of file cpu1_platform_cfg.h.

39.6.1.175 CFE_PLATFORM_TIME_MAX_LOCAL_SUBS

```
#define CFE_PLATFORM_TIME_MAX_LOCAL_SUBS 0
```

Definition at line 438 of file cpu1_platform_cfg.h.

39.6.1.176 CFE_PLATFORM_TIME_START_TASK_PRIORITY

```
#define CFE_PLATFORM_TIME_START_TASK_PRIORITY 60
```

Purpose Define TIME Task Priorities

Description:

Defines the cFE_TIME Task priority. Defines the cFE_TIME Tone Task priority. Defines the cFE_TIME 1HZ Task priority.

Limits

There is a lower limit of zero and an upper limit of 255 on these configuration paramaters. Remember that the meaning of each task priority is inverted – a "lower" number has a "higher" priority.

Definition at line 1157 of file cpu1_platform_cfg.h.

39.6.1.177 CFE_PLATFORM_TIME_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_TIME_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
```

Purpose Define TIME Task Stack Sizes

Description:

Defines the cFE_TIME Main Task Stack Size Defines the cFE_TIME Tone Task Stack Size Defines the cFE_TIME 1HZ Task Stack Size

Limits

There is a lower limit of 2048 on these configuration paramaters. There are no restrictions on the upper limit however, the maximum stack size size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 1176 of file cpu1_platform_cfg.h.

39.6.1.178 CFE_PLATFORM_TIME_TONE_TASK_PRIORITY

```
#define CFE_PLATFORM_TIME_TONE_TASK_PRIORITY 25
```

Definition at line 1158 of file cpu1_platform_cfg.h.

39.6.1.179 CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE

```
#define CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE 4096
```

Definition at line 1177 of file cpu1_platform_cfg.h.

39.7 sample_mission_cfg.h File Reference

Macros

- #define CFE_MISSION_SB_MAX_SB_MSG_SIZE 32768
- #define CFE_MISSION_TIME_CFG_DEFAULT_TAI true
- #define CFE_MISSION_TIME_CFG_DEFAULT_UTC false
- #define CFE_MISSION_TIME_CFG_FAKE_TONE true
- #define CFE_MISSION_TIME_AT_TONE_WAS true
- #define CFE_MISSION_TIME_AT_TONE_WILL_BE false
- #define CFE_MISSION_TIME_MIN_ELAPSED 0
- #define CFE_MISSION_TIME_MAX_ELAPSED 200000
- #define CFE_MISSION_TIME_DEF_MET_SECS 1000

- #define CFE_MISSION_TIME_DEF_MET_SUBS 0
- #define CFE_MISSION_TIME_DEF_STCF_SECS 1000000
- #define CFE_MISSION_TIME_DEF_STCF_SUBS 0
- #define CFE_MISSION_TIME_DEF_LEAPS 37
- #define CFE_MISSION_TIME_DEF_DELAY_SECS 0
- #define CFE_MISSION_TIME_DEF_DELAY_SUBS 1000
- #define CFE_MISSION_TIME_EPOCH_YEAR 1980
- #define CFE_MISSION_TIME_EPOCH_DAY 1
- #define CFE_MISSION_TIME_EPOCH_HOUR 0
- #define CFE_MISSION_TIME_EPOCH_MINUTE 0
- #define CFE_MISSION_TIME_EPOCH_SECOND 0
- #define CFE_MISSION_TIME_FS_FACTOR 789004800
- #define CFE_MISSION_ES_CDS_MAX_NAME_LENGTH 16
- #define CFE_MISSION_EVS_MAX_MESSAGE_LENGTH 122
- #define CFE_MISSION_ES_DEFAULT_CRC CFE_MISSION_ES_CRC_16
- #define CFE_MISSION_TBL_MAX_NAME_LENGTH 16
- #define CFE_MISSION_EVS_CMD_MSG 1
- #define CFE_MISSION_SB_CMD_MSG 3
- #define CFE_MISSION_TBL_CMD_MSG 4
- #define CFE_MISSION_TIME_CMD_MSG 5
- #define CFE_MISSION_ES_CMD_MSG 6
- #define CFE_MISSION_ES_SEND_HK_MSG 8
- #define CFE_MISSION_EVS_SEND_HK_MSG 9
- #define CFE_MISSION_SB_SEND_HK_MSG 11
- #define CFE_MISSION_TBL_SEND_HK_MSG 12
- #define CFE_MISSION_TIME_SEND_HK_MSG 13
- #define CFE_MISSION_SB_SUB_RPT_CTRL_MSG 14
- #define CFE_MISSION_TIME_TONE_CMD_MSG 16
- #define CFE_MISSION_TIME_1HZ_CMD_MSG 17
- #define CFE_MISSION_TIME_DATA_CMD_MSG 0
- #define CFE_MISSION_TIME_SEND_CMD_MSG 2
- #define CFE_MISSION_ES_HK_TLM_MSG 0
- #define CFE_MISSION_EVS_HK_TLM_MSG 1
- #define CFE_MISSION_SB_HK_TLM_MSG 3
- #define CFE_MISSION_TBL_HK_TLM_MSG 4
- #define CFE_MISSION_TIME_HK_TLM_MSG 5
- #define CFE_MISSION_TIME_DIAG_TLM_MSG 6
- #define CFE_MISSION_EVS_LONG_EVENT_MSG_MSG 8
- #define CFE_MISSION_EVS_SHORT_EVENT_MSG_MSG 9
- #define CFE_MISSION_SB_STATS_TLM_MSG 10
- #define CFE_MISSION_ES_APP_TLM_MSG 11
- #define CFE_MISSION_TBL_REG_TLM_MSG 12
- #define CFE_MISSION_SB_ALLSUBS_TLM_MSG 13
- #define CFE_MISSION_SB_ONESUB_TLM_MSG 14
- #define CFE_MISSION_ES_MEMSTATS_TLM_MSG 16
- #define CFE_MISSION_ES_MAX_APPLICATIONS 16
- #define CFE_MISSION_ES_PERF_MAX_IDS 128
- #define CFE_MISSION_ES_POOL_MAX_BUCKETS 17
- #define CFE_MISSION_TBL_MAX_FULL_NAME_LEN (CFE_MISSION_TBL_MAX_NAME_LENGTH + CFE_↩ MISSION_MAX_API_LEN + 4)
- #define CFE_MISSION_SB_MAX_PIPES 64

- #define CFE_MISSION_MAX_PATH_LEN 64
- #define CFE_MISSION_MAX_FILE_LEN 20
- #define CFE_MISSION_MAX_API_LEN 20
- #define CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN (CFE_MISSION_ES_CDS_MAX_NAME_LENGTH + CFE_MISSION_MAX_API_LEN + 4)

Checksum/CRC algorithm identifiers

- #define CFE_MISSION_ES_CRC_8 1
  CRC (8 bit additive - returns 32 bit total) (Currently not implemented)
- #define CFE_MISSION_ES_CRC_16 2
  CRC (16 bit additive - returns 32 bit total)
- #define CFE_MISSION_ES_CRC_32 3
  CRC (32 bit additive - returns 32 bit total) (Currently not implemented) .

39.7.1 Macro Definition Documentation

39.7.1.1 CFE_MISSION_ES_APP_TLM_MSG

```
#define CFE_MISSION_ES_APP_TLM_MSG 11
```

Definition at line 369 of file sample_mission_cfg.h.

39.7.1.2 CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN

```
#define CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN (CFE_MISSION_ES_CDS_MAX_NAME_LENGTH + CFE_MISSION_M↩
AX_API_LEN + 4)
```

Purpose Maximum Length of Full CDS Name in messages

Description:

Indicates the maximum length (in characters) of the entire CDS name of the following form: "ApplicationName.C↩ DSName"

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 564 of file sample_mission_cfg.h.

39.7.1.3 CFE_MISSION_ES_CDS_MAX_NAME_LENGTH

```
#define CFE_MISSION_ES_CDS_MAX_NAME_LENGTH 16
```

Purpose Maximum Length of CDS Name

Description:

Indicates the maximum length (in characters) of the CDS name ('CDSName') portion of a Full CDS Name of the following form: "ApplicationName.CDSName"

This length does not need to include an extra character for NULL termination.

Limits

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 248 of file sample_mission_cfg.h.

39.7.1.4 CFE_MISSION_ES_CMD_MSG

```
#define CFE_MISSION_ES_CMD_MSG 6
```

Definition at line 319 of file sample_mission_cfg.h.

39.7.1.5 CFE_MISSION_ES_CRC_16

```
#define CFE_MISSION_ES_CRC_16 2
```

CRC (16 bit additive - returns 32 bit total)

Definition at line 267 of file sample_mission_cfg.h.

39.7.1.6 CFE_MISSION_ES_CRC_32

```
#define CFE_MISSION_ES_CRC_32 3
```

CRC (32 bit additive - returns 32 bit total) (Currently not implemented) .

Definition at line 268 of file sample_mission_cfg.h.

39.7.1.7 CFE_MISSION_ES_CRC_8

```
#define CFE_MISSION_ES_CRC_8 1
```

CRC (8 bit additive - returns 32 bit total) (Currently not implemented)

Definition at line 266 of file sample_mission_cfg.h.

39.7.1.8 CFE_MISSION_ES_DEFAULT_CRC

```
#define CFE_MISSION_ES_DEFAULT_CRC CFE_MISSION_ES_CRC_16
```

Purpose Mission Default CRC algorithm

Description:

Indicates the which CRC algorithm should be used as the default for verifying the contents of Critical Data Stores and when calculating Table Image data integrity values.

Limits

Currently only CFE_MISSION_ES_CRC_16 is supported (see CFE_MISSION_ES_CRC_16)

Definition at line 285 of file sample_mission_cfg.h.

39.7.1.9 CFE_MISSION_ES_HK_TLM_MSG

```
#define CFE_MISSION_ES_HK_TLM_MSG 0
```

Purpose cFE Portable Message Numbers for Telemetry

Description:

Portable message numbers for the cFE telemetry messages NOTE: cFE MsgIds are the sum of the base numbers and the portable msg numbers.

Limits

Not Applicable

Definition at line 358 of file sample_mission_cfg.h.

39.7.1.10 CFE_MISSION_ES_MAX_APPLICATIONS

```
#define CFE_MISSION_ES_MAX_APPLICATIONS 16
```

Purpose Mission Max Apps in a message

Description:

Indicates the maximum number of apps in a telemetry housekeeping message

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

Definition at line 389 of file sample_mission_cfg.h.

39.7.1.11 CFE_MISSION_ES_MEMSTATS_TLM_MSG

```
#define CFE_MISSION_ES_MEMSTATS_TLM_MSG 16
```

Definition at line 373 of file sample_mission_cfg.h.

39.7.1.12 CFE_MISSION_ES_PERF_MAX_IDS

```
#define CFE_MISSION_ES_PERF_MAX_IDS 128
```

Purpose Define Max Number of Performance IDs for messages

Description:

Defines the maximum number of perf ids allowed in command/telemetry messages

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

Definition at line 406 of file sample_mission_cfg.h.

39.7.1.13 CFE_MISSION_ES_POOL_MAX_BUCKETS

```
#define CFE_MISSION_ES_POOL_MAX_BUCKETS 17
```

Purpose Maximum number of block sizes in pool structures

Description:

The upper limit for the number of block sizes supported in the generic pool implementation, which in turn implements the memory pools and CDS. This definition is used as the array size with the pool stats structure, and therefore should be consistent across all CPUs in a mission, as well as with the ground station.

There is also a platform-specific limit which may be fewer than this value.

Limits:

Must be at least one. No specific upper limit, but the number is anticipated to be reasonably small (i.e. tens, not hundreds). Large values have not been tested.

Definition at line 427 of file sample_mission_cfg.h.

39.7.1.14 CFE_MISSION_ES_SEND_HK_MSG

```
#define CFE_MISSION_ES_SEND_HK_MSG 8
```

Definition at line 321 of file sample_mission_cfg.h.

39.7.1.15 CFE_MISSION_EVS_CMD_MSG

```
#define CFE_MISSION_EVS_CMD_MSG 1
```

Purpose cFE Portable Message Numbers for Commands

Description:

Portable message numbers for the cFE command messages NOTE: cFE MsgIds are the sum of the base numbers and the portable msg numbers.

Limits

Not Applicable

Definition at line 314 of file sample_mission_cfg.h.

39.7.1.16 CFE_MISSION_EVS_HK_TLM_MSG

```
#define CFE_MISSION_EVS_HK_TLM_MSG 1
```

Definition at line 359 of file sample_mission_cfg.h.

39.7.1.17 CFE_MISSION_EVS_LONG_EVENT_MSG_MSG

```
#define CFE_MISSION_EVS_LONG_EVENT_MSG_MSG 8
```

Definition at line 366 of file sample_mission_cfg.h.

39.7.1.18 CFE_MISSION_EVS_MAX_MESSAGE_LENGTH

```
#define CFE_MISSION_EVS_MAX_MESSAGE_LENGTH 122
```

Purpose Maximum Event Message Length

Description:

Indicates the maximum length (in characters) of the formatted text string portion of an event message

This length does not need to include an extra character for NULL termination.

Limits

Not Applicable

Definition at line 262 of file sample_mission_cfg.h.

39.7.1.19 CFE_MISSION_EVS_SEND_HK_MSG

```
#define CFE_MISSION_EVS_SEND_HK_MSG 9
```

Definition at line 322 of file sample_mission_cfg.h.

39.7.1.20 CFE_MISSION_EVS_SHORT_EVENT_MSG_MSG

```
#define CFE_MISSION_EVS_SHORT_EVENT_MSG_MSG 9
```

Definition at line 367 of file sample_mission_cfg.h.

39.7.1.21 CFE_MISSION_MAX_API_LEN

```
#define CFE_MISSION_MAX_API_LEN 20
```

Purpose cFE Maximum length for API names within data exchange structures

Description:

The value of this constant dictates the size of filenames within all structures used for external data exchange, such as Software bus messages and table definitions. This is typically the same as OS_MAX_API_LEN but that is OSAL dependent – and as such it definable on a per-processor/OS basis and hence may be different across multiple processors. It is recommended to set this to the value of the largest OS_MAX_API_LEN in use on any CPU on the mission.

This affects only the layout of command/telemetry messages and table definitions; internal allocation may use the platform-specific OS_MAX_API_LEN value.

This length must include an extra character for NULL termination.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 544 of file sample_mission_cfg.h.

39.7.1.22 CFE_MISSION_MAX_FILE_LEN

```
#define CFE_MISSION_MAX_FILE_LEN 20
```

Purpose cFE Maximum length for filenames within data exchange structures

Description:

The value of this constant dictates the size of filenames within all structures used for external data exchange, such as Software bus messages and table definitions. This is typically the same as OS_MAX_FILE_LEN but that is O↩ SAL dependent – and as such it definable on a per-processor/OS basis and hence may be different across multiple processors. It is recommended to set this to the value of the largest OS_MAX_FILE_LEN in use on any CPU on the mission.

This affects only the layout of command/telemetry messages and table definitions; internal allocation may use the platform-specific OS_MAX_FILE_LEN value.

This length must include an extra character for NULL termination.

Limits

All CPUs within the same SB domain (mission) and ground tools must share the same definition. Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 518 of file sample_mission_cfg.h.

39.7.1.23 CFE_MISSION_MAX_PATH_LEN

```
#define CFE_MISSION_MAX_PATH_LEN 64
```

Purpose cFE Maximum length for pathnames within data exchange structures

Description:

The value of this constant dictates the size of pathnames within all structures used for external data exchange, such as Software bus messages and table definitions. This is typically the same as OS_MAX_PATH_LEN but that is OSAL dependent – and as such it definable on a per-processor/OS basis and hence may be different across multiple processors. It is recommended to set this to the value of the largest OS_MAX_PATH_LEN in use on any CPU on the mission.

This affects only the layout of command/telemetry messages and table definitions; internal allocation may use the platform-specific OS_MAX_PATH_LEN value.

This length must include an extra character for NULL termination.

Limits

All CPUs within the same SB domain (mission) and ground tools must share the same definition. Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 491 of file sample_mission_cfg.h.

39.7.1.24 CFE_MISSION_SB_ALLSUBS_TLM_MSG

```
#define CFE_MISSION_SB_ALLSUBS_TLM_MSG 13
```

Definition at line 371 of file sample_mission_cfg.h.

39.7.1.25 CFE_MISSION_SB_CMD_MSG

```
#define CFE_MISSION_SB_CMD_MSG 3
```

Definition at line 316 of file sample_mission_cfg.h.

39.7.1.26 CFE_MISSION_SB_HK_TLM_MSG

```
#define CFE_MISSION_SB_HK_TLM_MSG 3
```

Definition at line 361 of file sample_mission_cfg.h.

39.7.1.27 CFE_MISSION_SB_MAX_PIPES

```
#define CFE_MISSION_SB_MAX_PIPES 64
```

Purpose Maximum Number of pipes that SB command/telemetry messages may hold

Description:

Dictates the maximum number of unique Pipes the SB message defintions will hold.

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

Definition at line 464 of file sample_mission_cfg.h.

39.7.1.28 CFE_MISSION_SB_MAX_SB_MSG_SIZE

```
#define CFE_MISSION_SB_MAX_SB_MSG_SIZE 32768
```

Purpose Maximum SB Message Size

Description:

The following definition dictates the maximum message size allowed on the software bus. SB checks the pkt length field in the header of all messages sent. If the pkt length field indicates the message is larger than this define, SB sends an event and rejects the send.

Limits

This parameter has a lower limit of 6 (CCSDS primary header size). There are no restrictions on the upper limit however, the maximum message size is system dependent and should be verified. Total message size values that are checked against this configuration are defined by a 16 bit data word.

Definition at line 55 of file sample_mission_cfg.h.

39.7.1.29 CFE_MISSION_SB_ONESUB_TLM_MSG

```
#define CFE_MISSION_SB_ONESUB_TLM_MSG 14
```

Definition at line 372 of file sample_mission_cfg.h.

39.7.1.30 CFE_MISSION_SB_SEND_HK_MSG

```
#define CFE_MISSION_SB_SEND_HK_MSG 11
```

Definition at line 324 of file sample_mission_cfg.h.

39.7.1.31 CFE_MISSION_SB_STATS_TLM_MSG

```
#define CFE_MISSION_SB_STATS_TLM_MSG 10
```

Definition at line 368 of file sample_mission_cfg.h.

39.7.1.32 CFE_MISSION_SB_SUB_RPT_CTRL_MSG

```
#define CFE_MISSION_SB_SUB_RPT_CTRL_MSG 14
```

Definition at line 328 of file sample_mission_cfg.h.

39.7.1.33 CFE_MISSION_TBL_CMD_MSG

```
#define CFE_MISSION_TBL_CMD_MSG 4
```

Definition at line 317 of file sample_mission_cfg.h.

39.7.1.34 CFE_MISSION_TBL_HK_TLM_MSG

```
#define CFE_MISSION_TBL_HK_TLM_MSG 4
```

Definition at line 362 of file sample_mission_cfg.h.

39.7.1.35 CFE_MISSION_TBL_MAX_FULL_NAME_LEN

```
#define CFE_MISSION_TBL_MAX_FULL_NAME_LEN (CFE_MISSION_TBL_MAX_NAME_LENGTH + CFE_MISSION_MAX_API↩
_LEN + 4)
```

Purpose Maximum Length of Full Table Name in messages

Description:

Indicates the maximum length (in characters) of the entire table name within software bus messages, in "App↩ Name.TableName" notation.

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 447 of file sample_mission_cfg.h.

39.7.1.36 CFE_MISSION_TBL_MAX_NAME_LENGTH

```
#define CFE_MISSION_TBL_MAX_NAME_LENGTH 16
```

Purpose Maximum Table Name Length

Description:

Indicates the maximum length (in characers) of the table name ('TblName') portion of a Full Table Name of the following form: "ApplicationName.TblName"

This length does not need to include an extra character for NULL termination.

Limits

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 301 of file sample_mission_cfg.h.

39.7.1.37 CFE_MISSION_TBL_REG_TLM_MSG

```
#define CFE_MISSION_TBL_REG_TLM_MSG 12
```

Definition at line 370 of file sample_mission_cfg.h.

39.7.1.38 CFE_MISSION_TBL_SEND_HK_MSG

```
#define CFE_MISSION_TBL_SEND_HK_MSG 12
```

Definition at line 325 of file sample_mission_cfg.h.

39.7.1.39 CFE_MISSION_TIME_1HZ_CMD_MSG

```
#define CFE_MISSION_TIME_1HZ_CMD_MSG 17
```

Definition at line 331 of file sample_mission_cfg.h.

39.7.1.40 CFE_MISSION_TIME_AT_TONE_WAS

```
#define CFE_MISSION_TIME_AT_TONE_WAS true
```

Purpose Default Time and Tone Order

Description:

Time Services may be configured to expect the time at the tone data packet to either precede or follow the tone signal. If the time at the tone data packet follows the tone signal, then the data within the packet describes what the time "was" at the tone. If the time at the tone data packet precedes the tone signal, then the data within the packet describes what the time "will be" at the tone. One, and only one, of the following symbols must be set to true:

- CFE_MISSION_TIME_AT_TONE_WAS
- CFE_MISSION_TIME_AT_TONE_WILL_BE Note: If Time Services is defined as using a simulated tone signal (see CFE_MISSION_TIME_CFG_FAKE_TONE above), then the tone data packet must follow the tone signal.

Limits

Either CFE_MISSION_TIME_AT_TONE_WAS or CFE_MISSION_TIME_AT_TONE_WILL_BE must be set to true. They may not both be true and they may not both be false.

Definition at line 112 of file sample_mission_cfg.h.

39.7.1.41 CFE_MISSION_TIME_AT_TONE_WILL_BE

```
#define CFE_MISSION_TIME_AT_TONE_WILL_BE false
```

Definition at line 113 of file sample_mission_cfg.h.

39.7.1.42 CFE_MISSION_TIME_CFG_DEFAULT_TAI

```
#define CFE_MISSION_TIME_CFG_DEFAULT_TAI true
```

Purpose Default Time Format

Description:

The following definitions select either UTC or TAI as the default (mission specific) time format. Although it is possible for an application to request time in a specific format, most callers should use CFE_TIME_GetTime(), which returns time in the default format. This avoids having to modify each individual caller when the default choice is changed.

Limits

if CFE_MISSION_TIME_CFG_DEFAULT_TAI is defined as true then CFE_MISSION_TIME_CFG_DEFAULT_UTC must be defined as false. if CFE_MISSION_TIME_CFG_DEFAULT_TAI is defined as false then CFE_MISSION↩_TIME_CFG_DEFAULT_UTC must be defined as true.

Definition at line 75 of file sample_mission_cfg.h.

39.7.1.43 CFE_MISSION_TIME_CFG_DEFAULT_UTC

```
#define CFE_MISSION_TIME_CFG_DEFAULT_UTC false
```

Definition at line 76 of file sample_mission_cfg.h.

39.7.1.44 CFE_MISSION_TIME_CFG_FAKE_TONE

```
#define CFE_MISSION_TIME_CFG_FAKE_TONE true
```

Purpose Default Time Format

Description:

The following definition enables the use of a simulated time at the tone signal using a software bus message.

Limits

Not Applicable

Definition at line 89 of file sample_mission_cfg.h.

39.7.1.45 CFE_MISSION_TIME_CMD_MSG

```
#define CFE_MISSION_TIME_CMD_MSG 5
```

Definition at line 318 of file sample_mission_cfg.h.

39.7.1.46 CFE_MISSION_TIME_DATA_CMD_MSG

```
#define CFE_MISSION_TIME_DATA_CMD_MSG 0
```

Purpose cFE Portable Message Numbers for Global Messages

Description:

Portable message numbers for the cFE global messages NOTE: cFE MsgIds are the sum of the base numbers and the portable msg numbers.

Limits

Not Applicable

Definition at line 344 of file sample_mission_cfg.h.

39.7.1.47 CFE_MISSION_TIME_DEF_DELAY_SECS

```
#define CFE_MISSION_TIME_DEF_DELAY_SECS 0
```

Definition at line 172 of file sample_mission_cfg.h.

39.7.1.48 CFE_MISSION_TIME_DEF_DELAY_SUBS

```
#define CFE_MISSION_TIME_DEF_DELAY_SUBS 1000
```

Definition at line 173 of file sample_mission_cfg.h.

39.7.1.49 CFE_MISSION_TIME_DEF_LEAPS

```
#define CFE_MISSION_TIME_DEF_LEAPS 37
```

Definition at line 170 of file sample_mission_cfg.h.

39.7.1.50 CFE_MISSION_TIME_DEF_MET_SECS

```
#define CFE_MISSION_TIME_DEF_MET_SECS 1000
```

Purpose Default Time Values

Description:

Default time values are provided to avoid problems due to time calculations performed after startup but before commands can be processed. For example, if the default time format is UTC then it is important that the sum of MET and STCF always exceed the value of Leap Seconds to prevent the UTC time calculation `(time = MET + STCF - Leap Seconds)` from resulting in a negative (very large) number.

Some past missions have also created known (albeit wrong) default timestamps. For example, assume the epoch is defined as Jan 1, 1970 and further assume the default time values are set to create a timestamp of Jan 1, 2000. Even though the year 2000 timestamps are wrong, it may be of value to keep the time within some sort of bounds acceptable to the software.

Note: Sub-second units are in micro-seconds (0 to 999,999) and all values must be defined

Limits

Not Applicable

Definition at line 164 of file sample_mission_cfg.h.

39.7.1.51 CFE_MISSION_TIME_DEF_MET_SUBS

```
#define CFE_MISSION_TIME_DEF_MET_SUBS 0
```

Definition at line 165 of file sample_mission_cfg.h.

39.7.1.52 CFE_MISSION_TIME_DEF_STCF_SECS

```
#define CFE_MISSION_TIME_DEF_STCF_SECS 1000000
```

Definition at line 167 of file sample_mission_cfg.h.

39.7.1.53 CFE_MISSION_TIME_DEF_STCF_SUBS

```
#define CFE_MISSION_TIME_DEF_STCF_SUBS 0
```

Definition at line 168 of file sample_mission_cfg.h.

39.7.1.54 CFE_MISSION_TIME_DIAG_TLM_MSG

```
#define CFE_MISSION_TIME_DIAG_TLM_MSG 6
```

Definition at line 364 of file sample_mission_cfg.h.

39.7.1.55 CFE_MISSION_TIME_EPOCH_DAY

```
#define CFE_MISSION_TIME_EPOCH_DAY 1
```

Definition at line 190 of file sample_mission_cfg.h.

39.7.1.56 CFE_MISSION_TIME_EPOCH_HOUR

```
#define CFE_MISSION_TIME_EPOCH_HOUR 0
```

Definition at line 191 of file sample_mission_cfg.h.

39.7.1.57 CFE_MISSION_TIME_EPOCH_MINUTE

```
#define CFE_MISSION_TIME_EPOCH_MINUTE 0
```

Definition at line 192 of file sample_mission_cfg.h.

39.7.1.58 CFE_MISSION_TIME_EPOCH_SECOND

```
#define CFE_MISSION_TIME_EPOCH_SECOND 0
```

Definition at line 193 of file sample_mission_cfg.h.

39.7.1.59 CFE_MISSION_TIME_EPOCH_YEAR

```
#define CFE_MISSION_TIME_EPOCH_YEAR 1980
```

Purpose Default EPOCH Values

Description:

Default ground time epoch values Note: these values are used only by the CFE_TIME_Print() API function

Limits

Year - must be within 136 years Day - Jan 1 = 1, Feb 1 = 32, etc. Hour - 0 to 23 Minute - 0 to 59 Second - 0 to 59

Definition at line 189 of file sample_mission_cfg.h.

39.7.1.60 CFE_MISSION_TIME_FS_FACTOR

```
#define CFE_MISSION_TIME_FS_FACTOR 789004800
```

Purpose Time File System Factor

Description:

Define the s/c vs file system time conversion constant...

Note: this value is intended for use only by CFE TIME API functions to convert time values based on the ground system epoch (s/c time) to and from time values based on the file system epoch (fs time).

FS time = S/C time + factor S/C time = FS time - factor

Worksheet:

S/C epoch = Jan 1, 2005 (LRO ground system epoch) FS epoch = Jan 1, 1980 (vxWorks DOS file system epoch)

Delta = 25 years, 0 days, 0 hours, 0 minutes, 0 seconds

Leap years = 1980, 1984, 1988, 1992, 1996, 2000, 2004 (divisible by 4 – except if by 100 – unless also by 400)

1 year = 31,536,000 seconds 1 day = 86,400 seconds 1 hour = 3,600 seconds 1 minute = 60 seconds

25 years = 788,400,000 seconds 7 extra leap days = 604,800 seconds

total delta = 789,004,800 seconds

Limits

Not Applicable

Definition at line 231 of file sample_mission_cfg.h.

39.7.1.61 CFE_MISSION_TIME_HK_TLM_MSG

```
#define CFE_MISSION_TIME_HK_TLM_MSG 5
```

Definition at line 363 of file sample_mission_cfg.h.

39.7.1.62 CFE_MISSION_TIME_MAX_ELAPSED

```
#define CFE_MISSION_TIME_MAX_ELAPSED 200000
```

Definition at line 139 of file sample_mission_cfg.h.

39.7.1.63 CFE_MISSION_TIME_MIN_ELAPSED

```
#define CFE_MISSION_TIME_MIN_ELAPSED 0
```

Purpose Min and Max Time Elapsed

Description:

Based on the definition of Time and Tone Order (CFE_MISSION_TIME_AT_TONE_WAS/WILL_BE) either the "time at the tone" signal or data packet will follow the other. This definition sets the valid window of time for the second of the pair to lag behind the first. Time Services will invalidate both the tone and packet if the second does not arrive within this window following the first.

For example, if the data packet follows the tone, it might be valid for the data packet to arrive between zero and 100,000 micro-seconds after the tone. But, if the tone follows the the packet, it might be valid only if the packet arrived between 200,000 and 700,000 micro-seconds before the tone.

Note: units are in micro-seconds

Limits

0 to 999,999 decimal

Definition at line 138 of file sample_mission_cfg.h.

39.7.1.64 CFE_MISSION_TIME_SEND_CMD_MSG

```
#define CFE_MISSION_TIME_SEND_CMD_MSG 2
```

Definition at line 345 of file sample_mission_cfg.h.

39.7.1.65 CFE_MISSION_TIME_SEND_HK_MSG

```
#define CFE_MISSION_TIME_SEND_HK_MSG 13
```

Definition at line 326 of file sample_mission_cfg.h.

39.7.1.66 CFE_MISSION_TIME_TONE_CMD_MSG

```
#define CFE_MISSION_TIME_TONE_CMD_MSG 16
```

Definition at line 330 of file sample_mission_cfg.h.

39.8 sample_perfids.h File Reference

Macros

- #define CFE_MISSION_ES_PERF_EXIT_BIT 31

  bit (31) is reserved by the perf utilities

cFE Performance Monitor IDs (Reserved IDs 0-31)

- #define CFE_MISSION_ES_MAIN_PERF_ID 1

  Performance ID for Executive Services Task.
- #define CFE_MISSION_EVS_MAIN_PERF_ID 2

  Performance ID for Events Services Task.
- #define CFE_MISSION_TBL_MAIN_PERF_ID 3

  Performance ID for Table Services Task.
- #define CFE_MISSION_SB_MAIN_PERF_ID 4

  Performance ID for Software Bus Services Task.
- #define CFE_MISSION_SB_MSG_LIM_PERF_ID 5

  Performance ID for Software Bus Msg Limit Errors.
- #define CFE_MISSION_SB_PIPE_OFLOW_PERF_ID 27

  Performance ID for Software Bus Pipe Overflow Errors.
- #define CFE_MISSION_TIME_MAIN_PERF_ID 6

  Performance ID for Time Services Task.
- #define CFE_MISSION_TIME_TONE1HZISR_PERF_ID 7

  Performance ID for 1 Hz Tone ISR.
- #define CFE_MISSION_TIME_LOCAL1HZISR_PERF_ID 8

  Performance ID for 1 Hz Local ISR.
- #define CFE_MISSION_TIME_SENDMET_PERF_ID 9

  Performance ID for Time ToneSendMET.
- #define CFE_MISSION_TIME_LOCAL1HZTASK_PERF_ID 10

  Performance ID for 1 Hz Local Task.
- #define CFE_MISSION_TIME_TONE1HZTASK_PERF_ID 11

  Performance ID for 1 Hz Tone Task.

39.8.1 Macro Definition Documentation

39.8.1.1 CFE_MISSION_ES_MAIN_PERF_ID

```
#define CFE_MISSION_ES_MAIN_PERF_ID 1
```

Performance ID for Executive Services Task.

Definition at line 44 of file sample_perfids.h.

39.8.1.2 CFE_MISSION_ES_PERF_EXIT_BIT

```
#define CFE_MISSION_ES_PERF_EXIT_BIT 31
```

bit (31) is reserved by the perf utilities

Definition at line 40 of file sample_perfids.h.

39.8.1.3 CFE_MISSION_EVS_MAIN_PERF_ID

```
#define CFE_MISSION_EVS_MAIN_PERF_ID 2
```

Performance ID for Events Services Task.

Definition at line 45 of file sample_perfids.h.

39.8.1.4 CFE_MISSION_SB_MAIN_PERF_ID

```
#define CFE_MISSION_SB_MAIN_PERF_ID 4
```

Performance ID for Software Bus Services Task.

Definition at line 47 of file sample_perfids.h.

39.8.1.5 CFE_MISSION_SB_MSG_LIM_PERF_ID

```
#define CFE_MISSION_SB_MSG_LIM_PERF_ID 5
```

Performance ID for Software Bus Msg Limit Errors.

Definition at line 48 of file sample_perfids.h.

39.8.1.6 CFE_MISSION_SB_PIPE_OFLOW_PERF_ID

```
#define CFE_MISSION_SB_PIPE_OFLOW_PERF_ID 27
```

Performance ID for Software Bus Pipe Overflow Errors.

Definition at line 49 of file sample_perfids.h.

39.8.1.7 CFE_MISSION_TBL_MAIN_PERF_ID

```
#define CFE_MISSION_TBL_MAIN_PERF_ID 3
```

Performance ID for Table Services Task.

Definition at line 46 of file sample_perfids.h.

39.8.1.8 CFE_MISSION_TIME_LOCAL1HZISR_PERF_ID

```
#define CFE_MISSION_TIME_LOCAL1HZISR_PERF_ID 8
```

Performance ID for 1 Hz Local ISR.

Definition at line 53 of file sample_perfids.h.

39.8.1.9 CFE_MISSION_TIME_LOCAL1HZTASK_PERF_ID

```
#define CFE_MISSION_TIME_LOCAL1HZTASK_PERF_ID 10
```

Performance ID for 1 Hz Local Task.

Definition at line 56 of file sample_perfids.h.

39.8.1.10 CFE_MISSION_TIME_MAIN_PERF_ID

```
#define CFE_MISSION_TIME_MAIN_PERF_ID 6
```

Performance ID for Time Services Task.

Definition at line 51 of file sample_perfids.h.

39.8.1.11 CFE_MISSION_TIME_SENDMET_PERF_ID

```
#define CFE_MISSION_TIME_SENDMET_PERF_ID 9
```

Performance ID for Time ToneSendMET.

Definition at line 55 of file sample_perfids.h.

39.8.1.12 CFE_MISSION_TIME_TONE1HZISR_PERF_ID

```
#define CFE_MISSION_TIME_TONE1HZISR_PERF_ID 7
```

Performance ID for 1 Hz Tone ISR.

Definition at line 52 of file sample_perfids.h.

39.8.1.13 CFE_MISSION_TIME_TONE1HZTASK_PERF_ID

```
#define CFE_MISSION_TIME_TONE1HZTASK_PERF_ID 11
```

Performance ID for 1 Hz Tone Task.

Definition at line 57 of file sample_perfids.h.

39.9 cfe/docs/src/cfe_api.dox File Reference

39.10 cfe/docs/src/cfe_es.dox File Reference

39.11 cfe/docs/src/cfe_evs.dox File Reference

39.12 cfe/docs/src/cfe_glossary.dox File Reference

39.13 cfe/docs/src/cfe_sb.dox File Reference

39.14 cfe/docs/src/cfe_tbl.dox File Reference

39.15 cfe/docs/src/cfe_time.dox File Reference

39.16 cfe/docs/src/cfe_xref.dox File Reference

39.17 cfe/docs/src/cfs_versions.dox File Reference

39.18 cfe/docs/src/main.dox File Reference

39.19 cfe/modules/core_api/fsw/inc/cfe.h File Reference

```
#include "common_types.h"
#include "osapi.h"
#include "cfe_mission_cfg.h"
#include "cfe_error.h"
#include "cfe_es.h"
#include "cfe_evs.h"
#include "cfe_fs.h"
#include "cfe_sb.h"
#include "cfe_time.h"
#include "cfe_tbl.h"
#include "cfe_msg.h"
#include "cfe_resourceid.h"
#include "cfe_psp.h"
```

39.20 cfe/modules/core_api/fsw/inc/cfe_endian.h File Reference

```
#include "common_types.h"
```

Macros

- #define CFE_MAKE_BIG16(n) ((((n) << 8) & 0xFF00) | (((n) >> 8) & 0x00FF))
- #define CFE_MAKE_BIG32(n) ((((n) << 24) & 0xFF000000) | (((n) << 8) & 0x00FF0000) | (((n) >> 8) & 0x0000FF00) | (((n) >> 24) & 0x000000FF))

39.20.1 Macro Definition Documentation

39.20.1.1 CFE_MAKE_BIG16

```
#define CFE_MAKE_BIG16(
            n ) ((((n) << 8) & 0xFF00) | (((n) >> 8) & 0x00FF))
```

Definition at line 66 of file cfe_endian.h.

39.20.1.2 CFE_MAKE_BIG32

```
#define CFE_MAKE_BIG32(
            n ) ((((n) << 24) & 0xFF000000) | (((n) << 8) & 0x00FF0000) | (((n) >> 8) & 0x0000↩
FF00) | (((n) >> 24) & 0x000000FF))
```

Definition at line 67 of file cfe_endian.h.

39.21 cfe/modules/core_api/fsw/inc/cfe_error.h File Reference

```
#include "osapi.h"
```

Macros

- #define CFE_SEVERITY_BITMASK ((CFE_Status_t)0xc0000000)

  Error Severity Bitmask.
- #define CFE_SEVERITY_SUCCESS ((CFE_Status_t)0x00000000)

  Severity Success.
- #define CFE_SEVERITY_INFO ((CFE_Status_t)0x40000000)

  Severity Info.
- #define CFE_SEVERITY_ERROR ((CFE_Status_t)0xc0000000)

  Severity Error.
- #define CFE_SERVICE_BITMASK ((CFE_Status_t)0x0e000000)

  Error Service Bitmask.
- #define CFE_EVENTS_SERVICE ((CFE_Status_t)0x02000000)

  Event Service.
- #define CFE_EXECUTIVE_SERVICE ((CFE_Status_t)0x04000000)

  Executive Service.
- #define CFE_FILE_SERVICE ((CFE_Status_t)0x06000000)

  File Service.
- #define CFE_GENERIC_SERVICE ((CFE_Status_t)0x08000000)

  Generic Service.
- #define CFE_SOFTWARE_BUS_SERVICE ((CFE_Status_t)0x0a000000)

  Software Bus Service.
- #define CFE_TABLE_SERVICE ((CFE_Status_t)0x0c000000)

  Table Service.
- #define CFE_TIME_SERVICE ((CFE_Status_t)0x0e000000)

  Time Service.
- #define CFE_SUCCESS (0)

  Sucessful execution.
- #define CFE_STATUS_NO_COUNTER_INCREMENT ((CFE_Status_t)0x48000001)

  No Counter Increment.
- #define CFE_STATUS_WRONG_MSG_LENGTH ((CFE_Status_t)0xc8000002)

  Wrong Message Length.
- #define CFE_STATUS_UNKNOWN_MSG_ID ((CFE_Status_t)0xc8000003)

  Unknown Message ID.
- #define CFE_STATUS_BAD_COMMAND_CODE ((CFE_Status_t)0xc8000004)

  Bad Command Code.
- #define CFE_STATUS_EXTERNAL_RESOURCE_FAIL ((CFE_Status_t)0xc8000005)

  External failure.
- #define CFE_STATUS_REQUEST_ALREADY_PENDING ((int32)0xc8000006)

  Request already pending.
- #define CFE_STATUS_NOT_IMPLEMENTED ((CFE_Status_t)0xc800ffff)

  Not Implemented.
- #define CFE_EVS_UNKNOWN_FILTER ((CFE_Status_t)0xc2000001)

  Unknown Filter.
- #define CFE_EVS_APP_NOT_REGISTERED ((CFE_Status_t)0xc2000002)

  Application Not Registered.
- #define CFE_EVS_APP_ILLEGAL_APP_ID ((CFE_Status_t)0xc2000003)

Illegal Application ID.

- #define CFE_EVS_APP_FILTER_OVERLOAD ((CFE_Status_t)0xc2000004)

  Application Filter Overload.
- #define CFE_EVS_RESET_AREA_POINTER ((CFE_Status_t)0xc2000005)

  Reset Area Pointer Failure.
- #define CFE_EVS_EVT_NOT_REGISTERED ((CFE_Status_t)0xc2000006)

  Event Not Registered.
- #define CFE_EVS_FILE_WRITE_ERROR ((CFE_Status_t)0xc2000007)

  File Write Error.
- #define CFE_EVS_INVALID_PARAMETER ((CFE_Status_t)0xc2000008)

  Invalid Pointer.
- #define CFE_EVS_NOT_IMPLEMENTED ((CFE_Status_t)0xc200ffff)

  Not Implemented.
- #define CFE_ES_ERR_RESOURCEID_NOT_VALID ((CFE_Status_t)0xc4000001)

  Resource ID is not valid.
- #define CFE_ES_ERR_NAME_NOT_FOUND ((CFE_Status_t)0xc4000002)

  Resource Name Error.
- #define CFE_ES_ERR_BUFFER ((CFE_Status_t)0xc4000003)

  Invalid Pointer.
- #define CFE_ES_ERR_APP_CREATE ((CFE_Status_t)0xc4000004)

  Application Create Error.
- #define CFE_ES_ERR_CHILD_TASK_CREATE ((CFE_Status_t)0xc4000005)

  Child Task Create Error.
- #define CFE_ES_ERR_SYS_LOG_FULL ((CFE_Status_t)0xc4000006)

  System Log Full.
- #define CFE_ES_ERR_MEM_BLOCK_SIZE ((CFE_Status_t)0xc4000008)

  Memory Block Size Error.
- #define CFE_ES_ERR_LOAD_LIB ((CFE_Status_t)0xc4000009)

  Load Library Error.
- #define CFE_ES_BAD_ARGUMENT ((CFE_Status_t)0xc400000a)

  Bad Argument.
- #define CFE_ES_ERR_CHILD_TASK_REGISTER ((CFE_Status_t)0xc400000b)

  Child Task Register Error.
- #define CFE_ES_ERR_SHELL_CMD ((CFE_Status_t)0xc400000c)

  Shell Command Error.
- #define CFE_ES_CDS_ALREADY_EXISTS ((CFE_Status_t)0x4400000d)

  CDS Already Exists.
- #define CFE_ES_CDS_INSUFFICIENT_MEMORY ((CFE_Status_t)0xc400000e)

  CDS Insufficient Memory.
- #define CFE_ES_CDS_INVALID_NAME ((CFE_Status_t)0xc400000f)

  CDS Invalid Name.
- #define CFE_ES_CDS_INVALID_SIZE ((CFE_Status_t)0xc4000010)

  CDS Invalid Size.
- #define CFE_ES_CDS_INVALID ((CFE_Status_t)0xc4000012)

  CDS Invalid.
- #define CFE_ES_CDS_ACCESS_ERROR ((CFE_Status_t)0xc4000013)

  CDS Access Error.

- #define CFE_ES_FILE_IO_ERR ((CFE_Status_t)0xc4000014)

  File IO Error.
- #define CFE_ES_RST_ACCESS_ERR ((CFE_Status_t)0xc4000015)

  Reset Area Access Error.
- #define CFE_ES_ERR_APP_REGISTER ((CFE_Status_t)0xc4000017)

  Application Register Error.
- #define CFE_ES_ERR_CHILD_TASK_DELETE ((CFE_Status_t)0xc4000018)

  Child Task Delete Error.
- #define CFE_ES_ERR_CHILD_TASK_DELETE_MAIN_TASK ((CFE_Status_t)0xc4000019)

  Child Task Delete Passed Main Task.
- #define CFE_ES_CDS_BLOCK_CRC_ERR ((CFE_Status_t)0xc400001A)

  CDS Block CRC Error.
- #define CFE_ES_MUT_SEM_DELETE_ERR ((CFE_Status_t)0xc400001B)

  Mutex Semaphore Delete Error.
- #define CFE_ES_BIN_SEM_DELETE_ERR ((CFE_Status_t)0xc400001C)

  Binary Semaphore Delete Error.
- #define CFE_ES_COUNT_SEM_DELETE_ERR ((CFE_Status_t)0xc400001D)

  Counte Semaphore Delete Error.
- #define CFE_ES_QUEUE_DELETE_ERR ((CFE_Status_t)0xc400001E)

  Queue Delete Error.
- #define CFE_ES_FILE_CLOSE_ERR ((CFE_Status_t)0xc400001F)

  File Close Error.
- #define CFE_ES_CDS_WRONG_TYPE_ERR ((CFE_Status_t)0xc4000020)

  CDS Wrong Type Error.
- #define CFE_ES_CDS_OWNER_ACTIVE_ERR ((CFE_Status_t)0xc4000022)

  CDS Owner Active Error.
- #define CFE_ES_APP_CLEANUP_ERR ((CFE_Status_t)0xc4000023)

  Application Cleanup Error.
- #define CFE_ES_TIMER_DELETE_ERR ((CFE_Status_t)0xc4000024)

  Timer Delete Error.
- #define CFE_ES_BUFFER_NOT_IN_POOL ((CFE_Status_t)0xc4000025)

  Buffer Not In Pool.
- #define CFE_ES_TASK_DELETE_ERR ((CFE_Status_t)0xc4000026)

  Task Delete Error.
- #define CFE_ES_OPERATION_TIMED_OUT ((CFE_Status_t)0xc4000027)

  Operation Timed Out.
- #define CFE_ES_LIB_ALREADY_LOADED ((CFE_Status_t)0x44000028)

  Library Already Loaded.
- #define CFE_ES_ERR_SYS_LOG_TRUNCATED ((CFE_Status_t)0x44000029)

  System Log Message Truncated.
- #define CFE_ES_NO_RESOURCE_IDS_AVAILABLE ((CFE_Status_t)0xc400002B)

  Resource ID is not available.
- #define CFE_ES_POOL_BLOCK_INVALID ((CFE_Status_t)0xc400002C)

  Invalid pool block.
- #define CFE_ES_POOL_BOUNDS_ERROR ((CFE_Status_t)0xc400002D)

  Invalid pool size or buffer address.
- #define CFE_ES_ERR_DUPLICATE_NAME ((CFE_Status_t)0xc400002E)

Duplicate Name Error.

- #define CFE_ES_NOT_IMPLEMENTED ((CFE_Status_t)0xc400ffff)

  Not Implemented.
- #define CFE_FS_BAD_ARGUMENT ((CFE_Status_t)0xc6000001)

  Bad Argument.
- #define CFE_FS_INVALID_PATH ((CFE_Status_t)0xc6000002)

  Invalid Path.
- #define CFE_FS_FNAME_TOO_LONG ((CFE_Status_t)0xc6000003)

  Filename Too Long.
- #define CFE_FS_NOT_IMPLEMENTED ((CFE_Status_t)0xc600ffff)

  Not Implemented.
- #define CFE_SB_TIME_OUT ((CFE_Status_t)0xca000001)

  Time Out.
- #define CFE_SB_NO_MESSAGE ((CFE_Status_t)0xca000002)

  No Message.
- #define CFE_SB_BAD_ARGUMENT ((CFE_Status_t)0xca000003)

  Bad Argument.
- #define CFE_SB_MAX_PIPES_MET ((CFE_Status_t)0xca000004)

  Max Pipes Met.
- #define CFE_SB_PIPE_CR_ERR ((CFE_Status_t)0xca000005)

  Pipe Create Error.
- #define CFE_SB_PIPE_RD_ERR ((CFE_Status_t)0xca000006)

  Pipe Read Error.
- #define CFE_SB_MSG_TOO_BIG ((CFE_Status_t)0xca000007)

  Message Too Big.
- #define CFE_SB_BUF_ALOC_ERR ((CFE_Status_t)0xca000008)

  Buffer Allocation Error.
- #define CFE_SB_MAX_MSGS_MET ((CFE_Status_t)0xca000009)

  Max Messages Met.
- #define CFE_SB_MAX_DESTS_MET ((CFE_Status_t)0xca00000a)

  Max Destinations Met.
- #define CFE_SB_NO_SUBSCRIBERS ((CFE_Status_t)0xca00000b)

  No Subscribers.
- #define CFE_SB_INTERNAL_ERR ((CFE_Status_t)0xca00000c)

  Internal Error.
- #define CFE_SB_WRONG_MSG_TYPE ((CFE_Status_t)0xca00000d)

  Wrong Message Type.
- #define CFE_SB_BUFFER_INVALID ((CFE_Status_t)0xca00000e)

  Buffer Invalid.
- #define CFE_SB_NO_MSG_RECV ((CFE_Status_t)0xca00000f)

  No Message Recieved.
- #define CFE_SB_NOT_IMPLEMENTED ((CFE_Status_t)0xca00ffff)

  Not Implemented.
- #define CFE_TBL_ERR_INVALID_HANDLE ((CFE_Status_t)0xcc000001)

  Invalid Handle.
- #define CFE_TBL_ERR_INVALID_NAME ((CFE_Status_t)0xcc000002)

  Invalid Name.

- #define CFE_TBL_ERR_INVALID_SIZE ((CFE_Status_t)0xcc000003)

  Invalid Size.
- #define CFE_TBL_INFO_UPDATE_PENDING ((CFE_Status_t)0x4c000004)

  Update Pending.
- #define CFE_TBL_ERR_NEVER_LOADED ((CFE_Status_t)0xcc000005)

  Never Loaded.
- #define CFE_TBL_ERR_REGISTRY_FULL ((CFE_Status_t)0xcc000006)

  Registry Full.
- #define CFE_TBL_WARN_DUPLICATE ((CFE_Status_t)0x4c000007)

  Duplicate Warning.
- #define CFE_TBL_ERR_NO_ACCESS ((CFE_Status_t)0xcc000008)

  No Access.
- #define CFE_TBL_ERR_UNREGISTERED ((CFE_Status_t)0xcc000009)

  Unregistered.
- #define CFE_TBL_ERR_BAD_APP_ID ((CFE_Status_t)0xcc00000A)

  Bad Application ID.
- #define CFE_TBL_ERR_HANDLES_FULL ((CFE_Status_t)0xcc00000B)

  Handles Full.
- #define CFE_TBL_ERR_DUPLICATE_DIFF_SIZE ((CFE_Status_t)0xcc00000C)

  Duplicate Table With Different Size.
- #define CFE_TBL_ERR_DUPLICATE_NOT_OWNED ((CFE_Status_t)0xcc00000D)

  Dupicate Table And Not Owned.
- #define CFE_TBL_INFO_UPDATED ((CFE_Status_t)0x4c00000E)

  Updated.
- #define CFE_TBL_ERR_NO_BUFFER_AVAIL ((CFE_Status_t)0xcc00000F)

  No Buffer Available.
- #define CFE_TBL_ERR_DUMP_ONLY ((CFE_Status_t)0xcc000010)

  Dump Only Error.
- #define CFE_TBL_ERR_ILLEGAL_SRC_TYPE ((CFE_Status_t)0xcc000011)

  Illegal Source Type.
- #define CFE_TBL_ERR_LOAD_IN_PROGRESS ((CFE_Status_t)0xcc000012)

  Load In Progress.
- #define CFE_TBL_ERR_FILE_NOT_FOUND ((CFE_Status_t)0xcc000013)

  File Not Found.
- #define CFE_TBL_ERR_FILE_TOO_LARGE ((CFE_Status_t)0xcc000014)

  File Too Large.
- #define CFE_TBL_WARN_SHORT_FILE ((CFE_Status_t)0x4c000015)

  Short File Warning.
- #define CFE_TBL_ERR_BAD_CONTENT_ID ((CFE_Status_t)0xcc000016)

  Bad Content ID.
- #define CFE_TBL_INFO_NO_UPDATE_PENDING ((CFE_Status_t)0x4c000017)

  No Update Pending.
- #define CFE_TBL_INFO_TABLE_LOCKED ((CFE_Status_t)0x4c000018)

  Table Locked.
- #define CFE_TBL_INFO_VALIDATION_PENDING ((CFE_Status_t)0x4c000019)
- #define CFE_TBL_INFO_NO_VALIDATION_PENDING ((CFE_Status_t)0x4c00001A)
- #define CFE_TBL_ERR_BAD_SUBTYPE_ID ((CFE_Status_t)0xcc00001B)

Bad Subtype ID.

- #define CFE_TBL_ERR_FILE_SIZE_INCONSISTENT ((CFE_Status_t)0xcc00001C)

  File Size Inconsistent.
- #define CFE_TBL_ERR_NO_STD_HEADER ((CFE_Status_t)0xcc00001D)

  No Standard Header.
- #define CFE_TBL_ERR_NO_TBL_HEADER ((CFE_Status_t)0xcc00001E)

  No Table Header.
- #define CFE_TBL_ERR_FILENAME_TOO_LONG ((CFE_Status_t)0xcc00001F)

  Filename Too Long.
- #define CFE_TBL_ERR_FILE_FOR_WRONG_TABLE ((CFE_Status_t)0xcc000020)

  File For Wrong Table.
- #define CFE_TBL_ERR_LOAD_INCOMPLETE ((CFE_Status_t)0xcc000021)

  Load Incomplete.
- #define CFE_TBL_WARN_PARTIAL_LOAD ((CFE_Status_t)0x4c000022)

  Partial Load Warning.
- #define CFE_TBL_ERR_PARTIAL_LOAD ((CFE_Status_t)0xcc000023)

  Partial Load Error.
- #define CFE_TBL_INFO_DUMP_PENDING ((CFE_Status_t)0x4c000024)

  Dump Pending.
- #define CFE_TBL_ERR_INVALID_OPTIONS ((CFE_Status_t)0xcc000025)

  Invalid Options.
- #define CFE_TBL_WARN_NOT_CRITICAL ((CFE_Status_t)0x4c000026)

  Not Critical Warning.
- #define CFE_TBL_INFO_RECOVERED_TBL ((CFE_Status_t)0x4c000027)

  Recovered Table.
- #define CFE_TBL_ERR_BAD_SPACECRAFT_ID ((CFE_Status_t)0xcc000028)

  Bad Spacecraft ID.
- #define CFE_TBL_ERR_BAD_PROCESSOR_ID ((CFE_Status_t)0xcc000029)

  Bad Processor ID.
- #define CFE_TBL_MESSAGE_ERROR ((CFE_Status_t)0xcc00002a)

  Message Error.
- #define CFE_TBL_ERR_SHORT_FILE ((CFE_Status_t)0xcc00002b)
- #define CFE_TBL_ERR_ACCESS ((CFE_Status_t)0xcc00002c)
- #define CFE_TBL_BAD_ARGUMENT ((CFE_Status_t)0xcc00002d)

  Bad Argument.
- #define CFE_TBL_NOT_IMPLEMENTED ((CFE_Status_t)0xcc00ffff)

  Not Implemented.
- #define CFE_TIME_NOT_IMPLEMENTED ((CFE_Status_t)0xce00ffff)

  Not Implemented.
- #define CFE_TIME_INTERNAL_ONLY ((CFE_Status_t)0xce000001)

  Internal Only.
- #define CFE_TIME_OUT_OF_RANGE ((CFE_Status_t)0xce000002)

  Out Of Range.
- #define CFE_TIME_TOO_MANY_SYNCH_CALLBACKS ((CFE_Status_t)0xce000003)

  Too Many Sync Callbacks.
- #define CFE_TIME_CALLBACK_NOT_REGISTERED ((CFE_Status_t)0xce000004)

  Callback Not Registered.

- #define CFE_TIME_BAD_ARGUMENT ((CFE_Status_t)0xce000005)

  Bad Argument.
- #define CFE_ES_ERR_TASKID CFE_ES_ERR_RESOURCEID_NOT_VALID
- #define CFE_ES_ERR_APPID CFE_ES_ERR_RESOURCEID_NOT_VALID
- #define CFE_ES_ERR_MEM_HANDLE CFE_ES_ERR_RESOURCEID_NOT_VALID
- #define CFE_ES_ERR_APPNAME CFE_ES_ERR_NAME_NOT_FOUND
- #define CFE_ES_CDS_NOT_FOUND_ERR CFE_ES_ERR_NAME_NOT_FOUND
- #define CFE_ES_CDS_REGISTRY_FULL CFE_ES_NO_RESOURCE_IDS_AVAILABLE

Typedefs

- typedef int32 CFE_Status_t

39.21.1 Macro Definition Documentation

39.21.1.1 CFE_ES_CDS_NOT_FOUND_ERR

```
#define CFE_ES_CDS_NOT_FOUND_ERR CFE_ES_ERR_NAME_NOT_FOUND
```

Definition at line 1380 of file cfe_error.h.

39.21.1.2 CFE_ES_CDS_REGISTRY_FULL

```
#define CFE_ES_CDS_REGISTRY_FULL CFE_ES_NO_RESOURCE_IDS_AVAILABLE
```

Definition at line 1381 of file cfe_error.h.

39.21.1.3 CFE_ES_ERR_APPID

```
#define CFE_ES_ERR_APPID CFE_ES_ERR_RESOURCEID_NOT_VALID
```

Definition at line 1377 of file cfe_error.h.

39.21.1.4 CFE_ES_ERR_APPNAME

```
#define CFE_ES_ERR_APPNAME CFE_ES_ERR_NAME_NOT_FOUND
```

Definition at line 1379 of file cfe_error.h.

39.21.1.5 CFE_ES_ERR_MEM_HANDLE

```
#define CFE_ES_ERR_MEM_HANDLE CFE_ES_ERR_RESOURCEID_NOT_VALID
```

Definition at line 1378 of file cfe_error.h.

39.21.1.6 CFE_ES_ERR_TASKID

```
#define CFE_ES_ERR_TASKID CFE_ES_ERR_RESOURCEID_NOT_VALID
```

Definition at line 1376 of file cfe_error.h.

39.21.1.7 CFE_EVENTS_SERVICE

```
#define CFE_EVENTS_SERVICE ((CFE_Status_t)0x02000000)
```

Event Service.

Definition at line 104 of file cfe_error.h.

39.21.1.8 CFE_EXECUTIVE_SERVICE

```
#define CFE_EXECUTIVE_SERVICE ((CFE_Status_t)0x04000000)
```

Executive Service.

Definition at line 105 of file cfe_error.h.

39.21.1.9 CFE_FILE_SERVICE

```
#define CFE_FILE_SERVICE ((CFE_Status_t)0x06000000)
```

File Service.

Definition at line 106 of file cfe_error.h.

39.21.1.10 CFE_GENERIC_SERVICE

```
#define CFE_GENERIC_SERVICE ((CFE_Status_t)0x08000000)
```

Generic Service.

Definition at line 107 of file cfe_error.h.

39.21.1.11 CFE_SERVICE_BITMASK

```
#define CFE_SERVICE_BITMASK ((CFE_Status_t)0x0e000000)
```

Error Service Bitmask.

Definition at line 102 of file cfe_error.h.

39.21.1.12 CFE_SEVERITY_BITMASK

```
#define CFE_SEVERITY_BITMASK ((CFE_Status_t)0xc0000000)
```

Error Severity Bitmask.

Definition at line 93 of file cfe_error.h.

39.21.1.13 CFE_SEVERITY_ERROR

```
#define CFE_SEVERITY_ERROR ((CFE_Status_t)0xc0000000)
```

Severity Error.

Definition at line 97 of file cfe_error.h.

39.21.1.14 CFE_SEVERITY_INFO

```
#define CFE_SEVERITY_INFO ((CFE_Status_t)0x40000000)
```

Severity Info.

Definition at line 96 of file cfe_error.h.

39.21.1.15 CFE_SEVERITY_SUCCESS

```
#define CFE_SEVERITY_SUCCESS ((CFE_Status_t)0x00000000)
```

Severity Success.

Definition at line 95 of file cfe_error.h.

39.21.1.16 CFE_SOFTWARE_BUS_SERVICE

```
#define CFE_SOFTWARE_BUS_SERVICE ((CFE_Status_t)0x0a000000)
```

Software Bus Service.

Definition at line 108 of file cfe_error.h.

39.21.1.17 CFE_TABLE_SERVICE

```
#define CFE_TABLE_SERVICE ((CFE_Status_t)0x0c000000)
```

Table Service.

Definition at line 109 of file cfe_error.h.

39.21.1.18 CFE_TIME_SERVICE

```
#define CFE_TIME_SERVICE ((CFE_Status_t)0x0e000000)
```

Time Service.

Definition at line 110 of file cfe_error.h.

39.21.2 Typedef Documentation

39.21.2.1 CFE_Status_t

```
typedef int32 CFE_Status_t
```

Definition at line 49 of file cfe_error.h.

39.22 cfe/modules/core_api/fsw/inc/cfe_es.h File Reference

```
#include "common_types.h"
#include "cfe_error.h"
#include "cfe_es_api_typedefs.h"
```

Macros

- #define OS_PRINTF(m, n)
- #define CFE_ES_DBIT(x) (1L << (x)) /* Places a one at bit positions 0 thru 31 */
- #define CFE_ES_DTEST(i, x) (((i)&CFE_ES_DBIT(x)) != 0) /* true iff bit x of i is set */
- #define CFE_ES_TEST_LONG_MASK(m, s) (CFE_ES_DTEST(m[(s) / 32], (s) % 32)) /* Test a bit within an array of 32-bit integers. */
- #define CFE_ES_PerfLogEntry(id) (CFE_ES_PerfLogAdd(id, 0))

  Entry marker for use with Software Performance Analysis Tool.
- #define CFE_ES_PerfLogExit(id) (CFE_ES_PerfLogAdd(id, 1))

  Exit marker for use with Software Performance Analysis Tool.

Functions

- CFE_Status_t CFE_ES_AppID_ToIndex (CFE_ES_AppId_t AppID, uint32 *Idx)

  Obtain an index value correlating to an ES Application ID.
- int32 CFE_ES_LibID_ToIndex (CFE_ES_LibId_t LibID, uint32 *Idx)

  Obtain an index value correlating to an ES Library ID.
- CFE_Status_t CFE_ES_TaskID_ToIndex (CFE_ES_TaskId_t TaskID, uint32 *Idx)

  Obtain an index value correlating to an ES Task ID.
- CFE_Status_t CFE_ES_CounterID_ToIndex (CFE_ES_CounterId_t CounterID, uint32 *Idx)

  Obtain an index value correlating to an ES Counter ID.
- void CFE_ES_Main (uint32 StartType, uint32 StartSubtype, uint32 ModeId, const char *StartFilePath)

  cFE Main Entry Point used by Board Support Package to start cFE
- CFE_Status_t CFE_ES_ResetCFE (uint32 ResetType)

  Reset the cFE Core and all cFE Applications.
- CFE_Status_t CFE_ES_RestartApp (CFE_ES_AppId_t AppID)

  Restart a single cFE Application.
- CFE_Status_t CFE_ES_ReloadApp (CFE_ES_AppId_t AppID, const char *AppFileName)

  Reload a single cFE Application.
- CFE_Status_t CFE_ES_DeleteApp (CFE_ES_AppId_t AppID)

  Delete a cFE Application.
- void CFE_ES_ExitApp (uint32 ExitStatus)

  Exit a cFE Application.
- bool CFE_ES_RunLoop (uint32 *ExitStatus)

  Check for Exit, Restart, or Reload commands.
- CFE_Status_t CFE_ES_WaitForSystemState (uint32 MinSystemState, uint32 TimeOutMilliseconds)

  Allow an Application to Wait for a minimum global system state.
- void CFE_ES_WaitForStartupSync (uint32 TimeOutMilliseconds)

Allow an Application to Wait for the "OPERATIONAL" global system state.

- CFE_Status_t CFE_ES_RegisterApp (void)

  Registers a cFE Application with the Executive Services.
- void CFE_ES_IncrementTaskCounter (void)

  Increments the execution counter for the calling task.
- int32 CFE_ES_GetResetType (uint32 ∗ResetSubtypePtr)

  Return the most recent Reset Type.
- CFE_Status_t CFE_ES_GetAppID (CFE_ES_AppId_t ∗AppIdPtr)

  Get an Application ID for the calling Application.
- CFE_Status_t CFE_ES_GetTaskID (CFE_ES_TaskId_t ∗TaskIdPtr)

  Get the task ID of the calling context.
- CFE_Status_t CFE_ES_GetAppIDByName (CFE_ES_AppId_t ∗AppIdPtr, const char ∗AppName)

  Get an Application ID associated with a specified Application name.
- CFE_Status_t CFE_ES_GetLibIDByName (CFE_ES_LibId_t ∗LibIdPtr, const char ∗LibName)

  Get a Library ID associated with a specified Library name.
- CFE_Status_t CFE_ES_GetAppName (char ∗AppName, CFE_ES_AppId_t AppId, size_t BufferLength)

  Get an Application name for a specified Application ID.
- CFE_Status_t CFE_ES_GetLibName (char ∗LibName, CFE_ES_LibId_t LibId, size_t BufferLength)

  Get a Library name for a specified Library ID.
- CFE_Status_t CFE_ES_GetAppInfo (CFE_ES_AppInfo_t ∗AppInfo, CFE_ES_AppId_t AppId)

  Get Application Information given a specified App ID.
- CFE_Status_t CFE_ES_GetTaskInfo (CFE_ES_TaskInfo_t ∗TaskInfo, CFE_ES_TaskId_t TaskId)

  Get Task Information given a specified Task ID.
- int32 CFE_ES_GetLibInfo (CFE_ES_AppInfo_t ∗LibInfo, CFE_ES_LibId_t LibId)

  Get Library Information given a specified Resource ID.
- int32 CFE_ES_GetModuleInfo (CFE_ES_AppInfo_t ∗ModuleInfo, CFE_ResourceId_t ResourceId)

  Get Information given a specified Resource ID.
- CFE_Status_t CFE_ES_RegisterChildTask (void)

  Registers a cFE Child task associated with a cFE Application.
- CFE_Status_t CFE_ES_CreateChildTask (CFE_ES_TaskId_t ∗TaskIdPtr, const char ∗TaskName, CFE_ES↩_ChildTaskMainFuncPtr_t FunctionPtr, CFE_ES_StackPointer_t StackPtr, size_t StackSize, CFE_ES_Task↩Priority_Atom_t Priority, uint32 Flags)

  Creates a new task under an existing Application.
- CFE_Status_t CFE_ES_GetTaskIDByName (CFE_ES_TaskId_t ∗TaskIdPtr, const char ∗TaskName)

  Get a Task ID associated with a specified Task name.
- CFE_Status_t CFE_ES_GetTaskName (char ∗TaskName, CFE_ES_TaskId_t TaskId, size_t BufferLength)

  Get a Task name for a specified Task ID.
- CFE_Status_t CFE_ES_DeleteChildTask (CFE_ES_TaskId_t TaskId)

  Deletes a task under an existing Application.
- void CFE_ES_ExitChildTask (void)

  Exits a child task.
- void CFE_ES_BackgroundWakeup (void)

  Wakes up the CFE background task.
- CFE_Status_t CFE_ES_WriteToSysLog (const char ∗SpecStringPtr,...) OS_PRINTF(1

  Write a string to the cFE System Log.
- CFE_Status_t uint32 CFE_ES_CalculateCRC (const void ∗DataPtr, size_t DataLength, uint32 InputCRC, uint32 TypeCRC)

Calculate a CRC on a block of memory.

- void CFE_ES_ProcessAsyncEvent (void)

  Notification that an asynchronous event was detected by the underlying OS/PSP.
- CFE_Status_t CFE_ES_RegisterCDS (CFE_ES_CDSHandle_t ∗HandlePtr, size_t BlockSize, const char ∗Name)

  Reserve space (or re-obtain previously reserved space) in the Critical Data Store (CDS)
- CFE_Status_t CFE_ES_GetCDSBlockIDByName (CFE_ES_CDSHandle_t ∗BlockIdPtr, const char ∗BlockName)

  Get a CDS Block ID associated with a specified CDS Block name.
- CFE_Status_t CFE_ES_GetCDSBlockName (char ∗BlockName, CFE_ES_CDSHandle_t BlockId, size_t Buffer↩ Length)

  Get a Block name for a specified Block ID.
- CFE_Status_t CFE_ES_CopyToCDS (CFE_ES_CDSHandle_t Handle, void ∗DataToCopy)

  Save a block of data in the Critical Data Store (CDS)
- CFE_Status_t CFE_ES_RestoreFromCDS (void ∗RestoreToMemory, CFE_ES_CDSHandle_t Handle)

  Recover a block of data from the Critical Data Store (CDS)
- CFE_Status_t CFE_ES_PoolCreateNoSem (CFE_ES_MemHandle_t ∗PoolID, uint8 ∗MemPtr, size_t Size)

  Initializes a memory pool created by an application without using a semaphore during processing.
- CFE_Status_t CFE_ES_PoolCreate (CFE_ES_MemHandle_t ∗PoolID, uint8 ∗MemPtr, size_t Size)

  Initializes a memory pool created by an application while using a semaphore during processing.
- CFE_Status_t CFE_ES_PoolCreateEx (CFE_ES_MemHandle_t ∗PoolID, uint8 ∗MemPtr, size_t Size, uint16 NumBlockSizes, const size_t ∗BlockSizes, bool UseMutex)

  Initializes a memory pool created by an application with application specified block sizes.
- int32 CFE_ES_PoolDelete (CFE_ES_MemHandle_t PoolID)

  Deletes a memory pool that was previously created.
- int32 CFE_ES_GetPoolBuf (CFE_ES_MemPoolBuf_t ∗BufPtr, CFE_ES_MemHandle_t PoolID, size_t Size)

  Gets a buffer from the memory pool created by CFE_ES_PoolCreate or CFE_ES_PoolCreateNoSem.
- CFE_Status_t CFE_ES_GetPoolBufInfo (CFE_ES_MemHandle_t PoolID, CFE_ES_MemPoolBuf_t BufPtr)

  Gets info on a buffer previously allocated via CFE_ES_GetPoolBuf.
- int32 CFE_ES_PutPoolBuf (CFE_ES_MemHandle_t PoolID, CFE_ES_MemPoolBuf_t BufPtr)

  Releases a buffer from the memory pool that was previously allocated via CFE_ES_GetPoolBuf.
- CFE_Status_t CFE_ES_GetMemPoolStats (CFE_ES_MemPoolStats_t ∗BufPtr, CFE_ES_MemHandle_t Handle)

  Extracts the statistics maintained by the memory pool software.
- void CFE_ES_PerfLogAdd (uint32 Marker, uint32 EntryExit)

  Function called by CFE_ES_PerfLogEntry and CFE_ES_PerfLogExit macros.
- CFE_Status_t CFE_ES_RegisterGenCounter (CFE_ES_CounterId_t ∗CounterIdPtr, const char ∗CounterName)

  Register a generic counter.
- CFE_Status_t CFE_ES_DeleteGenCounter (CFE_ES_CounterId_t CounterId)

  Delete a generic counter.
- CFE_Status_t CFE_ES_IncrementGenCounter (CFE_ES_CounterId_t CounterId)

  Increments the specified generic counter.
- CFE_Status_t CFE_ES_SetGenCount (CFE_ES_CounterId_t CounterId, uint32 Count)

  Set the specified generic counter.
- CFE_Status_t CFE_ES_GetGenCount (CFE_ES_CounterId_t CounterId, uint32 ∗Count)

  Get the specified generic counter count.
- CFE_Status_t CFE_ES_GetGenCounterIDByName (CFE_ES_CounterId_t ∗CounterIdPtr, const char ∗CounterName)

  Get the Id associated with a generic counter name.
- CFE_Status_t CFE_ES_GetGenCounterName (char ∗CounterName, CFE_ES_CounterId_t CounterId, size_↩ t BufferLength)

  Get a Counter name for a specified Counter ID.

39.22.1 Macro Definition Documentation

39.22.1.1 CFE_ES_DBIT

```
#define CFE_ES_DBIT(
              x ) (1L << (x)) /* Places a one at bit positions 0 thru 31 */
```

Definition at line 58 of file cfe_es.h.

39.22.1.2 CFE_ES_DTEST

```
#define CFE_ES_DTEST(
              i,
              x ) (((i)&CFE_ES_DBIT(x)) != 0) /* true iff bit x of i is set */
```

Definition at line 59 of file cfe_es.h.

39.22.1.3 CFE_ES_TEST_LONG_MASK

```
#define CFE_ES_TEST_LONG_MASK(
              m,
              s ) (CFE_ES_DTEST(m[(s) / 32], (s) % 32)) /* Test a bit within an array of 32-bit
integers.  */
```

Definition at line 60 of file cfe_es.h.

39.22.1.4 OS_PRINTF

```
#define OS_PRINTF(
              m,
              n )
```

Definition at line 51 of file cfe_es.h.

39.23 cfe/modules/core_api/fsw/inc/cfe_es_api_typedefs.h File Reference

```
#include "common_types.h"
#include "cfe_es_extern_typedefs.h"
```

Data Structures

- union CFE_ES_PoolAlign

  Pool Alignement.

Macros

- #define CFE_ES_STATIC_POOL_TYPE(size)

  Static Pool Type.
- #define CFE_ES_MEMPOOLBUF_C(x) ((CFE_ES_MemPoolBuf_t)(x))

  Conversion macro to create buffer pointer from another type.
- #define CFE_ES_NO_MUTEX false

  Indicates that the memory pool selection will not use a semaphore.
- #define CFE_ES_USE_MUTEX true

  Indicates that the memory pool selection will use a semaphore.

Reset Type extensions

- #define CFE_ES_APP_RESTART CFE_PSP_RST_TYPE_MAX

Conversions for ES resource IDs

- #define CFE_ES_APPID_C(val) ((CFE_ES_AppId_t)CFE_RESOURCEID_WRAP(val))
- #define CFE_ES_TASKID_C(val) ((CFE_ES_TaskId_t)CFE_RESOURCEID_WRAP(val))
- #define CFE_ES_LIBID_C(val) ((CFE_ES_LibId_t)CFE_RESOURCEID_WRAP(val))
- #define CFE_ES_COUNTERID_C(val) ((CFE_ES_CounterId_t)CFE_RESOURCEID_WRAP(val))
- #define CFE_ES_MEMHANDLE_C(val) ((CFE_ES_MemHandle_t)CFE_RESOURCEID_WRAP(val))
- #define CFE_ES_CDSHANDLE_C(val) ((CFE_ES_CDSHandle_t)CFE_RESOURCEID_WRAP(val))

Type-specific initalizers for "undefined" resource IDs

- #define CFE_ES_APPID_UNDEFINED CFE_ES_APPID_C(CFE_RESOURCEID_UNDEFINED)
- #define CFE_ES_TASKID_UNDEFINED CFE_ES_TASKID_C(CFE_RESOURCEID_UNDEFINED)
- #define CFE_ES_LIBID_UNDEFINED CFE_ES_LIBID_C(CFE_RESOURCEID_UNDEFINED)
- #define CFE_ES_COUNTERID_UNDEFINED CFE_ES_COUNTERID_C(CFE_RESOURCEID_UNDEFIN↩ED)
- #define CFE_ES_MEMHANDLE_UNDEFINED CFE_ES_MEMHANDLE_C(CFE_RESOURCEID_UNDEFI↩NED)
- #define CFE_ES_CDS_BAD_HANDLE CFE_ES_CDSHANDLE_C(CFE_RESOURCEID_UNDEFINED)

Task Stack Constants

- #define CFE_ES_TASK_STACK_ALLOCATE NULL /∗ aka OS_TASK_STACK_ALLOCATE in proposed O↩SAL change ∗/

  Indicates that the stack for the child task should be dynamically allocated.

Typedefs

- typedef void(∗ CFE_ES_TaskEntryFuncPtr_t) (void)

  Required Prototype of Task Main Functions.
- typedef int32(∗ CFE_ES_LibraryEntryFuncPtr_t) (CFE_ES_LibId_t LibId)

  Required Prototype of Library Initialization Functions.
- typedef CFE_ES_TaskEntryFuncPtr_t CFE_ES_ChildTaskMainFuncPtr_t

  Compatible typedef for ES child task entry point.
- typedef void ∗ CFE_ES_StackPointer_t

  Type for the stack pointer of tasks.
- typedef union CFE_ES_PoolAlign CFE_ES_PoolAlign_t

  Pool Alignement.
- typedef void ∗ CFE_ES_MemPoolBuf_t

  Pointer type used for memory pool API.

39.23.1 Macro Definition Documentation

39.23.1.1 CFE_ES_APP_RESTART

```
#define CFE_ES_APP_RESTART CFE_PSP_RST_TYPE_MAX
```

Application only was reset (extend the PSP enumeration here)

Definition at line 59 of file cfe_es_api_typedefs.h.

39.23.1.2 CFE_ES_APPID_C

```
#define CFE_ES_APPID_C(
              val ) ((CFE_ES_AppId_t)CFE_RESOURCEID_WRAP(val))
```

Definition at line 158 of file cfe_es_api_typedefs.h.

39.23.1.3 CFE_ES_APPID_UNDEFINED

```
#define CFE_ES_APPID_UNDEFINED CFE_ES_APPID_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 170 of file cfe_es_api_typedefs.h.

39.23.1.4 CFE_ES_CDS_BAD_HANDLE

```
#define CFE_ES_CDS_BAD_HANDLE CFE_ES_CDSHANDLE_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 175 of file cfe_es_api_typedefs.h.

39.23.1.5 CFE_ES_CDSHANDLE_C

```
#define CFE_ES_CDSHANDLE_C(
              val ) ((CFE_ES_CDSHandle_t)CFE_RESOURCEID_WRAP(val))
```

Definition at line 163 of file cfe_es_api_typedefs.h.

39.23.1.6 CFE_ES_COUNTERID_C

```
#define CFE_ES_COUNTERID_C(
              val ) ((CFE_ES_CounterId_t)CFE_RESOURCEID_WRAP(val))
```

Definition at line 161 of file cfe_es_api_typedefs.h.

39.23.1.7 CFE_ES_COUNTERID_UNDEFINED

```
#define CFE_ES_COUNTERID_UNDEFINED CFE_ES_COUNTERID_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 173 of file cfe_es_api_typedefs.h.

39.23.1.8 CFE_ES_LIBID_C

```
#define CFE_ES_LIBID_C(
              val ) ((CFE_ES_LibId_t)CFE_RESOURCEID_WRAP(val))
```

Definition at line 160 of file cfe_es_api_typedefs.h.

39.23.1.9 CFE_ES_LIBID_UNDEFINED

```
#define CFE_ES_LIBID_UNDEFINED CFE_ES_LIBID_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 172 of file cfe_es_api_typedefs.h.

39.23.1.10 CFE_ES_MEMHANDLE_C

```
#define CFE_ES_MEMHANDLE_C(
              val ) ((CFE_ES_MemHandle_t)CFE_RESOURCEID_WRAP(val))
```

Definition at line 162 of file cfe_es_api_typedefs.h.

39.23.1.11 CFE_ES_MEMHANDLE_UNDEFINED

```
#define CFE_ES_MEMHANDLE_UNDEFINED CFE_ES_MEMHANDLE_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 174 of file cfe_es_api_typedefs.h.

39.23.1.12 CFE_ES_MEMPOOLBUF_C

```
#define CFE_ES_MEMPOOLBUF_C(
              x ) ((CFE_ES_MemPoolBuf_t)(x))
```

Conversion macro to create buffer pointer from another type.

In cases where the actual buffer pointer is computed, this macro aids in converting the computed address (typically an OSAL "cpuaddr" type) into a buffer pointer.

Note

Any address calculation needs to take machine alignment requirements into account.

Definition at line 143 of file cfe_es_api_typedefs.h.

39.23.1.13 CFE_ES_NO_MUTEX

```
#define CFE_ES_NO_MUTEX false
```

Indicates that the memory pool selection will not use a semaphore.

Definition at line 190 of file cfe_es_api_typedefs.h.

39.23.1.14 CFE_ES_STATIC_POOL_TYPE

```
#define CFE_ES_STATIC_POOL_TYPE(
              size )
```

Value:

```
union                                \
    {                                     \
        CFE_ES_PoolAlign_t Align;         \
        uint8              Data[size]; \
    }
```

Static Pool Type.

A macro to help instantiate static memory pools that are correctly aligned. This resolves to a union type that contains a member called "Data" that will be correctly aligned to be a memory pool and sized according to the argument.

Definition at line 110 of file cfe_es_api_typedefs.h.

39.23.1.15 CFE_ES_TASK_STACK_ALLOCATE

```
#define CFE_ES_TASK_STACK_ALLOCATE NULL /* aka OS_TASK_STACK_ALLOCATE in proposed OSAL change */
```

Indicates that the stack for the child task should be dynamically allocated.

This value may be supplied as the Stack Pointer argument to CFE_ES_ChildTaskCreate() to indicate that the stack should be dynamically allocated.

Definition at line 187 of file cfe_es_api_typedefs.h.

39.23.1.16 CFE_ES_TASKID_C

```
#define CFE_ES_TASKID_C(
              val ) ((CFE_ES_TaskId_t)CFE_RESOURCEID_WRAP(val))
```

Definition at line 159 of file cfe_es_api_typedefs.h.

39.23.1.17 CFE_ES_TASKID_UNDEFINED

```
#define CFE_ES_TASKID_UNDEFINED CFE_ES_TASKID_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 171 of file cfe_es_api_typedefs.h.

39.23.1.18 CFE_ES_USE_MUTEX

```
#define CFE_ES_USE_MUTEX true
```

Indicates that the memory pool selection will use a semaphore.

Definition at line 191 of file cfe_es_api_typedefs.h.

39.23.2 Typedef Documentation

39.23.2.1 CFE_ES_ChildTaskMainFuncPtr_t

```
typedef CFE_ES_TaskEntryFuncPtr_t CFE_ES_ChildTaskMainFuncPtr_t
```

Compatible typedef for ES child task entry point.

All ES task functions (main + child) use the same entry point type.

Definition at line 79 of file cfe_es_api_typedefs.h.

39.23.2.2 CFE_ES_LibraryEntryFuncPtr_t

```
typedef int32(* CFE_ES_LibraryEntryFuncPtr_t) (CFE_ES_LibId_t LibId)
```

Required Prototype of Library Initialization Functions.

Definition at line 71 of file cfe_es_api_typedefs.h.

39.23.2.3 CFE_ES_MemPoolBuf_t

```
typedef void* CFE_ES_MemPoolBuf_t
```

Pointer type used for memory pool API.

This is used in the Get/Put API calls to refer to a pool buffer.

This pointer is expected to be type cast to the real object type after getting a new buffer. Using void∗ allows this type conversion to occur easily.

Note

> Older versions of CFE implemented the API using a uint32∗, which required explicit type casting everywhere it was called. Although the API type is now void∗ to make usage easier, the pool buffers are aligned to machine requirements - typically 64 bits.

Definition at line 131 of file cfe_es_api_typedefs.h.

39.23.2.4 CFE_ES_PoolAlign_t

```
typedef union CFE_ES_PoolAlign CFE_ES_PoolAlign_t
```

Pool Alignement.

Union that can be used for minimum memory alignment of ES memory pools on the target. It contains the longest native data types such that the alignment of this structure should reflect the largest possible alignment requirements for any data on this processor.

39.23.2.5 CFE_ES_StackPointer_t

```
typedef void* CFE_ES_StackPointer_t
```

Type for the stack pointer of tasks.

This type is used in the CFE ES task API.

Definition at line 86 of file cfe_es_api_typedefs.h.

39.23.2.6 CFE_ES_TaskEntryFuncPtr_t

```
typedef void(* CFE_ES_TaskEntryFuncPtr_t) (void)
```

Required Prototype of Task Main Functions.

Definition at line 70 of file cfe_es_api_typedefs.h.

39.24 cfe/modules/core_api/fsw/inc/cfe_es_core_internal.h File Reference

```
#include "common_types.h"
#include "cfe_es_extern_typedefs.h"
```

Functions

- void CFE_ES_TaskMain (void)

  Entry Point for cFE Core Application.
- int32 CFE_ES_CDS_EarlyInit (void)

  Initializes the cFE core module API Library.
- int32 CFE_ES_RegisterCDSEx (CFE_ES_CDSHandle_t ∗HandlePtr, size_t UserBlockSize, const char ∗Name, bool CriticalTbl)

  Reserve space (or re-obtain previously reserved space) in the Critical Data Store (CDS)
- int32 CFE_ES_DeleteCDS (const char ∗CDSName, bool CalledByTblServices)

  Deletes the specified CDS from the CDS Registry and frees CDS Memory.

39.25 cfe/modules/core_api/fsw/inc/cfe_es_extern_typedefs.h File Reference

```
#include "common_types.h"
#include "cfe_resourceid_typedef.h"
#include "cfe_mission_cfg.h"
```

Data Structures

- struct CFE_ES_AppInfo

  Application Information.
- struct CFE_ES_TaskInfo

  Task Information.
- struct CFE_ES_CDSRegDumpRec

  CDS Register Dump Record.
- struct CFE_ES_BlockStats

  Block statistics.
- struct CFE_ES_MemPoolStats

  Memory Pool Statistics.

Macros

- #define CFE_ES_MEMOFFSET_C(x) ((CFE_ES_MemOffset_t)(x))
- #define CFE_ES_MEMADDRESS_C(x) ((CFE_ES_MemAddress_t)((cpuaddr)(x)&0xFFFFFFFF))

Typedefs

- typedef uint8 CFE_ES_LogMode_Enum_t

  Identifies handling of log messages after storage is filled.
- typedef uint8 CFE_ES_ExceptionAction_Enum_t

  Identifies action to take if exception occurs.
- typedef uint8 CFE_ES_AppType_Enum_t

  Identifies type of CFE application.
- typedef uint32 CFE_ES_RunStatus_Enum_t

  Run Status and Exit Status identifiers.
- typedef uint32 CFE_ES_SystemState_Enum_t

  The overall cFE System State.
- typedef uint8 CFE_ES_LogEntryType_Enum_t

  Type of entry in the Error and Reset (ER) Log.
- typedef uint32 CFE_ES_AppState_Enum_t

  Application Run State.
- typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_AppId_t

  A type for Application IDs.
- typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_TaskId_t

  A type for Task IDs.
- typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_LibId_t

A type for Library IDs.

- typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_CounterId_t

  A type for Counter IDs.
- typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_MemHandle_t

  Memory Handle type.
- typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_CDSHandle_t

  CDS Handle type.
- typedef uint16 CFE_ES_TaskPriority_Atom_t

  Type used for task priority in CFE ES as including the commands/telemetry messages.
- typedef uint32 CFE_ES_MemOffset_t

  Type used for memory sizes and offsets in commands and telemetry.
- typedef uint32 CFE_ES_MemAddress_t

  Type used for memory addresses in command and telemetry messages.
- typedef struct CFE_ES_AppInfo CFE_ES_AppInfo_t

  Application Information.
- typedef struct CFE_ES_TaskInfo CFE_ES_TaskInfo_t

  Task Information.
- typedef struct CFE_ES_CDSRegDumpRec CFE_ES_CDSRegDumpRec_t

  CDS Register Dump Record.
- typedef struct CFE_ES_BlockStats CFE_ES_BlockStats_t

  Block statistics.
- typedef struct CFE_ES_MemPoolStats CFE_ES_MemPoolStats_t

  Memory Pool Statistics.

Enumerations

- enum CFE_ES_LogMode { CFE_ES_LogMode_OVERWRITE = 0, CFE_ES_LogMode_DISCARD = 1 }

  Label definitions associated with CFE_ES_LogMode_Enum_t.
- enum CFE_ES_ExceptionAction { CFE_ES_ExceptionAction_RESTART_APP = 0, CFE_ES_ExceptionAction↩_PROC_RESTART = 1 }

  Label definitions associated with CFE_ES_ExceptionAction_Enum_t.
- enum CFE_ES_AppType { CFE_ES_AppType_CORE = 1, CFE_ES_AppType_EXTERNAL = 2, CFE_ES_App↩Type_LIBRARY = 3 }

  Label definitions associated with CFE_ES_AppType_Enum_t.
- enum CFE_ES_RunStatus {
  CFE_ES_RunStatus_UNDEFINED = 0, CFE_ES_RunStatus_APP_RUN = 1, CFE_ES_RunStatus_APP_EXIT = 2, CFE_ES_RunStatus_APP_ERROR = 3,
  CFE_ES_RunStatus_SYS_EXCEPTION = 4, CFE_ES_RunStatus_SYS_RESTART = 5, CFE_ES_RunStatus↩_SYS_RELOAD = 6, CFE_ES_RunStatus_SYS_DELETE = 7,
  CFE_ES_RunStatus_CORE_APP_INIT_ERROR = 8, CFE_ES_RunStatus_CORE_APP_RUNTIME_ERROR = 9, CFE_ES_RunStatus_MAX }

  Label definitions associated with CFE_ES_RunStatus_Enum_t.
- enum CFE_ES_SystemState {
  CFE_ES_SystemState_UNDEFINED = 0, CFE_ES_SystemState_EARLY_INIT = 1, CFE_ES_SystemState_C↩ORE_STARTUP = 2, CFE_ES_SystemState_CORE_READY = 3,
  CFE_ES_SystemState_APPS_INIT = 4, CFE_ES_SystemState_OPERATIONAL = 5, CFE_ES_SystemState_↩SHUTDOWN = 6, CFE_ES_SystemState_MAX }

  Label definitions associated with CFE_ES_SystemState_Enum_t.

- enum CFE_ES_LogEntryType { CFE_ES_LogEntryType_CORE = 1, CFE_ES_LogEntryType_APPLICATION = 2 }

  Label definitions associated with CFE_ES_LogEntryType_Enum_t.
- enum CFE_ES_AppState { CFE_ES_AppState_UNDEFINED = 0, CFE_ES_AppState_EARLY_INIT = 1, CFE_ES_AppState_LATE_INIT = 2, CFE_ES_AppState_RUNNING = 3, CFE_ES_AppState_WAITING = 4, CFE_ES_AppState_STOPPED = 5, CFE_ES_AppState_MAX }

  Label definitions associated with CFE_ES_AppState_Enum_t.

39.25.1 Macro Definition Documentation

39.25.1.1 CFE_ES_MEMADDRESS_C

```
#define CFE_ES_MEMADDRESS_C(
            x ) ((CFE_ES_MemAddress_t)((cpuaddr)(x)&0xFFFFFFFF))
```

Definition at line 429 of file cfe_es_extern_typedefs.h.

39.25.1.2 CFE_ES_MEMOFFSET_C

```
#define CFE_ES_MEMOFFSET_C(
            x ) ((CFE_ES_MemOffset_t)(x))
```

Definition at line 398 of file cfe_es_extern_typedefs.h.

39.25.2 Typedef Documentation

39.25.2.1 CFE_ES_AppId_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_AppId_t
```

A type for Application IDs.

This is the type that is used for any API accepting or returning an App ID

Definition at line 332 of file cfe_es_extern_typedefs.h.

39.25.2.2 CFE_ES_AppInfo_t

```
typedef struct CFE_ES_AppInfo CFE_ES_AppInfo_t
```

Application Information.

Structure that is used to provide information about an app. It is primarily used for the QueryOne and QueryAll Commands.

While this structure is primarily intended for Application info, it can also represent Library information where only a subset of the information applies.

39.25.2.3 CFE_ES_AppState_Enum_t

```
typedef uint32 CFE_ES_AppState_Enum_t
```

Application Run State.

The normal progression of APP states: UNDEFINED -> EARLY_INIT -> LATE_INIT -> RUNNING -> WAITING -> STOPPED

Note

These are defined in order so that relational comparisons e.g. if (STATEA < STATEB) are possible

See also

enum CFE_ES_AppState

Definition at line 325 of file cfe_es_extern_typedefs.h.

39.25.2.4 CFE_ES_AppType_Enum_t

```
typedef uint8 CFE_ES_AppType_Enum_t
```

Identifies type of CFE application.

See also

enum CFE_ES_AppType

Definition at line 116 of file cfe_es_extern_typedefs.h.

39.25.2.5 CFE_ES_BlockStats_t

```
typedef struct CFE_ES_BlockStats CFE_ES_BlockStats_t
```

Block statistics.

Sub-Structure that is used to provide information about a specific block size/bucket within a memory pool.

39.25.2.6 CFE_ES_CDSHandle_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_CDSHandle_t
```

CDS Handle type.

Data type used to hold Handles of Critical Data Stores. See CFE_ES_RegisterCDS

Definition at line 368 of file cfe_es_extern_typedefs.h.

39.25.2.7 CFE_ES_CDSRegDumpRec_t

```
typedef struct CFE_ES_CDSRegDumpRec CFE_ES_CDSRegDumpRec_t
```

CDS Register Dump Record.

Structure that is used to provide information about a critical data store. It is primarily used for the Dump CDS registry (CFE_ES_DUMP_CDS_REGISTRY_CC) command.

Note

There is not currently a telemetry message directly containing this data structure, but it does define the format of the data file generated by the Dump CDS registry command. Therefore it should be considered part of the overall telemetry interface.

39.25.2.8 CFE_ES_CounterId_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_CounterId_t
```

A type for Counter IDs.

This is the type that is used for any API accepting or returning an Counter ID

Definition at line 353 of file cfe_es_extern_typedefs.h.

39.25.2.9 CFE_ES_ExceptionAction_Enum_t

```
typedef uint8 CFE_ES_ExceptionAction_Enum_t
```

Identifies action to take if exception occurs.

See also

enum CFE_ES_ExceptionAction

Definition at line 86 of file cfe_es_extern_typedefs.h.

39.25.2.10 CFE_ES_LibId_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_LibId_t
```

A type for Library IDs.

This is the type that is used for any API accepting or returning a Lib ID

Definition at line 346 of file cfe_es_extern_typedefs.h.

39.25.2.11 CFE_ES_LogEntryType_Enum_t

```
typedef uint8 CFE_ES_LogEntryType_Enum_t
```

Type of entry in the Error and Reset (ER) Log.

See also

enum CFE_ES_LogEntryType

Definition at line 270 of file cfe_es_extern_typedefs.h.

39.25.2.12 CFE_ES_LogMode_Enum_t

```
typedef uint8 CFE_ES_LogMode_Enum_t
```

Identifies handling of log messages after storage is filled.

See also

enum CFE_ES_LogMode

Definition at line 61 of file cfe_es_extern_typedefs.h.

39.25.2.13 CFE_ES_MemAddress_t

```
typedef uint32 CFE_ES_MemAddress_t
```

Type used for memory addresses in command and telemetry messages.

For backward compatibility with existing CFE code this should be uint32, but if running on a 64-bit platform, addresses in telemetry will be truncated to 32 bits and therefore will not be valid.

On 64-bit platforms this can be a 64-bit address which will allow the full memory address in commands and telemetry, but this will break compatibility with existing control systems, and may also change the alignment/padding of messages.

In either case this must be an unsigned type.

FSW code should access this value via the macros provided, which converts to the native "cpuaddr" type provided by OSAL. This macro provides independence between the message representation and local representation of a memory address.

Definition at line 419 of file cfe_es_extern_typedefs.h.

39.25.2.14 CFE_ES_MemHandle_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_MemHandle_t
```

Memory Handle type.

Data type used to hold Handles of Memory Pools created via CFE_ES_PoolCreate and CFE_ES_PoolCreateNoSem

Definition at line 361 of file cfe_es_extern_typedefs.h.

39.25.2.15 CFE_ES_MemOffset_t

```
typedef uint32 CFE_ES_MemOffset_t
```

Type used for memory sizes and offsets in commands and telemetry.

For backward compatibility with existing CFE code this should be uint32, but all telemetry information will be limited to 4GB in size as a result.

On 64-bit platforms this can be a 64-bit value which will allow larger memory objects, but this will break compatibility with existing control systems, and may also change the alignment/padding of messages.

In either case this must be an unsigned type.

Definition at line 392 of file cfe_es_extern_typedefs.h.

39.25.2.16 CFE_ES_MemPoolStats_t

```
typedef struct CFE_ES_MemPoolStats CFE_ES_MemPoolStats_t
```

Memory Pool Statistics.

Structure that is used to provide information about a memory pool. Used by the Memory Pool Stats telemetry message.

See also

CFE_ES_SEND_MEM_POOL_STATS_CC

39.25.2.17 CFE_ES_RunStatus_Enum_t

```
typedef uint32 CFE_ES_RunStatus_Enum_t
```

Run Status and Exit Status identifiers.

See also

enum CFE_ES_RunStatus

Definition at line 186 of file cfe_es_extern_typedefs.h.

39.25.2.18 CFE_ES_SystemState_Enum_t

```
typedef uint32 CFE_ES_SystemState_Enum_t
```

The overall cFE System State.

These values are used with the CFE_ES_WaitForSystemState API call to synchronize application startup.

Note

These are defined in order so that relational comparisons e.g. if (STATEA $<$ STATEB) are possible

See also

enum CFE_ES_SystemState

Definition at line 245 of file cfe_es_extern_typedefs.h.

39.25.2.19 CFE_ES_TaskId_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_TaskId_t
```

A type for Task IDs.

This is the type that is used for any API accepting or returning a Task ID

Definition at line 339 of file cfe_es_extern_typedefs.h.

39.25.2.20 CFE_ES_TaskInfo_t

```
typedef struct CFE_ES_TaskInfo CFE_ES_TaskInfo_t
```

Task Information.

Structure that is used to provide information about a task. It is primarily used for the Query All Tasks (CFE_ES_QUE↩RY_ALL_TASKS_CC) command.

Note

There is not currently a telemetry message directly containing this data structure, but it does define the format of the data file generated by the Query All Tasks command. Therefore it should be considered part of the overall telemetry interface.

39.25.2.21 CFE_ES_TaskPriority_Atom_t

```
typedef uint16 CFE_ES_TaskPriority_Atom_t
```

Type used for task priority in CFE ES as including the commands/telemetry messages.

Note

the valid range is only 0-255 (same as OSAL) but a wider type is used for backward compatibility in binary formats of messages.

Definition at line 378 of file cfe_es_extern_typedefs.h.

39.25.3 Enumeration Type Documentation

39.25.3.1 CFE_ES_AppState

```
enum CFE_ES_AppState
```

Label definitions associated with CFE_ES_AppState_Enum_t.

Enumerator

CFE_ES_AppState_UNDEFINED	Initial state before app thread is started.
CFE_ES_AppState_EARLY_INIT	App thread has started, app performing early initialization of its own data.
CFE_ES_AppState_LATE_INIT	Early/Local initialization is complete. First sync point.
CFE_ES_AppState_RUNNING	All initialization is complete. Second sync point.
CFE_ES_AppState_WAITING	Application is waiting on a Restart/Reload/Delete request.
CFE_ES_AppState_STOPPED	Application is stopped.
CFE_ES_AppState_MAX	Reserved entry, marker for the maximum state.

Definition at line 275 of file cfe_es_extern_typedefs.h.

39.25.3.2 CFE_ES_AppType

`enum CFE_ES_AppType`

Label definitions associated with CFE_ES_AppType_Enum_t.

Enumerator

CFE_ES_AppType_CORE	CFE core application.
CFE_ES_AppType_EXTERNAL	CFE external application.
CFE_ES_AppType_LIBRARY	CFE library.

Definition at line 91 of file cfe_es_extern_typedefs.h.

39.25.3.3 CFE_ES_ExceptionAction

`enum CFE_ES_ExceptionAction`

Label definitions associated with CFE_ES_ExceptionAction_Enum_t.

Enumerator

CFE_ES_ExceptionAction_RESTART_APP	Restart application if exception occurs.
CFE_ES_ExceptionAction_PROC_RESTART	Restart processor if exception occurs.

Definition at line 66 of file cfe_es_extern_typedefs.h.

39.25.3.4 CFE_ES_LogEntryType

enum CFE_ES_LogEntryType

Label definitions associated with CFE_ES_LogEntryType_Enum_t.

Enumerator

CFE_ES_LogEntryType_CORE	Log entry from a core subsystem.
CFE_ES_LogEntryType_APPLICATION	Log entry from an application.

Definition at line 250 of file cfe_es_extern_typedefs.h.

39.25.3.5 CFE_ES_LogMode

enum CFE_ES_LogMode

Label definitions associated with CFE_ES_LogMode_Enum_t.

Enumerator

CFE_ES_LogMode_OVERWRITE	Overwrite Log Mode.
CFE_ES_LogMode_DISCARD	Discard Log Mode.

Definition at line 41 of file cfe_es_extern_typedefs.h.

39.25.3.6 CFE_ES_RunStatus

enum CFE_ES_RunStatus

Label definitions associated with CFE_ES_RunStatus_Enum_t.

Enumerator

CFE_ES_RunStatus_UNDEFINED	Reserved value, should not be used.
CFE_ES_RunStatus_APP_RUN	Indicates that the Application should continue to run.
CFE_ES_RunStatus_APP_EXIT	Indicates that the Application wants to exit normally.
CFE_ES_RunStatus_APP_ERROR	Indicates that the Application is quitting with an error.
CFE_ES_RunStatus_SYS_EXCEPTION	The cFE App caused an exception.
CFE_ES_RunStatus_SYS_RESTART	The system is requesting a restart of the cFE App.
CFE_ES_RunStatus_SYS_RELOAD	The system is requesting a reload of the cFE App.
CFE_ES_RunStatus_SYS_DELETE	The system is requesting that the cFE App is stopped.
CFE_ES_RunStatus_CORE_APP_INIT_ERROR	Indicates that the Core Application could not Init.
CFE_ES_RunStatus_CORE_APP_RUNTIME_ERROR	Indicates that the Core Application had a runtime failure.
CFE_ES_RunStatus_MAX	Reserved value, marker for the maximum state.

Definition at line 121 of file cfe_es_extern_typedefs.h.

39.25.3.7 CFE_ES_SystemState

```
enum CFE_ES_SystemState
```

Label definitions associated with CFE_ES_SystemState_Enum_t.

Enumerator

CFE_ES_SystemState_UNDEFINED	reserved
CFE_ES_SystemState_EARLY_INIT	single threaded mode while setting up CFE itself
CFE_ES_SystemState_CORE_STARTUP	core apps (CFE_ES_ObjectTable) are starting (multi-threaded)
CFE_ES_SystemState_CORE_READY	core is ready, starting other external apps/libraries (if any)
CFE_ES_SystemState_APPS_INIT	startup apps have all completed their early init, but not necessarily operational yet
CFE_ES_SystemState_OPERATIONAL	normal operation mode; all apps are RUNNING
CFE_ES_SystemState_SHUTDOWN	reserved for future use, all apps would be STOPPED
CFE_ES_SystemState_MAX	Reserved value, marker for the maximum state.

Definition at line 191 of file cfe_es_extern_typedefs.h.

39.26 cfe/modules/core_api/fsw/inc/cfe_evs.h File Reference

```
#include "common_types.h"
#include "cfe_error.h"
#include "cfe_evs_api_typedefs.h"
#include "cfe_es_api_typedefs.h"
#include "cfe_time_api_typedefs.h"
```

Macros

- #define CFE_EVS_Send(E, T, ...) CFE_EVS_SendEvent((E), CFE_EVS_EventType_##T, __VA_ARGS__)
- #define CFE_EVS_SendDbg(E, ...) CFE_EVS_Send(E, DEBUG, __VA_ARGS__)
- #define CFE_EVS_SendInfo(E, ...) CFE_EVS_Send(E, INFORMATION, __VA_ARGS__)
- #define CFE_EVS_SendErr(E, ...) CFE_EVS_Send(E, ERROR, __VA_ARGS__)
- #define CFE_EVS_SendCrit(E, ...) CFE_EVS_Send(E, CRITICAL, __VA_ARGS__)

Functions

- CFE_Status_t CFE_EVS_Register (const void ∗Filters, uint16 NumFilteredEvents, uint16 FilterScheme)

  Register an application for receiving event services.
- CFE_Status_t CFE_EVS_Unregister (void)

  Cleanup internal structures used by the event manager for the calling Application.
- CFE_Status_t CFE_EVS_SendEvent (uint16 EventID, uint16 EventType, const char ∗Spec,...) OS_PRINTF(3

  Generate a software event.
- CFE_Status_t CFE_Status_t CFE_EVS_SendEventWithAppID (uint16 EventID, uint16 EventType, CFE_ES_↩ AppId_t AppID, const char ∗Spec,...) OS_PRINTF(4

  Generate a software event given the specified Application ID.
- CFE_Status_t CFE_Status_t CFE_Status_t CFE_EVS_SendTimedEvent (CFE_TIME_SysTime_t Time, uint16 EventID, uint16 EventType, const char ∗Spec,...) OS_PRINTF(4

  Generate a software event with a specific time tag.
- CFE_Status_t CFE_EVS_ResetFilter (int16 EventID)

  Resets the calling application's event filter for a single event ID.
- CFE_Status_t CFE_EVS_ResetAllFilters (void)

  Resets all of the calling application's event filters.

39.26.1 Macro Definition Documentation

39.26.1.1 CFE_EVS_Send

```
#define CFE_EVS_Send(
            E,
            T,
            ...  ) CFE_EVS_SendEvent((E), CFE_EVS_EventType_##T, __VA_ARGS__)
```

Definition at line 48 of file cfe_evs.h.

39.26.1.2 CFE_EVS_SendCrit

```
#define CFE_EVS_SendCrit(
            E,
            ...  ) CFE_EVS_Send(E, CRITICAL, __VA_ARGS__)
```

Definition at line 52 of file cfe_evs.h.

39.26.1.3 CFE_EVS_SendDbg

```
#define CFE_EVS_SendDbg(
            E,
            ...  ) CFE_EVS_Send(E, DEBUG, __VA_ARGS__)
```

Definition at line 49 of file cfe_evs.h.

39.26.1.4 CFE_EVS_SendErr

```
#define CFE_EVS_SendErr(
            E,
            ...  ) CFE_EVS_Send(E, ERROR, __VA_ARGS__)
```

Definition at line 51 of file cfe_evs.h.

39.26.1.5 CFE_EVS_SendInfo

```
#define CFE_EVS_SendInfo(
            E,
            ...  ) CFE_EVS_Send(E, INFORMATION, __VA_ARGS__)
```

Definition at line 50 of file cfe_evs.h.

39.27 cfe/modules/core_api/fsw/inc/cfe_evs_api_typedefs.h File Reference

```
#include "common_types.h"
#include "cfe_evs_extern_typedefs.h"
```

Data Structures

- struct CFE_EVS_BinFilter

  Event message filter defintion structure.

Macros

Common Event Filter Mask Values

- #define CFE_EVS_NO_FILTER 0x0000

  Stops any filtering. All messages are sent.
- #define CFE_EVS_FIRST_ONE_STOP 0xFFFF

  Sends the first event. All remaining messages are filtered.
- #define CFE_EVS_FIRST_TWO_STOP 0xFFFE

  Sends the first 2 events. All remaining messages are filtered.
- #define CFE_EVS_FIRST_4_STOP 0xFFFC

  Sends the first 4 events. All remaining messages are filtered.
- #define CFE_EVS_FIRST_8_STOP 0xFFF8

  Sends the first 8 events. All remaining messages are filtered.
- #define CFE_EVS_FIRST_16_STOP 0xFFF0

  Sends the first 16 events. All remaining messages are filtered.
- #define CFE_EVS_FIRST_32_STOP 0xFFE0

  Sends the first 32 events. All remaining messages are filtered.
- #define CFE_EVS_FIRST_64_STOP 0xFFC0

  Sends the first 64 events. All remaining messages are filtered.
- #define CFE_EVS_EVERY_OTHER_ONE 0x0001

  Sends every other event.
- #define CFE_EVS_EVERY_OTHER_TWO 0x0002

  Sends two, filters one, sends two, filters one, etc.
- #define CFE_EVS_EVERY_FOURTH_ONE 0x0003

  Sends every fourth event message. All others are filtered.

Typedefs

- typedef struct CFE_EVS_BinFilter CFE_EVS_BinFilter_t

  Event message filter defintion structure.

39.27.1 Macro Definition Documentation

39.27.1.1 CFE_EVS_EVERY_FOURTH_ONE

```
#define CFE_EVS_EVERY_FOURTH_ONE 0x0003
```

Sends every fourth event message. All others are filtered.

Definition at line 54 of file cfe_evs_api_typedefs.h.

39.27.1.2 CFE_EVS_EVERY_OTHER_ONE

```
#define CFE_EVS_EVERY_OTHER_ONE 0x0001
```

Sends every other event.

Definition at line 52 of file cfe_evs_api_typedefs.h.

39.27.1.3 CFE_EVS_EVERY_OTHER_TWO

```
#define CFE_EVS_EVERY_OTHER_TWO 0x0002
```

Sends two, filters one, sends two, filters one, etc.

Definition at line 53 of file cfe_evs_api_typedefs.h.

39.27.1.4 CFE_EVS_FIRST_16_STOP

```
#define CFE_EVS_FIRST_16_STOP 0xFFF0
```

Sends the first 16 events. All remaining messages are filtered.

Definition at line 49 of file cfe_evs_api_typedefs.h.

39.27.1.5 CFE_EVS_FIRST_32_STOP

```
#define CFE_EVS_FIRST_32_STOP 0xFFE0
```

Sends the first 32 events. All remaining messages are filtered.

Definition at line 50 of file cfe_evs_api_typedefs.h.

39.27.1.6 CFE_EVS_FIRST_4_STOP

```
#define CFE_EVS_FIRST_4_STOP 0xFFFC
```

Sends the first 4 events. All remaining messages are filtered.

Definition at line 47 of file cfe_evs_api_typedefs.h.

39.27.1.7 CFE_EVS_FIRST_64_STOP

```
#define CFE_EVS_FIRST_64_STOP 0xFFC0
```

Sends the first 64 events. All remaining messages are filtered.

Definition at line 51 of file cfe_evs_api_typedefs.h.

39.27.1.8 CFE_EVS_FIRST_8_STOP

```
#define CFE_EVS_FIRST_8_STOP 0xFFF8
```

Sends the first 8 events. All remaining messages are filtered.

Definition at line 48 of file cfe_evs_api_typedefs.h.

39.27.1.9 CFE_EVS_FIRST_ONE_STOP

```
#define CFE_EVS_FIRST_ONE_STOP 0xFFFF
```

Sends the first event. All remaining messages are filtered.

Definition at line 45 of file cfe_evs_api_typedefs.h.

39.27.1.10 CFE_EVS_FIRST_TWO_STOP

```
#define CFE_EVS_FIRST_TWO_STOP 0xFFFE
```

Sends the first 2 events. All remaining messages are filtered.

Definition at line 46 of file cfe_evs_api_typedefs.h.

39.27.1.11 CFE_EVS_NO_FILTER

```
#define CFE_EVS_NO_FILTER 0x0000
```

Stops any filtering. All messages are sent.

Definition at line 44 of file cfe_evs_api_typedefs.h.

39.27.2 Typedef Documentation

39.27.2.1 CFE_EVS_BinFilter_t

```
typedef struct CFE_EVS_BinFilter CFE_EVS_BinFilter_t
```

Event message filter defintion structure.

39.28 cfe/modules/core_api/fsw/inc/cfe_evs_core_internal.h File Reference

```
#include "common_types.h"
#include "cfe_es_extern_typedefs.h"
```

Functions

- void CFE_EVS_TaskMain (void)

  Entry Point for cFE Core Application.
- int32 CFE_EVS_EarlyInit (void)

  Initializes the cFE core module API Library.
- int32 CFE_EVS_CleanUpApp (CFE_ES_AppId_t AppId)

  Removes EVS resources associated with specified Application.

39.29 cfe/modules/core_api/fsw/inc/cfe_evs_extern_typedefs.h File Reference

```
#include "common_types.h"
```

Typedefs

- typedef uint8 CFE_EVS_MsgFormat_Enum_t

  Identifies format of log messages.
- typedef uint8 CFE_EVS_LogMode_Enum_t

  Identifies handling of log messages after storage is filled.
- typedef uint16 CFE_EVS_EventType_Enum_t

  Identifies type of event message.
- typedef uint8 CFE_EVS_EventFilter_Enum_t

  Identifies event filter schemes.
- typedef uint8 CFE_EVS_EventOutput_Enum_t

  Identifies event output port.

Enumerations

- enum CFE_EVS_MsgFormat { CFE_EVS_MsgFormat_SHORT = 0, CFE_EVS_MsgFormat_LONG = 1 }

  Label definitions associated with CFE_EVS_MsgFormat_Enum_t.
- enum CFE_EVS_LogMode { CFE_EVS_LogMode_OVERWRITE = 0, CFE_EVS_LogMode_DISCARD = 1 }

  Label definitions associated with CFE_EVS_LogMode_Enum_t.
- enum CFE_EVS_EventType { CFE_EVS_EventType_DEBUG = 1, CFE_EVS_EventType_INFORMATION = 2, CFE_EVS_EventType_ERROR = 3, CFE_EVS_EventType_CRITICAL = 4 }

  Label definitions associated with CFE_EVS_EventType_Enum_t.
- enum CFE_EVS_EventFilter { CFE_EVS_EventFilter_BINARY = 0 }

  Label definitions associated with CFE_EVS_EventFilter_Enum_t.
- enum CFE_EVS_EventOutput { CFE_EVS_EventOutput_PORT1 = 1, CFE_EVS_EventOutput_PORT2 = 2, CFE_EVS_EventOutput_PORT3 = 3, CFE_EVS_EventOutput_PORT4 = 4 }

  Label definitions associated with CFE_EVS_EventOutput_Enum_t.

39.29.1 Typedef Documentation

39.29.1.1 CFE_EVS_EventFilter_Enum_t

```
typedef uint8 CFE_EVS_EventFilter_Enum_t
```

Identifies event filter schemes.

See also

enum CFE_EVS_EventFilter

Definition at line 139 of file cfe_evs_extern_typedefs.h.

39.29.1.2 CFE_EVS_EventOutput_Enum_t

```
typedef uint8 CFE_EVS_EventOutput_Enum_t
```

Identifies event output port.

See also

enum CFE_EVS_EventOutput

Definition at line 174 of file cfe_evs_extern_typedefs.h.

39.29.1.3 CFE_EVS_EventType_Enum_t

```
typedef uint16 CFE_EVS_EventType_Enum_t
```

Identifies type of event message.

See also

enum CFE_EVS_EventType

Definition at line 119 of file cfe_evs_extern_typedefs.h.

39.29.1.4 CFE_EVS_LogMode_Enum_t

```
typedef uint8 CFE_EVS_LogMode_Enum_t
```

Identifies handling of log messages after storage is filled.

See also

enum CFE_EVS_LogMode

Definition at line 84 of file cfe_evs_extern_typedefs.h.

39.29.1.5 CFE_EVS_MsgFormat_Enum_t

```
typedef uint8 CFE_EVS_MsgFormat_Enum_t
```

Identifies format of log messages.

See also

enum CFE_EVS_MsgFormat

Definition at line 59 of file cfe_evs_extern_typedefs.h.

39.29.2 Enumeration Type Documentation

39.29.2.1 CFE_EVS_EventFilter

```
enum CFE_EVS_EventFilter
```

Label definitions associated with CFE_EVS_EventFilter_Enum_t.

Enumerator

CFE_EVS_EventFilter_BINARY	Binary event filter.

Definition at line 124 of file cfe_evs_extern_typedefs.h.

39.29.2.2 CFE_EVS_EventOutput

`enum CFE_EVS_EventOutput`

Label definitions associated with CFE_EVS_EventOutput_Enum_t.

Enumerator

CFE_EVS_EventOutput_PORT1	Output Port 1.
CFE_EVS_EventOutput_PORT2	Output Port 2.
CFE_EVS_EventOutput_PORT3	Output Port 3.
CFE_EVS_EventOutput_PORT4	Output Port 4.

Definition at line 144 of file cfe_evs_extern_typedefs.h.

39.29.2.3 CFE_EVS_EventType

`enum CFE_EVS_EventType`

Label definitions associated with CFE_EVS_EventType_Enum_t.

Enumerator

CFE_EVS_EventType_DEBUG	Events that are intended only for debugging, not nominal operations.
CFE_EVS_EventType_INFORMATION	Events that identify a state change or action that is not an error.
CFE_EVS_EventType_ERROR	Events that identify an error but are not catastrophic (e.g. - bad command.
CFE_EVS_EventType_CRITICAL	Events that identify errors that are unrecoverable autonomously.

Definition at line 89 of file cfe_evs_extern_typedefs.h.

39.29.2.4 CFE_EVS_LogMode

`enum CFE_EVS_LogMode`

Label definitions associated with CFE_EVS_LogMode_Enum_t.

Enumerator

CFE_EVS_LogMode_OVERWRITE	Overwrite Log Mode.
CFE_EVS_LogMode_DISCARD	Discard Log Mode.

Definition at line 64 of file cfe_evs_extern_typedefs.h.

39.29.2.5 CFE_EVS_MsgFormat

`enum CFE_EVS_MsgFormat`

Label definitions associated with CFE_EVS_MsgFormat_Enum_t.

Enumerator

CFE_EVS_MsgFormat_SHORT	Short Format Messages.
CFE_EVS_MsgFormat_LONG	Long Format Messages.

Definition at line 39 of file cfe_evs_extern_typedefs.h.

39.30 cfe/modules/core_api/fsw/inc/cfe_fs.h File Reference

```
#include "common_types.h"
#include "osconfig.h"
#include "cfe_platform_cfg.h"
#include "cfe_error.h"
#include "cfe_fs_api_typedefs.h"
#include "cfe_fs_extern_typedefs.h"
```

Functions

- CFE_Status_t CFE_FS_ReadHeader (CFE_FS_Header_t *Hdr, osal_id_t FileDes)

  Read the contents of the Standard cFE File Header.
- void CFE_FS_InitHeader (CFE_FS_Header_t *Hdr, const char *Description, uint32 SubType)

  Initializes the contents of the Standard cFE File Header.
- CFE_Status_t CFE_FS_WriteHeader (osal_id_t FileDes, CFE_FS_Header_t *Hdr)

  Write the specified Standard cFE File Header to the specified file.
- CFE_Status_t CFE_FS_SetTimestamp (osal_id_t FileDes, CFE_TIME_SysTime_t NewTimestamp)

  Modifies the Time Stamp field in the Standard cFE File Header for the specified file.
- const char * CFE_FS_GetDefaultMountPoint (CFE_FS_FileCategory_t FileCategory)

  Get the default virtual mount point for a file category.
- const char * CFE_FS_GetDefaultExtension (CFE_FS_FileCategory_t FileCategory)

Get the default filename extension for a file category.

- int32 CFE_FS_ParseInputFileNameEx (char ∗OutputBuffer, const char ∗InputBuffer, size_t OutputBufSize, size↩_t InputBufSize, const char ∗DefaultInput, const char ∗DefaultPath, const char ∗DefaultExtension)

  Parse a filename input from an input buffer into a local buffer.
- int32 CFE_FS_ParseInputFileName (char ∗OutputBuffer, const char ∗InputName, size_t OutputBufSize, CFE_↩FS_FileCategory_t FileCategory)

  Parse a filename string from the user into a local buffer.
- CFE_Status_t CFE_FS_ExtractFilenameFromPath (const char ∗OriginalPath, char ∗FileNameOnly)

  Extracts the filename from a unix style path and filename string.
- int32 CFE_FS_BackgroundFileDumpRequest (CFE_FS_FileWriteMetaData_t ∗Meta)

  Register a background file dump request.
- bool CFE_FS_BackgroundFileDumpIsPending (const CFE_FS_FileWriteMetaData_t ∗Meta)

  Query if a background file write request is currently pending.
- bool CFE_FS_RunBackgroundFileDump (uint32 ElapsedTime, void ∗Arg)

  Execute the background file write job(s)

39.31 cfe/modules/core_api/fsw/inc/cfe_fs_api_typedefs.h File Reference

```
#include "common_types.h"
#include "osconfig.h"
#include "cfe_fs_extern_typedefs.h"
```

Data Structures

- struct CFE_FS_FileWriteMetaData

  External Metadata/State object associated with background file writes.

Typedefs

- typedef bool(∗ CFE_FS_FileWriteGetData_t) (void ∗Meta, uint32 RecordNum, void ∗∗Buffer, size_t ∗BufSize)
- typedef void(∗ CFE_FS_FileWriteOnEvent_t) (void ∗Meta, CFE_FS_FileWriteEvent_t Event, int32 Status, uint32 RecordNum, size_t BlockSize, size_t Position)
- typedef struct CFE_FS_FileWriteMetaData CFE_FS_FileWriteMetaData_t

  External Metadata/State object associated with background file writes.

Enumerations

- enum CFE_FS_FileCategory_t {
  CFE_FS_FileCategory_UNKNOWN, CFE_FS_FileCategory_DYNAMIC_MODULE, CFE_FS_FileCategory_BI↩NARY_DATA_DUMP, CFE_FS_FileCategory_TEXT_LOG,
  CFE_FS_FileCategory_SCRIPT, CFE_FS_FileCategory_TEMP, CFE_FS_FileCategory_MAX }

  Generalized file types/categories known to FS.
- enum CFE_FS_FileWriteEvent_t {
  CFE_FS_FileWriteEvent_UNDEFINED, CFE_FS_FileWriteEvent_COMPLETE, CFE_FS_FileWriteEvent_CR↩EATE_ERROR, CFE_FS_FileWriteEvent_HEADER_WRITE_ERROR,
  CFE_FS_FileWriteEvent_RECORD_WRITE_ERROR, CFE_FS_FileWriteEvent_MAX }

39.31.1 Typedef Documentation

39.31.1.1 CFE_FS_FileWriteGetData_t

```
typedef bool(* CFE_FS_FileWriteGetData_t) (void *Meta, uint32 RecordNum, void **Buffer, size_↩
t *BufSize)
```

Data Getter routine provided by requester

Outputs a data block. Should return true if the file is complete (last record/EOF), otherwise return false.

Definition at line 91 of file cfe_fs_api_typedefs.h.

39.31.1.2 CFE_FS_FileWriteMetaData_t

```
typedef struct CFE_FS_FileWriteMetaData CFE_FS_FileWriteMetaData_t
```

External Metadata/State object associated with background file writes.

Applications intending to schedule background file write jobs should instantiate this object in static/global data memory. This keeps track of the state of the file write request(s).

39.31.1.3 CFE_FS_FileWriteOnEvent_t

```
typedef void(* CFE_FS_FileWriteOnEvent_t) (void *Meta, CFE_FS_FileWriteEvent_t Event, int32 Status,
uint32 RecordNum, size_t BlockSize, size_t Position)
```

Event generator routine provided by requester

Invoked from certain points in the file write process. Implementation may invoke CFE_EVS_SendEvent() appropriately to inform of progress.

Definition at line 99 of file cfe_fs_api_typedefs.h.

39.31.2 Enumeration Type Documentation

39.31.2.1 CFE_FS_FileCategory_t

```
enum CFE_FS_FileCategory_t
```

Generalized file types/categories known to FS.

This defines different categories of files, where they may reside in different default locations of the virtualized file system.

This is different from, and should not be confused with, the "SubType" field in the FS header. This value is only used at runtime for FS APIs and should not actually appear in any output file or message.

Enumerator

CFE_FS_FileCategory_UNKNOWN	Placeholder, unknown file category
CFE_FS_FileCategory_DYNAMIC_MODULE	Dynamically loadable apps/libraries (e.g. .so, .o, .dll, etc)
CFE_FS_FileCategory_BINARY_DATA_DUMP	Binary log file generated by various data dump commands
CFE_FS_FileCategory_TEXT_LOG	Text-based log file generated by various commands
CFE_FS_FileCategory_SCRIPT	Text-based Script files (e.g. ES startup script)
CFE_FS_FileCategory_TEMP	Temporary/Ephemeral files
CFE_FS_FileCategory_MAX	Placeholder, keep last

Definition at line 53 of file cfe_fs_api_typedefs.h.

39.31.2.2 CFE_FS_FileWriteEvent_t

```
enum CFE_FS_FileWriteEvent_t
```

Enumerator

CFE_FS_FileWriteEvent_UNDEFINED	
CFE_FS_FileWriteEvent_COMPLETE	File is completed successfully
CFE_FS_FileWriteEvent_CREATE_ERROR	Unable to create/open file
CFE_FS_FileWriteEvent_HEADER_WRITE_ERROR	Unable to write FS header
CFE_FS_FileWriteEvent_RECORD_WRITE_ERROR	Unable to write data record
CFE_FS_FileWriteEvent_MAX	

Definition at line 73 of file cfe_fs_api_typedefs.h.

39.32 cfe/modules/core_api/fsw/inc/cfe_fs_core_internal.h File Reference

```
#include "common_types.h"
```

Functions

- int32 CFE_FS_EarlyInit (void)

  Initializes the cFE core module API Library.

39.33 cfe/modules/core_api/fsw/inc/cfe_fs_extern_typedefs.h File Reference

```
#include "common_types.h"
```

Data Structures

- struct CFE_FS_Header

  Standard cFE File header structure definition.

Macros

- #define CFE_FS_HDR_DESC_MAX_LEN 32

  Max length of description field in a standard cFE File Header.
- #define CFE_FS_FILE_CONTENT_ID 0x63464531

  Magic Number for cFE compliant files (= 'cFE1')

Typedefs

- typedef uint32 CFE_FS_SubType_Enum_t

  Content descriptor for File Headers.
- typedef struct CFE_FS_Header CFE_FS_Header_t

  Standard cFE File header structure definition.

Enumerations

- enum CFE_FS_SubType {
  CFE_FS_SubType_ES_ERLOG = 1, CFE_FS_SubType_ES_SYSLOG = 2, CFE_FS_SubType_ES_QUERYALL = 3, CFE_FS_SubType_ES_PERFDATA = 4,
  CFE_FS_SubType_ES_SHELL = 5, CFE_FS_SubType_ES_CDS_REG = 6, CFE_FS_SubType_TBL_REG = 9, CFE_FS_SubType_TBL_IMG = 8,
  CFE_FS_SubType_EVS_APPDATA = 15, CFE_FS_SubType_EVS_EVENTLOG = 16, CFE_FS_SubType_SB↩_PIPEDATA = 20, CFE_FS_SubType_SB_ROUTEDATA = 21,
  CFE_FS_SubType_SB_MAPDATA = 22, CFE_FS_SubType_ES_QUERYALLTASKS = 23 }

  Label definitions associated with CFE_FS_SubType_Enum_t.

39.33.1 Macro Definition Documentation

39.33.1.1 CFE_FS_FILE_CONTENT_ID

```
#define CFE_FS_FILE_CONTENT_ID 0x63464531
```

Magic Number for cFE compliant files (= 'cFE1')

Definition at line 47 of file cfe_fs_extern_typedefs.h.

39.33.1.2 CFE_FS_HDR_DESC_MAX_LEN

```
#define CFE_FS_HDR_DESC_MAX_LEN 32
```

Max length of description field in a standard cFE File Header.

Definition at line 45 of file cfe_fs_extern_typedefs.h.

39.33.2 Typedef Documentation

39.33.2.1 CFE_FS_Header_t

```
typedef struct CFE_FS_Header CFE_FS_Header_t
```

Standard cFE File header structure definition.

39.33.2.2 CFE_FS_SubType_Enum_t

```
typedef uint32 CFE_FS_SubType_Enum_t
```

Content descriptor for File Headers.

See also

enum CFE_FS_SubType

Definition at line 214 of file cfe_fs_extern_typedefs.h.

39.33.3 Enumeration Type Documentation

39.33.3.1 CFE_FS_SubType

```
enum CFE_FS_SubType
```

Label definitions associated with CFE_FS_SubType_Enum_t.

Enumerator

CFE_FS_SubType_ES_ERLOG	Executive Services Exception/Reset Log Type. Executive Services Exception/Reset Log File which is generated in response to a $sc_$cpu_ES_WriteERLog2File command.
CFE_FS_SubType_ES_SYSLOG	Executive Services System Log Type. Executive Services System Log File which is generated in response to a $sc_$cpu_ES_WriteSysLog2File command.
CFE_FS_SubType_ES_QUERYALL	Executive Services Information on All Applications File. Executive Services Information on All Applications File which is generated in response to a $sc_$cpu_ES_WriteAppInfo2File command.
CFE_FS_SubType_ES_PERFDATA	Executive Services Performance Data File. Executive Services Performance Analyzer Data File which is generated in response to a $sc_$cpu_ES_StopLAData command.
CFE_FS_SubType_ES_SHELL	Executive Services Shell Response File. Executive Services Shell Response Data File which is generated in response to a shell command.
CFE_FS_SubType_ES_CDS_REG	Executive Services Critical Data Store Registry Dump File. Executive Services Critical Data Store Registry Dump File which is generated in response to a $sc_$cpu_ES_WriteCDS2File command.
CFE_FS_SubType_TBL_REG	Table Services Registry Dump File. Table Services Registry Dump File which is generated in response to a $sc_$cpu_TBL_WriteReg2File command.
CFE_FS_SubType_TBL_IMG	Table Services Table Image File. Table Services Table Image File which is generated either on the ground or in response to a $sc_$cpu_TBL_DUMP command.
CFE_FS_SubType_EVS_APPDATA	Event Services Application Data Dump File. Event Services Application Data Dump File which is generated in response to a $sc_$cpu_EVS_WriteAppData2File command.
CFE_FS_SubType_EVS_EVENTLOG	Event Services Local Event Log Dump File. Event Services Local Event Log Dump File which is generated in response to a $sc_$cpu_EVS_WriteLog2File command.
CFE_FS_SubType_SB_PIPEDATA	Software Bus Pipe Data Dump File. Software Bus Pipe Data Dump File which is generated in response to a $sc_$cpu_SB_WritePipe2File command.
CFE_FS_SubType_SB_ROUTEDATA	Software Bus Message Routing Data Dump File. Software Bus Message Routing Data Dump File which is generated in response to a $sc_$cpu_SB_WriteRouting2File command.
CFE_FS_SubType_SB_MAPDATA	Software Bus Message Mapping Data Dump File. Software Bus Message Mapping Data Dump File which is generated in response to a $sc_$cpu_SB_WriteMap2File command.
CFE_FS_SubType_ES_QUERYALLTASKS	Executive Services Query All Tasks Data File. Executive Services Query All Tasks Data File which is generated in response to a $sc_$cpu_ES_WriteTaskInfo2File command.

Definition at line 52 of file cfe_fs_extern_typedefs.h.

39.34 cfe/modules/core_api/fsw/inc/cfe_msg.h File Reference

```
#include "common_types.h"
#include "cfe_error.h"
#include "cfe_msg_hdr.h"
#include "cfe_msg_api_typedefs.h"
#include "cfe_es_api_typedefs.h"
#include "cfe_sb_api_typedefs.h"
#include "cfe_time_api_typedefs.h"
```

Functions

- CFE_Status_t CFE_MSG_Init (CFE_MSG_Message_t *MsgPtr, CFE_SB_MsgId_t MsgId, CFE_MSG_Size_t Size)

  Initialize a message.
- CFE_Status_t CFE_MSG_GetSize (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_Size_t *Size)

  Gets the total size of a message.
- CFE_Status_t CFE_MSG_SetSize (CFE_MSG_Message_t *MsgPtr, CFE_MSG_Size_t Size)

  Sets the total size of a message.
- CFE_Status_t CFE_MSG_GetType (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_Type_t *Type)

  Gets the message type.
- CFE_Status_t CFE_MSG_SetType (CFE_MSG_Message_t *MsgPtr, CFE_MSG_Type_t Type)

  Sets the message type.
- CFE_Status_t CFE_MSG_GetHeaderVersion (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_HeaderVersion_t *Version)

  Gets the message header version.
- CFE_Status_t CFE_MSG_SetHeaderVersion (CFE_MSG_Message_t *MsgPtr, CFE_MSG_HeaderVersion_t Version)

  Sets the message header version.
- CFE_Status_t CFE_MSG_GetHasSecondaryHeader (const CFE_MSG_Message_t *MsgPtr, bool *HasSecondary)

  Gets the message secondary header boolean.
- CFE_Status_t CFE_MSG_SetHasSecondaryHeader (CFE_MSG_Message_t *MsgPtr, bool HasSecondary)

  Sets the message secondary header boolean.
- CFE_Status_t CFE_MSG_GetApId (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_ApId_t *ApId)

  Gets the message application ID.
- CFE_Status_t CFE_MSG_SetApId (CFE_MSG_Message_t *MsgPtr, CFE_MSG_ApId_t ApId)

  Sets the message application ID.
- CFE_Status_t CFE_MSG_GetSegmentationFlag (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_SegmentationFlag_t *SegFlag)

  Gets the message segmentation flag.
- CFE_Status_t CFE_MSG_SetSegmentationFlag (CFE_MSG_Message_t *MsgPtr, CFE_MSG_SegmentationFlag_t SegFlag)

  Sets the message segmentation flag.
- CFE_Status_t CFE_MSG_GetSequenceCount (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_SequenceCount_t *SeqCnt)

  Gets the message sequence count.

- CFE_Status_t CFE_MSG_SetSequenceCount (CFE_MSG_Message_t *MsgPtr, CFE_MSG_SequenceCount_t SeqCnt)

  Sets the message sequence count.
- CFE_Status_t CFE_MSG_GetEDSVersion (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_EDSVersion_t *Version)

  Gets the message EDS version.
- CFE_Status_t CFE_MSG_SetEDSVersion (CFE_MSG_Message_t *MsgPtr, CFE_MSG_EDSVersion_t Version)

  Sets the message EDS version.
- CFE_Status_t CFE_MSG_GetEndian (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_Endian_t *Endian)

  Gets the message endian.
- CFE_Status_t CFE_MSG_SetEndian (CFE_MSG_Message_t *MsgPtr, CFE_MSG_Endian_t Endian)

  Sets the message endian.
- CFE_Status_t CFE_MSG_GetPlaybackFlag (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_PlaybackFlag_t *PlayFlag)

  Gets the message playback flag.
- CFE_Status_t CFE_MSG_SetPlaybackFlag (CFE_MSG_Message_t *MsgPtr, CFE_MSG_PlaybackFlag_t PlayFlag)

  Sets the message playback flag.
- CFE_Status_t CFE_MSG_GetSubsystem (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_Subsystem_t *Subsystem)

  Gets the message subsystem.
- CFE_Status_t CFE_MSG_SetSubsystem (CFE_MSG_Message_t *MsgPtr, CFE_MSG_Subsystem_t Subsystem)

  Sets the message subsystem.
- CFE_Status_t CFE_MSG_GetSystem (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_System_t *System)

  Gets the message system.
- CFE_Status_t CFE_MSG_SetSystem (CFE_MSG_Message_t *MsgPtr, CFE_MSG_System_t System)

  Sets the message system.
- CFE_Status_t CFE_MSG_GenerateChecksum (CFE_MSG_Message_t *MsgPtr)

  Calculates and sets the checksum of a message.
- CFE_Status_t CFE_MSG_ValidateChecksum (const CFE_MSG_Message_t *MsgPtr, bool *IsValid)

  Validates the checksum of a message.
- CFE_Status_t CFE_MSG_SetFcnCode (CFE_MSG_Message_t *MsgPtr, CFE_MSG_FcnCode_t FcnCode)

  Sets the function code field in a message.
- CFE_Status_t CFE_MSG_GetFcnCode (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_FcnCode_t *FcnCode)

  Gets the function code field from a message.
- CFE_Status_t CFE_MSG_GetMsgTime (const CFE_MSG_Message_t *MsgPtr, CFE_TIME_SysTime_t *Time)

  Gets the time field from a message.
- CFE_Status_t CFE_MSG_SetMsgTime (CFE_MSG_Message_t *MsgPtr, CFE_TIME_SysTime_t Time)

  Sets the time field in a message.
- CFE_Status_t CFE_MSG_GetMsgId (const CFE_MSG_Message_t *MsgPtr, CFE_SB_MsgId_t *MsgId)

  Gets the message id from a message.
- CFE_Status_t CFE_MSG_SetMsgId (CFE_MSG_Message_t *MsgPtr, CFE_SB_MsgId_t MsgId)

  Sets the message id bits in a message.
- CFE_Status_t CFE_MSG_GetTypeFromMsgId (CFE_SB_MsgId_t MsgId, CFE_MSG_Type_t *Type)

  Gets message type using message ID.

39.35 cfe/modules/core_api/fsw/inc/cfe_msg_api_typedefs.h File Reference

```
#include "common_types.h"
#include "cfe_error.h"
```

Macros

- #define CFE_MSG_BAD_ARGUMENT CFE_SB_BAD_ARGUMENT

  Error - bad argument.
- #define CFE_MSG_NOT_IMPLEMENTED CFE_SB_NOT_IMPLEMENTED

  Error - not implemented.
- #define CFE_MSG_WRONG_MSG_TYPE CFE_SB_WRONG_MSG_TYPE

  Error - wrong type.

Typedefs

- typedef size_t CFE_MSG_Size_t

  Message size (CCSDS needs uint32 for max size)
- typedef uint32 CFE_MSG_Checksum_t

  Message checksum (Oversized to avoid redefine)
- typedef uint16 CFE_MSG_FcnCode_t

  Message function code.
- typedef uint16 CFE_MSG_HeaderVersion_t

  Message header version.
- typedef uint16 CFE_MSG_ApId_t

  Message application ID.
- typedef uint16 CFE_MSG_SequenceCount_t

  Message sequence count.
- typedef uint16 CFE_MSG_EDSVersion_t

  Message EDS version.
- typedef uint16 CFE_MSG_Subsystem_t

  Message subsystem.
- typedef uint16 CFE_MSG_System_t

  Message system.
- typedef enum CFE_MSG_Type CFE_MSG_Type_t

  Message type.
- typedef enum CFE_MSG_SegmentationFlag CFE_MSG_SegmentationFlag_t

  Segmentation flags.
- typedef enum CFE_MSG_Endian CFE_MSG_Endian_t

  Endian flag.
- typedef enum CFE_MSG_PlaybackFlag CFE_MSG_PlaybackFlag_t

  Playback flag.
- typedef union CFE_MSG_Message CFE_MSG_Message_t

  cFS generic base message
- typedef struct CFE_MSG_CommandHeader CFE_MSG_CommandHeader_t

  cFS command header
- typedef struct CFE_MSG_TelemetryHeader CFE_MSG_TelemetryHeader_t

  cFS telemetry header

Enumerations

- enum CFE_MSG_Type { CFE_MSG_Type_Invalid, CFE_MSG_Type_Cmd, CFE_MSG_Type_Tlm }

  Message type.
- enum CFE_MSG_SegmentationFlag { CFE_MSG_SegFlag_Invalid, CFE_MSG_SegFlag_Continue, CFE_MSG_SegFlag_First, CFE_MSG_SegFlag↩_Last, CFE_MSG_SegFlag_Unsegmented }

  Segmentation flags.
- enum CFE_MSG_Endian { CFE_MSG_Endian_Invalid, CFE_MSG_Endian_Big, CFE_MSG_Endian_Little }

  Endian flag.
- enum CFE_MSG_PlaybackFlag { CFE_MSG_PlayFlag_Invalid, CFE_MSG_PlayFlag_Original, CFE_MSG_↩PlayFlag_Playback }

  Playback flag.

39.35.1 Macro Definition Documentation

39.35.1.1 CFE_MSG_BAD_ARGUMENT

```
#define CFE_MSG_BAD_ARGUMENT CFE_SB_BAD_ARGUMENT
```

Error - bad argument.

Definition at line 39 of file cfe_msg_api_typedefs.h.

39.35.1.2 CFE_MSG_NOT_IMPLEMENTED

```
#define CFE_MSG_NOT_IMPLEMENTED CFE_SB_NOT_IMPLEMENTED
```

Error - not implemented.

Definition at line 40 of file cfe_msg_api_typedefs.h.

39.35.1.3 CFE_MSG_WRONG_MSG_TYPE

```
#define CFE_MSG_WRONG_MSG_TYPE CFE_SB_WRONG_MSG_TYPE
```

Error - wrong type.

Definition at line 41 of file cfe_msg_api_typedefs.h.

39.35.2 Typedef Documentation

39.35.2.1 CFE_MSG_ApId_t

```
typedef uint16 CFE_MSG_ApId_t
```

Message application ID.

Definition at line 50 of file cfe_msg_api_typedefs.h.

39.35.2.2 CFE_MSG_Checksum_t

```
typedef uint32 CFE_MSG_Checksum_t
```

Message checksum (Oversized to avoid redefine)

Definition at line 47 of file cfe_msg_api_typedefs.h.

39.35.2.3 CFE_MSG_CommandHeader_t

```
typedef struct CFE_MSG_CommandHeader CFE_MSG_CommandHeader_t
```

cFS command header

Definition at line 107 of file cfe_msg_api_typedefs.h.

39.35.2.4 CFE_MSG_EDSVersion_t

```
typedef uint16 CFE_MSG_EDSVersion_t
```

Message EDS version.

Definition at line 52 of file cfe_msg_api_typedefs.h.

39.35.2.5 CFE_MSG_Endian_t

```
typedef enum CFE_MSG_Endian CFE_MSG_Endian_t
```

Endian flag.

39.35.2.6 CFE_MSG_FcnCode_t

```
typedef uint16 CFE_MSG_FcnCode_t
```

Message function code.

Definition at line 48 of file cfe_msg_api_typedefs.h.

39.35.2.7 CFE_MSG_HeaderVersion_t

```
typedef uint16 CFE_MSG_HeaderVersion_t
```

Message header version.

Definition at line 49 of file cfe_msg_api_typedefs.h.

39.35.2.8 CFE_MSG_Message_t

```
typedef union CFE_MSG_Message CFE_MSG_Message_t
```

cFS generic base message

Definition at line 102 of file cfe_msg_api_typedefs.h.

39.35.2.9 CFE_MSG_PlaybackFlag_t

```
typedef enum CFE_MSG_PlaybackFlag CFE_MSG_PlaybackFlag_t
```

Playback flag.

39.35.2.10 CFE_MSG_SegmentationFlag_t

```
typedef enum CFE_MSG_SegmentationFlag CFE_MSG_SegmentationFlag_t
```

Segmentation flags.

39.35.2.11 CFE_MSG_SequenceCount_t

```
typedef uint16 CFE_MSG_SequenceCount_t
```

Message sequence count.

Definition at line 51 of file cfe_msg_api_typedefs.h.

39.35.2.12 CFE_MSG_Size_t

```
typedef size_t CFE_MSG_Size_t
```

Message size (CCSDS needs uint32 for max size)

Definition at line 46 of file cfe_msg_api_typedefs.h.

39.35.2.13 CFE_MSG_Subsystem_t

```
typedef uint16 CFE_MSG_Subsystem_t
```

Message subsystem.

Definition at line 53 of file cfe_msg_api_typedefs.h.

39.35.2.14 CFE_MSG_System_t

```
typedef uint16 CFE_MSG_System_t
```

Message system.

Definition at line 54 of file cfe_msg_api_typedefs.h.

39.35.2.15 CFE_MSG_TelemetryHeader_t

```
typedef struct CFE_MSG_TelemetryHeader CFE_MSG_TelemetryHeader_t
```

cFS telemetry header

Definition at line 112 of file cfe_msg_api_typedefs.h.

39.35.2.16 CFE_MSG_Type_t

```
typedef enum CFE_MSG_Type CFE_MSG_Type_t
```

Message type.

39.35.3 Enumeration Type Documentation

39.35.3.1 CFE_MSG_Endian

```
enum CFE_MSG_Endian
```

Endian flag.

Enumerator

CFE_MSG_Endian_Invalid	Invalid endian setting.
CFE_MSG_Endian_Big	Big endian.
CFE_MSG_Endian_Little	Little endian.

Definition at line 75 of file cfe_msg_api_typedefs.h.

39.35.3.2 CFE_MSG_PlaybackFlag

`enum CFE_MSG_PlaybackFlag`

Playback flag.

Enumerator

CFE_MSG_PlayFlag_Invalid	Invalid playback setting.
CFE_MSG_PlayFlag_Original	Original.
CFE_MSG_PlayFlag_Playback	Playback.

Definition at line 83 of file cfe_msg_api_typedefs.h.

39.35.3.3 CFE_MSG_SegmentationFlag

`enum CFE_MSG_SegmentationFlag`

Segmentation flags.

Enumerator

CFE_MSG_SegFlag_Invalid	Invalid segmentation flag.
CFE_MSG_SegFlag_Continue	Continuation segment of User Data.
CFE_MSG_SegFlag_First	First segment of User Data.
CFE_MSG_SegFlag_Last	Last segment of User Data.
CFE_MSG_SegFlag_Unsegmented	Unsegemented data.

Definition at line 65 of file cfe_msg_api_typedefs.h.

39.35.3.4 CFE_MSG_Type

`enum CFE_MSG_Type`

Message type.

Enumerator

CFE_MSG_Type_Invalid	Message type invalid, undefined, not implemented.
CFE_MSG_Type_Cmd	Command message type.
CFE_MSG_Type_Tlm	Telemetry message type.

Definition at line 57 of file cfe_msg_api_typedefs.h.

39.36 cfe/modules/core_api/fsw/inc/cfe_resourceid.h File Reference

```
#include "cfe_resourceid_api_typedefs.h"
```

Functions

- uint32 CFE_ResourceId_GetBase (CFE_ResourceId_t ResourceId)

  Get the Base value (type/category) from a resource ID value.
- uint32 CFE_ResourceId_GetSerial (CFE_ResourceId_t ResourceId)

  Get the Serial Number (sequential ID) from a resource ID value.
- CFE_ResourceId_t CFE_ResourceId_FindNext (CFE_ResourceId_t StartId, uint32 TableSize, bool(∗Check↩Func)(CFE_ResourceId_t))

  Locate the next resource ID which does not map to an in-use table entry.
- int32 CFE_ResourceId_ToIndex (CFE_ResourceId_t Id, uint32 BaseValue, uint32 TableSize, uint32 ∗Idx)

  Internal routine to aid in converting an ES resource ID to an array index.

Resource ID test/conversion macros and inline functions

- #define CFE_RESOURCEID_TO_ULONG(id) CFE_ResourceId_ToInteger(CFE_RESOURCEID_UNWRAP(id))

  Convert a derived (app-specific) ID directly into an "unsigned long".
- #define CFE_RESOURCEID_TEST_DEFINED(id) CFE_ResourceId_IsDefined(CFE_RESOURCEID_UNWR↩AP(id))

  Determine if a derived (app-specific) ID is defined or not.
- #define CFE_RESOURCEID_TEST_EQUAL(id1, id2) CFE_ResourceId_Equal(CFE_RESOURCEID_UNWR↩AP(id1), CFE_RESOURCEID_UNWRAP(id2))

  Determine if two derived (app-specific) IDs are equal.
- static unsigned long CFE_ResourceId_ToInteger (CFE_ResourceId_t id)

  Convert a resource ID to an integer.
- static CFE_ResourceId_t CFE_ResourceId_FromInteger (unsigned long Value)

  Convert an integer to a resource ID.
- static bool CFE_ResourceId_Equal (CFE_ResourceId_t id1, CFE_ResourceId_t id2)

  Compare two Resource ID values for equality.
- static bool CFE_ResourceId_IsDefined (CFE_ResourceId_t id)

  Check if a resource ID value is defined.

39.36.1 Detailed Description

Contains global prototypes and definitions related to resource management and related CFE resource IDs.

A CFE ES Resource ID is a common way to identify CFE-managed resources such as apps, tasks, counters, memory pools, CDS blocks, and other entities.

Simple operations are provided as inline functions, which should alleviate the need to do direct manipulation of resource IDs:

- Check for undefined ID value
- Check for equality of two ID values
- Convert ID to simple integer (typically for printing/logging)
- Convert simple integer to ID (inverse of above)

39.36.2 Macro Definition Documentation

39.36.2.1 CFE_RESOURCEID_TEST_DEFINED

```
#define CFE_RESOURCEID_TEST_DEFINED(
              id ) CFE_ResourceId_IsDefined(CFE_RESOURCEID_UNWRAP(id))
```

Determine if a derived (app-specific) ID is defined or not.

This generic routine is implemented as a macro so it is agnostic to the actual argument type, and it will evaluate correctly so long as the argument type is based on the CFE_RESOURCEID_BASE_TYPE.

Definition at line 72 of file cfe_resourceid.h.

39.36.2.2 CFE_RESOURCEID_TEST_EQUAL

```
#define CFE_RESOURCEID_TEST_EQUAL(
              id1,
              id2 ) CFE_ResourceId_Equal(CFE_RESOURCEID_UNWRAP(id1), CFE_RESOURCEID_UNWRAP(id2))
```

Determine if two derived (app-specific) IDs are equal.

This generic routine is implemented as a macro so it is agnostic to the actual argument type, and it will evaluate correctly so long as the argument type is based on the CFE_RESOURCEID_BASE_TYPE.

Definition at line 80 of file cfe_resourceid.h.

39.36.2.3 CFE_RESOURCEID_TO_ULONG

```
#define CFE_RESOURCEID_TO_ULONG(
            id ) CFE_ResourceId_ToInteger(CFE_RESOURCEID_UNWRAP(id))
```

Convert a derived (app-specific) ID directly into an "unsigned long".

This generic routine is implemented as a macro so it is agnostic to the actual argument type, and it will evaluate correctly so long as the argument type is based on the CFE_RESOURCEID_BASE_TYPE.

There is no inverse of this macro, as it depends on the actual derived type desired. Applications needing to recreate an ID from an integer should use CFE_ResourceId_FromInteger() combined with a cast/conversion to the correct/intended derived type, as needed.

Note

This evaluates as an "unsigned long" such that it can be used in printf()-style functions with the "%lx" modifier without extra casting, as this is the most typical use-case for representing an ID as an integer.

Definition at line 64 of file cfe_resourceid.h.

39.36.3 Function Documentation

39.36.3.1 CFE_ResourceId_Equal()

```
static bool CFE_ResourceId_Equal (
            CFE_ResourceId_t id1,
            CFE_ResourceId_t id2 )  [inline], [static]
```

Compare two Resource ID values for equality.

Parameters

in	*id1*	Resource ID to check
in	*id2*	Resource ID to check

Returns

true if id1 and id2 are equal, false otherwise.

Definition at line 135 of file cfe_resourceid.h.

39.36.3.2 CFE_ResourceId_FindNext()

```
CFE_ResourceId_t CFE_ResourceId_FindNext (
            CFE_ResourceId_t StartId,
            uint32 TableSize,
            bool(*)(CFE_ResourceId_t) CheckFunc )
```

Locate the next resource ID which does not map to an in-use table entry.

This begins searching from StartId which should be the most recently issued ID for the resource category. This will then search for the next ID which does *not* map to a table entry that is in use. That is, it does not alias any valid ID when converted to an array index.

returns an undefined ID value if no open slots are available

Parameters

in	*StartId*	the last issued ID for the resource category (app, lib, etc).
in	*TableSize*	the maximum size of the target table
in	*CheckFunc*	a function to check if the given ID is available

Returns

Next ID value which does not map to a valid entry

Return values

CFE_RESOURCEID_UNDEFINED	if no open slots.

Referenced by CFE_ResourceId_IsDefined().

39.36.3.3 CFE_ResourceId_FromInteger()

```
static CFE_ResourceId_t CFE_ResourceId_FromInteger (
            unsigned long Value )  [inline], [static]
```

Convert an integer to a resource ID.

This is the inverse of CFE_ResourceId_ToInteger(), and reconstitutes the original CFE_ResourceId_t value from the integer representation.

This may be used, for instance, where an ID value is parsed from a text file or message using C library APIs such as scanf() or strtoul().

See also

CFE_ResourceId_ToInteger()

Parameters

in	*Value*	Integer value to convert

Returns

ID value corresponding to integer

Definition at line 123 of file cfe_resourceid.h.

39.36.3.4 CFE_ResourceId_GetBase()

```
uint32 CFE_ResourceId_GetBase (
            CFE_ResourceId_t ResourceId )
```

Get the Base value (type/category) from a resource ID value.

This masks out the ID serial number to obtain the base value, which is different for each resource type.

Note

The value is NOT shifted or otherwise adjusted.

Parameters

in	*ResourceId*	the resource ID to decode

Returns

The base value associated with that ID

Referenced by CFE_ResourceId_IsDefined().

39.36.3.5 CFE_ResourceId_GetSerial()

```
uint32 CFE_ResourceId_GetSerial (
            CFE_ResourceId_t ResourceId )
```

Get the Serial Number (sequential ID) from a resource ID value.

This masks out the ID base value to obtain the serial number, which is different for each entity created.

Parameters

in	*Resource↩ Id*	the resource ID to decode

Returns

The serial number associated with that ID

Referenced by CFE_ResourceId_IsDefined().

39.36.3.6 CFE_ResourceId_IsDefined()

```
static bool CFE_ResourceId_IsDefined (
            CFE_ResourceId_t id )  [inline], [static]
```

Check if a resource ID value is defined.

The constant CFE_RESOURCEID_UNDEFINED represents an undefined ID value, such that the expression:

```
CFE_ResourceId_IsDefined(CFE_RESOURCEID_UNDEFINED)
```

Always returns false.

Parameters

in	*id*	Resource ID to check

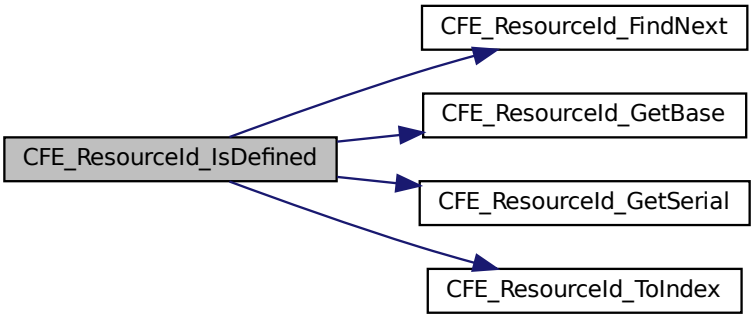
Returns

True if the ID may refer to a defined entity, false if invalid/undefined.

Definition at line 153 of file cfe_resourceid.h.

References CFE_ResourceId_FindNext(), CFE_ResourceId_GetBase(), CFE_ResourceId_GetSerial(), and CFE_↩ ResourceId_ToIndex().

Here is the call graph for this function:



39.36.3.7 CFE_ResourceId_ToIndex()

```
int32 CFE_ResourceId_ToIndex (
            CFE_ResourceId_t Id,
            uint32 BaseValue,
            uint32 TableSize,
            uint32 * Idx )
```

Internal routine to aid in converting an ES resource ID to an array index.

Parameters

in	*Id*	The resource ID
in	*BaseValue*	The respective ID base value corresponding to the ID type
in	*TableSize*	The actual size of the internal table (MAX index value + 1)
out	*Idx*	The output index

Returns

Status code, CFE_SUCCESS if successful.

Referenced by CFE_ResourceId_IsDefined().

39.36.3.8 CFE_ResourceId_ToInteger()

```
static unsigned long CFE_ResourceId_ToInteger (
            CFE_ResourceId_t id )  [inline], [static]
```

Convert a resource ID to an integer.

This is primarily intended for logging purposes, such was writing to debug console, event messages, or log files, using printf-like APIs.

For compatibility with C library APIs, this returns an "unsigned long" type and should be used with the "%lx" format specifier in a printf format string.

Note

No assumptions should be made about the actual integer value, such as its base/range. It may be printed, but should not be modified or tested/compared using other arithmetic ops, and should never be used as the index to an array or table. See the related function CFE_ResourceId_ToIndex() for cases where a zero-based array/table index is needed.

See also

CFE_ResourceId_FromInteger()

Parameters

in	*id*	Resource ID to convert

Returns

Integer value corresponding to ID

Definition at line 104 of file cfe_resourceid.h.

39.37 cfe/modules/core_api/fsw/inc/cfe_resourceid_api_typedefs.h File Reference

```
#include "cfe_resourceid_typedef.h"
```

Macros

Resource ID predefined values

- #define CFE_RESOURCEID_UNDEFINED ((CFE_ResourceId_t)CFE_RESOURCEID_WRAP(0))
  A resource ID value that represents an undefined/unused resource.
- #define CFE_RESOURCEID_RESERVED ((CFE_ResourceId_t)CFE_RESOURCEID_WRAP(0xFFFFFFFF))
  A resource ID value that represents a reserved entry.

39.37.1 Detailed Description

Contains global prototypes and definitions related to resource management and related CFE resource IDs.

A CFE ES Resource ID is a common way to identify CFE-managed resources such as apps, tasks, counters, memory pools, CDS blocks, and other entities.

Simple operations are provided as inline functions, which should alleviate the need to do direct manipulation of resource IDs:

- Check for undefined ID value
- Check for equality of two ID values
- Convert ID to simple integer (typically for printing/logging)
- Convert simple integer to ID (inverse of above)

39.37.2 Macro Definition Documentation

39.37.2.1 CFE_RESOURCEID_RESERVED

```
#define CFE_RESOURCEID_RESERVED ((CFE_ResourceId_t)CFE_RESOURCEID_WRAP(0xFFFFFFFF))
```

A resource ID value that represents a reserved entry.

This is not a valid value for any resource type, but is used to mark table entries that are not available for use. For instance, this may be used while setting up an entry initially.

Definition at line 76 of file cfe_resourceid_api_typedefs.h.

39.37.2.2 CFE_RESOURCEID_UNDEFINED

```
#define CFE_RESOURCEID_UNDEFINED ((CFE_ResourceId_t)CFE_RESOURCEID_WRAP(0))
```

A resource ID value that represents an undefined/unused resource.

This constant may be used to initialize local variables of the CFE_ResourceId_t type to a safe value that will not alias a valid ID.

By design, this value is also the result of zeroing a CFE_ResourceId_t type via standard functions like memset(), such that objects initialized using this method will also be set to safe values.

Definition at line 67 of file cfe_resourceid_api_typedefs.h.

39.38 cfe/modules/core_api/fsw/inc/cfe_sb.h File Reference

```
#include "common_types.h"
#include "cfe_error.h"
#include "cfe_sb_api_typedefs.h"
#include "cfe_es_api_typedefs.h"
#include "cfe_time_api_typedefs.h"
```

Macros

- #define CFE_BIT(x) (1 << (x))

  Places a one at bit positions 0 - 31.
- #define CFE_SET(i, x) ((i) |= CFE_BIT(x))

  Sets bit x of i.
- #define CFE_CLR(i, x) ((i) &= ~CFE_BIT(x))

  Clears bit x of i.
- #define CFE_TST(i, x) (((i)&CFE_BIT(x)) != 0)

  true(non zero) if bit x of i is set

Functions

- CFE_Status_t CFE_SB_CreatePipe (CFE_SB_PipeId_t *PipeIdPtr, uint16 Depth, const char *PipeName)

  Creates a new software bus pipe.
- CFE_Status_t CFE_SB_DeletePipe (CFE_SB_PipeId_t PipeId)

  Delete a software bus pipe.
- CFE_Status_t CFE_SB_PipeId_ToIndex (CFE_SB_PipeId_t PipeID, uint32 *Idx)

  Obtain an index value correlating to an SB Pipe ID.
- CFE_Status_t CFE_SB_SetPipeOpts (CFE_SB_PipeId_t PipeId, uint8 Opts)

  Set options on a pipe.
- CFE_Status_t CFE_SB_GetPipeOpts (CFE_SB_PipeId_t PipeId, uint8 *OptPtr)

  Get options on a pipe.
- CFE_Status_t CFE_SB_GetPipeName (char *PipeNameBuf, size_t PipeNameSize, CFE_SB_PipeId_t PipeId)

  Get the pipe name for a given id.
- CFE_Status_t CFE_SB_GetPipeIdByName (CFE_SB_PipeId_t *PipeIdPtr, const char *PipeName)

  Get pipe id by pipe name.
- CFE_Status_t CFE_SB_SubscribeEx (CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId, CFE_SB_Qos_↩t Quality, uint16 MsgLim)

  Subscribe to a message on the software bus.
- CFE_Status_t CFE_SB_Subscribe (CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId)

  Subscribe to a message on the software bus with default parameters.
- CFE_Status_t CFE_SB_SubscribeLocal (CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId, uint16 MsgLim)

  Subscribe to a message while keeping the request local to a cpu.
- CFE_Status_t CFE_SB_Unsubscribe (CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId)

  Remove a subscription to a message on the software bus.
- CFE_Status_t CFE_SB_UnsubscribeLocal (CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId)

Remove a subscription to a message on the software bus on the current CPU.

- CFE_Status_t CFE_SB_TransmitMsg (CFE_MSG_Message_t *MsgPtr, bool IncrementSequenceCount)

  Transmit a message.
- CFE_Status_t CFE_SB_SendMsg (CFE_MSG_Message_t *MsgPtr)

  DEPRECATED: Send a software bus message.
- CFE_Status_t CFE_SB_PassMsg (CFE_MSG_Message_t *MsgPtr)

  DEPRECATED: Passes a software bus message.
- CFE_Status_t CFE_SB_ReceiveBuffer (CFE_SB_Buffer_t **BufPtr, CFE_SB_PipeId_t PipeId, int32 TimeOut)

  Receive a message from a software bus pipe.
- CFE_Status_t CFE_SB_RcvMsg (CFE_SB_Buffer_t **BufPtr, CFE_SB_PipeId_t PipeId, int32 TimeOut)

  DEPRECATED: receive buffer.
- CFE_SB_Buffer_t * CFE_SB_ZeroCopyGetPtr (size_t MsgSize, CFE_SB_ZeroCopyHandle_t *BufferHandle)

  Get a buffer pointer to use for "zero copy" SB sends.
- CFE_Status_t CFE_SB_ZeroCopyReleasePtr (CFE_SB_Buffer_t *Ptr2Release, CFE_SB_ZeroCopyHandle_t ZeroCopyHandle)

  Release an unused "zero copy" buffer pointer.
- CFE_Status_t CFE_SB_TransmitBuffer (CFE_SB_Buffer_t *BufPtr, CFE_SB_ZeroCopyHandle_t ZeroCopyHandle, bool IncrementSequenceCount)

  Transmit a buffer.
- CFE_Status_t CFE_SB_ZeroCopySend (CFE_SB_Buffer_t *BufPtr, CFE_SB_ZeroCopyHandle_t BufferHandle)

  DEPRECATED: Send an SB message in "zero copy" mode.
- CFE_Status_t CFE_SB_ZeroCopyPass (CFE_SB_Buffer_t *BufPtr, CFE_SB_ZeroCopyHandle_t BufferHandle)

  DEPRECATED: Pass an SB message in "zero copy" mode.
- void CFE_SB_InitMsg (void *MsgPtr, CFE_SB_MsgId_t MsgId, size_t Length, bool Clear)

  DEPRECATED - Initialize a buffer for a software bus message.
- void CFE_SB_SetMsgId (CFE_MSG_Message_t *MsgPtr, CFE_SB_MsgId_t MsgId)

  DEPRECATED - Sets the message ID of a software bus message.
- void CFE_SB_SetUserDataLength (CFE_MSG_Message_t *MsgPtr, size_t DataLength)

  Sets the length of user data in a software bus message.
- void CFE_SB_SetTotalMsgLength (CFE_MSG_Message_t *MsgPtr, size_t TotalLength)

  DEPRECATED: Sets the total length of a software bus message.
- CFE_Status_t CFE_SB_SetMsgTime (CFE_MSG_Message_t *MsgPtr, CFE_TIME_SysTime_t Time)

  Sets the time field in a software bus message.
- void CFE_SB_TimeStampMsg (CFE_MSG_Message_t *MsgPtr)

  Sets the time field in a software bus message with the current spacecraft time.
- CFE_Status_t CFE_SB_SetCmdCode (CFE_MSG_Message_t *MsgPtr, uint16 CmdCode)

  DEPRECATED:Sets the command code field in a software bus message.
- int32 CFE_SB_MessageStringSet (char *DestStringPtr, const char *SourceStringPtr, size_t DestMaxSize, size_t SourceMaxSize)

  Copies a string into a software bus message.
- void * CFE_SB_GetUserData (CFE_MSG_Message_t *MsgPtr)

  Get a pointer to the user data portion of a software bus message.
- CFE_SB_MsgId_t CFE_SB_GetMsgId (const CFE_MSG_Message_t *MsgPtr)

  DEPRECATED:Get the message ID of a software bus message.
- size_t CFE_SB_GetUserDataLength (const CFE_MSG_Message_t *MsgPtr)

  Gets the length of user data in a software bus message.
- size_t CFE_SB_GetTotalMsgLength (const CFE_MSG_Message_t *MsgPtr)

DEPRECATED: Gets the total length of a software bus message.

- uint16 CFE_SB_GetCmdCode (CFE_MSG_Message_t *MsgPtr)

  DEPRECATED: Gets the command code field from a software bus message.
- CFE_TIME_SysTime_t CFE_SB_GetMsgTime (CFE_MSG_Message_t *MsgPtr)

  DEPRECATED: Gets the time field from a software bus message.
- int32 CFE_SB_MessageStringGet (char *DestStringPtr, const char *SourceStringPtr, const char *DefaultString, size_t DestMaxSize, size_t SourceMaxSize)

  Copies a string out of a software bus message.
- uint16 CFE_SB_GetChecksum (CFE_MSG_Message_t *MsgPtr)

  DEPRECATED:Gets the checksum field from a software bus message.
- void CFE_SB_GenerateChecksum (CFE_MSG_Message_t *MsgPtr)

  DEPRECATED:Calculates and sets the checksum of a software bus message.
- bool CFE_SB_ValidateChecksum (CFE_MSG_Message_t *MsgPtr)

  DEPRECATED:Validates the checksum of a software bus message.
- bool CFE_SB_IsValidMsgId (CFE_SB_MsgId_t MsgId)

  Identifies whether a given CFE_SB_MsgId_t is valid.
- static bool CFE_SB_MsgId_Equal (CFE_SB_MsgId_t MsgId1, CFE_SB_MsgId_t MsgId2)

  Identifies whether two CFE_SB_MsgId_t values are equal.
- static CFE_SB_MsgId_Atom_t CFE_SB_MsgIdToValue (CFE_SB_MsgId_t MsgId)

  Converts a CFE_SB_MsgId_t to a normal integer.
- static CFE_SB_MsgId_t CFE_SB_ValueToMsgId (CFE_SB_MsgId_Atom_t MsgIdValue)

  Converts a normal integer into a CFE_SB_MsgId_t.
- uint32 CFE_SB_GetPktType (CFE_SB_MsgId_t MsgId)

  Identifies packet type given message ID.

39.38.1 Macro Definition Documentation

39.38.1.1 CFE_BIT

```
#define CFE_BIT(
            x ) (1 << (x))
```

Places a one at bit positions 0 - 31.

Definition at line 51 of file cfe_sb.h.

39.38.1.2 CFE_CLR

```
#define CFE_CLR(
            i,
            x ) ((i) &= ~CFE_BIT(x))
```

Clears bit x of i.

Definition at line 53 of file cfe_sb.h.

39.38.1.3 CFE_SET

```
#define CFE_SET(
            i,
            x ) ((i) |= CFE_BIT(x))
```

Sets bit x of i.

Definition at line 52 of file cfe_sb.h.

39.38.1.4 CFE_TST

```
#define CFE_TST(
            i,
            x ) (((i)&CFE_BIT(x)) != 0)
```

true(non zero) if bit x of i is set

Definition at line 54 of file cfe_sb.h.

39.39 cfe/modules/core_api/fsw/inc/cfe_sb_api_typedefs.h File Reference

```
#include "common_types.h"
#include "cfe_sb_extern_typedefs.h"
#include "cfe_msg_api_typedefs.h"
#include "cfe_resourceid_api_typedefs.h"
#include "cfe_msg_hdr.h"
```

Data Structures

- union CFE_SB_Msg

  Software Bus generic message.
- struct CFE_SB_ZeroCopyHandle_t

  CFE_SB_ZeroCopyHandle_t to primitive type definition.

Macros

- #define CFE_SB_POLL 0

  Option used with CFE_SB_ReceiveBuffer to request immediate pipe status.
- #define CFE_SB_PEND_FOREVER -1

  Option used with CFE_SB_ReceiveBuffer to force a wait for next message.
- #define CFE_SB_SUBSCRIPTION 0

  Subtype specifier used in CFE_SB_SingleSubscriptionTlm_t by SBN App.
- #define CFE_SB_UNSUBSCRIPTION 1

  Subtype specified used in CFE_SB_SingleSubscriptionTlm_t by SBN App.
- #define CFE_SB_MSGID_WRAP_VALUE(val) ((CFE_SB_MsgId_t)(val))

  Translation macro to convert from MsgId integer values to opaque/abstract API values.
- #define CFE_SB_MSGID_UNWRAP_VALUE(mid) ((CFE_SB_MsgId_Atom_t)(mid))

  Translation macro to convert to MsgId integer values from opaque/abstract API values.
- #define CFE_SB_MSGID_RESERVED CFE_SB_MSGID_WRAP_VALUE(-1)

  Reserved value for CFE_SB_MsgId_t that will not match any valid MsgId.
- #define CFE_SB_INVALID_MSG_ID CFE_SB_MSGID_RESERVED

  A literal of the CFE_SB_MsgId_t type representing an invalid ID.
- #define CFE_SB_PKTTYPE_INVALID CFE_MSG_Type_Invalid

  CFE_SB_GetPktType response if message type can not be determined
- #define CFE_SB_PKTTYPE_CMD CFE_MSG_Type_Cmd

  CFE_SB_GetPktType response for command packets
- #define CFE_SB_PKTTYPE_TLM CFE_MSG_Type_Tlm

  CFE_SB_GetPktType response for telemetry packets
- #define CFE_SB_PIPEID_C(val) ((CFE_SB_PipeId_t)CFE_RESOURCEID_WRAP(val))

  Cast/Convert a generic CFE_ResourceId_t to a CFE_SB_PipeId_t.
- #define CFE_SB_INVALID_PIPE CFE_SB_PIPEID_C(CFE_RESOURCEID_UNDEFINED)

  A CFE_SB_PipeId_t value which is always invalid.
- #define CFE_SB_PIPEOPTS_IGNOREMINE 0x00000001

  Messages sent by the app that owns this pipe will not be sent to this pipe.
- #define CFE_SB_DEFAULT_QOS ((CFE_SB_Qos_t) {0})

  Default Qos macro.
- #define CFE_SB_CMD_HDR_SIZE (sizeof(CFE_MSG_CommandHeader_t))

  Size of command header.
- #define CFE_SB_TLM_HDR_SIZE (sizeof(CFE_MSG_TelemetryHeader_t))

  Size of telemetry header.
- #define CFE_SB_Default_Qos CFE_SB_DEFAULT_QOS
- #define CFE_SB_CMD_HDR_SIZE (sizeof(CFE_MSG_CommandHeader_t))

  Size of command header.
- #define CFE_SB_TLM_HDR_SIZE (sizeof(CFE_MSG_TelemetryHeader_t))

  Size of telemetry header.

Typedefs

- typedef union CFE_SB_Msg CFE_SB_Buffer_t

  Software Bus generic message.
- typedef CFE_SB_Buffer_t CFE_SB_Msg_t

  Deperecated type to minimize required changes.
- typedef CFE_MSG_CommandHeader_t CFE_SB_CmdHdr_t

  Deperecated type to minimize required changes.
- typedef CFE_MSG_TelemetryHeader_t CFE_SB_TlmHdr_t

  Deperecated type to minimize required changes.
- typedef CFE_MSG_Message_t * CFE_SB_MsgPtr_t

  Pointer to an SB Message.
- typedef uint8 * CFE_SB_MsgPayloadPtr_t

  CFE_SB_MsgPayloadPtr_t defined as an opaque pointer to a message Payload portion.

39.39.1 Macro Definition Documentation

39.39.1.1 CFE_SB_CMD_HDR_SIZE [1/2]

```
#define CFE_SB_CMD_HDR_SIZE (sizeof(CFE_MSG_CommandHeader_t))
```

Size of command header.

Definition at line 180 of file cfe_sb_api_typedefs.h.

39.39.1.2 CFE_SB_CMD_HDR_SIZE [2/2]

```
#define CFE_SB_CMD_HDR_SIZE (sizeof(CFE_MSG_CommandHeader_t))
```

Size of command header.

Definition at line 180 of file cfe_sb_api_typedefs.h.

39.39.1.3 CFE_SB_DEFAULT_QOS

```
#define CFE_SB_DEFAULT_QOS ((CFE_SB_Qos_t) {0})
```

Default Qos macro.

Definition at line 131 of file cfe_sb_api_typedefs.h.

39.39.1.4 CFE_SB_Default_Qos

```
#define CFE_SB_Default_Qos CFE_SB_DEFAULT_QOS
```

Deprecated use CFE_SB_DEFAULT_QOS

Definition at line 178 of file cfe_sb_api_typedefs.h.

39.39.1.5 CFE_SB_INVALID_MSG_ID

```
#define CFE_SB_INVALID_MSG_ID CFE_SB_MSGID_RESERVED
```

A literal of the CFE_SB_MsgId_t type representing an invalid ID.

This value should be used for runtime initialization of CFE_SB_MsgId_t values.

Note

This may be a compound literal in a future revision. Per C99, compound literals are lvalues, not rvalues, so this value should not be used in static/compile-time data initialization. For static data initialization purposes (rvalue), CFE_SB_MSGID_RESERVED should be used instead. However, in the current implementation, they are equivalent.

Definition at line 99 of file cfe_sb_api_typedefs.h.

39.39.1.6 CFE_SB_INVALID_PIPE

```
#define CFE_SB_INVALID_PIPE CFE_SB_PIPEID_C(CFE_RESOURCEID_UNDEFINED)
```

A CFE_SB_PipeId_t value which is always invalid.

This may be used as a safe initializer for CFE_SB_PipeId_t values

Definition at line 123 of file cfe_sb_api_typedefs.h.

39.39.1.7 CFE_SB_MSGID_RESERVED

```
#define CFE_SB_MSGID_RESERVED CFE_SB_MSGID_WRAP_VALUE(-1)
```

Reserved value for CFE_SB_MsgId_t that will not match any valid MsgId.

This rvalue macro can be used for static/compile-time data initialization to ensure that the initialized value does not alias to a valid MsgId object.

Definition at line 86 of file cfe_sb_api_typedefs.h.

39.39.1.8 CFE_SB_MSGID_UNWRAP_VALUE

```
#define CFE_SB_MSGID_UNWRAP_VALUE(
              mid ) ((CFE_SB_MsgId_Atom_t)(mid))
```

Translation macro to convert to MsgId integer values from opaque/abstract API values.

This conversion exists in macro form to allow compile-time evaluation for constants, and should not be used directly in application code.

For applications, use the CFE_SB_MsgIdToValue() inline function instead.

See also

CFE_SB_MsgIdToValue()

Definition at line 78 of file cfe_sb_api_typedefs.h.

Referenced by CFE_SB_MsgId_Equal(), and CFE_SB_MsgIdToValue().

39.39.1.9 CFE_SB_MSGID_WRAP_VALUE

```
#define CFE_SB_MSGID_WRAP_VALUE(
              val ) ((CFE_SB_MsgId_t)(val))
```

Translation macro to convert from MsgId integer values to opaque/abstract API values.

This conversion exists in macro form to allow compile-time evaluation for constants, and should not be used directly in application code.

For applications, use the CFE_SB_ValueToMsgId() inline function instead.

See also

CFE_SB_ValueToMsgId()

Definition at line 66 of file cfe_sb_api_typedefs.h.

Referenced by CFE_SB_ValueToMsgId().

39.39.1.10 CFE_SB_PEND_FOREVER

```
#define CFE_SB_PEND_FOREVER -1
```

Option used with CFE_SB_ReceiveBuffer to force a wait for next message.

Definition at line 48 of file cfe_sb_api_typedefs.h.

39.39.1.11 CFE_SB_PIPEID_C

```
#define CFE_SB_PIPEID_C(
              val ) ((CFE_SB_PipeId_t)CFE_RESOURCEID_WRAP(val))
```

Cast/Convert a generic CFE_ResourceId_t to a CFE_SB_PipeId_t.

Definition at line 116 of file cfe_sb_api_typedefs.h.

39.39.1.12 CFE_SB_PIPEOPTS_IGNOREMINE

```
#define CFE_SB_PIPEOPTS_IGNOREMINE 0x00000001
```

Messages sent by the app that owns this pipe will not be sent to this pipe.

Definition at line 128 of file cfe_sb_api_typedefs.h.

39.39.1.13 CFE_SB_POLL

```
#define CFE_SB_POLL 0
```

Option used with CFE_SB_ReceiveBuffer to request immediate pipe status.

Definition at line 47 of file cfe_sb_api_typedefs.h.

39.39.1.14 CFE_SB_SUBSCRIPTION

```
#define CFE_SB_SUBSCRIPTION 0
```

Subtype specifier used in CFE_SB_SingleSubscriptionTlm_t by SBN App.

Definition at line 49 of file cfe_sb_api_typedefs.h.

39.39.1.15 CFE_SB_TLM_HDR_SIZE [1/2]

```
#define CFE_SB_TLM_HDR_SIZE (sizeof(CFE_MSG_TelemetryHeader_t))
```

Size of telemetry header.

Definition at line 181 of file cfe_sb_api_typedefs.h.

39.39.1.16 CFE_SB_TLM_HDR_SIZE [2/2]

```
#define CFE_SB_TLM_HDR_SIZE (sizeof(CFE_MSG_TelemetryHeader_t))
```

Size of telemetry header.

Definition at line 181 of file cfe_sb_api_typedefs.h.

39.39.1.17 CFE_SB_UNSUBSCRIPTION

```
#define CFE_SB_UNSUBSCRIPTION 1
```

Subtype specified used in CFE_SB_SingleSubscriptionTlm_t by SBN App.

Definition at line 50 of file cfe_sb_api_typedefs.h.

39.39.2 Typedef Documentation

39.39.2.1 CFE_SB_Buffer_t

```
typedef union CFE_SB_Msg CFE_SB_Buffer_t
```

Software Bus generic message.

39.39.2.2 CFE_SB_CmdHdr_t

```
typedef CFE_MSG_CommandHeader_t CFE_SB_CmdHdr_t
```

Deperecated type to minimize required changes.

Definition at line 151 of file cfe_sb_api_typedefs.h.

39.39.2.3 CFE_SB_Msg_t

```
typedef CFE_SB_Buffer_t CFE_SB_Msg_t
```

Deperecated type to minimize required changes.

Definition at line 148 of file cfe_sb_api_typedefs.h.

39.39.2.4 CFE_SB_MsgPayloadPtr_t

```
typedef uint8* CFE_SB_MsgPayloadPtr_t
```

CFE_SB_MsgPayloadPtr_t defined as an opaque pointer to a message Payload portion.

Definition at line 163 of file cfe_sb_api_typedefs.h.

39.39.2.5 CFE_SB_MsgPtr_t

```
typedef CFE_MSG_Message_t* CFE_SB_MsgPtr_t
```

Pointer to an SB Message.

Definition at line 160 of file cfe_sb_api_typedefs.h.

39.39.2.6 CFE_SB_TlmHdr_t

```
typedef CFE_MSG_TelemetryHeader_t CFE_SB_TlmHdr_t
```

Deperecated type to minimize required changes.

Definition at line 154 of file cfe_sb_api_typedefs.h.

39.40 cfe/modules/core_api/fsw/inc/cfe_sb_core_internal.h File Reference

```
#include "common_types.h"
#include "cfe_es_extern_typedefs.h"
```

Functions

- void CFE_SB_TaskMain (void)

  Entry Point for cFE Core Application.
- int32 CFE_SB_EarlyInit (void)

  Initializes the cFE core module API Library.
- int32 CFE_SB_CleanUpApp (CFE_ES_AppId_t AppId)

  Removes SB resources associated with specified Application.

39.41 cfe/modules/core_api/fsw/inc/cfe_sb_extern_typedefs.h File Reference

```
#include "common_types.h"
#include "cfe_mission_cfg.h"
#include "cfe_resourceid_typedef.h"
```

Data Structures

- struct CFE_SB_Qos_t

  Quality Of Service Type Definition.

Macros

- #define CFE_SB_SUB_ENTRIES_PER_PKT 20

  Configuration parameter used by SBN App.

Typedefs

- typedef uint8 CFE_SB_QosPriority_Enum_t

  Selects the priorty level for message routing.
- typedef uint8 CFE_SB_QosReliability_Enum_t

  Selects the reliability level for message routing.
- typedef uint16 CFE_SB_RouteId_Atom_t

  An integer type that should be used for indexing into the Routing Table.
- typedef uint32 CFE_SB_MsgId_Atom_t

  CFE_SB_MsgId_Atom_t primitive type definition.
- typedef CFE_SB_MsgId_Atom_t CFE_SB_MsgId_t

  CFE_SB_MsgId_t type definition.
- typedef CFE_RESOURCEID_BASE_TYPE CFE_SB_PipeId_t

  CFE_SB_PipeId_t to primitive type definition.

Enumerations

- enum CFE_SB_QosPriority { CFE_SB_QosPriority_LOW = 0, CFE_SB_QosPriority_HIGH = 1 }

  Label definitions associated with CFE_SB_QosPriority_Enum_t.
- enum CFE_SB_QosReliability { CFE_SB_QosReliability_LOW = 0, CFE_SB_QosReliability_HIGH = 1 }

  Label definitions associated with CFE_SB_QosReliability_Enum_t.

39.41.1 Macro Definition Documentation

39.41.1.1 CFE_SB_SUB_ENTRIES_PER_PKT

```
#define CFE_SB_SUB_ENTRIES_PER_PKT 20
```

Configuration parameter used by SBN App.

Definition at line 38 of file cfe_sb_extern_typedefs.h.

39.41.2 Typedef Documentation

39.41.2.1 CFE_SB_MsgId_Atom_t

```
typedef uint32 CFE_SB_MsgId_Atom_t
```

CFE_SB_MsgId_Atom_t primitive type definition.

This is an integer type capable of holding any Message ID value Note: This value is limited via CFE_PLATFORM_S↩B_HIGHEST_VALID_MSGID

Definition at line 101 of file cfe_sb_extern_typedefs.h.

39.41.2.2 CFE_SB_MsgId_t

```
typedef CFE_SB_MsgId_Atom_t CFE_SB_MsgId_t
```

CFE_SB_MsgId_t type definition.

Software Bus message identifier used in many SB APIs

Currently this is directly mapped to the underlying holding type (not wrapped) for compatibility with existing usage semantics in apps (mainly switch/case statements)

Note

In a future version it could become a type-safe wrapper similar to the route index, to avoid message IDs getting mixed between other integer values.

Definition at line 114 of file cfe_sb_extern_typedefs.h.

39.41.2.3 CFE_SB_PipeId_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_SB_PipeId_t
```

CFE_SB_PipeId_t to primitive type definition.

Software Bus pipe identifier used in many SB APIs, as well as SB Telemetry messages and data files.

Definition at line 121 of file cfe_sb_extern_typedefs.h.

39.41.2.4 CFE_SB_QosPriority_Enum_t

```
typedef uint8 CFE_SB_QosPriority_Enum_t
```

Selects the priorty level for message routing.

See also

enum CFE_SB_QosPriority

Definition at line 63 of file cfe_sb_extern_typedefs.h.

39.41.2.5 CFE_SB_QosReliability_Enum_t

```
typedef uint8 CFE_SB_QosReliability_Enum_t
```

Selects the reliability level for message routing.

See also

enum CFE_SB_QosReliability

Definition at line 88 of file cfe_sb_extern_typedefs.h.

39.41.2.6 CFE_SB_RouteId_Atom_t

```
typedef uint16 CFE_SB_RouteId_Atom_t
```

An integer type that should be used for indexing into the Routing Table.

Definition at line 93 of file cfe_sb_extern_typedefs.h.

39.41.3 Enumeration Type Documentation

39.41.3.1 CFE_SB_QosPriority

```
enum CFE_SB_QosPriority
```

Label definitions associated with CFE_SB_QosPriority_Enum_t.

Enumerator

CFE_SB_QosPriority_LOW	Normal priority level.
CFE_SB_QosPriority_HIGH	High priority.

Definition at line 43 of file cfe_sb_extern_typedefs.h.

39.41.3.2 CFE_SB_QosReliability

```
enum CFE_SB_QosReliability
```

Label definitions associated with CFE_SB_QosReliability_Enum_t.

Enumerator

CFE_SB_QosReliability_LOW	Normal (best-effort) reliability.
CFE_SB_QosReliability_HIGH	High reliability.

Definition at line 68 of file cfe_sb_extern_typedefs.h.

39.42 cfe/modules/core_api/fsw/inc/cfe_tbl.h File Reference

```
#include "common_types.h"
#include "cfe_error.h"
#include "cfe_tbl_api_typedefs.h"
#include "cfe_sb_api_typedefs.h"
```

Functions

- CFE_Status_t CFE_TBL_Register (CFE_TBL_Handle_t ∗TblHandlePtr, const char ∗Name, size_t Size, uint16 TblOptionFlags, CFE_TBL_CallbackFuncPtr_t TblValidationFuncPtr)

  Register a table with cFE to obtain Table Management Services.
- CFE_Status_t CFE_TBL_Share (CFE_TBL_Handle_t ∗TblHandlePtr, const char ∗TblName)

  Obtain handle of table registered by another application.
- CFE_Status_t CFE_TBL_Unregister (CFE_TBL_Handle_t TblHandle)

  Unregister a previously registered table and free associated resources.
- CFE_Status_t CFE_TBL_Load (CFE_TBL_Handle_t TblHandle, CFE_TBL_SrcEnum_t SrcType, const void ∗SrcDataPtr)

  Load a specified table with data from specified source.
- CFE_Status_t CFE_TBL_Update (CFE_TBL_Handle_t TblHandle)

  Update contents of a specified table, if an update is pending.
- CFE_Status_t CFE_TBL_Validate (CFE_TBL_Handle_t TblHandle)

Perform steps to validate the contents of a table image.

- CFE_Status_t CFE_TBL_Manage (CFE_TBL_Handle_t TblHandle)

  Perform standard operations to maintain a table.
- CFE_Status_t CFE_TBL_DumpToBuffer (CFE_TBL_Handle_t TblHandle)

  Copies the contents of a Dump Only Table to a shared buffer.
- CFE_Status_t CFE_TBL_Modified (CFE_TBL_Handle_t TblHandle)

  Notify cFE Table Services that table contents have been modified by the Application.
- CFE_Status_t CFE_TBL_GetAddress (void **TblPtr, CFE_TBL_Handle_t TblHandle)

  Obtain the current address of the contents of the specified table.
- CFE_Status_t CFE_TBL_ReleaseAddress (CFE_TBL_Handle_t TblHandle)

  Release previously obtained pointer to the contents of the specified table.
- CFE_Status_t CFE_TBL_GetAddresses (void **TblPtrs[], uint16 NumTables, const CFE_TBL_Handle_t TblHandles[])

  Obtain the current addresses of an array of specified tables.
- CFE_Status_t CFE_TBL_ReleaseAddresses (uint16 NumTables, const CFE_TBL_Handle_t TblHandles[])

  Release the addresses of an array of specified tables.
- CFE_Status_t CFE_TBL_GetStatus (CFE_TBL_Handle_t TblHandle)

  Obtain current status of pending actions for a table.
- CFE_Status_t CFE_TBL_GetInfo (CFE_TBL_Info_t *TblInfoPtr, const char *TblName)

  Obtain characteristics/information of/about a specified table.
- CFE_Status_t CFE_TBL_NotifyByMessage (CFE_TBL_Handle_t TblHandle, CFE_SB_MsgId_t MsgId, CFE_MSG_FcnCode_t CommandCode, uint32 Parameter)

  Instruct cFE Table Services to notify Application via message when table requires management.

39.43 cfe/modules/core_api/fsw/inc/cfe_tbl_api_typedefs.h File Reference

```
#include "common_types.h"
#include "cfe_tbl_extern_typedefs.h"
#include "cfe_time_extern_typedefs.h"
```

Data Structures

- struct CFE_TBL_Info

  Table Info.

Macros

- #define CFE_TBL_OPT_BUFFER_MSK (0x0001)

  Table buffer mask.
- #define CFE_TBL_OPT_SNGL_BUFFER (0x0000)

  Single buffer table.
- #define CFE_TBL_OPT_DBL_BUFFER (0x0001)

  Double buffer table.
- #define CFE_TBL_OPT_LD_DMP_MSK (0x0002)

Table load/dump mask.

- #define CFE_TBL_OPT_LOAD_DUMP (0x0000)

  Load/Dump table.
- #define CFE_TBL_OPT_DUMP_ONLY (0x0002)

  Dump only table.
- #define CFE_TBL_OPT_USR_DEF_MSK (0x0004)

  Table user defined mask.
- #define CFE_TBL_OPT_NOT_USR_DEF (0x0000)

  Not user defined table.
- #define CFE_TBL_OPT_USR_DEF_ADDR (0x0006)

  User Defined table,.
- #define CFE_TBL_OPT_CRITICAL_MSK (0x0008)

  Table critical mask.
- #define CFE_TBL_OPT_NOT_CRITICAL (0x0000)

  Not critical table.
- #define CFE_TBL_OPT_CRITICAL (0x0008)

  Critical table.
- #define CFE_TBL_OPT_DEFAULT (CFE_TBL_OPT_SNGL_BUFFER | CFE_TBL_OPT_LOAD_DUMP)

  Default table options.
- #define CFE_TBL_MAX_FULL_NAME_LEN (CFE_MISSION_TBL_MAX_FULL_NAME_LEN)

  Table maximum full name length.
- #define CFE_TBL_BAD_TABLE_HANDLE (CFE_TBL_Handle_t)0xFFFF

  Bad table handle.

Typedefs

- typedef int32(∗ CFE_TBL_CallbackFuncPtr_t) (void ∗TblPtr)

  Table Callback Function.
- typedef int16 CFE_TBL_Handle_t

  Table Handle primitive.
- typedef enum CFE_TBL_SrcEnum CFE_TBL_SrcEnum_t

  Table Source.
- typedef struct CFE_TBL_Info CFE_TBL_Info_t

  Table Info.

Enumerations

- enum CFE_TBL_SrcEnum { CFE_TBL_SRC_FILE = 0, CFE_TBL_SRC_ADDRESS }

  Table Source.

39.43.1 Macro Definition Documentation

39.43.1.1 CFE_TBL_BAD_TABLE_HANDLE

```
#define CFE_TBL_BAD_TABLE_HANDLE (CFE_TBL_Handle_t)0xFFFF
```

Bad table handle.

Definition at line 80 of file cfe_tbl_api_typedefs.h.

39.43.1.2 CFE_TBL_MAX_FULL_NAME_LEN

```
#define CFE_TBL_MAX_FULL_NAME_LEN (CFE_MISSION_TBL_MAX_FULL_NAME_LEN)
```

Table maximum full name length.

The full length of table names is defined at the mission scope. This is defined here to support applications that depend on cfe_tbl.h providing this value.

Definition at line 77 of file cfe_tbl_api_typedefs.h.

39.43.2 Typedef Documentation

39.43.2.1 CFE_TBL_CallbackFuncPtr_t

```
typedef int32(* CFE_TBL_CallbackFuncPtr_t) (void *TblPtr)
```

Table Callback Function.

Definition at line 85 of file cfe_tbl_api_typedefs.h.

39.43.2.2 CFE_TBL_Handle_t

```
typedef int16 CFE_TBL_Handle_t
```

Table Handle primitive.

Definition at line 88 of file cfe_tbl_api_typedefs.h.

39.43.2.3 CFE_TBL_Info_t

```
typedef struct CFE_TBL_Info CFE_TBL_Info_t
```

Table Info.

39.43.2.4 CFE_TBL_SrcEnum_t

```
typedef enum CFE_TBL_SrcEnum CFE_TBL_SrcEnum_t
```

Table Source.

39.43.3 Enumeration Type Documentation

39.43.3.1 CFE_TBL_SrcEnum

```
enum CFE_TBL_SrcEnum
```

Table Source.

Enumerator

CFE_TBL_SRC_FILE	File source When this option is selected, the `SrcDataPtr` will be interpreted as a pointer to a null terminated character string. The string should specify the full path and filename of the file containing the initial data contents of the table.
CFE_TBL_SRC_ADDRESS	Address source When this option is selected, the `SrcDataPtr` will be interpreted as a pointer to a memory location that is the beginning of the initialization data for loading the table OR, in the case of a "user defined" dump only table, the address of the active table itself. The block of memory is assumed to be of the same size specified in the CFE_TBL_Register function Size parameter.

Definition at line 91 of file cfe_tbl_api_typedefs.h.

39.44 cfe/modules/core_api/fsw/inc/cfe_tbl_core_internal.h File Reference

```
#include "common_types.h"
#include "cfe_es_extern_typedefs.h"
```

Functions

- void CFE_TBL_TaskMain (void)

  Entry Point for cFE Table Services Core Application.
- int32 CFE_TBL_EarlyInit (void)

  Initializes the Table Services API Library.
- int32 CFE_TBL_CleanUpApp (CFE_ES_AppId_t AppId)

  Removes TBL resources associated with specified Application.

39.45 cfe/modules/core_api/fsw/inc/cfe_tbl_extern_typedefs.h File Reference

```
#include "common_types.h"
#include "cfe_es_extern_typedefs.h"
#include "cfe_mission_cfg.h"
```

Data Structures

- struct CFE_TBL_File_Hdr

  The definition of the header fields that are included in CFE Table Data files.

Typedefs

- typedef uint16 CFE_TBL_BufferSelect_Enum_t

  Selects the buffer to operate on for validate or dump commands.
- typedef struct CFE_TBL_File_Hdr CFE_TBL_File_Hdr_t

  The definition of the header fields that are included in CFE Table Data files.

Enumerations

- enum CFE_TBL_BufferSelect { CFE_TBL_BufferSelect_INACTIVE = 0, CFE_TBL_BufferSelect_ACTIVE = 1 }

  Label definitions associated with CFE_TBL_BufferSelect_Enum_t.

39.45.1 Typedef Documentation

39.45.1.1 CFE_TBL_BufferSelect_Enum_t

```
typedef uint16 CFE_TBL_BufferSelect_Enum_t
```

Selects the buffer to operate on for validate or dump commands.

See also

enum CFE_TBL_BufferSelect

Definition at line 61 of file cfe_tbl_extern_typedefs.h.

39.45.1.2 CFE_TBL_File_Hdr_t

```
typedef struct CFE_TBL_File_Hdr CFE_TBL_File_Hdr_t
```

The definition of the header fields that are included in CFE Table Data files.

This header follows the CFE_FS header and precedes the the actual table data.

39.45.2 Enumeration Type Documentation

39.45.2.1 CFE_TBL_BufferSelect

```
enum CFE_TBL_BufferSelect
```

Label definitions associated with CFE_TBL_BufferSelect_Enum_t.

Enumerator

CFE_TBL_BufferSelect_INACTIVE	Select the Inactive buffer for validate or dump.
CFE_TBL_BufferSelect_ACTIVE	Select the Active buffer for validate or dump.

Definition at line 41 of file cfe_tbl_extern_typedefs.h.

39.46 cfe/modules/core_api/fsw/inc/cfe_tbl_filedef.h File Reference

```
#include "cfe_mission_cfg.h"
#include "common_types.h"
#include "cfe_tbl_extern_typedefs.h"
#include "cfe_fs_extern_typedefs.h"
```

Data Structures

- struct CFE_TBL_FileDef

Macros

- #define CFE_TBL_FILEDEF(ObjName, TblName, Desc, Filename)

Typedefs

- typedef struct CFE_TBL_FileDef CFE_TBL_FileDef_t

39.46.1 Macro Definition Documentation

39.46.1.1 CFE_TBL_FILEDEF

```
#define CFE_TBL_FILEDEF(
              ObjName,
              TblName,
              Desc,
              Filename )
```

Value:

```
static OS_USED CFE_TBL_FileDef_t CFE_TBL_FileDef = {#ObjName "\0", #
      TblName "\0", #Desc "\0", #Filename "\0", \
                                                    sizeof(ObjName)};
```

The CFE_TBL_FILEDEF macro can be used to simplify the declaration of a table image when using the elf2cfetbl utility.

```
Note that the macro adds a NULL at the end to ensure that it is null-terminated. (C allows
a struct to be statically initialized with a string exactly the length of the array, which
loses the null terminator.) This means the actual length limit of the fields are the above
LEN - 1.

An example of the source code and how this macro would be used is as follows:
```

```
 #include "cfe_tbl_filedef.h"

typedef struct MyTblStruct
{
    int     Int1;
    int     Int2;
    int     Int3;
    char    Char1;
} MyTblStruct_t;

MyTblStruct_t MyTblStruct = { 0x01020304, 0x05060708, 0x090A0B0C, 0x0D };

CFE_TBL_FILEDEF(MyTblStruct, MyApp.TableName, Table Utility Test Table, MyTblDefault.bin )
```

Definition at line 100 of file cfe_tbl_filedef.h.

39.46.2 Typedef Documentation

39.46.2.1 CFE_TBL_FileDef_t

```
typedef struct CFE_TBL_FileDef CFE_TBL_FileDef_t
```

39.47 cfe/modules/core_api/fsw/inc/cfe_time.h File Reference

```
#include "common_types.h"
#include "cfe_error.h"
#include "cfe_time_api_typedefs.h"
#include "cfe_es_api_typedefs.h"
```

Macros

- #define CFE_TIME_Copy(m, t)

  Time Copy.

Functions

- CFE_TIME_SysTime_t CFE_TIME_GetTime (void)

  Get the current spacecraft time.
- CFE_TIME_SysTime_t CFE_TIME_GetTAI (void)

  Get the current TAI (MET + SCTF) time.
- CFE_TIME_SysTime_t CFE_TIME_GetUTC (void)

  Get the current UTC (MET + SCTF - Leap Seconds) time.
- CFE_TIME_SysTime_t CFE_TIME_GetMET (void)

  Get the current value of the Mission Elapsed Time (MET).
- uint32 CFE_TIME_GetMETseconds (void)

  Get the current seconds count of the mission-elapsed time.
- uint32 CFE_TIME_GetMETsubsecs (void)

  Get the current sub-seconds count of the mission-elapsed time.
- CFE_TIME_SysTime_t CFE_TIME_GetSTCF (void)

  Get the current value of the spacecraft time correction factor (STCF).
- int16 CFE_TIME_GetLeapSeconds (void)

  Get the current value of the leap seconds counter.
- CFE_TIME_ClockState_Enum_t CFE_TIME_GetClockState (void)

  Get the current state of the spacecraft clock.
- uint16 CFE_TIME_GetClockInfo (void)

  Provides information about the spacecraft clock.
- CFE_TIME_SysTime_t CFE_TIME_Add (CFE_TIME_SysTime_t Time1, CFE_TIME_SysTime_t Time2)

  Adds two time values.
- CFE_TIME_SysTime_t CFE_TIME_Subtract (CFE_TIME_SysTime_t Time1, CFE_TIME_SysTime_t Time2)

  Subtracts two time values.
- CFE_TIME_Compare_t CFE_TIME_Compare (CFE_TIME_SysTime_t TimeA, CFE_TIME_SysTime_t TimeB)

  Compares two time values.
- CFE_TIME_SysTime_t CFE_TIME_MET2SCTime (CFE_TIME_SysTime_t METTime)

  Convert specified MET into Spacecraft Time.
- uint32 CFE_TIME_Sub2MicroSecs (uint32 SubSeconds)

  Converts a sub-seconds count to an equivalent number of microseconds.
- uint32 CFE_TIME_Micro2SubSecs (uint32 MicroSeconds)

  Converts a number of microseconds to an equivalent sub-seconds count.
- void CFE_TIME_ExternalTone (void)

  Provides the 1 Hz signal from an external source.
- void CFE_TIME_ExternalMET (CFE_TIME_SysTime_t NewMET)

  Provides the Mission Elapsed Time from an external source.
- void CFE_TIME_ExternalGPS (CFE_TIME_SysTime_t NewTime, int16 NewLeaps)

  Provide the time from an external source that has data common to GPS receivers.
- void CFE_TIME_ExternalTime (CFE_TIME_SysTime_t NewTime)

  Provide the time from an external source that measures time relative to a known epoch.
- CFE_Status_t CFE_TIME_RegisterSynchCallback (CFE_TIME_SynchCallbackPtr_t CallbackFuncPtr)

  Registers a callback function that is called whenever time synchronization occurs.
- CFE_Status_t CFE_TIME_UnregisterSynchCallback (CFE_TIME_SynchCallbackPtr_t CallbackFuncPtr)

  Unregisters a callback function that is called whenever time synchronization occurs.
- void CFE_TIME_Print (char *PrintBuffer, CFE_TIME_SysTime_t TimeToPrint)

  Print a time value as a string.
- void CFE_TIME_Local1HzISR (void)

  This function should be called from the system PSP layer once per second.

39.47.1 Macro Definition Documentation

39.47.1.1 CFE_TIME_Copy

```
#define CFE_TIME_Copy(
            m,
            t )
```

Value:

```
{                                       \
        (m)->Seconds    = (t)->Seconds;     \
        (m)->Subseconds = (t)->Subseconds; \
    }
```

Time Copy.

Macro to copy systime into another systime. Preferred to use this macro as it does not require the two arguments to be exactly the same type, it will work with any two structures that define "Seconds" and "Subseconds" members.

Definition at line 53 of file cfe_time.h.

39.48 cfe/modules/core_api/fsw/inc/cfe_time_api_typedefs.h File Reference

```
#include "common_types.h"
#include "cfe_time_extern_typedefs.h"
```

Macros

- #define CFE_TIME_PRINTED_STRING_SIZE 24

  Required size of buffer to be passed into CFE_TIME_Print (includes null terminator)

Typedefs

- typedef enum CFE_TIME_Compare CFE_TIME_Compare_t

  Enumerated types identifying the relative relationships of two times.
- typedef int32(∗ CFE_TIME_SynchCallbackPtr_t) (void)

  Time Synchronization Callback Function Ptr Type.

Enumerations

- enum CFE_TIME_Compare { CFE_TIME_A_LT_B = -1, CFE_TIME_EQUAL = 0, CFE_TIME_A_GT_B = 1 }

  Enumerated types identifying the relative relationships of two times.

39.48.1 Macro Definition Documentation

39.48.1.1 CFE_TIME_PRINTED_STRING_SIZE

```
#define CFE_TIME_PRINTED_STRING_SIZE 24
```

Required size of buffer to be passed into CFE_TIME_Print (includes null terminator)

Definition at line 49 of file cfe_time_api_typedefs.h.

39.48.2 Typedef Documentation

39.48.2.1 CFE_TIME_Compare_t

```
typedef enum CFE_TIME_Compare CFE_TIME_Compare_t
```

Enumerated types identifying the relative relationships of two times.

Description

Since time fields contain numbers that are relative to an epoch time, then it is possible for a time value to be "negative". This can lead to some confusion about what relationship exists between two time values. To resolve this confusion, the cFE provides the API CFE_TIME_Compare which returns these enumerated values.

39.48.2.2 CFE_TIME_SynchCallbackPtr_t

```
typedef int32(* CFE_TIME_SynchCallbackPtr_t) (void)
```

Time Synchronization Callback Function Ptr Type.

Description

Applications that wish to get direct notification of the receipt of the cFE Time Synchronization signal (typically a 1 Hz signal), must register a callback function with the following prototype via the CFE_TIME_RegisterSynchCallback API.

Definition at line 80 of file cfe_time_api_typedefs.h.

39.48.3 Enumeration Type Documentation

39.48.3.1 CFE_TIME_Compare

```
enum CFE_TIME_Compare
```

Enumerated types identifying the relative relationships of two times.

Description

Since time fields contain numbers that are relative to an epoch time, then it is possible for a time value to be "negative". This can lead to some confusion about what relationship exists between two time values. To resolve this confusion, the cFE provides the API CFE_TIME_Compare which returns these enumerated values.

Enumerator

CFE_TIME_A_LT_B	The first specified time is considered to be before the second specified time.
CFE_TIME_EQUAL	The two specified times are considered to be equal.
CFE_TIME_A_GT↩ _B	The first specified time is considered to be after the second specified time.

Definition at line 65 of file cfe_time_api_typedefs.h.

39.49 cfe/modules/core_api/fsw/inc/cfe_time_core_internal.h File Reference

```
#include "common_types.h"
#include "cfe_es_extern_typedefs.h"
```

Functions

- void CFE_TIME_TaskMain (void)

  Entry Point for cFE Core Application.
- int32 CFE_TIME_EarlyInit (void)

  Initializes the cFE core module API Library.
- int32 CFE_TIME_CleanUpApp (CFE_ES_AppId_t AppId)

  Removes TIME resources associated with specified Application.

39.50 cfe/modules/core_api/fsw/inc/cfe_time_extern_typedefs.h File Reference

```
#include "common_types.h"
```

Data Structures

- struct CFE_TIME_SysTime

  Data structure used to hold system time values.

Typedefs

- typedef struct CFE_TIME_SysTime CFE_TIME_SysTime_t

  Data structure used to hold system time values.
- typedef uint8 CFE_TIME_FlagBit_Enum_t

  Bit positions of the various clock state flags.
- typedef int16 CFE_TIME_ClockState_Enum_t

  Enumerated types identifying the quality of the current time.
- typedef uint8 CFE_TIME_SourceSelect_Enum_t

  Clock Source Selection Parameters.
- typedef uint8 CFE_TIME_ToneSignalSelect_Enum_t

  Tone Signal Selection Parameters.
- typedef uint8 CFE_TIME_AdjustDirection_Enum_t

  STCF adjustment direction (for both one-time and 1Hz adjustments)
- typedef uint8 CFE_TIME_FlywheelState_Enum_t

  Fly-wheel status values.
- typedef uint8 CFE_TIME_SetState_Enum_t

  Clock status values (has the clock been set to correct time)

Enumerations

- enum CFE_TIME_FlagBit { CFE_TIME_FlagBit_CLKSET = 0, CFE_TIME_FlagBit_FLYING = 1, CFE_TIME_FlagBit_SRCINT = 2, CFE_TIME_FlagBit_SIGPRI = 3, CFE_TIME_FlagBit_SRVFLY = 4, CFE_TIME_FlagBit_CMDFLY = 5, CFE_TIME_FlagBit_ADDADJ = 6, CFE_TIME_FlagBit_ADD1HZ = 7, CFE_TIME_FlagBit_ADDTCL = 8, CFE_TIME_FlagBit_SERVER = 9, CFE_TIME_FlagBit_GDTONE = 10 }

  Label definitions associated with CFE_TIME_FlagBit_Enum_t.
- enum CFE_TIME_ClockState { CFE_TIME_ClockState_INVALID = -1, CFE_TIME_ClockState_VALID = 0, CFE_TIME_ClockState_FLYWHEEL = 1 }

  Label definitions associated with CFE_TIME_ClockState_Enum_t.
- enum CFE_TIME_SourceSelect { CFE_TIME_SourceSelect_INTERNAL = 1, CFE_TIME_SourceSelect_EXTERNAL = 2 }

  Label definitions associated with CFE_TIME_SourceSelect_Enum_t.
- enum CFE_TIME_ToneSignalSelect { CFE_TIME_ToneSignalSelect_PRIMARY = 1, CFE_TIME_ToneSignalSelect_REDUNDANT = 2 }

  Label definitions associated with CFE_TIME_ToneSignalSelect_Enum_t.
- enum CFE_TIME_AdjustDirection { CFE_TIME_AdjustDirection_ADD = 1, CFE_TIME_AdjustDirection_SUBTRACT = 2 }

  Label definitions associated with CFE_TIME_AdjustDirection_Enum_t.
- enum CFE_TIME_FlywheelState { CFE_TIME_FlywheelState_NO_FLY = 0, CFE_TIME_FlywheelState_IS_FLY = 1 }

  Label definitions associated with CFE_TIME_FlywheelState_Enum_t.
- enum CFE_TIME_SetState { CFE_TIME_SetState_NOT_SET = 0, CFE_TIME_SetState_WAS_SET = 1 }

  Label definitions associated with CFE_TIME_SetState_Enum_t.

39.50.1 Typedef Documentation

39.50.1.1 CFE_TIME_AdjustDirection_Enum_t

```
typedef uint8 CFE_TIME_AdjustDirection_Enum_t
```

STCF adjustment direction (for both one-time and 1Hz adjustments)

See also

enum CFE_TIME_AdjustDirection

Definition at line 250 of file cfe_time_extern_typedefs.h.

39.50.1.2 CFE_TIME_ClockState_Enum_t

```
typedef int16 CFE_TIME_ClockState_Enum_t
```

Enumerated types identifying the quality of the current time.

Description

The CFE_TIME_ClockState_Enum_t enumerations identify the three recognized states of the current time. If the clock has never been successfully synchronized with the primary onboard clock source, the time is conisdered to be CFE_TIME_ClockState_INVALID. If the time is currently synchronized (i.e. - the primary synchronization mechanism has not been dropped for any significant amount of time), then the current time is considered to be CFE_TIME_ClockState_VALID. If the time had, at some point in the past, been synchronized, but the synchronization with the primary onboard clock has since been lost, then the time is considered to be CFE_TIME_ClockState_FLYWHEEL. Since different clocks drift at different rates from one another, the accuracy of the time while in CFE_TIME_ClockState_FLYWHEEL is dependent upon the time spent in that state.

See also

enum CFE_TIME_ClockState

Definition at line 175 of file cfe_time_extern_typedefs.h.

39.50.1.3 CFE_TIME_FlagBit_Enum_t

```
typedef uint8 CFE_TIME_FlagBit_Enum_t
```

Bit positions of the various clock state flags.

See also

enum CFE_TIME_FlagBit

Definition at line 121 of file cfe_time_extern_typedefs.h.

39.50.1.4 CFE_TIME_FlywheelState_Enum_t

```
typedef uint8 CFE_TIME_FlywheelState_Enum_t
```

Fly-wheel status values.

See also

enum CFE_TIME_FlywheelState

Definition at line 275 of file cfe_time_extern_typedefs.h.

39.50.1.5 CFE_TIME_SetState_Enum_t

```
typedef uint8 CFE_TIME_SetState_Enum_t
```

Clock status values (has the clock been set to correct time)

See also

enum CFE_TIME_SetState

Definition at line 300 of file cfe_time_extern_typedefs.h.

39.50.1.6 CFE_TIME_SourceSelect_Enum_t

```
typedef uint8 CFE_TIME_SourceSelect_Enum_t
```

Clock Source Selection Parameters.

See also

enum CFE_TIME_SourceSelect

Definition at line 200 of file cfe_time_extern_typedefs.h.

39.50.1.7 CFE_TIME_SysTime_t

```
typedef struct CFE_TIME_SysTime CFE_TIME_SysTime_t
```

Data structure used to hold system time values.

Description

The CFE_TIME_SysTime_t data structure is used to hold time values. Time is referred to as the elapsed time (in seconds and subseconds) since a specified epoch time. The subseconds field contains the number of 2^(-32) second intervals that have elapsed since the epoch.

39.50.1.8 CFE_TIME_ToneSignalSelect_Enum_t

```
typedef uint8 CFE_TIME_ToneSignalSelect_Enum_t
```

Tone Signal Selection Parameters.

See also

enum CFE_TIME_ToneSignalSelect

Definition at line 225 of file cfe_time_extern_typedefs.h.

39.50.2 Enumeration Type Documentation

39.50.2.1 CFE_TIME_AdjustDirection

```
enum CFE_TIME_AdjustDirection
```

Label definitions associated with CFE_TIME_AdjustDirection_Enum_t.

Enumerator

CFE_TIME_AdjustDirection_ADD	Add time adjustment.
CFE_TIME_AdjustDirection_SUBTRACT	Subtract time adjustment.

Definition at line 230 of file cfe_time_extern_typedefs.h.

39.50.2.2 CFE_TIME_ClockState

`enum CFE_TIME_ClockState`

Label definitions associated with CFE_TIME_ClockState_Enum_t.

Enumerator

CFE_TIME_ClockState_INVALID	The spacecraft time has not been set since the last clock reset. Times returned by clock routines have no relationship to any ground-based time reference.
CFE_TIME_ClockState_VALID	The spacecraft time has been set at least once since the last clock reset, and it is synchronized with the primary on-board time base. Times returned by clock routines can be trusted.
CFE_TIME_ClockState_FLYWHEEL	The spacecraft time has been set at least once since the last clock reset, but it is not currently synchronized with the primary on-board time base. Times returned by clock routines are a "best guess" based on a non-optimal oscillator.

Definition at line 126 of file cfe_time_extern_typedefs.h.

39.50.2.3 CFE_TIME_FlagBit

`enum CFE_TIME_FlagBit`

Label definitions associated with CFE_TIME_FlagBit_Enum_t.

Enumerator

CFE_TIME_FlagBit_CLKSET	The spacecraft time has been set.
CFE_TIME_FlagBit_FLYING	This instance of Time Services is flywheeling.
CFE_TIME_FlagBit_SRCINT	The clock source is set to internal.
CFE_TIME_FlagBit_SIGPRI	The clock signal is set to primary.
CFE_TIME_FlagBit_SRVFLY	The Time Server is in flywheel mode.
CFE_TIME_FlagBit_CMDFLY	This instance of Time Services was commanded into flywheel mode.
CFE_TIME_FlagBit_ADDADJ	One time STCF Adjustment is to be done in positive direction.
CFE_TIME_FlagBit_ADD1HZ	1 Hz STCF Adjustment is to be done in a positive direction
CFE_TIME_FlagBit_ADDTCL	Time Client Latency is applied in a positive direction.
CFE_TIME_FlagBit_SERVER	This instance of Time Services is a Time Server.
CFE_TIME_FlagBit_GDTONE	The tone received is good compared to the last tone received.

Definition at line 56 of file cfe_time_extern_typedefs.h.

39.50.2.4 CFE_TIME_FlywheelState

`enum CFE_TIME_FlywheelState`

Label definitions associated with CFE_TIME_FlywheelState_Enum_t.

Enumerator

CFE_TIME_FlywheelState_NO_FLY	Not in flywheel state.
CFE_TIME_FlywheelState_IS_FLY	In flywheel state.

Definition at line 255 of file cfe_time_extern_typedefs.h.

39.50.2.5 CFE_TIME_SetState

`enum CFE_TIME_SetState`

Label definitions associated with CFE_TIME_SetState_Enum_t.

Enumerator

CFE_TIME_SetState_NOT_SET	Spacecraft time has not been set.
CFE_TIME_SetState_WAS_SET	Spacecraft time has been set.

Definition at line 280 of file cfe_time_extern_typedefs.h.

39.50.2.6 CFE_TIME_SourceSelect

`enum CFE_TIME_SourceSelect`

Label definitions associated with CFE_TIME_SourceSelect_Enum_t.

Enumerator

CFE_TIME_SourceSelect_INTERNAL	Use Internal Source.
CFE_TIME_SourceSelect_EXTERNAL	Use External Source.

Definition at line 180 of file cfe_time_extern_typedefs.h.

39.50.2.7 CFE_TIME_ToneSignalSelect

`enum CFE_TIME_ToneSignalSelect`

Label definitions associated with CFE_TIME_ToneSignalSelect_Enum_t.

Enumerator

CFE_TIME_ToneSignalSelect_PRIMARY	Primary Source.
CFE_TIME_ToneSignalSelect_REDUNDANT	Redundant Source.

Definition at line 205 of file cfe_time_extern_typedefs.h.

39.51 cfe/modules/core_api/fsw/inc/cfe_version.h File Reference

Purpose: Provide version identifiers for the cFE core.

Macros

- #define CFE_BUILD_NUMBER 436
- #define CFE_BUILD_BASELINE "v6.8.0-rc1"
- #define CFE_MAJOR_VERSION 6

  ONLY APPLY for OFFICIAL releases. Major version number.
- #define CFE_MINOR_VERSION 7

  ONLY APPLY for OFFICIAL releases. Minor version number.
- #define CFE_REVISION 99

  ONLY APPLY for OFFICIAL releases. Revision version number. A value of "99" indicates a development \ version.
- #define CFE_MISSION_REV 0

  ONLY USED by MISSION Implementations. Mission revision.
- #define CFE_STR_HELPER(x) #x

  Helper function to concatenate strings from integer macros.
- #define CFE_STR(x) CFE_STR_HELPER(x)

  Helper function to concatenate strings from integer macros.
- #define CFE_SRC_VERSION CFE_BUILD_BASELINE "+dev" CFE_STR(CFE_BUILD_NUMBER)

  Development Build Version Number.
- #define CFE_VERSION_STRING

  Development Build Version String.

39.51.1 Detailed Description

Purpose: Provide version identifiers for the cFE core.

target_config.h contains extended version information within it. This information is generated automatically by the build using git to determine the most recent tag and commit id.

39.51.2 Macro Definition Documentation

39.51.2.1 CFE_BUILD_BASELINE

```
#define CFE_BUILD_BASELINE "v6.8.0-rc1"
```

Development Build: git tag that is the base for the current development \

Definition at line 36 of file cfe_version.h.

39.51.2.2 CFE_BUILD_NUMBER

```
#define CFE_BUILD_NUMBER 436
```

Development Build: Number of commits since baseline

Definition at line 35 of file cfe_version.h.

39.51.2.3 CFE_MAJOR_VERSION

```
#define CFE_MAJOR_VERSION 6
```

ONLY APPLY for OFFICIAL releases. Major version number.

Definition at line 42 of file cfe_version.h.

39.51.2.4 CFE_MINOR_VERSION

```
#define CFE_MINOR_VERSION 7
```

ONLY APPLY for OFFICIAL releases. Minor version number.

Definition at line 43 of file cfe_version.h.

39.51.2.5 CFE_MISSION_REV

```
#define CFE_MISSION_REV 0
```

ONLY USED by MISSION Implementations. Mission revision.

Definition at line 48 of file cfe_version.h.

39.51.2.6 CFE_REVISION

```
#define CFE_REVISION 99
```

ONLY APPLY for OFFICIAL releases. Revision version number. A value of "99" indicates a development \ version.

Definition at line 44 of file cfe_version.h.

39.51.2.7 CFE_SRC_VERSION

```
#define CFE_SRC_VERSION CFE_BUILD_BASELINE "+dev" CFE_STR(CFE_BUILD_NUMBER)
```

Development Build Version Number.

Baseline git tag + Number of commits since baseline.
See Version Numbers for format differences between development and release versions.

Definition at line 57 of file cfe_version.h.

39.51.2.8 CFE_STR

```
#define CFE_STR(
              x ) CFE_STR_HELPER(x)
```

Helper function to concatenate strings from integer macros.

Definition at line 51 of file cfe_version.h.

39.51.2.9 CFE_STR_HELPER

```
#define CFE_STR_HELPER(
              x ) #x
```

Helper function to concatenate strings from integer macros.

Definition at line 50 of file cfe_version.h.

39.51.2.10 CFE_VERSION_STRING

```
#define CFE_VERSION_STRING
```

Value:

```
" cFE DEVELOPMENT BUILD " CFE_SRC_VERSION " (Codename: Bootes)" /* Codename for current
      development */ \
    ", Last Official Release: cfe v6.7.0"                          /* For full support please use this
      version */
```

Development Build Version String.

Reports the current development build's baseline, number, and name. Also includes a note about the latest official version.
See Version Numbers for format differences between development and release versions.

Definition at line 63 of file cfe_version.h.

39.52 cfe/modules/es/fsw/inc/cfe_es_events.h File Reference

Macros

- #define CFE_ES_MAX_EID 92
- #define CFE_ES_INIT_INF_EID 1 /∗ start up message "informational" ∗/

  'cFE ES Initialized'
- #define CFE_ES_INITSTATS_INF_EID 2

  'cFE Version %d.%d.%d chksm %d, OSAL Version %d.%d'
- #define CFE_ES_NOOP_INF_EID 3 /∗ processed command "informational" ∗/

  'No-op command'
- #define CFE_ES_RESET_INF_EID 4

  'Reset Counters command'
- #define CFE_ES_START_INF_EID 6

  'Started %s from %s, AppID = %d'
- #define CFE_ES_STOP_DBG_EID 7

  'Stop Application %s Initiated.'
- #define CFE_ES_STOP_INF_EID 8

  'Stop Application %s Completed.'
- #define CFE_ES_RESTART_APP_DBG_EID 9

  'Restart Application %s Initiated.'
- #define CFE_ES_RESTART_APP_INF_EID 10

  'Restart Application %s Completed, AppID=lu'
- #define CFE_ES_RELOAD_APP_DBG_EID 11

  'Reload Application %s Initiated.'
- #define CFE_ES_RELOAD_APP_INF_EID 12

  'Reload Application %s Completed, AppID=lu'
- #define CFE_ES_EXIT_APP_INF_EID 13

  'Exit Application %s Completed.'

- #define CFE_ES_ERREXIT_APP_INF_EID 14

  `'Exit Application %s Completed.'`
- #define CFE_ES_ONE_APP_EID 15

  `'Sent %s application data'`
- #define CFE_ES_ALL_APPS_EID 16

  `'App Info file written to %s, Entries=%d, FileSize=%d'`
- #define CFE_ES_SYSLOG1_INF_EID 17

  `'Cleared Executive Services log data'`
- #define CFE_ES_SYSLOG2_EID 18

  `'%s written:Size=%d,Entries=%d'`
- #define CFE_ES_ERLOG1_INF_EID 19

  `'Cleared mode log data'`
- #define CFE_ES_ERLOG2_EID 20

  `'%s written:Size=%d'`
- #define CFE_ES_MID_ERR_EID 21 /∗ invalid command packet "error" ∗/

  `'Invalid command pipe message ID: 0x%X'`
- #define CFE_ES_CC1_ERR_EID 22

  `'Invalid ground command code:  ID = 0x%X, CC = %d'`
- #define CFE_ES_LEN_ERR_EID 23

  `'Invalid cmd length:  ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'`
- #define CFE_ES_BOOT_ERR_EID 24 /∗ command specific "error" ∗/

  `'Invalid cFE restart type %d'`
- #define CFE_ES_START_ERR_EID 26

  `'Failed to start %s from %s, RC = %08X'`
- #define CFE_ES_START_INVALID_FILENAME_ERR_EID 27

  `'CFE_ES_StartAppCmd:  invalid filename:  %s'`
- #define CFE_ES_START_INVALID_ENTRY_POINT_ERR_EID 28

  `'CFE_ES_StartAppCmd:  App Entry Point is NULL.'`
- #define CFE_ES_START_NULL_APP_NAME_ERR_EID 29

  `'CFE_ES_StartAppCmd:  App Name is NULL.'`
- #define CFE_ES_START_PRIORITY_ERR_EID 31

  `'CFE_ES_StartAppCmd:  Priority is too large:  %d.'`
- #define CFE_ES_START_EXC_ACTION_ERR_EID 32

  `'CFE_ES_StartAppCmd:  Invalid Exception Action:  %d.'`
- #define CFE_ES_ERREXIT_APP_ERR_EID 33

  `'Exit Application %s on Error Failed:  CleanUpApp Error 0x%08X.'`
- #define CFE_ES_STOP_ERR1_EID 35

  `'Stop Application %s Failed, RC = 0x%08X'`
- #define CFE_ES_STOP_ERR2_EID 36

  `'Stop Application %s, GetAppIDByName failed.  RC = 0x%08X.'`
- #define CFE_ES_STOP_ERR3_EID 37

  `'Stop Application %s Failed:  CleanUpApp Error 0x%08X.'`
- #define CFE_ES_RESTART_APP_ERR1_EID 38

  `'Restart Application %s Failed, RC = 0x%08X'`
- #define CFE_ES_RESTART_APP_ERR2_EID 39

  `'Restart Application %s, GetAppIDByName failed.  RC = 0x%08X.'`
- #define CFE_ES_RESTART_APP_ERR3_EID 40

'Restart Application %s Failed: AppCreate Error 0x%08X.'

- #define CFE_ES_RESTART_APP_ERR4_EID 41

  'Restart Application %s Failed: CleanUpApp Error 0x%08X.'

- #define CFE_ES_RELOAD_APP_ERR1_EID 42

  'Failed to reload Application %s, rc = %08X'

- #define CFE_ES_RELOAD_APP_ERR2_EID 43

  'Reload Application %s, GetAppIDByName failed. RC = 0x%08X.'

- #define CFE_ES_RELOAD_APP_ERR3_EID 44

  'Reload Application %s Failed: AppCreate Error 0x%08X.'

- #define CFE_ES_RELOAD_APP_ERR4_EID 45

  'Reload Application %s Failed: CleanUpApp Error 0x%08X.'

- #define CFE_ES_EXIT_APP_ERR_EID 46

  'Exit Application %s Failed: CleanUpApp Error 0x%08X.'

- #define CFE_ES_PCR_ERR1_EID 47

  'ES_ProcControlReq: Invalid State (EXCEPTION) Application %s.'

- #define CFE_ES_PCR_ERR2_EID 48

  'ES_ProcControlReq: Unknown State (%d) Application %s.'

- #define CFE_ES_ONE_ERR_EID 49

  'Failed to send %s application data, RC = %08X'

- #define CFE_ES_ONE_APPID_ERR_EID 50

  'Failed to send %s application data: GetAppIDByName Failed, RC = 0x%08X'

- #define CFE_ES_OSCREATE_ERR_EID 51

  'Failed to write App Info file, OS_OpenCreate returned %d'

- #define CFE_ES_WRHDR_ERR_EID 52

  'Failed to write App Info file, WriteHdr rtnd %08X, exp %d'

- #define CFE_ES_TASKWR_ERR_EID 53

  'Failed to write App Info file, Task write RC = 0x%08X, exp %d'

- #define CFE_ES_SYSLOG2_ERR_EID 55

  'Error creating file %s, stat=0x%x'

- #define CFE_ES_ERLOG2_ERR_EID 56

  'Error creating file %s, stat=0x%x'

- #define CFE_ES_PERF_STARTCMD_EID 57

  'Start collecting performance data command, trigger mode = d'

- #define CFE_ES_PERF_STARTCMD_ERR_EID 58

  'Cannot start collecting performance data,perf data write in progress'

- #define CFE_ES_PERF_STARTCMD_TRIG_ERR_EID 59

  'Cannot start collecting performance data, trigger mode (d) out of range (d to d)'

- #define CFE_ES_PERF_STOPCMD_EID 60

  'Perf Stop Cmd Rcvd,%s will write %d entries.%dmS dly every %d entries'

- #define CFE_ES_PERF_STOPCMD_ERR2_EID 62

  'Stop performance data cmd ignored,perf data write in progress'

- #define CFE_ES_PERF_FILTMSKCMD_EID 63

  'Set Performance Filter Mask command'

- #define CFE_ES_PERF_FILTMSKERR_EID 64

  'Performance Filter Mask Cmd Error,Index(u)out of range(u)'

- #define CFE_ES_PERF_TRIGMSKCMD_EID 65

'Set Performance Trigger Mask command'

- #define CFE_ES_PERF_TRIGMSKERR_EID 66

  'Performance Trigger Mask Cmd Error,Index(u)out of range(u)'

- #define CFE_ES_PERF_LOG_ERR_EID 67

  'Error creating file %s, stat=%d'

- #define CFE_ES_PERF_DATAWRITTEN_EID 68

  '%s written:Size=%d,EntryCount=%d'

- #define CFE_ES_CDS_REGISTER_ERR_EID 69

  '%s Failed to Register CDS '%s', Status=0x%08X'

- #define CFE_ES_SYSLOGMODE_EID 70

  'Set OverWriteSysLog Command Received with Mode setting = %d'

- #define CFE_ES_ERR_SYSLOGMODE_EID 71

  'Set OverWriteSysLog Command: Invalid Mode setting = %d'

- #define CFE_ES_RESET_PR_COUNT_EID 72

  'Reset Processor Reset Count to Zero'

- #define CFE_ES_SET_MAX_PR_COUNT_EID 73

  'Maximum Processor Reset Count set to: %d'

- #define CFE_ES_FILEWRITE_ERR_EID 74

  'File write,byte cnt err,file %s,request=%d,actual=%d'

- #define CFE_ES_CDS_DELETE_ERR_EID 76

  'Error while deleting '%s' from CDS, See SysLog.(Err=0x%08X)'

- #define CFE_ES_CDS_NAME_ERR_EID 77

  'Unable to locate '%s' in CDS Registry'

- #define CFE_ES_CDS_DELETED_INFO_EID 78

  'Successfully removed '%s' from CDS'

- #define CFE_ES_CDS_DELETE_TBL_ERR_EID 79

  'CDS '%s' is a Critical Table CDS. Must be deleted via TBL Command'

- #define CFE_ES_CDS_OWNER_ACTIVE_EID 80

  'CDS '%s' not deleted because owning app is active'

- #define CFE_ES_TLM_POOL_STATS_INFO_EID 81

  'Successfully telemetered memory pool stats for 0x%08X'

- #define CFE_ES_INVALID_POOL_HANDLE_ERR_EID 82

  'Cannot telemeter memory pool stats. Illegal Handle (0x%08X)'

- #define CFE_ES_CDS_REG_DUMP_INF_EID 83

  'Successfully dumped CDS Registry to '%s':Size=%d,Entries=%d'

- #define CFE_ES_CDS_DUMP_ERR_EID 84

  'Error writing CDS Registry to '%s', Status=0x%08X'

- #define CFE_ES_WRITE_CFE_HDR_ERR_EID 85

  'Error writing cFE File Header to '%s', Status=0x%08X'

- #define CFE_ES_CREATING_CDS_DUMP_ERR_EID 86

  'Error creating CDS dump file '%s', Status=0x%08X'

- #define CFE_ES_TASKINFO_EID 87

  'Task Info file written to %s, Entries=%d, FileSize=%d'

- #define CFE_ES_TASKINFO_OSCREATE_ERR_EID 88

  'Failed to write Task Info file, OS_OpenCreate returned %d'

- #define CFE_ES_TASKINFO_WRHDR_ERR_EID 89

  'Failed to write Task Info file, WriteHdr rtnd %08X, exp %d'

- #define CFE_ES_TASKINFO_WR_ERR_EID 90

  'Failed to write Task Info file, Task write RC = 0x%08X, exp %d'
- #define CFE_ES_VERSION_INF_EID 91

  'Version Info: s, s'
- #define CFE_ES_BUILD_INF_EID 92

  'Build s by s%s, config s'
- #define CFE_ES_ERLOG_PENDING_ERR_EID 93

  'Error log write to file %s already in progress'

39.52.1 Macro Definition Documentation

39.52.1.1 CFE_ES_ALL_APPS_EID

```
#define CFE_ES_ALL_APPS_EID 16
```

`'App Info file written to %s, Entries=%d, FileSize=%d'`

Event Message `'App Info file written to %s, Entries=%d, FileSize=%d'`

Type: DEBUG

Cause:

This event message is issued upon successful completion of the cFE Executive Services Query All Applications command

The `'s'` field identifies the name of the file to which all Executive Services Application data has been written. The `Entries` field identifies, in decimal, the number of Applications whose data was written and the `FileSize` field gives the total number of bytes written to the file.

Definition at line 286 of file cfe_es_events.h.

39.52.1.2 CFE_ES_BOOT_ERR_EID

```
#define CFE_ES_BOOT_ERR_EID 24 /* command specific "error" */
```

`'Invalid cFE restart type %d'`

Event Message `'Invalid cFE restart type %d'`

Type: ERROR

Cause:

This event message is issued when the cFE Executive Services receives a cFE Restart Command whose parameter identifying the restart type is not equal to either CFE_PSP_RST_TYPE_PROCESSOR or CFE_PSP_RST_TYPE_P↩OWERON.

The 'd' field identifies the numeric, in decimal, of the restart type found in the received cFE Restart Command Packet.

Definition at line 418 of file cfe_es_events.h.

39.52.1.3 CFE_ES_BUILD_INF_EID

```
#define CFE_ES_BUILD_INF_EID 92
```

`'Build s by s%s, config s'`

Event Message `'Build s by s%s, config s'`

Type: INFORMATION

Cause:

This event message is always automatically issued when the Executive Services Task completes its Initialization, and as part of the Noop command.

The `Build` field identifies the build date, time, hostname and user identifier of the build host machine for the current running binary. The first string is the build date/time, and the second string is formatted as "user@hostname"

This additionally reports the configuration name that was selected by the user, which may affect various platform/mission limits.

By default, if not specified/overridden, the default values of these variables will be: BUILDDATE ==> the output of "date +%Y%m%d%H%M" HOSTNAME ==> the output of "hostname" USER ==> the output of "whoami"

The values can be overridden by setting an environment variable with the names above to the value desired for the field when running "make".

Definition at line 1437 of file cfe_es_events.h.

39.52.1.4 CFE_ES_CC1_ERR_EID

```
#define CFE_ES_CC1_ERR_EID 22
```

`'Invalid ground command code:  ID = 0x%X, CC = %d'`

Event Message `'Invalid ground command code:  ID = 0x%X, CC = %d'`

Type: ERROR

Cause:

This event message is generated when a message with the CFE_ES_CMD_MID message ID has arrived but whose Command Code is not one of the command codes specified in cfe_es.h . This problem is most likely to occur when:

1. A Message ID meant for another Application became corrupted and was set equal to CFE_ES_CMD_MID.
2. The Command Code field in the Message became corrupted.
3. The command database at the ground station has been corrupted.

The `ID` field in the event message specifies the Message ID (in hex) and the `CC` field specifies the Command Code (in decimal) found in the message.

Definition at line 383 of file cfe_es_events.h.

39.52.1.5 CFE_ES_CDS_DELETE_ERR_EID

```
#define CFE_ES_CDS_DELETE_ERR_EID 76
```

`'Error while deleting '%s' from CDS, See SysLog.(Err=0x%08X)'`

Event Message `'Error while deleting '%s' from CDS, See SysLog.(Err=0x%08X)'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Delete CDS Command fails to cleanly remove the specified CDS.

The `'s'` field identifies the name of the CDS that was attempted to be deleted the `Err` field specifies, in hex, the error code.

Definition at line 1160 of file cfe_es_events.h.

39.52.1.6 CFE_ES_CDS_DELETE_TBL_ERR_EID

```
#define CFE_ES_CDS_DELETE_TBL_ERR_EID 79
```

`'CDS '%s' is a Critical Table CDS. Must be deleted via TBL Command'`

Event Message `'CDS '%s' is a Critical Table CDS. Must be deleted via TBL Command'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Delete CDS Command specifies a name for a CDS that is a Critical Table image. Critical Table images can only be deleted via a Table Services command (CFE_TBL_DELE↩TE_CDS_CC).

The `'s'` field identifies the name of the CDS that was attempted to be deleted.

Definition at line 1207 of file cfe_es_events.h.

39.52.1.7 CFE_ES_CDS_DELETED_INFO_EID

```
#define CFE_ES_CDS_DELETED_INFO_EID 78
```

`'Successfully removed '%s' from CDS'`

Event Message `'Successfully removed '%s' from CDS'`

Type: INFORMATION

Cause:

This event message is generated when an Executive Services Delete CDS Command is successfully completed.

The `'s'` field identifies the name of the CDS that was deleted.

Definition at line 1190 of file cfe_es_events.h.

39.52.1.8 CFE_ES_CDS_DUMP_ERR_EID

```
#define CFE_ES_CDS_DUMP_ERR_EID 84
```

`'Error writing CDS Registry to '%s', Status=0x%08X'`

Event Message `'Error writing CDS Registry to '%s', Status=0x%08X'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Dump Critical Data Store Registry Command was being performed and it encountered a filesystem write error while writing a CDS Registry record.

The `'s'` field identifies the CDS Registry Dump Filename. The `'08X'` field identifies the error code returned from OS_write that caused the command to abort.

Definition at line 1293 of file cfe_es_events.h.

39.52.1.9 CFE_ES_CDS_NAME_ERR_EID

```
#define CFE_ES_CDS_NAME_ERR_EID 77
```

`'Unable to locate '%s' in CDS Registry'`

Event Message `'Unable to locate '%s' in CDS Registry'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Delete CDS Command specifies a name for a CDS that cannot be found in the CDS Registry.

The `'s'` field identifies the name of the CDS that was attempted to be deleted.

Definition at line 1175 of file cfe_es_events.h.

39.52.1.10 CFE_ES_CDS_OWNER_ACTIVE_EID

```
#define CFE_ES_CDS_OWNER_ACTIVE_EID 80
```

`'CDS '%s' not deleted because owning app is active'`

Event Message `'CDS '%s' not deleted because owning app is active'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Delete CDS Command specifies a name for a CDS whose prefix name identifies an application that is still registered in the system. CDSs can only be deleted when their owning applications have been removed from the system.

The `'s'` field identifies the name of the CDS that was attempted to be deleted.

Definition at line 1225 of file cfe_es_events.h.

39.52.1.11 CFE_ES_CDS_REG_DUMP_INF_EID

```
#define CFE_ES_CDS_REG_DUMP_INF_EID 83
```

`'Successfully dumped CDS Registry to '%s':Size=%d,Entries=%d'`

Event Message `'Successfully dumped CDS Registry to '%s':Size=%d,Entries=%d'`

Type: DEBUG

Cause:

This event message is generated when an Executive Services Dump Critical Data Store Registry Command is successfully executed. The specified file should have been created and contains the CDS Registry Entries.

The `'s'` field identifies the CDS Registry Dump Filename. The first `'d'` field specifies the size of the file (in bytes) The second `'d'` field specifies the number of CDS Registry Records that were written

Definition at line 1276 of file cfe_es_events.h.

39.52.1.12 CFE_ES_CDS_REGISTER_ERR_EID

```
#define CFE_ES_CDS_REGISTER_ERR_EID 69
```

`'%s Failed to Register CDS '%s', Status=0x%08X'`

Event Message `'%s Failed to Register CDS '%s', Status=0x%08X'`

Type: ERROR

Cause:

This event message is generated whenever an Application calls the CFE_ES_RegisterCDS API and fails to successfully create the desired CDS.

The first `'s'` field identifies the name of the Application which made the API call, the second `'s'` field specifies the name of the CDS as requested by the Application and the `Status` field provides the error code which identifies in more detail the nature of the failure (See return codes for the CFE_ES_RegisterCDS API).

Definition at line 1069 of file cfe_es_events.h.

39.52.1.13 CFE_ES_CREATING_CDS_DUMP_ERR_EID

```
#define CFE_ES_CREATING_CDS_DUMP_ERR_EID 86
```

`'Error creating CDS dump file '%s', Status=0x%08X'`

Event Message `'Error creating CDS dump file '%s', Status=0x%08X'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Dump Critical Data Store Registry Command is unable to create the specified file on the onboard filesystem.

The `'s'` field identifies the CDS Registry Dump Filename. The `'08X'` field identifies error code returned by the API OS_OpenCreate.

Definition at line 1326 of file cfe_es_events.h.

39.52.1.14 CFE_ES_ERLOG1_INF_EID

```
#define CFE_ES_ERLOG1_INF_EID 19
```

`'Cleared mode log data'`

Event Message `'Cleared mode log data'`

Type: INFORMATION

Cause:

This event message is generated upon successful completion of the cFE Executive Services Clear Exception Reset Log command

Definition at line 327 of file cfe_es_events.h.

39.52.1.15 CFE_ES_ERLOG2_EID

```
#define CFE_ES_ERLOG2_EID 20
```

`'%s written:Size=%d'`

Event Message `'%s written:Size=%d'`

Type: DEBUG

Cause:

This event message is generated when the Exception Reset Log has been successfully written to a file after receiving the cFE Executive Services Write Executive Services Exception Reset Log command

The `'s'` field identifies the name of the file written to and the `Size` field specifies, in decimal, the number of bytes written to the file.

Definition at line 343 of file cfe_es_events.h.

39.52.1.16 CFE_ES_ERLOG2_ERR_EID

```
#define CFE_ES_ERLOG2_ERR_EID 56
```

`'Error creating file %s, stat=0x%x'`

Event Message `'Error creating file %s, stat=0x%x'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Dump Exception Reset Log Command fails while attempting to create the specified file.

The `'s'` field identifies the name of the file that was attempted to be created and the `stat` field specifies, in hex, the error code returned by the OS_OpenCreate API.

Definition at line 892 of file cfe_es_events.h.

39.52.1.17 CFE_ES_ERLOG_PENDING_ERR_EID

```
#define CFE_ES_ERLOG_PENDING_ERR_EID 93
```

`'Error log write to file %s already in progress'`

Event Message `'Error log write to file %s already in progress'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Dump Exception Reset Log Command is received before a previously-issued command has finished executing

Definition at line 1450 of file cfe_es_events.h.

39.52.1.18 CFE_ES_ERR_SYSLOGMODE_EID

```
#define CFE_ES_ERR_SYSLOGMODE_EID 71
```

`'Set OverWriteSysLog Command:  Invalid Mode setting = %d'`

Event Message `'Set OverWriteSysLog Command:  Invalid Mode setting = %d'`

Type: ERROR

Cause:

This event message is generated upon unsuccessful completion of an Executive Services Set System Log Overwrite Mode Command .

The `setting` field identifies the illegal Overwrite Mode found in the command message. The mode must be either CFE_ES_LogMode_OVERWRITE (0) or CFE_ES_LogMode_DISCARD (1).

Definition at line 1101 of file cfe_es_events.h.

39.52.1.19 CFE_ES_ERREXIT_APP_ERR_EID

```
#define CFE_ES_ERREXIT_APP_ERR_EID 33
```

`'Exit Application %s on Error Failed:  CleanUpApp Error 0x%08X.'`

Event Message `'Exit Application %s on Error Failed:  CleanUpApp Error 0x%08X.'`

Type: ERROR

Cause:

This event message is generated when ES is completing the processing of the CFE_ES_ExitApp API call with the CFE_ES_RunStatus_APP_ERROR parameter and the call to CFE_ES_CleanUpApp fails. At this point the Application will likely be stopped or deleted, but it may be in an unknown state.

The `'s'` field identifies the name of the Application which was attempted to be reloaded and the `RC` field identifies the error code, in hex, that will identify the precise reason for the failure.

Definition at line 539 of file cfe_es_events.h.

39.52.1.20 CFE_ES_ERREXIT_APP_INF_EID

```
#define CFE_ES_ERREXIT_APP_INF_EID 14
```

`'Exit Application %s Completed.'`

Event Message `'Exit Application %s Completed.'`

Type: INFORMATION

Cause:

This event message is issued when the cFE finishes exiting/cleaning up an application that called the CFE_ES_ExitApp API with an ERROR condition. When an App calls this API, with the CFE_ES_RunStatus_APP_ERROR parameter, it indicates that the Application exited due to an error condition. The details of the error that occurred should be given by the Application through an event message, System Log entry, or both. The request is recorded and the Executive Services App will actually delete cFE Application before issuing this event message.

The `'s'` field identifies the name of the Application that was exited.

Definition at line 254 of file cfe_es_events.h.

39.52.1.21 CFE_ES_EXIT_APP_ERR_EID

```
#define CFE_ES_EXIT_APP_ERR_EID 46
```

`'Exit Application %s Failed:  CleanUpApp Error 0x%08X.'`

Event Message `'Exit Application %s Failed:  CleanUpApp Error 0x%08X.'`

Type: ERROR

Cause:

This event message is generated when ES is completing the processing of the CFE_ES_ExitApp API call and the call to CFE_ES_CleanUpApp fails. At this point the Application will likely be stopped or deleted, but it may be in an unknown state.

The `'s'` field identifies the name of the Application which was attempted to be reloaded and the `RC` field identifies the error code, in hex, that will identify the precise reason for the failure.

Definition at line 756 of file cfe_es_events.h.

39.52.1.22 CFE_ES_EXIT_APP_INF_EID

```
#define CFE_ES_EXIT_APP_INF_EID 13
```

`'Exit Application %s Completed.'`

Event Message `'Exit Application %s Completed.'`

Type: INFORMATION

Cause:

This event message is issued when the cFE finishes exiting/cleaning up an application that called the CFE_ES_ExitApp API with the CFE_ES_RunStatus_APP_EXIT parameter. When an App calls this API, the request is recorded and the Executive Services App will actually delete cFE Application before issuing this event message.

The `'s'` field identifies the name of the Application that was exited.

Definition at line 234 of file cfe_es_events.h.

39.52.1.23 CFE_ES_FILEWRITE_ERR_EID

```
#define CFE_ES_FILEWRITE_ERR_EID 74
```

`'File write,byte cnt err,file %s,request=%d,actual=%d'`

Event Message `'File write,byte cnt err,file %s,request=%d,actual=%d'`

Type: ERROR

Cause:

This event message is generated in response to any command requesting information to be written to a file and whose data is not completely written to the specified file.

The `file` field identifies the filename of the file to which the data failed to write completely, the `request` field specifies, in decimal, the number of bytes that were attempted to be written and the `actual` field indicates, in decimal, the actual number of bytes written to the file.

Definition at line 1144 of file cfe_es_events.h.

39.52.1.24 CFE_ES_INIT_INF_EID

```
#define CFE_ES_INIT_INF_EID 1 /* start up message "informational" */
```

`'cFE ES Initialized'`

Event Message `'cFE ES Initialized'`

Type: INFORMATION

Cause:

This event message is always automatically issued when the Executive Services Task completes its Initialization.

Definition at line 62 of file cfe_es_events.h.

39.52.1.25 CFE_ES_INITSTATS_INF_EID

```
#define CFE_ES_INITSTATS_INF_EID 2
```

`'cFE Version %d.%d.%d chksm %d, OSAL Version %d.%d'`

Event Message `'cFE Version %d.%d.%d chksm %d, OSAL Version %d.%d'`

Type: INFORMATION

Cause:

This event message is always automatically issued when the Executive Services Task completes its Initialization.

The `Version` field identifies the tagged version for the cFE Build, the `chksm` field provides the 16-bit checksum of the cFE Build and the `OSAL Version` field identifies the version of the OS Abstraction Layer on which this particular version of the cFE was built.

Definition at line 78 of file cfe_es_events.h.

39.52.1.26 CFE_ES_INVALID_POOL_HANDLE_ERR_EID

```
#define CFE_ES_INVALID_POOL_HANDLE_ERR_EID 82
```

`'Cannot telemeter memory pool stats.  Illegal Handle (0x%08X)'`

Event Message `'Cannot telemeter memory pool stats.  Illegal Handle (0x%08X)'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Telemeter Memory Statistics Command specifies a memory pool handle that is invalid. A handle is determined to be invalid when any of the following are true:

1. The handle does not contain a value that is an integral multiple of 4
2. The handle does not specify a valid area of memory
3. The handle does not point to an area of memory that contains the handle itself
4. The handle does not point to an area of memory whose Size field is an integral multiple of 4
5. The handle does not point to an area of memory whose End field is equal to the Start plus the Size

The `'08X'` field identifies the handle that was found in the command.

Definition at line 1258 of file cfe_es_events.h.

39.52.1.27 CFE_ES_LEN_ERR_EID

```
#define CFE_ES_LEN_ERR_EID 23
```

`'Invalid cmd length:  ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'`

Event Message `'Invalid cmd length:  ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'`

Type: ERROR

Cause:

This event message is generated when a message with the CFE_ES_CMD_MID message ID has arrived but whose packet length does not match the expected length for the specified command code.

The `ID` field in the event message specifies the Message ID (in hex), the `CC` field specifies the Command Code (in decimal), the `Exp` Len field specified the Expected Length (in decimal), and `Len` specifies the message Length (in decimal) found in the message.

Definition at line 401 of file cfe_es_events.h.

39.52.1.28 CFE_ES_MAX_EID

```
#define CFE_ES_MAX_EID 92
```

Definition at line 46 of file cfe_es_events.h.

39.52.1.29 CFE_ES_MID_ERR_EID

```
#define CFE_ES_MID_ERR_EID 21 /* invalid command packet "error" */
```

`'Invalid command pipe message ID: 0x%X'`

Event Message `'Invalid command pipe message ID: 0x%X'`

Type: ERROR

Cause:

This event message is generated when a message has arrived on the cFE Executive Services Application's Message Pipe that has a Message ID that is neither CFE_ES_SEND_HK_MID or CFE_ES_CMD_MID. Most likely, the cFE Software Bus routing table has become corrupt and is sending messages targeted for other Applications to the cFE Executive Services Application.

The `ID` field in the event message identifies the message ID (in hex) that was found in the message.

Definition at line 362 of file cfe_es_events.h.

39.52.1.30 CFE_ES_NOOP_INF_EID

```
#define CFE_ES_NOOP_INF_EID 3 /* processed command "informational" */
```

`'No-op command'`

Event Message `'No-op command'`

Type: INFORMATION

Cause:

This event message is always automatically issued in response to a cFE Executive Services NO-OP command

Definition at line 90 of file cfe_es_events.h.

39.52.1.31 CFE_ES_ONE_APP_EID

```
#define CFE_ES_ONE_APP_EID 15
```

`'Sent %s application data'`

Event Message `'Sent %s application data'`

Type: DEBUG

Cause:

This event message is issued upon successful completion of the cFE Executive Services Query One Application command

The `'s'` field identifies the name of the Application whose Executive Services Application information has been telemetered.

Definition at line 269 of file cfe_es_events.h.

39.52.1.32 CFE_ES_ONE_APPID_ERR_EID

```
#define CFE_ES_ONE_APPID_ERR_EID 50
```

`'Failed to send %s application data:  GetAppIDByName Failed, RC = 0x%08X'`

Event Message `'Failed to send %s application data:  GetAppIDByName Failed, RC = 0x%08X'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Request Application Data Command failed.

The `'s'` field identifies the name of the Application whose data was attempted to be telemetered and the `rc` field identifies the error code, in hex, that may identify the precise reason for the failure.

Definition at line 817 of file cfe_es_events.h.

39.52.1.33 CFE_ES_ONE_ERR_EID

```
#define CFE_ES_ONE_ERR_EID 49
```

`'Failed to send %s application data, RC = %08X'`

Event Message `'Failed to send %s application data, RC = %08X'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Request Application Data Command failed.

The `'s'` field identifies the name of the Application whose data was attempted to be telemetered and the `rc` field identifies the error code, in hex, that may identify the precise reason for the failure.

Definition at line 801 of file cfe_es_events.h.

39.52.1.34 CFE_ES_OSCREATE_ERR_EID

```
#define CFE_ES_OSCREATE_ERR_EID 51
```

`'Failed to write App Info file, OS_OpenCreate returned %d'`

Event Message `'Failed to write App Info file, OS_OpenCreate returned %d'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Dump Application Data Command fails to create the dump file.

The `'d'` parameter identifies, in decimal, the error code returned by OS_OpenCreate when the attempt was made to create the file.

Definition at line 832 of file cfe_es_events.h.

39.52.1.35 CFE_ES_PCR_ERR1_EID

```
#define CFE_ES_PCR_ERR1_EID 47
```

`'ES_ProcControlReq:  Invalid State (EXCEPTION) Application %s.'`

Event Message `'ES_ProcControlReq:  Invalid State (EXCEPTION) Application %s.'`

Type: ERROR

Cause:

This event message is generated when ES is processing it's internal Application table and encounters an App with the EXCEPTION state. Because exceptions are supposed to be processed immediately, this is an invalid state and should not happen. It may indicate some sort of memory corruption or other problem.

Definition at line 769 of file cfe_es_events.h.

39.52.1.36 CFE_ES_PCR_ERR2_EID

```
#define CFE_ES_PCR_ERR2_EID 48
```

`'ES_ProcControlReq:  Unknown State ( %d ) Application %s.'`

Event Message `'ES_ProcControlReq:  Unknown State ( %d ) Application %s.'`

Type: ERROR

Cause:

This event message is generated when ES is processing it's internal Application table and encounters an App with an unknown state. If this message occurs, it might be an indication of a memory corruption or other problem.

Definition at line 785 of file cfe_es_events.h.

39.52.1.37 CFE_ES_PERF_DATAWRITTEN_EID

```
#define CFE_ES_PERF_DATAWRITTEN_EID 68
```

`'%s written:Size=%d,EntryCount=%d'`

Event Message `'%s written:Size=%d,EntryCount=%d'`

Type: DEBUG

Cause:

This event message is generated when the Performance Log has been successfully written to a file after receiving the cFE Executive Services Stop Performance Analyzer Data Collection Command

The `'s'` field identifies the name of the file written to, the `Size` field specifies, in decimal, the number of bytes written to the file and the `EntryCount` field identifies the number of data entries that were written.

Definition at line 1052 of file cfe_es_events.h.

39.52.1.38 CFE_ES_PERF_FILTMSKCMD_EID

```
#define CFE_ES_PERF_FILTMSKCMD_EID 63
```

`'Set Performance Filter Mask command'`

Event Message `'Set Performance Filter Mask command'`

Type: DEBUG

Cause:

This event message is generated in response to receiving an Executive Services Set Performance Analyzer Filter Mask Command .

Definition at line 980 of file cfe_es_events.h.

39.52.1.39 CFE_ES_PERF_FILTMSKERR_EID

```
#define CFE_ES_PERF_FILTMSKERR_EID 64
```

`'Performance Filter Mask Cmd Error,Index(u)out of range(u)'`

Event Message `'Performance Filter Mask Cmd Error,Index(u)out of range(u)'`

Type: ERROR

Cause:

This event message is generated in response to receiving an Executive Services Set Performance Analyzer Filter Mask Command .

Definition at line 993 of file cfe_es_events.h.

39.52.1.40 CFE_ES_PERF_LOG_ERR_EID

```
#define CFE_ES_PERF_LOG_ERR_EID 67
```

`'Error creating file %s, stat=%d'`

Event Message `'Error creating file %s, stat=%d'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Stop Performance Analyzer Data Collection Command fails to create the associated logic analyzer dump file.

The `'s'` field identifies the name of the file that was attempted to be created and the `stat` field specifies, in decimal, the error code returned by the OS_OpenCreate API.

Definition at line 1035 of file cfe_es_events.h.

39.52.1.41 CFE_ES_PERF_STARTCMD_EID

```
#define CFE_ES_PERF_STARTCMD_EID 57
```

`'Start collecting performance data command, trigger mode = d'`

Event Message `'Start collecting performance data command, trigger mode = d'`

Type: DEBUG

Cause:

This event message is generated in response to receiving an Executive Services Start Performance Analyzer Data Collection Command

The `'d'` field identifies the requested trigger mode as defined by CFE_ES_PerfMode_t.

Definition at line 906 of file cfe_es_events.h.

39.52.1.42 CFE_ES_PERF_STARTCMD_ERR_EID

```
#define CFE_ES_PERF_STARTCMD_ERR_EID 58
```

`'Cannot start collecting performance data,perf data write in progress'`

Event Message `'Cannot start collecting performance data,perf data write in progress'`

Type: ERROR

Cause:

This event message is generated in response to receiving an Executive Services Start Performance Analyzer Data Collection Command

Definition at line 918 of file cfe_es_events.h.

39.52.1.43 CFE_ES_PERF_STARTCMD_TRIG_ERR_EID

```
#define CFE_ES_PERF_STARTCMD_TRIG_ERR_EID 59
```

`'Cannot start collecting performance data, trigger mode (d) out of range (d to d)'`

Event Message `'Cannot start collecting performance data, trigger mode (d) out of range (d to d)'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Start Performance Analyzer Data Collection Command command is received with a bad value for the requested trigger mode.

The first `'d'` field identifies the received trigger mode value as defined by CFE_ES_PerfMode_t. The second and third `'d'` fields specify the valid range of values for the trigger mode.

Definition at line 934 of file cfe_es_events.h.

39.52.1.44 CFE_ES_PERF_STOPCMD_EID

```
#define CFE_ES_PERF_STOPCMD_EID 60
```

`'Perf Stop Cmd Rcvd,%s will write %d entries.%dmS dly every %d entries'`

Event Message `'Perf Stop Cmd Rcvd,%s will write %d entries.%dmS dly every %d entries'`

Type: DEBUG

Cause:

This event message is generated upon receipt of a successful Performance Data Stop Command after receiving the cFE Executive Services Stop Performance Analyzer Data Collection Command

The `'s'` field identifies the name of the file write task that has begun execution. The first `'d'` identifies the total number of performance entries(in decimal) that will be written to the file. A performance data entry is defined by an unsigned 32 bit data point and an unsigned 64 bit time stamp. The second `'d'` identifies the millisecond delay between writes and the third `'d'` identifies the number of entries written (in decimal) between delays.

Definition at line 953 of file cfe_es_events.h.

39.52.1.45 CFE_ES_PERF_STOPCMD_ERR2_EID

```
#define CFE_ES_PERF_STOPCMD_ERR2_EID 62
```

`'Stop performance data cmd ignored,perf data write in progress'`

Event Message `'Stop performance data cmd ignored,perf data write in progress'`

Type: ERROR

Cause:

This event message is generated upon receipt of an unsuccessful Performance Data Stop Command after receiving the cFE Executive Services Stop Performance Analyzer Data Collection Command

Definition at line 967 of file cfe_es_events.h.

39.52.1.46 CFE_ES_PERF_TRIGMSKCMD_EID

```
#define CFE_ES_PERF_TRIGMSKCMD_EID 65
```

`'Set Performance Trigger Mask command'`

Event Message `'Set Performance Trigger Mask command'`

Type: DEBUG

Cause:

This event message is generated in response to receiving an Executive Services Set Performance Analyzer Trigger Mask Command .

Definition at line 1006 of file cfe_es_events.h.

39.52.1.47 CFE_ES_PERF_TRIGMSKERR_EID

```
#define CFE_ES_PERF_TRIGMSKERR_EID 66
```

`'Performance Trigger Mask Cmd Error,Index(u)out of range(u)'`

Event Message `'Performance Trigger Mask Cmd Error,Index(u)out of range(u)'`

Type: ERROR

Cause:

This event message is generated in response to receiving an Executive Services Set Performance Analyzer Trigger Mask Command .

Definition at line 1019 of file cfe_es_events.h.

39.52.1.48 CFE_ES_RELOAD_APP_DBG_EID

```
#define CFE_ES_RELOAD_APP_DBG_EID 11
```

`'Reload Application %s Initiated.'`

Event Message `'Reload Application %s Initiated.'`

Type: DEBUG

Cause:

This event message is issued upon successful processing of the cFE Executive Services Reload Application command Note that when this event is displayed, the Application is not reloaded. ES has accepted the request to reload the application, and it will be reloaded after the app exits it's main loop, or times out.

The `'s'` field identifies the name of the Application that will be reloaded.

Definition at line 202 of file cfe_es_events.h.

39.52.1.49 CFE_ES_RELOAD_APP_ERR1_EID

```
#define CFE_ES_RELOAD_APP_ERR1_EID 42
```

`'Failed to reload Application %s, rc = %08X'`

Event Message `'Failed to reload Application %s, rc = %08X'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Reload Application Command fails.

The `'s'` field identifies the name of the Application which was attempted to be reloaded and the `rc` field identifies the error code, in hex, that may identify the precise reason for the failure.

Definition at line 685 of file cfe_es_events.h.

39.52.1.50 CFE_ES_RELOAD_APP_ERR2_EID

```
#define CFE_ES_RELOAD_APP_ERR2_EID 43
```

`'Reload Application %s, GetAppIDByName failed.  RC = 0x%08X.'`

Event Message `'Reload Application %s, GetAppIDByName failed.  RC = 0x%08X.'`

Type: ERROR

Cause:

This event message is generated upon receipt of an Executive Services Reload Application Command which fails. This message is for a specific failure when the call to CFE_ES_GetAppIDByName fails. The application will not be reloaded at this point.

The `'s'` field identifies the name of the Application which was attempted to be reloaded and the `RC` field identifies the error code, in hex, that will identify the precise reason for the failure.

Definition at line 703 of file cfe_es_events.h.

39.52.1.51 CFE_ES_RELOAD_APP_ERR3_EID

```
#define CFE_ES_RELOAD_APP_ERR3_EID 44
```

`'Reload Application %s Failed:  AppCreate Error 0x%08X.'`

Event Message `'Reload Application %s Failed:  AppCreate Error 0x%08X.'`

Type: ERROR

Cause:

This event message is generated upon receipt of an Executive Services Reload Application Command which fails. This message is for a specific failure when the call to CFE_ES_AppCreate fails. The application will not be reloaded at this point.

The `'s'` field identifies the name of the Application which was attempted to be reloaded and the `RC` field identifies the error code, in hex, that will identify the precise reason for the failure.

Definition at line 721 of file cfe_es_events.h.

39.52.1.52 CFE_ES_RELOAD_APP_ERR4_EID

```
#define CFE_ES_RELOAD_APP_ERR4_EID 45
```

`'Reload Application %s Failed:  CleanUpApp Error 0x%08X.'`

Event Message `'Reload Application %s Failed:  CleanUpApp Error 0x%08X.'`

Type: ERROR

Cause:

This event message is generated upon receipt of an Executive Services Reload Application Command which fails. This message is for a specific failure when the call to CFE_ES_CleanUpApp fails. The application will not be reloaded at this point, and will likely be deleted or in an unknown state.

The `'s'` field identifies the name of the Application which was attempted to be reloaded and the `RC` field identifies the error code, in hex, that will identify the precise reason for the failure.

Definition at line 739 of file cfe_es_events.h.

39.52.1.53 CFE_ES_RELOAD_APP_INF_EID

```
#define CFE_ES_RELOAD_APP_INF_EID 12
```

`'Reload Application %s Completed, AppID=lu'`

Event Message `'Reload Application %s Completed, AppID=lu'`

Type: INFORMATION

Cause:

This event message is issued when the cFE finishes Reloading the cFE Application That was started when the Restart Application command was issued.

The `'s'` field identifies the name of the Application that was reloaded, and the lu field identifies the new Application ID

Definition at line 218 of file cfe_es_events.h.

39.52.1.54 CFE_ES_RESET_INF_EID

```
#define CFE_ES_RESET_INF_EID 4
```

`'Reset Counters command'`

Event Message `'Reset Counters command'`

Type: INFORMATION

Cause:

This event message is always automatically issued in response to a cFE Executive Services Reset Counters command

Definition at line 102 of file cfe_es_events.h.

39.52.1.55 CFE_ES_RESET_PR_COUNT_EID

```
#define CFE_ES_RESET_PR_COUNT_EID 72
```

`'Reset Processor Reset Count to Zero'`

Event Message `'Reset Processor Reset Count to Zero'`

Type: INFORMATION

Cause:

This event message is always generated in response to the Executive Services Set Processor Reset Counter to Zero Command .

Definition at line 1113 of file cfe_es_events.h.

39.52.1.56 CFE_ES_RESTART_APP_DBG_EID

```
#define CFE_ES_RESTART_APP_DBG_EID 9
```

`'Restart Application %s Initiated.'`

Event Message `'Restart Application %s Initiated.'`

Type: DEBUG

Cause:

This event message is issued upon successful processing of the cFE Executive Services Restart Application command Note that when this event is displayed, the Application is not restarted. ES has accepted the request to restart the application, and it will be restarted after the app exits it's main loop, or times out.

The `'s'` field identifies the name of the Application that will be restarted.

Definition at line 168 of file cfe_es_events.h.

39.52.1.57 CFE_ES_RESTART_APP_ERR1_EID

```
#define CFE_ES_RESTART_APP_ERR1_EID 38
```

`'Restart Application %s Failed, RC = 0x%08X'`

Event Message `'Restart Application %s Failed, RC = 0x%08X'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Restart Application Command fails.

The `'s'` field identifies the name of the Application which was attempted to be reset and the `rc` field identifies the error code, in hex, that may identify the precise reason for the failure.

Definition at line 611 of file cfe_es_events.h.

39.52.1.58 CFE_ES_RESTART_APP_ERR2_EID

```
#define CFE_ES_RESTART_APP_ERR2_EID 39
```

`'Restart Application %s, GetAppIDByName failed.  RC = 0x%08X.'`

Event Message `'Restart Application %s, GetAppIDByName failed.  RC = 0x%08X.'`

Type: ERROR

Cause:

This event message is generated upon receipt of an Executive Services Restart Application Command which fails. This message is for a specific failure when the call to CFE_ES_GetAppIDByName fails. The application will not be restarted at this point.

The `'s'` field identifies the name of the Application which was attempted to be restarted and the `RC` field identifies the error code, in hex, that will identify the precise reason for the failure.

Definition at line 629 of file cfe_es_events.h.

39.52.1.59 CFE_ES_RESTART_APP_ERR3_EID

```
#define CFE_ES_RESTART_APP_ERR3_EID 40
```

`'Restart Application %s Failed:  AppCreate Error 0x%08X.'`

Event Message `'Restart Application %s Failed:  AppCreate Error 0x%08X.'`

Type: ERROR

Cause:

This event message is generated upon receipt of an Executive Services Restart Application Command which fails. This message is for a specific failure when the call to CFE_ES_AppCreate fails. The application will not be restarted at this point.

The `'s'` field identifies the name of the Application which was attempted to be restarted and the `RC` field identifies the error code, in hex, that will identify the precise reason for the failure.

Definition at line 651 of file cfe_es_events.h.

39.52.1.60 CFE_ES_RESTART_APP_ERR4_EID

```
#define CFE_ES_RESTART_APP_ERR4_EID 41
```

`'Restart Application %s Failed:  CleanUpApp Error 0x%08X.'`

Event Message `'Restart Application %s Failed:  CleanUpApp Error 0x%08X.'`

Type: ERROR

Cause:

This event message is generated upon receipt of an Executive Services Restart Application Command which fails. This message is for a specific failure when the call to CFE_ES_CleanUpApp fails. The application will not be restarted at this point, but will likely be deleted or in an unknown state.

The `'s'` field identifies the name of the Application which was attempted to be restarted and the `RC` field identifies the error code, in hex, that will identify the precise reason for the failure.

Definition at line 669 of file cfe_es_events.h.

39.52.1.61 CFE_ES_RESTART_APP_INF_EID

```
#define CFE_ES_RESTART_APP_INF_EID 10
```

`'Restart Application %s Completed, AppID=lu'`

Event Message `'Restart Application %s Completed, AppID=lu'`

Type: INFORMATION

Cause:

This event message is issued when the cFE finishes Restarting the cFE Application That was started when the Restart Application command was issued.

The `'s'` field identifies the name of the Application that was restarted, and the lu field identifies the new Application ID

Definition at line 184 of file cfe_es_events.h.

39.52.1.62 CFE_ES_SET_MAX_PR_COUNT_EID

```
#define CFE_ES_SET_MAX_PR_COUNT_EID 73
```

`'Maximum Processor Reset Count set to:  %d'`

Event Message `'Maximum Processor Reset Count set to:  %d'`

Type: INFORMATION

Cause:

This event message is always generated in response to the Executive Services Set Maximum Processor Reset Limit Command .

The `'d'` field identifies, in decimal, the number of Processor Resets that will need to occur before a Power-On Reset is automatically performed.

Definition at line 1128 of file cfe_es_events.h.

39.52.1.63 CFE_ES_START_ERR_EID

```
#define CFE_ES_START_ERR_EID 26
```

`'Failed to start %s from %s, RC = %08X'`

Event Message `'Failed to start %s from %s, RC = %08X'`

Type: ERROR

Cause:

This event message is generated for an error encountered in response to an Executive Services Start Application Command .

This message is a general failure when the command passes the parameter validation, but fails when a call to CFE_↩ ES_AppCreate is called.

The `'s'` term identifies the name of the Application that was attempted to start. The second `'s'` field specifies the file from which the Application was loaded. The `'X'` field is the return code returned by the CFE_ES_AppCreate.

Definition at line 437 of file cfe_es_events.h.

39.52.1.64 CFE_ES_START_EXC_ACTION_ERR_EID

```
#define CFE_ES_START_EXC_ACTION_ERR_EID 32
```

`'CFE_ES_StartAppCmd:  Invalid Exception Action:  %d.'`

Event Message `'CFE_ES_StartAppCmd:  Invalid Exception Action:  %d.'`

Type: ERROR

Cause:

This event message is generated for an error encountered in response to an Executive Services Start Application Command .

This message reports a command failure when the Application Exception Action parameter is invalid. The valid options for this parameter are: 0 = Application will restart on an exception 1 = Application cause a processor restart on exception.

The `'d'` term identifies the Exception Action parameter that was given in the command.

Definition at line 522 of file cfe_es_events.h.

39.52.1.65 CFE_ES_START_INF_EID

```
#define CFE_ES_START_INF_EID 6
```

`'Started %s from %s, AppID = %d'`

Event Message `'Started %s from %s, AppID = %d'`

Type: INFORMATION

Cause:

This event message is automatically issued upon successful completion of a cFE Executive Services Start Application command

The first `'s'` string identifies the name of the started Application, the second `'s'` string identifies the filename from which the Application was loaded and the `AppId` field specifies the Application ID assigned to the newly started Application by the cFE Executive Services.

Definition at line 119 of file cfe_es_events.h.

39.52.1.66 CFE_ES_START_INVALID_ENTRY_POINT_ERR_EID

```
#define CFE_ES_START_INVALID_ENTRY_POINT_ERR_EID 28
```

`'CFE_ES_StartAppCmd:  App Entry Point is NULL.'`

Event Message `'CFE_ES_StartAppCmd:  App Entry Point is NULL.'`

Type: ERROR

Cause:

This event message is generated for an error encountered in response to an Executive Services Start Application Command .

This message reports a command failure when the Start Appplication Command is given a NULL Application Entry Point parameter. The command must contain an application entry point string. (Example: "SC_AppMain").

Definition at line 471 of file cfe_es_events.h.

39.52.1.67 CFE_ES_START_INVALID_FILENAME_ERR_EID

```
#define CFE_ES_START_INVALID_FILENAME_ERR_EID 27
```

`'CFE_ES_StartAppCmd: invalid filename: %s'`

Event Message `'CFE_ES_StartAppCmd: invalid filename: %s'`

Type: ERROR

Cause:

This event message is generated for an error encountered in response to an Executive Services Start Application Command .

This message reports a command failure when the Start Appplication Command is given an invalid filename. (Either NULL or too short to be a valid cFE file name).

The `'s'` term identifies the invalid filename that was sent with the command.

Definition at line 454 of file cfe_es_events.h.

39.52.1.68 CFE_ES_START_NULL_APP_NAME_ERR_EID

```
#define CFE_ES_START_NULL_APP_NAME_ERR_EID 29
```

`'CFE_ES_StartAppCmd: App Name is NULL.'`

Event Message `'CFE_ES_StartAppCmd: App Name is NULL.'`

Type: ERROR

Cause:

This event message is generated for an error encountered in response to an Executive Services Start Application Command .

This message reports a command failure when the Start Appplication Command is given a NULL Application Name parameter. The command must contain an application name string.

Definition at line 486 of file cfe_es_events.h.

39.52.1.69 CFE_ES_START_PRIORITY_ERR_EID

```
#define CFE_ES_START_PRIORITY_ERR_EID 31
```

`'CFE_ES_StartAppCmd:  Priority is too large:  %d.'`

Event Message `'CFE_ES_StartAppCmd:  Priority is too large:  %d.'`

Type: ERROR

Cause:

This event message is generated for an error encountered in response to an Executive Services Start Application Command .

This message reports a command failure when the Application priority greater than the maximum priority for a Task defined by the OS Abstraction Layer (256).

The `'d'` term identifies the priority that was given in the command.

Definition at line 503 of file cfe_es_events.h.

39.52.1.70 CFE_ES_STOP_DBG_EID

```
#define CFE_ES_STOP_DBG_EID 7
```

`'Stop Application %s Initiated.'`

Event Message `'Stop Application %s Initiated.'`

Type: DEBUG

Cause:

This event message is issued upon successful processing of the cFE Executive Services Stop Application command Note that when this event is displayed, the Application is not deleted. ES has accepted the request to delete the application, and it will be deleted after the app exits it's main loop, or times out.

The `'s'` field identifies the name of the Application that will be stopped.

Definition at line 136 of file cfe_es_events.h.

39.52.1.71 CFE_ES_STOP_ERR1_EID

```
#define CFE_ES_STOP_ERR1_EID 35
```

`'Stop Application %s Failed, RC = 0x%08X'`

Event Message `'Stop Application %s Failed, RC = 0x%08X'`

Type: ERROR

Cause:

This event message is generated upon receipt of an Executive Services Stop Application Command which fails.

The `'s'` field identifies the name of the Application which was attempted to be stopped and the `rc` field identifies the error code, in hex, that may identify the precise reason for the failure.

Definition at line 555 of file cfe_es_events.h.

39.52.1.72 CFE_ES_STOP_ERR2_EID

```
#define CFE_ES_STOP_ERR2_EID 36
```

`'Stop Application %s, GetAppIDByName failed.  RC = 0x%08X.'`

Event Message `'Stop Application %s, GetAppIDByName failed.  RC = 0x%08X.'`

Type: ERROR

Cause:

This event message is generated upon receipt of an Executive Services Stop Application Command which fails. This message is for a specific failure when the call to CFE_ES_GetAppIDByName fails. The application will not be deleted at this point.

The `'s'` field identifies the name of the Application which was attempted to be stopped and the `RC` field identifies the error code, in hex, that will identify the precise reason for the failure.

Definition at line 573 of file cfe_es_events.h.

39.52.1.73 CFE_ES_STOP_ERR3_EID

```
#define CFE_ES_STOP_ERR3_EID 37
```

`'Stop Application %s Failed:  CleanUpApp Error 0x%08X.'`

Event Message `'Stop Application %s Failed:  CleanUpApp Error 0x%08X.'`

Type: ERROR

Cause:

This event message is generated upon receipt of an Executive Services Stop Application Command which fails. This message is for a specific failure when the call to CFE_ES_GetAppIDByName fails. The application will not be deleted at this point.

The `'s'` field identifies the name of the Application which was attempted to be stopped and the `RC` field identifies the error code, in hex, that will identify the precise reason for the failure.

Definition at line 595 of file cfe_es_events.h.

39.52.1.74 CFE_ES_STOP_INF_EID

```
#define CFE_ES_STOP_INF_EID 8
```

`'Stop Application %s Completed.'`

Event Message `'Stop Application %s Completed.'`

Type: INFORMATION

Cause:

This event message is issued when the cFE finishes deleting the cFE Application That was started when the Stop Application command was issued.

The `'s'` field identifies the name of the Application that was stopped.

Definition at line 151 of file cfe_es_events.h.

39.52.1.75 CFE_ES_SYSLOG1_INF_EID

```
#define CFE_ES_SYSLOG1_INF_EID 17
```

`'Cleared Executive Services log data'`

Event Message `'Cleared Executive Services log data'`

Type: INFORMATION

Cause:

This event message is generated upon successful completion of the cFE Executive Services Clear System Log command

Definition at line 298 of file cfe_es_events.h.

39.52.1.76 CFE_ES_SYSLOG2_EID

```
#define CFE_ES_SYSLOG2_EID 18
```

`'%s written:Size=%d,Entries=%d'`

Event Message `'%s written:Size=%d,Entries=%d'`

Type: DEBUG

Cause:

This event message is generated when the System Log has been successfully written to a file after receiving the cFE Executive Services Write Executive Services System Log command

The `'s'` field identifies the name of the file written to, the `Size` field specifies, in decimal, the number of bytes written to the file and the `Entries` field identifies the number of System Log messages that were written.

Definition at line 315 of file cfe_es_events.h.

39.52.1.77 CFE_ES_SYSLOG2_ERR_EID

```
#define CFE_ES_SYSLOG2_ERR_EID 55
```

`'Error creating file %s, stat=0x%x'`

Event Message `'Error creating file %s, stat=0x%x'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Dump System Log Command fails while attempting to create the specified file.

The `'s'` field identifies the name of the file that was attempted to be created and the `stat` field specifies, in hex, the error code returned by the OS_OpenCreate API.

Definition at line 877 of file cfe_es_events.h.

39.52.1.78 CFE_ES_SYSLOGMODE_EID

```
#define CFE_ES_SYSLOGMODE_EID 70
```

`'Set OverWriteSysLog Command Received with Mode setting = %d'`

Event Message `'Set OverWriteSysLog Command Received with Mode setting = %d'`

Type: DEBUG

Cause:

This event message is generated upon successful completion of an Executive Services Set System Log Overwrite Mode Command .

The `setting` field identifies the newly chosen Overwrite Mode and should be equal to either CFE_ES_LogMode_↩ OVERWRITE or CFE_ES_LogMode_DISCARD.

Definition at line 1085 of file cfe_es_events.h.

39.52.1.79 CFE_ES_TASKINFO_EID

```
#define CFE_ES_TASKINFO_EID 87
```

`'Task Info file written to %s, Entries=%d, FileSize=%d'`

Event Message `'Task Info file written to %s, Entries=%d, FileSize=%d'`

Type: DEBUG

Cause:

This event message is issued upon successful completion of the cFE Executive Services Query All Tasks command

The `'s'` field identifies the name of the file to which all Executive Services Task data has been written. The `Entries` field identifies, in decimal, the number of Tasks whose data was written and the `FileSize` field gives the total number of bytes written to the file.

Definition at line 1343 of file cfe_es_events.h.

39.52.1.80 CFE_ES_TASKINFO_OSCREATE_ERR_EID

```
#define CFE_ES_TASKINFO_OSCREATE_ERR_EID 88
```

`'Failed to write Task Info file, OS_OpenCreate returned %d'`

Event Message `'Failed to write Task Info file, OS_OpenCreate returned %d'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Dump Task Data Command fails to create the dump file.

The `'d'` parameter identifies, in decimal, the error code returned by OS_OpenCreate when the attempt was made to create the file.

Definition at line 1358 of file cfe_es_events.h.

39.52.1.81 CFE_ES_TASKINFO_WR_ERR_EID

```
#define CFE_ES_TASKINFO_WR_ERR_EID 90
```

`'Failed to write Task Info file, Task write RC = 0x%08X, exp %d'`

Event Message `'Failed to write Task Info file, Task write RC = 0x%08X, exp %d'`

Type: ERROR

Cause:

This event message is generated whenever an Executive Services Dump Tasks Data Command fails while writing Tasks data to the specified file.

The `rtnd` field contains, in hex, the error code returned from the OS_write API. The expected return value is identified, in decimal, in the `exp` field.

Definition at line 1388 of file cfe_es_events.h.

39.52.1.82 CFE_ES_TASKINFO_WRHDR_ERR_EID

```
#define CFE_ES_TASKINFO_WRHDR_ERR_EID 89
```

`'Failed to write Task Info file, WriteHdr rtnd %08X, exp %d'`

Event Message `'Failed to write Task Info file, WriteHdr rtnd %08X, exp %d'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Dump Task Data Command fails while writing the cFE Standard File Header.

The `rtnd` field contains the error code returned by the CFE_FS_WriteHeader API. Nominally, the returned result should have been equal to the `exp` field (i.e. - sizeof(CFE_FS_Header_t)).

Definition at line 1373 of file cfe_es_events.h.

39.52.1.83 CFE_ES_TASKWR_ERR_EID

```
#define CFE_ES_TASKWR_ERR_EID 53
```

`'Failed to write App Info file, Task write RC = 0x%08X, exp %d'`

Event Message `'Failed to write App Info file, Task write RC = 0x%08X, exp %d'`

Type: ERROR

Cause:

This event message is generated whenever an Executive Services Dump Application Data Command fails while writing Application data to the specified file.

The `rtnd` field contains, in hex, the error code returned from the OS_write API. The expected return value is identified, in decimal, in the `exp` field.

Definition at line 862 of file cfe_es_events.h.

39.52.1.84 CFE_ES_TLM_POOL_STATS_INFO_EID

```
#define CFE_ES_TLM_POOL_STATS_INFO_EID 81
```

`'Successfully telemetered memory pool stats for 0x%08X'`

Event Message `'Successfully telemetered memory pool stats for 0x%08X'`

Type: DEBUG

Cause:

This event message is generated following successful execution of the Telemeter Memory Statistics Command .

Definition at line 1237 of file cfe_es_events.h.

39.52.1.85 CFE_ES_VERSION_INF_EID

```
#define CFE_ES_VERSION_INF_EID 91
```

`'Version Info:  s, s'`

Event Message `'Version Info:  s, s'`

Type: INFORMATION

Cause:

This event message is always automatically issued when the Executive Services Task completes its Initialization and as part of the Noop command

A separate version info event will be generated for every module which is statically linked into the CFE core executable (e.g. OSAL, PSP, MSG, SBR, etc).

The version information reported in this event is derived from the source revision control system at build time, as opposed to manually-assigned semantic version numbers. It is intendended to uniquely identify the actual source code that is currently running, to the extent this is possible.

The `Mission` version information also identifies the build configuration name, if available.

Definition at line 1410 of file cfe_es_events.h.

39.52.1.86 CFE_ES_WRHDR_ERR_EID

```
#define CFE_ES_WRHDR_ERR_EID 52
```

`'Failed to write App Info file, WriteHdr rtnd %08X, exp %d'`

Event Message `'Failed to write App Info file, WriteHdr rtnd %08X, exp %d'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Dump Application Data Command fails while writing the cFE Standard File Header.

The `rtnd` field contains the error code returned by the CFE_FS_WriteHeader API. Nominally, the returned result should have been equal to the `exp` field (i.e. - sizeof(CFE_FS_Header_t)).

Definition at line 847 of file cfe_es_events.h.

39.52.1.87 CFE_ES_WRITE_CFE_HDR_ERR_EID

```
#define CFE_ES_WRITE_CFE_HDR_ERR_EID 85
```

`'Error writing cFE File Header to '%s', Status=0x%08X'`

Event Message `'Error writing cFE File Header to '%s', Status=0x%08X'`

Type: ERROR

Cause:

This event message is generated when an Executive Services Dump Critical Data Store Registry Command command successfully created the CDS Dump File onboard but encountered an error while writing the standard cFE File Header to the file.

The `'s'` field identifies the CDS Registry Dump Filename. The `'08X'` field identifies error code returned by the API CFE_FS_WriteHeader.

Definition at line 1310 of file cfe_es_events.h.

39.53 cfe/modules/es/fsw/inc/cfe_es_msg.h File Reference

```
#include "common_types.h"
#include "cfe_msg_hdr.h"
#include "cfe_es_extern_typedefs.h"
```

Data Structures

- struct CFE_ES_NoArgsCmd

  Generic "no arguments" command.
- struct CFE_ES_RestartCmd_Payload

  Restart cFE Command Payload.
- struct CFE_ES_RestartCmd

  Restart cFE Command.
- struct CFE_ES_FileNameCmd_Payload

  Generic file name command payload.
- struct CFE_ES_FileNameCmd

  Generic file name command.
- struct CFE_ES_OverWriteSysLogCmd_Payload

  Overwrite/Discard System Log Configuration Command Payload.
- struct CFE_ES_OverWriteSysLogCmd

Overwrite/Discard System Log Configuration Command Payload.

- struct CFE_ES_StartAppCmd_Payload

  Start Application Command Payload.
- struct CFE_ES_StartApp

  Start Application Command.
- struct CFE_ES_AppNameCmd_Payload

  Generic application name command payload.
- struct CFE_ES_AppNameCmd

  Generic application name command.
- struct CFE_ES_AppReloadCmd_Payload

  Reload Application Command Payload.
- struct CFE_ES_ReloadAppCmd

  Reload Application Command.
- struct CFE_ES_SetMaxPRCountCmd_Payload

  Set Maximum Processor Reset Count Command Payload.
- struct CFE_ES_SetMaxPRCountCmd

  Set Maximum Processor Reset Count Command.
- struct CFE_ES_DeleteCDSCmd_Payload

  Delete Critical Data Store Command Payload.
- struct CFE_ES_DeleteCDSCmd

  Delete Critical Data Store Command.
- struct CFE_ES_StartPerfCmd_Payload

  Start Performance Analyzer Command Payload.
- struct CFE_ES_StartPerfDataCmd

  Start Performance Analyzer Command.
- struct CFE_ES_StopPerfCmd_Payload

  Stop Performance Analyzer Command Payload.
- struct CFE_ES_StopPerfDataCmd

  Stop Performance Analyzer Command.
- struct CFE_ES_SetPerfFilterMaskCmd_Payload

  Set Performance Analyzer Filter Mask Command Payload.
- struct CFE_ES_SetPerfFilterMaskCmd

  Set Performance Analyzer Filter Mask Command.
- struct CFE_ES_SetPerfTrigMaskCmd_Payload

  Set Performance Analyzer Trigger Mask Command Payload.
- struct CFE_ES_SetPerfTriggerMaskCmd

  Set Performance Analyzer Trigger Mask Command.
- struct CFE_ES_SendMemPoolStatsCmd_Payload

  Send Memory Pool Statistics Command Payload.
- struct CFE_ES_SendMemPoolStatsCmd

  Send Memory Pool Statistics Command.
- struct CFE_ES_DumpCDSRegistryCmd_Payload

  Dump CDS Registry Command Payload.
- struct CFE_ES_DumpCDSRegistryCmd

  Dump CDS Registry Command.
- struct CFE_ES_OneAppTlm_Payload
- struct CFE_ES_OneAppTlm

- struct CFE_ES_PoolStatsTlm_Payload
- struct CFE_ES_MemStatsTlm
- struct CFE_ES_HousekeepingTlm_Payload
- struct CFE_ES_HousekeepingTlm

Macros

Executive Services Command Codes

- #define CFE_ES_NOOP_CC 0
- #define CFE_ES_RESET_COUNTERS_CC 1
- #define CFE_ES_RESTART_CC 2
- #define CFE_ES_START_APP_CC 4
- #define CFE_ES_STOP_APP_CC 5
- #define CFE_ES_RESTART_APP_CC 6
- #define CFE_ES_RELOAD_APP_CC 7
- #define CFE_ES_QUERY_ONE_CC 8
- #define CFE_ES_QUERY_ALL_CC 9
- #define CFE_ES_CLEAR_SYSLOG_CC 10
- #define CFE_ES_WRITE_SYSLOG_CC 11
- #define CFE_ES_CLEAR_ER_LOG_CC 12
- #define CFE_ES_WRITE_ER_LOG_CC 13
- #define CFE_ES_START_PERF_DATA_CC 14
- #define CFE_ES_STOP_PERF_DATA_CC 15
- #define CFE_ES_SET_PERF_FILTER_MASK_CC 16
- #define CFE_ES_SET_PERF_TRIGGER_MASK_CC 17
- #define CFE_ES_OVER_WRITE_SYSLOG_CC 18
- #define CFE_ES_RESET_PR_COUNT_CC 19
- #define CFE_ES_SET_MAX_PR_COUNT_CC 20
- #define CFE_ES_DELETE_CDS_CC 21
- #define CFE_ES_SEND_MEM_POOL_STATS_CC 22
- #define CFE_ES_DUMP_CDS_REGISTRY_CC 23
- #define CFE_ES_QUERY_ALL_TASKS_CC 24

Typedefs

- typedef struct CFE_ES_NoArgsCmd CFE_ES_NoArgsCmd_t

  Generic "no arguments" command.
- typedef CFE_ES_NoArgsCmd_t CFE_ES_NoopCmd_t
- typedef CFE_ES_NoArgsCmd_t CFE_ES_ResetCountersCmd_t
- typedef CFE_ES_NoArgsCmd_t CFE_ES_ClearSysLogCmd_t
- typedef CFE_ES_NoArgsCmd_t CFE_ES_ClearERLogCmd_t
- typedef CFE_ES_NoArgsCmd_t CFE_ES_ResetPRCountCmd_t
- typedef struct CFE_ES_RestartCmd_Payload CFE_ES_RestartCmd_Payload_t

  Restart cFE Command Payload.
- typedef struct CFE_ES_RestartCmd CFE_ES_RestartCmd_t

  Restart cFE Command.
- typedef struct CFE_ES_FileNameCmd_Payload CFE_ES_FileNameCmd_Payload_t

  Generic file name command payload.
- typedef struct CFE_ES_FileNameCmd CFE_ES_FileNameCmd_t

  Generic file name command.
- typedef CFE_ES_FileNameCmd_t CFE_ES_QueryAllCmd_t

- typedef CFE_ES_FileNameCmd_t CFE_ES_QueryAllTasksCmd_t
- typedef CFE_ES_FileNameCmd_t CFE_ES_WriteSysLogCmd_t
- typedef CFE_ES_FileNameCmd_t CFE_ES_WriteERLogCmd_t
- typedef struct CFE_ES_OverWriteSysLogCmd_Payload CFE_ES_OverWriteSysLogCmd_Payload_t

  Overwrite/Discard System Log Configuration Command Payload.
- typedef struct CFE_ES_OverWriteSysLogCmd CFE_ES_OverWriteSysLogCmd_t

  Overwrite/Discard System Log Configuration Command Payload.
- typedef struct CFE_ES_StartAppCmd_Payload CFE_ES_StartAppCmd_Payload_t

  Start Application Command Payload.
- typedef struct CFE_ES_StartApp CFE_ES_StartAppCmd_t

  Start Application Command.
- typedef struct CFE_ES_AppNameCmd_Payload CFE_ES_AppNameCmd_Payload_t

  Generic application name command payload.
- typedef struct CFE_ES_AppNameCmd CFE_ES_AppNameCmd_t

  Generic application name command.
- typedef CFE_ES_AppNameCmd_t CFE_ES_StopAppCmd_t
- typedef CFE_ES_AppNameCmd_t CFE_ES_RestartAppCmd_t
- typedef CFE_ES_AppNameCmd_t CFE_ES_QueryOneCmd_t
- typedef struct CFE_ES_AppReloadCmd_Payload CFE_ES_AppReloadCmd_Payload_t

  Reload Application Command Payload.
- typedef struct CFE_ES_ReloadAppCmd CFE_ES_ReloadAppCmd_t

  Reload Application Command.
- typedef struct CFE_ES_SetMaxPRCountCmd_Payload CFE_ES_SetMaxPRCountCmd_Payload_t

  Set Maximum Processor Reset Count Command Payload.
- typedef struct CFE_ES_SetMaxPRCountCmd CFE_ES_SetMaxPRCountCmd_t

  Set Maximum Processor Reset Count Command.
- typedef struct CFE_ES_DeleteCDSCmd_Payload CFE_ES_DeleteCDSCmd_Payload_t

  Delete Critical Data Store Command Payload.
- typedef struct CFE_ES_DeleteCDSCmd CFE_ES_DeleteCDSCmd_t

  Delete Critical Data Store Command.
- typedef struct CFE_ES_StartPerfCmd_Payload CFE_ES_StartPerfCmd_Payload_t

  Start Performance Analyzer Command Payload.
- typedef struct CFE_ES_StartPerfDataCmd CFE_ES_StartPerfDataCmd_t

  Start Performance Analyzer Command.
- typedef struct CFE_ES_StopPerfCmd_Payload CFE_ES_StopPerfCmd_Payload_t

  Stop Performance Analyzer Command Payload.
- typedef struct CFE_ES_StopPerfDataCmd CFE_ES_StopPerfDataCmd_t

  Stop Performance Analyzer Command.
- typedef struct CFE_ES_SetPerfFilterMaskCmd_Payload CFE_ES_SetPerfFilterMaskCmd_Payload_t

  Set Performance Analyzer Filter Mask Command Payload.
- typedef struct CFE_ES_SetPerfFilterMaskCmd CFE_ES_SetPerfFilterMaskCmd_t

  Set Performance Analyzer Filter Mask Command.
- typedef struct CFE_ES_SetPerfTrigMaskCmd_Payload CFE_ES_SetPerfTrigMaskCmd_Payload_t

  Set Performance Analyzer Trigger Mask Command Payload.
- typedef struct CFE_ES_SetPerfTriggerMaskCmd CFE_ES_SetPerfTriggerMaskCmd_t

  Set Performance Analyzer Trigger Mask Command.
- typedef struct CFE_ES_SendMemPoolStatsCmd_Payload CFE_ES_SendMemPoolStatsCmd_Payload_t

Send Memory Pool Statistics Command Payload.

- typedef struct CFE_ES_SendMemPoolStatsCmd CFE_ES_SendMemPoolStatsCmd_t

  Send Memory Pool Statistics Command.
- typedef struct CFE_ES_DumpCDSRegistryCmd_Payload CFE_ES_DumpCDSRegistryCmd_Payload_t

  Dump CDS Registry Command Payload.
- typedef struct CFE_ES_DumpCDSRegistryCmd CFE_ES_DumpCDSRegistryCmd_t

  Dump CDS Registry Command.
- typedef struct CFE_ES_OneAppTlm_Payload CFE_ES_OneAppTlm_Payload_t
- typedef struct CFE_ES_OneAppTlm CFE_ES_OneAppTlm_t
- typedef struct CFE_ES_PoolStatsTlm_Payload CFE_ES_PoolStatsTlm_Payload_t
- typedef struct CFE_ES_MemStatsTlm CFE_ES_MemStatsTlm_t
- typedef struct CFE_ES_HousekeepingTlm_Payload CFE_ES_HousekeepingTlm_Payload_t
- typedef struct CFE_ES_HousekeepingTlm CFE_ES_HousekeepingTlm_t

39.53.1 Macro Definition Documentation

39.53.1.1 CFE_ES_CLEAR_ER_LOG_CC

```
#define CFE_ES_CLEAR_ER_LOG_CC 12
```

Name Clears the contents of the Exeception and Reset Log

Description

This command causes the contents of the Executive Services Exception and Reset Log to be cleared.

Command Mnemonic(s) $sc_$cpu_ES_ClearERLog

Command Structure

CFE_ES_ClearERLogCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- The CFE_ES_ERLOG1_INF_EID informational event message will be generated.
- **`$sc_$cpu_ES_ERLOGINDEX`** - Index into Exception Reset Log goes to zero

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not dangerous. However, any previously logged data will be lost.

See also

CFE_ES_CLEAR_SYSLOG_CC, CFE_ES_WRITE_SYSLOG_CC, CFE_ES_WRITE_ER_LOG_CC

Definition at line 560 of file cfe_es_msg.h.

39.53.1.2 CFE_ES_CLEAR_SYSLOG_CC

```
#define CFE_ES_CLEAR_SYSLOG_CC 10
```

Name Clear Executive Services System Log

Description

This command clears the contents of the Executive Services System Log.

Command Mnemonic(s) $sc_$cpu_ES_ClearSysLog

Command Structure

CFE_ES_ClearSysLogCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- The CFE_ES_SYSLOG1_INF_EID informational event message will be generated.
- **`$sc_$cpu_ES_SYSLOGBYTEUSED`** - System Log Bytes Used will go to zero
- **`$sc_$cpu_ES_SYSLOGENTRIES`** - Number of System Log Entries will go to zero

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not dangerous. However, any previously logged data will be lost.

See also

CFE_ES_WRITE_SYSLOG_CC, CFE_ES_CLEAR_ER_LOG_CC, CFE_ES_WRITE_ER_LOG_CC, CFE_ES↩_OVER_WRITE_SYSLOG_CC

Definition at line 480 of file cfe_es_msg.h.

39.53.1.3 CFE_ES_DELETE_CDS_CC

```
#define CFE_ES_DELETE_CDS_CC 21
```

Name Delete Critical Data Store

Description

This command allows the user to delete a Critical Data Store that was created by an Application that is now no longer executing.

Command Mnemonic(s) $sc_$cpu_ES_DeleteCDS

Command Structure

CFE_ES_DeleteCDSCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **$sc_$cpu_ES_CMDPC** - command execution counter will increment
- The CFE_ES_CDS_DELETED_INFO_EID informational event message will be generated.
- The specified CDS should no longer appear in a CDS Registry dump generated upon receipt of the CFE_↩ ES_DUMP_CDS_REGISTRY_CC command

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- The specified CDS is the CDS portion of a Critical Table. See CFE_TBL_DELETE_CDS_CC.
- The specified CDS is not found in the CDS Registry
- The specified CDS is associated with an Application that is still active
- An error occurred while accessing the CDS memory (see the System Log for more details)

Evidence of failure may be found in the following telemetry:

- **$sc_$cpu_ES_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not critical because it is not possible to delete a CDS that is associated with an active application. However, deleting a CDS does eliminate any "history" that an application may be wishing to keep.

See also

CFE_ES_DUMP_CDS_REGISTRY_CC, CFE_TBL_DELETE_CDS_CC

Definition at line 932 of file cfe_es_msg.h.

39.53.1.4 CFE_ES_DUMP_CDS_REGISTRY_CC

```
#define CFE_ES_DUMP_CDS_REGISTRY_CC 23
```

Name Dump Critical Data Store Registry to a File

Description

This command allows the user to dump the Critical Data Store Registry to an onboard file.

Command Mnemonic(s) $sc_$cpu_ES_WriteCDS2File

Command Structure

CFE_ES_DumpCDSRegistryCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **$sc_$cpu_ES_CMDPC** - command execution counter will increment
- The CFE_ES_CDS_REG_DUMP_INF_EID debug event message will be generated.
- The file specified in the command (or the default specified by the CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE configuration parameter) will be updated with the lastest information.

Error Conditions

This command may fail for the following reason(s):

- Error occurred while trying to create the dump file

Evidence of failure may be found in the following telemetry:

- **$sc_$cpu_ES_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous. It will create a new file in the file system (or overwrite an existing one) and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

See also

CFE_ES_DELETE_CDS_CC, CFE_TBL_DELETE_CDS_CC

Definition at line 1015 of file cfe_es_msg.h.

39.53.1.5 CFE_ES_NOOP_CC

```
#define CFE_ES_NOOP_CC 0
```

Name Executive Services No-Op

Description

This command performs no other function than to increment the command execution counter. The command may be used to verify general aliveness of the Executive Services task.

Command Mnemonic(s) $sc_$cpu_ES_NOOP

Command Structure

CFE_ES_NoopCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- The CFE_ES_NOOP_INF_EID informational event message will be generated

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- the CFE_ES_LEN_ERR_EID error event message will be generated

Criticality

None

See also

Definition at line 83 of file cfe_es_msg.h.

39.53.1.6 CFE_ES_OVER_WRITE_SYSLOG_CC

```
#define CFE_ES_OVER_WRITE_SYSLOG_CC 18
```

Name Set Executive Services System Log Mode to Discard/Overwrite

Description

This command allows the user to configure the Executive Services to either discard new System Log messages when it is full or to overwrite the oldest messages.

Command Mnemonic(s) $sc_$cpu_ES_OverwriteSysLogMode

Command Structure

CFE_ES_OverWriteSysLogCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_ES_SYSLOGMODE`** - Current System Log Mode should reflect the commanded value
- The CFE_ES_SYSLOGMODE_EID debug event message will be generated.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- The desired mode is neither CFE_ES_LogMode_OVERWRITE or CFE_ES_LogMode_DISCARD

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

None. (It should be noted that "Overwrite" mode would allow a message identifying the cause of a problem to be lost by a subsequent flood of additional messages).

See also

CFE_ES_CLEAR_SYSLOG_CC, CFE_ES_WRITE_SYSLOG_CC

Definition at line 808 of file cfe_es_msg.h.

39.53.1.7 CFE_ES_QUERY_ALL_CC

```
#define CFE_ES_QUERY_ALL_CC 9
```

Name Writes all Executive Services Information on All Applications to a File

Description

This command takes the information kept by Executive Services on all of the registered applications and writes it to the specified file.

Command Mnemonic(s) $sc_$cpu_ES_WriteAppInfo2File

Command Structure

CFE_ES_QueryAllCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- The CFE_ES_ALL_APPS_EID debug event message will be generated.
- The file specified in the command (or the default specified by the CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE configuration parameter) will be updated with the lastest information.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- An Error occurs while trying to write to the file

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous. It will create a new file in the file system (or overwrite an existing one) and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

See also

CFE_ES_QUERY_ONE_CC, CFE_ES_QUERY_ALL_TASKS_CC

Definition at line 442 of file cfe_es_msg.h.

39.53.1.8 CFE_ES_QUERY_ALL_TASKS_CC

```
#define CFE_ES_QUERY_ALL_TASKS_CC 24
```

Name Writes a list of All Executive Services Tasks to a File

Description

This command takes the information kept by Executive Services on all of the registered tasks and writes it to the specified file.

Command Mnemonic(s) $sc_$cpu_ES_WriteTaskInfo2File

Command Structure

CFE_ES_QueryAllTasksCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- The CFE_ES_TASKINFO_EID debug event message will be generated.
- The file specified in the command (or the default specified by the CFE_PLATFORM_ES_DEFAULT_TASK↩_LOG_FILE configuration parameter) will be updated with the lastest information.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- An Error occurs while trying to write to the file

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous. It will create a new file in the file system (or overwrite an existing one) and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

See also

CFE_ES_QUERY_ALL_CC, CFE_ES_QUERY_ONE_CC

Definition at line 1057 of file cfe_es_msg.h.

39.53.1.9 CFE_ES_QUERY_ONE_CC

```
#define CFE_ES_QUERY_ONE_CC 8
```

Name Request Executive Services Information on a Specified Application

Description

This command takes the information kept by Executive Services on the specified application and telemeters it to the ground.

Command Mnemonic(s) $sc_$cpu_ES_QueryApp

Command Structure

CFE_ES_QueryOneCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- The CFE_ES_ONE_APP_EID debug event message will be generated. NOTE: This event message only identifies that the act of stopping the application has begun, not that is has completed.
- Receipt of the CFE_ES_OneAppTlm_t telemetry packet

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- The specified application name is not recognized as an active application

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

None

See also

CFE_ES_QUERY_ALL_CC, CFE_ES_QUERY_ALL_TASKS_CC

Definition at line 400 of file cfe_es_msg.h.

39.53.1.10 CFE_ES_RELOAD_APP_CC

```
#define CFE_ES_RELOAD_APP_CC 7
```

Name Stops, Unloads, Loads from the command specfied File and Restarts an Application

Description

This command halts and removes the specified Application from the system. Then it immediately loads the Application from the command specified file and restarts it. This command is especially useful for restarting a Command Ingest Application since once it has been stopped, no further commands can come in to restart it.

Command Mnemonic(s) $sc_$cpu_ES_ReloadApp

Command Structure

CFE_ES_ReloadAppCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- The CFE_ES_RELOAD_APP_DBG_EID debug event message will be generated. NOTE: This event message only identifies that the act of stopping the application has begun, not that is has completed.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- The reload file is missing
- The specified application name is not recognized as an active application
- The specified application is one of the cFE's Core applications (ES, EVS, SB, TBL, TIME)

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases
- Additional information on the reason for command failure may be found in the System Log

Criticality

This command is not inherently dangerous, however the restarting of certain applications (e.g. - Spacecraft Attitude and Control) may have a detrimental effect on the spacecraft.

See also

CFE_ES_START_APP_CC, CFE_ES_STOP_APP_CC, CFE_ES_RESTART_APP_CC

Definition at line 362 of file cfe_es_msg.h.

39.53.1.11 CFE_ES_RESET_COUNTERS_CC

```
#define CFE_ES_RESET_COUNTERS_CC 1
```

Name Executive Services Reset Counters

Description

This command resets the following counters within the Executive Services housekeeping telemetry:

- Command Execution Counter
- Command Error Counter

Command Mnemonic(s) $sc_$cpu_ES_ResetCtrs

Command Structure

CFE_ES_ResetCountersCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- The CFE_ES_RESET_INF_EID informational event message will be generated

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- the CFE_ES_LEN_ERR_EID error event message will be generated

Criticality

This command is not inherently dangerous. However, it is possible for ground systems and on-board safing procedures to be designed such that they react to changes in the counter values that are reset by this command.

See also

CFE_ES_RESET_PR_COUNT_CC

Definition at line 122 of file cfe_es_msg.h.

39.53.1.12 CFE_ES_RESET_PR_COUNT_CC

```
#define CFE_ES_RESET_PR_COUNT_CC 19
```

Name Resets the Processor Reset Counter to Zero

Description

This command allows the user to reset the Processor Reset Counter to zero. The Processor Reset Counter counts the number of Processor Resets that have occurred so as to identify when a Processor Reset should automatically be upgraded to a full Power-On Reset.

Command Mnemonic(s) $sc_$cpu_ES_ResetPRCnt

Command Structure

CFE_ES_ResetPRCountCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_ES_ProcResetCnt`** - Current number of processor resets will go to zero
- The CFE_ES_RESET_PR_COUNT_EID informational event message will be generated.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not critical. The only impact would be that the system would have to have more processor resets before an automatic power-on reset occurred.

See also

CFE_ES_SET_MAX_PR_COUNT_CC, CFE_ES_RESET_COUNTERS_CC

Definition at line 848 of file cfe_es_msg.h.

39.53.1.13 CFE_ES_RESTART_APP_CC

```
#define CFE_ES_RESTART_APP_CC 6
```

Name Stops, Unloads, Loads using the previous File name, and Restarts an Application

Description

This command halts and removes the specified Application from the system. Then it immediately loads the Application from the same filename last used to start. This command is especially useful for restarting a Command Ingest Application since once it has been stopped, no further commands can come in to restart it.

Command Mnemonic(s) $sc_$cpu_ES_ResetApp

Command Structure

CFE_ES_RestartAppCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- The CFE_ES_RESTART_APP_DBG_EID debug event message will be generated. NOTE: This event message only identifies that the act of stopping the application has begun, not that is has completed.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- The original file is missing
- The specified application name is not recognized as an active application
- The specified application is one of the cFE's Core applications (ES, EVS, SB, TBL, TIME)

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases
- Additional information on the reason for command failure may be found in the System Log

Criticality

This command is not inherently dangerous, however the restarting of certain applications (e.g. - Spacecraft Attitude and Control) may have a detrimental effect on the spacecraft.

See also

CFE_ES_START_APP_CC, CFE_ES_STOP_APP_CC, CFE_ES_RELOAD_APP_CC

Definition at line 315 of file cfe_es_msg.h.

39.53.1.14 CFE_ES_RESTART_CC

```
#define CFE_ES_RESTART_CC 2
```

Name Executive Services Processor / Power-On Reset

Description

This command restarts the cFE in one of two modes. The Power-On Reset will cause the cFE to restart as though the power were first applied to the processor. The Processor Reset will attempt to retain the contents of the volatile disk and the contents of the Critical Data Store. NOTE: If a requested Processor Reset should cause the Processor Reset Counter (**$sc_$cpu_ES_ProcResetCnt)** to exceed OR EQUAL the limit CFE_PLATFORM_ES_MAX↩_PROCESSOR_RESETS (which is reported in housekeeping telemetry as **$sc_$cpu_ES_MaxProcResets)**, the command is **AUTOMATICALLY** upgraded to a Power-On Reset.

Command Mnemonic(s) $sc_$cpu_ES_ProcessorReset, $sc_$cpu_ES_PowerOnReset

Command Structure

CFE_ES_RestartCmd_t

Command Verification

Successful execution of this command (as a Processor Reset) may be verified with the following telemetry:

- **$sc_$cpu_ES_ProcResetCnt -** processor reset counter will increment
- New entries in the Exception Reset Log and System Log can be found
  NOTE: Verification of a Power-On Reset is shown through the loss of data nominally retained through a Processor Reset
  NOTE: Since the reset of the processor resets the command execution counter (**$sc_$cpu_ES_CMDPC)**, this counter **CANNOT** be used to verify command execution.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- The Restart Type was not a recognized value.

Evidence of failure may be found in the following telemetry:

- **$sc_$cpu_ES_CMDEC -** command error counter will increment
- the CFE_ES_BOOT_ERR_EID error event message will be generated

Criticality

This command is, by definition, dangerous. Significant loss of data will occur. All processes and the cFE itself will be stopped and restarted. With the Power-On reset option, all data on the volatile disk and the contents of the Critical Data Store will be lost.

See also

CFE_ES_RESET_PR_COUNT_CC, CFE_ES_SET_MAX_PR_COUNT_CC

Definition at line 172 of file cfe_es_msg.h.

39.53.1.15 CFE_ES_SEND_MEM_POOL_STATS_CC

```
#define CFE_ES_SEND_MEM_POOL_STATS_CC 22
```

Name Telemeter Memory Pool Statistics

Description

This command allows the user to obtain a snapshot of the statistics maintained for a specified memory pool.

Command Mnemonic(s) $sc_$cpu_ES_PoolStats

Command Structure

CFE_ES_SendMemPoolStatsCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- The CFE_ES_TLM_POOL_STATS_INFO_EID debug event message will be generated.
- The Memory Pool Statistics Telemetry Packet is produced

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- The specified handle is not associated with a known memory pool
- The specified handle caused a processor exception because it improperly identified a segment of memory

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

An incorrect Memory Pool Handle value can cause a system crash. Extreme care should be taken to ensure the memory handle value used in the command is correct.

See also

Definition at line 974 of file cfe_es_msg.h.

39.53.1.16 CFE_ES_SET_MAX_PR_COUNT_CC

```
#define CFE_ES_SET_MAX_PR_COUNT_CC 20
```

Name Configure the Maximum Number of Processor Resets before a Power-On Reset

Description

This command allows the user to specify the number of Processor Resets that are allowed before the next Processor Reset is upgraded to a Power-On Reset.

Command Mnemonic(s) $sc_$cpu_ES_SetMaxPRCnt

Command Structure

CFE_ES_SetMaxPRCountCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_ES_MaxProcResets`** - Current maximum number of processor resets before an automatic power-on reset will go to the command specified value.
- The CFE_ES_SET_MAX_PR_COUNT_EID informational event message will be generated.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

If the operator were to set the Maximum Processor Reset Count to too high a value, the processor would require an inordinate number of consecutive processor resets before an automatic power-on reset would occur. This could potentially leave the spacecraft without any control for a significant amount of time if a processor reset fails to clear a problem.

See also

CFE_ES_RESET_PR_COUNT_CC

Definition at line 889 of file cfe_es_msg.h.

39.53.1.17 CFE_ES_SET_PERF_FILTER_MASK_CC

```
#define CFE_ES_SET_PERF_FILTER_MASK_CC 16
```

Name Set Performance Analyzer's Filter Masks

Description

This command sets the Performance Analyzer's Filter Masks.

Command Mnemonic(s) $sc_$cpu_ES_LAFilterMask

Command Structure

CFE_ES_SetPerfFilterMaskCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **$sc_$cpu_ES_CMDPC -** command execution counter will increment
- **$sc_$cpu_ES_PerfFltrMask[MaskCnt]** - the current performance filter mask value(s) should reflect the commanded value
- The CFE_ES_PERF_FILTMSKCMD_EID debug event message will be generated.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- The Filter Mask ID number is out of range

Evidence of failure may be found in the following telemetry:

- **$sc_$cpu_ES_CMDEC -** command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

Changing the filter masks may cause a small change in the Performance Analyzer's CPU utilization.

See also

CFE_ES_START_PERF_DATA_CC, CFE_ES_STOP_PERF_DATA_CC, CFE_ES_SET_PERF_TRIGGER_M↩ASK_CC

Definition at line 729 of file cfe_es_msg.h.

39.53.1.18 CFE_ES_SET_PERF_TRIGGER_MASK_CC

```
#define CFE_ES_SET_PERF_TRIGGER_MASK_CC 17
```

Name Set Performance Analyzer's Trigger Masks

Description

This command sets the Performance Analyzer's Trigger Masks.

Command Mnemonic(s) $sc_$cpu_ES_LATriggerMask

Command Structure

CFE_ES_SetPerfTriggerMaskCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **$sc_$cpu_ES_CMDPC** - command execution counter will increment
- **$sc_$cpu_ES_PerfTrigMask[MaskCnt]** - the current performance trigger mask value(s) should reflect the commanded value
- The CFE_ES_PERF_TRIGMSKCMD_EID debug event message will be generated.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- The Trigger Mask ID number is out of range

Evidence of failure may be found in the following telemetry:

- **$sc_$cpu_ES_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

Changing the trigger masks may cause a small change in the Performance Analyzer's CPU utilization.

See also

CFE_ES_START_PERF_DATA_CC, CFE_ES_STOP_PERF_DATA_CC, CFE_ES_SET_PERF_FILTER_MA↩SK_CC

Definition at line 767 of file cfe_es_msg.h.

39.53.1.19 CFE_ES_START_APP_CC

```
#define CFE_ES_START_APP_CC 4
```

Name Load and Start an Application

Description

This command starts the specified application with the specified start address, stack size, etc options.

Command Mnemonic(s) $sc_$cpu_ES_StartApp

Command Structure

CFE_ES_StartAppCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- The CFE_ES_START_INF_EID informational event message will be generated

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- The specified application filename string is either a NULL string or less than four characters in length
- The specified application entry point is a NULL string
- The specified application name is a NULL string
- The specified stack size is less than CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
- The specified priority is greater than MAX_PRIORITY (as defined in osapi.c)
- The specified exception action is neither CFE_ES_ExceptionAction_RESTART_APP (0) or CFE_ES_↩ ExceptionAction_PROC_RESTART (1)
- The Operating System was unable to load the specified application file

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous although system resources could be taxed beyond their limits with the starting of erroneous or invalid applications.

See also

CFE_ES_STOP_APP_CC, CFE_ES_RESTART_APP_CC, CFE_ES_RELOAD_APP_CC

Definition at line 218 of file cfe_es_msg.h.

39.53.1.20 CFE_ES_START_PERF_DATA_CC

```
#define CFE_ES_START_PERF_DATA_CC 14
```

Name Start Performance Analyzer

Description

This command causes the Performance Analyzer to begin collecting data using the specified trigger mode.

Command Mnemonic(s) $sc_$cpu_ES_StartLAData

Command Structure

CFE_ES_StartPerfDataCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **$sc_$cpu_ES_CMDPC -** command execution counter will increment
- **$sc_$cpu_ES_PerfState -** Current performance analyzer state will change to either WAITING FOR TRIGGER or, if conditions are appropriate fast enough, TRIGGERED.
- **$sc_$cpu_ES_PerfMode -** Performance Analyzer Mode will change to the commanded trigger mode (TRIGGER START, TRIGGER CENTER, or TRIGGER END).
- **$sc_$cpu_ES_PerfTrigCnt -** Performance Trigger Count will go to zero
- **$sc_$cpu_ES_PerfDataStart -** Data Start Index will go to zero
- **$sc_$cpu_ES_PerfDataEnd -** Data End Index will go to zero
- **$sc_$cpu_ES_PerfDataCnt -** Performance Data Counter will go to zero
- The CFE_ES_PERF_STARTCMD_EID debug event message will be generated.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- A previous CFE_ES_STOP_PERF_DATA_CC command has not completely finished.
- An invalid trigger mode is requested.

Evidence of failure may be found in the following telemetry:

- **$sc_$cpu_ES_CMDEC -** command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous but may cause a small increase in CPU utilization as the performance analyzer data is collected.

See also

CFE_ES_STOP_PERF_DATA_CC, CFE_ES_SET_PERF_FILTER_MASK_CC, CFE_ES_SET_PERF_TRIGGER_MASK_CC

Definition at line 648 of file cfe_es_msg.h.

39.53.1.21 CFE_ES_STOP_APP_CC

```
#define CFE_ES_STOP_APP_CC 5
```

Name Stop and Unload Application

Description

This command halts and removes the specified Application from the system. **NOTE:** This command should never be used on the Command Ingest application. This would prevent further commands from entering the system. If Command Ingest needs to be stopped and restarted, use CFE_ES_RESTART_APP_CC or CFE_ES_RELOAD↩_APP_CC.

Command Mnemonic(s) $sc_$cpu_ES_StopApp

Command Structure

CFE_ES_StopAppCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- The CFE_ES_STOP_DBG_EID debug event message will be generated. NOTE: This event message only identifies that the stop has been started, not that is has completed.
- Once the stop has successfully completed, the list of Applications and Tasks created in response to the **`$sc_$cpu_ES_WriteAppInfo2File`**, **`$sc_$cpu_ES_WriteTaskInfo2File`** **should** no longer contain the specified application.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- The specified application name is not recognized as an active application
- The specified application is one of the cFE's Core applications (ES, EVS, SB, TBL, TIME)

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases
- Additional information on the reason for command failure may be found in the System Log

Criticality

This command is not inherently dangerous, however the removal of certain applications (e.g. - Spacecraft Attitude and Control) may have a detrimental effect on the spacecraft.

See also

CFE_ES_START_APP_CC, CFE_ES_RESTART_APP_CC, CFE_ES_RELOAD_APP_CC

Definition at line 268 of file cfe_es_msg.h.

39.53.1.22 CFE_ES_STOP_PERF_DATA_CC

```
#define CFE_ES_STOP_PERF_DATA_CC 15
```

Name Stop Performance Analyzer

Description

This command stops the Performance Analyzer from collecting any more data.

Command Mnemonic(s) $sc_$cpu_ES_StopLAData

Command Structure

CFE_ES_StopPerfDataCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_ES_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_ES_PerfState`** - Current performance analyzer state will change to IDLE.
- The CFE_ES_PERF_STOPCMD_EID debug event message will be generated.
- The file specified in the command (or the default specified by the CFE_PLATFORM_ES_DEFAULT_PERF↩_DUMP_FILENAME configuration parameter) will be updated with the lastest information.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- A previous Stop Performance Analyzer command is still in process
- An error occurred while spawning the child task responsible for dumping the Performance Analyzer data to a file

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_ES_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous. An additional low priority child task will be spawned, however, to dump the performance analyzer data to a file.

See also

CFE_ES_START_PERF_DATA_CC, CFE_ES_SET_PERF_FILTER_MASK_CC, CFE_ES_SET_PERF_TRIG↩GER_MASK_CC

Definition at line 691 of file cfe_es_msg.h.

39.53.1.23 CFE_ES_WRITE_ER_LOG_CC

```
#define CFE_ES_WRITE_ER_LOG_CC 13
```

Name Writes Exeception and Reset Log to a File

Description

This command causes the contents of the Executive Services Exception and Reset Log to be written to the specified file.

Command Mnemonic(s) $sc_$cpu_ES_WriteERLog2File

Command Structure

CFE_ES_WriteERLogCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **$sc_$cpu_ES_CMDPC** - command execution counter will increment
- The CFE_ES_ERLOG2_EID debug event message will be generated.
- The file specified in the command (or the default specified by the CFE_PLATFORM_ES_DEFAULT_ER_L↩OG_FILE configuration parameter) will be updated with the lastest information.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- An Error occurs while trying to write to the file

Evidence of failure may be found in the following telemetry:

- **$sc_$cpu_ES_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous. It will create a new file in the file system (or overwrite an existing one) and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

See also

CFE_ES_CLEAR_SYSLOG_CC, CFE_ES_WRITE_SYSLOG_CC, CFE_ES_CLEAR_ER_LOG_CC

Definition at line 602 of file cfe_es_msg.h.

39.53.1.24 CFE_ES_WRITE_SYSLOG_CC

```
#define CFE_ES_WRITE_SYSLOG_CC 11
```

Name Writes contents of Executive Services System Log to a File

Description

This command causes the contents of the Executive Services System Log to be written to a log file.

Command Mnemonic(s) $sc_$cpu_ES_WriteSysLog2File

Command Structure

CFE_ES_WriteSysLogCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **$sc_$cpu_ES_CMDPC** - command execution counter will increment
- The CFE_ES_SYSLOG2_EID debug event message will be generated.
- The file specified in the command (or the default specified by the CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE configuration parameter) will be updated with the lastest information.

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect
- An Error occurs while trying to write to the file

Evidence of failure may be found in the following telemetry:

- **$sc_$cpu_ES_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous. It will create a new file in the file system (or overwrite an existing one) and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

See also

CFE_ES_CLEAR_SYSLOG_CC, CFE_ES_CLEAR_ER_LOG_CC, CFE_ES_WRITE_ER_LOG_CC, CFE_ES_OVER_WRITE_SYSLOG_CC

Definition at line 523 of file cfe_es_msg.h.

39.53.2 Typedef Documentation

39.53.2.1 CFE_ES_AppNameCmd_Payload_t

```
typedef struct CFE_ES_AppNameCmd_Payload CFE_ES_AppNameCmd_Payload_t
```

Generic application name command payload.

For command details, see CFE_ES_STOP_APP_CC, CFE_ES_RESTART_APP_CC, CFE_ES_QUERY_ONE_CC

39.53.2.2 CFE_ES_AppNameCmd_t

```
typedef struct CFE_ES_AppNameCmd CFE_ES_AppNameCmd_t
```

Generic application name command.

39.53.2.3 CFE_ES_AppReloadCmd_Payload_t

```
typedef struct CFE_ES_AppReloadCmd_Payload CFE_ES_AppReloadCmd_Payload_t
```

Reload Application Command Payload.

For command details, see CFE_ES_RELOAD_APP_CC

39.53.2.4 CFE_ES_ClearERLogCmd_t

```
typedef CFE_ES_NoArgsCmd_t CFE_ES_ClearERLogCmd_t
```

Definition at line 1089 of file cfe_es_msg.h.

39.53.2.5 CFE_ES_ClearSysLogCmd_t

```
typedef CFE_ES_NoArgsCmd_t CFE_ES_ClearSysLogCmd_t
```

Definition at line 1088 of file cfe_es_msg.h.

39.53.2.6 CFE_ES_DeleteCDSCmd_Payload_t

```
typedef struct CFE_ES_DeleteCDSCmd_Payload CFE_ES_DeleteCDSCmd_Payload_t
```

Delete Critical Data Store Command Payload.

For command details, see CFE_ES_DELETE_CDS_CC

39.53.2.7 CFE_ES_DeleteCDSCmd_t

```
typedef struct CFE_ES_DeleteCDSCmd CFE_ES_DeleteCDSCmd_t
```

Delete Critical Data Store Command.

39.53.2.8 CFE_ES_DumpCDSRegistryCmd_Payload_t

```
typedef struct CFE_ES_DumpCDSRegistryCmd_Payload CFE_ES_DumpCDSRegistryCmd_Payload_t
```

Dump CDS Registry Command Payload.

For command details, see CFE_ES_DUMP_CDS_REGISTRY_CC

39.53.2.9 CFE_ES_DumpCDSRegistryCmd_t

```
typedef struct CFE_ES_DumpCDSRegistryCmd CFE_ES_DumpCDSRegistryCmd_t
```

Dump CDS Registry Command.

39.53.2.10 CFE_ES_FileNameCmd_Payload_t

```
typedef struct CFE_ES_FileNameCmd_Payload CFE_ES_FileNameCmd_Payload_t
```

Generic file name command payload.

This format is shared by several executive services commands. For command details, see CFE_ES_QUERY_ALL_CC, CFE_ES_QUERY_ALL_TASKS_CC, CFE_ES_WRITE_SYSLOG_CC, and CFE_ES_WRITE_ER_LOG_CC

39.53.2.11 CFE_ES_FileNameCmd_t

```
typedef struct CFE_ES_FileNameCmd CFE_ES_FileNameCmd_t
```

Generic file name command.

39.53.2.12 CFE_ES_HousekeepingTlm_Payload_t

```
typedef struct CFE_ES_HousekeepingTlm_Payload CFE_ES_HousekeepingTlm_Payload_t
```

Name Executive Services Housekeeping Packet

39.53.2.13 CFE_ES_HousekeepingTlm_t

```
typedef struct CFE_ES_HousekeepingTlm CFE_ES_HousekeepingTlm_t
```

39.53.2.14 CFE_ES_MemStatsTlm_t

```
typedef struct CFE_ES_MemStatsTlm CFE_ES_MemStatsTlm_t
```

39.53.2.15 CFE_ES_NoArgsCmd_t

```
typedef struct CFE_ES_NoArgsCmd CFE_ES_NoArgsCmd_t
```

Generic "no arguments" command.

This command structure is used for commands that do not have any parameters. This includes:

1. The Housekeeping Request Message
2. The No-Op Command (For details, see CFE_ES_NOOP_CC)
3. The Reset Counters Command (For details, see CFE_ES_RESET_COUNTERS_CC)

39.53.2.16 CFE_ES_NoopCmd_t

```
typedef CFE_ES_NoArgsCmd_t CFE_ES_NoopCmd_t
```

Definition at line 1086 of file cfe_es_msg.h.

39.53.2.17 CFE_ES_OneAppTlm_Payload_t

```
typedef struct CFE_ES_OneAppTlm_Payload CFE_ES_OneAppTlm_Payload_t
```

Name Single Application Information Packet

39.53.2.18 CFE_ES_OneAppTlm_t

```
typedef struct CFE_ES_OneAppTlm CFE_ES_OneAppTlm_t
```

39.53.2.19 CFE_ES_OverWriteSysLogCmd_Payload_t

```
typedef struct CFE_ES_OverWriteSysLogCmd_Payload CFE_ES_OverWriteSysLogCmd_Payload_t
```

Overwrite/Discard System Log Configuration Command Payload.

For command details, see CFE_ES_OVER_WRITE_SYSLOG_CC

39.53.2.20 CFE_ES_OverWriteSysLogCmd_t

```
typedef struct CFE_ES_OverWriteSysLogCmd CFE_ES_OverWriteSysLogCmd_t
```

Overwrite/Discard System Log Configuration Command Payload.

39.53.2.21 CFE_ES_PoolStatsTlm_Payload_t

```
typedef struct CFE_ES_PoolStatsTlm_Payload CFE_ES_PoolStatsTlm_Payload_t
```

Name Memory Pool Statistics Packet

39.53.2.22 CFE_ES_QueryAllCmd_t

```
typedef CFE_ES_FileNameCmd_t CFE_ES_QueryAllCmd_t
```

Definition at line 1140 of file cfe_es_msg.h.

39.53.2.23 CFE_ES_QueryAllTasksCmd_t

```
typedef CFE_ES_FileNameCmd_t CFE_ES_QueryAllTasksCmd_t
```

Definition at line 1141 of file cfe_es_msg.h.

39.53.2.24 CFE_ES_QueryOneCmd_t

```
typedef CFE_ES_AppNameCmd_t CFE_ES_QueryOneCmd_t
```

Definition at line 1226 of file cfe_es_msg.h.

39.53.2.25 CFE_ES_ReloadAppCmd_t

```
typedef struct CFE_ES_ReloadAppCmd CFE_ES_ReloadAppCmd_t
```

Reload Application Command.

39.53.2.26 CFE_ES_ResetCountersCmd_t

```
typedef CFE_ES_NoArgsCmd_t CFE_ES_ResetCountersCmd_t
```

Definition at line 1087 of file cfe_es_msg.h.

39.53.2.27 CFE_ES_ResetPRCountCmd_t

```
typedef CFE_ES_NoArgsCmd_t CFE_ES_ResetPRCountCmd_t
```

Definition at line 1090 of file cfe_es_msg.h.

39.53.2.28 CFE_ES_RestartAppCmd_t

```
typedef CFE_ES_AppNameCmd_t CFE_ES_RestartAppCmd_t
```

Definition at line 1225 of file cfe_es_msg.h.

39.53.2.29 CFE_ES_RestartCmd_Payload_t

```
typedef struct CFE_ES_RestartCmd_Payload CFE_ES_RestartCmd_Payload_t
```

Restart cFE Command Payload.

For command details, see CFE_ES_RESTART_CC

39.53.2.30 CFE_ES_RestartCmd_t

```
typedef struct CFE_ES_RestartCmd CFE_ES_RestartCmd_t
```

Restart cFE Command.

39.53.2.31 CFE_ES_SendMemPoolStatsCmd_Payload_t

```
typedef struct CFE_ES_SendMemPoolStatsCmd_Payload CFE_ES_SendMemPoolStatsCmd_Payload_t
```

Send Memory Pool Statistics Command Payload.

For command details, see CFE_ES_SEND_MEM_POOL_STATS_CC

39.53.2.32 CFE_ES_SendMemPoolStatsCmd_t

```
typedef struct CFE_ES_SendMemPoolStatsCmd CFE_ES_SendMemPoolStatsCmd_t
```

Send Memory Pool Statistics Command.

39.53.2.33 CFE_ES_SetMaxPRCountCmd_Payload_t

```
typedef struct CFE_ES_SetMaxPRCountCmd_Payload CFE_ES_SetMaxPRCountCmd_Payload_t
```

Set Maximum Processor Reset Count Command Payload.

For command details, see CFE_ES_SET_MAX_PR_COUNT_CC

39.53.2.34 CFE_ES_SetMaxPRCountCmd_t

```
typedef struct CFE_ES_SetMaxPRCountCmd CFE_ES_SetMaxPRCountCmd_t
```

Set Maximum Processor Reset Count Command.

39.53.2.35 CFE_ES_SetPerfFilterMaskCmd_Payload_t

```
typedef struct CFE_ES_SetPerfFilterMaskCmd_Payload CFE_ES_SetPerfFilterMaskCmd_Payload_t
```

Set Performance Analyzer Filter Mask Command Payload.

For command details, see CFE_ES_SET_PERF_FILTER_MASK_CC

39.53.2.36 CFE_ES_SetPerfFilterMaskCmd_t

```
typedef struct CFE_ES_SetPerfFilterMaskCmd CFE_ES_SetPerfFilterMaskCmd_t
```

Set Performance Analyzer Filter Mask Command.

39.53.2.37 CFE_ES_SetPerfTriggerMaskCmd_t

```
typedef struct CFE_ES_SetPerfTriggerMaskCmd CFE_ES_SetPerfTriggerMaskCmd_t
```

Set Performance Analyzer Trigger Mask Command.

39.53.2.38 CFE_ES_SetPerfTrigMaskCmd_Payload_t

```
typedef struct CFE_ES_SetPerfTrigMaskCmd_Payload CFE_ES_SetPerfTrigMaskCmd_Payload_t
```

Set Performance Analyzer Trigger Mask Command Payload.

For command details, see CFE_ES_SET_PERF_TRIGGER_MASK_CC

39.53.2.39 CFE_ES_StartAppCmd_Payload_t

```
typedef struct CFE_ES_StartAppCmd_Payload CFE_ES_StartAppCmd_Payload_t
```

Start Application Command Payload.

For command details, see CFE_ES_START_APP_CC

39.53.2.40 CFE_ES_StartAppCmd_t

```
typedef struct CFE_ES_StartApp CFE_ES_StartAppCmd_t
```

Start Application Command.

39.53.2.41 CFE_ES_StartPerfCmd_Payload_t

```
typedef struct CFE_ES_StartPerfCmd_Payload CFE_ES_StartPerfCmd_Payload_t
```

Start Performance Analyzer Command Payload.

For command details, see CFE_ES_START_PERF_DATA_CC

39.53.2.42 CFE_ES_StartPerfDataCmd_t

```
typedef struct CFE_ES_StartPerfDataCmd CFE_ES_StartPerfDataCmd_t
```

Start Performance Analyzer Command.

39.53.2.43 CFE_ES_StopAppCmd_t

```
typedef CFE_ES_AppNameCmd_t CFE_ES_StopAppCmd_t
```

Definition at line 1224 of file cfe_es_msg.h.

39.53.2.44 CFE_ES_StopPerfCmd_Payload_t

```
typedef struct CFE_ES_StopPerfCmd_Payload CFE_ES_StopPerfCmd_Payload_t
```

Stop Performance Analyzer Command Payload.

For command details, see CFE_ES_STOP_PERF_DATA_CC

39.53.2.45 CFE_ES_StopPerfDataCmd_t

```
typedef struct CFE_ES_StopPerfDataCmd CFE_ES_StopPerfDataCmd_t
```

Stop Performance Analyzer Command.

39.53.2.46 CFE_ES_WriteERLogCmd_t

```
typedef CFE_ES_FileNameCmd_t CFE_ES_WriteERLogCmd_t
```

Definition at line 1143 of file cfe_es_msg.h.

39.53.2.47 CFE_ES_WriteSysLogCmd_t

```
typedef CFE_ES_FileNameCmd_t CFE_ES_WriteSysLogCmd_t
```

Definition at line 1142 of file cfe_es_msg.h.

39.54 cfe/modules/evs/fsw/inc/cfe_evs_events.h File Reference

Macros

- #define CFE_EVS_MAX_EID 43
- #define CFE_EVS_NOOP_EID 0 /* Noop event identifier */

  'No-op command'
- #define CFE_EVS_STARTUP_EID 1

  'cFE EVS Initialized'
- #define CFE_EVS_ERR_WRLOGFILE_EID 2

  'Write Log File Command Error: OS_write = 0x%08X, filename = %s'
- #define CFE_EVS_ERR_CRLOGFILE_EID 3

  'Write Log File Command Error: OS_OpenCreate = 0x%08X, filename = %s'
- #define CFE_EVS_ERR_MSGID_EID 5

  'Invalid command packet, Message ID = 0x%08X'
- #define CFE_EVS_ERR_EVTIDNOREGS_EID 6

  '%s Event ID %d not registered for filtering: CC = %lu'
- #define CFE_EVS_ERR_APPNOREGS_EID 7

  '%s not registered with EVS: CC = %lu'
- #define CFE_EVS_ERR_ILLAPPIDRANGE_EID 8

  'Illegal application ID %d retrieved for %s: CC = %lu'
- #define CFE_EVS_ERR_NOAPPIDFOUND_EID 9

  'Unable to retrieve application ID for %s: CC = %lu'
- #define CFE_EVS_ERR_ILLEGALFMTMOD_EID 10

  'Set Event Format Mode Command: Invalid Event Format Mode = 0x%02x'
- #define CFE_EVS_ERR_MAXREGSFILTER_EID 11

  'Add Filter Command: number of registered filters has reached max = %d'
- #define CFE_EVS_ERR_WRDATFILE_EID 12

  'Write App Data Command Error: OS_write = 0x%08X, filename = %s'
- #define CFE_EVS_ERR_CRDATFILE_EID 13

  'Write App Data Command Error: OS_OpenCreate = 0x%08X, filename = %s'
- #define CFE_EVS_ERR_CC_EID 15

  'Invalid command code - ID = 0x%08x, CC = %d'
- #define CFE_EVS_RSTCNT_EID 16

  'Reset Counters Command Received'
- #define CFE_EVS_SETFILTERMSK_EID 17

  'Set Filter Mask Command Received with AppName=%s, EventID=0x%08x, Mask=0x%04x'
- #define CFE_EVS_ENAPORT_EID 18

  'Enable Ports Command Received with Port Bit Mask = 0x%02x'
- #define CFE_EVS_DISPORT_EID 19

  'Disable Ports Command Received with Port Bit Mask = 0x%02x'
- #define CFE_EVS_ENAEVTTYPE_EID 20

  'Enable Event Type Command Received with Event Type Bit Mask = 0x%02x'
- #define CFE_EVS_DISEVTTYPE_EID 21

  'Disable Event Type Command Received with Event Type Bit Mask = 0x%02x'
- #define CFE_EVS_SETEVTFMTMOD_EID 22

  'Set Event Format Mode Command Received with Mode = 0x%02x'

- #define CFE_EVS_ENAAPPEVTTYPE_EID 23

  'Enable App Event Type Command Received with AppName = %s, EventType Bit Mask = 0x%02x'
- #define CFE_EVS_DISAPPENTTYPE_EID 24

  'Disable App Event Type Command Received with AppName = %s, EventType Bit Mask = 0x%02x'
- #define CFE_EVS_ENAAPPEVT_EID 25

  'Enable App Events Command Received with AppName = %s'
- #define CFE_EVS_DISAPPEVT_EID 26

  'Disable App Events Command Received with AppName = %s'
- #define CFE_EVS_RSTEVTCNT_EID 27

  'Reset Event Counter Command Received with AppName = %s'
- #define CFE_EVS_RSTFILTER_EID 28

  'Reset Filter Command Received with AppName = %s, EventID = 0x%08x'
- #define CFE_EVS_RSTALLFILTER_EID 29

  'Reset All Filters Command Received with AppName = %s'
- #define CFE_EVS_ADDFILTER_EID 30

  'Add Filter Command Received with AppName = %s, EventID = 0x%08x, Mask = 0x%04x'
- #define CFE_EVS_DELFILTER_EID 31

  'Delete Filter Command Received with AppName = %s, EventID = 0x%08x'
- #define CFE_EVS_WRDAT_EID 32

  'Write App Data Command: %d application data entries written to %s'
- #define CFE_EVS_WRLOG_EID 33

  'Write Log File Command: %d event log entries written to %s'
- #define CFE_EVS_EVT_FILTERED_EID 37

  'Add Filter Command:AppName = %s, EventID = 0x%08x is already registered for filtering'
- #define CFE_EVS_LOGMODE_EID 38

  'Set Log Mode Command Error: Log Mode = %d'
- #define CFE_EVS_ERR_LOGMODE_EID 39

  'Set Log Mode Command Error: Log Mode = %d'
- #define CFE_EVS_ERR_INVALID_BITMASK_EID 40

  'Bit Mask = 0x%X out of range: CC = %lu'
- #define CFE_EVS_ERR_UNREGISTERED_EVS_APP 41

  'App %s not registered with Event Services. Unable to send event'
- #define CFE_EVS_FILTER_MAX_EID 42

  'Max filter count reached, AppName = %s, EventID = 0x%08x: Filter locked until reset'
- #define CFE_EVS_LEN_ERR_EID 43

  'Invalid cmd length: ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'

39.54.1 Macro Definition Documentation

39.54.1.1 CFE_EVS_ADDFILTER_EID

```
#define CFE_EVS_ADDFILTER_EID 30
```

`'Add Filter Command Received with AppName = %s, EventID = 0x%08x, Mask = 0x%04x'`

Event Message `'Add Filter Command Received with AppName = %s, EventID = 0x%08x, Mask = 0x%04x'`

Type: DEBUG

Cause:

This event message is generated upon successful completion of the "Add Filter" command.

The `AppName` field identifies the Application who is getting the new filter, the `EventID` field identifies the Event Identifier, in hex, that is getting the filter, and the `Mask` field specifies, in hex, what the binary filter mask has been set to.

Definition at line 493 of file cfe_evs_events.h.

39.54.1.2 CFE_EVS_DELFILTER_EID

```
#define CFE_EVS_DELFILTER_EID 31
```

`'Delete Filter Command Received with AppName = %s, EventID = 0x%08x'`

Event Message `'Delete Filter Command Received with AppName = %s, EventID = 0x%08x'`

Type: DEBUG

Cause:

This event message is generated upon successful completion of the "Delete Filter" command.

The `AppName` field identifies the Application who is getting the filter removed, the `EventID` field identifies the Event Identifier, in hex, whose filter is being deleted.

Definition at line 507 of file cfe_evs_events.h.

39.54.1.3 CFE_EVS_DISAPPENTTYPE_EID

```
#define CFE_EVS_DISAPPENTTYPE_EID 24
```

```
'Disable App Event Type Command Received with AppName = %s, EventType Bit Mask
= 0x%02x'
```

Event Message `'Disable App Event Type Command Received with AppName = %s, Event↩ Type Bit Mask = 0x%02x'`

Type: DEBUG

Cause:

This event message is generated upon successful completion of the "Disable Application Event Types" command.

The `AppName` field identifies the Application whose Event Type Disable status has changed and the `Mask` field specifies (in hex) the Event Types that have been disabled. Mask bits are defined by CFE_EVS_DEBUG_BIT, CFE_EVS_↩ INFORMATION_BIT, CFE_EVS_ERROR_BIT and CFE_EVS_CRITICAL_BIT.

Definition at line 412 of file cfe_evs_events.h.

39.54.1.4 CFE_EVS_DISAPPEVT_EID

```
#define CFE_EVS_DISAPPEVT_EID 26
```

```
'Disable App Events Command Received with AppName = %s'
```

Event Message `'Disable App Events Command Received with AppName = %s'`

Type: DEBUG

Cause:

This event message is generated upon successful completion of the "Disable Application Events" command.

The `AppName` field identifies the Application whose Events have been Disabled.

Definition at line 438 of file cfe_evs_events.h.

39.54.1.5 CFE_EVS_DISEVTTYPE_EID

```
#define CFE_EVS_DISEVTTYPE_EID 21
```

`'Disable Event Type Command Received with Event Type Bit Mask = 0x%02x'`

Event Message `'Disable Event Type Command Received with Event Type Bit Mask = 0x%02x'`

Type: DEBUG

Cause:

This event message is issued upon successful processing of the "Disable Event Type" command.

The `Mask` field identifies the Event Types that are disabled. Mask bits are defined by CFE_EVS_DEBUG_BIT, CFE_EVS_INFORMATION_BIT, CFE_EVS_ERROR_BIT and CFE_EVS_CRITICAL_BIT.

Definition at line 366 of file cfe_evs_events.h.

39.54.1.6 CFE_EVS_DISPORT_EID

```
#define CFE_EVS_DISPORT_EID 19
```

`'Disable Ports Command Received with Port Bit Mask = 0x%02x'`

Event Message `'Disable Ports Command Received with Port Bit Mask = 0x%02x'`

Type: DEBUG

Cause:

This event message is issued upon successful processing of the "Disable Ports" command.

The `Mask` field identifies (in hex) the ports are to be disabled. Mask bits are defined by CFE_EVS_PORT1_BIT, CFE_EVS_PORT2_BIT, CFE_EVS_PORT3_BIT and CFE_EVS_PORT4_BIT.

Definition at line 336 of file cfe_evs_events.h.

39.54.1.7 CFE_EVS_ENAAPPEVT_EID

```
#define CFE_EVS_ENAAPPEVT_EID 25
```

`'Enable App Events Command Received with AppName = %s'`

Event Message `'Enable App Events Command Received with AppName = %s'`

Type: DEBUG

Cause:

This event message is generated upon successful completion of the "Enable Application Events" command.

The `AppName` field identifies the Application whose Events have been Enabled.

Definition at line 425 of file cfe_evs_events.h.

39.54.1.8 CFE_EVS_ENAAPPEVTTYPE_EID

```
#define CFE_EVS_ENAAPPEVTTYPE_EID 23
```

`'Enable App Event Type Command Received with AppName = %s, EventType Bit Mask = 0x%02x'`

Event Message `'Enable App Event Type Command Received with AppName = %s, EventType Bit Mask = 0x%02x'`

Type: DEBUG

Cause:

This event message is generated upon successful completion of the "Enable Application Event Types" command.

The `AppName` field identifies the Application whose Event Type Enable status has changed and the `Mask` field specifies (in hex) the Event Types that have been enabled. Mask bits are defined by CFE_EVS_DEBUG_BIT, CFE_EVS_INFORMATION_BIT, CFE_EVS_ERROR_BIT and CFE_EVS_CRITICAL_BIT.

Definition at line 396 of file cfe_evs_events.h.

39.54.1.9 CFE_EVS_ENAEVTTYPE_EID

```
#define CFE_EVS_ENAEVTTYPE_EID 20
```

`'Enable Event Type Command Received with Event Type Bit Mask = 0x%02x'`

Event Message `'Enable Event Type Command Received with Event Type Bit Mask = 0x%02x'`

Type: DEBUG

Cause:

This event message is issued upon successful processing of the "Enable Event Type" command.

The `Mask` field identifies the Event Types that are enabled. Mask bits are defined by CFE_EVS_DEBUG_BIT, CFE_EVS_INFORMATION_BIT, CFE_EVS_ERROR_BIT and CFE_EVS_CRITICAL_BIT.

Definition at line 351 of file cfe_evs_events.h.

39.54.1.10 CFE_EVS_ENAPORT_EID

```
#define CFE_EVS_ENAPORT_EID 18
```

`'Enable Ports Command Received with Port Bit Mask = 0x%02x'`

Event Message `'Enable Ports Command Received with Port Bit Mask = 0x%02x'`

Type: DEBUG

Cause:

This event message is issued upon successful processing of the "Enable Ports" command.

The `Mask` field identifies the ports that are enabled. Mask bits are defined by CFE_EVS_PORT1_BIT, CFE_EVS_PORT2_BIT, CFE_EVS_PORT3_BIT and CFE_EVS_PORT4_BIT.

Definition at line 322 of file cfe_evs_events.h.

39.54.1.11 CFE_EVS_ERR_APPNOREGS_EID

```
#define CFE_EVS_ERR_APPNOREGS_EID 7
```

`'%s not registered with EVS: CC = %lu'`

Event Message `'%s not registered with EVS: CC = %lu'`

Type: ERROR

Cause:

This event message is generated when the specified command identifies an Application that has not been registered with the cFE Event Services.

The `CC` field contains the Command Code whose processing resulted in the generation of the event message. Possible values are CFE_EVS_SET_FILTER_CC, CFE_EVS_ENABLE_APP_EVENT_TYPE_CC, CFE_EVS_DISABLE_APP↩_EVENT_TYPE_CC, CFE_EVS_ENABLE_APP_EVENTS_CC, CFE_EVS_DISABLE_APP_EVENTS_CC, CFE_EV↩S_RESET_APP_COUNTER_CC, CFE_EVS_RESET_FILTER_CC, CFE_EVS_RESET_ALL_FILTERS_CC, CFE_E↩VS_ADD_EVENT_FILTER_CC, or CFE_EVS_DELETE_EVENT_FILTER_CC.

Definition at line 158 of file cfe_evs_events.h.

39.54.1.12 CFE_EVS_ERR_CC_EID

```
#define CFE_EVS_ERR_CC_EID 15
```

`'Invalid command code - ID = 0x%08x, CC = %d'`

Event Message `'Invalid command code - ID = 0x%08x, CC = %d'`

Type: ERROR

Cause:

This event message is generated when a message with the CFE_EVS_CMD_MID message ID has arrived but whose Command Code is not one of the specified accepted command codes specified. This problem is most likely to occur when:

1. A Message ID meant for another Application became corrupted and was set equal to CFE_EVS_CMD_MID.
2. The Command Code field in the Message became corrupted.
3. The command database at the ground station has been corrupted.

The `ID` field in the event message specifies the Message ID (in hex) and the `CC` field specifies the Command Code (in decimal) found in the message.

Definition at line 281 of file cfe_evs_events.h.

39.54.1.13 CFE_EVS_ERR_CRDATFILE_EID

```
#define CFE_EVS_ERR_CRDATFILE_EID 13
```

```
'Write App Data Command Error:  OS_OpenCreate = 0x%08X, filename = %s'
```

Event Message `'Write App Data Command Error:  OS_OpenCreate = 0x%08X, filename = %s'`

Type: ERROR

Cause:

This event message is generated when a filesystem error occurred when attempting to create the file that is to hold the event registry data.

The message text identifies the registry filename and specifies the return value, in hex, from the system function call. The expected return value is a file handle, which in this case should be a relatively small positive number. Error codes are negative.

Definition at line 261 of file cfe_evs_events.h.

39.54.1.14 CFE_EVS_ERR_CRLOGFILE_EID

```
#define CFE_EVS_ERR_CRLOGFILE_EID 3
```

```
'Write Log File Command Error:  OS_OpenCreate = 0x%08X, filename = %s'
```

Event Message `'Write Log File Command Error:  OS_OpenCreate = 0x%08X, filename = %s'`

Type: ERROR

Cause:

This event message is generated when a filesystem error occurred when attempting to create the file that is to hold the event message log.

The message text identifies the event log filename and specifies the return value, in hex, from the system function call. The expected return value is a file handle, which in this case should be a relatively small positive number. Error codes are negative.

Definition at line 104 of file cfe_evs_events.h.

39.54.1.15 CFE_EVS_ERR_EVTIDNOREGS_EID

```
#define CFE_EVS_ERR_EVTIDNOREGS_EID 6
```

`'%s Event ID %d not registered for filtering:  CC = %lu'`

Event Message `'%s Event ID %d not registered for filtering:  CC = %lu'`

Type: ERROR

Cause:

This event message is generated when the specified command identifies an Application and Event ID combination that is not found in the Events Registry.

The `%s` string contains the command specified Application Name the `Event ID` field identifies the command specified EventID (in decimal) that was not found in the Events Registry. The `CC` field specifies the Command Code whose processing generated the event message. It can be equal to either CFE_EVS_SET_FILTER_CC, CFE_EVS_RESET_FILTER_CC, or CFE_EVS_DELETE_EVENT_FILTER_CC.

Definition at line 140 of file cfe_evs_events.h.

39.54.1.16 CFE_EVS_ERR_ILLAPPIDRANGE_EID

```
#define CFE_EVS_ERR_ILLAPPIDRANGE_EID 8
```

`'Illegal application ID %d retrieved for %s:  CC = %lu'`

Event Message `'Illegal application ID %d retrieved for %s:  CC = %lu'`

Type: ERROR

Cause:

This event message is generated when the specified command identifies an Application whose name is found in the Events Registry but does not appear to be properly registered with the cFE Executive Services.

The `CC` field contains the Command Code whose processing resulted in the generation of the event message. Possible values are CFE_EVS_SET_FILTER_CC, CFE_EVS_ENABLE_APP_EVENT_TYPE_CC, CFE_EVS_DISABLE_APP_EVENT_TYPE_CC, CFE_EVS_ENABLE_APP_EVENTS_CC, CFE_EVS_DISABLE_APP_EVENTS_CC, CFE_EVS_RESET_APP_COUNTER_CC, CFE_EVS_RESET_FILTER_CC, CFE_EVS_RESET_ALL_FILTERS_CC, CFE_EVS_ADD_EVENT_FILTER_CC, or CFE_EVS_DELETE_EVENT_FILTER_CC.

Definition at line 176 of file cfe_evs_events.h.

39.54.1.17 CFE_EVS_ERR_ILLEGALFMTMOD_EID

```
#define CFE_EVS_ERR_ILLEGALFMTMOD_EID 10
```

`'Set Event Format Mode Command:  Invalid Event Format Mode = 0x%02x'`

Event Message `'Set Event Format Mode Command:  Invalid Event Format Mode = 0x%02x'`

Type: ERROR

Cause:

This event message is generated when a "Set Event Format Mode" command message has arrived and the CFE_EVS_SetLogMode_Payload_t::LogMode field is equal to neither CFE_EVS_MsgFormat_SHORT or CFE_EVS_MsgFormat_LONG. These are the only allowed values for the mode field.

The `Mode` field in the event message identifies the Mode value (in hex) that was found in the message.

Definition at line 211 of file cfe_evs_events.h.

39.54.1.18 CFE_EVS_ERR_INVALID_BITMASK_EID

```
#define CFE_EVS_ERR_INVALID_BITMASK_EID 40
```

`'Bit Mask = 0x%X out of range:  CC = %lu'`

Event Message `'Bit Mask = 0x%X out of range:  CC = %lu'`

Type: ERROR

Cause:

This event message is generated when the bit mask passed in is equal to zero or greater than 0x0F, because a bit mask of zero does nothing, and a bitmask of greater than 0x0F is invalid.

Definition at line 593 of file cfe_evs_events.h.

39.54.1.19 CFE_EVS_ERR_LOGMODE_EID

```
#define CFE_EVS_ERR_LOGMODE_EID 39
```

```
'Set Log Mode Command Error:  Log Mode = %d'
```

Event Message `'Set Log Mode Command Error:  Log Mode = %d`

Type: ERROR

Cause:

This event message is generated when a "Set Log Mode" command is received that specifies an invalid Log Mode command argument.

The event text identifies the invalid Log Mode command argument. Valid Log Mode command arguments are: CFE_EVS_LOG_OVERWRITE or CFE_EVS_LOG_DISCARD.

Definition at line 581 of file cfe_evs_events.h.

39.54.1.20 CFE_EVS_ERR_MAXREGSFILTER_EID

```
#define CFE_EVS_ERR_MAXREGSFILTER_EID 11
```

```
'Add Filter Command:  number of registered filters has reached max = %d'
```

Event Message `'Add Filter Command:  number of registered filters has reached max = %d'`

Type: ERROR

Cause:

This event message is generated upon receipt of an "Add Filter" command and the specified Application has already reached the maximum number of filters allowed (CFE_PLATFORM_EVS_MAX_EVENT_FILTERS).

The `max` field in the event message identifies the maximum number of event filters allowed per Application. This value should be equal to the configuration parameter CFE_PLATFORM_EVS_MAX_EVENT_FILTERS.

Definition at line 228 of file cfe_evs_events.h.

39.54.1.21 CFE_EVS_ERR_MSGID_EID

```
#define CFE_EVS_ERR_MSGID_EID 5
```

`'Invalid command packet, Message ID = 0x%08X'`

Event Message `'Invalid command packet, Message ID = 0x%08X'`

Type: ERROR

Cause:

This event message is generated when a message has arrived on the cFE Event Services Application's Message Pipe that has a Message ID that is neither CFE_EVS_CMD_MID or CFE_EVS_SEND_HK_MID. Most likely, the cFE Software Bus routing table has become corrupt and is sending messages targeted for other Applications to the cFE Event Services Application.

The `ID` field in the event message identifies the message ID (in hex) that was found in the message.

Definition at line 123 of file cfe_evs_events.h.

39.54.1.22 CFE_EVS_ERR_NOAPPIDFOUND_EID

```
#define CFE_EVS_ERR_NOAPPIDFOUND_EID 9
```

`'Unable to retrieve application ID for %s:  CC = %lu'`

Event Message `'Unable to retrieve application ID for %s:  CC = %lu'`

Type: ERROR

Cause:

This event message is generated when the specified command contains an Application name that is apparently found in the Events Registry but does not appear to be registered with the cFE Executive Services.

The `CC` field contains the Command Code whose processing resulted in the generation of the event message. Possible values are CFE_EVS_SET_FILTER_CC, CFE_EVS_ENABLE_APP_EVENT_TYPE_CC, CFE_EVS_DISABLE_APP↩_EVENT_TYPE_CC, CFE_EVS_ENABLE_APP_EVENTS_CC, CFE_EVS_DISABLE_APP_EVENTS_CC, CFE_EV↩S_RESET_APP_COUNTER_CC, CFE_EVS_RESET_FILTER_CC, CFE_EVS_RESET_ALL_FILTERS_CC, CFE_E↩VS_ADD_EVENT_FILTER_CC, or CFE_EVS_DELETE_EVENT_FILTER_CC.

Definition at line 194 of file cfe_evs_events.h.

39.54.1.23 CFE_EVS_ERR_UNREGISTERED_EVS_APP

```
#define CFE_EVS_ERR_UNREGISTERED_EVS_APP 41
```

```
'App %s not registered with Event Services.  Unable to send event'
```

Event Message `'App %s not registered with Event Services.  Unable to send event'`

Type: ERROR

Cause:

This event message is generated when an event message has been requested to be sent by an Application that has not registered itself with cFE Event Services.

Definition at line 605 of file cfe_evs_events.h.

39.54.1.24 CFE_EVS_ERR_WRDATFILE_EID

```
#define CFE_EVS_ERR_WRDATFILE_EID 12
```

```
'Write App Data Command Error:  OS_write = 0x%08X, filename = %s'
```

Event Message `'Write App Data Command Error:  OS_write = 0x%08X, filename = %s'`

Type: ERROR

Cause:

This event message is generated when a filesystem error occurred while writing the contents of the event registry to a file.

The message text identifies the registry filename and specifies the return value, in hex, from the system function call. The expected return value is the number of bytes written, which in this case should be equal to the size of a CFE_EV↩S_AppDataFile_t structure. Error codes are negative.

Definition at line 245 of file cfe_evs_events.h.

39.54.1.25 CFE_EVS_ERR_WRLOGFILE_EID

```
#define CFE_EVS_ERR_WRLOGFILE_EID 2
```

```
'Write Log File Command Error:  OS_write = 0x%08X, filename = %s'
```

Event Message `'Write Log File Command Error:  OS_write = 0x%08X, filename = %s'`

Type: ERROR

Cause:

This event message is generated when a filesystem error occurred while writing the contents of the event message log to a file.

The message text identifies the event log filename and specifies the return value, in hex, from the system function call. The expected return value is the number of bytes written, which in this case should be equal to the size of a CFE_EVS_LongEventTlm_t structure. Error codes are negative.

Definition at line 88 of file cfe_evs_events.h.

39.54.1.26 CFE_EVS_EVT_FILTERED_EID

```
#define CFE_EVS_EVT_FILTERED_EID 37
```

```
'Add Filter Command:AppName = %s, EventID = 0x%08x is already registered for
filtering'
```

Event Message `'Add Filter Command:AppName = %s, EventID = 0x%08x is already registered for filtering'`

Type: ERROR

Cause:

This event message is generated when an "Add Filter" command was received specifying an Event ID that has already had a filter added.

The `AppName` field identifies the Application whose filter was to be added and the `EventID` field identifies, in hex, the Event ID that the command was trying to add a filter for.

Definition at line 552 of file cfe_evs_events.h.

39.54.1.27 CFE_EVS_FILTER_MAX_EID

```
#define CFE_EVS_FILTER_MAX_EID 42
```

`'Max filter count reached, AppName = %s, EventID = 0x%08x:  Filter locked until reset'`

Event Message `'Max filter count reached, AppName = %s, EventID = 0x%08x:  Filter locked until reset'`

Type: INFORMATIONAL

Cause:

This event message is generated when the filtering count for a specific App and Event ID reaches CFE_EVS_MAX_F↩ ILTER_COUNT. The filtered event will no longer be received until the reset counter is reset via a "Reset an Event Filter for an Application" or a "Reset All Event Filters for an Application"

The `AppName` field identifies the Application and the `EventID` field identifies, in hex, the Event ID for the filter whose maxium was reached.

Definition at line 622 of file cfe_evs_events.h.

39.54.1.28 CFE_EVS_LEN_ERR_EID

```
#define CFE_EVS_LEN_ERR_EID 43
```

`'Invalid cmd length:  ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'`

Event Message `'Invalid cmd length:  ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'`

Type: ERROR

Cause:

This event message is generated when a message with the CFE_EVS_CMD_MID message ID has arrived but whose packet length does not match the expected length for the specified command code.

The `ID` field in the event message specifies the Message ID (in hex), the `CC` field specifies the Command Code (in decimal), the `Exp` Len field specified the Expected Length (in decimal), and `Len` specifies the message Length (in decimal) found in the message.

Definition at line 640 of file cfe_evs_events.h.

39.54.1.29 CFE_EVS_LOGMODE_EID

```
#define CFE_EVS_LOGMODE_EID 38
```

```
'Set Log Mode Command Error:  Log Mode = %d'
```

Event Message `'Set Log Mode Command Error:  Log Mode = %d'`

Type: DEBUG

Cause:

This event message is generated when a "Set Log Mode" command is completed successfully.

The event text identifies the Log Mode command argument. Valid Log Mode command arguments are: CFE_EVS_LOG_OVERWRITE or CFE_EVS_LOG_DISCARD.

Definition at line 566 of file cfe_evs_events.h.

39.54.1.30 CFE_EVS_MAX_EID

```
#define CFE_EVS_MAX_EID 43
```

Definition at line 45 of file cfe_evs_events.h.

39.54.1.31 CFE_EVS_NOOP_EID

```
#define CFE_EVS_NOOP_EID 0 /* Noop event identifier */
```

```
'No-op command'
```

Event Message `'No-op command'`

Type: INFORMATION

Cause:

This event message is always automatically issued in response to a cFE Event Services NO-OP command

Definition at line 59 of file cfe_evs_events.h.

39.54.1.32 CFE_EVS_RSTALLFILTER_EID

```
#define CFE_EVS_RSTALLFILTER_EID 29
```

`'Reset All Filters Command Received with AppName = %s'`

Event Message `'Reset All Filters Command Received with AppName = %s'`

Type: DEBUG

Cause:

This event message is generated upon successful completion of the "Reset Application Event Message Filters" command.

The `AppName` field identifies the Application whose entire set of Event Filters has been reset.

Definition at line 478 of file cfe_evs_events.h.

39.54.1.33 CFE_EVS_RSTCNT_EID

```
#define CFE_EVS_RSTCNT_EID 16
```

`'Reset Counters Command Received'`

Event Message `'Reset Counters Command Received'`

Type: DEBUG

Cause:

This event message is always automatically issued in response to a cFE Event Services Reset Counters command

Definition at line 293 of file cfe_evs_events.h.

39.54.1.34 CFE_EVS_RSTEVTCNT_EID

```
#define CFE_EVS_RSTEVTCNT_EID 27
```

`'Reset Event Counter Command Received with AppName = %s'`

Event Message `'Reset Event Counter Command Received with AppName = %s'`

Type: DEBUG

Cause:

This event message is generated upon successful completion of the "Reset Application Event Counter" command.

The `AppName` field identifies the Application whose Event Counter has been reset.

Definition at line 451 of file cfe_evs_events.h.

39.54.1.35 CFE_EVS_RSTFILTER_EID

```
#define CFE_EVS_RSTFILTER_EID 28
```

`'Reset Filter Command Received with AppName = %s, EventID = 0x%08x'`

Event Message `'Reset Filter Command Received with AppName = %s, EventID = 0x%08x'`

Type: DEBUG

Cause:

This event message is generated upon successful completion of the "Reset Application Event Message Filter" command.

The `AppName` field identifies the Application whose Event Message Filter has been reset and the `EventID` field identifies the specific event message whose filter has been reset.

Definition at line 465 of file cfe_evs_events.h.

39.54.1.36 CFE_EVS_SETEVTFMTMOD_EID

```
#define CFE_EVS_SETEVTFMTMOD_EID 22
```

`'Set Event Format Mode Command Received with Mode = 0x%02x'`

Event Message `'Set Event Format Mode Command Received with Mode = 0x%02x'`

Type: DEBUG

Cause:

This event message is generated upon successful completion of the "Set Event Format Mode" command.

The `Mode` field contains the newly chosen Event Format Mode (specified in hex). Acceptable values for this parameter are: CFE_EVS_MsgFormat_SHORT or CFE_EVS_MsgFormat_LONG

Definition at line 380 of file cfe_evs_events.h.

39.54.1.37 CFE_EVS_SETFILTERMSK_EID

```
#define CFE_EVS_SETFILTERMSK_EID 17
```

`'Set Filter Mask Command Received with AppName=%s, EventID=0x%08x, Mask=0x%04x'`

Event Message `'Set Filter Mask Command Received with AppName=%s, EventID=0x%08x, Mask=0x%04x'`

Type: DEBUG

Cause:

This event message is issued upon successful processing of a Set Filter Mask command.

The `AppName` field identifies the Application whose Filter Mask has been changed. The `EventID` field identifies the Event whose Filter Mask has been changed. The `Mask` field identifies the new Mask value associated with the specified event.

Definition at line 308 of file cfe_evs_events.h.

39.54.1.38 CFE_EVS_STARTUP_EID

```
#define CFE_EVS_STARTUP_EID 1
```

'cFE EVS Initialized'

Event Message 'cFE EVS Initialized'

Type: INFORMATION

Cause:

This event message is always automatically issued when the Event Services Task completes its Initialization.

Definition at line 71 of file cfe_evs_events.h.

39.54.1.39 CFE_EVS_WRDAT_EID

```
#define CFE_EVS_WRDAT_EID 32
```

'Write App Data Command: %d application data entries written to %s'

Event Message 'Write App Data Command: %d application data entries written to %s'

Type: DEBUG

Cause:

This event message is generated upon successful completion of the "Write Event Services Application Information to File" command.

The message text identifies the event log filename and specifies the number, in decimal, of events written to the file.

Definition at line 522 of file cfe_evs_events.h.

39.54.1.40 CFE_EVS_WRLOG_EID

```
#define CFE_EVS_WRLOG_EID 33
```

`'Write Log File Command:  %d event log entries written to %s'`

Event Message `'Write Log File Command:  %d event log entries written to %s'`

Type: DEBUG

Cause:

This event message is generated upon successful completion of the "Write Event Log to File" command.

The message text identifies the event log filename and specifies the number, in decimal, of events written to the file.

Definition at line 537 of file cfe_evs_events.h.

39.55 cfe/modules/evs/fsw/inc/cfe_evs_msg.h File Reference

```
#include "common_types.h"
#include "cfe_msg_hdr.h"
#include "cfe_evs_extern_typedefs.h"
#include "cfe_es_extern_typedefs.h"
```

Data Structures

- struct CFE_EVS_NoArgsCmd

  Command with no additional arguments.
- struct CFE_EVS_LogFileCmd_Payload

  Write Event Log to File Command Payload.
- struct CFE_EVS_WriteLogDataFileCmd

  Write Event Log to File Command.
- struct CFE_EVS_AppDataCmd_Payload

  Write Event Services Application Information to File Command Payload.
- struct CFE_EVS_WriteAppDataFileCmd

  Write Event Services Application Information to File Command.
- struct CFE_EVS_SetLogMode_Payload

  Set Log Mode Command Payload.
- struct CFE_EVS_SetLogModeCmd

  Set Log Mode Command.
- struct CFE_EVS_SetEventFormatCode_Payload

Set Event Format Mode Command Payload.

- struct CFE_EVS_SetEventFormatModeCmd

  Set Event Format Mode Command.
- struct CFE_EVS_BitMaskCmd_Payload

  Generic Bitmask Command Payload.
- struct CFE_EVS_BitMaskCmd

  Generic Bitmask Command.
- struct CFE_EVS_AppNameCmd_Payload

  Generic App Name Command Payload.
- struct CFE_EVS_AppNameCmd

  Generic App Name Command.
- struct CFE_EVS_AppNameEventIDCmd_Payload

  Generic App Name and Event ID Command Payload.
- struct CFE_EVS_AppNameEventIDCmd

  Generic App Name and Event ID Command.
- struct CFE_EVS_AppNameBitMaskCmd_Payload

  Generic App Name and Bitmask Command Payload.
- struct CFE_EVS_AppNameBitMaskCmd

  Generic App Name and Bitmask Command.
- struct CFE_EVS_AppNameEventIDMaskCmd_Payload

  Generic App Name, Event ID, Mask Command Payload.
- struct CFE_EVS_AppNameEventIDMaskCmd

  Generic App Name, Event ID, Mask Command.
- struct CFE_EVS_AppTlmData
- struct CFE_EVS_HousekeepingTlm_Payload
- struct CFE_EVS_HousekeepingTlm
- struct CFE_EVS_PacketID
- struct CFE_EVS_LongEventTlm_Payload
- struct CFE_EVS_ShortEventTlm_Payload
- struct CFE_EVS_LongEventTlm
- struct CFE_EVS_ShortEventTlm

Macros

- #define CFE_EVS_DEBUG_BIT 0x0001
- #define CFE_EVS_INFORMATION_BIT 0x0002
- #define CFE_EVS_ERROR_BIT 0x0004
- #define CFE_EVS_CRITICAL_BIT 0x0008
- #define CFE_EVS_PORT1_BIT 0x0001
- #define CFE_EVS_PORT2_BIT 0x0002
- #define CFE_EVS_PORT3_BIT 0x0004
- #define CFE_EVS_PORT4_BIT 0x0008
- #define CFE_EVS_LOG_OVERWRITE 0
- #define CFE_EVS_LOG_DISCARD 1

Event Services Command Codes

- #define CFE_EVS_NOOP_CC 0

- #define CFE_EVS_RESET_COUNTERS_CC 1
- #define CFE_EVS_ENABLE_EVENT_TYPE_CC 2
- #define CFE_EVS_DISABLE_EVENT_TYPE_CC 3
- #define CFE_EVS_SET_EVENT_FORMAT_MODE_CC 4
- #define CFE_EVS_ENABLE_APP_EVENT_TYPE_CC 5
- #define CFE_EVS_DISABLE_APP_EVENT_TYPE_CC 6
- #define CFE_EVS_ENABLE_APP_EVENTS_CC 7
- #define CFE_EVS_DISABLE_APP_EVENTS_CC 8
- #define CFE_EVS_RESET_APP_COUNTER_CC 9
- #define CFE_EVS_SET_FILTER_CC 10
- #define CFE_EVS_ENABLE_PORTS_CC 11
- #define CFE_EVS_DISABLE_PORTS_CC 12
- #define CFE_EVS_RESET_FILTER_CC 13
- #define CFE_EVS_RESET_ALL_FILTERS_CC 14
- #define CFE_EVS_ADD_EVENT_FILTER_CC 15
- #define CFE_EVS_DELETE_EVENT_FILTER_CC 16
- #define CFE_EVS_WRITE_APP_DATA_FILE_CC 17
- #define CFE_EVS_WRITE_LOG_DATA_FILE_CC 18
- #define CFE_EVS_SET_LOG_MODE_CC 19
- #define CFE_EVS_CLEAR_LOG_CC 20

Typedefs

- typedef struct CFE_EVS_NoArgsCmd CFE_EVS_NoArgsCmd_t

  Command with no additional arguments.
- typedef CFE_EVS_NoArgsCmd_t CFE_EVS_NoopCmd_t
- typedef CFE_EVS_NoArgsCmd_t CFE_EVS_ResetCountersCmd_t
- typedef CFE_EVS_NoArgsCmd_t CFE_EVS_ClearLogCmd_t
- typedef struct CFE_EVS_LogFileCmd_Payload CFE_EVS_LogFileCmd_Payload_t

  Write Event Log to File Command Payload.
- typedef struct CFE_EVS_WriteLogDataFileCmd CFE_EVS_WriteLogDataFileCmd_t

  Write Event Log to File Command.
- typedef struct CFE_EVS_AppDataCmd_Payload CFE_EVS_AppDataCmd_Payload_t

  Write Event Services Application Information to File Command Payload.
- typedef struct CFE_EVS_WriteAppDataFileCmd CFE_EVS_WriteAppDataFileCmd_t

  Write Event Services Application Information to File Command.
- typedef struct CFE_EVS_SetLogMode_Payload CFE_EVS_SetLogMode_Payload_t

  Set Log Mode Command Payload.
- typedef struct CFE_EVS_SetLogModeCmd CFE_EVS_SetLogModeCmd_t

  Set Log Mode Command.
- typedef struct CFE_EVS_SetEventFormatCode_Payload CFE_EVS_SetEventFormatMode_Payload_t

  Set Event Format Mode Command Payload.
- typedef struct CFE_EVS_SetEventFormatModeCmd CFE_EVS_SetEventFormatModeCmd_t

  Set Event Format Mode Command.
- typedef struct CFE_EVS_BitMaskCmd_Payload CFE_EVS_BitMaskCmd_Payload_t

  Generic Bitmask Command Payload.
- typedef struct CFE_EVS_BitMaskCmd CFE_EVS_BitMaskCmd_t

  Generic Bitmask Command.
- typedef CFE_EVS_BitMaskCmd_t CFE_EVS_EnablePortsCmd_t
- typedef CFE_EVS_BitMaskCmd_t CFE_EVS_DisablePortsCmd_t
- typedef CFE_EVS_BitMaskCmd_t CFE_EVS_EnableEventTypeCmd_t

- typedef CFE_EVS_BitMaskCmd_t CFE_EVS_DisableEventTypeCmd_t
- typedef struct CFE_EVS_AppNameCmd_Payload CFE_EVS_AppNameCmd_Payload_t

  Generic App Name Command Payload.
- typedef struct CFE_EVS_AppNameCmd CFE_EVS_AppNameCmd_t

  Generic App Name Command.
- typedef CFE_EVS_AppNameCmd_t CFE_EVS_EnableAppEventsCmd_t
- typedef CFE_EVS_AppNameCmd_t CFE_EVS_DisableAppEventsCmd_t
- typedef CFE_EVS_AppNameCmd_t CFE_EVS_ResetAppCounterCmd_t
- typedef CFE_EVS_AppNameCmd_t CFE_EVS_ResetAllFiltersCmd_t
- typedef struct CFE_EVS_AppNameEventIDCmd_Payload CFE_EVS_AppNameEventIDCmd_Payload_t

  Generic App Name and Event ID Command Payload.
- typedef struct CFE_EVS_AppNameEventIDCmd CFE_EVS_AppNameEventIDCmd_t

  Generic App Name and Event ID Command.
- typedef CFE_EVS_AppNameEventIDCmd_t CFE_EVS_ResetFilterCmd_t
- typedef CFE_EVS_AppNameEventIDCmd_t CFE_EVS_DeleteEventFilterCmd_t
- typedef struct CFE_EVS_AppNameBitMaskCmd_Payload CFE_EVS_AppNameBitMaskCmd_Payload_t

  Generic App Name and Bitmask Command Payload.
- typedef struct CFE_EVS_AppNameBitMaskCmd CFE_EVS_AppNameBitMaskCmd_t

  Generic App Name and Bitmask Command.
- typedef CFE_EVS_AppNameBitMaskCmd_t CFE_EVS_EnableAppEventTypeCmd_t
- typedef CFE_EVS_AppNameBitMaskCmd_t CFE_EVS_DisableAppEventTypeCmd_t
- typedef struct CFE_EVS_AppNameEventIDMaskCmd_Payload CFE_EVS_AppNameEventIDMaskCmd_↩ Payload_t

  Generic App Name, Event ID, Mask Command Payload.
- typedef struct CFE_EVS_AppNameEventIDMaskCmd CFE_EVS_AppNameEventIDMaskCmd_t

  Generic App Name, Event ID, Mask Command.
- typedef CFE_EVS_AppNameEventIDMaskCmd_t CFE_EVS_AddEventFilterCmd_t
- typedef CFE_EVS_AppNameEventIDMaskCmd_t CFE_EVS_SetFilterCmd_t
- typedef struct CFE_EVS_AppTlmData CFE_EVS_AppTlmData_t
- typedef struct CFE_EVS_HousekeepingTlm_Payload CFE_EVS_HousekeepingTlm_Payload_t
- typedef struct CFE_EVS_HousekeepingTlm CFE_EVS_HousekeepingTlm_t
- typedef struct CFE_EVS_PacketID CFE_EVS_PacketID_t
- typedef struct CFE_EVS_LongEventTlm_Payload CFE_EVS_LongEventTlm_Payload_t
- typedef struct CFE_EVS_ShortEventTlm_Payload CFE_EVS_ShortEventTlm_Payload_t
- typedef struct CFE_EVS_LongEventTlm CFE_EVS_LongEventTlm_t
- typedef struct CFE_EVS_ShortEventTlm CFE_EVS_ShortEventTlm_t

39.55.1 Macro Definition Documentation

39.55.1.1 CFE_EVS_ADD_EVENT_FILTER_CC

```
#define CFE_EVS_ADD_EVENT_FILTER_CC 15
```

Name Add Application Event Filter

Description

This command adds the given filter for the given application identifier and event identifier. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) $sc_$cpu_EVS_AddEvtFltr

Command Structure

CFE_EVS_AddEventFilterCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **$sc_$cpu_EVS_CMDPC** - command execution counter will increment
- The generation of CFE_EVS_ADDFILTER_EID debug event message

Error Conditions

This command may fail for the following reason(s):

- Invalid SB message (command) length
- Application selected is not registered to receive Event Service
- Application ID is out of range
  Evidence of failure may be found in the following telemetry:
  - **$sc_$cpu_EVS_CMDEC** - command error counter will increment
  - An Error specific event message

Criticality

None.

See also

CFE_EVS_SET_FILTER_CC, CFE_EVS_RESET_FILTER_CC, CFE_EVS_RESET_ALL_FILTERS_CC, CFE↩_EVS_DELETE_EVENT_FILTER_CC

Definition at line 720 of file cfe_evs_msg.h.

39.55.1.2 CFE_EVS_CLEAR_LOG_CC

```
#define CFE_EVS_CLEAR_LOG_CC 20
```

Name Clear Event Log

Description

This command clears the contents of the local event log.

Command Mnemonic(s) $sc_$cpu_EVS_ClrLog

Command Structure

CFE_EVS_ClearLogCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment

Error Conditions

This command may fail for the following reason(s):

- Invalid SB message (command) length
  Evidence of failure may be found in the following telemetry:
  - **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
  - An Error specific event message

Criticality

Clearing the local event log is not particularly hazardous, as the result may be making available space to record valuable event data. However, inappropriately clearing the local event log could result in a loss of critical information. Note: the event log is a back-up log to the on-board recorder.

See also

CFE_EVS_WRITE_LOG_DATA_FILE_CC, CFE_EVS_SET_LOG_MODE_CC

Definition at line 897 of file cfe_evs_msg.h.

39.55.1.3 CFE_EVS_CRITICAL_BIT

```
#define CFE_EVS_CRITICAL_BIT 0x0008
```

Definition at line 904 of file cfe_evs_msg.h.

39.55.1.4 CFE_EVS_DEBUG_BIT

```
#define CFE_EVS_DEBUG_BIT 0x0001
```

Definition at line 901 of file cfe_evs_msg.h.

39.55.1.5 CFE_EVS_DELETE_EVENT_FILTER_CC

```
#define CFE_EVS_DELETE_EVENT_FILTER_CC 16
```

Name Delete Application Event Filter

Description

This command removes the given filter for the given application identifier and event identifier. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) $sc_$cpu_EVS_DelEvtFltr

Command Structure

CFE_EVS_DeleteEventFilterCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_DELFILTER_EID debug event message

Error Conditions

This command may fail for the following reason(s):

- Invalid SB message (command) length
- Application selected is not registered to receive Event Service
- Application ID is out of range
  Evidence of failure may be found in the following telemetry:
  - **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
  - An Error specific event message

Criticality

None.

See also

CFE_EVS_SET_FILTER_CC, CFE_EVS_RESET_FILTER_CC, CFE_EVS_RESET_ALL_FILTERS_CC, CFE↩_EVS_ADD_EVENT_FILTER_CC

Definition at line 756 of file cfe_evs_msg.h.

39.55.1.6 CFE_EVS_DISABLE_APP_EVENT_TYPE_CC

```
#define CFE_EVS_DISABLE_APP_EVENT_TYPE_CC 6
```

Name Disable Application Event Type

Description

This command disables the command specified event type for the command specified application, preventing the application from sending event messages of the command specified event type through Event Service. An Event Type is defined to be a classification of an Event Message such as debug, informational, critical, and error. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) $sc_$cpu_EVS_DisAppEvtType, $sc_$cpu_EVS_DisAppEvtTypeMask

Command Structure

CFE_EVS_DisableAppEventTypeCmd_t The following bit positions apply to structure member named 'BitMask'. Bit 0 - Debug Bit 1 - Informational Bit 2 - Error Bit 3 - Critical A one in a bit position means the event type will be disabled (or filtered) for the specified application. A zero in a bit position means the filtering state is unchanged for the specified application.

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_DISAPPENTTYPE_EID debug event message
- The clearing of the Event Type Active Flag in The Event Type Active Flag in EVS App Data File

Error Conditions

This command may fail for the following reason(s):

- Invalid Event Type Selection
  Evidence of failure may be found in the following telemetry:
  - **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
  - An Error specific event message

Criticality

Disabling an application's event type is not particularly hazardous, as the result may be shutting off unnecessary event messages and possible event flooding of the system. However, inappropriately disabling an application's event type could result in a loss of critical information and missed behavior for the ground system.

See also

CFE_EVS_ENABLE_EVENT_TYPE_CC, CFE_EVS_DISABLE_EVENT_TYPE_CC, CFE_EVS_ENABLE_AP↩P_EVENT_TYPE_CC, CFE_EVS_ENABLE_APP_EVENTS_CC, CFE_EVS_DISABLE_APP_EVENTS_CC

Definition at line 368 of file cfe_evs_msg.h.

39.55.1.7 CFE_EVS_DISABLE_APP_EVENTS_CC

```
#define CFE_EVS_DISABLE_APP_EVENTS_CC 8
```

Name Disable Event Services for an Application

Description

This command disables the command specified application from sending events through Event Service. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) $sc_$cpu_EVS_DisAppEvGen

Command Structure

CFE_EVS_DisableAppEventsCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_DISAPPEVT_EID debug event message

Error Conditions

This command may fail for the following reason(s):

- Invalid SB message (command) length
- Application selected is not registered to receive Event Service
- Application ID is out of range
  Evidence of failure may be found in the following telemetry:
  - **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
  - An Error specific event message

Criticality

Disabling an application's events is not particularly hazardous, as the result may be shutting off unnecessary event messages and possible event flooding of the system. However, inappropriately disabling an application's events could result in a loss of critical information and missed behavior for the ground system.

See also

CFE_EVS_ENABLE_EVENT_TYPE_CC, CFE_EVS_DISABLE_EVENT_TYPE_CC, CFE_EVS_ENABLE_AP↩P_EVENT_TYPE_CC, CFE_EVS_DISABLE_APP_EVENT_TYPE_CC, CFE_EVS_ENABLE_APP_EVENTS_CC

Definition at line 450 of file cfe_evs_msg.h.

39.55.1.8 CFE_EVS_DISABLE_EVENT_TYPE_CC

```
#define CFE_EVS_DISABLE_EVENT_TYPE_CC 3
```

Name Disable Event Type

Description

This command disables the command specified Event Type preventing event messages of this type to be sent through Event Service. An Event Type is defined to be a classification of an Event Message such as debug, informational, error and critical. This command is a global disable of a particular event type, it applies to all applications.

Command Mnemonic(s) $sc_$cpu_EVS_DisEventType, $sc_$cpu_EVS_DisEventTypeMask

Command Structure

CFE_EVS_DisableEventTypeCmd_t The following bit positions apply to structure member named 'BitMask'. Bit 0 - Debug Bit 1 - Informational Bit 2 - Error Bit 3 - Critical A one in a bit position means the event type will be disabled (or filtered). A zero in a bit position means the filtering state is unchanged.

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_DISEVTTYPE_EID debug message

Error Conditions

This command may fail for the following reason(s):

- Invalid Event Type selection

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
- An Error specific event message

Criticality

Disabling an event type is not particularly hazardous, as the result may be shutting off unnecessary event messages and possible event flooding of the system. However, inappropriately disabling an event type could result in a loss of critical information and missed behavior for the ground system.

See also

CFE_EVS_ENABLE_EVENT_TYPE_CC, CFE_EVS_ENABLE_APP_EVENT_TYPE_CC, CFE_EVS_DISABL↩E_APP_EVENT_TYPE_CC, CFE_EVS_ENABLE_APP_EVENTS_CC, CFE_EVS_DISABLE_APP_EVENTS_CC

Definition at line 214 of file cfe_evs_msg.h.

39.55.1.9 CFE_EVS_DISABLE_PORTS_CC

```
#define CFE_EVS_DISABLE_PORTS_CC 12
```

Name Disable Event Services Output Ports

Description

This command disables the specified port from outputting event messages.

Command Mnemonic(s) $sc_$cpu_EVS_DisPort, $sc_$cpu_EVS_DisPortMask

Command Structure

CFE_EVS_DisablePortsCmd_t The following bit positions apply to structure member named 'BitMask'. Bit 0 - Port 1 Bit 1 - Port 2 Bit 2 - Port 3 Bit 3 - Port 4 A one in a bit position means the port will be disabled. A zero in a bit position means the port state is unchanged.

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_DISPORT_EID debug event message

Error Conditions

This command may fail for the following reason(s):

- Invalid SB message (command) length
- Invalid PORT selection

  Evidence of failure may be found in the following telemetry:

  - **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
  - An Error specific event message

Criticality

None.

See also

CFE_EVS_ENABLE_PORTS_CC

Definition at line 611 of file cfe_evs_msg.h.

39.55.1.10 CFE_EVS_ENABLE_APP_EVENT_TYPE_CC

```
#define CFE_EVS_ENABLE_APP_EVENT_TYPE_CC 5
```

Name Enable Application Event Type

Description

This command enables the command specified event type for the command specified application, allowing the application to send event messages of the command specified event type through Event Service. An Event Type is defined to be a classification of an Event Message such as debug, informational, critical, and error. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) $sc_$cpu_EVS_EnaAppEvtType, $sc_$cpu_EVS_EnaAppEvtTypeMask

Command Structure

CFE_EVS_EnableAppEventTypeCmd_t The following bit positions apply to structure member named 'BitMask'. Bit 0 - Debug Bit 1 - Informational Bit 2 - Error Bit 3 - Critical A one in a bit position means the event type will be enabled (or unfiltered) for the specified application. A zero in a bit position means the filtering state is unchanged for the specified application.

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_ENAAPPEVTTYPE_EID debug event message

Error Conditions

This command may fail for the following reason(s):

- Invalid Event Type Selection
  Evidence of failure may be found in the following telemetry:
  - **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
  - An Error specific event message

Criticality

Enabling an application event type is not particularly hazardous, as the result may be turning on necessary event messages and communication to the ground system. However, inappropriately enabling an application's event type could result in flooding of the ground system.

See also

CFE_EVS_ENABLE_EVENT_TYPE_CC, CFE_EVS_DISABLE_EVENT_TYPE_CC, CFE_EVS_DISABLE_AP↩P_EVENT_TYPE_CC, CFE_EVS_ENABLE_APP_EVENTS_CC, CFE_EVS_DISABLE_APP_EVENTS_CC

Definition at line 316 of file cfe_evs_msg.h.

39.55.1.11 CFE_EVS_ENABLE_APP_EVENTS_CC

```
#define CFE_EVS_ENABLE_APP_EVENTS_CC 7
```

Name Enable Event Services for an Application

Description

This command enables the command specified application to send events through the Event Service. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) $sc_$cpu_EVS_EnaAppEvGen

Command Structure

CFE_EVS_EnableAppEventsCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_ENAAPPEVT_EID debug event message
- The setting of the Active Flag in The Active Flag in EVS App Data File

Error Conditions

This command may fail for the following reason(s):

- Invalid SB message (command) length
- Application selected is not registered to receive Event Service
- Application ID is out of range
  Evidence of failure may be found in the following telemetry:
  - **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
  - An Error specific event message

Criticality

Enabling an application events is not particularly hazardous, as the result may be turning on necessary event messages and communication to the ground system. However, inappropriately enabling an application's events could result in flooding of the ground system.

See also

CFE_EVS_ENABLE_EVENT_TYPE_CC, CFE_EVS_DISABLE_EVENT_TYPE_CC, CFE_EVS_ENABLE_AP↩P_EVENT_TYPE_CC, CFE_EVS_DISABLE_APP_EVENT_TYPE_CC, CFE_EVS_DISABLE_APP_EVENTS_↩CC

Definition at line 409 of file cfe_evs_msg.h.

39.55.1.12 CFE_EVS_ENABLE_EVENT_TYPE_CC

```
#define CFE_EVS_ENABLE_EVENT_TYPE_CC 2
```

Name Enable Event Type

Description

This command enables the command specified Event Type allowing event messages of this type to be sent through Event Service. An Event Type is defined to be a classification of an Event Message such as debug, informational, error and critical. This command is a global enable of a particular event type, it applies to all applications.

Command Mnemonic(s) $sc_$cpu_EVS_EnaEventType, $sc_$cpu_EVS_EnaEventTypeMask

Command Structure

CFE_EVS_EnableEventTypeCmd_t The following bit positions apply to structure member named 'BitMask'. Bit 0 - Debug Bit 1 - Informational Bit 2 - Error Bit 3 - Critical A one in a bit position means the event type will be enabled (or unfiltered). A zero in a bit position means the filtering state is unchanged.

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_ENAEVTTYPE_EID debug message

Error Conditions

This command may fail for the following reason(s):

Invalid Event Type selection

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
- An Error specific event message

Criticality

Enabling an event type is not particularly hazardous, as the result may be turning on necessary event messages and communication to the ground system. However, inappropriately enabling an event type could result in flooding of the system.

See also

CFE_EVS_DISABLE_EVENT_TYPE_CC, CFE_EVS_ENABLE_APP_EVENT_TYPE_CC, CFE_EVS_DISABL↩E_APP_EVENT_TYPE_CC, CFE_EVS_ENABLE_APP_EVENTS_CC, CFE_EVS_DISABLE_APP_EVENTS_CC

Definition at line 164 of file cfe_evs_msg.h.

39.55.1.13 CFE_EVS_ENABLE_PORTS_CC

```
#define CFE_EVS_ENABLE_PORTS_CC 11
```

Name Enable Event Services Output Ports

Description

This command enables the command specified port to output event messages

Command Mnemonic(s) $sc_$cpu_EVS_EnaPort, $sc_$cpu_EVS_EnaPortMask

Command Structure

CFE_EVS_EnablePortsCmd_t The following bit positions apply to structure member named 'BitMask'. Bit 0 - Port 1 Bit 1 - Port 2 Bit 2 - Port 3 Bit 3 - Port 4 A one in a bit position means the port will be enabled. A zero in a bit position means the port state is unchanged.

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_ENAPORT_EID debug event message

Error Conditions

This command may fail for the following reason(s):

- Invalid SB message (command) length
- Invalid PORT selection

  Evidence of failure may be found in the following telemetry:

  - **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
  - An Error specific event message

Criticality

None.

See also

CFE_EVS_DISABLE_PORTS_CC

Definition at line 571 of file cfe_evs_msg.h.

39.55.1.14 CFE_EVS_ERROR_BIT

```
#define CFE_EVS_ERROR_BIT 0x0004
```

Definition at line 903 of file cfe_evs_msg.h.

39.55.1.15 CFE_EVS_INFORMATION_BIT

```
#define CFE_EVS_INFORMATION_BIT 0x0002
```

Definition at line 902 of file cfe_evs_msg.h.

39.55.1.16 CFE_EVS_LOG_DISCARD

```
#define CFE_EVS_LOG_DISCARD 1
```

Definition at line 914 of file cfe_evs_msg.h.

39.55.1.17 CFE_EVS_LOG_OVERWRITE

```
#define CFE_EVS_LOG_OVERWRITE 0
```

Definition at line 913 of file cfe_evs_msg.h.

39.55.1.18 CFE_EVS_NOOP_CC

```
#define CFE_EVS_NOOP_CC 0
```

Name Event Services No-Op

Description

This command performs no other function than to increment the command execution counter. The command may be used to verify general aliveness of the Event Services task.

Command Mnemonic(s) $sc_$cpu_EVS_NOOP

Command Structure

CFE_EVS_NoopCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The CFE_EVS_NOOP_EID informational event message will be generated

Error Conditions

There are no error conditions for this command. If the Event Services receives the command, the event is sent (although it may be filtered by EVS itself) and the counter is incremented unconditionally.

Criticality

None

See also

Definition at line 78 of file cfe_evs_msg.h.

39.55.1.19 CFE_EVS_PORT1_BIT

```
#define CFE_EVS_PORT1_BIT 0x0001
```

Definition at line 907 of file cfe_evs_msg.h.

39.55.1.20 CFE_EVS_PORT2_BIT

```
#define CFE_EVS_PORT2_BIT 0x0002
```

Definition at line 908 of file cfe_evs_msg.h.

39.55.1.21 CFE_EVS_PORT3_BIT

```
#define CFE_EVS_PORT3_BIT 0x0004
```

Definition at line 909 of file cfe_evs_msg.h.

39.55.1.22 CFE_EVS_PORT4_BIT

```
#define CFE_EVS_PORT4_BIT 0x0008
```

Definition at line 910 of file cfe_evs_msg.h.

39.55.1.23 CFE_EVS_RESET_ALL_FILTERS_CC

```
#define CFE_EVS_RESET_ALL_FILTERS_CC 14
```

Name Reset All Event Filters for an Application

Description

This command resets all of the command specified applications event filters. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) $sc_$cpu_EVS_RstAllFltrs

Command Structure

CFE_EVS_ResetAllFiltersCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_RSTALLFILTER_EID debug event message

Error Conditions

This command may fail for the following reason(s):

- Invalid SB message (command) length
- Application selected is not registered to receive Event Service
- Application ID is out of range
  Evidence of failure may be found in the following telemetry:
  - **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
  - An Error specific event message

Criticality

None.

See also

CFE_EVS_SET_FILTER_CC, CFE_EVS_RESET_FILTER_CC, CFE_EVS_ADD_EVENT_FILTER_CC, CFE_EVS_DELETE_EVENT_FILTER_CC

Definition at line 684 of file cfe_evs_msg.h.

39.55.1.24 CFE_EVS_RESET_APP_COUNTER_CC

```
#define CFE_EVS_RESET_APP_COUNTER_CC 9
```

Name Reset Application Event Counters

Description

This command sets the command specified application's event counter to zero. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) $sc_$cpu_EVS_RstAppCtrs

Command Structure

CFE_EVS_ResetAppCounterCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **$sc_$cpu_EVS_CMDPC** - command execution counter will increment
- The generation of CFE_EVS_RSTEVTCNT_EID debug event message

Error Conditions

This command may fail for the following reason(s):

- Invalid SB message (command) length
- Application selected is not registered to receive Event Service
- Application ID is out of range
  Evidence of failure may be found in the following telemetry:
  - **$sc_$cpu_EVS_CMDEC** - command error counter will increment
  - An Error specific event message

Criticality

This command is not inherently dangerous. However, it is possible for ground systems and on-board safing procedures to be designed such that they react to changes in the counter value that is reset by this command.

See also

CFE_EVS_RESET_COUNTERS_CC

Definition at line 488 of file cfe_evs_msg.h.

39.55.1.25 CFE_EVS_RESET_COUNTERS_CC

```
#define CFE_EVS_RESET_COUNTERS_CC 1
```

Name Event Services Reset Counters

Description

This command resets the following counters within the Event Services housekeeping telemetry:

- Command Execution Counter ($sc_$cpu_EVS_CMDPC)
- Command Error Counter ($sc_$cpu_EVS_CMDEC)

Command Mnemonic(s) $sc_$cpu_EVS_ResetCtrs

Command Structure

CFE_EVS_ResetCountersCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The CFE_EVS_RSTCNT_EID debug event message will be generated

Error Conditions

There are no error conditions for this command. If the Event Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

This command is not inherently dangerous. However, it is possible for ground systems and on-board safing procedures to be designed such that they react to changes in the counter values that are reset by this command.

See also

CFE_EVS_RESET_APP_COUNTER_CC

Definition at line 115 of file cfe_evs_msg.h.

39.55.1.26 CFE_EVS_RESET_FILTER_CC

```
#define CFE_EVS_RESET_FILTER_CC 13
```

Name Reset an Event Filter for an Application

Description

This command resets the command specified application's event filter for the command specified event ID. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) $sc_$cpu_EVS_RstBinFltrCtr

Command Structure

CFE_EVS_ResetFilterCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_RSTFILTER_EID debug event message

Error Conditions

This command may fail for the following reason(s):

- Invalid SB message (command) length
- Application selected is not registered to receive Event Service
- Application ID is out of range
  Evidence of failure may be found in the following telemetry:
  - **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
  - An Error specific event message

Criticality

None.

See also

CFE_EVS_SET_FILTER_CC, CFE_EVS_RESET_ALL_FILTERS_CC, CFE_EVS_ADD_EVENT_FILTER_CC, CFE_EVS_DELETE_EVENT_FILTER_CC

Definition at line 648 of file cfe_evs_msg.h.

39.55.1.27 CFE_EVS_SET_EVENT_FORMAT_MODE_CC

```
#define CFE_EVS_SET_EVENT_FORMAT_MODE_CC 4
```

Name Set Event Format Mode

Description

This command sets the event format mode to the command specified value. The event format mode may be either short or long. A short event format detaches the Event Data from the event message and only includes the following information in the event packet: Processor ID, Application ID, Event ID, and Event Type. Refer to section 5.3.3.4 for a description of the Event Service event packet contents. Event Data is defined to be data describing an Event that is supplied to the cFE Event Service. ASCII text strings are used as the primary format for Event Data because heritage ground systems use string compares as the basis for their automated alert systems. Two systems, ANSR and SERS were looked at for interface definitions. The short event format is used to accommodate experiences with limited telemetry bandwidth. The long event format includes all event information included within the short format along with the Event Data.

Command Mnemonic(s) $sc_$cpu_EVS_SetEvtFmt

Command Structure

CFE_EVS_SetEventFormatModeCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_SETEVTFMTMOD_EID debug message

Error Conditions

This command may fail for the following reason(s): Invalid SB message (command) length Invalid MODE selection

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
- An Error specific event message

Criticality

Setting the event format mode is not particularly hazardous, as the result may be saving necessary bandwidth. However, inappropriately setting the event format mode could result in a loss of information and missed behavior for the ground system

See also

Definition at line 263 of file cfe_evs_msg.h.

39.55.1.28 CFE_EVS_SET_FILTER_CC

```
#define CFE_EVS_SET_FILTER_CC 10
```

Name Set Application Event Filter

Description

This command sets the command specified application's event filter mask to the command specified value for the command specified event. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) $sc_$cpu_EVS_SetBinFltrMask

Command Structure

CFE_EVS_SetFilterCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_SETFILTERMSK_EID debug event message

Error Conditions

This command may fail for the following reason(s):

- Invalid SB message (command) length
- Application selected is not registered to receive Event Service
- Application ID is out of range
  Evidence of failure may be found in the following telemetry:
  - **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
  - An Error specific event message

Criticality

Setting an application event filter mask is not particularly hazardous, as the result may be shutting off unnecessary event messages and possible event flooding of the system. However, inappropriately setting an application's event filter mask could result in a loss of critical information and missed behavior for the ground system or flooding of the ground system.

See also

CFE_EVS_RESET_FILTER_CC, CFE_EVS_RESET_ALL_FILTERS_CC, CFE_EVS_ADD_EVENT_FILTER_CC, CFE_EVS_DELETE_EVENT_FILTER_CC

Definition at line 531 of file cfe_evs_msg.h.

39.55.1.29 CFE_EVS_SET_LOG_MODE_CC

```
#define CFE_EVS_SET_LOG_MODE_CC 19
```

Name Set Logging Mode

Description

This command sets the logging mode to the command specified value.

Command Mnemonic(s) $sc_$cpu_EVS_SetLogMode

Command Structure

CFE_EVS_SetLogModeCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_LOGMODE_EID debug event message

Error Conditions

This command may fail for the following reason(s):

- Invalid SB message (command) length
- Invalid MODE selected
  Evidence of failure may be found in the following telemetry:
  - **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
  - An Error specific event message

Criticality

Setting the event logging mode is not particularly hazardous, as the result may be saving valuable event data. However, inappropriately setting the log mode could result in a loss of critical information. Note: the event log is a back-up log to the on-board recorder.

See also

CFE_EVS_WRITE_LOG_DATA_FILE_CC, CFE_EVS_CLEAR_LOG_CC

Definition at line 862 of file cfe_evs_msg.h.

39.55.1.30 CFE_EVS_WRITE_APP_DATA_FILE_CC

```
#define CFE_EVS_WRITE_APP_DATA_FILE_CC 17
```

Name Write Event Services Application Information to File

Description

This command writes all application data to a file for all applications that have registered with the EVS. The application data includes the Application ID, Active Flag, Event Count, Event Types Active Flag, and Filter Data.

Command Mnemonic(s) $sc_$cpu_EVS_WriteAppData2File

Command Structure

CFE_EVS_WriteAppDataFileCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **$sc_$cpu_EVS_CMDPC -** command execution counter will increment
- The generation of CFE_EVS_WRDAT_EID debug event message
- The generation of the file written to

Error Conditions

This command may fail for the following reason(s):

- Invalid SB message (command) length
  Evidence of failure may be found in the following telemetry:
  - **$sc_$cpu_EVS_CMDEC -** command error counter will increment
  - An Error specific event message

Criticality

Writing a file is not particularly hazardous, but if proper file management is not taken, then the file system can fill up if this command is used repeatedly.

See also

CFE_EVS_WRITE_LOG_DATA_FILE_CC, CFE_EVS_SET_LOG_MODE_CC

Definition at line 792 of file cfe_evs_msg.h.

39.55.1.31 CFE_EVS_WRITE_LOG_DATA_FILE_CC

```
#define CFE_EVS_WRITE_LOG_DATA_FILE_CC 18
```

Name Write Event Log to File

Description

This command requests the Event Service to generate a file containing the contents of the local event log.

Command Mnemonic(s) $sc_$cpu_EVS_WriteLog2File

Command Structure

CFE_EVS_WriteLogDataFileCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_EVS_CMDPC`** - command execution counter will increment
- The generation of CFE_EVS_WRLOG_EID debug event message

Error Conditions

This command may fail for the following reason(s):

- Invalid SB message (command) length
  Evidence of failure may be found in the following telemetry:
  - **`$sc_$cpu_EVS_CMDEC`** - command error counter will increment
  - An Error specific event message

Criticality

Writing a file is not particularly hazardous, but if proper file management is not taken, then the file system can fill up if this command is used repeatedly.

See also

CFE_EVS_WRITE_APP_DATA_FILE_CC, CFE_EVS_SET_LOG_MODE_CC, CFE_EVS_CLEAR_LOG_CC

Definition at line 826 of file cfe_evs_msg.h.

39.55.2 Typedef Documentation

39.55.2.1 CFE_EVS_AddEventFilterCmd_t

```
typedef CFE_EVS_AppNameEventIDMaskCmd_t CFE_EVS_AddEventFilterCmd_t
```

Definition at line 1167 of file cfe_evs_msg.h.

39.55.2.2 CFE_EVS_AppDataCmd_Payload_t

```
typedef struct CFE_EVS_AppDataCmd_Payload CFE_EVS_AppDataCmd_Payload_t
```

Write Event Services Application Information to File Command Payload.

For command details, see CFE_EVS_WRITE_APP_DATA_FILE_CC

39.55.2.3 CFE_EVS_AppNameBitMaskCmd_Payload_t

```
typedef struct CFE_EVS_AppNameBitMaskCmd_Payload CFE_EVS_AppNameBitMaskCmd_Payload_t
```

Generic App Name and Bitmask Command Payload.

For command details, see CFE_EVS_ENABLE_APP_EVENT_TYPE_CC and/or CFE_EVS_DISABLE_APP_EVENT↩_TYPE_CC

39.55.2.4 CFE_EVS_AppNameBitMaskCmd_t

```
typedef struct CFE_EVS_AppNameBitMaskCmd CFE_EVS_AppNameBitMaskCmd_t
```

Generic App Name and Bitmask Command.

39.55.2.5 CFE_EVS_AppNameCmd_Payload_t

```
typedef struct CFE_EVS_AppNameCmd_Payload CFE_EVS_AppNameCmd_Payload_t
```

Generic App Name Command Payload.

For command details, see CFE_EVS_ENABLE_APP_EVENTS_CC, CFE_EVS_DISABLE_APP_EVENTS_CC, CFE↩_EVS_RESET_APP_COUNTER_CC and/or CFE_EVS_RESET_ALL_FILTERS_CC

39.55.2.6 CFE_EVS_AppNameCmd_t

```
typedef struct CFE_EVS_AppNameCmd CFE_EVS_AppNameCmd_t
```

Generic App Name Command.

39.55.2.7 CFE_EVS_AppNameEventIDCmd_Payload_t

```
typedef struct CFE_EVS_AppNameEventIDCmd_Payload CFE_EVS_AppNameEventIDCmd_Payload_t
```

Generic App Name and Event ID Command Payload.

For command details, see CFE_EVS_RESET_FILTER_CC and CFE_EVS_DELETE_EVENT_FILTER_CC

39.55.2.8 CFE_EVS_AppNameEventIDCmd_t

```
typedef struct CFE_EVS_AppNameEventIDCmd CFE_EVS_AppNameEventIDCmd_t
```

Generic App Name and Event ID Command.

39.55.2.9 CFE_EVS_AppNameEventIDMaskCmd_Payload_t

```
typedef struct CFE_EVS_AppNameEventIDMaskCmd_Payload CFE_EVS_AppNameEventIDMaskCmd_Payload_t
```

Generic App Name, Event ID, Mask Command Payload.

For command details, see CFE_EVS_SET_FILTER_CC, CFE_EVS_ADD_EVENT_FILTER_CC and/or CFE_EVS_↩ DELETE_EVENT_FILTER_CC

39.55.2.10 CFE_EVS_AppNameEventIDMaskCmd_t

```
typedef struct CFE_EVS_AppNameEventIDMaskCmd CFE_EVS_AppNameEventIDMaskCmd_t
```

Generic App Name, Event ID, Mask Command.

39.55.2.11 CFE_EVS_AppTlmData_t

```
typedef struct CFE_EVS_AppTlmData CFE_EVS_AppTlmData_t
```

39.55.2.12 CFE_EVS_BitMaskCmd_Payload_t

```
typedef struct CFE_EVS_BitMaskCmd_Payload CFE_EVS_BitMaskCmd_Payload_t
```

Generic Bitmask Command Payload.

For command details, see CFE_EVS_ENABLE_EVENT_TYPE_CC, CFE_EVS_DISABLE_EVENT_TYPE_CC, CFE↩ _EVS_ENABLE_PORTS_CC and/or CFE_EVS_DISABLE_PORTS_CC

39.55.2.13 CFE_EVS_BitMaskCmd_t

```
typedef struct CFE_EVS_BitMaskCmd CFE_EVS_BitMaskCmd_t
```

Generic Bitmask Command.

39.55.2.14 CFE_EVS_ClearLogCmd_t

```
typedef CFE_EVS_NoArgsCmd_t CFE_EVS_ClearLogCmd_t
```

Definition at line 933 of file cfe_evs_msg.h.

39.55.2.15 CFE_EVS_DeleteEventFilterCmd_t

```
typedef CFE_EVS_AppNameEventIDCmd_t CFE_EVS_DeleteEventFilterCmd_t
```

Definition at line 1107 of file cfe_evs_msg.h.

39.55.2.16 CFE_EVS_DisableAppEventsCmd_t

```
typedef CFE_EVS_AppNameCmd_t CFE_EVS_DisableAppEventsCmd_t
```

Definition at line 1076 of file cfe_evs_msg.h.

39.55.2.17 CFE_EVS_DisableAppEventTypeCmd_t

```
typedef CFE_EVS_AppNameBitMaskCmd_t CFE_EVS_DisableAppEventTypeCmd_t
```

Definition at line 1137 of file cfe_evs_msg.h.

39.55.2.18 CFE_EVS_DisableEventTypeCmd_t

```
typedef CFE_EVS_BitMaskCmd_t CFE_EVS_DisableEventTypeCmd_t
```

Definition at line 1047 of file cfe_evs_msg.h.

39.55.2.19 CFE_EVS_DisablePortsCmd_t

```
typedef CFE_EVS_BitMaskCmd_t CFE_EVS_DisablePortsCmd_t
```

Definition at line 1045 of file cfe_evs_msg.h.

39.55.2.20 CFE_EVS_EnableAppEventsCmd_t

```
typedef CFE_EVS_AppNameCmd_t CFE_EVS_EnableAppEventsCmd_t
```

Definition at line 1075 of file cfe_evs_msg.h.

39.55.2.21 CFE_EVS_EnableAppEventTypeCmd_t

```
typedef CFE_EVS_AppNameBitMaskCmd_t CFE_EVS_EnableAppEventTypeCmd_t
```

Definition at line 1136 of file cfe_evs_msg.h.

39.55.2.22 CFE_EVS_EnableEventTypeCmd_t

```
typedef CFE_EVS_BitMaskCmd_t CFE_EVS_EnableEventTypeCmd_t
```

Definition at line 1046 of file cfe_evs_msg.h.

39.55.2.23 CFE_EVS_EnablePortsCmd_t

```
typedef CFE_EVS_BitMaskCmd_t CFE_EVS_EnablePortsCmd_t
```

Definition at line 1044 of file cfe_evs_msg.h.

39.55.2.24 CFE_EVS_HousekeepingTlm_Payload_t

```
typedef struct CFE_EVS_HousekeepingTlm_Payload CFE_EVS_HousekeepingTlm_Payload_t
```

Name Event Services Housekeeping Telemetry Packet

39.55.2.25 CFE_EVS_HousekeepingTlm_t

```
typedef struct CFE_EVS_HousekeepingTlm CFE_EVS_HousekeepingTlm_t
```

39.55.2.26 CFE_EVS_LogFileCmd_Payload_t

```
typedef struct CFE_EVS_LogFileCmd_Payload CFE_EVS_LogFileCmd_Payload_t
```

Write Event Log to File Command Payload.

For command details, see CFE_EVS_WRITE_LOG_DATA_FILE_CC

39.55.2.27 CFE_EVS_LongEventTlm_Payload_t

```
typedef struct CFE_EVS_LongEventTlm_Payload CFE_EVS_LongEventTlm_Payload_t
```

Name Event Message Telemetry Packet (Long format)

39.55.2.28 CFE_EVS_LongEventTlm_t

```
typedef struct CFE_EVS_LongEventTlm CFE_EVS_LongEventTlm_t
```

39.55.2.29 CFE_EVS_NoArgsCmd_t

```
typedef struct CFE_EVS_NoArgsCmd CFE_EVS_NoArgsCmd_t
```

Command with no additional arguments.

39.55.2.30 CFE_EVS_NoopCmd_t

```
typedef CFE_EVS_NoArgsCmd_t CFE_EVS_NoopCmd_t
```

Definition at line 931 of file cfe_evs_msg.h.

39.55.2.31 CFE_EVS_PacketID_t

```
typedef struct CFE_EVS_PacketID CFE_EVS_PacketID_t
```

Telemetry packet structures

39.55.2.32 CFE_EVS_ResetAllFiltersCmd_t

```
typedef CFE_EVS_AppNameCmd_t CFE_EVS_ResetAllFiltersCmd_t
```

Definition at line 1078 of file cfe_evs_msg.h.

39.55.2.33 CFE_EVS_ResetAppCounterCmd_t

```
typedef CFE_EVS_AppNameCmd_t CFE_EVS_ResetAppCounterCmd_t
```

Definition at line 1077 of file cfe_evs_msg.h.

39.55.2.34 CFE_EVS_ResetCountersCmd_t

```
typedef CFE_EVS_NoArgsCmd_t CFE_EVS_ResetCountersCmd_t
```

Definition at line 932 of file cfe_evs_msg.h.

39.55.2.35 CFE_EVS_ResetFilterCmd_t

```
typedef CFE_EVS_AppNameEventIDCmd_t CFE_EVS_ResetFilterCmd_t
```

Definition at line 1106 of file cfe_evs_msg.h.

39.55.2.36 CFE_EVS_SetEventFormatMode_Payload_t

```
typedef struct CFE_EVS_SetEventFormatCode_Payload CFE_EVS_SetEventFormatMode_Payload_t
```

Set Event Format Mode Command Payload.

For command details, see CFE_EVS_SET_EVENT_FORMAT_MODE_CC

39.55.2.37 CFE_EVS_SetEventFormatModeCmd_t

```
typedef struct CFE_EVS_SetEventFormatModeCmd CFE_EVS_SetEventFormatModeCmd_t
```

Set Event Format Mode Command.

39.55.2.38 CFE_EVS_SetFilterCmd_t

```
typedef CFE_EVS_AppNameEventIDMaskCmd_t CFE_EVS_SetFilterCmd_t
```

Definition at line 1168 of file cfe_evs_msg.h.

39.55.2.39 CFE_EVS_SetLogMode_Payload_t

```
typedef struct CFE_EVS_SetLogMode_Payload CFE_EVS_SetLogMode_Payload_t
```

Set Log Mode Command Payload.

For command details, see CFE_EVS_SET_LOG_MODE_CC

39.55.2.40 CFE_EVS_SetLogModeCmd_t

```
typedef struct CFE_EVS_SetLogModeCmd CFE_EVS_SetLogModeCmd_t
```

Set Log Mode Command.

39.55.2.41 CFE_EVS_ShortEventTlm_Payload_t

```
typedef struct CFE_EVS_ShortEventTlm_Payload CFE_EVS_ShortEventTlm_Payload_t
```

Name Event Message Telemetry Packet (Short format)

39.55.2.42 CFE_EVS_ShortEventTlm_t

```
typedef struct CFE_EVS_ShortEventTlm CFE_EVS_ShortEventTlm_t
```

39.55.2.43 CFE_EVS_WriteAppDataFileCmd_t

```
typedef struct CFE_EVS_WriteAppDataFileCmd CFE_EVS_WriteAppDataFileCmd_t
```

Write Event Services Application Information to File Command.

39.55.2.44 CFE_EVS_WriteLogDataFileCmd_t

```
typedef struct CFE_EVS_WriteLogDataFileCmd CFE_EVS_WriteLogDataFileCmd_t
```

Write Event Log to File Command.

39.56 cfe/modules/msg/fsw/inc/ccsds_hdr.h File Reference

```
#include "common_types.h"
```

Data Structures

- struct CCSDS_PrimaryHeader

  CCSDS packet primary header.
- struct CCSDS_ExtendedHeader

  CCSDS packet extended header.

Typedefs

- typedef struct CCSDS_PrimaryHeader CCSDS_PrimaryHeader_t

  CCSDS packet primary header.
- typedef struct CCSDS_ExtendedHeader CCSDS_ExtendedHeader_t

  CCSDS packet extended header.

39.56.1 Typedef Documentation

39.56.1.1 CCSDS_ExtendedHeader_t

```
typedef struct CCSDS_ExtendedHeader CCSDS_ExtendedHeader_t
```

CCSDS packet extended header.

39.56.1.2 CCSDS_PrimaryHeader_t

```
typedef struct CCSDS_PrimaryHeader CCSDS_PrimaryHeader_t
```

CCSDS packet primary header.

39.57 cfe/modules/sb/fsw/inc/cfe_sb_events.h File Reference

Macros

- #define CFE_SB_MAX_EID 67
- #define CFE_SB_INIT_EID 1

  `'cFE SB Initialized'`
- #define CFE_SB_CR_PIPE_BAD_ARG_EID 2

  `'CreatePipeErr:Bad Input Arg:app=%s,ptr=0x%x,depth=%d,maxdepth=%d'`
- #define CFE_SB_MAX_PIPES_MET_EID 3

  `'CreatePipeErr:Max Pipes(%d)In Use.app %s'`
- #define CFE_SB_CR_PIPE_ERR_EID 4

  `'CreatePipeErr:OS_QueueCreate returned %d,app %s'`
- #define CFE_SB_PIPE_ADDED_EID 5

  `'Pipe Created:name %s,id %d,app %s'`
- #define CFE_SB_SETPIPEOPTS_ID_ERR_EID 55

  `'SetPipeOptsErr:Invalid pipe id (%d).app %s'`
- #define CFE_SB_SETPIPEOPTS_OWNER_ERR_EID 56

  `'SetPipeOptsErr:Caller not owner (%d).app %s'`
- #define CFE_SB_SETPIPEOPTS_EID 57

  `'SetPipeOpts:  Options set (%d).  app %s'`
- #define CFE_SB_GETPIPEOPTS_ID_ERR_EID 58

  `'GetPipeOptsErr:Invalid pipe id (%d).app %s'`
- #define CFE_SB_GETPIPEOPTS_PTR_ERR_EID 59

  `'GetPipeOptsErr:Invalid opts ptr.app %s'`
- #define CFE_SB_GETPIPEOPTS_EID 60

  `'GetPipeOpts:  Options retrieved.  app %s'`
- #define CFE_SB_GETPIPENAME_EID 62

  `'GetPipeName:  Name retrieved.  NameOut %s,Id %d, app %s'`
- #define CFE_SB_GETPIPENAME_NULL_PTR_EID 63

  `'GetPipeName:  Null ptr error.  Id %d, app %s'`
- #define CFE_SB_GETPIPENAME_ID_ERR_EID 64

  `'GetPipeName:  Id error.  NameOut %s,Id %d, app %s'`
- #define CFE_SB_GETPIPEIDBYNAME_EID 65

  `'GetPipeIdByName:  ID retrieved.  Name %s,IdOut 0x%x, app %s'`
- #define CFE_SB_GETPIPEIDBYNAME_NULL_ERR_EID 66

  `'GetPipeIdByName Err:Bad input argument,Name 0x%x,IdOut 0xx,App %s'`
- #define CFE_SB_GETPIPEIDBYNAME_NAME_ERR_EID 67

  `'GetPipeIdByName Err:Name not found,Name %s,IdOut 0xx,App %s'`
- #define CFE_SB_SUB_ARG_ERR_EID 6

  `'Subscribe Err:Bad Arg,MsgId 0x%x,PipeId %d,app %s,scope %d'`
- #define CFE_SB_DUP_SUBSCRIP_EID 7

  `'Duplicate Subscription,MsgId 0x%x on %s pipe,app %s'`
- #define CFE_SB_MAX_MSGS_MET_EID 8

  `'Subscribe Err:Max Msgs(%d)In Use,MsgId 0x%x,pipe %s,app %s'`
- #define CFE_SB_MAX_DESTS_MET_EID 9

  `'Subscribe Err:Max Dests(%d)In Use For Msg 0x%x,pipe %s,app %s'`

- #define CFE_SB_SUBSCRIPTION_RCVD_EID 10

  `'Subscription Rcvd:MsgId 0x%x on %s(%d),app %s'`
- #define CFE_SB_UNSUB_ARG_ERR_EID 11

  `'UnSubscribe Err:Bad Arg,MsgId 0x%x,PipeId %d,app %s,scope %d'`
- #define CFE_SB_UNSUB_NO_SUBS_EID 12

  `'Unsubscribe Err:No subs for Msg 0x%x on %s,app %s'`
- #define CFE_SB_SEND_BAD_ARG_EID 13

  `'Send Err:Bad input argument,Arg 0x%x,App %s'`
- #define CFE_SB_SEND_NO_SUBS_EID 14

  `'No subscribers for MsgId 0x%x,sender %s'`
- #define CFE_SB_MSG_TOO_BIG_EID 15

  `'Send Err:Msg Too Big MsgId=0x%x,app=%s,size=%d,MaxSz=%d'`
- #define CFE_SB_GET_BUF_ERR_EID 16

  `'Send Err:Request for Buffer Failed.  MsgId 0x%x,app %s,size %d'`
- #define CFE_SB_MSGID_LIM_ERR_EID 17

  `'Send Err:Msg Limit Err MsgId 0x%x,pipe %s,sender %s'`
- #define CFE_SB_RCV_BAD_ARG_EID 18

  `'Rcv Err:Bad Input Arg:BufPtr 0x%x,pipe %d,t/o %d,app %s'`
- #define CFE_SB_BAD_PIPEID_EID 19

  `'Rcv Err:PipeId %d does not exist,app %s'`
- #define CFE_SB_DEST_BLK_ERR_EID 20

  `'Subscribe Err:Request for Destination Blk failed for Msg 0x%x,Pipe %s'`
- #define CFE_SB_SEND_INV_MSGID_EID 21

  `'Send Err:Invalid msgid in msg,MsgId 0x%x,App %s'`
- #define CFE_SB_SUBSCRIPTION_RPT_EID 22

  `'Sending Subscription Report Msg=0x%x,Pipe=%d,Stat=0x%x'`
- #define CFE_SB_HASHCOLLISION_EID 23

  `'Msg hash collision:  MsgId = 0x%x, collisions = %u'`
- #define CFE_SB_Q_FULL_ERR_EID 25

  `'Pipe Overflow,MsgId 0x%x,pipe %s,stat 0x%x,app %s'`
- #define CFE_SB_Q_WR_ERR_EID 26

  `'Pipe Write Err,MsgId 0x%x,pipe %s,stat 0x%x,app %s'`
- #define CFE_SB_Q_RD_ERR_EID 27

  `'Pipe Read Err,pipe %s,app %s,stat 0x%x'`
- #define CFE_SB_CMD0_RCVD_EID 28

  `'No-op Cmd Rcvd'`
- #define CFE_SB_CMD1_RCVD_EID 29

  `'Reset Counters Cmd Rcvd'`
- #define CFE_SB_SND_STATS_EID 32

  `'Software Bus Statistics packet sent'`
- #define CFE_SB_ENBL_RTE1_EID 33

  `'Enbl Route Cmd:Route does not exist.Msg 0x%x,Pipe %d'`
- #define CFE_SB_ENBL_RTE2_EID 34

  `'Enabling Route,Msg 0x%x,Pipe %d'`
- #define CFE_SB_ENBL_RTE3_EID 35

  `'Enbl Route Cmd:Invalid Param.Msg 0x%x,Pipe %d'`
- #define CFE_SB_DSBL_RTE1_EID 36

```
'Disable Route Cmd:Route does not exist,Msg 0x%x,Pipe %d'
```

- #define CFE_SB_DSBL_RTE2_EID 37

```
'Route Disabled,Msg 0x%x,Pipe %d'
```

- #define CFE_SB_DSBL_RTE3_EID 38

```
'Disable Route Cmd:Invalid Param.Msg 0x%x,Pipe %d'
```

- #define CFE_SB_SND_RTG_EID 39

```
'%s written:Size=%d,Entries=%d'
```

- #define CFE_SB_SND_RTG_ERR1_EID 40

```
'Error creating file %s, stat=0x%x'
```

- #define CFE_SB_BAD_CMD_CODE_EID 42

```
'Invalid Cmd, Unexpected Command Code %d'
```

- #define CFE_SB_BAD_MSGID_EID 43

```
'Invalid Cmd, Unexpected Msg Id:  0x%x'
```

- #define CFE_SB_FULL_SUB_PKT_EID 44

```
'Full Sub Pkt %d Sent,Entries=%d,Stat=0x%x
'
```

- #define CFE_SB_PART_SUB_PKT_EID 45

```
'Partial Sub Pkt %d Sent,Entries=%d,Stat=0x%x'
```

- #define CFE_SB_DEL_PIPE_ERR1_EID 46

```
'Pipe Delete Error:Bad Argument,PipedId %d,Requestor %s,Idx %d,Stat %d'
```

- #define CFE_SB_PIPE_DELETED_EID 47

```
'Pipe Deleted:id %d,owner %s'
```

- #define CFE_SB_SUBSCRIPTION_REMOVED_EID 48

```
'Subscription Removed:Msg 0x%x on pipe %d,app %s'
```

- #define CFE_SB_FILEWRITE_ERR_EID 49

```
'File write,byte cnt err,file %s,request=%d,actual=%d'
```

- #define CFE_SB_SUB_INV_PIPE_EID 50

```
'Subscribe Err:Invalid Pipe Id,Msg=0x%x,PipeId=%d,App %s'
```

- #define CFE_SB_SUB_INV_CALLER_EID 51

```
'Subscribe Err:Caller(%s) is not the owner of pipe %d, Msg=0x%x'
```

- #define CFE_SB_UNSUB_INV_PIPE_EID 52

```
'Unsubscribe Err:Invalid Pipe Id Msg=0x%x,Pipe=%d,app=%s'
```

- #define CFE_SB_UNSUB_INV_CALLER_EID 53

```
'Unsubscribe Err:Caller(%s) is not the owner of pipe %d,Msg=0x%x'
```

- #define CFE_SB_DEL_PIPE_ERR2_EID 54

```
'Pipe Delete Error:Caller(%s) is not the owner of pipe %d'
```

- #define CFE_SB_LEN_ERR_EID 61

```
'Invalid cmd length:  ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'
```

- #define CFE_SB_CR_PIPE_NAME_TAKEN_EID 62

```
'CreatePipeErr:Name Taken:app=%s,ptr=0x%x,depth=%d,maxdepth=%d'
```

- #define CFE_SB_CR_PIPE_NO_FREE_EID 63

```
'CreatePipeErr:No Free:app=%s,ptr=0x%x,depth=%d,maxdepth=%d'
```

39.57.1 Macro Definition Documentation

39.57.1.1 CFE_SB_BAD_CMD_CODE_EID

```
#define CFE_SB_BAD_CMD_CODE_EID 42
```

`'Invalid Cmd, Unexpected Command Code %d'`

Event Message `'Invalid Cmd, Unexpected Command Code %d'`

Type: ERROR

Cause:

This error event message is issued when the SB receives a cmd that has an unexpected cmd code.

Definition at line 709 of file cfe_sb_events.h.

39.57.1.2 CFE_SB_BAD_MSGID_EID

```
#define CFE_SB_BAD_MSGID_EID 43
```

`'Invalid Cmd, Unexpected Msg Id:  0x%x'`

Event Message `'Invalid Cmd, Unexpected Msg Id:  0x%x'`

Type: ERROR

Cause:

This error event message is issued when the SB receives a msg that has an unexpected msg id.

Definition at line 721 of file cfe_sb_events.h.

39.57.1.3 CFE_SB_BAD_PIPEID_EID

```
#define CFE_SB_BAD_PIPEID_EID 19
```

```
'Rcv Err:PipeId %d does not exist,app %s'
```

Event Message `'Rcv Err:PipeId %d does not exist,app %s'`

Type: ERROR

Cause:

This error event message is issued when an invalid PipeId is passed into the CFE_SB_ReceiveBuffer API. The SB Pipe Table shows all valid PipeIds and may be viewed for verification.

Definition at line 457 of file cfe_sb_events.h.

39.57.1.4 CFE_SB_CMD0_RCVD_EID

```
#define CFE_SB_CMD0_RCVD_EID 28
```

```
'No-op Cmd Rcvd'
```

Event Message `'No-op Cmd Rcvd'`

Type: INFORMATION

Cause:

This info event message is issued in response an SB NO-OP command

Definition at line 569 of file cfe_sb_events.h.

39.57.1.5 CFE_SB_CMD1_RCVD_EID

```
#define CFE_SB_CMD1_RCVD_EID 29
```

`'Reset Counters Cmd Rcvd'`

Event Message `'Reset Counters Cmd Rcvd'`

Type: DEBUG

Cause:

This debug event message is issued in response an SB Reset Counters command

Definition at line 580 of file cfe_sb_events.h.

39.57.1.6 CFE_SB_CR_PIPE_BAD_ARG_EID

```
#define CFE_SB_CR_PIPE_BAD_ARG_EID 2
```

`'CreatePipeErr:Bad Input Arg:app=%s,ptr=0x%x,depth=%d,maxdepth=%d'`

Event Message `'CreatePipeErr:Bad Input Arg:app=%s,ptr=0x%x,depth=%d,maxdepth=%d'`

Type: ERROR

Cause:

This error event message is issued when the CFE_SB_CreatePipe API receives a bad argument. In this case, a bad argument is defined by the following: A NULL PipeIdPtr, PipeDepth = 0 and PipeDepth > maximum pipe depth

Definition at line 75 of file cfe_sb_events.h.

39.57.1.7 CFE_SB_CR_PIPE_ERR_EID

```
#define CFE_SB_CR_PIPE_ERR_EID 4
```

`'CreatePipeErr:OS_QueueCreate returned %d,app %s'`

Event Message `'CreatePipeErr:OS_QueueCreate returned %d,app %s'`

Type: ERROR

Cause:

This error event message is issued when the CFE_SB_CreatePipe API is called and the OS returns an error when the OS returns an error from the OS_QueueCreate API. The error status returned by the OS is displayed in the event. Most commonly, this event is displayed as a result of trying to create pipes with the same name.

Definition at line 101 of file cfe_sb_events.h.

39.57.1.8 CFE_SB_CR_PIPE_NAME_TAKEN_EID

```
#define CFE_SB_CR_PIPE_NAME_TAKEN_EID 62
```

`'CreatePipeErr:Name Taken:app=%s,ptr=0x%x,depth=%d,maxdepth=%d'`

Event Message `'CreatePipeErr:Name Taken:app=%s,ptr=0x%x,depth=%d,maxdepth=%d'`

Type: ERROR

Cause:

This error event message is issued when the CFE_SB_CreatePipe API tries to create a pipe with a name that is in use.

Definition at line 891 of file cfe_sb_events.h.

39.57.1.9 CFE_SB_CR_PIPE_NO_FREE_EID

```
#define CFE_SB_CR_PIPE_NO_FREE_EID 63
```

`'CreatePipeErr:No Free:app=%s,ptr=0x%x,depth=%d,maxdepth=%d'`

Event Message `'CreatePipeErr:No Free:app=%s,ptr=0x%x,depth=%d,maxdepth=%d'`

Type: ERROR

Cause:

This error event message is issued when the CFE_SB_CreatePipe API is unable to create a queue because there are no queues free.

Definition at line 903 of file cfe_sb_events.h.

39.57.1.10 CFE_SB_DEL_PIPE_ERR1_EID

```
#define CFE_SB_DEL_PIPE_ERR1_EID 46
```

`'Pipe Delete Error:Bad Argument,PipedId %d,Requestor %s,Idx %d,Stat %d'`

Event Message `'Pipe Delete Error:Bad Argument,PipedId %d,Requestor %s,Idx %d,Stat %d'`

Type: ERROR

Cause:

This error event message is issued from CFE_SB_DeletePipeFull when an invalid pipe ID is passed in

Definition at line 757 of file cfe_sb_events.h.

39.57.1.11 CFE_SB_DEL_PIPE_ERR2_EID

```
#define CFE_SB_DEL_PIPE_ERR2_EID 54
```

```
'Pipe Delete Error:Caller(%s) is not the owner of pipe %d'
```

Event Message `'Pipe Delete Error:Caller(%s) is not the owner of pipe %d'`

Type: ERROR

Cause:

This error event message is issued when the CFE_SB_DeletePipe API is called by a task that is not the owner of the pipe. Pipes may be deleted only by the task that created the pipe or ES(for cleanup purposes).

Definition at line 861 of file cfe_sb_events.h.

39.57.1.12 CFE_SB_DEST_BLK_ERR_EID

```
#define CFE_SB_DEST_BLK_ERR_EID 20
```

```
'Subscribe Err:Request for Destination Blk failed for Msg 0x%x,Pipe %s'
```

Event Message `'Subscribe Err:Request for Destination Blk failed for Msg 0x%x,Pipe %s'`

Type: ERROR

Cause:

This error event message is issued when the SB receives an error from the memory pool in the attempt to obtain a new destination block. Then memory pool statistics may be viewed by sending the related ES command.

Definition at line 470 of file cfe_sb_events.h.

39.57.1.13 CFE_SB_DSBL_RTE1_EID

```
#define CFE_SB_DSBL_RTE1_EID 36
```

`'Disable Route Cmd:Route does not exist,Msg 0x%x,Pipe %d'`

Event Message `'Disable Route Cmd:Route does not exist,Msg 0x%x,Pipe %d'`

Type: ERROR

Cause:

This error event message is issued when SB receives a cmd to disable a route that does not exist in the routing table. A route is defined by a MsgId, PipeId pair.

Definition at line 643 of file cfe_sb_events.h.

39.57.1.14 CFE_SB_DSBL_RTE2_EID

```
#define CFE_SB_DSBL_RTE2_EID 37
```

`'Route Disabled,Msg 0x%x,Pipe %d'`

Event Message `'Route Disabled,Msg 0x%x,Pipe %d'`

Type: DEBUG

Cause:

This debug event message is issued when SB receives a cmd to disable a route and the request is successfully executed.

Definition at line 655 of file cfe_sb_events.h.

39.57.1.15 CFE_SB_DSBL_RTE3_EID

```
#define CFE_SB_DSBL_RTE3_EID 38
```

```
'Disable Route Cmd:Invalid Param.Msg 0x%x,Pipe %d'
```

Event Message `'Disable Route Cmd:Invalid Param.Msg 0x%x,Pipe %d'`

Type: ERROR

Cause:

This error event message is issued when SB receives a cmd to disable a route and the MsgId or PipeId does not pass the validation checks. The MsgId must be less than cfg param CFE_PLATFORM_SB_HIGHEST_VALID_MSGID. The PipeId must exist and be less than cfg param CFE_PLATFORM_SB_MAX_PIPES. The SB pipe table may be viewed to verify the PipeId existence.

Definition at line 670 of file cfe_sb_events.h.

39.57.1.16 CFE_SB_DUP_SUBSCRIP_EID

```
#define CFE_SB_DUP_SUBSCRIP_EID 7
```

```
'Duplicate Subscription,MsgId 0x%x on %s pipe,app %s'
```

Event Message `'Duplicate Subscription,MsgId 0x%x on %s pipe,app %s'`

Type: INFORMATION

Cause:

This info event message is issued when a subscription request is received that already exists in the routing table. A duplicate subscription is defined by a matching MsgId and PipeId. No other parameters are used in detecting a duplicate subscription. NOTE: By default, SB filters this event. The EVS filter algorithm allows the first event to pass through the filter, but all subsequent events with this event id will be filtered. A command must be sent to unfilter this event if the user desires to see it.

Definition at line 283 of file cfe_sb_events.h.

39.57.1.17 CFE_SB_ENBL_RTE1_EID

```
#define CFE_SB_ENBL_RTE1_EID 33
```

`'Enbl Route Cmd:Route does not exist.Msg 0x%x,Pipe %d'`

Event Message `'Enbl Route Cmd:Route does not exist.Msg 0x%x,Pipe %d'`

Type: ERROR

Cause:

This error event message is issued when SB receives a cmd to enable a route that does not exist in the routing table. A route is defined by a MsgId, PipeId pair.

Definition at line 604 of file cfe_sb_events.h.

39.57.1.18 CFE_SB_ENBL_RTE2_EID

```
#define CFE_SB_ENBL_RTE2_EID 34
```

`'Enabling Route,Msg 0x%x,Pipe %d'`

Event Message `'Enabling Route,Msg 0x%x,Pipe %d'`

Type: DEBUG

Cause:

This debug event message is issued when SB receives a cmd to enable a route and the request is successfully executed.

Definition at line 616 of file cfe_sb_events.h.

39.57.1.19 CFE_SB_ENBL_RTE3_EID

```
#define CFE_SB_ENBL_RTE3_EID 35
```

`'Enbl Route Cmd:Invalid Param.Msg 0x%x,Pipe %d'`

Event Message `'Enbl Route Cmd:Invalid Param.Msg 0x%x,Pipe %d'`

Type: ERROR

Cause:

This error event message is issued when SB receives a cmd to enable a route and the MsgId or PipeId does not pass the validation checks. The MsgId must be less than cfg param CFE_PLATFORM_SB_HIGHEST_VALID_MSGID. The PipeId must exist and be less than cfg param CFE_PLATFORM_SB_MAX_PIPES. The SB pipe table may be viewed to verify the PipeId existence.

Definition at line 631 of file cfe_sb_events.h.

39.57.1.20 CFE_SB_FILEWRITE_ERR_EID

```
#define CFE_SB_FILEWRITE_ERR_EID 49
```

`'File write,byte cnt err,file %s,request=%d,actual=%d'`

Event Message `'File write,byte cnt err,file %s,request=%d,actual=%d'`

Type: ERROR

Cause:

This error event message is issued when one of many SB's file write operations is unsuccessful. This event is a result of CFE_FS_WriteHeader or OS_write returning something other than the number of bytes requested to be written. The requested value and the return value are displayed in the event.

Definition at line 795 of file cfe_sb_events.h.

39.57.1.21 CFE_SB_FULL_SUB_PKT_EID

```
#define CFE_SB_FULL_SUB_PKT_EID 44
```

```
'Full Sub Pkt %d Sent,Entries=%d,Stat=0x%x
'
```

Event Message `'Full Sub Pkt %d Sent,Entries=%d,Stat=0x%x
'`

Type: DEBUG

Cause:

This debug event message is issued in response to the 'Send Previous Subscriptions' command and a full pkt segment is sent.

Definition at line 733 of file cfe_sb_events.h.

39.57.1.22 CFE_SB_GET_BUF_ERR_EID

```
#define CFE_SB_GET_BUF_ERR_EID 16
```

```
'Send Err:Request for Buffer Failed.  MsgId 0x%x,app %s,size %d'
```

Event Message `'Send Err:Request for Buffer Failed.  MsgId 0x%x,app %s,size %d'`

Type: ERROR

Cause:

This error event message is issued when a transmit API fails to receive the necessary buffer memory from the ES memory pool. This could be an indication that the cfg param CFE_PLATFORM_SB_BUF_MEMORY_BYTES is set too low. To check this, send SB cmd to dump the SB statistics pkt and view the buffer memory parameters.

Definition at line 412 of file cfe_sb_events.h.

39.57.1.23 CFE_SB_GETPIPEIDBYNAME_EID

```
#define CFE_SB_GETPIPEIDBYNAME_EID 65
```

`'GetPipeIdByName:  ID retrieved.  Name %s,IdOut 0x%x, app %s'`

Event Message `'GetPipeIdByName:  ID retrieved.  Name %s,IdOut 0x%x, app %s'`

Type: DEBUG

Cause:

This debug event is generated when id is retrieved by name.

Definition at line 227 of file cfe_sb_events.h.

39.57.1.24 CFE_SB_GETPIPEIDBYNAME_NAME_ERR_EID

```
#define CFE_SB_GETPIPEIDBYNAME_NAME_ERR_EID 67
```

`'GetPipeIdByName Err:Name not found,Name %s,IdOut 0xx,App %s'`

Event Message `'GetPipeIdByName Err:Name not found,Name %s,IdOut 0xx,App %s'`

Type: ERROR

Cause:

This error event message is issued when the CFE_SB_GetPipeIdByName API receives an invalid name.

Definition at line 251 of file cfe_sb_events.h.

39.57.1.25 CFE_SB_GETPIPEIDBYNAME_NULL_ERR_EID

```
#define CFE_SB_GETPIPEIDBYNAME_NULL_ERR_EID 66
```

`'GetPipeIdByName Err:Bad input argument,Name 0x%x,IdOut 0xx,App %s'`

Event Message `'GetPipeIdByName Err:Bad input argument,Name 0x%x,IdOut 0xx,App %s'`

Type: ERROR

Cause:

This error event message is issued when the CFE_SB_GetPipeIdByName API receives a NULL ptr as an argument.

Definition at line 239 of file cfe_sb_events.h.

39.57.1.26 CFE_SB_GETPIPENAME_EID

```
#define CFE_SB_GETPIPENAME_EID 62
```

`'GetPipeName:  Name retrieved.  NameOut %s,Id %d, app %s'`

Event Message `'GetPipeName:  Name retrieved.  NameOut %s,Id %d, app %s'`

Type: DEBUG

Cause:

This debug event is generated when name is retrieved by id.

Definition at line 194 of file cfe_sb_events.h.

39.57.1.27 CFE_SB_GETPIPENAME_ID_ERR_EID

```
#define CFE_SB_GETPIPENAME_ID_ERR_EID 64
```

`'GetPipeName:  Id error.  NameOut %s,Id %d, app %s'`

Event Message `'GetPipeName:  Id error.  NameOut %s,Id %d, app %s'`

Type: ERROR

Cause:

This debug event is generated when name is retrieved by id.

Definition at line 216 of file cfe_sb_events.h.

39.57.1.28 CFE_SB_GETPIPENAME_NULL_PTR_EID

```
#define CFE_SB_GETPIPENAME_NULL_PTR_EID 63
```

`'GetPipeName:  Null ptr error.  Id %d, app %s'`

Event Message `'GetPipeName:  Null ptr error.  Id %d, app %s'`

Type: ERROR

Cause:

This debug event is generated when the name buffer ptr is null.

Definition at line 205 of file cfe_sb_events.h.

39.57.1.29 CFE_SB_GETPIPEOPTS_EID

```
#define CFE_SB_GETPIPEOPTS_EID 60
```

`'GetPipeOpts:  Options retrieved.  app %s'`

Event Message `'GetPipeOpts:  Options retrieved.  app %s'`

Type: DEBUG

Cause:

This debug event is generated when options are retrieved.

Definition at line 183 of file cfe_sb_events.h.

39.57.1.30 CFE_SB_GETPIPEOPTS_ID_ERR_EID

```
#define CFE_SB_GETPIPEOPTS_ID_ERR_EID 58
```

`'GetPipeOptsErr:Invalid pipe id (%d).app %s'`

Event Message `'GetPipeOptsErr:Invalid pipe id (%d).app %s'`

Type: ERROR

Cause:

This error event message is issued when the CFE_SB_GetPipeOpts API is called and the PipeID is invalid.

Definition at line 160 of file cfe_sb_events.h.

39.57.1.31 CFE_SB_GETPIPEOPTS_PTR_ERR_EID

```
#define CFE_SB_GETPIPEOPTS_PTR_ERR_EID 59
```

```
'GetPipeOptsErr:Invalid opts ptr.app %s'
```

Event Message `'GetPipeOptsErr:Invalid opts ptr.app %s'`

Type: ERROR

Cause:

This error event message is issued when the CFE_SB_GetPipeOpts API is called and the pointer is invalid.

Definition at line 172 of file cfe_sb_events.h.

39.57.1.32 CFE_SB_HASHCOLLISION_EID

```
#define CFE_SB_HASHCOLLISION_EID 23
```

```
'Msg hash collision:  MsgId = 0x%x, collisions = %u'
```

Event Message `'Msg hash collision:  MsgId = 0x%x, collisions = %u'`

Type: DEBUG

Cause:

This event message is generated when a message id hash collision occurs when subscribing to a message. Collisions indicate how many slots were incremented to find an opening.

Number of collisions will directly impact software bus performance. These can be resolved by adjusting MsgId values or increasing CFE_PLATFORM_SB_MAX_MSG_IDS.

Definition at line 510 of file cfe_sb_events.h.

39.57.1.33 CFE_SB_INIT_EID

```
#define CFE_SB_INIT_EID 1
```

`'cFE SB Initialized'`

Event Message `'cFE SB Initialized'`

Type: INFORMATION

Cause:

This event message is issued when the Software Bus Task completes its initialization.

Definition at line 62 of file cfe_sb_events.h.

39.57.1.34 CFE_SB_LEN_ERR_EID

```
#define CFE_SB_LEN_ERR_EID 61
```

`'Invalid cmd length:  ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'`

Event Message `'Invalid cmd length:  ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'`

Type: ERROR

Cause:

This event message is generated when a message with the CFE_SB_CMD_MID message ID has arrived but whose packet length does not match the expected length for the specified command code.

The `ID` field in the event message specifies the Message ID (in hex), the `CC` field specifies the Command Code (in decimal), the `Exp` Len field specified the Expected Length (in decimal), and `Len` specifies the message Length (in decimal) found in the message.

Definition at line 879 of file cfe_sb_events.h.

39.57.1.35 CFE_SB_MAX_DESTS_MET_EID

```
#define CFE_SB_MAX_DESTS_MET_EID 9
```

`'Subscribe Err:Max Dests(%d)In Use For Msg 0x%x,pipe %s,app %s'`

Event Message `'Subscribe Err:Max Dests(%d)In Use For Msg 0x%x,pipe %s,app %s'`

Type: ERROR

Cause:

This error event message is issued when a subscription request is received and all destinations for that MsgId are in use. The number of destinations per msgid is a configuration parameter named CFE_PLATFORM_SB_MAX_DEST↩_PER_PKT. A destination is defined as a pipe.

Definition at line 314 of file cfe_sb_events.h.

39.57.1.36 CFE_SB_MAX_EID

```
#define CFE_SB_MAX_EID 67
```

Definition at line 43 of file cfe_sb_events.h.

39.57.1.37 CFE_SB_MAX_MSGS_MET_EID

```
#define CFE_SB_MAX_MSGS_MET_EID 8
```

`'Subscribe Err:Max Msgs(%d)In Use,MsgId 0x%x,pipe %s,app %s'`

Event Message `'Subscribe Err:Max Msgs(%d)In Use,MsgId 0x%x,pipe %s,app %s'`

Type: ERROR

Cause:

This error event message is issued when one of the SB subscribe APIs are called with a new MsgId, and SB cannot accommodate the new MsgId because the maximum number of MsgIds are in use. The maximum number of MsgIds is defined by cfg param CFE_PLATFORM_SB_MAX_MSG_IDS. This cfg param dictates the number of elements in the SB routing table. There is one element per MsgId. The user may monitor the routing table utilization figures (msgids currently in use, high water mark and max allowed) by sending the SB cmd to dump the SB statistics data.

Definition at line 300 of file cfe_sb_events.h.

39.57.1.38 CFE_SB_MAX_PIPES_MET_EID

```
#define CFE_SB_MAX_PIPES_MET_EID 3
```

`'CreatePipeErr:Max Pipes(%d)In Use.app %s'`

Event Message `'CreatePipeErr:Max Pipes(%d)In Use.app %s'`

Type: ERROR

Cause:

This error event message is issued when the CFE_SB_CreatePipe API is called and the maximum number of pipes (defined by cfg param CFE_PLATFORM_SB_MAX_PIPES) are in use.

Definition at line 87 of file cfe_sb_events.h.

39.57.1.39 CFE_SB_MSG_TOO_BIG_EID

```
#define CFE_SB_MSG_TOO_BIG_EID 15
```

`'Send Err:Msg Too Big MsgId=0x%x,app=%s,size=%d,MaxSz=%d'`

Event Message `'Send Err:Msg Too Big MsgId=0x%x,app=%s,size=%d,MaxSz=%d'`

Type: ERROR

Cause:

This error event message is issued when a transmit API is called and the packet length field in the message header implies that the message size exceeds the max size defined by mission cfg param CFE_MISSION_SB_MAX_SB_M↩SG_SIZE. The request to send the message is denied, there is no partial packet sent.

Definition at line 398 of file cfe_sb_events.h.

39.57.1.40 CFE_SB_MSGID_LIM_ERR_EID

```
#define CFE_SB_MSGID_LIM_ERR_EID 17
```

```
'Send Err:Msg Limit Err MsgId 0x%x,pipe %s,sender %s'
```

Event Message `'Send Err:Msg Limit Err MsgId 0x%x,pipe %s,sender %s'`

Type: ERROR

Cause:

This error event message is issued when a transmit API cannot route the MsgId (displayed in event) to the pipe (displayed in the event) because the pipe currently contains the maximum number of messages of this type (MsgId). This is typically an indication that the receiver is not reading its pipe fast enough, or at all. A less typical scenerio is that the sender is sending a burst of pkts of this type (or MsgId) and the receiver (owner of 'pipe') cannot keep up. The subscriber of the message dictates this limit count in the 'MsgLim' parameter of the CFE_SB_SubscribeEx API or uses the default value of 4 if using the CFE_SB_Subscribe API.

Definition at line 431 of file cfe_sb_events.h.

39.57.1.41 CFE_SB_PART_SUB_PKT_EID

```
#define CFE_SB_PART_SUB_PKT_EID 45
```

```
'Partial Sub Pkt %d Sent,Entries=%d,Stat=0x%x'
```

Event Message `'Partial Sub Pkt %d Sent,Entries=%d,Stat=0x%x'`

Type: DEBUG

Cause:

This debug event message is issued in response to the 'Send Previous Subscriptions' command and a partial pkt segment is sent.

Definition at line 745 of file cfe_sb_events.h.

39.57.1.42 CFE_SB_PIPE_ADDED_EID

```
#define CFE_SB_PIPE_ADDED_EID 5
```

`'Pipe Created:name %s,id %d,app %s'`

Event Message `'Pipe Created:name %s,id %d,app %s'`

Type: DEBUG

Cause:

This debug event message is issued when a pipe was successfully created in the CFE_SB_CreatePipe API.

Definition at line 113 of file cfe_sb_events.h.

39.57.1.43 CFE_SB_PIPE_DELETED_EID

```
#define CFE_SB_PIPE_DELETED_EID 47
```

`'Pipe Deleted:id %d,owner %s'`

Event Message `'Pipe Deleted:id %d,owner %s'`

Type: DEBUG

Cause:

This debug event message is issued when the CFE_SB_DeletePipe API is called and the request is successfully completed.

Definition at line 769 of file cfe_sb_events.h.

39.57.1.44 CFE_SB_Q_FULL_ERR_EID

```
#define CFE_SB_Q_FULL_ERR_EID 25
```

`'Pipe Overflow,MsgId 0x%x,pipe %s,stat 0x%x,app %s'`

Event Message `'Pipe Overflow,MsgId 0x%x,pipe %s,stat 0x%x,app %s'`

Type: ERROR

Cause:

This error event message is issued when a transmit API is called and encounters an error when attempting to write the msg to the destination pipe (which is an underlying queue). This could indicate that the owner of the pipe is not readings its messages fast enough or at all. It may also mean that the pipe depth is not deep enough. The pipe depth is an input parameter to the CFE_SB_CreatePipe API.

Definition at line 526 of file cfe_sb_events.h.

39.57.1.45 CFE_SB_Q_RD_ERR_EID

```
#define CFE_SB_Q_RD_ERR_EID 27
```

`'Pipe Read Err,pipe %s,app %s,stat 0x%x'`

Event Message `'Pipe Read Err,pipe %s,app %s,stat 0x%x'`

Type: ERROR

Cause:

This error event message is issued when a transmit API is called and encounters an error when attempting to read the msg from the destination pipe (which is an underlying queue). More precisely, the OS API OS_QueueGet has returned an unexpected error. The return code is displayed in the event. For more information, the user may look up the return code in the OSAL documention or source code.

Definition at line 558 of file cfe_sb_events.h.

39.57.1.46 CFE_SB_Q_WR_ERR_EID

```
#define CFE_SB_Q_WR_ERR_EID 26
```

`'Pipe Write Err,MsgId 0x%x,pipe %s,stat 0x%x,app %s'`

Event Message `'Pipe Write Err,MsgId 0x%x,pipe %s,stat 0x%x,app %s'`

Type: ERROR

Cause:

This error event message is issued when a transmit API is called and encounters an error when attempting to write the msg to the destination pipe (which is an underlying queue). More precisely, the OS API OS_QueuePut has returned an unexpected error. The return code is displayed in the event. For more information, the user may look up the return code in the OSAL documention or source code.

Definition at line 542 of file cfe_sb_events.h.

39.57.1.47 CFE_SB_RCV_BAD_ARG_EID

```
#define CFE_SB_RCV_BAD_ARG_EID 18
```

`'Rcv Err:Bad Input Arg:BufPtr 0x%x,pipe %d,t/o %d,app %s'`

Event Message `'Rcv Err:Bad Input Arg:BufPtr 0x%x,pipe %d,t/o %d,app %s'`

Type: ERROR

Cause:

This error event message is issued when an invalid paramter is passed into the CFE_SB_ReceiveBuffer API. Two possibile problems would be the first parameter (∗BufPtr) being NULL or the third paramter (TimeOut) being less than -1.

Definition at line 444 of file cfe_sb_events.h.

39.57.1.48 CFE_SB_SEND_BAD_ARG_EID

```
#define CFE_SB_SEND_BAD_ARG_EID 13
```

```
'Send Err:Bad input argument,Arg 0x%x,App %s'
```

Event Message `'Send Err:Bad input argument,Arg 0x%x,App %s'`

Type: ERROR

Cause:

This error event message is issued when a transmit API receives an invalid (possibly NULL) ptr as an argument.

Definition at line 366 of file cfe_sb_events.h.

39.57.1.49 CFE_SB_SEND_INV_MSGID_EID

```
#define CFE_SB_SEND_INV_MSGID_EID 21
```

```
'Send Err:Invalid msgid in msg,MsgId 0x%x,App %s'
```

Event Message `'Send Err:Invalid msgid in msg,MsgId 0x%x,App %s'`

Type: ERROR

Cause:

This error event message is issued when a transmit API is called and the SB discovers that the message to send has a msg id that is invalid. It may be due to a msg id that is greater than cfg parameter CFE_PLATFORM_SB_HIGHEST_↩ VALID_MSGID

Definition at line 483 of file cfe_sb_events.h.

39.57.1.50 CFE_SB_SEND_NO_SUBS_EID

```
#define CFE_SB_SEND_NO_SUBS_EID 14
```

`'No subscribers for MsgId 0x%x,sender %s'`

Event Message `'No subscribers for MsgId 0x%x,sender %s'`

Type: INFORMATION

Cause:

This info event message is issued when a transmit API is called and there are no subscribers (therefore no destinations) for the message to be sent. Each time the SB detects this situation, the corresponding SB telemetry point is incremented.. NOTE: By default, SB filters this event. The EVS filter algorithm allows the first event to pass through the filter, but all subsequent events with this event id will be filtered. A command must be sent to unfilter this event if the user desires to see it.

Definition at line 384 of file cfe_sb_events.h.

39.57.1.51 CFE_SB_SETPIPEOPTS_EID

```
#define CFE_SB_SETPIPEOPTS_EID 57
```

`'SetPipeOpts:  Options set (%d).  app %s'`

Event Message `'SetPipeOpts:  Options set (%d).  app %s'`

Type: DEBUG

Cause:

This debug event is generated when options are set.

Definition at line 148 of file cfe_sb_events.h.

39.57.1.52 CFE_SB_SETPIPEOPTS_ID_ERR_EID

```
#define CFE_SB_SETPIPEOPTS_ID_ERR_EID 55
```

`'SetPipeOptsErr:Invalid pipe id (%d).app %s'`

Event Message `'SetPipeOptsErr:Invalid pipe id (%d).app %s'`

Type: ERROR

Cause:

This error event message is issued when the CFE_SB_SetPipeOpts API is called and the PipeID is invalid.

Definition at line 125 of file cfe_sb_events.h.

39.57.1.53 CFE_SB_SETPIPEOPTS_OWNER_ERR_EID

```
#define CFE_SB_SETPIPEOPTS_OWNER_ERR_EID 56
```

`'SetPipeOptsErr:Caller not owner (%d).app %s'`

Event Message `'SetPipeOptsErr:Caller not owner (%d).app %s'`

Type: ERROR

Cause:

This error event message is issued when the CFE_SB_SetPipeOpts API is called and the pipe is owned by another app ID.

Definition at line 137 of file cfe_sb_events.h.

39.57.1.54 CFE_SB_SND_RTG_EID

```
#define CFE_SB_SND_RTG_EID 39
```

`'%s written:Size=%d,Entries=%d'`

Event Message `'%s written:Size=%d,Entries=%d'`

Type: DEBUG

Cause:

This debug event message is issued after the SB routing info file, pipe info file or the map info file is written and closed. This is done is response to the SB 'Send Routing Info' cmd, the SB 'Send pipe Info' cmd or the SB 'Send Map Info' cmd, respectively.

Definition at line 684 of file cfe_sb_events.h.

39.57.1.55 CFE_SB_SND_RTG_ERR1_EID

```
#define CFE_SB_SND_RTG_ERR1_EID 40
```

`'Error creating file %s, stat=0x%x'`

Event Message `'Error creating file %s, stat=0x%x'`

Type: ERROR

Cause:

This error event message is issued when the SB 'Send Routing Info' cmd is received and the file create fails. The event displays the status received from the OS.

Definition at line 697 of file cfe_sb_events.h.

39.57.1.56 CFE_SB_SND_STATS_EID

```
#define CFE_SB_SND_STATS_EID 32
```

`'Software Bus Statistics packet sent'`

Event Message `'Software Bus Statistics packet sent'`

Type: DEBUG

Cause:

This debug event message is issued when SB receives a cmd to send the SB statistics pkt.

Definition at line 592 of file cfe_sb_events.h.

39.57.1.57 CFE_SB_SUB_ARG_ERR_EID

```
#define CFE_SB_SUB_ARG_ERR_EID 6
```

`'Subscribe Err:Bad Arg,MsgId 0x%x,PipeId %d,app %s,scope %d'`

Event Message `'Subscribe Err:Bad Arg,MsgId 0x%x,PipeId %d,app %s,scope %d'`

Type: ERROR

Cause:

This error event message is issued when one of the Subscribe API's are called with an invalid MsgId. An invalid MsgId is defined as being greater than the cfg param CFE_PLATFORM_SB_HIGHEST_VALID_MSGID.

Definition at line 265 of file cfe_sb_events.h.

39.57.1.58 CFE_SB_SUB_INV_CALLER_EID

```
#define CFE_SB_SUB_INV_CALLER_EID 51
```

`'Subscribe Err:Caller(%s) is not the owner of pipe %d, Msg=0x%x'`

Event Message `'Subscribe Err:Caller(%s) is not the owner of pipe %d, Msg=0x%x'`

Type: ERROR

Cause:

This error event message is issued when one of the SB subscribe API's are called and the requestor is not the owner of the pipe. Only the owner of the pipe may subscribe to messages on the pipe.

Definition at line 821 of file cfe_sb_events.h.

39.57.1.59 CFE_SB_SUB_INV_PIPE_EID

```
#define CFE_SB_SUB_INV_PIPE_EID 50
```

`'Subscribe Err:Invalid Pipe Id,Msg=0x%x,PipeId=%d,App %s'`

Event Message `'Subscribe Err:Invalid Pipe Id,Msg=0x%x,PipeId=%d,App %s'`

Type: ERROR

Cause:

This error event message is issued when the input PipeId has a value that is not listed in the pipe table. This typically means that the pipe does not exist. The pipe table may be viewed for verification.

Definition at line 808 of file cfe_sb_events.h.

39.57.1.60 CFE_SB_SUBSCRIPTION_RCVD_EID

```
#define CFE_SB_SUBSCRIPTION_RCVD_EID 10
```

`'Subscription Rcvd:MsgId 0x%x on %s(%d),app %s'`

Event Message `'Subscription Rcvd:MsgId 0x%x on %s(%d),app %s'`

Type: DEBUG

Cause:

This debug event message is issued when a subscription is successfully made through one of the SB Subscribe API's

Definition at line 326 of file cfe_sb_events.h.

39.57.1.61 CFE_SB_SUBSCRIPTION_REMOVED_EID

```
#define CFE_SB_SUBSCRIPTION_REMOVED_EID 48
```

`'Subscription Removed:Msg 0x%x on pipe %d,app %s'`

Event Message `'Subscription Removed:Msg 0x%x on pipe %d,app %s'`

Type: DEBUG

Cause:

This debug event message is issued when CFE_SB_Unsubscribe API is called and the request is successfully completed.

Definition at line 781 of file cfe_sb_events.h.

39.57.1.62 CFE_SB_SUBSCRIPTION_RPT_EID

```
#define CFE_SB_SUBSCRIPTION_RPT_EID 22
```

`'Sending Subscription Report Msg=0x%x,Pipe=%d,Stat=0x%x'`

Event Message `'Sending Subscription Report Msg=0x%x,Pipe=%d,Stat=0x%x'`

Type: DEBUG

Cause:

This debug event message is issued when SB subscription reporting is enabled, (which is disabled by default) and a subscription is successfully received.

Definition at line 495 of file cfe_sb_events.h.

39.57.1.63 CFE_SB_UNSUB_ARG_ERR_EID

```
#define CFE_SB_UNSUB_ARG_ERR_EID 11
```

`'UnSubscribe Err:Bad Arg,MsgId 0x%x,PipeId %d,app %s,scope %d'`

Event Message `'UnSubscribe Err:Bad Arg,MsgId 0x%x,PipeId %d,app %s,scope %d'`

Type: ERROR

Cause:

This error event message is issued when a request to unsubscribe fails due to an invalid msgid or an invalid pipeid in one of SB's unsubscribe API's. The msgid must be less than cfg param CFE_PLATFORM_SB_HIGHEST_VALID_MSGID and the pipeid must have been created and have a value less than cfg param CFE_PLATFORM_SB_MAX_PIPES. The SB pipe table may be viewed to verify its value or existence.

Definition at line 341 of file cfe_sb_events.h.

39.57.1.64 CFE_SB_UNSUB_INV_CALLER_EID

```
#define CFE_SB_UNSUB_INV_CALLER_EID 53
```

`'Unsubscribe Err:Caller(%s) is not the owner of pipe %d,Msg=0x%x'`

Event Message `'Unsubscribe Err:Caller(%s) is not the owner of pipe %d,Msg=0x%x'`

Type: ERROR

Cause:

This error event message is issued when one of the SB unsubscribe API's are called and the requestor is not the owner of the pipe (or ES). Only the owner of the pipe(or ES for cleanup purposes)may unsubscribe messages from a pipe.

Definition at line 848 of file cfe_sb_events.h.

39.57.1.65 CFE_SB_UNSUB_INV_PIPE_EID

```
#define CFE_SB_UNSUB_INV_PIPE_EID 52
```

`'Unsubscribe Err:Invalid Pipe Id Msg=0x%x,Pipe=%d,app=%s'`

Event Message `'Unsubscribe Err:Invalid Pipe Id Msg=0x%x,Pipe=%d,app=%s'`

Type: ERROR

Cause:

This error event message is issued when one of the SB unsubscribe API's are called and the input parameter PipeId is not listed in the pipe table. This typically means that the pipe does not exist. The pipe table may be viewed for verification.

Definition at line 835 of file cfe_sb_events.h.

39.57.1.66 CFE_SB_UNSUB_NO_SUBS_EID

```
#define CFE_SB_UNSUB_NO_SUBS_EID 12
```

`'Unsubscribe Err:No subs for Msg 0x%x on %s,app %s'`

Event Message `'Unsubscribe Err:No subs for Msg 0x%x on %s,app %s'`

Type: INFORMATION

Cause:

This info event message is issued when a request to unsubscribe fails due to a non existent msgid/pipeid combination in the SB routing table. The SB routing table may be viewed to see a list of valid msgid/pipeid combinations.

Definition at line 354 of file cfe_sb_events.h.

39.58 cfe/modules/sb/fsw/inc/cfe_sb_msg.h File Reference

```
#include "common_types.h"
#include "cfe_msg_hdr.h"
#include "cfe_sb_extern_typedefs.h"
#include "cfe_es_extern_typedefs.h"
```

Data Structures

- struct CFE_SB_WriteFileInfoCmd_Payload

  Write File Info Command Payload.
- struct CFE_SB_WriteFileInfoCmd

  Write File Info Command.
- struct CFE_SB_RouteCmd_Payload

  Enable/Disable Route Command Payload.
- struct CFE_SB_RouteCmd

  Enable/Disable Route Command.
- struct CFE_SB_HousekeepingTlm_Payload
- struct CFE_SB_HousekeepingTlm
- struct CFE_SB_PipeDepthStats

  SB Pipe Depth Statistics.
- struct CFE_SB_PipeInfoEntry

  SB Pipe Information File Entry.
- struct CFE_SB_StatsTlm_Payload
- struct CFE_SB_StatsTlm

- struct CFE_SB_RoutingFileEntry

  SB Routing File Entry.
- struct CFE_SB_MsgMapFileEntry

  SB Map File Entry.
- struct CFE_SB_SingleSubscriptionTlm_Payload
- struct CFE_SB_SingleSubscriptionTlm
- struct CFE_SB_SubEntries

  SB Previous Subscriptions Entry.
- struct CFE_SB_AllSubscriptionsTlm_Payload
- struct CFE_SB_AllSubscriptionsTlm

Macros

- #define CFE_SB_NOOP_CC 0
- #define CFE_SB_RESET_COUNTERS_CC 1
- #define CFE_SB_SEND_SB_STATS_CC 2
- #define CFE_SB_WRITE_ROUTING_INFO_CC 3
- #define CFE_SB_ENABLE_ROUTE_CC 4
- #define CFE_SB_DISABLE_ROUTE_CC 5
- #define CFE_SB_WRITE_PIPE_INFO_CC 7
- #define CFE_SB_WRITE_MAP_INFO_CC 8
- #define CFE_SB_ENABLE_SUB_REPORTING_CC 9
- #define CFE_SB_DISABLE_SUB_REPORTING_CC 10
- #define CFE_SB_SEND_PREV_SUBS_CC 11

Typedefs

- typedef CFE_MSG_CommandHeader_t CFE_SB_NoopCmd_t
- typedef CFE_MSG_CommandHeader_t CFE_SB_ResetCountersCmd_t
- typedef CFE_MSG_CommandHeader_t CFE_SB_EnableSubReportingCmd_t
- typedef CFE_MSG_CommandHeader_t CFE_SB_DisableSubReportingCmd_t
- typedef CFE_MSG_CommandHeader_t CFE_SB_SendSbStatsCmd_t
- typedef CFE_MSG_CommandHeader_t CFE_SB_SendPrevSubsCmd_t
- typedef struct CFE_SB_WriteFileInfoCmd_Payload CFE_SB_WriteFileInfoCmd_Payload_t

  Write File Info Command Payload.
- typedef struct CFE_SB_WriteFileInfoCmd CFE_SB_WriteFileInfoCmd_t

  Write File Info Command.
- typedef CFE_SB_WriteFileInfoCmd_t CFE_SB_WriteRoutingInfoCmd_t
- typedef CFE_SB_WriteFileInfoCmd_t CFE_SB_WritePipeInfoCmd_t
- typedef CFE_SB_WriteFileInfoCmd_t CFE_SB_WriteMapInfoCmd_t
- typedef struct CFE_SB_RouteCmd_Payload CFE_SB_RouteCmd_Payload_t

  Enable/Disable Route Command Payload.
- typedef struct CFE_SB_RouteCmd CFE_SB_RouteCmd_t

  Enable/Disable Route Command.
- typedef CFE_SB_RouteCmd_t CFE_SB_EnableRouteCmd_t
- typedef CFE_SB_RouteCmd_t CFE_SB_DisableRouteCmd_t
- typedef struct CFE_SB_HousekeepingTlm_Payload CFE_SB_HousekeepingTlm_Payload_t
- typedef struct CFE_SB_HousekeepingTlm CFE_SB_HousekeepingTlm_t

- typedef struct CFE_SB_PipeDepthStats CFE_SB_PipeDepthStats_t

  SB Pipe Depth Statistics.
- typedef struct CFE_SB_PipeInfoEntry CFE_SB_PipeInfoEntry_t

  SB Pipe Information File Entry.
- typedef struct CFE_SB_StatsTlm_Payload CFE_SB_StatsTlm_Payload_t
- typedef struct CFE_SB_StatsTlm CFE_SB_StatsTlm_t
- typedef struct CFE_SB_RoutingFileEntry CFE_SB_RoutingFileEntry_t

  SB Routing File Entry.
- typedef struct CFE_SB_MsgMapFileEntry CFE_SB_MsgMapFileEntry_t

  SB Map File Entry.
- typedef struct CFE_SB_SingleSubscriptionTlm_Payload CFE_SB_SingleSubscriptionTlm_Payload_t
- typedef struct CFE_SB_SingleSubscriptionTlm CFE_SB_SingleSubscriptionTlm_t
- typedef struct CFE_SB_SubEntries CFE_SB_SubEntries_t

  SB Previous Subscriptions Entry.
- typedef struct CFE_SB_AllSubscriptionsTlm_Payload CFE_SB_AllSubscriptionsTlm_Payload_t
- typedef struct CFE_SB_AllSubscriptionsTlm CFE_SB_AllSubscriptionsTlm_t

39.58.1 Macro Definition Documentation

39.58.1.1 CFE_SB_DISABLE_ROUTE_CC

```
#define CFE_SB_DISABLE_ROUTE_CC 5
```

Name Disable Software Bus Route

Description

This command will disable a particular destination. The destination is specified in terms of MsgID and PipeID. The MsgId and PipeID are parmaters in the command. All destinations are enabled by default.

Command Mnemonic(s) $sc_$cpu_SB_DisRoute

Command Structure

CFE_SB_DisableRouteCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_SB_CMDPC`** - command execution counter will increment
- View routing information CFE_SB_WRITE_ROUTING_INFO_CC to verify enable/disable state change
- The CFE_SB_DSBL_RTE2_EID debug event message will be generated. All debug events are filtered by default.
- Destination will stop receiving messages.

Error Conditions

An Error may occur if the MsgId or PipeId parmaters do not pass validation or the destination does not exist.

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_SB_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases. See CFE_SB_DSBL_RTE1_EID or CFE_SB_DSBL_RTE3_EID

Criticality

This command is not intended to be used in nominal conditions. It is possible to get into a state where a destination cannot be re-enabled without reseting the processor. For instance, sending this command with CFE_SB_CMD_MID and the SB_Cmd_Pipe would inhibit any ground commanding to the software bus until the processor was reset. There are similar problems that may occur when using this command.

Definition at line 271 of file cfe_sb_msg.h.

39.58.1.2 CFE_SB_DISABLE_SUB_REPORTING_CC

```
#define CFE_SB_DISABLE_SUB_REPORTING_CC 10
```

Name Disable Subscription Reporting Command

Description

This command will disable subscription reporting and is intended to be used only by the CFS SBN (Software Bus Networking) Application. It is not intended to be sent from the ground or used by operations. When subscription reporting is enabled, SB will generate and send a software bus packet for each subscription received. The software bus packet that is sent contains the information received in the subscription API. This subscription report is neeeded by SBN if offboard routing is required.

Command Mnemonic(s) $sc_$cpu_SB_DisSubRptg

Command Structure

CFE_SB_DisableSubReportingCmd_t

Command Verification

Successful execution of this command will result in the suppression of packets (with the CFE_SB_ONESUB_TLM_MID MsgId) for each subscription received by SB through the subscription APIs.

Error Conditions

None

Criticality

None

See also

CFE_SB_SingleSubscriptionTlm_t, CFE_SB_ENABLE_SUB_REPORTING_CC, CFE_SB_SEND_PREV_SUBS_CC

Definition at line 422 of file cfe_sb_msg.h.

39.58.1.3 CFE_SB_ENABLE_ROUTE_CC

```
#define CFE_SB_ENABLE_ROUTE_CC 4
```

Name Enable Software Bus Route

Description

This command will enable a particular destination. The destination is specified in terms of MsgID and PipeID. The MsgId and PipeID are parmaters in the command. All destinations are enabled by default. This command is needed only after a CFE_SB_DISABLE_ROUTE_CC command is used.

Command Mnemonic(s) $sc_$cpu_SB_EnaRoute

Command Structure

CFE_SB_EnableRouteCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **$sc_$cpu_SB_CMDPC** - command execution counter will increment
- View routing information CFE_SB_WRITE_ROUTING_INFO_CC to verify enable/disable state change
- The CFE_SB_ENBL_RTE2_EID debug event message will be generated. All debug events are filtered by default.
- Destination will begin receiving messages.

Error Conditions

An Error may occur if the MsgId or PipeId parmaters do not pass validation or the destination does not exist.

Evidence of failure may be found in the following telemetry:

- **$sc_$cpu_SB_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases. See CFE_SB_ENBL_RTE1_EID or CFE_SB_ENBL_RTE3_EID

Criticality

This command is not inherently dangerous.

Definition at line 230 of file cfe_sb_msg.h.

39.58.1.4 CFE_SB_ENABLE_SUB_REPORTING_CC

```
#define CFE_SB_ENABLE_SUB_REPORTING_CC 9
```

Name Enable Subscription Reporting Command

Description

This command will enable subscription reporting and is intended to be used only by the CFS SBN (Software Bus Networking) Application. It is not intended to be sent from the ground or used by operations. When subscription reporting is enabled, SB will generate and send a software bus packet for each subscription received. The software bus packet that is sent contains the information received in the subscription API. This subscription report is neeeded by SBN if offboard routing is required.

Command Mnemonic(s) $sc_$cpu_SB_EnaSubRptg

Command Structure

CFE_SB_EnableSubReportingCmd_t

Command Verification

Successful execution of this command will result in the sending of a packet (with the CFE_SB_ONESUB_TLM_MID MsgId) for each subscription received by SB through the subscription APIs.

Error Conditions

None

Criticality

None

See also

CFE_SB_SingleSubscriptionTlm_t, CFE_SB_DISABLE_SUB_REPORTING_CC, CFE_SB_SEND_PREV_SUBS_CC

Definition at line 389 of file cfe_sb_msg.h.

39.58.1.5 CFE_SB_NOOP_CC

```
#define CFE_SB_NOOP_CC 0
```

Name Software Bus No-Op

Description

This command performs no other function than to increment the command execution counter. The command may be used to verify general aliveness of the Software Bus task.

Command Mnemonic(s) $sc_$cpu_SB_NOOP

Command Structure

CFE_SB_NoopCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_SB_CMDPC`** - command execution counter will increment
- The CFE_SB_CMD0_RCVD_EID informational event message will be generated

Error Conditions

There are no error conditions for this command. If the Software Bus receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

None

See also

Definition at line 78 of file cfe_sb_msg.h.

39.58.1.6 CFE_SB_RESET_COUNTERS_CC

```
#define CFE_SB_RESET_COUNTERS_CC 1
```

Name Software Bus Reset Counters

Description

This command resets the following counters within the Software Bus housekeeping telemetry:

- Command Execution Counter ($sc_$cpu_SB_CMDPC)
- Command Error Counter ($sc_$cpu_SB_CMDEC)

Command Mnemonic(s) $sc_$cpu_SB_ResetCtrs

Command Structure

CFE_SB_ResetCountersCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_SB_CMDPC`** - command execution counter will increment
- The CFE_SB_CMD1_RCVD_EID informational event message will be generated

Error Conditions

There are no error conditions for this command. If the Software Bus receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

This command is not inherently dangerous. However, it is possible for ground systems and on-board safing procedures to be designed such that they react to changes in the counter values that are reset by this command.

See also

Definition at line 115 of file cfe_sb_msg.h.

39.58.1.7 CFE_SB_SEND_PREV_SUBS_CC

```
#define CFE_SB_SEND_PREV_SUBS_CC 11
```

Name Send Previous Subscriptions Command

This command generates a series of packets that contain information

regarding all subscriptions previously received by SB. This command is intended to be used only by the CFS S↩BN(Software Bus Networking) Application. It is not intended to be sent from the ground or used by operations. When this command is received the software bus will generate and send a series of packets containing information about all subscription previously received.

Command Mnemonic(s) $sc_$cpu_SB_SendPrevSubs

Command Structure

CFE_SB_SendPrevSubsCmd_t

Command Verification

Successful execution of this command will result in a series of packets (with the CFE_SB_ALLSUBS_TLM_MID MsgId) being sent on the software bus.

Error Conditions

None

Criticality

None

See also

CFE_SB_AllSubscriptionsTlm_t, CFE_SB_ENABLE_SUB_REPORTING_CC, CFE_SB_DISABLE_SUB_REP↩ORTING_CC

Definition at line 454 of file cfe_sb_msg.h.

39.58.1.8 CFE_SB_SEND_SB_STATS_CC

```
#define CFE_SB_SEND_SB_STATS_CC 2
```

Name Send Software Bus Statistics

Description

This command will cause the SB task to send a statistics packet containing current utilization figures and high water marks which may be useful for checking the margin of the SB platform configuration settings.

Command Mnemonic(s) $sc_$cpu_SB_DumpStats

Command Structure

CFE_SB_SendSbStatsCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_SB_CMDPC`** - command execution counter will increment
- Receipt of statistics packet with MsgId CFE_SB_STATS_TLM_MID
- The CFE_SB_SND_STATS_EID debug event message will be generated. All debug events are filtered by default.

Error Conditions

There are no error conditions for this command. If the Software Bus receives the command, the debug event is sent and the counter is incremented unconditionally.

Criticality

This command is not inherently dangerous. It will create and send a message on the software bus. If performed repeatedly, it is possible that receiver pipes may overflow.

See also

Definition at line 150 of file cfe_sb_msg.h.

39.58.1.9 CFE_SB_WRITE_MAP_INFO_CC

```
#define CFE_SB_WRITE_MAP_INFO_CC 8
```

Name Write Map Info to a File

This command will create a file containing the software bus message

map information. The message map is a lookup table (an array of uint16s)that allows fast access to the correct routing table element during a softeware bus send operation. This is diasgnostic information that may be needed due to the dynamic nature of the cFE software bus. An abosulte path and filename may be specified in the command. If this command field contains an empty string (NULL terminator as the first character) the default file path and name is used. The default file path and name is defined in the platform configuration file as CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME.

Command Mnemonic(s) $sc_$cpu_SB_WriteMap2File

Command Structure

CFE_SB_WriteMapInfoCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_SB_CMDPC`** - command execution counter will increment.
- Specified filename created at specified location. See description.
- The CFE_SB_SND_RTG_EID debug event message will be generated. All debug events are filtered by default.

Error Conditions

- Errors may occur during write operations to the file. Possible causes might be insufficient space in the file system or the filename or file path is improperly specified.

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_SB_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases. See CFE_SB_SND_RTG_ERR1_EID and CFE_SB_FILEWRITE_ERR_EID

Criticality

This command is not inherently dangerous. It will create a new file in the file system and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

Definition at line 356 of file cfe_sb_msg.h.

39.58.1.10 CFE_SB_WRITE_PIPE_INFO_CC

```
#define CFE_SB_WRITE_PIPE_INFO_CC 7
```

Name Write Pipe Info to a File

Description

This command will create a file containing the software bus pipe information. The pipe information contains information about every pipe that has been created through the CFE_SB_CreatePipe API. An abosulte path and filename may be specified in the command. If this command field contains an empty string (NULL terminator as the first character) the default file path and name is used. The default file path and name is defined in the platform configuration file as CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME.

Command Mnemonic(s) $sc_$cpu_SB_WritePipe2File

Command Structure

CFE_SB_WritePipeInfoCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_SB_CMDPC`** - command execution counter will increment.
- Specified filename created at specified location. See description.
- The CFE_SB_SND_RTG_EID debug event message will be generated. All debug events are filtered by default.

Error Conditions

- Errors may occur during write operations to the file. Possible causes might be insufficient space in the file system or the filename or file path is improperly specified.

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_SB_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases. See CFE_SB_SND_RTG_ERR1_EID and CFE_SB_FILEWRITE_ERR_EID

Criticality

This command is not inherently dangerous. It will create a new file in the file system and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

Definition at line 313 of file cfe_sb_msg.h.

39.58.1.11 CFE_SB_WRITE_ROUTING_INFO_CC

```
#define CFE_SB_WRITE_ROUTING_INFO_CC 3
```

Name Write Software Bus Routing Info to a File

Description

This command will create a file containing the software bus routing information. The routing information contains information about every subscription that has been received through the SB subscription APIs. An abosulte path and filename may be specified in the command. If this command field contains an empty string (NULL terminator as the first character) the default file path and name is used. The default file path and name is defined in the platform configuration file as CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME.

Command Mnemonic(s) $sc_$cpu_SB_WriteRouting2File

Command Structure

CFE_SB_WriteRoutingInfoCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_SB_CMDPC`** - command execution counter will increment.
- Specified filename created at specified location. See description.
- The CFE_SB_SND_RTG_EID debug event message will be generated. All debug events are filtered by default.

Error Conditions

- Errors may occur during write operations to the file. Possible causes might be insufficient space in the file system or the filename or file path is improperly specified.

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_SB_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases. See CFE_SB_SND_RTG_ERR1_EID and CFE_SB_FILEWRITE_ERR_EID

Criticality

This command is not inherently dangerous. It will create a new file in the file system and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

Definition at line 192 of file cfe_sb_msg.h.

39.58.2 Typedef Documentation

39.58.2.1 CFE_SB_AllSubscriptionsTlm_Payload_t

```
typedef struct CFE_SB_AllSubscriptionsTlm_Payload CFE_SB_AllSubscriptionsTlm_Payload_t
```

Name SB Previous Subscriptions Packet

This structure defines the pkt(s) sent by SB that contains a list of all current subscriptions. This pkt is generated on cmd and intended to be used primarily by the Software Bus Networking Application (SBN). Typically, when the cmd is received there are more subscriptions than can fit in one pkt. The complete list of subscriptions is sent via a series of segmented pkts.

39.58.2.2 CFE_SB_AllSubscriptionsTlm_t

```
typedef struct CFE_SB_AllSubscriptionsTlm CFE_SB_AllSubscriptionsTlm_t
```

39.58.2.3 CFE_SB_DisableRouteCmd_t

```
typedef CFE_SB_RouteCmd_t CFE_SB_DisableRouteCmd_t
```

Definition at line 529 of file cfe_sb_msg.h.

39.58.2.4 CFE_SB_DisableSubReportingCmd_t

```
typedef CFE_MSG_CommandHeader_t CFE_SB_DisableSubReportingCmd_t
```

Definition at line 470 of file cfe_sb_msg.h.

39.58.2.5 CFE_SB_EnableRouteCmd_t

```
typedef CFE_SB_RouteCmd_t CFE_SB_EnableRouteCmd_t
```

Definition at line 528 of file cfe_sb_msg.h.

39.58.2.6 CFE_SB_EnableSubReportingCmd_t

```
typedef CFE_MSG_CommandHeader_t CFE_SB_EnableSubReportingCmd_t
```

Definition at line 469 of file cfe_sb_msg.h.

39.58.2.7 CFE_SB_HousekeepingTlm_Payload_t

```
typedef struct CFE_SB_HousekeepingTlm_Payload CFE_SB_HousekeepingTlm_Payload_t
```

Name Software Bus task housekeeping Packet

39.58.2.8 CFE_SB_HousekeepingTlm_t

```
typedef struct CFE_SB_HousekeepingTlm CFE_SB_HousekeepingTlm_t
```

39.58.2.9 CFE_SB_MsgMapFileEntry_t

```
typedef struct CFE_SB_MsgMapFileEntry CFE_SB_MsgMapFileEntry_t
```

SB Map File Entry.

Structure of one element of the map information in response to CFE_SB_WRITE_MAP_INFO_CC

39.58.2.10 CFE_SB_NoopCmd_t

```
typedef CFE_MSG_CommandHeader_t CFE_SB_NoopCmd_t
```

Definition at line 467 of file cfe_sb_msg.h.

39.58.2.11 CFE_SB_PipeDepthStats_t

```
typedef struct CFE_SB_PipeDepthStats CFE_SB_PipeDepthStats_t
```

SB Pipe Depth Statistics.

Used in SB Statistics Telemetry Packet CFE_SB_StatsTlm_t

39.58.2.12 CFE_SB_PipeInfoEntry_t

```
typedef struct CFE_SB_PipeInfoEntry CFE_SB_PipeInfoEntry_t
```

SB Pipe Information File Entry.

This statistics structure is output as part of the CFE SB "Send Pipe Info" command (CFE_SB_SEND_PIPE_INFO_CC).

Previous versions of CFE simply wrote the internal CFE_SB_PipeD_t object to the file, but this also contains information such as pointers which are not relevant outside the running CFE process.

By defining the pipe info structure separately, it also provides some independence, such that the internal CFE_SB_↩ PipeD_t definition can evolve without changing the binary format of the information file.

39.58.2.13 CFE_SB_ResetCountersCmd_t

```
typedef CFE_MSG_CommandHeader_t CFE_SB_ResetCountersCmd_t
```

Definition at line 468 of file cfe_sb_msg.h.

39.58.2.14 CFE_SB_RouteCmd_Payload_t

```
typedef struct CFE_SB_RouteCmd_Payload CFE_SB_RouteCmd_Payload_t
```

Enable/Disable Route Command Payload.

This structure contains a definition used by two SB commands, 'Enable Route' CFE_SB_ENABLE_ROUTE_CC and 'Disable Route' CFE_SB_DISABLE_ROUTE_CC. A route is the destination pipe for a particular message and is therefore defined as a MsgId and PipeId combination.

39.58.2.15 CFE_SB_RouteCmd_t

```
typedef struct CFE_SB_RouteCmd CFE_SB_RouteCmd_t
```

Enable/Disable Route Command.

39.58.2.16 CFE_SB_RoutingFileEntry_t

```
typedef struct CFE_SB_RoutingFileEntry CFE_SB_RoutingFileEntry_t
```

SB Routing File Entry.

Structure of one element of the routing information in response to CFE_SB_WRITE_ROUTING_INFO_CC

39.58.2.17 CFE_SB_SendPrevSubsCmd_t

```
typedef CFE_MSG_CommandHeader_t CFE_SB_SendPrevSubsCmd_t
```

Definition at line 472 of file cfe_sb_msg.h.

39.58.2.18 CFE_SB_SendSbStatsCmd_t

```
typedef CFE_MSG_CommandHeader_t CFE_SB_SendSbStatsCmd_t
```

Definition at line 471 of file cfe_sb_msg.h.

39.58.2.19 CFE_SB_SingleSubscriptionTlm_Payload_t

```
typedef struct CFE_SB_SingleSubscriptionTlm_Payload CFE_SB_SingleSubscriptionTlm_Payload_t
```

Name SB Subscription Report Packet

This structure defines the pkt sent by SB when a subscription or a request to unsubscribe is received while subscription reporting is enabled. By default subscription reporting is disabled. This feature is intended to be used primarily by Software Bus Networking Application (SBN)

See also

CFE_SB_ENABLE_SUB_REPORTING_CC, CFE_SB_DISABLE_SUB_REPORTING_CC

39.58.2.20 CFE_SB_SingleSubscriptionTlm_t

```
typedef struct CFE_SB_SingleSubscriptionTlm CFE_SB_SingleSubscriptionTlm_t
```

39.58.2.21 CFE_SB_StatsTlm_Payload_t

```
typedef struct CFE_SB_StatsTlm_Payload CFE_SB_StatsTlm_Payload_t
```

Name SB Statistics Telemetry Packet

SB Statistics packet sent in response to CFE_SB_SEND_SB_STATS_CC

39.58.2.22 CFE_SB_StatsTlm_t

```
typedef struct CFE_SB_StatsTlm CFE_SB_StatsTlm_t
```

39.58.2.23 CFE_SB_SubEntries_t

```
typedef struct CFE_SB_SubEntries CFE_SB_SubEntries_t
```

SB Previous Subscriptions Entry.

This structure defines an entry used in the CFE_SB_PrevSubsPkt_t Intended to be used primarily by Software Bus Networking Application (SBN)

Used in structure definition CFE_SB_AllSubscriptionsTlm_t

39.58.2.24 CFE_SB_WriteFileInfoCmd_Payload_t

```
typedef struct CFE_SB_WriteFileInfoCmd_Payload CFE_SB_WriteFileInfoCmd_Payload_t
```

Write File Info Command Payload.

This structure contains a generic definition used by SB commands that write to a file

39.58.2.25 CFE_SB_WriteFileInfoCmd_t

```
typedef struct CFE_SB_WriteFileInfoCmd CFE_SB_WriteFileInfoCmd_t
```

Write File Info Command.

39.58.2.26 CFE_SB_WriteMapInfoCmd_t

```
typedef CFE_SB_WriteFileInfoCmd_t CFE_SB_WriteMapInfoCmd_t
```

Definition at line 498 of file cfe_sb_msg.h.

39.58.2.27 CFE_SB_WritePipeInfoCmd_t

```
typedef CFE_SB_WriteFileInfoCmd_t CFE_SB_WritePipeInfoCmd_t
```

Definition at line 497 of file cfe_sb_msg.h.

39.58.2.28 CFE_SB_WriteRoutingInfoCmd_t

```
typedef CFE_SB_WriteFileInfoCmd_t CFE_SB_WriteRoutingInfoCmd_t
```

Definition at line 496 of file cfe_sb_msg.h.

39.59 cfe/modules/tbl/fsw/inc/cfe_tbl_events.h File Reference

Macros

- #define CFE_TBL_MAX_EID 103

Informational Event Message IDs

- #define CFE_TBL_INIT_INF_EID 1
  'Task Initialized'

Command Response Informational Event Message IDs

- #define CFE_TBL_NOOP_INF_EID 10
  'No-op command'
- #define CFE_TBL_RESET_INF_EID 11
  'Reset Counters command'
- #define CFE_TBL_FILE_LOADED_INF_EID 12
  'Successful load of '%s' into '%s' working buffer'
- #define CFE_TBL_OVERWRITE_DUMP_INF_EID 13
  'Successfully overwrote '%s' with Table '%s''
- #define CFE_TBL_WRITE_DUMP_INF_EID 14
  'Successfully dumped Table '%s' to '%s''
- #define CFE_TBL_OVERWRITE_REG_DUMP_INF_EID 15
  'Successfully overwrote '%s' with Table Registry'
- #define CFE_TBL_VAL_REQ_MADE_INF_EID 16
  'Tbl Services issued validation request for '%s''
- #define CFE_TBL_LOAD_PEND_REQ_INF_EID 17
  'Tbl Services notifying App that '%s' has a load pending'
- #define CFE_TBL_TLM_REG_CMD_INF_EID 18
  'Table Registry entry for '%s' will be telemetered'
- #define CFE_TBL_LOAD_ABORT_INF_EID 21
  'Table Load Aborted for '%s''
- #define CFE_TBL_WRITE_REG_DUMP_INF_EID 22
  'Successfully dumped Table Registry to '%s':Size=%d,Entries=%d'
- #define CFE_TBL_ASSUMED_VALID_INF_EID 23
  'Tbl Services assumes '%s' is valid. No Validation Function has been registered'

Command Error Event Message IDs

- #define CFE_TBL_MID_ERR_EID 50
  'Invalid message ID - ID = 0x%X'
- #define CFE_TBL_CC1_ERR_EID 51
  'Invalid command code - ID = 0x%X, CC = %d'
- #define CFE_TBL_LEN_ERR_EID 52

'Invalid cmd pkt - ID = 0x%X, CC = %d, Len = %d'

- #define CFE_TBL_FILE_ACCESS_ERR_EID 53

  'Unable to open file '%s' for table load, Status = 0x%08X'
- #define CFE_TBL_FILE_STD_HDR_ERR_EID 54

  'Unable to read std header for '%s', Status = 0x%08X'
- #define CFE_TBL_FILE_TBL_HDR_ERR_EID 55

  'Unable to read tbl header for '%s', Status = 0x%08X'
- #define CFE_TBL_FAIL_HK_SEND_ERR_EID 56

  'Unable to send Hk Packet (Status=0x%08X)'
- #define CFE_TBL_NO_SUCH_TABLE_ERR_EID 57

  'Unable to locate '%s' in Table Registry'
- #define CFE_TBL_FILE_TYPE_ERR_EID 58

  'File '%s' is not a cFE file type, ContentType = 0x%08X'
- #define CFE_TBL_FILE_SUBTYPE_ERR_EID 59

  'File subtype for '%s' is wrong. Subtype = 0x%08X'
- #define CFE_TBL_NO_WORK_BUFFERS_ERR_EID 60

  'No working buffers available for table '%s''
- #define CFE_TBL_INTERNAL_ERROR_ERR_EID 61

  'Internal Error (Status=0x%08X)'
- #define CFE_TBL_CREATING_DUMP_FILE_ERR_EID 62

  'Error creating dump file '%s', Status=0x%08X'
- #define CFE_TBL_WRITE_CFE_HDR_ERR_EID 63

  'Error writing cFE File Header to '%s', Status=0x%08X'
- #define CFE_TBL_WRITE_TBL_HDR_ERR_EID 64

  'Error writing Tbl image File Header to '%s', Status=0x%08X'
- #define CFE_TBL_WRITE_TBL_IMG_ERR_EID 65

  'Error writing Tbl image to '%s', Status=0x%08X'
- #define CFE_TBL_NO_INACTIVE_BUFFER_ERR_EID 66

  'No Inactive Buffer for Table '%s' present'
- #define CFE_TBL_TOO_MANY_VALIDATIONS_ERR_EID 67

  'Too many Table Validations have been requested'
- #define CFE_TBL_WRITE_TBL_REG_ERR_EID 68

  'Error writing Registry to '%s', Status=0x%08X'
- #define CFE_TBL_LOAD_ABORT_ERR_EID 69

  'Cannot abort load of '%s'. No load started.'
- #define CFE_TBL_ACTIVATE_ERR_EID 70

  'Cannot activate table '%s'. No Inactive image available'
- #define CFE_TBL_FILE_INCOMPLETE_ERR_EID 71

  'Incomplete load of '%s' into '%s' working buffer'
- #define CFE_TBL_LOAD_EXCEEDS_SIZE_ERR_EID 72

  'Cannot load '%s' (%d) at offset %d in '%s' (%d)'
- #define CFE_TBL_ZERO_LENGTH_LOAD_ERR_EID 73

  'Table Hdr in '%s' indicates no data in file'
- #define CFE_TBL_PARTIAL_LOAD_ERR_EID 74

  ''%s' has partial load for uninitialized table '%s''
- #define CFE_TBL_FILE_TOO_BIG_ERR_EID 75

  'File '%s' has more data than Tbl Hdr indicates (%d)'
- #define CFE_TBL_TOO_MANY_DUMPS_ERR_EID 76

  'Too many Dump Only Table Dumps have been requested'
- #define CFE_TBL_DUMP_PENDING_ERR_EID 77

  'A dump for '%s' is already pending'
- #define CFE_TBL_ACTIVATE_DUMP_ONLY_ERR_EID 78

  'Illegal attempt to activate dump-only table '%s''
- #define CFE_TBL_LOADING_A_DUMP_ONLY_ERR_EID 79

  'Attempted to load DUMP-ONLY table '%s' from '%s''

- #define CFE_TBL_ILLEGAL_BUFF_PARAM_ERR_EID 80

  'Cmd for Table '%s' had illegal buffer parameter (0x%08X)'
- #define CFE_TBL_UNVALIDATED_ERR_EID 81

  'Cannot activate table '%s'. Inactive image not Validated'
- #define CFE_TBL_IN_REGISTRY_ERR_EID 82

  ''%s' found in Table Registry. CDS cannot be deleted until table is unregistered'
- #define CFE_TBL_NOT_CRITICAL_TBL_ERR_EID 83

  'Table '%s' is in Critical Table Registry but CDS is not tagged as a table'
- #define CFE_TBL_NOT_IN_CRIT_REG_ERR_EID 84

  'Table '%s' is not found in Critical Table Registry'
- #define CFE_TBL_CDS_NOT_FOUND_ERR_EID 85

  'Unable to locate '%s' in CDS Registry'
- #define CFE_TBL_CDS_DELETE_ERR_EID 86

  'Error while deleting '%s' from CDS, See SysLog.(Err=0x%08X)'
- #define CFE_TBL_CDS_OWNER_ACTIVE_ERR_EID 87

  'CDS '%s' owning app is still active'
- #define CFE_TBL_LOADING_PENDING_ERR_EID 88

  'Attempted to load table '%s' while previous load is still pending'
- #define CFE_TBL_FAIL_NOTIFY_SEND_ERR_EID 89

  'Manage Notification Pkt Error(MsgId=0x%08X, CC=0x%04X, Param=0x%08X, Status=0x%08X)'

API Informational Event Message IDs

- #define CFE_TBL_LOAD_SUCCESS_INF_EID 35

  'Successfully loaded '%s' from '%s''
- #define CFE_TBL_VALIDATION_INF_EID 36

  '%s validation successful for Inactive '%s''
- #define CFE_TBL_UPDATE_SUCCESS_INF_EID 37

  '%s Successfully Updated '%s''
- #define CFE_TBL_CDS_DELETED_INFO_EID 38

  'Successfully removed '%s' from CDS'

API Error Event Message IDs

- #define CFE_TBL_REGISTER_ERR_EID 90

  '%s Failed to Register '%s', Status=0x%08X'
- #define CFE_TBL_SHARE_ERR_EID 91

  '%s Failed to Share '%s', Status=0x%08X'
- #define CFE_TBL_UNREGISTER_ERR_EID 92

  '%s Failed to Unregister '%s', Status=0x%08X'
- #define CFE_TBL_LOAD_VAL_ERR_EID 93
- #define CFE_TBL_LOAD_TYPE_ERR_EID 94

  '%s Failed to Load '%s' (Invalid Source Type)"
- #define CFE_TBL_UPDATE_ERR_EID 95

  '%s Failed to Update '%s', Status=0x%08X"
- #define CFE_TBL_VALIDATION_ERR_EID 96

  '%s validation failed for Inactive '%s', Status=0x%08X'
- #define CFE_TBL_SPACECRAFT_ID_ERR_EID 97

  'Unable to verify Spacecraft ID for '%s', ID = 0x%08X'
- #define CFE_TBL_PROCESSOR_ID_ERR_EID 98

  'Unable to verify Processor ID for '%s', ID = 0x%08X'
- #define CFE_TBL_LOAD_DUMPONLY_ERR_EID 99

  Attempted to load Dump Only Tbl 's'
- #define CFE_TBL_LOAD_IN_PROGRESS_ERR_EID 100

Load already in progress for 's'

- #define CFE_TBL_LOAD_FILENAME_LONG_ERR_EID 101

  Filename is too long ('s' (lu) > lu)

- #define CFE_TBL_LOAD_TBLNAME_MISMATCH_ERR_EID 102

  table name mismatch (exp=s, tblfilhdr=s)

- #define CFE_TBL_HANDLE_ACCESS_ERR_EID 103

  No access to Tbl handle=d

39.59.1 Macro Definition Documentation

39.59.1.1 CFE_TBL_ACTIVATE_DUMP_ONLY_ERR_EID

```
#define CFE_TBL_ACTIVATE_DUMP_ONLY_ERR_EID 78
```

`'Illegal attempt to activate dump-only table '%s''`

Event Message `'Illegal attempt to activate dump-only table '%s''`

Type: ERROR

Cause:

This event message is generated when a Table Activate command for a Dump-Only Table was received. By definition, Dump-Only tables are not allowed to be loaded with any new data.

Definition at line 692 of file cfe_tbl_events.h.

39.59.1.2 CFE_TBL_ACTIVATE_ERR_EID

```
#define CFE_TBL_ACTIVATE_ERR_EID 70
```

`'Cannot activate table '%s'.  No Inactive image available'`

Event Message `'Cannot activate table '%s'.  No Inactive image available'`

Type: ERROR

Cause:

This event message is generated when an Activate Table command is received and the command specified table does not currently have an inactive buffer associated with it.

Definition at line 585 of file cfe_tbl_events.h.

39.59.1.3 CFE_TBL_ASSUMED_VALID_INF_EID

```
#define CFE_TBL_ASSUMED_VALID_INF_EID 23
```

```
'Tbl Services assumes '%s' is valid.  No Validation Function has been registered'
```

Event Message `'Tbl Services assumes '%s' is valid.  No Validation Function has been registered'`

Type: INFORMATION

Cause:

This event message is generated when Table Services has received a Validation Command for a table that never specified a Validation Function when it was registered via the CFE_TBL_Register API.

Definition at line 240 of file cfe_tbl_events.h.

39.59.1.4 CFE_TBL_CC1_ERR_EID

```
#define CFE_TBL_CC1_ERR_EID 51
```

```
'Invalid command code - ID = 0x%X, CC = %d'
```

Event Message `'Invalid command code - ID = 0x%X, CC = %d'`

Type: ERROR

Cause:

This event message is generated when a message with the CFE_TBL_CMD_MID message ID has arrived but whose Command Code is not one of the command codes specified in cfe_tbl_msg.h . This problem is most likely to occur when:

1. A Message ID meant for another Application became corrupted and was set equal to CFE_TBL_CMD_MID.
2. The Command Code field in the Message became corrupted.
3. The command database at the ground station has been corrupted.

The `ID` field in the event message specifies the Message ID (in hex) and the `CC` field specifies the Command Code (in decimal) found in the message.

Definition at line 283 of file cfe_tbl_events.h.

39.59.1.5 CFE_TBL_CDS_DELETE_ERR_EID

```
#define CFE_TBL_CDS_DELETE_ERR_EID 86
```

```
'Error while deleting '%s' from CDS, See SysLog.(Err=0x%08X)'
```

Event Message `'Error while deleting '%s' from CDS, See SysLog.(Err=0x%08X)'`

Type: ERROR

Cause:

This event message is generated when an unexpected error was encountered during the deletion of the CDS. The System Log should have more precise information on the nature of the error.

Definition at line 798 of file cfe_tbl_events.h.

39.59.1.6 CFE_TBL_CDS_DELETED_INFO_EID

```
#define CFE_TBL_CDS_DELETED_INFO_EID 38
```

```
'Successfully removed '%s' from CDS'
```

Event Message `'Successfully removed '%s' from CDS'`

Type: INFORMATION

Cause:

This event message is generated when a Critical Table's CDS has been successfully deleted.

Definition at line 894 of file cfe_tbl_events.h.

39.59.1.7 CFE_TBL_CDS_NOT_FOUND_ERR_EID

```
#define CFE_TBL_CDS_NOT_FOUND_ERR_EID 85
```

`'Unable to locate '%s' in CDS Registry'`

Event Message `'Unable to locate '%s' in CDS Registry'`

Type: ERROR

Cause:

This event message is generated when a Table Delete Critical Data Store command is received specifying a table name that WAS found in the Critical Table Registry but its associated entry in the Critical Data Store Registry was not found. Somehow the two entities have become out of synch.

Definition at line 786 of file cfe_tbl_events.h.

39.59.1.8 CFE_TBL_CDS_OWNER_ACTIVE_ERR_EID

```
#define CFE_TBL_CDS_OWNER_ACTIVE_ERR_EID 87
```

`'CDS '%s' owning app is still active'`

Event Message `'CDS '%s' owning app is still active'`

Type: ERROR

Cause:

This event message is generated when an attempt is made to delete a CDS while an application with the same name as the CDS Prefix is still registered in the system. Owning applications must not be active before an associated CDS can be deleted.

Definition at line 811 of file cfe_tbl_events.h.

39.59.1.9 CFE_TBL_CREATING_DUMP_FILE_ERR_EID

```
#define CFE_TBL_CREATING_DUMP_FILE_ERR_EID 62
```

`'Error creating dump file '%s', Status=0x%08X'`

Event Message `'Error creating dump file '%s', Status=0x%08X'`

Type: ERROR

Cause:

This event message is generated when a Table Dump or Table Registry Dump command was received and the cFE Table Services is unable to create the specified file.

The `Status` field provides the return status from the OS_OpenCreate function call.

Definition at line 471 of file cfe_tbl_events.h.

39.59.1.10 CFE_TBL_DUMP_PENDING_ERR_EID

```
#define CFE_TBL_DUMP_PENDING_ERR_EID 77
```

`'A dump for '%s' is already pending'`

Event Message `'A dump for '%s' is already pending'`

Type: ERROR

Cause:

This event message is generated when a Table Dump command for a Dump-Only Table was received and Table Services hasn't finished processing the previous Table Dump command for the same Table.

Definition at line 680 of file cfe_tbl_events.h.

39.59.1.11 CFE_TBL_FAIL_HK_SEND_ERR_EID

```
#define CFE_TBL_FAIL_HK_SEND_ERR_EID 56
```

`'Unable to send Hk Packet (Status=0x%08X)'`

Event Message `'Unable to send Hk Packet (Status=0x%08X)'`

Type: ERROR

Cause:

This event message is generated when failure occurs while attempting to send the Housekeeping Message over the Software Bus.

The `Status` field of the event message contains the error code.

Definition at line 368 of file cfe_tbl_events.h.

39.59.1.12 CFE_TBL_FAIL_NOTIFY_SEND_ERR_EID

```
#define CFE_TBL_FAIL_NOTIFY_SEND_ERR_EID 89
```

`'Manage Notification Pkt Error(MsgId=0x%08X, CC=0x%04X, Param=0x%08X, Status=0x%08X)'`

Event Message `'Manage Notification Pkt Error(MsgId=0x%08X, CC=0x%04X, Param=0x%08X, Status=0x%08X)'`

Type: ERROR

Cause:

This event message is generated when a table management notification message fails to be sent via the software bus.

The `MsgId` is the message ID of the table management notification message that was attempted to be sent, the `CC` is the command code, the `Param` is the application specified command parameter and the `Status` is the error code returned.

Definition at line 841 of file cfe_tbl_events.h.

39.59.1.13 CFE_TBL_FILE_ACCESS_ERR_EID

```
#define CFE_TBL_FILE_ACCESS_ERR_EID 53
```

`'Unable to open file '%s' for table load, Status = 0x%08X'`

Event Message `'Unable to open file '%s' for table load, Status = 0x%08X'`

Type: ERROR

Cause:

This event message is generated upon receipt of a Load Table command when the specified file containing the table image to be loaded cannot be opened. Possible causes for this are:

1. The filename was misspelled
2. The path to the file was incorrect
3. The length (including terminator) of the filename and/or path exceeds the allowable length (see OS_MAX_PATH_LEN and OS_MAX_FILE_NAME, respectively)

The `Status` field in the event message indicates the error code returned by the OS_OpenCreate API.

Definition at line 322 of file cfe_tbl_events.h.

39.59.1.14 CFE_TBL_FILE_INCOMPLETE_ERR_EID

```
#define CFE_TBL_FILE_INCOMPLETE_ERR_EID 71
```

`'Incomplete load of '%s' into '%s' working buffer'`

Event Message `'Incomplete load of '%s' into '%s' working buffer'`

Type: ERROR

Cause:

This event message is generated when a Load Table command is received and the Table Services is unable to load the number of bytes specified in the Table Image Header of the command specified file from the file into the Inactive Buffer.

Definition at line 598 of file cfe_tbl_events.h.

39.59.1.15 CFE_TBL_FILE_LOADED_INF_EID

```
#define CFE_TBL_FILE_LOADED_INF_EID 12
```

`'Successful load of '%s' into '%s' working buffer'`

Event Message `'Successful load of '%s' into '%s' working buffer'`

Type: INFORMATION

Cause:

This event message is always generated after a successful execution of a cFE Table Services Load Table command

Definition at line 108 of file cfe_tbl_events.h.

39.59.1.16 CFE_TBL_FILE_STD_HDR_ERR_EID

```
#define CFE_TBL_FILE_STD_HDR_ERR_EID 54
```

`'Unable to read std header for '%s', Status = 0x%08X'`

Event Message `'Unable to read std header for '%s', Status = 0x%08X'`

Type: ERROR

Cause:

This event message is generated when a read failure occurs during the reading of the cFE Standard File Header of a table image file specified either by an Application calling the CFE_TBL_Load API or in response to a command to Table Services requesting a table image file be loaded into an inactive buffer.

The `Status` field of the event message contains the error code returned by CFE_FS_ReadHeader.

Definition at line 338 of file cfe_tbl_events.h.

39.59.1.17 CFE_TBL_FILE_SUBTYPE_ERR_EID

```
#define CFE_TBL_FILE_SUBTYPE_ERR_EID 59
```

`'File subtype for '%s' is wrong.  Subtype = 0x%08X'`

Event Message `'File subtype for '%s' is wrong.  Subtype = 0x%08X'`

Type: ERROR

Cause:

This event message is generated when either an Application calls the CFE_TBL_Load API or a Table Load command has been received and the specified file has a cFE Standard File Header whose Sub Type is not equal to the expected CFE_FS_SubType_TBL_IMG. Most likely causes for this are:

1. The specified file is not a cFE table image file.
2. The specified file has been created with bad "endianess" (headers should always conform to a big endian format).
3. The specified file has become corrupted.

The `SubType` field specified in the event message contains the sub type that was found in the specified file.

Definition at line 427 of file cfe_tbl_events.h.

39.59.1.18 CFE_TBL_FILE_TBL_HDR_ERR_EID

```
#define CFE_TBL_FILE_TBL_HDR_ERR_EID 55
```

`'Unable to read tbl header for '%s', Status = 0x%08X'`

Event Message `'Unable to read tbl header for '%s', Status = 0x%08X'`

Type: ERROR

Cause:

This event message is generated when a read failure occurs during the reading of the cFE Table File Secondary Header of a table image file specified either by an Application calling the CFE_TBL_Load API or in response to a command to Table Services requesting a table image file be loaded into an inactive buffer.

The `Status` field of the event message contains the error code returned by OS_read.

Definition at line 354 of file cfe_tbl_events.h.

39.59.1.19 CFE_TBL_FILE_TOO_BIG_ERR_EID

```
#define CFE_TBL_FILE_TOO_BIG_ERR_EID 75
```

`'File '%s' has more data than Tbl Hdr indicates (%d)'`

Event Message `'File '%s' has more data than Tbl Hdr indicates (%d)'`

Type: ERROR

Cause:

This event message is generated when a Load Table command is received and Table Services is able to locate more data in the specified Table Image file than the Table Header claims is present.

Definition at line 654 of file cfe_tbl_events.h.

39.59.1.20 CFE_TBL_FILE_TYPE_ERR_EID

```
#define CFE_TBL_FILE_TYPE_ERR_EID 58
```

`'File '%s' is not a cFE file type, ContentType = 0x%08X'`

Event Message `'File '%s' is not a cFE file type, ContentType = 0x%08X'`

Type: ERROR

Cause:

This event message is generated when either an Application calls the CFE_TBL_Load API or a Table Load command has been received and the specified file has a cFE Standard File Header whose Content Type is not equal to the expected CFE_FS_FILE_CONTENT_ID. Most likely causes for this are:

1. The specified file is not a cFE compatible file.
2. The specified file has been created with bad "endianess" (headers should always conform to a big endian format).
3. The specified file has become corrupted.

The `ContentType` field specified in the event message contains the content type that was found in the specified file.

Definition at line 406 of file cfe_tbl_events.h.

39.59.1.21 CFE_TBL_HANDLE_ACCESS_ERR_EID

```
#define CFE_TBL_HANDLE_ACCESS_ERR_EID 103
```

```
No access to Tbl handle=d
```

Event Message `No access to Tbl handle=d`

Type: ERROR

Cause:

The application ID does not have access to the table handle.

Definition at line 1096 of file cfe_tbl_events.h.

39.59.1.22 CFE_TBL_ILLEGAL_BUFF_PARAM_ERR_EID

```
#define CFE_TBL_ILLEGAL_BUFF_PARAM_ERR_EID 80
```

```
'Cmd for Table '%s' had illegal buffer parameter (0x%08X)'
```

Event Message `'Cmd for Table '%s' had illegal buffer parameter (0x%08X)'`

Type: ERROR

Cause:

This event message is generated when either a Table Validate command or a Table Dump Command contains a buffer identifier that does not equal either of the valid values (see CFE_TBL_DumpCmd_Payload_t::ActiveTableFlag or CFE_TBL_ValidateCmd_Payload_t::ActiveTableFlag)

The parameter in the Event Message indicates (in hex) the value found for the ActiveTableFlag in the command.

Definition at line 719 of file cfe_tbl_events.h.

39.59.1.23 CFE_TBL_IN_REGISTRY_ERR_EID

```
#define CFE_TBL_IN_REGISTRY_ERR_EID 82
```

```
''%s' found in Table Registry.  CDS cannot be deleted until table is unregistered'
```

Event Message `''%s' found in Table Registry.  CDS cannot be deleted until table is unregistered'`

Type: ERROR

Cause:

This event message is generated when a Table Delete Critical Data Store command is received specifying a Table Image that is still registered. Critical Table Images cannot be removed from the CDS until the table is first removed from the Registry. Unload the owning application and try again.

Definition at line 746 of file cfe_tbl_events.h.

39.59.1.24 CFE_TBL_INIT_INF_EID

```
#define CFE_TBL_INIT_INF_EID 1
```

```
'Task Initialized'
```

Event Message `'Task Initialized'`

Type: INFORMATION

Cause:

This event message is always automatically issued when the Table Services Task completes its Initialization.

Definition at line 68 of file cfe_tbl_events.h.

39.59.1.25 CFE_TBL_INTERNAL_ERROR_ERR_EID

```
#define CFE_TBL_INTERNAL_ERROR_ERR_EID 61
```

`'Internal Error (Status=0x%08X)'`

Event Message `'Internal Error (Status=0x%08X)'`

Type: ERROR

Cause:

This event message is generated when a Table Load command was issued and the cFE Table Services is unable to allocate a working table buffer for an unexpected reason.

The `Status` field provides the return status from the function that was to provide a working buffer.

Definition at line 457 of file cfe_tbl_events.h.

39.59.1.26 CFE_TBL_LEN_ERR_EID

```
#define CFE_TBL_LEN_ERR_EID 52
```

`'Invalid cmd pkt - ID = 0x%X, CC = %d, Len = %d'`

Event Message `'Invalid cmd pkt - ID = 0x%X, CC = %d, Len = %d'`

Type: ERROR

Cause:

This event message is generated when a message with the CFE_TBL_CMD_MID message ID has arrived but whose packet length does not match the expected length for the specified command code.

The `ID` field in the event message specifies the Message ID (in hex), the `CC` field specifies the Command Code (in decimal) and `Len` specifies the message Length (in decimal) found in the message.

Definition at line 300 of file cfe_tbl_events.h.

39.59.1.27 CFE_TBL_LOAD_ABORT_ERR_EID

```
#define CFE_TBL_LOAD_ABORT_ERR_EID 69
```

```
'Cannot abort load of '%s'.  No load started.'
```

Event Message `'Cannot abort load of '%s'.  No load started.'`

Type: ERROR

Cause:

This event message is generated when an Abort Load command is received and the command specified table is not currently in the process of being loaded.

Definition at line 573 of file cfe_tbl_events.h.

39.59.1.28 CFE_TBL_LOAD_ABORT_INF_EID

```
#define CFE_TBL_LOAD_ABORT_INF_EID 21
```

```
'Table Load Aborted for '%s''
```

Event Message `'Table Load Aborted for '%s''`

Type: INFORMATION

Cause:

This event message is generated upon successful execution of a cFE Table Services Abort Table Load command .

Definition at line 212 of file cfe_tbl_events.h.

39.59.1.29 CFE_TBL_LOAD_DUMPONLY_ERR_EID

```
#define CFE_TBL_LOAD_DUMPONLY_ERR_EID 99
```

```
Attempted to load Dump Only Tbl 's'
```

Event Message `Attempted to load Dump Only Tbl 's'`

Type: ERROR

Cause:

This event message is generated when an application attempts to load a dump-only table.

Definition at line 1051 of file cfe_tbl_events.h.

39.59.1.30 CFE_TBL_LOAD_EXCEEDS_SIZE_ERR_EID

```
#define CFE_TBL_LOAD_EXCEEDS_SIZE_ERR_EID 72
```

```
'Cannot load '%s' (%d) at offset %d in '%s' (%d)'
```

Event Message `'Cannot load '%s' (%d) at offset %d in '%s' (%d)'`

Type: ERROR

Cause:

This event message is generated when a Load Table command is received and the Table Header in the specified Table Image file identifies a number of bytes with a specified starting offset that would exceed the size of the specified table. For example, if a table had 10 bytes and the Table Header indicated that the Table Image in the file contains 7 bytes that starts at offset 5, then the data content would have exceeded the 10 byte limit of the table.

The numbers in parenthesis in the event message text indicate the data size (in bytes) for the specified load file and the registered size for the specified table.

Definition at line 616 of file cfe_tbl_events.h.

39.59.1.31 CFE_TBL_LOAD_FILENAME_LONG_ERR_EID

```
#define CFE_TBL_LOAD_FILENAME_LONG_ERR_EID 101
```

`Filename is too long ('s' (lu) > lu)`

Event Message `Filename is too long ('s' (lu) > lu)`

Type: ERROR

Cause:

The filename provided for the table file is too long (exceeding OS_MAX_PATH_LEN - 1).

Definition at line 1074 of file cfe_tbl_events.h.

39.59.1.32 CFE_TBL_LOAD_IN_PROGRESS_ERR_EID

```
#define CFE_TBL_LOAD_IN_PROGRESS_ERR_EID 100
```

`Load already in progress for 's'`

Event Message `Load already in progress for 's'`

Type: ERROR

Cause:

This event message is generated when an application attempts to load a table already in progress. Likely due to a race condition.

Definition at line 1063 of file cfe_tbl_events.h.

39.59.1.33 CFE_TBL_LOAD_PEND_REQ_INF_EID

```
#define CFE_TBL_LOAD_PEND_REQ_INF_EID 17
```

```
'Tbl Services notifying App that '%s' has a load pending'
```

Event Message `'Tbl Services notifying App that '%s' has a load pending'`

Type: DEBUG

Cause:

This event message is generated upon successful execution of a cFE Table Services Activate Table command . It should be noted, however, that this Event Message does *NOT* indicate completion of the Table Activation. It is *ONLY* indicating that the appropriate flag has been set to *NOTIFY* the table's owning Application that an Update has been requested. Completion of the Update is indicated by either the CFE_TBL_UPDATE_SUCCESS_INF_EID or CFE_TBL_UPDATE_ERR_EID event messages.

Definition at line 187 of file cfe_tbl_events.h.

39.59.1.34 CFE_TBL_LOAD_SUCCESS_INF_EID

```
#define CFE_TBL_LOAD_SUCCESS_INF_EID 35
```

```
'Successfully loaded '%s' from '%s''
```

Event Message `'Successfully loaded '%s' from '%s''`

Type: DEBUG (the first time) and INFORMATION (normally)

Cause:

This event message is generated when a Table is successfully updated by its owning Application with the contents of the Application specified file or memory area. This Event Message only appears when an Application successfully calls the CFE_TBL_Load API.

Definition at line 858 of file cfe_tbl_events.h.

39.59.1.35 CFE_TBL_LOAD_TBLNAME_MISMATCH_ERR_EID

```
#define CFE_TBL_LOAD_TBLNAME_MISMATCH_ERR_EID 102
```

`table name mismatch (exp=s, tblfilhdr=s)`

Event Message `table name mismatch (exp=s, tblfilhdr=s)`

Type: ERROR

Cause:

The table name in the table file header does not match the specified table name.

Definition at line 1085 of file cfe_tbl_events.h.

39.59.1.36 CFE_TBL_LOAD_TYPE_ERR_EID

```
#define CFE_TBL_LOAD_TYPE_ERR_EID 94
```

`'%s Failed to Load '%s' (Invalid Source Type)"`

Event Message `'%s Failed to Load '%s' (Invalid Source Type)"`

Type: ERROR

Cause:

This event message is generated when an Application calls CFE_TBL_Load with a bad value for the `SrcType` parameter. The `SrcType` must be one of the values specified by CFE_TBL_SrcEnum_t.

Definition at line 954 of file cfe_tbl_events.h.

39.59.1.37 CFE_TBL_LOAD_VAL_ERR_EID

```
#define CFE_TBL_LOAD_VAL_ERR_EID 93
```

Definition at line 942 of file cfe_tbl_events.h.

39.59.1.38 CFE_TBL_LOADING_A_DUMP_ONLY_ERR_EID

```
#define CFE_TBL_LOADING_A_DUMP_ONLY_ERR_EID 79
```

```
'Attempted to load DUMP-ONLY table '%s' from '%s''
```

Event Message `'Attempted to load DUMP-ONLY table '%s' from '%s''`

Type: ERROR

Cause:

This event message is generated when a Table Load command for a Dump-Only Table was received. By definition, Dump-Only tables are not allowed to be loaded with any new data.

Definition at line 704 of file cfe_tbl_events.h.

39.59.1.39 CFE_TBL_LOADING_PENDING_ERR_EID

```
#define CFE_TBL_LOADING_PENDING_ERR_EID 88
```

```
'Attempted to load table '%s' while previous load is still pending'
```

Event Message `'Attempted to load table '%s' while previous load is still pending'`

Type: ERROR

Cause:

This event message is generated when an attempt is made to load a table while a previous load is still pending. The most likely cause of this is the owning application is waiting for an appropriate time to load the table with the specified contents. In order to override this load, the user would be required to issue the Abort Load Command .

Definition at line 825 of file cfe_tbl_events.h.

39.59.1.40 CFE_TBL_MAX_EID

```
#define CFE_TBL_MAX_EID 103
```

Definition at line 49 of file cfe_tbl_events.h.

39.59.1.41 CFE_TBL_MID_ERR_EID

```
#define CFE_TBL_MID_ERR_EID 50
```

`'Invalid message ID - ID = 0x%X'`

Event Message `'Invalid message ID - ID = 0x%X'`

Type: ERROR

Cause:

This event message is generated when a message has arrived on the cFE Table Services Application's Message Pipe that has a Message ID that is neither CFE_TBL_SEND_HK_MID or CFE_TBL_CMD_MID. Most likely, the cFE Software Bus routing table has become corrupt and is sending messages targeted for other Applications to the cFE Table Services Application.

The `ID` field in the event message identifies the message ID (in hex) that was found in the message.

Definition at line 262 of file cfe_tbl_events.h.

39.59.1.42 CFE_TBL_NO_INACTIVE_BUFFER_ERR_EID

```
#define CFE_TBL_NO_INACTIVE_BUFFER_ERR_EID 66
```

`'No Inactive Buffer for Table '%s' present'`

Event Message `'No Inactive Buffer for Table '%s' present'`

Type: ERROR

Cause:

This event message is generated when a Table Dump or a Table Validate command for an Inactive Table Buffer was received and there isn't an Inactive Table Buffer associated with the specified Table.

Definition at line 528 of file cfe_tbl_events.h.

39.59.1.43 CFE_TBL_NO_SUCH_TABLE_ERR_EID

```
#define CFE_TBL_NO_SUCH_TABLE_ERR_EID 57
```

`'Unable to locate '%s' in Table Registry'`

Event Message `'Unable to locate '%s' in Table Registry'`

Type: ERROR

Cause:

This event message is generated when a command that specifies a table name has a table name that is not found in the Table Registry. Most likely causes for this are:

1. Table name was misspelled in the command.
2. The Application that Registered the Table has either failed to run or has been terminated thus removing the Table from the Registry.
3. The Table Registry has become corrupted.

Definition at line 385 of file cfe_tbl_events.h.

39.59.1.44 CFE_TBL_NO_WORK_BUFFERS_ERR_EID

```
#define CFE_TBL_NO_WORK_BUFFERS_ERR_EID 60
```

`'No working buffers available for table '%s''`

Event Message `'No working buffers available for table '%s''`

Type: ERROR

Cause:

This event message is generated when either a Table Load Command for a Single Buffered Table or a Table Dump Command for a Dump Only Table has been sent AND there are no Shared Buffers available to hold either the load image or the dump image. To free a Shared Buffer, either a previously loaded table image must be activated or aborted OR the operator has to wait for previously dumped Dump Only tables have had a chance to be written to a file (which occurs whenever the cFE Table Services receives a Housekeeping Request).

Definition at line 443 of file cfe_tbl_events.h.

39.59.1.45 CFE_TBL_NOOP_INF_EID

```
#define CFE_TBL_NOOP_INF_EID 10
```

`'No-op command'`

Event Message `'No-op command'`

Type: INFORMATION

Cause:

This event message is always automatically issued in response to a cFE Table Services NO-OP command

Definition at line 84 of file cfe_tbl_events.h.

39.59.1.46 CFE_TBL_NOT_CRITICAL_TBL_ERR_EID

```
#define CFE_TBL_NOT_CRITICAL_TBL_ERR_EID 83
```

`'Table '%s' is in Critical Table Registry but CDS is not tagged as a table'`

Event Message `'Table '%s' is in Critical Table Registry but CDS is not tagged as a table'`

Type: ERROR

Cause:

This event message is generated when a Table Delete Critical Data Store command is received specifying a CDS name for a Critical Data Store that is NOT a critical table image. To delete CDSs that are not Critical Table Images, the Executive Services command CFE_ES_DELETE_CDS_CC must be used.

Definition at line 759 of file cfe_tbl_events.h.

39.59.1.47 CFE_TBL_NOT_IN_CRIT_REG_ERR_EID

```
#define CFE_TBL_NOT_IN_CRIT_REG_ERR_EID 84
```

`'Table '%s' is not found in Critical Table Registry'`

Event Message `'Table '%s' is not found in Critical Table Registry'`

Type: ERROR

Cause:

This event message is generated when a Table Delete Critical Data Store command is received specifying a table name that cannot be found in the Critical Table Registry. If a Critical Data Store exists with the specified name, then the Critical Table Registry has somehow gotten out of sync with the CDS. Otherwise, the likely cause of this error is a misspelled table name in the command.

Definition at line 773 of file cfe_tbl_events.h.

39.59.1.48 CFE_TBL_OVERWRITE_DUMP_INF_EID

```
#define CFE_TBL_OVERWRITE_DUMP_INF_EID 13
```

`'Successfully overwrote '%s' with Table '%s''`

Event Message `'Successfully overwrote '%s' with Table '%s''`

Type: INFORMATION

Cause:

This event message is always generated after a successful execution of a cFE Table Services Dump Table command where the command specified target filename was the same as a file already present in the onboard filesystem. If the specified file did not exist, the event message would have been CFE_TBL_WRITE_DUMP_INF_EID.

Definition at line 123 of file cfe_tbl_events.h.

39.59.1.49 CFE_TBL_OVERWRITE_REG_DUMP_INF_EID

```
#define CFE_TBL_OVERWRITE_REG_DUMP_INF_EID 15
```

```
'Successfully overwrote '%s' with Table Registry'
```

Event Message `'Successfully overwrote '%s' with Table Registry'`

Type: DEBUG

Cause:

This event message is always generated after a successful execution of a cFE Table Services Dump Table Registry command where the command specified target filename was the same as a file already present in the onboard filesystem. If the specified file did not exist, the event message would have been CFE_TBL_WRITE_REG_DUMP_INF_EID.

Definition at line 153 of file cfe_tbl_events.h.

39.59.1.50 CFE_TBL_PARTIAL_LOAD_ERR_EID

```
#define CFE_TBL_PARTIAL_LOAD_ERR_EID 74
```

```
''%s' has partial load for uninitialized table '%s''
```

Event Message `''%s' has partial load for uninitialized table '%s''`

Type: ERROR

Cause:

This event message is generated when a Load Table command is received and the Table Header in the specified Table Image file indicates the starting offset for the table is non-zero and the table has never been previously, completely loaded. Partial Table loads are only allowed after the table has had a successful load.

Definition at line 642 of file cfe_tbl_events.h.

39.59.1.51 CFE_TBL_PROCESSOR_ID_ERR_EID

```
#define CFE_TBL_PROCESSOR_ID_ERR_EID 98
```

`'Unable to verify Processor ID for '%s', ID = 0x%08X'`

Event Message `'Unable to verify Processor ID for '%s', ID = 0x%08X'`

Type: ERROR

Cause:

This event message is generated when either an Application calls the CFE_TBL_Load API or a Table Load command has been received and the specified table file has failed Processor ID validation. Verification of Processor ID in table files is enabled/disabled via CFE_PLATFORM_TBL_VALID_PRID_COUNT, defined in the platform configuration header file. This event message can only be generated if CFE_PLATFORM_TBL_VALID_PRID_COUNT has a non-zero value and the table file has a cFE Standard File Header whose Processor ID does not match one of the values defined for Processor ID verification in the platform config file. The most likely causes for this error are:

1. The specified table file is not intended for this processor.
2. The specified table file has been created with bad "endianess" (headers should always conform to a big endian format).
3. The specified table file has become corrupted.
4. The definition for CFE_PLATFORM_TBL_VALID_PRID_COUNT is not large enough to include all of the valid Processor ID entries in the platform config file.
5. There is no entry for this Processor ID in the platform config file list of valid Processor ID's.

The `ID` field specified in the event message contains the Processor ID that was found in the specified table file.

Definition at line 1040 of file cfe_tbl_events.h.

39.59.1.52 CFE_TBL_REGISTER_ERR_EID

```
#define CFE_TBL_REGISTER_ERR_EID 90
```

`'%s Failed to Register '%s', Status=0x%08X'`

Event Message `'%s Failed to Register '%s', Status=0x%08X'`

Type: ERROR

Cause:

This event message is generated when an Application calls CFE_TBL_Register unsuccessfully.

The `Status` field of the Event Message can be used to identify the reason for the failure by looking it up in the cfe_error.h file

Definition at line 911 of file cfe_tbl_events.h.

39.59.1.53 CFE_TBL_RESET_INF_EID

```
#define CFE_TBL_RESET_INF_EID 11
```

`'Reset Counters command'`

Event Message `'Reset Counters command'`

Type: INFORMATION

Cause:

This event message is always automatically issued in response to a cFE Table Services Reset Counters command

Definition at line 96 of file cfe_tbl_events.h.

39.59.1.54 CFE_TBL_SHARE_ERR_EID

```
#define CFE_TBL_SHARE_ERR_EID 91
```

`'%s Failed to Share '%s', Status=0x%08X'`

Event Message `'%s Failed to Share '%s', Status=0x%08X'`

Type: ERROR

Cause:

This event message is generated when an Application calls CFE_TBL_Share unsuccessfully.

The `Status` field of the Event Message can be used to identify the reason for the failure by looking it up in the cfe_error.h file

Definition at line 925 of file cfe_tbl_events.h.

39.59.1.55 CFE_TBL_SPACECRAFT_ID_ERR_EID

```
#define CFE_TBL_SPACECRAFT_ID_ERR_EID 97
```

`'Unable to verify Spacecraft ID for '%s', ID = 0x%08X'`

Event Message `'Unable to verify Spacecraft ID for '%s', ID = 0x%08X'`

Type: ERROR

Cause:

This event message is generated when either an Application calls the CFE_TBL_Load API or a Table Load command has been received and the specified table file has failed Spacecraft ID validation. Verification of Spacecraft ID in table files is enabled/disabled via CFE_PLATFORM_TBL_VALID_SCID_COUNT, defined in the platform configuration header file. This event message can only be generated if CFE_PLATFORM_TBL_VALID_SCID_COUNT has a non-zero value and the table file has a cFE Standard File Header whose Spacecraft ID does not match one of the values defined for Spacecraft ID verification in the platform config file. The most likely causes for this error are:

1. The specified table file is not intended for this spacecraft.
2. The specified table file has been created with bad "endianess" (headers should always conform to a big endian format).
3. The specified table file has become corrupted.
4. The definition for CFE_PLATFORM_TBL_VALID_SCID_COUNT is not large enough to include all of the valid Spacecraft ID entries in the platform config file.
5. There is no entry for this Spacecraft ID in the platform config file list of valid Spacecraft ID's.

The `ID` field specified in the event message contains the Spacecraft ID that was found in the specified table file.

Definition at line 1012 of file cfe_tbl_events.h.

39.59.1.56 CFE_TBL_TLM_REG_CMD_INF_EID

```
#define CFE_TBL_TLM_REG_CMD_INF_EID 18
```

`'Table Registry entry for '%s' will be telemetered'`

Event Message `'Table Registry entry for '%s' will be telemetered'`

Type: DEBUG

Cause:

This event message is generated upon successful execution of a cFE Table Services Telemeter Table Registry Entry command . Subsequent Table Services Housekeeping Telemetry should contain the desired Table Registry Entry data.

Definition at line 200 of file cfe_tbl_events.h.

39.59.1.57 CFE_TBL_TOO_MANY_DUMPS_ERR_EID

```
#define CFE_TBL_TOO_MANY_DUMPS_ERR_EID 76
```

`'Too many Dump Only Table Dumps have been requested'`

Event Message `'Too many Dump Only Table Dumps have been requested'`

Type: ERROR

Cause:

This event message is generated when a Table Dump command for a Dump-Only Table was received and there are no more free Dump Only Control Blocks available. The number of simultaneous Dump Only Tables that can be pending is specified by the configuration parameter CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS which is found in the cfe_platform_cfg.h file.

Definition at line 668 of file cfe_tbl_events.h.

39.59.1.58 CFE_TBL_TOO_MANY_VALIDATIONS_ERR_EID

```
#define CFE_TBL_TOO_MANY_VALIDATIONS_ERR_EID 67
```

`'Too many Table Validations have been requested'`

Event Message `'Too many Table Validations have been requested'`

Type: ERROR

Cause:

This event message is generated when a Table Validate command was received and there are no more free Validation Result Blocks available. The number of simultaneous validations that can be pending is specified by the configuration parameter CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS which is found in the cfe_platform_cfg.h file.

Validation Commands lock one of the Validation Result Blocks upon receipt of the validation command until the result of the Validation, performed by the table's owning Application, has been reported in a Table Services Housekeeping Request Message.

Definition at line 546 of file cfe_tbl_events.h.

39.59.1.59 CFE_TBL_UNREGISTER_ERR_EID

```
#define CFE_TBL_UNREGISTER_ERR_EID 92
```

```
'%s Failed to Unregister '%s', Status=0x%08X'
```

Event Message `'%s Failed to Unregister '%s', Status=0x%08X'`

Type: ERROR

Cause:

This event message is generated when an Application calls CFE_TBL_Unregister unsuccessfully.

The `Status` field of the Event Message can be used to identify the reason for the failure by looking it up in the cfe_error.h file

Definition at line 939 of file cfe_tbl_events.h.

39.59.1.60 CFE_TBL_UNVALIDATED_ERR_EID

```
#define CFE_TBL_UNVALIDATED_ERR_EID 81
```

```
'Cannot activate table '%s'.  Inactive image not Validated'
```

Event Message `'Cannot activate table '%s'.  Inactive image not Validated'`

Type: ERROR

Cause:

This event message is generated when a Table Activate command is received specifying a Table Image that has not been Validated. If a table has a validation function associated with it (as defined by the owning Application when the Table is first Registered), then the Inactive Image MUST be successfully Validated prior to Activation.

Definition at line 733 of file cfe_tbl_events.h.

39.59.1.61 CFE_TBL_UPDATE_ERR_EID

```
#define CFE_TBL_UPDATE_ERR_EID 95
```

```
'%s Failed to Update '%s', Status=0x%08X"
```

Event Message `'%s Failed to Update '%s', Status=0x%08X"`

Type: ERROR

Cause:

This event message is generated when an Application calls CFE_TBL_Update (or, via an internal call, the CFE_TBL↩_Manage) API and the Table fails to properly update.

The `Status` parameter in the Event Message can be used to identify the reason for the failure by looking it up in the cfe_error.h file.

Definition at line 969 of file cfe_tbl_events.h.

39.59.1.62 CFE_TBL_UPDATE_SUCCESS_INF_EID

```
#define CFE_TBL_UPDATE_SUCCESS_INF_EID 37
```

```
'%s Successfully Updated '%s''
```

Event Message `'%s Successfully Updated '%s''`

Type: INFORMATION

Cause:

This event message is generated when a Table's Active Buffer is successfully updated with the contents of its Inactive Buffer.

Definition at line 883 of file cfe_tbl_events.h.

39.59.1.63 CFE_TBL_VAL_REQ_MADE_INF_EID

```
#define CFE_TBL_VAL_REQ_MADE_INF_EID 16
```

```
'Tbl Services issued validation request for '%s''
```

Event Message `'Tbl Services issued validation request for '%s''`

Type: DEBUG

Cause:

This event message is generated upon successful execution of a cFE Table Services Validate Table command . It should be noted, however, that this Event Message does *NOT* indicate completion of the Table Validation. It is *ONLY* indicating that the appropriate flag has been set to *NOTIFY* the table's owning Application that a Validation has been requested. Completion of the Validation is indicated by either the CFE_TBL_VALIDATION_INF_EID or CFE_TBL_VALIDATION↩_ERR_EID event messages.

Definition at line 170 of file cfe_tbl_events.h.

39.59.1.64 CFE_TBL_VALIDATION_ERR_EID

```
#define CFE_TBL_VALIDATION_ERR_EID 96
```

```
'%s validation failed for Inactive '%s', Status=0x%08X'
```

Event Message `'%s validation failed for Inactive '%s', Status=0x%08X'`

Type: ERROR

Cause:

This event message is generated when an Application calls CFE_TBL_Validate (or, via an internal call, the CFE_TBL↩_Manage) API and the Table fails its Validation.

The `Status` parameter in the Event Message contains the status code returned by the Table's Validation function as defined by the owning Application when the Table was Registered.

Definition at line 984 of file cfe_tbl_events.h.

39.59.1.65 CFE_TBL_VALIDATION_INF_EID

```
#define CFE_TBL_VALIDATION_INF_EID 36
```

```
'%s validation successful for Inactive '%s''
```

Event Message `'%s validation successful for Inactive '%s''`

Type: INFORMATION

Cause:

This event message is generated when a Table Image is successfully validated by its owning Application via the Validation function specified by the owning Application when the table was first registered.

Definition at line 871 of file cfe_tbl_events.h.

39.59.1.66 CFE_TBL_WRITE_CFE_HDR_ERR_EID

```
#define CFE_TBL_WRITE_CFE_HDR_ERR_EID 63
```

```
'Error writing cFE File Header to '%s', Status=0x%08X'
```

Event Message `'Error writing cFE File Header to '%s', Status=0x%08X'`

Type: ERROR

Cause:

This event message is generated when a Table Dump or Table Registry Dump command was received and the cFE Table Services is unable to write the standard cFE File Header to the specified file.

The `Status` field provides the return status from the CFE_FS_WriteHeader function call.

Definition at line 486 of file cfe_tbl_events.h.

39.59.1.67 CFE_TBL_WRITE_DUMP_INF_EID

```
#define CFE_TBL_WRITE_DUMP_INF_EID 14
```

```
'Successfully dumped Table '%s' to '%s''
```

Event Message `'Successfully dumped Table '%s' to '%s''`

Type: INFORMATION

Cause:

This event message is always generated after a successful execution of a cFE Table Services Dump Table command where the command specified target filename was a currently non-existent file. If the file did already exist, the event message would have been CFE_TBL_OVERWRITE_DUMP_INF_EID.

Definition at line 138 of file cfe_tbl_events.h.

39.59.1.68 CFE_TBL_WRITE_REG_DUMP_INF_EID

```
#define CFE_TBL_WRITE_REG_DUMP_INF_EID 22
```

```
'Successfully dumped Table Registry to '%s':Size=%d,Entries=%d'
```

Event Message `'Successfully dumped Table Registry to '%s':Size=%d,Entries=%d'`

Type: DEBUG

Cause:

This event message is always generated after a successful execution of a cFE Table Services Dump Table Registry command where the command specified target filename was a currently non-existent file. If the file did already exist, the event message would have been CFE_TBL_OVERWRITE_REG_DUMP_INF_EID.

Definition at line 227 of file cfe_tbl_events.h.

39.59.1.69 CFE_TBL_WRITE_TBL_HDR_ERR_EID

```
#define CFE_TBL_WRITE_TBL_HDR_ERR_EID 64
```

`'Error writing Tbl image File Header to '%s', Status=0x%08X'`

Event Message `'Error writing Tbl image File Header to '%s', Status=0x%08X'`

Type: ERROR

Cause:

This event message is generated when a Table Dump command was received and the cFE Table Services is unable to write the standard cFE Table Image Header to the specified file.

The `Status` field provides the return status from the OS_write function call.

Definition at line 500 of file cfe_tbl_events.h.

39.59.1.70 CFE_TBL_WRITE_TBL_IMG_ERR_EID

```
#define CFE_TBL_WRITE_TBL_IMG_ERR_EID 65
```

`'Error writing Tbl image to '%s', Status=0x%08X'`

Event Message `'Error writing Tbl image to '%s', Status=0x%08X'`

Type: ERROR

Cause:

This event message is generated when a Table Dump command was received and the cFE Table Services is unable to write the contents of the specified Table image to the specified file.

The `Status` field provides the return status from the OS_write function call.

Definition at line 515 of file cfe_tbl_events.h.

39.59.1.71 CFE_TBL_WRITE_TBL_REG_ERR_EID

```
#define CFE_TBL_WRITE_TBL_REG_ERR_EID 68
```

`'Error writing Registry to '%s', Status=0x%08X'`

Event Message `'Error writing Registry to '%s', Status=0x%08X'`

Type: ERROR

Cause:

This event message is generated when a Table Registry Dump command was received and the cFE Table Services is unable to write the entire contents of the Table Registry to the specified file.

The `Status` field provides the return status from the OS_write function call.

Definition at line 561 of file cfe_tbl_events.h.

39.59.1.72 CFE_TBL_ZERO_LENGTH_LOAD_ERR_EID

```
#define CFE_TBL_ZERO_LENGTH_LOAD_ERR_EID 73
```

`'Table Hdr in '%s' indicates no data in file'`

Event Message `'Table Hdr in '%s' indicates no data in file'`

Type: ERROR

Cause:

This event message is generated when a Load Table command is received and the Table Header in the specified Table Image file claims the file contains no data.

Definition at line 628 of file cfe_tbl_events.h.

39.60 cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h File Reference

```
#include "common_types.h"
#include "cfe_msg_hdr.h"
#include "cfe_tbl_extern_typedefs.h"
#include "cfe_time_extern_typedefs.h"
```

Data Structures

- struct CFE_TBL_NoArgsCmd

  Generic "no arguments" command.
- struct CFE_TBL_LoadCmd_Payload

  Load Table Command Payload.
- struct CFE_TBL_LoadCmd

  Load Table Command.
- struct CFE_TBL_DumpCmd_Payload

  Dump Table Command Payload.
- struct CFE_TBL_DumpCmd
- struct CFE_TBL_ValidateCmd_Payload

  Validate Table Command Payload.
- struct CFE_TBL_ValidateCmd

  Validate Table Command.
- struct CFE_TBL_ActivateCmd_Payload

  Activate Table Command Payload.
- struct CFE_TBL_ActivateCmd

  Activate Table Command.
- struct CFE_TBL_DumpRegistryCmd_Payload

  Dump Registry Command Payload.
- struct CFE_TBL_DumpRegistryCmd

  Dump Registry Command.
- struct CFE_TBL_SendRegistryCmd_Payload

  Send Table Registry Command Payload.
- struct CFE_TBL_SendRegistryCmd

  Send Table Registry Command.
- struct CFE_TBL_DelCDSCmd_Payload

  Delete Critical Table CDS Command Payload.
- struct CFE_TBL_DeleteCDSCmd

  Delete Critical Table CDS Command.
- struct CFE_TBL_AbortLoadCmd_Payload

  Abort Load Command Payload.
- struct CFE_TBL_AbortLoadCmd

  Abort Load Command.
- struct CFE_TBL_NotifyCmd_Payload

  Table Management Notification Command Payload.
- struct CFE_TBL_NotifyCmd
- struct CFE_TBL_HousekeepingTlm_Payload
- struct CFE_TBL_HousekeepingTlm
- struct CFE_TBL_TblRegPacket_Payload
- struct CFE_TBL_TableRegistryTlm

Macros

Table Services Command Codes

- #define CFE_TBL_NOOP_CC 0
- #define CFE_TBL_RESET_COUNTERS_CC 1
- #define CFE_TBL_LOAD_CC 2
- #define CFE_TBL_DUMP_CC 3
- #define CFE_TBL_VALIDATE_CC 4
- #define CFE_TBL_ACTIVATE_CC 5
- #define CFE_TBL_DUMP_REGISTRY_CC 6
- #define CFE_TBL_SEND_REGISTRY_CC 7
- #define CFE_TBL_DELETE_CDS_CC 8
- #define CFE_TBL_ABORT_LOAD_CC 9

Typedefs

- typedef struct CFE_TBL_NoArgsCmd CFE_TBL_NoArgsCmd_t

  Generic "no arguments" command.
- typedef CFE_TBL_NoArgsCmd_t CFE_TBL_NoopCmd_t
- typedef CFE_TBL_NoArgsCmd_t CFE_TBL_ResetCountersCmd_t
- typedef struct CFE_TBL_LoadCmd_Payload CFE_TBL_LoadCmd_Payload_t

  Load Table Command Payload.
- typedef struct CFE_TBL_LoadCmd CFE_TBL_LoadCmd_t

  Load Table Command.
- typedef struct CFE_TBL_DumpCmd_Payload CFE_TBL_DumpCmd_Payload_t

  Dump Table Command Payload.
- typedef struct CFE_TBL_DumpCmd CFE_TBL_DumpCmd_t
- typedef struct CFE_TBL_ValidateCmd_Payload CFE_TBL_ValidateCmd_Payload_t

  Validate Table Command Payload.
- typedef struct CFE_TBL_ValidateCmd CFE_TBL_ValidateCmd_t

  Validate Table Command.
- typedef struct CFE_TBL_ActivateCmd_Payload CFE_TBL_ActivateCmd_Payload_t

  Activate Table Command Payload.
- typedef struct CFE_TBL_ActivateCmd CFE_TBL_ActivateCmd_t

  Activate Table Command.
- typedef struct CFE_TBL_DumpRegistryCmd_Payload CFE_TBL_DumpRegistryCmd_Payload_t

  Dump Registry Command Payload.
- typedef struct CFE_TBL_DumpRegistryCmd CFE_TBL_DumpRegistryCmd_t

  Dump Registry Command.
- typedef struct CFE_TBL_SendRegistryCmd_Payload CFE_TBL_SendRegistryCmd_Payload_t

  Send Table Registry Command Payload.
- typedef struct CFE_TBL_SendRegistryCmd CFE_TBL_SendRegistryCmd_t

  Send Table Registry Command.
- typedef struct CFE_TBL_DelCDSCmd_Payload CFE_TBL_DelCDSCmd_Payload_t

  Delete Critical Table CDS Command Payload.
- typedef struct CFE_TBL_DeleteCDSCmd CFE_TBL_DeleteCDSCmd_t

  Delete Critical Table CDS Command.
- typedef struct CFE_TBL_AbortLoadCmd_Payload CFE_TBL_AbortLoadCmd_Payload_t

Abort Load Command Payload.

- typedef struct CFE_TBL_AbortLoadCmd CFE_TBL_AbortLoadCmd_t

  Abort Load Command.

- typedef struct CFE_TBL_NotifyCmd_Payload CFE_TBL_NotifyCmd_Payload_t

  Table Management Notification Command Payload.

- typedef struct CFE_TBL_NotifyCmd CFE_TBL_NotifyCmd_t
- typedef struct CFE_TBL_HousekeepingTlm_Payload CFE_TBL_HousekeepingTlm_Payload_t
- typedef struct CFE_TBL_HousekeepingTlm CFE_TBL_HousekeepingTlm_t
- typedef struct CFE_TBL_TblRegPacket_Payload CFE_TBL_TblRegPacket_Payload_t
- typedef struct CFE_TBL_TableRegistryTlm CFE_TBL_TableRegistryTlm_t

39.60.1 Macro Definition Documentation

39.60.1.1 CFE_TBL_ABORT_LOAD_CC

```
#define CFE_TBL_ABORT_LOAD_CC 9
```

Name Abort Table Load

Description

This command will cause Table Services to discard the contents of a table buffer that was previously loaded with the data in a file as specified by a Table Load command. For single buffered tables, the allocated shared working buffer is freed and becomes available for other Table Load commands.

Command Mnemonic(s) $sc_$cpu_TBL_LOADABORT

Command Structure

CFE_TBL_AbortLoadCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TBL_CMDPC`** - command execution counter will increment
- The CFE_TBL_LOAD_ABORT_INF_EID informational event message is generated
- If the load was aborted for a single buffered table, the **`$sc_$cpu_TBL_NumFreeShrBuf` telemetry** point should increment

Error Conditions

This command may fail for the following reason(s):

- The specified table name was not found in the table registry.
- The specified table did not have a load in progress to be aborted.

Evidence of failure may be found in the following telemetry:

- **$sc_$cpu_TBL_CMDEC** - command error counter will increment
- Error specific event message

Criticality

This command will cause the loss of data put into an inactive table buffer.

See also

CFE_TBL_LOAD_CC, CFE_TBL_DUMP_CC, CFE_TBL_VALIDATE_CC, CFE_TBL_ACTIVATE_CC

Definition at line 479 of file cfe_tbl_msg.h.

39.60.1.2 CFE_TBL_ACTIVATE_CC

```
#define CFE_TBL_ACTIVATE_CC 5
```

Name Activate Table

Description

This command will cause Table Services to notify a table's owner that an update is pending. The owning application will then update the contents of the active table buffer with the contents of the associated inactive table buffer at a time of their convenience.

Command Mnemonic(s) $sc_$cpu_TBL_ACTIVATE

Command Structure

CFE_TBL_ActivateCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **$sc_$cpu_TBL_CMDPC** - command execution counter will increment
- The CFE_TBL_UPDATE_SUCCESS_INF_EID informational event message will be generated

Error Conditions

This command may fail for the following reason(s):

- A single buffered table's inactive buffer was requested to be dumped and no such buffer is currently allocated.
- The specified table name was not found in the table registry.

Evidence of failure may be found in the following telemetry:

- **$sc_$cpu_TBL_CMDEC** - command error counter will increment
- Command specific error event message are issued for all error cases

Criticality

This command will cause the contents of the specified table to be updated with the contents in the inactive table buffer.

See also

CFE_TBL_LOAD_CC, CFE_TBL_DUMP_CC, CFE_TBL_VALIDATE_CC, CFE_TBL_ABORT_LOAD_CC

Definition at line 319 of file cfe_tbl_msg.h.

39.60.1.3 CFE_TBL_DELETE_CDS_CC

```
#define CFE_TBL_DELETE_CDS_CC 8
```

Name Delete Critical Table from Critical Data Store

Description

This command will delete the Critical Data Store (CDS) associated with the specified Critical Table. Note that any table still present in the Table Registry is unable to be deleted from the Critical Data Store. All Applications that are accessing the critical table must release and unregister their access before the CDS can be deleted.

Command Mnemonic(s) $sc_$cpu_TBL_DeleteCDS

Command Structure

CFE_TBL_DeleteCDSCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TBL_CMDPC`** - command execution counter will increment
- The CFE_TBL_CDS_DELETED_INFO_EID informational event message will be generated

Error Conditions

This command may fail for the following reason(s):

- The specified table name was not found in the critical data store registry
- The specified table name WAS found in the table registry (all registrations/sharing of the table must be unregistered before the table's CDS can be deleted)
- The table's owning application is still active

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_TBL_CMDEC`** - command error counter will increment
- Error specific event message

Criticality

This command will cause the loss of the specified table's contents before the owning Application was terminated.

See also

CFE_ES_DUMP_CDS_REGISTRY_CC, CFE_ES_DELETE_CDS_CC

Definition at line 440 of file cfe_tbl_msg.h.

39.60.1.4 CFE_TBL_DUMP_CC

```
#define CFE_TBL_DUMP_CC 3
```

Name Dump Table

Description

This command will cause the Table Services to put the contents of the specified table buffer into the command specified file.

Command Mnemonic(s) $sc_$cpu_TBL_DUMP

Command Structure

CFE_TBL_DumpCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TBL_CMDPC`** - command execution counter will increment
- Either the CFE_TBL_OVERWRITE_DUMP_INF_EID OR the CFE_TBL_WRITE_DUMP_INF_EID informational event message will be generated

Error Conditions

This command may fail for the following reason(s):

- A single buffered table's inactive buffer was requested to be dumped and no such buffer is currently allocated.
- Error occurred during write operation to file. Possible causes might be insufficient space in the file system or the filename or file path is improperly specified.
- The specified table name was not found in the table registry.

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_TBL_CMDEC`** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous. It will create a new file in the file system and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

See also

CFE_TBL_LOAD_CC, CFE_TBL_VALIDATE_CC, CFE_TBL_ACTIVATE_CC, CFE_TBL_ABORT_LOAD_CC

Definition at line 222 of file cfe_tbl_msg.h.

39.60.1.5 CFE_TBL_DUMP_REGISTRY_CC

```
#define CFE_TBL_DUMP_REGISTRY_CC 6
```

Name Dump Table Registry

Description

This command will cause Table Services to write some of the contents of the Table Registry to the command specified file. This allows the operator to see the current state and configuration of all tables that have been registered with the cFE.

Command Mnemonic(s) $sc_$cpu_TBL_WriteReg2File

Command Structure

CFE_TBL_DumpRegistryCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **$sc_$cpu_TBL_CMDPC** - command execution counter will increment
- The generation of either CFE_TBL_OVERWRITE_REG_DUMP_INF_EID or CFE_TBL_WRITE_REG_DUMP_INF_EID debug event messages
- The specified file should appear (or be updated) at the specified location in the file system

Error Conditions

This command may fail for the following reason(s):

- Error occurred during write operation to file. Possible causes might be insufficient space in the file system or the filename or file path is improperly specified.

Evidence of failure may be found in the following telemetry:

- **$sc_$cpu_TBL_CMDEC** - command error counter will increment
- An Error specific event message

Criticality

This command is not inherently dangerous. It will create a new file in the file system and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

See also

CFE_TBL_SEND_REGISTRY_CC

Definition at line 361 of file cfe_tbl_msg.h.

39.60.1.6 CFE_TBL_LOAD_CC

```
#define CFE_TBL_LOAD_CC 2
```

Name Load Table

Description

This command loads the contents of the specified file into an inactive buffer for the table specified within the file.

Command Mnemonic(s) $sc_$cpu_TBL_Load

Command Structure

CFE_TBL_LoadCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TBL_CMDPC`** - command execution counter will increment
- The CFE_TBL_FILE_LOADED_INF_EID informational event message will be generated

Error Conditions

This command can fail for the following reasons:

- Table name found in table image file's table header is not found in table registry (ie - The table associated with the table image in the file has not been registered by an application).
- The table image file's header indicates the file contains 'x' number of bytes of data but the file contains less.
- No working buffers are available for the load. This would indicate that too many single-buffered table loads are in progress at the same time.
- The table image file's header indicates the data to be loaded is beyond the size of the table. Either the number of bytes in the file are too many or the starting offset into the table is too high.
- The table image file's header indicates there is no data in the file (ie - Number of bytes to load is zero).
- An attempt is being made to load an uninitialized table with a file containing only a partial table image.
- The table image file was unable to be opened. Either the file does not exist at the specified location, the filename is in error, or the file system has been corrupted.

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_TBL_CMDEC`** - command error counter will increment
- Command specific error event messages are issued for all error cases

Criticality

This command is not inherently dangerous. It is performing the first step of loading a table and can be aborted (using the Abort Table Load command described below) without affecting the contents of the active table image.

See also

CFE_TBL_DUMP_CC, CFE_TBL_VALIDATE_CC, CFE_TBL_ACTIVATE_CC, CFE_TBL_ABORT_LOAD_CC

Definition at line 179 of file cfe_tbl_msg.h.

39.60.1.7 CFE_TBL_NOOP_CC

```
#define CFE_TBL_NOOP_CC 0
```

Name Table No-Op

Description

This command performs no other function than to increment the command execution counter. The command may be used to verify general aliveness of the Table Services task.

Command Mnemonic(s) $sc_$cpu_TBL_NOOP

Command Structure

CFE_TBL_NoopCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TBL_CMDPC`** - command execution counter will increment
- The CFE_TBL_NOOP_INF_EID informational event message will be generated

Error Conditions

There are no error conditions for this command. If the Table Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

None

See also

Definition at line 85 of file cfe_tbl_msg.h.

39.60.1.8 CFE_TBL_RESET_COUNTERS_CC

```
#define CFE_TBL_RESET_COUNTERS_CC 1
```

Name Table Reset Counters

Description

This command resets the following counters within the Table Services housekeeping telemetry:

- Command Execution Counter ($sc_$cpu_TBL_CMDPC)
- Command Error Counter ($sc_$cpu_TBL_CMDEC)
- Successful Table Validations Counter ($sc_$cpu_TBL_ValSuccessCtr)
- Failed Table Validations Counter ($sc_$cpu_TBL_ValFailedCtr)
- Number of Table Validations Requested ($sc_$cpu_TBL_ValReqCtr)

Command Mnemonic(s) $sc_$cpu_TBL_ResetCtrs

Command Structure

CFE_TBL_ResetCountersCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TBL_CMDPC`** - command execution counter will increment
- The CFE_TBL_RESET_INF_EID debug event message will be generated

Error Conditions

There are no error conditions for this command. If the Table Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

This command is not inherently dangerous. However, it is possible for ground systems and on-board safing procedures to be designed such that they react to changes in the counter values that are reset by this command.

See also

Definition at line 125 of file cfe_tbl_msg.h.

39.60.1.9 CFE_TBL_SEND_REGISTRY_CC

```
#define CFE_TBL_SEND_REGISTRY_CC 7
```

Name Telemeter One Table Registry Entry

Description

This command will cause Table Services to telemeter the contents of the Table Registry for the command specified table.

Command Mnemonic(s) $sc_$cpu_TBL_TLMReg

Command Structure

CFE_TBL_SendRegistryCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TBL_CMDPC`** - command execution counter will increment
- Receipt of a Table Registry Info Packet (see CFE_TBL_TableRegistryTlm_t)
- The CFE_TBL_TLM_REG_CMD_INF_EID debug event message will be generated

Error Conditions

This command may fail for the following reason(s):

- The specified table name was not found in the table registry.

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_TBL_CMDEC`** - command error counter will increment
- Error specific event message

Criticality

This command is not inherently dangerous. It will generate additional telemetry.

See also

CFE_TBL_DUMP_REGISTRY_CC

Definition at line 396 of file cfe_tbl_msg.h.

39.60.1.10 CFE_TBL_VALIDATE_CC

```
#define CFE_TBL_VALIDATE_CC 4
```

Name Validate Table

Description

This command will cause Table Services to calculate the Data Integrity Value for the specified table and to notify the owning application that the table's validation function should be executed. The results of both the Data Integrity Value computation and the validation function are reported in Table Services Housekeeping Telemetry.

Command Mnemonic(s) $sc_$cpu_TBL_VALIDATE

Command Structure

CFE_TBL_ValidateCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TBL_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_TBL_ValReqCtr`** - table validation request counter will increment
- **`$sc_$cpu_TBL_LastValCRC`** - calculated data integrity value will be updated
- The CFE_TBL_VAL_REQ_MADE_INF_EID debug event message (indicating the application is being notified of a validation request)

If the specified table has an associated validation function, then the following telemetry will also change:

- Either **`$sc_$cpu_TBL_ValSuccessCtr`** **OR** **`$sc_$cpu_TBL_ValFailedCtr`** **will** increment
- **`$sc_$cpu_TBL_ValCompltdCtr`** - table validations performed counter will increment
- **`$sc_$cpu_TBl_LastValS`** - table validation function return status will update
- The CFE_TBL_VALIDATION_INF_EID informational event message (indicating the validation function return status) will be generated

Error Conditions

This command may fail for the following reason(s):

- A single buffered table's inactive buffer was requested to be dumped and no such buffer is currently allocated.
- Too many validations have been requested simultaneously. The operator must wait for one or more applications to perform their table validation functions before trying again.
- The specified table name was not found in the table registry.

Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_TBL_CMDEC`** - command error counter will increment
- Command specific error event message are issued for all error cases

Criticality

The success or failure of a table validation does not have any immediate impact on table contents. The results are sent to the operator in telemetry and the operator must determine whether the results are acceptable and send a command to activate the validated table image.

See also

CFE_TBL_LOAD_CC, CFE_TBL_DUMP_CC, CFE_TBL_ACTIVATE_CC, CFE_TBL_ABORT_LOAD_CC

Definition at line 279 of file cfe_tbl_msg.h.

39.60.2 Typedef Documentation

39.60.2.1 CFE_TBL_AbortLoadCmd_Payload_t

```
typedef struct CFE_TBL_AbortLoadCmd_Payload CFE_TBL_AbortLoadCmd_Payload_t
```

Abort Load Command Payload.

For command details, see CFE_TBL_ABORT_LOAD_CC

39.60.2.2 CFE_TBL_AbortLoadCmd_t

```
typedef struct CFE_TBL_AbortLoadCmd CFE_TBL_AbortLoadCmd_t
```

Abort Load Command.

39.60.2.3 CFE_TBL_ActivateCmd_Payload_t

```
typedef struct CFE_TBL_ActivateCmd_Payload CFE_TBL_ActivateCmd_Payload_t
```

Activate Table Command Payload.

For command details, see CFE_TBL_ACTIVATE_CC

39.60.2.4 CFE_TBL_ActivateCmd_t

```
typedef struct CFE_TBL_ActivateCmd CFE_TBL_ActivateCmd_t
```

Activate Table Command.

39.60.2.5 CFE_TBL_DelCDSCmd_Payload_t

```
typedef struct CFE_TBL_DelCDSCmd_Payload CFE_TBL_DelCDSCmd_Payload_t
```

Delete Critical Table CDS Command Payload.

For command details, see CFE_TBL_DELETE_CDS_CC

39.60.2.6 CFE_TBL_DeleteCDSCmd_t

```
typedef struct CFE_TBL_DeleteCDSCmd CFE_TBL_DeleteCDSCmd_t
```

Delete Critical Table CDS Command.

39.60.2.7 CFE_TBL_DumpCmd_Payload_t

```
typedef struct CFE_TBL_DumpCmd_Payload CFE_TBL_DumpCmd_Payload_t
```

Dump Table Command Payload.

For command details, see CFE_TBL_DUMP_CC

39.60.2.8 CFE_TBL_DumpCmd_t

```
typedef struct CFE_TBL_DumpCmd CFE_TBL_DumpCmd_t
```

/brief Dump Table Command

39.60.2.9 CFE_TBL_DumpRegistryCmd_Payload_t

```
typedef struct CFE_TBL_DumpRegistryCmd_Payload CFE_TBL_DumpRegistryCmd_Payload_t
```

Dump Registry Command Payload.

For command details, see CFE_TBL_DUMP_REGISTRY_CC

39.60.2.10 CFE_TBL_DumpRegistryCmd_t

```
typedef struct CFE_TBL_DumpRegistryCmd CFE_TBL_DumpRegistryCmd_t
```

Dump Registry Command.

39.60.2.11 CFE_TBL_HousekeepingTlm_Payload_t

```
typedef struct CFE_TBL_HousekeepingTlm_Payload CFE_TBL_HousekeepingTlm_Payload_t
```

Name Table Services Housekeeping Packet

39.60.2.12 CFE_TBL_HousekeepingTlm_t

```
typedef struct CFE_TBL_HousekeepingTlm CFE_TBL_HousekeepingTlm_t
```

39.60.2.13 CFE_TBL_LoadCmd_Payload_t

```
typedef struct CFE_TBL_LoadCmd_Payload CFE_TBL_LoadCmd_Payload_t
```

Load Table Command Payload.

For command details, see CFE_TBL_LOAD_CC

39.60.2.14 CFE_TBL_LoadCmd_t

```
typedef struct CFE_TBL_LoadCmd CFE_TBL_LoadCmd_t
```

Load Table Command.

39.60.2.15 CFE_TBL_NoArgsCmd_t

```
typedef struct CFE_TBL_NoArgsCmd CFE_TBL_NoArgsCmd_t
```

Generic "no arguments" command.

This command structure is used for commands that do not have any parameters. This includes:

1. The Housekeeping Request Message
2. The No-Op Command (For details, see CFE_TBL_NOOP_CC)
3. The Reset Counters Command (For details, see CFE_TBL_RESET_COUNTERS_CC)

39.60.2.16 CFE_TBL_NoopCmd_t

```
typedef CFE_TBL_NoArgsCmd_t CFE_TBL_NoopCmd_t
```

Definition at line 506 of file cfe_tbl_msg.h.

39.60.2.17 CFE_TBL_NotifyCmd_Payload_t

```
typedef struct CFE_TBL_NotifyCmd_Payload CFE_TBL_NotifyCmd_Payload_t
```

Table Management Notification Command Payload.

Description

Whenever an application that owns a table calls the CFE_TBL_NotifyByMessage API following the table registration, Table services will generate the following command message with the application specified message ID, command code and parameter whenever the table requires management (e.g. - loads and validations).

39.60.2.18 CFE_TBL_NotifyCmd_t

```
typedef struct CFE_TBL_NotifyCmd CFE_TBL_NotifyCmd_t
```

/brief Table Management Notification Command

39.60.2.19 CFE_TBL_ResetCountersCmd_t

```
typedef CFE_TBL_NoArgsCmd_t CFE_TBL_ResetCountersCmd_t
```

Definition at line 507 of file cfe_tbl_msg.h.

39.60.2.20 CFE_TBL_SendRegistryCmd_Payload_t

```
typedef struct CFE_TBL_SendRegistryCmd_Payload CFE_TBL_SendRegistryCmd_Payload_t
```

Send Table Registry Command Payload.

For command details, see CFE_TBL_SEND_REGISTRY_CC

39.60.2.21 CFE_TBL_SendRegistryCmd_t

```
typedef struct CFE_TBL_SendRegistryCmd CFE_TBL_SendRegistryCmd_t
```

Send Table Registry Command.

39.60.2.22 CFE_TBL_TableRegistryTlm_t

```
typedef struct CFE_TBL_TableRegistryTlm CFE_TBL_TableRegistryTlm_t
```

39.60.2.23 CFE_TBL_TblRegPacket_Payload_t

```
typedef struct CFE_TBL_TblRegPacket_Payload CFE_TBL_TblRegPacket_Payload_t
```

Name Table Registry Info Packet

39.60.2.24 CFE_TBL_ValidateCmd_Payload_t

```
typedef struct CFE_TBL_ValidateCmd_Payload CFE_TBL_ValidateCmd_Payload_t
```

Validate Table Command Payload.

For command details, see CFE_TBL_VALIDATE_CC

39.60.2.25 CFE_TBL_ValidateCmd_t

```
typedef struct CFE_TBL_ValidateCmd CFE_TBL_ValidateCmd_t
```

Validate Table Command.

39.61 cfe/modules/time/fsw/inc/cfe_time_events.h File Reference

Macros

- #define CFE_TIME_MAX_EID 49
- #define CFE_TIME_INIT_EID 1 /∗ start up message "informational" ∗/

  'cFE TIME Initialized'
- #define CFE_TIME_NOOP_EID 4 /∗ processed command "informational" ∗/

  'No-op command'
- #define CFE_TIME_RESET_EID 5

  'Reset Counters command'
- #define CFE_TIME_DIAG_EID 6

  'Request diagnostics command'
- #define CFE_TIME_STATE_EID 7

  'Set Clock State = %s'
- #define CFE_TIME_SOURCE_EID 8

  'Set Time Source = %s'
- #define CFE_TIME_SIGNAL_EID 9

  'Set Tone Source = %s'
- #define CFE_TIME_DELAY_EID 11

  'Set Tone Delay - secs = %d, usecs = %d, ssecs = 0x%X, dir = %d'
- #define CFE_TIME_TIME_EID 12

  'Set Time - secs = %d, usecs = %d, ssecs = 0x%X'
- #define CFE_TIME_MET_EID 13

`'Set MET - secs = %d, usecs = %d, ssecs = 0x%X'`

- #define CFE_TIME_STCF_EID 14

  `'Set STCF - secs = %d, usecs = %d, ssecs = 0x%X'`

- #define CFE_TIME_DELTA_EID 15

  `'STCF Adjust - secs = %d, usecs = %d, ssecs = 0x%X, dir[1=Positive, 2=Negative] = %d'`

- #define CFE_TIME_1HZ_EID 16

  `'STCF 1Hz Adjust - secs = %d, ssecs = 0x%X, dir = %d'`

- #define CFE_TIME_LEAPS_EID 17

  `'Set Leap Seconds = %d'`

- #define CFE_TIME_FLY_ON_EID 20 /∗ flywheel state "informational" ∗/

  `'Start FLYWHEEL'`

- #define CFE_TIME_FLY_OFF_EID 21

  `'Stop FLYWHEEL'`

- #define CFE_TIME_EXIT_ERR_EID 25 /∗ task termination "error" ∗/
- #define CFE_TIME_ID_ERR_EID 26 /∗ invalid command packet "error" ∗/

  `'Invalid message ID - ID = 0x%X'`

- #define CFE_TIME_CC_ERR_EID 27

  `'Invalid command code - ID = 0x%X, CC = %d'`

- #define CFE_TIME_STATE_ERR_EID 30 /∗ processed command "error" ∗/

  `'Invalid Clock State = 0x%X'`

- #define CFE_TIME_SOURCE_ERR_EID 31

  `'Invalid Time Source = 0x%X'`

- #define CFE_TIME_SIGNAL_ERR_EID 32

  `'Invalid Tone Source = 0x%X'`

- #define CFE_TIME_DELAY_ERR_EID 33

  `'Invalid Tone Delay - secs = %d, usecs = %d'`

- #define CFE_TIME_TIME_ERR_EID 34

  `'Invalid Time - secs = %d, usecs = %d'`

- #define CFE_TIME_MET_ERR_EID 35

  `'Invalid MET - secs = %d, usecs = %d'`

- #define CFE_TIME_STCF_ERR_EID 36

  `'Invalid STCF - secs = %d, usecs = %d'`

- #define CFE_TIME_DELTA_ERR_EID 37

  `'Invalid STCF Adjust - secs = %d, usecs = %d, dir[1=Positive, 2=Negative] = %d'`

- #define CFE_TIME_1HZ_ERR_EID 38
- #define CFE_TIME_SOURCE_CFG_EID 40 /∗ cmd disabled per cfg "error" ∗/

  `'Set Source commands invalid without CFE_PLATFORM_TIME_CFG_SOURCE set to true'`

- #define CFE_TIME_SIGNAL_CFG_EID 41

  `'Set Signal commands invalid without CFE_PLATFORM_TIME_CFG_SIGNAL set to true'`

- #define CFE_TIME_DELAY_CFG_EID 42

  `'Set Delay commands invalid without CFE_PLATFORM_TIME_CFG_CLIENT set to true'`

- #define CFE_TIME_TIME_CFG_EID 43

  `'Set Time commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'`

- #define CFE_TIME_MET_CFG_EID 44

  `'Set MET commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'`

- #define CFE_TIME_STCF_CFG_EID 45

'Set STCF commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'

- #define CFE_TIME_LEAPS_CFG_EID 46

  'Set Leaps commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'

- #define CFE_TIME_DELTA_CFG_EID 47

  'STCF Adjust commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'

- #define CFE_TIME_1HZ_CFG_EID 48

  '1Hz Adjust commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'

- #define CFE_TIME_LEN_ERR_EID 49

  'Invalid cmd length: ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'

39.61.1 Macro Definition Documentation

39.61.1.1 CFE_TIME_1HZ_CFG_EID

```
#define CFE_TIME_1HZ_CFG_EID 48
```

`'1Hz Adjust commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'`

Event Message `'1Hz Adjust commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives either a Add STCF Adjustment each second Command OR a Subtract STCF Adjustment each second command and the Time Services configuration parameter CFE_PLATFORM_TIME_CFG_SERVER has not been set to true in the cfe_platform_cfg.h file.

Definition at line 602 of file cfe_time_events.h.

39.61.1.2 CFE_TIME_1HZ_EID

```
#define CFE_TIME_1HZ_EID 16
```

`'STCF 1Hz Adjust - secs = %d, ssecs = 0x%X, dir = %d'`

Event Message `'STCF 1Hz Adjust - secs = %d, ssecs = 0x%X, dir = %d'`

Type: INFORMATION

Cause:

This event message is generated upon successful completion of any of the following cFE Time Services STCF Adjustment Commands:

- Add STCF Adjustment each second command
- Subtract STCF Adjustment each second command

The `secs` field specifies the number of seconds the STCF is to be adjusted by, the `ssecs` field specifies the number of sub-seconds (1/2^32 seconds) the STCF is to be adjusted by and the `dir` field identifies whether the adjustment was added or subtracted. The direction value can be either CFE_TIME_AdjustDirection_ADD or CFE_TIME_Adjust↩Direction_SUBTRACT.

Definition at line 250 of file cfe_time_events.h.

39.61.1.3 CFE_TIME_1HZ_ERR_EID

```
#define CFE_TIME_1HZ_ERR_EID 38
```

(obsolete - unused)

Definition at line 473 of file cfe_time_events.h.

39.61.1.4 CFE_TIME_CC_ERR_EID

```
#define CFE_TIME_CC_ERR_EID 27
```

`'Invalid command code - ID = 0x%X, CC = %d'`

Event Message `'Invalid command code - ID = 0x%X, CC = %d'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives a message from the software bus that contains a unrecognized command code in its header..

The `ID` field specifies, in hex, the message ID of the message containing the unrecognized command code, identified, in decimal, by the `CC` field.

Definition at line 321 of file cfe_time_events.h.

39.61.1.5 CFE_TIME_DELAY_CFG_EID

```
#define CFE_TIME_DELAY_CFG_EID 42
```

`'Set Delay commands invalid without CFE_PLATFORM_TIME_CFG_CLIENT set to true'`

Event Message `'Set Delay commands invalid without CFE_PLATFORM_TIME_CFG_CLIENT set to true'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives either a Add Tone Delay Command OR a Subtract Tone Delay Command and the Time Services configuration parameter CFE_PLATFORM_TIME_CFG_CLIENT has not been set to true in the cfe_platform_cfg.h file.

Definition at line 516 of file cfe_time_events.h.

39.61.1.6 CFE_TIME_DELAY_EID

```
#define CFE_TIME_DELAY_EID 11
```

`'Set Tone Delay - secs = %d, usecs = %d, ssecs = 0x%X, dir = %d'`

Event Message `'Set Tone Delay - secs = %d, usecs = %d, ssecs = 0x%X, dir = %d'`

Type: INFORMATION

Cause:

This event message is generated upon successful completion of either a cFE Time Services Add Time Delay OR a Subtract Time Delay command

The `secs` field specifies the new delay (in seconds), the `usecs` field specifies the delay in micro-seconds, the `ssecs` field is the micro-seconds field converted to Spacecraft Time sub-seconds and the `dir` field identifies the direction of the delay. The direction can be either CFE_TIME_AdjustDirection_ADD or CFE_TIME_AdjustDirection_SUBTRACT.

Definition at line 161 of file cfe_time_events.h.

39.61.1.7 CFE_TIME_DELAY_ERR_EID

```
#define CFE_TIME_DELAY_ERR_EID 33
```

`'Invalid Tone Delay - secs = %d, usecs = %d'`

Event Message `'Invalid Tone Delay - secs = %d, usecs = %d'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives either a Add Tone Delay Command OR a Subtract Tone Delay Command that contains a microsecond field that is greater than or equal to 1000000.

The `secs` field specifies, in decimal, the tone signal delay in seconds and the `usecs` field specifies, in decimal, the micro-second delay that was in error.

Definition at line 395 of file cfe_time_events.h.

39.61.1.8 CFE_TIME_DELTA_CFG_EID

```
#define CFE_TIME_DELTA_CFG_EID 47
```

```
'STCF Adjust commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'
```

Event Message `'STCF Adjust commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives either a Add Single STCF Adjustment Command OR a Subtract Single STCF Adjustment command and the Time Services configuration parameter CFE_PLATFORM_TIME_CFG_SERVER has not been set to true in the cfe_platform_cfg.h file.

Definition at line 587 of file cfe_time_events.h.

39.61.1.9 CFE_TIME_DELTA_EID

```
#define CFE_TIME_DELTA_EID 15
```

```
'STCF Adjust - secs = %d, usecs = %d, ssecs = 0x%X, dir[1=Positive, 2=Negative] = %d'
```

Event Message `'STCF Adjust - secs = %d, usecs = %d, ssecs = 0x%X, dir[1=Positive, 2=Negative] = %d'`

Type: INFORMATION

Cause:

This event message is generated upon successful completion of any of the following cFE Time Services STCF Adjustment Commands:

- Add Single STCF Adjustment command
- Subtract Single STCF Adjustment command

The `secs` field specifies the number of seconds the STCF is to be adjusted by, the `usecs` field specifies the number of micro-seconds, the `ssecs` field is the micro-seconds field converted to Spacecraft Time sub-seconds and the `dir` field identifies whether the adjustment was added or subtracted. The direction can be either CFE_TIME_AdjustDirection_ADD or CFE_TIME_AdjustDirection_SUBTRACT.

Definition at line 230 of file cfe_time_events.h.

39.61.1.10 CFE_TIME_DELTA_ERR_EID

```
#define CFE_TIME_DELTA_ERR_EID 37
```

```
'Invalid STCF Adjust - secs = %d, usecs = %d, dir[1=Positive, 2=Negative] =
%d'
```

Event Message `'Invalid STCF Adjust - secs = %d, usecs = %d, dir[1=Positive, 2=Negative] = %d'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives either a Add Single STCF Adjustment Command OR a Subtract Single STCF Adjustment command that contains a microsecond field that is greater than or equal to 1,000,000.

The `secs` field specifies the number of seconds the STCF is to be adjusted by, the `usecs` field specifies the number of micro-seconds that was in error, the `dir` field identifies whether the adjustment was to be added or subtracted. The direction can be either CFE_TIME_AdjustDirection_ADD or CFE_TIME_AdjustDirection_SUBTRACT.

Definition at line 469 of file cfe_time_events.h.

39.61.1.11 CFE_TIME_DIAG_EID

```
#define CFE_TIME_DIAG_EID 6
```

```
'Request diagnostics command'
```

Event Message `'Request diagnostics command'`

Type: DEBUG

Cause:

This event message is always automatically issued in response to a cFE Time Services Request Diagnostics command

Definition at line 96 of file cfe_time_events.h.

39.61.1.12 CFE_TIME_EXIT_ERR_EID

```
#define CFE_TIME_EXIT_ERR_EID 25 /* task termination "error" */
```

Definition at line 289 of file cfe_time_events.h.

39.61.1.13 CFE_TIME_FLY_OFF_EID

```
#define CFE_TIME_FLY_OFF_EID 21
```

`'Stop FLYWHEEL'`

Event Message `'Stop FLYWHEEL'`

Type: INFORMATION

Cause:

This event message is generated whenever the Time Services exits FLYWHEEL mode.

Definition at line 287 of file cfe_time_events.h.

39.61.1.14 CFE_TIME_FLY_ON_EID

```
#define CFE_TIME_FLY_ON_EID 20 /* flywheel state "informational" */
```

`'Start FLYWHEEL'`

Event Message `'Start FLYWHEEL'`

Type: INFORMATION

Cause:

This event message is generated whenever the Time Services enters FLYWHEEL mode.

Definition at line 276 of file cfe_time_events.h.

39.61.1.15 CFE_TIME_ID_ERR_EID

```
#define CFE_TIME_ID_ERR_EID 26 /* invalid command packet "error" */
```

`'Invalid message ID - ID = 0x%X'`

Event Message `'Invalid message ID - ID = 0x%X'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives a message from the software bus that is not one of Time Services recognized messages.

The `ID` field specifies, in hex, the message ID of the inappropriately received message.

Definition at line 305 of file cfe_time_events.h.

39.61.1.16 CFE_TIME_INIT_EID

```
#define CFE_TIME_INIT_EID 1 /* start up message "informational" */
```

`'cFE TIME Initialized'`

Event Message `'cFE TIME Initialized'`

Type: INFORMATION

Cause:

This event message is always automatically issued when the Time Services Task completes its Initialization.

Definition at line 60 of file cfe_time_events.h.

39.61.1.17 CFE_TIME_LEAPS_CFG_EID

```
#define CFE_TIME_LEAPS_CFG_EID 46
```

```
'Set Leaps commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'
```

Event Message `'Set Leaps commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives a Set Leap Seconds Command and the Time Services configuration parameter CFE_PLATFORM_TIME_CFG_SERVER has not been set to true in the cfe_platform_cfg.h file.

Definition at line 572 of file cfe_time_events.h.

39.61.1.18 CFE_TIME_LEAPS_EID

```
#define CFE_TIME_LEAPS_EID 17
```

```
'Set Leap Seconds = %d'
```

Event Message `'Set Leap Seconds = %d'`

Type: INFORMATION

Cause:

This event message is generated upon successful completion of the Set Leap Seconds command

The `%d` field contains the number of seconds the Spacecraft's Leap Seconds has been set to.

Definition at line 265 of file cfe_time_events.h.

39.61.1.19 CFE_TIME_LEN_ERR_EID

```
#define CFE_TIME_LEN_ERR_EID 49
```

`'Invalid cmd length:  ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'`

Event Message `'Invalid cmd length:  ID = 0x%X, CC = %d, Exp Len = %d, Len = %d'`

Type: ERROR

Cause:

This event message is generated when a message with the CFE_TIME_CMD_MID message ID has arrived but whose packet length does not match the expected length for the specified command code.

The `ID` field in the event message specifies the Message ID (in hex), the `CC` field specifies the Command Code (in decimal), the `Exp` Len field specified the Expected Length (in decimal), and `Len` specifies the message Length (in decimal) found in the message.

Definition at line 620 of file cfe_time_events.h.

39.61.1.20 CFE_TIME_MAX_EID

```
#define CFE_TIME_MAX_EID 49
```

Definition at line 45 of file cfe_time_events.h.

39.61.1.21 CFE_TIME_MET_CFG_EID

```
#define CFE_TIME_MET_CFG_EID 44
```

`'Set MET commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'`

Event Message `'Set MET commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives a Set Mission Elapsed Time Command and the Time Services configuration parameter CFE_PLATFORM_TIME_CFG_SERVER has not been set to true in the cfe_platform_cfg.h file.

Definition at line 544 of file cfe_time_events.h.

39.61.1.22 CFE_TIME_MET_EID

```
#define CFE_TIME_MET_EID 13
```

`'Set MET - secs = %d, usecs = %d, ssecs = 0x%X'`

Event Message `'Set MET - secs = %d, usecs = %d, ssecs = 0x%X'`

Type: INFORMATION

Cause:

This event message is generated upon successful completion of a cFE Time Services Set Mission Elapsed Time command

The `secs` field specifies the new MET (in seconds), the `usecs` field specifies the MET micro-seconds, the `ssecs` field is the micro-seconds field converted to Spacecraft Time sub-seconds

Definition at line 193 of file cfe_time_events.h.

39.61.1.23 CFE_TIME_MET_ERR_EID

```
#define CFE_TIME_MET_ERR_EID 35
```

`'Invalid MET - secs = %d, usecs = %d'`

Event Message `'Invalid MET - secs = %d, usecs = %d'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives a Set Mission Elapsed Time Command that contains a microsecond field that is greater than or equal to 1,000,000.

The `secs` field specifies, in decimal, the MET in seconds and the `usecs` field specifies, in decimal, the micro-second field of the MET that was in error.

Definition at line 431 of file cfe_time_events.h.

39.61.1.24 CFE_TIME_NOOP_EID

```
#define CFE_TIME_NOOP_EID 4 /* processed command "informational" */
```

'No-op command'

Event Message 'No-op command'

Type: INFORMATION

Cause:

This event message is always automatically issued in response to a cFE Time Services NO-OP command

Definition at line 72 of file cfe_time_events.h.

39.61.1.25 CFE_TIME_RESET_EID

```
#define CFE_TIME_RESET_EID 5
```

'Reset Counters command'

Event Message 'Reset Counters command'

Type: DEBUG

Cause:

This event message is always automatically issued in response to a cFE Time Services Reset Counters command

Definition at line 84 of file cfe_time_events.h.

39.61.1.26 CFE_TIME_SIGNAL_CFG_EID

```
#define CFE_TIME_SIGNAL_CFG_EID 41
```

`'Set Signal commands invalid without CFE_PLATFORM_TIME_CFG_SIGNAL set to true'`

Event Message `'Set Signal commands invalid without CFE_PLATFORM_TIME_CFG_SIGNAL set to true'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives a Set Clock Signal Command and the Time Services configuration parameter CFE_PLATFORM_TIME_CFG_SIGNAL has not been set to true in the cfe_platform_cfg.h file.

Definition at line 501 of file cfe_time_events.h.

39.61.1.27 CFE_TIME_SIGNAL_EID

```
#define CFE_TIME_SIGNAL_EID 9
```

`'Set Tone Source = %s'`

Event Message `'Set Tone Source = %s'`

Type: INFORMATION

Cause:

This event message is generated upon successful completion of a cFE Time Services Set Clock Signal command

The `'%s'` field will identify whether the command specified `PRIMARY`, or `REDUNDANT`.

Definition at line 141 of file cfe_time_events.h.

39.61.1.28 CFE_TIME_SIGNAL_ERR_EID

```
#define CFE_TIME_SIGNAL_ERR_EID 32
```

`'Invalid Tone Source = 0x%X'`

Event Message `'Invalid Tone Source = 0x%X'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives a Set Clock Signal Command that contains a desired clock source that is none of the following:

- CFE_TIME_ToneSignalSelect_PRIMARY
- CFE_TIME_ToneSignalSelect_REDUNDANT

The `Source` field specifies, in hex, the signal source value received in the command message.

Definition at line 376 of file cfe_time_events.h.

39.61.1.29 CFE_TIME_SOURCE_CFG_EID

```
#define CFE_TIME_SOURCE_CFG_EID 40 /* cmd disabled per cfg "error" */
```

`'Set Source commands invalid without CFE_PLATFORM_TIME_CFG_SOURCE set to true'`

Event Message `'Set Source commands invalid without CFE_PLATFORM_TIME_CFG_SOURCE set to true'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives a Set Clock Source Command and the Time Services configuration parameter CFE_PLATFORM_TIME_CFG_SOURCE has not been set to true in the cfe_platform_cfg.h file.

Definition at line 487 of file cfe_time_events.h.

39.61.1.30 CFE_TIME_SOURCE_EID

```
#define CFE_TIME_SOURCE_EID 8
```

`'Set Time Source = %s'`

Event Message `'Set Time Source = %s'`

Type: INFORMATION

Cause:

This event message is generated upon successful completion of a cFE Time Services Set Time Source command

The `'%s'` field will identify whether the command specified `INTERNAL`, or `EXTERNAL`.

Definition at line 126 of file cfe_time_events.h.

39.61.1.31 CFE_TIME_SOURCE_ERR_EID

```
#define CFE_TIME_SOURCE_ERR_EID 31
```

`'Invalid Time Source = 0x%X'`

Event Message `'Invalid Time Source = 0x%X'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives a Set Clock Source Command that contains a desired clock source that is none of the following:

- CFE_TIME_SourceSelect_INTERNAL
- CFE_TIME_SourceSelect_EXTERNAL

The `Source` field specifies, in hex, the source value received in the command message.

Definition at line 358 of file cfe_time_events.h.

39.61.1.32 CFE_TIME_STATE_EID

```
#define CFE_TIME_STATE_EID 7
```

`'Set Clock State = %s'`

Event Message `'Set Clock State = %s'`

Type: INFORMATION

Cause:

This event message is generated upon successful completion of a cFE Time Services Set Time State command

The `'%s'` field will identify whether the command specified `VALID`, `INVALID`, or `FLYWHEEL`.

Definition at line 111 of file cfe_time_events.h.

39.61.1.33 CFE_TIME_STATE_ERR_EID

```
#define CFE_TIME_STATE_ERR_EID 30 /* processed command "error" */
```

`'Invalid Clock State = 0x%X'`

Event Message `'Invalid Clock State = 0x%X'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives a Set Clock State Command that contains a desired clock state that is none of the following:

- CFE_TIME_ClockState_INVALID
- CFE_TIME_ClockState_VALID
- CFE_TIME_ClockState_FLYWHEEL

The `State` field specifies, in hex, the state value received in the command message.

Definition at line 340 of file cfe_time_events.h.

39.61.1.34 CFE_TIME_STCF_CFG_EID

```
#define CFE_TIME_STCF_CFG_EID 45
```

`'Set STCF commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'`

Event Message `'Set STCF commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives a Set Spacecraft Time Correlation Factor Command and the Time Services configuration parameter CFE_PLATFORM_TIME_CFG_SERVER has not been set to true in the cfe_platform_cfg.h file.

Definition at line 558 of file cfe_time_events.h.

39.61.1.35 CFE_TIME_STCF_EID

```
#define CFE_TIME_STCF_EID 14
```

`'Set STCF - secs = %d, usecs = %d, ssecs = 0x%X'`

Event Message `'Set STCF - secs = %d, usecs = %d, ssecs = 0x%X'`

Type: INFORMATION

Cause:

This event message is generated upon successful completion of a cFE Time Services Set Spacecraft Time Correlation Factor command

The `secs` field specifies the new STCF (in seconds), the `usecs` field specifies the STCF micro-seconds, the `ssecs` field is the micro-seconds field converted to Spacecraft Time sub-seconds.

Definition at line 210 of file cfe_time_events.h.

39.61.1.36 CFE_TIME_STCF_ERR_EID

```
#define CFE_TIME_STCF_ERR_EID 36
```

`'Invalid STCF - secs = %d, usecs = %d'`

Event Message `'Invalid STCF - secs = %d, usecs = %d'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives a Set Spacecraft Time Correlation Factor Command that contains a microsecond field that is greater than or equal to 1,000,000.

The `secs` field specifies, in decimal, the STCF in seconds and the `usecs` field specifies, in decimal, the micro-second field of the STCF that was in error.

Definition at line 449 of file cfe_time_events.h.

39.61.1.37 CFE_TIME_TIME_CFG_EID

```
#define CFE_TIME_TIME_CFG_EID 43
```

`'Set Time commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'`

Event Message `'Set Time commands invalid without CFE_PLATFORM_TIME_CFG_SERVER set to true'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives a Set Spacecraft Time Command and the Time Services configuration parameter CFE_PLATFORM_TIME_CFG_SERVER has not been set to true in the cfe_platform_cfg.h file.

Definition at line 530 of file cfe_time_events.h.

39.61.1.38 CFE_TIME_TIME_EID

```
#define CFE_TIME_TIME_EID 12
```

`'Set Time - secs = %d, usecs = %d, ssecs = 0x%X'`

Event Message `'Set Time - secs = %d, usecs = %d, ssecs = 0x%X'`

Type: INFORMATION

Cause:

This event message is generated upon successful completion of a cFE Time Services Set Time command

The `secs` field specifies the new spacecraft time (in seconds), the `usecs` field specifies the spacecraft time micro-seconds, the `ssecs` field is the micro-seconds field converted to Spacecraft Time sub-seconds

Definition at line 177 of file cfe_time_events.h.

39.61.1.39 CFE_TIME_TIME_ERR_EID

```
#define CFE_TIME_TIME_ERR_EID 34
```

`'Invalid Time - secs = %d, usecs = %d'`

Event Message `'Invalid Time - secs = %d, usecs = %d'`

Type: ERROR

Cause:

This event message is generated whenever Time Services receives a Set Spacecraft Time Command that contains a microsecond field that is greater than or equal to 1,000,000.

The `secs` field specifies, in decimal, the spacecraft time in seconds and the `usecs` field specifies, in decimal, the micro-second field of the spacecraft time that was in error.

Definition at line 413 of file cfe_time_events.h.

39.62 cfe/modules/time/fsw/inc/cfe_time_msg.h File Reference

```
#include "common_types.h"
#include "cfe_msg_hdr.h"
#include "cfe_time_extern_typedefs.h"
```

Data Structures

- struct CFE_TIME_NoArgsCmd

  Generic no argument command.
- struct CFE_TIME_LeapsCmd_Payload

  Set leap seconds command payload.
- struct CFE_TIME_SetLeapSecondsCmd

  Set leap seconds command.
- struct CFE_TIME_StateCmd_Payload

  Set clock state command payload.
- struct CFE_TIME_SetStateCmd

  Set clock state command.
- struct CFE_TIME_SourceCmd_Payload

  Set time data source command payload.
- struct CFE_TIME_SetSourceCmd

  Set time data source command.
- struct CFE_TIME_SignalCmd_Payload

  Set tone signal source command payload.
- struct CFE_TIME_SetSignalCmd

  Set tone signal source command.
- struct CFE_TIME_TimeCmd_Payload

  Generic seconds, microseconds command payload.
- struct CFE_TIME_TimeCmd

  Generic seconds, microseconds argument command.
- struct CFE_TIME_OneHzAdjustmentCmd_Payload

  Generic seconds, subseconds command payload.
- struct CFE_TIME_OneHzAdjustmentCmd

  Generic seconds, subseconds adjustment command.
- struct CFE_TIME_ToneDataCmd_Payload

  Time at tone data command payload.
- struct CFE_TIME_ToneDataCmd

  Time at tone data command.
- struct CFE_TIME_HousekeepingTlm_Payload
- struct CFE_TIME_HousekeepingTlm
- struct CFE_TIME_DiagnosticTlm_Payload
- struct CFE_TIME_DiagnosticTlm

Macros

- #define CFE_TIME_FLAG_CLKSET 0x8000

  The spacecraft time has been set.
- #define CFE_TIME_FLAG_FLYING 0x4000

  This instance of Time Services is flywheeling.
- #define CFE_TIME_FLAG_SRCINT 0x2000

  The clock source is set to "internal".
- #define CFE_TIME_FLAG_SIGPRI 0x1000

  The clock signal is set to "primary".
- #define CFE_TIME_FLAG_SRVFLY 0x0800

  The Time Server is in flywheel mode.
- #define CFE_TIME_FLAG_CMDFLY 0x0400

  This instance of Time Services was commanded into flywheel mode.
- #define CFE_TIME_FLAG_ADDADJ 0x0200

  One time STCF Adjustment is to be done in positive direction.
- #define CFE_TIME_FLAG_ADD1HZ 0x0100

  1 Hz STCF Adjustment is to be done in a positive direction
- #define CFE_TIME_FLAG_ADDTCL 0x0080

  Time Client Latency is applied in a positive direction.
- #define CFE_TIME_FLAG_SERVER 0x0040

  This instance of Time Services is a Time Server.
- #define CFE_TIME_FLAG_GDTONE 0x0020

  The tone received is good compared to the last tone received.
- #define CFE_TIME_FLAG_UNUSED 0x001F

  Reserved flags - should be zero.

Time Services Command Codes

- #define CFE_TIME_NOOP_CC 0 /∗ no-op command ∗/
- #define CFE_TIME_RESET_COUNTERS_CC 1 /∗ reset counters ∗/
- #define CFE_TIME_SEND_DIAGNOSTIC_TLM_CC 2 /∗ request diagnostic hk telemetry ∗/
- #define CFE_TIME_SET_SOURCE_CC 3 /∗ set clock source (int vs ext) ∗/
- #define CFE_TIME_SET_STATE_CC 4 /∗ set clock state ∗/
- #define CFE_TIME_ADD_DELAY_CC 5 /∗ add tone delay value ∗/
- #define CFE_TIME_SUB_DELAY_CC 6 /∗ sub tone delay value ∗/
- #define CFE_TIME_SET_TIME_CC 7 /∗ set time ∗/
- #define CFE_TIME_SET_MET_CC 8 /∗ set MET ∗/
- #define CFE_TIME_SET_STCF_CC 9 /∗ set STCF ∗/
- #define CFE_TIME_SET_LEAP_SECONDS_CC 10 /∗ set Leap Seconds ∗/
- #define CFE_TIME_ADD_ADJUST_CC 11 /∗ add one time STCF adjustment ∗/
- #define CFE_TIME_SUB_ADJUST_CC 12 /∗ subtract one time STCF adjustment ∗/
- #define CFE_TIME_ADD_1HZ_ADJUSTMENT_CC 13 /∗ add 1Hz STCF adjustment ∗/
- #define CFE_TIME_SUB_1HZ_ADJUSTMENT_CC 14 /∗ subtract 1Hz STCF adjustment ∗/
- #define CFE_TIME_SET_SIGNAL_CC 15 /∗ set clock signal (pri vs red) ∗/

Typedefs

- typedef struct CFE_TIME_NoArgsCmd CFE_TIME_NoArgsCmd_t

  Generic no argument command.
- typedef CFE_TIME_NoArgsCmd_t CFE_TIME_NoopCmd_t
- typedef CFE_TIME_NoArgsCmd_t CFE_TIME_ResetCountersCmd_t
- typedef CFE_TIME_NoArgsCmd_t CFE_TIME_SendDiagnosticCmd_t
- typedef CFE_TIME_NoArgsCmd_t CFE_TIME_1HzCmd_t
- typedef CFE_TIME_NoArgsCmd_t CFE_TIME_ToneSignalCmd_t
- typedef CFE_TIME_NoArgsCmd_t CFE_TIME_FakeToneCmd_t
- typedef struct CFE_TIME_LeapsCmd_Payload CFE_TIME_LeapsCmd_Payload_t

  Set leap seconds command payload.
- typedef struct CFE_TIME_SetLeapSecondsCmd CFE_TIME_SetLeapSecondsCmd_t

  Set leap seconds command.
- typedef struct CFE_TIME_StateCmd_Payload CFE_TIME_StateCmd_Payload_t

  Set clock state command payload.
- typedef struct CFE_TIME_SetStateCmd CFE_TIME_SetStateCmd_t

  Set clock state command.
- typedef struct CFE_TIME_SourceCmd_Payload CFE_TIME_SourceCmd_Payload_t

  Set time data source command payload.
- typedef struct CFE_TIME_SetSourceCmd CFE_TIME_SetSourceCmd_t

  Set time data source command.
- typedef struct CFE_TIME_SignalCmd_Payload CFE_TIME_SignalCmd_Payload_t

  Set tone signal source command payload.
- typedef struct CFE_TIME_SetSignalCmd CFE_TIME_SetSignalCmd_t

  Set tone signal source command.
- typedef struct CFE_TIME_TimeCmd_Payload CFE_TIME_TimeCmd_Payload_t

  Generic seconds, microseconds command payload.
- typedef struct CFE_TIME_TimeCmd CFE_TIME_TimeCmd_t

  Generic seconds, microseconds argument command.
- typedef CFE_TIME_TimeCmd_t CFE_TIME_AddDelayCmd_t
- typedef CFE_TIME_TimeCmd_t CFE_TIME_SubDelayCmd_t
- typedef CFE_TIME_TimeCmd_t CFE_TIME_SetMETCmd_t
- typedef CFE_TIME_TimeCmd_t CFE_TIME_SetSTCFCmd_t
- typedef CFE_TIME_TimeCmd_t CFE_TIME_AddAdjustCmd_t
- typedef CFE_TIME_TimeCmd_t CFE_TIME_SubAdjustCmd_t
- typedef CFE_TIME_TimeCmd_t CFE_TIME_SetTimeCmd_t
- typedef struct CFE_TIME_OneHzAdjustmentCmd_Payload CFE_TIME_OneHzAdjustmentCmd_Payload_t

  Generic seconds, subseconds command payload.
- typedef struct CFE_TIME_OneHzAdjustmentCmd CFE_TIME_OneHzAdjustmentCmd_t

  Generic seconds, subseconds adjustment command.
- typedef CFE_TIME_OneHzAdjustmentCmd_t CFE_TIME_Add1HZAdjustmentCmd_t
- typedef CFE_TIME_OneHzAdjustmentCmd_t CFE_TIME_Sub1HZAdjustmentCmd_t
- typedef struct CFE_TIME_ToneDataCmd_Payload CFE_TIME_ToneDataCmd_Payload_t

  Time at tone data command payload.
- typedef struct CFE_TIME_ToneDataCmd CFE_TIME_ToneDataCmd_t

  Time at tone data command.
- typedef struct CFE_TIME_HousekeepingTlm_Payload CFE_TIME_HousekeepingTlm_Payload_t
- typedef struct CFE_TIME_HousekeepingTlm CFE_TIME_HousekeepingTlm_t
- typedef struct CFE_TIME_DiagnosticTlm_Payload CFE_TIME_DiagnosticTlm_Payload_t
- typedef struct CFE_TIME_DiagnosticTlm CFE_TIME_DiagnosticTlm_t

39.62.1 Macro Definition Documentation

39.62.1.1 CFE_TIME_ADD_1HZ_ADJUSTMENT_CC

```
#define CFE_TIME_ADD_1HZ_ADJUSTMENT_CC 13 /* add 1Hz STCF adjustment */
```

Name Add Delta to Spacecraft Time Correlation Factor each 1Hz

Description

This command has been updated to take actual sub-seconds (1/2^32 seconds) rather than micro-seconds as an input argument. This change occurred after the determination was made that one micro-second is too large an increment for a constant 1Hz adjustment.

This command continuously adjusts the Spacecraft Time Correlation Factor (STCF) every second, by adding the specified value. The adjustment to the STCF is applied in the Time Service local 1Hz interrupt handler. As the local 1Hz interrupt is not synchronized to the tone signal, one cannot say when the adjustment will occur, other than once a second, at about the same time relative to the tone.

There was some debate about whether the maximum 1Hz clock drift correction factor would ever need to exceed some small fraction of a second. But, the decision was made to provide the capability to make 1Hz adjustments greater than one second and leave it to the ground system to provide mission specific limits.

Command Mnemonic(s) $sc_$cpu_TIME_Add1HzSTCF

Command Structure

CFE_TIME_Add1HZAdjustmentCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_TIME_STCFSecs`** - Housekeeping Telemetry point indicating new STCF seconds value
- **`$sc_$cpu_TIME_STCFSubsecs`** - Housekeeping Telemetry point indicating new STCF subseconds value
- The CFE_TIME_1HZ_EID informational event message will be generated

Error Conditions

- Platorm receiving the command is not a Time Server

  Evidence of Failure may be found in the following telemetry:
- **`$sc_$cpu_TIME_CMDEC`** - command error counter will increment
- Error specific event message will be issued (CFE_TIME_1HZ_CFG_EID)

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

CFE_TIME_ADD_ADJUST_CC, CFE_TIME_SUB_ADJUST_CC, CFE_TIME_SUB_1HZ_ADJUSTMENT_CC

Definition at line 616 of file cfe_time_msg.h.

39.62.1.2 CFE_TIME_ADD_ADJUST_CC

```
#define CFE_TIME_ADD_ADJUST_CC 11 /* add one time STCF adjustment */
```

Name Add Delta to Spacecraft Time Correlation Factor

Description

This command adjusts the Spacecraft Time Correlation Factor (STCF) by adding the specified value. The new STCF takes effect immediately upon execution of this command.

Command Mnemonic(s) $sc_$cpu_TIME_AddSTCFAdj

Command Structure

CFE_TIME_AddAdjustCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_TIME_STCFSecs`** - Housekeeping Telemetry point indicating new STCF seconds value
- **`$sc_$cpu_TIME_STCFSubsecs`** - Housekeeping Telemetry point indicating new STCF subseconds value
- The CFE_TIME_DELTA_EID informational event message will be generated

Error Conditions

- An invalid number of microseconds was specified (must be less than 1 million)
- Platorm receiving the command is not a Time Server

  Evidence of Failure may be found in the following telemetry:
- **`$sc_$cpu_TIME_CMDEC`** - command error counter will increment
- Error specific event messages will be issued (CFE_TIME_DELTA_ERR_EID or CFE_TIME_DELTA_CFG_↩EID)

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

CFE_TIME_ADD_ADJUST_CC, CFE_TIME_SUB_ADJUST_CC, CFE_TIME_ADD_1HZ_ADJUSTMENT_CC, CFE_TIME_SUB_1HZ_ADJUSTMENT_CC

Definition at line 536 of file cfe_time_msg.h.

39.62.1.3 CFE_TIME_ADD_DELAY_CC

```
#define CFE_TIME_ADD_DELAY_CC 5 /* add tone delay value */
```

Name Add Time to Tone Time Delay

Description

This command is used to factor out a known, predictable latency between the Time Server and a particular Time Client. The correction is applied (added) to the current time calculation for Time Clients, so this command has no meaning for Time Servers. Each Time Client can have a unique latency setting. The latency value is a positive number of seconds and microseconds that represent the deviation from the time maintained by the Time Server.

Command Mnemonic(s) $sc_$cpu_TIME_AddClockLat

Command Structure

CFE_TIME_AddDelayCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **$sc_$cpu_TIME_CMDPC -** command execution counter will increment
- **$sc_$cpu_TIME_DLatentS,** $sc_$cpu_TIME_DLatentSs - Housekeeping Telemetry point indicating command specified values
- **$sc_$cpu_TIME_DLatentDir -** Diagnostic Telemetry point indicating commanded latency direction
- The CFE_TIME_DELAY_EID informational event message will be generated

Error Conditions

- An invalid number of microseconds was specified (must be less than 1 million)
- Platorm receiving the command is not a Time Client

  Evidence of Failure may be found in the following telemetry:
- **$sc_$cpu_TIME_CMDEC -** command error counter will increment
- Error specific event messages will be issued (CFE_TIME_DELAY_CFG_EID or CFE_TIME_DELAY_ERR↩_EID)

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

CFE_TIME_SUB_DELAY_CC

Definition at line 305 of file cfe_time_msg.h.

39.62.1.4 CFE_TIME_NOOP_CC

```
#define CFE_TIME_NOOP_CC 0 /* no-op command */
```

Name Time No-Op

Description

This command performs no other function than to increment the command execution counter. The command may be used to verify general aliveness of the Time Services task.

Command Mnemonic(s) $sc_$cpu_TIME_NOOP

Command Structure

CFE_TIME_NoopCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- The CFE_TIME_NOOP_EID informational event message will be generated

Error Conditions

There are no error conditions for this command. If the Time Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

None

See also

Definition at line 82 of file cfe_time_msg.h.

39.62.1.5 CFE_TIME_RESET_COUNTERS_CC

```
#define CFE_TIME_RESET_COUNTERS_CC 1 /* reset counters */
```

Name Time Reset Counters

Description

This command resets the following counters within the Time Services Housekeeping Telemetry :

- Command Execution Counter ($sc_$cpu_TIME_CMDPC)
- Command Error Counter ($sc_$cpu_TIME_CMDEC) This command also resets the following counters within the Time Services Diagnostic Telemetry :
- Tone Signal Detected Software Bus Message Counter ($sc_$cpu_TIME_DTSDetCNT)
- Time at the Tone Data Software Bus Message Counter ($sc_$cpu_TIME_DTatTCNT)
- Tone Signal/Data Verify Counter ($sc_$cpu_TIME_DVerifyCNT)
- Tone Signal/Data Error Counter ($sc_$cpu_TIME_DVerifyER)
- Tone Signal Interrupt Counter ($sc_$cpu_TIME_DTsISRCNT)
- Tone Signal Interrupt Error Counter ($sc_$cpu_TIME_DTsISRERR)
- Tone Signal Task Counter ($sc_$cpu_TIME_DTsTaskCNT)
- Local 1 Hz Interrupt Counter ($sc_$cpu_TIME_D1HzISRCNT)
- Local 1 Hz Task Counter ($sc_$cpu_TIME_D1HzTaskCNT)
- Reference Time Version Counter ($sc_$cpu_TIME_DVersionCNT)

Command Mnemonic(s) $sc_$cpu_TIME_ResetCtrs

Command Structure

CFE_TIME_ResetCountersCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- The CFE_TIME_RESET_EID informational event message will be generated

Error Conditions

There are no error conditions for this command. If the Time Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

None

See also

Definition at line 126 of file cfe_time_msg.h.

39.62.1.6 CFE_TIME_SEND_DIAGNOSTIC_TLM_CC

```
#define CFE_TIME_SEND_DIAGNOSTIC_TLM_CC 2 /* request diagnostic hk telemetry */
```

Name Request TIME Diagnostic Telemetry

Description

This command requests that the Time Service generate a message containing various data values not included in the normal Time Service housekeeping message. The command requests only a single copy of the diagnostic message. Refer to CFE_TIME_DiagnosticTlm_t for a description of the Time Service diagnostic message contents.

Command Mnemonic(s) $sc_$cpu_TIME_RequestDiag

Command Structure

CFE_TIME_SendDiagnosticCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- Sequence Counter for CFE_TIME_DiagnosticTlm_t will increment
- The CFE_TIME_DIAG_EID debug event message will be generated

Error Conditions

There are no error conditions for this command. If the Time Services receives the command, the event and telemetry is sent (although one or both may be filtered by EVS and TO) and the counter is incremented unconditionally.

Criticality

None

See also

Definition at line 160 of file cfe_time_msg.h.

39.62.1.7 CFE_TIME_SET_LEAP_SECONDS_CC

```
#define CFE_TIME_SET_LEAP_SECONDS_CC 10 /* set Leap Seconds */
```

Name Set Leap Seconds

Description

This command sets the spacecraft Leap Seconds to the specified value. Leap Seconds may be positive or negative, and there is no limit to the value except, of course, the limit imposed by the 16 bit signed integer data type. The new Leap Seconds value takes effect immediately upon execution of this command.

Command Mnemonic(s) $sc_$cpu_TIME_SetClockLeap

Command Structure

CFE_TIME_SetLeapSecondsCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_TIME_LeapSecs`** - Housekeeping Telemetry point indicating new Leap seconds value
- The CFE_TIME_LEAPS_EID informational event message will be generated

Error Conditions

- Platorm receiving the command is not a Time Server

  Evidence of Failure may be found in the following telemetry:
- **`$sc_$cpu_TIME_CMDEC`** - command error counter will increment
- Error specific event messages will be issued (CFE_TIME_LEAPS_CFG_EID)

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

CFE_TIME_SET_TIME_CC, CFE_TIME_SET_MET_CC, CFE_TIME_SET_STCF_CC

Definition at line 500 of file cfe_time_msg.h.

39.62.1.8 CFE_TIME_SET_MET_CC

```
#define CFE_TIME_SET_MET_CC 8 /* set MET */
```

Name Set Mission Elapsed Time

Description

This command sets the Mission Elapsed Timer (MET) to the specified value.

Note that the MET (as implemented for cFE Time Service) is a logical representation and not a physical timer. Thus, setting the MET is not dependent on whether the hardware supports a MET register that can be written to.

Note also that Time Service "assumes" that during normal operation, the MET is synchronized to the tone signal. Therefore, unless operating in FLYWHEEL mode, the sub-seconds portion of the MET will be set to zero at the next tone signal interrupt.

The new MET takes effect immediately upon execution of this command.

Command Mnemonic(s) $sc_$cpu_TIME_SetClockMET

Command Structure

CFE_TIME_SetMETCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_TIME_METSecs`** - Housekeeping Telemetry point indicating new MET seconds value
- **`$sc_$cpu_TIME_METSubsecs`** - Housekeeping Telemetry point indicating new MET subseconds value
- The CFE_TIME_MET_EID informational event message will be generated

Error Conditions

- An invalid number of microseconds was specified (must be less than 1 million)
- Platorm receiving the command is not a Time Server

  Evidence of Failure may be found in the following telemetry:
- **`$sc_$cpu_TIME_CMDEC`** - command error counter will increment
- Error specific event messages will be issued (CFE_TIME_MET_CFG_EID or CFE_TIME_MET_ERR_EID)

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

CFE_TIME_SET_TIME_CC, CFE_TIME_SET_STCF_CC, CFE_TIME_SET_LEAP_SECONDS_CC

Definition at line 428 of file cfe_time_msg.h.

39.62.1.9 CFE_TIME_SET_SIGNAL_CC

```
#define CFE_TIME_SET_SIGNAL_CC 15 /* set clock signal (pri vs red) */
```

Name Set Tone Signal Source

Description

This command selects the Time Service tone signal source. Although the list of potential tone signal sources is mission specific, a common choice is the selection of primary or redundant tone signal. The selection may be available to both the Time Server and Time Clients, depending on hardware configuration.

Notes:

- This command is only valid when the CFE_PLATFORM_TIME_CFG_SIGNAL configuration parameter in the cfe_platform_cfg.h file has been set to true.

Command Mnemonic(s) $sc_$cpu_TIME_SetSignal

Command Structure

CFE_TIME_SetSignalCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_TIME_DSignal`** - Diagnostic Telemetry point will indicate the command specified value
- The CFE_TIME_SIGNAL_EID informational event message will be generated

Error Conditions

- Invalid Signal selection (a value other than CFE_TIME_ToneSignalSelect_PRIMARY or CFE_TIME_Tone↩SignalSelect_REDUNDANT was specified)
- Multiple Tone Signal Sources not available on this platform

  Evidence of failure may be found in the following telemetry:
- **`$sc_$cpu_TIME_CMDEC`** - Command Error counter will increment
- Error specific event message (either CFE_TIME_SIGNAL_CFG_EID or CFE_TIME_SIGNAL_ERR_EID)

Criticality

Although tone signal source selection is important, this command is not critical

See also

CFE_TIME_SET_STATE_CC, CFE_TIME_SET_SOURCE_CC

Definition at line 706 of file cfe_time_msg.h.

39.62.1.10 CFE_TIME_SET_SOURCE_CC

```
#define CFE_TIME_SET_SOURCE_CC 3 /* set clock source (int vs ext) */
```

Name Set Time Source

Description

This command selects the Time Service clock source. Although the list of potential clock sources is mission specific and defined via configuration parameters, this command provides a common method for switching between the local processor clock and an external source for time data.

When commanded to accept external time data (GPS, MET, spacecraft time, etc.), the Time Server will enable input via an API function specific to the configuration definitions for the particular source. When commanded to use internal time data, the Time Server will ignore the external data. However, the Time Server will continue to use the API function as the trigger to generate a "time at the tone" command packet regardless of the internal/external command selection.

Notes:

- Operating in FLYWHEEL mode is not considered a choice related to clock source, but rather an element of the clock state. See below for a description of the CFE_TIME_SET_STATE_CC command.
- This command is only valid when the CFE_PLATFORM_TIME_CFG_SOURCE configuration parameter in the cfe_platform_cfg.h file has been set to true.

Command Mnemonic(s) $sc_$cpu_TIME_SetSource

Command Structure

CFE_TIME_SetSourceCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_TIME_DSource`** - Diagnostic Telemetry point will indicate the command specified value
- The CFE_TIME_SOURCE_EID informational event message will be generated

Error Conditions

- Invalid Source selection (a value other than CFE_TIME_SourceSelect_INTERNAL or CFE_TIME_Source↩Select_EXTERNAL was specified)
- Time source selection not allowed on this platform

  Evidence of failure may be found in the following telemetry:
- **`$sc_$cpu_TIME_CMDEC`** - Command Error counter will increment
- Error specific event message (either CFE_TIME_SOURCE_CFG_EID or CFE_TIME_SOURCE_ERR_EID)

Criticality

Although clock source selection is important, this command is not critical.

See also

CFE_TIME_SET_STATE_CC, CFE_TIME_SET_SIGNAL_CC

Definition at line 210 of file cfe_time_msg.h.

39.62.1.11 CFE_TIME_SET_STATE_CC

```
#define CFE_TIME_SET_STATE_CC 4 /* set clock state */
```

Name Set Time State

Description

This command indirectly affects the Time Service on-board determination of clock state. Clock state is a combination of factors, most significantly whether the spacecraft time has been accurately set, and whether Time Service is operating in FLYWHEEL mode.

This command may be used to notify the Time Server that spacecraft time is now correct, or that time is no longer correct. This information will be distributed to Time Clients, and in turn, to any interested sub-systems.

Also, this command may be used to force a Time Server or Time Client into FLYWHEEL mode. Use of FLYWHEEL mode is mainly for debug purposes although in extreme circumstances, it may be of value to force Time Service not to rely on normal time updates. Note that when commanded into FLYWHEEL mode, the Time Service will remain so until receipt of another "set state" command setting the state into a mode other than FLYWHEEL.

Note also that setting the clock state to VALID or INVALID on a Time Client that is currently getting time updates from the Time Server will have very limited effect. As soon as the Time Client receives the next time update, the VALID/INVALID selection will be set to that of the Time Server. However, setting a Time Client to FLYWHEEL cannot be overridden by the Time Server since the Time Client will ignore time updates from the Time Server while in FLYWHEEL mode.

Command Mnemonic(s) $sc_$cpu_TIME_SetState

Command Structure

CFE_TIME_SetStateCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_TIME_StateFlg`**, $sc_$cpu_TIME_FlagSet, $sc_$cpu_TIME_FlagFly, $sc_$cpu_TIME_FlagSrc, $sc_$cpu_TIME_FlagPri, $sc_$cpu_TIME_FlagSfly, $sc_$cpu_TIME_FlagCfly, $sc_$cpu_TIME_FlagAdjd, $sc_$cpu_TIME_Flag1Hzd, $sc_$cpu_TIME_FlagClat, $sc_$cpu_TIME_FlagSorC, $sc_$cpu_TIME_FlagNIU - Housekeeping Telemetry point "may"l indicate the command specified value (see above)
- The CFE_TIME_STATE_EID informational event message will be generated

Error Conditions

- Invalid State selection (a value other than CFE_TIME_ClockState_INVALID, CFE_TIME_ClockState_VALID or CFE_TIME_ClockState_FLYWHEEL was specified)
- Time source selection not allowed on this platform

  Evidence of failure may be found in the following telemetry:

- **`$sc_$cpu_TIME_CMDEC`** - Command Error counter will increment

- Error specific event message (CFE_TIME_STATE_ERR_EID)

Criticality

Setting Time Service into FLYWHEEL mode is not particularly hazardous, as the result may be that the calculation of spacecraft time is done using a less than optimal timer. However, inappropriately setting the clock state to V↩ALID (indicating that spacecraft time is accurate) may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

CFE_TIME_SET_SOURCE_CC, CFE_TIME_SET_SIGNAL_CC

Definition at line 267 of file cfe_time_msg.h.

39.62.1.12 CFE_TIME_SET_STCF_CC

```
#define CFE_TIME_SET_STCF_CC 9 /* set STCF */
```

Name Set Spacecraft Time Correlation Factor

Description

This command sets the Spacecraft Time Correlation Factor (STCF) to the specified value. This command differs from the previously described SET CLOCK in the nature of the command argument. This command sets the STCF value directly, rather than extracting the STCF from a value representing the total of MET, STCF and optionally, Leap Seconds. The new STCF takes effect immediately upon execution of this command.

Command Mnemonic(s) $sc_$cpu_TIME_SetClockSTCF

Command Structure

CFE_TIME_SetSTCFCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_TIME_STCFSecs`** - Housekeeping Telemetry point indicating new STCF seconds value
- **`$sc_$cpu_TIME_STCFSubsecs`** - Housekeeping Telemetry point indicating new STCF subseconds value
- The CFE_TIME_STCF_EID informational event message will be generated

Error Conditions

- An invalid number of microseconds was specified (must be less than 1 million)

- Platorm receiving the command is not a Time Server

  Evidence of Failure may be found in the following telemetry:
- **`$sc_$cpu_TIME_CMDEC`** - command error counter will increment
- Error specific event messages will be issued (CFE_TIME_STCF_CFG_EID or CFE_TIME_STCF_ERR_EID)

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

CFE_TIME_SET_TIME_CC, CFE_TIME_SET_MET_CC, CFE_TIME_SET_LEAP_SECONDS_CC

Definition at line 465 of file cfe_time_msg.h.

39.62.1.13 CFE_TIME_SET_TIME_CC

```
#define CFE_TIME_SET_TIME_CC 7 /* set time */
```

Name Set Spacecraft Time

Description

This command sets the spacecraft clock to a new value, regardless of the current setting (time jam). The new time value represents the desired offset from the mission-defined time epoch and takes effect immediately upon execution of this command. Time Service will calculate a new STCF value based on the current MET and the desired new time using one of the following:

If Time Service is configured to compute current time as TAI

- **STCF = (new time) - (current MET)**
- **(current time) = (current MET) + STCF**

  If Time Service is configured to compute current time as UTC
- **STCF = ((new time) - (current MET)) + (Leap Seconds)**
- **(current time) = ((curent MET) + STCF) - (Leap Seconds)**

Command Mnemonic(s) $sc_$cpu_TIME_SetClock

Command Structure

CFE_TIME_SetTimeCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_TIME_STCFSecs`** - Housekeeping Telemetry point indicating newly calculated STCF seconds value
- **`$sc_$cpu_TIME_STCFSubsecs`** - Housekeeping Telemetry point indicating newly calculated STCF sub-seconds value
- The CFE_TIME_TIME_EID informational event message will be generated

Error Conditions

- An invalid number of microseconds was specified (must be less than 1 million)
- Platorm receiving the command is not a Time Server

  Evidence of Failure may be found in the following telemetry:

- **`$sc_$cpu_TIME_CMDEC`** - command error counter will increment
- Error specific event messages will be issued (CFE_TIME_TIME_CFG_EID or CFE_TIME_TIME_ERR_EID)

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

CFE_TIME_SET_MET_CC, CFE_TIME_SET_STCF_CC, CFE_TIME_SET_LEAP_SECONDS_CC

Definition at line 388 of file cfe_time_msg.h.

39.62.1.14 CFE_TIME_SUB_1HZ_ADJUSTMENT_CC

```
#define CFE_TIME_SUB_1HZ_ADJUSTMENT_CC 14 /* subtract 1Hz STCF adjustment */
```

Name Subtract Delta from Spacecraft Time Correlation Factor each 1Hz

Description

This command has been updated to take actual sub-seconds (1/2^32 seconds) rather than micro-seconds as an input argument. This change occurred after the determination was made that one micro-second is too large an increment for a constant 1Hz adjustment.

This command continuously adjusts the Spacecraft Time Correlation Factor (STCF) every second, by subtracting the specified value. The adjustment to the STCF is applied in the Time Service local 1Hz interrupt handler. As the local 1Hz interrupt is not synchronized to the tone signal, one cannot say when the adjustment will occur, other than once a second, at about the same time relative to the tone.

There was some debate about whether the maximum 1Hz clock drift correction factor would ever need to exceed some small fraction of a second. But, the decision was made to provide the capability to make 1Hz adjustments greater than one second and leave it to the ground system to provide mission specific limits.

Command Mnemonic(s) $sc_$cpu_TIME_Sub1HzSTCF

Command Structure

CFE_TIME_Sub1HZAdjustmentCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry: Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_TIME_STCFSecs`** - Housekeeping Telemetry point indicating new STCF seconds value
- **`$sc_$cpu_TIME_STCFSubsecs`** - Housekeeping Telemetry point indicating new STCF subseconds value
- The CFE_TIME_1HZ_EID informational event message will be generated

Error Conditions

- Platorm receiving the command is not a Time Server

  Evidence of Failure may be found in the following telemetry:
- **`$sc_$cpu_TIME_CMDEC`** - command error counter will increment
- Error specific event message will be issued (CFE_TIME_1HZ_CFG_EID)

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

CFE_TIME_ADD_ADJUST_CC, CFE_TIME_SUB_ADJUST_CC, CFE_TIME_ADD_1HZ_ADJUSTMENT_CC

Definition at line 664 of file cfe_time_msg.h.

39.62.1.15 CFE_TIME_SUB_ADJUST_CC

```
#define CFE_TIME_SUB_ADJUST_CC 12 /* subtract one time STCF adjustment */
```

Name Subtract Delta from Spacecraft Time Correlation Factor

Description

This command adjusts the Spacecraft Time Correlation Factor (STCF) by subtracting the specified value. The new STCF takes effect immediately upon execution of this command.

Command Mnemonic(s) $sc_$cpu_TIME_SubSTCFAdj

Command Structure

CFE_TIME_SubAdjustCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_TIME_STCFSecs`** - Housekeeping Telemetry point indicating new STCF seconds value
- **`$sc_$cpu_TIME_STCFSubsecs`** - Housekeeping Telemetry point indicating new STCF subseconds value
- The CFE_TIME_DELTA_EID informational event message will be generated

Error Conditions

- An invalid number of microseconds was specified (must be less than 1 million)
- Platorm receiving the command is not a Time Server

  Evidence of Failure may be found in the following telemetry:
- **`$sc_$cpu_TIME_CMDEC`** - command error counter will increment
- Error specific event messages will be issued (CFE_TIME_DELTA_ERR_EID or CFE_TIME_DELTA_CFG_↩EID)

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

CFE_TIME_ADD_ADJUST_CC, CFE_TIME_ADD_1HZ_ADJUSTMENT_CC, CFE_TIME_SUB_1HZ_ADJUST↩MENT_CC

Definition at line 570 of file cfe_time_msg.h.

39.62.1.16 CFE_TIME_SUB_DELAY_CC

```
#define CFE_TIME_SUB_DELAY_CC 6 /* sub tone delay value */
```

Name Subtract Time from Tone Time Delay

Description

This command is used to factor out a known, predictable latency between the Time Server and a particular Time Client. The correction is applied (subtracted) to the current time calculation for Time Clients, so this command has no meaning for Time Servers. Each Time Client can have a unique latency setting. The latency value is a positive number of seconds and microseconds that represent the deviation from the time maintained by the Time Server.

Note that it is unimaginable that the seconds value will ever be anything but zero.

Command Mnemonic(s) $sc_$cpu_TIME_SubClockLat

Command Structure

CFE_TIME_SubDelayCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **`$sc_$cpu_TIME_CMDPC`** - command execution counter will increment
- **`$sc_$cpu_TIME_DLatentS`**, $sc_$cpu_TIME_DLatentSs - Housekeeping Telemetry point indicating command specified values
- **`$sc_$cpu_TIME_DLatentDir`** - Diagnostic Telemetry point indicating commanded latency direction
- The CFE_TIME_DELAY_EID informational event message will be generated

Error Conditions

- An invalid number of microseconds was specified (must be less than 1 million)
- Platorm receiving the command is not a Time Client

  Evidence of Failure may be found in the following telemetry:
- **`$sc_$cpu_TIME_CMDEC`** - command error counter will increment
- Error specific event messages will be issued (CFE_TIME_DELAY_CFG_EID or CFE_TIME_DELAY_ERR↩_EID)

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

CFE_TIME_ADD_DELAY_CC

Definition at line 343 of file cfe_time_msg.h.

39.62.2 Typedef Documentation

39.62.2.1 CFE_TIME_1HzCmd_t

```
typedef CFE_TIME_NoArgsCmd_t CFE_TIME_1HzCmd_t
```

Definition at line 745 of file cfe_time_msg.h.

39.62.2.2 CFE_TIME_Add1HZAdjustmentCmd_t

```
typedef CFE_TIME_OneHzAdjustmentCmd_t CFE_TIME_Add1HZAdjustmentCmd_t
```

Definition at line 879 of file cfe_time_msg.h.

39.62.2.3 CFE_TIME_AddAdjustCmd_t

```
typedef CFE_TIME_TimeCmd_t CFE_TIME_AddAdjustCmd_t
```

Definition at line 851 of file cfe_time_msg.h.

39.62.2.4 CFE_TIME_AddDelayCmd_t

```
typedef CFE_TIME_TimeCmd_t CFE_TIME_AddDelayCmd_t
```

Definition at line 847 of file cfe_time_msg.h.

39.62.2.5 CFE_TIME_DiagnosticTlm_Payload_t

```
typedef struct CFE_TIME_DiagnosticTlm_Payload CFE_TIME_DiagnosticTlm_Payload_t
```

Name Time Services Diagnostics Packet

39.62.2.6 CFE_TIME_DiagnosticTlm_t

```
typedef struct CFE_TIME_DiagnosticTlm CFE_TIME_DiagnosticTlm_t
```

39.62.2.7 CFE_TIME_FakeToneCmd_t

```
typedef CFE_TIME_NoArgsCmd_t CFE_TIME_FakeToneCmd_t
```

Definition at line 747 of file cfe_time_msg.h.

39.62.2.8 CFE_TIME_HousekeepingTlm_Payload_t

```
typedef struct CFE_TIME_HousekeepingTlm_Payload CFE_TIME_HousekeepingTlm_Payload_t
```

Name Time Services Housekeeping Packet

39.62.2.9 CFE_TIME_HousekeepingTlm_t

```
typedef struct CFE_TIME_HousekeepingTlm CFE_TIME_HousekeepingTlm_t
```

39.62.2.10 CFE_TIME_LeapsCmd_Payload_t

```
typedef struct CFE_TIME_LeapsCmd_Payload CFE_TIME_LeapsCmd_Payload_t
```

Set leap seconds command payload.

39.62.2.11 CFE_TIME_NoArgsCmd_t

```
typedef struct CFE_TIME_NoArgsCmd CFE_TIME_NoArgsCmd_t
```

Generic no argument command.

39.62.2.12 CFE_TIME_NoopCmd_t

```
typedef CFE_TIME_NoArgsCmd_t CFE_TIME_NoopCmd_t
```

Definition at line 742 of file cfe_time_msg.h.

39.62.2.13 CFE_TIME_OneHzAdjustmentCmd_Payload_t

```
typedef struct CFE_TIME_OneHzAdjustmentCmd_Payload CFE_TIME_OneHzAdjustmentCmd_Payload_t
```

Generic seconds, subseconds command payload.

39.62.2.14 CFE_TIME_OneHzAdjustmentCmd_t

```
typedef struct CFE_TIME_OneHzAdjustmentCmd CFE_TIME_OneHzAdjustmentCmd_t
```

Generic seconds, subseconds adjustment command.

39.62.2.15 CFE_TIME_ResetCountersCmd_t

```
typedef CFE_TIME_NoArgsCmd_t CFE_TIME_ResetCountersCmd_t
```

Definition at line 743 of file cfe_time_msg.h.

39.62.2.16 CFE_TIME_SendDiagnosticCmd_t

```
typedef CFE_TIME_NoArgsCmd_t CFE_TIME_SendDiagnosticCmd_t
```

Definition at line 744 of file cfe_time_msg.h.

39.62.2.17 CFE_TIME_SetLeapSecondsCmd_t

```
typedef struct CFE_TIME_SetLeapSecondsCmd CFE_TIME_SetLeapSecondsCmd_t
```

Set leap seconds command.

39.62.2.18 CFE_TIME_SetMETCmd_t

```
typedef CFE_TIME_TimeCmd_t CFE_TIME_SetMETCmd_t
```

Definition at line 849 of file cfe_time_msg.h.

39.62.2.19 CFE_TIME_SetSignalCmd_t

```
typedef struct CFE_TIME_SetSignalCmd CFE_TIME_SetSignalCmd_t
```

Set tone signal source command.

39.62.2.20 CFE_TIME_SetSourceCmd_t

```
typedef struct CFE_TIME_SetSourceCmd CFE_TIME_SetSourceCmd_t
```

Set time data source command.

39.62.2.21 CFE_TIME_SetStateCmd_t

```
typedef struct CFE_TIME_SetStateCmd CFE_TIME_SetStateCmd_t
```

Set clock state command.

39.62.2.22 CFE_TIME_SetSTCFCmd_t

```
typedef CFE_TIME_TimeCmd_t CFE_TIME_SetSTCFCmd_t
```

Definition at line 850 of file cfe_time_msg.h.

39.62.2.23 CFE_TIME_SetTimeCmd_t

```
typedef CFE_TIME_TimeCmd_t CFE_TIME_SetTimeCmd_t
```

Definition at line 853 of file cfe_time_msg.h.

39.62.2.24 CFE_TIME_SignalCmd_Payload_t

```
typedef struct CFE_TIME_SignalCmd_Payload CFE_TIME_SignalCmd_Payload_t
```

Set tone signal source command payload.

39.62.2.25 CFE_TIME_SourceCmd_Payload_t

```
typedef struct CFE_TIME_SourceCmd_Payload CFE_TIME_SourceCmd_Payload_t
```

Set time data source command payload.

39.62.2.26 CFE_TIME_StateCmd_Payload_t

```
typedef struct CFE_TIME_StateCmd_Payload CFE_TIME_StateCmd_Payload_t
```

Set clock state command payload.

39.62.2.27 CFE_TIME_Sub1HZAdjustmentCmd_t

```
typedef CFE_TIME_OneHzAdjustmentCmd_t CFE_TIME_Sub1HZAdjustmentCmd_t
```

Definition at line 880 of file cfe_time_msg.h.

39.62.2.28 CFE_TIME_SubAdjustCmd_t

```
typedef CFE_TIME_TimeCmd_t CFE_TIME_SubAdjustCmd_t
```

Definition at line 852 of file cfe_time_msg.h.

39.62.2.29 CFE_TIME_SubDelayCmd_t

```
typedef CFE_TIME_TimeCmd_t CFE_TIME_SubDelayCmd_t
```

Definition at line 848 of file cfe_time_msg.h.

39.62.2.30 CFE_TIME_TimeCmd_Payload_t

```
typedef struct CFE_TIME_TimeCmd_Payload CFE_TIME_TimeCmd_Payload_t
```

Generic seconds, microseconds command payload.

39.62.2.31 CFE_TIME_TimeCmd_t

```
typedef struct CFE_TIME_TimeCmd CFE_TIME_TimeCmd_t
```

Generic seconds, microseconds argument command.

39.62.2.32 CFE_TIME_ToneDataCmd_Payload_t

```
typedef struct CFE_TIME_ToneDataCmd_Payload CFE_TIME_ToneDataCmd_Payload_t
```

Time at tone data command payload.

39.62.2.33 CFE_TIME_ToneDataCmd_t

```
typedef struct CFE_TIME_ToneDataCmd CFE_TIME_ToneDataCmd_t
```

Time at tone data command.

39.62.2.34 CFE_TIME_ToneSignalCmd_t

```
typedef CFE_TIME_NoArgsCmd_t CFE_TIME_ToneSignalCmd_t
```

Definition at line 746 of file cfe_time_msg.h.

39.63 osal/src/os/inc/common_types.h File Reference

```
#include <stdint.h>
#include <stddef.h>
#include <stdbool.h>
```

Macros

- #define CompileTimeAssert(Condition, Message) typedef char Message[(Condition) ? 1 : -1]
- #define _EXTENSION_
- #define OS_USED
- #define OS_PRINTF(n, m)
- #define OSAL_SIZE_C(X) ((size_t)(X))
- #define OSAL_BLOCKCOUNT_C(X) ((osal_blockcount_t)(X))
- #define OSAL_INDEX_C(X) ((osal_index_t)(X))
- #define OSAL_OBJTYPE_C(X) ((osal_objtype_t)(X))

Typedefs

- typedef int8_t int8
- typedef int16_t int16
- typedef int32_t int32
- typedef int64_t int64
- typedef uint8_t uint8
- typedef uint16_t uint16
- typedef uint32_t uint32
- typedef uint64_t uint64
- typedef intptr_t intptr
- typedef uintptr_t cpuaddr
- typedef size_t cpusize
- typedef ptrdiff_t cpudiff
- typedef uint32_t osal_id_t
- typedef size_t osal_blockcount_t
- typedef uint32 osal_index_t
- typedef uint32 osal_objtype_t
- typedef void(∗ OS_ArgCallback_t) (osal_id_t object_id, void ∗arg)

  General purpose OSAL callback function.

Functions

- CompileTimeAssert (sizeof(uint8)==1, TypeUint8WrongSize)
- CompileTimeAssert (sizeof(uint16)==2, TypeUint16WrongSize)
- CompileTimeAssert (sizeof(uint32)==4, TypeUint32WrongSize)
- CompileTimeAssert (sizeof(uint64)==8, TypeUint64WrongSize)
- CompileTimeAssert (sizeof(int8)==1, Typeint8WrongSize)
- CompileTimeAssert (sizeof(int16)==2, Typeint16WrongSize)
- CompileTimeAssert (sizeof(int32)==4, Typeint32WrongSize)
- CompileTimeAssert (sizeof(int64)==8, Typeint64WrongSize)
- CompileTimeAssert (sizeof(cpuaddr) >=sizeof(void ∗), TypePtrWrongSize)

39.63.1 Macro Definition Documentation

39.63.1.1 _EXTENSION_

```
#define _EXTENSION_
```

Definition at line 67 of file common_types.h.

39.63.1.2 CompileTimeAssert

```
#define CompileTimeAssert(
            Condition,
            Message ) typedef char Message[(Condition) ?  1 :  -1]
```

Definition at line 50 of file common_types.h.

39.63.1.3 OS_PRINTF

```
#define OS_PRINTF(
            n,
            m )
```

Definition at line 69 of file common_types.h.

39.63.1.4 OS_USED

```
#define OS_USED
```

Definition at line 68 of file common_types.h.

39.63.1.5 OSAL_BLOCKCOUNT_C

```
#define OSAL_BLOCKCOUNT_C(
            X ) ((osal_blockcount_t)(X))
```

Definition at line 183 of file common_types.h.

39.63.1.6 OSAL_INDEX_C

```
#define OSAL_INDEX_C(
            X ) ((osal_index_t)(X))
```

Definition at line 184 of file common_types.h.

39.63.1.7 OSAL_OBJTYPE_C

```
#define OSAL_OBJTYPE_C(
            X ) ((osal_objtype_t)(X))
```

Definition at line 185 of file common_types.h.

39.63.1.8 OSAL_SIZE_C

```
#define OSAL_SIZE_C(
              X ) ((size_t)(X))
```

Definition at line 182 of file common_types.h.

39.63.2 Typedef Documentation

39.63.2.1 cpuaddr

```
typedef uintptr_t cpuaddr
```

Definition at line 90 of file common_types.h.

39.63.2.2 cpudiff

```
typedef ptrdiff_t cpudiff
```

Definition at line 92 of file common_types.h.

39.63.2.3 cpusize

```
typedef size_t cpusize
```

Definition at line 91 of file common_types.h.

39.63.2.4 int16

```
typedef int16_t int16
```

Definition at line 82 of file common_types.h.

39.63.2.5 int32

```
typedef int32_t int32
```

Definition at line 83 of file common_types.h.

39.63.2.6 int64

```
typedef int64_t int64
```

Definition at line 84 of file common_types.h.

39.63.2.7 int8

```
typedef int8_t int8
```

Definition at line 81 of file common_types.h.

39.63.2.8 intptr

```
typedef intptr_t intptr
```

Definition at line 89 of file common_types.h.

39.63.2.9 OS_ArgCallback_t

```
typedef void(* OS_ArgCallback_t) (osal_id_t object_id, void *arg)
```

General purpose OSAL callback function.

This may be used by multiple APIS

Definition at line 126 of file common_types.h.

39.63.2.10 osal_blockcount_t

```
typedef size_t osal_blockcount_t
```

A type used to represent a number of blocks or buffers

This is used with file system and queue implementations.

Definition at line 104 of file common_types.h.

39.63.2.11 osal_id_t

```
typedef uint32_t osal_id_t
```

A type to be used for OSAL resource identifiers.

Definition at line 97 of file common_types.h.

39.63.2.12 osal_index_t

```
typedef uint32 osal_index_t
```

A type used to represent an index into a table structure

This is used when referring directly to a table index as opposed to an object ID. It is primarily intended for internal use, but is also output from public APIs such as OS_ObjectIdToArrayIndex().

Definition at line 114 of file common_types.h.

39.63.2.13 osal_objtype_t

```
typedef uint32 osal_objtype_t
```

A type used to represent the runtime type or category of an OSAL object

Definition at line 119 of file common_types.h.

39.63.2.14 uint16

```
typedef uint16_t uint16
```

Definition at line 86 of file common_types.h.

39.63.2.15 uint32

```
typedef uint32_t uint32
```

Definition at line 87 of file common_types.h.

39.63.2.16 uint64

```
typedef uint64_t uint64
```

Definition at line 88 of file common_types.h.

39.63.2.17 uint8

```
typedef uint8_t uint8
```

Definition at line 85 of file common_types.h.

39.63.3 Function Documentation

39.63.3.1 CompileTimeAssert() [1/9]

```
CompileTimeAssert (
            sizeof(uint8)  = =1,
            TypeUint8WrongSize  )
```

39.63.3.2 CompileTimeAssert() [2/9]

```
CompileTimeAssert (
            sizeof(uint16)  = =2,
            TypeUint16WrongSize  )
```

39.63.3.3 CompileTimeAssert() [3/9]

```
CompileTimeAssert (
            sizeof(uint32)  = =4,
            TypeUint32WrongSize  )
```

39.63.3.4 CompileTimeAssert() [4/9]

```
CompileTimeAssert (
            sizeof(uint64)  = =8,
            TypeUint64WrongSize  )
```

39.63.3.5 CompileTimeAssert() [5/9]

```
CompileTimeAssert (
            sizeof(int8)  = =1,
            Typeint8WrongSize  )
```

39.63.3.6 CompileTimeAssert() [6/9]

```
CompileTimeAssert (
            sizeof(int16)  = =2,
            Typeint16WrongSize  )
```

39.63.3.7 CompileTimeAssert() [7/9]

```
CompileTimeAssert (
            sizeof(int32)  = =4,
            Typeint32WrongSize  )
```

39.63.3.8 CompileTimeAssert() [8/9]

```
CompileTimeAssert (
            sizeof(int64)  = =8,
            Typeint64WrongSize  )
```

39.63.3.9 CompileTimeAssert() [9/9]

```
CompileTimeAssert (
            sizeof(cpuaddr) >=sizeof(void *) ,
            TypePtrWrongSize  )
```

39.64 osal/src/os/inc/osapi-binsem.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- struct OS_bin_sem_prop_t

  OSAL binary semaphore properties.

Macros

- #define OS_SEM_FULL 1

  Semaphore full state.
- #define OS_SEM_EMPTY 0

  Semaphore empty state.

Functions

- int32 OS_BinSemCreate (osal_id_t *sem_id, const char *sem_name, uint32 sem_initial_value, uint32 options)

  Creates a binary semaphore.
- int32 OS_BinSemFlush (osal_id_t sem_id)

  Unblock all tasks pending on the specified semaphore.
- int32 OS_BinSemGive (osal_id_t sem_id)

  Increment the semaphore value.
- int32 OS_BinSemTake (osal_id_t sem_id)

  Decrement the semaphore value.
- int32 OS_BinSemTimedWait (osal_id_t sem_id, uint32 msecs)

  Decrement the semaphore value with a timeout.
- int32 OS_BinSemDelete (osal_id_t sem_id)

  Deletes the specified Binary Semaphore.
- int32 OS_BinSemGetIdByName (osal_id_t *sem_id, const char *sem_name)

  Find an existing semaphore ID by name.
- int32 OS_BinSemGetInfo (osal_id_t sem_id, OS_bin_sem_prop_t *bin_prop)

  Fill a property object buffer with details regarding the resource.

39.65 osal/src/os/inc/osapi-bsp.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Functions

- uint32 OS_BSP_GetArgC (void)
- char *const * OS_BSP_GetArgV (void)
- void OS_BSP_SetExitCode (int32 code)

39.66 osal/src/os/inc/osapi-clock.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- struct OS_time_t

  OSAL time interval structure.

Enumerations

- enum { OS_TIME_TICK_RESOLUTION_NS = 100, OS_TIME_TICKS_PER_SECOND = 1000000000 / OS_TI↩ME_TICK_RESOLUTION_NS, OS_TIME_TICKS_PER_MSEC = 1000000 / OS_TIME_TICK_RESOLUTION_↩NS, OS_TIME_TICKS_PER_USEC = 1000 / OS_TIME_TICK_RESOLUTION_NS }

  Multipliers/divisors to convert ticks into standardized units.

Functions

- int32 OS_GetLocalTime (OS_time_t *time_struct)

  Get the local time.
- int32 OS_SetLocalTime (const OS_time_t *time_struct)

  Set the local time.
- static int64 OS_TimeGetTotalSeconds (OS_time_t tm)

  Get interval from an OS_time_t object normalized to whole number of seconds.
- static int64 OS_TimeGetTotalMilliseconds (OS_time_t tm)

  Get interval from an OS_time_t object normalized to millisecond units.
- static int64 OS_TimeGetTotalMicroseconds (OS_time_t tm)

  Get interval from an OS_time_t object normalized to microsecond units.
- static int64 OS_TimeGetTotalNanoseconds (OS_time_t tm)

  Get interval from an OS_time_t object normalized to nanosecond units.
- static int64 OS_TimeGetFractionalPart (OS_time_t tm)

  Get subseconds portion (fractional part only) from an OS_time_t object.
- static uint32 OS_TimeGetSubsecondsPart (OS_time_t tm)

  Get 32-bit normalized subseconds (fractional part only) from an OS_time_t object.
- static uint32 OS_TimeGetMillisecondsPart (OS_time_t tm)

  Get milliseconds portion (fractional part only) from an OS_time_t object.
- static uint32 OS_TimeGetMicrosecondsPart (OS_time_t tm)

  Get microseconds portion (fractional part only) from an OS_time_t object.
- static uint32 OS_TimeGetNanosecondsPart (OS_time_t tm)

  Get nanoseconds portion (fractional part only) from an OS_time_t object.
- static OS_time_t OS_TimeAssembleFromNanoseconds (int64 seconds, uint32 nanoseconds)

  Assemble/Convert a number of seconds + nanoseconds into an OS_time_t interval.
- static OS_time_t OS_TimeAssembleFromMicroseconds (int64 seconds, uint32 microseconds)

  Assemble/Convert a number of seconds + microseconds into an OS_time_t interval.
- static OS_time_t OS_TimeAssembleFromMilliseconds (int64 seconds, uint32 milliseconds)

  Assemble/Convert a number of seconds + milliseconds into an OS_time_t interval.
- static OS_time_t OS_TimeAssembleFromSubseconds (int64 seconds, uint32 subseconds)

  Assemble/Convert a number of seconds + subseconds into an OS_time_t interval.
- static OS_time_t OS_TimeAdd (OS_time_t time1, OS_time_t time2)

  Computes the sum of two time intervals.
- static OS_time_t OS_TimeSubtract (OS_time_t time1, OS_time_t time2)

  Computes the difference between two time intervals.

39.66.1 Enumeration Type Documentation

39.66.1.1 anonymous enum

```
anonymous enum
```

Multipliers/divisors to convert ticks into standardized units.

Various fixed conversion factor constants used by the conversion routines

A 100ns tick time allows max intervals of about +/- 14000 years in a 64-bit signed integer value.

Note

Applications should not directly use these values, but rather use conversion routines below to obtain standardized units (seconds/microseconds/etc).

Enumerator

OS_TIME_TICK_RESOLUTION_NS	
OS_TIME_TICKS_PER_SECOND	
OS_TIME_TICKS_PER_MSEC	
OS_TIME_TICKS_PER_USEC	

Definition at line 61 of file osapi-clock.h.

39.67 osal/src/os/inc/osapi-common.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Typedefs

- typedef int32(∗ OS_EventHandler_t) (OS_Event_t event, osal_id_t object_id, void ∗data)

  A callback routine for event handling.

Enumerations

- enum OS_Event_t {
  OS_EVENT_RESERVED = 0, OS_EVENT_RESOURCE_ALLOCATED, OS_EVENT_RESOURCE_CREATED, OS_EVENT_RESOURCE_DELETED,
  OS_EVENT_TASK_STARTUP, OS_EVENT_MAX }

  A set of events that can be used with BSP event callback routines.

Functions

- void OS_Application_Startup (void)

  Application startup.
- void OS_Application_Run (void)

  Application run.
- int32 OS_API_Init (void)

  Initialization of API.
- void OS_IdleLoop (void)

  Background thread implementation - waits forever for events to occur.
- void OS_DeleteAllObjects (void)

  delete all resources created in OSAL.
- void OS_ApplicationShutdown (uint8 flag)

  Initiate orderly shutdown.
- void OS_ApplicationExit (int32 Status)

  Exit/Abort the application.
- int32 OS_RegisterEventHandler (OS_EventHandler_t handler)

  Callback routine registration.

39.67.1 Typedef Documentation

39.67.1.1 OS_EventHandler_t

```
typedef int32(* OS_EventHandler_t) (OS_Event_t event, osal_id_t object_id, void *data)
```

A callback routine for event handling.

Parameters

in	*event*	The event that occurred
in	*object↩ _id*	The associated object_id, or 0 if not associated with an object
in,out	*data*	An abstract data/context object associated with the event, or NULL.

Returns

status Execution status, see OSAL Return Code Defines.

Definition at line 98 of file osapi-common.h.

39.67.2 Enumeration Type Documentation

39.67.2.1 OS_Event_t

```
enum OS_Event_t
```

A set of events that can be used with BSP event callback routines.

Enumerator

OS_EVENT_RESERVED	no-op/reserved event id value
OS_EVENT_RESOURCE_ALLOCATED	resource/id has been newly allocated but not yet created. This event is invoked from WITHIN the locked region, in the context of the task which is allocating the resource. If the handler returns non-success, the error will be returned to the caller and the creation process is aborted.
OS_EVENT_RESOURCE_CREATED	resource/id has been fully created/finalized. Invoked outside locked region, in the context of the task which created the resource. Data object is not used, passed as NULL. Return value is ignored - this is for information purposes only.
OS_EVENT_RESOURCE_DELETED	resource/id has been deleted. Invoked outside locked region, in the context of the task which deleted the resource. Data object is not used, passed as NULL. Return value is ignored - this is for information purposes only.
OS_EVENT_TASK_STARTUP	New task is starting. Invoked outside locked region, in the context of the task which is currently starting, before the entry point is called. Data object is not used, passed as NULL. If the handler returns non-success, task startup is aborted and the entry point is not called.
OS_EVENT_MAX	placeholder for end of enum, not used

Definition at line 34 of file osapi-common.h.

39.68 osal/src/os/inc/osapi-constants.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Macros

- #define OS_PEND (-1)
- #define OS_CHECK (0)
- #define OS_OBJECT_ID_UNDEFINED ((osal_id_t) {0})

  Initializer for the osal_id_t type which will not match any valid value.
- #define OS_OBJECT_CREATOR_ANY OS_OBJECT_ID_UNDEFINED

  Constant that may be passed to OS_ForEachObject()/OS_ForEachObjectOfType() to match any creator (i.e. get all objects)
- #define OS_MAX_LOCAL_PATH_LEN (OS_MAX_PATH_LEN + OS_FS_PHYS_NAME_LEN)

  Maximum length of a local/native path name string.

39.68.1 Macro Definition Documentation

39.68.1.1 OS_CHECK

```
#define OS_CHECK (0)
```

Definition at line 35 of file osapi-constants.h.

39.68.1.2 OS_MAX_LOCAL_PATH_LEN

```
#define OS_MAX_LOCAL_PATH_LEN (OS_MAX_PATH_LEN + OS_FS_PHYS_NAME_LEN)
```

Maximum length of a local/native path name string.

This is a concatenation of the OSAL virtual path with the system mount point or device name

Definition at line 54 of file osapi-constants.h.

39.68.1.3 OS_OBJECT_CREATOR_ANY

```
#define OS_OBJECT_CREATOR_ANY OS_OBJECT_ID_UNDEFINED
```

Constant that may be passed to OS_ForEachObject()/OS_ForEachObjectOfType() to match any creator (i.e. get all objects)

Definition at line 46 of file osapi-constants.h.

39.68.1.4 OS_OBJECT_ID_UNDEFINED

```
#define OS_OBJECT_ID_UNDEFINED ((osal_id_t) {0})
```

Initializer for the osal_id_t type which will not match any valid value.

Definition at line 40 of file osapi-constants.h.

Referenced by OS_Application_Startup().

39.68.1.5 OS_PEND

```
#define OS_PEND (-1)
```

Definition at line 34 of file osapi-constants.h.

39.69 osal/src/os/inc/osapi-countsem.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- struct OS_count_sem_prop_t

  OSAL counting semaphore properties.

Functions

- int32 OS_CountSemCreate (osal_id_t *sem_id, const char *sem_name, uint32 sem_initial_value, uint32 options)

  Creates a counting semaphore.
- int32 OS_CountSemGive (osal_id_t sem_id)

  Increment the semaphore value.
- int32 OS_CountSemTake (osal_id_t sem_id)

  Decrement the semaphore value.
- int32 OS_CountSemTimedWait (osal_id_t sem_id, uint32 msecs)

  Decrement the semaphore value with timeout.
- int32 OS_CountSemDelete (osal_id_t sem_id)

  Deletes the specified counting Semaphore.
- int32 OS_CountSemGetIdByName (osal_id_t *sem_id, const char *sem_name)

  Find an existing semaphore ID by name.
- int32 OS_CountSemGetInfo (osal_id_t sem_id, OS_count_sem_prop_t *count_prop)

  Fill a property object buffer with details regarding the resource.

39.70 osal/src/os/inc/osapi-dir.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- struct os_dirent_t

  Directory entry.

Macros

- #define OS_DIRENTRY_NAME(x) ((x).FileName)

  Access filename part of the dirent structure.

Functions

- int32 OS_DirectoryOpen (osal_id_t *dir_id, const char *path)

  Opens a directory.
- int32 OS_DirectoryClose (osal_id_t dir_id)

  Closes an open directory.
- int32 OS_DirectoryRewind (osal_id_t dir_id)

  Rewinds an open directory.
- int32 OS_DirectoryRead (osal_id_t dir_id, os_dirent_t *dirent)

  Reads the next name in the directory.
- int32 OS_mkdir (const char *path, uint32 access)

  Makes a new directory.
- int32 OS_rmdir (const char *path)

  Removes a directory from the file system.

39.70.1 Macro Definition Documentation

39.70.1.1 OS_DIRENTRY_NAME

```
#define OS_DIRENTRY_NAME(
              x ) ((x).FileName)
```

Access filename part of the dirent structure.

Definition at line 38 of file osapi-dir.h.

39.71 osal/src/os/inc/osapi-error.h File Reference

```
#include "common_types.h"
```

Macros

- #define OS_ERROR_NAME_LENGTH 35

  Error string name length.
- #define OS_SUCCESS (0)

  Successful execution.
- #define OS_ERROR (-1)

  Failed execution.
- #define OS_INVALID_POINTER (-2)

  Invalid pointer.
- #define OS_ERROR_ADDRESS_MISALIGNED (-3)

  Address misalignment.

- #define OS_ERROR_TIMEOUT (-4)

  Error timeout.
- #define OS_INVALID_INT_NUM (-5)

  Invalid Interrupt number.
- #define OS_SEM_FAILURE (-6)

  Semaphore failure.
- #define OS_SEM_TIMEOUT (-7)

  Semaphore timeout.
- #define OS_QUEUE_EMPTY (-8)

  Queue empty.
- #define OS_QUEUE_FULL (-9)

  Queue full.
- #define OS_QUEUE_TIMEOUT (-10)

  Queue timeout.
- #define OS_QUEUE_INVALID_SIZE (-11)

  Queue invalid size.
- #define OS_QUEUE_ID_ERROR (-12)

  Queue ID error.
- #define OS_ERR_NAME_TOO_LONG (-13)

  name length including null terminator greater than OS_MAX_API_NAME
- #define OS_ERR_NO_FREE_IDS (-14)

  No free IDs.
- #define OS_ERR_NAME_TAKEN (-15)

  Name taken.
- #define OS_ERR_INVALID_ID (-16)

  Invalid ID.
- #define OS_ERR_NAME_NOT_FOUND (-17)

  Name not found.
- #define OS_ERR_SEM_NOT_FULL (-18)

  Semaphore not full.
- #define OS_ERR_INVALID_PRIORITY (-19)

  Invalid priority.
- #define OS_INVALID_SEM_VALUE (-20)

  Invalid semaphore value.
- #define OS_ERR_FILE (-27)

  File error.
- #define OS_ERR_NOT_IMPLEMENTED (-28)

  Not implemented.
- #define OS_TIMER_ERR_INVALID_ARGS (-29)

  Timer invalid arguments.
- #define OS_TIMER_ERR_TIMER_ID (-30)

  Timer ID error.
- #define OS_TIMER_ERR_UNAVAILABLE (-31)

  Timer unavailable.
- #define OS_TIMER_ERR_INTERNAL (-32)

  Timer internal error.
- #define OS_ERR_OBJECT_IN_USE (-33)

Object in use.

- #define OS_ERR_BAD_ADDRESS (-34)

  Bad address.
- #define OS_ERR_INCORRECT_OBJ_STATE (-35)

  Incorrect object state.
- #define OS_ERR_INCORRECT_OBJ_TYPE (-36)

  Incorrect object type.
- #define OS_ERR_STREAM_DISCONNECTED (-37)

  Stream disconnected.
- #define OS_ERR_OPERATION_NOT_SUPPORTED (-38)

  Requested operation not support on supplied object(s)
- #define OS_ERR_INVALID_SIZE (-40)

  Invalid Size.
- #define OS_FS_ERR_PATH_TOO_LONG (-103)

  FS path too long.
- #define OS_FS_ERR_NAME_TOO_LONG (-104)

  FS name too long.
- #define OS_FS_ERR_DRIVE_NOT_CREATED (-106)

  FS drive not created.
- #define OS_FS_ERR_DEVICE_NOT_FREE (-107)

  FS device not free.
- #define OS_FS_ERR_PATH_INVALID (-108)

  FS path invalid.

Typedefs

- typedef char os_err_name_t[OS_ERROR_NAME_LENGTH]

  For the OS_GetErrorName() function, to ensure everyone is making an array of the same length.

Functions

- int32 OS_GetErrorName (int32 error_num, os_err_name_t *err_name)

  Convert an error number to a string.

39.71.1 Macro Definition Documentation

39.71.1.1 OS_ERROR_NAME_LENGTH

```
#define OS_ERROR_NAME_LENGTH 35
```

Error string name length.

The sizes of strings in OSAL functions are built with this limit in mind. Always check the uses of os_err_name_t when changing this value.

Definition at line 35 of file osapi-error.h.

39.71.2 Typedef Documentation

39.71.2.1 os_err_name_t

```
typedef char os_err_name_t[OS_ERROR_NAME_LENGTH]
```

For the OS_GetErrorName() function, to ensure everyone is making an array of the same length.

Implementation note for developers:

The sizes of strings in OSAL functions are built with this OS_ERROR_NAME_LENGTH limit in mind. Always check the uses of os_err_name_t when changing this value.

Definition at line 47 of file osapi-error.h.

39.72 osal/src/os/inc/osapi-file.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
#include "osapi-clock.h"
```

Data Structures

- struct OS_file_prop_t

  OSAL file properties.
- struct os_fstat_t

  File system status.

Macros

- #define OS_READ_ONLY 0
- #define OS_WRITE_ONLY 1
- #define OS_READ_WRITE 2
- #define OS_SEEK_SET 0
- #define OS_SEEK_CUR 1
- #define OS_SEEK_END 2
- #define OS_FILESTAT_MODE(x) ((x).FileModeBits)

  Access file stat mode bits.
- #define OS_FILESTAT_ISDIR(x) ((x).FileModeBits & OS_FILESTAT_MODE_DIR)

  File stat is directory logical.
- #define OS_FILESTAT_EXEC(x) ((x).FileModeBits & OS_FILESTAT_MODE_EXEC)

  File stat is executable logical.
- #define OS_FILESTAT_WRITE(x) ((x).FileModeBits & OS_FILESTAT_MODE_WRITE)

  File stat is write enabled logical.
- #define OS_FILESTAT_READ(x) ((x).FileModeBits & OS_FILESTAT_MODE_READ)

  File stat is read enabled logical.
- #define OS_FILESTAT_SIZE(x) ((x).FileSize)

  Access file stat size field.
- #define OS_FILESTAT_TIME(x) (OS_TimeGetTotalSeconds((x).FileTime))

  Access file stat time field as a whole number of seconds.

Enumerations

- enum { OS_FILESTAT_MODE_EXEC = 0x00001, OS_FILESTAT_MODE_WRITE = 0x00002, OS_FILESTAT↩_MODE_READ = 0x00004, OS_FILESTAT_MODE_DIR = 0x10000 }

  File stat mode bits.
- enum OS_file_flag_t { OS_FILE_FLAG_NONE = 0x00, OS_FILE_FLAG_CREATE = 0x01, OS_FILE_FLAG_T↩RUNCATE = 0x02 }

  Flags that can be used with opening of a file (bitmask)

Functions

- int32 OS_OpenCreate (osal_id_t *filedes, const char *path, int32 flags, int32 access)

  Open or create a file.
- int32 OS_close (osal_id_t filedes)

  Closes an open file handle.
- int32 OS_read (osal_id_t filedes, void *buffer, size_t nbytes)

  Read from a file handle.
- int32 OS_write (osal_id_t filedes, const void *buffer, size_t nbytes)

  Write to a file handle.
- int32 OS_TimedRead (osal_id_t filedes, void *buffer, size_t nbytes, int32 timeout)

  File/Stream input read with a timeout.
- int32 OS_TimedWrite (osal_id_t filedes, const void *buffer, size_t nbytes, int32 timeout)

  File/Stream output write with a timeout.
- int32 OS_chmod (const char *path, uint32 access)

  Changes the permissions of a file.
- int32 OS_stat (const char *path, os_fstat_t *filestats)

  Obtain information about a file or directory.
- int32 OS_lseek (osal_id_t filedes, int32 offset, uint32 whence)

  Seeks to the specified position of an open file.
- int32 OS_remove (const char *path)

  Removes a file from the file system.
- int32 OS_rename (const char *old_filename, const char *new_filename)

  Renames a file.
- int32 OS_cp (const char *src, const char *dest)

  Copies a single file from src to dest.
- int32 OS_mv (const char *src, const char *dest)

  Move a single file from src to dest.
- int32 OS_FDGetInfo (osal_id_t filedes, OS_file_prop_t *fd_prop)

  Obtain information about an open file.
- int32 OS_FileOpenCheck (const char *Filename)

  Checks to see if a file is open.
- int32 OS_CloseAllFiles (void)

  Close all open files.
- int32 OS_CloseFileByName (const char *Filename)

  Close a file by filename.

39.72.1 Macro Definition Documentation

39.72.1.1 OS_FILESTAT_EXEC

```
#define OS_FILESTAT_EXEC(
            x ) ((x).FileModeBits & OS_FILESTAT_MODE_EXEC)
```

File stat is executable logical.

Definition at line 92 of file osapi-file.h.

39.72.1.2 OS_FILESTAT_ISDIR

```
#define OS_FILESTAT_ISDIR(
            x ) ((x).FileModeBits & OS_FILESTAT_MODE_DIR)
```

File stat is directory logical.

Definition at line 90 of file osapi-file.h.

39.72.1.3 OS_FILESTAT_MODE

```
#define OS_FILESTAT_MODE(
            x ) ((x).FileModeBits)
```

Access file stat mode bits.

Definition at line 88 of file osapi-file.h.

39.72.1.4 OS_FILESTAT_READ

```
#define OS_FILESTAT_READ(
            x ) ((x).FileModeBits & OS_FILESTAT_MODE_READ)
```

File stat is read enabled logical.

Definition at line 96 of file osapi-file.h.

39.72.1.5 OS_FILESTAT_SIZE

```
#define OS_FILESTAT_SIZE(
            x ) ((x).FileSize)
```

Access file stat size field.

Definition at line 98 of file osapi-file.h.

39.72.1.6 OS_FILESTAT_TIME

```
#define OS_FILESTAT_TIME(
            x ) (OS_TimeGetTotalSeconds((x).FileTime))
```

Access file stat time field as a whole number of seconds.

Definition at line 100 of file osapi-file.h.

39.72.1.7 OS_FILESTAT_WRITE

```
#define OS_FILESTAT_WRITE(
            x ) ((x).FileModeBits & OS_FILESTAT_MODE_WRITE)
```

File stat is write enabled logical.

Definition at line 94 of file osapi-file.h.

39.72.2 Enumeration Type Documentation

39.72.2.1 anonymous enum

```
anonymous enum
```

File stat mode bits.

We must also define replacements for the stat structure's mode bits. This is currently just a small subset since the OSAL just presents a very simplified view of the filesystem to the upper layers. And since not all OS'es are POSIX, the more POSIX-specific bits are not relevant anyway.

Enumerator

OS_FILESTAT_MODE_EXEC	
OS_FILESTAT_MODE_WRITE	
OS_FILESTAT_MODE_READ	
OS_FILESTAT_MODE_DIR	

Definition at line 79 of file osapi-file.h.

39.72.2.2 OS_file_flag_t

```
enum OS_file_flag_t
```

Flags that can be used with opening of a file (bitmask)

Enumerator

OS_FILE_FLAG_NONE	
OS_FILE_FLAG_CREATE	
OS_FILE_FLAG_TRUNCATE	

Definition at line 105 of file osapi-file.h.

39.73 osal/src/os/inc/osapi-filesys.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- struct os_fsinfo_t

  OSAL file system info.
- struct OS_statvfs_t

Macros

- #define OS_CHK_ONLY 0
- #define OS_REPAIR 1

Functions

- int32 OS_FileSysAddFixedMap (osal_id_t ∗filesys_id, const char ∗phys_path, const char ∗virt_path)

  Create a fixed mapping between an existing directory and a virtual OSAL mount point.
- int32 OS_mkfs (char ∗address, const char ∗devname, const char ∗volname, size_t blocksize, osal_blockcount_t numblocks)

  Makes a file system on the target.
- int32 OS_mount (const char ∗devname, const char ∗mountpoint)

  Mounts a file system.

- int32 OS_initfs (char *address, const char *devname, const char *volname, size_t blocksize, osal_blockcount_t numblocks)

  Initializes an existing file system.
- int32 OS_rmfs (const char *devname)

  Removes a file system.
- int32 OS_unmount (const char *mountpoint)

  Unmounts a mounted file system.
- int32 OS_FileSysStatVolume (const char *name, OS_statvfs_t *statbuf)

  Obtains information about size and free space in a volume.
- int32 OS_chkfs (const char *name, bool repair)

  Checks the health of a file system and repairs it if necessary.
- int32 OS_FS_GetPhysDriveName (char *PhysDriveName, const char *MountPoint)

  Obtains the physical drive name associated with a mount point.
- int32 OS_TranslatePath (const char *VirtualPath, char *LocalPath)

  Translates a OSAL Virtual file system path to a host Local path.
- int32 OS_GetFsInfo (os_fsinfo_t *filesys_info)

  Returns information about the file system.

39.73.1 Macro Definition Documentation

39.73.1.1 OS_CHK_ONLY

```
#define OS_CHK_ONLY 0
```

Unused, API takes bool

Definition at line 31 of file osapi-filesys.h.

39.73.1.2 OS_REPAIR

```
#define OS_REPAIR 1
```

Unused, API takes bool

Definition at line 32 of file osapi-filesys.h.

39.74 osal/src/os/inc/osapi-heap.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- struct OS_heap_prop_t

  OSAL heap properties.

Functions

- int32 OS_HeapGetInfo (OS_heap_prop_t *heap_prop)

  Return current info on the heap.

39.75 osal/src/os/inc/osapi-idmap.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Macros

- #define OS_OBJECT_INDEX_MASK 0xFFFF

  Object index mask.
- #define OS_OBJECT_TYPE_SHIFT 16

  Object type shift.
- #define OS_OBJECT_TYPE_UNDEFINED 0x00

  Object type undefined.
- #define OS_OBJECT_TYPE_OS_TASK 0x01

  Object task type.
- #define OS_OBJECT_TYPE_OS_QUEUE 0x02

  Object queue type.
- #define OS_OBJECT_TYPE_OS_COUNTSEM 0x03

  Object counting semaphore type.
- #define OS_OBJECT_TYPE_OS_BINSEM 0x04

  Object binary semaphore type.
- #define OS_OBJECT_TYPE_OS_MUTEX 0x05

  Object mutex type.
- #define OS_OBJECT_TYPE_OS_STREAM 0x06

  Object stream type.
- #define OS_OBJECT_TYPE_OS_DIR 0x07

  Object directory type.
- #define OS_OBJECT_TYPE_OS_TIMEBASE 0x08

  Object timebase type.
- #define OS_OBJECT_TYPE_OS_TIMECB 0x09

  Object timer callback type.
- #define OS_OBJECT_TYPE_OS_MODULE 0x0A

  Object module type.
- #define OS_OBJECT_TYPE_OS_FILESYS 0x0B

  Object file system type.
- #define OS_OBJECT_TYPE_OS_CONSOLE 0x0C

  Object console type.
- #define OS_OBJECT_TYPE_USER 0x10

  Object user type.

Functions

- static unsigned long OS_ObjectIdToInteger (osal_id_t object_id)

  Obtain an integer value corresponding to an object ID.
- static osal_id_t OS_ObjectIdFromInteger (unsigned long value)

  Obtain an osal ID corresponding to an integer value.
- static bool OS_ObjectIdEqual (osal_id_t object_id1, osal_id_t object_id2)

  Check two OSAL object ID values for equality.
- static bool OS_ObjectIdDefined (osal_id_t object_id)

  Check if an object ID is defined.
- int32 OS_GetResourceName (osal_id_t object_id, char *buffer, size_t buffer_size)

  Obtain the name of an object given an arbitrary object ID.
- osal_objtype_t OS_IdentifyObject (osal_id_t object_id)

  Obtain the type of an object given an arbitrary object ID.
- int32 OS_ConvertToArrayIndex (osal_id_t object_id, osal_index_t *ArrayIndex)

  Converts an abstract ID into a number suitable for use as an array index.
- int32 OS_ObjectIdToArrayIndex (osal_objtype_t idtype, osal_id_t object_id, osal_index_t *ArrayIndex)

  Converts an abstract ID into a number suitable for use as an array index.
- void OS_ForEachObject (osal_id_t creator_id, OS_ArgCallback_t callback_ptr, void *callback_arg)

  call the supplied callback function for all valid object IDs
- void OS_ForEachObjectOfType (osal_objtype_t objtype, osal_id_t creator_id, OS_ArgCallback_t callback_ptr, void *callback_arg)

  call the supplied callback function for valid object IDs of a specific type

39.75.1 Macro Definition Documentation

39.75.1.1 OS_OBJECT_INDEX_MASK

```
#define OS_OBJECT_INDEX_MASK 0xFFFF
```

Object index mask.

Definition at line 32 of file osapi-idmap.h.

39.75.1.2 OS_OBJECT_TYPE_SHIFT

```
#define OS_OBJECT_TYPE_SHIFT 16
```

Object type shift.

Definition at line 33 of file osapi-idmap.h.

39.76 osal/src/os/inc/osapi-macros.h File Reference

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include "osconfig.h"
#include "common_types.h"
#include "osapi-printf.h"
```

Macros

- #define BUGREPORT(...) OS_printf(__VA_ARGS__)
- #define BUGCHECK(cond, errcode)

  Basic Bug-Checking macro.
- #define ARGCHECK(cond, errcode)

  Generic argument checking macro for non-critical values.
- #define LENGTHCHECK(str, len, errcode) ARGCHECK(memchr(str, '\0', len), errcode)

  String length limit check macro.

39.76.1 Macro Definition Documentation

39.76.1.1 ARGCHECK

```
#define ARGCHECK(
              cond,
              errcode )
```

Value:

```
if (!(cond))                \
    {                           \
        return errcode;         \
    }
```

Generic argument checking macro for non-critical values.

This macro checks a conditional that is expected to be true, and return a value if it evaluates false.

ARGCHECK can be used to check for out of range or other invalid argument conditions which may (validly) occur at runtime and do not necessarily indicate bugs in the application.

These argument checks are NOT considered a fatal errors. The application continues to run normally. This does not report the error on the console.

As such, ARGCHECK actions are always compiled in - not selectable at compile-time.

See also

BUGCHECK for checking critical values that indicate bugs

Definition at line 122 of file osapi-macros.h.

39.76.1.2 BUGCHECK

```
#define BUGCHECK(
            cond,
            errcode )
```

Value:

```
if (!(cond))                                                                                    \
    {
        \
         BUGREPORT("\n**BUG** %s():%d:check \'%s\' FAILED --> %s\n\n", __func__, __LINE__, #cond, #errcode);
        \
         return errcode;
        \
    }
```

Basic Bug-Checking macro.

This macro checks a conditional, and if it is FALSE, then it generates a report - which may in turn contain additional actions.

BUGCHECK should only be used for conditions which are critical and must always be true. If such a condition is ever false then it indicates a bug in the application which must be resolved. It may or may not be possible to continue operation if a bugcheck fails.

See also

ARGCHECK for checking non-critical values

Definition at line 96 of file osapi-macros.h.

39.76.1.3 BUGREPORT

```
#define BUGREPORT(
            ...  ) OS_printf(__VA_ARGS__)
```

Definition at line 79 of file osapi-macros.h.

39.76.1.4 LENGTHCHECK

```
#define LENGTHCHECK(
            str,
            len,
            errcode ) ARGCHECK(memchr(str, '\0', len), errcode)
```

String length limit check macro.

This macro is a specialized version of ARGCHECK that confirms a string will fit into a buffer of the specified length, and return an error code if it will not.

Note

this uses ARGCHECK, thus treating a string too long as a normal runtime (i.e. non-bug) error condition with a typical error return to the caller.

Definition at line 137 of file osapi-macros.h.

39.77 osal/src/os/inc/osapi-module.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- struct OS_module_address_t

  OSAL module address properties.
- struct OS_module_prop_t

  OSAL module properties.
- struct OS_static_symbol_record_t

  Associates a single symbol name with a memory address.

Macros

- #define OS_MODULE_FLAG_GLOBAL_SYMBOLS 0x00

  Requests OS_ModuleLoad() to add the symbols to the global symbol table.
- #define OS_MODULE_FLAG_LOCAL_SYMBOLS 0x01

  Requests OS_ModuleLoad() to keep the symbols local/private to this module.

Functions

- int32 OS_SymbolLookup (cpuaddr ∗symbol_address, const char ∗symbol_name)

  Find the Address of a Symbol.
- int32 OS_ModuleSymbolLookup (osal_id_t module_id, cpuaddr ∗symbol_address, const char ∗symbol_name)

  Find the Address of a Symbol within a module.
- int32 OS_SymbolTableDump (const char ∗filename, size_t size_limit)

  Dumps the system symbol table to a file.
- int32 OS_ModuleLoad (osal_id_t ∗module_id, const char ∗module_name, const char ∗filename, uint32 flags)

  Loads an object file.
- int32 OS_ModuleUnload (osal_id_t module_id)

  Unloads the module file.
- int32 OS_ModuleInfo (osal_id_t module_id, OS_module_prop_t ∗module_info)

  Obtain information about a module.

39.77.1 Macro Definition Documentation

39.77.1.1 OS_MODULE_FLAG_GLOBAL_SYMBOLS

```
#define OS_MODULE_FLAG_GLOBAL_SYMBOLS 0x00
```

Requests OS_ModuleLoad() to add the symbols to the global symbol table.

When supplied as the "flags" argument to OS_ModuleLoad(), this indicates that the symbols in the loaded module should be added to the global symbol table. This will make symbols in this library available for use when resolving symbols in future module loads.

This is the default mode of operation for OS_ModuleLoad().

Note

On some operating systems, use of this option may make it difficult to unload the module in the future, if the symbols are in use by other entities.

Definition at line 49 of file osapi-module.h.

39.77.1.2 OS_MODULE_FLAG_LOCAL_SYMBOLS

```
#define OS_MODULE_FLAG_LOCAL_SYMBOLS 0x01
```

Requests OS_ModuleLoad() to keep the symbols local/private to this module.

When supplied as the "flags" argument to OS_ModuleLoad(), this indicates that the symbols in the loaded module should NOT be added to the global symbol table. This means the symbols in the loaded library will not available to for use by other modules.

Use this option is recommended for cases where no other entities will need to reference symbols within this module. This helps ensure that the module can be more safely unloaded in the future, by preventing other modules from binding to it. It also helps reduce the likelihood of symbol name conflicts among modules.

Note

To look up symbols within a module loaded with this flag, use OS_SymbolLookupInModule() instead of OS_SymbolLookup(). Also note that references obtained using this method are not tracked by the OS; the application must ensure that all references obtained in this manner have been cleaned up/released before unloading the module.

Definition at line 71 of file osapi-module.h.

39.78 osal/src/os/inc/osapi-mutex.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- struct OS_mut_sem_prop_t

  OSAL mutex properties.

Functions

- int32 OS_MutSemCreate (osal_id_t *sem_id, const char *sem_name, uint32 options)

  Creates a mutex semaphore.
- int32 OS_MutSemGive (osal_id_t sem_id)

  Releases the mutex object referenced by sem_id.
- int32 OS_MutSemTake (osal_id_t sem_id)

  Acquire the mutex object referenced by sem_id.
- int32 OS_MutSemDelete (osal_id_t sem_id)

  Deletes the specified Mutex Semaphore.
- int32 OS_MutSemGetIdByName (osal_id_t *sem_id, const char *sem_name)

  Find an existing mutex ID by name.
- int32 OS_MutSemGetInfo (osal_id_t sem_id, OS_mut_sem_prop_t *mut_prop)

  Fill a property object buffer with details regarding the resource.

39.79 osal/src/os/inc/osapi-network.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Functions

- int32 OS_NetworkGetID (void)

  Gets the network ID of the local machine.
- int32 OS_NetworkGetHostName (char *host_name, size_t name_len)

  Gets the local machine network host name.

39.80 osal/src/os/inc/osapi-printf.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Functions

- void OS_printf (const char *string,...) OS_PRINTF(1

  Abstraction for the system printf() call.
- void void OS_printf_disable (void)

  This function disables the output from OS_printf.
- void OS_printf_enable (void)

  This function enables the output from OS_printf.

39.81 osal/src/os/inc/osapi-queue.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- struct OS_queue_prop_t

  OSAL queue properties.

Functions

- int32 OS_QueueCreate (osal_id_t *queue_id, const char *queue_name, osal_blockcount_t queue_depth, size↩_t data_size, uint32 flags)

  Create a message queue.
- int32 OS_QueueDelete (osal_id_t queue_id)

  Deletes the specified message queue.
- int32 OS_QueueGet (osal_id_t queue_id, void *data, size_t size, size_t *size_copied, int32 timeout)

  Receive a message on a message queue.
- int32 OS_QueuePut (osal_id_t queue_id, const void *data, size_t size, uint32 flags)

  Put a message on a message queue.
- int32 OS_QueueGetIdByName (osal_id_t *queue_id, const char *queue_name)

  Find an existing queue ID by name.
- int32 OS_QueueGetInfo (osal_id_t queue_id, OS_queue_prop_t *queue_prop)

  Fill a property object buffer with details regarding the resource.

39.82 osal/src/os/inc/osapi-select.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- struct OS_FdSet

  An abstract structure capable of holding several OSAL IDs.

Enumerations

- enum OS_StreamState_t { OS_STREAM_STATE_BOUND = 0x01, OS_STREAM_STATE_CONNECTED = 0x02, OS_STREAM_STATE_READABLE = 0x04, OS_STREAM_STATE_WRITABLE = 0x08 }

  For the OS_SelectSingle() function's in/out StateFlags parameter, the state(s) of the stream and the result of the select is a combination of one or more of these states.

Functions

- int32 OS_SelectMultiple (OS_FdSet *ReadSet, OS_FdSet *WriteSet, int32 msecs)

  Wait for events across multiple file handles.
- int32 OS_SelectSingle (osal_id_t objid, uint32 *StateFlags, int32 msecs)

  Wait for events on a single file handle.
- int32 OS_SelectFdZero (OS_FdSet *Set)

  Clear a FdSet structure.
- int32 OS_SelectFdAdd (OS_FdSet *Set, osal_id_t objid)

  Add an ID to an FdSet structure.
- int32 OS_SelectFdClear (OS_FdSet *Set, osal_id_t objid)

  Clear an ID from an FdSet structure.
- bool OS_SelectFdIsSet (OS_FdSet *Set, osal_id_t objid)

  Check if an FdSet structure contains a given ID.

39.82.1 Enumeration Type Documentation

39.82.1.1 OS_StreamState_t

```
enum OS_StreamState_t
```

For the OS_SelectSingle() function's in/out StateFlags parameter, the state(s) of the stream and the result of the select is a combination of one or more of these states.

See also

OS_SelectSingle()

Enumerator

OS_STREAM_STATE_BOUND	whether the stream is bound
OS_STREAM_STATE_CONNECTED	whether the stream is connected
OS_STREAM_STATE_READABLE	whether the stream is readable
OS_STREAM_STATE_WRITABLE	whether the stream is writable

Definition at line 51 of file osapi-select.h.

39.83 osal/src/os/inc/osapi-shell.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Functions

- int32 OS_ShellOutputToFile (const char *Cmd, osal_id_t filedes)

  Executes the command and sends output to a file.

39.84 osal/src/os/inc/osapi-sockets.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- union OS_SockAddrData_t

  Storage buffer for generic network address.
- struct OS_SockAddr_t

  Encapsulates a generic network address.
- struct OS_socket_prop_t

  Encapsulates socket properties.

Macros

- #define OS_SOCKADDR_MAX_LEN 28

Enumerations

- enum OS_SocketDomain_t { OS_SocketDomain_INVALID, OS_SocketDomain_INET, OS_SocketDomain_IN↩ET6, OS_SocketDomain_MAX }

  Socket domain.
- enum OS_SocketType_t { OS_SocketType_INVALID, OS_SocketType_DATAGRAM, OS_SocketType_STREAM, OS_SocketType_MAX }

  Socket type.

Functions

- int32 OS_SocketAddrInit (OS_SockAddr_t *Addr, OS_SocketDomain_t Domain)

  Initialize a socket address structure to hold an address of the given family.
- int32 OS_SocketAddrToString (char *buffer, size_t buflen, const OS_SockAddr_t *Addr)

  Get a string representation of a network host address.
- int32 OS_SocketAddrFromString (OS_SockAddr_t *Addr, const char *string)

  Set a network host address from a string representation.
- int32 OS_SocketAddrGetPort (uint16 *PortNum, const OS_SockAddr_t *Addr)

  Get the port number of a network address.
- int32 OS_SocketAddrSetPort (OS_SockAddr_t *Addr, uint16 PortNum)

Set the port number of a network address.

- int32 OS_SocketOpen (osal_id_t *sock_id, OS_SocketDomain_t Domain, OS_SocketType_t Type)

  Opens a socket.
- int32 OS_SocketBind (osal_id_t sock_id, const OS_SockAddr_t *Addr)

  Binds a socket to a given local address.
- int32 OS_SocketConnect (osal_id_t sock_id, const OS_SockAddr_t *Addr, int32 timeout)

  Connects a socket to a given remote address.
- int32 OS_SocketAccept (osal_id_t sock_id, osal_id_t *connsock_id, OS_SockAddr_t *Addr, int32 timeout)

  Waits for and accept the next incoming connection on the given socket.
- int32 OS_SocketRecvFrom (osal_id_t sock_id, void *buffer, size_t buflen, OS_SockAddr_t *RemoteAddr, int32 timeout)

  Reads data from a message-oriented (datagram) socket.
- int32 OS_SocketSendTo (osal_id_t sock_id, const void *buffer, size_t buflen, const OS_SockAddr_t *Remote↩ Addr)

  Sends data to a message-oriented (datagram) socket.
- int32 OS_SocketGetIdByName (osal_id_t *sock_id, const char *sock_name)

  Gets an OSAL ID from a given name.
- int32 OS_SocketGetInfo (osal_id_t sock_id, OS_socket_prop_t *sock_prop)

  Gets information about an OSAL Socket ID.

39.84.1 Macro Definition Documentation

39.84.1.1 OS_SOCKADDR_MAX_LEN

```
#define OS_SOCKADDR_MAX_LEN 28
```

Definition at line 45 of file osapi-sockets.h.

39.84.2 Enumeration Type Documentation

39.84.2.1 OS_SocketDomain_t

```
enum OS_SocketDomain_t
```

Socket domain.

Enumerator

OS_SocketDomain_INVALID	Invalid.
OS_SocketDomain_INET	IPv4 address family, most commonly used)
OS_SocketDomain_INET6	IPv6 address family, depends on OS/network stack support.
OS_SocketDomain_MAX	Maximum.

Definition at line 60 of file osapi-sockets.h.

39.84.2.2 OS_SocketType_t

```
enum OS_SocketType_t
```

Socket type.

Enumerator

OS_SocketType_INVALID	Invalid.
OS_SocketType_DATAGRAM	A connectionless, message-oriented socket.
OS_SocketType_STREAM	A stream-oriented socket with the concept of a connection.
OS_SocketType_MAX	Maximum.

Definition at line 69 of file osapi-sockets.h.

39.85 osal/src/os/inc/osapi-task.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- struct OS_task_prop_t

  OSAL task properties.

Macros

- #define OS_MAX_TASK_PRIORITY 255

  Upper limit for OSAL task priorities.
- #define OS_FP_ENABLED 1

  Floating point enabled state for a task.
- #define OSAL_PRIORITY_C(X) ((osal_priority_t) {X})
- #define OSAL_STACKPTR_C(X) ((osal_stackptr_t) {X})
- #define OSAL_TASK_STACK_ALLOCATE OSAL_STACKPTR_C(NULL)

Typedefs

- typedef uint8_t osal_priority_t

  Type to be used for OSAL task priorities.
- typedef void * osal_stackptr_t

  Type to be used for OSAL stack pointer.
- typedef void osal_task

  For task entry point.

Functions

- typedef osal_task ((∗osal_task_entry)(void))

  For task entry point.
- int32 OS_TaskCreate (osal_id_t ∗task_id, const char ∗task_name, osal_task_entry function_pointer, osal_stackptr_t stack_pointer, size_t stack_size, osal_priority_t priority, uint32 flags)

  Creates a task and starts running it.
- int32 OS_TaskDelete (osal_id_t task_id)

  Deletes the specified Task.
- void OS_TaskExit (void)

  Exits the calling task.
- int32 OS_TaskInstallDeleteHandler (osal_task_entry function_pointer)

  Installs a handler for when the task is deleted.
- int32 OS_TaskDelay (uint32 millisecond)

  Delay a task for specified amount of milliseconds.
- int32 OS_TaskSetPriority (osal_id_t task_id, osal_priority_t new_priority)

  Sets the given task to a new priority.
- int32 OS_TaskRegister (void)

  Obsolete.
- osal_id_t OS_TaskGetId (void)

  Obtain the task id of the calling task.
- int32 OS_TaskGetIdByName (osal_id_t ∗task_id, const char ∗task_name)

  Find an existing task ID by name.
- int32 OS_TaskGetInfo (osal_id_t task_id, OS_task_prop_t ∗task_prop)

  Fill a property object buffer with details regarding the resource.
- int32 OS_TaskFindIdBySystemData (osal_id_t ∗task_id, const void ∗sysdata, size_t sysdata_size)

  Reverse-lookup the OSAL task ID from an operating system ID.

39.85.1 Macro Definition Documentation

39.85.1.1 OS_FP_ENABLED

```
#define OS_FP_ENABLED 1
```

Floating point enabled state for a task.

Definition at line 35 of file osapi-task.h.

39.85.1.2 OS_MAX_TASK_PRIORITY

```
#define OS_MAX_TASK_PRIORITY 255
```

Upper limit for OSAL task priorities.

Definition at line 32 of file osapi-task.h.

39.85.1.3 OSAL_PRIORITY_C

```
#define OSAL_PRIORITY_C(
            X ) ((osal_priority_t) {X})
```

Definition at line 46 of file osapi-task.h.

39.85.1.4 OSAL_STACKPTR_C

```
#define OSAL_STACKPTR_C(
            X ) ((osal_stackptr_t) {X})
```

Definition at line 53 of file osapi-task.h.

39.85.1.5 OSAL_TASK_STACK_ALLOCATE

```
#define OSAL_TASK_STACK_ALLOCATE OSAL_STACKPTR_C(NULL)
```

Definition at line 54 of file osapi-task.h.

39.85.2 Typedef Documentation

39.85.2.1 osal_priority_t

```
typedef uint8_t osal_priority_t
```

Type to be used for OSAL task priorities.

OSAL priorities are in reverse order, and range from 0 (highest; will preempt all other tasks) to 255 (lowest; will not preempt any other task).

Definition at line 44 of file osapi-task.h.

39.85.2.2 osal_stackptr_t

```
typedef void* osal_stackptr_t
```

Type to be used for OSAL stack pointer.

Definition at line 51 of file osapi-task.h.

39.85.2.3 osal_task

```
typedef void osal_task
```

For task entry point.

Definition at line 68 of file osapi-task.h.

39.85.3 Function Documentation

39.85.3.1 osal_task()

```
typedef osal_task (
            (*)(void) osal_task_entry )
```

For task entry point.

39.86 osal/src/os/inc/osapi-timebase.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- struct OS_timebase_prop_t

  Time base properties.

Typedefs

- typedef uint32(∗ OS_TimerSync_t) (osal_id_t timer_id)

  Timer sync.

Functions

- int32 OS_TimeBaseCreate (osal_id_t ∗timebase_id, const char ∗timebase_name, OS_TimerSync_t external_↩sync)

  Create an abstract Time Base resource.
- int32 OS_TimeBaseSet (osal_id_t timebase_id, uint32 start_time, uint32 interval_time)

  Sets the tick period for simulated time base objects.
- int32 OS_TimeBaseDelete (osal_id_t timebase_id)

  Deletes a time base object.
- int32 OS_TimeBaseGetIdByName (osal_id_t ∗timebase_id, const char ∗timebase_name)

  Find the ID of an existing time base resource.
- int32 OS_TimeBaseGetInfo (osal_id_t timebase_id, OS_timebase_prop_t ∗timebase_prop)

  Obtain information about a timebase resource.
- int32 OS_TimeBaseGetFreeRun (osal_id_t timebase_id, uint32 ∗freerun_val)

  Read the value of the timebase free run counter.

39.86.1 Typedef Documentation

39.86.1.1 OS_TimerSync_t

```
typedef uint32(* OS_TimerSync_t) (osal_id_t timer_id)
```

Timer sync.

Definition at line 34 of file osapi-timebase.h.

39.87 osal/src/os/inc/osapi-timer.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- struct OS_timer_prop_t

  Timer properties.

Typedefs

- typedef void(∗ OS_TimerCallback_t) (osal_id_t timer_id)

  Timer callback.

Functions

- int32 OS_TimerCreate (osal_id_t ∗timer_id, const char ∗timer_name, uint32 ∗clock_accuracy, OS_Timer↩Callback_t callback_ptr)

  Create a timer object.
- int32 OS_TimerAdd (osal_id_t ∗timer_id, const char ∗timer_name, osal_id_t timebase_id, OS_ArgCallback_↩t callback_ptr, void ∗callback_arg)

  Add a timer object based on an existing TimeBase resource.
- int32 OS_TimerSet (osal_id_t timer_id, uint32 start_time, uint32 interval_time)

  Configures a periodic or one shot timer.
- int32 OS_TimerDelete (osal_id_t timer_id)

  Deletes a timer resource.
- int32 OS_TimerGetIdByName (osal_id_t ∗timer_id, const char ∗timer_name)

  Locate an existing timer resource by name.
- int32 OS_TimerGetInfo (osal_id_t timer_id, OS_timer_prop_t ∗timer_prop)

  Gets information about an existing timer.

39.87.1 Typedef Documentation

39.87.1.1 OS_TimerCallback_t

```
typedef void(* OS_TimerCallback_t) (osal_id_t timer_id)
```

Timer callback.

Definition at line 34 of file osapi-timer.h.

39.88 osal/src/os/inc/osapi-version.h File Reference

Purpose:

```
#include "common_types.h"
```

Macros

- #define OS_BUILD_NUMBER 350
- #define OS_BUILD_BASELINE "v5.1.0-rc1"
- #define OS_MAJOR_VERSION 5

  ONLY APPLY for OFFICIAL releases. Major version number.
- #define OS_MINOR_VERSION 0

  ONLY APPLY for OFFICIAL releases. Minor version number.
- #define OS_REVISION 0

  ONLY APPLY for OFFICIAL releases. Revision number.
- #define OS_MISSION_REV 0xFF

  Mission revision.
- #define OS_STR_HELPER(x) #x

  Helper function to concatenate strings from integer.
- #define OS_STR(x) OS_STR_HELPER(x)

  Helper function to concatenate strings from integer.
- #define OS_VERSION OS_BUILD_BASELINE "+dev" OS_STR(OS_BUILD_NUMBER)

  Development Build Version Number.
- #define OS_VERSION_CODENAME "Bootes"

  Version code name All modular components which are tested/validated together should share the same code name.
- #define OS_VERSION_STRING

  Development Build Version String.
- #define OSAL_API_VERSION ((OS_MAJOR_VERSION ∗ 10000) + (OS_MINOR_VERSION ∗ 100) + OS_RE↩VISION)

  Combines the revision components into a single value.

Functions

- const char * OS_GetVersionString (void)
- const char * OS_GetVersionCodeName (void)
- void OS_GetVersionNumber (uint8 VersionNumbers[4])

  Obtain the OSAL numeric version number.
- uint32 OS_GetBuildNumber (void)

  Obtain the OSAL library numeric build number.

39.88.1 Detailed Description

Purpose:

Provide version identifiers for cFS' Operating System Abstraction Layer See Version Numbers for version and build number and description

39.88.2 Macro Definition Documentation

39.88.2.1 OS_BUILD_BASELINE

```
#define OS_BUILD_BASELINE "v5.1.0-rc1"
```

Definition at line 36 of file osapi-version.h.

39.88.2.2 OS_BUILD_NUMBER

```
#define OS_BUILD_NUMBER 350
```

Definition at line 35 of file osapi-version.h.

39.88.2.3 OS_MAJOR_VERSION

```
#define OS_MAJOR_VERSION 5
```

ONLY APPLY for OFFICIAL releases. Major version number.

Definition at line 41 of file osapi-version.h.

39.88.2.4 OS_MINOR_VERSION

```
#define OS_MINOR_VERSION 0
```

ONLY APPLY for OFFICIAL releases. Minor version number.

Definition at line 42 of file osapi-version.h.

39.88.2.5 OS_MISSION_REV

```
#define OS_MISSION_REV 0xFF
```

Mission revision.

Set to 0 on OFFIFICIAL releases, and set to 255 (0xFF) on development versions. Values 1-254 are reserved for mission use to denote patches/customizations as needed.

Definition at line 51 of file osapi-version.h.

39.88.2.6 OS_REVISION

```
#define OS_REVISION 0
```

ONLY APPLY for OFFICIAL releases. Revision number.

Definition at line 43 of file osapi-version.h.

39.88.2.7 OS_STR

```
#define OS_STR(
            x ) OS_STR_HELPER(x)
```

Helper function to concatenate strings from integer.

Definition at line 57 of file osapi-version.h.

39.88.2.8 OS_STR_HELPER

```
#define OS_STR_HELPER(
            x ) #x
```

Helper function to concatenate strings from integer.

Definition at line 56 of file osapi-version.h.

39.88.2.9 OS_VERSION

```
#define OS_VERSION OS_BUILD_BASELINE "+dev" OS_STR(OS_BUILD_NUMBER)
```

Development Build Version Number.

Baseline git tag + Number of commits since baseline.
See Version Numbers for format differences between development and release versions.

Definition at line 63 of file osapi-version.h.

39.88.2.10 OS_VERSION_CODENAME

```
#define OS_VERSION_CODENAME "Bootes"
```

Version code name All modular components which are tested/validated together should share the same code name.

Definition at line 68 of file osapi-version.h.

39.88.2.11 OS_VERSION_STRING

```
#define OS_VERSION_STRING
```

Value:

```
" OSAL Development Build\n"                                                                    \
    " " OS_VERSION " (Codename: " OS_VERSION_CODENAME ")\n" /* Codename for
       current development */ \
    " Latest Official Version: osal v5.0.0"                 /* For full support please use official release
       version */
```

Development Build Version String.

Reports the current development build's baseline, number, and name. Also includes a note about the latest official version.
See Version Numbers for format differences between development and release versions.

Definition at line 74 of file osapi-version.h.

39.88.2.12 OSAL_API_VERSION

```
#define OSAL_API_VERSION ((OS_MAJOR_VERSION * 10000) + (OS_MINOR_VERSION * 100) + OS_REVISION)
```

Combines the revision components into a single value.

Applications can check against this number
e.g. "#if OSAL_API_VERSION >= 40100" would check if some feature added in OSAL 4.1 is present.

Definition at line 84 of file osapi-version.h.

39.88.3 Function Documentation

39.88.3.1 OS_GetBuildNumber()

```
uint32 OS_GetBuildNumber (
            void  )
```

Obtain the OSAL library numeric build number.

The build number is a monotonically increasing number that (coarsely) reflects the number of commits/changes that have been merged since the epoch release. During development cycles this number should increase after each subsequent merge/modification.

Like other version information, this is a fixed number assigned at compile time.

Returns

The OSAL library build number

39.88.3.2 OS_GetVersionCodeName()

```
const char* OS_GetVersionCodeName (
            void  )
```

Gets the OSAL version code name

All NASA CFE/CFS components (including CFE framework, OSAL and PSP) that work together will share the same code name.

Returns

OSAL code name. This is a fixed value string and is never NULL.

39.88.3.3 OS_GetVersionNumber()

```
void OS_GetVersionNumber (
            uint8 VersionNumbers[4] )
```

Obtain the OSAL numeric version number.

This retrieves the numeric OSAL version identifier as an array of 4 uint8 values.

The array of numeric values is in order of precedence: [0] = Major Number [1] = Minor Number [2] = Revision Number [3] = Mission Revision

The "Mission Revision" (last output) also indicates whether this is an official release, a patched release, or a development version. 0 indicates an official release 1-254 local patch level (reserved for mission use) 255 indicates a development build

Parameters

out	*VersionNumbers*	A fixed-size array to be filled with the version numbers

39.88.3.4 OS_GetVersionString()

```
const char* OS_GetVersionString (
            void  )
```

Gets the OSAL version/baseline ID as a string

This returns the content of the OS_VERSION macro defined above, and is specifically just the baseline and development build ID (if applicable), without any extra info.

Returns

Basic version identifier. This is a fixed value string and is never NULL.

39.89 osal/src/os/inc/osapi.h File Reference

```
#include <stdio.h>
#include <stdlib.h>
#include <stdarg.h>
#include "common_types.h"
#include "osapi-version.h"
#include "osconfig.h"
#include "osapi-binsem.h"
#include "osapi-clock.h"
#include "osapi-common.h"
#include "osapi-constants.h"
#include "osapi-countsem.h"
#include "osapi-dir.h"
#include "osapi-error.h"
#include "osapi-file.h"
#include "osapi-filesys.h"
#include "osapi-heap.h"
#include "osapi-macros.h"
#include "osapi-idmap.h"
#include "osapi-module.h"
#include "osapi-mutex.h"
#include "osapi-network.h"
#include "osapi-printf.h"
#include "osapi-queue.h"
#include "osapi-select.h"
#include "osapi-shell.h"
#include "osapi-sockets.h"
#include "osapi-task.h"
#include "osapi-timebase.h"
#include "osapi-timer.h"
#include "osapi-bsp.h"
```

39.90 psp/fsw/inc/cfe_psp.h File Reference

```
#include "common_types.h"
#include "osapi.h"
```

Data Structures

- struct CFE_PSP_MemTable_t

Macros

- #define CFE_PSP_SUCCESS (0)
- #define CFE_PSP_ERROR (-1)
- #define CFE_PSP_INVALID_POINTER (-2)
- #define CFE_PSP_ERROR_ADDRESS_MISALIGNED (-3)
- #define CFE_PSP_ERROR_TIMEOUT (-4)
- #define CFE_PSP_INVALID_INT_NUM (-5)
- #define CFE_PSP_INVALID_MEM_ADDR (-21)
- #define CFE_PSP_INVALID_MEM_TYPE (-22)
- #define CFE_PSP_INVALID_MEM_RANGE (-23)
- #define CFE_PSP_INVALID_MEM_WORDSIZE (-24)
- #define CFE_PSP_INVALID_MEM_SIZE (-25)
- #define CFE_PSP_INVALID_MEM_ATTR (-26)
- #define CFE_PSP_ERROR_NOT_IMPLEMENTED (-27)
- #define CFE_PSP_INVALID_MODULE_NAME (-28)
- #define CFE_PSP_INVALID_MODULE_ID (-29)
- #define CFE_PSP_NO_EXCEPTION_DATA (-30)
- #define CFE_PSP_PANIC_STARTUP 1
- #define CFE_PSP_PANIC_VOLATILE_DISK 2
- #define CFE_PSP_PANIC_MEMORY_ALLOC 3
- #define CFE_PSP_PANIC_NONVOL_DISK 4
- #define CFE_PSP_PANIC_STARTUP_SEM 5
- #define CFE_PSP_PANIC_CORE_APP 6
- #define CFE_PSP_PANIC_GENERAL_FAILURE 7
- #define BUFF_SIZE 256
- #define SIZE_BYTE 1
- #define SIZE_HALF 2
- #define SIZE_WORD 3
- #define CFE_PSP_MEM_RAM 1
- #define CFE_PSP_MEM_EEPROM 2
- #define CFE_PSP_MEM_ANY 3
- #define CFE_PSP_MEM_INVALID 4
- #define CFE_PSP_MEM_ATTR_WRITE 0x01
- #define CFE_PSP_MEM_ATTR_READ 0x02
- #define CFE_PSP_MEM_ATTR_READWRITE 0x03
- #define CFE_PSP_MEM_SIZE_BYTE 0x01
- #define CFE_PSP_MEM_SIZE_WORD 0x02
- #define CFE_PSP_MEM_SIZE_DWORD 0x04

- #define CFE_PSP_MAJOR_VERSION (GLOBAL_PSP_CONFIGDATA.PSP_VersionInfo.MajorVersion)
- #define CFE_PSP_MINOR_VERSION (GLOBAL_PSP_CONFIGDATA.PSP_VersionInfo.MinorVersion)
- #define CFE_PSP_REVISION (GLOBAL_PSP_CONFIGDATA.PSP_VersionInfo.Revision)
- #define CFE_PSP_MISSION_REV (GLOBAL_PSP_CONFIGDATA.PSP_VersionInfo.MissionRev)
- #define CFE_PSP_VERSION (GLOBAL_PSP_CONFIGDATA.PSP_VersionInfo.VersionString)

Reset Types

- #define CFE_PSP_RST_TYPE_PROCESSOR 1
- #define CFE_PSP_RST_TYPE_POWERON 2
- #define CFE_PSP_RST_TYPE_MAX 3

Reset Sub-Types

- #define CFE_PSP_RST_SUBTYPE_POWER_CYCLE 1

  Reset caused by power having been removed and restored.
- #define CFE_PSP_RST_SUBTYPE_PUSH_BUTTON 2

  Reset caused by reset button on the board.
- #define CFE_PSP_RST_SUBTYPE_HW_SPECIAL_COMMAND 3

  Reset was caused by a reset line having been stimulated by a hardware special command.
- #define CFE_PSP_RST_SUBTYPE_HW_WATCHDOG 4

  Reset was caused by a watchdog timer expiring.
- #define CFE_PSP_RST_SUBTYPE_RESET_COMMAND 5

  Reset was caused by cFE ES processing a Reset Command .
- #define CFE_PSP_RST_SUBTYPE_EXCEPTION 6

  Reset was caused by a Processor Exception.
- #define CFE_PSP_RST_SUBTYPE_UNDEFINED_RESET 7

  Reset was caused in an unknown manner.
- #define CFE_PSP_RST_SUBTYPE_HWDEBUG_RESET 8

  Reset was caused by a JTAG or BDM connection.
- #define CFE_PSP_RST_SUBTYPE_BANKSWITCH_RESET 9

  Reset reverted to a cFE POWERON due to a boot bank switch.
- #define CFE_PSP_RST_SUBTYPE_MAX 10

  Placeholder to indicate 1+ the maximum value that the PSP will ever use.

Functions

- void CFE_PSP_Main (void)
- void CFE_PSP_GetTime (OS_time_t *LocalTime)
- void CFE_PSP_Restart (uint32 resetType)
- uint32 CFE_PSP_GetRestartType (uint32 *restartSubType)
- void CFE_PSP_FlushCaches (uint32 type, void *address, uint32 size)
- uint32 CFE_PSP_GetProcessorId (void)
- uint32 CFE_PSP_GetSpacecraftId (void)
- const char * CFE_PSP_GetProcessorName (void)
- uint32 CFE_PSP_Get_Timer_Tick (void)
- uint32 CFE_PSP_GetTimerTicksPerSecond (void)
- uint32 CFE_PSP_GetTimerLow32Rollover (void)
- void CFE_PSP_Get_Timebase (uint32 *Tbu, uint32 *Tbl)
- uint32 CFE_PSP_Get_Dec (void)
- int32 CFE_PSP_GetCDSSize (uint32 *SizeOfCDS)

- int32 CFE_PSP_WriteToCDS (const void ∗PtrToDataToWrite, uint32 CDSOffset, uint32 NumBytes)
- int32 CFE_PSP_ReadFromCDS (void ∗PtrToDataToRead, uint32 CDSOffset, uint32 NumBytes)
- int32 CFE_PSP_GetResetArea (cpuaddr ∗PtrToResetArea, uint32 ∗SizeOfResetArea)
- int32 CFE_PSP_GetUserReservedArea (cpuaddr ∗PtrToUserArea, uint32 ∗SizeOfUserArea)
- int32 CFE_PSP_GetVolatileDiskMem (cpuaddr ∗PtrToVolDisk, uint32 ∗SizeOfVolDisk)
- int32 CFE_PSP_GetKernelTextSegmentInfo (cpuaddr ∗PtrToKernelSegment, uint32 ∗SizeOfKernelSegment)
- int32 CFE_PSP_GetCFETextSegmentInfo (cpuaddr ∗PtrToCFESegment, uint32 ∗SizeOfCFESegment)
- void CFE_PSP_WatchdogInit (void)
- void CFE_PSP_WatchdogEnable (void)
- void CFE_PSP_WatchdogDisable (void)
- void CFE_PSP_WatchdogService (void)
- uint32 CFE_PSP_WatchdogGet (void)
- void CFE_PSP_WatchdogSet (uint32 WatchdogValue)
- void CFE_PSP_Panic (int32 ErrorCode)
- int32 CFE_PSP_InitSSR (uint32 bus, uint32 device, char ∗DeviceName)
- int32 CFE_PSP_Decompress (char ∗srcFileName, char ∗dstFileName)
- void CFE_PSP_AttachExceptions (void)
- void CFE_PSP_SetDefaultExceptionEnvironment (void)
- uint32 CFE_PSP_Exception_GetCount (void)
- int32 CFE_PSP_Exception_GetSummary (uint32 ∗ContextLogId, osal_id_t ∗TaskId, char ∗ReasonBuf, uint32 ReasonSize)
- int32 CFE_PSP_Exception_CopyContext (uint32 ContextLogId, void ∗ContextBuf, uint32 ContextSize)
- int32 CFE_PSP_PortRead8 (cpuaddr PortAddress, uint8 ∗ByteValue)
- int32 CFE_PSP_PortWrite8 (cpuaddr PortAddress, uint8 ByteValue)
- int32 CFE_PSP_PortRead16 (cpuaddr PortAddress, uint16 ∗uint16Value)
- int32 CFE_PSP_PortWrite16 (cpuaddr PortAddress, uint16 uint16Value)
- int32 CFE_PSP_PortRead32 (cpuaddr PortAddress, uint32 ∗uint32Value)
- int32 CFE_PSP_PortWrite32 (cpuaddr PortAddress, uint32 uint32Value)
- int32 CFE_PSP_MemRead8 (cpuaddr MemoryAddress, uint8 ∗ByteValue)
- int32 CFE_PSP_MemWrite8 (cpuaddr MemoryAddress, uint8 ByteValue)
- int32 CFE_PSP_MemRead16 (cpuaddr MemoryAddress, uint16 ∗uint16Value)
- int32 CFE_PSP_MemWrite16 (cpuaddr MemoryAddress, uint16 uint16Value)
- int32 CFE_PSP_MemRead32 (cpuaddr MemoryAddress, uint32 ∗uint32Value)
- int32 CFE_PSP_MemWrite32 (cpuaddr MemoryAddress, uint32 uint32Value)
- int32 CFE_PSP_MemCpy (void ∗dest, const void ∗src, uint32 n)
- int32 CFE_PSP_MemSet (void ∗dest, uint8 value, uint32 n)
- int32 CFE_PSP_MemValidateRange (cpuaddr Address, uint32 Size, uint32 MemoryType)
- uint32 CFE_PSP_MemRanges (void)
- int32 CFE_PSP_MemRangeSet (uint32 RangeNum, uint32 MemoryType, cpuaddr StartAddr, uint32 Size, uint32 WordSize, uint32 Attributes)
- int32 CFE_PSP_MemRangeGet (uint32 RangeNum, uint32 ∗MemoryType, cpuaddr ∗StartAddr, uint32 ∗Size, uint32 ∗WordSize, uint32 ∗Attributes)
- int32 CFE_PSP_EepromWrite8 (cpuaddr MemoryAddress, uint8 ByteValue)
- int32 CFE_PSP_EepromWrite16 (cpuaddr MemoryAddress, uint16 uint16Value)
- int32 CFE_PSP_EepromWrite32 (cpuaddr MemoryAddress, uint32 uint32Value)
- int32 CFE_PSP_EepromWriteEnable (uint32 Bank)
- int32 CFE_PSP_EepromWriteDisable (uint32 Bank)
- int32 CFE_PSP_EepromPowerUp (uint32 Bank)
- int32 CFE_PSP_EepromPowerDown (uint32 Bank)
- const char ∗ CFE_PSP_GetVersionString (void)

  Obtain the PSP version/baseline identifier string.

- const char * CFE_PSP_GetVersionCodeName (void)

  Obtain the version code name.
- void CFE_PSP_GetVersionNumber (uint8 VersionNumbers[4])

  Obtain the PSP numeric version numbers as uint8 values.
- uint32 CFE_PSP_GetBuildNumber (void)

  Obtain the PSP library numeric build number.

39.90.1 Macro Definition Documentation

39.90.1.1 BUFF_SIZE

```
#define BUFF_SIZE 256
```

Definition at line 84 of file cfe_psp.h.

39.90.1.2 CFE_PSP_ERROR

```
#define CFE_PSP_ERROR (-1)
```

Definition at line 54 of file cfe_psp.h.

Referenced by CFE_PSP_GetCDSSize(), CFE_PSP_GetCFETextSegmentInfo(), CFE_PSP_GetKernelTextSegment↩Info(), CFE_PSP_GetResetArea(), CFE_PSP_GetUserReservedArea(), CFE_PSP_GetVolatileDiskMem(), CFE_PS↩P_InitSSR(), CFE_PSP_ReadFromCDS(), and CFE_PSP_WriteToCDS().

39.90.1.3 CFE_PSP_ERROR_ADDRESS_MISALIGNED

```
#define CFE_PSP_ERROR_ADDRESS_MISALIGNED (-3)
```

Definition at line 56 of file cfe_psp.h.

39.90.1.4 CFE_PSP_ERROR_NOT_IMPLEMENTED

```
#define CFE_PSP_ERROR_NOT_IMPLEMENTED (-27)
```

Definition at line 65 of file cfe_psp.h.

Referenced by CFE_PSP_GetKernelTextSegmentInfo().

39.90.1.5 CFE_PSP_ERROR_TIMEOUT

```
#define CFE_PSP_ERROR_TIMEOUT (-4)
```

Definition at line 57 of file cfe_psp.h.

39.90.1.6 CFE_PSP_INVALID_INT_NUM

```
#define CFE_PSP_INVALID_INT_NUM (-5)
```

Definition at line 58 of file cfe_psp.h.

39.90.1.7 CFE_PSP_INVALID_MEM_ADDR

```
#define CFE_PSP_INVALID_MEM_ADDR (-21)
```

Definition at line 59 of file cfe_psp.h.

39.90.1.8 CFE_PSP_INVALID_MEM_ATTR

```
#define CFE_PSP_INVALID_MEM_ATTR (-26)
```

Definition at line 64 of file cfe_psp.h.

39.90.1.9 CFE_PSP_INVALID_MEM_RANGE

```
#define CFE_PSP_INVALID_MEM_RANGE (-23)
```

Definition at line 61 of file cfe_psp.h.

39.90.1.10 CFE_PSP_INVALID_MEM_SIZE

```
#define CFE_PSP_INVALID_MEM_SIZE (-25)
```

Definition at line 63 of file cfe_psp.h.

39.90.1.11 CFE_PSP_INVALID_MEM_TYPE

```
#define CFE_PSP_INVALID_MEM_TYPE (-22)
```

Definition at line 60 of file cfe_psp.h.

39.90.1.12 CFE_PSP_INVALID_MEM_WORDSIZE

```
#define CFE_PSP_INVALID_MEM_WORDSIZE (-24)
```

Definition at line 62 of file cfe_psp.h.

39.90.1.13 CFE_PSP_INVALID_MODULE_ID

```
#define CFE_PSP_INVALID_MODULE_ID (-29)
```

Definition at line 67 of file cfe_psp.h.

39.90.1.14 CFE_PSP_INVALID_MODULE_NAME

```
#define CFE_PSP_INVALID_MODULE_NAME (-28)
```

Definition at line 66 of file cfe_psp.h.

39.90.1.15 CFE_PSP_INVALID_POINTER

```
#define CFE_PSP_INVALID_POINTER (-2)
```

Definition at line 55 of file cfe_psp.h.

39.90.1.16 CFE_PSP_MAJOR_VERSION

```
#define CFE_PSP_MAJOR_VERSION (GLOBAL_PSP_CONFIGDATA.PSP_VersionInfo.MajorVersion)
```

Definition at line 150 of file cfe_psp.h.

39.90.1.17 CFE_PSP_MEM_ANY

```
#define CFE_PSP_MEM_ANY 3
```

Definition at line 94 of file cfe_psp.h.

39.90.1.18 CFE_PSP_MEM_ATTR_READ

```
#define CFE_PSP_MEM_ATTR_READ 0x02
```

Definition at line 101 of file cfe_psp.h.

39.90.1.19 CFE_PSP_MEM_ATTR_READWRITE

```
#define CFE_PSP_MEM_ATTR_READWRITE 0x03
```

Definition at line 102 of file cfe_psp.h.

39.90.1.20 CFE_PSP_MEM_ATTR_WRITE

```
#define CFE_PSP_MEM_ATTR_WRITE 0x01
```

Definition at line 100 of file cfe_psp.h.

39.90.1.21 CFE_PSP_MEM_EEPROM

```
#define CFE_PSP_MEM_EEPROM 2
```

Definition at line 93 of file cfe_psp.h.

39.90.1.22 CFE_PSP_MEM_INVALID

```
#define CFE_PSP_MEM_INVALID 4
```

Definition at line 95 of file cfe_psp.h.

39.90.1.23 CFE_PSP_MEM_RAM

```
#define CFE_PSP_MEM_RAM 1
```

Definition at line 92 of file cfe_psp.h.

39.90.1.24 CFE_PSP_MEM_SIZE_BYTE

```
#define CFE_PSP_MEM_SIZE_BYTE 0x01
```

Definition at line 107 of file cfe_psp.h.

39.90.1.25 CFE_PSP_MEM_SIZE_DWORD

```
#define CFE_PSP_MEM_SIZE_DWORD 0x04
```

Definition at line 109 of file cfe_psp.h.

39.90.1.26 CFE_PSP_MEM_SIZE_WORD

```
#define CFE_PSP_MEM_SIZE_WORD 0x02
```

Definition at line 108 of file cfe_psp.h.

39.90.1.27 CFE_PSP_MINOR_VERSION

```
#define CFE_PSP_MINOR_VERSION (GLOBAL_PSP_CONFIGDATA.PSP_VersionInfo.MinorVersion)
```

Definition at line 151 of file cfe_psp.h.

39.90.1.28 CFE_PSP_MISSION_REV

```
#define CFE_PSP_MISSION_REV (GLOBAL_PSP_CONFIGDATA.PSP_VersionInfo.MissionRev)
```

Definition at line 153 of file cfe_psp.h.

39.90.1.29 CFE_PSP_NO_EXCEPTION_DATA

```
#define CFE_PSP_NO_EXCEPTION_DATA (-30)
```

Definition at line 68 of file cfe_psp.h.

39.90.1.30 CFE_PSP_PANIC_CORE_APP

```
#define CFE_PSP_PANIC_CORE_APP 6
```

Definition at line 78 of file cfe_psp.h.

39.90.1.31 CFE_PSP_PANIC_GENERAL_FAILURE

```
#define CFE_PSP_PANIC_GENERAL_FAILURE 7
```

Definition at line 79 of file cfe_psp.h.

39.90.1.32 CFE_PSP_PANIC_MEMORY_ALLOC

```
#define CFE_PSP_PANIC_MEMORY_ALLOC 3
```

Definition at line 75 of file cfe_psp.h.

39.90.1.33 CFE_PSP_PANIC_NONVOL_DISK

```
#define CFE_PSP_PANIC_NONVOL_DISK 4
```

Definition at line 76 of file cfe_psp.h.

39.90.1.34 CFE_PSP_PANIC_STARTUP

```
#define CFE_PSP_PANIC_STARTUP 1
```

Definition at line 73 of file cfe_psp.h.

39.90.1.35 CFE_PSP_PANIC_STARTUP_SEM

```
#define CFE_PSP_PANIC_STARTUP_SEM 5
```

Definition at line 77 of file cfe_psp.h.

39.90.1.36 CFE_PSP_PANIC_VOLATILE_DISK

```
#define CFE_PSP_PANIC_VOLATILE_DISK 2
```

Definition at line 74 of file cfe_psp.h.

39.90.1.37 CFE_PSP_REVISION

```
#define CFE_PSP_REVISION (GLOBAL_PSP_CONFIGDATA.PSP_VersionInfo.Revision)
```

Definition at line 152 of file cfe_psp.h.

39.90.1.38 CFE_PSP_RST_SUBTYPE_BANKSWITCH_RESET

```
#define CFE_PSP_RST_SUBTYPE_BANKSWITCH_RESET 9
```

Reset reverted to a cFE POWERON due to a boot bank switch.

Definition at line 144 of file cfe_psp.h.

39.90.1.39 CFE_PSP_RST_SUBTYPE_EXCEPTION

```
#define CFE_PSP_RST_SUBTYPE_EXCEPTION 6
```

Reset was caused by a Processor Exception.

Definition at line 138 of file cfe_psp.h.

39.90.1.40 CFE_PSP_RST_SUBTYPE_HW_SPECIAL_COMMAND

```
#define CFE_PSP_RST_SUBTYPE_HW_SPECIAL_COMMAND 3
```

Reset was caused by a reset line having been stimulated by a hardware special command.

Definition at line 132 of file cfe_psp.h.

39.90.1.41 CFE_PSP_RST_SUBTYPE_HW_WATCHDOG

```
#define CFE_PSP_RST_SUBTYPE_HW_WATCHDOG 4
```

Reset was caused by a watchdog timer expiring.

Definition at line 134 of file cfe_psp.h.

39.90.1.42 CFE_PSP_RST_SUBTYPE_HWDEBUG_RESET

```
#define CFE_PSP_RST_SUBTYPE_HWDEBUG_RESET 8
```

Reset was caused by a JTAG or BDM connection.

Definition at line 142 of file cfe_psp.h.

39.90.1.43 CFE_PSP_RST_SUBTYPE_MAX

```
#define CFE_PSP_RST_SUBTYPE_MAX 10
```

Placeholder to indicate 1+ the maximum value that the PSP will ever use.

Definition at line 146 of file cfe_psp.h.

39.90.1.44 CFE_PSP_RST_SUBTYPE_POWER_CYCLE

```
#define CFE_PSP_RST_SUBTYPE_POWER_CYCLE 1
```

Reset caused by power having been removed and restored.

Definition at line 128 of file cfe_psp.h.

39.90.1.45 CFE_PSP_RST_SUBTYPE_PUSH_BUTTON

```
#define CFE_PSP_RST_SUBTYPE_PUSH_BUTTON 2
```

Reset caused by reset button on the board.

Definition at line 130 of file cfe_psp.h.

39.90.1.46 CFE_PSP_RST_SUBTYPE_RESET_COMMAND

```
#define CFE_PSP_RST_SUBTYPE_RESET_COMMAND 5
```

Reset was caused by cFE ES processing a Reset Command .

Definition at line 136 of file cfe_psp.h.

39.90.1.47 CFE_PSP_RST_SUBTYPE_UNDEFINED_RESET

```
#define CFE_PSP_RST_SUBTYPE_UNDEFINED_RESET 7
```

Reset was caused in an unknown manner.

Definition at line 140 of file cfe_psp.h.

39.90.1.48 CFE_PSP_RST_TYPE_MAX

```
#define CFE_PSP_RST_TYPE_MAX 3
```

Placeholder to indicate 1+ the maximum value that the PSP will ever use.

Definition at line 118 of file cfe_psp.h.

39.90.1.49 CFE_PSP_RST_TYPE_POWERON

```
#define CFE_PSP_RST_TYPE_POWERON 2
```

All memory has been cleared

Definition at line 117 of file cfe_psp.h.

Referenced by CFE_PSP_InitProcessorReservedMemory(), CFE_PSP_Restart(), and OS_Application_Startup().

39.90.1.50 CFE_PSP_RST_TYPE_PROCESSOR

```
#define CFE_PSP_RST_TYPE_PROCESSOR 1
```

Volatile disk, CDS and User Reserved memory may be valid

Definition at line 116 of file cfe_psp.h.

Referenced by CFE_PSP_InitProcessorReservedMemory(), and OS_Application_Startup().

39.90.1.51 CFE_PSP_SUCCESS

```
#define CFE_PSP_SUCCESS (0)
```

Definition at line 53 of file cfe_psp.h.

Referenced by CFE_PSP_ExceptionGetSummary_Impl(), CFE_PSP_GetCDSSize(), CFE_PSP_GetCFEText↩SegmentInfo(), CFE_PSP_GetResetArea(), CFE_PSP_GetUserReservedArea(), CFE_PSP_GetVolatileDiskMem(), CFE_PSP_InitProcessorReservedMemory(), CFE_PSP_ReadFromCDS(), CFE_PSP_WriteToCDS(), and OS_↩Application_Startup().

39.90.1.52 CFE_PSP_VERSION

```
#define CFE_PSP_VERSION (GLOBAL_PSP_CONFIGDATA.PSP_VersionInfo.VersionString)
```

Definition at line 154 of file cfe_psp.h.

39.90.1.53 SIZE_BYTE

```
#define SIZE_BYTE 1
```

Definition at line 85 of file cfe_psp.h.

39.90.1.54 SIZE_HALF

```
#define SIZE_HALF 2
```

Definition at line 86 of file cfe_psp.h.

39.90.1.55 SIZE_WORD

```
#define SIZE_WORD 3
```

Definition at line 87 of file cfe_psp.h.

39.90.2 Function Documentation

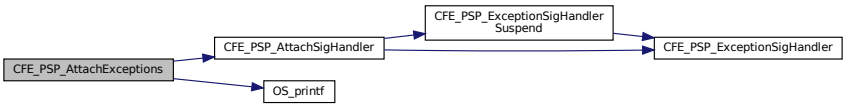
39.90.2.1 CFE_PSP_AttachExceptions()

```
void CFE_PSP_AttachExceptions (
            void  )
```

Definition at line 204 of file cfe_psp_exception.c.

References CFE_PSP_AsyncMask, CFE_PSP_AttachSigHandler(), and OS_printf().

Here is the call graph for this function:



39.90.2.2 CFE_PSP_Decompress()

```
int32 CFE_PSP_Decompress (
            char * srcFileName,
            char * dstFileName )
```

39.90.2.3 CFE_PSP_EepromPowerDown()

```
int32 CFE_PSP_EepromPowerDown (
            uint32 Bank )
```

39.90.2.4 CFE_PSP_EepromPowerUp()

```
int32 CFE_PSP_EepromPowerUp (
            uint32 Bank )
```

39.90.2.5 CFE_PSP_EepromWrite16()

```
int32 CFE_PSP_EepromWrite16 (
            cpuaddr MemoryAddress,
            uint16 uint16Value )
```

39.90.2.6 CFE_PSP_EepromWrite32()

```
int32 CFE_PSP_EepromWrite32 (
            cpuaddr MemoryAddress,
            uint32 uint32Value )
```

39.90.2.7 CFE_PSP_EepromWrite8()

```
int32 CFE_PSP_EepromWrite8 (
            cpuaddr MemoryAddress,
            uint8 ByteValue )
```

39.90.2.8 CFE_PSP_EepromWriteDisable()

```
int32 CFE_PSP_EepromWriteDisable (
            uint32 Bank )
```

39.90.2.9 CFE_PSP_EepromWriteEnable()

```
int32 CFE_PSP_EepromWriteEnable (
            uint32 Bank )
```

39.90.2.10 CFE_PSP_Exception_CopyContext()

```
int32 CFE_PSP_Exception_CopyContext (
            uint32 ContextLogId,
            void * ContextBuf,
            uint32 ContextSize )
```

39.90.2.11 CFE_PSP_Exception_GetCount()

```
uint32 CFE_PSP_Exception_GetCount (
            void  )
```

39.90.2.12 CFE_PSP_Exception_GetSummary()

```
int32 CFE_PSP_Exception_GetSummary (
            uint32 * ContextLogId,
            osal_id_t * TaskId,
            char * ReasonBuf,
            uint32 ReasonSize )
```

39.90.2.13 CFE_PSP_FlushCaches()

```
void CFE_PSP_FlushCaches (
            uint32 type,
            void * address,
            uint32 size )
```

Definition at line 153 of file cfe_psp_support.c.

39.90.2.14 CFE_PSP_Get_Dec()

```
uint32 CFE_PSP_Get_Dec (
            void  )
```

Definition at line 188 of file cfe_psp_timer.c.

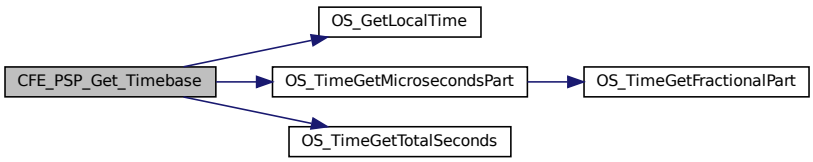
39.90.2.15 CFE_PSP_Get_Timebase()

```
void CFE_PSP_Get_Timebase (
            uint32 * Tbu,
            uint32 * Tbl )
```

Definition at line 165 of file cfe_psp_timer.c.

References OS_GetLocalTime(), OS_TimeGetMicrosecondsPart(), and OS_TimeGetTotalSeconds().

Here is the call graph for this function:



39.90.2.16 CFE_PSP_Get_Timer_Tick()

```
uint32 CFE_PSP_Get_Timer_Tick (
            void  )
```

Definition at line 105 of file cfe_psp_timer.c.

39.90.2.17 CFE_PSP_GetBuildNumber()

```
uint32 CFE_PSP_GetBuildNumber (
            void  )
```

Obtain the PSP library numeric build number.

The build number is a monotonically increasing number that (coarsely) reflects the number of commits/changes that have been merged since the epoch release. During development cycles this number should increase after each subsequent merge/modification.

Like other version information, this is a fixed number assigned at compile time.

Returns

The OSAL library build number

39.90.2.18 CFE_PSP_GetCDSSize()

```
int32 CFE_PSP_GetCDSSize (
            uint32 * SizeOfCDS )
```

Definition at line 215 of file cfe_psp_memory.c.

References CFE_PSP_CDS_SIZE, CFE_PSP_ERROR, and CFE_PSP_SUCCESS.

39.90.2.19 CFE_PSP_GetCFETextSegmentInfo()

```
int32 CFE_PSP_GetCFETextSegmentInfo (
            cpuaddr * PtrToCFESegment,
            uint32 * SizeOfCFESegment )
```

Definition at line 799 of file cfe_psp_memory.c.

References _fini, _init, CFE_PSP_ERROR, and CFE_PSP_SUCCESS.

39.90.2.20 CFE_PSP_GetKernelTextSegmentInfo()

```
int32 CFE_PSP_GetKernelTextSegmentInfo (
            cpuaddr * PtrToKernelSegment,
            uint32 * SizeOfKernelSegment )
```

Definition at line 773 of file cfe_psp_memory.c.

References CFE_PSP_ERROR, and CFE_PSP_ERROR_NOT_IMPLEMENTED.

39.90.2.21 CFE_PSP_GetProcessorId()

```
uint32 CFE_PSP_GetProcessorId (
            void  )
```

Definition at line 175 of file cfe_psp_support.c.

References CFE_PSP_CpuId.

39.90.2.22 CFE_PSP_GetProcessorName()

```
const char* CFE_PSP_GetProcessorName (
            void  )
```

Definition at line 215 of file cfe_psp_support.c.

References CFE_PSP_CpuName.

39.90.2.23 CFE_PSP_GetResetArea()

```
int32 CFE_PSP_GetResetArea (
            cpuaddr * PtrToResetArea,
            uint32 * SizeOfResetArea )
```

Definition at line 441 of file cfe_psp_memory.c.

References CFE_PSP_ERROR, CFE_PSP_ReservedMemoryMap, and CFE_PSP_SUCCESS.

39.90.2.24 CFE_PSP_GetRestartType()

```
uint32 CFE_PSP_GetRestartType (
            uint32 * restartSubType )
```

39.90.2.25 CFE_PSP_GetSpacecraftId()

```
uint32 CFE_PSP_GetSpacecraftId (
            void  )
```

Definition at line 195 of file cfe_psp_support.c.

References CFE_PSP_SpacecraftId.

39.90.2.26 CFE_PSP_GetTime()

```
void CFE_PSP_GetTime (
            OS_time_t * LocalTime )
```

Definition at line 82 of file cfe_psp_timer.c.

References OS_GetLocalTime().

Here is the call graph for this function:

39.90.2.27 CFE_PSP_GetTimerLow32Rollover()

```
uint32 CFE_PSP_GetTimerLow32Rollover (
            void  )
```

Definition at line 147 of file cfe_psp_timer.c.

References CFE_PSP_TIMER_LOW32_ROLLOVER.

39.90.2.28 CFE_PSP_GetTimerTicksPerSecond()

```
uint32 CFE_PSP_GetTimerTicksPerSecond (
            void  )
```

Definition at line 126 of file cfe_psp_timer.c.

References CFE_PSP_TIMER_TICKS_PER_SECOND.

39.90.2.29 CFE_PSP_GetUserReservedArea()

```
int32 CFE_PSP_GetUserReservedArea (
            cpuaddr * PtrToUserArea,
            uint32 * SizeOfUserArea )
```

Definition at line 556 of file cfe_psp_memory.c.

References CFE_PSP_ERROR, CFE_PSP_ReservedMemoryMap, and CFE_PSP_SUCCESS.

39.90.2.30 CFE_PSP_GetVersionCodeName()

```
const char* CFE_PSP_GetVersionCodeName (
            void  )
```

Obtain the version code name.

This retrieves the PSP code name. This is a compatibility indicator for the overall NASA CFS ecosystem. All modular components which are intended to interoperate should report the same code name.

Returns

Code name. This is a fixed string and cannot be NULL.

39.90.2.31 CFE_PSP_GetVersionNumber()

```
void CFE_PSP_GetVersionNumber (
            uint8 VersionNumbers[4] )
```

Obtain the PSP numeric version numbers as uint8 values.

This retrieves the numeric PSP version identifier as an array of 4 uint8 values.

The array of numeric values is in order of precedence: [0] = Major Number [1] = Minor Number [2] = Revision Number [3] = Mission Revision

The "Mission Revision" (last output) also indicates whether this is an official release, a patched release, or a development version. 0 indicates an official release 1-254 local patch level (reserved for mission use) 255 indicates a development build

Parameters

out	*VersionNumbers*	A fixed-size array to be filled with the version numbers

39.90.2.32 CFE_PSP_GetVersionString()

```
const char* CFE_PSP_GetVersionString (
            void  )
```

Obtain the PSP version/baseline identifier string.

This retrieves the PSP version identifier string without extra info

Returns

Version string. This is a fixed string and cannot be NULL.

39.90.2.33 CFE_PSP_GetVolatileDiskMem()

```
int32 CFE_PSP_GetVolatileDiskMem (
            cpuaddr * PtrToVolDisk,
            uint32 * SizeOfVolDisk )
```

Definition at line 615 of file cfe_psp_memory.c.

References CFE_PSP_ERROR, CFE_PSP_ReservedMemoryMap, and CFE_PSP_SUCCESS.

39.90.2.34 CFE_PSP_InitSSR()

```
int32 CFE_PSP_InitSSR (
            uint32 bus,
            uint32 device,
            char * DeviceName )
```

Definition at line 66 of file cfe_psp_ssr.c.

References CFE_PSP_ERROR.

39.90.2.35 CFE_PSP_Main()

```
void CFE_PSP_Main (
            void  )
```

39.90.2.36 CFE_PSP_MemCpy()

```
int32 CFE_PSP_MemCpy (
            void * dest,
            const void * src,
            uint32 n )
```

39.90.2.37 CFE_PSP_MemRangeGet()

```
int32 CFE_PSP_MemRangeGet (
            uint32 RangeNum,
            uint32 * MemoryType,
            cpuaddr * StartAddr,
            uint32 * Size,
            uint32 * WordSize,
            uint32 * Attributes )
```

39.90.2.38 CFE_PSP_MemRanges()

```
uint32 CFE_PSP_MemRanges (
            void  )
```

39.90.2.39 CFE_PSP_MemRangeSet()

```
int32 CFE_PSP_MemRangeSet (
            uint32 RangeNum,
            uint32 MemoryType,
            cpuaddr StartAddr,
            uint32 Size,
            uint32 WordSize,
            uint32 Attributes )
```

39.90.2.40 CFE_PSP_MemRead16()

```
int32 CFE_PSP_MemRead16 (
            cpuaddr MemoryAddress,
            uint16 * uint16Value )
```

39.90.2.41 CFE_PSP_MemRead32()

```
int32 CFE_PSP_MemRead32 (
            cpuaddr MemoryAddress,
            uint32 * uint32Value )
```

39.90.2.42 CFE_PSP_MemRead8()

```
int32 CFE_PSP_MemRead8 (
            cpuaddr MemoryAddress,
            uint8 * ByteValue )
```

39.90.2.43 CFE_PSP_MemSet()

```
int32 CFE_PSP_MemSet (
            void * dest,
            uint8 value,
            uint32 n )
```

39.90.2.44 CFE_PSP_MemValidateRange()

```
int32 CFE_PSP_MemValidateRange (
            cpuaddr Address,
            uint32 Size,
            uint32 MemoryType )
```

39.90.2.45 CFE_PSP_MemWrite16()

```
int32 CFE_PSP_MemWrite16 (
            cpuaddr MemoryAddress,
            uint16 uint16Value )
```

39.90.2.46 CFE_PSP_MemWrite32()

```
int32 CFE_PSP_MemWrite32 (
            cpuaddr MemoryAddress,
            uint32 uint32Value )
```

39.90.2.47 CFE_PSP_MemWrite8()

```
int32 CFE_PSP_MemWrite8 (
            cpuaddr MemoryAddress,
            uint8 ByteValue )
```

39.90.2.48 CFE_PSP_Panic()

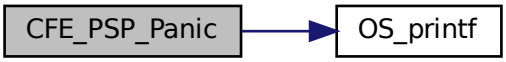
```
void CFE_PSP_Panic (
            int32 ErrorCode )
```

Definition at line 133 of file cfe_psp_support.c.

References OS_printf().

Referenced by OS_Application_Startup().

Here is the call graph for this function:



39.90.2.49 CFE_PSP_PortRead16()

```
int32 CFE_PSP_PortRead16 (
            cpuaddr PortAddress,
            uint16 * uint16Value )
```

39.90.2.50 CFE_PSP_PortRead32()

```
int32 CFE_PSP_PortRead32 (
            cpuaddr PortAddress,
            uint32 * uint32Value )
```

39.90.2.51 CFE_PSP_PortRead8()

```
int32 CFE_PSP_PortRead8 (
            cpuaddr PortAddress,
            uint8 * ByteValue )
```

39.90.2.52 CFE_PSP_PortWrite16()

```
int32 CFE_PSP_PortWrite16 (
            cpuaddr PortAddress,
            uint16 uint16Value )
```

39.90.2.53 CFE_PSP_PortWrite32()

```
int32 CFE_PSP_PortWrite32 (
            cpuaddr PortAddress,
            uint32 uint32Value )
```

39.90.2.54 CFE_PSP_PortWrite8()

```
int32 CFE_PSP_PortWrite8 (
            cpuaddr PortAddress,
            uint8 ByteValue )
```

39.90.2.55 CFE_PSP_ReadFromCDS()

```
int32 CFE_PSP_ReadFromCDS (
            void * PtrToDataToRead,
            uint32 CDSOffset,
            uint32 NumBytes )
```

Definition at line 287 of file cfe_psp_memory.c.

References CFE_PSP_CDS_SIZE, CFE_PSP_ERROR, CFE_PSP_ReservedMemoryMap, and CFE_PSP_SUCCE↩SS.

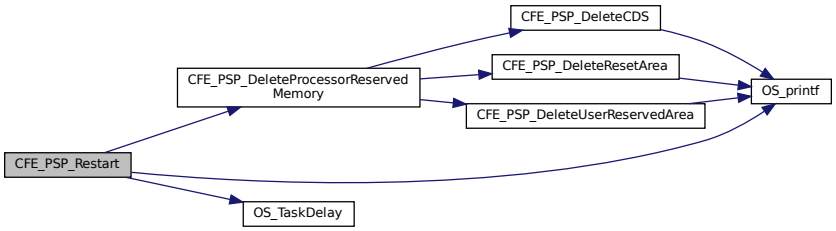
39.90.2.56 CFE_PSP_Restart()

```
void CFE_PSP_Restart (
            uint32 resetType )
```

Definition at line 69 of file cfe_psp_support.c.

References CFE_PSP_DeleteProcessorReservedMemory(), CFE_PSP_IdleTaskState, CFE_PSP_Reserved↩MemoryMap, CFE_PSP_RST_TYPE_POWERON, OS_printf(), and OS_TaskDelay().

Here is the call graph for this function:



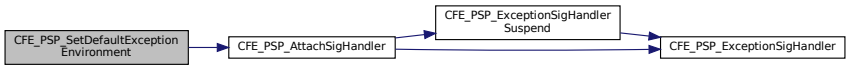
39.90.2.57 CFE_PSP_SetDefaultExceptionEnvironment()

```
void CFE_PSP_SetDefaultExceptionEnvironment (
            void  )
```

Definition at line 263 of file cfe_psp_exception.c.

References CFE_PSP_AttachSigHandler().

Here is the call graph for this function:



39.90.2.58 CFE_PSP_WatchdogDisable()

```
void CFE_PSP_WatchdogDisable (
            void  )
```

Definition at line 107 of file cfe_psp_watchdog.c.

39.90.2.59 CFE_PSP_WatchdogEnable()

```
void CFE_PSP_WatchdogEnable (
            void  )
```

Definition at line 95 of file cfe_psp_watchdog.c.

39.90.2.60 CFE_PSP_WatchdogGet()

```
uint32 CFE_PSP_WatchdogGet (
            void  )
```

Definition at line 142 of file cfe_psp_watchdog.c.

References CFE_PSP_WatchdogValue.

39.90.2.61 CFE_PSP_WatchdogInit()

```
void CFE_PSP_WatchdogInit (
            void  )
```

Definition at line 74 of file cfe_psp_watchdog.c.

References CFE_PSP_WatchdogValue.

39.90.2.62 CFE_PSP_WatchdogService()

```
void CFE_PSP_WatchdogService (
            void  )
```

Definition at line 125 of file cfe_psp_watchdog.c.

39.90.2.63 CFE_PSP_WatchdogSet()

```
void CFE_PSP_WatchdogSet (
            uint32 WatchdogValue )
```

Definition at line 162 of file cfe_psp_watchdog.c.

References CFE_PSP_WatchdogValue.

39.90.2.64 CFE_PSP_WriteToCDS()

```
int32 CFE_PSP_WriteToCDS (
            const void * PtrToDataToWrite,
            uint32 CDSOffset,
            uint32 NumBytes )
```

Definition at line 245 of file cfe_psp_memory.c.

References CFE_PSP_CDS_SIZE, CFE_PSP_ERROR, CFE_PSP_ReservedMemoryMap, and CFE_PSP_SUCCE↩SS.

39.91 psp/fsw/inc/cfe_psp_configdata.h File Reference

```
#include "osapi.h"
#include "cfe_psp.h"
```

Data Structures

- struct CFE_PSP_VersionInfo_t
- struct Target_PspConfigData

Variables

- Target_PspConfigData GLOBAL_PSP_CONFIGDATA
- CFE_PSP_MemTable_t CFE_PSP_MemoryTable [ ]

39.91.1 Detailed Description

Created on: Dec 31, 2014 Author: joseph.p.hickey@nasa.gov

39.91.2 Variable Documentation

39.91.2.1 CFE_PSP_MemoryTable

```
CFE_PSP_MemTable_t CFE_PSP_MemoryTable[]
```

Extern reference to the psp memory table Allows the actual instantiation to be done outside this module

Definition at line 44 of file cfe_psp_memtab.c.

39.91.2.2 GLOBAL_PSP_CONFIGDATA

```
Target_PspConfigData GLOBAL_PSP_CONFIGDATA
```

Extern reference to psp config struct. Allows the actual instantiation to be done outside this module

39.92 psp/fsw/pc-linux/src/cfe_psp_exception.c File Reference

```
#include <stdio.h>
#include <string.h>
#include <pthread.h>
#include "common_types.h"
#include "osapi.h"
#include "cfe_psp.h"
#include "cfe_psp_config.h"
#include "cfe_psp_exceptionstorage_types.h"
#include "cfe_psp_exceptionstorage_api.h"
#include <execinfo.h>
#include <signal.h>
```

Functions

- void CFE_PSP_ExceptionSigHandler (int signo, siginfo_t ∗si, void ∗ctxt)
- void CFE_PSP_ExceptionSigHandlerSuspend (int signo, siginfo_t ∗si, void ∗ctxt)
- void CFE_PSP_AttachSigHandler (int signo)
- void CFE_PSP_AttachExceptions (void)
- void CFE_PSP_SetDefaultExceptionEnvironment (void)
- int32 CFE_PSP_ExceptionGetSummary_Impl (const CFE_PSP_Exception_LogData_t ∗Buffer, char ∗Reason↩ Buf, uint32 ReasonSize)

Variables

- sigset_t CFE_PSP_AsyncMask

39.92.1 Function Documentation

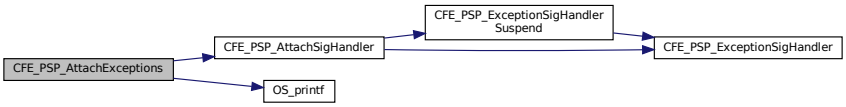
39.92.1.1 CFE_PSP_AttachExceptions()

```
void CFE_PSP_AttachExceptions (
            void  )
```

Definition at line 204 of file cfe_psp_exception.c.

References CFE_PSP_AsyncMask, CFE_PSP_AttachSigHandler(), and OS_printf().

Here is the call graph for this function:



39.92.1.2 CFE_PSP_AttachSigHandler()

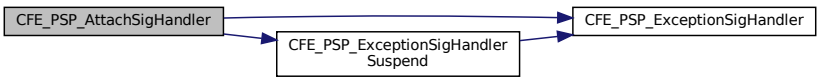
```
void CFE_PSP_AttachSigHandler (
            int signo )
```

Definition at line 150 of file cfe_psp_exception.c.

References CFE_PSP_AsyncMask, CFE_PSP_ExceptionSigHandler(), and CFE_PSP_ExceptionSigHandler↩ Suspend().

Referenced by CFE_PSP_AttachExceptions(), and CFE_PSP_SetDefaultExceptionEnvironment().

Here is the call graph for this function:



39.92.1.3 CFE_PSP_ExceptionGetSummary_Impl()

```
int32 CFE_PSP_ExceptionGetSummary_Impl (
            const CFE_PSP_Exception_LogData_t * Buffer,
            char * ReasonBuf,
            uint32 ReasonSize )
```

Definition at line 279 of file cfe_psp_exception.c.

References CFE_PSP_SUCCESS.

39.92.1.4 CFE_PSP_ExceptionSigHandler()

```
void CFE_PSP_ExceptionSigHandler (
            int signo,
            siginfo_t * si,
            void * ctxt )
```

Definition at line 70 of file cfe_psp_exception.c.

References CFE_PSP_IdleTaskState.

Referenced by CFE_PSP_AttachSigHandler(), and CFE_PSP_ExceptionSigHandlerSuspend().

39.92.1.5 CFE_PSP_ExceptionSigHandlerSuspend()

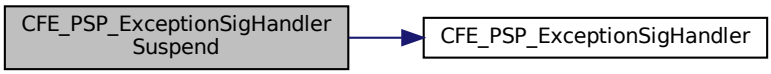
```
void CFE_PSP_ExceptionSigHandlerSuspend (
            int signo,
            siginfo_t * si,
            void * ctxt )
```

Definition at line 127 of file cfe_psp_exception.c.

References CFE_PSP_AsyncMask, and CFE_PSP_ExceptionSigHandler().

Referenced by CFE_PSP_AttachSigHandler().

Here is the call graph for this function:



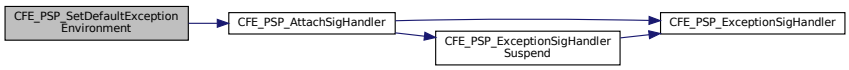
39.92.1.6 CFE_PSP_SetDefaultExceptionEnvironment()

```
void CFE_PSP_SetDefaultExceptionEnvironment (
            void  )
```

Definition at line 263 of file cfe_psp_exception.c.

References CFE_PSP_AttachSigHandler().

Here is the call graph for this function:



39.92.2 Variable Documentation

39.92.2.1 CFE_PSP_AsyncMask

```
sigset_t CFE_PSP_AsyncMask
```

Definition at line 58 of file cfe_psp_exception.c.

Referenced by CFE_PSP_AttachExceptions(), CFE_PSP_AttachSigHandler(), and CFE_PSP_ExceptionSigHandler↩Suspend().

39.93 psp/fsw/pc-linux/src/cfe_psp_memory.c File Reference

```
#include <stdio.h>
#include <stdlib.h>
#include <sys/types.h>
#include <unistd.h>
#include <string.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <fcntl.h>
#include "common_types.h"
#include "osapi.h"
#include "cfe_psp.h"
#include "cfe_psp_config.h"
#include "cfe_psp_memory.h"
#include "target_config.h"
```

Data Structures

- struct CFE_PSP_LinuxReservedAreaFixedLayout_t

Macros

- #define CFE_PSP_CDS_KEY_FILE ".cdskeyfile"
- #define CFE_PSP_RESET_KEY_FILE ".resetkeyfile"
- #define CFE_PSP_RESERVED_KEY_FILE ".reservedkeyfile"
- #define CFE_PSP_CDS_SIZE (GLOBAL_CONFIGDATA.CfeConfig->CdsSize)
- #define CFE_PSP_RESET_AREA_SIZE (GLOBAL_CONFIGDATA.CfeConfig->ResetAreaSize)
- #define CFE_PSP_USER_RESERVED_SIZE (GLOBAL_CONFIGDATA.CfeConfig->UserReservedSize)

Functions

- void CFE_PSP_InitCDS (void)
- void CFE_PSP_InitResetArea (void)
- void CFE_PSP_InitVolatileDiskMem (void)
- void CFE_PSP_InitUserReservedArea (void)
- void CFE_PSP_DeleteCDS (void)
- int32 CFE_PSP_GetCDSSize (uint32 ∗SizeOfCDS)
- int32 CFE_PSP_WriteToCDS (const void ∗PtrToDataToWrite, uint32 CDSOffset, uint32 NumBytes)
- int32 CFE_PSP_ReadFromCDS (void ∗PtrToDataToRead, uint32 CDSOffset, uint32 NumBytes)
- void CFE_PSP_DeleteResetArea (void)
- int32 CFE_PSP_GetResetArea (cpuaddr ∗PtrToResetArea, uint32 ∗SizeOfResetArea)
- void CFE_PSP_DeleteUserReservedArea (void)
- int32 CFE_PSP_GetUserReservedArea (cpuaddr ∗PtrToUserArea, uint32 ∗SizeOfUserArea)
- int32 CFE_PSP_GetVolatileDiskMem (cpuaddr ∗PtrToVolDisk, uint32 ∗SizeOfVolDisk)
- void CFE_PSP_SetupReservedMemoryMap (void)
- int32 CFE_PSP_InitProcessorReservedMemory (uint32 RestartType)
- void CFE_PSP_DeleteProcessorReservedMemory (void)
- int32 CFE_PSP_GetKernelTextSegmentInfo (cpuaddr ∗PtrToKernelSegment, uint32 ∗SizeOfKernelSegment)
- int32 CFE_PSP_GetCFETextSegmentInfo (cpuaddr ∗PtrToCFESegment, uint32 ∗SizeOfCFESegment)

Variables

- unsigned int _init
- unsigned int _fini
- int ResetAreaShmId
- int CDSShmId
- int UserShmId
- CFE_PSP_ReservedMemoryMap_t CFE_PSP_ReservedMemoryMap

39.93.1 Macro Definition Documentation

39.93.1.1 CFE_PSP_CDS_KEY_FILE

```
#define CFE_PSP_CDS_KEY_FILE ".cdskeyfile"
```

Definition at line 67 of file cfe_psp_memory.c.

Referenced by CFE_PSP_InitCDS(), and CFE_PSP_SetupReservedMemoryMap().

39.93.1.2 CFE_PSP_CDS_SIZE

```
#define CFE_PSP_CDS_SIZE (GLOBAL_CONFIGDATA.CfeConfig->CdsSize)
```

Definition at line 77 of file cfe_psp_memory.c.

Referenced by CFE_PSP_GetCDSSize(), CFE_PSP_InitCDS(), CFE_PSP_InitProcessorReservedMemory(), CFE_↩ PSP_ReadFromCDS(), and CFE_PSP_WriteToCDS().

39.93.1.3 CFE_PSP_RESERVED_KEY_FILE

```
#define CFE_PSP_RESERVED_KEY_FILE ".reservedkeyfile"
```

Definition at line 69 of file cfe_psp_memory.c.

Referenced by CFE_PSP_InitUserReservedArea(), and CFE_PSP_SetupReservedMemoryMap().

39.93.1.4 CFE_PSP_RESET_AREA_SIZE

```
#define CFE_PSP_RESET_AREA_SIZE (GLOBAL_CONFIGDATA.CfeConfig->ResetAreaSize)
```

Definition at line 78 of file cfe_psp_memory.c.

Referenced by CFE_PSP_InitProcessorReservedMemory(), and CFE_PSP_InitResetArea().

39.93.1.5 CFE_PSP_RESET_KEY_FILE

```
#define CFE_PSP_RESET_KEY_FILE ".resetkeyfile"
```

Definition at line 68 of file cfe_psp_memory.c.

Referenced by CFE_PSP_InitResetArea(), and CFE_PSP_SetupReservedMemoryMap().

39.93.1.6 CFE_PSP_USER_RESERVED_SIZE

```
#define CFE_PSP_USER_RESERVED_SIZE (GLOBAL_CONFIGDATA.CfeConfig->UserReservedSize)
```

Definition at line 79 of file cfe_psp_memory.c.

Referenced by CFE_PSP_InitProcessorReservedMemory(), and CFE_PSP_InitUserReservedArea().

39.93.2 Function Documentation

39.93.2.1 CFE_PSP_DeleteCDS()

```
void CFE_PSP_DeleteCDS (
            void  )
```

Definition at line 183 of file cfe_psp_memory.c.

References CDSShmId, and OS_printf().

Referenced by CFE_PSP_DeleteProcessorReservedMemory().

Here is the call graph for this function:

39.93.2.2 CFE_PSP_DeleteProcessorReservedMemory()

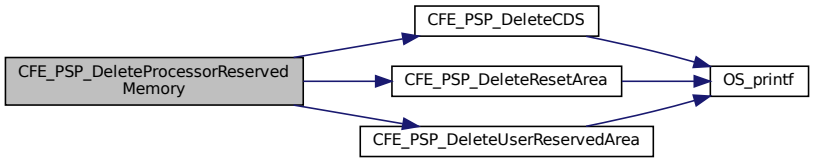
```
void CFE_PSP_DeleteProcessorReservedMemory (
            void  )
```

Definition at line 746 of file cfe_psp_memory.c.

References CFE_PSP_DeleteCDS(), CFE_PSP_DeleteResetArea(), and CFE_PSP_DeleteUserReservedArea().

Referenced by CFE_PSP_Restart().

Here is the call graph for this function:



39.93.2.3 CFE_PSP_DeleteResetArea()

```
void CFE_PSP_DeleteResetArea (
            void  )
```

Definition at line 407 of file cfe_psp_memory.c.

References OS_printf(), and ResetAreaShmId.

Referenced by CFE_PSP_DeleteProcessorReservedMemory().

Here is the call graph for this function:

39.93.2.4 CFE_PSP_DeleteUserReservedArea()

```
void CFE_PSP_DeleteUserReservedArea (
            void  )
```

Definition at line 525 of file cfe_psp_memory.c.

References OS_printf(), and UserShmId.

Referenced by CFE_PSP_DeleteProcessorReservedMemory().

Here is the call graph for this function:

39.93.2.5 CFE_PSP_GetCDSSize()

```
int32 CFE_PSP_GetCDSSize (
            uint32 * SizeOfCDS )
```

Definition at line 215 of file cfe_psp_memory.c.

References CFE_PSP_CDS_SIZE, CFE_PSP_ERROR, and CFE_PSP_SUCCESS.

39.93.2.6 CFE_PSP_GetCFETextSegmentInfo()

```
int32 CFE_PSP_GetCFETextSegmentInfo (
            cpuaddr * PtrToCFESegment,
            uint32 * SizeOfCFESegment )
```

Definition at line 799 of file cfe_psp_memory.c.

References _fini, _init, CFE_PSP_ERROR, and CFE_PSP_SUCCESS.

39.93.2.7 CFE_PSP_GetKernelTextSegmentInfo()

```
int32 CFE_PSP_GetKernelTextSegmentInfo (
            cpuaddr * PtrToKernelSegment,
            uint32 * SizeOfKernelSegment )
```

Definition at line 773 of file cfe_psp_memory.c.

References CFE_PSP_ERROR, and CFE_PSP_ERROR_NOT_IMPLEMENTED.

39.93.2.8 CFE_PSP_GetResetArea()

```
int32 CFE_PSP_GetResetArea (
            cpuaddr * PtrToResetArea,
            uint32 * SizeOfResetArea )
```

Definition at line 441 of file cfe_psp_memory.c.

References CFE_PSP_ERROR, CFE_PSP_ReservedMemoryMap, and CFE_PSP_SUCCESS.

39.93.2.9 CFE_PSP_GetUserReservedArea()

```
int32 CFE_PSP_GetUserReservedArea (
            cpuaddr * PtrToUserArea,
            uint32 * SizeOfUserArea )
```

Definition at line 556 of file cfe_psp_memory.c.

References CFE_PSP_ERROR, CFE_PSP_ReservedMemoryMap, and CFE_PSP_SUCCESS.

39.93.2.10 CFE_PSP_GetVolatileDiskMem()

```
int32 CFE_PSP_GetVolatileDiskMem (
            cpuaddr * PtrToVolDisk,
            uint32 * SizeOfVolDisk )
```

Definition at line 615 of file cfe_psp_memory.c.

References CFE_PSP_ERROR, CFE_PSP_ReservedMemoryMap, and CFE_PSP_SUCCESS.

39.93.2.11 CFE_PSP_InitCDS()

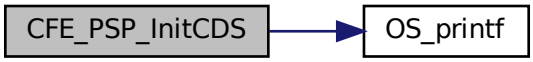
```
void CFE_PSP_InitCDS (
            void  )
```

Definition at line 136 of file cfe_psp_memory.c.

References CDSShmId, CFE_PSP_CDS_KEY_FILE, CFE_PSP_CDS_SIZE, CFE_PSP_ReservedMemoryMap, and OS_printf().

Referenced by CFE_PSP_SetupReservedMemoryMap().

Here is the call graph for this function:



39.93.2.12 CFE_PSP_InitProcessorReservedMemory()

```
int32 CFE_PSP_InitProcessorReservedMemory (
            uint32 RestartType )
```

Definition at line 679 of file cfe_psp_memory.c.

References CFE_PSP_CDS_SIZE, CFE_PSP_ReservedMemoryMap, CFE_PSP_RESET_AREA_SIZE, CFE_PSP↩_RST_TYPE_POWERON, CFE_PSP_RST_TYPE_PROCESSOR, CFE_PSP_SUCCESS, CFE_PSP_USER_RESE↩RVED_SIZE, and OS_printf().

Referenced by OS_Application_Startup().

Here is the call graph for this function:

39.93.2.13 CFE_PSP_InitResetArea()

```
void CFE_PSP_InitResetArea (
            void  )
```

Definition at line 335 of file cfe_psp_memory.c.

References CFE_PSP_LinuxReservedAreaFixedLayout_t::BootRecord, CFE_PSP_ReservedMemoryMap, CFE_P↩SP_RESET_AREA_SIZE, CFE_PSP_RESET_KEY_FILE, CFE_PSP_LinuxReservedAreaFixedLayout_t::Exception↩Storage, OS_printf(), and ResetAreaShmId.

Referenced by CFE_PSP_SetupReservedMemoryMap().

Here is the call graph for this function:

39.93.2.14 CFE_PSP_InitUserReservedArea()

```
void CFE_PSP_InitUserReservedArea (
            void  )
```

Definition at line 478 of file cfe_psp_memory.c.

References CFE_PSP_RESERVED_KEY_FILE, CFE_PSP_ReservedMemoryMap, CFE_PSP_USER_RESERVED↩_SIZE, OS_printf(), and UserShmId.

Referenced by CFE_PSP_SetupReservedMemoryMap().

Here is the call graph for this function:

39.93.2.15 CFE_PSP_InitVolatileDiskMem()

```
void CFE_PSP_InitVolatileDiskMem (
            void  )
```

Definition at line 594 of file cfe_psp_memory.c.

Referenced by CFE_PSP_SetupReservedMemoryMap().

39.93.2.16 CFE_PSP_ReadFromCDS()

```
int32 CFE_PSP_ReadFromCDS (
            void * PtrToDataToRead,
            uint32 CDSOffset,
            uint32 NumBytes )
```

Definition at line 287 of file cfe_psp_memory.c.

References CFE_PSP_CDS_SIZE, CFE_PSP_ERROR, CFE_PSP_ReservedMemoryMap, and CFE_PSP_SUCCE↩SS.

39.93.2.17 CFE_PSP_SetupReservedMemoryMap()

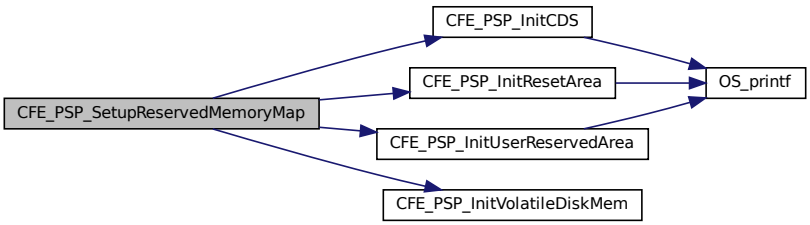
```
void CFE_PSP_SetupReservedMemoryMap (
            void  )
```

Definition at line 651 of file cfe_psp_memory.c.

References CFE_PSP_CDS_KEY_FILE, CFE_PSP_InitCDS(), CFE_PSP_InitResetArea(), CFE_PSP_InitUser↩ ReservedArea(), CFE_PSP_InitVolatileDiskMem(), CFE_PSP_RESERVED_KEY_FILE, and CFE_PSP_RESET_KE↩ Y_FILE.

Referenced by OS_Application_Startup().

Here is the call graph for this function:



39.93.2.18 CFE_PSP_WriteToCDS()

```
int32 CFE_PSP_WriteToCDS (
            const void * PtrToDataToWrite,
            uint32 CDSOffset,
            uint32 NumBytes )
```

Definition at line 245 of file cfe_psp_memory.c.

References CFE_PSP_CDS_SIZE, CFE_PSP_ERROR, CFE_PSP_ReservedMemoryMap, and CFE_PSP_SUCCE↩ SS.

39.93.3 Variable Documentation

39.93.3.1 _fini

```
unsigned int _fini
```

Referenced by CFE_PSP_GetCFETextSegmentInfo().

39.93.3.2 _init

```
unsigned int _init
```

Referenced by CFE_PSP_GetCFETextSegmentInfo().

39.93.3.3 CDSShmId

```
int CDSShmId
```

Definition at line 105 of file cfe_psp_memory.c.

Referenced by CFE_PSP_DeleteCDS(), and CFE_PSP_InitCDS().

39.93.3.4 CFE_PSP_ReservedMemoryMap

```
CFE_PSP_ReservedMemoryMap_t CFE_PSP_ReservedMemoryMap
```

Definition at line 112 of file cfe_psp_memory.c.

Referenced by CFE_PSP_GetResetArea(), CFE_PSP_GetUserReservedArea(), CFE_PSP_GetVolatileDiskMem(), CFE_PSP_InitCDS(), CFE_PSP_InitProcessorReservedMemory(), CFE_PSP_InitResetArea(), CFE_PSP_InitUser↩ ReservedArea(), CFE_PSP_ReadFromCDS(), CFE_PSP_Restart(), CFE_PSP_WriteToCDS(), and OS_Application↩ _Startup().

39.93.3.5 ResetAreaShmId

```
int ResetAreaShmId
```

Definition at line 104 of file cfe_psp_memory.c.

Referenced by CFE_PSP_DeleteResetArea(), and CFE_PSP_InitResetArea().

39.93.3.6 UserShmId

```
int UserShmId
```

Definition at line 106 of file cfe_psp_memory.c.

Referenced by CFE_PSP_DeleteUserReservedArea(), and CFE_PSP_InitUserReservedArea().

39.94 psp/fsw/pc-linux/src/cfe_psp_memtab.c File Reference

```
#include "common_types.h"
#include "osapi.h"
#include "cfe_psp.h"
#include "cfe_psp_config.h"
```

Variables

- CFE_PSP_MemTable_t CFE_PSP_MemoryTable [CFE_PSP_MEM_TABLE_SIZE]

39.94.1 Variable Documentation

39.94.1.1 CFE_PSP_MemoryTable

```
CFE_PSP_MemTable_t CFE_PSP_MemoryTable[CFE_PSP_MEM_TABLE_SIZE]
```

Initial value:

```
= {
    {CFE_PSP_MEM_RAM, CFE_PSP_MEM_SIZE_DWORD, 0, 0xFFFFFFFF,
      CFE_PSP_MEM_ATTR_READWRITE},
    {CFE_PSP_MEM_INVALID, 0, 0, 0, CFE_PSP_MEM_ATTR_READWRITE},
    {CFE_PSP_MEM_INVALID, 0, 0, 0, CFE_PSP_MEM_ATTR_READWRITE},
    {CFE_PSP_MEM_INVALID, 0, 0, 0, CFE_PSP_MEM_ATTR_READWRITE},
    {CFE_PSP_MEM_INVALID, 0, 0, 0, CFE_PSP_MEM_ATTR_READWRITE},
    {CFE_PSP_MEM_INVALID, 0, 0, 0, CFE_PSP_MEM_ATTR_READWRITE},
    {CFE_PSP_MEM_INVALID, 0, 0, 0, CFE_PSP_MEM_ATTR_READWRITE},
    {CFE_PSP_MEM_INVALID, 0, 0, 0, CFE_PSP_MEM_ATTR_READWRITE},
    {CFE_PSP_MEM_INVALID, 0, 0, 0, CFE_PSP_MEM_ATTR_READWRITE},
    {CFE_PSP_MEM_INVALID, 0, 0, 0, CFE_PSP_MEM_ATTR_READWRITE},
}
```

Extern reference to the psp memory table Allows the actual instantiation to be done outside this module

Definition at line 44 of file cfe_psp_memtab.c.

39.95 psp/fsw/pc-linux/src/cfe_psp_ssr.c File Reference

```
#include "common_types.h"
#include "osapi.h"
#include "cfe_psp.h"
#include "cfe_psp_config.h"
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
```

Functions

- int32 CFE_PSP_InitSSR (uint32 bus, uint32 device, char *DeviceName)

39.95.1 Function Documentation

39.95.1.1 CFE_PSP_InitSSR()

```
int32 CFE_PSP_InitSSR (
            uint32 bus,
            uint32 device,
            char * DeviceName )
```

Definition at line 66 of file cfe_psp_ssr.c.

References CFE_PSP_ERROR.

39.96 psp/fsw/pc-linux/src/cfe_psp_start.c File Reference

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/wait.h>
#include <sys/types.h>
#include <unistd.h>
#include <signal.h>
#include <sys/time.h>
#include <getopt.h>
#include <limits.h>
#include <pthread.h>
#include <sched.h>
#include <errno.h>
#include "common_types.h"
#include "osapi.h"
#include "cfe_psp.h"
#include "cfe_psp_memory.h"
#include "target_config.h"
#include "cfe_psp_module.h"
```

Data Structures

- struct CFE_PSP_CommandData_t

Macros

- #define CFE_PSP_MAIN_FUNCTION (∗GLOBAL_CONFIGDATA.CfeConfig->SystemMain)
- #define CFE_PSP_1HZ_FUNCTION (∗GLOBAL_CONFIGDATA.CfeConfig->System1HzISR)
- #define CFE_PSP_NONVOL_STARTUP_FILE (GLOBAL_CONFIGDATA.CfeConfig->NonvolStartupFile)
- #define CFE_PSP_CPU_ID (GLOBAL_CONFIGDATA.Default_CpuId)
- #define CFE_PSP_CPU_NAME (GLOBAL_CONFIGDATA.Default_CpuName)
- #define CFE_PSP_SPACECRAFT_ID (GLOBAL_CONFIGDATA.Default_SpacecraftId)
- #define CFE_PSP_CPU_NAME_LENGTH 32
- #define CFE_PSP_RESET_NAME_LENGTH 10
- #define CFE_PSP_KERNEL_NAME_LENGTH_MAX 16

Functions

- void CFE_PSP_TimerHandler (int signum)
- void CFE_PSP_DisplayUsage (char ∗Name)
- void CFE_PSP_ProcessArgumentDefaults (CFE_PSP_CommandData_t ∗CommandDataDefault)
- void CFE_PSP_SetupLocal1Hz (void)
- int32 CFE_PSP_OS_EventHandler (OS_Event_t event, osal_id_t object_id, void ∗data)
- void OS_Application_Startup (void)

  Application startup.
- void OS_Application_Run (void)

  Application run.

Variables

- uint32 TimerCounter
- CFE_PSP_CommandData_t CommandData
- uint32 CFE_PSP_SpacecraftId
- uint32 CFE_PSP_CpuId
- char CFE_PSP_CpuName [CFE_PSP_CPU_NAME_LENGTH]
- CFE_PSP_IdleTaskState_t CFE_PSP_IdleTaskState
- static const char ∗ optString = "R:S:C:I:N:h"
- static const struct option longOpts [ ]

39.96.1 Macro Definition Documentation

39.96.1.1 CFE_PSP_1HZ_FUNCTION

```
#define CFE_PSP_1HZ_FUNCTION (*GLOBAL_CONFIGDATA.CfeConfig->System1HzISR)
```

Definition at line 68 of file cfe_psp_start.c.

Referenced by CFE_PSP_TimerHandler().

39.96.1.2 CFE_PSP_CPU_ID

```
#define CFE_PSP_CPU_ID (GLOBAL_CONFIGDATA.Default_CpuId)
```

Definition at line 70 of file cfe_psp_start.c.

Referenced by CFE_PSP_DisplayUsage(), and CFE_PSP_ProcessArgumentDefaults().

39.96.1.3 CFE_PSP_CPU_NAME

```
#define CFE_PSP_CPU_NAME (GLOBAL_CONFIGDATA.Default_CpuName)
```

Definition at line 71 of file cfe_psp_start.c.

Referenced by CFE_PSP_DisplayUsage(), and CFE_PSP_ProcessArgumentDefaults().

39.96.1.4 CFE_PSP_CPU_NAME_LENGTH

```
#define CFE_PSP_CPU_NAME_LENGTH 32
```

Definition at line 78 of file cfe_psp_start.c.

Referenced by CFE_PSP_ProcessArgumentDefaults(), and OS_Application_Startup().

39.96.1.5 CFE_PSP_KERNEL_NAME_LENGTH_MAX

```
#define CFE_PSP_KERNEL_NAME_LENGTH_MAX 16
```

Definition at line 85 of file cfe_psp_start.c.

Referenced by CFE_PSP_OS_EventHandler().

39.96.1.6 CFE_PSP_MAIN_FUNCTION

```
#define CFE_PSP_MAIN_FUNCTION (*GLOBAL_CONFIGDATA.CfeConfig->SystemMain)
```

Definition at line 67 of file cfe_psp_start.c.

Referenced by OS_Application_Startup().

39.96.1.7 CFE_PSP_NONVOL_STARTUP_FILE

```
#define CFE_PSP_NONVOL_STARTUP_FILE (GLOBAL_CONFIGDATA.CfeConfig->NonvolStartupFile)
```

Definition at line 69 of file cfe_psp_start.c.

Referenced by OS_Application_Startup().

39.96.1.8 CFE_PSP_RESET_NAME_LENGTH

```
#define CFE_PSP_RESET_NAME_LENGTH 10
```

Definition at line 79 of file cfe_psp_start.c.

Referenced by OS_Application_Startup().

39.96.1.9 CFE_PSP_SPACECRAFT_ID

```
#define CFE_PSP_SPACECRAFT_ID (GLOBAL_CONFIGDATA.Default_SpacecraftId)
```

Definition at line 72 of file cfe_psp_start.c.

Referenced by CFE_PSP_DisplayUsage(), and CFE_PSP_ProcessArgumentDefaults().

39.96.2 Function Documentation

39.96.2.1 CFE_PSP_DisplayUsage()

```
void CFE_PSP_DisplayUsage (
            char * Name )
```

Definition at line 562 of file cfe_psp_start.c.

References CFE_PSP_CPU_ID, CFE_PSP_CPU_NAME, and CFE_PSP_SPACECRAFT_ID.

Referenced by OS_Application_Startup().

39.96.2.2 CFE_PSP_OS_EventHandler()

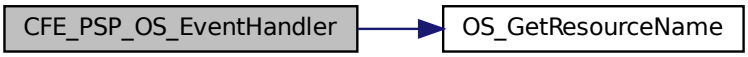
```
int32 CFE_PSP_OS_EventHandler (
            OS_Event_t event,
            osal_id_t object_id,
            void * data )
```

Definition at line 156 of file cfe_psp_start.c.

References CFE_PSP_KERNEL_NAME_LENGTH_MAX, OS_EVENT_RESOURCE_ALLOCATED, OS_EVENT_RESOURCE_CREATED, OS_EVENT_RESOURCE_DELETED, OS_EVENT_TASK_STARTUP, OS_GetResourceName(), OS_MAX_API_NAME, and OS_SUCCESS.

Referenced by OS_Application_Startup().

Here is the call graph for this function:



39.96.2.3 CFE_PSP_ProcessArgumentDefaults()

```
void CFE_PSP_ProcessArgumentDefaults (
            CFE_PSP_CommandData_t * CommandDataDefault )
```

Definition at line 611 of file cfe_psp_start.c.

References CFE_PSP_CPU_ID, CFE_PSP_CPU_NAME, CFE_PSP_CPU_NAME_LENGTH, CFE_PSP_SPACECRAFT_ID, CFE_PSP_CommandData_t::CpuId, CFE_PSP_CommandData_t::CpuName, CFE_PSP_CommandData_t::GotCpuId, CFE_PSP_CommandData_t::GotCpuName, CFE_PSP_CommandData_t::GotSpacecraftId, CFE_PSP_CommandData_t::GotSubType, CFE_PSP_CommandData_t::SpacecraftId, and CFE_PSP_CommandData_t::SubType.

Referenced by OS_Application_Startup().

39.96.2.4 CFE_PSP_SetupLocal1Hz()

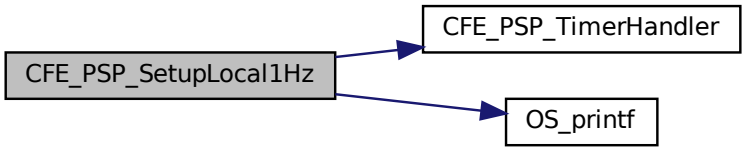
```
void CFE_PSP_SetupLocal1Hz (
            void  )
```

Definition at line 667 of file cfe_psp_start.c.

References CFE_PSP_TimerHandler(), OS_printf(), and TimerCounter.

Referenced by OS_Application_Startup().

Here is the call graph for this function:



39.96.2.5 CFE_PSP_TimerHandler()

```
void CFE_PSP_TimerHandler (
            int signum )
```

Definition at line 538 of file cfe_psp_start.c.

References CFE_PSP_1HZ_FUNCTION, and TimerCounter.

Referenced by CFE_PSP_SetupLocal1Hz().

39.96.3 Variable Documentation

39.96.3.1 CFE_PSP_CpuId

```
uint32 CFE_PSP_CpuId
```

Definition at line 127 of file cfe_psp_start.c.

Referenced by CFE_PSP_GetProcessorId(), and OS_Application_Startup().

39.96.3.2 CFE_PSP_CpuName

```
char CFE_PSP_CpuName[CFE_PSP_CPU_NAME_LENGTH]
```

Definition at line 128 of file cfe_psp_start.c.

Referenced by CFE_PSP_GetProcessorName(), and OS_Application_Startup().

39.96.3.3 CFE_PSP_IdleTaskState

```
CFE_PSP_IdleTaskState_t CFE_PSP_IdleTaskState
```

Definition at line 130 of file cfe_psp_start.c.

Referenced by CFE_PSP_ExceptionSigHandler(), CFE_PSP_Restart(), OS_Application_Run(), and OS_Application↩_Startup().

39.96.3.4 CFE_PSP_SpacecraftId

```
uint32 CFE_PSP_SpacecraftId
```

Definition at line 126 of file cfe_psp_start.c.

Referenced by CFE_PSP_GetSpacecraftId(), and OS_Application_Startup().

39.96.3.5 CommandData

```
CFE_PSP_CommandData_t CommandData
```

Definition at line 125 of file cfe_psp_start.c.

39.96.3.6 longOpts

```
const struct option longOpts[]  [static]
```

Initial value:

```
= {{"reset", required_argument, NULL, 'R'},
                                        {"subtype", required_argument, NULL, 'S'},
                                        {"cpuid", required_argument, NULL, 'C'},
                                        {"scid", required_argument, NULL, 'I'},
                                        {"cpuname", required_argument, NULL, 'N'},
                                        {"help", no_argument, NULL, 'h'},
                                        {NULL, no_argument, NULL, 0}}
```

Definition at line 140 of file cfe_psp_start.c.

Referenced by OS_Application_Startup().

39.96.3.7 optString

```
const char* optString = "R:S:C:I:N:h"  [static]
```

Definition at line 135 of file cfe_psp_start.c.

Referenced by OS_Application_Startup().

39.96.3.8 TimerCounter

```
uint32 TimerCounter
```

Definition at line 124 of file cfe_psp_start.c.

Referenced by CFE_PSP_SetupLocal1Hz(), and CFE_PSP_TimerHandler().

39.97 psp/fsw/pc-linux/src/cfe_psp_support.c File Reference

```
#include <stdio.h>
#include <stdlib.h>
#include <sys/types.h>
#include <unistd.h>
#include "common_types.h"
#include "osapi.h"
#include "cfe_psp.h"
#include "cfe_psp_config.h"
#include "cfe_psp_memory.h"
```

Functions

- void CFE_PSP_Restart (uint32 reset_type)
- void CFE_PSP_Panic (int32 ErrorCode)
- void CFE_PSP_FlushCaches (uint32 type, void *address, uint32 size)
- uint32 CFE_PSP_GetProcessorId (void)
- uint32 CFE_PSP_GetSpacecraftId (void)
- const char * CFE_PSP_GetProcessorName (void)

Variables

- uint32 CFE_PSP_SpacecraftId
- uint32 CFE_PSP_CpuId
- char CFE_PSP_CpuName [ ]

39.97.1 Function Documentation

39.97.1.1 CFE_PSP_FlushCaches()

```
void CFE_PSP_FlushCaches (
            uint32 type,
            void * address,
            uint32 size )
```

Definition at line 153 of file cfe_psp_support.c.

39.97.1.2 CFE_PSP_GetProcessorId()

```
uint32 CFE_PSP_GetProcessorId (
            void  )
```

Definition at line 175 of file cfe_psp_support.c.

References CFE_PSP_CpuId.

39.97.1.3 CFE_PSP_GetProcessorName()

```
const char* CFE_PSP_GetProcessorName (
            void  )
```

Definition at line 215 of file cfe_psp_support.c.

References CFE_PSP_CpuName.

39.97.1.4 CFE_PSP_GetSpacecraftId()

```
uint32 CFE_PSP_GetSpacecraftId (
            void  )
```

Definition at line 195 of file cfe_psp_support.c.

References CFE_PSP_SpacecraftId.

39.97.1.5 CFE_PSP_Panic()

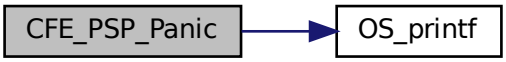
```
void CFE_PSP_Panic (
            int32 ErrorCode )
```

Definition at line 133 of file cfe_psp_support.c.

References OS_printf().

Referenced by OS_Application_Startup().

Here is the call graph for this function:



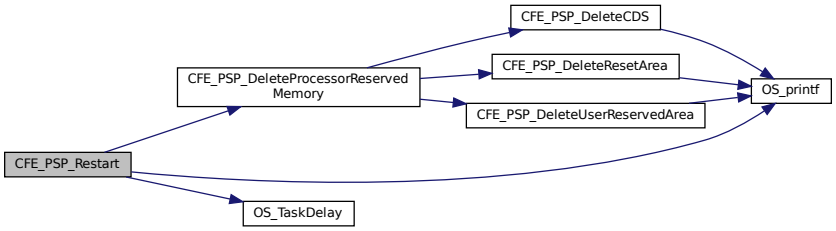
39.97.1.6 CFE_PSP_Restart()

```
void CFE_PSP_Restart (
            uint32 reset_type )
```

Definition at line 69 of file cfe_psp_support.c.

References CFE_PSP_DeleteProcessorReservedMemory(), CFE_PSP_IdleTaskState, CFE_PSP_Reserved↩MemoryMap, CFE_PSP_RST_TYPE_POWERON, OS_printf(), and OS_TaskDelay().

Here is the call graph for this function:



39.97.2 Variable Documentation

39.97.2.1 CFE_PSP_CpuId

```
uint32 CFE_PSP_CpuId
```

Definition at line 127 of file cfe_psp_start.c.

Referenced by CFE_PSP_GetProcessorId(), and OS_Application_Startup().

39.97.2.2 CFE_PSP_CpuName

```
char CFE_PSP_CpuName[]
```

Definition at line 128 of file cfe_psp_start.c.

Referenced by CFE_PSP_GetProcessorName(), and OS_Application_Startup().

39.97.2.3 CFE_PSP_SpacecraftId

```
uint32 CFE_PSP_SpacecraftId
```

Definition at line 126 of file cfe_psp_start.c.

Referenced by CFE_PSP_GetSpacecraftId(), and OS_Application_Startup().

39.98 psp/fsw/pc-linux/src/cfe_psp_timer.c File Reference

```
#include "common_types.h"
#include "osapi.h"
#include <stdio.h>
#include <stdlib.h>
#include "cfe_psp.h"
```

Macros

- #define CFE_PSP_TIMER_TICKS_PER_SECOND 1000000

  Resolution of the least significant 32 bits of the 64 bit time stamp returned by OS_BSPGet_Timebase in timer ticks per second. The timer resolution for accuracy should not be any slower than 1000000 ticks per second or 1 us per tick.

- #define CFE_PSP_TIMER_LOW32_ROLLOVER 1000000

  The number that the least significant 32 bits of the 64 bit time stamp returned by OS_BSPGet_Timebase rolls over. If the lower 32 bits rolls at 1 second, then the OS_BSP_TIMER_LOW32_ROLLOVER will be 1000000. if the lower 32 bits rolls at its maximum value (2^32) then OS_BSP_TIMER_LOW32_ROLLOVER will be 0.

Functions

- void CFE_PSP_GetTime (OS_time_t *LocalTime)
- uint32 CFE_PSP_Get_Timer_Tick (void)
- uint32 CFE_PSP_GetTimerTicksPerSecond (void)
- uint32 CFE_PSP_GetTimerLow32Rollover (void)
- void CFE_PSP_Get_Timebase (uint32 *Tbu, uint32 *Tbl)
- uint32 CFE_PSP_Get_Dec (void)

39.98.1 Macro Definition Documentation

39.98.1.1 CFE_PSP_TIMER_LOW32_ROLLOVER

```
#define CFE_PSP_TIMER_LOW32_ROLLOVER 1000000
```

The number that the least significant 32 bits of the 64 bit time stamp returned by OS_BSPGet_Timebase rolls over. If the lower 32 bits rolls at 1 second, then the OS_BSP_TIMER_LOW32_ROLLOVER will be 1000000. if the lower 32 bits rolls at its maximum value (2^32) then OS_BSP_TIMER_LOW32_ROLLOVER will be 0.

Definition at line 72 of file cfe_psp_timer.c.

Referenced by CFE_PSP_GetTimerLow32Rollover().

39.98.1.2 CFE_PSP_TIMER_TICKS_PER_SECOND

```
#define CFE_PSP_TIMER_TICKS_PER_SECOND 1000000
```

Resolution of the least significant 32 bits of the 64 bit time stamp returned by OS_BSPGet_Timebase in timer ticks per second. The timer resolution for accuracy should not be any slower than 1000000 ticks per second or 1 us per tick.

Definition at line 63 of file cfe_psp_timer.c.

Referenced by CFE_PSP_GetTimerTicksPerSecond().

39.98.2 Function Documentation

39.98.2.1 CFE_PSP_Get_Dec()

```
uint32 CFE_PSP_Get_Dec (
            void  )
```

Definition at line 188 of file cfe_psp_timer.c.

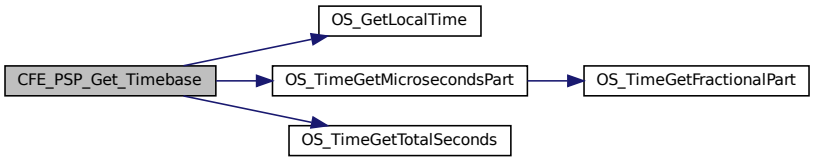
39.98.2.2 CFE_PSP_Get_Timebase()

```
void CFE_PSP_Get_Timebase (
            uint32 * Tbu,
            uint32 * Tbl )
```

Definition at line 165 of file cfe_psp_timer.c.

References OS_GetLocalTime(), OS_TimeGetMicrosecondsPart(), and OS_TimeGetTotalSeconds().

Here is the call graph for this function:



39.98.2.3 CFE_PSP_Get_Timer_Tick()

```
uint32 CFE_PSP_Get_Timer_Tick (
            void )
```

Definition at line 105 of file cfe_psp_timer.c.

39.98.2.4 CFE_PSP_GetTime()

```
void CFE_PSP_GetTime (
            OS_time_t * LocalTime )
```

Definition at line 82 of file cfe_psp_timer.c.

References OS_GetLocalTime().

Here is the call graph for this function:

39.98.2.5 CFE_PSP_GetTimerLow32Rollover()

```
uint32 CFE_PSP_GetTimerLow32Rollover (
            void  )
```

Definition at line 147 of file cfe_psp_timer.c.

References CFE_PSP_TIMER_LOW32_ROLLOVER.

39.98.2.6 CFE_PSP_GetTimerTicksPerSecond()

```
uint32 CFE_PSP_GetTimerTicksPerSecond (
            void  )
```

Definition at line 126 of file cfe_psp_timer.c.

References CFE_PSP_TIMER_TICKS_PER_SECOND.

39.99 psp/fsw/pc-linux/src/cfe_psp_watchdog.c File Reference

```
#include "common_types.h"
#include "osapi.h"
#include <stdio.h>
#include <stdlib.h>
#include "cfe_psp.h"
#include "cfe_psp_config.h"
```

Functions

- void CFE_PSP_WatchdogInit (void)
- void CFE_PSP_WatchdogEnable (void)
- void CFE_PSP_WatchdogDisable (void)
- void CFE_PSP_WatchdogService (void)
- uint32 CFE_PSP_WatchdogGet (void)
- void CFE_PSP_WatchdogSet (uint32 WatchdogValue)

Variables

- uint32 CFE_PSP_WatchdogValue = CFE_PSP_WATCHDOG_MAX

39.99.1 Function Documentation

39.99.1.1 CFE_PSP_WatchdogDisable()

```
void CFE_PSP_WatchdogDisable (
            void  )
```

Definition at line 107 of file cfe_psp_watchdog.c.

39.99.1.2 CFE_PSP_WatchdogEnable()

```
void CFE_PSP_WatchdogEnable (
            void  )
```

Definition at line 95 of file cfe_psp_watchdog.c.

39.99.1.3 CFE_PSP_WatchdogGet()

```
uint32 CFE_PSP_WatchdogGet (
            void  )
```

Definition at line 142 of file cfe_psp_watchdog.c.

References CFE_PSP_WatchdogValue.

39.99.1.4 CFE_PSP_WatchdogInit()

```
void CFE_PSP_WatchdogInit (
            void  )
```

Definition at line 74 of file cfe_psp_watchdog.c.

References CFE_PSP_WatchdogValue.

39.99.1.5 CFE_PSP_WatchdogService()

```
void CFE_PSP_WatchdogService (
            void  )
```

Definition at line 125 of file cfe_psp_watchdog.c.

39.99.1.6 CFE_PSP_WatchdogSet()

```
void CFE_PSP_WatchdogSet (
            uint32 WatchdogValue )
```

Definition at line 162 of file cfe_psp_watchdog.c.

References CFE_PSP_WatchdogValue.

39.99.2 Variable Documentation

39.99.2.1 CFE_PSP_WatchdogValue

```
uint32 CFE_PSP_WatchdogValue = CFE_PSP_WATCHDOG_MAX
```

Definition at line 63 of file cfe_psp_watchdog.c.

Referenced by CFE_PSP_WatchdogGet(), CFE_PSP_WatchdogInit(), and CFE_PSP_WatchdogSet().

Index

EXTENSION
common_types.h, 1253
_fini
cfe_psp_memory.c, 1338
_init
cfe_psp_memory.c, 1338

ARGCHECK
osapi-macros.h, 1278
accuracy
OS_timebase_prop_t, 745
OS_timer_prop_t, 747
ActiveBuffer
CFE_TBL_HousekeepingTlm_Payload, 666
ActiveBufferAddr
CFE_TBL_TblRegPacket_Payload, 683
ActiveTableFlag
CFE_TBL_DumpCmd_Payload, 658
CFE_TBL_ValidateCmd_Payload, 689
ActualLength
OS_SockAddr_t, 738
addr
OS_module_prop_t, 734
AddrData
OS_SockAddr_t, 738
Address
OS_static_symbol_record_t, 741
AddressesAreValid
CFE_ES_AppInfo, 515
AlignPtr
OS_SockAddrData_t, 739
AlignU32
OS_SockAddrData_t, 739
AppData
CFE_EVS_HousekeepingTlm_Payload, 589
AppDataFilename
CFE_EVS_AppDataCmd_Payload, 574
AppEnableStatus
CFE_EVS_AppTlmData, 583
AppEntryPoint
CFE_ES_StartAppCmd_Payload, 566
AppFileName
CFE_ES_AppReloadCmd_Payload, 523
CFE_ES_StartAppCmd_Payload, 566
AppID
CFE_EVS_AppTlmData, 583
AppId
CFE_ES_TaskInfo, 572
CFE_SB_PipeInfoEntry, 631
AppInfo
CFE_ES_OneAppTlm_Payload, 550
AppMessageSentCounter
CFE_EVS_AppTlmData, 584
AppName
CFE_ES_TaskInfo, 572
CFE_EVS_AppNameBitMaskCmd_Payload, 576
CFE_EVS_AppNameCmd_Payload, 578
CFE_EVS_AppNameEventIDCmd_Payload, 580
CFE_EVS_AppNameEventIDMaskCmd_Payload, 582
CFE_EVS_PacketID, 598
CFE_SB_PipeInfoEntry, 631
CFE_SB_RoutingFileEntry, 637
Application
CFE_ES_AppNameCmd_Payload, 522
CFE_ES_AppReloadCmd_Payload, 523
CFE_ES_SendMemPoolStatsCmd_Payload, 558
CFE_ES_StartAppCmd_Payload, 566
ApplicationID
CFE_FS_Header, 609
AtToneDelay
CFE_TIME_DiagnosticTlm_Payload, 693
AtToneLatch
CFE_TIME_DiagnosticTlm_Payload, 693
AtToneLeapSeconds
CFE_TIME_DiagnosticTlm_Payload, 693
CFE_TIME_ToneDataCmd_Payload, 723
AtToneMET
CFE_TIME_DiagnosticTlm_Payload, 693
CFE_TIME_ToneDataCmd_Payload, 723
AtToneSTCF
CFE_TIME_DiagnosticTlm_Payload, 693
CFE_TIME_ToneDataCmd_Payload, 723
AtToneState
CFE_TIME_ToneDataCmd_Payload, 723
Attributes
CFE_PSP_MemTable_t, 615

BSSAddress
CFE_ES_AppInfo, 515
BSSSize
CFE_ES_AppInfo, 516
BUFF_SIZE
cfe_psp.h, 1301
BUGCHECK
osapi-macros.h, 1278
BUGREPORT
osapi-macros.h, 1279
BitMask
CFE_EVS_AppNameBitMaskCmd_Payload, 576
CFE_EVS_BitMaskCmd_Payload, 587
block_size

OS_statvfs_t, 742
BlockSize
CFE_ES_BlockStats, 524
BlockStats
CFE_ES_MemPoolStats, 545
blocks_free
OS_statvfs_t, 742
BootRecord
CFE_PSP_LinuxReservedAreaFixedLayout_t, 614
BootSource
CFE_ES_HousekeepingTlm_Payload, 534
bss_address
OS_module_address_t, 732
bss_size
OS_module_address_t, 733
BufDscPtr
CFE_SB_ZeroCopyHandle_t, 651
Buffer
OS_SockAddrData_t, 739
build/doc/osconfig-example.h, 750
build/native/default_cpu1/inc/cfe_msgids.h, 759
build/native/default_cpu1/inc/cfe_platform_cfg.h, 759
build/native/default_cpu1/inc/osconfig.h, 759
BuildNumber
CFE_PSP_VersionInfo_t, 616
ByteAlign4
CFE_TBL_TblRegPacket_Payload, 683
ByteAlignPad1
CFE_TBL_HousekeepingTlm_Payload, 666
ByteAlignSpare
CFE_ES_CDSRegDumpRec, 525

CCSDS_ExtendedHeader, 512
Subsystem, 512
SystemId, 512
CCSDS_ExtendedHeader_t
ccsds_hdr.h, 1098
CCSDS_PrimaryHeader, 513
Length, 513
Sequence, 513
StreamId, 513
CCSDS_PrimaryHeader_t
ccsds_hdr.h, 1098
CDSShmId
cfe_psp_memory.c, 1339
cFE Access Table Content APIs, 339
CFE_TBL_GetAddress, 339
CFE_TBL_GetAddresses, 340
CFE_TBL_ReleaseAddress, 341
CFE_TBL_ReleaseAddresses, 343
cFE Application Behavior APIs, 183
CFE_ES_ExitApp, 183
CFE_ES_IncrementTaskCounter, 184
CFE_ES_RegisterApp, 184
CFE_ES_RunLoop, 185
CFE_ES_WaitForStartupSync, 186
CFE_ES_WaitForSystemState, 186
cFE Application Control APIs, 180
CFE_ES_DeleteApp, 180
CFE_ES_ReloadApp, 180
CFE_ES_RestartApp, 181
cFE Checksum Control APIs, 316
CFE_SB_GenerateChecksum, 316
CFE_SB_GetChecksum, 316
CFE_SB_ValidateChecksum, 317
cFE Child Task APIs, 198
CFE_ES_CreateChildTask, 198
CFE_ES_DeleteChildTask, 199
CFE_ES_ExitChildTask, 200
CFE_ES_GetTaskIDByName, 200
CFE_ES_GetTaskName, 201
CFE_ES_RegisterChildTask, 202
cFE Clock State Flag Defines, 376
CFE_TIME_FLAG_ADD1HZ, 376
CFE_TIME_FLAG_ADDADJ, 376
CFE_TIME_FLAG_ADDTCL, 377
CFE_TIME_FLAG_CLKSET, 377
CFE_TIME_FLAG_CMDFLY, 377
CFE_TIME_FLAG_FLYING, 377
CFE_TIME_FLAG_GDTONE, 377
CFE_TIME_FLAG_SERVER, 378
CFE_TIME_FLAG_SIGPRI, 378
CFE_TIME_FLAG_SRCINT, 378
CFE_TIME_FLAG_SRVFLY, 378
CFE_TIME_FLAG_UNUSED, 378
cFE Critical Data Store APIs, 208
CFE_ES_CopyToCDS, 208
CFE_ES_GetCDSBlockIDByName, 209
CFE_ES_GetCDSBlockName, 210
CFE_ES_RegisterCDS, 210
CFE_ES_RestoreFromCDS, 211
cFE Entry/Exit APIs, 178
CFE_ES_Main, 178
CFE_ES_ResetCFE, 179
cFE External Time Source APIs, 367
CFE_TIME_ExternalGPS, 367
CFE_TIME_ExternalMET, 368
CFE_TIME_ExternalTime, 368
CFE_TIME_ExternalTone, 369
CFE_TIME_RegisterSynchCallback, 370
CFE_TIME_UnregisterSynchCallback, 370
cFE File Header Management APIs, 246
CFE_FS_InitHeader, 246
CFE_FS_ReadHeader, 246
CFE_FS_SetTimestamp, 247
CFE_FS_WriteHeader, 248
cFE File Utility APIs, 250
CFE_FS_BackgroundFileDumpIsPending, 250

CFE_FS_BackgroundFileDumpRequest, 251
CFE_FS_ExtractFilenameFromPath, 251
CFE_FS_GetDefaultExtension, 252
CFE_FS_GetDefaultMountPoint, 252
CFE_FS_ParseInputFileName, 252
CFE_FS_ParseInputFileNameEx, 253
CFE_FS_RunBackgroundFileDump, 254
cFE Generic Counter APIs, 224
CFE_ES_DeleteGenCounter, 224
CFE_ES_GetGenCount, 225
CFE_ES_GetGenCounterIDByName, 226
CFE_ES_GetGenCounterName, 226
CFE_ES_IncrementGenCounter, 227
CFE_ES_RegisterGenCounter, 228
CFE_ES_SetGenCount, 229
cFE Get Current Time APIs, 354
CFE_TIME_GetMETseconds, 354
CFE_TIME_GetMETsubsecs, 355
CFE_TIME_GetMET, 354
CFE_TIME_GetTAI, 355
CFE_TIME_GetTime, 356
CFE_TIME_GetUTC, 357
cFE Get Table Information APIs, 345
CFE_TBL_GetInfo, 345
CFE_TBL_GetStatus, 346
CFE_TBL_NotifyByMessage, 347
cFE Get Time Information APIs, 358
CFE_TIME_GetClockInfo, 358
CFE_TIME_GetClockState, 358
CFE_TIME_GetLeapSeconds, 359
CFE_TIME_GetSTCF, 359
cFE Getting Message Characteristics APIs, 310
CFE_SB_GetCmdCode, 310
CFE_SB_GetMsgId, 311
CFE_SB_GetMsgTime, 311
CFE_SB_GetTotalMsgLength, 312
CFE_SB_GetUserData, 313
CFE_SB_GetUserDataLength, 313
CFE_SB_MessageStringGet, 314
cFE Information APIs, 188
CFE_ES_GetAppIDByName, 189
CFE_ES_GetAppID, 188
CFE_ES_GetAppInfo, 190
CFE_ES_GetAppName, 191
CFE_ES_GetLibIDByName, 191
CFE_ES_GetLibInfo, 192
CFE_ES_GetLibName, 193
CFE_ES_GetModuleInfo, 194
CFE_ES_GetResetType, 195
CFE_ES_GetTaskID, 196
CFE_ES_GetTaskInfo, 196
cFE Internal Event Services APIs, internal to CFE core, 244
CFE_EVS_CleanUpApp, 244
CFE_EVS_EarlyInit, 244
CFE_EVS_TaskMain, 244
cFE Internal Executive Service APIs, internal to CFE core, 231
CFE_ES_CDS_EarlyInit, 231
CFE_ES_DeleteCDS, 231
CFE_ES_RegisterCDSEx, 232
CFE_ES_TaskMain, 233
cFE Internal File Service APIs, internal to CFE core, 256
CFE_FS_EarlyInit, 256
cFE Internal Software Bus APIs, internal to CFE core, 324
CFE_SB_CleanUpApp, 324
CFE_SB_EarlyInit, 324
CFE_SB_TaskMain, 324
cFE Internal Table Service APIs, internal to CFE core, 352
CFE_TBL_CleanUpApp, 352
CFE_TBL_EarlyInit, 352
CFE_TBL_TaskMain, 352
cFE Internal Time APIs, internal to CFE core, 374
CFE_TIME_CleanUpApp, 374
CFE_TIME_EarlyInit, 374
CFE_TIME_TaskMain, 374
cFE Manage Table Content APIs, 332
CFE_TBL_DumpToBuffer, 332
CFE_TBL_Load, 333
CFE_TBL_Manage, 334
CFE_TBL_Modified, 335
CFE_TBL_Update, 336
CFE_TBL_Validate, 337
cFE Memory Manager APIs, 213
CFE_ES_GetMemPoolStats, 213
CFE_ES_GetPoolBuf, 214
CFE_ES_GetPoolBufInfo, 215
CFE_ES_PoolCreate, 216
CFE_ES_PoolCreateEx, 217
CFE_ES_PoolCreateNoSem, 218
CFE_ES_PoolDelete, 219
CFE_ES_PutPoolBuf, 219
cFE Message header APIs, 257
CFE_MSG_GenerateChecksum, 258
CFE_MSG_GetApId, 259
CFE_MSG_GetEDSVersion, 260
CFE_MSG_GetEndian, 260
CFE_MSG_GetFcnCode, 261
CFE_MSG_GetHasSecondaryHeader, 262
CFE_MSG_GetHeaderVersion, 262
CFE_MSG_GetMsgTime, 263
CFE_MSG_GetPlaybackFlag, 264
CFE_MSG_GetSegmentationFlag, 264
CFE_MSG_GetSequenceCount, 265
CFE_MSG_GetSize, 265
CFE_MSG_GetSubsystem, 266
CFE_MSG_GetSystem, 267
CFE_MSG_GetType, 267

CFE_MSG_Init, 268
CFE_MSG_SetApId, 269
CFE_MSG_SetEDSVersion, 269
CFE_MSG_SetEndian, 270
CFE_MSG_SetFcnCode, 270
CFE_MSG_SetHasSecondaryHeader, 271
CFE_MSG_SetHeaderVersion, 272
CFE_MSG_SetMsgTime, 272
CFE_MSG_SetPlaybackFlag, 273
CFE_MSG_SetSegmentationFlag, 274
CFE_MSG_SetSequenceCount, 274
CFE_MSG_SetSize, 275
CFE_MSG_SetSubsystem, 276
CFE_MSG_SetSystem, 276
CFE_MSG_SetType, 277
CFE_MSG_ValidateChecksum, 278
cFE Message ID APIs, 319
CFE_SB_GetPktType, 319
CFE_SB_IsValidMsgId, 319
CFE_SB_MsgId_Equal, 320
CFE_SB_MsgIdToValue, 321
CFE_SB_ValueToMsgId, 321
cFE Message Id APIs, 279
CFE_MSG_GetMsgId, 279
CFE_MSG_GetTypeFromMsgId, 279
CFE_MSG_SetMsgId, 280
cFE Message Subscription Control APIs, 289
CFE_SB_Subscribe, 289
CFE_SB_SubscribeEx, 290
CFE_SB_SubscribeLocal, 291
CFE_SB_Unsubscribe, 292
CFE_SB_UnsubscribeLocal, 293
cFE Miscellaneous APIs, 204
CFE_ES_BackgroundWakeup, 204
CFE_ES_CalculateCRC, 204
CFE_ES_ProcessAsyncEvent, 206
CFE_ES_WriteToSysLog, 206
cFE Miscellaneous Time APIs, 372
CFE_TIME_Local1HzISR, 372
CFE_TIME_Print, 372
cFE Performance Monitor APIs, 221
CFE_ES_PerfLogAdd, 222
CFE_ES_PerfLogEntry, 221
CFE_ES_PerfLogExit, 221
cFE Pipe Management APIs, 282
CFE_SB_CreatePipe, 282
CFE_SB_DeletePipe, 283
CFE_SB_GetPipeIdByName, 284
CFE_SB_GetPipeName, 285
CFE_SB_GetPipeOpts, 285
CFE_SB_PipeId_ToIndex, 287
CFE_SB_SetPipeOpts, 288
cFE Registration APIs, 234, 326
CFE_EVS_Register, 234
CFE_EVS_Unregister, 235
CFE_TBL_Register, 326
CFE_TBL_Share, 329
CFE_TBL_Unregister, 330
cFE Reset Event Filter APIs, 242
CFE_EVS_ResetAllFilters, 242
CFE_EVS_ResetFilter, 242
cFE Resource ID APIs, 174
CFE_ES_AppID_ToIndex, 174
CFE_ES_CounterID_ToIndex, 175
CFE_ES_LibID_ToIndex, 175
CFE_ES_TaskID_ToIndex, 176
cFE Return Code Defines, 134
CFE_ES_APP_CLEANUP_ERR, 140
CFE_ES_BAD_ARGUMENT, 140
CFE_ES_BIN_SEM_DELETE_ERR, 140
CFE_ES_BUFFER_NOT_IN_POOL, 141
CFE_ES_CDS_ACCESS_ERROR, 141
CFE_ES_CDS_ALREADY_EXISTS, 141
CFE_ES_CDS_BLOCK_CRC_ERR, 141
CFE_ES_CDS_INSUFFICIENT_MEMORY, 142
CFE_ES_CDS_INVALID_NAME, 142
CFE_ES_CDS_INVALID_SIZE, 142
CFE_ES_CDS_INVALID, 142
CFE_ES_CDS_OWNER_ACTIVE_ERR, 143
CFE_ES_CDS_WRONG_TYPE_ERR, 143
CFE_ES_COUNT_SEM_DELETE_ERR, 143
CFE_ES_ERR_APP_CREATE, 143
CFE_ES_ERR_APP_REGISTER, 144
CFE_ES_ERR_BUFFER, 144
CFE_ES_ERR_CHILD_TASK_CREATE, 144
CFE_ES_ERR_CHILD_TASK_DELETE_MAIN_T↩ ASK, 145
CFE_ES_ERR_CHILD_TASK_DELETE, 144
CFE_ES_ERR_CHILD_TASK_REGISTER, 145
CFE_ES_ERR_DUPLICATE_NAME, 145
CFE_ES_ERR_LOAD_LIB, 145
CFE_ES_ERR_MEM_BLOCK_SIZE, 146
CFE_ES_ERR_NAME_NOT_FOUND, 146
CFE_ES_ERR_RESOURCEID_NOT_VALID, 146
CFE_ES_ERR_SHELL_CMD, 146
CFE_ES_ERR_SYS_LOG_FULL, 147
CFE_ES_ERR_SYS_LOG_TRUNCATED, 147
CFE_ES_FILE_CLOSE_ERR, 147
CFE_ES_FILE_IO_ERR, 147
CFE_ES_LIB_ALREADY_LOADED, 148
CFE_ES_MUT_SEM_DELETE_ERR, 148
CFE_ES_NO_RESOURCE_IDS_AVAILABLE, 148
CFE_ES_NOT_IMPLEMENTED, 148
CFE_ES_OPERATION_TIMED_OUT, 149
CFE_ES_POOL_BLOCK_INVALID, 149
CFE_ES_POOL_BOUNDS_ERROR, 149
CFE_ES_QUEUE_DELETE_ERR, 149
CFE_ES_RST_ACCESS_ERR, 150

CFE_ES_TASK_DELETE_ERR, 150
CFE_ES_TIMER_DELETE_ERR, 150
CFE_EVS_APP_FILTER_OVERLOAD, 150
CFE_EVS_APP_ILLEGAL_APP_ID, 151
CFE_EVS_APP_NOT_REGISTERED, 151
CFE_EVS_EVT_NOT_REGISTERED, 151
CFE_EVS_FILE_WRITE_ERROR, 151
CFE_EVS_INVALID_PARAMETER, 152
CFE_EVS_NOT_IMPLEMENTED, 152
CFE_EVS_RESET_AREA_POINTER, 152
CFE_EVS_UNKNOWN_FILTER, 152
CFE_FS_BAD_ARGUMENT, 153
CFE_FS_FNAME_TOO_LONG, 153
CFE_FS_INVALID_PATH, 153
CFE_FS_NOT_IMPLEMENTED, 153
CFE_SB_BAD_ARGUMENT, 154
CFE_SB_BUF_ALOC_ERR, 154
CFE_SB_BUFFER_INVALID, 154
CFE_SB_INTERNAL_ERR, 154
CFE_SB_MAX_DESTS_MET, 155
CFE_SB_MAX_MSGS_MET, 155
CFE_SB_MAX_PIPES_MET, 155
CFE_SB_MSG_TOO_BIG, 155
CFE_SB_NO_MESSAGE, 156
CFE_SB_NO_MSG_RECV, 156
CFE_SB_NO_SUBSCRIBERS, 156
CFE_SB_NOT_IMPLEMENTED, 156
CFE_SB_PIPE_CR_ERR, 157
CFE_SB_PIPE_RD_ERR, 157
CFE_SB_TIME_OUT, 157
CFE_SB_WRONG_MSG_TYPE, 157
CFE_STATUS_BAD_COMMAND_CODE, 158
CFE_STATUS_EXTERNAL_RESOURCE_FAIL, 158
CFE_STATUS_NO_COUNTER_INCREMENT, 158
CFE_STATUS_NOT_IMPLEMENTED, 158
CFE_STATUS_REQUEST_ALREADY_PENDING, 159
CFE_STATUS_UNKNOWN_MSG_ID, 159
CFE_STATUS_WRONG_MSG_LENGTH, 159
CFE_SUCCESS, 159
CFE_TBL_BAD_ARGUMENT, 160
CFE_TBL_ERR_ACCESS, 160
CFE_TBL_ERR_BAD_APP_ID, 160
CFE_TBL_ERR_BAD_CONTENT_ID, 160
CFE_TBL_ERR_BAD_PROCESSOR_ID, 161
CFE_TBL_ERR_BAD_SPACECRAFT_ID, 161
CFE_TBL_ERR_BAD_SUBTYPE_ID, 161
CFE_TBL_ERR_DUMP_ONLY, 161
CFE_TBL_ERR_DUPLICATE_DIFF_SIZE, 162
CFE_TBL_ERR_DUPLICATE_NOT_OWNED, 162
CFE_TBL_ERR_FILE_FOR_WRONG_TABLE, 162
CFE_TBL_ERR_FILE_NOT_FOUND, 162
CFE_TBL_ERR_FILE_SIZE_INCONSISTENT, 163
CFE_TBL_ERR_FILE_TOO_LARGE, 163
CFE_TBL_ERR_FILENAME_TOO_LONG, 163
CFE_TBL_ERR_HANDLES_FULL, 163
CFE_TBL_ERR_ILLEGAL_SRC_TYPE, 164
CFE_TBL_ERR_INVALID_HANDLE, 164
CFE_TBL_ERR_INVALID_NAME, 164
CFE_TBL_ERR_INVALID_OPTIONS, 164
CFE_TBL_ERR_INVALID_SIZE, 165
CFE_TBL_ERR_LOAD_IN_PROGRESS, 165
CFE_TBL_ERR_LOAD_INCOMPLETE, 165
CFE_TBL_ERR_NEVER_LOADED, 166
CFE_TBL_ERR_NO_ACCESS, 166
CFE_TBL_ERR_NO_BUFFER_AVAIL, 166
CFE_TBL_ERR_NO_STD_HEADER, 166
CFE_TBL_ERR_NO_TBL_HEADER, 167
CFE_TBL_ERR_PARTIAL_LOAD, 167
CFE_TBL_ERR_REGISTRY_FULL, 167
CFE_TBL_ERR_SHORT_FILE, 167
CFE_TBL_ERR_UNREGISTERED, 168
CFE_TBL_INFO_DUMP_PENDING, 168
CFE_TBL_INFO_NO_UPDATE_PENDING, 168
CFE_TBL_INFO_NO_VALIDATION_PENDING, 168
CFE_TBL_INFO_RECOVERED_TBL, 169
CFE_TBL_INFO_TABLE_LOCKED, 169
CFE_TBL_INFO_UPDATE_PENDING, 169
CFE_TBL_INFO_UPDATED, 169
CFE_TBL_INFO_VALIDATION_PENDING, 170
CFE_TBL_MESSAGE_ERROR, 170
CFE_TBL_NOT_IMPLEMENTED, 170
CFE_TBL_WARN_DUPLICATE, 170
CFE_TBL_WARN_NOT_CRITICAL, 171
CFE_TBL_WARN_PARTIAL_LOAD, 171
CFE_TBL_WARN_SHORT_FILE, 171
CFE_TIME_BAD_ARGUMENT, 171
CFE_TIME_CALLBACK_NOT_REGISTERED, 172
CFE_TIME_INTERNAL_ONLY, 172
CFE_TIME_NOT_IMPLEMENTED, 172
CFE_TIME_OUT_OF_RANGE, 172
CFE_TIME_TOO_MANY_SYNCH_CALLBACKS, 173
cFE SB Packet Type Defines, 323
CFE_SB_PKTTYPE_CMD, 323
CFE_SB_PKTTYPE_INVALID, 323
CFE_SB_PKTTYPE_TLM, 323
cFE Send Event APIs, 237
CFE_EVS_SendEvent, 237
CFE_EVS_SendEventWithAppID, 238
CFE_EVS_SendTimedEvent, 239
cFE Send/Receive Message APIs, 294
CFE_SB_PassMsg, 294
CFE_SB_RcvMsg, 295
CFE_SB_ReceiveBuffer, 295
CFE_SB_SendMsg, 296
CFE_SB_TransmitMsg, 297
cFE Setting Message Characteristics APIs, 304

CFE_SB_InitMsg, 304
CFE_SB_MessageStringSet, 305
CFE_SB_SetCmdCode, 306
CFE_SB_SetMsgId, 306
CFE_SB_SetMsgTime, 307
CFE_SB_SetTotalMsgLength, 308
CFE_SB_SetUserDataLength, 308
CFE_SB_TimeStampMsg, 309
cFE Table Type Defines, 349
CFE_TBL_OPT_BUFFER_MSK, 349
CFE_TBL_OPT_CRITICAL_MSK, 350
CFE_TBL_OPT_CRITICAL, 349
CFE_TBL_OPT_DBL_BUFFER, 350
CFE_TBL_OPT_DEFAULT, 350
CFE_TBL_OPT_DUMP_ONLY, 350
CFE_TBL_OPT_LD_DMP_MSK, 350
CFE_TBL_OPT_LOAD_DUMP, 351
CFE_TBL_OPT_NOT_CRITICAL, 351
CFE_TBL_OPT_NOT_USR_DEF, 351
CFE_TBL_OPT_SNGL_BUFFER, 351
CFE_TBL_OPT_USR_DEF_ADDR, 351
CFE_TBL_OPT_USR_DEF_MSK, 351
cFE Time Arithmetic APIs, 361
CFE_TIME_Add, 361
CFE_TIME_Compare, 362
CFE_TIME_Subtract, 363
cFE Time Conversion APIs, 364
CFE_TIME_MET2SCTime, 364
CFE_TIME_Micro2SubSecs, 364
CFE_TIME_Sub2MicroSecs, 365
cFE Zero Copy APIs, 299
CFE_SB_TransmitBuffer, 299
CFE_SB_ZeroCopyGetPtr, 300
CFE_SB_ZeroCopyPass, 301
CFE_SB_ZeroCopyReleasePtr, 302
CFE_SB_ZeroCopySend, 302
CFE_BIT
cfe_sb.h, 927
CFE_BUILD_BASELINE
cfe_version.h, 958
CFE_BUILD_NUMBER
cfe_version.h, 959
CFE_CLR
cfe_sb.h, 927
CFE_ES_ALL_APPS_EID
cfe_es_events.h, 965
CFE_ES_APP_CLEANUP_ERR
cFE Return Code Defines, 140
CFE_ES_APP_RESTART
cfe_es_api_typedefs.h, 873
CFE_ES_APP_TLM_MID
cpu1_msgids.h, 770
CFE_ES_APPID_UNDEFINED
cfe_es_api_typedefs.h, 873
CFE_ES_APPID_C
cfe_es_api_typedefs.h, 873
CFE_ES_AppID_ToIndex
cFE Resource ID APIs, 174
CFE_ES_AppId_t
cfe_es_extern_typedefs.h, 881
CFE_ES_AppInfo, 514
AddressesAreValid, 515
BSSAddress, 515
BSSSize, 516
CodeAddress, 516
CodeSize, 516
DataAddress, 516
DataSize, 517
EntryPoint, 517
ExceptionAction, 517
ExecutionCounter, 517
FileName, 518
MainTaskId, 518
MainTaskName, 518
Name, 518
NumOfChildTasks, 519
Priority, 519
ResourceId, 519
StackSize, 519
StartAddress, 520
Type, 520
CFE_ES_AppInfo_t
cfe_es_extern_typedefs.h, 881
CFE_ES_AppNameCmd, 520
CmdHeader, 521
Payload, 521
CFE_ES_AppNameCmd_Payload, 521
Application, 522
CFE_ES_AppNameCmd_Payload_t
cfe_es_msg.h, 1036
CFE_ES_AppNameCmd_t
cfe_es_msg.h, 1036
CFE_ES_AppReloadCmd_Payload, 522
AppFileName, 523
Application, 523
CFE_ES_AppReloadCmd_Payload_t
cfe_es_msg.h, 1036
CFE_ES_AppState
cfe_es_extern_typedefs.h, 887
CFE_ES_AppState_Enum_t
cfe_es_extern_typedefs.h, 882
CFE_ES_AppType
cfe_es_extern_typedefs.h, 888
CFE_ES_AppType_Enum_t
cfe_es_extern_typedefs.h, 882
CFE_ES_BAD_ARGUMENT
cFE Return Code Defines, 140
CFE_ES_BIN_SEM_DELETE_ERR

cFE Return Code Defines, 140
CFE_ES_BOOT_ERR_EID
cfe_es_events.h, 965
CFE_ES_BUFFER_NOT_IN_POOL
cFE Return Code Defines, 141
CFE_ES_BUILD_INF_EID
cfe_es_events.h, 966
CFE_ES_BackgroundWakeup
cFE Miscellaneous APIs, 204
CFE_ES_BlockStats, 523
BlockSize, 524
NumCreated, 524
NumFree, 524
CFE_ES_BlockStats_t
cfe_es_extern_typedefs.h, 882
CFE_ES_CC1_ERR_EID
cfe_es_events.h, 966
CFE_ES_CDS_ACCESS_ERROR
cFE Return Code Defines, 141
CFE_ES_CDS_ALREADY_EXISTS
cFE Return Code Defines, 141
CFE_ES_CDS_BAD_HANDLE
cfe_es_api_typedefs.h, 873
CFE_ES_CDS_BLOCK_CRC_ERR
cFE Return Code Defines, 141
CFE_ES_CDS_DELETE_ERR_EID
cfe_es_events.h, 967
CFE_ES_CDS_DELETE_TBL_ERR_EID
cfe_es_events.h, 967
CFE_ES_CDS_DELETED_INFO_EID
cfe_es_events.h, 968
CFE_ES_CDS_DUMP_ERR_EID
cfe_es_events.h, 968
CFE_ES_CDS_EarlyInit
cFE Internal Executive Service APIs, internal to CFE core, 231
CFE_ES_CDS_INSUFFICIENT_MEMORY
cFE Return Code Defines, 142
CFE_ES_CDS_INVALID_NAME
cFE Return Code Defines, 142
CFE_ES_CDS_INVALID_SIZE
cFE Return Code Defines, 142
CFE_ES_CDS_INVALID
cFE Return Code Defines, 142
CFE_ES_CDS_NAME_ERR_EID
cfe_es_events.h, 969
CFE_ES_CDS_NOT_FOUND_ERR
cfe_error.h, 864
CFE_ES_CDS_OWNER_ACTIVE_EID
cfe_es_events.h, 969
CFE_ES_CDS_OWNER_ACTIVE_ERR
cFE Return Code Defines, 143
CFE_ES_CDS_REG_DUMP_INF_EID
cfe_es_events.h, 970
CFE_ES_CDS_REGISTER_ERR_EID
cfe_es_events.h, 970
CFE_ES_CDS_REGISTRY_FULL
cfe_error.h, 864
CFE_ES_CDS_WRONG_TYPE_ERR
cFE Return Code Defines, 143
CFE_ES_CDSHANDLE_C
cfe_es_api_typedefs.h, 874
CFE_ES_CDSHandle_t
cfe_es_extern_typedefs.h, 883
CFE_ES_CDSRegDumpRec, 524
ByteAlignSpare, 525
Handle, 525
Name, 525
Size, 526
Table, 526
CFE_ES_CDSRegDumpRec_t
cfe_es_extern_typedefs.h, 883
CFE_ES_CLEAR_ER_LOG_CC
cfe_es_msg.h, 1012
CFE_ES_CLEAR_SYSLOG_CC
cfe_es_msg.h, 1013
CFE_ES_CMD_MID
cpu1_msgids.h, 770
CFE_ES_COUNT_SEM_DELETE_ERR
cFE Return Code Defines, 143
CFE_ES_COUNTERID_UNDEFINED
cfe_es_api_typedefs.h, 874
CFE_ES_COUNTERID_C
cfe_es_api_typedefs.h, 874
CFE_ES_CREATING_CDS_DUMP_ERR_EID
cfe_es_events.h, 971
CFE_ES_CalculateCRC
cFE Miscellaneous APIs, 204
CFE_ES_ChildTaskMainFuncPtr_t
cfe_es_api_typedefs.h, 877
CFE_ES_ClearERLogCmd_t
cfe_es_msg.h, 1036
CFE_ES_ClearSysLogCmd_t
cfe_es_msg.h, 1036
CFE_ES_CopyToCDS
cFE Critical Data Store APIs, 208
CFE_ES_CounterID_ToIndex
cFE Resource ID APIs, 175
CFE_ES_CounterId_t
cfe_es_extern_typedefs.h, 883
CFE_ES_CreateChildTask
cFE Child Task APIs, 198
CFE_ES_DBIT
cfe_es.h, 871
CFE_ES_DELETE_CDS_CC
cfe_es_msg.h, 1013
CFE_ES_DTEST
cfe_es.h, 871

CFE_ES_DUMP_CDS_REGISTRY_CC
cfe_es_msg.h, 1014
CFE_ES_DeleteApp
cFE Application Control APIs, 180
CFE_ES_DeleteCDSCmd, 526
CmdHeader, 527
Payload, 527
CFE_ES_DeleteCDSCmd_Payload, 527
CdsName, 528
CFE_ES_DeleteCDSCmd_Payload_t
cfe_es_msg.h, 1036
CFE_ES_DeleteCDSCmd_t
cfe_es_msg.h, 1036
CFE_ES_DeleteCDS
cFE Internal Executive Service APIs, internal to CFE core, 231
CFE_ES_DeleteChildTask
cFE Child Task APIs, 199
CFE_ES_DeleteGenCounter
cFE Generic Counter APIs, 224
CFE_ES_DumpCDSRegistryCmd, 528
CmdHeader, 529
Payload, 529
CFE_ES_DumpCDSRegistryCmd_Payload, 529
DumpFilename, 530
CFE_ES_DumpCDSRegistryCmd_Payload_t
cfe_es_msg.h, 1037
CFE_ES_DumpCDSRegistryCmd_t
cfe_es_msg.h, 1037
CFE_ES_ERLOG1_INF_EID
cfe_es_events.h, 971
CFE_ES_ERLOG2_EID
cfe_es_events.h, 972
CFE_ES_ERLOG2_ERR_EID
cfe_es_events.h, 972
CFE_ES_ERLOG_PENDING_ERR_EID
cfe_es_events.h, 973
CFE_ES_ERR_APP_CREATE
cFE Return Code Defines, 143
CFE_ES_ERR_APP_REGISTER
cFE Return Code Defines, 144
CFE_ES_ERR_APPID
cfe_error.h, 864
CFE_ES_ERR_APPNAME
cfe_error.h, 864
CFE_ES_ERR_BUFFER
cFE Return Code Defines, 144
CFE_ES_ERR_CHILD_TASK_CREATE
cFE Return Code Defines, 144
CFE_ES_ERR_CHILD_TASK_DELETE_MAIN_TASK
cFE Return Code Defines, 145
CFE_ES_ERR_CHILD_TASK_DELETE
cFE Return Code Defines, 144
CFE_ES_ERR_CHILD_TASK_REGISTER
cFE Return Code Defines, 145
CFE_ES_ERR_DUPLICATE_NAME
cFE Return Code Defines, 145
CFE_ES_ERR_LOAD_LIB
cFE Return Code Defines, 145
CFE_ES_ERR_MEM_BLOCK_SIZE
cFE Return Code Defines, 146
CFE_ES_ERR_MEM_HANDLE
cfe_error.h, 864
CFE_ES_ERR_NAME_NOT_FOUND
cFE Return Code Defines, 146
CFE_ES_ERR_RESOURCEID_NOT_VALID
cFE Return Code Defines, 146
CFE_ES_ERR_SHELL_CMD
cFE Return Code Defines, 146
CFE_ES_ERR_SYS_LOG_FULL
cFE Return Code Defines, 147
CFE_ES_ERR_SYS_LOG_TRUNCATED
cFE Return Code Defines, 147
CFE_ES_ERR_SYSLOGMODE_EID
cfe_es_events.h, 973
CFE_ES_ERR_TASKID
cfe_error.h, 865
CFE_ES_ERREXIT_APP_ERR_EID
cfe_es_events.h, 974
CFE_ES_ERREXIT_APP_INF_EID
cfe_es_events.h, 974
CFE_ES_EXIT_APP_ERR_EID
cfe_es_events.h, 975
CFE_ES_EXIT_APP_INF_EID
cfe_es_events.h, 975
CFE_ES_ExceptionAction
cfe_es_extern_typedefs.h, 888
CFE_ES_ExceptionAction_Enum_t
cfe_es_extern_typedefs.h, 883
CFE_ES_ExitApp
cFE Application Behavior APIs, 183
CFE_ES_ExitChildTask
cFE Child Task APIs, 200
CFE_ES_FILE_CLOSE_ERR
cFE Return Code Defines, 147
CFE_ES_FILE_IO_ERR
cFE Return Code Defines, 147
CFE_ES_FILEWRITE_ERR_EID
cfe_es_events.h, 976
CFE_ES_FileNameCmd, 530
CmdHeader, 530
Payload, 530
CFE_ES_FileNameCmd_Payload, 531
FileName, 531
CFE_ES_FileNameCmd_Payload_t
cfe_es_msg.h, 1037
CFE_ES_FileNameCmd_t
cfe_es_msg.h, 1037

CFE_ES_GetAppIDByName
cFE Information APIs, 189
CFE_ES_GetAppID
cFE Information APIs, 188
CFE_ES_GetAppInfo
cFE Information APIs, 190
CFE_ES_GetAppName
cFE Information APIs, 191
CFE_ES_GetCDSBlockIDByName
cFE Critical Data Store APIs, 209
CFE_ES_GetCDSBlockName
cFE Critical Data Store APIs, 210
CFE_ES_GetGenCount
cFE Generic Counter APIs, 225
CFE_ES_GetGenCounterIDByName
cFE Generic Counter APIs, 226
CFE_ES_GetGenCounterName
cFE Generic Counter APIs, 226
CFE_ES_GetLibIDByName
cFE Information APIs, 191
CFE_ES_GetLibInfo
cFE Information APIs, 192
CFE_ES_GetLibName
cFE Information APIs, 193
CFE_ES_GetMemPoolStats
cFE Memory Manager APIs, 213
CFE_ES_GetModuleInfo
cFE Information APIs, 194
CFE_ES_GetPoolBuf
cFE Memory Manager APIs, 214
CFE_ES_GetPoolBufInfo
cFE Memory Manager APIs, 215
CFE_ES_GetResetType
cFE Information APIs, 195
CFE_ES_GetTaskIDByName
cFE Child Task APIs, 200
CFE_ES_GetTaskID
cFE Information APIs, 196
CFE_ES_GetTaskInfo
cFE Information APIs, 196
CFE_ES_GetTaskName
cFE Child Task APIs, 201
CFE_ES_HK_TLM_MID
cpu1_msgids.h, 770
CFE_ES_HousekeepingTlm, 532
Payload, 532
TlmHeader, 532
CFE_ES_HousekeepingTlm_Payload, 532
BootSource, 534
CFECoreChecksum, 535
CFEMajorVersion, 535
CFEMinorVersion, 535
CFEMissionRevision, 535
CFERevision, 536
CommandCounter, 536
CommandErrorCounter, 536
ERLogEntries, 536
ERLogIndex, 537
HeapBlocksFree, 537
HeapBytesFree, 537
HeapMaxBlockSize, 537
MaxProcessorResets, 538
OSALMajorVersion, 538
OSALMinorVersion, 538
OSALMissionRevision, 538
OSALRevision, 539
PerfDataCount, 539
PerfDataEnd, 539
PerfDataStart, 539
PerfDataToWrite, 540
PerfFilterMask, 540
PerfMode, 540
PerfState, 540
PerfTriggerCount, 541
PerfTriggerMask, 541
ProcessorResets, 541
RegisteredCoreApps, 541
RegisteredExternalApps, 542
RegisteredLibs, 542
RegisteredTasks, 542
ResetSubtype, 542
ResetType, 543
SysLogBytesUsed, 543
SysLogEntries, 543
SysLogMode, 543
SysLogSize, 544
CFE_ES_HousekeepingTlm_Payload_t
cfe_es_msg.h, 1037
CFE_ES_HousekeepingTlm_t
cfe_es_msg.h, 1037
CFE_ES_INIT_INF_EID
cfe_es_events.h, 976
CFE_ES_INITSTATS_INF_EID
cfe_es_events.h, 977
CFE_ES_INVALID_POOL_HANDLE_ERR_EID
cfe_es_events.h, 977
CFE_ES_IncrementGenCounter
cFE Generic Counter APIs, 227
CFE_ES_IncrementTaskCounter
cFE Application Behavior APIs, 184
CFE_ES_LEN_ERR_EID
cfe_es_events.h, 978
CFE_ES_LIB_ALREADY_LOADED
cFE Return Code Defines, 148
CFE_ES_LIBID_UNDEFINED
cfe_es_api_typedefs.h, 874
CFE_ES_LIBID_C
cfe_es_api_typedefs.h, 874

CFE_ES_LibID_ToIndex
cFE Resource ID APIs, 175
CFE_ES_LibId_t
cfe_es_extern_typedefs.h, 884
CFE_ES_LibraryEntryFuncPtr_t
cfe_es_api_typedefs.h, 877
CFE_ES_LogEntryType
cfe_es_extern_typedefs.h, 888
CFE_ES_LogEntryType_Enum_t
cfe_es_extern_typedefs.h, 884
CFE_ES_LogMode
cfe_es_extern_typedefs.h, 889
CFE_ES_LogMode_Enum_t
cfe_es_extern_typedefs.h, 884
CFE_ES_MAX_EID
cfe_es_events.h, 978
CFE_ES_MEMADDRESS_C
cfe_es_extern_typedefs.h, 881
CFE_ES_MEMHANDLE_UNDEFINED
cfe_es_api_typedefs.h, 875
CFE_ES_MEMHANDLE_C
cfe_es_api_typedefs.h, 874
CFE_ES_MEMOFFSET_C
cfe_es_extern_typedefs.h, 881
CFE_ES_MEMPOOLBUF_C
cfe_es_api_typedefs.h, 875
CFE_ES_MEMSTATS_TLM_MID
cpu1_msgids.h, 770
CFE_ES_MID_ERR_EID
cfe_es_events.h, 979
CFE_ES_MUT_SEM_DELETE_ERR
cFE Return Code Defines, 148
CFE_ES_Main
cFE Entry/Exit APIs, 178
CFE_ES_MemAddress_t
cfe_es_extern_typedefs.h, 884
CFE_ES_MemHandle_t
cfe_es_extern_typedefs.h, 885
CFE_ES_MemOffset_t
cfe_es_extern_typedefs.h, 885
CFE_ES_MemPoolBuf_t
cfe_es_api_typedefs.h, 877
CFE_ES_MemPoolStats, 544
BlockStats, 545
CheckErrCtr, 545
NumBlocksRequested, 545
NumFreeBytes, 546
PoolSize, 546
CFE_ES_MemPoolStats_t
cfe_es_extern_typedefs.h, 885
CFE_ES_MemStatsTlm, 546
Payload, 547
TlmHeader, 547
CFE_ES_MemStatsTlm_t
cfe_es_msg.h, 1038
CFE_ES_NO_MUTEX
cfe_es_api_typedefs.h, 875
CFE_ES_NO_RESOURCE_IDS_AVAILABLE
cFE Return Code Defines, 148
CFE_ES_NOOP_CC
cfe_es_msg.h, 1015
CFE_ES_NOOP_INF_EID
cfe_es_events.h, 979
CFE_ES_NOT_IMPLEMENTED
cFE Return Code Defines, 148
CFE_ES_NoArgsCmd, 547
CmdHeader, 548
CFE_ES_NoArgsCmd_t
cfe_es_msg.h, 1038
CFE_ES_NoopCmd_t
cfe_es_msg.h, 1038
CFE_ES_ONE_APP_EID
cfe_es_events.h, 979
CFE_ES_ONE_APPID_ERR_EID
cfe_es_events.h, 980
CFE_ES_ONE_ERR_EID
cfe_es_events.h, 980
CFE_ES_OPERATION_TIMED_OUT
cFE Return Code Defines, 149
CFE_ES_OSCREATE_ERR_EID
cfe_es_events.h, 981
CFE_ES_OVER_WRITE_SYSLOG_CC
cfe_es_msg.h, 1016
CFE_ES_OneAppTlm, 548
Payload, 549
TlmHeader, 549
CFE_ES_OneAppTlm_Payload, 549
AppInfo, 550
CFE_ES_OneAppTlm_Payload_t
cfe_es_msg.h, 1038
CFE_ES_OneAppTlm_t
cfe_es_msg.h, 1038
CFE_ES_OverWriteSysLogCmd, 550
CmdHeader, 550
Payload, 550
CFE_ES_OverWriteSysLogCmd_Payload, 551
Mode, 551
CFE_ES_OverWriteSysLogCmd_Payload_t
cfe_es_msg.h, 1038
CFE_ES_OverWriteSysLogCmd_t
cfe_es_msg.h, 1039
CFE_ES_PCR_ERR1_EID
cfe_es_events.h, 981
CFE_ES_PCR_ERR2_EID
cfe_es_events.h, 982
CFE_ES_PERF_DATAWRITTEN_EID
cfe_es_events.h, 982
CFE_ES_PERF_FILTMSKCMD_EID

cfe_es_events.h, 983
CFE_ES_PERF_FILTMSKERR_EID
cfe_es_events.h, 983
CFE_ES_PERF_LOG_ERR_EID
cfe_es_events.h, 984
CFE_ES_PERF_STARTCMD_EID
cfe_es_events.h, 984
CFE_ES_PERF_STARTCMD_ERR_EID
cfe_es_events.h, 985
CFE_ES_PERF_STARTCMD_TRIG_ERR_EID
cfe_es_events.h, 985
CFE_ES_PERF_STOPCMD_EID
cfe_es_events.h, 986
CFE_ES_PERF_STOPCMD_ERR2_EID
cfe_es_events.h, 986
CFE_ES_PERF_TRIGMSKCMD_EID
cfe_es_events.h, 987
CFE_ES_PERF_TRIGMSKERR_EID
cfe_es_events.h, 987
CFE_ES_POOL_BLOCK_INVALID
cFE Return Code Defines, 149
CFE_ES_POOL_BOUNDS_ERROR
cFE Return Code Defines, 149
CFE_ES_PerfLogAdd
cFE Performance Monitor APIs, 222
CFE_ES_PerfLogEntry
cFE Performance Monitor APIs, 221
CFE_ES_PerfLogExit
cFE Performance Monitor APIs, 221
CFE_ES_PoolAlign, 552
LongDouble, 552
LongInt, 552
Ptr, 552
CFE_ES_PoolAlign_t
cfe_es_api_typedefs.h, 877
CFE_ES_PoolCreate
cFE Memory Manager APIs, 216
CFE_ES_PoolCreateEx
cFE Memory Manager APIs, 217
CFE_ES_PoolCreateNoSem
cFE Memory Manager APIs, 218
CFE_ES_PoolDelete
cFE Memory Manager APIs, 219
CFE_ES_PoolStatsTlm_Payload, 553
PoolHandle, 553
PoolStats, 553
CFE_ES_PoolStatsTlm_Payload_t
cfe_es_msg.h, 1039
CFE_ES_ProcessAsyncEvent
cFE Miscellaneous APIs, 206
CFE_ES_PutPoolBuf
cFE Memory Manager APIs, 219
CFE_ES_QUERY_ALL_CC
cfe_es_msg.h, 1017
CFE_ES_QUERY_ALL_TASKS_CC
cfe_es_msg.h, 1018
CFE_ES_QUERY_ONE_CC
cfe_es_msg.h, 1019
CFE_ES_QUEUE_DELETE_ERR
cFE Return Code Defines, 149
CFE_ES_QueryAllCmd_t
cfe_es_msg.h, 1039
CFE_ES_QueryAllTasksCmd_t
cfe_es_msg.h, 1039
CFE_ES_QueryOneCmd_t
cfe_es_msg.h, 1039
CFE_ES_RELOAD_APP_CC
cfe_es_msg.h, 1020
CFE_ES_RELOAD_APP_DBG_EID
cfe_es_events.h, 988
CFE_ES_RELOAD_APP_ERR1_EID
cfe_es_events.h, 988
CFE_ES_RELOAD_APP_ERR2_EID
cfe_es_events.h, 989
CFE_ES_RELOAD_APP_ERR3_EID
cfe_es_events.h, 989
CFE_ES_RELOAD_APP_ERR4_EID
cfe_es_events.h, 990
CFE_ES_RELOAD_APP_INF_EID
cfe_es_events.h, 990
CFE_ES_RESET_COUNTERS_CC
cfe_es_msg.h, 1021
CFE_ES_RESET_INF_EID
cfe_es_events.h, 991
CFE_ES_RESET_PR_COUNT_CC
cfe_es_msg.h, 1022
CFE_ES_RESET_PR_COUNT_EID
cfe_es_events.h, 991
CFE_ES_RESTART_APP_CC
cfe_es_msg.h, 1023
CFE_ES_RESTART_APP_DBG_EID
cfe_es_events.h, 992
CFE_ES_RESTART_APP_ERR1_EID
cfe_es_events.h, 992
CFE_ES_RESTART_APP_ERR2_EID
cfe_es_events.h, 993
CFE_ES_RESTART_APP_ERR3_EID
cfe_es_events.h, 993
CFE_ES_RESTART_APP_ERR4_EID
cfe_es_events.h, 994
CFE_ES_RESTART_APP_INF_EID
cfe_es_events.h, 994
CFE_ES_RESTART_CC
cfe_es_msg.h, 1024
CFE_ES_RST_ACCESS_ERR
cFE Return Code Defines, 150
CFE_ES_RegisterApp
cFE Application Behavior APIs, 184

CFE_ES_RegisterCDSEx
cFE Internal Executive Service APIs, internal to CFE core, 232
CFE_ES_RegisterCDS
cFE Critical Data Store APIs, 210
CFE_ES_RegisterChildTask
cFE Child Task APIs, 202
CFE_ES_RegisterGenCounter
cFE Generic Counter APIs, 228
CFE_ES_ReloadApp
cFE Application Control APIs, 180
CFE_ES_ReloadAppCmd, 554
CmdHeader, 554
Payload, 554
CFE_ES_ReloadAppCmd_t
cfe_es_msg.h, 1039
CFE_ES_ResetCFE
cFE Entry/Exit APIs, 179
CFE_ES_ResetCountersCmd_t
cfe_es_msg.h, 1040
CFE_ES_ResetPRCountCmd_t
cfe_es_msg.h, 1040
CFE_ES_RestartApp
cFE Application Control APIs, 181
CFE_ES_RestartAppCmd_t
cfe_es_msg.h, 1040
CFE_ES_RestartCmd, 555
CmdHeader, 555
Payload, 555
CFE_ES_RestartCmd_Payload, 556
RestartType, 556
CFE_ES_RestartCmd_Payload_t
cfe_es_msg.h, 1040
CFE_ES_RestartCmd_t
cfe_es_msg.h, 1040
CFE_ES_RestoreFromCDS
cFE Critical Data Store APIs, 211
CFE_ES_RunLoop
cFE Application Behavior APIs, 185
CFE_ES_RunStatus
cfe_es_extern_typedefs.h, 889
CFE_ES_RunStatus_Enum_t
cfe_es_extern_typedefs.h, 886
CFE_ES_SEND_HK_MID
cpu1_msgids.h, 770
CFE_ES_SEND_MEM_POOL_STATS_CC
cfe_es_msg.h, 1025
CFE_ES_SET_MAX_PR_COUNT_CC
cfe_es_msg.h, 1026
CFE_ES_SET_MAX_PR_COUNT_EID
cfe_es_events.h, 995
CFE_ES_SET_PERF_FILTER_MASK_CC
cfe_es_msg.h, 1027
CFE_ES_SET_PERF_TRIGGER_MASK_CC
cfe_es_msg.h, 1028
CFE_ES_START_APP_CC
cfe_es_msg.h, 1029
CFE_ES_START_ERR_EID
cfe_es_events.h, 995
CFE_ES_START_EXC_ACTION_ERR_EID
cfe_es_events.h, 996
CFE_ES_START_INF_EID
cfe_es_events.h, 996
CFE_ES_START_INVALID_ENTRY_POINT_ERR_EID
cfe_es_events.h, 997
CFE_ES_START_INVALID_FILENAME_ERR_EID
cfe_es_events.h, 997
CFE_ES_START_NULL_APP_NAME_ERR_EID
cfe_es_events.h, 998
CFE_ES_START_PERF_DATA_CC
cfe_es_msg.h, 1030
CFE_ES_START_PRIORITY_ERR_EID
cfe_es_events.h, 998
CFE_ES_STATIC_POOL_TYPE
cfe_es_api_typedefs.h, 875
CFE_ES_STOP_APP_CC
cfe_es_msg.h, 1031
CFE_ES_STOP_DBG_EID
cfe_es_events.h, 999
CFE_ES_STOP_ERR1_EID
cfe_es_events.h, 999
CFE_ES_STOP_ERR2_EID
cfe_es_events.h, 1000
CFE_ES_STOP_ERR3_EID
cfe_es_events.h, 1000
CFE_ES_STOP_INF_EID
cfe_es_events.h, 1001
CFE_ES_STOP_PERF_DATA_CC
cfe_es_msg.h, 1032
CFE_ES_SYSLOG1_INF_EID
cfe_es_events.h, 1001
CFE_ES_SYSLOG2_EID
cfe_es_events.h, 1002
CFE_ES_SYSLOG2_ERR_EID
cfe_es_events.h, 1002
CFE_ES_SYSLOGMODE_EID
cfe_es_events.h, 1003
CFE_ES_SendMemPoolStatsCmd, 557
CmdHeader, 557
Payload, 557
CFE_ES_SendMemPoolStatsCmd_Payload, 558
Application, 558
PoolHandle, 558
CFE_ES_SendMemPoolStatsCmd_Payload_t
cfe_es_msg.h, 1040
CFE_ES_SendMemPoolStatsCmd_t
cfe_es_msg.h, 1041
CFE_ES_SetGenCount

cFE Generic Counter APIs, 229
CFE_ES_SetMaxPRCountCmd, 559
CmdHeader, 559
Payload, 559
CFE_ES_SetMaxPRCountCmd_Payload, 560
MaxPRCount, 560
CFE_ES_SetMaxPRCountCmd_Payload_t
cfe_es_msg.h, 1041
CFE_ES_SetMaxPRCountCmd_t
cfe_es_msg.h, 1041
CFE_ES_SetPerfFilterMaskCmd, 560
CmdHeader, 561
Payload, 561
CFE_ES_SetPerfFilterMaskCmd_Payload, 561
FilterMask, 562
FilterMaskNum, 562
CFE_ES_SetPerfFilterMaskCmd_Payload_t
cfe_es_msg.h, 1041
CFE_ES_SetPerfFilterMaskCmd_t
cfe_es_msg.h, 1041
CFE_ES_SetPerfTrigMaskCmd_Payload, 563
TriggerMask, 564
TriggerMaskNum, 564
CFE_ES_SetPerfTrigMaskCmd_Payload_t
cfe_es_msg.h, 1042
CFE_ES_SetPerfTriggerMaskCmd, 562
CmdHeader, 563
Payload, 563
CFE_ES_SetPerfTriggerMaskCmd_t
cfe_es_msg.h, 1041
CFE_ES_StackPointer_t
cfe_es_api_typedefs.h, 878
CFE_ES_StartApp, 564
CmdHeader, 565
Payload, 565
CFE_ES_StartAppCmd_Payload, 565
AppEntryPoint, 566
AppFileName, 566
Application, 566
ExceptionAction, 567
Priority, 567
StackSize, 567
CFE_ES_StartAppCmd_Payload_t
cfe_es_msg.h, 1042
CFE_ES_StartAppCmd_t
cfe_es_msg.h, 1042
CFE_ES_StartPerfCmd_Payload, 567
TriggerMode, 568
CFE_ES_StartPerfCmd_Payload_t
cfe_es_msg.h, 1042
CFE_ES_StartPerfDataCmd, 568
CmdHeader, 569
Payload, 569
CFE_ES_StartPerfDataCmd_t
cfe_es_msg.h, 1042
CFE_ES_StopAppCmd_t
cfe_es_msg.h, 1042
CFE_ES_StopPerfCmd_Payload, 569
DataFileName, 570
CFE_ES_StopPerfCmd_Payload_t
cfe_es_msg.h, 1043
CFE_ES_StopPerfDataCmd, 570
CmdHeader, 571
Payload, 571
CFE_ES_StopPerfDataCmd_t
cfe_es_msg.h, 1043
CFE_ES_SystemState
cfe_es_extern_typedefs.h, 890
CFE_ES_SystemState_Enum_t
cfe_es_extern_typedefs.h, 886
CFE_ES_TASK_DELETE_ERR
cFE Return Code Defines, 150
CFE_ES_TASK_STACK_ALLOCATE
cfe_es_api_typedefs.h, 876
CFE_ES_TASKID_UNDEFINED
cfe_es_api_typedefs.h, 876
CFE_ES_TASKID_C
cfe_es_api_typedefs.h, 876
CFE_ES_TASKINFO_EID
cfe_es_events.h, 1003
CFE_ES_TASKINFO_OSCREATE_ERR_EID
cfe_es_events.h, 1004
CFE_ES_TASKINFO_WR_ERR_EID
cfe_es_events.h, 1004
CFE_ES_TASKINFO_WRHDR_ERR_EID
cfe_es_events.h, 1005
CFE_ES_TASKWR_ERR_EID
cfe_es_events.h, 1005
CFE_ES_TEST_LONG_MASK
cfe_es.h, 871
CFE_ES_TIMER_DELETE_ERR
cFE Return Code Defines, 150
CFE_ES_TLM_POOL_STATS_INFO_EID
cfe_es_events.h, 1006
CFE_ES_TaskEntryFuncPtr_t
cfe_es_api_typedefs.h, 878
CFE_ES_TaskID_ToIndex
cFE Resource ID APIs, 176
CFE_ES_TaskId_t
cfe_es_extern_typedefs.h, 886
CFE_ES_TaskInfo, 571
AppId, 572
AppName, 572
ExecutionCounter, 572
Priority, 572
Spare, 573
StackSize, 573
TaskId, 573

TaskName, 573
CFE_ES_TaskInfo_t
cfe_es_extern_typedefs.h, 887
CFE_ES_TaskMain
cFE Internal Executive Service APIs, internal to CFE core, 233
CFE_ES_TaskPriority_Atom_t
cfe_es_extern_typedefs.h, 887
CFE_ES_USE_MUTEX
cfe_es_api_typedefs.h, 876
CFE_ES_VERSION_INF_EID
cfe_es_events.h, 1006
CFE_ES_WRHDR_ERR_EID
cfe_es_events.h, 1007
CFE_ES_WRITE_CFE_HDR_ERR_EID
cfe_es_events.h, 1007
CFE_ES_WRITE_ER_LOG_CC
cfe_es_msg.h, 1033
CFE_ES_WRITE_SYSLOG_CC
cfe_es_msg.h, 1034
CFE_ES_WaitForStartupSync
cFE Application Behavior APIs, 186
CFE_ES_WaitForSystemState
cFE Application Behavior APIs, 186
CFE_ES_WriteERLogCmd_t
cfe_es_msg.h, 1043
CFE_ES_WriteSysLogCmd_t
cfe_es_msg.h, 1043
CFE_ES_WriteToSysLog
cFE Miscellaneous APIs, 206
CFE_EVENTS_SERVICE
cfe_error.h, 865
CFE_EVS_ADD_EVENT_FILTER_CC
cfe_evs_msg.h, 1068
CFE_EVS_ADDFILTER_EID
cfe_evs_events.h, 1045
CFE_EVS_APP_FILTER_OVERLOAD
cFE Return Code Defines, 150
CFE_EVS_APP_ILLEGAL_APP_ID
cFE Return Code Defines, 151
CFE_EVS_APP_NOT_REGISTERED
cFE Return Code Defines, 151
CFE_EVS_AddEventFilterCmd_t
cfe_evs_msg.h, 1090
CFE_EVS_AppDataCmd_Payload, 574
AppDataFilename, 574
CFE_EVS_AppDataCmd_Payload_t
cfe_evs_msg.h, 1091
CFE_EVS_AppNameBitMaskCmd, 574
CmdHeader, 575
Payload, 575
CFE_EVS_AppNameBitMaskCmd_Payload, 575
AppName, 576
BitMask, 576
Spare, 576
CFE_EVS_AppNameBitMaskCmd_Payload_t
cfe_evs_msg.h, 1091
CFE_EVS_AppNameBitMaskCmd_t
cfe_evs_msg.h, 1091
CFE_EVS_AppNameCmd, 577
CmdHeader, 577
Payload, 577
CFE_EVS_AppNameCmd_Payload, 578
AppName, 578
CFE_EVS_AppNameCmd_Payload_t
cfe_evs_msg.h, 1091
CFE_EVS_AppNameCmd_t
cfe_evs_msg.h, 1091
CFE_EVS_AppNameEventIDCmd, 579
CmdHeader, 579
Payload, 579
CFE_EVS_AppNameEventIDCmd_Payload, 580
AppName, 580
EventID, 580
CFE_EVS_AppNameEventIDCmd_Payload_t
cfe_evs_msg.h, 1091
CFE_EVS_AppNameEventIDCmd_t
cfe_evs_msg.h, 1092
CFE_EVS_AppNameEventIDMaskCmd, 581
CmdHeader, 581
Payload, 581
CFE_EVS_AppNameEventIDMaskCmd_Payload, 582
AppName, 582
EventID, 582
Mask, 582
CFE_EVS_AppNameEventIDMaskCmd_Payload_t
cfe_evs_msg.h, 1092
CFE_EVS_AppNameEventIDMaskCmd_t
cfe_evs_msg.h, 1092
CFE_EVS_AppTlmData, 583
AppEnableStatus, 583
AppID, 583
AppMessageSentCounter, 584
Padding, 584
CFE_EVS_AppTlmData_t
cfe_evs_msg.h, 1092
CFE_EVS_BinFilter, 584
EventID, 585
Mask, 585
CFE_EVS_BinFilter_t
cfe_evs_api_typedefs.h, 896
CFE_EVS_BitMaskCmd, 585
CmdHeader, 586
Payload, 586
CFE_EVS_BitMaskCmd_Payload, 586
BitMask, 587
Spare, 587
CFE_EVS_BitMaskCmd_Payload_t

cfe_evs_msg.h, 1092
CFE_EVS_BitMaskCmd_t
cfe_evs_msg.h, 1092
CFE_EVS_CLEAR_LOG_CC
cfe_evs_msg.h, 1069
CFE_EVS_CMD_MID
cpu1_msgids.h, 771
CFE_EVS_CRITICAL_BIT
cfe_evs_msg.h, 1070
CFE_EVS_CleanUpApp
cFE Internal Event Services APIs, internal to CFE core, 244
CFE_EVS_ClearLogCmd_t
cfe_evs_msg.h, 1093
CFE_EVS_DEBUG_BIT
cfe_evs_msg.h, 1070
CFE_EVS_DELETE_EVENT_FILTER_CC
cfe_evs_msg.h, 1071
CFE_EVS_DELFILTER_EID
cfe_evs_events.h, 1046
CFE_EVS_DISABLE_APP_EVENT_TYPE_CC
cfe_evs_msg.h, 1071
CFE_EVS_DISABLE_APP_EVENTS_CC
cfe_evs_msg.h, 1072
CFE_EVS_DISABLE_EVENT_TYPE_CC
cfe_evs_msg.h, 1073
CFE_EVS_DISABLE_PORTS_CC
cfe_evs_msg.h, 1074
CFE_EVS_DISAPPENTTYPE_EID
cfe_evs_events.h, 1046
CFE_EVS_DISAPPEVT_EID
cfe_evs_events.h, 1047
CFE_EVS_DISEVTTYPE_EID
cfe_evs_events.h, 1047
CFE_EVS_DISPORT_EID
cfe_evs_events.h, 1048
CFE_EVS_DeleteEventFilterCmd_t
cfe_evs_msg.h, 1093
CFE_EVS_DisableAppEventTypeCmd_t
cfe_evs_msg.h, 1093
CFE_EVS_DisableAppEventsCmd_t
cfe_evs_msg.h, 1093
CFE_EVS_DisableEventTypeCmd_t
cfe_evs_msg.h, 1093
CFE_EVS_DisablePortsCmd_t
cfe_evs_msg.h, 1093
CFE_EVS_ENAAPPEVT_EID
cfe_evs_events.h, 1048
CFE_EVS_ENAAPPEVTTYPE_EID
cfe_evs_events.h, 1049
CFE_EVS_ENABLE_APP_EVENT_TYPE_CC
cfe_evs_msg.h, 1075
CFE_EVS_ENABLE_APP_EVENTS_CC
cfe_evs_msg.h, 1076
CFE_EVS_ENABLE_EVENT_TYPE_CC
cfe_evs_msg.h, 1077
CFE_EVS_ENABLE_PORTS_CC
cfe_evs_msg.h, 1078
CFE_EVS_ENAEVTTYPE_EID
cfe_evs_events.h, 1049
CFE_EVS_ENAPORT_EID
cfe_evs_events.h, 1050
CFE_EVS_ERR_APPNOREGS_EID
cfe_evs_events.h, 1050
CFE_EVS_ERR_CC_EID
cfe_evs_events.h, 1051
CFE_EVS_ERR_CRDATFILE_EID
cfe_evs_events.h, 1051
CFE_EVS_ERR_CRLOGFILE_EID
cfe_evs_events.h, 1052
CFE_EVS_ERR_EVTIDNOREGS_EID
cfe_evs_events.h, 1052
CFE_EVS_ERR_ILLAPPIDRANGE_EID
cfe_evs_events.h, 1053
CFE_EVS_ERR_ILLEGALFMTMOD_EID
cfe_evs_events.h, 1053
CFE_EVS_ERR_INVALID_BITMASK_EID
cfe_evs_events.h, 1054
CFE_EVS_ERR_LOGMODE_EID
cfe_evs_events.h, 1054
CFE_EVS_ERR_MAXREGSFILTER_EID
cfe_evs_events.h, 1055
CFE_EVS_ERR_MSGID_EID
cfe_evs_events.h, 1055
CFE_EVS_ERR_NOAPPIDFOUND_EID
cfe_evs_events.h, 1056
CFE_EVS_ERR_UNREGISTERED_EVS_APP
cfe_evs_events.h, 1056
CFE_EVS_ERR_WRDATFILE_EID
cfe_evs_events.h, 1057
CFE_EVS_ERR_WRLOGFILE_EID
cfe_evs_events.h, 1057
CFE_EVS_ERROR_BIT
cfe_evs_msg.h, 1079
CFE_EVS_EVERY_FOURTH_ONE
cfe_evs_api_typedefs.h, 893
CFE_EVS_EVERY_OTHER_ONE
cfe_evs_api_typedefs.h, 893
CFE_EVS_EVERY_OTHER_TWO
cfe_evs_api_typedefs.h, 894
CFE_EVS_EVT_FILTERED_EID
cfe_evs_events.h, 1058
CFE_EVS_EVT_NOT_REGISTERED
cFE Return Code Defines, 151
CFE_EVS_EarlyInit
cFE Internal Event Services APIs, internal to CFE core, 244
CFE_EVS_EnableAppEventTypeCmd_t

cfe_evs_msg.h, 1094
CFE_EVS_EnableAppEventsCmd_t
cfe_evs_msg.h, 1094
CFE_EVS_EnableEventTypeCmd_t
cfe_evs_msg.h, 1094
CFE_EVS_EnablePortsCmd_t
cfe_evs_msg.h, 1094
CFE_EVS_EventFilter
cfe_evs_extern_typedefs.h, 898
CFE_EVS_EventFilter_Enum_t
cfe_evs_extern_typedefs.h, 897
CFE_EVS_EventOutput
cfe_evs_extern_typedefs.h, 899
CFE_EVS_EventOutput_Enum_t
cfe_evs_extern_typedefs.h, 897
CFE_EVS_EventType
cfe_evs_extern_typedefs.h, 899
CFE_EVS_EventType_Enum_t
cfe_evs_extern_typedefs.h, 897
CFE_EVS_FILE_WRITE_ERROR
cFE Return Code Defines, 151
CFE_EVS_FILTER_MAX_EID
cfe_evs_events.h, 1058
CFE_EVS_FIRST_16_STOP
cfe_evs_api_typedefs.h, 894
CFE_EVS_FIRST_32_STOP
cfe_evs_api_typedefs.h, 894
CFE_EVS_FIRST_4_STOP
cfe_evs_api_typedefs.h, 894
CFE_EVS_FIRST_64_STOP
cfe_evs_api_typedefs.h, 894
CFE_EVS_FIRST_8_STOP
cfe_evs_api_typedefs.h, 895
CFE_EVS_FIRST_ONE_STOP
cfe_evs_api_typedefs.h, 895
CFE_EVS_FIRST_TWO_STOP
cfe_evs_api_typedefs.h, 895
CFE_EVS_HK_TLM_MID
cpu1_msgids.h, 771
CFE_EVS_HousekeepingTlm, 587
Payload, 588
TlmHeader, 588
CFE_EVS_HousekeepingTlm_Payload, 588
AppData, 589
CommandCounter, 590
CommandErrorCounter, 590
LogEnabled, 590
LogFullFlag, 590
LogMode, 591
LogOverflowCounter, 591
MessageFormatMode, 591
MessageSendCounter, 591
MessageTruncCounter, 592
OutputPort, 592
Spare1, 592
Spare2, 592
Spare3, 593
UnregisteredAppCounter, 593
CFE_EVS_HousekeepingTlm_Payload_t
cfe_evs_msg.h, 1094
CFE_EVS_HousekeepingTlm_t
cfe_evs_msg.h, 1094
CFE_EVS_INFORMATION_BIT
cfe_evs_msg.h, 1080
CFE_EVS_INVALID_PARAMETER
cFE Return Code Defines, 152
CFE_EVS_LEN_ERR_EID
cfe_evs_events.h, 1059
CFE_EVS_LOG_DISCARD
cfe_evs_msg.h, 1080
CFE_EVS_LOG_OVERWRITE
cfe_evs_msg.h, 1080
CFE_EVS_LOGMODE_EID
cfe_evs_events.h, 1059
CFE_EVS_LONG_EVENT_MSG_MID
cpu1_msgids.h, 771
CFE_EVS_LogFileCmd_Payload, 593
LogFilename, 594
CFE_EVS_LogFileCmd_Payload_t
cfe_evs_msg.h, 1095
CFE_EVS_LogMode
cfe_evs_extern_typedefs.h, 899
CFE_EVS_LogMode_Enum_t
cfe_evs_extern_typedefs.h, 898
CFE_EVS_LongEventTlm, 594
Payload, 595
TlmHeader, 595
CFE_EVS_LongEventTlm_Payload, 595
Message, 596
PacketID, 596
Spare1, 596
Spare2, 596
CFE_EVS_LongEventTlm_Payload_t
cfe_evs_msg.h, 1095
CFE_EVS_LongEventTlm_t
cfe_evs_msg.h, 1095
CFE_EVS_MAX_EID
cfe_evs_events.h, 1060
CFE_EVS_MsgFormat
cfe_evs_extern_typedefs.h, 900
CFE_EVS_MsgFormat_Enum_t
cfe_evs_extern_typedefs.h, 898
CFE_EVS_NO_FILTER
cfe_evs_api_typedefs.h, 895
CFE_EVS_NOOP_CC
cfe_evs_msg.h, 1080
CFE_EVS_NOOP_EID
cfe_evs_events.h, 1060

CFE_EVS_NOT_IMPLEMENTED
cFE Return Code Defines, 152
CFE_EVS_NoArgsCmd, 597
CmdHeader, 597
CFE_EVS_NoArgsCmd_t
cfe_evs_msg.h, 1095
CFE_EVS_NoopCmd_t
cfe_evs_msg.h, 1095
CFE_EVS_PORT1_BIT
cfe_evs_msg.h, 1081
CFE_EVS_PORT2_BIT
cfe_evs_msg.h, 1081
CFE_EVS_PORT3_BIT
cfe_evs_msg.h, 1081
CFE_EVS_PORT4_BIT
cfe_evs_msg.h, 1081
CFE_EVS_PacketID_t
cfe_evs_msg.h, 1095
CFE_EVS_PacketID, 597
AppName, 598
EventID, 598
EventType, 598
ProcessorID, 598
SpacecraftID, 599
CFE_EVS_RESET_ALL_FILTERS_CC
cfe_evs_msg.h, 1082
CFE_EVS_RESET_APP_COUNTER_CC
cfe_evs_msg.h, 1082
CFE_EVS_RESET_AREA_POINTER
cFE Return Code Defines, 152
CFE_EVS_RESET_COUNTERS_CC
cfe_evs_msg.h, 1083
CFE_EVS_RESET_FILTER_CC
cfe_evs_msg.h, 1084
CFE_EVS_RSTALLFILTER_EID
cfe_evs_events.h, 1060
CFE_EVS_RSTCNT_EID
cfe_evs_events.h, 1061
CFE_EVS_RSTEVTCNT_EID
cfe_evs_events.h, 1061
CFE_EVS_RSTFILTER_EID
cfe_evs_events.h, 1062
CFE_EVS_Register
cFE Registration APIs, 234
CFE_EVS_ResetAllFilters
cFE Reset Event Filter APIs, 242
CFE_EVS_ResetAllFiltersCmd_t
cfe_evs_msg.h, 1095
CFE_EVS_ResetAppCounterCmd_t
cfe_evs_msg.h, 1096
CFE_EVS_ResetCountersCmd_t
cfe_evs_msg.h, 1096
CFE_EVS_ResetFilter
cFE Reset Event Filter APIs, 242
CFE_EVS_ResetFilterCmd_t
cfe_evs_msg.h, 1096
CFE_EVS_SEND_HK_MID
cpu1_msgids.h, 771
CFE_EVS_SET_EVENT_FORMAT_MODE_CC
cfe_evs_msg.h, 1085
CFE_EVS_SET_FILTER_CC
cfe_evs_msg.h, 1086
CFE_EVS_SET_LOG_MODE_CC
cfe_evs_msg.h, 1087
CFE_EVS_SETEVTFMTMOD_EID
cfe_evs_events.h, 1062
CFE_EVS_SETFILTERMSK_EID
cfe_evs_events.h, 1063
CFE_EVS_SHORT_EVENT_MSG_MID
cpu1_msgids.h, 771
CFE_EVS_STARTUP_EID
cfe_evs_events.h, 1063
CFE_EVS_Send
cfe_evs.h, 891
CFE_EVS_SendCrit
cfe_evs.h, 891
CFE_EVS_SendDbg
cfe_evs.h, 891
CFE_EVS_SendErr
cfe_evs.h, 892
CFE_EVS_SendEvent
cFE Send Event APIs, 237
CFE_EVS_SendEventWithAppID
cFE Send Event APIs, 238
CFE_EVS_SendInfo
cfe_evs.h, 892
CFE_EVS_SendTimedEvent
cFE Send Event APIs, 239
CFE_EVS_SetEventFormatCode_Payload, 599
MsgFormat, 600
Spare, 600
CFE_EVS_SetEventFormatMode_Payload_t
cfe_evs_msg.h, 1096
CFE_EVS_SetEventFormatModeCmd, 600
CmdHeader, 601
Payload, 601
CFE_EVS_SetEventFormatModeCmd_t
cfe_evs_msg.h, 1096
CFE_EVS_SetFilterCmd_t
cfe_evs_msg.h, 1096
CFE_EVS_SetLogMode_Payload, 601
LogMode, 602
Spare, 602
CFE_EVS_SetLogMode_Payload_t
cfe_evs_msg.h, 1097
CFE_EVS_SetLogModeCmd, 602
CmdHeader, 603
Payload, 603

CFE_EVS_SetLogModeCmd_t
cfe_evs_msg.h, 1097
CFE_EVS_ShortEventTlm, 603
Payload, 603
TlmHeader, 604
CFE_EVS_ShortEventTlm_Payload, 604
PacketID, 604
CFE_EVS_ShortEventTlm_Payload_t
cfe_evs_msg.h, 1097
CFE_EVS_ShortEventTlm_t
cfe_evs_msg.h, 1097
CFE_EVS_TaskMain
cFE Internal Event Services APIs, internal to CFE core, 244
CFE_EVS_UNKNOWN_FILTER
cFE Return Code Defines, 152
CFE_EVS_Unregister
cFE Registration APIs, 235
CFE_EVS_WRDAT_EID
cfe_evs_events.h, 1064
CFE_EVS_WRITE_APP_DATA_FILE_CC
cfe_evs_msg.h, 1088
CFE_EVS_WRITE_LOG_DATA_FILE_CC
cfe_evs_msg.h, 1089
CFE_EVS_WRLOG_EID
cfe_evs_events.h, 1064
CFE_EVS_WriteAppDataFileCmd, 605
CmdHeader, 605
Payload, 605
CFE_EVS_WriteAppDataFileCmd_t
cfe_evs_msg.h, 1097
CFE_EVS_WriteLogDataFileCmd, 606
CmdHeader, 606
Payload, 606
CFE_EVS_WriteLogDataFileCmd_t
cfe_evs_msg.h, 1097
CFE_EXECUTIVE_SERVICE
cfe_error.h, 865
CFE_FILE_SERVICE
cfe_error.h, 865
CFE_FS_BAD_ARGUMENT
cFE Return Code Defines, 153
CFE_FS_BackgroundFileDumpIsPending
cFE File Utility APIs, 250
CFE_FS_BackgroundFileDumpRequest
cFE File Utility APIs, 251
CFE_FS_EarlyInit
cFE Internal File Service APIs, internal to CFE core, 256
CFE_FS_ExtractFilenameFromPath
cFE File Utility APIs, 251
CFE_FS_FILE_CONTENT_ID
cfe_fs_extern_typedefs.h, 904
CFE_FS_FNAME_TOO_LONG
cFE Return Code Defines, 153
CFE_FS_FileCategory_t
cfe_fs_api_typedefs.h, 902
CFE_FS_FileWriteEvent_t
cfe_fs_api_typedefs.h, 903
CFE_FS_FileWriteGetData_t
cfe_fs_api_typedefs.h, 902
CFE_FS_FileWriteMetaData, 607
Description, 607
FileName, 607
FileSubType, 608
GetData, 608
IsPending, 608
OnEvent, 608
CFE_FS_FileWriteMetaData_t
cfe_fs_api_typedefs.h, 902
CFE_FS_FileWriteOnEvent_t
cfe_fs_api_typedefs.h, 902
CFE_FS_GetDefaultExtension
cFE File Utility APIs, 252
CFE_FS_GetDefaultMountPoint
cFE File Utility APIs, 252
CFE_FS_HDR_DESC_MAX_LEN
cfe_fs_extern_typedefs.h, 904
CFE_FS_Header, 609
ApplicationID, 609
ContentType, 609
Description, 610
Length, 610
ProcessorID, 610
SpacecraftID, 610
SubType, 610
TimeSeconds, 611
TimeSubSeconds, 611
CFE_FS_Header_t
cfe_fs_extern_typedefs.h, 905
CFE_FS_INVALID_PATH
cFE Return Code Defines, 153
CFE_FS_InitHeader
cFE File Header Management APIs, 246
CFE_FS_NOT_IMPLEMENTED
cFE Return Code Defines, 153
CFE_FS_ParseInputFileName
cFE File Utility APIs, 252
CFE_FS_ParseInputFileNameEx
cFE File Utility APIs, 253
CFE_FS_ReadHeader
cFE File Header Management APIs, 246
CFE_FS_RunBackgroundFileDump
cFE File Utility APIs, 254
CFE_FS_SetTimestamp
cFE File Header Management APIs, 247
CFE_FS_SubType
cfe_fs_extern_typedefs.h, 905

CFE_FS_SubType_Enum_t
cfe_fs_extern_typedefs.h, 905
CFE_FS_WriteHeader
cFE File Header Management APIs, 248
CFE_GENERIC_SERVICE
cfe_error.h, 865
CFE_MAJOR_VERSION
cfe_version.h, 959
CFE_MAKE_BIG16
cfe_endian.h, 857
CFE_MAKE_BIG32
cfe_endian.h, 857
CFE_MINOR_VERSION
cfe_version.h, 959
CFE_MISSION_ES_APP_TLM_MSG
sample_mission_cfg.h, 835
CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN
sample_mission_cfg.h, 835
CFE_MISSION_ES_CDS_MAX_NAME_LENGTH
sample_mission_cfg.h, 835
CFE_MISSION_ES_CMD_MSG
sample_mission_cfg.h, 836
CFE_MISSION_ES_CRC_16
sample_mission_cfg.h, 836
CFE_MISSION_ES_CRC_32
sample_mission_cfg.h, 836
CFE_MISSION_ES_CRC_8
sample_mission_cfg.h, 836
CFE_MISSION_ES_DEFAULT_CRC
sample_mission_cfg.h, 837
CFE_MISSION_ES_HK_TLM_MSG
sample_mission_cfg.h, 837
CFE_MISSION_ES_MAIN_PERF_ID
sample_perfids.h, 853
CFE_MISSION_ES_MAX_APPLICATIONS
sample_mission_cfg.h, 837
CFE_MISSION_ES_MEMSTATS_TLM_MSG
sample_mission_cfg.h, 838
CFE_MISSION_ES_PERF_EXIT_BIT
sample_perfids.h, 853
CFE_MISSION_ES_PERF_MAX_IDS
sample_mission_cfg.h, 838
CFE_MISSION_ES_POOL_MAX_BUCKETS
sample_mission_cfg.h, 838
CFE_MISSION_ES_SEND_HK_MSG
sample_mission_cfg.h, 839
CFE_MISSION_EVS_CMD_MSG
sample_mission_cfg.h, 839
CFE_MISSION_EVS_HK_TLM_MSG
sample_mission_cfg.h, 839
CFE_MISSION_EVS_LONG_EVENT_MSG_MSG
sample_mission_cfg.h, 840
CFE_MISSION_EVS_MAIN_PERF_ID
sample_perfids.h, 854
CFE_MISSION_EVS_MAX_MESSAGE_LENGTH
sample_mission_cfg.h, 840
CFE_MISSION_EVS_SEND_HK_MSG
sample_mission_cfg.h, 840
CFE_MISSION_EVS_SHORT_EVENT_MSG_MSG
sample_mission_cfg.h, 840
CFE_MISSION_MAX_API_LEN
sample_mission_cfg.h, 840
CFE_MISSION_MAX_FILE_LEN
sample_mission_cfg.h, 841
CFE_MISSION_MAX_PATH_LEN
sample_mission_cfg.h, 841
CFE_MISSION_REV
cfe_version.h, 959
CFE_MISSION_SB_ALLSUBS_TLM_MSG
sample_mission_cfg.h, 842
CFE_MISSION_SB_CMD_MSG
sample_mission_cfg.h, 842
CFE_MISSION_SB_HK_TLM_MSG
sample_mission_cfg.h, 842
CFE_MISSION_SB_MAIN_PERF_ID
sample_perfids.h, 854
CFE_MISSION_SB_MAX_PIPES
sample_mission_cfg.h, 843
CFE_MISSION_SB_MAX_SB_MSG_SIZE
sample_mission_cfg.h, 843
CFE_MISSION_SB_MSG_LIM_PERF_ID
sample_perfids.h, 854
CFE_MISSION_SB_ONESUB_TLM_MSG
sample_mission_cfg.h, 843
CFE_MISSION_SB_PIPE_OFLOW_PERF_ID
sample_perfids.h, 854
CFE_MISSION_SB_SEND_HK_MSG
sample_mission_cfg.h, 844
CFE_MISSION_SB_STATS_TLM_MSG
sample_mission_cfg.h, 844
CFE_MISSION_SB_SUB_RPT_CTRL_MSG
sample_mission_cfg.h, 844
CFE_MISSION_TBL_CMD_MSG
sample_mission_cfg.h, 844
CFE_MISSION_TBL_HK_TLM_MSG
sample_mission_cfg.h, 844
CFE_MISSION_TBL_MAIN_PERF_ID
sample_perfids.h, 854
CFE_MISSION_TBL_MAX_FULL_NAME_LEN
sample_mission_cfg.h, 844
CFE_MISSION_TBL_MAX_NAME_LENGTH
sample_mission_cfg.h, 845
CFE_MISSION_TBL_REG_TLM_MSG
sample_mission_cfg.h, 845
CFE_MISSION_TBL_SEND_HK_MSG
sample_mission_cfg.h, 846
CFE_MISSION_TIME_1HZ_CMD_MSG
sample_mission_cfg.h, 846

CFE_MISSION_TIME_AT_TONE_WAS
sample_mission_cfg.h, 846
CFE_MISSION_TIME_AT_TONE_WILL_BE
sample_mission_cfg.h, 846
CFE_MISSION_TIME_CFG_DEFAULT_TAI
sample_mission_cfg.h, 847
CFE_MISSION_TIME_CFG_DEFAULT_UTC
sample_mission_cfg.h, 847
CFE_MISSION_TIME_CFG_FAKE_TONE
sample_mission_cfg.h, 847
CFE_MISSION_TIME_CMD_MSG
sample_mission_cfg.h, 847
CFE_MISSION_TIME_DATA_CMD_MSG
sample_mission_cfg.h, 848
CFE_MISSION_TIME_DEF_DELAY_SECS
sample_mission_cfg.h, 848
CFE_MISSION_TIME_DEF_DELAY_SUBS
sample_mission_cfg.h, 848
CFE_MISSION_TIME_DEF_LEAPS
sample_mission_cfg.h, 848
CFE_MISSION_TIME_DEF_MET_SECS
sample_mission_cfg.h, 848
CFE_MISSION_TIME_DEF_MET_SUBS
sample_mission_cfg.h, 849
CFE_MISSION_TIME_DEF_STCF_SECS
sample_mission_cfg.h, 849
CFE_MISSION_TIME_DEF_STCF_SUBS
sample_mission_cfg.h, 849
CFE_MISSION_TIME_DIAG_TLM_MSG
sample_mission_cfg.h, 849
CFE_MISSION_TIME_EPOCH_DAY
sample_mission_cfg.h, 850
CFE_MISSION_TIME_EPOCH_HOUR
sample_mission_cfg.h, 850
CFE_MISSION_TIME_EPOCH_MINUTE
sample_mission_cfg.h, 850
CFE_MISSION_TIME_EPOCH_SECOND
sample_mission_cfg.h, 850
CFE_MISSION_TIME_EPOCH_YEAR
sample_mission_cfg.h, 850
CFE_MISSION_TIME_FS_FACTOR
sample_mission_cfg.h, 850
CFE_MISSION_TIME_HK_TLM_MSG
sample_mission_cfg.h, 851
CFE_MISSION_TIME_LOCAL1HZISR_PERF_ID
sample_perfids.h, 855
CFE_MISSION_TIME_LOCAL1HZTASK_PERF_ID
sample_perfids.h, 855
CFE_MISSION_TIME_MAIN_PERF_ID
sample_perfids.h, 855
CFE_MISSION_TIME_MAX_ELAPSED
sample_mission_cfg.h, 851
CFE_MISSION_TIME_MIN_ELAPSED
sample_mission_cfg.h, 851
CFE_MISSION_TIME_SEND_CMD_MSG
sample_mission_cfg.h, 852
CFE_MISSION_TIME_SEND_HK_MSG
sample_mission_cfg.h, 852
CFE_MISSION_TIME_SENDMET_PERF_ID
sample_perfids.h, 855
CFE_MISSION_TIME_TONE1HZISR_PERF_ID
sample_perfids.h, 855
CFE_MISSION_TIME_TONE1HZTASK_PERF_ID
sample_perfids.h, 856
CFE_MISSION_TIME_TONE_CMD_MSG
sample_mission_cfg.h, 852
CFE_MSG_ApId_t
cfe_msg_api_typedefs.h, 911
CFE_MSG_BAD_ARGUMENT
cfe_msg_api_typedefs.h, 910
CFE_MSG_Checksum_t
cfe_msg_api_typedefs.h, 911
CFE_MSG_CommandHeader_t
cfe_msg_api_typedefs.h, 911
CFE_MSG_EDSVersion_t
cfe_msg_api_typedefs.h, 911
CFE_MSG_Endian
cfe_msg_api_typedefs.h, 913
CFE_MSG_Endian_t
cfe_msg_api_typedefs.h, 911
CFE_MSG_FcnCode_t
cfe_msg_api_typedefs.h, 911
CFE_MSG_GenerateChecksum
cFE Message header APIs, 258
CFE_MSG_GetApId
cFE Message header APIs, 259
CFE_MSG_GetEDSVersion
cFE Message header APIs, 260
CFE_MSG_GetEndian
cFE Message header APIs, 260
CFE_MSG_GetFcnCode
cFE Message header APIs, 261
CFE_MSG_GetHasSecondaryHeader
cFE Message header APIs, 262
CFE_MSG_GetHeaderVersion
cFE Message header APIs, 262
CFE_MSG_GetMsgId
cFE Message Id APIs, 279
CFE_MSG_GetMsgTime
cFE Message header APIs, 263
CFE_MSG_GetPlaybackFlag
cFE Message header APIs, 264
CFE_MSG_GetSegmentationFlag
cFE Message header APIs, 264
CFE_MSG_GetSequenceCount
cFE Message header APIs, 265
CFE_MSG_GetSize
cFE Message header APIs, 265

CFE_MSG_GetSubsystem
cFE Message header APIs, 266
CFE_MSG_GetSystem
cFE Message header APIs, 267
CFE_MSG_GetType
cFE Message header APIs, 267
CFE_MSG_GetTypeFromMsgId
cFE Message Id APIs, 279
CFE_MSG_HeaderVersion_t
cfe_msg_api_typedefs.h, 912
CFE_MSG_Init
cFE Message header APIs, 268
CFE_MSG_Message_t
cfe_msg_api_typedefs.h, 912
CFE_MSG_NOT_IMPLEMENTED
cfe_msg_api_typedefs.h, 910
CFE_MSG_PlaybackFlag
cfe_msg_api_typedefs.h, 914
CFE_MSG_PlaybackFlag_t
cfe_msg_api_typedefs.h, 912
CFE_MSG_SegmentationFlag
cfe_msg_api_typedefs.h, 914
CFE_MSG_SegmentationFlag_t
cfe_msg_api_typedefs.h, 912
CFE_MSG_SequenceCount_t
cfe_msg_api_typedefs.h, 912
CFE_MSG_SetApId
cFE Message header APIs, 269
CFE_MSG_SetEDSVersion
cFE Message header APIs, 269
CFE_MSG_SetEndian
cFE Message header APIs, 270
CFE_MSG_SetFcnCode
cFE Message header APIs, 270
CFE_MSG_SetHasSecondaryHeader
cFE Message header APIs, 271
CFE_MSG_SetHeaderVersion
cFE Message header APIs, 272
CFE_MSG_SetMsgId
cFE Message Id APIs, 280
CFE_MSG_SetMsgTime
cFE Message header APIs, 272
CFE_MSG_SetPlaybackFlag
cFE Message header APIs, 273
CFE_MSG_SetSegmentationFlag
cFE Message header APIs, 274
CFE_MSG_SetSequenceCount
cFE Message header APIs, 274
CFE_MSG_SetSize
cFE Message header APIs, 275
CFE_MSG_SetSubsystem
cFE Message header APIs, 276
CFE_MSG_SetSystem
cFE Message header APIs, 276
CFE_MSG_SetType
cFE Message header APIs, 277
CFE_MSG_Size_t
cfe_msg_api_typedefs.h, 913
CFE_MSG_Subsystem_t
cfe_msg_api_typedefs.h, 913
CFE_MSG_System_t
cfe_msg_api_typedefs.h, 913
CFE_MSG_TelemetryHeader_t
cfe_msg_api_typedefs.h, 913
CFE_MSG_Type
cfe_msg_api_typedefs.h, 914
CFE_MSG_Type_t
cfe_msg_api_typedefs.h, 913
CFE_MSG_ValidateChecksum
cFE Message header APIs, 278
CFE_MSG_WRONG_MSG_TYPE
cfe_msg_api_typedefs.h, 910
CFE_PLATFORM_CMD_MID_BASE_GLOB
cpu1_msgids.h, 772
CFE_PLATFORM_CMD_MID_BASE
cpu1_msgids.h, 771
CFE_PLATFORM_CORE_MAX_STARTUP_MSEC
cpu1_platform_cfg.h, 779
CFE_PLATFORM_ENDIAN
cpu1_platform_cfg.h, 780
CFE_PLATFORM_ES_APP_KILL_TIMEOUT
cpu1_platform_cfg.h, 780
CFE_PLATFORM_ES_APP_SCAN_RATE
cpu1_platform_cfg.h, 781
CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE
cpu1_platform_cfg.h, 781
CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES
cpu1_platform_cfg.h, 782
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01
cpu1_platform_cfg.h, 782
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02
cpu1_platform_cfg.h, 782
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03
cpu1_platform_cfg.h, 783
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04
cpu1_platform_cfg.h, 783
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05
cpu1_platform_cfg.h, 783
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06
cpu1_platform_cfg.h, 783
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07
cpu1_platform_cfg.h, 783
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08
cpu1_platform_cfg.h, 783
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09
cpu1_platform_cfg.h, 784
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10
cpu1_platform_cfg.h, 784

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11
cpu1_platform_cfg.h, 784
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12
cpu1_platform_cfg.h, 784
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13
cpu1_platform_cfg.h, 784
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14
cpu1_platform_cfg.h, 784
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15
cpu1_platform_cfg.h, 785
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16
cpu1_platform_cfg.h, 785
CFE_PLATFORM_ES_CDS_SIZE
cpu1_platform_cfg.h, 785
CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE
cpu1_platform_cfg.h, 785
CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_↩
FILE
cpu1_platform_cfg.h, 786
CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE
cpu1_platform_cfg.h, 786
CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILE↩
NAME
cpu1_platform_cfg.h, 786
CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MO↩
DE
cpu1_platform_cfg.h, 787
CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE
cpu1_platform_cfg.h, 787
CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
cpu1_platform_cfg.h, 788
CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE
cpu1_platform_cfg.h, 788
CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE
cpu1_platform_cfg.h, 789
CFE_PLATFORM_ES_ER_LOG_ENTRIES
cpu1_platform_cfg.h, 789
CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE
cpu1_platform_cfg.h, 790
CFE_PLATFORM_ES_MAX_APPLICATIONS
cpu1_platform_cfg.h, 790
CFE_PLATFORM_ES_MAX_BLOCK_SIZE
cpu1_platform_cfg.h, 790
CFE_PLATFORM_ES_MAX_GEN_COUNTERS
cpu1_platform_cfg.h, 791
CFE_PLATFORM_ES_MAX_LIBRARIES
cpu1_platform_cfg.h, 791
CFE_PLATFORM_ES_MAX_MEMORY_POOLS
cpu1_platform_cfg.h, 791
CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS
cpu1_platform_cfg.h, 792
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01
cpu1_platform_cfg.h, 792
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02
cpu1_platform_cfg.h, 793
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03
cpu1_platform_cfg.h, 793
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04
cpu1_platform_cfg.h, 793
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05
cpu1_platform_cfg.h, 793
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06
cpu1_platform_cfg.h, 794
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07
cpu1_platform_cfg.h, 794
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08
cpu1_platform_cfg.h, 794
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09
cpu1_platform_cfg.h, 794
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10
cpu1_platform_cfg.h, 794
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11
cpu1_platform_cfg.h, 794
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12
cpu1_platform_cfg.h, 795
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13
cpu1_platform_cfg.h, 795
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14
cpu1_platform_cfg.h, 795
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15
cpu1_platform_cfg.h, 795
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16
cpu1_platform_cfg.h, 795
CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN
cpu1_platform_cfg.h, 795
CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRI↩
NG
cpu1_platform_cfg.h, 796
CFE_PLATFORM_ES_NONVOL_STARTUP_FILE
cpu1_platform_cfg.h, 796
CFE_PLATFORM_ES_OBJECT_TABLE_SIZE
cpu1_platform_cfg.h, 796
CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY
cpu1_platform_cfg.h, 797
CFE_PLATFORM_ES_PERF_CHILD_PRIORITY
cpu1_platform_cfg.h, 797
CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE
cpu1_platform_cfg.h, 797
CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE
cpu1_platform_cfg.h, 798
CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS
cpu1_platform_cfg.h, 798
CFE_PLATFORM_ES_PERF_FILTMASK_ALL
cpu1_platform_cfg.h, 798
CFE_PLATFORM_ES_PERF_FILTMASK_INIT
cpu1_platform_cfg.h, 799
CFE_PLATFORM_ES_PERF_FILTMASK_NONE
cpu1_platform_cfg.h, 799

CFE_PLATFORM_ES_PERF_TRIGMASK_ALL
cpu1_platform_cfg.h, 799
CFE_PLATFORM_ES_PERF_TRIGMASK_INIT
cpu1_platform_cfg.h, 800
CFE_PLATFORM_ES_PERF_TRIGMASK_NONE
cpu1_platform_cfg.h, 800
CFE_PLATFORM_ES_POOL_MAX_BUCKETS
cpu1_platform_cfg.h, 800
CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING
cpu1_platform_cfg.h, 801
CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS
cpu1_platform_cfg.h, 801
CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESE↩
RVED
cpu1_platform_cfg.h, 801
CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE
cpu1_platform_cfg.h, 802
CFE_PLATFORM_ES_RESET_AREA_SIZE
cpu1_platform_cfg.h, 802
CFE_PLATFORM_ES_START_TASK_PRIORITY
cpu1_platform_cfg.h, 803
CFE_PLATFORM_ES_START_TASK_STACK_SIZE
cpu1_platform_cfg.h, 803
CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT↩
_MSEC
cpu1_platform_cfg.h, 804
CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC
cpu1_platform_cfg.h, 804
CFE_PLATFORM_ES_SYSTEM_LOG_SIZE
cpu1_platform_cfg.h, 805
CFE_PLATFORM_ES_USER_RESERVED_SIZE
cpu1_platform_cfg.h, 805
CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE
cpu1_platform_cfg.h, 806
CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE
cpu1_platform_cfg.h, 806
CFE_PLATFORM_EVS_DEFAULT_LOG_FILE
cpu1_platform_cfg.h, 807
CFE_PLATFORM_EVS_DEFAULT_LOG_MODE
cpu1_platform_cfg.h, 807
CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_↩
MODE
cpu1_platform_cfg.h, 807
CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG
cpu1_platform_cfg.h, 808
CFE_PLATFORM_EVS_LOG_MAX
cpu1_platform_cfg.h, 808
CFE_PLATFORM_EVS_MAX_EVENT_FILTERS
cpu1_platform_cfg.h, 808
CFE_PLATFORM_EVS_PORT_DEFAULT
cpu1_platform_cfg.h, 809
CFE_PLATFORM_EVS_START_TASK_PRIORITY
cpu1_platform_cfg.h, 809
CFE_PLATFORM_EVS_START_TASK_STACK_SIZE
cpu1_platform_cfg.h, 809
CFE_PLATFORM_SB_BUF_MEMORY_BYTES
cpu1_platform_cfg.h, 810
CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME
cpu1_platform_cfg.h, 810
CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT
cpu1_platform_cfg.h, 811
CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME
cpu1_platform_cfg.h, 811
CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENA↩
ME
cpu1_platform_cfg.h, 812
CFE_PLATFORM_SB_FILTER_MASK1
cpu1_platform_cfg.h, 812
CFE_PLATFORM_SB_FILTER_MASK2
cpu1_platform_cfg.h, 812
CFE_PLATFORM_SB_FILTER_MASK3
cpu1_platform_cfg.h, 813
CFE_PLATFORM_SB_FILTER_MASK4
cpu1_platform_cfg.h, 813
CFE_PLATFORM_SB_FILTER_MASK5
cpu1_platform_cfg.h, 813
CFE_PLATFORM_SB_FILTER_MASK6
cpu1_platform_cfg.h, 813
CFE_PLATFORM_SB_FILTER_MASK7
cpu1_platform_cfg.h, 813
CFE_PLATFORM_SB_FILTER_MASK8
cpu1_platform_cfg.h, 813
CFE_PLATFORM_SB_FILTERED_EVENT1
cpu1_platform_cfg.h, 814
CFE_PLATFORM_SB_FILTERED_EVENT2
cpu1_platform_cfg.h, 814
CFE_PLATFORM_SB_FILTERED_EVENT3
cpu1_platform_cfg.h, 814
CFE_PLATFORM_SB_FILTERED_EVENT4
cpu1_platform_cfg.h, 814
CFE_PLATFORM_SB_FILTERED_EVENT5
cpu1_platform_cfg.h, 814
CFE_PLATFORM_SB_FILTERED_EVENT6
cpu1_platform_cfg.h, 815
CFE_PLATFORM_SB_FILTERED_EVENT7
cpu1_platform_cfg.h, 815
CFE_PLATFORM_SB_FILTERED_EVENT8
cpu1_platform_cfg.h, 815
CFE_PLATFORM_SB_HIGHEST_VALID_MSGID
cpu1_platform_cfg.h, 815
CFE_PLATFORM_SB_MAX_BLOCK_SIZE
cpu1_platform_cfg.h, 815
CFE_PLATFORM_SB_MAX_DEST_PER_PKT
cpu1_platform_cfg.h, 816
CFE_PLATFORM_SB_MAX_MSG_IDS
cpu1_platform_cfg.h, 816
CFE_PLATFORM_SB_MAX_PIPES
cpu1_platform_cfg.h, 816

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01
cpu1_platform_cfg.h, 817
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02
cpu1_platform_cfg.h, 817
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03
cpu1_platform_cfg.h, 817
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04
cpu1_platform_cfg.h, 818
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05
cpu1_platform_cfg.h, 818
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06
cpu1_platform_cfg.h, 818
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07
cpu1_platform_cfg.h, 818
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08
cpu1_platform_cfg.h, 818
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09
cpu1_platform_cfg.h, 818
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10
cpu1_platform_cfg.h, 819
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11
cpu1_platform_cfg.h, 819
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12
cpu1_platform_cfg.h, 819
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13
cpu1_platform_cfg.h, 819
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14
cpu1_platform_cfg.h, 819
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15
cpu1_platform_cfg.h, 819
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16
cpu1_platform_cfg.h, 820
CFE_PLATFORM_SB_START_TASK_PRIORITY
cpu1_platform_cfg.h, 820
CFE_PLATFORM_SB_START_TASK_STACK_SIZE
cpu1_platform_cfg.h, 820
CFE_PLATFORM_TBL_BUF_MEMORY_BYTES
cpu1_platform_cfg.h, 820
CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE
cpu1_platform_cfg.h, 821
CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES
cpu1_platform_cfg.h, 821
CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE
cpu1_platform_cfg.h, 821
CFE_PLATFORM_TBL_MAX_NUM_HANDLES
cpu1_platform_cfg.h, 822
CFE_PLATFORM_TBL_MAX_NUM_TABLES
cpu1_platform_cfg.h, 822
CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS
cpu1_platform_cfg.h, 822
CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS
cpu1_platform_cfg.h, 823
CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE
cpu1_platform_cfg.h, 823
CFE_PLATFORM_TBL_START_TASK_PRIORITY
cpu1_platform_cfg.h, 824
CFE_PLATFORM_TBL_START_TASK_STACK_SIZE
cpu1_platform_cfg.h, 824
CFE_PLATFORM_TBL_U32FROM4CHARS
cpu1_platform_cfg.h, 824
CFE_PLATFORM_TBL_VALID_PRID_1
cpu1_platform_cfg.h, 825
CFE_PLATFORM_TBL_VALID_PRID_2
cpu1_platform_cfg.h, 825
CFE_PLATFORM_TBL_VALID_PRID_3
cpu1_platform_cfg.h, 825
CFE_PLATFORM_TBL_VALID_PRID_4
cpu1_platform_cfg.h, 825
CFE_PLATFORM_TBL_VALID_PRID_COUNT
cpu1_platform_cfg.h, 826
CFE_PLATFORM_TBL_VALID_SCID_1
cpu1_platform_cfg.h, 826
CFE_PLATFORM_TBL_VALID_SCID_2
cpu1_platform_cfg.h, 826
CFE_PLATFORM_TBL_VALID_SCID_COUNT
cpu1_platform_cfg.h, 827
CFE_PLATFORM_TIME_1HZ_TASK_PRIORITY
cpu1_platform_cfg.h, 827
CFE_PLATFORM_TIME_1HZ_TASK_STACK_SIZE
cpu1_platform_cfg.h, 827
CFE_PLATFORM_TIME_CFG_CLIENT
cpu1_platform_cfg.h, 827
CFE_PLATFORM_TIME_CFG_LATCH_FLY
cpu1_platform_cfg.h, 827
CFE_PLATFORM_TIME_CFG_SERVER
cpu1_platform_cfg.h, 828
CFE_PLATFORM_TIME_CFG_SIGNAL
cpu1_platform_cfg.h, 828
CFE_PLATFORM_TIME_CFG_SOURCE
cpu1_platform_cfg.h, 828
CFE_PLATFORM_TIME_CFG_SRC_GPS
cpu1_platform_cfg.h, 829
CFE_PLATFORM_TIME_CFG_SRC_MET
cpu1_platform_cfg.h, 829
CFE_PLATFORM_TIME_CFG_SRC_TIME
cpu1_platform_cfg.h, 829
CFE_PLATFORM_TIME_CFG_START_FLY
cpu1_platform_cfg.h, 830
CFE_PLATFORM_TIME_CFG_TONE_LIMIT
cpu1_platform_cfg.h, 830
CFE_PLATFORM_TIME_CFG_VIRTUAL
cpu1_platform_cfg.h, 830
CFE_PLATFORM_TIME_MAX_DELTA_SECS
cpu1_platform_cfg.h, 831
CFE_PLATFORM_TIME_MAX_DELTA_SUBS
cpu1_platform_cfg.h, 831
CFE_PLATFORM_TIME_MAX_LOCAL_SECS
cpu1_platform_cfg.h, 832

CFE_PLATFORM_TIME_MAX_LOCAL_SUBS
cpu1_platform_cfg.h, 832
CFE_PLATFORM_TIME_START_TASK_PRIORITY
cpu1_platform_cfg.h, 832
CFE_PLATFORM_TIME_START_TASK_STACK_SIZE
cpu1_platform_cfg.h, 832
CFE_PLATFORM_TIME_TONE_TASK_PRIORITY
cpu1_platform_cfg.h, 833
CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE
cpu1_platform_cfg.h, 833
CFE_PLATFORM_TLM_MID_BASE
cpu1_msgids.h, 772
CFE_PSP_1HZ_FUNCTION
cfe_psp_start.c, 1342
CFE_PSP_AsyncMask
cfe_psp_exception.c, 1329
CFE_PSP_AttachExceptions
cfe_psp.h, 1310
cfe_psp_exception.c, 1326
CFE_PSP_AttachSigHandler
cfe_psp_exception.c, 1327
CFE_PSP_CDS_KEY_FILE
cfe_psp_memory.c, 1330
CFE_PSP_CDS_SIZE
cfe_psp_memory.c, 1330
CFE_PSP_CPU_ID
cfe_psp_start.c, 1342
CFE_PSP_CPU_NAME_LENGTH
cfe_psp_start.c, 1343
CFE_PSP_CPU_NAME
cfe_psp_start.c, 1343
CFE_PSP_CommandData_t, 611
CpuId, 612
CpuName, 612
GotCpuId, 612
GotCpuName, 612
GotResetType, 612
GotSpacecraftId, 613
GotSubType, 613
ResetType, 613
SpacecraftId, 613
SubType, 613
CFE_PSP_CpuId
cfe_psp_start.c, 1346
cfe_psp_support.c, 1350
CFE_PSP_CpuName
cfe_psp_start.c, 1346
cfe_psp_support.c, 1351
CFE_PSP_Decompress
cfe_psp.h, 1311
CFE_PSP_DeleteCDS
cfe_psp_memory.c, 1332
CFE_PSP_DeleteProcessorReservedMemory
cfe_psp_memory.c, 1332
CFE_PSP_DeleteResetArea
cfe_psp_memory.c, 1332
CFE_PSP_DeleteUserReservedArea
cfe_psp_memory.c, 1333
CFE_PSP_DisplayUsage
cfe_psp_start.c, 1344
CFE_PSP_ERROR_ADDRESS_MISALIGNED
cfe_psp.h, 1301
CFE_PSP_ERROR_NOT_IMPLEMENTED
cfe_psp.h, 1301
CFE_PSP_ERROR_TIMEOUT
cfe_psp.h, 1301
CFE_PSP_ERROR
cfe_psp.h, 1301
CFE_PSP_EepromPowerDown
cfe_psp.h, 1311
CFE_PSP_EepromPowerUp
cfe_psp.h, 1311
CFE_PSP_EepromWrite16
cfe_psp.h, 1311
CFE_PSP_EepromWrite32
cfe_psp.h, 1311
CFE_PSP_EepromWrite8
cfe_psp.h, 1312
CFE_PSP_EepromWriteDisable
cfe_psp.h, 1312
CFE_PSP_EepromWriteEnable
cfe_psp.h, 1312
CFE_PSP_Exception_CopyContext
cfe_psp.h, 1312
CFE_PSP_Exception_GetCount
cfe_psp.h, 1312
CFE_PSP_Exception_GetSummary
cfe_psp.h, 1312
CFE_PSP_ExceptionGetSummary_Impl
cfe_psp_exception.c, 1327
CFE_PSP_ExceptionSigHandler
cfe_psp_exception.c, 1327
CFE_PSP_ExceptionSigHandlerSuspend
cfe_psp_exception.c, 1328
CFE_PSP_FlushCaches
cfe_psp.h, 1313
cfe_psp_support.c, 1349
CFE_PSP_Get_Dec
cfe_psp.h, 1313
cfe_psp_timer.c, 1352
CFE_PSP_Get_Timebase
cfe_psp.h, 1313
cfe_psp_timer.c, 1352
CFE_PSP_Get_Timer_Tick
cfe_psp.h, 1313
cfe_psp_timer.c, 1353
CFE_PSP_GetBuildNumber
cfe_psp.h, 1314

CFE_PSP_GetCDSSize
cfe_psp.h, 1314
cfe_psp_memory.c, 1333
CFE_PSP_GetCFETextSegmentInfo
cfe_psp.h, 1314
cfe_psp_memory.c, 1334
CFE_PSP_GetKernelTextSegmentInfo
cfe_psp.h, 1314
cfe_psp_memory.c, 1334
CFE_PSP_GetProcessorId
cfe_psp.h, 1315
cfe_psp_support.c, 1349
CFE_PSP_GetProcessorName
cfe_psp.h, 1315
cfe_psp_support.c, 1349
CFE_PSP_GetResetArea
cfe_psp.h, 1315
cfe_psp_memory.c, 1334
CFE_PSP_GetRestartType
cfe_psp.h, 1315
CFE_PSP_GetSpacecraftId
cfe_psp.h, 1315
cfe_psp_support.c, 1349
CFE_PSP_GetTime
cfe_psp.h, 1316
cfe_psp_timer.c, 1353
CFE_PSP_GetTimerLow32Rollover
cfe_psp.h, 1316
cfe_psp_timer.c, 1353
CFE_PSP_GetTimerTicksPerSecond
cfe_psp.h, 1316
cfe_psp_timer.c, 1354
CFE_PSP_GetUserReservedArea
cfe_psp.h, 1316
cfe_psp_memory.c, 1334
CFE_PSP_GetVersionCodeName
cfe_psp.h, 1317
CFE_PSP_GetVersionNumber
cfe_psp.h, 1317
CFE_PSP_GetVersionString
cfe_psp.h, 1317
CFE_PSP_GetVolatileDiskMem
cfe_psp.h, 1318
cfe_psp_memory.c, 1335
CFE_PSP_INVALID_INT_NUM
cfe_psp.h, 1302
CFE_PSP_INVALID_MEM_ADDR
cfe_psp.h, 1302
CFE_PSP_INVALID_MEM_ATTR
cfe_psp.h, 1302
CFE_PSP_INVALID_MEM_RANGE
cfe_psp.h, 1302
CFE_PSP_INVALID_MEM_SIZE
cfe_psp.h, 1302
CFE_PSP_INVALID_MEM_TYPE
cfe_psp.h, 1302
CFE_PSP_INVALID_MEM_WORDSIZE
cfe_psp.h, 1303
CFE_PSP_INVALID_MODULE_ID
cfe_psp.h, 1303
CFE_PSP_INVALID_MODULE_NAME
cfe_psp.h, 1303
CFE_PSP_INVALID_POINTER
cfe_psp.h, 1303
CFE_PSP_IdleTaskState
cfe_psp_start.c, 1347
CFE_PSP_InitCDS
cfe_psp_memory.c, 1335
CFE_PSP_InitProcessorReservedMemory
cfe_psp_memory.c, 1335
CFE_PSP_InitResetArea
cfe_psp_memory.c, 1336
CFE_PSP_InitSSR
cfe_psp.h, 1318
cfe_psp_ssr.c, 1341
CFE_PSP_InitUserReservedArea
cfe_psp_memory.c, 1336
CFE_PSP_InitVolatileDiskMem
cfe_psp_memory.c, 1337
CFE_PSP_KERNEL_NAME_LENGTH_MAX
cfe_psp_start.c, 1343
CFE_PSP_LinuxReservedAreaFixedLayout_t, 614
BootRecord, 614
ExceptionStorage, 614
CFE_PSP_MAIN_FUNCTION
cfe_psp_start.c, 1343
CFE_PSP_MAJOR_VERSION
cfe_psp.h, 1303
CFE_PSP_MEM_ANY
cfe_psp.h, 1303
CFE_PSP_MEM_ATTR_READWRITE
cfe_psp.h, 1304
CFE_PSP_MEM_ATTR_READ
cfe_psp.h, 1304
CFE_PSP_MEM_ATTR_WRITE
cfe_psp.h, 1304
CFE_PSP_MEM_EEPROM
cfe_psp.h, 1304
CFE_PSP_MEM_INVALID
cfe_psp.h, 1304
CFE_PSP_MEM_RAM
cfe_psp.h, 1304
CFE_PSP_MEM_SIZE_BYTE
cfe_psp.h, 1305
CFE_PSP_MEM_SIZE_DWORD
cfe_psp.h, 1305
CFE_PSP_MEM_SIZE_WORD
cfe_psp.h, 1305

CFE_PSP_MINOR_VERSION
cfe_psp.h, 1305
CFE_PSP_MISSION_REV
cfe_psp.h, 1305
CFE_PSP_Main
cfe_psp.h, 1318
CFE_PSP_MemCpy
cfe_psp.h, 1318
CFE_PSP_MemRangeGet
cfe_psp.h, 1319
CFE_PSP_MemRangeSet
cfe_psp.h, 1319
CFE_PSP_MemRanges
cfe_psp.h, 1319
CFE_PSP_MemRead16
cfe_psp.h, 1319
CFE_PSP_MemRead32
cfe_psp.h, 1319
CFE_PSP_MemRead8
cfe_psp.h, 1320
CFE_PSP_MemSet
cfe_psp.h, 1320
CFE_PSP_MemTable_t, 615
Attributes, 615
MemoryType, 615
Size, 615
StartAddr, 615
WordSize, 616
CFE_PSP_MemValidateRange
cfe_psp.h, 1320
CFE_PSP_MemWrite16
cfe_psp.h, 1320
CFE_PSP_MemWrite32
cfe_psp.h, 1320
CFE_PSP_MemWrite8
cfe_psp.h, 1320
CFE_PSP_MemoryTable
cfe_psp_configdata.h, 1325
cfe_psp_memtab.c, 1340
CFE_PSP_NO_EXCEPTION_DATA
cfe_psp.h, 1305
CFE_PSP_NONVOL_STARTUP_FILE
cfe_psp_start.c, 1343
CFE_PSP_OS_EventHandler
cfe_psp_start.c, 1344
CFE_PSP_PANIC_CORE_APP
cfe_psp.h, 1306
CFE_PSP_PANIC_GENERAL_FAILURE
cfe_psp.h, 1306
CFE_PSP_PANIC_MEMORY_ALLOC
cfe_psp.h, 1306
CFE_PSP_PANIC_NONVOL_DISK
cfe_psp.h, 1306
CFE_PSP_PANIC_STARTUP_SEM
cfe_psp.h, 1306
CFE_PSP_PANIC_STARTUP
cfe_psp.h, 1306
CFE_PSP_PANIC_VOLATILE_DISK
cfe_psp.h, 1307
CFE_PSP_Panic
cfe_psp.h, 1321
cfe_psp_support.c, 1349
CFE_PSP_PortRead16
cfe_psp.h, 1321
CFE_PSP_PortRead32
cfe_psp.h, 1321
CFE_PSP_PortRead8
cfe_psp.h, 1321
CFE_PSP_PortWrite16
cfe_psp.h, 1322
CFE_PSP_PortWrite32
cfe_psp.h, 1322
CFE_PSP_PortWrite8
cfe_psp.h, 1322
CFE_PSP_ProcessArgumentDefaults
cfe_psp_start.c, 1345
CFE_PSP_RESERVED_KEY_FILE
cfe_psp_memory.c, 1331
CFE_PSP_RESET_AREA_SIZE
cfe_psp_memory.c, 1331
CFE_PSP_RESET_KEY_FILE
cfe_psp_memory.c, 1331
CFE_PSP_RESET_NAME_LENGTH
cfe_psp_start.c, 1344
CFE_PSP_REVISION
cfe_psp.h, 1307
CFE_PSP_RST_SUBTYPE_BANKSWITCH_RESET
cfe_psp.h, 1307
CFE_PSP_RST_SUBTYPE_EXCEPTION
cfe_psp.h, 1307
CFE_PSP_RST_SUBTYPE_HW_SPECIAL_COMMAND
cfe_psp.h, 1307
CFE_PSP_RST_SUBTYPE_HW_WATCHDOG
cfe_psp.h, 1307
CFE_PSP_RST_SUBTYPE_HWDEBUG_RESET
cfe_psp.h, 1308
CFE_PSP_RST_SUBTYPE_MAX
cfe_psp.h, 1308
CFE_PSP_RST_SUBTYPE_POWER_CYCLE
cfe_psp.h, 1308
CFE_PSP_RST_SUBTYPE_PUSH_BUTTON
cfe_psp.h, 1308
CFE_PSP_RST_SUBTYPE_RESET_COMMAND
cfe_psp.h, 1308
CFE_PSP_RST_SUBTYPE_UNDEFINED_RESET
cfe_psp.h, 1309
CFE_PSP_RST_TYPE_MAX
cfe_psp.h, 1309

CFE_PSP_RST_TYPE_POWERON
cfe_psp.h, 1309
CFE_PSP_RST_TYPE_PROCESSOR
cfe_psp.h, 1309
CFE_PSP_ReadFromCDS
cfe_psp.h, 1322
cfe_psp_memory.c, 1337
CFE_PSP_ReservedMemoryMap
cfe_psp_memory.c, 1339
CFE_PSP_Restart
cfe_psp.h, 1322
cfe_psp_support.c, 1350
CFE_PSP_SPACECRAFT_ID
cfe_psp_start.c, 1344
CFE_PSP_SUCCESS
cfe_psp.h, 1309
CFE_PSP_SetDefaultExceptionEnvironment
cfe_psp.h, 1323
cfe_psp_exception.c, 1328
CFE_PSP_SetupLocal1Hz
cfe_psp_start.c, 1345
CFE_PSP_SetupReservedMemoryMap
cfe_psp_memory.c, 1337
CFE_PSP_SpacecraftId
cfe_psp_start.c, 1347
cfe_psp_support.c, 1351
CFE_PSP_TIMER_LOW32_ROLLOVER
cfe_psp_timer.c, 1352
CFE_PSP_TIMER_TICKS_PER_SECOND
cfe_psp_timer.c, 1352
CFE_PSP_TimerHandler
cfe_psp_start.c, 1346
CFE_PSP_USER_RESERVED_SIZE
cfe_psp_memory.c, 1331
CFE_PSP_VERSION
cfe_psp.h, 1310
CFE_PSP_VersionInfo_t, 616
BuildNumber, 616
MajorVersion, 616
MinorVersion, 617
MissionRev, 617
Revision, 617
VersionCodeName, 617
VersionString, 617
CFE_PSP_WatchdogDisable
cfe_psp.h, 1323
cfe_psp_watchdog.c, 1354
CFE_PSP_WatchdogEnable
cfe_psp.h, 1323
cfe_psp_watchdog.c, 1355
CFE_PSP_WatchdogGet
cfe_psp.h, 1324
cfe_psp_watchdog.c, 1355
CFE_PSP_WatchdogInit
cfe_psp.h, 1324
cfe_psp_watchdog.c, 1355
CFE_PSP_WatchdogService
cfe_psp.h, 1324
cfe_psp_watchdog.c, 1355
CFE_PSP_WatchdogSet
cfe_psp.h, 1324
cfe_psp_watchdog.c, 1355
CFE_PSP_WatchdogValue
cfe_psp_watchdog.c, 1356
CFE_PSP_WriteToCDS
cfe_psp.h, 1324
cfe_psp_memory.c, 1338
CFE_RESOURCEID_RESERVED
cfe_resourceid_api_typedefs.h, 924
CFE_RESOURCEID_TEST_DEFINED
cfe_resourceid.h, 917
CFE_RESOURCEID_TEST_EQUAL
cfe_resourceid.h, 917
CFE_RESOURCEID_TO_ULONG
cfe_resourceid.h, 917
CFE_RESOURCEID_UNDEFINED
cfe_resourceid_api_typedefs.h, 924
CFE_REVISION
cfe_version.h, 959
CFE_ResourceId_Equal
cfe_resourceid.h, 918
CFE_ResourceId_FindNext
cfe_resourceid.h, 918
CFE_ResourceId_FromInteger
cfe_resourceid.h, 919
CFE_ResourceId_GetBase
cfe_resourceid.h, 920
CFE_ResourceId_GetSerial
cfe_resourceid.h, 920
CFE_ResourceId_IsDefined
cfe_resourceid.h, 921
CFE_ResourceId_ToIndex
cfe_resourceid.h, 922
CFE_ResourceId_ToInteger
cfe_resourceid.h, 922
CFE_SB_ALLSUBS_TLM_MID
cpu1_msgids.h, 772
CFE_SB_AllSubscriptionsTlm, 618
Hdr, 618
Payload, 618
CFE_SB_AllSubscriptionsTlm_Payload, 618
Entries, 619
Entry, 619
PktSegment, 619
TotalSegments, 620
CFE_SB_AllSubscriptionsTlm_Payload_t
cfe_sb_msg.h, 1147
CFE_SB_AllSubscriptionsTlm_t

cfe_sb_msg.h, 1147
CFE_SB_BAD_ARGUMENT
cFE Return Code Defines, 154
CFE_SB_BAD_CMD_CODE_EID
cfe_sb_events.h, 1101
CFE_SB_BAD_MSGID_EID
cfe_sb_events.h, 1102
CFE_SB_BAD_PIPEID_EID
cfe_sb_events.h, 1102
CFE_SB_BUF_ALOC_ERR
cFE Return Code Defines, 154
CFE_SB_BUFFER_INVALID
cFE Return Code Defines, 154
CFE_SB_Buffer_t
cfe_sb_api_typedefs.h, 934
CFE_SB_CMD0_RCVD_EID
cfe_sb_events.h, 1103
CFE_SB_CMD1_RCVD_EID
cfe_sb_events.h, 1103
CFE_SB_CMD_HDR_SIZE
cfe_sb_api_typedefs.h, 930
CFE_SB_CMD_MID
cpu1_msgids.h, 773
CFE_SB_CR_PIPE_BAD_ARG_EID
cfe_sb_events.h, 1104
CFE_SB_CR_PIPE_ERR_EID
cfe_sb_events.h, 1104
CFE_SB_CR_PIPE_NAME_TAKEN_EID
cfe_sb_events.h, 1105
CFE_SB_CR_PIPE_NO_FREE_EID
cfe_sb_events.h, 1105
CFE_SB_CleanUpApp
cFE Internal Software Bus APIs, internal to CFE core, 324
CFE_SB_CmdHdr_t
cfe_sb_api_typedefs.h, 934
CFE_SB_CreatePipe
cFE Pipe Management APIs, 282
CFE_SB_DEFAULT_QOS
cfe_sb_api_typedefs.h, 930
CFE_SB_DEL_PIPE_ERR1_EID
cfe_sb_events.h, 1106
CFE_SB_DEL_PIPE_ERR2_EID
cfe_sb_events.h, 1106
CFE_SB_DEST_BLK_ERR_EID
cfe_sb_events.h, 1107
CFE_SB_DISABLE_ROUTE_CC
cfe_sb_msg.h, 1136
CFE_SB_DISABLE_SUB_REPORTING_CC
cfe_sb_msg.h, 1137
CFE_SB_DSBL_RTE1_EID
cfe_sb_events.h, 1107
CFE_SB_DSBL_RTE2_EID
cfe_sb_events.h, 1108
CFE_SB_DSBL_RTE3_EID
cfe_sb_events.h, 1108
CFE_SB_DUP_SUBSCRIP_EID
cfe_sb_events.h, 1109
CFE_SB_Default_Qos
cfe_sb_api_typedefs.h, 930
CFE_SB_DeletePipe
cFE Pipe Management APIs, 283
CFE_SB_DisableRouteCmd_t
cfe_sb_msg.h, 1147
CFE_SB_DisableSubReportingCmd_t
cfe_sb_msg.h, 1147
CFE_SB_ENABLE_ROUTE_CC
cfe_sb_msg.h, 1137
CFE_SB_ENABLE_SUB_REPORTING_CC
cfe_sb_msg.h, 1138
CFE_SB_ENBL_RTE1_EID
cfe_sb_events.h, 1109
CFE_SB_ENBL_RTE2_EID
cfe_sb_events.h, 1110
CFE_SB_ENBL_RTE3_EID
cfe_sb_events.h, 1110
CFE_SB_EarlyInit
cFE Internal Software Bus APIs, internal to CFE core, 324
CFE_SB_EnableRouteCmd_t
cfe_sb_msg.h, 1147
CFE_SB_EnableSubReportingCmd_t
cfe_sb_msg.h, 1147
CFE_SB_FILEWRITE_ERR_EID
cfe_sb_events.h, 1111
CFE_SB_FULL_SUB_PKT_EID
cfe_sb_events.h, 1111
CFE_SB_GET_BUF_ERR_EID
cfe_sb_events.h, 1112
CFE_SB_GETPIPEIDBYNAME_EID
cfe_sb_events.h, 1112
CFE_SB_GETPIPEIDBYNAME_NAME_ERR_EID
cfe_sb_events.h, 1113
CFE_SB_GETPIPEIDBYNAME_NULL_ERR_EID
cfe_sb_events.h, 1113
CFE_SB_GETPIPENAME_EID
cfe_sb_events.h, 1114
CFE_SB_GETPIPENAME_ID_ERR_EID
cfe_sb_events.h, 1114
CFE_SB_GETPIPENAME_NULL_PTR_EID
cfe_sb_events.h, 1115
CFE_SB_GETPIPEOPTS_EID
cfe_sb_events.h, 1115
CFE_SB_GETPIPEOPTS_ID_ERR_EID
cfe_sb_events.h, 1116
CFE_SB_GETPIPEOPTS_PTR_ERR_EID
cfe_sb_events.h, 1116
CFE_SB_GenerateChecksum

cFE Checksum Control APIs, 316
CFE_SB_GetChecksum
cFE Checksum Control APIs, 316
CFE_SB_GetCmdCode
cFE Getting Message Characteristics APIs, 310
CFE_SB_GetMsgId
cFE Getting Message Characteristics APIs, 311
CFE_SB_GetMsgTime
cFE Getting Message Characteristics APIs, 311
CFE_SB_GetPipeIdByName
cFE Pipe Management APIs, 284
CFE_SB_GetPipeName
cFE Pipe Management APIs, 285
CFE_SB_GetPipeOpts
cFE Pipe Management APIs, 285
CFE_SB_GetPktType
cFE Message ID APIs, 319
CFE_SB_GetTotalMsgLength
cFE Getting Message Characteristics APIs, 312
CFE_SB_GetUserData
cFE Getting Message Characteristics APIs, 313
CFE_SB_GetUserDataLength
cFE Getting Message Characteristics APIs, 313
CFE_SB_HASHCOLLISION_EID
cfe_sb_events.h, 1117
CFE_SB_HK_TLM_MID
cpu1_msgids.h, 773
CFE_SB_HousekeepingTlm, 620
Hdr, 620
Payload, 621
CFE_SB_HousekeepingTlm_Payload, 621
CommandCounter, 622
CommandErrorCounter, 622
CreatePipeErrorCounter, 622
DuplicateSubscriptionsCounter, 623
GetPipeIdByNameErrorCounter, 623
InternalErrorCounter, 623
MemInUse, 623
MemPoolHandle, 624
MsgLimitErrorCounter, 624
MsgReceiveErrorCounter, 624
MsgSendErrorCounter, 624
NoSubscribersCounter, 625
PipeOptsErrorCounter, 625
PipeOverflowErrorCounter, 625
Spare2Align, 625
SubscribeErrorCounter, 626
UnmarkedMem, 626
CFE_SB_HousekeepingTlm_Payload_t
cfe_sb_msg.h, 1148
CFE_SB_HousekeepingTlm_t
cfe_sb_msg.h, 1148
CFE_SB_INIT_EID
cfe_sb_events.h, 1117
CFE_SB_INTERNAL_ERR
cFE Return Code Defines, 154
CFE_SB_INVALID_MSG_ID
cfe_sb_api_typedefs.h, 931
CFE_SB_INVALID_PIPE
cfe_sb_api_typedefs.h, 931
CFE_SB_InitMsg
cFE Setting Message Characteristics APIs, 304
CFE_SB_IsValidMsgId
cFE Message ID APIs, 319
CFE_SB_LEN_ERR_EID
cfe_sb_events.h, 1118
CFE_SB_MAX_DESTS_MET_EID
cfe_sb_events.h, 1118
CFE_SB_MAX_DESTS_MET
cFE Return Code Defines, 155
CFE_SB_MAX_EID
cfe_sb_events.h, 1119
CFE_SB_MAX_MSGS_MET_EID
cfe_sb_events.h, 1119
CFE_SB_MAX_MSGS_MET
cFE Return Code Defines, 155
CFE_SB_MAX_PIPES_MET_EID
cfe_sb_events.h, 1119
CFE_SB_MAX_PIPES_MET
cFE Return Code Defines, 155
CFE_SB_MSG_TOO_BIG_EID
cfe_sb_events.h, 1120
CFE_SB_MSG_TOO_BIG
cFE Return Code Defines, 155
CFE_SB_MSGID_LIM_ERR_EID
cfe_sb_events.h, 1120
CFE_SB_MSGID_RESERVED
cfe_sb_api_typedefs.h, 931
CFE_SB_MSGID_UNWRAP_VALUE
cfe_sb_api_typedefs.h, 931
CFE_SB_MSGID_WRAP_VALUE
cfe_sb_api_typedefs.h, 932
CFE_SB_MessageStringGet
cFE Getting Message Characteristics APIs, 314
CFE_SB_MessageStringSet
cFE Setting Message Characteristics APIs, 305
CFE_SB_Msg, 626
LongDouble, 627
LongInt, 627
Msg, 627
CFE_SB_Msg_t
cfe_sb_api_typedefs.h, 934
CFE_SB_MsgId_Atom_t
cfe_sb_extern_typedefs.h, 937
CFE_SB_MsgId_Equal
cFE Message ID APIs, 320
CFE_SB_MsgId_t
cfe_sb_extern_typedefs.h, 937

CFE_SB_MsgIdToValue
cFE Message ID APIs, 321
CFE_SB_MsgMapFileEntry, 628
Index, 628
MsgId, 628
CFE_SB_MsgMapFileEntry_t
cfe_sb_msg.h, 1148
CFE_SB_MsgPayloadPtr_t
cfe_sb_api_typedefs.h, 934
CFE_SB_MsgPtr_t
cfe_sb_api_typedefs.h, 935
CFE_SB_NO_MESSAGE
cFE Return Code Defines, 156
CFE_SB_NO_MSG_RECV
cFE Return Code Defines, 156
CFE_SB_NO_SUBSCRIBERS
cFE Return Code Defines, 156
CFE_SB_NOOP_CC
cfe_sb_msg.h, 1139
CFE_SB_NOT_IMPLEMENTED
cFE Return Code Defines, 156
CFE_SB_NoopCmd_t
cfe_sb_msg.h, 1148
CFE_SB_ONESUB_TLM_MID
cpu1_msgids.h, 773
CFE_SB_PART_SUB_PKT_EID
cfe_sb_events.h, 1121
CFE_SB_PEND_FOREVER
cfe_sb_api_typedefs.h, 932
CFE_SB_PIPE_ADDED_EID
cfe_sb_events.h, 1121
CFE_SB_PIPE_CR_ERR
cFE Return Code Defines, 157
CFE_SB_PIPE_DELETED_EID
cfe_sb_events.h, 1122
CFE_SB_PIPE_RD_ERR
cFE Return Code Defines, 157
CFE_SB_PIPEID_C
cfe_sb_api_typedefs.h, 932
CFE_SB_PIPEOPTS_IGNOREMINE
cfe_sb_api_typedefs.h, 933
CFE_SB_PKTTYPE_CMD
cFE SB Packet Type Defines, 323
CFE_SB_PKTTYPE_INVALID
cFE SB Packet Type Defines, 323
CFE_SB_PKTTYPE_TLM
cFE SB Packet Type Defines, 323
CFE_SB_POLL
cfe_sb_api_typedefs.h, 933
CFE_SB_PassMsg
cFE Send/Receive Message APIs, 294
CFE_SB_PipeDepthStats, 629
CurrentQueueDepth, 629
MaxQueueDepth, 629
PeakQueueDepth, 630
PipeId, 630
Spare, 630
CFE_SB_PipeDepthStats_t
cfe_sb_msg.h, 1148
CFE_SB_PipeId_ToIndex
cFE Pipe Management APIs, 287
CFE_SB_PipeId_t
cfe_sb_extern_typedefs.h, 937
CFE_SB_PipeInfoEntry, 631
AppId, 631
AppName, 631
CurrentQueueDepth, 632
MaxQueueDepth, 632
Opts, 632
PeakQueueDepth, 632
PipeId, 632
PipeName, 633
SendErrors, 633
Spare, 633
CFE_SB_PipeInfoEntry_t
cfe_sb_msg.h, 1148
CFE_SB_Q_FULL_ERR_EID
cfe_sb_events.h, 1122
CFE_SB_Q_RD_ERR_EID
cfe_sb_events.h, 1123
CFE_SB_Q_WR_ERR_EID
cfe_sb_events.h, 1123
CFE_SB_Qos_t, 633
Priority, 634
Reliability, 634
CFE_SB_QosPriority
cfe_sb_extern_typedefs.h, 938
CFE_SB_QosPriority_Enum_t
cfe_sb_extern_typedefs.h, 937
CFE_SB_QosReliability
cfe_sb_extern_typedefs.h, 939
CFE_SB_QosReliability_Enum_t
cfe_sb_extern_typedefs.h, 938
CFE_SB_RCV_BAD_ARG_EID
cfe_sb_events.h, 1124
CFE_SB_RESET_COUNTERS_CC
cfe_sb_msg.h, 1140
CFE_SB_RcvMsg
cFE Send/Receive Message APIs, 295
CFE_SB_ReceiveBuffer
cFE Send/Receive Message APIs, 295
CFE_SB_ResetCountersCmd_t
cfe_sb_msg.h, 1149
CFE_SB_RouteCmd, 634
Hdr, 635
Payload, 635
CFE_SB_RouteCmd_Payload, 635
MsgId, 636

Pipe, 636
Spare, 636
CFE_SB_RouteCmd_Payload_t
cfe_sb_msg.h, 1149
CFE_SB_RouteCmd_t
cfe_sb_msg.h, 1149
CFE_SB_RouteId_Atom_t
cfe_sb_extern_typedefs.h, 938
CFE_SB_RoutingFileEntry, 637
AppName, 637
MsgCnt, 638
MsgId, 638
PipeId, 638
PipeName, 638
State, 638
CFE_SB_RoutingFileEntry_t
cfe_sb_msg.h, 1149
CFE_SB_SEND_BAD_ARG_EID
cfe_sb_events.h, 1124
CFE_SB_SEND_HK_MID
cpu1_msgids.h, 773
CFE_SB_SEND_INV_MSGID_EID
cfe_sb_events.h, 1125
CFE_SB_SEND_NO_SUBS_EID
cfe_sb_events.h, 1125
CFE_SB_SEND_PREV_SUBS_CC
cfe_sb_msg.h, 1141
CFE_SB_SEND_SB_STATS_CC
cfe_sb_msg.h, 1142
CFE_SB_SETPIPEOPTS_EID
cfe_sb_events.h, 1126
CFE_SB_SETPIPEOPTS_ID_ERR_EID
cfe_sb_events.h, 1126
CFE_SB_SETPIPEOPTS_OWNER_ERR_EID
cfe_sb_events.h, 1127
CFE_SB_SND_RTG_EID
cfe_sb_events.h, 1127
CFE_SB_SND_RTG_ERR1_EID
cfe_sb_events.h, 1128
CFE_SB_SND_STATS_EID
cfe_sb_events.h, 1128
CFE_SB_STATS_TLM_MID
cpu1_msgids.h, 773
CFE_SB_SUB_ARG_ERR_EID
cfe_sb_events.h, 1129
CFE_SB_SUB_ENTRIES_PER_PKT
cfe_sb_extern_typedefs.h, 936
CFE_SB_SUB_INV_CALLER_EID
cfe_sb_events.h, 1129
CFE_SB_SUB_INV_PIPE_EID
cfe_sb_events.h, 1130
CFE_SB_SUB_RPT_CTRL_MID
cpu1_msgids.h, 773
CFE_SB_SUBSCRIPTION_RCVD_EID
cfe_sb_events.h, 1130
CFE_SB_SUBSCRIPTION_REMOVED_EID
cfe_sb_events.h, 1131
CFE_SB_SUBSCRIPTION_RPT_EID
cfe_sb_events.h, 1131
CFE_SB_SUBSCRIPTION
cfe_sb_api_typedefs.h, 933
CFE_SB_SendMsg
cFE Send/Receive Message APIs, 296
CFE_SB_SendPrevSubsCmd_t
cfe_sb_msg.h, 1149
CFE_SB_SendSbStatsCmd_t
cfe_sb_msg.h, 1150
CFE_SB_SetCmdCode
cFE Setting Message Characteristics APIs, 306
CFE_SB_SetMsgId
cFE Setting Message Characteristics APIs, 306
CFE_SB_SetMsgTime
cFE Setting Message Characteristics APIs, 307
CFE_SB_SetPipeOpts
cFE Pipe Management APIs, 288
CFE_SB_SetTotalMsgLength
cFE Setting Message Characteristics APIs, 308
CFE_SB_SetUserDataLength
cFE Setting Message Characteristics APIs, 308
CFE_SB_SingleSubscriptionTlm, 639
Hdr, 639
Payload, 639
CFE_SB_SingleSubscriptionTlm_Payload, 640
MsgId, 640
Pipe, 641
Qos, 641
SubType, 641
CFE_SB_SingleSubscriptionTlm_Payload_t
cfe_sb_msg.h, 1150
CFE_SB_SingleSubscriptionTlm_t
cfe_sb_msg.h, 1150
CFE_SB_StatsTlm, 641
Hdr, 642
Payload, 642
CFE_SB_StatsTlm_Payload, 642
MaxMemAllowed, 643
MaxMsgIdsAllowed, 644
MaxPipeDepthAllowed, 644
MaxPipesAllowed, 644
MaxSubscriptionsAllowed, 644
MemInUse, 645
MsgIdsInUse, 645
PeakMemInUse, 645
PeakMsgIdsInUse, 645
PeakPipesInUse, 646
PeakSBBuffersInUse, 646
PeakSubscriptionsInUse, 646
PipeDepthStats, 646

PipesInUse, 647
SBBuffersInUse, 647
SubscriptionsInUse, 647
CFE_SB_StatsTlm_Payload_t
cfe_sb_msg.h, 1150
CFE_SB_StatsTlm_t
cfe_sb_msg.h, 1150
CFE_SB_SubEntries, 648
MsgId, 648
Pipe, 648
Qos, 648
CFE_SB_SubEntries_t
cfe_sb_msg.h, 1151
CFE_SB_Subscribe
cFE Message Subscription Control APIs, 289
CFE_SB_SubscribeEx
cFE Message Subscription Control APIs, 290
CFE_SB_SubscribeLocal
cFE Message Subscription Control APIs, 291
CFE_SB_TIME_OUT
cFE Return Code Defines, 157
CFE_SB_TLM_HDR_SIZE
cfe_sb_api_typedefs.h, 933
CFE_SB_TaskMain
cFE Internal Software Bus APIs, internal to CFE core, 324
CFE_SB_TimeStampMsg
cFE Setting Message Characteristics APIs, 309
CFE_SB_TlmHdr_t
cfe_sb_api_typedefs.h, 935
CFE_SB_TransmitBuffer
cFE Zero Copy APIs, 299
CFE_SB_TransmitMsg
cFE Send/Receive Message APIs, 297
CFE_SB_UNSUB_ARG_ERR_EID
cfe_sb_events.h, 1132
CFE_SB_UNSUB_INV_CALLER_EID
cfe_sb_events.h, 1132
CFE_SB_UNSUB_INV_PIPE_EID
cfe_sb_events.h, 1133
CFE_SB_UNSUB_NO_SUBS_EID
cfe_sb_events.h, 1133
CFE_SB_UNSUBSCRIPTION
cfe_sb_api_typedefs.h, 934
CFE_SB_Unsubscribe
cFE Message Subscription Control APIs, 292
CFE_SB_UnsubscribeLocal
cFE Message Subscription Control APIs, 293
CFE_SB_ValidateChecksum
cFE Checksum Control APIs, 317
CFE_SB_ValueToMsgId
cFE Message ID APIs, 321
CFE_SB_WRITE_MAP_INFO_CC
cfe_sb_msg.h, 1143
CFE_SB_WRITE_PIPE_INFO_CC
cfe_sb_msg.h, 1144
CFE_SB_WRITE_ROUTING_INFO_CC
cfe_sb_msg.h, 1145
CFE_SB_WRONG_MSG_TYPE
cFE Return Code Defines, 157
CFE_SB_WriteFileInfoCmd, 649
Hdr, 649
Payload, 649
CFE_SB_WriteFileInfoCmd_Payload, 650
Filename, 650
CFE_SB_WriteFileInfoCmd_Payload_t
cfe_sb_msg.h, 1151
CFE_SB_WriteFileInfoCmd_t
cfe_sb_msg.h, 1151
CFE_SB_WriteMapInfoCmd_t
cfe_sb_msg.h, 1151
CFE_SB_WritePipeInfoCmd_t
cfe_sb_msg.h, 1151
CFE_SB_WriteRoutingInfoCmd_t
cfe_sb_msg.h, 1151
CFE_SB_ZeroCopyGetPtr
cFE Zero Copy APIs, 300
CFE_SB_ZeroCopyHandle_t, 651
BufDscPtr, 651
CFE_SB_ZeroCopyPass
cFE Zero Copy APIs, 301
CFE_SB_ZeroCopyReleasePtr
cFE Zero Copy APIs, 302
CFE_SB_ZeroCopySend
cFE Zero Copy APIs, 302
CFE_SERVICE_BITMASK
cfe_error.h, 866
CFE_SEVERITY_BITMASK
cfe_error.h, 866
CFE_SEVERITY_ERROR
cfe_error.h, 866
CFE_SEVERITY_INFO
cfe_error.h, 866
CFE_SEVERITY_SUCCESS
cfe_error.h, 866
CFE_SET
cfe_sb.h, 927
CFE_SOFTWARE_BUS_SERVICE
cfe_error.h, 867
CFE_SRC_VERSION
cfe_version.h, 960
CFE_STATUS_BAD_COMMAND_CODE
cFE Return Code Defines, 158
CFE_STATUS_EXTERNAL_RESOURCE_FAIL
cFE Return Code Defines, 158
CFE_STATUS_NO_COUNTER_INCREMENT
cFE Return Code Defines, 158
CFE_STATUS_NOT_IMPLEMENTED

cFE Return Code Defines, 158
CFE_STATUS_REQUEST_ALREADY_PENDING
cFE Return Code Defines, 159
CFE_STATUS_UNKNOWN_MSG_ID
cFE Return Code Defines, 159
CFE_STATUS_WRONG_MSG_LENGTH
cFE Return Code Defines, 159
CFE_STR_HELPER
cfe_version.h, 960
CFE_STR
cfe_version.h, 960
CFE_SUCCESS
cFE Return Code Defines, 159
CFE_Status_t
cfe_error.h, 867
CFE_TABLE_SERVICE
cfe_error.h, 867
CFE_TBL_ABORT_LOAD_CC
cfe_tbl_msg.h, 1192
CFE_TBL_ACTIVATE_CC
cfe_tbl_msg.h, 1193
CFE_TBL_ACTIVATE_DUMP_ONLY_ERR_EID
cfe_tbl_events.h, 1155
CFE_TBL_ACTIVATE_ERR_EID
cfe_tbl_events.h, 1155
CFE_TBL_ASSUMED_VALID_INF_EID
cfe_tbl_events.h, 1155
CFE_TBL_AbortLoadCmd, 651
CmdHeader, 652
Payload, 652
CFE_TBL_AbortLoadCmd_Payload, 652
TableName, 653
CFE_TBL_AbortLoadCmd_Payload_t
cfe_tbl_msg.h, 1202
CFE_TBL_AbortLoadCmd_t
cfe_tbl_msg.h, 1202
CFE_TBL_ActivateCmd, 653
CmdHeader, 654
Payload, 654
CFE_TBL_ActivateCmd_Payload, 654
TableName, 655
CFE_TBL_ActivateCmd_Payload_t
cfe_tbl_msg.h, 1202
CFE_TBL_ActivateCmd_t
cfe_tbl_msg.h, 1202
CFE_TBL_BAD_ARGUMENT
cFE Return Code Defines, 160
CFE_TBL_BAD_TABLE_HANDLE
cfe_tbl_api_typedefs.h, 941
CFE_TBL_BufferSelect
cfe_tbl_extern_typedefs.h, 945
CFE_TBL_BufferSelect_Enum_t
cfe_tbl_extern_typedefs.h, 944
CFE_TBL_CC1_ERR_EID
cfe_tbl_events.h, 1156
CFE_TBL_CDS_DELETE_ERR_EID
cfe_tbl_events.h, 1156
CFE_TBL_CDS_DELETED_INFO_EID
cfe_tbl_events.h, 1157
CFE_TBL_CDS_NOT_FOUND_ERR_EID
cfe_tbl_events.h, 1157
CFE_TBL_CDS_OWNER_ACTIVE_ERR_EID
cfe_tbl_events.h, 1158
CFE_TBL_CMD_MID
cpu1_msgids.h, 774
CFE_TBL_CREATING_DUMP_FILE_ERR_EID
cfe_tbl_events.h, 1158
CFE_TBL_CallbackFuncPtr_t
cfe_tbl_api_typedefs.h, 942
CFE_TBL_CleanUpApp
cFE Internal Table Service APIs, internal to CFE core, 352
CFE_TBL_DELETE_CDS_CC
cfe_tbl_msg.h, 1194
CFE_TBL_DUMP_CC
cfe_tbl_msg.h, 1194
CFE_TBL_DUMP_PENDING_ERR_EID
cfe_tbl_events.h, 1159
CFE_TBL_DUMP_REGISTRY_CC
cfe_tbl_msg.h, 1195
CFE_TBL_DelCDSCmd_Payload, 655
TableName, 655
CFE_TBL_DelCDSCmd_Payload_t
cfe_tbl_msg.h, 1202
CFE_TBL_DeleteCDSCmd, 656
CmdHeader, 656
Payload, 656
CFE_TBL_DeleteCDSCmd_t
cfe_tbl_msg.h, 1202
CFE_TBL_DumpCmd, 657
CmdHeader, 657
Payload, 657
CFE_TBL_DumpCmd_Payload, 658
ActiveTableFlag, 658
DumpFilename, 659
TableName, 659
CFE_TBL_DumpCmd_Payload_t
cfe_tbl_msg.h, 1203
CFE_TBL_DumpCmd_t
cfe_tbl_msg.h, 1203
CFE_TBL_DumpRegistryCmd, 659
CmdHeader, 660
Payload, 660
CFE_TBL_DumpRegistryCmd_Payload, 660
DumpFilename, 661
CFE_TBL_DumpRegistryCmd_Payload_t
cfe_tbl_msg.h, 1203
CFE_TBL_DumpRegistryCmd_t

cfe_tbl_msg.h, 1203
CFE_TBL_DumpToBuffer
cFE Manage Table Content APIs, 332
CFE_TBL_ERR_ACCESS
cFE Return Code Defines, 160
CFE_TBL_ERR_BAD_APP_ID
cFE Return Code Defines, 160
CFE_TBL_ERR_BAD_CONTENT_ID
cFE Return Code Defines, 160
CFE_TBL_ERR_BAD_PROCESSOR_ID
cFE Return Code Defines, 161
CFE_TBL_ERR_BAD_SPACECRAFT_ID
cFE Return Code Defines, 161
CFE_TBL_ERR_BAD_SUBTYPE_ID
cFE Return Code Defines, 161
CFE_TBL_ERR_DUMP_ONLY
cFE Return Code Defines, 161
CFE_TBL_ERR_DUPLICATE_DIFF_SIZE
cFE Return Code Defines, 162
CFE_TBL_ERR_DUPLICATE_NOT_OWNED
cFE Return Code Defines, 162
CFE_TBL_ERR_FILE_FOR_WRONG_TABLE
cFE Return Code Defines, 162
CFE_TBL_ERR_FILE_NOT_FOUND
cFE Return Code Defines, 162
CFE_TBL_ERR_FILE_SIZE_INCONSISTENT
cFE Return Code Defines, 163
CFE_TBL_ERR_FILE_TOO_LARGE
cFE Return Code Defines, 163
CFE_TBL_ERR_FILENAME_TOO_LONG
cFE Return Code Defines, 163
CFE_TBL_ERR_HANDLES_FULL
cFE Return Code Defines, 163
CFE_TBL_ERR_ILLEGAL_SRC_TYPE
cFE Return Code Defines, 164
CFE_TBL_ERR_INVALID_HANDLE
cFE Return Code Defines, 164
CFE_TBL_ERR_INVALID_NAME
cFE Return Code Defines, 164
CFE_TBL_ERR_INVALID_OPTIONS
cFE Return Code Defines, 164
CFE_TBL_ERR_INVALID_SIZE
cFE Return Code Defines, 165
CFE_TBL_ERR_LOAD_IN_PROGRESS
cFE Return Code Defines, 165
CFE_TBL_ERR_LOAD_INCOMPLETE
cFE Return Code Defines, 165
CFE_TBL_ERR_NEVER_LOADED
cFE Return Code Defines, 166
CFE_TBL_ERR_NO_ACCESS
cFE Return Code Defines, 166
CFE_TBL_ERR_NO_BUFFER_AVAIL
cFE Return Code Defines, 166
CFE_TBL_ERR_NO_STD_HEADER
cFE Return Code Defines, 166
CFE_TBL_ERR_NO_TBL_HEADER
cFE Return Code Defines, 167
CFE_TBL_ERR_PARTIAL_LOAD
cFE Return Code Defines, 167
CFE_TBL_ERR_REGISTRY_FULL
cFE Return Code Defines, 167
CFE_TBL_ERR_SHORT_FILE
cFE Return Code Defines, 167
CFE_TBL_ERR_UNREGISTERED
cFE Return Code Defines, 168
CFE_TBL_EarlyInit
cFE Internal Table Service APIs, internal to CFE core, 352
CFE_TBL_FAIL_HK_SEND_ERR_EID
cfe_tbl_events.h, 1159
CFE_TBL_FAIL_NOTIFY_SEND_ERR_EID
cfe_tbl_events.h, 1160
CFE_TBL_FILE_ACCESS_ERR_EID
cfe_tbl_events.h, 1160
CFE_TBL_FILE_INCOMPLETE_ERR_EID
cfe_tbl_events.h, 1161
CFE_TBL_FILE_LOADED_INF_EID
cfe_tbl_events.h, 1161
CFE_TBL_FILE_STD_HDR_ERR_EID
cfe_tbl_events.h, 1162
CFE_TBL_FILE_SUBTYPE_ERR_EID
cfe_tbl_events.h, 1162
CFE_TBL_FILE_TBL_HDR_ERR_EID
cfe_tbl_events.h, 1163
CFE_TBL_FILE_TOO_BIG_ERR_EID
cfe_tbl_events.h, 1163
CFE_TBL_FILE_TYPE_ERR_EID
cfe_tbl_events.h, 1164
CFE_TBL_FILEDEF
cfe_tbl_filedef.h, 946
CFE_TBL_File_Hdr, 661
NumBytes, 662
Offset, 662
Reserved, 662
TableName, 662
CFE_TBL_File_Hdr_t
cfe_tbl_extern_typedefs.h, 944
CFE_TBL_FileDef, 662
Description, 663
ObjectName, 663
ObjectSize, 663
TableName, 663
TgtFilename, 664
CFE_TBL_FileDef_t
cfe_tbl_filedef.h, 947
CFE_TBL_GetAddress
cFE Access Table Content APIs, 339
CFE_TBL_GetAddresses

cFE Access Table Content APIs, 340
CFE_TBL_GetInfo
cFE Get Table Information APIs, 345
CFE_TBL_GetStatus
cFE Get Table Information APIs, 346
CFE_TBL_HANDLE_ACCESS_ERR_EID
cfe_tbl_events.h, 1164
CFE_TBL_HK_TLM_MID
cpu1_msgids.h, 774
CFE_TBL_Handle_t
cfe_tbl_api_typedefs.h, 942
CFE_TBL_HousekeepingTlm, 664
Payload, 664
TlmHeader, 665
CFE_TBL_HousekeepingTlm_Payload, 665
ActiveBuffer, 666
ByteAlignPad1, 666
CommandCounter, 667
CommandErrorCounter, 667
FailedValCounter, 667
LastFileDumped, 667
LastFileLoaded, 668
LastTableLoaded, 668
LastUpdateTime, 668
LastUpdatedTable, 668
LastValCrc, 669
LastValStatus, 669
LastValTableName, 669
MemPoolHandle, 669
NumFreeSharedBufs, 670
NumLoadPending, 670
NumTables, 670
NumValRequests, 670
SuccessValCounter, 671
ValidationCounter, 671
CFE_TBL_HousekeepingTlm_Payload_t
cfe_tbl_msg.h, 1203
CFE_TBL_HousekeepingTlm_t
cfe_tbl_msg.h, 1203
CFE_TBL_ILLEGAL_BUFF_PARAM_ERR_EID
cfe_tbl_events.h, 1165
CFE_TBL_IN_REGISTRY_ERR_EID
cfe_tbl_events.h, 1165
CFE_TBL_INFO_DUMP_PENDING
cFE Return Code Defines, 168
CFE_TBL_INFO_NO_UPDATE_PENDING
cFE Return Code Defines, 168
CFE_TBL_INFO_NO_VALIDATION_PENDING
cFE Return Code Defines, 168
CFE_TBL_INFO_RECOVERED_TBL
cFE Return Code Defines, 169
CFE_TBL_INFO_TABLE_LOCKED
cFE Return Code Defines, 169
CFE_TBL_INFO_UPDATE_PENDING
cFE Return Code Defines, 169
CFE_TBL_INFO_UPDATED
cFE Return Code Defines, 169
CFE_TBL_INFO_VALIDATION_PENDING
cFE Return Code Defines, 170
CFE_TBL_INIT_INF_EID
cfe_tbl_events.h, 1166
CFE_TBL_INTERNAL_ERROR_ERR_EID
cfe_tbl_events.h, 1166
CFE_TBL_Info, 671
Crc, 672
Critical, 672
DoubleBuffered, 673
DumpOnly, 673
FileCreateTimeSecs, 673
FileCreateTimeSubSecs, 673
LastFileLoaded, 673
NumUsers, 674
Size, 674
TableLoadedOnce, 674
TimeOfLastUpdate, 674
UserDefAddr, 674
CFE_TBL_Info_t
cfe_tbl_api_typedefs.h, 942
CFE_TBL_LEN_ERR_EID
cfe_tbl_events.h, 1167
CFE_TBL_LOAD_ABORT_ERR_EID
cfe_tbl_events.h, 1167
CFE_TBL_LOAD_ABORT_INF_EID
cfe_tbl_events.h, 1168
CFE_TBL_LOAD_CC
cfe_tbl_msg.h, 1196
CFE_TBL_LOAD_DUMPONLY_ERR_EID
cfe_tbl_events.h, 1168
CFE_TBL_LOAD_EXCEEDS_SIZE_ERR_EID
cfe_tbl_events.h, 1169
CFE_TBL_LOAD_FILENAME_LONG_ERR_EID
cfe_tbl_events.h, 1169
CFE_TBL_LOAD_IN_PROGRESS_ERR_EID
cfe_tbl_events.h, 1170
CFE_TBL_LOAD_PEND_REQ_INF_EID
cfe_tbl_events.h, 1170
CFE_TBL_LOAD_SUCCESS_INF_EID
cfe_tbl_events.h, 1171
CFE_TBL_LOAD_TBLNAME_MISMATCH_ERR_EID
cfe_tbl_events.h, 1171
CFE_TBL_LOAD_TYPE_ERR_EID
cfe_tbl_events.h, 1172
CFE_TBL_LOAD_VAL_ERR_EID
cfe_tbl_events.h, 1172
CFE_TBL_LOADING_A_DUMP_ONLY_ERR_EID
cfe_tbl_events.h, 1172
CFE_TBL_LOADING_PENDING_ERR_EID
cfe_tbl_events.h, 1173

CFE_TBL_Load
cFE Manage Table Content APIs, 333
CFE_TBL_LoadCmd, 675
CmdHeader, 675
Payload, 675
CFE_TBL_LoadCmd_Payload, 676
LoadFilename, 676
CFE_TBL_LoadCmd_Payload_t
cfe_tbl_msg.h, 1204
CFE_TBL_LoadCmd_t
cfe_tbl_msg.h, 1204
CFE_TBL_MAX_EID
cfe_tbl_events.h, 1173
CFE_TBL_MAX_FULL_NAME_LEN
cfe_tbl_api_typedefs.h, 942
CFE_TBL_MESSAGE_ERROR
cFE Return Code Defines, 170
CFE_TBL_MID_ERR_EID
cfe_tbl_events.h, 1174
CFE_TBL_Manage
cFE Manage Table Content APIs, 334
CFE_TBL_Modified
cFE Manage Table Content APIs, 335
CFE_TBL_NO_INACTIVE_BUFFER_ERR_EID
cfe_tbl_events.h, 1174
CFE_TBL_NO_SUCH_TABLE_ERR_EID
cfe_tbl_events.h, 1174
CFE_TBL_NO_WORK_BUFFERS_ERR_EID
cfe_tbl_events.h, 1175
CFE_TBL_NOOP_CC
cfe_tbl_msg.h, 1197
CFE_TBL_NOOP_INF_EID
cfe_tbl_events.h, 1175
CFE_TBL_NOT_CRITICAL_TBL_ERR_EID
cfe_tbl_events.h, 1176
CFE_TBL_NOT_IMPLEMENTED
cFE Return Code Defines, 170
CFE_TBL_NOT_IN_CRIT_REG_ERR_EID
cfe_tbl_events.h, 1176
CFE_TBL_NoArgsCmd, 677
CmdHeader, 677
CFE_TBL_NoArgsCmd_t
cfe_tbl_msg.h, 1204
CFE_TBL_NoopCmd_t
cfe_tbl_msg.h, 1204
CFE_TBL_NotifyByMessage
cFE Get Table Information APIs, 347
CFE_TBL_NotifyCmd, 677
CmdHeader, 678
Payload, 678
CFE_TBL_NotifyCmd_Payload, 678
Parameter, 679
CFE_TBL_NotifyCmd_Payload_t
cfe_tbl_msg.h, 1204
CFE_TBL_NotifyCmd_t
cfe_tbl_msg.h, 1205
CFE_TBL_OPT_BUFFER_MSK
cFE Table Type Defines, 349
CFE_TBL_OPT_CRITICAL_MSK
cFE Table Type Defines, 350
CFE_TBL_OPT_CRITICAL
cFE Table Type Defines, 349
CFE_TBL_OPT_DBL_BUFFER
cFE Table Type Defines, 350
CFE_TBL_OPT_DEFAULT
cFE Table Type Defines, 350
CFE_TBL_OPT_DUMP_ONLY
cFE Table Type Defines, 350
CFE_TBL_OPT_LD_DMP_MSK
cFE Table Type Defines, 350
CFE_TBL_OPT_LOAD_DUMP
cFE Table Type Defines, 351
CFE_TBL_OPT_NOT_CRITICAL
cFE Table Type Defines, 351
CFE_TBL_OPT_NOT_USR_DEF
cFE Table Type Defines, 351
CFE_TBL_OPT_SNGL_BUFFER
cFE Table Type Defines, 351
CFE_TBL_OPT_USR_DEF_ADDR
cFE Table Type Defines, 351
CFE_TBL_OPT_USR_DEF_MSK
cFE Table Type Defines, 351
CFE_TBL_OVERWRITE_DUMP_INF_EID
cfe_tbl_events.h, 1177
CFE_TBL_OVERWRITE_REG_DUMP_INF_EID
cfe_tbl_events.h, 1177
CFE_TBL_PARTIAL_LOAD_ERR_EID
cfe_tbl_events.h, 1178
CFE_TBL_PROCESSOR_ID_ERR_EID
cfe_tbl_events.h, 1178
CFE_TBL_REG_TLM_MID
cpu1_msgids.h, 774
CFE_TBL_REGISTER_ERR_EID
cfe_tbl_events.h, 1179
CFE_TBL_RESET_COUNTERS_CC
cfe_tbl_msg.h, 1198
CFE_TBL_RESET_INF_EID
cfe_tbl_events.h, 1179
CFE_TBL_Register
cFE Registration APIs, 326
CFE_TBL_ReleaseAddress
cFE Access Table Content APIs, 341
CFE_TBL_ReleaseAddresses
cFE Access Table Content APIs, 343
CFE_TBL_ResetCountersCmd_t
cfe_tbl_msg.h, 1205
CFE_TBL_SEND_HK_MID
cpu1_msgids.h, 774

CFE_TBL_SEND_REGISTRY_CC
cfe_tbl_msg.h, 1199
CFE_TBL_SHARE_ERR_EID
cfe_tbl_events.h, 1180
CFE_TBL_SPACECRAFT_ID_ERR_EID
cfe_tbl_events.h, 1180
CFE_TBL_SendRegistryCmd, 679
CmdHeader, 680
Payload, 680
CFE_TBL_SendRegistryCmd_Payload, 680
TableName, 681
CFE_TBL_SendRegistryCmd_Payload_t
cfe_tbl_msg.h, 1205
CFE_TBL_SendRegistryCmd_t
cfe_tbl_msg.h, 1205
CFE_TBL_Share
cFE Registration APIs, 329
CFE_TBL_SrcEnum
cfe_tbl_api_typedefs.h, 943
CFE_TBL_SrcEnum_t
cfe_tbl_api_typedefs.h, 942
CFE_TBL_TLM_REG_CMD_INF_EID
cfe_tbl_events.h, 1181
CFE_TBL_TOO_MANY_DUMPS_ERR_EID
cfe_tbl_events.h, 1181
CFE_TBL_TOO_MANY_VALIDATIONS_ERR_EID
cfe_tbl_events.h, 1182
CFE_TBL_TableRegistryTlm, 681
Payload, 682
TlmHeader, 682
CFE_TBL_TableRegistryTlm_t
cfe_tbl_msg.h, 1205
CFE_TBL_TaskMain
cFE Internal Table Service APIs, internal to CFE core, 352
CFE_TBL_TblRegPacket_Payload, 682
ActiveBufferAddr, 683
ByteAlign4, 683
Crc, 684
Critical, 684
DoubleBuffered, 684
DumpOnly, 684
FileCreateTimeSecs, 685
FileCreateTimeSubSecs, 685
InactiveBufferAddr, 685
LastFileLoaded, 685
LoadPending, 686
Name, 686
OwnerAppName, 686
Size, 686
TableLoadedOnce, 687
TimeOfLastUpdate, 687
ValidationFuncPtr, 687
CFE_TBL_TblRegPacket_Payload_t
cfe_tbl_msg.h, 1205
CFE_TBL_UNREGISTER_ERR_EID
cfe_tbl_events.h, 1182
CFE_TBL_UNVALIDATED_ERR_EID
cfe_tbl_events.h, 1183
CFE_TBL_UPDATE_ERR_EID
cfe_tbl_events.h, 1183
CFE_TBL_UPDATE_SUCCESS_INF_EID
cfe_tbl_events.h, 1184
CFE_TBL_Unregister
cFE Registration APIs, 330
CFE_TBL_Update
cFE Manage Table Content APIs, 336
CFE_TBL_VAL_REQ_MADE_INF_EID
cfe_tbl_events.h, 1184
CFE_TBL_VALIDATE_CC
cfe_tbl_msg.h, 1200
CFE_TBL_VALIDATION_ERR_EID
cfe_tbl_events.h, 1185
CFE_TBL_VALIDATION_INF_EID
cfe_tbl_events.h, 1185
CFE_TBL_Validate
cFE Manage Table Content APIs, 337
CFE_TBL_ValidateCmd, 688
CmdHeader, 688
Payload, 688
CFE_TBL_ValidateCmd_Payload, 689
ActiveTableFlag, 689
TableName, 689
CFE_TBL_ValidateCmd_Payload_t
cfe_tbl_msg.h, 1206
CFE_TBL_ValidateCmd_t
cfe_tbl_msg.h, 1206
CFE_TBL_WARN_DUPLICATE
cFE Return Code Defines, 170
CFE_TBL_WARN_NOT_CRITICAL
cFE Return Code Defines, 171
CFE_TBL_WARN_PARTIAL_LOAD
cFE Return Code Defines, 171
CFE_TBL_WARN_SHORT_FILE
cFE Return Code Defines, 171
CFE_TBL_WRITE_CFE_HDR_ERR_EID
cfe_tbl_events.h, 1186
CFE_TBL_WRITE_DUMP_INF_EID
cfe_tbl_events.h, 1186
CFE_TBL_WRITE_REG_DUMP_INF_EID
cfe_tbl_events.h, 1187
CFE_TBL_WRITE_TBL_HDR_ERR_EID
cfe_tbl_events.h, 1187
CFE_TBL_WRITE_TBL_IMG_ERR_EID
cfe_tbl_events.h, 1188
CFE_TBL_WRITE_TBL_REG_ERR_EID
cfe_tbl_events.h, 1188
CFE_TBL_ZERO_LENGTH_LOAD_ERR_EID

cfe_tbl_events.h, 1189
CFE_TIME_1HZ_CFG_EID
cfe_time_events.h, 1208
CFE_TIME_1HZ_CMD_MID
cpu1_msgids.h, 774
CFE_TIME_1HZ_EID
cfe_time_events.h, 1208
CFE_TIME_1HZ_ERR_EID
cfe_time_events.h, 1209
CFE_TIME_1HzCmd_t
cfe_time_msg.h, 1246
CFE_TIME_ADD_1HZ_ADJUSTMENT_CC
cfe_time_msg.h, 1230
CFE_TIME_ADD_ADJUST_CC
cfe_time_msg.h, 1230
CFE_TIME_ADD_DELAY_CC
cfe_time_msg.h, 1231
CFE_TIME_Add
cFE Time Arithmetic APIs, 361
CFE_TIME_Add1HZAdjustmentCmd_t
cfe_time_msg.h, 1247
CFE_TIME_AddAdjustCmd_t
cfe_time_msg.h, 1247
CFE_TIME_AddDelayCmd_t
cfe_time_msg.h, 1247
CFE_TIME_AdjustDirection
cfe_time_extern_typedefs.h, 955
CFE_TIME_AdjustDirection_Enum_t
cfe_time_extern_typedefs.h, 953
CFE_TIME_BAD_ARGUMENT
cFE Return Code Defines, 171
CFE_TIME_CALLBACK_NOT_REGISTERED
cFE Return Code Defines, 172
CFE_TIME_CC_ERR_EID
cfe_time_events.h, 1209
CFE_TIME_CMD_MID
cpu1_msgids.h, 774
CFE_TIME_CleanUpApp
cFE Internal Time APIs, internal to CFE core, 374
CFE_TIME_ClockState
cfe_time_extern_typedefs.h, 955
CFE_TIME_ClockState_Enum_t
cfe_time_extern_typedefs.h, 953
CFE_TIME_Compare
cFE Time Arithmetic APIs, 362
cfe_time_api_typedefs.h, 950
CFE_TIME_Compare_t
cfe_time_api_typedefs.h, 950
CFE_TIME_Copy
cfe_time.h, 949
CFE_TIME_DATA_CMD_MID
cpu1_msgids.h, 775
CFE_TIME_DELAY_CFG_EID
cfe_time_events.h, 1210
CFE_TIME_DELAY_EID
cfe_time_events.h, 1210
CFE_TIME_DELAY_ERR_EID
cfe_time_events.h, 1211
CFE_TIME_DELTA_CFG_EID
cfe_time_events.h, 1211
CFE_TIME_DELTA_EID
cfe_time_events.h, 1212
CFE_TIME_DELTA_ERR_EID
cfe_time_events.h, 1212
CFE_TIME_DIAG_EID
cfe_time_events.h, 1213
CFE_TIME_DIAG_TLM_MID
cpu1_msgids.h, 775
CFE_TIME_DiagnosticTlm, 690
Payload, 690
TlmHeader, 690
CFE_TIME_DiagnosticTlm_Payload, 690
AtToneDelay, 693
AtToneLatch, 693
AtToneLeapSeconds, 693
AtToneMET, 693
AtToneSTCF, 693
ClockFlyState, 694
ClockSetState, 694
ClockSignal, 694
ClockSource, 694
ClockStateAPI, 695
ClockStateFlags, 695
CurrentLatch, 695
CurrentMET, 695
CurrentTAI, 696
CurrentUTC, 696
DataStoreStatus, 696
DelayDirection, 696
Forced2Fly, 697
LocalIntCounter, 697
LocalTaskCounter, 697
MaxElapsed, 697
MaxLocalClock, 698
MinElapsed, 698
OneHzAdjust, 698
OneHzDirection, 698
OneTimeAdjust, 699
OneTimeDirection, 699
ServerFlyState, 699
TimeSinceTone, 699
ToneDataCounter, 700
ToneDataLatch, 700
ToneIntCounter, 700
ToneIntErrorCounter, 700
ToneMatchCounter, 701
ToneMatchErrorCounter, 701
ToneOverLimit, 701

ToneSignalCounter, 701
ToneSignalLatch, 702
ToneTaskCounter, 702
ToneUnderLimit, 702
VersionCounter, 702
VirtualMET, 703
CFE_TIME_DiagnosticTlm_Payload_t
cfe_time_msg.h, 1247
CFE_TIME_DiagnosticTlm_t
cfe_time_msg.h, 1247
CFE_TIME_EXIT_ERR_EID
cfe_time_events.h, 1213
CFE_TIME_EarlyInit
cFE Internal Time APIs, internal to CFE core, 374
CFE_TIME_ExternalGPS
cFE External Time Source APIs, 367
CFE_TIME_ExternalMET
cFE External Time Source APIs, 368
CFE_TIME_ExternalTime
cFE External Time Source APIs, 368
CFE_TIME_ExternalTone
cFE External Time Source APIs, 369
CFE_TIME_FLAG_ADD1HZ
cFE Clock State Flag Defines, 376
CFE_TIME_FLAG_ADDADJ
cFE Clock State Flag Defines, 376
CFE_TIME_FLAG_ADDTCL
cFE Clock State Flag Defines, 377
CFE_TIME_FLAG_CLKSET
cFE Clock State Flag Defines, 377
CFE_TIME_FLAG_CMDFLY
cFE Clock State Flag Defines, 377
CFE_TIME_FLAG_FLYING
cFE Clock State Flag Defines, 377
CFE_TIME_FLAG_GDTONE
cFE Clock State Flag Defines, 377
CFE_TIME_FLAG_SERVER
cFE Clock State Flag Defines, 378
CFE_TIME_FLAG_SIGPRI
cFE Clock State Flag Defines, 378
CFE_TIME_FLAG_SRCINT
cFE Clock State Flag Defines, 378
CFE_TIME_FLAG_SRVFLY
cFE Clock State Flag Defines, 378
CFE_TIME_FLAG_UNUSED
cFE Clock State Flag Defines, 378
CFE_TIME_FLY_OFF_EID
cfe_time_events.h, 1214
CFE_TIME_FLY_ON_EID
cfe_time_events.h, 1214
CFE_TIME_FakeToneCmd_t
cfe_time_msg.h, 1247
CFE_TIME_FlagBit
cfe_time_extern_typedefs.h, 956
CFE_TIME_FlagBit_Enum_t
cfe_time_extern_typedefs.h, 953
CFE_TIME_FlywheelState
cfe_time_extern_typedefs.h, 956
CFE_TIME_FlywheelState_Enum_t
cfe_time_extern_typedefs.h, 954
CFE_TIME_GetClockInfo
cFE Get Time Information APIs, 358
CFE_TIME_GetClockState
cFE Get Time Information APIs, 358
CFE_TIME_GetLeapSeconds
cFE Get Time Information APIs, 359
CFE_TIME_GetMETseconds
cFE Get Current Time APIs, 354
CFE_TIME_GetMETsubsecs
cFE Get Current Time APIs, 355
CFE_TIME_GetMET
cFE Get Current Time APIs, 354
CFE_TIME_GetSTCF
cFE Get Time Information APIs, 359
CFE_TIME_GetTAI
cFE Get Current Time APIs, 355
CFE_TIME_GetTime
cFE Get Current Time APIs, 356
CFE_TIME_GetUTC
cFE Get Current Time APIs, 357
CFE_TIME_HK_TLM_MID
cpu1_msgids.h, 775
CFE_TIME_HousekeepingTlm, 703
Payload, 704
TlmHeader, 704
CFE_TIME_HousekeepingTlm_Payload, 704
ClockStateAPI, 705
ClockStateFlags, 705
CommandCounter, 705
CommandErrorCounter, 706
LeapSeconds, 706
Seconds1HzAdj, 706
SecondsDelay, 706
SecondsMET, 707
SecondsSTCF, 707
Subsecs1HzAdj, 707
SubsecsDelay, 707
SubsecsMET, 708
SubsecsSTCF, 708
CFE_TIME_HousekeepingTlm_Payload_t
cfe_time_msg.h, 1248
CFE_TIME_HousekeepingTlm_t
cfe_time_msg.h, 1248
CFE_TIME_ID_ERR_EID
cfe_time_events.h, 1214
CFE_TIME_INIT_EID
cfe_time_events.h, 1215
CFE_TIME_INTERNAL_ONLY

cFE Return Code Defines, 172
CFE_TIME_LEAPS_CFG_EID
cfe_time_events.h, 1215
CFE_TIME_LEAPS_EID
cfe_time_events.h, 1216
CFE_TIME_LEN_ERR_EID
cfe_time_events.h, 1216
CFE_TIME_LeapsCmd_Payload, 708
LeapSeconds, 709
CFE_TIME_LeapsCmd_Payload_t
cfe_time_msg.h, 1248
CFE_TIME_Local1HzISR
cFE Miscellaneous Time APIs, 372
CFE_TIME_MAX_EID
cfe_time_events.h, 1217
CFE_TIME_MET2SCTime
cFE Time Conversion APIs, 364
CFE_TIME_MET_CFG_EID
cfe_time_events.h, 1217
CFE_TIME_MET_EID
cfe_time_events.h, 1217
CFE_TIME_MET_ERR_EID
cfe_time_events.h, 1218
CFE_TIME_Micro2SubSecs
cFE Time Conversion APIs, 364
CFE_TIME_NOOP_CC
cfe_time_msg.h, 1232
CFE_TIME_NOOP_EID
cfe_time_events.h, 1218
CFE_TIME_NOT_IMPLEMENTED
cFE Return Code Defines, 172
CFE_TIME_NoArgsCmd, 709
CmdHeader, 710
CFE_TIME_NoArgsCmd_t
cfe_time_msg.h, 1248
CFE_TIME_NoopCmd_t
cfe_time_msg.h, 1248
CFE_TIME_OUT_OF_RANGE
cFE Return Code Defines, 172
CFE_TIME_OneHzAdjustmentCmd, 710
CmdHeader, 710
Payload, 710
CFE_TIME_OneHzAdjustmentCmd_Payload, 711
Seconds, 711
Subseconds, 711
CFE_TIME_OneHzAdjustmentCmd_Payload_t
cfe_time_msg.h, 1248
CFE_TIME_OneHzAdjustmentCmd_t
cfe_time_msg.h, 1249
CFE_TIME_PRINTED_STRING_SIZE
cfe_time_api_typedefs.h, 950
CFE_TIME_Print
cFE Miscellaneous Time APIs, 372
CFE_TIME_RESET_COUNTERS_CC
cfe_time_msg.h, 1233
CFE_TIME_RESET_EID
cfe_time_events.h, 1219
CFE_TIME_RegisterSynchCallback
cFE External Time Source APIs, 370
CFE_TIME_ResetCountersCmd_t
cfe_time_msg.h, 1249
CFE_TIME_SEND_CMD_MID
cpu1_msgids.h, 775
CFE_TIME_SEND_DIAGNOSTIC_TLM_CC
cfe_time_msg.h, 1234
CFE_TIME_SEND_HK_MID
cpu1_msgids.h, 775
CFE_TIME_SERVICE
cfe_error.h, 867
CFE_TIME_SET_LEAP_SECONDS_CC
cfe_time_msg.h, 1235
CFE_TIME_SET_MET_CC
cfe_time_msg.h, 1236
CFE_TIME_SET_SIGNAL_CC
cfe_time_msg.h, 1237
CFE_TIME_SET_SOURCE_CC
cfe_time_msg.h, 1238
CFE_TIME_SET_STATE_CC
cfe_time_msg.h, 1239
CFE_TIME_SET_STCF_CC
cfe_time_msg.h, 1241
CFE_TIME_SET_TIME_CC
cfe_time_msg.h, 1242
CFE_TIME_SIGNAL_CFG_EID
cfe_time_events.h, 1219
CFE_TIME_SIGNAL_EID
cfe_time_events.h, 1220
CFE_TIME_SIGNAL_ERR_EID
cfe_time_events.h, 1220
CFE_TIME_SOURCE_CFG_EID
cfe_time_events.h, 1221
CFE_TIME_SOURCE_EID
cfe_time_events.h, 1221
CFE_TIME_SOURCE_ERR_EID
cfe_time_events.h, 1222
CFE_TIME_STATE_EID
cfe_time_events.h, 1222
CFE_TIME_STATE_ERR_EID
cfe_time_events.h, 1223
CFE_TIME_STCF_CFG_EID
cfe_time_events.h, 1223
CFE_TIME_STCF_EID
cfe_time_events.h, 1224
CFE_TIME_STCF_ERR_EID
cfe_time_events.h, 1224
CFE_TIME_SUB_1HZ_ADJUSTMENT_CC
cfe_time_msg.h, 1243
CFE_TIME_SUB_ADJUST_CC

cfe_time_msg.h, 1244
CFE_TIME_SUB_DELAY_CC
cfe_time_msg.h, 1245
CFE_TIME_SendDiagnosticCmd_t
cfe_time_msg.h, 1249
CFE_TIME_SetLeapSecondsCmd, 712
CmdHeader, 712
Payload, 712
CFE_TIME_SetLeapSecondsCmd_t
cfe_time_msg.h, 1249
CFE_TIME_SetMETCmd_t
cfe_time_msg.h, 1249
CFE_TIME_SetSTCFCmd_t
cfe_time_msg.h, 1250
CFE_TIME_SetSignalCmd, 713
CmdHeader, 713
Payload, 713
CFE_TIME_SetSignalCmd_t
cfe_time_msg.h, 1249
CFE_TIME_SetSourceCmd, 714
CmdHeader, 714
Payload, 714
CFE_TIME_SetSourceCmd_t
cfe_time_msg.h, 1250
CFE_TIME_SetState
cfe_time_extern_typedefs.h, 957
CFE_TIME_SetState_Enum_t
cfe_time_extern_typedefs.h, 954
CFE_TIME_SetStateCmd, 715
CmdHeader, 715
Payload, 715
CFE_TIME_SetStateCmd_t
cfe_time_msg.h, 1250
CFE_TIME_SetTimeCmd_t
cfe_time_msg.h, 1250
CFE_TIME_SignalCmd_Payload, 716
ToneSource, 716
CFE_TIME_SignalCmd_Payload_t
cfe_time_msg.h, 1250
CFE_TIME_SourceCmd_Payload, 717
TimeSource, 717
CFE_TIME_SourceCmd_Payload_t
cfe_time_msg.h, 1250
CFE_TIME_SourceSelect
cfe_time_extern_typedefs.h, 957
CFE_TIME_SourceSelect_Enum_t
cfe_time_extern_typedefs.h, 954
CFE_TIME_StateCmd_Payload, 717
ClockState, 718
CFE_TIME_StateCmd_Payload_t
cfe_time_msg.h, 1251
CFE_TIME_Sub1HZAdjustmentCmd_t
cfe_time_msg.h, 1251
CFE_TIME_Sub2MicroSecs
cFE Time Conversion APIs, 365
CFE_TIME_SubAdjustCmd_t
cfe_time_msg.h, 1251
CFE_TIME_SubDelayCmd_t
cfe_time_msg.h, 1251
CFE_TIME_Subtract
cFE Time Arithmetic APIs, 363
CFE_TIME_SynchCallbackPtr_t
cfe_time_api_typedefs.h, 950
CFE_TIME_SysTime, 718
Seconds, 719
Subseconds, 719
CFE_TIME_SysTime_t
cfe_time_extern_typedefs.h, 954
CFE_TIME_TIME_CFG_EID
cfe_time_events.h, 1225
CFE_TIME_TIME_EID
cfe_time_events.h, 1225
CFE_TIME_TIME_ERR_EID
cfe_time_events.h, 1226
CFE_TIME_TONE_CMD_MID
cpu1_msgids.h, 775
CFE_TIME_TOO_MANY_SYNCH_CALLBACKS
cFE Return Code Defines, 173
CFE_TIME_TaskMain
cFE Internal Time APIs, internal to CFE core, 374
CFE_TIME_TimeCmd, 719
CmdHeader, 720
Payload, 720
CFE_TIME_TimeCmd_Payload, 720
MicroSeconds, 720
Seconds, 721
CFE_TIME_TimeCmd_Payload_t
cfe_time_msg.h, 1251
CFE_TIME_TimeCmd_t
cfe_time_msg.h, 1251
CFE_TIME_ToneDataCmd, 721
CmdHeader, 721
Payload, 722
CFE_TIME_ToneDataCmd_Payload, 722
AtToneLeapSeconds, 723
AtToneMET, 723
AtToneSTCF, 723
AtToneState, 723
CFE_TIME_ToneDataCmd_Payload_t
cfe_time_msg.h, 1252
CFE_TIME_ToneDataCmd_t
cfe_time_msg.h, 1252
CFE_TIME_ToneSignalCmd_t
cfe_time_msg.h, 1252
CFE_TIME_ToneSignalSelect
cfe_time_extern_typedefs.h, 957
CFE_TIME_ToneSignalSelect_Enum_t
cfe_time_extern_typedefs.h, 955

CFE_TIME_UnregisterSynchCallback
cFE External Time Source APIs, 370
CFE_TST
cfe_sb.h, 928
CFE_VERSION_STRING
cfe_version.h, 960
CFECoreChecksum
CFE_ES_HousekeepingTlm_Payload, 535
CFEMajorVersion
CFE_ES_HousekeepingTlm_Payload, 535
CFEMinorVersion
CFE_ES_HousekeepingTlm_Payload, 535
CFEMissionRevision
CFE_ES_HousekeepingTlm_Payload, 535
CFERevision
CFE_ES_HousekeepingTlm_Payload, 536
ccsds_hdr.h
CCSDS_ExtendedHeader_t, 1098
CCSDS_PrimaryHeader_t, 1098
CdsName
CFE_ES_DeleteCDSCmd_Payload, 528
cfe/docs/src/cfe_api.dox, 856
cfe/docs/src/cfe_es.dox, 856
cfe/docs/src/cfe_evs.dox, 856
cfe/docs/src/cfe_glossary.dox, 856
cfe/docs/src/cfe_sb.dox, 856
cfe/docs/src/cfe_tbl.dox, 856
cfe/docs/src/cfe_time.dox, 856
cfe/docs/src/cfe_xref.dox, 856
cfe/docs/src/cfs_versions.dox, 856
cfe/docs/src/main.dox, 856
cfe/modules/core_api/fsw/inc/cfe.h, 856
cfe/modules/core_api/fsw/inc/cfe_endian.h, 857
cfe/modules/core_api/fsw/inc/cfe_error.h, 857
cfe/modules/core_api/fsw/inc/cfe_es.h, 868
cfe/modules/core_api/fsw/inc/cfe_es_api_typedefs.h, 871
cfe/modules/core_api/fsw/inc/cfe_es_core_internal.h, 878
cfe/modules/core_api/fsw/inc/cfe_es_extern_typedefs.h, 879
cfe/modules/core_api/fsw/inc/cfe_evs.h, 890
cfe/modules/core_api/fsw/inc/cfe_evs_api_typedefs.h, 892
cfe/modules/core_api/fsw/inc/cfe_evs_core_internal.h, 896
cfe/modules/core_api/fsw/inc/cfe_evs_extern_typedefs.h, 896
cfe/modules/core_api/fsw/inc/cfe_fs.h, 900
cfe/modules/core_api/fsw/inc/cfe_fs_api_typedefs.h, 901
cfe/modules/core_api/fsw/inc/cfe_fs_core_internal.h, 903
cfe/modules/core_api/fsw/inc/cfe_fs_extern_typedefs.h, 903
cfe/modules/core_api/fsw/inc/cfe_msg.h, 907
cfe/modules/core_api/fsw/inc/cfe_msg_api_typedefs.h, 909
cfe/modules/core_api/fsw/inc/cfe_resourceid.h, 916
cfe/modules/core_api/fsw/inc/cfe_resourceid_api_typedefs.h, 923
cfe/modules/core_api/fsw/inc/cfe_sb.h, 925
cfe/modules/core_api/fsw/inc/cfe_sb_api_typedefs.h, 928
cfe/modules/core_api/fsw/inc/cfe_sb_core_internal.h, 935
cfe/modules/core_api/fsw/inc/cfe_sb_extern_typedefs.h, 935
cfe/modules/core_api/fsw/inc/cfe_tbl.h, 939
cfe/modules/core_api/fsw/inc/cfe_tbl_api_typedefs.h, 940
cfe/modules/core_api/fsw/inc/cfe_tbl_core_internal.h, 943
cfe/modules/core_api/fsw/inc/cfe_tbl_extern_typedefs.h, 944
cfe/modules/core_api/fsw/inc/cfe_tbl_filedef.h, 946
cfe/modules/core_api/fsw/inc/cfe_time.h, 947
cfe/modules/core_api/fsw/inc/cfe_time_api_typedefs.h, 949
cfe/modules/core_api/fsw/inc/cfe_time_core_internal.h, 951
cfe/modules/core_api/fsw/inc/cfe_time_extern_typedefs.h, 951
cfe/modules/core_api/fsw/inc/cfe_version.h, 958
cfe/modules/es/fsw/inc/cfe_es_events.h, 961
cfe/modules/es/fsw/inc/cfe_es_msg.h, 1008
cfe/modules/evs/fsw/inc/cfe_evs_events.h, 1044
cfe/modules/evs/fsw/inc/cfe_evs_msg.h, 1065
cfe/modules/msg/fsw/inc/ccsds_hdr.h, 1098
cfe/modules/sb/fsw/inc/cfe_sb_events.h, 1099
cfe/modules/sb/fsw/inc/cfe_sb_msg.h, 1134
cfe/modules/tbl/fsw/inc/cfe_tbl_events.h, 1152
cfe/modules/tbl/fsw/inc/cfe_tbl_msg.h, 1190
cfe/modules/time/fsw/inc/cfe_time_events.h, 1206
cfe/modules/time/fsw/inc/cfe_time_msg.h, 1227
cfe_endian.h
CFE_MAKE_BIG16, 857
CFE_MAKE_BIG32, 857
cfe_error.h
CFE_ES_CDS_NOT_FOUND_ERR, 864
CFE_ES_CDS_REGISTRY_FULL, 864
CFE_ES_ERR_APPID, 864
CFE_ES_ERR_APPNAME, 864
CFE_ES_ERR_MEM_HANDLE, 864
CFE_ES_ERR_TASKID, 865
CFE_EVENTS_SERVICE, 865
CFE_EXECUTIVE_SERVICE, 865
CFE_FILE_SERVICE, 865
CFE_GENERIC_SERVICE, 865
CFE_SERVICE_BITMASK, 866
CFE_SEVERITY_BITMASK, 866
CFE_SEVERITY_ERROR, 866
CFE_SEVERITY_INFO, 866
CFE_SEVERITY_SUCCESS, 866
CFE_SOFTWARE_BUS_SERVICE, 867
CFE_Status_t, 867
CFE_TABLE_SERVICE, 867

CFE_TIME_SERVICE, 867
cfe_es.h
CFE_ES_DBIT, 871
CFE_ES_DTEST, 871
CFE_ES_TEST_LONG_MASK, 871
OS_PRINTF, 871
cfe_es_api_typedefs.h
CFE_ES_APP_RESTART, 873
CFE_ES_APPID_UNDEFINED, 873
CFE_ES_APPID_C, 873
CFE_ES_CDS_BAD_HANDLE, 873
CFE_ES_CDSHANDLE_C, 874
CFE_ES_COUNTERID_UNDEFINED, 874
CFE_ES_COUNTERID_C, 874
CFE_ES_ChildTaskMainFuncPtr_t, 877
CFE_ES_LIBID_UNDEFINED, 874
CFE_ES_LIBID_C, 874
CFE_ES_LibraryEntryFuncPtr_t, 877
CFE_ES_MEMHANDLE_UNDEFINED, 875
CFE_ES_MEMHANDLE_C, 874
CFE_ES_MEMPOOLBUF_C, 875
CFE_ES_MemPoolBuf_t, 877
CFE_ES_NO_MUTEX, 875
CFE_ES_PoolAlign_t, 877
CFE_ES_STATIC_POOL_TYPE, 875
CFE_ES_StackPointer_t, 878
CFE_ES_TASK_STACK_ALLOCATE, 876
CFE_ES_TASKID_UNDEFINED, 876
CFE_ES_TASKID_C, 876
CFE_ES_TaskEntryFuncPtr_t, 878
CFE_ES_USE_MUTEX, 876
cfe_es_events.h
CFE_ES_ALL_APPS_EID, 965
CFE_ES_BOOT_ERR_EID, 965
CFE_ES_BUILD_INF_EID, 966
CFE_ES_CC1_ERR_EID, 966
CFE_ES_CDS_DELETE_ERR_EID, 967
CFE_ES_CDS_DELETE_TBL_ERR_EID, 967
CFE_ES_CDS_DELETED_INFO_EID, 968
CFE_ES_CDS_DUMP_ERR_EID, 968
CFE_ES_CDS_NAME_ERR_EID, 969
CFE_ES_CDS_OWNER_ACTIVE_EID, 969
CFE_ES_CDS_REG_DUMP_INF_EID, 970
CFE_ES_CDS_REGISTER_ERR_EID, 970
CFE_ES_CREATING_CDS_DUMP_ERR_EID, 971
CFE_ES_ERLOG1_INF_EID, 971
CFE_ES_ERLOG2_EID, 972
CFE_ES_ERLOG2_ERR_EID, 972
CFE_ES_ERLOG_PENDING_ERR_EID, 973
CFE_ES_ERR_SYSLOGMODE_EID, 973
CFE_ES_ERREXIT_APP_ERR_EID, 974
CFE_ES_ERREXIT_APP_INF_EID, 974
CFE_ES_EXIT_APP_ERR_EID, 975
CFE_ES_EXIT_APP_INF_EID, 975
CFE_ES_FILEWRITE_ERR_EID, 976
CFE_ES_INIT_INF_EID, 976
CFE_ES_INITSTATS_INF_EID, 977
CFE_ES_INVALID_POOL_HANDLE_ERR_EID, 977
CFE_ES_LEN_ERR_EID, 978
CFE_ES_MAX_EID, 978
CFE_ES_MID_ERR_EID, 979
CFE_ES_NOOP_INF_EID, 979
CFE_ES_ONE_APP_EID, 979
CFE_ES_ONE_APPID_ERR_EID, 980
CFE_ES_ONE_ERR_EID, 980
CFE_ES_OSCREATE_ERR_EID, 981
CFE_ES_PCR_ERR1_EID, 981
CFE_ES_PCR_ERR2_EID, 982
CFE_ES_PERF_DATAWRITTEN_EID, 982
CFE_ES_PERF_FILTMSKCMD_EID, 983
CFE_ES_PERF_FILTMSKERR_EID, 983
CFE_ES_PERF_LOG_ERR_EID, 984
CFE_ES_PERF_STARTCMD_EID, 984
CFE_ES_PERF_STARTCMD_ERR_EID, 985
CFE_ES_PERF_STARTCMD_TRIG_ERR_EID, 985
CFE_ES_PERF_STOPCMD_EID, 986
CFE_ES_PERF_STOPCMD_ERR2_EID, 986
CFE_ES_PERF_TRIGMSKCMD_EID, 987
CFE_ES_PERF_TRIGMSKERR_EID, 987
CFE_ES_RELOAD_APP_DBG_EID, 988
CFE_ES_RELOAD_APP_ERR1_EID, 988
CFE_ES_RELOAD_APP_ERR2_EID, 989
CFE_ES_RELOAD_APP_ERR3_EID, 989
CFE_ES_RELOAD_APP_ERR4_EID, 990
CFE_ES_RELOAD_APP_INF_EID, 990
CFE_ES_RESET_INF_EID, 991
CFE_ES_RESET_PR_COUNT_EID, 991
CFE_ES_RESTART_APP_DBG_EID, 992
CFE_ES_RESTART_APP_ERR1_EID, 992
CFE_ES_RESTART_APP_ERR2_EID, 993
CFE_ES_RESTART_APP_ERR3_EID, 993
CFE_ES_RESTART_APP_ERR4_EID, 994
CFE_ES_RESTART_APP_INF_EID, 994
CFE_ES_SET_MAX_PR_COUNT_EID, 995
CFE_ES_START_ERR_EID, 995
CFE_ES_START_EXC_ACTION_ERR_EID, 996
CFE_ES_START_INF_EID, 996
CFE_ES_START_INVALID_ENTRY_POINT_ERR↩_EID, 997
CFE_ES_START_INVALID_FILENAME_ERR_EID, 997
CFE_ES_START_NULL_APP_NAME_ERR_EID, 998
CFE_ES_START_PRIORITY_ERR_EID, 998
CFE_ES_STOP_DBG_EID, 999
CFE_ES_STOP_ERR1_EID, 999
CFE_ES_STOP_ERR2_EID, 1000
CFE_ES_STOP_ERR3_EID, 1000

CFE_ES_STOP_INF_EID, 1001
CFE_ES_SYSLOG1_INF_EID, 1001
CFE_ES_SYSLOG2_EID, 1002
CFE_ES_SYSLOG2_ERR_EID, 1002
CFE_ES_SYSLOGMODE_EID, 1003
CFE_ES_TASKINFO_EID, 1003
CFE_ES_TASKINFO_OSCREATE_ERR_EID, 1004
CFE_ES_TASKINFO_WR_ERR_EID, 1004
CFE_ES_TASKINFO_WRHDR_ERR_EID, 1005
CFE_ES_TASKWR_ERR_EID, 1005
CFE_ES_TLM_POOL_STATS_INFO_EID, 1006
CFE_ES_VERSION_INF_EID, 1006
CFE_ES_WRHDR_ERR_EID, 1007
CFE_ES_WRITE_CFE_HDR_ERR_EID, 1007
cfe_es_extern_typedefs.h
CFE_ES_AppId_t, 881
CFE_ES_AppInfo_t, 881
CFE_ES_AppState, 887
CFE_ES_AppState_Enum_t, 882
CFE_ES_AppType, 888
CFE_ES_AppType_Enum_t, 882
CFE_ES_BlockStats_t, 882
CFE_ES_CDSHandle_t, 883
CFE_ES_CDSRegDumpRec_t, 883
CFE_ES_CounterId_t, 883
CFE_ES_ExceptionAction, 888
CFE_ES_ExceptionAction_Enum_t, 883
CFE_ES_LibId_t, 884
CFE_ES_LogEntryType, 888
CFE_ES_LogEntryType_Enum_t, 884
CFE_ES_LogMode, 889
CFE_ES_LogMode_Enum_t, 884
CFE_ES_MEMADDRESS_C, 881
CFE_ES_MEMOFFSET_C, 881
CFE_ES_MemAddress_t, 884
CFE_ES_MemHandle_t, 885
CFE_ES_MemOffset_t, 885
CFE_ES_MemPoolStats_t, 885
CFE_ES_RunStatus, 889
CFE_ES_RunStatus_Enum_t, 886
CFE_ES_SystemState, 890
CFE_ES_SystemState_Enum_t, 886
CFE_ES_TaskId_t, 886
CFE_ES_TaskInfo_t, 887
CFE_ES_TaskPriority_Atom_t, 887
cfe_es_msg.h
CFE_ES_AppNameCmd_Payload_t, 1036
CFE_ES_AppNameCmd_t, 1036
CFE_ES_AppReloadCmd_Payload_t, 1036
CFE_ES_CLEAR_ER_LOG_CC, 1012
CFE_ES_CLEAR_SYSLOG_CC, 1013
CFE_ES_ClearERLogCmd_t, 1036
CFE_ES_ClearSysLogCmd_t, 1036
CFE_ES_DELETE_CDS_CC, 1013
CFE_ES_DUMP_CDS_REGISTRY_CC, 1014
CFE_ES_DeleteCDSCmd_Payload_t, 1036
CFE_ES_DeleteCDSCmd_t, 1036
CFE_ES_DumpCDSRegistryCmd_Payload_t, 1037
CFE_ES_DumpCDSRegistryCmd_t, 1037
CFE_ES_FileNameCmd_Payload_t, 1037
CFE_ES_FileNameCmd_t, 1037
CFE_ES_HousekeepingTlm_Payload_t, 1037
CFE_ES_HousekeepingTlm_t, 1037
CFE_ES_MemStatsTlm_t, 1038
CFE_ES_NOOP_CC, 1015
CFE_ES_NoArgsCmd_t, 1038
CFE_ES_NoopCmd_t, 1038
CFE_ES_OVER_WRITE_SYSLOG_CC, 1016
CFE_ES_OneAppTlm_Payload_t, 1038
CFE_ES_OneAppTlm_t, 1038
CFE_ES_OverWriteSysLogCmd_Payload_t, 1038
CFE_ES_OverWriteSysLogCmd_t, 1039
CFE_ES_PoolStatsTlm_Payload_t, 1039
CFE_ES_QUERY_ALL_CC, 1017
CFE_ES_QUERY_ALL_TASKS_CC, 1018
CFE_ES_QUERY_ONE_CC, 1019
CFE_ES_QueryAllCmd_t, 1039
CFE_ES_QueryAllTasksCmd_t, 1039
CFE_ES_QueryOneCmd_t, 1039
CFE_ES_RELOAD_APP_CC, 1020
CFE_ES_RESET_COUNTERS_CC, 1021
CFE_ES_RESET_PR_COUNT_CC, 1022
CFE_ES_RESTART_APP_CC, 1023
CFE_ES_RESTART_CC, 1024
CFE_ES_ReloadAppCmd_t, 1039
CFE_ES_ResetCountersCmd_t, 1040
CFE_ES_ResetPRCountCmd_t, 1040
CFE_ES_RestartAppCmd_t, 1040
CFE_ES_RestartCmd_Payload_t, 1040
CFE_ES_RestartCmd_t, 1040
CFE_ES_SEND_MEM_POOL_STATS_CC, 1025
CFE_ES_SET_MAX_PR_COUNT_CC, 1026
CFE_ES_SET_PERF_FILTER_MASK_CC, 1027
CFE_ES_SET_PERF_TRIGGER_MASK_CC, 1028
CFE_ES_START_APP_CC, 1029
CFE_ES_START_PERF_DATA_CC, 1030
CFE_ES_STOP_APP_CC, 1031
CFE_ES_STOP_PERF_DATA_CC, 1032
CFE_ES_SendMemPoolStatsCmd_Payload_t, 1040
CFE_ES_SendMemPoolStatsCmd_t, 1041
CFE_ES_SetMaxPRCountCmd_Payload_t, 1041
CFE_ES_SetMaxPRCountCmd_t, 1041
CFE_ES_SetPerfFilterMaskCmd_Payload_t, 1041
CFE_ES_SetPerfFilterMaskCmd_t, 1041
CFE_ES_SetPerfTrigMaskCmd_Payload_t, 1042
CFE_ES_SetPerfTriggerMaskCmd_t, 1041
CFE_ES_StartAppCmd_Payload_t, 1042
CFE_ES_StartAppCmd_t, 1042

CFE_ES_StartPerfCmd_Payload_t, 1042
CFE_ES_StartPerfDataCmd_t, 1042
CFE_ES_StopAppCmd_t, 1042
CFE_ES_StopPerfCmd_Payload_t, 1043
CFE_ES_StopPerfDataCmd_t, 1043
CFE_ES_WRITE_ER_LOG_CC, 1033
CFE_ES_WRITE_SYSLOG_CC, 1034
CFE_ES_WriteERLogCmd_t, 1043
CFE_ES_WriteSysLogCmd_t, 1043
cfe_evs.h
CFE_EVS_Send, 891
CFE_EVS_SendCrit, 891
CFE_EVS_SendDbg, 891
CFE_EVS_SendErr, 892
CFE_EVS_SendInfo, 892
cfe_evs_api_typedefs.h
CFE_EVS_BinFilter_t, 896
CFE_EVS_EVERY_FOURTH_ONE, 893
CFE_EVS_EVERY_OTHER_ONE, 893
CFE_EVS_EVERY_OTHER_TWO, 894
CFE_EVS_FIRST_16_STOP, 894
CFE_EVS_FIRST_32_STOP, 894
CFE_EVS_FIRST_4_STOP, 894
CFE_EVS_FIRST_64_STOP, 894
CFE_EVS_FIRST_8_STOP, 895
CFE_EVS_FIRST_ONE_STOP, 895
CFE_EVS_FIRST_TWO_STOP, 895
CFE_EVS_NO_FILTER, 895
cfe_evs_events.h
CFE_EVS_ADDFILTER_EID, 1045
CFE_EVS_DELFILTER_EID, 1046
CFE_EVS_DISAPPENTTYPE_EID, 1046
CFE_EVS_DISAPPEVT_EID, 1047
CFE_EVS_DISEVTTYPE_EID, 1047
CFE_EVS_DISPORT_EID, 1048
CFE_EVS_ENAAPPEVT_EID, 1048
CFE_EVS_ENAAPPEVTTYPE_EID, 1049
CFE_EVS_ENAEVTTYPE_EID, 1049
CFE_EVS_ENAPORT_EID, 1050
CFE_EVS_ERR_APPNOREGS_EID, 1050
CFE_EVS_ERR_CC_EID, 1051
CFE_EVS_ERR_CRDATFILE_EID, 1051
CFE_EVS_ERR_CRLOGFILE_EID, 1052
CFE_EVS_ERR_EVTIDNOREGS_EID, 1052
CFE_EVS_ERR_ILLAPPIDRANGE_EID, 1053
CFE_EVS_ERR_ILLEGALFMTMOD_EID, 1053
CFE_EVS_ERR_INVALID_BITMASK_EID, 1054
CFE_EVS_ERR_LOGMODE_EID, 1054
CFE_EVS_ERR_MAXREGSFILTER_EID, 1055
CFE_EVS_ERR_MSGID_EID, 1055
CFE_EVS_ERR_NOAPPIDFOUND_EID, 1056
CFE_EVS_ERR_UNREGISTERED_EVS_APP, 1056
CFE_EVS_ERR_WRDATFILE_EID, 1057
CFE_EVS_ERR_WRLOGFILE_EID, 1057
CFE_EVS_EVT_FILTERED_EID, 1058
CFE_EVS_FILTER_MAX_EID, 1058
CFE_EVS_LEN_ERR_EID, 1059
CFE_EVS_LOGMODE_EID, 1059
CFE_EVS_MAX_EID, 1060
CFE_EVS_NOOP_EID, 1060
CFE_EVS_RSTALLFILTER_EID, 1060
CFE_EVS_RSTCNT_EID, 1061
CFE_EVS_RSTEVTCNT_EID, 1061
CFE_EVS_RSTFILTER_EID, 1062
CFE_EVS_SETEVTFMTMOD_EID, 1062
CFE_EVS_SETFILTERMSK_EID, 1063
CFE_EVS_STARTUP_EID, 1063
CFE_EVS_WRDAT_EID, 1064
CFE_EVS_WRLOG_EID, 1064
cfe_evs_extern_typedefs.h
CFE_EVS_EventFilter, 898
CFE_EVS_EventFilter_Enum_t, 897
CFE_EVS_EventOutput, 899
CFE_EVS_EventOutput_Enum_t, 897
CFE_EVS_EventType, 899
CFE_EVS_EventType_Enum_t, 897
CFE_EVS_LogMode, 899
CFE_EVS_LogMode_Enum_t, 898
CFE_EVS_MsgFormat, 900
CFE_EVS_MsgFormat_Enum_t, 898
cfe_evs_msg.h
CFE_EVS_ADD_EVENT_FILTER_CC, 1068
CFE_EVS_AddEventFilterCmd_t, 1090
CFE_EVS_AppDataCmd_Payload_t, 1091
CFE_EVS_AppNameBitMaskCmd_Payload_t, 1091
CFE_EVS_AppNameBitMaskCmd_t, 1091
CFE_EVS_AppNameCmd_Payload_t, 1091
CFE_EVS_AppNameCmd_t, 1091
CFE_EVS_AppNameEventIDCmd_Payload_t, 1091
CFE_EVS_AppNameEventIDCmd_t, 1092
CFE_EVS_AppNameEventIDMaskCmd_Payload_t, 1092
CFE_EVS_AppNameEventIDMaskCmd_t, 1092
CFE_EVS_AppTlmData_t, 1092
CFE_EVS_BitMaskCmd_Payload_t, 1092
CFE_EVS_BitMaskCmd_t, 1092
CFE_EVS_CLEAR_LOG_CC, 1069
CFE_EVS_CRITICAL_BIT, 1070
CFE_EVS_ClearLogCmd_t, 1093
CFE_EVS_DEBUG_BIT, 1070
CFE_EVS_DELETE_EVENT_FILTER_CC, 1071
CFE_EVS_DISABLE_APP_EVENT_TYPE_CC, 1071
CFE_EVS_DISABLE_APP_EVENTS_CC, 1072
CFE_EVS_DISABLE_EVENT_TYPE_CC, 1073
CFE_EVS_DISABLE_PORTS_CC, 1074
CFE_EVS_DeleteEventFilterCmd_t, 1093

CFE_EVS_DisableAppEventTypeCmd_t, 1093
CFE_EVS_DisableAppEventsCmd_t, 1093
CFE_EVS_DisableEventTypeCmd_t, 1093
CFE_EVS_DisablePortsCmd_t, 1093
CFE_EVS_ENABLE_APP_EVENT_TYPE_CC, 1075
CFE_EVS_ENABLE_APP_EVENTS_CC, 1076
CFE_EVS_ENABLE_EVENT_TYPE_CC, 1077
CFE_EVS_ENABLE_PORTS_CC, 1078
CFE_EVS_ERROR_BIT, 1079
CFE_EVS_EnableAppEventTypeCmd_t, 1094
CFE_EVS_EnableAppEventsCmd_t, 1094
CFE_EVS_EnableEventTypeCmd_t, 1094
CFE_EVS_EnablePortsCmd_t, 1094
CFE_EVS_HousekeepingTlm_Payload_t, 1094
CFE_EVS_HousekeepingTlm_t, 1094
CFE_EVS_INFORMATION_BIT, 1080
CFE_EVS_LOG_DISCARD, 1080
CFE_EVS_LOG_OVERWRITE, 1080
CFE_EVS_LogFileCmd_Payload_t, 1095
CFE_EVS_LongEventTlm_Payload_t, 1095
CFE_EVS_LongEventTlm_t, 1095
CFE_EVS_NOOP_CC, 1080
CFE_EVS_NoArgsCmd_t, 1095
CFE_EVS_NoopCmd_t, 1095
CFE_EVS_PORT1_BIT, 1081
CFE_EVS_PORT2_BIT, 1081
CFE_EVS_PORT3_BIT, 1081
CFE_EVS_PORT4_BIT, 1081
CFE_EVS_PacketID_t, 1095
CFE_EVS_RESET_ALL_FILTERS_CC, 1082
CFE_EVS_RESET_APP_COUNTER_CC, 1082
CFE_EVS_RESET_COUNTERS_CC, 1083
CFE_EVS_RESET_FILTER_CC, 1084
CFE_EVS_ResetAllFiltersCmd_t, 1095
CFE_EVS_ResetAppCounterCmd_t, 1096
CFE_EVS_ResetCountersCmd_t, 1096
CFE_EVS_ResetFilterCmd_t, 1096
CFE_EVS_SET_EVENT_FORMAT_MODE_CC, 1085
CFE_EVS_SET_FILTER_CC, 1086
CFE_EVS_SET_LOG_MODE_CC, 1087
CFE_EVS_SetEventFormatMode_Payload_t, 1096
CFE_EVS_SetEventFormatModeCmd_t, 1096
CFE_EVS_SetFilterCmd_t, 1096
CFE_EVS_SetLogMode_Payload_t, 1097
CFE_EVS_SetLogModeCmd_t, 1097
CFE_EVS_ShortEventTlm_Payload_t, 1097
CFE_EVS_ShortEventTlm_t, 1097
CFE_EVS_WRITE_APP_DATA_FILE_CC, 1088
CFE_EVS_WRITE_LOG_DATA_FILE_CC, 1089
CFE_EVS_WriteAppDataFileCmd_t, 1097
CFE_EVS_WriteLogDataFileCmd_t, 1097
cfe_fs_api_typedefs.h
CFE_FS_FileCategory_t, 902
CFE_FS_FileWriteEvent_t, 903
CFE_FS_FileWriteGetData_t, 902
CFE_FS_FileWriteMetaData_t, 902
CFE_FS_FileWriteOnEvent_t, 902
cfe_fs_extern_typedefs.h
CFE_FS_FILE_CONTENT_ID, 904
CFE_FS_HDR_DESC_MAX_LEN, 904
CFE_FS_Header_t, 905
CFE_FS_SubType, 905
CFE_FS_SubType_Enum_t, 905
cfe_msg_api_typedefs.h
CFE_MSG_ApId_t, 911
CFE_MSG_BAD_ARGUMENT, 910
CFE_MSG_Checksum_t, 911
CFE_MSG_CommandHeader_t, 911
CFE_MSG_EDSVersion_t, 911
CFE_MSG_Endian, 913
CFE_MSG_Endian_t, 911
CFE_MSG_FcnCode_t, 911
CFE_MSG_HeaderVersion_t, 912
CFE_MSG_Message_t, 912
CFE_MSG_NOT_IMPLEMENTED, 910
CFE_MSG_PlaybackFlag, 914
CFE_MSG_PlaybackFlag_t, 912
CFE_MSG_SegmentationFlag, 914
CFE_MSG_SegmentationFlag_t, 912
CFE_MSG_SequenceCount_t, 912
CFE_MSG_Size_t, 913
CFE_MSG_Subsystem_t, 913
CFE_MSG_System_t, 913
CFE_MSG_TelemetryHeader_t, 913
CFE_MSG_Type, 914
CFE_MSG_Type_t, 913
CFE_MSG_WRONG_MSG_TYPE, 910
cfe_psp.h
BUFF_SIZE, 1301
CFE_PSP_AttachExceptions, 1310
CFE_PSP_Decompress, 1311
CFE_PSP_ERROR_ADDRESS_MISALIGNED, 1301
CFE_PSP_ERROR_NOT_IMPLEMENTED, 1301
CFE_PSP_ERROR_TIMEOUT, 1301
CFE_PSP_ERROR, 1301
CFE_PSP_EepromPowerDown, 1311
CFE_PSP_EepromPowerUp, 1311
CFE_PSP_EepromWrite16, 1311
CFE_PSP_EepromWrite32, 1311
CFE_PSP_EepromWrite8, 1312
CFE_PSP_EepromWriteDisable, 1312
CFE_PSP_EepromWriteEnable, 1312
CFE_PSP_Exception_CopyContext, 1312
CFE_PSP_Exception_GetCount, 1312
CFE_PSP_Exception_GetSummary, 1312
CFE_PSP_FlushCaches, 1313

CFE_PSP_Get_Dec, 1313
CFE_PSP_Get_Timebase, 1313
CFE_PSP_Get_Timer_Tick, 1313
CFE_PSP_GetBuildNumber, 1314
CFE_PSP_GetCDSSize, 1314
CFE_PSP_GetCFETextSegmentInfo, 1314
CFE_PSP_GetKernelTextSegmentInfo, 1314
CFE_PSP_GetProcessorId, 1315
CFE_PSP_GetProcessorName, 1315
CFE_PSP_GetResetArea, 1315
CFE_PSP_GetRestartType, 1315
CFE_PSP_GetSpacecraftId, 1315
CFE_PSP_GetTime, 1316
CFE_PSP_GetTimerLow32Rollover, 1316
CFE_PSP_GetTimerTicksPerSecond, 1316
CFE_PSP_GetUserReservedArea, 1316
CFE_PSP_GetVersionCodeName, 1317
CFE_PSP_GetVersionNumber, 1317
CFE_PSP_GetVersionString, 1317
CFE_PSP_GetVolatileDiskMem, 1318
CFE_PSP_INVALID_INT_NUM, 1302
CFE_PSP_INVALID_MEM_ADDR, 1302
CFE_PSP_INVALID_MEM_ATTR, 1302
CFE_PSP_INVALID_MEM_RANGE, 1302
CFE_PSP_INVALID_MEM_SIZE, 1302
CFE_PSP_INVALID_MEM_TYPE, 1302
CFE_PSP_INVALID_MEM_WORDSIZE, 1303
CFE_PSP_INVALID_MODULE_ID, 1303
CFE_PSP_INVALID_MODULE_NAME, 1303
CFE_PSP_INVALID_POINTER, 1303
CFE_PSP_InitSSR, 1318
CFE_PSP_MAJOR_VERSION, 1303
CFE_PSP_MEM_ANY, 1303
CFE_PSP_MEM_ATTR_READWRITE, 1304
CFE_PSP_MEM_ATTR_READ, 1304
CFE_PSP_MEM_ATTR_WRITE, 1304
CFE_PSP_MEM_EEPROM, 1304
CFE_PSP_MEM_INVALID, 1304
CFE_PSP_MEM_RAM, 1304
CFE_PSP_MEM_SIZE_BYTE, 1305
CFE_PSP_MEM_SIZE_DWORD, 1305
CFE_PSP_MEM_SIZE_WORD, 1305
CFE_PSP_MINOR_VERSION, 1305
CFE_PSP_MISSION_REV, 1305
CFE_PSP_Main, 1318
CFE_PSP_MemCpy, 1318
CFE_PSP_MemRangeGet, 1319
CFE_PSP_MemRangeSet, 1319
CFE_PSP_MemRanges, 1319
CFE_PSP_MemRead16, 1319
CFE_PSP_MemRead32, 1319
CFE_PSP_MemRead8, 1320
CFE_PSP_MemSet, 1320
CFE_PSP_MemValidateRange, 1320
CFE_PSP_MemWrite16, 1320
CFE_PSP_MemWrite32, 1320
CFE_PSP_MemWrite8, 1320
CFE_PSP_NO_EXCEPTION_DATA, 1305
CFE_PSP_PANIC_CORE_APP, 1306
CFE_PSP_PANIC_GENERAL_FAILURE, 1306
CFE_PSP_PANIC_MEMORY_ALLOC, 1306
CFE_PSP_PANIC_NONVOL_DISK, 1306
CFE_PSP_PANIC_STARTUP_SEM, 1306
CFE_PSP_PANIC_STARTUP, 1306
CFE_PSP_PANIC_VOLATILE_DISK, 1307
CFE_PSP_Panic, 1321
CFE_PSP_PortRead16, 1321
CFE_PSP_PortRead32, 1321
CFE_PSP_PortRead8, 1321
CFE_PSP_PortWrite16, 1322
CFE_PSP_PortWrite32, 1322
CFE_PSP_PortWrite8, 1322
CFE_PSP_REVISION, 1307
CFE_PSP_RST_SUBTYPE_BANKSWITCH_RESET, 1307
CFE_PSP_RST_SUBTYPE_EXCEPTION, 1307
CFE_PSP_RST_SUBTYPE_HW_SPECIAL_COMMAND, 1307
CFE_PSP_RST_SUBTYPE_HW_WATCHDOG, 1307
CFE_PSP_RST_SUBTYPE_HWDEBUG_RESET, 1308
CFE_PSP_RST_SUBTYPE_MAX, 1308
CFE_PSP_RST_SUBTYPE_POWER_CYCLE, 1308
CFE_PSP_RST_SUBTYPE_PUSH_BUTTON, 1308
CFE_PSP_RST_SUBTYPE_RESET_COMMAND, 1308
CFE_PSP_RST_SUBTYPE_UNDEFINED_RESET, 1309
CFE_PSP_RST_TYPE_MAX, 1309
CFE_PSP_RST_TYPE_POWERON, 1309
CFE_PSP_RST_TYPE_PROCESSOR, 1309
CFE_PSP_ReadFromCDS, 1322
CFE_PSP_Restart, 1322
CFE_PSP_SUCCESS, 1309
CFE_PSP_SetDefaultExceptionEnvironment, 1323
CFE_PSP_VERSION, 1310
CFE_PSP_WatchdogDisable, 1323
CFE_PSP_WatchdogEnable, 1323
CFE_PSP_WatchdogGet, 1324
CFE_PSP_WatchdogInit, 1324
CFE_PSP_WatchdogService, 1324
CFE_PSP_WatchdogSet, 1324
CFE_PSP_WriteToCDS, 1324
SIZE_BYTE, 1310
SIZE_HALF, 1310
SIZE_WORD, 1310
cfe_psp_configdata.h

CFE_PSP_MemoryTable, 1325
GLOBAL_PSP_CONFIGDATA, 1325
cfe_psp_exception.c
CFE_PSP_AsyncMask, 1329
CFE_PSP_AttachExceptions, 1326
CFE_PSP_AttachSigHandler, 1327
CFE_PSP_ExceptionGetSummary_Impl, 1327
CFE_PSP_ExceptionSigHandler, 1327
CFE_PSP_ExceptionSigHandlerSuspend, 1328
CFE_PSP_SetDefaultExceptionEnvironment, 1328
cfe_psp_memory.c
_fini, 1338
_init, 1338
CDSShmId, 1339
CFE_PSP_CDS_KEY_FILE, 1330
CFE_PSP_CDS_SIZE, 1330
CFE_PSP_DeleteCDS, 1332
CFE_PSP_DeleteProcessorReservedMemory, 1332
CFE_PSP_DeleteResetArea, 1332
CFE_PSP_DeleteUserReservedArea, 1333
CFE_PSP_GetCDSSize, 1333
CFE_PSP_GetCFETextSegmentInfo, 1334
CFE_PSP_GetKernelTextSegmentInfo, 1334
CFE_PSP_GetResetArea, 1334
CFE_PSP_GetUserReservedArea, 1334
CFE_PSP_GetVolatileDiskMem, 1335
CFE_PSP_InitCDS, 1335
CFE_PSP_InitProcessorReservedMemory, 1335
CFE_PSP_InitResetArea, 1336
CFE_PSP_InitUserReservedArea, 1336
CFE_PSP_InitVolatileDiskMem, 1337
CFE_PSP_RESERVED_KEY_FILE, 1331
CFE_PSP_RESET_AREA_SIZE, 1331
CFE_PSP_RESET_KEY_FILE, 1331
CFE_PSP_ReadFromCDS, 1337
CFE_PSP_ReservedMemoryMap, 1339
CFE_PSP_SetupReservedMemoryMap, 1337
CFE_PSP_USER_RESERVED_SIZE, 1331
CFE_PSP_WriteToCDS, 1338
ResetAreaShmId, 1339
UserShmId, 1339
cfe_psp_memtab.c
CFE_PSP_MemoryTable, 1340
cfe_psp_ssr.c
CFE_PSP_InitSSR, 1341
cfe_psp_start.c
CFE_PSP_1HZ_FUNCTION, 1342
CFE_PSP_CPU_ID, 1342
CFE_PSP_CPU_NAME_LENGTH, 1343
CFE_PSP_CPU_NAME, 1343
CFE_PSP_CpuId, 1346
CFE_PSP_CpuName, 1346
CFE_PSP_DisplayUsage, 1344
CFE_PSP_IdleTaskState, 1347
CFE_PSP_KERNEL_NAME_LENGTH_MAX, 1343
CFE_PSP_MAIN_FUNCTION, 1343
CFE_PSP_NONVOL_STARTUP_FILE, 1343
CFE_PSP_OS_EventHandler, 1344
CFE_PSP_ProcessArgumentDefaults, 1345
CFE_PSP_RESET_NAME_LENGTH, 1344
CFE_PSP_SPACECRAFT_ID, 1344
CFE_PSP_SetupLocal1Hz, 1345
CFE_PSP_SpacecraftId, 1347
CFE_PSP_TimerHandler, 1346
CommandData, 1347
longOpts, 1347
optString, 1347
TimerCounter, 1348
cfe_psp_support.c
CFE_PSP_CpuId, 1350
CFE_PSP_CpuName, 1351
CFE_PSP_FlushCaches, 1349
CFE_PSP_GetProcessorId, 1349
CFE_PSP_GetProcessorName, 1349
CFE_PSP_GetSpacecraftId, 1349
CFE_PSP_Panic, 1349
CFE_PSP_Restart, 1350
CFE_PSP_SpacecraftId, 1351
cfe_psp_timer.c
CFE_PSP_Get_Dec, 1352
CFE_PSP_Get_Timebase, 1352
CFE_PSP_Get_Timer_Tick, 1353
CFE_PSP_GetTime, 1353
CFE_PSP_GetTimerLow32Rollover, 1353
CFE_PSP_GetTimerTicksPerSecond, 1354
CFE_PSP_TIMER_LOW32_ROLLOVER, 1352
CFE_PSP_TIMER_TICKS_PER_SECOND, 1352
cfe_psp_watchdog.c
CFE_PSP_WatchdogDisable, 1354
CFE_PSP_WatchdogEnable, 1355
CFE_PSP_WatchdogGet, 1355
CFE_PSP_WatchdogInit, 1355
CFE_PSP_WatchdogService, 1355
CFE_PSP_WatchdogSet, 1355
CFE_PSP_WatchdogValue, 1356
cfe_resourceid.h
CFE_RESOURCEID_TEST_DEFINED, 917
CFE_RESOURCEID_TEST_EQUAL, 917
CFE_RESOURCEID_TO_ULONG, 917
CFE_ResourceId_Equal, 918
CFE_ResourceId_FindNext, 918
CFE_ResourceId_FromInteger, 919
CFE_ResourceId_GetBase, 920
CFE_ResourceId_GetSerial, 920
CFE_ResourceId_IsDefined, 921
CFE_ResourceId_ToIndex, 922
CFE_ResourceId_ToInteger, 922
cfe_resourceid_api_typedefs.h

CFE_RESOURCEID_RESERVED, 924
CFE_RESOURCEID_UNDEFINED, 924
cfe_sb.h
CFE_BIT, 927
CFE_CLR, 927
CFE_SET, 927
CFE_TST, 928
cfe_sb_api_typedefs.h
CFE_SB_Buffer_t, 934
CFE_SB_CMD_HDR_SIZE, 930
CFE_SB_CmdHdr_t, 934
CFE_SB_DEFAULT_QOS, 930
CFE_SB_Default_Qos, 930
CFE_SB_INVALID_MSG_ID, 931
CFE_SB_INVALID_PIPE, 931
CFE_SB_MSGID_RESERVED, 931
CFE_SB_MSGID_UNWRAP_VALUE, 931
CFE_SB_MSGID_WRAP_VALUE, 932
CFE_SB_Msg_t, 934
CFE_SB_MsgPayloadPtr_t, 934
CFE_SB_MsgPtr_t, 935
CFE_SB_PEND_FOREVER, 932
CFE_SB_PIPEID_C, 932
CFE_SB_PIPEOPTS_IGNOREMINE, 933
CFE_SB_POLL, 933
CFE_SB_SUBSCRIPTION, 933
CFE_SB_TLM_HDR_SIZE, 933
CFE_SB_TlmHdr_t, 935
CFE_SB_UNSUBSCRIPTION, 934
cfe_sb_events.h
CFE_SB_BAD_CMD_CODE_EID, 1101
CFE_SB_BAD_MSGID_EID, 1102
CFE_SB_BAD_PIPEID_EID, 1102
CFE_SB_CMD0_RCVD_EID, 1103
CFE_SB_CMD1_RCVD_EID, 1103
CFE_SB_CR_PIPE_BAD_ARG_EID, 1104
CFE_SB_CR_PIPE_ERR_EID, 1104
CFE_SB_CR_PIPE_NAME_TAKEN_EID, 1105
CFE_SB_CR_PIPE_NO_FREE_EID, 1105
CFE_SB_DEL_PIPE_ERR1_EID, 1106
CFE_SB_DEL_PIPE_ERR2_EID, 1106
CFE_SB_DEST_BLK_ERR_EID, 1107
CFE_SB_DSBL_RTE1_EID, 1107
CFE_SB_DSBL_RTE2_EID, 1108
CFE_SB_DSBL_RTE3_EID, 1108
CFE_SB_DUP_SUBSCRIP_EID, 1109
CFE_SB_ENBL_RTE1_EID, 1109
CFE_SB_ENBL_RTE2_EID, 1110
CFE_SB_ENBL_RTE3_EID, 1110
CFE_SB_FILEWRITE_ERR_EID, 1111
CFE_SB_FULL_SUB_PKT_EID, 1111
CFE_SB_GET_BUF_ERR_EID, 1112
CFE_SB_GETPIPEIDBYNAME_EID, 1112
CFE_SB_GETPIPEIDBYNAME_NAME_ERR_EID, 1113
CFE_SB_GETPIPEIDBYNAME_NULL_ERR_EID, 1113
CFE_SB_GETPIPENAME_EID, 1114
CFE_SB_GETPIPENAME_ID_ERR_EID, 1114
CFE_SB_GETPIPENAME_NULL_PTR_EID, 1115
CFE_SB_GETPIPEOPTS_EID, 1115
CFE_SB_GETPIPEOPTS_ID_ERR_EID, 1116
CFE_SB_GETPIPEOPTS_PTR_ERR_EID, 1116
CFE_SB_HASHCOLLISION_EID, 1117
CFE_SB_INIT_EID, 1117
CFE_SB_LEN_ERR_EID, 1118
CFE_SB_MAX_DESTS_MET_EID, 1118
CFE_SB_MAX_EID, 1119
CFE_SB_MAX_MSGS_MET_EID, 1119
CFE_SB_MAX_PIPES_MET_EID, 1119
CFE_SB_MSG_TOO_BIG_EID, 1120
CFE_SB_MSGID_LIM_ERR_EID, 1120
CFE_SB_PART_SUB_PKT_EID, 1121
CFE_SB_PIPE_ADDED_EID, 1121
CFE_SB_PIPE_DELETED_EID, 1122
CFE_SB_Q_FULL_ERR_EID, 1122
CFE_SB_Q_RD_ERR_EID, 1123
CFE_SB_Q_WR_ERR_EID, 1123
CFE_SB_RCV_BAD_ARG_EID, 1124
CFE_SB_SEND_BAD_ARG_EID, 1124
CFE_SB_SEND_INV_MSGID_EID, 1125
CFE_SB_SEND_NO_SUBS_EID, 1125
CFE_SB_SETPIPEOPTS_EID, 1126
CFE_SB_SETPIPEOPTS_ID_ERR_EID, 1126
CFE_SB_SETPIPEOPTS_OWNER_ERR_EID, 1127
CFE_SB_SND_RTG_EID, 1127
CFE_SB_SND_RTG_ERR1_EID, 1128
CFE_SB_SND_STATS_EID, 1128
CFE_SB_SUB_ARG_ERR_EID, 1129
CFE_SB_SUB_INV_CALLER_EID, 1129
CFE_SB_SUB_INV_PIPE_EID, 1130
CFE_SB_SUBSCRIPTION_RCVD_EID, 1130
CFE_SB_SUBSCRIPTION_REMOVED_EID, 1131
CFE_SB_SUBSCRIPTION_RPT_EID, 1131
CFE_SB_UNSUB_ARG_ERR_EID, 1132
CFE_SB_UNSUB_INV_CALLER_EID, 1132
CFE_SB_UNSUB_INV_PIPE_EID, 1133
CFE_SB_UNSUB_NO_SUBS_EID, 1133
cfe_sb_extern_typedefs.h
CFE_SB_MsgId_Atom_t, 937
CFE_SB_MsgId_t, 937
CFE_SB_PipeId_t, 937
CFE_SB_QosPriority, 938
CFE_SB_QosPriority_Enum_t, 937
CFE_SB_QosReliability, 939
CFE_SB_QosReliability_Enum_t, 938
CFE_SB_RouteId_Atom_t, 938

CFE_SB_SUB_ENTRIES_PER_PKT, 936
cfe_sb_msg.h
CFE_SB_AllSubscriptionsTlm_Payload_t, 1147
CFE_SB_AllSubscriptionsTlm_t, 1147
CFE_SB_DISABLE_ROUTE_CC, 1136
CFE_SB_DISABLE_SUB_REPORTING_CC, 1137
CFE_SB_DisableRouteCmd_t, 1147
CFE_SB_DisableSubReportingCmd_t, 1147
CFE_SB_ENABLE_ROUTE_CC, 1137
CFE_SB_ENABLE_SUB_REPORTING_CC, 1138
CFE_SB_EnableRouteCmd_t, 1147
CFE_SB_EnableSubReportingCmd_t, 1147
CFE_SB_HousekeepingTlm_Payload_t, 1148
CFE_SB_HousekeepingTlm_t, 1148
CFE_SB_MsgMapFileEntry_t, 1148
CFE_SB_NOOP_CC, 1139
CFE_SB_NoopCmd_t, 1148
CFE_SB_PipeDepthStats_t, 1148
CFE_SB_PipeInfoEntry_t, 1148
CFE_SB_RESET_COUNTERS_CC, 1140
CFE_SB_ResetCountersCmd_t, 1149
CFE_SB_RouteCmd_Payload_t, 1149
CFE_SB_RouteCmd_t, 1149
CFE_SB_RoutingFileEntry_t, 1149
CFE_SB_SEND_PREV_SUBS_CC, 1141
CFE_SB_SEND_SB_STATS_CC, 1142
CFE_SB_SendPrevSubsCmd_t, 1149
CFE_SB_SendSbStatsCmd_t, 1150
CFE_SB_SingleSubscriptionTlm_Payload_t, 1150
CFE_SB_SingleSubscriptionTlm_t, 1150
CFE_SB_StatsTlm_Payload_t, 1150
CFE_SB_StatsTlm_t, 1150
CFE_SB_SubEntries_t, 1151
CFE_SB_WRITE_MAP_INFO_CC, 1143
CFE_SB_WRITE_PIPE_INFO_CC, 1144
CFE_SB_WRITE_ROUTING_INFO_CC, 1145
CFE_SB_WriteFileInfoCmd_Payload_t, 1151
CFE_SB_WriteFileInfoCmd_t, 1151
CFE_SB_WriteMapInfoCmd_t, 1151
CFE_SB_WritePipeInfoCmd_t, 1151
CFE_SB_WriteRoutingInfoCmd_t, 1151
cfe_tbl_api_typedefs.h
CFE_TBL_BAD_TABLE_HANDLE, 941
CFE_TBL_CallbackFuncPtr_t, 942
CFE_TBL_Handle_t, 942
CFE_TBL_Info_t, 942
CFE_TBL_MAX_FULL_NAME_LEN, 942
CFE_TBL_SrcEnum, 943
CFE_TBL_SrcEnum_t, 942
cfe_tbl_events.h
CFE_TBL_ACTIVATE_DUMP_ONLY_ERR_EID, 1155
CFE_TBL_ACTIVATE_ERR_EID, 1155
CFE_TBL_ASSUMED_VALID_INF_EID, 1155
CFE_TBL_CC1_ERR_EID, 1156
CFE_TBL_CDS_DELETE_ERR_EID, 1156
CFE_TBL_CDS_DELETED_INFO_EID, 1157
CFE_TBL_CDS_NOT_FOUND_ERR_EID, 1157
CFE_TBL_CDS_OWNER_ACTIVE_ERR_EID, 1158
CFE_TBL_CREATING_DUMP_FILE_ERR_EID, 1158
CFE_TBL_DUMP_PENDING_ERR_EID, 1159
CFE_TBL_FAIL_HK_SEND_ERR_EID, 1159
CFE_TBL_FAIL_NOTIFY_SEND_ERR_EID, 1160
CFE_TBL_FILE_ACCESS_ERR_EID, 1160
CFE_TBL_FILE_INCOMPLETE_ERR_EID, 1161
CFE_TBL_FILE_LOADED_INF_EID, 1161
CFE_TBL_FILE_STD_HDR_ERR_EID, 1162
CFE_TBL_FILE_SUBTYPE_ERR_EID, 1162
CFE_TBL_FILE_TBL_HDR_ERR_EID, 1163
CFE_TBL_FILE_TOO_BIG_ERR_EID, 1163
CFE_TBL_FILE_TYPE_ERR_EID, 1164
CFE_TBL_HANDLE_ACCESS_ERR_EID, 1164
CFE_TBL_ILLEGAL_BUFF_PARAM_ERR_EID, 1165
CFE_TBL_IN_REGISTRY_ERR_EID, 1165
CFE_TBL_INIT_INF_EID, 1166
CFE_TBL_INTERNAL_ERROR_ERR_EID, 1166
CFE_TBL_LEN_ERR_EID, 1167
CFE_TBL_LOAD_ABORT_ERR_EID, 1167
CFE_TBL_LOAD_ABORT_INF_EID, 1168
CFE_TBL_LOAD_DUMPONLY_ERR_EID, 1168
CFE_TBL_LOAD_EXCEEDS_SIZE_ERR_EID, 1169
CFE_TBL_LOAD_FILENAME_LONG_ERR_EID, 1169
CFE_TBL_LOAD_IN_PROGRESS_ERR_EID, 1170
CFE_TBL_LOAD_PEND_REQ_INF_EID, 1170
CFE_TBL_LOAD_SUCCESS_INF_EID, 1171
CFE_TBL_LOAD_TBLNAME_MISMATCH_ERR_↩ EID, 1171
CFE_TBL_LOAD_TYPE_ERR_EID, 1172
CFE_TBL_LOAD_VAL_ERR_EID, 1172
CFE_TBL_LOADING_A_DUMP_ONLY_ERR_EID, 1172
CFE_TBL_LOADING_PENDING_ERR_EID, 1173
CFE_TBL_MAX_EID, 1173
CFE_TBL_MID_ERR_EID, 1174
CFE_TBL_NO_INACTIVE_BUFFER_ERR_EID, 1174
CFE_TBL_NO_SUCH_TABLE_ERR_EID, 1174
CFE_TBL_NO_WORK_BUFFERS_ERR_EID, 1175
CFE_TBL_NOOP_INF_EID, 1175
CFE_TBL_NOT_CRITICAL_TBL_ERR_EID, 1176
CFE_TBL_NOT_IN_CRIT_REG_ERR_EID, 1176
CFE_TBL_OVERWRITE_DUMP_INF_EID, 1177
CFE_TBL_OVERWRITE_REG_DUMP_INF_EID, 1177
CFE_TBL_PARTIAL_LOAD_ERR_EID, 1178

CFE_TBL_PROCESSOR_ID_ERR_EID, 1178
CFE_TBL_REGISTER_ERR_EID, 1179
CFE_TBL_RESET_INF_EID, 1179
CFE_TBL_SHARE_ERR_EID, 1180
CFE_TBL_SPACECRAFT_ID_ERR_EID, 1180
CFE_TBL_TLM_REG_CMD_INF_EID, 1181
CFE_TBL_TOO_MANY_DUMPS_ERR_EID, 1181
CFE_TBL_TOO_MANY_VALIDATIONS_ERR_EID, 1182
CFE_TBL_UNREGISTER_ERR_EID, 1182
CFE_TBL_UNVALIDATED_ERR_EID, 1183
CFE_TBL_UPDATE_ERR_EID, 1183
CFE_TBL_UPDATE_SUCCESS_INF_EID, 1184
CFE_TBL_VAL_REQ_MADE_INF_EID, 1184
CFE_TBL_VALIDATION_ERR_EID, 1185
CFE_TBL_VALIDATION_INF_EID, 1185
CFE_TBL_WRITE_CFE_HDR_ERR_EID, 1186
CFE_TBL_WRITE_DUMP_INF_EID, 1186
CFE_TBL_WRITE_REG_DUMP_INF_EID, 1187
CFE_TBL_WRITE_TBL_HDR_ERR_EID, 1187
CFE_TBL_WRITE_TBL_IMG_ERR_EID, 1188
CFE_TBL_WRITE_TBL_REG_ERR_EID, 1188
CFE_TBL_ZERO_LENGTH_LOAD_ERR_EID, 1189
cfe_tbl_extern_typedefs.h
CFE_TBL_BufferSelect, 945
CFE_TBL_BufferSelect_Enum_t, 944
CFE_TBL_File_Hdr_t, 944
cfe_tbl_filedef.h
CFE_TBL_FILEDEF, 946
CFE_TBL_FileDef_t, 947
cfe_tbl_msg.h
CFE_TBL_ABORT_LOAD_CC, 1192
CFE_TBL_ACTIVATE_CC, 1193
CFE_TBL_AbortLoadCmd_Payload_t, 1202
CFE_TBL_AbortLoadCmd_t, 1202
CFE_TBL_ActivateCmd_Payload_t, 1202
CFE_TBL_ActivateCmd_t, 1202
CFE_TBL_DELETE_CDS_CC, 1194
CFE_TBL_DUMP_CC, 1194
CFE_TBL_DUMP_REGISTRY_CC, 1195
CFE_TBL_DelCDSCmd_Payload_t, 1202
CFE_TBL_DeleteCDSCmd_t, 1202
CFE_TBL_DumpCmd_Payload_t, 1203
CFE_TBL_DumpCmd_t, 1203
CFE_TBL_DumpRegistryCmd_Payload_t, 1203
CFE_TBL_DumpRegistryCmd_t, 1203
CFE_TBL_HousekeepingTlm_Payload_t, 1203
CFE_TBL_HousekeepingTlm_t, 1203
CFE_TBL_LOAD_CC, 1196
CFE_TBL_LoadCmd_Payload_t, 1204
CFE_TBL_LoadCmd_t, 1204
CFE_TBL_NOOP_CC, 1197
CFE_TBL_NoArgsCmd_t, 1204
CFE_TBL_NoopCmd_t, 1204
CFE_TBL_NotifyCmd_Payload_t, 1204
CFE_TBL_NotifyCmd_t, 1205
CFE_TBL_RESET_COUNTERS_CC, 1198
CFE_TBL_ResetCountersCmd_t, 1205
CFE_TBL_SEND_REGISTRY_CC, 1199
CFE_TBL_SendRegistryCmd_Payload_t, 1205
CFE_TBL_SendRegistryCmd_t, 1205
CFE_TBL_TableRegistryTlm_t, 1205
CFE_TBL_TblRegPacket_Payload_t, 1205
CFE_TBL_VALIDATE_CC, 1200
CFE_TBL_ValidateCmd_Payload_t, 1206
CFE_TBL_ValidateCmd_t, 1206
cfe_time.h
CFE_TIME_Copy, 949
cfe_time_api_typedefs.h
CFE_TIME_Compare, 950
CFE_TIME_Compare_t, 950
CFE_TIME_PRINTED_STRING_SIZE, 950
CFE_TIME_SynchCallbackPtr_t, 950
cfe_time_events.h
CFE_TIME_1HZ_CFG_EID, 1208
CFE_TIME_1HZ_EID, 1208
CFE_TIME_1HZ_ERR_EID, 1209
CFE_TIME_CC_ERR_EID, 1209
CFE_TIME_DELAY_CFG_EID, 1210
CFE_TIME_DELAY_EID, 1210
CFE_TIME_DELAY_ERR_EID, 1211
CFE_TIME_DELTA_CFG_EID, 1211
CFE_TIME_DELTA_EID, 1212
CFE_TIME_DELTA_ERR_EID, 1212
CFE_TIME_DIAG_EID, 1213
CFE_TIME_EXIT_ERR_EID, 1213
CFE_TIME_FLY_OFF_EID, 1214
CFE_TIME_FLY_ON_EID, 1214
CFE_TIME_ID_ERR_EID, 1214
CFE_TIME_INIT_EID, 1215
CFE_TIME_LEAPS_CFG_EID, 1215
CFE_TIME_LEAPS_EID, 1216
CFE_TIME_LEN_ERR_EID, 1216
CFE_TIME_MAX_EID, 1217
CFE_TIME_MET_CFG_EID, 1217
CFE_TIME_MET_EID, 1217
CFE_TIME_MET_ERR_EID, 1218
CFE_TIME_NOOP_EID, 1218
CFE_TIME_RESET_EID, 1219
CFE_TIME_SIGNAL_CFG_EID, 1219
CFE_TIME_SIGNAL_EID, 1220
CFE_TIME_SIGNAL_ERR_EID, 1220
CFE_TIME_SOURCE_CFG_EID, 1221
CFE_TIME_SOURCE_EID, 1221
CFE_TIME_SOURCE_ERR_EID, 1222
CFE_TIME_STATE_EID, 1222
CFE_TIME_STATE_ERR_EID, 1223
CFE_TIME_STCF_CFG_EID, 1223

CFE_TIME_STCF_EID, 1224
CFE_TIME_STCF_ERR_EID, 1224
CFE_TIME_TIME_CFG_EID, 1225
CFE_TIME_TIME_EID, 1225
CFE_TIME_TIME_ERR_EID, 1226
cfe_time_extern_typedefs.h
CFE_TIME_AdjustDirection, 955
CFE_TIME_AdjustDirection_Enum_t, 953
CFE_TIME_ClockState, 955
CFE_TIME_ClockState_Enum_t, 953
CFE_TIME_FlagBit, 956
CFE_TIME_FlagBit_Enum_t, 953
CFE_TIME_FlywheelState, 956
CFE_TIME_FlywheelState_Enum_t, 954
CFE_TIME_SetState, 957
CFE_TIME_SetState_Enum_t, 954
CFE_TIME_SourceSelect, 957
CFE_TIME_SourceSelect_Enum_t, 954
CFE_TIME_SysTime_t, 954
CFE_TIME_ToneSignalSelect, 957
CFE_TIME_ToneSignalSelect_Enum_t, 955
cfe_time_msg.h
CFE_TIME_1HzCmd_t, 1246
CFE_TIME_ADD_1HZ_ADJUSTMENT_CC, 1230
CFE_TIME_ADD_ADJUST_CC, 1230
CFE_TIME_ADD_DELAY_CC, 1231
CFE_TIME_Add1HZAdjustmentCmd_t, 1247
CFE_TIME_AddAdjustCmd_t, 1247
CFE_TIME_AddDelayCmd_t, 1247
CFE_TIME_DiagnosticTlm_Payload_t, 1247
CFE_TIME_DiagnosticTlm_t, 1247
CFE_TIME_FakeToneCmd_t, 1247
CFE_TIME_HousekeepingTlm_Payload_t, 1248
CFE_TIME_HousekeepingTlm_t, 1248
CFE_TIME_LeapsCmd_Payload_t, 1248
CFE_TIME_NOOP_CC, 1232
CFE_TIME_NoArgsCmd_t, 1248
CFE_TIME_NoopCmd_t, 1248
CFE_TIME_OneHzAdjustmentCmd_Payload_t, 1248
CFE_TIME_OneHzAdjustmentCmd_t, 1249
CFE_TIME_RESET_COUNTERS_CC, 1233
CFE_TIME_ResetCountersCmd_t, 1249
CFE_TIME_SEND_DIAGNOSTIC_TLM_CC, 1234
CFE_TIME_SET_LEAP_SECONDS_CC, 1235
CFE_TIME_SET_MET_CC, 1236
CFE_TIME_SET_SIGNAL_CC, 1237
CFE_TIME_SET_SOURCE_CC, 1238
CFE_TIME_SET_STATE_CC, 1239
CFE_TIME_SET_STCF_CC, 1241
CFE_TIME_SET_TIME_CC, 1242
CFE_TIME_SUB_1HZ_ADJUSTMENT_CC, 1243
CFE_TIME_SUB_ADJUST_CC, 1244
CFE_TIME_SUB_DELAY_CC, 1245
CFE_TIME_SendDiagnosticCmd_t, 1249
CFE_TIME_SetLeapSecondsCmd_t, 1249
CFE_TIME_SetMETCmd_t, 1249
CFE_TIME_SetSTCFCmd_t, 1250
CFE_TIME_SetSignalCmd_t, 1249
CFE_TIME_SetSourceCmd_t, 1250
CFE_TIME_SetStateCmd_t, 1250
CFE_TIME_SetTimeCmd_t, 1250
CFE_TIME_SignalCmd_Payload_t, 1250
CFE_TIME_SourceCmd_Payload_t, 1250
CFE_TIME_StateCmd_Payload_t, 1251
CFE_TIME_Sub1HZAdjustmentCmd_t, 1251
CFE_TIME_SubAdjustCmd_t, 1251
CFE_TIME_SubDelayCmd_t, 1251
CFE_TIME_TimeCmd_Payload_t, 1251
CFE_TIME_TimeCmd_t, 1251
CFE_TIME_ToneDataCmd_Payload_t, 1252
CFE_TIME_ToneDataCmd_t, 1252
CFE_TIME_ToneSignalCmd_t, 1252
cfe_version.h
CFE_BUILD_BASELINE, 958
CFE_BUILD_NUMBER, 959
CFE_MAJOR_VERSION, 959
CFE_MINOR_VERSION, 959
CFE_MISSION_REV, 959
CFE_REVISION, 959
CFE_SRC_VERSION, 960
CFE_STR_HELPER, 960
CFE_STR, 960
CFE_VERSION_STRING, 960
CheckErrCtr
CFE_ES_MemPoolStats, 545
ClockFlyState
CFE_TIME_DiagnosticTlm_Payload, 694
ClockSetState
CFE_TIME_DiagnosticTlm_Payload, 694
ClockSignal
CFE_TIME_DiagnosticTlm_Payload, 694
ClockSource
CFE_TIME_DiagnosticTlm_Payload, 694
ClockState
CFE_TIME_StateCmd_Payload, 718
ClockStateAPI
CFE_TIME_DiagnosticTlm_Payload, 695
CFE_TIME_HousekeepingTlm_Payload, 705
ClockStateFlags
CFE_TIME_DiagnosticTlm_Payload, 695
CFE_TIME_HousekeepingTlm_Payload, 705
CmdHeader
CFE_ES_AppNameCmd, 521
CFE_ES_DeleteCDSCmd, 527
CFE_ES_DumpCDSRegistryCmd, 529
CFE_ES_FileNameCmd, 530
CFE_ES_NoArgsCmd, 548
CFE_ES_OverWriteSysLogCmd, 550

CFE_ES_ReloadAppCmd, 554
CFE_ES_RestartCmd, 555
CFE_ES_SendMemPoolStatsCmd, 557
CFE_ES_SetMaxPRCountCmd, 559
CFE_ES_SetPerfFilterMaskCmd, 561
CFE_ES_SetPerfTriggerMaskCmd, 563
CFE_ES_StartApp, 565
CFE_ES_StartPerfDataCmd, 569
CFE_ES_StopPerfDataCmd, 571
CFE_EVS_AppNameBitMaskCmd, 575
CFE_EVS_AppNameCmd, 577
CFE_EVS_AppNameEventIDCmd, 579
CFE_EVS_AppNameEventIDMaskCmd, 581
CFE_EVS_BitMaskCmd, 586
CFE_EVS_NoArgsCmd, 597
CFE_EVS_SetEventFormatModeCmd, 601
CFE_EVS_SetLogModeCmd, 603
CFE_EVS_WriteAppDataFileCmd, 605
CFE_EVS_WriteLogDataFileCmd, 606
CFE_TBL_AbortLoadCmd, 652
CFE_TBL_ActivateCmd, 654
CFE_TBL_DeleteCDSCmd, 656
CFE_TBL_DumpCmd, 657
CFE_TBL_DumpRegistryCmd, 660
CFE_TBL_LoadCmd, 675
CFE_TBL_NoArgsCmd, 677
CFE_TBL_NotifyCmd, 678
CFE_TBL_SendRegistryCmd, 680
CFE_TBL_ValidateCmd, 688
CFE_TIME_NoArgsCmd, 710
CFE_TIME_OneHzAdjustmentCmd, 710
CFE_TIME_SetLeapSecondsCmd, 712
CFE_TIME_SetSignalCmd, 713
CFE_TIME_SetSourceCmd, 714
CFE_TIME_SetStateCmd, 715
CFE_TIME_TimeCmd, 720
CFE_TIME_ToneDataCmd, 721
code_address
OS_module_address_t, 733
code_size
OS_module_address_t, 733
CodeAddress
CFE_ES_AppInfo, 516
CodeSize
CFE_ES_AppInfo, 516
CommandCounter
CFE_ES_HousekeepingTlm_Payload, 536
CFE_EVS_HousekeepingTlm_Payload, 590
CFE_SB_HousekeepingTlm_Payload, 622
CFE_TBL_HousekeepingTlm_Payload, 667
CFE_TIME_HousekeepingTlm_Payload, 705
CommandData
cfe_psp_start.c, 1347
CommandErrorCounter
CFE_ES_HousekeepingTlm_Payload, 536
CFE_EVS_HousekeepingTlm_Payload, 590
CFE_SB_HousekeepingTlm_Payload, 622
CFE_TBL_HousekeepingTlm_Payload, 667
CFE_TIME_HousekeepingTlm_Payload, 706
common_types.h
EXTENSION, 1253
CompileTimeAssert, 1253, 1258, 1259
cpuaddr, 1255
cpudiff, 1255
cpusize, 1255
int16, 1255
int32, 1255
int64, 1255
int8, 1256
intptr, 1256
OS_ArgCallback_t, 1256
OS_PRINTF, 1254
OS_USED, 1254
OSAL_BLOCKCOUNT_C, 1254
OSAL_INDEX_C, 1254
OSAL_OBJTYPE_C, 1254
OSAL_SIZE_C, 1254
osal_blockcount_t, 1256
osal_id_t, 1256
osal_index_t, 1257
osal_objtype_t, 1257
uint16, 1257
uint32, 1257
uint64, 1257
uint8, 1258
CompileTimeAssert
common_types.h, 1253, 1258, 1259
ContentType
CFE_FS_Header, 609
cpu1_msgids.h, 768
CFE_ES_APP_TLM_MID, 770
CFE_ES_CMD_MID, 770
CFE_ES_HK_TLM_MID, 770
CFE_ES_MEMSTATS_TLM_MID, 770
CFE_ES_SEND_HK_MID, 770
CFE_EVS_CMD_MID, 771
CFE_EVS_HK_TLM_MID, 771
CFE_EVS_LONG_EVENT_MSG_MID, 771
CFE_EVS_SEND_HK_MID, 771
CFE_EVS_SHORT_EVENT_MSG_MID, 771
CFE_PLATFORM_CMD_MID_BASE_GLOB, 772
CFE_PLATFORM_CMD_MID_BASE, 771
CFE_PLATFORM_TLM_MID_BASE, 772
CFE_SB_ALLSUBS_TLM_MID, 772
CFE_SB_CMD_MID, 773
CFE_SB_HK_TLM_MID, 773
CFE_SB_ONESUB_TLM_MID, 773
CFE_SB_SEND_HK_MID, 773

CFE_SB_STATS_TLM_MID, 773
CFE_SB_SUB_RPT_CTRL_MID, 773
CFE_TBL_CMD_MID, 774
CFE_TBL_HK_TLM_MID, 774
CFE_TBL_REG_TLM_MID, 774
CFE_TBL_SEND_HK_MID, 774
CFE_TIME_1HZ_CMD_MID, 774
CFE_TIME_CMD_MID, 774
CFE_TIME_DATA_CMD_MID, 775
CFE_TIME_DIAG_TLM_MID, 775
CFE_TIME_HK_TLM_MID, 775
CFE_TIME_SEND_CMD_MID, 775
CFE_TIME_SEND_HK_MID, 775
CFE_TIME_TONE_CMD_MID, 775
cpu1_platform_cfg.h, 776
CFE_PLATFORM_CORE_MAX_STARTUP_MSEC, 779
CFE_PLATFORM_ENDIAN, 780
CFE_PLATFORM_ES_APP_KILL_TIMEOUT, 780
CFE_PLATFORM_ES_APP_SCAN_RATE, 781
CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE, 781
CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES, 782
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_01, 782
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_02, 782
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_03, 783
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_04, 783
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_05, 783
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_06, 783
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_07, 783
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_08, 783
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_09, 784
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_10, 784
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_11, 784
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_12, 784
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_13, 784
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_14, 784
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_15, 785
CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE↩_16, 785
CFE_PLATFORM_ES_CDS_SIZE, 785
CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE, 785
CFE_PLATFORM_ES_DEFAULT_CDS_REG_DU↩MP_FILE, 786
CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE, 786
CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_↩FILENAME, 786
CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG↩_MODE, 787
CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_↩MODE, 787
CFE_PLATFORM_ES_DEFAULT_STACK_SIZE, 788
CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE, 788
CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FI↩LE, 789
CFE_PLATFORM_ES_ER_LOG_ENTRIES, 789
CFE_PLATFORM_ES_ER_LOG_MAX_CONTEX↩T_SIZE, 790
CFE_PLATFORM_ES_MAX_APPLICATIONS, 790
CFE_PLATFORM_ES_MAX_BLOCK_SIZE, 790
CFE_PLATFORM_ES_MAX_GEN_COUNTERS, 791
CFE_PLATFORM_ES_MAX_LIBRARIES, 791
CFE_PLATFORM_ES_MAX_MEMORY_POOLS, 791
CFE_PLATFORM_ES_MAX_PROCESSOR_RES↩ETS, 792
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01, 792
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02, 793
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03, 793
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04, 793
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05, 793
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06, 794
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07, 794
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08, 794
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09, 794
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10, 794
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11, 794
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12, 795
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13, 795
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14, 795
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15, 795
CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16, 795
CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE↩_MIN, 795
CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_↩STRING, 796
CFE_PLATFORM_ES_NONVOL_STARTUP_FILE,

796
CFE_PLATFORM_ES_OBJECT_TABLE_SIZE, 796
CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY, 797
CFE_PLATFORM_ES_PERF_CHILD_PRIORITY, 797
CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE, 797
CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE, 798
CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS, 798
CFE_PLATFORM_ES_PERF_FILTMASK_ALL, 798
CFE_PLATFORM_ES_PERF_FILTMASK_INIT, 799
CFE_PLATFORM_ES_PERF_FILTMASK_NONE, 799
CFE_PLATFORM_ES_PERF_TRIGMASK_ALL, 799
CFE_PLATFORM_ES_PERF_TRIGMASK_INIT, 800
CFE_PLATFORM_ES_PERF_TRIGMASK_NONE, 800
CFE_PLATFORM_ES_POOL_MAX_BUCKETS, 800
CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING, 801
CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS, 801
CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED, 801
CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE, 802
CFE_PLATFORM_ES_RESET_AREA_SIZE, 802
CFE_PLATFORM_ES_START_TASK_PRIORITY, 803
CFE_PLATFORM_ES_START_TASK_STACK_SIZE, 803
CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC, 804
CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC, 804
CFE_PLATFORM_ES_SYSTEM_LOG_SIZE, 805
CFE_PLATFORM_ES_USER_RESERVED_SIZE, 805
CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE, 806
CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE, 806
CFE_PLATFORM_EVS_DEFAULT_LOG_FILE, 807
CFE_PLATFORM_EVS_DEFAULT_LOG_MODE, 807
CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE, 807
CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG, 808
CFE_PLATFORM_EVS_LOG_MAX, 808
CFE_PLATFORM_EVS_MAX_EVENT_FILTERS, 808
CFE_PLATFORM_EVS_PORT_DEFAULT, 809
CFE_PLATFORM_EVS_START_TASK_PRIORITY, 809
CFE_PLATFORM_EVS_START_TASK_STACK_SIZE, 809
CFE_PLATFORM_SB_BUF_MEMORY_BYTES, 810
CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME, 810
CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT, 811
CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME, 811
CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME, 812
CFE_PLATFORM_SB_FILTER_MASK1, 812
CFE_PLATFORM_SB_FILTER_MASK2, 812
CFE_PLATFORM_SB_FILTER_MASK3, 813
CFE_PLATFORM_SB_FILTER_MASK4, 813
CFE_PLATFORM_SB_FILTER_MASK5, 813
CFE_PLATFORM_SB_FILTER_MASK6, 813
CFE_PLATFORM_SB_FILTER_MASK7, 813
CFE_PLATFORM_SB_FILTER_MASK8, 813
CFE_PLATFORM_SB_FILTERED_EVENT1, 814
CFE_PLATFORM_SB_FILTERED_EVENT2, 814
CFE_PLATFORM_SB_FILTERED_EVENT3, 814
CFE_PLATFORM_SB_FILTERED_EVENT4, 814
CFE_PLATFORM_SB_FILTERED_EVENT5, 814
CFE_PLATFORM_SB_FILTERED_EVENT6, 815
CFE_PLATFORM_SB_FILTERED_EVENT7, 815
CFE_PLATFORM_SB_FILTERED_EVENT8, 815
CFE_PLATFORM_SB_HIGHEST_VALID_MSGID, 815
CFE_PLATFORM_SB_MAX_BLOCK_SIZE, 815
CFE_PLATFORM_SB_MAX_DEST_PER_PKT, 816
CFE_PLATFORM_SB_MAX_MSG_IDS, 816
CFE_PLATFORM_SB_MAX_PIPES, 816
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01, 817
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02, 817
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03, 817
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04, 818
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05, 818
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06, 818
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07, 818
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08, 818
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09, 818
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10, 819
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11, 819
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12, 819
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13, 819
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14, 819
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15, 819
CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16, 820

CFE_PLATFORM_SB_START_TASK_PRIORITY, 820
CFE_PLATFORM_SB_START_TASK_STACK_SIZE, 820
CFE_PLATFORM_TBL_BUF_MEMORY_BYTES, 820
CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE, 821
CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES, 821
CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE, 821
CFE_PLATFORM_TBL_MAX_NUM_HANDLES, 822
CFE_PLATFORM_TBL_MAX_NUM_TABLES, 822
CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS, 822
CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS, 823
CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE, 823
CFE_PLATFORM_TBL_START_TASK_PRIORITY, 824
CFE_PLATFORM_TBL_START_TASK_STACK_SIZE, 824
CFE_PLATFORM_TBL_U32FROM4CHARS, 824
CFE_PLATFORM_TBL_VALID_PRID_1, 825
CFE_PLATFORM_TBL_VALID_PRID_2, 825
CFE_PLATFORM_TBL_VALID_PRID_3, 825
CFE_PLATFORM_TBL_VALID_PRID_4, 825
CFE_PLATFORM_TBL_VALID_PRID_COUNT, 826
CFE_PLATFORM_TBL_VALID_SCID_1, 826
CFE_PLATFORM_TBL_VALID_SCID_2, 826
CFE_PLATFORM_TBL_VALID_SCID_COUNT, 827
CFE_PLATFORM_TIME_1HZ_TASK_PRIORITY, 827
CFE_PLATFORM_TIME_1HZ_TASK_STACK_SIZE, 827
CFE_PLATFORM_TIME_CFG_CLIENT, 827
CFE_PLATFORM_TIME_CFG_LATCH_FLY, 827
CFE_PLATFORM_TIME_CFG_SERVER, 828
CFE_PLATFORM_TIME_CFG_SIGNAL, 828
CFE_PLATFORM_TIME_CFG_SOURCE, 828
CFE_PLATFORM_TIME_CFG_SRC_GPS, 829
CFE_PLATFORM_TIME_CFG_SRC_MET, 829
CFE_PLATFORM_TIME_CFG_SRC_TIME, 829
CFE_PLATFORM_TIME_CFG_START_FLY, 830
CFE_PLATFORM_TIME_CFG_TONE_LIMIT, 830
CFE_PLATFORM_TIME_CFG_VIRTUAL, 830
CFE_PLATFORM_TIME_MAX_DELTA_SECS, 831
CFE_PLATFORM_TIME_MAX_DELTA_SUBS, 831
CFE_PLATFORM_TIME_MAX_LOCAL_SECS, 832
CFE_PLATFORM_TIME_MAX_LOCAL_SUBS, 832
CFE_PLATFORM_TIME_START_TASK_PRIORITY, 832
CFE_PLATFORM_TIME_START_TASK_STACK_SIZE, 832
CFE_PLATFORM_TIME_TONE_TASK_PRIORITY, 833
CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE, 833
CpuId
CFE_PSP_CommandData_t, 612
CpuName
CFE_PSP_CommandData_t, 612
cpuaddr
common_types.h, 1255
cpudiff
common_types.h, 1255
cpusize
common_types.h, 1255
Crc
CFE_TBL_Info, 672
CFE_TBL_TblRegPacket_Payload, 684
CreatePipeErrorCounter
CFE_SB_HousekeepingTlm_Payload, 622
creator
OS_bin_sem_prop_t, 724
OS_count_sem_prop_t, 725
OS_mut_sem_prop_t, 736
OS_queue_prop_t, 737
OS_socket_prop_t, 740
OS_task_prop_t, 743
OS_timebase_prop_t, 746
OS_timer_prop_t, 747
Critical
CFE_TBL_Info, 672
CFE_TBL_TblRegPacket_Payload, 684
CurrentLatch
CFE_TIME_DiagnosticTlm_Payload, 695
CurrentMET
CFE_TIME_DiagnosticTlm_Payload, 695
CurrentQueueDepth
CFE_SB_PipeDepthStats, 629
CFE_SB_PipeInfoEntry, 632
CurrentTAI
CFE_TIME_DiagnosticTlm_Payload, 696
CurrentUTC
CFE_TIME_DiagnosticTlm_Payload, 696

data_address
OS_module_address_t, 733
data_size
OS_module_address_t, 733
DataAddress
CFE_ES_AppInfo, 516
DataFileName
CFE_ES_StopPerfCmd_Payload, 570
DataSize

CFE_ES_AppInfo, 517
DataStoreStatus
CFE_TIME_DiagnosticTlm_Payload, 696
DelayDirection
CFE_TIME_DiagnosticTlm_Payload, 696
Description
CFE_FS_FileWriteMetaData, 607
CFE_FS_Header, 610
CFE_TBL_FileDef, 663
DoubleBuffered
CFE_TBL_Info, 673
CFE_TBL_TblRegPacket_Payload, 684
DumpFilename
CFE_ES_DumpCDSRegistryCmd_Payload, 530
CFE_TBL_DumpCmd_Payload, 659
CFE_TBL_DumpRegistryCmd_Payload, 661
DumpOnly
CFE_TBL_Info, 673
CFE_TBL_TblRegPacket_Payload, 684
DuplicateSubscriptionsCounter
CFE_SB_HousekeepingTlm_Payload, 623

ERLogEntries
CFE_ES_HousekeepingTlm_Payload, 536
ERLogIndex
CFE_ES_HousekeepingTlm_Payload, 537
Entries
CFE_SB_AllSubscriptionsTlm_Payload, 619
Entry
CFE_SB_AllSubscriptionsTlm_Payload, 619
entry_point
OS_module_prop_t, 735
EntryPoint
CFE_ES_AppInfo, 517
EventID
CFE_EVS_AppNameEventIDCmd_Payload, 580
CFE_EVS_AppNameEventIDMaskCmd_Payload, 582
CFE_EVS_BinFilter, 585
CFE_EVS_PacketID, 598
EventType
CFE_EVS_PacketID, 598
ExceptionAction
CFE_ES_AppInfo, 517
CFE_ES_StartAppCmd_Payload, 567
ExceptionStorage
CFE_PSP_LinuxReservedAreaFixedLayout_t, 614
ExecutionCounter
CFE_ES_AppInfo, 517
CFE_ES_TaskInfo, 572

FailedValCounter
CFE_TBL_HousekeepingTlm_Payload, 667
FileCreateTimeSecs
CFE_TBL_Info, 673
CFE_TBL_TblRegPacket_Payload, 685
FileCreateTimeSubSecs
CFE_TBL_Info, 673
CFE_TBL_TblRegPacket_Payload, 685
FileModeBits
os_fstat_t, 730
FileName
CFE_ES_AppInfo, 518
CFE_ES_FileNameCmd_Payload, 531
CFE_FS_FileWriteMetaData, 607
os_dirent_t, 726
FileSize
os_fstat_t, 730
FileSubType
CFE_FS_FileWriteMetaData, 608
FileTime
os_fstat_t, 730
Filename
CFE_SB_WriteFileInfoCmd_Payload, 650
filename
OS_module_prop_t, 735
FilterMask
CFE_ES_SetPerfFilterMaskCmd_Payload, 562
FilterMaskNum
CFE_ES_SetPerfFilterMaskCmd_Payload, 562
flags
OS_module_address_t, 733
Forced2Fly
CFE_TIME_DiagnosticTlm_Payload, 697
free_blocks
OS_heap_prop_t, 731
free_bytes
OS_heap_prop_t, 731
FreeFds
os_fsinfo_t, 729
FreeVolumes
os_fsinfo_t, 729
freerun_time
OS_timebase_prop_t, 746

GLOBAL_PSP_CONFIGDATA
cfe_psp_configdata.h, 1325
GetData
CFE_FS_FileWriteMetaData, 608
GetPipeIdByNameErrorCounter
CFE_SB_HousekeepingTlm_Payload, 623
GotCpuId
CFE_PSP_CommandData_t, 612
GotCpuName
CFE_PSP_CommandData_t, 612
GotResetType
CFE_PSP_CommandData_t, 612
GotSpacecraftId
CFE_PSP_CommandData_t, 613

GotSubType
CFE_PSP_CommandData_t, 613

HW_NumEepromBanks
Target_PspConfigData, 748
Handle
CFE_ES_CDSRegDumpRec, 525
Hdr
CFE_SB_AllSubscriptionsTlm, 618
CFE_SB_HousekeepingTlm, 620
CFE_SB_RouteCmd, 635
CFE_SB_SingleSubscriptionTlm, 639
CFE_SB_StatsTlm, 642
CFE_SB_WriteFileInfoCmd, 649
HeapBlocksFree
CFE_ES_HousekeepingTlm_Payload, 537
HeapBytesFree
CFE_ES_HousekeepingTlm_Payload, 537
HeapMaxBlockSize
CFE_ES_HousekeepingTlm_Payload, 537
host_module_id
OS_module_prop_t, 735

InactiveBufferAddr
CFE_TBL_TblRegPacket_Payload, 685
Index
CFE_SB_MsgMapFileEntry, 628
int16
common_types.h, 1255
int32
common_types.h, 1255
int64
common_types.h, 1255
int8
common_types.h, 1256
InternalErrorCounter
CFE_SB_HousekeepingTlm_Payload, 623
interval_time
OS_timer_prop_t, 747
intptr
common_types.h, 1256
IsPending
CFE_FS_FileWriteMetaData, 608
IsValid
OS_file_prop_t, 728

LENGTHCHECK
osapi-macros.h, 1279
largest_free_block
OS_heap_prop_t, 732
LastFileDumped
CFE_TBL_HousekeepingTlm_Payload, 667
LastFileLoaded
CFE_TBL_HousekeepingTlm_Payload, 668
CFE_TBL_Info, 673
CFE_TBL_TblRegPacket_Payload, 685
LastTableLoaded
CFE_TBL_HousekeepingTlm_Payload, 668
LastUpdateTime
CFE_TBL_HousekeepingTlm_Payload, 668
LastUpdatedTable
CFE_TBL_HousekeepingTlm_Payload, 668
LastValCrc
CFE_TBL_HousekeepingTlm_Payload, 669
LastValStatus
CFE_TBL_HousekeepingTlm_Payload, 669
LastValTableName
CFE_TBL_HousekeepingTlm_Payload, 669
LeapSeconds
CFE_TIME_HousekeepingTlm_Payload, 706
CFE_TIME_LeapsCmd_Payload, 709
Length
CCSDS_PrimaryHeader, 513
CFE_FS_Header, 610
LoadFilename
CFE_TBL_LoadCmd_Payload, 676
LoadPending
CFE_TBL_TblRegPacket_Payload, 686
LocalIntCounter
CFE_TIME_DiagnosticTlm_Payload, 697
LocalTaskCounter
CFE_TIME_DiagnosticTlm_Payload, 697
LogEnabled
CFE_EVS_HousekeepingTlm_Payload, 590
LogFilename
CFE_EVS_LogFileCmd_Payload, 594
LogFullFlag
CFE_EVS_HousekeepingTlm_Payload, 590
LogMode
CFE_EVS_HousekeepingTlm_Payload, 591
CFE_EVS_SetLogMode_Payload, 602
LogOverflowCounter
CFE_EVS_HousekeepingTlm_Payload, 591
LongDouble
CFE_ES_PoolAlign, 552
CFE_SB_Msg, 627
LongInt
CFE_ES_PoolAlign, 552
CFE_SB_Msg, 627
longOpts
cfe_psp_start.c, 1347

MainTaskId
CFE_ES_AppInfo, 518
MainTaskName
CFE_ES_AppInfo, 518
MajorVersion
CFE_PSP_VersionInfo_t, 616
Mask

CFE_EVS_AppNameEventIDMaskCmd_Payload, 582
CFE_EVS_BinFilter, 585
MaxElapsed
CFE_TIME_DiagnosticTlm_Payload, 697
MaxFds
os_fsinfo_t, 729
MaxLocalClock
CFE_TIME_DiagnosticTlm_Payload, 698
MaxMemAllowed
CFE_SB_StatsTlm_Payload, 643
MaxMsgIdsAllowed
CFE_SB_StatsTlm_Payload, 644
MaxPRCount
CFE_ES_SetMaxPRCountCmd_Payload, 560
MaxPipeDepthAllowed
CFE_SB_StatsTlm_Payload, 644
MaxPipesAllowed
CFE_SB_StatsTlm_Payload, 644
MaxProcessorResets
CFE_ES_HousekeepingTlm_Payload, 538
MaxQueueDepth
CFE_SB_PipeDepthStats, 629
CFE_SB_PipeInfoEntry, 632
MaxSubscriptionsAllowed
CFE_SB_StatsTlm_Payload, 644
MaxVolumes
os_fsinfo_t, 729
MemInUse
CFE_SB_HousekeepingTlm_Payload, 623
CFE_SB_StatsTlm_Payload, 645
MemPoolHandle
CFE_SB_HousekeepingTlm_Payload, 624
CFE_TBL_HousekeepingTlm_Payload, 669
MemoryType
CFE_PSP_MemTable_t, 615
Message
CFE_EVS_LongEventTlm_Payload, 596
MessageFormatMode
CFE_EVS_HousekeepingTlm_Payload, 591
MessageSendCounter
CFE_EVS_HousekeepingTlm_Payload, 591
MessageTruncCounter
CFE_EVS_HousekeepingTlm_Payload, 592
MicroSeconds
CFE_TIME_TimeCmd_Payload, 720
MinElapsed
CFE_TIME_DiagnosticTlm_Payload, 698
MinorVersion
CFE_PSP_VersionInfo_t, 617
MissionRev
CFE_PSP_VersionInfo_t, 617
Mode
CFE_ES_OverWriteSysLogCmd_Payload, 551
Module
OS_static_symbol_record_t, 741
Msg
CFE_SB_Msg, 627
MsgCnt
CFE_SB_RoutingFileEntry, 638
MsgFormat
CFE_EVS_SetEventFormatCode_Payload, 600
MsgId
CFE_SB_MsgMapFileEntry, 628
CFE_SB_RouteCmd_Payload, 636
CFE_SB_RoutingFileEntry, 638
CFE_SB_SingleSubscriptionTlm_Payload, 640
CFE_SB_SubEntries, 648
MsgIdsInUse
CFE_SB_StatsTlm_Payload, 645
MsgLimitErrorCounter
CFE_SB_HousekeepingTlm_Payload, 624
MsgReceiveErrorCounter
CFE_SB_HousekeepingTlm_Payload, 624
MsgSendErrorCounter
CFE_SB_HousekeepingTlm_Payload, 624

Name
CFE_ES_AppInfo, 518
CFE_ES_CDSRegDumpRec, 525
CFE_TBL_TblRegPacket_Payload, 686
OS_static_symbol_record_t, 741
name
OS_bin_sem_prop_t, 724
OS_count_sem_prop_t, 725
OS_module_prop_t, 735
OS_mut_sem_prop_t, 736
OS_queue_prop_t, 737
OS_socket_prop_t, 740
OS_task_prop_t, 743
OS_timebase_prop_t, 746
OS_timer_prop_t, 747
Network ID APIs, 469
OS_NetworkGetHostName, 469
OS_NetworkGetID, 469
NoSubscribersCounter
CFE_SB_HousekeepingTlm_Payload, 625
nominal_interval_time
OS_timebase_prop_t, 746
NumBlocksRequested
CFE_ES_MemPoolStats, 545
NumBytes
CFE_TBL_File_Hdr, 662
NumCreated
CFE_ES_BlockStats, 524
NumFree
CFE_ES_BlockStats, 524
NumFreeBytes

CFE_ES_MemPoolStats, 546
NumFreeSharedBufs
CFE_TBL_HousekeepingTlm_Payload, 670
NumLoadPending
CFE_TBL_HousekeepingTlm_Payload, 670
NumOfChildTasks
CFE_ES_AppInfo, 519
NumTables
CFE_TBL_HousekeepingTlm_Payload, 670
NumUsers
CFE_TBL_Info, 674
NumValRequests
CFE_TBL_HousekeepingTlm_Payload, 670

OS_API_Init
OSAL Core Operation APIs, 400
OS_Application_Run
OSAL Core Operation APIs, 401
OS_Application_Startup
OSAL Core Operation APIs, 401
OS_ApplicationExit
OSAL Core Operation APIs, 402
OS_ApplicationShutdown
OSAL Core Operation APIs, 403
OS_ArgCallback_t
common_types.h, 1256
OS_BSP_GetArgC
OSAL BSP low level access APIs, 386
OS_BSP_GetArgV
OSAL BSP low level access APIs, 386
OS_BSP_SetExitCode
OSAL BSP low level access APIs, 386
OS_BUFFER_MSG_DEPTH
osconfig-example.h, 751
osconfig.h, 760
OS_BUFFER_SIZE
osconfig-example.h, 751
osconfig.h, 760
OS_BUILD_BASELINE
osapi-version.h, 1293
OS_BUILD_NUMBER
osapi-version.h, 1293
OS_BinSemCreate
OSAL Binary Semaphore APIs, 380
OS_BinSemDelete
OSAL Binary Semaphore APIs, 381
OS_BinSemFlush
OSAL Binary Semaphore APIs, 381
OS_BinSemGetIdByName
OSAL Binary Semaphore APIs, 382
OS_BinSemGetInfo
OSAL Binary Semaphore APIs, 383
OS_BinSemGive
OSAL Binary Semaphore APIs, 383
OS_BinSemTake
OSAL Binary Semaphore APIs, 384
OS_BinSemTimedWait
OSAL Binary Semaphore APIs, 384
OS_CHECK
osapi-constants.h, 1265
OS_CHK_ONLY
osapi-filesys.h, 1275
OS_CloseAllFiles
OSAL Standard File APIs, 428
OS_CloseFileByName
OSAL Standard File APIs, 429
OS_ConvertToArrayIndex
OSAL Object ID Utility APIs, 452
OS_CountSemCreate
OSAL Counting Semaphore APIs, 405
OS_CountSemDelete
OSAL Counting Semaphore APIs, 406
OS_CountSemGetIdByName
OSAL Counting Semaphore APIs, 406
OS_CountSemGetInfo
OSAL Counting Semaphore APIs, 407
OS_CountSemGive
OSAL Counting Semaphore APIs, 407
OS_CountSemTake
OSAL Counting Semaphore APIs, 408
OS_CountSemTimedWait
OSAL Counting Semaphore APIs, 409
OS_DIRENTRY_NAME
osapi-dir.h, 1267
OS_DeleteAllObjects
OSAL Core Operation APIs, 403
OS_DirectoryClose
OSAL Directory APIs, 410
OS_DirectoryOpen
OSAL Directory APIs, 410
OS_DirectoryRead
OSAL Directory APIs, 411
OS_DirectoryRewind
OSAL Directory APIs, 411
OS_ERR_BAD_ADDRESS
OSAL Return Code Defines, 415
OS_ERR_FILE
OSAL Return Code Defines, 416
OS_ERR_INCORRECT_OBJ_STATE
OSAL Return Code Defines, 416
OS_ERR_INCORRECT_OBJ_TYPE
OSAL Return Code Defines, 416
OS_ERR_INVALID_ID
OSAL Return Code Defines, 416
OS_ERR_INVALID_PRIORITY
OSAL Return Code Defines, 416
OS_ERR_INVALID_SIZE
OSAL Return Code Defines, 417

OS_ERR_NAME_NOT_FOUND
OSAL Return Code Defines, 417
OS_ERR_NAME_TAKEN
OSAL Return Code Defines, 417
OS_ERR_NAME_TOO_LONG
OSAL Return Code Defines, 417
OS_ERR_NO_FREE_IDS
OSAL Return Code Defines, 417
OS_ERR_NOT_IMPLEMENTED
OSAL Return Code Defines, 418
OS_ERR_OBJECT_IN_USE
OSAL Return Code Defines, 418
OS_ERR_OPERATION_NOT_SUPPORTED
OSAL Return Code Defines, 418
OS_ERR_SEM_NOT_FULL
OSAL Return Code Defines, 418
OS_ERR_STREAM_DISCONNECTED
OSAL Return Code Defines, 418
OS_ERROR_ADDRESS_MISALIGNED
OSAL Return Code Defines, 419
OS_ERROR_NAME_LENGTH
osapi-error.h, 1269
OS_ERROR_TIMEOUT
OSAL Return Code Defines, 419
OS_ERROR
OSAL Return Code Defines, 419
OS_Event_t
osapi-common.h, 1263
OS_EventHandler_t
osapi-common.h, 1263
OS_FDGetInfo
OSAL Standard File APIs, 430
OS_FILESTAT_EXEC
osapi-file.h, 1272
OS_FILESTAT_ISDIR
osapi-file.h, 1272
OS_FILESTAT_MODE
osapi-file.h, 1272
OS_FILESTAT_READ
osapi-file.h, 1272
OS_FILESTAT_SIZE
osapi-file.h, 1272
OS_FILESTAT_TIME
osapi-file.h, 1273
OS_FILESTAT_WRITE
osapi-file.h, 1273
OS_FP_ENABLED
osapi-task.h, 1288
OS_FS_DEV_NAME_LEN
osconfig-example.h, 752
osconfig.h, 761
OS_FS_ERR_DEVICE_NOT_FREE
OSAL Return Code Defines, 419
OS_FS_ERR_DRIVE_NOT_CREATED
OSAL Return Code Defines, 419
OS_FS_ERR_NAME_TOO_LONG
OSAL Return Code Defines, 420
OS_FS_ERR_PATH_INVALID
OSAL Return Code Defines, 420
OS_FS_ERR_PATH_TOO_LONG
OSAL Return Code Defines, 420
OS_FS_GetPhysDriveName
OSAL File System Level APIs, 441
OS_FS_PHYS_NAME_LEN
osconfig-example.h, 752
osconfig.h, 761
OS_FS_VOL_NAME_LEN
osconfig-example.h, 752
osconfig.h, 761
OS_FdSet, 726
object_ids, 727
OS_FileOpenCheck
OSAL Standard File APIs, 430
OS_FileSysAddFixedMap
OSAL File System Level APIs, 440
OS_FileSysStatVolume
OSAL File System Level APIs, 441
OS_ForEachObject
OSAL Object ID Utility APIs, 453
OS_ForEachObjectOfType
OSAL Object ID Utility APIs, 453
OS_GetBuildNumber
osapi-version.h, 1296
OS_GetErrorName
OSAL Error Info APIs, 424
OS_GetFsInfo
OSAL File System Level APIs, 442
OS_GetLocalTime
OSAL Real Time Clock APIs, 387
OS_GetResourceName
OSAL Object ID Utility APIs, 454
OS_GetVersionCodeName
osapi-version.h, 1296
OS_GetVersionNumber
osapi-version.h, 1296
OS_GetVersionString
osapi-version.h, 1297
OS_HeapGetInfo
OSAL Heap APIs, 447
OS_INVALID_INT_NUM
OSAL Return Code Defines, 420
OS_INVALID_POINTER
OSAL Return Code Defines, 420
OS_INVALID_SEM_VALUE
OSAL Return Code Defines, 421
OS_IdentifyObject
OSAL Object ID Utility APIs, 454
OS_IdleLoop

OSAL Core Operation APIs, 403
OS_MAJOR_VERSION
osapi-version.h, 1293
OS_MAX_API_NAME
osconfig-example.h, 752
osconfig.h, 761
OS_MAX_BIN_SEMAPHORES
osconfig-example.h, 752
osconfig.h, 761
OS_MAX_CMD_LEN
osconfig-example.h, 753
osconfig.h, 762
OS_MAX_CONSOLES
osconfig-example.h, 753
osconfig.h, 762
OS_MAX_COUNT_SEMAPHORES
osconfig-example.h, 753
osconfig.h, 762
OS_MAX_FILE_NAME
osconfig-example.h, 753
osconfig.h, 762
OS_MAX_FILE_SYSTEMS
osconfig-example.h, 754
osconfig.h, 763
OS_MAX_LOCAL_PATH_LEN
osapi-constants.h, 1265
OS_MAX_MODULES
osconfig-example.h, 754
osconfig.h, 763
OS_MAX_MUTEXES
osconfig-example.h, 754
osconfig.h, 763
OS_MAX_NUM_OPEN_DIRS
osconfig-example.h, 754
osconfig.h, 763
OS_MAX_NUM_OPEN_FILES
osconfig-example.h, 755
osconfig.h, 764
OS_MAX_PATH_LEN
osconfig-example.h, 755
osconfig.h, 764
OS_MAX_QUEUES
osconfig-example.h, 755
osconfig.h, 764
OS_MAX_SYM_LEN
osconfig-example.h, 755
osconfig.h, 764
OS_MAX_TASK_PRIORITY
osapi-task.h, 1288
OS_MAX_TASKS
osconfig-example.h, 756
osconfig.h, 765
OS_MAX_TIMEBASES
osconfig-example.h, 756
osconfig.h, 765
OS_MAX_TIMERS
osconfig-example.h, 756
osconfig.h, 765
OS_MINOR_VERSION
osapi-version.h, 1293
OS_MISSION_REV
osapi-version.h, 1294
OS_MODULE_FILE_EXTENSION
osconfig-example.h, 757
osconfig.h, 765
OS_MODULE_FLAG_GLOBAL_SYMBOLS
osapi-module.h, 1280
OS_MODULE_FLAG_LOCAL_SYMBOLS
osapi-module.h, 1281
OS_ModuleInfo
OSAL Dynamic Loader and Symbol APIs, 459
OS_ModuleLoad
OSAL Dynamic Loader and Symbol APIs, 460
OS_ModuleSymbolLookup
OSAL Dynamic Loader and Symbol APIs, 460
OS_ModuleUnload
OSAL Dynamic Loader and Symbol APIs, 461
OS_MutSemCreate
OSAL Mutex APIs, 464
OS_MutSemDelete
OSAL Mutex APIs, 465
OS_MutSemGetIdByName
OSAL Mutex APIs, 465
OS_MutSemGetInfo
OSAL Mutex APIs, 466
OS_MutSemGive
OSAL Mutex APIs, 466
OS_MutSemTake
OSAL Mutex APIs, 467
OS_NetworkGetHostName
Network ID APIs, 469
OS_NetworkGetID
Network ID APIs, 469
OS_OBJECT_CREATOR_ANY
osapi-constants.h, 1265
OS_OBJECT_ID_UNDEFINED
osapi-constants.h, 1265
OS_OBJECT_INDEX_MASK
osapi-idmap.h, 1277
OS_OBJECT_TYPE_OS_BINSEM
OSAL Object Type Defines, 448
OS_OBJECT_TYPE_OS_CONSOLE
OSAL Object Type Defines, 448
OS_OBJECT_TYPE_OS_COUNTSEM
OSAL Object Type Defines, 449
OS_OBJECT_TYPE_OS_DIR
OSAL Object Type Defines, 449
OS_OBJECT_TYPE_OS_FILESYS

OSAL Object Type Defines, 449
OS_OBJECT_TYPE_OS_MODULE
OSAL Object Type Defines, 449
OS_OBJECT_TYPE_OS_MUTEX
OSAL Object Type Defines, 449
OS_OBJECT_TYPE_OS_QUEUE
OSAL Object Type Defines, 450
OS_OBJECT_TYPE_OS_STREAM
OSAL Object Type Defines, 450
OS_OBJECT_TYPE_OS_TASK
OSAL Object Type Defines, 450
OS_OBJECT_TYPE_OS_TIMEBASE
OSAL Object Type Defines, 450
OS_OBJECT_TYPE_OS_TIMECB
OSAL Object Type Defines, 450
OS_OBJECT_TYPE_SHIFT
osapi-idmap.h, 1277
OS_OBJECT_TYPE_UNDEFINED
OSAL Object Type Defines, 451
OS_OBJECT_TYPE_USER
OSAL Object Type Defines, 451
OS_ObjectIdDefined
OSAL Object ID Utility APIs, 455
OS_ObjectIdEqual
OSAL Object ID Utility APIs, 455
OS_ObjectIdFromInteger
OSAL Object ID Utility APIs, 456
OS_ObjectIdToArrayIndex
OSAL Object ID Utility APIs, 456
OS_ObjectIdToInteger
OSAL Object ID Utility APIs, 457
OS_OpenCreate
OSAL Standard File APIs, 432
OS_PEND
osapi-constants.h, 1265
OS_PRINTF_CONSOLE_NAME
osconfig-example.h, 757
osconfig.h, 766
OS_PRINTF
cfe_es.h, 871
common_types.h, 1254
OS_QUEUE_EMPTY
OSAL Return Code Defines, 421
OS_QUEUE_FULL
OSAL Return Code Defines, 421
OS_QUEUE_ID_ERROR
OSAL Return Code Defines, 421
OS_QUEUE_INVALID_SIZE
OSAL Return Code Defines, 421
OS_QUEUE_MAX_DEPTH
osconfig-example.h, 757
osconfig.h, 766
OS_QUEUE_TIMEOUT
OSAL Return Code Defines, 422
OS_QueueCreate
OSAL Message Queue APIs, 473
OS_QueueDelete
OSAL Message Queue APIs, 474
OS_QueueGet
OSAL Message Queue APIs, 475
OS_QueueGetIdByName
OSAL Message Queue APIs, 475
OS_QueueGetInfo
OSAL Message Queue APIs, 476
OS_QueuePut
OSAL Message Queue APIs, 476
OS_READ_ONLY
OSAL File Access Option Defines, 425
OS_READ_WRITE
OSAL File Access Option Defines, 425
OS_REPAIR
osapi-filesys.h, 1275
OS_REVISION
osapi-version.h, 1294
OS_RegisterEventHandler
OSAL Core Operation APIs, 404
OS_SEEK_CUR
OSAL Reference Point For Seek Offset Defines, 426
OS_SEEK_END
OSAL Reference Point For Seek Offset Defines, 426
OS_SEEK_SET
OSAL Reference Point For Seek Offset Defines, 426
OS_SEM_EMPTY
OSAL Semaphore State Defines, 379
OS_SEM_FAILURE
OSAL Return Code Defines, 422
OS_SEM_FULL
OSAL Semaphore State Defines, 379
OS_SEM_TIMEOUT
OSAL Return Code Defines, 422
OS_SHELL_CMD_INPUT_FILE_NAME
osconfig-example.h, 757
osconfig.h, 766
OS_SOCKADDR_MAX_LEN
osapi-sockets.h, 1286
osconfig-example.h, 758
osconfig.h, 766
OS_STR_HELPER
osapi-version.h, 1294
OS_STR
osapi-version.h, 1294
OS_SUCCESS
OSAL Return Code Defines, 422
OS_SelectFdAdd
OSAL Select APIs, 478
OS_SelectFdClear
OSAL Select APIs, 478
OS_SelectFdIsSet

OSAL Select APIs, 478
OS_SelectFdZero
OSAL Select APIs, 479
OS_SelectMultiple
OSAL Select APIs, 479
OS_SelectSingle
OSAL Select APIs, 480
OS_SetLocalTime
OSAL Real Time Clock APIs, 388
OS_ShellOutputToFile
OSAL Shell APIs, 481
OS_SockAddr_t, 737
ActualLength, 738
AddrData, 738
OS_SockAddrData_t, 738
AlignPtr, 739
AlignU32, 739
Buffer, 739
OS_SocketAccept
OSAL Socket Management APIs, 486
OS_SocketAddrFromString
OSAL Socket Address APIs, 482
OS_SocketAddrGetPort
OSAL Socket Address APIs, 483
OS_SocketAddrInit
OSAL Socket Address APIs, 483
OS_SocketAddrSetPort
OSAL Socket Address APIs, 484
OS_SocketAddrToString
OSAL Socket Address APIs, 484
OS_SocketBind
OSAL Socket Management APIs, 487
OS_SocketConnect
OSAL Socket Management APIs, 487
OS_SocketDomain_t
osapi-sockets.h, 1286
OS_SocketGetIdByName
OSAL Socket Management APIs, 489
OS_SocketGetInfo
OSAL Socket Management APIs, 490
OS_SocketOpen
OSAL Socket Management APIs, 490
OS_SocketRecvFrom
OSAL Socket Management APIs, 491
OS_SocketSendTo
OSAL Socket Management APIs, 491
OS_SocketType_t
osapi-sockets.h, 1287
OS_StreamState_t
osapi-select.h, 1284
OS_SymbolLookup
OSAL Dynamic Loader and Symbol APIs, 461
OS_SymbolTableDump
OSAL Dynamic Loader and Symbol APIs, 462
OS_TIMER_ERR_INTERNAL
OSAL Return Code Defines, 422
OS_TIMER_ERR_INVALID_ARGS
OSAL Return Code Defines, 423
OS_TIMER_ERR_TIMER_ID
OSAL Return Code Defines, 423
OS_TIMER_ERR_UNAVAILABLE
OSAL Return Code Defines, 423
OS_TaskCreate
OSAL Task APIs, 493
OS_TaskDelay
OSAL Task APIs, 494
OS_TaskDelete
OSAL Task APIs, 495
OS_TaskExit
OSAL Task APIs, 495
OS_TaskFindIdBySystemData
OSAL Task APIs, 495
OS_TaskGetId
OSAL Task APIs, 497
OS_TaskGetIdByName
OSAL Task APIs, 497
OS_TaskGetInfo
OSAL Task APIs, 498
OS_TaskInstallDeleteHandler
OSAL Task APIs, 498
OS_TaskRegister
OSAL Task APIs, 499
OS_TaskSetPriority
OSAL Task APIs, 499
OS_TimeAdd
OSAL Real Time Clock APIs, 388
OS_TimeAssembleFromMicroseconds
OSAL Real Time Clock APIs, 389
OS_TimeAssembleFromMilliseconds
OSAL Real Time Clock APIs, 390
OS_TimeAssembleFromNanoseconds
OSAL Real Time Clock APIs, 390
OS_TimeAssembleFromSubseconds
OSAL Real Time Clock APIs, 391
OS_TimeBaseCreate
OSAL Time Base APIs, 501
OS_TimeBaseDelete
OSAL Time Base APIs, 502
OS_TimeBaseGetFreeRun
OSAL Time Base APIs, 502
OS_TimeBaseGetIdByName
OSAL Time Base APIs, 503
OS_TimeBaseGetInfo
OSAL Time Base APIs, 504
OS_TimeBaseSet
OSAL Time Base APIs, 504
OS_TimeGetFractionalPart
OSAL Real Time Clock APIs, 391

OS_TimeGetMicrosecondsPart
OSAL Real Time Clock APIs, 392
OS_TimeGetMillisecondsPart
OSAL Real Time Clock APIs, 393
OS_TimeGetNanosecondsPart
OSAL Real Time Clock APIs, 394
OS_TimeGetSubsecondsPart
OSAL Real Time Clock APIs, 394
OS_TimeGetTotalMicroseconds
OSAL Real Time Clock APIs, 395
OS_TimeGetTotalMilliseconds
OSAL Real Time Clock APIs, 396
OS_TimeGetTotalNanoseconds
OSAL Real Time Clock APIs, 396
OS_TimeGetTotalSeconds
OSAL Real Time Clock APIs, 397
OS_TimeSubtract
OSAL Real Time Clock APIs, 397
OS_TimedRead
OSAL Standard File APIs, 436
OS_TimedWrite
OSAL Standard File APIs, 437
OS_TimerAdd
OSAL Timer APIs, 506
OS_TimerCallback_t
osapi-timer.h, 1292
OS_TimerCreate
OSAL Timer APIs, 507
OS_TimerDelete
OSAL Timer APIs, 508
OS_TimerGetIdByName
OSAL Timer APIs, 508
OS_TimerGetInfo
OSAL Timer APIs, 509
OS_TimerSet
OSAL Timer APIs, 510
OS_TimerSync_t
osapi-timebase.h, 1291
OS_TranslatePath
OSAL File System Level APIs, 445
OS_USED
common_types.h, 1254
OS_UTILITYTASK_PRIORITY
osconfig-example.h, 758
osconfig.h, 767
OS_UTILITYTASK_STACK_SIZE
osconfig-example.h, 758
osconfig.h, 767
OS_VERSION_CODENAME
osapi-version.h, 1295
OS_VERSION_STRING
osapi-version.h, 1295
OS_VERSION
osapi-version.h, 1294
OS_WRITE_ONLY
OSAL File Access Option Defines, 425
OS_bin_sem_prop_t, 724
creator, 724
name, 724
value, 724
OS_chkfs
OSAL File System Level APIs, 439
OS_chmod
OSAL Standard File APIs, 427
OS_close
OSAL Standard File APIs, 428
OS_count_sem_prop_t, 725
creator, 725
name, 725
value, 725
OS_cp
OSAL Standard File APIs, 429
OS_file_flag_t
osapi-file.h, 1274
OS_file_prop_t, 727
IsValid, 728
Path, 728
User, 728
OS_heap_prop_t, 731
free_blocks, 731
free_bytes, 731
largest_free_block, 732
OS_initfs
OSAL File System Level APIs, 442
OS_lseek
OSAL Standard File APIs, 431
OS_mkdir
OSAL Directory APIs, 412
OS_mkfs
OSAL File System Level APIs, 443
OS_module_address_t, 732
bss_address, 732
bss_size, 733
code_address, 733
code_size, 733
data_address, 733
data_size, 733
flags, 733
valid, 734
OS_module_prop_t, 734
addr, 734
entry_point, 735
filename, 735
host_module_id, 735
name, 735
OS_mount
OSAL File System Level APIs, 444
OS_mut_sem_prop_t, 735

creator, 736
name, 736
OS_mv
OSAL Standard File APIs, 432
OS_printf
OSAL Printf APIs, 471
OS_printf_disable
OSAL Printf APIs, 471
OS_printf_enable
OSAL Printf APIs, 472
OS_queue_prop_t, 736
creator, 737
name, 737
OS_read
OSAL Standard File APIs, 433
OS_remove
OSAL Standard File APIs, 434
OS_rename
OSAL Standard File APIs, 434
OS_rmdir
OSAL Directory APIs, 412
OS_rmfs
OSAL File System Level APIs, 444
OS_socket_prop_t, 740
creator, 740
name, 740
OS_stat
OSAL Standard File APIs, 435
OS_static_symbol_record_t, 741
Address, 741
Module, 741
Name, 741
OS_statvfs_t, 742
block_size, 742
blocks_free, 742
total_blocks, 742
OS_task_prop_t, 743
creator, 743
name, 743
priority, 744
stack_size, 744
OS_time_t, 744
ticks, 745
OS_timebase_prop_t, 745
accuracy, 745
creator, 746
freerun_time, 746
name, 746
nominal_interval_time, 746
OS_timer_prop_t, 746
accuracy, 747
creator, 747
interval_time, 747
name, 747
start_time, 747
OS_unmount
OSAL File System Level APIs, 446
OS_write
OSAL Standard File APIs, 437
OSAL BSP low level access APIs, 386
OS_BSP_GetArgC, 386
OS_BSP_GetArgV, 386
OS_BSP_SetExitCode, 386
OSAL Binary Semaphore APIs, 380
OS_BinSemCreate, 380
OS_BinSemDelete, 381
OS_BinSemFlush, 381
OS_BinSemGetIdByName, 382
OS_BinSemGetInfo, 383
OS_BinSemGive, 383
OS_BinSemTake, 384
OS_BinSemTimedWait, 384
OSAL Core Operation APIs, 400
OS_API_Init, 400
OS_Application_Run, 401
OS_Application_Startup, 401
OS_ApplicationExit, 402
OS_ApplicationShutdown, 403
OS_DeleteAllObjects, 403
OS_IdleLoop, 403
OS_RegisterEventHandler, 404
OSAL Counting Semaphore APIs, 405
OS_CountSemCreate, 405
OS_CountSemDelete, 406
OS_CountSemGetIdByName, 406
OS_CountSemGetInfo, 407
OS_CountSemGive, 407
OS_CountSemTake, 408
OS_CountSemTimedWait, 409
OSAL Directory APIs, 410
OS_DirectoryClose, 410
OS_DirectoryOpen, 410
OS_DirectoryRead, 411
OS_DirectoryRewind, 411
OS_mkdir, 412
OS_rmdir, 412
OSAL Dynamic Loader and Symbol APIs, 459
OS_ModuleInfo, 459
OS_ModuleLoad, 460
OS_ModuleSymbolLookup, 460
OS_ModuleUnload, 461
OS_SymbolLookup, 461
OS_SymbolTableDump, 462
OSAL Error Info APIs, 424
OS_GetErrorName, 424
OSAL File Access Option Defines, 425
OS_READ_ONLY, 425
OS_READ_WRITE, 425

OS_WRITE_ONLY, 425
OSAL File System Level APIs, 439
OS_FS_GetPhysDriveName, 441
OS_FileSysAddFixedMap, 440
OS_FileSysStatVolume, 441
OS_GetFsInfo, 442
OS_TranslatePath, 445
OS_chkfs, 439
OS_initfs, 442
OS_mkfs, 443
OS_mount, 444
OS_rmfs, 444
OS_unmount, 446
OSAL Heap APIs, 447
OS_HeapGetInfo, 447
OSAL Message Queue APIs, 473
OS_QueueCreate, 473
OS_QueueDelete, 474
OS_QueueGet, 475
OS_QueueGetIdByName, 475
OS_QueueGetInfo, 476
OS_QueuePut, 476
OSAL Mutex APIs, 464
OS_MutSemCreate, 464
OS_MutSemDelete, 465
OS_MutSemGetIdByName, 465
OS_MutSemGetInfo, 466
OS_MutSemGive, 466
OS_MutSemTake, 467
OSAL Object ID Utility APIs, 452
OS_ConvertToArrayIndex, 452
OS_ForEachObject, 453
OS_ForEachObjectOfType, 453
OS_GetResourceName, 454
OS_IdentifyObject, 454
OS_ObjectIdDefined, 455
OS_ObjectIdEqual, 455
OS_ObjectIdFromInteger, 456
OS_ObjectIdToArrayIndex, 456
OS_ObjectIdToInteger, 457
OSAL Object Type Defines, 448
OS_OBJECT_TYPE_OS_BINSEM, 448
OS_OBJECT_TYPE_OS_CONSOLE, 448
OS_OBJECT_TYPE_OS_COUNTSEM, 449
OS_OBJECT_TYPE_OS_DIR, 449
OS_OBJECT_TYPE_OS_FILESYS, 449
OS_OBJECT_TYPE_OS_MODULE, 449
OS_OBJECT_TYPE_OS_MUTEX, 449
OS_OBJECT_TYPE_OS_QUEUE, 450
OS_OBJECT_TYPE_OS_STREAM, 450
OS_OBJECT_TYPE_OS_TASK, 450
OS_OBJECT_TYPE_OS_TIMEBASE, 450
OS_OBJECT_TYPE_OS_TIMECB, 450
OS_OBJECT_TYPE_UNDEFINED, 451
OS_OBJECT_TYPE_USER, 451
OSAL Printf APIs, 471
OS_printf, 471
OS_printf_disable, 471
OS_printf_enable, 472
OSAL Real Time Clock APIs, 387
OS_GetLocalTime, 387
OS_SetLocalTime, 388
OS_TimeAdd, 388
OS_TimeAssembleFromMicroseconds, 389
OS_TimeAssembleFromMilliseconds, 390
OS_TimeAssembleFromNanoseconds, 390
OS_TimeAssembleFromSubseconds, 391
OS_TimeGetFractionalPart, 391
OS_TimeGetMicrosecondsPart, 392
OS_TimeGetMillisecondsPart, 393
OS_TimeGetNanosecondsPart, 394
OS_TimeGetSubsecondsPart, 394
OS_TimeGetTotalMicroseconds, 395
OS_TimeGetTotalMilliseconds, 396
OS_TimeGetTotalNanoseconds, 396
OS_TimeGetTotalSeconds, 397
OS_TimeSubtract, 397
OSAL Reference Point For Seek Offset Defines, 426
OS_SEEK_CUR, 426
OS_SEEK_END, 426
OS_SEEK_SET, 426
OSAL Return Code Defines, 414
OS_ERR_BAD_ADDRESS, 415
OS_ERR_FILE, 416
OS_ERR_INCORRECT_OBJ_STATE, 416
OS_ERR_INCORRECT_OBJ_TYPE, 416
OS_ERR_INVALID_ID, 416
OS_ERR_INVALID_PRIORITY, 416
OS_ERR_INVALID_SIZE, 417
OS_ERR_NAME_NOT_FOUND, 417
OS_ERR_NAME_TAKEN, 417
OS_ERR_NAME_TOO_LONG, 417
OS_ERR_NO_FREE_IDS, 417
OS_ERR_NOT_IMPLEMENTED, 418
OS_ERR_OBJECT_IN_USE, 418
OS_ERR_OPERATION_NOT_SUPPORTED, 418
OS_ERR_SEM_NOT_FULL, 418
OS_ERR_STREAM_DISCONNECTED, 418
OS_ERROR_ADDRESS_MISALIGNED, 419
OS_ERROR_TIMEOUT, 419
OS_ERROR, 419
OS_FS_ERR_DEVICE_NOT_FREE, 419
OS_FS_ERR_DRIVE_NOT_CREATED, 419
OS_FS_ERR_NAME_TOO_LONG, 420
OS_FS_ERR_PATH_INVALID, 420
OS_FS_ERR_PATH_TOO_LONG, 420
OS_INVALID_INT_NUM, 420
OS_INVALID_POINTER, 420

OS_INVALID_SEM_VALUE, 421
OS_QUEUE_EMPTY, 421
OS_QUEUE_FULL, 421
OS_QUEUE_ID_ERROR, 421
OS_QUEUE_INVALID_SIZE, 421
OS_QUEUE_TIMEOUT, 422
OS_SEM_FAILURE, 422
OS_SEM_TIMEOUT, 422
OS_SUCCESS, 422
OS_TIMER_ERR_INTERNAL, 422
OS_TIMER_ERR_INVALID_ARGS, 423
OS_TIMER_ERR_TIMER_ID, 423
OS_TIMER_ERR_UNAVAILABLE, 423
OSAL Select APIs, 478
OS_SelectFdAdd, 478
OS_SelectFdClear, 478
OS_SelectFdIsSet, 478
OS_SelectFdZero, 479
OS_SelectMultiple, 479
OS_SelectSingle, 480
OSAL Semaphore State Defines, 379
OS_SEM_EMPTY, 379
OS_SEM_FULL, 379
OSAL Shell APIs, 481
OS_ShellOutputToFile, 481
OSAL Socket Address APIs, 482
OS_SocketAddrFromString, 482
OS_SocketAddrGetPort, 483
OS_SocketAddrInit, 483
OS_SocketAddrSetPort, 484
OS_SocketAddrToString, 484
OSAL Socket Management APIs, 486
OS_SocketAccept, 486
OS_SocketBind, 487
OS_SocketConnect, 487
OS_SocketGetIdByName, 489
OS_SocketGetInfo, 490
OS_SocketOpen, 490
OS_SocketRecvFrom, 491
OS_SocketSendTo, 491
OSAL Standard File APIs, 427
OS_CloseAllFiles, 428
OS_CloseFileByName, 429
OS_FDGetInfo, 430
OS_FileOpenCheck, 430
OS_OpenCreate, 432
OS_TimedRead, 436
OS_TimedWrite, 437
OS_chmod, 427
OS_close, 428
OS_cp, 429
OS_lseek, 431
OS_mv, 432
OS_read, 433
OS_remove, 434
OS_rename, 434
OS_stat, 435
OS_write, 437
OSAL Task APIs, 493
OS_TaskCreate, 493
OS_TaskDelay, 494
OS_TaskDelete, 495
OS_TaskExit, 495
OS_TaskFindIdBySystemData, 495
OS_TaskGetId, 497
OS_TaskGetIdByName, 497
OS_TaskGetInfo, 498
OS_TaskInstallDeleteHandler, 498
OS_TaskRegister, 499
OS_TaskSetPriority, 499
OSAL Time Base APIs, 501
OS_TimeBaseCreate, 501
OS_TimeBaseDelete, 502
OS_TimeBaseGetFreeRun, 502
OS_TimeBaseGetIdByName, 503
OS_TimeBaseGetInfo, 504
OS_TimeBaseSet, 504
OSAL Timer APIs, 506
OS_TimerAdd, 506
OS_TimerCreate, 507
OS_TimerDelete, 508
OS_TimerGetIdByName, 508
OS_TimerGetInfo, 509
OS_TimerSet, 510
OSAL_API_VERSION
osapi-version.h, 1295
OSAL_BLOCKCOUNT_C
common_types.h, 1254
OSAL_CONFIG_DEBUG_PERMISSIVE_MODE
osconfig.h, 767
OSAL_CONFIG_DEBUG_PRINTF
osconfig.h, 767
OSAL_CONFIG_INCLUDE_DYNAMIC_LOADER
osconfig.h, 768
OSAL_CONFIG_INCLUDE_NETWORK
osconfig.h, 768
OSAL_CONFIG_INCLUDE_STATIC_LOADER
osconfig.h, 768
OSAL_INDEX_C
common_types.h, 1254
OSAL_OBJTYPE_C
common_types.h, 1254
OSAL_PRIORITY_C
osapi-task.h, 1288
OSAL_SIZE_C
common_types.h, 1254
OSAL_STACKPTR_C
osapi-task.h, 1289

OSAL_TASK_STACK_ALLOCATE
osapi-task.h, 1289
OSALMajorVersion
CFE_ES_HousekeepingTlm_Payload, 538
OSALMinorVersion
CFE_ES_HousekeepingTlm_Payload, 538
OSALMissionRevision
CFE_ES_HousekeepingTlm_Payload, 538
OSALRevision
CFE_ES_HousekeepingTlm_Payload, 539
object_ids
OS_FdSet, 727
ObjectName
CFE_TBL_FileDef, 663
ObjectSize
CFE_TBL_FileDef, 663
Offset
CFE_TBL_File_Hdr, 662
OnEvent
CFE_FS_FileWriteMetaData, 608
OneHzAdjust
CFE_TIME_DiagnosticTlm_Payload, 698
OneHzDirection
CFE_TIME_DiagnosticTlm_Payload, 698
OneTimeAdjust
CFE_TIME_DiagnosticTlm_Payload, 699
OneTimeDirection
CFE_TIME_DiagnosticTlm_Payload, 699
optString
cfe_psp_start.c, 1347
Opts
CFE_SB_PipeInfoEntry, 632
os_dirent_t, 726
FileName, 726
os_err_name_t
osapi-error.h, 1270
os_fsinfo_t, 728
FreeFds, 729
FreeVolumes, 729
MaxFds, 729
MaxVolumes, 729
os_fstat_t, 730
FileModeBits, 730
FileSize, 730
FileTime, 730
osal/src/os/inc/common_types.h, 1252
osal/src/os/inc/osapi-binsem.h, 1259
osal/src/os/inc/osapi-bsp.h, 1260
osal/src/os/inc/osapi-clock.h, 1260
osal/src/os/inc/osapi-common.h, 1262
osal/src/os/inc/osapi-constants.h, 1264
osal/src/os/inc/osapi-countsem.h, 1266
osal/src/os/inc/osapi-dir.h, 1266
osal/src/os/inc/osapi-error.h, 1267
osal/src/os/inc/osapi-file.h, 1270
osal/src/os/inc/osapi-filesys.h, 1274
osal/src/os/inc/osapi-heap.h, 1275
osal/src/os/inc/osapi-idmap.h, 1276
osal/src/os/inc/osapi-macros.h, 1278
osal/src/os/inc/osapi-module.h, 1280
osal/src/os/inc/osapi-mutex.h, 1281
osal/src/os/inc/osapi-network.h, 1282
osal/src/os/inc/osapi-printf.h, 1282
osal/src/os/inc/osapi-queue.h, 1283
osal/src/os/inc/osapi-select.h, 1283
osal/src/os/inc/osapi-shell.h, 1284
osal/src/os/inc/osapi-sockets.h, 1285
osal/src/os/inc/osapi-task.h, 1287
osal/src/os/inc/osapi-timebase.h, 1290
osal/src/os/inc/osapi-timer.h, 1291
osal/src/os/inc/osapi-version.h, 1292
osal/src/os/inc/osapi.h, 1297
osal_blockcount_t
common_types.h, 1256
osal_id_t
common_types.h, 1256
osal_index_t
common_types.h, 1257
osal_objtype_t
common_types.h, 1257
osal_priority_t
osapi-task.h, 1289
osal_stackptr_t
osapi-task.h, 1289
osal_task
osapi-task.h, 1289, 1290
osapi-common.h
OS_Event_t, 1263
OS_EventHandler_t, 1263
osapi-constants.h
OS_CHECK, 1265
OS_MAX_LOCAL_PATH_LEN, 1265
OS_OBJECT_CREATOR_ANY, 1265
OS_OBJECT_ID_UNDEFINED, 1265
OS_PEND, 1265
osapi-dir.h
OS_DIRENTRY_NAME, 1267
osapi-error.h
OS_ERROR_NAME_LENGTH, 1269
os_err_name_t, 1270
osapi-file.h
OS_FILESTAT_EXEC, 1272
OS_FILESTAT_ISDIR, 1272
OS_FILESTAT_MODE, 1272
OS_FILESTAT_READ, 1272
OS_FILESTAT_SIZE, 1272
OS_FILESTAT_TIME, 1273
OS_FILESTAT_WRITE, 1273

OS_file_flag_t, 1274
osapi-filesys.h
OS_CHK_ONLY, 1275
OS_REPAIR, 1275
osapi-idmap.h
OS_OBJECT_INDEX_MASK, 1277
OS_OBJECT_TYPE_SHIFT, 1277
osapi-macros.h
ARGCHECK, 1278
BUGCHECK, 1278
BUGREPORT, 1279
LENGTHCHECK, 1279
osapi-module.h
OS_MODULE_FLAG_GLOBAL_SYMBOLS, 1280
OS_MODULE_FLAG_LOCAL_SYMBOLS, 1281
osapi-select.h
OS_StreamState_t, 1284
osapi-sockets.h
OS_SOCKADDR_MAX_LEN, 1286
OS_SocketDomain_t, 1286
OS_SocketType_t, 1287
osapi-task.h
OS_FP_ENABLED, 1288
OS_MAX_TASK_PRIORITY, 1288
OSAL_PRIORITY_C, 1288
OSAL_STACKPTR_C, 1289
OSAL_TASK_STACK_ALLOCATE, 1289
osal_priority_t, 1289
osal_stackptr_t, 1289
osal_task, 1289, 1290
osapi-timebase.h
OS_TimerSync_t, 1291
osapi-timer.h
OS_TimerCallback_t, 1292
osapi-version.h
OS_BUILD_BASELINE, 1293
OS_BUILD_NUMBER, 1293
OS_GetBuildNumber, 1296
OS_GetVersionCodeName, 1296
OS_GetVersionNumber, 1296
OS_GetVersionString, 1297
OS_MAJOR_VERSION, 1293
OS_MINOR_VERSION, 1293
OS_MISSION_REV, 1294
OS_REVISION, 1294
OS_STR_HELPER, 1294
OS_STR, 1294
OS_VERSION_CODENAME, 1295
OS_VERSION_STRING, 1295
OS_VERSION, 1294
OSAL_API_VERSION, 1295
osconfig-example.h
OS_BUFFER_MSG_DEPTH, 751
OS_BUFFER_SIZE, 751
OS_FS_DEV_NAME_LEN, 752
OS_FS_PHYS_NAME_LEN, 752
OS_FS_VOL_NAME_LEN, 752
OS_MAX_API_NAME, 752
OS_MAX_BIN_SEMAPHORES, 752
OS_MAX_CMD_LEN, 753
OS_MAX_CONSOLES, 753
OS_MAX_COUNT_SEMAPHORES, 753
OS_MAX_FILE_NAME, 753
OS_MAX_FILE_SYSTEMS, 754
OS_MAX_MODULES, 754
OS_MAX_MUTEXES, 754
OS_MAX_NUM_OPEN_DIRS, 754
OS_MAX_NUM_OPEN_FILES, 755
OS_MAX_PATH_LEN, 755
OS_MAX_QUEUES, 755
OS_MAX_SYM_LEN, 755
OS_MAX_TASKS, 756
OS_MAX_TIMEBASES, 756
OS_MAX_TIMERS, 756
OS_MODULE_FILE_EXTENSION, 757
OS_PRINTF_CONSOLE_NAME, 757
OS_QUEUE_MAX_DEPTH, 757
OS_SHELL_CMD_INPUT_FILE_NAME, 757
OS_SOCKADDR_MAX_LEN, 758
OS_UTILITYTASK_PRIORITY, 758
OS_UTILITYTASK_STACK_SIZE, 758
osconfig.h
OS_BUFFER_MSG_DEPTH, 760
OS_BUFFER_SIZE, 760
OS_FS_DEV_NAME_LEN, 761
OS_FS_PHYS_NAME_LEN, 761
OS_FS_VOL_NAME_LEN, 761
OS_MAX_API_NAME, 761
OS_MAX_BIN_SEMAPHORES, 761
OS_MAX_CMD_LEN, 762
OS_MAX_CONSOLES, 762
OS_MAX_COUNT_SEMAPHORES, 762
OS_MAX_FILE_NAME, 762
OS_MAX_FILE_SYSTEMS, 763
OS_MAX_MODULES, 763
OS_MAX_MUTEXES, 763
OS_MAX_NUM_OPEN_DIRS, 763
OS_MAX_NUM_OPEN_FILES, 764
OS_MAX_PATH_LEN, 764
OS_MAX_QUEUES, 764
OS_MAX_SYM_LEN, 764
OS_MAX_TASKS, 765
OS_MAX_TIMEBASES, 765
OS_MAX_TIMERS, 765
OS_MODULE_FILE_EXTENSION, 765
OS_PRINTF_CONSOLE_NAME, 766
OS_QUEUE_MAX_DEPTH, 766
OS_SHELL_CMD_INPUT_FILE_NAME, 766

OS_SOCKADDR_MAX_LEN, 766
OS_UTILITYTASK_PRIORITY, 767
OS_UTILITYTASK_STACK_SIZE, 767
OSAL_CONFIG_DEBUG_PERMISSIVE_MODE, 767
OSAL_CONFIG_DEBUG_PRINTF, 767
OSAL_CONFIG_INCLUDE_DYNAMIC_LOADER, 768
OSAL_CONFIG_INCLUDE_NETWORK, 768
OSAL_CONFIG_INCLUDE_STATIC_LOADER, 768
OutputPort
CFE_EVS_HousekeepingTlm_Payload, 592
OwnerAppName
CFE_TBL_TblRegPacket_Payload, 686

PSP_ExceptionLogSize
Target_PspConfigData, 748
PSP_MemTableSize
Target_PspConfigData, 749
PSP_MemoryTable
Target_PspConfigData, 749
PSP_VersionInfo
Target_PspConfigData, 749
PSP_WatchdogMax
Target_PspConfigData, 749
PSP_WatchdogMin
Target_PspConfigData, 749
PacketID
CFE_EVS_LongEventTlm_Payload, 596
CFE_EVS_ShortEventTlm_Payload, 604
Padding
CFE_EVS_AppTlmData, 584
Parameter
CFE_TBL_NotifyCmd_Payload, 679
Path
OS_file_prop_t, 728
Payload
CFE_ES_AppNameCmd, 521
CFE_ES_DeleteCDSCmd, 527
CFE_ES_DumpCDSRegistryCmd, 529
CFE_ES_FileNameCmd, 530
CFE_ES_HousekeepingTlm, 532
CFE_ES_MemStatsTlm, 547
CFE_ES_OneAppTlm, 549
CFE_ES_OverWriteSysLogCmd, 550
CFE_ES_ReloadAppCmd, 554
CFE_ES_RestartCmd, 555
CFE_ES_SendMemPoolStatsCmd, 557
CFE_ES_SetMaxPRCountCmd, 559
CFE_ES_SetPerfFilterMaskCmd, 561
CFE_ES_SetPerfTriggerMaskCmd, 563
CFE_ES_StartApp, 565
CFE_ES_StartPerfDataCmd, 569
CFE_ES_StopPerfDataCmd, 571
CFE_EVS_AppNameBitMaskCmd, 575
CFE_EVS_AppNameCmd, 577
CFE_EVS_AppNameEventIDCmd, 579
CFE_EVS_AppNameEventIDMaskCmd, 581
CFE_EVS_BitMaskCmd, 586
CFE_EVS_HousekeepingTlm, 588
CFE_EVS_LongEventTlm, 595
CFE_EVS_SetEventFormatModeCmd, 601
CFE_EVS_SetLogModeCmd, 603
CFE_EVS_ShortEventTlm, 603
CFE_EVS_WriteAppDataFileCmd, 605
CFE_EVS_WriteLogDataFileCmd, 606
CFE_SB_AllSubscriptionsTlm, 618
CFE_SB_HousekeepingTlm, 621
CFE_SB_RouteCmd, 635
CFE_SB_SingleSubscriptionTlm, 639
CFE_SB_StatsTlm, 642
CFE_SB_WriteFileInfoCmd, 649
CFE_TBL_AbortLoadCmd, 652
CFE_TBL_ActivateCmd, 654
CFE_TBL_DeleteCDSCmd, 656
CFE_TBL_DumpCmd, 657
CFE_TBL_DumpRegistryCmd, 660
CFE_TBL_HousekeepingTlm, 664
CFE_TBL_LoadCmd, 675
CFE_TBL_NotifyCmd, 678
CFE_TBL_SendRegistryCmd, 680
CFE_TBL_TableRegistryTlm, 682
CFE_TBL_ValidateCmd, 688
CFE_TIME_DiagnosticTlm, 690
CFE_TIME_HousekeepingTlm, 704
CFE_TIME_OneHzAdjustmentCmd, 710
CFE_TIME_SetLeapSecondsCmd, 712
CFE_TIME_SetSignalCmd, 713
CFE_TIME_SetSourceCmd, 714
CFE_TIME_SetStateCmd, 715
CFE_TIME_TimeCmd, 720
CFE_TIME_ToneDataCmd, 722
PeakMemInUse
CFE_SB_StatsTlm_Payload, 645
PeakMsgIdsInUse
CFE_SB_StatsTlm_Payload, 645
PeakPipesInUse
CFE_SB_StatsTlm_Payload, 646
PeakQueueDepth
CFE_SB_PipeDepthStats, 630
CFE_SB_PipeInfoEntry, 632
PeakSBBuffersInUse
CFE_SB_StatsTlm_Payload, 646
PeakSubscriptionsInUse
CFE_SB_StatsTlm_Payload, 646
PerfDataCount
CFE_ES_HousekeepingTlm_Payload, 539
PerfDataEnd

CFE_ES_HousekeepingTlm_Payload, 539
PerfDataStart
CFE_ES_HousekeepingTlm_Payload, 539
PerfDataToWrite
CFE_ES_HousekeepingTlm_Payload, 540
PerfFilterMask
CFE_ES_HousekeepingTlm_Payload, 540
PerfMode
CFE_ES_HousekeepingTlm_Payload, 540
PerfState
CFE_ES_HousekeepingTlm_Payload, 540
PerfTriggerCount
CFE_ES_HousekeepingTlm_Payload, 541
PerfTriggerMask
CFE_ES_HousekeepingTlm_Payload, 541
Pipe
CFE_SB_RouteCmd_Payload, 636
CFE_SB_SingleSubscriptionTlm_Payload, 641
CFE_SB_SubEntries, 648
PipeDepthStats
CFE_SB_StatsTlm_Payload, 646
PipeId
CFE_SB_PipeDepthStats, 630
CFE_SB_PipeInfoEntry, 632
CFE_SB_RoutingFileEntry, 638
PipeName
CFE_SB_PipeInfoEntry, 633
CFE_SB_RoutingFileEntry, 638
PipeOptsErrorCounter
CFE_SB_HousekeepingTlm_Payload, 625
PipeOverflowErrorCounter
CFE_SB_HousekeepingTlm_Payload, 625
PipesInUse
CFE_SB_StatsTlm_Payload, 647
PktSegment
CFE_SB_AllSubscriptionsTlm_Payload, 619
PoolHandle
CFE_ES_PoolStatsTlm_Payload, 553
CFE_ES_SendMemPoolStatsCmd_Payload, 558
PoolSize
CFE_ES_MemPoolStats, 546
PoolStats
CFE_ES_PoolStatsTlm_Payload, 553
Priority
CFE_ES_AppInfo, 519
CFE_ES_StartAppCmd_Payload, 567
CFE_ES_TaskInfo, 572
CFE_SB_Qos_t, 634
priority
OS_task_prop_t, 744
ProcessorID
CFE_EVS_PacketID, 598
CFE_FS_Header, 610
ProcessorResets
CFE_ES_HousekeepingTlm_Payload, 541
psp/fsw/inc/cfe_psp.h, 1298
psp/fsw/inc/cfe_psp_configdata.h, 1325
psp/fsw/pc-linux/src/cfe_psp_exception.c, 1326
psp/fsw/pc-linux/src/cfe_psp_memory.c, 1329
psp/fsw/pc-linux/src/cfe_psp_memtab.c, 1340
psp/fsw/pc-linux/src/cfe_psp_ssr.c, 1340
psp/fsw/pc-linux/src/cfe_psp_start.c, 1341
psp/fsw/pc-linux/src/cfe_psp_support.c, 1348
psp/fsw/pc-linux/src/cfe_psp_timer.c, 1351
psp/fsw/pc-linux/src/cfe_psp_watchdog.c, 1354
Ptr
CFE_ES_PoolAlign, 552

Qos
CFE_SB_SingleSubscriptionTlm_Payload, 641
CFE_SB_SubEntries, 648

RegisteredCoreApps
CFE_ES_HousekeepingTlm_Payload, 541
RegisteredExternalApps
CFE_ES_HousekeepingTlm_Payload, 542
RegisteredLibs
CFE_ES_HousekeepingTlm_Payload, 542
RegisteredTasks
CFE_ES_HousekeepingTlm_Payload, 542
Reliability
CFE_SB_Qos_t, 634
Reserved
CFE_TBL_File_Hdr, 662
ResetAreaShmId
cfe_psp_memory.c, 1339
ResetSubtype
CFE_ES_HousekeepingTlm_Payload, 542
ResetType
CFE_ES_HousekeepingTlm_Payload, 543
CFE_PSP_CommandData_t, 613
ResourceId
CFE_ES_AppInfo, 519
RestartType
CFE_ES_RestartCmd_Payload, 556
Revision
CFE_PSP_VersionInfo_t, 617

SBBuffersInUse
CFE_SB_StatsTlm_Payload, 647
SIZE_BYTE
cfe_psp.h, 1310
SIZE_HALF
cfe_psp.h, 1310
SIZE_WORD
cfe_psp.h, 1310
sample_mission_cfg.h, 833
CFE_MISSION_ES_APP_TLM_MSG, 835

CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN, 835
CFE_MISSION_ES_CDS_MAX_NAME_LENGTH, 835
CFE_MISSION_ES_CMD_MSG, 836
CFE_MISSION_ES_CRC_16, 836
CFE_MISSION_ES_CRC_32, 836
CFE_MISSION_ES_CRC_8, 836
CFE_MISSION_ES_DEFAULT_CRC, 837
CFE_MISSION_ES_HK_TLM_MSG, 837
CFE_MISSION_ES_MAX_APPLICATIONS, 837
CFE_MISSION_ES_MEMSTATS_TLM_MSG, 838
CFE_MISSION_ES_PERF_MAX_IDS, 838
CFE_MISSION_ES_POOL_MAX_BUCKETS, 838
CFE_MISSION_ES_SEND_HK_MSG, 839
CFE_MISSION_EVS_CMD_MSG, 839
CFE_MISSION_EVS_HK_TLM_MSG, 839
CFE_MISSION_EVS_LONG_EVENT_MSG_MSG, 840
CFE_MISSION_EVS_MAX_MESSAGE_LENGTH, 840
CFE_MISSION_EVS_SEND_HK_MSG, 840
CFE_MISSION_EVS_SHORT_EVENT_MSG_MSG, 840
CFE_MISSION_MAX_API_LEN, 840
CFE_MISSION_MAX_FILE_LEN, 841
CFE_MISSION_MAX_PATH_LEN, 841
CFE_MISSION_SB_ALLSUBS_TLM_MSG, 842
CFE_MISSION_SB_CMD_MSG, 842
CFE_MISSION_SB_HK_TLM_MSG, 842
CFE_MISSION_SB_MAX_PIPES, 843
CFE_MISSION_SB_MAX_SB_MSG_SIZE, 843
CFE_MISSION_SB_ONESUB_TLM_MSG, 843
CFE_MISSION_SB_SEND_HK_MSG, 844
CFE_MISSION_SB_STATS_TLM_MSG, 844
CFE_MISSION_SB_SUB_RPT_CTRL_MSG, 844
CFE_MISSION_TBL_CMD_MSG, 844
CFE_MISSION_TBL_HK_TLM_MSG, 844
CFE_MISSION_TBL_MAX_FULL_NAME_LEN, 844
CFE_MISSION_TBL_MAX_NAME_LENGTH, 845
CFE_MISSION_TBL_REG_TLM_MSG, 845
CFE_MISSION_TBL_SEND_HK_MSG, 846
CFE_MISSION_TIME_1HZ_CMD_MSG, 846
CFE_MISSION_TIME_AT_TONE_WAS, 846
CFE_MISSION_TIME_AT_TONE_WILL_BE, 846
CFE_MISSION_TIME_CFG_DEFAULT_TAI, 847
CFE_MISSION_TIME_CFG_DEFAULT_UTC, 847
CFE_MISSION_TIME_CFG_FAKE_TONE, 847
CFE_MISSION_TIME_CMD_MSG, 847
CFE_MISSION_TIME_DATA_CMD_MSG, 848
CFE_MISSION_TIME_DEF_DELAY_SECS, 848
CFE_MISSION_TIME_DEF_DELAY_SUBS, 848
CFE_MISSION_TIME_DEF_LEAPS, 848
CFE_MISSION_TIME_DEF_MET_SECS, 848
CFE_MISSION_TIME_DEF_MET_SUBS, 849
CFE_MISSION_TIME_DEF_STCF_SECS, 849
CFE_MISSION_TIME_DEF_STCF_SUBS, 849
CFE_MISSION_TIME_DIAG_TLM_MSG, 849
CFE_MISSION_TIME_EPOCH_DAY, 850
CFE_MISSION_TIME_EPOCH_HOUR, 850
CFE_MISSION_TIME_EPOCH_MINUTE, 850
CFE_MISSION_TIME_EPOCH_SECOND, 850
CFE_MISSION_TIME_EPOCH_YEAR, 850
CFE_MISSION_TIME_FS_FACTOR, 850
CFE_MISSION_TIME_HK_TLM_MSG, 851
CFE_MISSION_TIME_MAX_ELAPSED, 851
CFE_MISSION_TIME_MIN_ELAPSED, 851
CFE_MISSION_TIME_SEND_CMD_MSG, 852
CFE_MISSION_TIME_SEND_HK_MSG, 852
CFE_MISSION_TIME_TONE_CMD_MSG, 852
sample_perfids.h, 853
CFE_MISSION_ES_MAIN_PERF_ID, 853
CFE_MISSION_ES_PERF_EXIT_BIT, 853
CFE_MISSION_EVS_MAIN_PERF_ID, 854
CFE_MISSION_SB_MAIN_PERF_ID, 854
CFE_MISSION_SB_MSG_LIM_PERF_ID, 854
CFE_MISSION_SB_PIPE_OFLOW_PERF_ID, 854
CFE_MISSION_TBL_MAIN_PERF_ID, 854
CFE_MISSION_TIME_LOCAL1HZISR_PERF_ID, 855
CFE_MISSION_TIME_LOCAL1HZTASK_PERF_ID, 855
CFE_MISSION_TIME_MAIN_PERF_ID, 855
CFE_MISSION_TIME_SENDMET_PERF_ID, 855
CFE_MISSION_TIME_TONE1HZISR_PERF_ID, 855
CFE_MISSION_TIME_TONE1HZTASK_PERF_ID, 856
Seconds
CFE_TIME_OneHzAdjustmentCmd_Payload, 711
CFE_TIME_SysTime, 719
CFE_TIME_TimeCmd_Payload, 721
Seconds1HzAdj
CFE_TIME_HousekeepingTlm_Payload, 706
SecondsDelay
CFE_TIME_HousekeepingTlm_Payload, 706
SecondsMET
CFE_TIME_HousekeepingTlm_Payload, 707
SecondsSTCF
CFE_TIME_HousekeepingTlm_Payload, 707
SendErrors
CFE_SB_PipeInfoEntry, 633
Sequence
CCSDS_PrimaryHeader, 513
ServerFlyState
CFE_TIME_DiagnosticTlm_Payload, 699
Size
CFE_ES_CDSRegDumpRec, 526

CFE_PSP_MemTable_t, 615
CFE_TBL_Info, 674
CFE_TBL_TblRegPacket_Payload, 686
SpacecraftID
CFE_EVS_PacketID, 599
CFE_FS_Header, 610
SpacecraftId
CFE_PSP_CommandData_t, 613
Spare
CFE_ES_TaskInfo, 573
CFE_EVS_AppNameBitMaskCmd_Payload, 576
CFE_EVS_BitMaskCmd_Payload, 587
CFE_EVS_SetEventFormatCode_Payload, 600
CFE_EVS_SetLogMode_Payload, 602
CFE_SB_PipeDepthStats, 630
CFE_SB_PipeInfoEntry, 633
CFE_SB_RouteCmd_Payload, 636
Spare1
CFE_EVS_HousekeepingTlm_Payload, 592
CFE_EVS_LongEventTlm_Payload, 596
Spare2
CFE_EVS_HousekeepingTlm_Payload, 592
CFE_EVS_LongEventTlm_Payload, 596
Spare2Align
CFE_SB_HousekeepingTlm_Payload, 625
Spare3
CFE_EVS_HousekeepingTlm_Payload, 593
stack_size
OS_task_prop_t, 744
StackSize
CFE_ES_AppInfo, 519
CFE_ES_StartAppCmd_Payload, 567
CFE_ES_TaskInfo, 573
start_time
OS_timer_prop_t, 747
StartAddr
CFE_PSP_MemTable_t, 615
StartAddress
CFE_ES_AppInfo, 520
State
CFE_SB_RoutingFileEntry, 638
StreamId
CCSDS_PrimaryHeader, 513
SubType
CFE_FS_Header, 610
CFE_PSP_CommandData_t, 613
CFE_SB_SingleSubscriptionTlm_Payload, 641
SubscribeErrorCounter
CFE_SB_HousekeepingTlm_Payload, 626
SubscriptionsInUse
CFE_SB_StatsTlm_Payload, 647
Subseconds
CFE_TIME_OneHzAdjustmentCmd_Payload, 711
CFE_TIME_SysTime, 719
Subsecs1HzAdj
CFE_TIME_HousekeepingTlm_Payload, 707
SubsecsDelay
CFE_TIME_HousekeepingTlm_Payload, 707
SubsecsMET
CFE_TIME_HousekeepingTlm_Payload, 708
SubsecsSTCF
CFE_TIME_HousekeepingTlm_Payload, 708
Subsystem
CCSDS_ExtendedHeader, 512
SuccessValCounter
CFE_TBL_HousekeepingTlm_Payload, 671
SysLogBytesUsed
CFE_ES_HousekeepingTlm_Payload, 543
SysLogEntries
CFE_ES_HousekeepingTlm_Payload, 543
SysLogMode
CFE_ES_HousekeepingTlm_Payload, 543
SysLogSize
CFE_ES_HousekeepingTlm_Payload, 544
SystemId
CCSDS_ExtendedHeader, 512

Table
CFE_ES_CDSRegDumpRec, 526
TableLoadedOnce
CFE_TBL_Info, 674
CFE_TBL_TblRegPacket_Payload, 687
TableName
CFE_TBL_AbortLoadCmd_Payload, 653
CFE_TBL_ActivateCmd_Payload, 655
CFE_TBL_DelCDSCmd_Payload, 655
CFE_TBL_DumpCmd_Payload, 659
CFE_TBL_File_Hdr, 662
CFE_TBL_FileDef, 663
CFE_TBL_SendRegistryCmd_Payload, 681
CFE_TBL_ValidateCmd_Payload, 689
Target_PspConfigData, 748
HW_NumEepromBanks, 748
PSP_ExceptionLogSize, 748
PSP_MemTableSize, 749
PSP_MemoryTable, 749
PSP_VersionInfo, 749
PSP_WatchdogMax, 749
PSP_WatchdogMin, 749
TaskId
CFE_ES_TaskInfo, 573
TaskName
CFE_ES_TaskInfo, 573
TgtFilename
CFE_TBL_FileDef, 664
ticks
OS_time_t, 745
TimeOfLastUpdate

CFE_TBL_Info, 674
CFE_TBL_TblRegPacket_Payload, 687
TimeSeconds
CFE_FS_Header, 611
TimeSinceTone
CFE_TIME_DiagnosticTlm_Payload, 699
TimeSource
CFE_TIME_SourceCmd_Payload, 717
TimeSubSeconds
CFE_FS_Header, 611
TimerCounter
cfe_psp_start.c, 1348
TlmHeader
CFE_ES_HousekeepingTlm, 532
CFE_ES_MemStatsTlm, 547
CFE_ES_OneAppTlm, 549
CFE_EVS_HousekeepingTlm, 588
CFE_EVS_LongEventTlm, 595
CFE_EVS_ShortEventTlm, 604
CFE_TBL_HousekeepingTlm, 665
CFE_TBL_TableRegistryTlm, 682
CFE_TIME_DiagnosticTlm, 690
CFE_TIME_HousekeepingTlm, 704
ToneDataCounter
CFE_TIME_DiagnosticTlm_Payload, 700
ToneDataLatch
CFE_TIME_DiagnosticTlm_Payload, 700
ToneIntCounter
CFE_TIME_DiagnosticTlm_Payload, 700
ToneIntErrorCounter
CFE_TIME_DiagnosticTlm_Payload, 700
ToneMatchCounter
CFE_TIME_DiagnosticTlm_Payload, 701
ToneMatchErrorCounter
CFE_TIME_DiagnosticTlm_Payload, 701
ToneOverLimit
CFE_TIME_DiagnosticTlm_Payload, 701
ToneSignalCounter
CFE_TIME_DiagnosticTlm_Payload, 701
ToneSignalLatch
CFE_TIME_DiagnosticTlm_Payload, 702
ToneSource
CFE_TIME_SignalCmd_Payload, 716
ToneTaskCounter
CFE_TIME_DiagnosticTlm_Payload, 702
ToneUnderLimit
CFE_TIME_DiagnosticTlm_Payload, 702
total_blocks
OS_statvfs_t, 742
TotalSegments
CFE_SB_AllSubscriptionsTlm_Payload, 620
TriggerMask
CFE_ES_SetPerfTrigMaskCmd_Payload, 564
TriggerMaskNum
CFE_ES_SetPerfTrigMaskCmd_Payload, 564
TriggerMode
CFE_ES_StartPerfCmd_Payload, 568
Type
CFE_ES_AppInfo, 520

uint16
common_types.h, 1257
uint32
common_types.h, 1257
uint64
common_types.h, 1257
uint8
common_types.h, 1258
UnmarkedMem
CFE_SB_HousekeepingTlm_Payload, 626
UnregisteredAppCounter
CFE_EVS_HousekeepingTlm_Payload, 593
User
OS_file_prop_t, 728
UserDefAddr
CFE_TBL_Info, 674
UserShmId
cfe_psp_memory.c, 1339

valid
OS_module_address_t, 734
ValidationCounter
CFE_TBL_HousekeepingTlm_Payload, 671
ValidationFuncPtr
CFE_TBL_TblRegPacket_Payload, 687
value
OS_bin_sem_prop_t, 724
OS_count_sem_prop_t, 725
VersionCodeName
CFE_PSP_VersionInfo_t, 617
VersionCounter
CFE_TIME_DiagnosticTlm_Payload, 702
VersionString
CFE_PSP_VersionInfo_t, 617
VirtualMET
CFE_TIME_DiagnosticTlm_Payload, 703

WordSize
CFE_PSP_MemTable_t, 616